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U.S. Department of Veterans Affairs

US Army Corps of Engineers ®
New England District

West Haven VA Medical Center

Chilled Water Plant Replacement

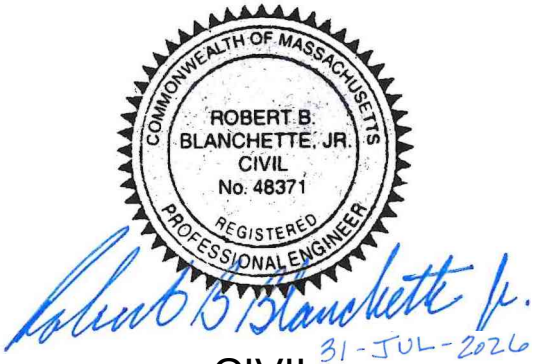
and Site Utility Improvements – Phase A2

Surgical and Clinical Tower

July 31, 2026

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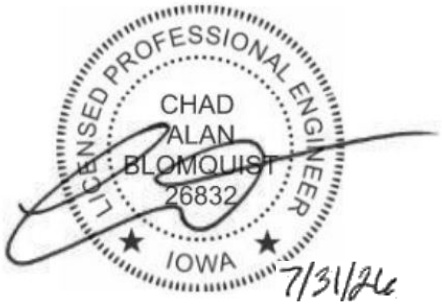
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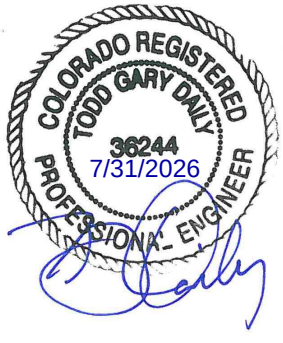
MECHANICAL &
PLUMBING



ELECTRICAL



TELECOMMUNICATIONS



FIRE PROTECTION

PROJECT TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 32 00	08/02	GEOTECHNICAL DATA

DIVISION 01 - GENERAL REQUIREMENTS

01 00 00	11/21	GENERAL REQUIREMENTS
01 11 00	08/15, CHG 2: 08/21	SUMMARY OF WORK
01 14 00		WORK RESTRICTIONS
01 22 00	04/94	MEASUREMENT AND PAYMENT
01 30 00	11/20, CHG 1: 08/21	ADMINISTRATIVE REQUIREMENTS
01 32 01.00 10	08/23, CHG 1: 08/24	PROJECT SCHEDULE
01 33 00	08/18, CHG 4: 02/21	SUBMITTAL PROCEDURES
01 33 21	02/20	CONSTRUCTION PHASE BIM, CIM, AND CAD REQUIREMENTS
01 35 26	05/24	GOVERNMENTAL SAFETY REQUIREMENTS
01 35 50	05/24	CONSTRUCTION INDOOR AIR QUALITY MANAGEMENT
01 42 00	02/19	SOURCES FOR REFERENCE PUBLICATIONS
01 43 30	01/25	MOCKUPS
01 45 00	08/23, CHG 1: 05/24	QUALITY CONTROL
01 45 00.15 10	11/16, CHG 2: 08/19	RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
01 45 29		TESTING LABORATORY SERVICES
01 45 35	11/20	SPECIAL INSPECTIONS
01 45 41		EXTERIOR ENCLOSURE PERFORMANCE REQUIREMENTS, MOCKUPS, AND TESTING
01 50 00	11/20, CHG 1: 08/21	TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
01 57 19		TEMPORARY ENVIRONMENTAL CONTROLS
01 58 00	08/19, CHG 3: 11/21	PROJECT IDENTIFICATION
01 58 16		TEMPORARY INTERIOR SIGNAGE
01 74 19	02/19, CHG 3: 11/21	CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
01 78 00	05/19, CHG 1: 08/21	CLOSEOUT SUBMITTALS
01 78 23	05/23	OPERATION AND MAINTENANCE DATA
01 81 13	10/17	SUSTAINABLE CONSTRUCTION REQUIREMENTS
01 81 13.02	4/23	SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED v4 BD+C
01 86 25		CYBERSECURITY FOR FACILITY RELATED CONTROL SYSTEMS
01 91 00.01	05/19, CHG 2: 08/20	DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS

DIVISION 02 - EXISTING CONDITIONS

02 41 00	08/22	DEMOLITION
02 82 11	1/21	TRADITIONAL ASBESTOS ABATEMENT
02 82 13.13	1/21	GLOVEBAG ASBESTOS ABATEMENT
02 83 33.13	1/21	LEAD-BASED PAINT REMOVAL AND DISPOSAL

DIVISION 03 - CONCRETE

03 30 00	1/21	CAST-IN-PLACE CONCRETE

DIVISION 04 - MASONRY

04 01 00	1/21	MAINTENANCE OF MASONRY
04 05 13	2/17	MASONRY MORTARING
04 20 00	8/17	UNIT MASONRY

DIVISION 05 - METALS

05 12 00	11/18	STRUCTURAL STEEL FRAMING
05 31 00	1/21	STEEL DECKING
05 36 00	1/21	COMPOSITE METAL DECKING
05 40 00	1/21	COLD-FORMED METAL FRAMING
05 50 00	8/18	METAL FABRICATIONS
05 51 00	1/21	METAL STAIRS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

06 10 00	10/17	ROUGH CARPENTRY
06 16 63	1/21	CEMENTITIOUS SHEATHING

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

07 01 50.19	1/21	PREPARATION FOR RE-ROOFING
07 08 00.01	8/19	DVA/USACE PROJECTS FACILITY EXTERIOR CLOSURE COMMISSIONING
07 13 52	1/21	MODIFIED BITUMINOUS SHEET WATERPROOFING
07 18 13	1/21	PEDESTRIAN TRAFFIC COATINGS
07 21 13	1/21	THERMAL INSULATION
07 21 23	1/21	LOOSE-FILL INSULATION
07 22 00	1/21	ROOF AND DECK INSULATION
07 27 27	1/21	FLUID-APPLIED MEMBRANE AIR BARRIERS, VAPOR RETARDING
07 42 13	08/25	METAL WALL PANELS
07 42 63	08/25	FABRICATED WALL PANEL ASSEMBLIES
07 54 23	1/21	THERMOPLASTIC POLYOLEFIN (TPO) ROOFING
07 60 00	1/21	FLASHING AND SHEET METAL
07 71 00	1/21	ROOF SPECIALTIES
07 72 00	1/21	ROOF ACCESSORIES
07 81 00	7/23	APPLIED FIREPROOFING
07 84 00	1/21	FIRESTOPPING
07 92 00	4/22	JOINT SEALANTS
07 95 13	1/21	EXPANSION JOINT COVER ASSEMBLIES

DIVISION 08 - OPENINGS

08 11 13	1/21	HOLLOW METAL DOORS AND FRAMES
08 31 13	06/25	ACCESS DOORS AND FRAMES
08 33 00	1/21	COILING DOORS
08 41 13	1/21	ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS
08 44 13	4/20	GLAZED ALUMINUM CURTAIN WALLS
08 71 00	3/24	DOOR HARDWARE
08 80 00	1/21	GLAZING
08 90 00	1/21	LOUVERS AND VENTS

DIVISION 09 - FINISHES

09 05 16	1/21	SUBSURFACE PREPARATION FOR FLOOR FINISHES
09 22 16	6/18	NON-STRUCTURAL METAL FRAMING
09 29 00	4/20	GYPSUM BOARD

09 30 13	1/21	CERAMIC/PORCELAIN TILING
09 51 00	12/18	ACOUSTICAL CEILINGS
09 62 38	08/17, CHG 1: 08/18	STATIC-CONTROL FLOORING
09 65 13	1/21	RESILIENT BASE AND ACCESSORIES
09 65 19	5/18	RESILIENT TILE FLOORING
09 67 23.20	12/21	RESINOUS (EPOXY BASE) WITH VINYL CHIP BROADCAST (RES-2)
09 91 00		PAINTING

DIVISION 10 - SPECIALTIES

10 14 00	1/21	SIGNAGE
10 22 13	1/21	WIRE MESH PARTITIONS
10 26 00	1/21	WALL AND DOOR PROTECTION
10 28 00	1/21	TOILET, BATH, AND LAUNDRY ACCESSORIES
10 44 13	8/18	FIRE EXTINGUISHER CABINETS

DIVISION 12 - FURNISHINGS

12 24 00	8/17	WINDOW SHADES
12 36 00	11/23	WINDOW STOOLS

DIVISION 14 - CONVEYING EQUIPMENT

14 21 00	4/24	TRACTION ELEVATOR - NEW

DIVISION 21 - FIRE SUPPRESSION

21 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF FIRE SUPPRESSION SYSTEMS
21 13 13	6/15	WET-PIPE SPRINKLER SYSTEMS
21 30 13	5/15	ELECTRIC-DRIVEN, FIRE PUMPS

DIVISION 22 - PLUMBING

22 05 11	1/23	COMMON WORK RESULTS FOR PLUMBING
22 05 12	9/20	GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT
22 05 19	9/20	METERS AND GAUGES FOR PLUMBING PIPING
22 05 23	11/22	GENERAL-DUTY VALVES FOR PLUMBING PIPING
22 05 33	9/20	HEAT TRACING FOR PLUMBING PIPING
22 07 11	9/15	PLUMBING INSULATION
22 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS
22 11 00	5/21	FACILITY WATER DISTRIBUTION
22 11 23	9/15	DOMESTIC WATER PUMPS
22 12 16	9/15	FACILITY ELEVATED, POTABLE-WATER STORAGE TANKS
22 13 00	9/20	FACILITY SANITARY AND VENT PIPING
22 14 00	9/15	FACILITY STORM DRAINAGE
22 14 29	9/15	SUMP PUMPS
22 14 36	9/15	PACKAGED, SUBMERSIBLE, DRAINAGE PUMP UNITS
22 15 00	9/20	GENERAL SERVICE COMPRESSED-AIR SYSTEMS
22 33 00	9/20	HEAT PUMP DOMESTIC WATER HEATER
22 40 00	5/24	PLUMBING FIXTURES
22 62 00	5/23	VACUUM SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES
22 63 00	5/23	GAS SYSTEMS FOR LABORATORY AND

HEALTHCARE FACILITIES

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

23 05 11	3/23	COMMON WORK RESULTS FOR HVAC
23 05 12	3/23	GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT
23 05 41	3/23	NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT
23 05 93.01	11/21	DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC
23 07 11	2/20	HVAC AND BOILER PLANT INSULATION
23 08 00	05/23, CHG 1: 08/24	COMMISSIONING OF MECHANICAL SYSTEMS
23 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS
23 09 23	3/23	DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC
23 10 00	4/20	FACILITY FUEL OIL SYSTEMS
23 21 11	3/24	BOILER PLANT PIPING SYSTEMS
23 21 13	06/25	HYDRONIC PIPING
23 21 23	3/23	HYDRONIC PUMPS
23 22 13	3/24	STEAM AND CONDENSATE HEATING PIPING
23 22 23	4/20	STEAM CONDENSATE PUMPS
23 25 00	3/23	HVAC WATER TREATMENT
23 31 00	3/23	HVAC DUCTS AND CASINGS
23 34 00	3/23	HVAC FANS
23 37 00	3/23	AIR OUTLETS AND INLETS
23 40 00	3/23	HVAC AIR CLEANING DEVICES
23 50 11	3/22	BOILER PLANT MECHANICAL EQUIPMENT
23 64 00	3/24	PACKAGED WATER CHILLERS
23 65 00	3/23	COOLING TOWERS
23 73 00	3/23	INDOOR CENTRAL-STATION AIR-HANDLING UNITS
23 81 00	3/24	DECENTRALIZED UNITARY HVAC EQUIPMENT
23 81 43	3/23	AIR-SOURCE UNITARY HEAT PUMPS
23 82 00	3/23	CONVECTION HEATING AND COOLING UNITS
23 82 16	3/24	AIR COILS

DIVISION 25 - INTEGRATED AUTOMATION

25 05 11.01	08/24	CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING AUTOMATION SYSTEMS (L-M-M)
25 05 11.02	08/24	CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - UTILITY MONITORING AND CONTROL SYSTEMS (L-L-L)
25 05 11.03	08/24	CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - ELECTRONIC SECURITY SYSTEMS (M-M-M)
25 05 11.04	08/24	CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - FIRE AND LIFE SAFETY SYSTEMS (L-L-L)
25 10 10	2/10	ADVANCED UTILITY METERING SYSTEM

DIVISION 26 - ELECTRICAL

26 05 11	11/22	REQUIREMENTS FOR ELECTRICAL INSTALLATIONS
26 05 13	06/25	MEDIUM-VOLTAGE CABLES
26 05 19	11/22	LOW-VOLTAGE ELECTRICAL POWER

		CONDUCTORS AND CABLES
26 05 26	11/22	GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS
26 05 33	11/22	RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS
26 05 41	06/25	UNDERGROUND ELECTRICAL CONSTRUCTION
26 05 73	11/22	OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY
26 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF ELECTRICAL SYSTEMS
26 09 23	11/22	LIGHTING CONTROLS
26 11 16	11/22	SECONDARY UNIT SUBSTATIONS
26 12 19	11/22	PAD-MOUNTED, MEDIUM-VOLTAGE TRANSFORMERS
26 13 01	05/24	PAD-MOUNTED DEAD-FRONT AIR INSULATED SWITCHGEAR
26 13 13	06/25	MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR
26 22 00	11/22	LOW-VOLTAGE TRANSFORMERS
26 23 00	11/22	LOW-VOLTAGE SWITCHGEAR
26 24 13	11/22	DISTRIBUTION SWITCHBOARDS
26 24 16	11/22	PANELBOARDS
26 27 26	11/22	WIRING DEVICES
26 29 11	11/22	MOTOR CONTROLLERS
26 29 21	11/22	ENCLOSED SWITCHES AND CIRCUIT BREAKERS
26 33 53	11/22	STATIC UNINTERRUPTIBLE POWER SUPPLY
26 36 23	11/22	AUTOMATIC TRANSFER SWITCHES
26 41 00	11/22	FACILITY LIGHTNING PROTECTION
26 43 13	11/22	SURGE PROTECTIVE DEVICES
26 51 00	3/24	INTERIOR LIGHTING
26 56 00	11/22	EXTERIOR LIGHTING

DIVISION 27 - COMMUNICATIONS

27 05 11	9/19	REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS
27 05 26	11/21	GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS
27 05 33	10/18	CONDUITS AND BACKBOXES FOR COMMUNICATIONS SYSTEMS
27 05 36	10/18	CABLE TRAYS FOR COMMUNICATIONS SYSTEMS
27 05 53	7/21	IDENTIFICATION FOR COMMUNICATIONS SYSTEMS
27 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF COMMUNICATIONS SYSTEMS
27 10 00	6/15	STRUCTURED CABLING
27 11 00	6/15	TELECOMMUNICATIONS ROOM FITTINGS
27 11 16		COMMUNICATIONS CABINETS, RACKS, FRAMES, AND ENCLOSURES
27 11 19		COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS
27 13 13		COMMUNICATIONS COPPER BACKBONE CABLING
27 13 13.13		COMMUNICATIONS COPPER CABLE SPLICING AND TERMINATIONS
27 13 23		COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING
27 13 23.13		COMMUNICATIONS OPTICAL FIBER SPLICING AND TERMINATIONS
27 15 00	1/16	COMMUNICATIONS STRUCTURED CABLING

27 15 13		COMMUNICATIONS COPPER HORIZONTAL CABLING
27 16 19		COMMUNICATIONS PATCH CORDS, STATION CORDS, AND CROSS CONNECT WIRE
27 51 16	10/18	PUBLIC ADDRESS
27 78 00	5/24	CLOSEOUT SUBMITTALS FOR COMMUNICATIONS

DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

28 05 00	4/18	COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY
28 05 13	10/18	CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY
28 05 26	9/11	GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY
28 05 28.33	9/11	CONDUITS AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY
28 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS
28 13 00	10/11	PHYSICAL ACCESS CONTROL SYSTEM
28 23 00	9/11	VIDEO SURVEILLANCE
28 31 00	10/11	FIRE DETECTION AND ALARM

DIVISION 31 - EARTHWORK

31 20 00	08/23	EARTHWORK
31 23 19	10/12	DEWATERING
31 23 23.33	10/12	FLOWABLE FILL

DIVISION 32 - EXTERIOR IMPROVEMENTS

32 05 23	8/16	CEMENT AND CONCRETE FOR EXTERIOR IMPROVEMENTS
32 12 16	9/15	ASPHALT PAVING
32 13 15.20	11/22	CONCRETE PAVEMENT FOR FUEL STORAGE CONTAINMENT DIKES
32 17 23	8/16	PAVEMENT MARKINGS
32 31 13	11/21	CHAIN LINK FENCES AND GATES
32 31 40	11/21	HIGH-SECURITY FENCES AND GATES

DIVISION 33 - UTILITIES

33 08 00.01	8/19	DVA/USACE PROJECTS COMMISSIONING OF SITE UTILITY SYSTEMS
33 10 00	3/17	WATER UTILITIES
33 30 00	05/18	SANITARY SEWERAGE
33 40 00	11/21	STORMWATER UTILITIES
33 46 13	05/18	FOUNDATION DRAINAGE
33 56 19	11/22	FUEL IMPERMEABLE LINER SYSTEM
33 63 00	2/24	STEAM ENERGY DISTRIBUTION
33 71 02	08/21	UNDERGROUND ELECTRICAL DISTRIBUTION

-- End of Project Table of Contents --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DOCUMENT 00 32 00

GEOTECHNICAL DATA

08/02

PART 1 GENERAL

1.1 SUMMARY
1.2 REFERENCES
1.3 EXAMINATION OF SAMPLES
1.4 SOIL CLASSIFICATION
1.5 INTERPRETATION

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION (Not Applicable)

-- End of Section Table of Contents --

DOCUMENT 00 32 00

GEOTECHNICAL DATA
08/02

PART 1 GENERAL

1.1 SUMMARY

The surface conditions indicated on the contract drawings and in the specifications are the result of site surveys and test pits. Copies of the logs of test pit explorations and locations at the site where subsurface investigations were performed are attached to the appendix binder and shown on the contract drawings.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D 2488	(2017) Description and Identification of Soils (Visual-Manual Procedure)

1.3 EXAMINATION OF SAMPLES

Subsurface explorations were performed by the U.S. Army Corps of Engineers. Pertinent subsurface information is attached to this section of the specifications.

1.4 SOIL CLASSIFICATION

The soil classification of the USACE soil samples was performed in accordance with ASTM D 2488 in the field.

1.5 INTERPRETATION

Subsurface investigation data are provided only for information purposes and for the convenience of the Contractor. The data shown on the boring logs is for the specific locations indicated only and no assurance is given that these conditions are representative of conditions between borings or areas adjacent thereto. The responsibility lies with the Contractor to interpret subsurface conditions that may affect his work.

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION (Not Applicable)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 00 00

GENERAL REQUIREMENTS

11/21

PART 1 GENERAL

1.1 REFERENCES
1.2 SAFETY REQUIREMENTS
1.3 GENERAL REQUIREMENTS
1.4 SUBMITTALS
1.5 STATEMENT OF BID ITEM(S)
1.6 OPERATIONS AND STORAGE AREAS (FAR 52.236-10)
1.7 ALTERATIONS
1.8 DISPOSAL AND RETENTION
1.9 PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS (FAR 52.236-9)
1.10 RESTORATION
1.11 PROFESSIONAL SURVEYING SERVICES
1.12 LAYOUT OF WORK
1.13 USE OF ROADWAYS
1.14 TEMPORARY USE OF MECHANICAL AND ELECTRICAL EQUIPMENT
1.15 TEMPORARY USE OF EXISTING ELEVATORS
1.16 TESTS
1.17 PHOTOGRAPHIC DOCUMENTATION
1.18 FINAL ELEVATION DIGITAL IMAGES
1.19 WEB-BASED PHOTOGRAPHIC DOCUMENTATION
1.20 HISTORIC PRESERVATION

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 00 00
11/21

GENERAL REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.2 SAFETY REQUIREMENTS

Refer to Section 01 35 26, SAFETY REQUIREMENTS for safety and infection control requirements.

1.3 GENERAL REQUIREMENTS

a. Contractor shall completely prepare site for building operations, including demolition and removal of existing structures, and furnish labor and materials and perform work for the Chilled Water Plant Replacement and Site Utility Improvements, Phase A2 Surgical and Clinical Tower project as required by drawings and specifications.

b. Offices of Leo A. Daly | Perkins & Will Joint Venture, as Architect-Engineers, will render certain technical services during construction. Such services shall be considered as advisory to the Government and shall not be construed as expressing or implying a contractual act of the Government without affirmations by Contracting Officer or his duly authorized representative.

c. All employees of general contractor and subcontractors shall comply with VA security management program and obtain permission of the VA police, be identified by project and employer, and restricted from unauthorized access.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Schedule Of Approximate Phasing Dates; G

Survey; G

Re-Survey; G

Professional Surveying Services; G

Temporary Use Of Mechanical And Electrical Equipment; G

Photographic Documentation; G

Commercial Photographer; G

Detailed Construction Schedule; G

Web-Based Photographic Documentation Qualifications; G

Work Plan, Phasing And Subphasing Plans; G

SD-02 Shop Drawings

Web-Based Photographic Documentation; G

SD-06 Test Reports

Written Testing And Commissioning Plan; G

Test Results; G

SD-11 Closeout Submittals

Final Elevation Digital Images; G

Web-Based Photographic Documentation Imagery; G

1.5 STATEMENT OF BID ITEM(S)

a. Refer to Section 01 22 00, MEASUREMENT AND PAYMENT for bidding requirements.

1.6 OPERATIONS AND STORAGE AREAS (FAR 52.236-10)

a. The Contractor shall confine all operations (including storage of materials) on Government premises to areas authorized or approved by the Contracting Officer. The Contractor shall hold and save the Government, its officers and agents, free and harmless from liability of any nature occasioned by the Contractor's performance.

b. Temporary buildings (e.g., storage sheds, shops, offices) and utilities may be erected by the Contractor only with the approval of the Contracting Officer and shall be built with labor and materials furnished by the Contractor without expense to the Government. The temporary buildings and utilities shall remain the property of the Contractor and shall be removed by the Contractor at its expense upon completion of the work.

c. The Contractor shall, under regulations prescribed by the Contracting

Officer, use only established roadways, or use temporary roadways constructed by the Contractor when and as authorized by the Contracting Officer. When materials are transported in prosecuting the work, vehicles shall not be loaded beyond the loading capacity recommended by the manufacturer of the vehicle or prescribed by any Federal, State, or local law or regulation. When it is necessary to cross curbs or sidewalks, the Contractor shall protect them from damage. The Contractor shall repair or pay for the repair of any damaged curbs, sidewalks, or roads.

d. Working space and space available for storing materials shall be as shown on the drawings.

e. Workers are subject to rules of Medical Center applicable to their conduct.

f. Execute work in such a manner as to interfere as little as possible with work being done by others. Keep roads clear of construction materials, debris, standing construction equipment and vehicles at all times. Execute all work such that road closures are limited to one lane at a time. Lane closure sequencing and durations shall be clearly identified in the Traffic Control Plan and Construction Phasing Plan submitted for review and approval.

g. Execute work so as to interfere as little as possible with normal functioning of Medical Center as a whole, including operations of utility services, fire protection systems and any existing equipment, and with work being done by others. Use of equipment and tools that transmit vibrations and noises through the building structure, are not permitted in buildings that are occupied, during construction, jointly by patients or medical personnel, and Contractor's personnel, except as permitted by Contracting Officer where required by limited working space.

(1) Do not store materials and equipment in other than assigned areas.

(2) Schedule delivery of materials and equipment to immediate construction working areas or assigned storage areas between the hours of 7pm to 6am. and limit deliveries so they do not impede Medical Center traffic or ongoing operations. Provide unobstructed access to Medical Center areas required to remain in operation.

(3) Where access by Medical Center personnel to vacated portions of buildings is not required, storage of Contractor's materials and equipment will be permitted subject to fire and safety requirements and Contracting Officer approval.

h. Phasing:

(1) The Medical Center must maintain its operation 24 hours a day 7 days a week. Therefore, any interruption in service must be scheduled and coordinated with the Contracting Officer, as identified in Section 01 14 00 WORK RESTRICTIONS and on the phasing drawings, to ensure that no lapses in operation occur. It is the CONTRACTOR'S responsibility to develop a work plan, phasing and subphasing plans and detailed construction schedule, at a minimum, the procedures to be employed, the equipment and materials to be used, the interim life safety measure to be used

during the work, and a schedule defining the duration of the work with milestone subtasks. The work to be outlined shall include, but not be limited to: Access and egress to and from existing buildings, fire truck access to all fire hydrants and buildings, utility shutdowns, off-hours work, sidewalk closures or re-routes/detours and road closures.

(2) To ensure such executions, Contractor shall furnish the Contracting Officer with a schedule of approximate phasing dates on which the Contractor intends to accomplish work in each specific area of site, building or portion thereof. In addition, Contractor shall notify the Contracting Officer 30 days in advance of the proposed date of starting work in each specific area of site, building or portion thereof. Arrange such phasing dates to ensure accomplishment of this work in successive phases mutually agreeable to Medical Center Director, Contracting Officer and Contractor, as indicated on the phasing plans.

i. When a building and/or construction site is turned over to Contractor, Contractor shall accept entire responsibility including upkeep and maintenance therefore:

(1) Contractor shall maintain a minimum temperature of (40 degrees F) at all times, except as otherwise specified.

(2) Contractor shall maintain in operating condition existing fire protection and alarm equipment. In connection with fire alarm equipment, Contractor shall make arrangements for pre-inspection of site with Fire Department or Company (Department of Veterans Affairs or municipal) whichever will be required to respond to an alarm from Contractor's employee or watchman.

j. Utilities Services: Maintain existing utility services for Medical Center at all times. Provide temporary facilities, labor, materials, equipment, connections, and utilities to assure uninterrupted services. Where necessary to cut existing water, steam, gases, sewer or air pipes, or conduits, wires, cables, etc. of utility services or of fire protection systems and communications systems (including telephone), they shall be cut and capped at suitable places where shown; back to the nearest main or, in absence of such indication, where directed by Contracting Officer.

k. To minimize interference of construction activities with flow of Medical Center traffic, comply with the following:

(1) Keep roads, walks and entrances to grounds, to parking and to occupied areas of buildings clear of construction materials, debris and standing construction equipment and vehicles. Wherever excavation for new utility lines cross existing roads, at least one lane must be open to traffic at all times with approval.

(2) Method and scheduling of required cutting, altering and removal of existing roads, walks and entrances must be approved by the Contracting Officer a minimum of 30 days in advance. Refer to Civil drawings for additional information.

l. Coordinate the work for this contract with other construction operations as directed by Contracting Officer. This includes the

scheduling of traffic and the use of roadways, as specified in Article, USE OF ROADWAYS.

(1) Contractor shall schedule, organize, and lead weekly coordination meetings with other contractors performing work on the medical campus.

(2) List of other Projects and Projected Construction Schedule:

(a) Auxiliary Chiller Plant - July 20, 2026 to April 21, 2027

(b) Boiler Plant - March 30, 2027 to June 15, 2029

(c) Electronic Health Record Modernization (EHRM) - June 25, 2027 to February 25, 2028

(d) Steam Upgrade Phase 3 - August 26, 2026 to August 30, 2028

(e) Steam Upgrade for Buildings 16, 21, 22, 24, 27, 34 - September 24, 2025 to March 31, 2027

(f) SPS Relocation - November 1, 2025 to October 1, 2027

(g) Pharmacy Relocation - May 21, 2025 to March 31, 2027

(h) Fire Protection Upgrades - September 30, 2025 to September 30, 2027

(i) Bulk Oxygen - March 27, 2025 to August 21, 2026

(j) OR Renovation - July 31, 2024 to August 1, 2027

(k) Phase 1A Non-Permanent Structures - August 15, 2025 to September 12, 2027

1.7 ALTERATIONS

a. Survey: Before any work is started, the Contractor shall make a thorough survey with the Contracting Officer and a representative of VA Facilities Management Service, of areas of buildings and site in which alterations occur and areas which are anticipated routes of access, and furnish a report, signed by all three (3), to the Contracting Officer. This report shall list site locations and by rooms and spaces of buildings:

(1) Existing condition and types of resilient flooring, doors, windows, walls and other surfaces not required to be altered throughout affected areas of buildings.

(2) Existence and conditions of items such as plumbing fixtures and accessories, electrical fixtures, equipment, venetian blinds, shades, etc., required by drawings to be either reused or relocated, or both.

(3) Existing Conditions of site, including roadways, sidewalks, lawns, trees, fencing, etc., as required.

(4) Shall note any discrepancies between drawings and existing

conditions at site.

(5) Shall designate areas for working space, materials storage and routes of access to areas within buildings where alterations occur and which have been agreed upon by Contractor and Contracting Officer.

b. Any items required by drawings to be either reused or relocated or both, found during this survey to be nonexistent, or in opinion of Contracting Officer , to be in such condition that their use is impossible or impractical, shall be furnished and/or replaced by Contractor with new items in accordance with specifications which will be furnished by Government.

c. Re-Survey: Thirty days before expected partial or final inspection date, the Contractor and Contracting Officer together shall make a thorough re-survey of the areas of buildings involved. They shall furnish a report on conditions then existing, of resilient flooring, doors, windows, walls and other surfaces as compared with conditions of same as noted in first condition survey report:

(1) Re-survey report shall also list any damage caused by Contractor to such flooring and other surfaces, despite protection measures; and, will form basis for determining extent of repair work required of Contractor to restore damage caused by Contractor's workers in executing work of this contract.

d. Protection: Provide the following protective measures:

(1) Wherever existing roof surfaces are disturbed they shall be protected against water infiltration. In case of leaks, they shall be repaired immediately upon discovery.

(2) Temporary protection against damage for portions of existing structures and grounds where work is to be done, materials handled and equipment moved and/or relocated.

(3) Protection of interior of existing structures at all times, from damage, dust and weather inclemency. Wherever work is performed, floor surfaces that are to remain in place shall be adequately protected prior to starting work, and this protection shall be maintained intact until all work in the area is completed.

1.8 DISPOSAL AND RETENTION

a. Materials and equipment accruing from work removed and from demolition of buildings or structures, or parts thereof, shall be disposed of as follows:

(1) Reserved items which are to remain property of the Government are identified by attached tags or noted on drawings or in specifications as items to be stored. Items that remain property of the Government shall be removed or dislodged from present locations in such a manner as to prevent damage which would be detrimental to re-installation and reuse. Store such items where directed by Contracting Officer.

(2) Items not reserved shall become property of the Contractor and be removed by Contractor from Medical Center .

(3) Items of portable equipment and furnishings located in rooms and spaces in which work is to be done under this contract shall remain the property of the Government. When rooms and spaces are vacated by the Department of Veterans Affairs during the alteration period, such items which are NOT required by drawings and specifications to be either relocated or reused will be removed by the Government in advance of work to avoid interfering with Contractor's operation.

1.9 PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS (FAR 52.236-9)

a. The Contractor shall preserve and protect all structures, equipment, and vegetation (such as trees, shrubs, and grass) on or adjacent to the work site, which are not to be removed and which do not unreasonably interfere with the work required under this contract. The Contractor shall only remove trees when specifically authorized to do so, and shall avoid damaging vegetation that will remain in place. If any limbs or branches of trees are broken during contract performance, or by the careless operation of equipment, or by workers, the Contractor shall trim those limbs or branches with a clean cut and paint the cut with a tree-pruning compound as directed by the Contracting Officer.

b. The Contractor shall protect from damage all existing improvements and utilities at or near the work site and on adjacent property of a third party, the locations of which are made known to or should be known by the Contractor. The Contractor shall repair any damage to those facilities, including those that are the property of a third party, resulting from failure to comply with the requirements of this contract or failure to exercise reasonable care in performing the work. If the Contractor fails or refuses to repair the damage promptly, the Contracting Officer may have the necessary work performed and charge the cost to the Contractor.

c. Refer to Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS, for additional requirements on protecting vegetation, soils and the environment. Refer to Articles, "Alterations", "Restoration", and "Operations and Storage Areas" for additional instructions concerning repair of damage to structures and site improvements.

d. A National Pollutant Discharge Elimination System (NPDES) permit is required for this project. The Contractor is considered an "operator" under the permit and has extensive responsibility for compliance with permit requirements. The contractor shall furnish all information and certifications that are required to comply with the permit process and permit requirements. Many of the permit requirements will be satisfied by completing construction as shown and specified. Some requirements involve the Contractor's method of operations and operations planning and the Contractor is responsible for employing best management practices. The affected activities often include, but are not limited to the following:

(1) Designating areas for equipment maintenance and repair;

(2) Providing waste receptacles at convenient locations and provide regular collection of wastes;

(3) Locating equipment wash down areas on site, and provide appropriate control of wash-waters;

(4) Providing protected storage areas for chemicals, paints, solvents, fertilizers, and other potentially toxic materials; and

(5) Providing adequately maintained sanitary facilities.

1.10 RESTORATION

a. Remove, cut, alter, replace, patch and repair existing work as necessary to install new work. Except as otherwise shown or specified, do not cut, alter or remove any structural work, and do not disturb any ducts, plumbing, steam, gas, or electric work without approval of the Contracting Officer. Existing work to be altered or extended and that is found to be defective in any way, shall be reported to the Contracting Officer before it is disturbed. Materials and workmanship used in restoring work, shall conform in type and quality to that of original existing construction, except as otherwise shown or specified.

b. Upon completion of contract, deliver work complete and undamaged. Existing work (walls, ceilings, partitions, floors, mechanical and electrical work, lawns, paving, roads, walks, etc.) disturbed or removed as a result of performing required new work, shall be patched, repaired, reinstalled, or replaced with new work, and refinished and left in as good condition as existed before commencing work.

c. At Contractor's own expense, Contractor shall immediately restore to service and repair any damage caused by Contractor's workers to existing piping and conduits, wires, cables, etc., of utility services or of fire protection systems and communications systems (including telephone) which are not scheduled for discontinuance or abandonment.

1.11 PROFESSIONAL SURVEYING SERVICES

A registered professional land surveyor or registered civil engineer whose services are retained and paid for by the Contractor shall perform services specified herein and in other specification sections. The Contractor shall certify that the land surveyor or civil engineer is not one who is a regular employee of the Contractor, and that the land surveyor or civil engineer has no financial interest in this contract.

1.12 LAYOUT OF WORK

a. The Contractor shall lay out the work from Government established base lines and bench marks, indicated on the drawings, and shall be responsible for all measurements in connection with the layout. The Contractor shall furnish, at Contractor's own expense, all stakes, templates, platforms, equipment, tools, materials, and labor required to lay out any part of the work. The Contractor shall be responsible for executing the work to the lines and grades that may be established or indicated by the Contracting Officer. The Contractor shall also be responsible for maintaining and preserving all stakes and other marks established by the Contracting Officer until authorized to remove them. If such marks are destroyed by the Contractor or through Contractor's negligence before their removal is authorized, the Contracting Officer may replace them and deduct the expense of the

replacement from any amounts due or to become due to the Contractor.

b. Establish and plainly mark center lines for each building and corner of column lines and/or addition to each existing building, lines for each gravesite control monument, and such other lines and grades that are reasonably necessary to properly assure that location, orientation, and elevations established for each such structure and/or addition, roads, parking lots, are in accordance with lines and elevations shown on contract drawings.

c. Following completion of general mass excavation and before any other permanent work is performed, establish and plainly mark (through use of appropriate batter boards or other means) sufficient additional survey control points or system of points as may be necessary to assure proper alignment, orientation, and grade of all major features of work. Survey shall include, but not be limited to, location of lines and grades of footings, exterior walls, center lines of columns in both directions, major utilities and elevations of floor slabs:

   (1) Such additional survey control points or system of points thus established shall be checked and certified by a registered land surveyor or registered civil engineer. Furnish such certification to the Contracting Officer before any work (such as footings, floor slabs, columns, walls, utilities and other major controlling features) is placed.

d. During progress of work, and particularly as work progresses from floor to floor, Contractor shall have line grades and plumbness of all major form work checked and certified by a registered land surveyor or registered civil engineer as meeting requirements of contract drawings. Furnish such certification to the Contracting Officer before any major items of concrete work are placed. In addition, Contractor shall also furnish to the Contracting Officer certificates from a registered land surveyor or registered civil engineer that the following work is complete in every respect as required by contract drawings.

   (1) Lines of each building and/or addition.

   (2) Elevations of bottoms of footings and tops of floors of each building and/or addition.

   (3) Lines and elevations of sewers and of all outside distribution systems.

   (4) Lines of elevations of all swales and interment areas.

   (5) Lines and elevations of roads, streets and parking lots.

e. Whenever changes from contract drawings are made in line or grading requiring certificates, record such changes on a reproducible drawing bearing the registered land surveyor or registered civil engineer seal, and forward these drawings upon completion of work to Contracting Officer.

f. Upon completion of the work, the Contractor shall furnish the Contracting Officer one electronic copy and reproducible drawings at the scale of the contract drawings, showing the finished grade on the

grid developed for constructing the work, including burial monuments and fifty foot stationing along new road centerlines. These drawings shall bear the seal of the registered land surveyor or registered civil engineer.

g. The Contractor shall perform the surveying and layout work of this and other articles and specifications in accordance with the provisions of Article "Professional Surveying Services".

1.13 USE OF ROADWAYS

a. For hauling, use only established public roads and roads on Medical Center property and, when authorized by the Contracting Officer, such temporary roads which are necessary in the performance of contract work. Temporary roads shall be constructed and restoration performed by the Contractor at Contractor's expense. When necessary to cross curbing, sidewalks, or similar construction, they must be protected by well-constructed bridges.

b. When new permanent roads are to be a part of this contract, Contractor may construct them immediately for use to facilitate building operations. These roads may be used by all who have business thereon within zone of building operations.

c. When certain buildings (or parts of certain buildings) are required to be completed in advance of general date of completion, all roads leading thereto must be completed and available for use at time set for completion of such buildings or parts thereof.

1.14 TEMPORARY USE OF MECHANICAL AND ELECTRICAL EQUIPMENT

a. Use of new installed mechanical and electrical equipment to provide heat, ventilation, plumbing, light and power will be permitted subject to written approval and compliance with the following provisions:

(1) Permission to use each unit or system must be given by Contracting Officer in writing. If the equipment is not installed and maintained in accordance with the written agreement and following provisions, the Contracting Officer will withdraw permission for use of the equipment.

(2) Electrical installations used by the equipment shall be completed in accordance with the drawings and specifications to prevent damage to the equipment and the electrical systems, i.e. transformers, relays, circuit breakers, fuses, conductors, motor controllers and their overload elements shall be properly sized, coordinated and adjusted. Installation of temporary electrical equipment or devices shall be in accordance with NFPA 70, National Electrical Code, (2014 Edition), Article 590, Temporary Installations. Voltage supplied to each item of equipment shall be verified to be correct and it shall be determined that motors are not overloaded. The electrical equipment shall be thoroughly cleaned before using it and again immediately before final inspection including vacuum cleaning and wiping clean interior and exterior surfaces.

(3) Units shall be properly lubricated, balanced, and aligned. Vibrations must be eliminated.

(4) Automatic temperature control systems for preheat coils shall function properly and all safety controls shall function to prevent coil freeze-up damage.

(5) The air filtering system utilized shall be that which is designed for the system when complete, and all filter elements shall be replaced at completion of construction and prior to testing and balancing of system.

(6) All components of heat production and distribution system, metering equipment, condensate returns, and other auxiliary facilities used in temporary service shall be cleaned prior to use; maintained to prevent corrosion internally and externally during use; and cleaned, maintained and inspected prior to acceptance by the Government.

b. Prior to final inspection, the equipment or parts used which show wear and tear beyond normal, shall be replaced with identical replacements, at no additional cost to the Government.

c. This paragraph shall not reduce the requirements of the mechanical and electrical specifications sections.

d. Any damage to the equipment or excessive wear due to prolonged use will be repaired replaced by the contractor at the contractor’s expense.

1.15 TEMPORARY USE OF EXISTING ELEVATORS

a. Use of existing elevators for handling building materials and Contractor's personnel will be permitted subject to following provisions:

(1) Contractor makes all arrangements with the Contracting Officer for use of elevators. The Contracting Officer will ascertain that elevators are in proper condition. Contractor may use service elevator in Building No. One for daily use between the hours of 7:00 AM and 4:00 PM. and for special nonrecurring time intervals when permission is granted. Personnel for operating elevators will not be provided by the Department of Veterans Affairs. The use of Building No. 15 elevators is prohibited.

(2) Contractor covers and provides maximum protection of following elevator components:

(a) Entrance jambs, heads soffits and threshold plates.

(b) Entrance columns, canopy, return panels and inside surfaces of car enclosure walls.

(c) Finish flooring.

(3) Government will accept hoisting ropes of elevator and rope of each speed governor if they are worn under normal operation. However, if these ropes are damaged by action of foreign matter such as sand, lime, grit, stones, etc., during temporary use, they shall be removed and replaced by new hoisting ropes at the contractors expense.

(4) Place elevator in condition equal, less normal wear, to that existing at time it was placed in service of Contractor as approved by Contracting Officer.

1.16 TESTS

a. The contractor shall provide a written testing and commissioning plan complete with component level, equipment level, sub-system level and system level breakdowns. The plan will provide a schedule and a written sequence of what will be tested, how and what the expected outcome will be. This document will be submitted for approval prior to commencing work. The contractor shall document the results of the approved plan and submit for approval with the as built documentation.

b. Pre-test mechanical and electrical equipment and systems and make corrections required for proper operation of such systems before requesting final tests. Final test will not be conducted unless pre-tested.

c. Conduct final tests required in various sections of specifications in presence of an authorized representative of the Contracting Officer. Contractor shall furnish all labor, materials, equipment, instruments, and forms, to conduct and record such tests.

d. Mechanical and electrical systems shall be balanced, controlled and coordinated. A system is defined as the entire system which must be coordinated to work together during normal operation to produce results for which the system is designed. For example, air conditioning supply air is only one part of entire system which provides comfort conditions for a building. Other related components are return air, exhaust air, steam, chilled water, refrigerant, hot water, controls and electricity, etc. Another example of a system which involves several components of different disciplines is a boiler installation. Efficient and acceptable boiler operation depends upon the coordination and proper operation of fuel, combustion air, controls, steam, feedwater, condensate and other related components.

e. All related components as defined above shall be functioning when any system component is tested. Tests shall be completed within a reasonably period of time during which operating and environmental conditions remain reasonably constant and are typical of the design conditions.

f. Individual test result of any component, where required, will only be accepted when submitted with the test results of related components and of the entire system.

1.17 PHOTOGRAPHIC DOCUMENTATION

a. During the construction period through completion, provide photographic documentation of construction progress and at selected milestones including electronic indexing, navigation, storage and remote access to the documentation, as per these specifications. The commercial photographer or the subcontractor used for this work shall meet the following qualifications:

(1) Demonstrable minimum experience of three (3) years in operation providing documentation and advanced indexing/navigation systems

including a representative portfolio of construction projects of similar type, size, duration and complexity as the Project.

(2) Demonstrable ability to service projects throughout North America, which shall be demonstrated by a representative portfolio of active projects of similar type, size, duration and complexity as the Project.

b. Photographic documentation elements:

(1) Each digital image shall be taken with a professional grade camera with minimum size of 6 megapixels (MP) capable of producing 8 x 10 inch prints with a minimum of 2272 x 1704 pixels and 16 x 20 inch prints with a minimum 2592 x 1944 pixels.

(2) Indexing and navigation system shall utilize actual AUTOCAD construction drawings, making such drawings interactive on an on-line interface. For all documentation referenced herein, indexing and navigation must be organized by both time (date-stamped) and location throughout the project.

(3) Documentation shall combine indexing and navigation system with inspection-grade digital photography designed to capture actual conditions throughout construction and at critical milestones. Documentation shall be accessible on-line through use of an internet connection. Documentation shall allow for secure multiple-user access, simultaneously, on-line.

(4) Before construction, the building pad, adjacent streets, roadways, parkways, driveways, curbs, sidewalks, landscaping, adjacent utilities and adjacent structures surrounding the building pad and site shall be documented. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings. If site work or pad preparation is extensive, this documentation may be required immediately before construction and at several pre-determined intervals before building work commences.

(5) Construction progress for all trades shall be tracked at pre-determined intervals, but not less than once every thirty (30) calendar days (“Progressions”). Progression documentation shall track both the exterior and interior construction of the building. Exterior Progressions shall track 360 degrees around the site and each building. Interior Progressions shall track interior improvements beginning when stud work commences and continuing until Project completion.

(6) As-built condition of pre-foundation utilities and site utilities shall be documented prior to pouring footers, placing concrete and/or backfilling. This process shall include all underground and in-slab utilities within the building(s) envelope(s) and utility runs in the immediate vicinity of the building(s) envelope(s). This may also include utilities enclosed in slab-on-deck in multi-story buildings. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive site utility plans.

(7) As-built conditions of mechanical, electrical, plumbing and all other systems shall be documented post-inspection and

pre-insulation, sheet rock or dry wall installation. This process shall include all finished systems located in the walls and ceilings of all buildings at the Project. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings.

(8) As-built conditions of exterior skin and elevations shall be documented with an increased concentration of digital photographs as directed by the Contracting Officer in order to capture pre-determined focal points, such as waterproofing, window flashing, radiused steel work, architectural or Exterior Insulation and Finish Systems (EIFS) detailing. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive elevations or elevation details.

(9) As-built finished conditions of the interior of each building including floors, ceilings and walls shall be documented at certificate of occupancy or equivalent, or just prior to occupancy, or both, as directed by the Contracting Officer. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings.

(10) Miscellaneous events that occur during any Contractor site visit shall be dated, labeled and inserted into a Section in the navigation structure entitled “Slideshows,” allowing this information to be stored in the same “place” as the formal scope.

(11) Customizable project-specific digital photographic documentation of other details or milestones. Indexing and navigation accomplished through interactive architectural plans.

(12) Monthly (64 min) exterior progressions (360 degrees around the project) and slideshows (all elevations and building envelope).

(13) Weekly (40 min) Site Progressions - Photographic documentation capturing the project at different stages of construction. These progressions shall capture underground utilities, excavation, grading, backfill, landscaping and road construction throughout the duration of the project.

(14) Monthly (80 min) interior progressions of all walls of the entire project to begin at time of substantial framed or as directed by the Contracting Officer through to completion.

(15) Detailed Exact-Built of all Slabs for all project slab pours just prior to placing concrete or as directed by the Contracting Officer.

(16) Detailed Interior exact built overlapping photos of the entire building to include documentation of all mechanical, electrical and plumbing systems in every wall and ceiling, to be conducted after rough-ins are complete, just prior to insulation and or drywall, or as directed by Contracting Officer.

(17) Finished detailed Interior exact built overlapping photos of all walls, ceilings, and floors to be scheduled by Contracting Officer prior to occupancy.

c. Images shall be taken by a commercial photographer and must show distinctly, at as large a scale as possible, all parts of work embraced in the picture.

d. Coordination of photo shoots is accomplished through Contracting Officer. Contractor shall also attend construction team meetings as necessary. Contractor's operations team shall provide regular updates regarding the status of the documentation, including photo shoots concluded, the availability of new Progressions or Exact-Builts viewable on-line and anticipated future shoot dates.

e. Contractor shall provide all on-line domain/web hosting, security measures, and redundant server back-up of the documentation.

f. Contractor shall provide technical support related to using the system or service.

g. Upon completion of the project, final copies of the documentation (the "Permanent Record") with the indexing and navigation system embedded (and active) shall be provided in an electronic media format, typically a DVD or external hard-drive. Permanent Record shall have Building Information Modeling (BIM) interface capabilities. On-line access terminates upon delivery of the Permanent Record.

1.18 FINAL ELEVATION DIGITAL IMAGES

a. A minimum of four (4) images of each elevation shall be taken with a minimum 6 MP camera, by a professional photographer with different settings to allow the Contracting Officer to select the image to be printed. All images are provided to the RE on a CD.

b. Photographs shall be taken upon completion, including landscaping. They shall be taken on a clear sunny day to obtain sufficient detail to show depth and to provide clear, sharp pictures. Pictures shall be (16 by 20 inches), printed on regular weight paper, matte finish archival grade photographic paper and produced by a RA4 process from the digital image with a minimum 300 PPI. Identifying data shall be carried on label affixed to back of photograph without damage to photograph and shall be similar to that provided for final construction photographs.

c. Furnish six (6) (16 by 20) inch color prints of the following buildings constructed under this project (elevations as selected by the RE from the images taken above). Photographs shall be artistically composed showing full front elevations. All images shall become property of the Government. Each of the selected six prints shall be place in a frame with a minimum of 2 inches of appropriate matting as a border. Provide a selection of a minimum of 3 different frames from which the SRE will select one style to frame all six prints. Photographs with frames shall be delivered to the Contracting Officer in boxes suitable for shipping.

(1) Chilled Water Plant Replacement.

1.19 WEB-BASED PHOTOGRAPHIC DOCUMENTATION

A. Web-Based Photographic Documentation Qualifications: A firm specializing in providing photographic equipment, web-based software and related services for construction projects, with record of providing satisfactory services similat to those required for the Project.

B. Web-based photographic documentation imagery

(1) Submit continuous live video recordings by posting to web-based photographic documentation service provider's website and monthly on video disc.

(2) Identification: For each recording, provide the following information:

a. Name of Project.
b. Date(s) and Time(s) video recording was recorded.
c. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.

C. Web-Based Photographic Documentation Equipment

(1) Project Camera: Provide fixed exterior camera installation, mounted to provide unobstructed view of construction site from location approved by Contracting Officer.

a. Provide 3 fixed-location cameras, with the following characteristics:

1. 180 degree panning view.
2. Capable of producing minimum 6 megapixel pictures.
3. Provide power supply, active high-speed data connection to service provider's network, and static public IP address for each camera.

(2) Web-Based Image Access: Password-protected access for Project team administered by Contractor, providing current image access and archival image access by date and time, with image downloadable to viewer's device.

D. Web-Based Construction Photographic Documentation

(1) Live streaming Construction Site Images: Provide web-accessible image of current site status from fixed location camera(s), updated at 1 minute intervals during daytime operation.

(2) Maintain cameras and web-based access in good working order according to web-based construction documentation service provider's written instructions until final complation. Provide for service of cameras and related networking devices and software.

1.20 HISTORIC PRESERVATION

Where the Contractor or any of the Contractor's employees, prior to, or during the construction work, are advised of or discover any possible archeological, historical and/or cultural resources, the Contractor shall

immediately notify the Contracting Officer verbally, and then with a written follow up.

PART 2 PRODUCTS

Not Used.

PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS
1.2 WORK COVERED BY CONTRACT DOCUMENTS
1.2.1 Project Description
1.2.2 Location
1.3 OCCUPANCY OF PREMISES
1.4 EXISTING WORK
1.5 ON-SITE PERMITS
1.5.1 Utility Outage Requests and Utility Connection Requests
1.5.2 Borrow, Excavation, Welding, and Hot Work Permits
1.6 LOCATION OF UNDERGROUND UTILITIES
1.6.1 Notification Prior to Excavation
1.7 PROJECT PHASING
1.8 HAZARDOUS MATERIALS
1.9 TESTING
1.10 SITE SECURITY
1.11 WORK SEQUENCE
1.12 CONTRACTOR USE OF PREMISES
1.12.1 Work Limits
1.12.2 Protection and Security
1.12.3 Damaged Property
1.12.4 Contractor's Receipt of Supplies
1.12.5 Daily Clean Up
1.13 PRE-CONSTRUCTION PHOTOGRAPHIC SURVEY
1.14 SAFETY AND INFECTION CONTROL

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 ATTACHMENTS TO THIS SECTION, REFER TO THE APPENDIX BINDER

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Utility Outage Requests; G

Utility Connection Requests; G

Borrow Permits; G

Hot Work Permits; G

Excavation Permits; G

Welding Permits; G

Phasing And Subphasing Plans; G

PCRA and ILSM Forms; G

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

The new Chilled Water Plant to be constructed at the Department of Veteran Affairs (DVA) campus in West Haven, Connecticut is intended to upgrade the existing chilled water system to current standards and requirements to include capacity for the planned Surgical and Clinical Tower and provide year round cooling.

The base scope of the project includes construction of new chiller plant and fuel oil tank farm (Base Bid 0001) and construction of new electrical ductbanks connecting west of the Fisher House, Building 19, and Building 1 (Base Bid 0002). The project includes four option bids which comprised of replacement of the existing campus domestic and fire protection water services from Campbell Avenue (Option Bid 0003), new above ground water tanks (Option Bid 0004), redundant utility main loops to Building 1 (Option Bid 0005), and new electrical service and redundant feeders by the electrical utility provider (Option Bid 0006).

1.2.2 Location

The work is located at the the VA Connecticut Healthcare System (VACHS) medical center located in West Haven, Connecticut, as indicated on the

Contract drawings.

1.3 OCCUPANCY OF PREMISES

Buildings 1, 15, 16, 19 and 24 will remain occupied during construction.

Only utility tie-ins to other locations and buildings on campus will be occupied.

1.4 EXISTING WORK

In addition to FAR 52.236-9 Protection of Existing Vegetation, Structures, Equipment, Utilities, and Improvements:

a. Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

b. Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

1.5 ON-SITE PERMITS

1.5.1 Utility Outage Requests and Utility Connection Requests

Schedule work to minimize outages. For utility outages and connections required during the execution of work that affect existing systems, schedule outside the regular working hours or on weekends, as approved by the Contracting Officer. Schedule utility outages and connections to minimize disruptions tothe Government. No additional payment will be provided for utility outages and connections required to be performed outside the regular work hours.

Submit requests for utility outages and connections in writing to the Contracting Officer for approval per the requirements in Section 01 14 00 WORK RESTRICTIONS. In each request, state the system involved, area involved, approximate duration of outage, and the nature of work involved.

1.5.2 Borrow, Excavation, Welding, and Hot Work Permits

ACTIVITY	SUBMISSION DATE	SUBMISSION FORM
Borrow permits	14 calendar days prior to work	Contractor Form
Hot work permits	14 calendar days prior to work	Contractor Form
Excavation permits	14 calendar days prior to work	Contractor Form
Welding permits	14 calendar days prior to work	Contractor Form

Refer to the Appendix Binder for form templates.

1.6 LOCATION OF UNDERGROUND UTILITIES

Contractor shall review, understand, and plan for requirements of PCRA (Pre-Construction Risk Assessment) (Refer to Appendix Binder)). Completion of the PCRA shall be coordinated with and will be done by the Medical Center prior to construction. The Project Geotechnical Report has been attached for reference.

Obtain excavation permits prior to start of excavation, and comply with Installation requirements for locating and marking underground utilities. See attached sample permit. Contact local utility locating service a minimum of 5 working days prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify existing utility locations indicated on contract drawings, within area of work.

1.6.1 Notification Prior to Excavation

Notify the Contracting Officer at least 21 days prior to starting excavation work. Refer to specification section 01 14 00 for additional excavation criteria.

1.7 PROJECT PHASING

The Chilled Water Plant will connect to Building 16 via direct addition which will require the demolition and removal of the existing chiller plant Building 16A. As part of the chiller plant a new fuel tank farm is located north of the new building and south of Building 35A. The water storage tanks are proposed west of Building 24 and partially in Parking Lot 6.
Underground utilities are direct buried with utility vaults where required. Major impacts to existing infrastructure include relocation and expansion of multiple utility distribution systems affected by the new construction and minor re-grading of the site as well as construction of retaining walls.

The proposed phasing approach is intended to follow:
1) Base Bid 0002 - Install redundant campus electrical feeder . Relocate existing campus feeder to north of the new fuel storage locations.

2) Base Bid 0001 - Fuel storage system and chilled water plant is constructed. New Fuel storage needs to be completed before the existing fuel oil tanks and chiller plant can be demolished.

3) Option Bid 0003- Install domestic and fire protection water services from Campbell Avenue to Building 16.

4) Option Bid 0004 - Construct water storage tanks and supporting infrastructure.

5) Option Bid 0005 - Install redundant utility feeds to Building 1.

6) Option Bid 0006 - New electrical service by UI

Phasing and subphasing plans shall show haul routes and maintain unobstructed access to existing fire hydrants, egress/ ingress to and from existing buildings. Refer to phasing drawings for additional information.

1.8 HAZARDOUS MATERIALS

Building materials containing asbestos, lead and other materials will be impacted during the execution of this Contract. Refer to the Hazardous Materials Survey Report for additional project-specific materials and the requirements of this specification.

1.9 TESTING

Before placement and installation of work subject to tests by the testing laboratory retained by the Contractor, the Contractor shall notify the Contracting Officer in sufficient time to enable witnessing of the testing laboratory personnel taking and testing of specimens and field inspection. Such prior notice shall be not less than two (2) weeks unless otherwise designated by the Contracting Officer.

1.10 SITE SECURITY

The Contractor shall report any vandalism, suspicious activities or devices to the Contracting Officer or local police as soon as possible. The Contracting Officer will notify the Contractor of any heightened security measures and will expect vigilant monitoring of equipment, grounds and security fencing while working. At a heightened security posture, there may be work areas that are restricted. The Contracting Officer will notify the Contractor of these restricted areas, and work may be delayed or restricted or will be performed in these areas under the direct supervision of U.S. Army Corps of Engineers personnel.

1.11 WORK SEQUENCE

There are certain essential criteria relative to the preparation of a work sequence and time schedule which the Contractor will be required to implement and follow during the prosecution of the work. Minor variations in the sequence of the items of work as specified may be made by the Contractor, provided such variations do not conflict with critical elements of the schedule. Proposed minor variations shall be noted on the progress schedule required by Section 01 32 01.00 10 PROJECT SCHEDULE. Variations shall be approved by the Contracting Officer prior to implementation.

1.12 CONTRACTOR USE OF PREMISES

1.12.1 Work Limits

Work shall be restricted to the limits of the work as indicated in Section 01 50 00 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS and as indicated by the Contracting Officer.

1.12.2 Protection and Security

a. Protection to Contractor personnel or their equipment cannot be provided at the work site by the Government.

b. The Contractor shall protect all its personnel, Government personnel, and the general public from injury.

c. The Contractor shall conduct all its work so as to prevent injury or unsafe conditions during construction.

1.12.3 Damaged Property

Work shall proceed in a manner which will minimize disturbance or risk of damage to structures and surrounding lands. The Contractor shall repair such items damaged in the course of carrying out the work at no additional cost to the Government within 3 weeks of damage occurrence unless deemed mission critical by Contracting Officer. All repairs shall match similar existing items in all aspects. All replacements shall be in kind.

1.12.4 Contractor's Receipt of Supplies

The Contractor shall be responsible for all arrangements for the receipt of materials and supplies at the job site. Government personnel are not permitted to receive or sign for items delivered to the site.

1.12.5 Daily Clean Up

The Contractor shall at all times keep rubbish from entering surrounding lands and water.

1.13 PRE-CONSTRUCTION PHOTOGRAPHIC SURVEY

The Contractor shall perform a pre-construction survey at the project site with the Contracting Officer. This pre-construction survey shall include still photographs and videos in sufficent quantity to fully document the existing condition of all features within the project site, both inside and outside of buildings, which may be impacted by contruction activities, such as structures, pavement, sidewalks, curbing, fencing, vegetation, trees and limbs, exterior lights, light poles, power poles, signs, floor finishes, wall finishes, ceiling finishes, mechanical and electrical features, elevators, furniture, and other pertinent features. A written narrative shall be prepared by the Contractor which organizes and summarizes the findings of the pre-construction survey photographs and videos. Each item shall be identified as "to remain", "to be removed and reset", "to be removed and returned to owner", or "to be demolished or discarded consistent with the contract drawings". The condition of each item shall also be documented. The written narrative, together with the photographs and videos, shall constitute a "Pre-construction Survey Report", which shall be submitted to the Contracting Officer for approval. No on-site work shall be performed until this "Pre-construction Survey Report" has been approved. This report will be used by the Contracting Officer to assess potential damage to property caused by Contractor operations. Damage to items remaining or to be reused caused by the Contractor's operations shall be repaired by the Contractor at no additional expense to the Government.

1.14 SAFETY AND INFECTION CONTROL

PCRA and ILSM forms, the Contractor shall review, understand, and incorporate the requirements identified in the Pre-Construction Risk Assessment (PCRA), Interim Life Safety Measures (ILSM) Requirement Assessment Worksheet, and the associated Infection Control Risk Assessment (ICRA) and ILSM exhibits included in the Appendix Binder.

Completion, submission, and approval of PCRA and ILSM documentation may be required separately for the Base Bids and each Bid Option, as directed by the Medical Center.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 ATTACHMENTS TO THIS SECTION, REFER TO THE APPENDIX BINDER

Attach No.	Attachment Description
Appendix Section - G6.0	VHA Pre-Construction Risk Assessment
Appendix Section - G6.0	VHA Interim Life Safety Assessment

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 14 00

WORK RESTRICTIONS

PART 1 GENERAL

1.1 SUBMITTALS
1.2 SPECIAL SCHEDULING REQUIREMENTS
1.3 CONTRACTOR ACCESS AND USE OF PREMISES
1.3.1 Medical Center Regulations
1.3.1.1 Contractor/Sub-contractor Parking
1.3.1.2 Subcontractors and Personnel Contacts
1.3.1.3 No Smoking Policy
1.3.2 Working Hours
1.3.3 Work Outside Regular Hours
1.3.4 Occupied Buildings
1.3.5 Utility Cutovers and Interruptions and Shutdowns
1.4 SECURITY REQUIREMENTS
1.4.1 Security Procedures
1.4.2 Key Control
1.4.3 Document Control
1.4.4 Identification (ID) Badges and Background Checks
1.4.5 Flash Badge
1.4.6 Person Identification Verification (PIV) Badge
1.4.7 Access to (WHVAMC) Restricted Areas

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 14 00

WORK RESTRICTIONS

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel

1.2 SPECIAL SCHEDULING REQUIREMENTS

The VA Medical Center will remain in operation during the entire construction period. The contractor must conduct his operations so as to cause the least possible interference with normal operations of the activity.

a. Have materials, equipment, and personnel required to perform the work at the site prior to the commencement of the work. Specific items of work to which this requirement applies include:

(1) Major utility interruptions or shutdowns.

(2) Roadway interruptions or shutdowns.

b. Utility Interruptions/Shutdowns/Tie-Ins:

(1) Requires minimum 120 calendar days advance notice. Contractor shall include all projected utility impacts and shutdowns on baseline schedules, each updated schedule and look aheads.

(2) Major utility shutdowns, including electrical, steam, and chilled water: Maximum duration – not to exceed Friday 7:00pm to Sunday 7:00pm. For steam shutdowns that require complete boiler plant shutdowns, cooldown and reheat times shall be included in this timeframe, approximately 2 hours each.

(3) Steam system shutdowns to be planned and occur only between May 1st through September 30th.

(4) Chilled water system shutdowns to be planned and occur only between November 1st through April 1st.

(5) Control of shutdown/LOTO Process:

a. Breakers, valves, shutoff devices, etc. within immediate

construction site or if outside construction site, but only impacts construction site, contractor shall have full control of process, including shutting down and locking out device or equipment.

b. Breakers, valves, shutoff devies, etc. outside of the immediate construction site and that will impact any other part or function of the medical center besides the construction site will be shut down by the medical center. Contractor is still required to follow medical center policies and procedures and provide their own lock out and tag out devices to lock out the devices and/or equipment.

c. Excavation work: Shall be planned/coordinated at least 3 weeks in advance of work.

(1) Excavations near major and critical utilities shall be completed during daylight hours. Contractor notification to Government shall be made a minimum of 21 days in advance. Government oversight will be required full time and so shall be coordinated.

(2) Excavations during dark hours shall be fully lighted.

(3) Excavations causing major road or other site interruptions shall be discussed and coordinated at least 30 days in advance of the work activity to minimize impact.

(4) Excavation work plan needs to have been reviewed and approved prior to utility notification and commencement of work.

1.3 CONTRACTOR ACCESS AND USE OF PREMISES

1.3.1 Medical Center Regulations

Ensure that Contractor personnel employed on the Medical Center become familiar with and obey Medical Center regulations including safety, fire, traffic and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE) in designated areas. Do not enter any restricted areas unless required to do so and until cleared for such entry. Ensure all Contractor equipment, include delivery vehicles, are clearly identified with their company name.

1.3.1.1 Contractor/Sub-contractor Parking

Contractor and sub-contractor personnel parking will NOT be provided on-site.

1.3.1.2 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.3.1.3 No Smoking Policy

Smoking is prohibited anywhere on the premises of the Medical Center.

1.3.2 Working Hours

The Contractor's normal work hours shall be from 6:00 AM to 4:00 PM. Normal work days shall be Monday through Friday, excluding weekends (Saturday and Sunday) and the observance of Federal holidays. The following are Federal Holiday Schedule observed:

New Year's Day (January 1st)
Birthday of Martin Luther King, Jr. (third Monday in January)
Washington's Birthday (third Monday in February)
Memorial Day (last Monday in May)
Juneteenth National Independence Day (June 19th)
Independence Day (July 4th)
Labor Day (first Monday in September)
Columbus Day (second Monday in October)
Veterans Day (November 11th)
Thanksgiving Day (fourth Thursday of November)
Christmas Day (December 25th)

1.3.3 Work Outside Regular Hours

Work outside regular working hours, weekends (Saturday and Sunday) and observed holidays requires Contracting Officer written approval. Make application 14 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer. Submit emergency work requests at least two calendar days before the scheduled work.

Work outside normal working hours is defined as work that interrupts facility operations, restricts access to buildings, or otherwise impacts normal occupancy of existing buildings. Such work shall be performed between 6:00 p.m. and 6:00 a.m., Monday through Friday, and between 6:00 p.m. on Friday and 6:00 a.m. on the following Monday

Construction activities that generate excessive noise or vibration shall be coordinated in advance with the Medical Center. As directed, such activities may be required to be performed during off-hours to minimize impacts to Medical Center operations, patients, and staff. Refer to section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS for reduction of noise criteria.

1.3.4 Occupied Buildings

The Contractor shall be working in and around existing buildings which are occupied. Do not enter the buildings without prior approval of the Contracting Officer.

The existing buildings and their contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer.

Provide dust covers or protective enclosures to protect existing work that remains, and Government material during the construction period.

The Contractor will remove and relocate other Government property in the areas of the buildings scheduled to receive work.

1.3.5 Utility Cutovers and Interruptions and Shutdowns

a. Utility cutovers and interruptions must be performed and completed between 7:00 PM on Friday through 7:00 PM on Sunday. Conform with procedures required in paragraph WORK OUTSIDE REGULAR HOURS.

b. Ensure that new utility lines are complete, except for the connection, before interrupting existing service.

c. Interruption to water, sanitary sewer, storm sewer, telephone service, electric service, air conditioning, heating, fire alarm, and compressed air are considered utility cutovers pursuant to the paragraph WORK OUTSIDE REGULAR HOURS. Such interruptions are further limited to the start of the work day until the end of the same work day. This time limit includes time for deactivation and reactivation.

d. Operation of Station Utilities: The Contractor must not operate nor disturb the setting of control devices in the station utilities system, including water, sewer, electrical, and steam services. The Government will operate the control devices as required for normal conduct of the work. The Contractor must notify the Contracting Officer a minimum of 120 days in advance of all projected utility impacts and shutdowns. All utility impacts and shutdowns shall be included on all baseline and updated project schedules. Complete boiler plant shutdowns shall include cool down and reheat times within the specified timeframe (2 hours each). Steam system shutdowns shall only occur between May 1st through September 30th. Chilled water system shutdowns shall only occur between November 1st through April 1st.

e. At a minimum of 21 days prior to the planned outage,interruption, or tie-in, contractor must provide a more detailed written work plan. Plan shall include, but not be limited to, confirmed planned interruption or tie-in date and time, duration of outage, details of what utilities are impacted, where, and what they serve, Medical Center's Hazardous Energy Policy 138-075 Lockout/Tagout Permit completed, and any road, pedestrian, or building impacts, along with mitigation efforts.

In addition, work on existing utilities that does not result in an interruption (investigations, tie-ins, operation of valves or switches, etc.) must also be requested in advance. At a minimum of 21 prior to this planned work, contractor shall also provide written work plan. Plan shall include, but not be limited to date and time, duration of work, details of what utilities are impacted, where, and what they serve, and any road, pedestrian, or building impacts, along with mitigation efforts.

1.4 SECURITY REQUIREMENTS

1.4.1 Security Procedures

a. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

b. The General Contractor's employees shall not enter the project site without an appropriate Identification (ID) badge. Along with ID badges, all workers must be able to speak and understand English during construction. They may also be subject to inspection of their personal effects when entering or leaving the project site. Access needed to non-construction areas (areas on the Medical Center outside of the Contractor's work limits) will be provided by the Contracting Officer or his/her representative upon written or verbal request.

c. Before starting on-site work, notify the Contracting Officer within 7 calendar days so that security arrangements can be provided for the employees. This notice is separate from any notices required for utility shutdowns described elsewhere in these specifications.

d. No photography or videography of VA patients or VA staff is allowed without written permission of the Contracting Officer.

e. The VA reserves the right to close down or shut down the project site and order General Contractor's employees off the premises in the event of a national emergency. The General Contractor may return to the site only with the written approval of the Contracting Officer.

1.4.2 Key Control

a. Provide duplicate keys and lock combinations to the Contracting Officer for the purpose of security inspections of every area of the project including fence gates and building/room access. Provide a Point-of-Contact, to accompany the Government personnel, that can provide access to tool boxes and parked equipment, and take any emergency action.

b. Turn over all permanent lock cylinders to the VA locksmith for permanent installation. See Section 08 71 00, DOOR HARDWARE and coordinate.

1.4.3 Document Control

a. Before starting any work, the General Contractor/Sub Contractors shall submit an electronic security memorandum describing the approach to following goals and maintaining confidentiality of "sensitive information".

b. The General Contractor is responsible for safekeeping of all drawings, project manual and other project information. This information shall be shared only with those with a specific need to accomplish the project.

c. Certain documents, sketches, videos or photographs and drawings may be marked "Law Enforcement Sensitive" or "Sensitive Unclassified" or "Controlled Unclassified Information". Secure such information in separate containers and limit the access to only those who will need it for the project. Return the information to the Contracting Officer upon request.

d. These security documents shall not be removed or transmitted from the project site without the written approval of Contracting Officer.

e. All paper waste or electronic media such as CD's and diskettes shall be shredded and destroyed in a manner acceptable to the Government.

f. Notify the Contracting Officer and Site Security Officer immediately when there is a loss or compromise of "sensitive information".

g. All electronic information shall be stored in specified location following Government standards and procedures using an Engineering Document Management Software (EDMS).

(1) Security, access and maintenance of all project drawings, both scanned and electronic shall be performed and tracked through the EDMS system.

(2) Sensitive information" including drawings and other documents may be attached to e-mail provided all Government encryption procedures are followed.

h. Designate submittals and delivery date for each product in Progress Schedule. Notify the Contracting Officer in writing within 180 calendar days of NTP the date on which the materials and equipment are required.

i. Use of Medical Center public parking spaces is prohibited.

1.4.4 Identification (ID) Badges and Background Checks

As the VA routinely reviews and updates policies and procedures covering contractor access, security requirements may change during the term of this contract and new policies and procedures may be implemented unilaterally during the term of this contract. Obtain a time-limited ID badge from the VA as follows:

1.4.5 Flash Badge

Obtain a flash badge from the West Haven VA Medical Center (WHVAMC) Police Service for all employees. Provide the following information within 24 hours in advance:

a. Full Name of the Individual seeking a flash badge

b. Name of the Company for which the individual works.

c. Location on (WHVAMC) campus where work is occurring: Coordinate with Resident Engineer.

d. Requested expiration Date of the badge: Maximum is 180 days.

(1) A valid Driver's License is required to pickup the badge. Badge must be worn at all times.

(2) A flash badge may be used for access while the individual is undergoing fingerprinting and a background investigation to obtain a PIV badge from the VHA Service Center (VSC) in Ohio, U.S.A., as described below.

1.4.6 Person Identification Verification (PIV) Badge

Requirements to obtain a PIV badge are as follows:

a. Complete Security Packet.

b. Two Identification (ID) cards compliant with FIPS 201-2 (section 2.7) that are validated using a method to validate authenticity of the documents. Validation will be done by the VSC.

c. Fingerprinting and background investigation. Fingerprinting can be done at the local VA Facility. The Electronic Fingerprint Verification Form must be submitted.

d. The Office of Security and Law Enforcement will notify the Contractor with instructions for the Contractor's employees, coordinate the background investigations, and notify the Contracting Officer and Contractor of the results of the investigations.

e. Background Investigation (BI) Requirements:

(1) Favorable Special Agreement Check (SAC)/FBI National Criminal History Check (NCHC).

(2) Favorable NACI.

(3) A PIV badge may be issued if NACI is initiated at the Office of Personnel Management.

1.4.7 Access to (WHVAMC) Restricted Areas

Notify the Contracting Officer within 72 hours in advance for access to any WHVAMC restricted areas. Coordinate with Contracting Officer the extent of restricted areas.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 22 00

MEASUREMENT AND PAYMENT

04/94

PART 1 GENERAL

1.1 CONTRACT COST BREAKDOWN
1.2 SUBMITTALS
1.3 JOB PAYMENT ITEMS
1.4 UNIT PRICE PAYMENT ITEMS
1.5 BIDDING SCHEDULE - PAYMENT ITEMS

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION (Not Applicable)

-- End of Section Table of Contents --

SECTION 01 22 00

MEASUREMENT AND PAYMENT
04/94

PART 1 GENERAL

1.1 CONTRACT COST BREAKDOWN

The Contractor must furnish within 30 days after the date of Notice to Proceed, and prior to the submission of its first partial payment estimate, a breakdown of its job pay items or items which will be reviewed by the Contracting Officer as to propriety of distribution of the total cost to the various accounts. Any unbalanced items as between early and late payment items or other discrepancies will be revised by the Contracting Officer to agree with a reasonable cost of the work included in the various items. This Contract cost breakdown will then be utilized as the basis for progress payments to the Contractor.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-05 Design Data

Quantity Surveys

Submit originals of all field notes and all other records relating to quantity surveys.

1.3 JOB PAYMENT ITEMS

Payment items for the work of this contract for which contract job payments will be made are listed in the BIDDING SCHEDULE and described below. The job price and payment made for each item listed shall constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for which separate payment is not otherwise provided.

1.4 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this contract on which the contract unit price payments will be made are listed in the BIDDING SCHEDULE and described below. The unit price and payment made for each item listed shall constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items. Submit originals of all field notes and all other records relating to Quantity Surveys.

1.5 BIDDING SCHEDULE - PAYMENT ITEMS

Payment items for the work of this contract on which the contract progress payments will be based are listed in the BIDDING SCHEDULE and are described below. All costs for items of work, which are not specifically mentioned to be included in a particular Bidding Schedule job or unit price payment item, shall be included in the listed job item most closely associated with the work involved.

Base Bid Items

A. Item Number 0001, "Chilled Water Plant Replacement".

All costs for labor, equipment, and materials for the construction of the chilled water plant to include, but not limited to all utilities, services, and appurtenances as shown on the contract drawings and in accordance with the specifications. This item also includes the fuel oil storage tank farm and enclosure and the fuel line supplies connecting to Building 16 and tap to future new boiler plant. A new electrical duct bank service connecting services from Building 19 to an existing electrical manhole. Demolition of the existing ice storage system, existing chilled water plant, existing fuel tank oil tanks, auxiliary chiller plant, and all associated utilities, services, and appurtenances are also included.

Unit of Measure: Job.

B. Item Number 0002, "Site Electrical Utilities Campus Redundant Feed".

All costs for labor, equipment, and materials for the new electrical duct banks connecting the electrical substation west of Fisher House, Building 19, and Building 1, as shown on the contract drawings and in accordance with the specifications.

Unit of Measure: Job.

C. Item Number 0007, "Removal of Asbestos Containing Material as part of the Existing Building Demolition".

All costs for labor, equipment, and materials for the complete removal of asbestos containing material as part of the existing building demolition, as shown on the contract drawings and in accordance with the specifications.

Unit of Measure: Job.

D. Item Number 0008, "Rock Excavation".

All costs for labor, equipment, and materials for the rock excavation as needed for the installation of all work on this project, as shown on the contract drawings and in accordance with the specifications. Payment will be made based on quantities determined from surveys of pre- and post- rock excavation of the foundation surface profile.

Unit of Measure: Cubic Yard (CD).

E. Item Number 0009, "Underground Asbestos Containing Material (ACM) Pipe Removal".

All costs for labor, equipment, and materials for the complete removal of underground asbestos containing material (ACM) pipe and/or pipe insulation for the installation of all work on this project, as shown on the contract drawings and in accordance with the specifications. Payment will be made based on the measured length of the underground ACM pipe removed.

Unit of Measure: Linear Feet (LF).

Optional Bid Items

D. Item Number 0003, "Campbell Avenue - Domestic Water and Fire Protection Service Piping".

All costs for labor, equipment, and materials for the installation of the domestic and fire protection water services and appurtenances from Campbell Avenue, as shown on the contract drawings and in accordance with the specifications. This item includes connections from Building 17 through Building 16 and into Building 15 to connect to the existing water service. It also includes the removal of the existing campus domestic water and fire protection services to Campbell Avenue.

Unit of Measure: Job.

E. Item Number 0004, "Water Storage Tanks and Piping".

All costs for labor, equipment, and materials for the installation of the above ground water storge tanks and supporting infrastructure and appurtenances, as shown on the contract drawings and in accordance with the specifications. This item includes all connections from the water tanks to Building 16 and Building 15. It also includes the electrical ductbank service routed from Building 17 to the water storage tanks.

Unit of Measure: Job.

F. Item Number 0005, "Site Utilities Backfeeds to Building 1".

All costs for labor, equipment, and materials for the utility feeds and connection to Building 1 to include, but not limited all associated services and appurtenances, as shown on the contract drawings and in accordance with the specifications. This item consists of the chilled water supply, chilled water return, domestic water, fire protection water, high pressure steam, and pumped condensate. This work includes the selective demolition and replacement of existing equipment and associated utilities and accessories in the sub-ground mechanical room for the backfeed connections.

Unit of Measure: Job.

G. Item Number 0006, "United Illuminating Electrical Service Allowance".

The allowance of $1,400,000 is to pay for the electrical utility

charges related to work to bring a second independent electrical service to the campus, and shall be included in the lump sum bid under this item. Payment for the electrical company charges shall be up to but not exceeding the $1,400,000 allowance. The Contractor must submit a copy of the invoice from the electrical utility company to the Contracting Officer to receive payment for this item.

Unit of Measure: Job.

PART 2 PRODUCTS (Not Applicable)

PART 3 EXECUTION (Not Applicable)

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

11/20, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES
1.2 SUBMITTALS
1.3 MINIMUM INSURANCE REQUIREMENTS
1.4 SUPERVISION
1.4.1 Superintendent Qualifications
1.4.2 Duties
1.4.3 Non-Compliance Actions
1.5 PRECONSTRUCTION
1.5.1 Attendees
1.6 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)
1.6.1 Red Zone Checklist
1.6.2 Meetings
1.7 PARTNERING
1.7.1 Facilitated (Formal) Partnering
1.8 GOVERNANCE STRUCTURE
1.8.1 Senior Advisory Group (SAG)

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS
11/20, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health Requirements (SOH)

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

View Location Map

Progress and Completion Pictures

1.3 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability, during the entire period of performance under this contract. Provide other insurance coverage as required by law.

1.4 SUPERVISION

1.4.1 Superintendent Qualifications

Provide project superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

1.4.2 Duties

The project superintendent is primarily responsible for managing

subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.4.3 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.5 PRECONSTRUCTION

Prior to commencing any work at the site, coordinate with the Contracting Officer a time and place to meet for the Preconstruction . The must take place prior to commencement of any work at the site. The purpose of this is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, Medical Center access, outage requests, hot work permits, schedule requirements, quality control, schedule of prices or earned value report, shop drawings, submittals, cybersecurity, prosecution of the work, government acceptance, final inspections and contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing.

1.5.1 Attendees

Contractor attendees must include the Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and major subcontractors.

1.6 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start planning the turnover process at the Pre-Construction Conference meeting with a discussion of the Red Zone process and convene at regularly scheduled NRZ Meetings beginning at approximately 75 percent of project completion. Include the following in the facility Turnover effort:

1.6.1 Red Zone Checklist

a. Contracting Officer's Technical Representative (COTR) will provide the Contractor a copy of the Red Zone Checklist template.

b. Prior to 75 percent completion, modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. Submit the modified Red Zone Checklist to the Contracting Officer. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary.

1.6.2 Meetings

a. Conduct regular Red Zone Meetings beginning at approximately 75 percent project completion, or three to six months prior to Beneficial Occupancy Date (BOD), whichever comes first.

b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.

c. Using the Red Zone Checklist as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.

d. During the Red Zone Meetings discuss with the COTR any upcoming activities that require Government involvement.

e. Maintain the Red Zone Checklist by documenting the actual completion dates as work is completed and update the Red Zone Checklist with revised planned completion dates as necessary to match progress. Distribute copies of the current Red Zone Checklist to attendees at each Red Zone Meeting.

1.7 PARTNERING

Contractor shall host the partnering session within 45 calendar days of contract award. To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the Medical Center, end user, , Contractor, key subcontractors and the Designer of Record are required to participate in the Partnering process.

1.7.1 Facilitated (Formal) Partnering

a. Within 35 calendar days after award and prior to the start of work, host a Formal Partnering session with key personnel from the project team including both Contractor and Government personnel. All costs associated with the Partnering session including the third-party independent Facilitator Consultant, meeting room and other incidental items are the responsibility of the Contractor.

b. Before the Facilitated (Formal) Partnering session, coordinate with the Facilitator all requirements for incidental items (such as audio-visual equipment, easels, flipchart paper, colored markers, note pads, pens/pencils, colored flash cards) and have these items available at the Partnering session. Provide copies of any documents required for distribution to all attendees. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.

c. The Initial Partnering Session must be a duration of one day and be held at a location off base as agreed to by the Contracting Officer. Partnering session may take place concurrently with the Pre-Construction Meeting.

d. Facilitator must be experienced in conducting corporate Partnering sessions and must be a third-party independent facilitating consultant - not an employee of the Contractor. The Facilitator is responsible for leading all aspects of the Partnering session necessary to achieve the Partnering goal.

e. An outcome of the Partnering session must be an escalation matrix agreed upon by both the Government and Contractor, which identifies key Government and Contractor decision makers by name and anticipated decision durations.

f. Host follow-on Partnering Sessions at three- to six-month intervals or more frequently if needed and lasting generally a half day or less. Attendees need only be those required to resolve current issues. The same Facilitator used in the Initial Partnering session must lead the follow-on sessions unless an alternative is permitted by the Contractor Officer. All costs associated with follow-on Partnering sessions are the responsibility of the Contractor.

1.8 GOVERNANCE STRUCTURE

An integrated governance structure is used to manage the project, shown below. There are members of each organization strategically and tactically placed within the Governance structure to facilitate communication and issue resolution at the appropriate level; including the ability to obtain support from each appropriate management structure to solve issues that are inherently the responsibility of separate agencies. The Contractor will participate as members of the Governance structure at the Senior Advisory Group level and below. Timely delivery cannot be accomplished by any one organization alone and requires an integrated team concept promoted via the "One Team" Governance concept.

The Objectives are:

(1) Effective and accountable project leadership;

(2) Open and effective communication within the team and with stakeholders; and the result is

(3) The team fosters trust, honesty, and shared values

1.8.1 Senior Advisory Group (SAG)

Meetings held quarterly after the formal partnering session. Members include:

a. USACE NAD Senior Executive, SES

b. USACE NAD MID Chief

c. USACE NAD Regional Chief of E&C

d. HQUSACE PMO

e. DVA CFM Executive Liaison

f. DVA CFM Eastern Regional Director

g. DVA CFM PMO

h. VAMC Director

i. VISN 1 Capital Asset Manager (CAM)

j. Contractor Executive(s)

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 32 01.00 10

PROJECT SCHEDULE

08/23, CHG 1: 08/24

the preparation and maintenance of the project schedule for construction projects or design-build construction projects

PART 1 GENERAL

1.1 REFERENCES
1.2 SUBMITTALS
1.3 PROJECT SCHEDULER QUALIFICATIONS

PART 2 PRODUCTS

2.1 SOFTWARE
2.1.1 Government Default Software
2.1.2 Contractor Software
2.1.2.1 Other Than Primavera

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS
3.1.1 Schedule Status Reports
3.1.2 Scheduling Expectations Kickoff Meeting(SEKO)
3.2 BASIS FOR PAYMENT AND COST LOADING
3.2.1 Activity Cost Loading
3.2.2 Cost Loading of Submittals
3.2.3 Withholdings / Payment Rejection
3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS
3.3.1 Level of Detail Required
3.3.2 Activity Durations
3.3.3 Procurement Activities
3.3.4 Mandatory Activities
3.3.5 Government Activities
3.3.6 Standard Activity Coding Dictionary
3.3.6.1 Workers Per Day (WRKP)
3.3.6.2 Responsible Party Coding (RESP)
3.3.6.3 Area of Work Coding (AREA)
3.3.6.4 Modification Number (MODF)
3.3.6.5 Bid Item Coding (BIDI)
3.3.6.6 Phase of Work Coding (PHAS)
3.3.6.7 Category of Work Coding (CATW)
3.3.6.8 Feature of Work Coding (FOW)
3.3.7 Contract Milestones and Constraints
3.3.7.1 Project Start Date Milestone and Constraint
3.3.7.2 End Project Finish Milestone and Constraint
3.3.7.3 Interim Completion Dates and Constraints
3.3.7.3.1 Start Phase

3.3.7.3.2 End Phase
3.3.8 Adverse Weather
3.3.9 Calendars
3.3.10 Open Ended Logic
3.3.11 Default Progress Data Disallowed
3.3.12 Out-of-Sequence Progress
3.3.13 Added and Deleted Activities
3.3.14 Original Durations
3.3.15 Leads, Lags, and Start to Finish Relationships
3.3.16 Retained Logic
3.3.17 Percent Complete
3.3.18 Remaining Duration
3.3.19 Cost Loading of Closeout Activities
3.3.19.1 As-Built Drawings
3.3.19.2 O & M Manuals
3.3.20 Early Completion Schedule and the Right to Finish Early
3.4 PROJECT SCHEDULE SUBMISSIONS
3.4.1 Preliminary Project Schedule Submission
3.4.2 Initial Project Schedule Submission
3.4.3 Periodic Schedule Updates
3.5 SUBMISSION REQUIREMENTS
3.5.1 Electronic Scheduling Data
3.5.2 Narrative Report
3.5.3 Schedule Reports
3.5.3.1 Activity Report
3.5.3.2 Logic Report
3.5.3.3 Total Float Report
3.5.3.4 Earnings Report by CLIN
3.5.3.5 Scheduling and Leveling Report
3.5.4 Network Diagram
3.5.4.1 Continuous Flow
3.5.4.2 Project Milestone Dates
3.5.4.3 Critical Path
3.5.4.4 [Enter Appropriate Subpart Title Here]
3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram
3.5.4.6 Feature of Work (FOW) Grouping
3.5.4.7 Approved Definable Feature of Work (DFOW) listing from QC Plan
3.6 PERIODIC SCHEDULE UPDATE
3.6.1 Periodic Schedule Update Meetings
3.6.2 Update Submission Following Progress Meeting
3.7 WEEKLY PROGRESS MEETINGS
3.8 REQUESTS FOR TIME EXTENSIONS
3.8.1 Justification of Delay
3.8.2 Changes That Do Not Cause Delay
3.8.3 Time Impact Analysis (Prospective Analysis)
3.8.4 Forensic Schedule Analysis (Retrospective Analysis)
3.8.5 Fragmentary Network (Fragnet)
3.8.6 Time Extension
3.8.7 Impact to Early Completion Schedule
3.9 FAILURE TO ACHIEVE PROGRESS
3.9.1 Artificially Improving Progress
3.9.2 Failure to Perform
3.9.3 Recovery Schedule
3.10 OWNERSHIP OF FLOAT
3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM
3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

-- End of Section Table of Contents --

SECTION 01 32 01.00 10

PROJECT SCHEDULE
08/23, CHG 1: 08/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AACE INTERNATIONAL (AACE)

AACE 52R-06	(2006) Time Impact Analysis - As Applied in Construction

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1-1-11	(2017) Administration -- Project Schedules

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Project Scheduler Qualifications; G,

Preliminary Project Schedule; G,

Initial Project Schedule; G,

Periodic Schedule Update; G,

1.3 PROJECT SCHEDULER QUALIFICATIONS

Designate an authorized representative to be responsible for the preparation of the schedule and all required updating and production of reports. Submit no later than 15 calendar days after Notice to Proceed (NTP).The authorized representative must have a minimum of 5 -years experience scheduling construction projects similar in size and nature to this project with scheduling software that meets the requirements of this specification. Representative must have a comprehensive knowledge of CPM scheduling principles and application. The authorized representative must be on the job site full-time, with no other duties. Failure to meet these requirements after government acceptance will require a replacement within 10 calendar days. Deficiencies warranting replacement include rejected schedule submissions, failure to address government review comments,

inadequate proficiency in Primavera P6, or inability to apply CPM principles effectively.

PART 2 PRODUCTS

2.1 SOFTWARE

The scheduling software utilized to produce and update the schedules required herein must be capable of meeting all requirements of this specification.

2.1.1 Government Default Software

The Government uses Primavera P6 Professional. Submit all files in the Primavera P6 XER format in a version of P6 importable by the Government system.

2.1.2 Contractor Software

Scheduling software used by the contractor must be commercially available from the software vendor for purchase with vendor software support agreements available. The software routine used to create the required sdef file must be created and supported by the software manufacturer.

2.1.2.1 Other Than Primavera

If the Contractor chooses software other than Primavera P6, that is compliant with this specification, provide for the Government's use two licenses, two computers, and training for mid-level proficiency for two Government employees in the use of the software. These computers will be stand-alone and not connected to Government network. Computers and licenses will be returned at project completion.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Prepare for approval a Project Schedule, as specified herein, pursuant to FAR Clause 52.236-15 Schedules for Construction Contracts. Show in the schedule the proposed sequence to perform the work and dates contemplated for starting and completing all schedule activities. The scheduling of the entire project is required. The scheduling of construction is the responsibility of the Contractor. Contractor management personnel must actively participate in its development. Subcontractors, and suppliers working on the project must also contribute in developing and maintaining an accurate Project Schedule. Provide a schedule that is a forward planning as well as a project monitoring tool. Use the Critical Path Method (CPM) of network calculation to generate all Project Schedules. Prepare each Project Schedule using the Precedence Diagram Method (PDM).

3.1.1 Schedule Status Reports

Provide a schedule status report on at least a monthly basis. If, in the opinion of the Contracting Officer, the Contractor falls behind the approved schedule, take steps necessary to improve progress, including those that may be required by the Contracting Officer, without additional cost to the Government. In this circumstance, the Contracting Officer may require the Contractor to increase the number of shifts, overtime operations, days of work, or the amount of construction plant or a

combination of these, and to submit for approval any supplementary schedule or schedules as the Contracting Officer deems necessary to get back on schedule.

3.1.2 Scheduling Expectations Kickoff Meeting(SEKO)

In conjunction with the post award conference or pre construction conference the Government and the Contractor will hold a separate and distinct Scheduling Expectations Kickoff Meeting(SEKO). The intent of this meeting is for the Government to review with the contractor the requirements of the scheduling specification, communicate the Government’s expectations regarding schedule development and management, and answer any questions the Contractor may have regarding the schedule requirements.

3.2 BASIS FOR PAYMENT AND COST LOADING

The schedule is the basis for determining contract earnings during each update period and therefore the amount of each progress payment. The aggregate value of all activities coded to a contract CLIN must equal the value of the CLIN. Match the progress payment "Pay Period Thru" date in RMS to the schedule data date.

3.2.1 Activity Cost Loading

Activity cost loading must be reasonable and without front-end loading. Activities with a negative cost loading is not allowed. Provide additional documentation to demonstrate reasonableness if requested by the Contracting Officer.

3.2.2 Cost Loading of Submittals

No costs are to be assigned to activities for the preparation, review, or approval of submittals, except as described under paragraph COST LOADING OF CLOSEOUT ACTIVITIES, paragraph AS-BUILT DRAWINGS, and paragraph O&M MANUALS.

3.2.3 Withholdings / Payment Rejection

Failure to meet the requirements of this specification may result in the disapproval of the preliminary, initial, or periodic schedule updates and subsequent rejection of payment requests until compliance is met.

In the event that the Contracting Officer directs schedule revisions and those revisions have not been included in subsequent Project Schedule revisions or updates, or if the contractor does not address government review comments,the Contracting Officer may reject the schedule and progress payment until such revisions to the project schedule have been made.

3.3 PROJECT SCHEDULE DETAILED REQUIREMENTS

3.3.1 Level of Detail Required

Develop the Project Schedule to the appropriate level of detail to address major milestones and to allow for satisfactory project planning and execution and minimize out-of-sequence conditions. The schedule must use

logic-driven sequencing, eliminate redundant logic, and accurately reflect the project's scope and constraints.. Failure to meet these requirements will result in its disapproval. The Contracting Officer will consider, but is not limited to, the following characteristics and requirements to determine appropriate level of detail:

3.3.2 Activity Durations

Reasonable activity durations are those that allow the progress of ongoing activities to be accurately determined between update periods. Non-procurement activities Original Durations (OD) are not to exceed 20 workdays or 30 calendar days.

3.3.3 Procurement Activities

Include in the schedule only procurement activities related to the critical path or long lead submittals. This includes their approvals, procurement, fabrication, and delivery of long lead materials, equipment, fabricated assemblies, and supplies. Long lead procurement activities are those with an anticipated fabrication and delivery sequences of over 90 calendar days.

3.3.4 Mandatory Activities

Include the following activities in the initial project schedule and all updates.

a. Individual activities for submission, and government review of Division 01 SD-01 Preconstruction Submittals. This includes, but is not limited to:

   - Phasing Plan
   - Salvage Plan
   - Accident Prevention Plan
   - Quality Control Plan
   - Environmental Protection Plan
   - Excavation Work Plan

b. Submission, review and acceptance of features requiring design completion

c. Submission of mechanical/electrical/information systems layout drawings.

d. Long procurement activities defined in section 3.3.3.

e. Individual activities for submission and government review of O & M manuals.

f. Individual activities for submission and government review of as-built drawings.

g. Individual activities for submission and government review of DD1354 data and installed equipment lists.

h. Individual activities for submission and government review of testing and air balance (TAB).

i. Individual activities for submission and government review of TAB specialist design review report.

j. Individual activities for submission and government review of fire protection specialist.

k. Air and water balancing.

l. Controls testing plan submission.

m. Controls testing.

n. Performance verification testing.

o. Other systems testing, if required.

p. Submission and approval of Building Commissioning Plan, Controls testing plan, test data, and reports per 01 91 00.01: Develop the schedule logic associated with testing and commissioning of mechanical systems to a level of detail consistent with the contract commissioning requirements. All tasks associated with building testing and commissioning will be completed prior to submission of building commissioning report and subsequent contract completion.

These activities may include but are not limited to: air and water balancing, building commissioning -functional performance testing, controls testing, and performance verification testing. Include other systems testing, if required. Include additional commissioning requirements noted elsewhere in the Contract as required in the schedule and discuss during the SEKO meeting, noted in paragraph SCHEDULING EXPECTATIONS KICKOFF MEETING (SEKO).

q. Contractor's pre-final inspection.

r. Contractor's Punch List Correction from Pre-Final Inspection.

s. Government's pre-final inspection.

t. Government's Punch List Correction from Pre-Final Inspection.

u. Final Inspection.

v. Activities to represent the preparation, coordination, and submission of the PCRA, followed by government review.

w. Activities to represent the preparation, coordination, and submission of the ILSM, followed by government review.

x. Individual utility shutdown and cutover activities (e.g. electrical, chilled water, domestic water, fire protection).

y. Individual activities for preparation and execution of the asbestos glovebag abatement and lead-based paint removal.

z. Individual mockup activities per 01 43 30.

aa. Individual activities for each system to be commissioned per 01 91 00.01, 1.6.

ab. Milestones for the following.

- Phasing Plan accepted
- PCRA / ILSM Approval
- Pre-Construction Photographic Survey Approved
- New Redundant Electrical Feeder Energized
- New Fuel Storage System Operational and Accepted
- New Chilled Water Plant Commissioning Complete
- Chilled Water Cutover Complete
- UI New Service Energized

3.3.5 Government Activities

Show in schedule, Government and other agency activities that could impact progress. These activities include, but are not limited to approvals, environmental permit approvals by State regulators, inspections, utility tie-in, Government Furnished Equipment (GFE), Notice to Proceed (NTP) for phasing requirements, UI design and installation of new service, review and approval of the PCRA and ILSM.

3.3.6 Standard Activity Coding Dictionary

Use the activity coding structure defined in the Standard Data Exchange Format (SDEF) in ER 1-1-11. This exact structure is mandatory. Develop and assign all Activity Codes to activities as detailed herein.

The SDEF format is as follows:

Field	Activity Code	Length	Description
1	WRKP	3	Workers per day
2	RESP	4	Responsible party
3	AREA	4	Area of work
4	MODF	6	Modification Number
5	BIDI	6	Bid Item (CLIN)
6	PHAS	2	Phase of work
7	CATW	1	Category of work
8	FOW	20	Feature of work*

*Some systems require that FEATURE OF WORK values be placed in several activity code fields. The notation shown is for Primavera P6. Refer to the specific software guidelines with respect to the FEATURE OF WORK field requirements.

3.3.6.1 Workers Per Day (WRKP)

Assign Workers per Day for all field construction or direct work activities, if directed by the Contracting Officer. Workers per day is based on the average number of workers expected each day to perform a task for the duration of that activity.

3.3.6.2 Responsible Party Coding (RESP)

Assign responsibility code for all activities to the Prime Contractor, Subcontractor(s) or Government agency(ies) responsible for performing the activity.

a. Activities coded with a Government Responsibility code include, but are not limited to: Government approvals, Government design reviews, environmental permit approvals by State regulators, Government Furnished Property/Equipment (GFP) and Notice to Proceed (NTP) for phasing requirements.

b. Activities cannot have more than one Responsibility Code, without exception. Examples of acceptable activity code values are: DOR (for the designer of record); ELEC (for the electrical subcontractor); MECH (for the mechanical subcontractor); and GOVT (for USACE).

3.3.6.3 Area of Work Coding (AREA)

Assign Work Area code to activities based upon the work area in which the activity occurs. Define work areas based on resource constraints or space constraints that would preclude a resource, such as a particular trade or craft work crew from working in more than one work area at a time due to restraints on resources or space. Examples of Work Area Coding include different areas within a floor of a building, different floors within a building, and different buildings within a complex of buildings. Activities cannot have more than one Work Area Code.

3.3.6.4 Modification Number (MODF)

Assign a Modification Number Code to any activity or sequence of activities added to the schedule as a result of a Contract Modification, when approved by Contracting Officer. Key all Code values to the Government's modification numbering system. An activity can have only one Modification Number Code.

3.3.6.5 Bid Item Coding (BIDI)

Assign a Bid Item Code to all activities using the Contract Line Item Number (CLIN) to which the activity belongs, even when an activity is not cost loaded. An activity can have only one BIDI Code.

3.3.6.6 Phase of Work Coding (PHAS)

Assign Phase of Work Code to all activities. Examples of phase of work are procurement phase and construction phase. Each activity can have only one Phase of Work code.

a. If the contract specifies phasing with separately defined performance periods, identify a Phase Code to allow filtering and organizing the schedule accordingly. Align with the project's phasing plan, or proposed phasing in 01 11 00, section 1.7.

3.3.6.7 Category of Work Coding (CATW)

Assign a Category of Work Code to all activities. Category of Work Codes are a greater level of detail than the Phase of Work coding. Category of Work Codes include, but are not limited to construction submittal,

procurement, fabrication, weather sensitive installation, non-weather sensitive installation, start-up, and testing activities. Each activity can have no more than one Category of Work Code.

3.3.6.8 Feature of Work Coding (FOW)

Assign a Feature of Work Code to appropriate activities based on the Definable Feature of Work identified in the approved QC plan. Each activity must have only one FOW code, without exception. Activities that encompass multiple Features of Work indicate insufficient schedule detail. Because the FOW code is critical to the Three-Phase Quality Control program, each activity must be detailed enough to correspond to a single Feature of Work.

Definable Feature of Work is defined in Section 01 45 00 QUALITY CONTROL. A single, standardized Definable Feature of Work list must be used consistently across the schedule and quality control processes, with identical code values. The P6 schedule's FOW coding structure must align exactly with the Definable Feature of Work coding in the approved Quality Control Plan, with no extraneous codes.

Ensure the schedule includes all necessary activities for each Feature of Work. This can be verified using the Feature of Work Report outlined in Section 3.5.4.6.

3.3.7 Contract Milestones and Constraints

Milestone activities are to be used for significant project events including, but not limited to, project phasing, project start and end activities, or interim completion dates. The use of artificial float constraints such as "zero free float" or "zero total float" are prohibited.

Mandatory constraints that ignore or affect network logic are prohibited. No constrained dates are allowed in the schedule other than those specified herein. Submit additional constraints to the Contracting Officer for approval on a case by case basis.

3.3.7.1 Project Start Date Milestone and Constraint

The first activity in the project schedule must be a start milestone titled "NTP Acknowledged," which must have a "Start On" constraint date equal to the date that the NTP is acknowledged.

3.3.7.2 End Project Finish Milestone and Constraint

The last activity in the schedule must be a finish milestone titled "End Project."

Constrain the project schedule to the Contract Completion Date in such a way that if the schedule calculates an early finish, then the float calculation for "End Project" milestone reflects positive float on the longest path. If the project schedule calculates a late finish, then the "End Project" milestone float calculation reflects negative float on the longest path.

3.3.7.3 Interim Completion Dates and Constraints

Constrain contractually specified interim completion dates to show

negative float when the calculated late finish date of the last activity in that phase is later than the specified interim completion date.

3.3.7.3.1 Start Phase

Use a start milestone as the first activity for a project phase. Call the start milestone "Start Phase X" where "X" refers to the phase of work. Add the following phase milestones:

- Start Phase Base Bid 0002 - Install Redundant Campus Electrical Feeder
- Start Phase Base Bid 0001 - Construct Fuel Storage and Chilled Water Plant
- Start Phase Option Bid 0003 - Install Domestic and Fire Protection Water Services
- Start Phase Option Bid 0004 - Construct Water Storage Tanks and Infrastructure
- Start Phase Option Bid 0005 - Install Redundant Utility Feeds to Building 1
- Start Phase Option Bid 0006 - UI Electrical Service Upgrade

3.3.7.3.2 End Phase

Use a finish milestone as the last activity for a project phase. Call the finish milestone "End Phase X" where "X" refers to the phase of work. Add the following phase milestones:

- End Phase Base Bid 0002 - Install Redundant Campus Electrical Feeder
- End Phase Base Bid 0001 - Construct Fuel Storage and Chilled Water Plant
- End Phase Option Bid 0003 - Install Domestic and Fire Protection Water Services
- End Phase Option Bid 0004 - Construct Water Storage Tanks and Infrastructure
- End Phase Option Bid 0005 - Install Redundant Utility Feeds to Building 1
- End Phase Option Bid 0006 - UI Electrical Service Upgrade

3.3.8 Adverse Weather

Ensure anticipated adverse weather is accounted for in the Project Schedule. AACE 84R-13 provides methodologies for techniques in planning for adverse weather. The preferred methodology is the use of a weather calendar. If a method other than a weather calendar is proposed, discuss the reason during the SEKO meeting.

Refer to ER 415-1-15 CONSTRUCTION TIME EXTENSIONS FOR WEATHER for guidance on weather delay application.

3.3.9 Calendars

Schedule activities on a Calendar to which the activity logically belongs. Develop calendars to accommodate any contract defined work period such as a 7-day calendar for Government Acceptance activities, concrete cure times, etc. Develop the default Calendar to match the physical work plan with non-work periods identified including weekends and holidays. Develop Seasonal Calendar(s) and assign to seasonally affected activities as applicable.

If an activity is weather sensitive it should be assigned to a calendar showing non-work days on a monthly basis, with the non-work days selected at random across the weeks of the calendar, using the anticipated adverse weather delay work days provided in the Special Contract Clauses. Assign non-work days over a seven-day week as weather records are compiled on

seven-day weeks, which may cause some of the weather related non-work days to fall on weekends.

Refer to ER 415-1-15 CONSTRUCTION TIME EXTENSIONS FOR WEATHER for guidance on weather delay application.

Ensure the schedule includes the following schedule calendars. Refer to 01 14 00, section 1.2:

- Chilled Water Shutdown Calendar (working time only 01-Nov to 01-Apr)
- Steam Shutdown Calendar (working time only 01-May to 30-Sep)
- Weekend Outage Calendar (working time only Friday 1900 to Sunday 1900)
- Standard Workday Calendar (Monday to Friday 0600 to 1600, excluding federal holidays)
- Weather Sensitive Calendar

3.3.10 Open Ended Logic

Only two open ended activities are allowed: the first activity "NTP Acknowledged" is to have no predecessor logic, and the last activity -"End Project" is to have no successor logic. Except as noted above, each activity must have at least a start-to-start or finish-to-start relationship with its predecessor and a finish-to-finish or finish-to-start relationship with its successor.

Predecessor open ended logic may be allowed in a time impact analyses upon the Contracting Officer's approval.

3.3.11 Default Progress Data Disallowed

Actual Start and Finish dates must not automatically update with default mechanisms included in the scheduling software. Updating the percent complete and the remaining duration of any activity must be independent functions. Disable program features that calculate one of these parameters from the other. Activity Actual Start (AS) and Actual Finish (AF) dates assigned during the updating process must match those dates provided in the Contractor Quality Control Reports. Failure to document the AS and AF dates in the Daily Quality Control report will result in disapproval of the Contractor's schedule.

3.3.12 Out-of-Sequence Progress

Activities that have progressed before all preceding logic has been satisfied (Out-of-Sequence Progress) will be allowed only on a case-by-case basis subject to approval by the Contracting Officer.

Out-of-sequence progress - where an activity is started or completed before all its predecessor logic has been satisfied - will only be permitted on a case-by-case basis, subject to approval by the Contracting Officer..

a. Corrective Action: Implement logic corrections to eliminate out-of-sequence progress. Corrections must adhere to established CPM definition of predecessor-successor relationships using appropriate CPM logic ties to ensure accurate schedule calculations and maintain schedule integrity.

b. b. Justification: If logic changes are not implemented, provide a written justification for maintaining the existing sequencing and obtain approval before submitting the updated project schedule.

c. c. Ongoing Monitoring: Out-of-sequence progress that has been approved to remain unchanged but persists beyond one schedule update cycle must be corrected through logic revisions, subject to approval by the Contracting Officer.

d. d. Documentation & Reporting: All occurrences of out-of-sequence progress and any related logic changes must be addressed in the Narrative Report and discussed in Periodic Schedule Update Meetings.

3.3.13 Added and Deleted Activities

Activities may be added or deleted in the schedule only as necessary to maintain an accurate and up-to-date project schedule and must be approved by the Contracting Officer.

All additions and deletions must be identified in the Narrative Report, including the Activity ID, Activity Name, and a justification for the change. Activity IDs and Activity Names cannot be changed to represent a different scope of work; any such changes must be submitted as new activities and require approval before implementation.

All schedule modifications are subject to government review and acceptance to ensure compliance with contract requirements and to maintain schedule integrity.

3.3.14 Original Durations

Activity Original Durations (OD) must be reasonable to perform the work item. OD changes are prohibited unless justification is provided and approved by the Contracting Officer.

3.3.15 Leads, Lags, and Start to Finish Relationships

Lags must be reasonable as determined by the Government. They must not be used to replace realistic activity durations, artificially create or absorb float, or circumvent proper schedule logic. All lags must be clearly identified in the Narrative Report, along with a justification explaining their necessity.

a. Leads (negative lags) are prohibited.

b. Start to Finish (SF) relationships are prohibited.

3.3.16 Retained Logic

Schedule calculations must retain the logic between predecessors and successors ("retained logic" mode) even when the successor activity(s) starts and the predecessor activity(s) has not finished (out-of-sequence progress). Software features that in effect sever the tie between predecessor and successor activities when the successor has started and the predecessor logic is not satisfied ("progress override") are not be

allowed.

3.3.17 Percent Complete

Update the percent complete for each activity started, based on the realistic assessment of earned value. Active activities are expected to experience work effort each day once started with no interruptions or suspended effort. If an activity's remaining duration accounts for suspended work periods, it indicates an insufficient level of detail and is prohibited.

Activities which are complete but for remaining minor punch list work and which do not restrain the initiation of successor activities may be declared 100 percent complete to allow for proper schedule management. Budgeted cost of activity is to be reduced by the amount needed to correct deficiency. The activity referenced in paragraph COST LOADING of CLOSEOUT ACTIVITIES must have its budgeted cost increased by this amount.

3.3.18 Remaining Duration

Update the remaining duration for each activity based on the realistic number of workdays required to complete the remaining scope. Remaining durations are an independent estimates and are not required to mathematically correlate with the percentage complete, as percent complete is based on earned cost rather than time elapsed.

The remaining duration for unstarted activities must not be less than the original duration, ensuring that planned work reflects a realistic execution timeline. Adjustments to remaining duration for in-progress activities must be based on observable progress and updated field conditions rather than arbitrary reductions or percentage-based calculations.

3.3.19 Cost Loading of Closeout Activities

Cost load the "Government's Punch List Correction from Pre-Final Inspection " activity(ies) not less than 1 percent of the present contract value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

3.3.19.1 As-Built Drawings

If there is no separate contract line item (CLIN) for as-built drawings, cost load the "Government Review of as-built drawings" activity not less than $35,000 or 1 percent of the present contract value, which ever is greater, up to $200,000. Activity will be declared 100 percent complete upon the Government's approval.

3.3.19.2 O & M Manuals

Cost load the "Government Review of O & M manuals" activity not less than $20,000. Activity will be declared 100 percent complete upon the Government's approval of all O & M manuals.

3.3.20 Early Completion Schedule and the Right to Finish Early

An Early Completion Schedule is an Initial Project Schedule (IPS) that

indicates all scope of the required contract work will be completed before the contractually required completion date.

a. No IPS indicating an Early Completion will be accepted without being fully resource-loaded (including crew sizes and hours) and the Government agreeing that the schedule is reasonable and achievable.

b. The Government is under no obligation to accelerate work items it is responsible for to ensure that the early completion is met nor is it responsible to modify incremental funding (if applicable) for the project to meet the contractor's accelerated work.

3.4 PROJECT SCHEDULE SUBMISSIONS

Provide the submissions as described below. The data reports and network diagrams required for each submission are contained in paragraph SUBMISSION REQUIREMENTS. If the Contractor fails or refuses to furnish the information and schedule updates as set forth herein, the Contractor will be deemed not to have provided an estimate upon which a progress payment can be made.

Review comments made by the Government on the schedule(s) do not relieve the Contractor from compliance with requirements of the Contract Documents.

Disapproved schedules must be resubmitted within 7 calendar days, incorporating all Government review comments. The resubmission must include a detailed response outlining the actions taken to address each comment

3.4.1 Preliminary Project Schedule Submission

Within 15 calendar days after the NTP is acknowledged, submit the Preliminary Project Schedule defining the planned operations detailed for the first 90 calendar days for approval. The approved Preliminary Project Schedule will be used for payment purposes not to exceed 90 calendar days after NTP. After this period, it will no longer be valid for payment. Completely cost load the Preliminary Project Schedule to balance the contract award CLINS shown on the Price Schedule. The Preliminary Project Schedule may be summary in nature for the remaining performance period. It must be early start and late finish constrained and logically tied as specified. The Preliminary Project Schedule forms the basis for the Initial Project Schedule specified herein and must include all of the required plan and program preparations, submissions and approvals identified in the contract (for example, Quality Control Plan, Safety Plan, and Environmental Protection Plan) as well as permitting activities, design review conference activities, and other non-construction activities intended to occur within the first 90 calendar days. Government acceptance of the associated design package(s) and all other specified Program and Plan approvals must occur prior to any planned construction activities. Code activities that are summary in nature, after the first 90 calendar days, with the following activity codes: Bid Item (CLIN) code (BIDI), Responsibility Code (RESP), and Feature of Work code (FOW)..

3.4.2 Initial Project Schedule Submission

Submit the Initial Project Schedule for approval within 42 calendar days after notice to proceed is issued. The schedule must demonstrate a reasonable and realistic sequence of activities which represent all work through the entire contract performance period. No payment will be made

for work items not fully detailed in the Project Schedule.

3.4.3 Periodic Schedule Updates

The Project Schedule must be updated regularly, at a minimum of once per month. The contractor shall provide a draft Periodic Schedule Update for review during the Periodic Schedule Update Meetings, as outlined in Section 3.6.1.

a. Pre-Update Status Review: Before performing the Periodic Schedule Update, the contractor must submit a Schedule Status Listing to the Government for review and approval. This listing must include columns for each schedule Activity ID, Activity Name, Calendar, Original Duration, Previous Remaining Duration, Current Remaining Duration, Previous Physical % Complete, Current Physical % Complete, Actual Start, and Actual Finish Dates. The Actual Start and Actual Finish Dates must match those recorded in the Contractor's Quality Control Reports.

b. Draft Periodic Schedule Update Submission: The contractor must submit the draft Periodic Schedule Update to the Government within two (2) workdays of the data date. This submission must include the P6 .xer file and a draft Narrative Report, prepared in accordance with Section 3.5.2.

c. Government Review Period: Upon receipt of the draft submission, the Government has three (3) workdays to review the Periodic Schedule Update.

d. Periodic Schedule Update Meeting: The contractor is responsible for coordinating and scheduling the Periodic Schedule Update Meeting within five (5) workdays of the data date. The contractor must organize the meeting, prepare detailed meeting minutes, and distribute them to all stakeholders following the meeting.

e. Final Schedule Update Submission: The contractor must incorporate all schedule changes discussed in the Periodic Schedule Update Meeting and formally submit the final Periodic Schedule Update within four (4) workdays after the Periodic Schedule Update Meeting. This submission is required for government acceptance per Section 3.5 and must be completed before the progress payment can be processed per Section 3.2.

3.5 SUBMISSION REQUIREMENTS

Submit the following items for the Preliminary Schedule, Initial Schedule, and every Periodic Schedule Update throughout the life of the project:

3.5.1 Electronic Scheduling Data

Provide submittals in electronic format through RMS per transmittal to the Contracting Officer the current project schedule and all previously submitted schedules in the format of the scheduling software (e.g. .xer). Also include the Narrative Report and all required Schedule Reports. The electronic file is to identify the type of schedule (Preliminary, Initial, Update), full contract number, Data Date and schedule file name. Each schedule must have a unique file name and use project specific settings. The file naming convention is to be determined at SEKO meeting.

3.5.2 Narrative Report

Provide a Narrative Report with each schedule submission. The Schedule Narrative Report is a "stand alone" report in PDF or Word format. Title the report "Schedule Narrative Report". Show the Report date, schedule name, and contract number on the first page. Indicate who is responsible for the schedule update and narrative report. Provide a brief description of the Project Scope. The Narrative Report is expected to communicate to the Government the thorough analysis of the schedule output and the plans to compensate for problems, either current or potential, which are revealed through that analysis. Include the following information at a minimum in the Narrative Report:

a. Identify and discuss the work scheduled to start in the next update period.

b. A description of activities along the two most critical paths where the total float is less than or equal to 20 work days.

c. Describe current and anticipated problem areas, delaying factors and their impact, and an explanation of corrective actions taken or required to be taken. Identify activities affected by Activity ID and Activity Name.

d. Identify and explain why activities, based on their calculated late dates, should have either started or finished during the update period but did not.

e. Identify and discuss all schedule changes by activity ID and activity name including what specifically was changed and why the change was needed. Include at a minimum new and deleted activities, logic changes, duration changes, calendar changes, lag changes, cost changes, and actual start and finish date changes.

f. Identify and discuss out-of-sequence work.

g. If the longest path changes from a monthly update to the next, provide an explanation of the factors driving the change.

h. Identify and discuss all stagnant activities.

i. Discuss the status of utility shutdowns, including steam and chilled water, and required notices.

j. Provide a detailed response to Government review comments from the previous update, clearly stating the actions taken to address each comment. Failure to adequately respond to Government review comments may result in schedule disapproval and withholding of progress payments per section 3.2.3.

3.5.3 Schedule Reports

The format, filtering, organizing, and sorting for each schedule report will be as directed by the Contracting Officer. Typically, reports contain Activity ID, Activity Name, Calendar, Original Duration, Remaining Duration, Physical % Complete, Early Start Date, Early Finish Date, Late Start Date, Late Finish Date, Actual Start Date, Actual Finish Date, and Total Float. Provide the reports electronically in .pdf format. The

following lists typical reports that will be requested:

3.5.3.1 Activity Report

List of all activities sorted according to activity number.

3.5.3.2 Logic Report

List of detailed predecessor and successor activities for every activity in ascending order by activity number.

3.5.3.3 Total Float Report

A list of all incomplete activities sorted in ascending order of total float. List activities which have the same amount of total float in ascending order of Early Start Dates. Do not show completed activities on this report.

3.5.3.4 Earnings Report by CLIN

A compilation of the Total Earnings on the project from the NTP to the data date, which reflects the earnings of activities based on the agreements made in the schedule update meeting defined herein. Provided a complete schedule update has been furnished, this report serves as the basis of determining progress payments. Group activities by CLIN number and sort by activity number. Provide a total CLIN percent earned value, CLIN percent complete, and project percent complete. For each individual CLIN and project as a whole, provide a total Cost % Complete. The printed report must contain the following for each activity: the Activity Number, Activity Description, Original Budgeted Amount, Earnings to Date, Earnings This Period, and Cost % Complete.

3.5.3.5 Scheduling and Leveling Report

Provide a Scheduling/Leveling Report generated from the current project schedule being submitted.

3.5.4 Network Diagram

The Network Diagram is required for the Preliminary, Initial and Periodic Updates. Depict and display the order and interdependence of activities and the sequence in which the work is to be accomplished. The Contracting Officer will use, but is not limited to, the following conditions to review compliance with this paragraph:

3.5.4.1 Continuous Flow

Show a continuous flow from left to right with no arrows from right to left. Show the activity number, description, duration, and estimated earned value on the diagram.

3.5.4.2 Project Milestone Dates

Show dates on the diagram for start of project, any contract required interim completion dates, and contract completion dates.

3.5.4.3 Critical Path

Show all activities on the critical path. The critical path is defined as

the longest path.

3.5.4.4 [Enter Appropriate Subpart Title Here]Banding

Organize the project using the WBS to assist in the understanding of the project sequence, following the guidance found in AACE 33R-15 DEVELOPING THE PROJECT WORK BREAKDOWN STRUCTURE. This flow will arrange group activities by major elements of work in a hierarchical level of increasing detail that decomposes the project elements into the lowest manageable level practical.

3.5.4.5 Cash Flow / Schedule Variance Control (SVC) Diagram

With each schedule submission, provide a SVC diagram showing 1) Cash Flow S-Curves indicating planned project cost based on projected early and late activity finish dates, and 2) Actual costs to-date.

The SVC diagram must also include a table displaying, for each month of the project's duration, the specific dollar values for Planned Early Costs, Planned Late Costs, Planned Average Costs, and Actual Costs. Additionally, the table must report the Planned Cost % Complete and Actual Cost % Complete for each month.

The SVC diagram shall incorporate all current project costs at the time of each submission, including base contract costs, exercised options, and modifications. Any modifications that impact contract time must also be reflected in the diagram to ensure an accurate representation of project cash flow and schedule variance.

3.5.4.6 Feature of Work (FOW) Grouping

With each schedule submission, provide a listing of all schedule activities grouped and sorted by their assigned Feature of Work (FOW) code, in accordance with Section 3.3.6.8. The listing must display the FOW ID/code number and title, sorted alphabetically by the ID/code. This listing is intended to facilitate the identification of any missing activities corresponding to the Definable Feature of Work (DFOW) listing in the contractor's Quality Control Plan.

3.5.4.7 Approved Definable Feature of Work (DFOW) listing from QC Plan

Attach the most recently approved Definable Feature of Work (DFOW) listing from the Quality Control Plan as part of each schedule submission. This ensures alignment between the project schedule and the contractor's approved quality control framework.

3.6 PERIODIC SCHEDULE UPDATE

3.6.1 Periodic Schedule Update Meetings

Conduct periodic schedule update meetings for the purpose of reviewing the proposed Periodic Schedule Update, Narrative Report, Schedule Reports, and progress payment. Conduct meetings at least monthly within five days of the proposed schedule data date. Provide a computer with the scheduling software loaded and a projector which allows all meeting participants to view the proposed schedule during the meeting. The Contractor's authorized scheduler must organize, group, sort, filter, perform schedule revisions as needed and review functions as requested by the Contractor and/or Government. The Contractor scheduler must attend in person unless

virtual attendance is approved by Contracting Officer. The meeting is a working interactive exchange which allows the Government and Contractor the opportunity to review the updated schedule on a real time and interactive basis. The meeting will last no longer than 8 hours. Provide a draft of the proposed narrative report and schedule data file to the Government a minimum of three (3) workdays in advance of the meeting. The Contractor's Project Manager must attend the meeting with the authorized representative of the Contracting Officer. Superintendents, foremen and major subcontractors must attend the meeting as required to discuss the project schedule and work. Following the schedule update meeting, make corrections to the schedule and resubmit, following the required schedule naming convention. Include only those changes approved by the Government in the submission and invoice for payment.

3.6.2 Update Submission Following Progress Meeting

Submit the complete Periodic Schedule Update of the Project Schedule containing all approved progress, revisions, and adjustments, pursuant to paragraph SUBMISSION REQUIREMENTS not later than 4 workdays after the periodic schedule update meeting.

3.7 WEEKLY PROGRESS MEETINGS

Conduct a weekly meeting with the Government (or as otherwise mutually agreed to) between the meetings described in paragraph entitled PERIODIC SCHEDULE UPDATE MEETINGS for the purpose of jointly reviewing the actual progress of the project as compared to the as planned progress and to review planned activities for the upcoming two weeks. Use the current approved schedule update for the purposes of this meeting and for the production and review of reports. At the weekly progress meeting, address the status of RFIs, RFPs and Submittals.

3.8 REQUESTS FOR TIME EXTENSIONS

ASCE 67-17 provides delay analysis guidelines. If there is a conflict between the contract and guidance provided in ASCE 67-17, the contract will govern. Provide justification of delay to the Contracting Officer in accordance with the contract provisions and clauses for approval, within 10 days of a delay occuring. Also prepare a time impact analysis for each Government request for proposal (RFP)regardless if the contractor is requesting a time extension. All activities included in a Time Impact Analysis (TIA) must meet the level of detail and logic requirements outlined in this specification. The fragmentary network (fragnet) developed for the TIA may only be inserted into the project schedule after Government review and acceptance to ensure accuracy and compliance with contract scheduling requirements.

3.8.1 Justification of Delay

Provide a description of the event(s) that caused the delay and/or impact to the work. As part of the description, identify all schedule activities impacted. Show that the event that caused the delay/impact was the responsibility of the Government. Provide a time impact analysis that demonstrates the effects of the delay or impact on the project completion date, or interim completion date(s). Evaluate multiple impacts chronologically; each with its own justification of delay. With multiple impacts consider any concurrency of delay. A time extension and the schedule fragnet becomes part of the project schedule and all future schedule updates upon approval by the Contracting Officer.

3.8.2 Changes That Do Not Cause Delay

If it is determined that a change does not constitute a schedule time extension, a fragnet as described in paragraph FRAGMENTARY NETWORK, is to be created and inserted into the Project schedule.

3.8.3 Time Impact Analysis (Prospective Analysis)

Prepare a time impact analysis for approval by the Contracting Officer based on industry recommended practice AACE 52R-06. Utilize a copy of the last approved schedule prior to the first day of the impact or delay for the time impact analysis. If Contracting Officer determines the time frame between the last approved schedule and the first day of impact is too great, prepare an interim updated schedule to perform the time impact analysis. Unless approved by the Contracting Officer, no other changes may be incorporated into the schedule being used to justify the time impact.

3.8.4 Forensic Schedule Analysis (Retrospective Analysis)

Prepare an analysis for approval by the Contracting Officer based on industry standard AACE 29R-03.

3.8.5 Fragmentary Network (Fragnet)

Prepare a proposed fragnet for time impact analysis consisting of a sequence of new activities that are proposed to be added to the project schedule to demonstrate the influence of the delay or impact to the project's contractual dates. Clearly show how the proposed fragnet is to be tied into the project schedule including all predecessors and successors to the fragnet activities. The proposed fragnet must be approved by the Contracting Officer prior to incorporation into the project schedule.

3.8.6 Time Extension

The Contracting Officer must approve the Justification of Delay including the time impact analysis before a time extension will be granted. No time extension will be granted unless the delay consumes all available Project Float and extends the projected finish date ("End Project" milestone) beyond the Contract Completion Date. The time extension will be in calendar days.

Actual delays that are found to be caused by the Contractor's own actions, which result in a calculated schedule delay, are not a cause for an extension to the performance period, completion date, or any interim milestone date.

3.8.7 Impact to Early Completion Schedule

No extended overhead will be paid for delay prior to the original Contract Completion Date for an Early Completion IPS unless the Contractor actually performed work in accordance with that Early Completion Schedule. The Contractor must show that an early completion was achievable had it not been for the impact.

3.9 FAILURE TO ACHIEVE PROGRESS

Should the progress fall behind the approved project schedule for reasons other than those that are excusable within the terms of the contract, the Contracting Officer may require provision of a written recovery plan for approval. The plan must detail how progress will be made-up to include which activities will be accelerated by adding additional crews, longer work hours, extra workdays, etc.

3.9.1 Artificially Improving Progress

Artificially improving progress by means such as, but not limited to, revising the schedule logic, modifying, or adding constraints, shortening activity durations, or changing calendars in the project schedule is prohibited. Indicate assumptions made and the basis for any logic, constraint, duration, and calendar changes used in the creation of the recovery plan. Any additional resources, manpower, or daily and weekly work hour changes proposed in the recovery plan must be evident at the work site and documented in the daily report along with the Schedule Narrative Report.

3.9.2 Failure to Perform

Failure to perform work and maintain progress in accordance with the supplemental recovery plan may result in an interim and final unsatisfactory performance rating and may result in corrective action directed by the Contracting Officer pursuant to FAR 52.236-15 Schedules for Construction Contracts, FAR 52.249-10 Default (Fixed-Price Construction), and other contract provisions.

3.9.3 Recovery Schedule

Should the Contracting Officer find it necessary, submit a recovery schedule pursuant to FAR 52.236-15 Schedules for Construction Contracts.

3.10 OWNERSHIP OF FLOAT

Except for the provision given in the paragraph IMPACT TO EARLY COMPLETION SCHEDULE, float available in the schedule, at any time, belongs to the Project and is available for Contractor and Government use. This includes activity and project float. Activity float is the number of workdays that an activity can be delayed without causing a delay to the "End Project" finish milestone. Project float (if applicable) is the number of work days between the projected early finish and the "End Project" finish.

Methods to limit or control ownership of float, such as unnecessary logic restraints, the inclusion of lag and constraint assignments to activities and logic, and excessive activity durations are not allowed.

3.11 TRANSFER OF SCHEDULE DATA INTO RESIDENT MANAGEMENT SYSTEM

Ensure schedule data is uploaded to RMS. This data is additional supporting data in a form and detail required by the Contracting Officer pursuant to FAR 52.232-5 Payments under Fixed-Price Construction Contracts. The receipt of a proper payment request pursuant to FAR 52.232-27 Prompt Payment for Construction Contracts is contingent upon the Government receiving both acceptable and approvable hard copies and matching electronic versions of the application for progress payment.

3.12 PRIMAVERA P6 MANDATORY REQUIREMENTS

Ensure Primavera P6 settings provide a schedule capable of fulfilling the requirements of the contract. The following settings are mandatory and required in all schedule submissions to the Government:

a. Activity Codes must be Project Level, not Global or EPS level.

b. Calendars must be Project Level, not Global or Resource level.

c. Activity Duration Types must be set to "Fixed Duration & Units".

d. Percent Complete Types must be set to "Physical".

e. Time Period Admin Preferences must remain the default "8.0 hr/day, 40 hr/week, 172 hr/month, 2000 hr/year". Set Calendar Work Hours/Day to 8.0 Hour days.

f. Set Schedule Option for defining Critical Activities to "Longest Path".

g. Set Schedule Option for defining progressed activities to "Retained Logic".

h. Set up cost loading using a single lump sum non-labor resource. The Price/Unit must be $1/hr, Default Units/Time must be "8h/d", and settings "Auto Compute Actuals" and "Calculate costs from units" un-selected.

i. Activity ID's must not exceed 10 characters.

j. Activity Names must have a verb-noun structure and contain the most defining and detailed description within the first 30 characters.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 00

SUBMITTAL PROCEDURES

08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 DEFINITIONS
1.1.1 Submittal Descriptions (SD)
1.1.2 Approving Authority
1.1.3 Work
1.2 SUBMITTALS
1.3 SUBMITTAL CLASSIFICATION
1.3.1 Government Approved (G)
1.3.1.1 APPROVAL CODES
1.3.2 For Information Only
1.4 PREPARATION
1.4.1 Transmittal Form
1.4.2 Submittal Format
1.4.2.1 Format of SD-01 Preconstruction Submittals
1.4.2.2 Format for SD-02 Shop Drawings
1.4.2.2.1 Drawing Identification
1.4.2.3 Format of SD-03 Product Data
1.4.2.3.1 Product Information
1.4.2.3.2 Standards
1.4.2.3.3 Data Submission
1.4.2.4 Format of SD-04 Samples
1.4.2.4.1 Sample Characteristics
1.4.2.4.2 Sample Incorporation
1.4.2.4.3 Comparison Sample
1.4.2.5 Format of SD-05 Design Data
1.4.2.6 Format of SD-06 Test Reports
1.4.2.7 Format of SD-07 Certificates
1.4.2.8 Format of SD-08 Manufacturer's Instructions
1.4.2.8.1 Standards
1.4.2.9 Format of SD-09 Manufacturer's Field Reports
1.4.2.10 Format of SD-10 Operation and Maintenance Data (O&M)
1.4.2.11 Format of SD-11 Closeout Submittals
1.4.3 Source Drawings for Shop Drawings
1.4.3.1 Source Drawings
1.4.3.2 Terms and Conditions
1.4.4 Electronic File Format
1.5 QUANTITY OF SUBMITTALS
1.5.1 Number of SD-01 Preconstruction Submittal Copies
1.5.2 Number of SD-02 Shop Drawing Copies
1.5.3 Number of SD-03 Product Data Copies
1.5.4 Number of SD-04 Samples
1.5.5 Number of SD-05 Design Data Copies
1.5.6 Number of SD-06 Test Report Copies
1.5.7 Number of SD-07 Certificate Copies
1.5.8 Number of SD-08 Manufacturer's Instructions Copies

1.5.9 Number of SD-09 Manufacturer's Field Report Copies
1.5.10 Number of SD-10 Operation and Maintenance Data Copies
1.5.11 Number of SD-11 Closeout Submittals Copies
1.6 INFORMATION ONLY SUBMITTALS
1.7 PROJECT SUBMITTAL REGISTER
1.7.1 Submittal Management
1.7.2 Preconstruction Use of Submittal Register
1.7.3 Contractor Use of Submittal Register
1.7.4 Approving Authority Use of Submittal Register
1.7.5 Action Codes
1.7.6 Delivery of Copies
1.8 VARIATIONS
1.8.1 Considering Variations
1.8.2 Proposing Variations
1.8.3 Warranting that Variations are Compatible
1.9 SCHEDULING
1.10 GOVERNMENT APPROVING AUTHORITY
1.10.1 Review Notations
1.11 DISAPPROVED SUBMITTALS
1.12 APPROVED SUBMITTALS
1.13 APPROVED SAMPLES

PART 2 PRODUCTS

PART 3 EXECUTION

ATTACHMENTS:

Appendix A - Submittal Register

-- End of Section Table of Contents --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS)concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.1.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.3 SUBMITTAL CLASSIFICATION

1.3.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.3.1.1 APPROVAL CODES

a. Construction Division {"AO" (Area Office), "RO" (Resident Office), or "PO" (Project Office) Reviewer}: An "AO", "RO", or "PO" in column "f" indicates that the submittal review action is by New England District Construction Division. Send all such submittals to the project Resident or Area Engineer, as applicable.

b. Engineering Division {"DO" (District Office) Reviewer): A "DO" on the attached submittal register, column "f" indicates that the submittal review action is by the New England District, Engineering Division, or other organization in the District Office. Send all such submittals to the project Resident or Area Engineer for distribution to the appropriate approving authority.

c. Architect-Engineer Firm {"AE" reviewer): An "AE" on the attached submittal register, column "f" indicates that the submittal review action is by the Architect-Engineer firm associated with the project.

d. Government acceptance is required for submittals with a "G, A" designation."

1.3.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.4 PREPARATION

1.4.1 Transmittal Form

1.4.2 Submittal Format

1.4.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.4.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.

b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Present shop drawings sized 8 1/2 by 11 inches as part of the bound volume for submittals. Present larger drawings in sets. Submit an electronic copy of drawings in PDF format.

1.4.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than two inches on the right-hand side of each sheet for the Government disposition stamp.

1.4.2.3 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it

pertains.

1.4.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

1.4.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.4.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. All submittals shall be bookmarked and bundled per specification section. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.4.2.4 Format of SD-04 Samples

1.4.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

a. Sample of Equipment or Device: Full size.

b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.

c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.

d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.

e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.

f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.

g. Sample Panel: 4 by 4 feet.

h. Sample Installation: 100 square feet.

1.4.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.4.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.4.2.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper. Provide a bound volume for submittals containing numerous pages.

1.4.2.6 Format of SD-06 Test Reports

Provide reports on 8 1/2 by 11 inch paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.4.2.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper. Provide a bound volume for submittals containing numerous pages.

1.4.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section as a complete, bound volume. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.4.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.4.2.9 Format of SD-09 Manufacturer's Field Reports

Provide reports on 8 1/2 by 11 inch paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.4.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.4.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.4.3 Source Drawings for Shop Drawings

1.4.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.4.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection

with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.4.4 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. In addition to the electronic submittal, provide three hard copies of the submittals. Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

1.5 QUANTITY OF SUBMITTALS

1.5.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit three sets of administrative submittals.

1.5.2 Number of SD-02 Shop Drawing Copies

Submit six copies of submittals of shop drawings requiring review and approval by a QC organization. Submit seven copies of shop drawings requiring review and approval by the Contracting Officer.

1.5.3 Number of SD-03 Product Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.5.4 Number of SD-04 Samples

a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.

b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.

c. Submit one sample installation, where directed.

d. Submit one sample of nonsolid materials.

1.5.5 Number of SD-05 Design Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.5.6 Number of SD-06 Test Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings, other than field test results that will be submitted with QC reports.

1.5.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.5.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.5.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC reports.

1.5.10 Number of SD-10 Operation and Maintenance Data Copies

Submit two paper copies and two copies of electronic file O&M data to the Contracting Officer for review and approval.

1.5.11 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit four paper sets of adminitrative submittals in addition to the required electronic format submittal above in paragraph "Electronic Fiel Format".

1.6 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Approval of the Contracting Officer is not required on information only

submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.7 PROJECT SUBMITTAL REGISTER

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.7.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.7.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.7.3 Contractor Use of Submittal Register

Update the following fields with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.7.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.7.5 Action Codes

1.7.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.8 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.8.1 Considering Variations

Discussion of variations with the Contracting Officer before submission of a variation submittal will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the

Government to require rejection and removal of such work at no additional cost to the Government.

1.8.2 Proposing Variations

When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk."

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.8.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals.

a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.

b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 21 days after the date of submission.

1.10 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

a. Note the date on which the submittal was received.

b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.

c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. Three copies of the submittal will be retained by the Contracting Officer and two copies of the submittal will be returned to the Contractor.

1.10.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.

b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.

c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.

d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.11 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of

such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.12 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained with in each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.13 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 21

CONSTRUCTION PHASE BIM, CIM, AND CAD REQUIREMENTS

02/20

PART 1 GENERAL

1.1 DEFINITIONS
1.1.1 BIM - Building Information Modeling
1.1.2 CIM - Civil Information Modeling
1.1.3 BIM Execution Plan (BxP)
1.1.4 CAD - Computer Aided Drafting
1.1.5 Model
1.1.6 Model Element
1.1.7 Facility/Site Data
1.1.8 IFC - Industry Foundation Class
1.1.9 Level of Development (LOD)
1.1.10 Record Drawings
1.1.11 Record Model
1.1.12 Construction Drawings
1.1.13 Construction Model
1.2 REFERENCES
1.2.1 Related Requirements:
1.2.2 Referenced Standards:
1.3 SUBMITTALS
1.4 DELIVERABLE FILES SOFTWARE VERSIONS
1.5 QUALITY CONTROL
1.5.1 Model Standards Checks
1.5.2 CAD Standards Checks
1.5.3 Visual Checks
1.5.4 Deliverable File Software Versions
1.6 CONTRACTOR COORDINATION

PART 2 PRODUCTS

2.1 GOVERNMENT FURNISHED MATERIAL
2.1.1 Variation with Contract Drawings
2.1.2 Model Completion and Quality
2.1.3 Data Loss, Corruption, and Error
2.2 CONSTRUCTION FILE CONTENT
2.2.1 IFC Support
2.2.2 Model Level of Development (LOD)
2.3 CONSTRUCTION MODEL
2.4 BIM EXECUTION PLAN
2.4.1 BIM/CAD PACKAGE
2.5 MINIMUM MODELING AND DATA REQUIREMENTS
2.5.1 VA Facility Management Data
2.5.2 Minimum Modeling Matrix
2.5.3 Additional Requirements
2.5.3.1 Classification
2.5.3.2 Spatial Data and Objects, and Data

2.5.3.3 Schedules
2.5.3.4 Dimensioning and Tolerances
2.5.3.5 Sections and Details
2.5.3.6 Legends
2.5.3.7 Drawing Indices
2.5.3.8 Project Equipment List
2.5.3.9 Room Contents List with Room Equipment Data
2.5.3.10 AS-Built FM Model
2.5.3.11 BIM Objects and Categories
2.5.3.12 Laser Scan Data
2.5.3.13 Additional VA BIM Standards
2.6 OWNERSHIP AND RIGHTS IN DATA

PART 3 EXECUTION

3.1 BIM EXECUTION PLAN (BXP)
3.2 CONSTRUCTION COORDINATION MEETING
3.3 INTERIM SUBMITTALS
3.3.1 General Document and File Review Requirements
3.3.2 Interim Construction BIM Model
3.3.3 Interim Construction CIM Model
3.3.4 Interim Construction CAD Drawings
3.3.5 IFC Coordination View and COBie
3.4 CONSTRUCTION CLOSEOUT SUBMITTALS
3.4.1 Final Construction BIM Model
3.4.2 Final Construction CIM Model
3.4.3 Final Construction CAD Drawings
3.4.4 VA Facility Management Data

-- End of Section Table of Contents --

SECTION 01 33 21

CONSTRUCTION PHASE BIM, CIM, AND CAD REQUIREMENTS
02/20

PART 1 GENERAL

1.1 DEFINITIONS

The following definitions apply specifically to the project BIM, CIM, and CAD requirements.

1.1.1 BIM - Building Information Modeling

Building Information Modeling (BIM) is a process involving the generation and management of digital representations of physical and functional characteristics of a facility. As used in the context of this specification, the BIM model refers to the digital model file(s) used to create the Contract and Construction As-built Drawings.

The A/E is employing Autodesk Revit software and its various discipline components for BIM modeling and producing the project Drawings.

1.1.2 CIM - Civil Information Modeling

Civil Information Modeling (CIM) refers to the process of creating digital representations within a civil model(s) for project site features and infrastructure. As used in the context of this specification, the CIM model refers to the digital civil model file(s) that used to create Contract and Construction drawings that reflect As-built site conditions.

1.1.3 BIM Execution Plan (BxP)

A BIM project management plan internal to the working teams, which outlines the management roles, division of work between Prime AE/ consultants or general contractor/subcontractors for modeling responsibilities, standards, software versions, etc. used for model creation on a project.

1.1.4 CAD - Computer Aided Drafting

Digital drawing files (e.g., plans, elevations, sections, schedules, details, etc.) derived from the BIM and CIM model(s) or drafted directly.

The A/E will furnish project drawings in Autodesk AutoCAD CAD software for producing the Contract drawing sheets, other than those produced by Revit.

1.1.5 Model

A digital representation of physical and functional characteristics of a facility or a part thereof, comprised of Model Elements with Facility/Site Data.

1.1.6 Model Element

A self-contained element with a unique identification, whose behavior and

properties are defined by Facility/Site Data and software processes. Model Elements can represent a physical entity such as a pump or a concrete wall and range from the simple to the complex.

1.1.7 Facility/Site Data

The non-graphical information attached to objects in the Model that defines various characteristics of the object. Facility/Site Data can include properties such as parametric values that drive physical sizes, material definitions, and characteristics (e.g. wood, metal), manufacturer data, industry standards (e.g. AISC steel properties), and project identification numbers. Facility/Site Data can also define supplementary physical entities that are not shown graphically in the Model, such as insulation around a duct, hardware on a door, content of conduit, or transformer properties.

1.1.8 IFC - Industry Foundation Class

A data model standard and file format used for the exchange of BIM data; for more information visit: www.buildingsmart-tech.org. Note: In the context of this specification section, IFC does not mean "Issued For Construction".

1.1.9 Level of Development (LOD)

Per the BIM Forum's Level of Development Specification: https://bimforum.org/resource/lod-level-of-development-lod-specification/

"…The Level of Development (LOD) Specification is a reference that enables practitioners in the AEC Industry to specify and articulate with a high level of clarity the content and reliability of Building Information Models (BIMs) at various stages in the design and construction process…"

Design models were developed to LOD 300 for Building Service Equipment. Exceptions were made for select elements where modeling to LOD 300 would result in excessive model density, reduced model legibility, or provide no additional benefit to design coordination. Such elements may be modeled to LOD 200 or 100 which was sufficient to convey the required size, location, orientation, and design intent.

Construction models maintained by the Contractor shall comply with the applicable Level of Development (LOD) requirements of the VA BIM Manual, Version 2.2.

1.1.10 Record Drawings

The Record Drawings are produced from the Record Model(s). The AE is responsible for conforming the Record drawings to the actual constructed building (the as-built). The Record Drawings will be updated on a regular basis to keep current with construction changes and include any associated plans, elevations, details, shop drawings, or other information pertinent for Facility Management. Construction details, or construction information that is not useful for managing and maintaining the facility (such as concrete pour sections, construction sequencing, etc.), will not be included in the Record Drawings. The GC is responsible to provide the A/E access to the evolving Construction Drawings and record of all field changes.

1.1.11 Record Model

The Record Model is the Design-Intent Model conformed to the actual constructed building (the as-built) and this responsibility of the AE. After construction is complete, the Record Model will be the Engineering Record of Construction used for on-going Facility Management. The Record Model will be updated on a regular basis to keep current with construction changes. The GC is responsible to provide the A/E access to the evolving Construction Models and record of all field changes. The GC is to coordinate with the AE and the VA/USACE to establish a Common Data Environment (CDE) to which all stakeholders have access to authored files and data sets.

1.1.12 Construction Drawings

The Construction drawings are produced from the Construction model(s). They do not include annotations from the as-built drawings.

1.1.13 Construction Model

The Construction Models (used for construction and reflecting the actual constructed facility at the end of the project) will be comprised of construction models and field changes, including, (but not limited to) RFIs, fabrication information, revision addenda, and construction change directives (CCD). The GC is responsible for delivering their construction BIMs to VA. This includes updated construction phase facility/site data for components.

1.2 REFERENCES

1.2.1 Related Requirements:

Division 1 section: 01 78 23, OPERATION AND MAINTENANCE DATA

1.2.2 Referenced Standards:

The VA BIM Standards described in the VA BIM Manual V2.2 or latest version will be the authoritative source for this project. All VA BIM information is located on the technical information library website.

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL INSTITUTE OF BUILDING SCIENCES (NIBS)

NCS V6	(2016) National CAD Standards Version 6 https://www.nationalcadstandard.org/ncs6/

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA BIM STANDARD V2.2	(2023) PG-18-13 Veterans Affairs BIM STANDARD

https://vatilms.va.gov/vatilms/reports/PG-18-13

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation;

submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES and VA BIM Standard Requirements:

SD-01 Preconstruction Submittals

BIM Execution Plan (BXP); G

SD-05 Design Data

BxP Review Demonstration

Interim Construction CAD Drawings; G

IFC Coordination View; G

3-D Interactive View; G

Submitted Files List; G

Construction Coordination Report

Construction Model

Interim Construction BIM Model; G

Interim Construction CIM Model; G

SD-07 Certificates

Contractor Quality Control Certification; G

SD-11 Closeout Submittals

VA Facility Management Data; G

Final Construction BIM Model; G

Final Construction CIM Model; G

Final Construction CAD Drawings; G

1.4 DELIVERABLE FILES SOFTWARE VERSIONS

Deliver Construction, As-Built FM BIM models in Autodesk Revit (RVT) software format, Construction CIM model files in Autodesk Civil 3D file format (DWG), and Construction CAD drawing files in Autodesk AutoCAD File accuracy (DWG). Interactive 3d model file to be delivered in Navisworks (NWD, NWC). For additional deliverable requirements see 3.9 in VA BIM STANDARD V2.2. Confirm with the Medical Center Chief Engineer for any variance in software versions of delivery.

1.5 QUALITY CONTROL

Implement quality control measures for the models and files, including but not limited to, the following activities:

1.5.1 Model Standards Checks

Ensure that the models have no undefined, incorrectly defined, or duplicated elements. Take corrective action to correct non-compliant elements. Provide the Government with detailed justification and request Government acceptance for any non-compliant element which the Contractor proposes to be allowed to remain in the models.

1.5.2 CAD Standards Checks

Use software content auditing tools to ensure that CAD drawing standards including fonts, dimensions, line styles, layers, and other formatting issues are compliant according to the National CAD Standards Version 6 (NCS V6). Copies of these documents are available at: https://www.nationalcadstandard.org/ncs6/

1.5.3 Visual Checks

Perform visual checks to ensure the files reflect recorded construction conditions and that there are no unintended elements and/or artifacts in the models and drawings.

1.5.4 Deliverable File Software Versions

Deliver Construction As-Built BIM models in Autodesk Revit, Construction As-Built CIM model files in Autodesk Civil 3D software format, and Construction As-Built CAD drawing files in Autodesk AutoCad File format. Use the same version, format provided in the Govt Furnished Material (GFM), or the version documented and approved in your BIM Execution Plan (B xP).

1.6 CONTRACTOR COORDINATION

Provide digital file updates to subcontractors. Also provide drawing files produced from the model in the format(s) required by the subcontractors. Integration of subcontractor models shall be an ongoing process during construction.

Establish and maintain a schedule of BIM/CIM/CAD coordination meetings for subcontractors and field personnel to ensure that all changes and corrections are incorporated into the as-built drawings and Construction model files.

BIM/CIM/CAD coordination meetings should be regularly held (generally weekly) to fully coordinate work. Prior to coordination meeting an interference report will be issued by the Team model Manager. Coordination meetings are to be followed up with a coordination report. The GC after each critical coordination meeting during construction until construction is complete, will provide VA with a Design or Construction Management Progress Summary Report of active coordination issues for VA Quality Assurance review, along with a trending graph showing progress.

Any deviations that impinge on space reservations as noted in VA BIM Standards must be approved by VA in writing.

Virtual Mockups The GC must use virtual mockups of all locations where analyzing the constructability of a system or assembly needs a very high level of detail (e.g. complex façade and roof system interfaces that are geometrically challenging, form work, glazing, tie-back, masonry,

flashing, etc.) in order to increase planning, sequencing, and coordination of the installation. For complicated systems with several trade subcontractors and materials, an image-based sequence of work (step-by-step instruction formatting) must be distributed to the trades and other team members.

The GC must provide access on demand to VA Senior/Resident Engineer to the models and to observe the Construction Coordination effort.

PART 2 PRODUCTS

2.1 GOVERNMENT FURNISHED MATERIAL

a. The Government will provide BIM, CIM Models, and CAD files as Government Furnished Material (GFM) for the Contractor's use. The GFM may include Autodesk Revit, Civil 3d, and AutoCAD digital files created for producing the contract drawings. The information and level of detail contained in the GFM shall be maintained in the Construction BIM/CIM models and Construction CAD drawing files. The Construction CAD files are to be derived from the Construction models.

b. The Contractor has the option of preparing their own BIM/CIM models and CAD drawings as a basis for Construction model(s), Construction CAD drawings, and interim submittals. If so, the models must incorporate the same level of detail, properties, functionality, and format that is included in the GFM.

2.1.1 Variation with Contract Drawings

The Government makes no warranty that the GFM models and CAD files are complete or fully consistent with the contract drawings. The digital BIM/CIM and CAD files provided are not part of the contract documents. If there is any discrepancy between the BIM/CIM models, the CAD drawings, and the contract drawings, the contract drawings shall govern. The Government has no responsibility to modify any GFMs due to changes in the design that occur after award.

2.1.2 Model Completion and Quality

The Government makes no guarantee that the BIM/CIM models and CAD files provide the level of completeness or quality required for closeout submittal. It is the Contractor's responsibility to complete the models and facility data to completely and correctly represent the as-built conditions of the facility and the site. At a minimum, the Contractor will provide a Level of Development in accordance with the GFM or with the USACE Minimum Modeling Matrix (M3) V1.3 20140919 (at https://cadbimcenter.erdc.dren.mil/DownloadFile.aspx?ID=121), whichever provides the greater LOD.

2.1.3 Data Loss, Corruption, and Error

Transfer of GFM files to the Contractor may result in corrupted files resulting in data loss and/or errors. Use of GFM files is at the Contractor's risk. Verify data integrity upon receipt and request a replacement if necessary.

Any adjustment of file structure, format, or software version required to make GFM compatible with the Contractor's computer systems and/or software is the responsibility of the Contractor.

2.2 CONSTRUCTION FILE CONTENT

The interim and final submittal digital Construction files at a minimum shall contain the content below.

2.2.1 IFC Support

The Contractor's BIM application(s) and software(s) must support the Industry Foundation Class (IFC) Standard. Submit any deviations from, or additions to, the IFC property sets for any new spaces, systems, and equipment for Government acceptance. For additional information regarding IFC visit: www.buildingsmart-tech.org.

BIM software authoring platforms used for documentation of VA capital asset projects MUST be:

- Certified to meet the most current Industry Foundation Classes (IFC) Coordination View, and FM Handover View (COBie) when FM Data is needed.
- Within one version of the current authoring software release at project turnover. This is to ensure that the Model is viable for facility management. (Ref VA BIM v2.2/Section 2.3)

2.2.2 Model Level of Development (LOD)

Refer to Sections 2.14.6 and 4.8.1 of the VA BIM STANDARD V2.2 for model development requirements.

2.3 CONSTRUCTION MODEL

As outlined in VA BIM STANDARD V2.2 the Construction model will be updated to keep current with constructions changes and include any associated plans, elevations, details, shop drawings, or other information for facility management. For further requirements see 3.5 of VA BIM STANDARD V2.2.

2.4 BIM EXECUTION PLAN

Submit a BIM Execution Plan (BxP), for work to be performed during the Construction Phase. The BxP shall include a description of the methods of coordination of Contractor and subcontractor models, model elements, and facility/site data. See 1.3.3 of VA BIM STANDARD V2.2. As outlined, utilize the latest version of the USACE BIM PROJECT EXECUTION PLAN (B xP)template to develop and submit for review to the regional VA BIM Manager.Make note that the USACE PxP template does not account for all VA BIM requirements and uses and must be developed further to meet those called for in the VA BIM STANDARD V2.2. The template can be downloaded from the CAD/BIM Technology Center website at: https://cadbimcenter.erdc.dren.mil/default.aspx?p=a&t=1&i=14.

2.4.1 BIM/CAD PACKAGE

a. BIM/CAD Kick-Off Meeting

(1) Conduct an BIM/CAD Kick-Off Meeting prior to submission of the BIM/CAD BxP, within 45 days after Notice to Proceed. Required meeting attendance includes, at a minimum, the DOR, the design

drawing and modeling specialist or BIM/CIM Manager, any sub-contractors who will be performing BIM/CAD trade coordination, and the Geographic District BIM Manager or delegate.

(2) The intent of this meeting is to coordinate the expectations for the BIM/CAD BxP.

b. Submit Construction Model Package and also provide the following items:

(1) BIM/CAD BxP: Provide an electronic copy of the most current approved version of the project BIM/CAD BxP.

(2) Electronic Files: Provide an electronic list (.txt file or similar), of all submitted electronic files including a description, directory, and file name for each file submitted. Identify which files have been produced from the Model and Facility Data. For all sheet files, include a list of the sheet titles and sheet numbers.

(3) BIM/CAD Files: Provide all native BIM/CAD files associated with the contract drawings and associated as-modeled drawings. Update and maintain in compliance with the BIM/CAD formatting, content requirement, and standards, in order to yield a complete and coordinated document package.

(4) Quality Control (QC) Reports: Provide electronic PDFs of all QC reports and checklist utilized to ensure full compliance with the contract requirements and standards.

(5) CAD Exports of BIM-Generated Sheets and Drawings: Provide supplemental 2D CAD exports from the project BIM model as needed to demonstrate compliance with contract requirements.

2.5 MINIMUM MODELING AND DATA REQUIREMENTS

2.5.1 VA Facility Management Data

As outlined in VA BIM STANDARD V2.2 deliver VA Facility Management Data as outlined in the VA FM Data spreadsheet. Can be obtained from https://www.cfm.va.gov/til/bim/VA-FM-Data.xlsx

2.5.2 Minimum Modeling Matrix

Develop a Minimum Modeling Matrix documenting elements included in the facility and site. The Minimum Modeling Matrix shall describe the minimum modeling and data requirements by defining the Level of Development (LOD) and Element Grade in Construction.

2.5.3 Additional Requirements

2.5.3.1 Classification

a. All modeled elements shall include Facility/Site Data as outlined in VA BIM STANDARD V2.2, submittal to include a completed FM Data File. The VA Planner to provide a BIM Export file specific to project from SEPS. See 2.12 of VA BIM STANDARD V2.2.

b. OmniClass Codes must be assigned to all spaces, building equipment, systems, and underground utilities.

c. Spatial Objects must maintain the codes provided in the SEPS BIM Export file. If a new Space Object has been created or does not have an OmniClass code in the PFD, then the appropriate OmniClass code must be applied. Use OmniClass Tables 13 and 14 in combination as appropriate for spaces and OmniClass Tables 21, 22, and 23 for building assets: Table 21 for systems, Table 22 to align modeled technical solutions with specification sections, and Table 23 for building equipment. (Ref VA BIM v2.2/Section 2.11)

2.5.3.2 Spatial Data and Objects, and Data

a. Coding noted in VA Spatial and Equipment Data Spreadsheet from the SEPS BIM File must be maintained.

b. The FM model shall include spatial data, and objects.

c. Spaces must be modeled for each functional room and space and to include actual net square footage and net volume and holding data to develop the room finish schedule including room names and numbers. Include program information to verify design space against programmed space, using this information to validate area quantities.

d. All Spatial objects must be generated with the BIM tool as outlined in 1.4 above and associated with the bounding elements (walls, doors, windows, floors, columns, ceilings.

e. Coding noted in the VA Spatial & Equipment Data Spreadsheet from SEPS must be maintained with the spatial object throughout project development.

2.5.3.3 Schedules

Produce schedules from the facility/site data within the model. Any exceptions should be documented in the BxP and submitted to the VA for review.

2.5.3.4 Dimensioning and Tolerances

All model objects and dimensioning tolerances to meet minimum referenced requirements in the VA BIM STANDARD V2.2.

2.5.3.5 Sections and Details

For those sections and details not derived directly from the model, verify that geometry and data depicting the sections and details are consistent with model elements.

2.5.3.6 Legends

Use model elements to produce representations shown in the legends and match graphical representations shown in plans, sections, and elevations.

2.5.3.7 Drawing Indices

When the BIM platform supports it, drawing indexes should be derived from the model.

2.5.3.8 Project Equipment List

a. As outlined in VA BIM Manual V2.2, submittal to include a final Project Equipment List (PEL) to be furnished to the VA. See 2.12 of the VA BIM Manual V2.2 on the WBDG.

b. Project Equipment List is to be provided to VA as a digital PDF file and an Excel file (.XLSX) and must include the individual equipment names, JSN (Joint Services Number), required data codes, VA Acquisition codes, utility codes for the equipment in each room as provided by original VA SEPS file.

2.5.3.9 Room Contents List with Room Equipment Data

The Project Equipment and Room Contents List is to be provided to VA as a digital PDF file and an Excel file (.XLSX) and must include the individual equipment names, JSN (Joint Services Number), required data codes, VA Acquisition codes, utility codes for the equipment in each room as provided by original VA SEPS file. See 2.14.7 in VA BIM STANDARD V2.2.

2.5.3.10 AS-Built FM Model

a. During construction, the contractor will update the Construction model and any associated plans, elevations, details, shop drawings or other documents to conform to the constructed building.

b. The final locations and sizes of the MEPFT equipment, distribution systems, fire walls and smoke partitions and interior partitions must reflect as-built conditions.

c. The AS-BUILT Construction Models must be submitted within 30 days of substantial Completion.

2.5.3.11 BIM Objects and Categories

All BIM Objects and categories must be named in accordance with FM Data Spreadsheet and VA space & equipment standards noted in Appendix 3 - VA Spatial & Equipment Data from the SEPS BIM Export file must be maintained with the Spatial Object throughout project development so that the specific space object can be tracked.

2.5.3.12 Laser Scan Data

All laser scan data obtained from field conditions prior to, during, or post completion of work shall be submitted within 30 days in native file format.

2.5.3.13 Additional VA BIM Standards

All additional BIM requirements and modeling standards for building structures, systems, equipment, utilities, major vegetation, site conditions, points of connection, and other elements are outlined within the VA BIM GUIDE V2.2, which must be followed.

2.6 OWNERSHIP AND RIGHTS IN DATA

Upon submittal, the Government retains all rights to ownership of all BIM and CIM models and CAD files developed for the project, and as modified by the Contractor, in accordance with FAR Part 27.4 and clauses incorporated

elsewhere in the Contract. The Government may make use of this data following any deliverable.

PART 3 EXECUTION

3.1 BIM EXECUTION PLAN (BXP)

Within thirty (30) days of contract award, submit a digital copy of the BIM Execution Plan. The A/E along with will review the plan and confirm acceptability of the Plan or advise the Contracting Officer as to additional processes or activities necessary to be incorporated. Within thirty (30) days after the acceptance of the Plan, perform a BxP Review Demonstration for clarification, and to verify the functionality of model workflow and processes. If modifications are required, complete the modifications and resubmit the plan and perform subsequent demonstration for Government acceptance.

The Government reserves the right to withhold payment for this portion of the work until the plan is acceptable to the Government. The Government may also withhold payment for unacceptable performance in executing the accepted plan.

3.2 CONSTRUCTION COORDINATION MEETING

Conduct regular (weekly) Construction Coordination meetings IAW VA BIM STANDARD V2.2. Prior to each scheduled Coordination meeting, an updated interference report will be issued to the respective team members by the Construction Contractor's Model Manager to help facilitate and manage issue resolution.

After each meeting submit a Construction Coordination Report. Identify any coordination issues, especially unresolved clashes. If no unresolved clashes exist, state that explicitly.

3.3 INTERIM SUBMITTALS

Within thirty days after the 50 percent construction progress milestone, submit the following items:

3.3.1 General Document and File Review Requirements

For each submittal provide the following:

a. All model files and CAD files in the software versions and formats as required for Interim Construction Submittal.

b. A 3-D Interactive View format of the BIM/CIM models in Autodesk Navisworks or Adobe 3D PDF, Google Earth KMZ, or other format per plan requirements. The file format and/or software versions for reviews may change between submittals.

c. A Submitted Files List. The list should include a description, directory, and file name for each file submitted. For all CAD sheet files, include the sheet number and sheet title. Identify files that have been produced from the submitted models.

d. Contractor Quality Control Certification confirming that all quality control checks as listed in Paragraph QUALITY CONTROL above have been completed. This certification shall be discussed as part of the

interim review process and shall address cross-discipline interferences, if any.

3.3.2 Interim Construction BIM Model

Submit the interim Construction BIM model files reflecting current facility conditions.

3.3.3 Interim Construction CIM Model

Submit the interim Construction CIM model files reflecting current site conditions.

3.3.4 Interim Construction CAD Drawings

Submit the interim Construction CAD files reflecting current site and facility conditions.

3.3.5 IFC Coordination View and COBie

Provide an IFC Coordination View in IFC Express format for each deliverable. Provide exported property set data for IFC supported named building elements.

3.4 CONSTRUCTION CLOSEOUT SUBMITTALS

See Section 01 78 00 CLOSEOUT SUBMITTALS for information regarding required submittals of BIM and CIM models and CAD drawings at completion of construction.

3.4.1 Final Construction BIM Model

Submit Final Construction BIM Model files reflecting final facility conditions.

3.4.2 Final Construction CIM Model

Submit Final Construction CIM model files reflecting final site conditions.

3.4.3 Final Construction CAD Drawings

Submit Final Construction CAD Drawing files reflecting final site and facility conditions.

3.4.4 VA Facility Management Data

Submit Facility Management (FM Data) in COBie format.
ref VA BIM v2.2, section 3.4.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

05/24

safety and occupational health requirements for the protection of Contractor and Government personnel, property, and resources

PART 1 GENERAL

1.1 REFERENCES
1.2 DEFINITIONS
1.2.1 Site Safety and Health Officer (SSHO)
1.2.1.1 Level One SSHO
1.2.1.2 Level Two SSHO
1.2.1.3 Level Three SSHO
1.2.1.4 Alternate SSHO
1.2.2 Competent Person (CP)
1.2.3 Qualified Person (QP)
1.3 SUBMITTALS
1.4 PUBLIC HEALTH EMERGENCIES
1.5 MONTHLY EXPOSURE REPORTS
1.6 REGULATORY REQUIREMENTS
1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS
1.7.1 Site Safety and Health Officer (SSHO)
1.7.1.1 Qualifications of SSHO
1.7.1.2 Duties of SSHO
1.7.1.3 Safety Meetings
1.7.2 Roles and Responsibilities of Prime Contractor and SSHO
1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements
1.7.4 Competent Person for Confined Space Entry
1.7.5 Preconstruction Conference
1.8 ACCIDENT PREVENTION PLAN (APP)
1.8.1 Accident Prevention Plan (APP) - Construction
1.8.2 Names and Qualifications
1.9 ACTIVITY HAZARD ANALYSIS (AHA)
1.9.1 AHA Management
1.9.2 AHA Signature Log
1.10 SITE SAFETY REFERENCE MATERIALS
1.11 EMERGENCY MEDICAL TREATMENT
1.12 NOTIFICATIONS AND REPORTS
1.12.1 Accident Notification
1.12.2 Accident Reports
1.12.3 LHE Inspection Reports
1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging
1.13 HOT WORK PERMIT
1.13.1 Permit and Personnel Requirements
1.13.2 Work Around Flammable Materials

1.14 DUST AND FUME MITIGATION
1.14.1 Work shall follow VA requirements and all PCRA measures required.
1.14.2 [Enter Appropriate Subpart Title Here]
1.14.3 Additional mitigation shall also be planned, such as:
1.15 CONFINED SPACE ENTRY REQUIREMENTS
1.15.1 Rescue Procedures and Coordination with Local Emergency Responders
1.16 HIGH NOISE LEVEL PROTECTION

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK
3.1.1 Worksite Communication
3.1.2 Hazardous Material Exclusions
3.1.3 Unforeseen Hazardous Material
3.2 UTILITY OUTAGE REQUIREMENTS
3.3 OUTAGE COORDINATION MEETING
3.4 UTILITY INVESTIGATION WORK PLAN
3.5 UTILITY OUTAGE, INTERRUPTION, TIE-IN WORK PLAN
3.6 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)
3.6.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility
3.6.2 Lockout/Tagout Isolation
3.6.3 Lockout/Tagout Removal
3.7 FALL PROTECTION PROGRAM
3.7.1 Fall Protection Equipment and Systems
3.7.1.1 Additional Personal Fall Protection Measures
3.7.1.2 Personal Fall Protection Equipment
3.8 EQUIPMENT
3.8.1 Material Handling Equipment (MHE)
3.8.2 Load Handling Equipment (LHE)
3.8.3 Machinery and Mechanized Equipment
3.8.4 Base Mounted Drum Hoists
3.8.5 Use of Explosives
3.8.6 EXCAVATIONS
3.8.6.1 Utility Locations
3.8.6.2 Utility Location Verification
3.8.6.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces
3.9 ELECTRICAL
3.9.1 Electrical Work
3.9.2 Qualifications
3.9.3 Arc Flash
3.9.4 Grounding
3.9.5 Testing

-- End of Section Table of Contents --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ANSI/ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2020) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1	(2014; R 2019) American National Standard for Industrial Head Protection

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2026) Standard for Portable Fire Extinguishers
NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023; ERTA 7 2023; TIA 23-15) National Electrical Code
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition

	Operations
NFPA 306	(2024) Standard for the Control of Gas Hazards on Vessels

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222	(2018H; Add 1 2019) Structural Standard for Antenna Supporting Structures and Antennas and Small Wind Turbine Support Structures

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Health Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1926	Safety and Health Regulations for Construction
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Site Demolition Plan; G

Final IAQ Management Plan; S

Indoor Air Quality (IAQ) Management Plan; G

Utility Investigation Work Plan; G

Utility Outage, Interruption, Tie-In Work Plan; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Crane Operators/Riggers

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Hot Work Permit

License Certificates

Standard Lift Plan; G

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) www.osha.gov, the Centers for Disease Control and Prevention (CDC) www.cdc.gov, and as required by federal, state and local requirements.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the progress payment.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and all applicable federal, state, and local laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs will be designated on an ENG Form 6282, provided by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are

not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

(1) Are designated on an ENG Form 6282.

(2) Meet minimum training and experience requirements identified in EM 385-1-1.

(3) Execute roles and responsibilities identified in EM 385-1-1.

1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1 for this project.

1.7.4 Competent Person for Confined Space Entry

Provide a CP for Confined Space Entry who meets the requirements of EM 385-1-1 and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Certified Industrial Hygienist. Confined space and

enclosed space work must comply with NFPA 306, or as applicable in 29 CFR 1910 for general industry, 29 CFR 1926 for construction.

1.7.5 Preconstruction Conference

a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).

b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP and Schedule will be created after being given Notice to Proceed.

c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Accident Prevention Plan (APP) - Construction

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.

b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

In the event of a medical emergency on the West Haven campus, if the personnel are located in Buildings 1 or 2, VA Police shall be immediately contacted at (203) 932-5711 ext. 4911. If the personnel are located outside of these buildings on campus, local medical emergency services shall be called by dialing 911. Once 911 has been called, the VA police will be informed that 911 has been called. VA Police will assist responders. Contractor will also notify project specific representative/contracting officer and local site project monitor.

However, if the contractor requiring medical assistance can safely be transported to the facility's Emergency Department, the contractor will receive emergency care. Generally, care is limited to the point that the individual is medically stable and can be discharged or safely transferred to a non-VA healthcare facility. For non-life-threatening injuries, contractor employees shall be transferred to non-VA healthcare facility for treatment when required.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	**Notify KO or Contracting Officer**	**Complete Final Accident Report on ENG 3394 and provide to KO or C ontracting Officer**
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over $600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable. The contractor shall submit within 14 days a Lessons-Learned document for all recordable injury, illnesses, and property damage incidents as defined in EM 385-1-1.

b. Near Misses: Report all "Near Misses" to the Contracting Officer, using local accident reporting procedures, within 24 hours. The Contracting Officer will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1 and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e., welding or cutting) or operating other flame-producing/spark producing devices, from the USACE RE/Contracting Officer on site. Contractors are required to meet all criteria before a permit is issued. Provide at least two 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of 1 hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency local fire department phone number. Report any fire, no matter how small, to the responsible local fire department immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1.

1.14 DUST AND FUME MITIGATION

1.14.1 Work shall follow VA requirements and all PCRA measures required.

1.14.2 All existing outside air HVAC intakes within 100 feet of work shall include plan for additional filtration installed by contractor, such as existing operating room suite, laboratory, sterile processing, etc.
Contractor shall assess most efficient filter types, to perform the mitigations, such as charcoal type, and review and coordinate with Government.
Any added filters shall also not hinder/ affect airflow to the existing space.

1.14.3 Additional mitigation shall also be planned, such as:

(1) No idling of vehicles within 100 foot radius of Building 1. Where not feasible, further coordination discussions/approval shall be made with the Government.

(2) Industrial fans to redirect or ducts to redirect and/or exhaust scrubbers.

(3) Rescheduling of work to minimize impact on hospital operations.

(4) Provide water misting of existing construction to be demolished during demolition activities.

1.15 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with EM 385-1-1, 29 CFR 1926, 29 CFR 1910, and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.15.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.16 HIGH NOISE LEVEL PROTECTION

Schedule operations that involve the use of equipment with output of high noise levels (i.e. jackhammers, air compressors and explosive-actuated devices) during the hours of 8:00 a.m. to 9:00 p.m. Use of any such equipment must be approved in writing by the Contracting Officer prior to commencement of work. If work is required to be completed outside the prescribed work hours stated in section 01 14 00, request must be coordinated 14 days in advance to the Contracting Officer for review and approval.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

a. Head Protection that meets ANSI/ISEA Z89.1

b. Long Pants

c. Appropriate Safety Footwear

d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations. Stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 120 days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1. In accordance with EM 385-1-1, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Public Utilities representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HECP training in accordance with EM 385-1-1. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 UTILITY INVESTIGATION WORK PLAN

Work plan must define the details of what is required to interrupt and connect to all existing Medical Center utilities. Plan shall provide the requested date and time, duration of request, details of what utilities are involved, where they are located along with tie-in point, what the utilities serve, and reason for the request. If any road, pedestrian, or building impacts are expected, they shall be defined and clearly shown, with drawings showing area of impact along with mitigation efforts to minimize the impacts, to include traffic control, detour plans, fencing, and signage plans, etc.

3.5 UTILITY OUTAGE, INTERRUPTION, TIE-IN WORK PLAN

Work plan must define the details of what is required to interrupt and connect to all existing Medical Center utilities. Plan shall provide the requested date and time, duration of request, details of what utilities are involved, where they are located along with tie-in point, what the utilities serve, and reason for the request. Completed Medical Center's Hazardous Energy Policy 138-075 Lockout/Tagout Permit shall be included. If any road, pedestrian, or building impacts are expected, they shall be defined and clearly shown, with drawings showing area of impact along with mitigation efforts to minimize the impacts, to include traffic control, detour plans, fencing, and signage plans, etc.

3.6 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.6.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must

occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.6.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

3.6.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.7 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1.

3.7.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.7.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing

work.

3.7.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

3.8 EQUIPMENT

3.8.1 Material Handling Equipment (MHE)

a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.

b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.

c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.8.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

a. Equip cranes and derricks as specified in EM 385-1-1.

b. Notify the Contracting Officer 15 working days in advance of any LHE entering the activity, in accordance with EM 385-1-1, so that necessary quality assurance spot checks can be coordinated.

Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, ASME B30.9 Standards safety standards.

c. OR a formal request may be submitted to the Contracting Officer for documentation, if available, indicating the roof of the building has been previously designed and constructed using a reinforced concrete roof slab or other means to provide overhead protection from construction activities specifically associated with the vertical or horizontal expansion of the same building and a conference and a Contractor provided engineering analysis, performed by Professional Engineer, demonstrates that overhead protection provided by the roof is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.

d. Comply with ASME B30.22for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware .

e. Crane operators shall not carry loads:

(1) over the general public or VAMC personnel

1. The top two or more floors are vacated under the lift affected area AND a Contractor provided engineering analysis, performed by a Professional Engineer, demonstrates that overhead protection provided by the roof and vacated floors is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater
2. OR Overhead protection is designed by a Professional Engineer and installed over the roof by the Contractor to protect the occupants below. The installed system shall be sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.
3. OR A formal request be submitted to the Contracting Officer for documentation, if available, indicating the roof of the building has been previously designed and constructed using a reinforced concrete roof slab or other means to provide overhead protections from construction activities specifically associated with the vertical or horizontal expansion of the same building and a Contractor provided engineering analysis, performed by Professional Engineer, demonstrates that overhead protection provided by the roof is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.

f. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Section 11, and ASME B30.5 or ASME B30.22 as applicable.

g. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.

h. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.

i. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.

j. Use cribbing when performing lifts on outriggers.

k. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.

l. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.

m. Maintain inspection records in accordance by EM 385-1-1, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.

n. Maintain written reports of operational and load testing in accordance with EM 385-1-1, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.

o. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).

p. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 20 mph, the operator, rigger and lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.

q. Follow FAA guidelines when required based on project location.

3.8.3 Machinery and Mechanized Equipment

a. Proof of qualifications for operator must be kept on the project site for review.

b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.8.4 Base Mounted Drum Hoists

a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ASSP A10.22.

b. Rigging gear must be in accordance with applicable ASME/OSHA standards.

c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.

d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1 and trained, at a minimum, in accordance with EM 385-1-1. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.

e. Material and personnel must not be hoisted simultaneously.

f. Personnel cage must be marked with the capacity (in number of persons) and load limit in pounds.

g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.8.5 Use of Explosives

The use of explosives is not required and is expressly forbidden for use on the project site.

3.8.6 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.8.6.1 Utility Locations

Provide a third party, independent, private utility locating company (hired by the Contractor) to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.8.6.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within 3 feet of the underground system.

3.8.6.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company (hired by the Contractor)must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings

does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1.

3.9.1 Electrical Work

As described in EM 385-1-1, electrical work is prohibited and is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work and pre-approval has been provided by the Medical Center and Contracting Officer. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 50

CONSTRUCTION INDOOR AIR QUALITY MANAGEMENT

05/24

PART 1 GENERAL

1.1 SUMMARY
1.2 PROJECT GOALS
1.3 REFERENCE SECTIONS
1.4 SUBMITTALS

PART 2 PRODUCTS

2.1 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN
2.2 CONSTRUCTION FILTERS

PART 3 EXECUTION

3.1 CONSTRUCTION PROCEDURES

-- End of Section Table of Contents --

SECTION 01 35 50

CONSTRUCTION INDOOR AIR QUALITY MANAGEMENT
05/24

PART 1 GENERAL

1.1 SUMMARY

This section addresses: Construction procedures to promote Enhanced Indoor Environmental Quality during and after construction.

1.2 PROJECT GOALS

Dust and Airborne Particulates: Prevent deposition of dust and other particulates in HVAC ducts and equipment. Contractor shall bear the cost of cleaning required due to failure to protect ducts and equipment from construction dust.

Develop and implement an indoor air quality policy that considers the following: moisture control, use of low emitting materials and products with low pollutant emissions, necessary protocols to protect indoor air quality during construction and in the finished building, prohibition of smoking in any form inside and within 50 feet of all building entrances, operable windows, and building ventilation intakes, and use of integrated pest management techniques.

1.3 REFERENCE SECTIONS

a. Section 01 45 00 QUALITY CONTROL

1.4 SUBMITTALS

Submit the following in accordance with Section 01 33 00SUBMITTAL PROCEDURES:

SD-03 Product Data

Construction Indoor Air Quality (IAQ) Management Plan; G

SD-06 Test Reports

SD-08 Manufacturer's Instructions

Construction Filters

Final Filters

SD-11 Closeout Submittals

Guiding Principles Documetation

Construction Photographs

Filtration Log

PART 2 PRODUCTS

2.1 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Provide a construction Indoor Air Quality (IAQ) Management Plan with all appropriate measures of SMACNA 1507 Chapter 3 and Guidelines for Sustainable Federal Buildings Enhanced Indoor Air Environmental Quality for New Construction highlighted and addressed. Plan shall include Construction Photographs.

a. Submit photographs and a catalog of photographs documenting implementation of Construction IAQ Management Plan. Catalog shall state date photo was taken, item photographed, and room number. Photographs and catalog to be submitted monthly.

2.2 CONSTRUCTION FILTERS

Medium efficiency, pleated, disposable type, consisting of media, media support grid and enclosing frame. Filter media: high density glass micro- fibers laminated to a glass woven mesh backing that is MERV 8 when rated in accordance with ASHRAE 52.2. Listed by UL as Class 1. Indicate minimum efficiency reporting value of (MERV) as determined by ASHRAE 52.2. Identify Final Filters meeting GBCI GP Assessment requirements, including Guiding Principles Documentation, and provide a Filtration Log. Maintain a log of all filtration media installed during construction. Log shall include filter manufacturer, model number, MERV rating, location of installed filter, and date final filters specified by equipment manufacturer were installed.

PART 3 EXECUTION

3.1 CONSTRUCTION PROCEDURES

Prevent the absorption of moisture and humidity by adsorptive materials by:

a. Sequencing the delivery of such materials so that they are not present in the building until work is completed and dry.

b. Delivery and storage of such materials in fully sealed moisture-impermeable packaging.

c. Provide sufficient ventilation for drying within reasonable time frame.

Begin construction ventilation when building is substantially enclosed.

If extremely dusty or dirty work must be conducted inside the building, shut down HVAC systems for the duration, remove dust and dirt completely before restarting systems.

When working in a portion of a clean or occupied building, prevent movement of air from construction area to clean or occupied area.

HVAC equipment and supply air ductwork may be used for ventilation during construction.

a. Ensure that MERV 8 air filters are correctly installed on all return air devices and in rooftop air handling units prior to start-up; replace filters with clean MERV 8 filters when they become loaded and

restrict airflow to spaces.

Do not store construction materials or waste in mechanical or electrical rooms.

Prior to use of return air ductwork without intake filters clean up and remove dust and debris generated by construction activities.

a. Inspect duct intakes, return air grilles, and terminal units for dust.

b. Clean plenum spaces, including top sides of lay-in ceilings, outsides of ducts, tops of pipes and conduit.

c. Clean tops of doors and frames.

d. Clean mechanical and electrical rooms, including tops of pipes, ducts, and conduit, equipment, and supports.

e. Clean return air plenums.

f. Remove intake filters last, after cleaning is complete.

Do not perform dusty or dirty work after start-up until air filters are installed.

Coils and fans shall be cleaned prior to testing and balancing procedures. Air filters shall be replaced prior to initiating testing and balancing.

Use other relevant recommendations of SMACNA IAQ Guideline for Occupied Buildings Under Construction for avoiding unnecessary contamination due to construction procedures.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19

PART 1 GENERAL

1.1 REFERENCES
1.2 ORDERING INFORMATION

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS
02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AACE INTERNATIONAL (AACE)
1265 Suncrest Towne Centre Drive
Morgantown, WV 26505-1876 USA
Ph: 304-296-8444
Fax: 304-291-5728
Internet: https://web.aacei.org/

ACOUSTICAL SOCIETY OF AMERICA (ASA)
1305 Walt Whitman Road, Suite 300
Melville, NY 11747-4300
Ph: 516-576-2360
Fax: 631-923-2875
E-mail: asa@acousticalsociety.org
Internet: https://acousticalsociety.org/

AIR BARRIER ASSOCIATION OF AMERICA (ABAA)
1600 Boston-Providence Hwy
Walpole, MA 02081
Ph: 1-866-956-5888
Fax: 1-866-956-5819
Internet: https://www.airbarrier.org/

AIR DUCT COUNCIL (ADC)
1901 N. Roselle Road, Suite 800
Schaumburg, IL 60195
Ph: 847-706-6750
Fax: 847-706-6751
E-mail: info@flexibleduct.org
Internet: https://flexibleduct.org/

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)
30 West University Drive
Arlington Heights, IL 60004-1893
Ph: 847-394-0150
Fax: 847-253-0088
E-mail: communications@amca.org
Internet: http://www.amca.org

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2111 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: http://www.ahrinet.org

ALUMINUM ASSOCIATION (AA)
1400 Crystal Drive
Suite 430
Arlington, VA 22202
Ph: 703-358-2960
E-Mail: info@aluminum.org
Internet: https://www.aluminum.org/

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)
1900 E Golf Rd, Suite 1250
Schaumburg, IL 60173
Ph: 847-303-5664
E-mail: customerservice@aamanet.org
Internet: https://aamanet.org/

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)
444 North Capital Street, NW, Suite 249
Washington, DC 20001
Ph: 202-624-5800
Fax: 202-624-5806
E-Mail: info@aashto.org
Internet: https://www.transportation.org/

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)
330 N. Wabash Ave., Suite 2000
Chicago, IL 60611
Ph: 202-367-1155
E-mail: info@americanbearings.org
Internet: https://www.americanbearings.org/

AMERICAN CONCRETE INSTITUTE (ACI)
38800 Country Club Drive
Farmington Hills, MI 48331-3439
Ph: 248-848-3700
Fax: 248-848-3701
Internet: https://www.concrete.org/

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)
1330 Kemper Meadow Drive
Cincinnati, OH 45240
Ph: 513-742-2020
Fax: 513-742-3355
Internet: https://www.acgih.org/

AMERICAN HARDBOARD ASSOCIATION (AHA)
1210 West Northwest Highway
Palatine, IL 60067
Ph: 847-934-8800
Fax: 847-934-8803
E-mail: aha@hardboard.org
Internet: http://domensino.com/AHA/

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-5444
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org
Internet: https://www.aisc.org/

AMERICAN INSTITUTE OF TIMBER CONSTRUCTION (AITC)
7012 South Revere Parkway, Suite 140
Centennial, CO 80112
Ph: 503-639-0651
Fax: 503-684-8928
E-mail: mschoen@wclib.org
Internet: http://www.aitc-glulam.org

AMERICAN IRON AND STEEL INSTITUTE (AISI)
25 Massachusetts Avenue, NW Suite 800
Washington, DC 20001
Ph: 202-452-7100
Internet: https://www.steel.org/

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)
7470 New Technology Way, Suite F
Frederick, MD 21703
Ph: 301-972-1700
Fax: 301-540-8004
E-mail: alsc@alsc.org
Internet: http://www.alsc.org

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
1899 L Street, NW,11th Floor
Washington, DC 20036
Ph: 202-293-8020
Fax: 202-293-9287
E-mail: storemanager@ansi.org
Internet: https://www.ansi.org/

AMERICAN PETROLEUM INSTITUTE (API)
1220 L Street, NW
Washington, DC 20005-4070
Ph: 202-682-8000
Internet: https://www.api.org/

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)
P.O. Box 28518
1711 Arlingate Lane
Columbus, OH 43228-0518
Ph: 800-222-2768 or 614-274-6003
Fax: 614-274-6899

E-mail: tjones@asnt.org
Internet: https://www.asnt.org/

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300
Internet: https://www.asce.org/

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: https://www.ashrae.org/

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: https://www.asme.org/

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: https://www.assp.org/

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)
18927 Hickory Creek Drive, Suite 220
Mokena, IL 60448
Ph: 708-995-3019
Fax: 708-479-6139
Internet: http://www.asse-plumbing.org

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: https://www.awwa.org/

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Internet: https://www.aws.org/

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)
P.O. Box 361784
Birmingham, AL 35236-1784
Ph: 205-733-4077
Fax: 205-733-4075
Internet: http://www.awpa.com

APA - THE ENGINEERED WOOD ASSOCIATION (APA)
7011 South 19th St.
Tacoma, WA 98466-5333
Ph: 253-565-6600
Fax: 253-565-7265
Internet: https://www.apawood.org/

ASSOCIATED AIR BALANCE COUNCIL (AABC)
1220 19th St NW, Suite 410
Washington, DC 20036
Ph: 202-737-0202
Fax: 202-315-0285
E-mail: info@aabc.com
Internet: https://www.aabc.com/

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)
600 North 18th Street
P.O. Box 2641
Birmingham, AL 35291
Ph: 205-257-3839
Fax: 205-257-2540
Internet: https://aeic.org/

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555
E-mail: service@astm.org
Internet: https://www.astm.org/

BICSI International Standards Program (BICSI)
8610 Hidden River Parkway
Tampa, FL 33637, USA
Ph: +1 813-979-1991 or 800-242-7405 (USA and Canada toll-free)
Fax: +1 813-971-4311
Email: bicsi@bicsi.org
Internet: https://www.bicsi.org/

BIFMA INTERNATIONAL (BIFMA)
678 Front Ave. NW, Suite 150
Grand Rapids, MI 49504-5368
Ph: 616-285-3963
E-mail: email@bifma.org
Internet: https://www.bifma.org/

BRICK INDUSTRY ASSOCIATION (BIA)
12007 Sunrise Valley Drive, Suite 430
Reston, VA 20191
Ph: 703-620-0010

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)
355 Lexington Avenue, 15th Floor
New York, NY 10017
Ph: 212-297-2122
Fax: 212-370-9047
Internet: https://www.buildershardware.com/

CALIFORNIA AIR RESOURCES BOARD (CARB)
1001 I Street
Sacramento, CA 95814
Ph: 800-242-4450
Email: helpline@arb.ca.gov
Internet: https://ww2.arb.ca.gov/

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)
PO Box 997377, MS 0500
Sacramento, CA 95899-7377
Ph: 916-558-1784
Internet: https://www.cdph.ca.gov/

CARPET AND RUG INSTITUTE (CRI)
P.O. Box 2048
Dalton, GA 30722-2048
Ph: 706-278-3176
Fax: 706-278-8835
Internet: https://carpet-rug.org/

CAST IRON SOIL PIPE INSTITUTE (CISPI)
2401 Fieldcrest Drive
Mundelein, IL 60060
Ph: 224-864-2910
Internet: https://www.cispi.org/

COMPRESSED GAS ASSOCIATION (CGA)
14501 George Carter Way, Suite 103
Chantilly, VA 20151-1788
Ph: 703-788-2700
Fax: 703-961-1831
E-mail: cga@cganet.com
Internet: https://www.cganet.com/

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)
933 North Plum Grove Road
Schaumburg, IL 60173-4758
Ph: 847-517-1200
Fax: 847-517-1206
Internet: http://www.crsi.org/

CONSUMER ELECTRONICS ASSOCIATION (CEA)
1919 South Eads St.
Arlington, VA 22202
Ph: 703-907-7600
E-mail: CTA@CTA.tech
Internet: https://www.cta.tech/

COOLING TECHNOLOGY INSTITUTE (CTI)
3845 Cypress Creek Parkway, Suite 420
Houston, TX 77068
Ph: 281-583-4087
E-mail: vmanser@cti.org
Internet: https://www.coolingtechnology.org/

COPPER DEVELOPMENT ASSOCIATION (CDA)
Internet: https://www.copper.org/

ELECTRONIC INDUSTRIES ALLIANCE (EIA)
EIA has become part of the ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ELECTROSTATIC DISCHARGE ASSOCIATION (ESD)
7900 Turin Road, Building 3
Rome, NY 13440-2069
Ph: 315-339-6937
Fax: 315-339-6793
E-mail: info@esda.org
https://www.esda.org/

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)
CEN-CENELEC Management Centre
Rue de la Science 23
B - 1040 Brussels, Belgium
Ph: 32-2-550-08-11
Fax: 32-2-550-08-19
Internet: https://www.cen.eu/

EXPANSION JOINT MANUFACTURERS ASSOCIATION (EJMA)
25 North Broadway
Tarrytown, NY 10591
Fax: 914-332-1541
E-mail: inquiries@ejma.org
Internet: http://www.ejma.org

FM GLOBAL (FM)
270 Central Avenue
Johnston, RI 02919-4949
Ph: 401-275-3000
Fax: 401-275-3029
Internet: https://www.fmglobal.com/

FOREST STEWARDSHIP COUNCIL (FSC)
708 First Street North, Suite 235
Minneapolis, MN 55401
Ph: 612-353-4511
E-mail: info@us.fcs.org
Internet: https://us.fsc.org/

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH (FCCCHR)
USC Foundation Office
Research Annex 219
Los Angeles, CA 90089-7700
Ph: 866-545-6340
Fax: 213-740-8399
E-mail: fccchr@usc.edu
Internet: https://fccchr.usc.edu/

GLASS ASSOCIATION OF NORTH AMERICA (GANA)
National Glass Association
1945 Old Gallows Rd., Suite 750
Vienna, VA 22182
Ph: 866-342-5642
Ph: 703-442-4890
Fax: 703-442-0630

Internet: http://www.glasswebsite.com

GREEN SEAL (GS)
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PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 43 30

MOCKUPS

01/25

PART 1 GENERAL

1.1 SUMMARY
1.2 DEFINITIONS
1.2.1 Pre-Installation Meetings
1.3 SUBMITTALS
1.4 INFORMATIONAL SUBMITTALS
1.5 QUALITY CONTROL
1.5.1 Regulatory Requirements
1.6 COORDINATION

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS
2.2 INTEGRATED EXTERIOR MOCKUPS

PART 3 EXECUTION

3.1 TESTING OF INTEGRATED EXTERIOR MOCKUPS

-- End of Section Table of Contents --

SECTION 01 43 30

MOCKUPS
01/25

PART 1 GENERAL

1.1 SUMMARY

Section includes:

a. Integrated exterior mockups for performance inspection.

b. Integrated exterior mockups for visual review.

Related requirements:

a. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS for testing building enclosure systems and assemblies as part of the exterior enclosure commissioning process.

1.2 DEFINITIONS

a. Integrated Exterior Mockups for Performance Inspection: Mockups of the exterior envelope constructed in a testing laboratory as freestanding temporary built elements, consisting of multiple products, assemblies and sub-assemblies. The testing laboratory will facilitate the substructure to which the tested specimen is attached or restrained to. The Structural EoR (Engineer of Record) will provide imposed loads to the testing laboratory.

b. Integrated Exterior Mockups for Visual Review: Mockups of the exterior envelope constructed on-site as freestanding temporary built elements, consisting of multiple products, assemblies, and subassemblies

1.2.1 Pre-Installation Meetings

a. Preinstallation Conference: Conduct conference at Project site.

(1) Meet with Owner, Construction Manager, Architect, testing and inspecting agency representative, and installers of major systems whose Work is included in integrated exterior and room mockups.

(2) Review coordination of equipment and furnishings provided by the Owner for room mockups.

(3) Review locations and extent of mockups.

(4) Review testing procedures to be performed performance on mockups.

(5) Review and finalize schedule for mockups, and verify availability of materials, personnel, equipment, and facilities needed to complete mockups and maintain schedule for the Work.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-02 Shop Drawings

a. Shop Drawings: For integrated exterior visual and performance mockups and room mockups; G, AE

(1) Include plans, elevations, sections, and mounting and support details.

(2) Indicate manufacturer and model number of individual components, subassemblies, and assemblies.

(3) Include site location drawing indicating orientation of mockup.

(4) Revise and resubmit Shop Drawings to reflect approved modifications in details and component interfaces resulting from changes made during testing procedures.

b. Delegated Design Submittal: For temporary structural supports for mockups not attached to building structure, including analysis data signed and sealed by the qualified professional engineer licensed in the State of California responsible for their preparation.

1.4 INFORMATIONAL SUBMITTALS

a. Qualification Data: For testing agency.

b. Preconstruction Test Reports: For integrated exterior mockups.

1.5 QUALITY CONTROL

1.5.1 Regulatory Requirements

a. Testing Agency Qualifications: Qualified in accordance with ASTM E699 for testing indicated meeting requirements per 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

b. Build visual mockups to do the following:

(1) Verify selections made under Sample submittals.

(2) Demonstrate aesthetic effects.

(3) Demonstrate the qualities of products and workmanship.

c. Build performance mockups to do the following:

(1) Demonstrate acceptable coordination between components and systems.

(2) Perform preconstruction testing, such as window air- and water-leakage testing as specified in Section 01 45 41, EXTERIOR ENCLOSURE PERFORMANCE REQUIREMENTS, MOCKUPS AND TESTING..

d. Fabrication: Before fabricating or installing portions of the Work

requiring mockups, build mockups for each form of construction and finish required. Use materials indicated and installation methods as required for the Work.

(1) Build mockups of size indicated on attached Mockup Drawings.

(2) Employ supervisory personnel who will oversee mockup construction. Employ workers who will be employed to perform same tasks during the construction at Project.

(3) Demonstrate the proposed range of aesthetic effects and workmanship.

(4) Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.

(5) Demolish and remove mockups when directed unless otherwise indicated.

e. Notifications:

(1) Notify Contracting Officer, Architect, Commissioning Authority (CxA), and Commissioning Manager (CxM) seven days in advance of the dates and times when mockups will be constructed.

(2) Notify Contracting Officer, Architect, Commissioning Authority (CxA), and Commissioning Manager (CxM) 14 days in advance of the dates and times when mockups will be tested.

(3) Allow 14 days for initial review and each re-review of each mockup.

f. Approval: Obtain Contracting Officer's and Architect's approval of mockups before starting fabrication or construction of corresponding Work.

(1) Unless otherwise indicated, approved mockups establish the standard by which the Work will be judged.

(2) Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Architect specifically approves such deviations in writing.

(3) Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.6 COORDINATION

Coordinate schedule for construction of mockups, so construction, testing, and review of mockups do not impact Project schedule.

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

a. Delegated Design: Engage a qualified professional engineer licensed in the State of Connecticut, to design support structure for any free-standing mockups.

b. Structural Performance:

(1) Seismic Performance: Mockups and support structure to withstand the effects of earthquake motions determined in accordance with ASCE 7ASCE/SEI 7.

(2) Wind Loads: As indicated on Drawings.

c. Mockup Testing Performance Requirements: Perform tests using design pressures and performance criteria indicated for assemblies and products that are specified in other Sections and incorporated into integrated exterior mockups. Refer to requirements in Section 01 91 00.01 DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

2.2 INTEGRATED EXTERIOR MOCKUPS

a. Construct integrated exterior mockups according to approved mockup Shop Drawings. Provide shop drawings designed by licensed specialty engineer as part of Delegated Design Submittal. Construct mockups to demonstrate constructability, coordination of trades, and sequencing of Work; and to ensure materials, components, subassemblies, assemblies, and interfaces integrate into a system complying with indicated performance and aesthetic requirements. Build mockups in place when an issue is not isolated in a stand alone mockup.

b. Design and construct foundation and superstructure to support free-standing integrated exterior mockups.

c. Build integrated exterior mockups using installers and construction methods that will be used in completed construction.

d. Use specified products that have been approved by Architect. Coordinate installation of materials and products specified in individual Specification Sections that include Work included in integrated exterior mockups.

e. The work of integrated exterior mockups are to include, but is not limited to, the following items at a minimum. Specific materials listed are subject for inclusion if applicable in building design to cover all different assemblies applicable in building design.

(1) Flashing and sheet metal trim.

(2) Joint sealants.

(3) Metal wall panels.

(4) Glazed curtain walls.

(5) Vertical aluminum shading fins.

(6) Glazing.

f. Photographic Documentation: Document construction of integrated exterior mockups with photographs in accordance with Section 01 00 00 "General Requirements." Provide photographs showing details of interface of different materials and assemblies.

(1) Document testing procedures, including water leakage and other

deficiencies. Photograph modifications to component interfaces intended to correct deficiencies.

g. Provide and document modifications to construction details and interfaces between components and systems required to properly sequence the Work, or to pass performance testing requirements. Obtain Architect's approval for modifications.

h. Retain approved mockups constructed in place. Incorporate fully into the Work.

PART 3 EXECUTION

3.1 TESTING OF INTEGRATED EXTERIOR MOCKUPS

a. Refer to Section 01 45 41, EXTERIOR ENCLOSURE PERFORMANCE REQUIREMENTS, MOCKUPS AND TESTING for all testing requirements for integrated exterior performance mockups.

b. Integrated Exterior Mockup Testing Agency: Engage a qualified testing agency to perform tests and inspections.

(1) Testing and inspecting agency will interpret tests and state in each report whether tested Work complies with or deviates from requirements.

c. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect components, assemblies, and installations, including connections, and also to observe testing for the following systems and assemblies.

(1) Curtain wall specified in Section 084413 "Glazed Aluminum Curtain Walls."

(2) List of systems and assemblies to be edited by Contractor or designer of record to reflect all different assemblies that are in the design.

d. Integrated exterior mockup will be considered defective if it does not pass tests and inspections.

e. Prepare test and inspection reports.

f. Review and approval of all exterior mockups shall be done by the Architect and Contracting Officer.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)

11/16, CHG 2: 08/19

PART 1 GENERAL

1.1 REFERENCES
1.2 MEASUREMENT AND PAYMENT
1.3 CONTRACT ADMINISTRATION
1.3.1 Correspondence and Electronic Communications
1.3.2 Other Factors
1.4 RMS SOFTWARE
1.4.1 SYSTEM REQUIREMENTS
1.5 CONTRACT DATABASE - GOVERNMENT
1.6 CONTRACT DATABASE - CONTRACTOR
1.6.1 Administration
1.6.1.1 Contractor Information
1.6.1.2 Subcontractor Information
1.6.1.3 Correspondence
1.6.1.4 Equipment
1.6.1.5 Reports
1.6.1.6 Request For Information (RFI)
1.6.2 Finances
1.6.2.1 Pay Activity Data
1.6.2.2 Payment Requests
1.6.3 Quality Control (QC)
1.6.3.1 Quality Control (QC) Reports
1.6.3.2 Deficiency Tracking.
1.6.3.3 Three-Phase Control Meetings
1.6.3.4 Labor and Equipment Hours
1.6.3.5 Accident/Safety Reporting
1.6.3.6 Definable Features of Work
1.6.3.7 Activity Hazard Analysis
1.6.4 Submittal Management
1.6.5 Schedule
1.6.6 Closeout
1.7 IMPLEMENTATION
1.8 NOTIFICATION OF NONCOMPLIANCE

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/16, CHG 2: 08/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

Administration
Finances
Quality Control
Submittal Monitoring
Scheduling
Closeout
Import/Export of Data

The Contractor must be prepared to transition from the Resident Management System (RMS) to the new Construction Management Platform (CMP) at any time during the period of performance of this contract, once the USACE enterprise implements this change. The Contractor's obligation to transition from RMS to the CMP system is a requirement of this contract. Therefore, additional time or cost increase to complete this transition will not be considered. In the event of the transition, RMS project information will be migrated to CMP by the government and available in the CMP system. After the transition, the Contractor must utilize the CMP system to perform all contract administration functions previously required in RMS, including all areas listed above. After the transition,

"RMS" and "CMP" will be synonymous for all "Resident Management System" and "RMS" references in this contract's specification. To facilitate the transition from RMS to CMP, online training for the CMP system will be made available to the Contractor with enough time to train staff, prior to the transition occurring.

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some correspondence, including pay requests and payrolls, are also to be provided in paper format with original signatures. Paper documents will govern, in the event of discrepancy with the electronic version.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under Fixed-Priced Construction Contracts; Section 01 32 01.00 10 PROJECT SCHEDULE; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 00.00 10 QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a Windows-based program that can be run on a Windows-based PC meeting the requirements as specified in paragraph SYSTEM REQUIREMENTS. Download, install and be able to utilize the latest version of the RMS software within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (https://rms.usace.army.mil). RMS customer service is available 24 hours per day, 7 days per week at (210) 473-8658. The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.4.1 SYSTEM REQUIREMENTS

The following is the recommended system configuration to run the Contractor Mode RMS for full utilization of all features for all types and sizes of contracts. Smaller, less complicated, projects may not require the configuration levels described below. Required confiduration also noted below.

Recommended RMS System Requirements	
Hardware	
Windows-Based PC	1.7 GHz i3; AMD A6 3650 GHz or higher processor (Required)
RAM	8 GB

Hard drive disk	100 GB specae for sole use by RMS system
Monitor	Screen resolution 1366 x 768
Mouse or other point device	
Windows compatible printer	Laser printer must have 4 MB+ of RAM
Connection to the Internet	Minimum 4 mbs per user
Software	
MS Windows	Windows 10 x 64 bit (RMS requires 64 O/S or newer (REQUIRED)
Work Processing software	Viewer for MS Word 2013, MS Excel 2013 or newer (REQUIRED)
E-mail	MAPI compatiable (REQUIRED)
Virus protection software	Reguarly upgraded with all issued Manufacturer's updates and is able to detect most zero day viruses (REQUIRED)

1.5 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.6 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.6.1 Administration

1.6.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.6.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within 7 calendar days of receiving access to the contract in RMS or within 7 calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade,

address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.6.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.6.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment inspection dates.

1.6.1.5 Reports

Track the status of the project utilizing the reports available in RMS. The value of these reports is reflective of the quality of the data input. These reports include the Progress Payment Request worksheet, Quality Control (QC) comments, Submittal Register Status, and Three-Phase Control worksheets.

1.6.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response. The Government shall have 21 days to review all RFI's.

1.6.2 Finances

1.6.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule. The sum of pay activities equals the total contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.6.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data either electronically or by hard copy. Unless waived by the Contracting Officer, a signed paper copy of the approved payment certification and request is also required and will govern in the event of discrepancy with the electronic version.

1.6.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS.

Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Insure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments of same within 7 calendar days of Government acceptance of the CQC Plan.

1.6.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.6.3.2 Deficiency Tracking.

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.6.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS. Worksheets for the three-phase control meetings are generated within RMS.

1.6.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.6.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.6.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.6.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC

Module utilizing the document package manager.

1.6.4 Submittal Management

Enter all current submittal register data and information into RMS within 7 calendar days of receiving access to the contract in RMS. The information shown on the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES will already be entered into the RMS database when access is granted. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, spare parts, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the contract. Submit hard copies of all submittals unless waived by the Contracting Officer.

1.6.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 01.00 13 PROJECT SCHEDULE.

1.6.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.7 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.8 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within 7 calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 00

QUALITY CONTROL

08/23, CHG 1: 05/24

PART 1 GENERAL

1.1 REFERENCES
1.2 PAYMENT
1.3 SUBMITTALS
1.4 GENERAL REQUIREMENTS
1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS
1.5.1 Meetings
1.5.1.1 Quality Control Plan Meeting
1.5.1.2 Coordination and Mutual Understanding Meeting
1.5.1.2.1 Purpose
1.5.1.2.2 Coordination of Activities as specified in Section 01 35 50 CONSTRUCTION INDOOR AIR QUALITY MANAGEMENT
1.5.1.2.3 Attendees
1.5.1.3 Quality Control (QC) Meetings
1.5.2 Contractor Quality Control (CQC) Plan
1.5.2.1 Content of Contractor Quality Control (CQC) Plan
1.5.3 Acceptance of the Quality Control (QC) Plan
1.5.4 Notification of Changes
1.6 QUALITY CONTROL (QC) ORGANIZATION
1.6.1 Quality Control (QC) Manager
1.6.1.1 Duties
1.6.1.2 Qualifications
1.6.1.3 Construction Quality Management Training
1.6.2 Organizational Changes
1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications
1.6.4 Quality Control (QC) Specialists
1.6.4.1 Fire Protection QC Specialist (FPQC)
1.6.4.1.1 Qualifications
1.6.4.1.2 Responsibilities
1.6.4.1.3 Schedule of Fire Protection System Inspections
1.6.5 Submittal Reviewers Duties and Qualifications
1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL
1.8 THREE PHASES OF CONTROL
1.8.1 Preparatory Phase
1.8.2 Initial Phase
1.8.3 Follow-Up Phase
1.8.4 Additional Preparatory and Initial Phases
1.8.5 Notification of Three Phases of Control for Off-Site Work
1.8.6 Deficiency/Rework Items List
1.9 TESTING
1.9.1 Capability Check
1.9.2 Test Results
1.10 COMPLETION INSPECTIONS
1.10.1 Punch-Out Inspection

1.10.2 Pre-Final Inspection
1.10.3 Final Acceptance Inspection
1.11 QUALITY CONTROL (QC) CERTIFICATIONS
1.11.1 Contractor Quality Control (CQC) Report Certification
1.11.2 Completion Certification
1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER
1.12.1 Construction Documentation
1.12.2 Quality Control Activities
1.12.3 Verification Statement
1.12.4 Quality Control Validation
1.12.5 Testing Plan and Log
1.12.6 As-Built Drawings
1.13 NOTIFICATION ON NON-COMPLIANCE
1.14 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 45 00

QUALITY CONTROL
08/23, CHG 1: 05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM E329	(2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Bid Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

SD-06 Test Reports

Verification Statement

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager,

Superintendent, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations that comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.5.1 Meetings

1.5.1.1 Quality Control Plan Meeting

Prior to submission of the QC Plan, the Contractor may request a meeting with the Contracting Officer to discuss the QC Plan requirements of this Contract.

The purpose of this meeting is to develop a mutual understanding of the QC Plan requirements prior to plan development and submission and to agree on the Contractor's list of Definable Feature of Work (DFOW).

1.5.1.2 Coordination and Mutual Understanding Meeting

After the before start of construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor and the Government. Provide a copy of the signed minutes to all attendees and include in the QC Plan. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

1.5.1.2.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site

work, design intent,environmental requirements and procedures, coordination of activities to be performed, Special Inspections, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

a. Waste Management Plan.

b. Procedures for noise and acoustics management.

c. Environmental Protection Plan.

d. Environmental regulatory requirements.

e. Indoor Air Quality (IAQ) Management Plan.

1.5.1.2.2 Coordination of Activities as specified in Section 01 35 50 CONSTRUCTION INDOOR AIR QUALITY MANAGEMENT

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation. Schedule construction operations with consideration for indoor air quality as specified in the IAQ Management Plan.Coordinate Special Inspections.

1.5.1.2.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

1.5.1.3 Quality Control (QC) Meetings

After the start of construction, conduct weekly QC meetings led by the QC Manager at the work site with the Project Superintendent, CxC, and the other personnel as necessary. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within 2 working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

a. Review the minutes of the previous meeting.

b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.

c. Review the status of submittals and Request For Information (RFIs).

d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph and all documentation required for that work.

e. Review Testing Plan and Log including status of tests performed since last QC Meeting.

f. Resolve QC and production problems. Discuss status of pending change orders.

g. Address items that may require revising the QC Plan.

h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.

i. Review environmental requirements and procedures.

j. Review Environmental Management Plan.

k. Review Waste Management Plan.

l. Review the status of training completion.

m. Review IAQ Management Plan.

1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 30 days, the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements

1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents, with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.

b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.

c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraph CONSTRUCTION QUALITY MANAGEMENT TRAINING.

d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.

e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.

f. APPOINTMENT LETTERS: Letters signed by an officer of the firm

appointing the QC Manager and Alternate QC Manager, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work that is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists or quality control representatives outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.

g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.

h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.

i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.

j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.

k. Reporting procedures, including proposed reporting formats.

l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.

m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination of Mutual Understanding Meeting.

n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the

Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.

o. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.

p. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.

1.5.3 Acceptance of the Quality Control (QC) Plan

The Contracting Officer's acceptance of the Contractor QC Planis required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.5.4 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel. Proposed changes are subject to acceptance by the Contracting Officer.

1.6 QUALITY CONTROL (QC) ORGANIZATION

1.6.1 Quality Control (QC) Manager

1.6.1.1 Duties

Provide a QC Manager at the work site to implement and manage the QC program. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the Coordination and Mutual Understanding Meeting, perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities. The QC manager is responsible for the quality control for perimeter construction.

1.6.1.2 Qualifications

The QC Manager must possess at least a Bachelor's degree in engineering, architecture, or construction management, with a minimum of 10 years construction experience as a Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction Contracts which included the major trades that are part of this Contract. The individual must have at least 4 years experience as a QC Manager. The individual must be familiar with the requirements of

EM 385-1-1 and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

1.6.1.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager and all members of the QC team must have completed the CQM for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, retake the course to remain current.

1.6.2 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6.4 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Matrix listed below, who must assist and report to the QC Manager and who may perform production related duties but must be allowed sufficient time to perform their assigned quality control duties. These individuals or specialized technical companies are employees of the Prime or subcontractor QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC Meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their area of responsibility. QC Specialist must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in Matrix listed herein.

Experience Matrix	
1. Area	**2-1. Qualification** **2-2. Experience**

Civil	2-1. Not applicable. 2-2. Technician with 5 years related experience
Mechanical	2-1. Not applicable 2-2. 5 years of experience supervising mechanical features of work in the field with a construction company
Electrical	2-1. Not applicable. 2-2. 5 years of experience supervising electrical features of work in the field with a construction company

1.6.4.1 Fire Protection QC Specialist (FPQC)

Provide a Fire Protection Quality Control Specialist (FPQC) within the QC organization to perform quality control related activities as specified herein on fire protection and life safety systems installed under this Contract.

1.6.4.1.1 Qualifications

The FPQC must have the following qualifications:

a. Be a registered Professional Engineer (P.E.) licensed by a Licensing Board in the United States, the District of Columbia, Guam or Puerto Rico, having passed the National Council of Examiners for Engineering and Surveying (NCEES) examination specifically in the discipline of Fire Protection Engineering.

b. Have a minimum of 5 years of Fire Protection Engineering experience on projects of similar relevance and complexity to the fire protection

work specified under this Contract.

c. Other than the contractual obligations with the Prime Contractor, the FPQC must have no other business relationship (i.e., employee, owner, partner, operating officer, distributor, salesman, technical representative, family relationship, or financial investment) with the Prime Contractor or subcontractors.

d. Be employed by an independent engineering firm or company. The firm may identify multiple, to a maximum of five, licensed Fire Protection Engineers for the performance of the duties under this Contract but must submit the names and qualifications for Government approval for all individuals identified prior to them performing any work under this Contract. These individuals may not be substituted without prior approval from the Contracting Officer.

1.6.4.1.2 Responsibilities

FPQC duties and responsibilities:

a. Assist in the development of the QC Plan including the Testing Plan and Log and executing the three phases of control for work involving the installation and testing of fire protection and life safety systems as an extension of the QC Manager.

b. Participate in project QC Meetings. Participate in Preparatory and Initial Phase meetings and perform and Follow-up inspections for work involving the installation and testing of fire protection and life safety systems.

c. Review and certify that all submittals pertaining to fire protection and life safety systems are complete and accurate prior to submission to the Government for approval. The FPQC Specialist is responsible for ensuring submittals are complete and accurate and all corrections have been made prior to submission to the Government. The Government reserves the right to reject any submittal that has not first been reviewed and certified by the FPQC and so marked, in writing, attesting to such review and completeness of the submittal.

d. The Government reserves the right to reject any submittal or construction that is not in compliance to Contract. Government reviews do not relieve the Contractor responsibility for providing adequate quality control measures and do not constitute or imply acceptance of Contract variation.

e. Perform construction surveillance in accordance with the Schedule of Fire Protection System Inspections. Construction surveillance includes but is not limited to performing periodic on-site inspections during construction at specified milestones, performing a pre-final inspection of installed systems and witnessing functional testing; and participating and documenting in an on-site final acceptance inspection of fire protection and life safety systems with the Government FPE.

f. Document inspection results on a FPQC report prepared each day inspections are performed. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the FPQC Specialist. In person inspection and Remote

inspections must be documented via video (.mp4) or photo (.jpeg) or both. Video/photographic documentation must include before and after conditions and physical measurements. Forward the FPQC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the FPQC must be documented on a FPQC daily report.

1.6.4.1.3 Schedule of Fire Protection System Inspections

A schedule, prepared by the Fire Protection DOR, which lists each of the required visual inspections and observations required by the FPQC. The schedule is included at the end of this UFGS section.

1.6.5 Submittal Reviewers Duties and Qualifications

Provide a Submittal Reviewer, other than the QC Manager, qualified in the disciplines being reviewed, to review and certify that the submittals meet the requirements of this Contract prior to certification or approval by the QC Manager.

Each submittal must be reviewed by an individual with 10 years of construction experience.

1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract.

1.8 THREE PHASES OF CONTROL

CQC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both on-site and off-site work for each definable feature of the construction work as follows:

1.8.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least five business days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by the QC Specialists, the Project Superintendent, and the foreman responsible for the DFOW. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. Perform the following prior to beginning work on each DFOW:

a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the prepatory

inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.

b. Review the Contract drawings.

c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.

d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.

e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.

f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer.

g. Review coordination of product/material delivery to designated prepared areas to execute the work.

h. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.

i. Check to assure that all materials and equipment have been tested, submitted, and approved.

j. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFOW. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.

k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

1.8.2 Initial Phase

Notify the Contracting Officer at least 2 business days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with the Project Superintendent, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases. Perform the following for each DFOW:

a. Check work to ensure that it is in full compliance with Contract requirements. Review minutes of the preparatory meeting.

b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.

c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

d. Resolve any workmanship issues.

e. Ensure that testing is performed by the approved laboratory.

f. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.

g. Review project specific work plans (i.e., HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.

1.8.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-Up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.

b. Maintain the quality of workmanship required.

c. Ensure that testing is performed by the approved laboratory.

d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.

e. Do not build upon nor conceal non-conforming work.

f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.

1.8.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

1.8.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least 2 weeks prior to the start of the preparatory and initial phases.

1.8.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.

b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.

c. Attach a copy of the "Deficiency/Rework Items List" to the last daily CQC Report of each month.

d. The Contractor is responsible for including those items identified by the Contracting Officer.

e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.

f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.

g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

1.9 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory. Perform the following activities and record and provide the following data:

a. Verify that testing procedures comply with Contract requirements.

b. Verify that facilities and testing equipment are available and comply with testing standards.

c. Check test instrument calibration data against certified standards.

d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.

e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

1.9.1 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

1.9.2 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager.

1.10 COMPLETION INSPECTIONS

1.10.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.10.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred by the Contracting Officer in a timely manner and be accomplished before the Contract completion date for the work, or any increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

1.10.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.11 QUALITY CONTROL (QC) CERTIFICATIONS

1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.11.2 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.12 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

1.12.1 Construction Documentation

Reports are required for each day that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities. Reports are required for each day work is performed. Account for each calendar day throughout the life of the

Contract.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

1.12.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

a. The name and area of responsibility of the Contractors and any subcontractors.

b. Operating plant/equipment with hours worked, idle, or down for repair.

c. Work performed each day, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.

d. Control phase activities performed. Preparatory, and Initial phase Checklists associated with the DFOW referenced to the construction schedule. Follow-up phase activities identified to the DFOW. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOW note this and include those reports.

e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.

f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.

g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.

h. Offsite surveillance activities, including actions taken.

i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.

j. Instructions given/received and conflicts in plans and specifications.

1.12.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Furnish the original and one copy of these records in report form to the Government by 10:00 AM the next working day after the date covered by the report. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

1.12.4 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.

b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.

c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.

d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES.

e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.

f. An up-to-date copy of the paragraph DEFICIENCY/REWORK ITEMS LIST.

g. Commence of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

1.12.5 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated

"Testing Plan and Log" to the last daily CQC Report of each month. Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.12.6 As-Built Drawings

The QC Manager must ensure the as-built drawings, required by Section 01 78 00 CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document commences with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the Contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager or QC Specialist assigned to an area of responsibility must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.13 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

1.14 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 29

TESTING LABORATORY SERVICES

PART 1 GENERAL

1.1 DESCRIPTION
1.2 REFERENCES
1.3 SUBMITTALS
1.4 REQUIREMENTS

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 EARTHWORK
3.2 FOUNDATION CAISSONS
3.3 LANDSCAPING
3.4 ASPHALT CONCRETE PAVING
3.5 SITE WORK CONCRETE
3.6 CONCRETE
3.7 REINFORCEMENT
3.8 SHOTCRETE
3.9 MASONRY
3.10 STRUCTURAL STEEL
3.11 STEEL DECKING
3.12 SHEAR CONNECTOR STUDS
3.13 SPRAYED-ON FIREPROOFING
3.14 TYPE OF TEST

-- End of Section Table of Contents --

SECTION 01 45 29

TESTING LABORATORY SERVICES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies materials testing activities and inspection services required during project construction to be provided by a Testing Laboratory retained by the General Contractor.

1.2 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 27	(2014) Standard Method of Test for Sieve Analysis of Fine and Coarse Aggregates
AASHTO T 96	(2022) Standard Method of Test for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
AASHTO T 104	(1999) Standard Method of Test for Soundness of Aggregate by Use of Sodium Sulfate or Magnesium Sulfate

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 506.4	Guide for the Evaluation of Shotcrete

ASTM INTERNATIONAL (ASTM)

ASTM A370	(2022) Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM C31/C31M	(2022) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C33/C33M	(2018) Standard Specification for Concrete Aggregates
ASTM C39/C39M	(2021) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM C138/C138M	(2017a) Standard Test Method for Density

	(Unit Weight), Yield, and Air Content (Gravimetric) of Concrete
ASTM C140/C140M	(2022c) Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C172	(2010) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	(2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C330	(2014) Standard Specification for Lightweight Aggregates for Structural Concrete
ASTM C567/C567M	(2019) Determining Density of Structural Lightweight Concrete
ASTM C780	(2020) Standard Test Method for Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry
ASTM C1019	(2019) Standard Test Method for Sampling and Testing Grout
ASTM C1064/C1064M	(2017) Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete
ASTM C1077	(2017) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation
ASTM C1314	(2014) Standard Test Method for Compressive Strength of Masonry Prisms
ASTM D1140	(2017) Standard Test Methods for Determining the Amount of Material Finer than 75-µm (No. 200) Sieve in Soils by Washing
ASTM D1188	(2015) Standard Test Method for Bulk Specific Gravity and Density of Compacted Bituminous Mixtures Using Coated Samples
ASTM D1556/D1556M	(2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
ASTM D1557	(2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of

	Soil Using Modified Effort (56,000 ft-lbf/ft3)
ASTM D2419	(2014) Sand Equivalent Value of Soils and Fine Aggregate
ASTM D2974	(2020; E 2020) Moisture, Ash, and Organic Matter of Peat and Other Organic Soils
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D3740	(2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction
ASTM D6913/D6913M	(2017) Standard Test Methods for Particle-Size Distribution (Gradation) of Soils Using Sieve Analysis
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)
ASTM E94/E94M	(2022) Standard Guide for Radiographic Examination Using Industrial Radiographic Film
ASTM E164	(2019) Standard Practice for Contact Ultrasonic Testing of Weldments
ASTM E329	(2021) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
ASTM E543	(2021) Standard Specification for Agencies Performing Non-Destructive Testing
ASTM E605	(1993; R 2011) Thickness and Density of Sprayed Fire-Resistive Material (SFRM) Applied to Structural Members
ASTM E709	(2021) Standard Guide for Magnetic Particle Testing
ASTM E1155	(2020) Standard Test Method for Determining Floor Flatness and Floor Levelness Numbers
ASTM F3125/F3125M	(2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.02 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Certificate Of Accreditation And Scope Of Accreditation; G

Test Reports; G

Load Test Report; G

Certified “Caisson Field Record”; G

Laboratory Test Report Of Topsoil; G

Record Of Welders And Their Certification; G

SD-06 Test Reports

Post-Tensioning Reports; G

Certified Compression Test Reports; G

Report Of Observations; G

1.4 REQUIREMENTS

a. Accreditation Requirements: Construction materials testing laboratories must be accredited by a laboratory accreditation authority and will be required to submit a copy of the Certificate of Accreditation and Scope of Accreditation. The laboratory’s scope of accreditation must include the appropriate ASTM standards (i.e.; ASTM E329, ASTM C1077, ASTM D3666, ASTM D3740, ASTM E543) listed in the technical sections of the specifications. Laboratories engaged in Hazardous Materials Testing shall meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the “Corporate Office.”

b. Inspection and Testing: Testing laboratory shall inspect materials and workmanship and perform tests described herein and additional tests requested by Resident Engineer. When it appears materials furnished, or work performed by Contractor fail to meet construction contract requirements, Testing Laboratory shall direct attention of Resident Engineer to such failure.

c. Written Reports: Testing laboratory shall submit test reports to

Resident Engineer, Contractor, unless other arrangements are agreed to in writing by the Resident Engineer. Submit reports of tests that fail to meet construction contract requirements on colored paper.

d. Verbal Reports: Give verbal notification to Resident Engineer immediately of any irregularity.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 EARTHWORK

a. General: The Testing Laboratory shall provide qualified personnel, materials, equipment, and transportation as required to perform the services identified/required herein, within the agreed to schedule and/or time frame. The work to be performed shall be as identified herein and shall include but not be limited to the following:

(1) Observe fill and subgrades during proof-rolling to evaluate suitability of surface material to receive fill or base course. Provide recommendations to the Resident Engineer regarding suitability or unsuitability of areas where proof-rolling was observed. Where unsuitable results are observed, witness excavation of unsuitable material and recommend to Resident Engineer extent of removal and replacement of unsuitable materials and observe proof-rolling of replaced areas until satisfactory results are obtained.

(2) Provide full time observation during placement and compaction to fill and structural fill associated with building earthwork, site earthwork, and trench earthwork to verify that earthwork compaction obtained is in accordance with contract documents.

(3) Provide supervised geotechnical technician to inspect excavation, subsurface preparation, and backfill for structural fill.

b. Testing Compaction:

(1) Determine maximum density and optimum moisture content for each type of fill, structural fill, backfill and subgrade material used, in compliance with ASTM D1557.

(2) Make field density tests in accordance with the primary testing method following ASTM D6938 and ASTM D1556/D1556M. Should the testing laboratory propose these alternative methods, they should provide satisfactory explanation to the Resident Engineer before the tests are conducted.

(a) General Fill: At least one test per every 1,000 cubic yards and every 2 vertical feet of fill placement.

(b) Structural Fill: At least one test per every 500 cubic yards and every 2 vertical feet of fill placement.

(c) Building Slab Subgrade: At least one test of subgrade for every 2000 square feet of building slab, but in no case fewer than three tests per structure.

(d) Foundation Wall Backfill: One test per each 100 lineal feet of wall per 2 vertical feet of compacted fill but in no case fewer than two tests.

(e) Pavement Subgrade: One test for each 4,000 square feet of subgrade, but in no case fewer than two tests.

(f) Pavement Aggregate Base and Asphalt Concrete: At least one test per each 4,000 square feet of pavement, but in no case fewer than two tests.

(g) Curb, Gutter, and Sidewalk: One test for each 200 lineal feet of curb, gutter, and sidewalk, or for each section less than 200 lineal feet, but in no case fewer than two tests.

(h) Trenches: One test at maximum 200 foot intervals per 2 feet of vertical lift and at changes in required density, but in no case fewer than two tests.

(i) Footing Subgrade: At least one test for each layer of soil on which footings will be placed. Subsequent verification and approval of each footing subgrade may be based on a visual comparison of each subgrade with related tested subgrade when acceptable to Resident Engineer. In each compacted fill layer below wall footings, perform one field density test for every 100 feet of wall. Verify subgrade is level, all loose or disturbed soils have been removed, and correlate actual soil conditions observed with those indicated by test borings.

c. Fill and Backfill Material Gradation: One test per 2000 yards stockpiled or in-place source material. Gradation of fill and backfill material shall be determined in accordance with ASTM C136/C136M, ASTM D6913/D6913M, ASTM D1140 and ASTM D2419 for sand equivalent tests.

d. Testing for Footing Bearing Capacity: Evaluate if suitable bearing capacity material is encountered in footing subgrade.

e. Testing Materials: Test suitability of on-site and off-site borrow as directed by Resident Engineer.

3.2 FOUNDATION CAISSONS

a. Concrete Testing: Test concrete including materials for concrete as required in Article, CONCRETE of this section, except make two test cylinders for each day's placement of concrete.

b. Maintain a record of concrete used in each caisson. Compare records with calculated volumes.

c. Inspect sides and bottom of each caisson for compliance with contract documents.

d. Submit a certified “Caisson Field Record” for each caisson, recording actual elevation at bottom of shaft; final center line location of top; variation of shaft from plumb; results of all tests performed; material types encountered during drilling; cleanness of bottom; seepage of water; still water level; variation of shaft (dimensions from the plans); location and size of reinforcement; presence and

degree of caving; and equipment and methods used to drill and install the caisson.

3.3 LANDSCAPING

a. Test topsoil for organic materials, pH, phosphate, potash content, and gradation of particles.

(1) Test for organic material by using ASTM D2974.

(2) Determine percent of silt, sand, clay, and foreign materials such as rock, roots, and vegetation.

b. Submit laboratory test report of topsoil to Resident Engineer.

3.4 ASPHALT CONCRETE PAVING

a. Aggregate Base Course:

(1) Determine maximum density and optimum moisture content for aggregate base material in accordance with ASTM D1557.

(2) Make a minimum of three field density tests on each day's final compaction on each aggregate course in accordance with ASTM D1556/D1556M or ASTM D6938.

(3) Sample and test aggregate as necessary to insure compliance with specification requirements for gradation, wear, and soundness as specified in the applicable state highway standards and specifications.

b. Asphalt Concrete:

(1) Aggregate: Sample and test aggregates in stock pile and hot-bins as necessary to insure compliance with specification requirements for gradation (AASHTO T 27, wear (AASHTO T 96), and soundness (AASHTO T 104).

(2) Temperature: Check temperature of each load of asphalt concrete at mixing plant and at site of paving operation.

(3) Density: Make a minimum of two field density tests in accordance with ASTM D1188 of asphalt base and surface course for each day's paving operation.

3.5 SITE WORK CONCRETE

Test site work concrete including materials for concrete as required in Article CONCRETE of this section.

3.6 CONCRETE

a. Batch Plant Inspection and Materials Testing:

(1) Perform continuous batch plant inspection until concrete quality is established to satisfaction of Resident Engineer with concurrence of Contracting Officer and perform periodic inspections thereafter as determined by Resident Engineer.

(2) Periodically inspect and test batch proportioning equipment for accuracy and report deficiencies to Resident Engineer.

(3) Sample and test mix ingredients as necessary to insure compliance with specifications.

(4) Sample and test aggregates daily and as necessary for moisture content. Test the dry rodded weight of the coarse aggregate whenever a sieve analysis is made, and when it appears there has been a change in the aggregate.

(5) Certify, in duplicate, ingredients and proportions and amounts of ingredients in concrete conform to approved trial mixes. When concrete is batched or mixed off immediate building site, certify (by signing, initialing or stamping thereon) on delivery slips (duplicate) that ingredients in truck-load mixes conform to proportions of aggregate weight, cement factor, and water-cement ratio of approved trial mixes.

b. Field Inspection and Materials Testing:

(1) Provide a technician at site of placement at all times to perform concrete sampling and testing.

(2) Review the delivery tickets of the ready-mix concrete trucks arriving on-site. Notify the Contractor if the concrete cannot be placed within the specified time limits or if the type of concrete delivered is incorrect. Reject any loads that do not comply with the Specification requirements. Rejected loads are to be removed from the site at the Contractor’s expense. Any rejected concrete that is placed will be subject to removal.

(3) Take concrete samples at point of placement in accordance with ASTM C172. Mold and cure compression test cylinders in accordance with ASTM C31/C31M. Make at least four cylinders for each (50 cubic yards) or less of each concrete type, and at least four cylinders for any one day's pour for each concrete type. After good concrete quality control has been established and maintained as determined by Resident Engineer make four cylinders for each (100 cubic yards) or less of each concrete type, and at least four cylinders from any one day's pour for each concrete type. Label each cylinder with an identification number. Resident Engineer may require additional cylinders to be molded and cured under job conditions.

(4) Perform slump tests in accordance with ASTM C143/C143M. Test the first truck each day, and every time test cylinders are made. Test pumped concrete at the hopper and at the discharge end of the hose at the beginning of each day’s pumping operations to determine change in slump.

(5) Determine the air content of concrete per ASTM C173/C173M. For concrete required to be air-entrained, test the first truck and every (25 cubic yards) thereafter each day. For concrete not required to be air-entrained, test every (100 cubic yards) at random. For pumped concrete, initially test concrete at both the hopper and the discharge end of the hose to determine change in air content.

(6) If slump or air content fall outside specified limits, make another test immediately from another portion of same batch.

(7) Perform unit weight tests in compliance with ASTM C138/C138M for normal weight concrete and ASTM C567/C567M for lightweight concrete. Test the first truck and each time cylinders are made.

(8) Notify laboratory technician at batch plant of mix irregularities and request materials and proportioning check.

(9) Verify that specified mixing has been accomplished.

(10) Environmental Conditions: Determine the temperature per ASTM C1064/C1064M for each truckload of concrete during hot weather and cold weather concreting operations:

(a) When ambient air temperature falls below (40 degrees F), record maximum and minimum air temperatures in each 24 hour period; record air temperature inside protective enclosure; record minimum temperature of surface of hardened concrete.

(b) When ambient air temperature rises above (85 degrees F), record maximum and minimum air temperature in each 24 hour period; record minimum relative humidity; record maximum wind velocity; record maximum temperature of surface of hardened concrete.

(11) Inspect the reinforcing steel placement, including bar size, bar spacing, top and bottom concrete cover, proper tie into the chairs, and grade of steel prior to concrete placement. Submit detailed report of observations.

(12) Observe conveying, placement, and consolidation of concrete for conformance to specifications.

(13) Observe condition of formed surfaces upon removal of formwork prior to repair of surface defects and observe repair of surface defects.

(14) Observe curing procedures for conformance with specifications, record dates of concrete placement, start of preliminary curing, start of final curing, end of curing period.

(15) Observe preparations for placement of concrete:

(a) Inspect handling, conveying, and placing equipment, inspect vibrating and compaction equipment.

(b) Inspect preparation of construction, expansion, and isolation joints.

(16) Observe preparations for protection from hot weather, cold weather, sun, and rain, and preparations for curing.

(17) Observe concrete mixing:

(a) Monitor and record amount of water added at project site.

(b) Observe minimum and maximum mixing times.

(18) Measure concrete flatwork for levelness and flatness as follows:

(a) Perform Floor Tolerance Measurements FF and FL in accordance with ASTM E1155. Calculate the actual overall F- numbers using the inferior/superior area method.

(b) Perform all floor tolerance measurements within 48 hours after slab installation and prior to removal of shoring and formwork.

(c) Provide the Contractor and the Resident Engineer with the results of all profile tests, including a running tabulation of the overall FF and FL values for all slabs installed to date, within 72 hours after each slab installation.

(19) Other inspections:

(a) Grouting under base plates.

(b) Grouting anchor bolts and reinforcing steel in hardened concrete.

c. Laboratory Tests of Field Samples:

(1) Test compression test cylinders for strength in accordance with ASTM C39/C39M. For each test series, test one cylinder at 7 days and two cylinders at 28 days. Use remaining cylinder as a spare tested as directed by Resident Engineer. Compile laboratory test reports as follows: Compressive strength test shall be result of one cylinder, except when one cylinder shows evidence of improper sampling, molding or testing, in which case it shall be discarded and strength of spare cylinder shall be used.

(2) Make weight tests of hardened lightweight structural concrete in accordance with ASTM C567/C567M.

(3) Furnish certified compression test reports (duplicate) to Resident Engineer. In test report, indicate the following information:

(a) Cylinder identification number and date cast.

(b) Specific location at which test samples were taken.

(c) Type of concrete, slump, and percent air.

(d) Compressive strength of concrete in (psi).

(e) Weight of lightweight structural concrete in (pounds per cubic feet).

(f) Weather conditions during placing.

(g) Temperature of concrete in each test cylinder when test cylinder was molded.

(h) Maximum and minimum ambient temperature during placing.

(i) Ambient temperature when concrete sample in test cylinder was taken.

(j) Date delivered to laboratory and date tested.

3.7 REINFORCEMENT

a. Review mill test reports furnished by Contractor.

b. Perform sampling at fabricating plant. Take two samples from each (25 tons) or fraction thereof of each size of reinforcing steel No. 10 thru No. 57 (No. 3 thru No. 18).

c. Make one tensile and one bend test in accordance with ASTM A370 from each pair of samples obtained.

d. Written report shall include, in addition to test results, heat number, manufacturer, type and grade of steel, and bar size.

e. Perform tension tests of mechanical and welded splices in accordance with ASTM A370.

3.8 SHOTCRETE

a. Inspection and Material Testing:

(1) Provide field inspection and testing service as required by Resident Engineer to certify that shotcrete has been applied in accordance with contract documents.

(2) Periodically inspect and test proportioning equipment for accuracy and report deficiencies to Resident Engineer.

(3) Sample and test mix ingredients as necessary to insure compliance with specifications.

(4) Sample and test aggregates daily and as necessary for moisture content. Report instances of excessive moisture to Resident Engineer.

(5) Certify, in duplicate, that ingredients and proportions and amounts of ingredients in shotcrete conform to approved trial mixes.

(6) Provide field inspection of the proper size and placement of the reinforcement in the shotcrete.

b. Shotcrete Sampling:

(1) Provide a technician at site of placement to perform shotcrete sampling.

(2) Take cores in accordance with ACI 506.4.

(3) Insure maintenance of water-cement ratio established by approved trial mix.

(4) Verify specified mixing has been accomplished.

c. Laboratory Tests of Field Sample Panels:

(1) Compression test core for strength in accordance with ACI 506.4.

For each test series of four cores, test one core at 7 days and two cores at 28 days. Use remaining core as a spare to be tested at either 7 or 28 days as required. Compile laboratory test reports as follows: Compressive strength test shall be result of one core, except when one core shows evidence of improper sampling or testing, in which case it shall be discarded and strength of spare core shall be used.

(2) Submit certified compression test reports (duplicate) to Resident Engineer. On test report, indicate following information:

(a) Core identification number and date cast.

(b) Specific location at which test samples were taken.

(c) Compressive strength of shotcrete in (psi).

(d) Weather conditions during placing.

(e) Temperature of shotcrete in each test core when test core was taken.

(f) Maximum and minimum ambient temperature during placing.

(g) Ambient temperature when shotcrete sample was taken.

(h) Date delivered to laboratory and date tested.

d. Submit inspection reports certification and instances of noncompliance to Resident Engineer.

3.9 MASONRY

a. Mortar Tests:

(1) Laboratory compressive strength test:

(a) Comply with ASTM C780.

(b) Obtain samples during or immediately after discharge from batch mixer.

(c) Furnish molds with (2 inch), 3 compartment gang cube.

(d) Test one sample at 7 days and 2 samples at 28 days.

(2) Two tests during first week of operation; one test per week after initial test until masonry completion.

b. Grout Tests:

(1) Laboratory compressive strength test:

(a) Comply with ASTM C1019.

(b) Test one sample at 7 days and 2 samples at 28 days.

(c) Perform test for each (2500 square feet) of masonry.

c. Masonry Unit Tests:

(1) Laboratory Compressive Strength Test:

(a) Comply with ASTM C140/C140M.

(b) Test 3 samples for each (5000 square feet) of wall area.

d. Prism Tests: For each type of wall construction indicated, test masonry prisms per ASTM C1314 for each (5000 square feet) of wall area. Prepare one set of prisms for testing at 7 days and one set for testing at 28 days.

3.10 STRUCTURAL STEEL

a. General: Provide shop and field inspection and testing services to certify structural steel work is done in accordance with contract documents. Welding shall conform to AWS D1.1/D1.1M Structural Welding Code.

b. Prefabrication Inspection:

(1) Review design and shop detail drawings for size, length, type and location of all welds to be made.

(2) Approve welding procedure qualifications either by pre-qualification or by witnessing qualifications tests.

(3) Approve welder qualifications by certification or retesting.

(4) Approve procedure for control of distortion and shrinkage stresses.

(5) Approve procedures for welding in accordance with applicable sections of AWS D1.1/D1.1M.

c. Fabrication and Erection:

(1) Weld Inspection:

(a) Inspect welding equipment for capacity, maintenance and working condition.

(b) Verify specified electrodes and handling and storage of electrodes in accordance with AWS D1.1/D1.1M.

(c) Inspect preparation and assembly of materials to be welded for conformance with AWS D1.1/D1.1M.

(d) Inspect preheating and interpass temperatures for conformance with AWS D1.1/D1.1M.

(e) Measure 25 percent of fillet welds.

(f) Welding Magnetic Particle Testing: Test in accordance with ASTM E709 for a minimum of:

1. 20 percent of all shear plate fillet welds at random, final pass only.

2. 20 percent of all continuity plate and bracing gusset plate fillet welds, at random, final pass only.

3. 100 percent of tension member fillet welds (i.e., hanger connection plates and other similar connections) for root and final passes.

4. 20 percent of length of built-up column member partial penetration and fillet welds at random for root and final passes.

5. 100 percent of length of built-up girder member partial penetration and fillet welds for root and final passes.

(g) Welding Ultrasonic Testing: Test in accordance with ASTM E164 and AWS D1.1/D1.1M for 100 percent of all full penetration welds, braced and moment frame column splices, and a minimum of 20 percent of all other partial penetration column splices, at random.

(h) Welding Radiographic Testing: Test in accordance with ASTM E94/E94M, and AWS D1.1/D1.1M for 5 percent of all full penetration welds at random.

(i) Verify that correction of rejected welds are made in accordance with AWS D1.1/D1.1M.

(j) Testing and inspection do not relieve the Contractor of the responsibility for providing materials and fabrication procedures in compliance with the specified requirements.

(2) Bolt Inspection:

(a) Inspect high-strength bolted connections in accordance AISC Specifications for Structural Joints Using ASTM F3125/F3125M Bolts.

(b) Slip-Critical Connections: Inspect 10 percent of bolts, but not less than 2 bolts, selected at random in each connection in accordance with AISC Specifications for Structural Joints Using ASTM F3125/F3125M Bolts. Inspect all bolts in connection when one or more are rejected.

(c) Fully Pre-tensioned Connections: Inspect 10 percent of bolts, but not less than 2 bolts, selected at random in 25 percent of connections in accordance with AISC Specification for Structural Joints Using ASTM F3125/F3125M Bolts. Inspect all bolts in connection when one or more are rejected.

(d) Bolts installed by turn-of-nut tightening may be inspected with calibrated wrench when visual inspection was not performed during tightening.

(e) Snug Tight Connections: Inspect 10 percent of connections verifying that plies of connected elements have been brought into snug contact.

(f) Inspect field erected assemblies; verify locations of structural steel for plumbness, level, and alignment.

d. Submit inspection reports, record of welders and their certification, and identification, and instances of noncompliance to Resident

Engineer.

3.11 STEEL DECKING

a. Provide field inspection of welds of metal deck to the supporting steel, and testing services to insure steel decking has been installed in accordance with contract documents and manufacturer's requirements.

b. Qualification of Field Welding: Qualify welding processes and welding operators in accordance with "Welder Qualification" procedures of AWS D1.1/D1.1M. Refer to the "Plug Weld Qualification Procedure" in Part 3 "Field Quality Control."

c. Submit inspection reports, certification, and instances of noncompliance to Resident Engineer.

3.12 SHEAR CONNECTOR STUDS

a. Provide field inspection and testing services required by AWS D1.1/D1.1M to insure shear connector studs have been installed in accordance with contract documents.

b. Tests: Test 20 percent of headed studs for fastening strength in accordance with AWS D1.1/D1.1M.

c. Submit inspection reports, certification, and instances of noncompliance to Resident Engineer.

3.13 SPRAYED-ON FIREPROOFING

a. Provide field inspection and testing services to certify sprayed-on fireproofing has been applied in accordance with contract documents.

b. Obtain a copy of approved submittals from Resident Engineer.

c. Use approved installation in test areas as criteria for inspection of work.

d. Test sprayed-on fireproofing for thickness and density in accordance with ASTM E605.

(1) Thickness gauge specified in ASTM E605 may be modified for pole extension so that overhead sprayed material can be reached from floor.

e. Location of test areas for field tests as follows:

(1) Thickness: Select one bay per floor, or one bay for each (10,000 square feet) of floor area, whichever provides for greater number of tests. Take thickness determinations from each of following locations: Metal deck, beam, and column.

(2) Density: Take density determinations from each floor, or one test from each (10,000 square feet) of floor area, whichever provides for greater number of tests, from each of the following areas: Underside of metal deck, beam flanges, and beam web.

f. Submit inspection reports, certification, and instances of noncompliance to Resident Engineer.

3.14 TYPE OF TEST

Approximate Number of Tests Required

a. Earthwork:

Laboratory Compaction Test, Soils:
(ASTM D1557)
Field Density, Soils _____

b. Landscaping:

Topsoil Test_____

c. Aggregate Base:

Laboratory Compaction, (ASTM D1557) _____
Field Density, (ASTM D1556/D1556M/ASTM D6938)_____
Aggregate, Base Course Gradation (AASHTO T 27) _____
Wear (AASHTO T 96) _____
Soundness (AASHTO T 104) _____

d. Asphalt Concrete:

Field Density, ASTM D1188 _____
Aggregate, Asphalt Concrete Gradation (AASHTO T 27) _____
Wear (AASHTO T 96) _____
Soundness (AASHTO T 104) _____

e. Concrete:

Making and Curing Concrete Test Cylinders (ASTM C31/C31M) _____
Compressive Strength, Test Cylinders (ASTM C39/C39M) _____
Concrete Slump Test (ASTM C143/C143M) _____
Concrete Air Content Test (ASTM C173/C173M) _____
Unit Weight, Lightweight Concrete (ASTM C567/C567M) _____
Aggregate, Normal Weight: Gradation (ASTM C33/C33M) _____
Deleterious Substances (ASTM C33/C33M) _____
Soundness (ASTM C33/C33M) _____
Abrasion (ASTM C33/C33M) _____
Aggregate, Lightweight Gradation (ASTM C330) _____
Deleterious Substances (ASTM C330) _____
Unit Weight (ASTM C330) _____
Flatness and Levelness Readings (ASTM E1155) (number of days) _____

f. Reinforcing Steel:

Tensile Test (ASTM A370)_____
Bend Test (ASTM A370) _____
Mechanical Splice (ASTM A370) _____
Welded Splice Test (ASTM A370)_____

g. Shotcrete:

Taking and Curing Test Cores (ACI 506.4) _____
Compressive Strength, Test Cores (ACI 506.4)_____

h. Structural Steel:

Ultrasonic Testing of Welds (ASTM E164) _____
Magnetic Particle Testing of Welds (ASTM E709) _____
Radiographic Testing of Welds (ASTM E94/E94M) _____

i. Sprayed-On Fireproofing:

Thickness and Density Tests (ASTM E605) _____

j. Inspection:

Technical Personnel (Man-days) 100

k. Technical Personnel: (Minimum 42 months)

(1) Technicians to perform tests and inspection listed above. Laboratory will be equipped with concrete cylinder storage facilities, compression machine, cube molds, proctor molds, balances, scales, moisture ovens, slump cones, air meter, and all necessary equipment for compaction control.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 35

SPECIAL INSPECTIONS

11/20

PART 1 GENERAL

1.1 REFERENCES
1.2 GENERAL REQUIREMENTS
1.3 DEFINITIONS
1.3.1 Continuous Special Inspections
1.3.2 Perform
1.3.3 Observe
1.3.4 Special Inspector (SI)
1.3.5 Associate Special Inspector (ASI)
1.3.6 Third Party
1.3.7 Special Inspector of Record (SIOR)
1.3.8 Contracting Officer
1.3.9 Contractor's Quality Control (QC) Manager
1.3.10 Structural Engineer of Record (SER)
1.3.11 Statement of Special Inspections (SSI)
1.3.12 Schedule of Special Inspections (SSI)
1.3.13 Designated Seismic Systems (DSS)
1.3.14 Definable Feature of Work (DFOW)
1.4 SUBMITTALS
1.5 SPECIAL INSPECTOR QUALIFICATIONS
1.5.1 Steel Construction and High Strength Bolting - CCC, CUP, LTH and Site
1.5.1.1 Special Inspector
1.5.1.2 Associate Special Inspector
1.5.2 Welding Structural Steel - CCC, CUP, LTH and Site
1.5.2.1 Special Inspector
1.5.2.2 Associate Special Inspector
1.5.3 Nondestructive Testing of Welds - CCC, CUP, LTH and Site
1.5.3.1 Special Inspector
1.5.3.2 Associate Special Inspector
1.5.4 Concrete Construction - CCC, CUP, LTH and Site
1.5.4.1 Special Inspector
1.5.4.2 Associate Special Inspector
1.5.5 Masonry Construction - CUP, LTH and Site
1.5.5.1 Special Inspector
1.5.5.2 Associate Special Inspector
1.5.6 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements - CCC, CUP, LTH and Site
1.5.6.1 Special Inspector
1.5.6.2 Associate Special Inspector
1.5.7 Deep Foundations - CCC and LTH
1.5.7.1 Special Inspector
1.5.7.2 Associate Special Inspector
1.5.8 Sprayed Fire Resistant Material - CCC and LTH
1.5.8.1 Special Inspector

1.5.8.2 Associate Special Inspector
1.5.9 Fire-Resistant Penetrations and Joints - CCC, CUP and LTH
1.5.9.1 Special Inspector
1.5.9.2 Associate Special Inspector
1.5.10 Vapor Mitigation Systems - Site
1.5.10.1 Special QC Inspector
1.5.10.2 QC Inspection Requirements
1.5.10.3 Smoke Testing
1.5.10.4 Concrete Placement
1.5.10.5 Vapor Barrier Repair
1.5.11 Special Inspector of Record (SIOR)

PART 2 PRODUCTS

2.1 FABRICATOR SPECIAL INSPECTIONS

PART 3 EXECUTION

3.1 RESPONSIBILITIES
3.1.1 Special Inspector of Record
3.1.2 Quality Control Manager
3.1.3 Special Inspectors
3.2 DEFECTIVE WORK

-- End of Section Table of Contents --

SECTION 01 45 35

SPECIAL INSPECTIONS
11/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22	(2022; Supp 1 2023; Supp 2 2023) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections and Chapter 17 of ICC IBC. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the Prime Contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.

Structural observations must be performed . The Contractor must provide notification to the Contracting Officer and Structural Engineer of Record 14 days prior to the following points of construction to allow for structural observation:

a. Placing of concrete

b. Placing of reinforcing

c. Placing of structural steel

d. Placing of metal deck

e. Placing of anchor bolts

f. Placing of mechanical and adhesive anchors

1.3 DEFINITIONS

1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.

1.3.2 Perform

Perform these Special Inspections tasks for each welded joint, slip-critical steel connection, demand critical elements and systems as designated by design code, statement of special inspections and Structural Engineer of Record requirements.

1.3.3 Observe

Observe these Special Inspections items on a periodic daily basis. Operations need not be delayed pending these inspections.

1.3.4 Special Inspector (SI)

A qualified person retained by the Contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

1.3.5 Associate Special Inspector (ASI)

A qualified person who assists the SI in performing Special Inspections but must perform inspection under the direct supervision of the SI and cannot perform inspections without the SI on site. The licensed engineer must be licensed in the state in which the project is constructed.

1.3.6 Third Party

A Special inspector must not be an employee of the Contractor or of any Sub-Contractor performing the work to be inspected.

1.3.7 Special Inspector of Record (SIOR)

A licensed engineer in responsible charge of supervision of all special inspectors for the project and approved by the Contracting Officer. The SIOR must be an independent third party entity hired directly by the Prime Contractor. Licensed engineer must be licensed in the state in which the project is constructed.

1.3.8 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (CO).

1.3.9 Contractor's Quality Control (QC) Manager

An individual retained by the Prime Contractor and qualified in accordance with the Section 01 45 00 QUALITY CONTROL having the overall responsibility for the Contractor's QC organization.

1.3.10 Structural Engineer of Record (SER)

A registered design professional contracted by the Government as an A/E responsible for the overall design and review of submittal documents prepared by the contractor. The SER is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in the state in which the design professional works. The SER is also referred to as the Engineer of Record (EOR) and Structural Engineer of Record (SEOR) in design code documents.

1.3.11 Statement of Special Inspections (SSI)

A document developed by the SER identifying the material, systems, components and work required to have Special Inspections. This statement is included at the end of this specification.

1.3.12 Schedule of Special Inspections (SSI)

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspection is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17. This schedule is included at the end of this specification.

1.3.13 Designated Seismic Systems (DSS)

Those nonstructural components that require design in accordance with ASCE 7-22 Chapter 13 and for which the component importance factor, Ip, is greater than 1.0. This designation applies to systems that are required to be operational following the Design Earthquake for RC I - IV structures and following the MCER for RC V structures. All systems in RC V facilities are considered part of the Designated Seismic Systems. Designated Seismic Systems will have an Importance Factor Ip = 1.5.

1.3.14 Definable Feature of Work (DFOW)

An inspection group that is separate and distinct from other inspection groups, having inspection requirements or inspectors that are unique.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

SIOR Letter of Acceptance; G

Special Inspections Project Manual; G

Special Inspections Agency's Written NDT Practices with method and evidence of regular equipment calibration where applicable

SD-06 Test Reports

Special Inspections Daily Reports

Special Inspections Biweekly Reports

Interim Report

SD-07 Certificates

AISC Certified Steel Fabricator

AC472 Accreditation

Certificate of Compliance

Special Inspector of Record Qualifications; G

Special Inspector Qualifications; G

Qualification Records for NDT technicians

SD-11 Closeout Submittals

Comprehensive Final Report of Special Inspections; G

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector and the special inspector of record.

1.5.1 Steel Construction and High Strength Bolting - CCC, CUP, LTH and Site

1.5.1.1 Special Inspector

a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or

b. Registered Professional Engineer with three years of related experience

1.5.1.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.2 Welding Structural Steel - CCC, CUP, LTH and Site

1.5.2.1 Special Inspector

a. ICC Structural Welding Special Inspector certificate with one year of related experience, or

b. AWS Certified Welding Inspector

1.5.2.2 Associate Special Inspector

AWS Certified Associate Welding Inspector

1.5.3 Nondestructive Testing of Welds - CCC, CUP, LTH and Site

1.5.3.1 Special Inspector

NDT Level III Certificate

1.5.3.2 Associate Special Inspector

NDT Level II Certificate plus one year of related experience

1.5.4 Concrete Construction - CCC, CUP, LTH and Site

1.5.4.1 Special Inspector

a. ICC Reinforced Concrete Special Inspector Certificate with one year of related experience, or

b. ACI Concrete Construction Special Inspector, or

c. Registered Professional Engineer with three years of related experience

1.5.4.2 Associate Special Inspector

a. ACI Concrete Construction Special Inspector in Training, or

b. Engineer-In-Training with one year of related experience

1.5.5 Masonry Construction - CUP, LTH and Site

1.5.5.1 Special Inspector

a. ICC Structural Masonry Special Inspector Certificate with one year of related experience, or

b. Registered Professional Engineer with three years of related experience

1.5.5.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.6 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements - CCC, CUP, LTH and Site

1.5.6.1 Special Inspector

a. ICC Soils Special Inspector Certificate with one year of related experience, or

b. NICET Soils Technician Level II Certificate in Construction Material Testing, or

c. Geologist-In-Training with three years of related experience, or

d. Registered Professional Engineer with three years of related experience

1.5.6.2 Associate Special Inspector

a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or

b. Engineer-In-Training with one year of related experience

1.5.7 Deep Foundations - CCC and LTH

1.5.7.1 Special Inspector

a. NICET Soils Technician Level II Certificate in Construction Material Testing, or

b. Geologist-In-Training with three years of related experience, or

c. Registered Professional Engineer with three years of related experience

1.5.7.2 Associate Special Inspector

a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or

b. NICET Geotechnical Engineering Technician Level I Construction or Generalist Certificate with one year of related experience, or

c. Engineer-In-Training with one year of related experience

1.5.8 Sprayed Fire Resistant Material - CCC and LTH

1.5.8.1 Special Inspector

a. ICC Spray-applied Fireproofing Special Inspector Certificate, or

b. ICC Fire Inspector I Certificate with one year of related experience, or

c. Registered Professional Engineer or Architect with related experience

1.5.8.2 Associate Special Inspector

Engineer-In-Training with one year of related experience

1.5.9 Fire-Resistant Penetrations and Joints - CCC, CUP and LTH

1.5.9.1 Special Inspector

a. Passed the UL Firestop Exam with one year of related experience, or

b. Passed the FM Firestop Exam with one year of related experience, or

c. Registered Professional Engineer with related experience

1.5.9.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.10 Vapor Mitigation Systems - Site

1.5.10.1 Special QC Inspector

The Special QC inspector shall have direct experience in the installation and QA/QC of vapor mitigation systems, and be a certified inspector with

the elected vapor barrier manufacturer (such as EPRO, CETCO, Land Science, etc.).

1.5.10.2 QC Inspection Requirements

a. Coupon samples will be collected a minimum of one every 500-2,500 sq. ft. to verify thickness of the vapor barrier. The minimum thickness of the coupon sample is 80-mil. Readings shall be recorded on the Coupon Sampling and Smoke Testing Log by the Special QC Inspector or Engineer.

b. The passive venting system will be inspected and approved by the Special QC Inspector prior to placement of the vapor barrier. The Contractor will leave horizontal vent lines and connections uncovered for inspection by the Special QC Inspector. After successful inspection, the horizontal vent lines can be covered with at least 2 inches of gravel.

c. The vapor barrier will be visually inspected by the Special QC Inspector. Any apparent deficiencies and/or installation problems will be corrected prior to Smoke Testing.

d. The Special QC Inspector shall verify that the substrate is properly prepared according to manufacturer's requirements and the methane mitigation system plans. The substrate will be free from materials and jagged edges that can potentially puncture the vapor barrier. All loose earth, cobbles, wire tacks and other foreign matters shall be completely removed.

1.5.10.3 Smoke Testing

Smoke testing of the vapor barrier shall be the responsibility of the Contractor and observed by the Special QC Inspector.

a. The date, time, testing reference area, temperature, wind speed/direction, and cloud cover shall be recorded on the Smoke Testing Record completed by the Special QC Inspector. The ambient air temperature at the time of testing should be in excess of 45° F, and the wind speed at ground level should be 10 mph or less. (Note: Visual identification of leaks becomes more difficult with increasing wind speed.)

b. Delineate a smoke testing area of 2,000-5,000 sq. ft. (maximum). Smoke test the distinct areas of the vapor barrier. only inert, non-toxic smoke is to be utilized for the Smoke Test.

c. The Contractor will cut openings (minimum 4 inches by 4 inches) in the vapor barrier at the testing control areas for the purpose of smoke testing. The coupons will be provided to the Quality Inspector for thickness verification.

d. Activate smoke generator/blower system (nominal 150-950 cfm). Apply sufficient smoke to ensure that smoke is permeating the entire designated test area. For verification, ensure that smoke is leaking through test control openings.

e. Pump smoke beneath the vapor barrier (minimum 1-2 minutes). Observe for leaks in the vapor barrier.

f. Thoroughly inspect the entire vapor barrier surface within the

delineated testing areas. Mark leak locations where smoke appears.

g. Repair leak locations and retest to confirm repair and integrity of the vapor barrier.

h. Once the smoke test has been successfully completed, the Special QC Inspector will sign-off on test and provide QC documentation for the VA.

1.5.10.4 Concrete Placement

Concrete placement can occur after QC Inspection and testing has been successfully performed and Quality Inspector approves placement of the concrete. The QC Inspector will be present during placement of the concrete to verify that the in-place vapor barrier is not damaged during concrete placement. Vehicle and equipment traffic, including Laser Screed and outriggers will not occur directly on the in-place vapor barrier. If the QC Inspector suspects or observes vapor barrier damage during concrete placement, the suspected damaged area will be blocked off until the vapor barrier is inspected, and if necessary, repaired according to the instructions provided below. Concrete placement will not occur in the suspected damaged areas until the QC Inspector verifies the vapor barrier is not damaged or is repaired accordingly.

1.5.10.5 Vapor Barrier Repair

If the Quality Inspector observes or suspects a breach (puncture, hole, tear) in the vapor barrier during vapor barrier installation, concrete placement over vapor barrier, or during use of the laser screed, the breach will be repaired prior to commencing construction. Repair activities will be coordinated with the GC and the vapor barrier installer. Repairs shall be executed within 24 hours. Vapor barrier repairs shall be conducted in accordance with the manufacturer's recommendations. The Quality Inspector will inspect and approve the repair after repair sequence is completed. A smoke test will be performed to verify the repair, if directed by the Quality Inspector.

1.5.11 Special Inspector of Record (SIOR)

Registered Professional Engineer with five years of related experience.

PART 2 PRODUCTS

2.1 FABRICATOR SPECIAL INSPECTIONS

Special Inspections of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following to the Contracting Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

AISC Certified Steel Fabricator.

International Accreditation Service, AC472 Accreditation

At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special Inspections,

stating that the materials supplied and work performed by the fabricator are in accordance with the construction documents.

PART 3 EXECUTION

3.1 RESPONSIBILITIES

3.1.1 Special Inspector of Record

a. Supervise all Special Inspectors required by the Contract Documents and the IBC.

b. Submit a SIOR Letter of Acceptance to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR.

c. Verify the qualifications of all of the Special Inspectors.

d. Verify the qualifications of fabricators.

e. Submit Special Inspections agency's written NDT practices for the monitoring and control of the agency's operations to include the following:

(1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.

(2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.

f. Submit qualification records for nondestructive testing (NDT) technicians designated for the project.

g. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.

h. Prepare a Special Inspections Project Manual, which must cover the following:

(1) Roles and responsibilities of the following individuals during Special Inspections: SIOR, SI, ASI, General Contractor's QC Manager and SER.

(2) Organizational chart or communication plan, indicating lines of communication.

(3) Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, who to contact for inspection requests, and availability of alternate inspectors.

(4) Indicate the Government reporting requirements.

(5) Propose forms or templates to be used by SI and SIOR to document inspections.

(6) Indicate procedures for tracking nonconforming work and verification that corrective work is complete.

(7) Indicate how the SIOR and SI will participate in weekly QC meetings.

(8) Indicate how Special Inspections of shop fabricated items will be handled when the fabricator's shop is not certified in accordance with paragraph FABRICATOR SPECIAL INSPECTIONS.

(9) Include a section in the manual that covers each specific item requiring Special Inspections that is indicated on the Schedule of Special Inspections. Provide names and qualifications of each special inspector who will be performing the Special Inspections for each specific item. Provide detail on how the Special Inspections are to be carried out for each item so that the expectations are clear for the General Contractor and the Subcontractor performing the work.

Make a copy of the Special Inspections Project Manual available on the job site during construction. Submit a copy of the Special Inspections Project Manual for approval.

i. Attend coordination and mutual understanding meeting where the information in the Special Inspections Project Manual will be reviewed to verify that all parties have a clear understanding of the Special Inspections provisions and the individual duties and responsibilities of each party.

j. Maintain a 3-ring binder for the Special Inspector's daily and biweekly reports and the Special Inspections Project Manual. This file must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the SER.

k. Submit a copy of the Special Inspector's daily reports to the QC Manager.

l. Discrepancies that are observed during Special Inspections must be reported to the QC Manager for correction. If discrepancies are not corrected before the special inspector leaves the site the observed discrepancies must be documented in the daily report. Deficiencies/discrepancies must be submitted in RMS as QC Deficiencies.

m. Submit a biweekly Special Inspections report until all work requiring Special Inspections is complete. A report is required for each biweekly period in which Special Inspections activity occurs, and must include the following:

(1) A brief summary of the work performed during the reporting time frame.

(2) Changes and discrepancies with the drawings, specifications and mechanical or electrical component certification, that were observed during the reporting period.

(3) Discrepancies which were resolved or corrected.

(4) A list of nonconforming items requiring resolution.

(5) All applicable test results including nondestructive testing reports.

(6) Deficiencies/discrepancies must be submitted in RMS as QC Deficiencies.

n. At the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an Interim report that documents the Special Inspections completed for that DFOW including corrections of all discrepancies noted in the daily reports. Interim reports of Special Inspections must be signed and dated by the SIOR.

o. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project including corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and bear the seal of the SIOR.

3.1.2 Quality Control Manager

a. Maintain a rework items list that includes discrepancies noted on the Special Inspectors daily report.

3.1.3 Special Inspectors

a. Inspect all elements of the project for which the special inspector is qualified to inspect and are identified in the Schedule of Special Inspections.

b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) for which the special inspector is qualified to inspect.

c. Submit Special Inspections agency's written NDT practices for the monitoring and control of the agency's operations to include the following:

(1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualifications and certification of inspection personnel.

(2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.

d. Submit qualification records for nondestructive testing (NDT) technicians designated for the project.

e. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.

f. Submit daily reports to the SIOR.

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept such work.

-- End of Section --

Project:	West Haven VAMC - Chilled Water Plant Replacement
Location:	West Haven, CT
Project #:	689-040
Date:	7/14/2026

STATEMENT OF SPECIAL INSPECTIONS

Seismic Design Category:	A
Risk Category:	IV
Number of Stories above Grade:	3
Height of Highest Occupied Roof or Floor	72

Hazardous Occupancy or attached to such? No Group H Occupancies

Special Inspector of Record (SIOR)

A Special Inspector of Record (SIOR) **IS** required (per UFGS 01 45 35, Section 1.3.7)

SIOR Name (Registered Professional):	Contractor to provide prior to start of construction
Professional Registration Number:	
Consulting Firm Name (if any):	
SIOR Office AND Mobile Phone Number:	

Lateral Force Resisting System (LFRS)

2024 IBC 1704.3.2 and 1704.3.3

Following is a listing of critical elements of the main wind/seismic force-resisting system for this structure. Carefully inspect these elements as part of the roles and responsiblities of the Special Inspector (reference the Schedule of Special Inspections for inspection checklists).

Vertical LFRS Elements	Notes
Steel Ordinary Braced Frames (NOT SPECIFICALLY DETAILED FOR SEISMIC R=3)	NORTH SOUTH AND EAST WEST. SEE ELEVATION ON PLAN
ORDINARY REINFORCED MASONRY SHEAR WALLS	GRID LINE 2

Horizontal LFRS Elements	Notes
COLLECTOR ELEMENTS	IDENTIFIED ON PLAN. SEE PLAN KEY FOR SYMBOLS
CONCRETE ON METAL DECK	1ST FLOOR, 2ND FLOOR AND ROOF

Project: West Haven VAMC - Chilled Water Plant Replace
Location: West Haven, CT
Project #: 689-040
Date: 7/14/2026

Designated Seismic Systems (DSS)

(2024 IBC 1705.13.4) (ASCE 7-22, 13.2.3, C13.2.2) (UFC 3-301-1, 2-5.3)

ELECTRICAL Components that are part of the Designated Seismic System (DSS) , which require a Certificate of Compliance
N/A
N/A
N/A
N/A
N/A

If additional space is required, append an additional sheet listing the remaining DSS

MECHANICAL/PLUMBING Components that are part of the Designated Seismic System (DSS), which require a Certificate of Compliance
N/A
N/A
N/A
N/A
N/A
N/A

If additional space is required, append an additional sheet listing the remaining DSS

OTHER Designated Seismic Systems (DSS) requiring a Certificate of Compliance
N/A
N/A
N/A
N/A
N/A
N/A

Final Walk-Down Inspection and Report

(UFC 3 301 01 SECTION 2-5.4)

Final Walk Down Inspection of non-structural Designated Seismic Systems does not apply to this project (no Designated Seismic Systems)

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SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 41

EXTERIOR ENCLOSURE PERFORMANCE REQUIREMENTS, MOCKUPS, AND TESTING

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 DEFINITIONS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 QUALITY ASSURANCE

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

PART 3 EXECUTION

3.1 MOCKUP INSTALLATION
3.2 FIELD QUALITY CONTROL TESTING
3.3 TEST REPORTS
3.4 REMEDIAL WORK BY CONTRACTOR

-- End of Section Table of Contents --

SECTION 01 45 41

EXTERIOR ENCLOSURE PERFORMANCE REQUIREMENTS, MOCKUPS, AND TESTING

PART 1 GENERAL

1.1 SUMMARY

a. Section Includes

(1) General exterior enclosure performance requirements.

(3) Exterior enclosure field performance testing requirements.

b. General Requirements

(1) Provide exterior enclosure in accordance with requirements of the Contract Documents. The following requirements apply generally to all exterior enclosure systems. Where a particular requirement applies to a given enclosure component or system, these are given in the relevant system section.

(2) Extent of Exterior Enclosure Work: Details for the exterior enclosure, as shown schematically and together with the systems to be developed, are intended to establish performance and material qualities required. The exterior enclosure shall be a complete system including all aluminum, glass, insulation, attachments, flashing, panels, support framing, sheet metal flashing, air and water barriers, waterproofing, sealants, drainage systems, and adjacent systems as required to form a high quality weatherproof enclosure. System shall provide two distinct lines of protection for water infiltration and one distinct line for air, and shall accommodate all wind, seismic, and building movements. Details not shown are similar in character to those detailed. Contractor must develop final details to accommodate the fabrication, erection, and installation of the work in accordance with the design intent shown. All proposed details and finishes for each type of enclosure system shall be reviewed and accepted by the Architect prior to fabrication.

(3) Alternatives proposed by Tenderers to the design nominated on the tender documents will be considered, provided they satisfy all the performance requirements of this specification and are equal to that material or design specified.

(4) Any discrepancies or contradictions in the drawings or specifications shall be brought to the attention of the Architect for clarification during the tender bid period by the tenderer.

(5) Drawings and general provisions of the Contract, including General and Supplemental General Conditions and Division 01 Specification Sections, apply to this Section.

(6) Refer to other Divisions of these Specifications to determine the type and extent of work therein affecting the Work of this trade, whether or not such work is specifically mentioned in this Section.

(7) Refer to the visual mockup and performance mockup drawings included in the drawing package.

1.2 RELATED WORK

a. Section 01 43 30, MOCKUPS

b. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS

c. Section 03 30 00, CAST-IN-PLACE CONCRETE k. Section 07 27 26, FLUID-APPLIED MEMBRANE AIR BARRIER, VAPOR PERMEABLE

d. Section 07 42 13, METAL WALL PANELS

e. Section 07 60 00, FLASHING AND SHEET METAL

f. Section 07 92 13, EXTERIOR JOINT SEALANTS

g. Section 08 41 13, ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

h. Section 08 44 13, GLAZED ALUMINUM CURTAIN WALLS

i. Section 08 80 00, GLAZING

1.3 REFERENCES

a. General: Comply with the applicable provisions of the referenced standards except as modified herein, by the governing codes, or by the Contract Documents. Where a recommendation or suggestion occurs in the referenced standards, such recommendation or suggestion shall be considered mandatory. In the event of conflict between referenced standards, this Specification, or within a document itself, the more stringent standard or requirement shall govern.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 501.2	(2015) Quality Assurance and Diagnostic Water Leakage Field Check of Installed Storefronts, Curtain Walls and Sloped Glazing Systems

ASTM INTERNATIONAL (ASTM)

ASTM E699	(2009) Standard Practice for Evaluation of Agencies Involved in Testing, Quality Assurance, and Evaluating of Building Components
ASTM E779	(2019) Standard Test Method for Determining Air Leakage Rate by Fan Pressurization

1.4 DEFINITIONS

a. Building Enclosure, Building Envelope, Exterior Enclosure, or Exterior Envelope: Materials, components, systems, and assemblies collectively intended to provide shelter or environmental separation between

interior and exterior or between two or more environmentally distinct interior spaces in a building.

b. Condensation: Water, ice, or frost occurring on more than 5 percent of the exposed interior surface of any one component or water that is not collected and positively drained to the exterior through the condensation drainage gutter system. Condensation on concealed surfaces that is not collected and positively drained to the exterior is not acceptable.

c. Failure: Water leakage or condensation constitute a failure.

d. Water Leakage: Water that accumulates at any location within or around the building enclosure system(s) that is not designed to accommodate, control, and direct such water to the building exterior. This includes any water past the interior plane of the glazing of the fenestration.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Architect/Engineer approval is required for submittals with an "AE" designation. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following for mockups and testing in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G, AE

Proposed Test Procedures; G, AE

Proposed Field Test Locations; G, AE

SD-02 Shop Drawings

As-Built Redline Markups; G, AE

SD-05 Design Data

Calculations; G, AE

SD-06 Test Reports

Enclosure Performance Test Report; G, AE

Mockup Visual Documentation; G,AE

1.6 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submit product data and other submittals in accordance with the related technical specification sections in addition to mockup and testing submittals outlined herein.

c. Qualifications: Independent testing agency and independent testing laboratory qualifications.

d. Proposed Test Procedures: Prior to construction of each mockup assembly, submit for Architect's review, and concurrent with the submission of mockup shop drawings, calculations, material submittals, and testing laboratory qualifications, a detailed description of each test procedure required for each mockup coordinated with specified performance criteria and statement of desired results. In addition, note proposed location, number, and type of instrumentation to measure deflections, rotations and sets.

e. Proposed Field Test Locations: Drawings indicating proposed test locations and designated test procedures. Include plans and elevations and independent testing agency qualifications.

f. Calculations: Calculations concurrent with shop and erection drawings to substantiate mockup(s) complies with specified design criteria. Calculations shall contain the seal of a licensed Professional Engineer registered in the State of Connecticut. Provide the following:

(1) Engineering calculations to show that maximum deflections do not exceed specified performance requirements under full design loading.

(2) Structural calculations for frames, connections, panels and anchorage assemblies.

(3) Data for glass showing that the probability of breakage at the design wind pressure will not exceed the specified probability of breakage for each type, size, and thickness of glass. Submit glass manufacturer's substantiating data.

(4) Calculations of expansion and contraction.

(5) Calculations for window washing equipment in contact with exterior wall.

h. Quality Control Submittals

(1) As-Built Redline Markups: The curtain wall manufacturer shall submit as-built redline markups of mockup(s) with actual dimensions and thicknesses of component parts. Include actual measured sizes of glass to 1/32 inches for height and width and 0.001 inches for thickness. Clearly identify changes in the fabrication of the mockup that occurred during erection of the mockup specimen. Include in this submission the actual method of attaching the mockup specimen(s) to the Testing Laboratory's chamber.

(2) Enclosure Performance Test Report: Test report(s), prepared by the Independent Testing Agency, for each specified test showing required performance criteria and test results. Arrange with the Independent Testing Agency to prepare test reports in accordance with testing procedure described in the reference test standards. Include in the test reports the following as applicable:

(a) The nature and objectives of the test, measurement, or chemical analysis.

(b) The credentials of the organization undertaking the test, including their accreditation for such testing, measurement or analysis.

(c) The dates, weather, and witnesses present.

(d) Details of the samples, the method used, and the limits of accuracy.

(e) A factual account of the results.

(f) Witnessing mockup installation and dismantling.

(g) Color photographs of test specimen(s) and relevant details.

(h) Detailed descriptions of modifications made to the test specimen prior to and during the tests, including color images of modifications.

(i) Any change to the method statement submitted with the Shop Drawings.

(j) Evidence of damage or displacement revealed during the mockup testing or dismantling.

(3) Mockup Visual Documentation: Digital color images of mockup before, during, and after preconstruction testing. Record the installation of each individual component or trades work incorporated into the mockup and various stages of the mockup testing sequence as required by the Architect. Give particular attention to deviations which arise during construction or during testing.

1.7 QUALITY ASSURANCE

a. Mockup Meetings

(1) Prior to the start of construction of each mockup assembly at the test facility, and at the Architect's direction, meet to review methods and sequence of each mockup construction. Meeting shall include all agenda requests made by the Architect. Attendees shall include the Contractor awarded the portion of the Contract Work related to each mockup, related Subcontractors, the Independent Testing Agency, and any others whose work on the Project requires coordination with this Work. Include in this meeting individuals supervising both the work of the mockups as well as the construction of the final Contract Work on the Project.

b. Contractor Responsibility

(1) Retain an Independent Testing Agency to perform specified tests to verify proposed system’s stipulated performance. The Contractor must be responsible for preliminary submissions, scheduling, and orderly execution of tests and retests, if necessary.

(2) The Contractor must cover the complete costs for the installation

of the mockup assemblies until the final completion of testing and related acceptance. Supervisory construction personnel for the Contractor and trade contractors shall be present for key erection sequences and testing.

(3) Provide labor, materials, and equipment required for testing by testing agency personnel. Provide powered scaffold, hose, water supply, and communication system to perform tests. Notify Architect and testing agency when wall construction is ready for testing.

(4) Provide safe access to the Architect/Consultant and testing agency to the exterior of test locations.

(5) Satisfactory test results do not relieve Contractor from conforming to requirements of Contract Documents, Shop Drawings, and Project Specifications. Installation work on remainder of building shall match area checked, unless otherwise instructed in writing.

(6) Pending actual performance, extent or absence of water leakage in initial water penetration test, and measures adopted by the Contractor to eliminate leakage, Architect will determine need and scope of additional tests and test methods. Remedial measures are subject to approval by Architect and shall maintain standards of quality and durability.

(7) If a specimen fails testing, the Contractor is responsible for all costs for retests. These costs include, but are not limited to: access, equipment, labor, materials and the Independent Testing Agency required to complete retesting.

(a) All costs associated with retests because of failures that are the result of contractor deficiencies will be charged to the contractor, including Architect and consultant costs of monitoring the retests.

(b) Remedial measures found necessary and effective in eliminating leakage in area checked shall be used in fabricating and installing remainder of the work.

c. Delegated Engineering Requirements

(1) Contract Documents: Drawings and Specifications indicate design intent for Work included in this Technical Section and do not necessarily indicate or specify the full extent of work that may be required.

(2) Delegated Engineering Responsibility: Contractor must employ a qualified structural engineer to provide engineering as required for testing mockup for exterior building enclosure system including support structure.

d. Testing Agency Qualifications for Field Testing

(1) Qualified according to ASTM E699 for testing indicated.

(a) Testing equipment shall be calibrated at maximum 12 calendar

month intervals by devices of accuracy traceable to the National Bureau of Standards or accepted values of natural physical constants. Testing agency shall furnish copy of certification.

e. Testing

(1) Arrange for and have tests completed sufficiently in advance to allow for incorporation of test findings and modifications into production of final system.

(2) Conduct tests of each specified mockup under the direction of an Independent Testing Agency in the presence of the Contractor, Architect, Government, Installer(s), component manufacturers, and fabricators for each specified system to be mocked up and tested. Proceed with testing only after approval of the test program and procedures.

(3) Contractor must not pretest mockups without prior approval from the Architect. If approval is granted, perform pretesting only in the presence of the Government, Architect, and/or their authorized representatives.

(4) Perform necessary corrections to the Work in the presence of the Government, Architect, and/or their authorized representatives.

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. Design Requirements

(1) General: Design, fabricate, and install exterior enclosure to provide shelter and continuous environmental separation between interior and exterior.

(2) Building Movement: Design, fabricate, and install exterior enclosure to withstand differential movements from wind, seismic, structural loading, thermal and moisture cycling, and similar movements.

(a) Service Level Drift: Building enclosure shall remain undamaged and air and watertight as specified in this Section at Service Level Drift.

(b) Design Level Drift: There shall be no failure or permanent distortion of anchors, frames, or concrete, and no panels or components shall fall from the building at Design Level Drift.

(c) Other Movements: Building enclosure shall remain undamaged and air and watertight as specified in this Section.

(3) Air and Water Control: Design, fabricate, and install component parts to prevent leakage of air and water into the building.

(a) Design and construct exterior wall and fenestration assemblies with rainwater management principles to prevent water accumulation within the enclosure assembly. Provide an exterior weather seal and primary air and water control layer separated by a free-draining cavity drained to the exterior at regular intervals

(minimum every story height). Provide dual seal continuity at interfaces with adjacent systems.

(b) Provide a dedicated air and water barrier behind rainscreen cladding. Materials outboard of the water control layer shall be moisture-insusceptible and shall accommodate repeated wetting and drying cycles.

(c) Provide tight joints and effectively seal component parts of exterior enclosure, including their joints with contiguous work, against water leakage and air infiltration.

(d) Whole building air leakage limit: 0.25 cfm/square feet at 1.57 pounds/square foot.

(4) Thermal Control: Design, fabricate, and install exterior enclosure with an integral continuous insulation barrier. The R-value or thickness shown on the drawings is the minimum effective thermal value or thickness for the insulation barrier. Provide additional insulation thickness and thermally broken cladding support to compensate for thermal bridging losses.

(5) Vapor Control: Provide vapor retarders as required by code to control vapor diffusion and prevent condensation and long-term moisture accumulation in the exterior enclosure assemblies or provide an approved design using an accepted engineering practice for hygrothermal analysis (e.g., ASHRAE Standard 160).

(6) Condensation: Design, fabricate, and install the exterior enclosure to prevent condensation inboard of the vapor control plane under the most onerous environmental conditions specified.

(7) Environmental Conditions

(a) Outdoor: 43.3 degrees F ambient air temperature.

(b) Indoor: Mechanical engineer to provide 70 degrees F, 50 percent relative humidity.

(8) Temperature Requirements: Design, fabricate, and install component parts to provide for expansion and contraction of the exterior wall over an ambient exterior temperature range and exterior metal surface temperature of -10 degrees F through +180 degrees F and an interior temperature range of +55 degrees F to 100 degrees F without buckling, sealed joint failure, glass breakage, undue stress on members or anchors, and other detrimental effects.

(a) Shadow Boxes: Shadow boxes shall be designed for an exposed surface metal temperature (including paint coating system) range of +20 degrees F to +235 degrees F. The metal back pan shall exhibit no distress (buckling or distortion), nor shall fastener failure occur as a result of temperature exposure.

b. Performance Requirements: Summary of performance requirements for systems included in the mockup (performance requirements apply to all systems, unless specified otherwise).

(1) Air Infiltration

(a) Fixed Glazed Aluminum Wall Systems: Maximum 0.06 cfm/square feet at a static pressure differential of 6.24 pounds/square feet.

(b) Operable Component Areas: Maximum 0.30 cfm/square feet at 6.24 pounds/square feet.

(2) Water Penetration: No water leakage at an air pressure difference equal to 20 percent of inward design wind load, but not less than 12 pounds/square feet, for all enclosure systems.

c. Testing Requirements

(1) Test building enclosure system mockups and field installations in accordance with Performance Requirements, without reduction, as indicated in Technical Sections and herein.

PART 3 EXECUTION

3.1 MOCKUP INSTALLATION

a. Install products and materials in accordance with specified requirements. Include flashings and anchors and representative interior trim and aluminum covers to demonstrate performance and engagement will not be affected by testing.

b. Simulate building conditions, including the arrangement of structural members and anchorages that will exist within the building.

c. Glass shall be the same type and manufacturer that will be used on the project, unless approved otherwise by substitution. Provide not less than one additional lite for each type and size of glass.

(1) Provide additional accessory materials including, but not limited to gaskets, removable covers, and similar items so as not to delay testing should these items need to be replaced during testing.

3.2 FIELD QUALITY CONTROL TESTING

a. General

(1) Performance Requirements: Test systems in accordance with definitions and at performance requirements, without reduction for field installation, as indicated in Technical Sections and as summarized in Performance Requirements Article.

(2) Field Quality Control: Testing and inspecting of representative areas of the Contract Work shall take place as installation proceeds to determine compliance of installed assemblies with specified requirements.

(3) Testing shall be performed after completing the installation of the Contract Work and before the installation of interior finishes has begun.

(4) Provide a minimum of two weeks' notice to all parties to allow them to attend all testing.

(5) Contract Work will be considered defective if it does not pass

tests and inspections. Repair or remove and replace Work that is considered defective, does not meet requirements, or that is damaged by testing; replace to conform to specified requirements. Perform retesting to confirm compliance of replaced or additional work with specified requirements.

(6) For each air and water test, there shall be at least four(4) initially successful tests performed on each type of wall assembly.

(7) Test Specimen

(a) Water Spray Nozzle Testing: Include 300 linear feet at locations designated by Architect or Government. Include fenestration system joints and interfaces, system perimeter condition, transitions between systems, and air barrier penetrations.

(b) Include perimeter joint sealants within pressure chambers. Field pressure chambers not required to include adjacent wall construction.

(c) Test all joints or other conditions within designated areas where leakage may occur.

(d) Contractor to propose and Architect and Government to approve test locations based on installation completeness and accessibility.

b. Water Spray Nozzle Testing: AAMA 501.2.

(1) Test minimum three areas up to 300 linear feet during construction before 10, 35, and 70 percent completion (nine specimens total).

c. Building Enclosure Air Leakage Testing.

(1) Quantitative Whole Building Air Leakage Testing: ASTM E779.

(a) Testing shall be performed once the air barrier is completely installed but not covered (e.g., by overburden, cladding) to allow for evaluation of leakage path and reduce repair efforts.

(b) The air leakage boundary for testing shall be calculated using the total building envelope area, derived from the building perimeter shown on the floor and roof plans and the vertical dimensions indicated on the elevation drawings.

(c) Conduct testing as follows:

1. Building envelope with HVAC-related openings and non-rated assemblies excluded.

2. Pressurization and depressurization with the results averaged.

3. Multi-point test in increments of 5 to 10 Pa with induced pressure difference range from 10 Pa to 75 Pa.

4. If the pressure test fails to meet the established criteria, use diagnostic test methods, including infrared

thermography, to identify the leak locations.

3.3 TEST REPORTS

a. Testing laboratory or agency shall be responsible for conducting and reporting tests, shall state in report whether test specimen conforms to requirements of Contract Documents and approved drawings, and shallspecifically note any deviations.

(1) Include written documentation describing remedial work, including repairs and modifications required.

(2) Required corrections shall be performed in presence of Architect.

b. Testing shall be witnessed by the Architect; acceptance of test assembly and test results shall be determined by the Architect.

c. Test reports shall be prepared according to testing standard indicated. Testing agency shall submit reports directly to Government and Architect.

3.4 REMEDIAL WORK BY CONTRACTOR

a. In event of failure to meet test requirements, Contractor and subcontractors shall redesign, rework, and/or re-fabricate, reship and re-erect assemblies as required until test requirements are met, at no additional cost to Owner. Contractor must provide written documentation clarifying repairs or modifications made.

(1) For each failed field test, Contractor must provide three additional consecutively successful tests on like specimens at locations selected by the Architect.

b. Wherever leakage has occurred, joints shall be made watertight in manner acceptable to Architect, and in compliance with specified performance requirements.

c. Remedial work involving use of curing-type compounds shall be allowed to cure for one week minimum, but not less than manufacturer's written recommendation, before re-checking for leakage.

d. After required remedial work and curing time are completed, retest repaired joints.

e. If leakage continues, additional remedial measures shall be performed; checking shall be repeated until joints in designated area are found to be acceptable.

(1) Rechecking, when required, shall be performed at the Contractor's expense by testing agency originally employed.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

11/20, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES
1.2 SUBMITTALS
1.3 CONSTRUCTION SITE PLAN
1.4 TEMPORARY POWER PLAN
1.5 STAGING AND WORK AREAS
1.5.1 Use of Site
1.5.1.1 Location of Temporary Facilities
1.5.2 Area Use Plan
1.6 BACKFLOW PREVENTERS CERTIFICATE
1.6.1 Backflow Tester Certificate
1.6.2 Backflow Prevention Training Certificate
1.7 CYBERSECURITY DURING CONSTRUCTION
1.7.1 Temporary IP Networks
1.7.1.1 Network Boundaries and Connections
1.7.2 Temporary Wireless IP Networks
1.7.3 Passwords and Passphrases
1.7.4 Contractor Temporary Network Cybersecurity Compliance Statements
1.8 TRAFFIC CONTROL
1.8.1 General Requirements
1.8.1.1 Definition
1.8.2 Beginning of Contractor's Responsibility
1.8.3 Interface With Other Work Forces
1.8.4 Traffic Control Plan
1.8.5 Regulatory Requirements
1.8.6 Traffic Control Measures
1.8.6.1 Safety
1.8.6.2 Responsibility
1.8.7 Haul Route Plan

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE
2.1.1 Bulletin Board
2.1.2 Project Identification Signs
2.1.3 Warning Signs
2.2 TEMPORARY TRAFFIC CONTROL
2.2.1 Haul Roads
2.2.2 Barricades
2.3 FENCING
2.3.1 Hybrid Type Jersey Type Barriers with Chain Link Fencing
2.3.2 Post-Driven Chain Link Fencing
2.4 TEMPORARY WIRING
2.5 BACKFLOW PREVENTERS

PART 3 EXECUTION

3.1 EMPLOYEE PARKING
3.2 AVAILABILITY AND USE OF UTILITY SERVICES
3.2.1 Temporary Utilities
3.2.1.1 Electricial
3.2.1.2 Heat
3.2.1.3 Water
3.2.2 Obstruction Lighting of Cranes
3.2.3 Fire Protection
3.3 SANITARY FACILITIES
3.3.1 Temporary Toilet Facilities
3.3.2 Disposal of Waste Material
3.4 TEMPORARY TELEPHONE/DATA SERVICE
3.5 INTERRUPTION OF EXISTING UTILITIES SERVICES
3.5.1 Medical Center Service Interruptions
3.6 ABANDONED LINES
3.7 REQUIREMENTS CONCERNING EXISTING UNDERGROUND UTILITIES
3.8 DEDICATED FIRE WATER SERVICE LINES
3.8.1 Fire Flow Data
3.9 TRAFFIC PROVISIONS
3.9.1 Construction in General
3.9.1.1 Delay to Traffic
3.9.2 Debris
3.9.2.1 Hauling
3.9.3 Ingress and Egress
3.9.4 Existing Signage
3.9.5 Parking
3.9.5.1 Hazards
3.9.5.1.1 Temporary Traffic Control Devices
3.9.5.1.1.1 Roadway Maintenance
3.9.5.1.2 Closures/Detours
3.9.5.2 Construction Signs
3.9.6 Barricades, Cones, And Drums
3.9.7 Flaggers
3.9.7.1 Liability
3.9.7.1.1 Maintenance of Traffic
3.9.7.1.2 Protection of Traffic
3.10 Dust Control
3.11 Snow Removal
3.12 CONTRACTOR'S TEMPORARY FACILITIES
3.12.1 Administrative Field Offices
3.12.2 Quality Control Manager Records and Field Office
3.12.3 Storage Area
3.12.4 Appearance of Trailers
3.12.5 Safety Systems
3.12.6 Weather Protection of Temporary Facilities and Stored Materials
3.12.6.1 Building and Site Storm Protection
3.13 PLANT COMMUNICATIONS
3.14 TEMPORARY PROJECT SAFETY FENCING
3.15 DUMPSTERS
3.16 CLEANUP
3.17 RESTORATION OF STORAGE AREA AND OTHER IMPACTED AREAS

-- End of Section Table of Contents --

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
11/20, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511	(2017) Reduced-Pressure Principle Backflow Prevention Assembly

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4) National Electrical Code
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1	(2016; Rev L; Change 2) Obstruction Marking and Lighting

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Temporary Power Plan; G

Traffic Control Plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Area Use Plan; G

SD-06 Test Reports

Backflow Preventer Tests

SD-07 Certificates

Backflow Tester Certification

Backflow Preventers Certificate of Full Approval

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 TEMPORARY POWER PLAN

Prepare and submit a Temporary Power Plan. Indicate the proposed temporary power connection location and detail any temporary power to be done.

1.5 STAGING AND WORK AREAS

1.5.1 Use of Site

The construction site indicated on the drawings for the work covered by these specifications will be furnished by the Government. The area within the work limits indicated on the drawings will be made available to the Contractor after the Contractor has submitted pre- construction submittals and the same have been approved by the Contracting Officer, thus allowing the Contractor to begin on-site operations.

1.5.1.1 Location of Temporary Facilities

The Contractor's field offices, staging areas, stockpiles, storage, and temporary buildings shall be placed in areas secured by the Contractor and as approved by the Contracting Officer. The Contractor staging area(s) shall be the location of all offices and storage trailers, storage of materials and equipment; storage and disbursement of fuel, oil and gas; maintenance of equipment and fabrication of materials. Temporary movement or relocation of Contractor facilities shall be made only on approval by the Contracting Officer. Construct a temporary chain-link security fence around all staging and work areas in accordance with this Section at no additional cost to the Government.

a. The Government will make available the area within the work limits

shown on the project drawings as stated in the paragraph above titled "USE OF SITE" in phases as shown based on progress of work and current project schedule. This area represents the total space that will be made available for the work covered by these project specifications and drawings and not be provided to the Contractor at the same time.

b. No Contractor motor vehicles shall be parked on site.

c. If the Contractor determines that it requires additional space in addition to the space within the work limits, it shall be the Contractor's responsibility to secure such space at the Contractor's expense.

d. Such off-VAHS campus space shall be addressed in the "AREA USE PLAN" described below in this specifications section.

e. Contractor shall provide off-site employee parking and a shuttle service to the work site.

1.5.2 Area Use Plan

Submit an area use plan to the Contracting Officer for approval within 30 calendar days after receipt of the Notice to Proceed. If applicable the plan may be phased to coordinate with the work but in no case submitted later than 30 calendar days before occupying a new area. Prepare an Area Use Plan designating proposed intended use of all areas at the project site and staging (areas), including all staging areas adjacent to the project site but outside the designated construction site. This plan shall include, but not be limited to the following: the locations and dimensions of temporary facilities, access roads to the various project sites; drawings showing proposed road detours; drawings showing temporary water and sewage disposal installations; construction plant and building installations, if any; concrete truck washout area; material storage area(s); and office or administration facilities, avenues of ingress/egress to the fenced area and details of the fence installation. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, and temporary sanitary facilities.

1.6 BACKFLOW PREVENTERS CERTIFICATE

1.6.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with a company participating in other phases of this Contract.

1.6.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

1.7 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.7.1 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.7.1.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.7.2 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.7.3 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.7.4 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.8 TRAFFIC CONTROL

1.8.1 General Requirements

This work consists of providing facilities and control measures for public traffic adjacent to and around the Project site for the life of the Contract.

1.8.1.1 Definition

Public traffic includes motor vehicles, bicycles, and pedestrians.

1.8.2 Beginning of Contractor's Responsibility

The Contractor's responsibilities for control of public traffic begin on the day any on-site work begins within the Project work limits and along the construction haul route as shown on the drawings. Traffic control measures shall be in place prior to other construction activity.

1.8.3 Interface With Other Work Forces

Cooperate with general site access controls which may be established by Department of Veterans Affairs (VA) and City of West Haven prior to award and during the period of this Contract.

1.8.4 Traffic Control Plan

The Plan shall be in written form and shall indicate the type and location of all signs, lights, barricades, striping, and barriers to be used for safe passage of public traffic on adjacent areas of the project and the VA Medical Center campus, and for the protection of construction traffic and workers through the project. The Plan shall include specific details addressing ingress and egress at all access points along haul roads and local roads on the VA Medical Center campus. The Plan shall indicate conditions and setups for each phase of the Contractor's activities. In no case shall the Contractor begin work until the Plan has been approved. Modifications to the Plan that become necessary shall be submitted in writing for approval. Except in an emergency, no modifications to the approved Plan shall be implemented on the project until the Contractor submits and receives approval of such modification.

1.8.5 Regulatory Requirements

Meet all conditions established for the use of existing roadways, pedestrian areas, and haul routes by those having jurisdiction there over, including seasonal or other limitations or restrictions, and the payment of excess size and weight fees. Comply with applicable local regulations for haul routes over public highways, roads, or bridges. At all times, coordinate the work under this contract with the authorities having jurisdiction over streets and highways prior to installing or implementing a new traffic control procedure.

1.8.6 Traffic Control Measures

Work includes the furnishing, erection, relocation, maintenance and eventual removal of temporary signs, barricades, pavement markings, lighting, signals, flaggers and other traffic control devices as may be required for the maintenance and protection of traffic.

1.8.6.1 Safety

Operations shall be conducted in a manner to insure the convenience and safety of travelers and abutting property owners as well as the safety of the Contractor's employees.

1.8.6.2 Responsibility

The Contractor has sole responsibility for providing the necessary traffic control devices and taking other appropriate measures for the protection of the public and contractor personnel. The fact that provisions may be included in the plans and other contract documents, or may be ordered, for the maintenance and protection of traffic during construction, does not relieve the Contractor of his responsibilities. In the absence of specific provisions in the contract plans and specifications, or instructions from the Contracting Officer, additional devices and operational measures as may be considered necessary for the protection of the public and his employees shall be provided.

1.8.7 Haul Route Plan

This plan shall include offices, material storage areas and structures, the turn-around area for tractor-trailers, and the access routes to these areas. Haul routes from the sites through the Medical Center campus to major highways shall be indicated. All required traffic signs, special limits, warning devices, lighting and other such safety devices required by EM 385-1-1, OSHA, and local cities (on state roads) shall be shown. Provisions shall be made in the plan for alternate routes when excavations block designated haul routes. The plan shall be revised and resubmitted if the haul routes being used are not as shown on the plan.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 Section 01 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer. All temporary signage shall be provided by the Contractor. The Contractor shall coordinate all verbiage with the VAMC.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as indicated and as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Section 04. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract in accordance with EM 385-1-1 Section 04. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control, although optional, must be adequate to ensure safe operation at all

times. Location, grade, width, and alignment of construction and haul roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 FENCING

Provide fencing along the construction site and at all open excavations, trenches and tunnels to control access by unauthorized personnel. Access to the construction site must be secured either by gates that are closed to prevent unauthorized access or manned by flagger to prevent unauthorized access at all times.Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1 listed below. Remove the fence upon completion and acceptance of the work.

According to EM 385-1-1, Chapter 25: Perimeter protection must be provided to prevent personnel, vehicles, and equipment from falling into excavations according to the following hierarchy:

(a) Class I perimeter protection is required if the excavation is exposed to members of the public or vehicles or equipment.

(a-1) Class I perimeter protection must meet the requirements of a guardrail system according to chapter 21.

(b) Class II perimeter protection, at minimum, is required if the excavation does not meet the requirements for Class I perimeter protection but is: routinely exposed to employees and is deeper than 6 feet or contains hazards (for example, impalement, hazardous substances).

(b-1) Class II perimeter protection consists of warning barricades or flagging placed not closer than 6 feet (1.8 m) from the edge of the excavation.

(c) Class III perimeter protection, at minimum, is required if the excavation does not meet the requirements for either Class I or Class II perimeter protection.

(c-1) Class III. Perimeter protection consists of warning barricades or flagging placed not closer than 6 inches (15.2 cm) or nor more than 6 feet (1.8 m) from the edge of the excavation.

To block public view of the construction and fully secure the work area, enclose the project work area and Contractor lay-down area with a 8 ft high chain link fence and gates with brown, UV light resistant, plastic fabric mesh netting (similar to tennis court or other screening)utilizing one of the types described in paragraphs 2.3.1 and 2.3.2 only. All project

work areas are required to be enclosed by fencing.

2.3.1 Hybrid Type Jersey Type Barriers with Chain Link Fencing

Temporary panel fencing must be galvanized steel chain link panels attached to water-filled jersey type barriers 6 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each jersey barrier must be weighted down with water in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

2.3.2 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 8 feet high supported by an tightly secured to galvanized steel posts driven below grade. Fence posts must be located on minimum 10 foot centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Section 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

2.5 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 150 pound flanged cast iron, mounted gate valve and strainer, 304 stainless steel or bronze, internal parts.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

There is no on-site parking available for the Construction Contract employees.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

The Contractor will provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable

codes and standards.

3.2.1.1 Electricial

The Contractor, at Contractor's expense and in a workmanlike manner, in compliance with code and as satisfactory to the Contracting Officer, shall install and maintain all necessary temporary connections and distribution lines, and all meters required to measure the amount of electricity used. Before final acceptance of the work by the Government, remove all the temporary connections, distribution lines, meters, and associated paraphernalia and repair and/or restore the infrastructure as required.

a. Provide construction power facilities in accordance with the safety requirements of the National Electrical Code NFPA 70 and the SAFETY AND HEALTH REQUIREMENTS MANUAL EM 385-1-1. All temporary electrical installations shall be acquired from the local electrical utility company, at the Contractor's expense, and shall be subject to the approval of the Contracting Officer.

b. If additional electrical power is required for construction provide the power using generators. Provide adequate electrical fault protection so that Contractor electrical equipment faults will not adversely affect the site electrical system or the utility system.

3.2.1.2 Heat

The Contractor shall furnish temporary heat necessary to prevent injury to work and materials through dampness and cold. Use of open salamanders or any temporary heating devices which may be fire hazards or may smoke and damage finished work, will not be permitted. Maintain minimum temperatures as specified for various materials.

3.2.1.3 Water

The Contractor, at Contractor's expense, shall acquire all temporary water services required for the Contractor's field office and all construction activities directly from a local source. Install and maintain necessary supply connections and piping for same, but only at such locations and in such manner as may be approved by the Contracting Officer. Connections to the medical campus potable water system is NOT permitted.

3.2.2 Obstruction Lighting of Cranes

Provide a minimum of 2 aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 100 feet above ground level. Light construction and installation must comply with FAA AC 70/7460-1. Lights must be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

3.2.3 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.3 SANITARY FACILITIES

3.3.1 Temporary Toilet Facilities

The Contractor, at Contractor's expense,shall provide where directed, (for use of all Contractor's workmen) ample temporary sanitary toilet accommodations with suitable sewer and water connections; or, when approved by Contracting Officer, provide suitable dry closets where directed. Keep such places clean and free from flies, and all connections and appliances connected therewith are to be removed prior to completion of contract, and premises left perfectly clean. Use of the toilet facilities in any existing building(s) is NOT permitted.

3.3.2 Disposal of Waste Material

Unless otherwise specified, waste material shall be disposed of by the Contractor at a licensed off-site sanitary landfill or sewage disposal plant. Permission to use the off-site sanitary landfill or sewage disposal plant shall be obtained by the Contractor and any costs attendant thereto shall be borne by the Contractor.

3.4 TEMPORARY TELEPHONE/DATA SERVICE

If temporary telephone/data service is needed, the Contractor is required to determine line/data service availability and to make arrangements for installation of telephone lines and instruments and/or data services with the telephone company/data services provider. All costs associated with telephone/data installation and service shall be the responsibility of the Contractor.

3.5 INTERRUPTION OF EXISTING UTILITIES SERVICES

Perform the work under this contract with a minimum of outage time for all utilities. The existing utilities services may be interrupted only when approved by the Contracting Officer. The interruption time shall be kept to a minimum. Depending upon the activities at the Medical Center, which require continuous service from the existing utility, an interruption may not be subject to schedule at the time desired by the Contractor. In such cases the interruption may have to be scheduled at a time of minimum requirement of demand for the utility, which may include weekends/evenings. The amount of time requested by the Contractor for interruption of existing utility service shall be as approved by the Contracting Officer

3.5.1 Medical Center Service Interruptions

Maintain existing utility services for Medical Center facilities at all times. Provide non-permanent structures, labor, materials, equipment, connections, and utilities to assure uninterrupted services. Where necessary to cut existing water, steam, gases, sewer or air pipes, or conduits, wires, cables, etc. of utility services or of fire protection systems and communications systems (including telephone), they shall be cut and capped at suitable places where shown; or, in absence of such indication, where directed by Resident Engineer.

a. No utility service such as water, gas, steam, sewers or electricity, or fire protection systems and communications systems may be interrupted without prior approval of the Contracting Officer. Electrical work shall be accomplished with all affected circuits or

equipment de-energized. When an electrical outage cannot be accomplished, work on any energized circuits or equipment shall not commence without a detailed work plan, the Medical Center Director's approval and written approval of the Contracting Officer.

b. Submit a request to interrupt any such services to the Contracting Officer in writing, 120 calendar days in advance of proposed interruption. Request shall state reason, date, exact time of, and approximate duration of such interruption. Furthermore, at a minimum of 21 days, prior to the planned outage, interruption, or tie-in, contractor must provide a more detailed written work plan. Plan shall include, but not be limited to, confirmed planned interruption or tie-in date and time, duration of outage, details of what utilities are impacted, where,and what they serve, Medical Center's Hazardous Energy Policy 138-075 Lockout/Tagout Permit completed, and any road, pedestrian, or building impacts, along with mitigation efforts. The Contracting Officer will generally provide a written reply to the Contractor's request approximately one week prior to the proposed interruption

c. Work on existing utilities that does not result in an interruption (such as investigations, tie-ins, operation of valves or switches, etc.) must also be requested in advance. At a minimum of 21 prior to the planned work, contractor shall provide written work plan. Plan shall include, but not be limited to date and time, duration of work, details of what utilities are impacted, where, and what they serve, and any road, pedestrian, or building impacts, along with mitigation efforts.

d. Contractor will be advised (in writing) of approval of request, or of which other date and/or time such interruption will cause least inconvenience to operations of Medical Center. The interruption time, approved by the Contracting Officer, may occur at other than Contractor's normal working hours.

e. In case of a contract construction emergency, service will be interrupted on approval of the Contracting Officer. Such approval will be confirmed in writing as soon as practical.

f. Whenever it is required that a connection fee be paid to a public utility provider for new permanent service to the construction project, for such items as water, sewer, electricity, gas or steam, payment of such fee shall be the responsibility of the Contractor.

3.6 ABANDONED LINES

All service lines such as wires, cables, conduits, ducts, pipes and the like, and their hangers or supports, which are to be abandoned but are not required to be entirely removed, shall be sealed, capped or plugged at the main, branch or panel they originate from. The lines shall not be capped in finished areas, but shall be removed and sealed, capped or plugged in ceilings, within furred spaces, in unfinished areas, or within walls or partitions; so that they are completely behind the finished surfaces.

3.7 REQUIREMENTS CONCERNING EXISTING UNDERGROUND UTILITIES

The drawings indicate known, existing, underground utilities which are located in areas requiring work under this contract. The locations shown are approximate and the depths are not indicated. Contact the utility

companies or agencies which furnish utilities in the area where the project is located to determine if there are other known, new, or suspected additional underground utilities in the work areas other than those shown on the drawings. Prior to commencement of any excavation in areas where there are known utilities, it shall be the responsibility of the Contractor to contact the owner of each known utility and request that the utility owner mark the location of the utility and also obtain information as to the depth of the utility. For all utilities within VA property boundary, it is fully the contractor's responsibility to perform necessary utility surveys and locations meeting quality level B requirements at a minimum. Submit proof to the Contracting Officer that the Contractor has contacted the owners of known utilities and has obtained the location, including depth below grade, for such utilities. Protect all existing utilities in accordance with the Contract Clause entitled, PROTECTION OF EXISTING VEGETATION, EQUIPMENT, STRUCTURES,UTILITIES, AND IMPROVEMENTS. In addition, take all the necessary precautions to prevent harm to personnel and damage to equipment and existing construction from utilities which conduct electricity or transmit combustible gases.

3.8 DEDICATED FIRE WATER SERVICE LINES

3.8.1 Fire Flow Data

For determination and documentation of fire protection, conduct and provide a main capacity test. Data to be included with the flow tests are static pressures, residual pressures, flow rates, date, domestic and fire pumps in operation, time tests were conducted, and name of personnel conducting the fire hydrant flow tests. The static pressures, residual pressures, flow rates, test (residual) hydrant and flow hydrants shall be shown on the appropriate contract drawings. Become familiar with the existing water system prior to conducting the hydrant flow tests. Fire hydrant flow tests required for fire protection design shall be made in accordance with the procedures specified in NFPA 291, Recommended Practice for Fire Flow Testing and Marking of Hydrants. Flow tests shall be performed in the project site and submit fire hydrant flow test data prior to commencing design of the sprinkler systems. Coordinate with the Contracting Officer prior to conducting such tests.

3.9 TRAFFIC PROVISIONS

3.9.1 Construction in General

The Contracting Officer shall be notified before starting any work that might inconvenience or endanger traffic, or where the Contractor must provide signs, barricades, flaggers, and/or other appurtenances for the maintenance and protection of traffic.

3.9.1.1 Delay to Traffic

Construction operations shall be conducted to insure a minimum of delay to traffic. Stopping traffic for more than 5 minutes shall not be permitted unless specifically authorized. Allow emergency vehicles immediate passage.

Wherever excavation for new utility lines cross existing roads or for any other reason/operation, at least one lane must be open to traffic at all times with approval.

3.9.2 Debris

The traveled way shall be maintained free of foreign objects such as spilled earth, rocks, timber, and other items that may fall from transporting vehicles. Materials spilled by or dropped from any vehicle used in the Contractor's hauling operations along or across any public traveled way, both within and outside the project work limits, shall be removed immediately.

3.9.2.1 Hauling

a. For hauling, use only established public roads and roads on Medical Center property and, when authorized by the Contracting Officer, such temporary roads which are necessary in the performance of contract work. Temporary roads shall be constructed and restoration performed by the Contractor at Contractor's expense. When necessary to cross curbing, sidewalks, or similar construction, they must be protected by well-constructed bridges.

b. When new permanent roads are to be a part of this contract, the Contractor may construct them immediately for use to facilitate building operations. These roads may be used by all who have business thereon within the zone of building operations.

3.9.3 Ingress and Egress

Provide and maintain safe and adequate ingress and egress to and from intersecting highways, private points of access, and other establishment at existing or new access points, consistent with the work, unless otherwise shown on the plans or authorized.

a. Keep walks and entrances to grounds, to parking and to occupied areas of buildings clear of construction materials, debris and standing construction equipment and vehicles.

b. The method and scheduling of the required cutting, altering and removal of existing roads, walks and entrances must be approved by the Contracting Officer.

3.9.4 Existing Signage

Existing traffic signs and other traffic control devices within the project limits which are to remain shall be protected from damage by construction operations. Replace any such items lost or damaged because of negligence on the part of the Contractor. Temporary relocation of any such items be required because of construction operations with direction of the Contracting Officer.

3.9.5 Parking

Do not park, store equipment or materials adjacent to the traveled way or public rights-of-way where it may be a hazard to the traffic. Normally, a clear distance of at least 30 feet from the edge of pavement shall be kept free of any such obstacles unless otherwise authorized. Parking on the Medical Center Campus is limited to certain areas as specified under this Section STAGING AREAS. This will be enforced by the VA Medical Center Police.

3.9.5.1 Hazards

Whenever construction operations near the traveled way result in a hazardous condition to the traveling public, the extent of such operations shall be limited to work that can be performed within a reasonable period of time. Once started, such work shall progress continuously to completion within a reasonable extent of time.

3.9.5.1.1 Temporary Traffic Control Devices

Temporary devices shall remain in operation only as long as they are needed for traffic guidance and safety. Only those devices shall be in place as apply to conditions and operations in existence at a particular time. Traffic control devices shall be cleaned, repainted, reflectorized or replaced as necessary to provide adequate visibility and legibility.

3.9.5.1.1.1 Roadway Maintenance

Existing and temporary pavements shall be kept in good repairs using the same or materials compatible with the pavement. The gross weight of each loaded vehicle using roadways and crossing shall not exceed State and local limits. If any pavement has been damaged as a result of the Contractor's operations, the pavement shall be reconstructed to the proper line and grade in conformance with CA Specifications

3.9.5.1.2 Closures/Detours

a. The project site shall be closed to non-construction public access during the construction period. Provide signage placed as needed and approved by the Contracting Officer to prevent non-construction public traffic entry onto the project area and to reroute public traffic as required.

b. Required Public Access. Except as specified, the existing public access to the VA Medical Center must remain open to public access during the Contract Period.

c. Temporary closure measures shall remain in place, until the public access areas are reopened to public access.

d. Notify, in writing, the Contracting Officer and VA Medical Center at least 14 calendar days in advance of the planned closure(s). Do not close the public access areas until the Contracting Officer provides written approval. After approval, provide approved 48 hour public notification signs at each closure point prior to closing the public access routes/areas.

e. If a traffic detour plan is required, the plan must be approved and all measures in place prior to rerouting of any traffic. The plan shall show location of all signage. The traffic detour plan is subject to the approval of the VA Medical Center.

3.9.5.2 Construction Signs

Signs shall be furnished, erected, maintained, moved and removed as required and as directed to adequately and safely inform and direct the traveling public. The number of signs indicated in the applicable MUTCD, are a minimum and the Contracting Officer may require additional signs. Provide adequate quantity of these signs on hand prior to starting

construction operations for use as required. Sign sizes and details shall conform to the applicable standard models. Signs and markers shall indicate actual existing conditions and shall be moved, removed, relocated, or changed as directed. Signs shall be kept clean and in good condition for the duration of the contract. They shall be mounted at the required height on the adequate supports and placed in proper position and alignment so as to give maximum visibility both day and night. During nonworking hours and following the completion of a particular construction operation, all warning signs, except those required to remain for the safety of the public, shall be removed or covered. All signage language shall be approved by the VA Medical Center.

3.9.6 Barricades, Cones, And Drums

Barricades, cones, and drums shall be furnished, erected, maintained, moved, replaced, and removed where indicated in the applicable MUTCD, or as directed. Where required or directed, construction barricades shall be supplemented either by approved flashing or steady burning lights as appropriate.

3.9.7 Flaggers

Flaggers shall be provided where vehicles are entering or leaving Highway right of way. Flaggers shall wear orange helmets and vests.

3.9.7.1 Liability

Operations shall be conducted in such a manner that no undue hazard will result due to the above requirements, and the procedures and policies described therein shall in no way act as a waiver of any of the terms of the liability of the Contractor or his surety.

3.9.7.1.1 Maintenance of Traffic

a. Conduct operations in a manner that will not close any thoroughfare or interfere in any way with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a Traffic Control Plan detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations. Make all notifications and obtain any permits required for modification to traffic movements outside Station's jurisdiction. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.

b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.

c. Provide, erect, and maintain, at contractors expense, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage, overhead protection authority having jurisdiction.

3.9.7.1.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction

period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of any damage to roads caused by construction operations.

3.10 Dust Control

Dusty conditions resulting from the Contractor's operations shall be prevented and/or corrected by the use of calcium chloride and/or water.

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS .

3.11 Snow Removal

Snow removal on existing campus roadways will be addressed by the Medical Center in a reasonable, timely fashion to maintain typical Medical Center operations. Any impacts to work efforts due to lack of snow removal will be evaluated as it relates to Time Extensions for Unusually Severe Weather under Divison 01 as appropriate. It is the responsibility of the Contractor to remove snow in all work limits once the work area has been provided to the Contractor.

3.12 CONTRACTOR'S TEMPORARY FACILITIES

Contractor-owned or -leased trailers must be identified by Government assigned numbers. Apply the number to the trailer within 14 calendar days of notification, or sooner, if directed by the Government. Temporary facilities must meet requirements as identified in EM 385-1-1 Section 04.

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1 Section 04. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.12.1 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

In the event a new building is constructed for the temporary project field office, it must be a minimum 12 feet in width, 16 feet in length and have a minimum of 7 feet headroom. Equip the building with approved electrical wiring, at least one double convenience outlet and the required switches and fuses to provide 110-120 volt power. Provide a work table with stool, desk with chair, two additional chairs, and one legal size file cabinet that can be locked. The building must be waterproof, supplied with a heater, have a minimum of two doors, electric lights, a telephone, a

battery-operated smoke detector alarm, a sufficient number of adjustable windows for adequate light and ventilation, and a supply of approved drinking water. Provide approved sanitary facilities. Screen the windows and doors and provide the doors with deadbolt type locking devices or a padlock and heavy-duty hasp bolted to the door. Door hinge pins must be non-removable. Arrange the windows to open and to be securely fastened from the inside. Protect glass panels in windows by bars or heavy mesh screens to prevent easy access. In warm weather, provide air conditioning capable of maintaining the office at 50 percent relative humidity and a room temperature 20 degrees F below the outside temperature when the outside temperature is 95 degrees F. Unless otherwise directed by the Contracting Officer, remove the building from the site upon completion and acceptance of the work.

3.12.2 Quality Control Manager Records and Field Office

Provide on the jobsite an office with approximately 200 square feet of useful floor area for the exclusive use of the QC Manager. Provide a weathertight structure with adequate heating and cooling, toilet facilities, lighting, ventilation, a 4 by 8 foot plan table, a standard size office desk and chair, computer station, and working communications facilities. Provide either a 1,500 watt radiant heater and a window-mounted air conditioner rated at 9,000 Btus minimum or a window-mounted heat pump of the same minimum heating and cooling ratings. Provide a door with a cylinder lock and windows with locking hardware. Make utility connections. Locate as directed. File quality control records in the office and make available at all times to the Government. After completion of the work, remove the entire structure from the site.

3.12.3 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Include plastic strip inserts, colored brown, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. Grassed or unpaved areas, which are not established roadways, and will be traversed with construction equipment or other vehicles, must be covered with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways, should the Contractor elect to traverse them with construction equipment or other vehicles. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

3.12.4 Appearance of Trailers

a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Installation property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.

b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.12.5 Safety Systems

Protect the integrity of all installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.12.6 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.12.6.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.13 PLANT COMMUNICATIONS

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

3.14 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.15 DUMPSTERS

Equip dumpsters with a secure cover and paint the standard installation color. Keep dumpster closed, except when being loaded with trash and

debris. Locate dumpsters behind the construction fence or out of the public view. Empty site dumpsters at least once a week, or as needed to keep the site free of debris and trash. If necessary, provide 55 gallon trash containers painted the darker installation color to collect debris in the construction site area. For large demolitions, large dumpsters without lids are acceptable, but must not have debris higher than the sides before emptying.

3.16 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.17 RESTORATION OF STORAGE AREA AND OTHER IMPACTED AREAS

Upon completion of the project remove barricades, temporary utilities, sanitation facilities, construction vehicles, materials and equipment, temporary chain-link fencing (if used), and any other temporary equipment, materials, or products from within and around the work areas and temporary storage areas. Clean and repair damage caused by installation or removal of temporary materials, equipment, or utilities. Areas used by the Contractor for the storage of equipment or material, or other use, shall be restored to the original or better condition, including pavement repair, or top soil and seeding as necessary. Gravel used to traverse grassed areas shall be removed and the area restored to its original condition, including top soil and seeding as necessary.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.1 DESCRIPTION
1.2 QUALITY CONTROL
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 PROTECTION OF ENVIRONMENTAL RESOURCES

-- End of Section Table of Contents --

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the control of environmental pollution and damage that the Contractor must consider for air, water, and land resources. It includes management of visual aesthetics, noise, solid waste, radiant energy, and radioactive materials, as well as other pollutants and resources encountered or generated by the Contractor. The Contractor is obligated to consider specified control measures with the costs included within the various contract items of work.

b. Environmental pollution and damage is defined as the presence of chemical, physical, or biological elements or agents which:

(1) Adversely effect human health or welfare,

(2) Unfavorably alter ecological balances of importance to human life,

(3) Effect other species of importance to humankind, or;

(4) Degrade the utility of the environment for aesthetic, cultural, and historical purposes.

c. Definitions of Pollutants:

(1) Chemical Waste: Petroleum products, bituminous materials, salts, acids, alkalis, herbicides, pesticides, organic chemicals, and inorganic wastes.

(2) Debris: Combustible and noncombustible wastes, such as leaves, tree trimmings, ashes, and waste materials resulting from construction or maintenance and repair work.

(3) Sediment: Soil and other debris that has been eroded and transported by runoff water.

(4) Solid Waste: Rubbish, debris, garbage, and other discarded solid materials resulting from industrial, commercial, and agricultural operations and from community activities.

(5) Surface Discharge: The term "Surface Discharge" implies that the water is discharged with possible sheeting action and subsequent soil erosion may occur. Waters that are surface discharged may terminate in drainage ditches, storm sewers, creeks, and/or "water of the United States" and would require a permit to discharge water from the governing agency.

(6) Rubbish: Combustible and noncombustible wastes such as paper, boxes, glass and crockery, metal and lumber scrap, tin cans, and bones.

(7) Sanitary Wastes:

(a) Sewage: Domestic sanitary sewage and human and animal waste.

(b) Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2 QUALITY CONTROL

a. Establish and maintain quality control for the environmental protection of all items set forth herein.

b. Record on daily reports any problems in complying with laws, regulations, and ordinances. Note any corrective action taken.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Environmental Protection Plan; G,

1.4 SUBMITTAL INSTRUCTIONS

a. In accordance with Section, 01 33 00, SUBMITTAL PROCEDURES furnish the following:

(1) Environmental Protection Plan: After the contract is awarded and prior to the commencement of the work, the Contractor shall meet with the Contracting Officer to discuss the proposed Environmental Protection Plan and to develop mutual understanding relative to details of environmental protection. Not more than 20 days after the meeting, the Contractor shall prepare and submit to the Contracting Officer for approval, a written and/or graphic Environmental Protection Plan including, but not limited to, the following:

(a) Name(s) of person(s) within the Contractor's organization who is (are) responsible for ensuring adherence to the Environmental Protection Plan.

(b) Name(s) and qualifications of person(s) responsible for manifesting hazardous waste to be removed from the site.

(c) Name(s) and qualifications of person(s) responsible for training the Contractor's environmental protection personnel.

(d) Description of the Contractor's environmental protection personnel training program.

(e) A list of Federal, State, and local laws, regulations, and permits concerning environmental protection, pollution control, noise control and abatement that are applicable to the Contractor's proposed operations and the requirements imposed by those laws, regulations, and permits.

(f) Methods for protection of features to be preserved within authorized work areas including trees, shrubs, vines, grasses, ground cover, landscape features, air and water quality, fish and wildlife, soil, historical, and archeological and cultural resources.

(g) Procedures to provide the environmental protection that comply with the applicable laws and regulations. Describe the procedures to correct pollution of the environment due to accident, natural causes, or failure to follow the procedures as described in the Environmental Protection Plan.

(h) Permits, licenses, and the location of the solid waste disposal area.

(i) Drawings showing locations of any proposed temporary excavations or embankments for haul roads, stream crossings, material storage areas, structures, sanitary facilities, and stockpiles of excess or spoil materials. Include as part of an Erosion Control Plan approved by the District Office of the U.S. Soil Conservation Service and the Department of Veterans Affairs.

(j) Environmental Monitoring Plans for the job site including land, water, air, and noise.

(k) Work Area Plan showing the proposed activity in each portion of the area and identifying the areas of limited use or nonuse. Plan should include measures for marking the limits of use areas. This plan may be incorporated within the Erosion Control Plan.

(l) Inclusion of “best management practices” and methodologies.

b. Approval of the Contractor's Environmental Protection Plan will not relieve the Contractor of responsibility for adequate and continued control of pollutants and other environmental protection measures.

1.5 PROTECTION OF ENVIRONMENTAL RESOURCES

a. Protect environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire period of this contract. Confine activities to areas defined by the specifications and drawings.

b. Protection of Land Resources: Prior to construction, identify all land resources to be preserved within the work area. Do not remove, cut, deface, injure, or destroy land resources including trees, shrubs, vines, grasses, topsoil, and landforms without permission from the Contracting Officer. Do not fasten or attach ropes, cables, or guys to trees for anchorage unless specifically authorized, or where special emergency use is permitted. Provide erosion control plans, in phases where required.

(1) Work Area Limits: Prior to any construction, mark the areas that require work to be performed under this contract. Mark or fence isolated areas within the general work area that are to be saved and protected. Protect monuments, works of art, and markers before construction operations begin. Convey to all personnel the purpose of marking and protecting all necessary objects.

(2) Protection of Landscape: Protect trees, shrubs, vines, grasses, land forms, and other landscape features shown on the drawings to be preserved by marking, fencing, or using any other approved techniques.

(a) Box and protect from damage existing trees and shrubs to remain on the construction site.

(b) Immediately repair all damage to existing trees and shrubs by trimming, cleaning, and painting with antiseptic tree paint.

(c) Do not store building materials or perform construction activities closer to existing trees or shrubs than the farthest extension of their limbs.

(d) Tree removal should be performed outside of bat activity period.

(3) Reduction of Exposure of Unprotected Erodible Soils: Plan and conduct earthwork to minimize the duration of exposure of unprotected soils. Clear areas in reasonably sized increments only as needed to use. Form earthwork to final grade as shown. Immediately protect side slopes and back slopes upon completion of rough grading.

(4) Temporary Protection of Disturbed Areas: Construct diversion ditches, benches, and berms to retard and divert runoff from the construction site to protected drainage areas approved under paragraph 208 of the Clean Water Act.

(a) Sediment Basins: Trap sediment from construction areas in temporary or permanent sediment basins that accommodate the runoff of a local 10 (design year) storm. After each storm, pump the basins dry and remove the accumulated sediment. Control overflow/drainage with paved weirs or by vertical overflow pipes, draining from the surface.

(b) Reuse or conserve the collected topsoil sediment as directed by the Contracting Officer.

(c) Institute effluent quality monitoring programs as required by Federal, State, and local environmental agencies.

(5) Erosion and Sedimentation Control Devices: The erosion and sediment controls selected and maintained by the Contractor shall be such that water quality standards are not violated as a result of the Contractor's activities. Construct or install all temporary and permanent erosion and sedimentation control features on the Environmental Protection Plan. Maintain temporary erosion and sediment control measures such as berms, dikes, drains, sedimentation basins, grassing, and mulching, until permanent drainage and erosion control facilities are completed and operative.

(6) Manage borrow areas on and off Government property to minimize erosion and to prevent sediment from entering nearby water courses or lakes.

(7) Manage and control spoil areas on and off Government property to limit spoil to areas shown and prevent erosion of soil or sediment from entering nearby water courses or lakes.

(8) Protect adjacent areas from despoilment by temporary excavations and embankments.

(9) Handle and dispose of solid wastes in such a manner that will prevent contamination of the environment. Place solid wastes (excluding clearing debris) in containers that are emptied on a regular schedule. Transport all solid waste off Government property and dispose of waste in compliance with Federal, State, and local requirements.

(10) Store chemical waste away from the work areas in corrosion resistant containers and dispose of waste in accordance with Federal, State, and local regulations. Chemical waste shall be stored within secondary containment to prevent spilling.

(11) Handle discarded materials other than those included in the solid waste category as directed by the Contracting Officer.

c. Protection of Water Resources: Keep construction activities under surveillance, management, and control to avoid pollution of surface and ground waters and sewer systems. Implement management techniques to control water pollution by the listed construction activities that are included in this contract.

(1) Washing and Curing Water: Do not allow wastewater directly derived from construction activities to enter water areas. Collect and place wastewater in retention ponds allowing the suspended material to settle, the pollutants to separate, or the water to evaporate.

(2) Control movement of materials and equipment at stream crossings during construction to prevent violation of water pollution control standards of the Federal, State, or local government.

(3) Monitor water areas affected by construction.

d. Protection of Fish and Wildlife Resources: Keep construction activities under surveillance, management, and control to minimize interference with, disturbance of, or damage to fish and wildlife. Prior to beginning construction operations, list species that require specific attention along with measures for their protection.

e. Protection of Air Resources: Keep construction activities under surveillance, management, and control to minimize pollution of air resources. Burning is not permitted on the job site. Keep activities, equipment, processes, and work operated or performed, in strict accordance with the State of Connecticut and all VA requirements and all PCRA measures required and Federal emission and performance laws and standards. Maintain ambient air quality standards set by the Environmental Protection Agency, for those construction operations and activities specified.

(1) Particulates: Control dust particles, aerosols, and gaseous by-products from all construction activities, processing, and preparation of materials (such as from asphaltic batch plants) at

all times, including weekends, holidays, and hours when work is not in progress.

(2) Particulates Control: Maintain all excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and all other work areas within or outside the project boundaries free from particulates which would cause a hazard or a nuisance. Sprinklering, chemical treatment of an approved type, light bituminous treatment, baghouse, scrubbers, electrostatic precipitators, or other methods are permitted to control particulates in the work area.

(3) Hydrocarbons and Carbon Monoxide: Control monoxide emissions from equipment to Federal and State allowable limits.

(4) Odors: Control odors, vapors and fumes from construction activities as identified on the Project's PCRA documentation/forms. The odors, vapors and fumes must be in compliance with state regulations and local ordinances and may not constitute a health hazard.

f. Reduction of Noise: Minimize noise using every action possible. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer. Maintain noise-produced work at or below the decibel levels and within the time periods specified.

(1) Perform construction activities involving repetitive, high-level impact noise only between 8:00 a.m. and 9:00 p.m unless otherwise permitted by local West Haven ordinance or the Contracting Officer. Repetitive impact noise on the property shall not exceed the following dB limitations:

Time Duration of Impact Noise	Sound Level in dB
More than 12 minutes in any hour	70
Less than 30 seconds of any hour	85
Less than three minutes of any hour	80
Less than 12 minutes of any hour	75

(2) Provide sound-deadening devices on equipment and take noise abatement measures that are necessary to comply with the requirements of this contract, consisting of, but not limited to, the following:

(a) Maintain maximum permissible construction equipment noise levels at 50 feet (dBA):

EARTHMOVING		MATERIALS HANDLING	
FRONT	75	CONCRETE MIXERS	75
BACKHOES	75	CONCRETE PUMPS	75
DOZERS	75	CRANES	75
TRACTORS	75	DERRICKS IMPACT	75
		PILE DRIVERS	95
GRADERS	75	JACK HAMMERS	75
TRUCKS	75	ROCK DRILLS	80
PAVERS,	80	PNEUMATIC TOOLS	80

PUMPS	75		
GENERATORS	75	SAWS	75
COMPRESSORS	75	VIBRATORS	75

(b) Use shields or other physical barriers to restrict noise transmission.

(c) Provide soundproof housings or enclosures for noise-producing machinery.

(d) Use efficient silencers on equipment air intakes.

(e) Use efficient intake and exhaust mufflers on internal combustion engines that are maintained so equipment performs below noise levels specified.

(f) Line hoppers and storage bins with sound deadening material.

(g) Conduct truck loading, unloading, and hauling operations so that noise is kept to a minimum.

(3) Measure sound level for noise exposure due to the construction at least once every five successive working days while work is being performed above 55 dB(A) noise level. Measure noise exposure at the property line or 50 feet from the noise source, whichever is greater. Measure the sound levels on the A weighing network of a General Purpose sound level meter at slow response. To minimize the effect of reflective sound waves at buildings, take measurements at (three to six feet) in front of any building face. Submit the recorded information to the Contracting Officer noting any problems and the alternatives for mitigating actions.

g. Restoration of Damaged Property: If any direct or indirect damage is done to public or private property resulting from any act, omission, neglect, or misconduct, the Contractor shall restore the damaged property to a condition equal to that existing before the damage at no additional cost to the Government. Repair, rebuild, or restore property as directed or make good such damage in an acceptable manner.

h. Final Clean-up: On completion of project and after removal of all debris, rubbish, and temporary construction, Contractor shall leave the construction area in a clean condition satisfactory to the Contracting Officer. Cleaning shall include off the station disposal of all items and materials not required to be salvaged, as well as all debris and rubbish resulting from demolition and new work operations.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 58 00

PROJECT IDENTIFICATION

08/19, CHG 3: 11/21

temporary signs for project identification

PART 1 GENERAL

1.1 REFERENCES
1.2 SUBMITTALS
1.3 PROJECT IDENTIFICATION SIGN
1.3.1 Construction Project Signs

PART 2 PRODUCTS

PART 3 EXECUTION

ATTACHMENTS:

project sign detail

-- End of Section Table of Contents --

SECTION 01 58 00

PROJECT IDENTIFICATION
08/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 310-1-6a	(2006; 2019 Change 2) Project Operation -- Sign Standards Manual, VOL 1
EP 310-1-6b	(2006) Sign Standards Manual, VOL 2, Appendices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Construction Sign Design Mockup; G

The requirements for the signs, their content, and location are as indicated and as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

1.3 PROJECT IDENTIFICATION SIGN

Prior to initiating any work on site, provide one project identification sign at the location designated. Construct the sign in accordance with project sign detail, per the attached requirements. Maintain sign throughout the life of the project. Upon completion of the project, remove the sign from the site.

1.3.1 Construction Project Signs

Furnish the construction project sign package, maintain the signs during construction, and remove the signs from the job site upon completion of the project. The construction project sign package consists of two signs: one for project identification and the other to show the on-the-job safety performance of the contractor. Ensure that the package conforms to the requirements of EP 310-1-6a and EP 310-1-6b, specifically Section 16. Submit the sign legend orders as described in Section 16 of EP 310-1-6a before erecting the signs.

Construction signage shall comply with the requirements identified in Joint Policy Flash 2022-04 (see attached).

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

CONSTRUCTION SIGN SCHEDULE



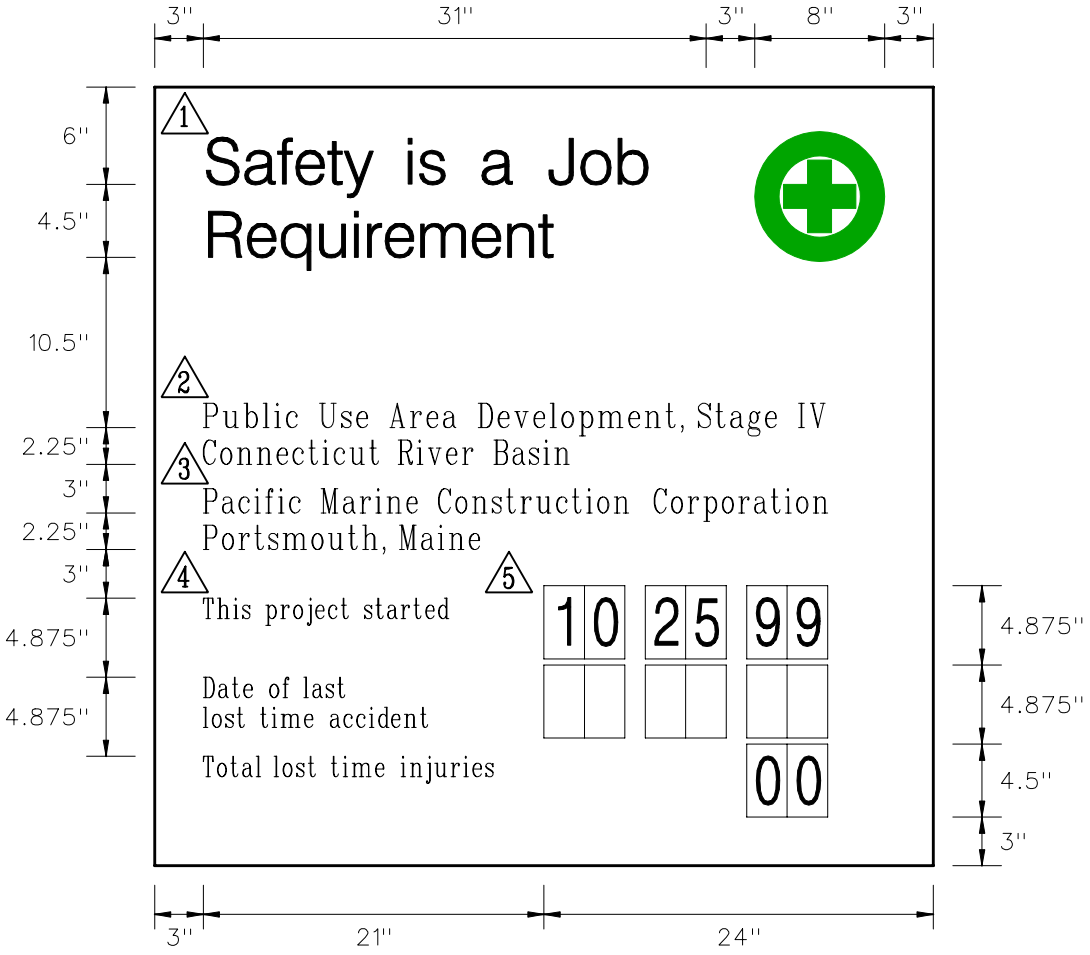
CONSTRUCTION SIGN SCHEDULE

Standardized Design for Trailer Sign

VA Livermore Realignment Resident Engineer Office

U.S. Army Corps of Engineers
Sacramento District

U.S. Department of Veterans Affairs
Office of Construction and Facilities Management, Western Region



NOTES: LEGEND GROUP

DESCRIPTION

1. Standard two-line title: "Safety is a Job Requirement" with (8"OD.) Safety Green First Aid logo. Color: To match PMS 347, Typeface: 3" Helvetica Bold, Color: Black.

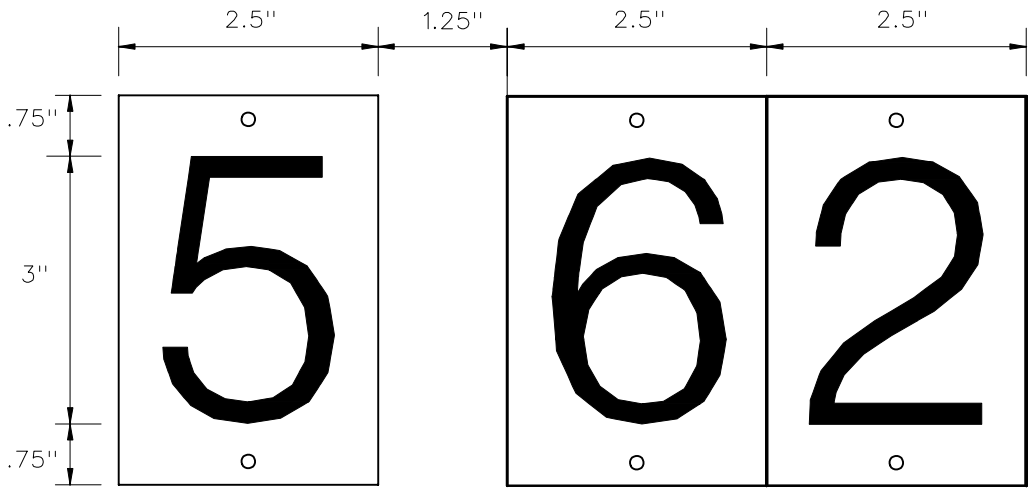
2. One-to two-line project title legend describes the work being done under this contract and name of host project. Color: Black, Typeface: 1.5" Helvetica Regular, Maximum line length: 42".

3. One-two-line identification: name of prime contractor and city, state address. Color: Black, Typeface: 1.5" Helvetica Regular, Maximum line length: 42".

4. Standard safety record captions as shown. Color: Black, Typeface: 1.25" Helvetica Regular.

5. Replaceable numbers are to be mounted on white .060 aluminum plates and screw-mounted to background. Color: Black, Typeface: 3" Helvetica Regular, Plate size: 2.5"x4.5".

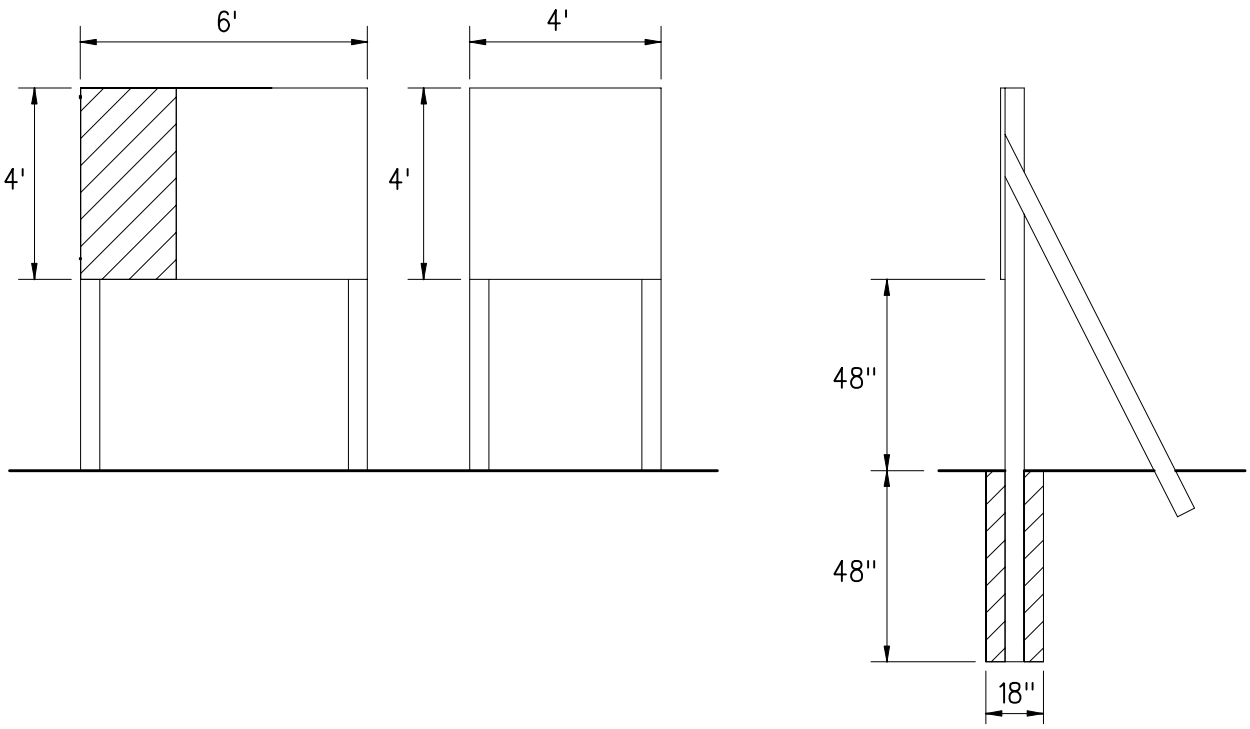
DEPARTMENT OF THE ARMY NEW ENGLAND DISTRICT CORPS OF ENGINEERS CONCORD, MASSACHUSETTS	STANDARD CONSTRUCTION DETAIL SAFETY PERFORMANCE SIGN DETAIL	1/2



NOTES:

1. Replace numbers are to be mounted on white .OLD aluminum plates and serew-mounted to background of Safety Performance Sign.

DEPARTMENT OF THE ARMY NEW ENGLAND DISTRICT CORPS OF ENGINEERS CONCORD, MASSACHUSETTS	STANDARD CONSTRUCTION DETAIL INDIVIDUAL METAL PLATES DETAIL	2 2



NOTES:

1 The sign panels are to be fabricated from .75" High Density Overlay Plywood. Panel preparation to follow HDO specifications provided in Appendix B.

2 Sign graphics to be prepared on a white non-reflective vinyl film with positionable adhesive backing.

3 All graphics except for the Communications Red background with Corps signature on the project sign are to be die-cut or computer-cut non-reflective vinyl, pre-spaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown on pages 16.2-3.

4 The 2'x4' Communications Red panel (to match PMS-032) with full Corps signature (reverse version) is to be screen printed on the white background. Identification of the district or division may be applied under the signature with white cut vinyl letters prepared to Corps standards. Large scale reprouduction artwork for the signature is provided on page 4.8 (photographically enlarge from 6.875" to 10.5").

5 Drill and insert six (6) .375" T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face.

6 Apply graphic panel to prepared HDO plywood panel following manufacturers' instructions.

7 Sign uprights to be structural grade 4"x4" treated Douglas Fir or Southern Yellow Pine, No.1 or better. Post to be 12' long. Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (.5") back of hole to accept socket head cap screw (4"x.375").

8 Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2"x4" struts on inside face of uprights to reinforce installation as shown.

DEPARTMENT OF THE ARMY NEW ENGLAND DISTRICT CORPS OF ENGINEERS CONCORD, MASSACHUSETTS	STANDARD CONSTRUCTION DETAIL **FABRICATION AND MOUNTING GUIDELINES** MILITARY PROJECT	1 1

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 58 16

TEMPORARY INTERIOR SIGNAGE

PART 1 GENERAL

1.1 SUBMITTALS

PART 2 PRODUCTS

2.1 TEMPORARY SIGNS

PART 3 EXECUTION/REMOVAL

3.1 INSTALLATION
3.2 LOCATION

-- End of Section Table of Contents --

SECTION 01 58 16

TEMPORARY INTERIOR SIGNAGE

PART 1 GENERAL

This section specifies temporary interior signs.

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Temporary Signs; G,

PART 2 PRODUCTS

2.1 TEMPORARY SIGNS

a. Fabricate from 3 mil mat finish white plastic.

b. Cut to (4-inch) wide by 12 inch long size tag.

c. Punch (1/8-inch) diameter hole centered on (4-inch) dimension of tag. Edge of Hole spaced approximately (1/2-inch) from one end on tag.

d. Ties: Steel wire (0.0120-inch) thick, attach to tag with twist tie, leaving (6-inch) long free ends.

PART 3 EXECUTION/REMOVAL

3.1 INSTALLATION

a. Install temporary signs attached to room door frame or room door knob, lever, or pull for doors on corridor openings.

b. Mark on signs with felt tip marker having approximately (1/8-inch) wide stroke for clearly legible numbers or letters.

c. Identify room with numbers as designated on floor plans.

d. All temporary signs must be removed prior to final cleaning.

3.2 LOCATION

a. Install on doors that have room, corridor, and space numbers shown.

b. Doors that do not require signs are as follows:

(1) Corridor barrier doors (cross-corridor) in corridor with same number.

(2) Folding doors or partitions.

(3) Toilet or bathroom doors within and between rooms.

(4) Communicating doors in partitions between rooms with corridor entrance doors.

(5) Closet doors within rooms.

c. Replace missing, damaged, or illegible signs.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES
1.2 DEFINITIONS
1.2.1 Co-mingle
1.2.2 Construction Waste
1.2.3 Demolition Debris/Waste
1.2.4 Disposal
1.2.5 Diversion
1.2.6 Final Construction Waste Diversion Report
1.2.7 Recycling
1.2.8 Reuse
1.2.9 Salvage
1.2.10 Source Separation
1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)
1.4 CONSTRUCTION WASTE MANAGEMENT
1.4.1 Implementation of Construction Waste Management Program
1.4.2 Oversight
1.4.3 Special Programs
1.4.4 Special Instructions
1.4.5 Waste Streams
1.5 SUBMITTALS
1.6 MEETINGS
1.7 CONSTRUCTION WASTE MANAGEMENT PLAN
1.8 RECORDS (DOCUMENTATION)
1.8.1 General
1.8.2 Accumulated
1.9 REPORTS
1.9.1 General
1.9.2 Quarterly Reporting
1.9.3 Annual Reporting
1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT
1.11 COLLECTION
1.11.1 Source Separation Method
1.11.2 Co-Mingled Method
1.11.3 Other Methods
1.12 DISPOSAL
1.12.1 Recycle
1.12.2 Waste

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 273	Standards for Universal Waste Management
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
49 CFR 178	Specifications for Packagings

1.2 DEFINITIONS

1.2.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.2.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.2.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.2.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.2.5 Diversion

The practice of diverting waste from disposal in a landfill or incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.2.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.2.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.2.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.2.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.2.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 75 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.4 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.4.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work,

as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.4.2 Oversight

The Quality Control Manager, as specified in Section 01 45 00 QUALITY CONTROL, is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project.

1.4.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.4.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.4.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste steams not listed:

a. Land Clearing Debris

b. Asphalt

c. Masonry and CMU

d. Concrete

e. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)

f. Wood (nails and staples allowed)

g. Glass

h. Paper

i. Plastics (PET, HDPE,PVC,LDPE,PP,PS, Other)

j. Gypsum

k. Non-hazardous paint and paint cans

l. Carpet

m. Ceiling Tiles

n. Insulation

o. Beverage Containers

1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-06 Test Reports

Quarterly Reports

Annual Report

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; S

1.6 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00 QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

a. Preconstruction meeting.

b. Regular site Quality Control meetings.

c. Work safety meeting (if applicable).

1.7 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan not less than 60 calendar days before scheduled final site or building design approval. Revise and resubmit Construction Waste Management Plan until it receives final approval from the Contracting Officer, in order for construction to begin. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project..

b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.

c. Description of the regular meetings to be held to address waste management.

d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.

e. Name of landfill and incinerator to be used.

f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.

g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.

h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.

i. Description of the means by which materials identified in item (g) above will be protected from contamination.

j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).

k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.

m. Detailed plan and distribution of waste diversion between buildings, when project is a part of a campus.

n. Facilities or subcontractors offering construction waste transport on-site or off-site must ensure that proper shipping orders, bill of lading, manifests, or other shipping documents containing waste diversion information meet requirements of 40 CFR 273 Universal Waste

Management, 49 CFR 173 Shippers - General Requirements for Shipments and Packagings, and 49 CFR 178 Specifications for Packaging. Individuals signing manifests or other shipping documents should meet the minimum training requirements.

o. List each supplier who deliver construction materials, in bulk, or package products in returnable containers or returnable packaging, or have take-back programs. List each program and the applicable material to actively monitor and track to assist in meeting waste diversion requirements on the project.

1.8 RECORDS (DOCUMENTATION)

1.8.1 General

Maintain records to document the types and quantities of waste generated and diverted though re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.8.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.9 REPORTS

1.9.1 General

Maintain current construction waste diversion information on site for periodic inspection by the Contracting Officer. Include in the quarterly reports, annual reports and final reports: the project name, contract information, information for waste generated, diverted and disposed of for the current reporting period and show cumulative weight (pounds) totals for the project. Reports must identify quantifies of waste by weight (pounds) type and disposal method. Also include in each report, supporting documentation to include manifests, weigh tickets, receipts, and invoices specifically identifying the project and waste material type and weighted sum.

1.9.2 Quarterly Reporting

Provide cumulative reports at the end of each quarter (December, March, June, and September, corresponding with the federal fiscal year for reporting purposes). Submit quarterly reports not later than 15 calendar days after the preceding quarter has ended. Submit Quarterly Reports to the appropriate office or identified point of contact.

1.9.3 Annual Reporting

Provide a cumulative construction waste diversion report annually. Submit annual report not later than 30 calendar days after the preceding fourth

quarter has ended. Provide copy of annual construction waste diversion report to the installation POC.

1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide Final Construction Waste Diversion Report 60 days prior to the Beneficial Occupancy Date (BOD).

1.11 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS. Separate materials by one of the following methods described herein:

1.11.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.11.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to an authorized recycling facility, which meets all applicable requirements to accept and dispose of recyclable materials in accordance with all applicable local, state and federal regulations. The Co-mingled materials must be sorted and processed in accordance with the approved Construction Waste Management Plan.

1.11.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.12 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.12.1 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.12.2 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used. -- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 00

CLOSEOUT SUBMITTALS

05/19, CHG 1: 08/21

PART 1 GENERAL

1.1 DEFINITIONS
1.1.1 As-Built Drawings
1.1.2 Record Drawings
1.2 SOURCE DRAWING FILES
1.2.1 Terms and Conditions
1.3 PROJECT RECORD DOCUMENTS
1.3.1 Preliminary Record Drawings
1.3.2 Final Record Drawings
1.3.3 Post-Record Drawing Work
1.3.4 Payment for Final Record Drawings
1.4 SUBMITTALS
1.5 SPARE PARTS DATA
1.6 WARRANTY MANAGEMENT
1.6.1 Warranty Management Plan
1.6.2 Performance Bond
1.6.3 Pre-Warranty Conference
1.6.4 Warranty Tags

PART 2 PRODUCTS

2.1 Equipment Data
2.2 CERTIFICATION OF EPA DESIGNATED ITEMS
2.3 CERTIFICATION OF USDA DESIGNATED ITEMS

PART 3 EXECUTION

3.1 RED ZONE MEETING
3.2 AS-BUILT DRAWINGS
3.2.1 Markup Guidelines
3.2.2 As-Built Drawings Content
3.3 RECORD DRAWINGS
3.4 OPERATION AND MAINTENANCE MANUALS
3.4.1 Submittal Requirements
3.4.1.1 Field Training
3.4.1.2 Draft O & M Manuels
3.4.1.3 Commencement of Warranty of Construction
3.4.2 Government Possession of Work
3.4.3 Payment
3.4.4 Preparation and Submission of Operation and Maintenance Manuals
3.4.4.1 General Requirements
3.4.4.2 Format
3.4.4.3 Equipment, Operating, Maintenace, and Repair Manuals
3.5 CLEANUP

-- End of Section Table of Contents --

SECTION 01 78 00

CLOSEOUT SUBMITTALS
05/19, CHG 1: 08/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as. These files serve as the basis for the creation of the record drawings.

1.1.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.2 SOURCE DRAWING FILES

Request the full set of electronic drawings, in the source format, for Record Drawing preparation, after award and at least 30 days prior to required use.

1.2.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction drawings and data for the referenced project. Any other use or reuse shall be at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or cause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any

conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction drawings and data related to this contract, remove all previous indicia of ownership (seals, logos, signatures, initials and dates).

1.3 PROJECT RECORD DOCUMENTS

1.3.1 Preliminary Record Drawings

The Contractor shall mark up both a reproducible set and a set of prints to show as-built conditions. These two sets, hereafter called preliminary record drawings, or singly, reproducibles or prints, shall be kept current and available on the jobsite at all times, except as noted below. For drawings contained within the Specifications, the Contractor shall mark up copies of these drawings to show as-built conditions; these copies will be considered the preliminary record drawings and shall be kept current and available on the jobsite at all times, except as noted below. Assign a member of the Contractor's Quality Control Organization to be responsibility for the maintenance and currency of the preliminary record drawings. This assignment and any reassignment of duties concerning the maintenance of the record drawings shall be promptly reported to the Contracting Officer's representative for approval. All changes from the contract drawings which are made in the work or additional information which might be uncovered in the course of construction, including uncharted utilities, shall be accurately and neatly recorded as they occur
by means of details and notes. Clearly identify all changes and/or required additions to the preliminary record drawings in a contrasting color and which is compatible with reproduction of the preliminary record drawings. Update preliminary record drawings by Friday of each week. During periods when the reproducibles are being copied and are therefore not available at the jobsite, continue posting all required data to the prints. Minimize the time that the reproducibles are away from the jobsite and update them with all as-built data immediately upon their return. The preliminary record drawings will be jointly inspected for accuracy and completeness by the Contracting Officer's representative and the assigned representative of the Contractor's Quality Control Organization prior to submission of each monthly pay estimate. See paragraph, "Withholding for Preliminary Record Drawings." The record drawings shall show the following information, but not be limited thereto:

a. The location and description of utility lines or other installation of any kind or description known to or found to exist within the construction area. The location of exterior utilities includes actual measured horizontal distances from utilities to permanent facilities/ features. These measurements shall be within an accuracy range of 6 inches and shall be shown at sufficient points to permit easy location of utilities for future maintenance purposes. Show measurements for all change of direction points and all surface or underground components such as valves, manholes, drop inlets, cleanouts, meter, etc. Indicate the general depth range of each underground utility line (i.e., 3 to 4 feet in depth). The description of exterior utilities includes the actual quantity, size, and material of utility lines.

b. The location and size of all uncharted existing utilities encountered.

c. The location and dimensions of any changes within the building or structure.

d. Correct grade or alinement of roads, structures or utilities if any changes were made from contract drawings.

e. Correct elevations if changes were made in site grading.

f. Changes in details of design or additional information obtained from working drawings specified to be prepared and/or furnished by the Contractor including but not limited to fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment foundations, etc.

g. The topography and grades of all drainage installed or affected as a part of the project construction.

h. Options

Where contract drawings or specifications allow options, only the option selected for construction shall be shown on the record drawings.

i. Blue Line or Black Line Prints

Blue-line or black-line prints shall be full size. All blue or black line prints shall exhibit good readable print with clear, sharp, dark lines, and shall not be smeared, faded, double imaged, or have torn or ragged edges.

j. Prefinal Inspection For Each Item of Work

As part of the prefinal inspection for each item of work, the current updated preliminary marked-up record drawings and as-built CADD drawing files will be reviewed. They shall comply with this specification prior to scheduling the final inspection, and/or prior to substantial completion of the item of work.

k. Preliminary Record Drawing Final Submittal

Prior to scheduling the red zone meeting and the final acceptance inspection of the last or only bid schedule item of work, the updated preliminary marked-up record drawings and the updated as-built CADD drawing files shall be completed and delivered to the Contracting Officer's Representative for review and approval. If upon review, the drawings and files are found to contain errors and/or omissions, they will be returned to the Contractor for corrections. Failure of the Contractor to make timely delivery of the preliminary record drawings and files on any or all items of work will be cause for the Government to delay substantial completion and to withhold the amount indicated in paragraph "Withholding for Preliminary Record Drawings," in accordance with the terms and conditions of the contract.

l. Withholding for Preliminary Record Drawings

Failure by the Contractor to maintain current and satisfactory preliminary record drawings in accordance with these requirements will result in withholding from progress payments 10 percent of the

progress payment amount until such time as the record drawings are brought into
compliance. This withheld amount will be indicated on monthly payment estimates until the Contractor has fulfilled these contract requirements.

m. Final Inspection

For each interim item of work, furnish a copy of the preliminary record drawings for that item, which the Contractor has reproduced from the approved preliminary record drawing reproducibles, to the Contracting Officer's representative at the time of final inspection for that item. At the time of final inspection on the last or only item of work, the Contractor shall deliver a copy of the complete set of the approved preliminary record drawings to the Contracting Officer's Representative.

1.3.2 Final Record Drawings

Upon approval of the preliminary record drawings, the Contracting Officer will return the approved preliminary record drawing prints back to the Contractor. The Contractor will then modify the CADD files as may be necessary to correctly show all the features of the project as it was constructed by bringing the contract set into agreement with the preliminary record drawings, including adding additional drawings and CADD files as may be necessary. The Contractor shall furnish the as-built drawings in the same file format specified. The CADD files will be furnished to the Contractor on a separate hard drive. These CADD files are part of the permanent records of this project and the Contractor shall be responsible for the protection and safety thereof until returned to the Contracting Officer. Drawings, tracings, or CADD files damaged or lost by the Contractor shall be satisfactorily replaced by the Contractor at the Contractor's expense. CADD files will be audited by the Contracting Officer and for accuracy and conformance to the above specified drafting and CADD standards. Final Record Drawings must have the professional stamp of the Designer of Record (DOR).

a. Drafting

Only personnel proficient in the preparation of engineering drawings and CADD shall be employed to modify the original contract drawings, prepare additional new drawings, and modify the CADD files. Additions and corrections to the contract drawings shall be neat, clean, and legible and shall match the adjacent existing line work and/or lettering being annotated in type, density, size and style. All modifications, CADD information, and new drawings shall conform to applicable requirements specified in the paragraph "CADD Standards." The Contractor shall ensure that all delivered CADD digital files and data (e.g., sheet files, model files, cell/block libraries) are compatible with the Government's target CADD system and operating system, and adhere to the standards and requirements specified. The term "compatible" means that data is in native digital format i.e., .dgn (MicroStation) or .dwg (AutoCAD). It is the responsibility of the Contractor to ensure this level of compatibility.

b. CADD Standards

The Contractor shall submit a written request for approval of any deviations from the Government's established CADD standards.

Deviations will not be permitted unless prior written approval of such deviations has been received from the Government.

c. Final Revisions

When final revisions have been completed, place the words "REVISED RECORD DRAWING," in letters at least 3/16 inch high, and the date of completion in the revision block above the latest existing revision notation on each drawing CADD file.

d. Border Sheets

The border sheet to be used for any new record drawings shall be the same as used on the original drawings.

e. Copies of the Final Record Drawings

f. Submittal Requirements

The Contractor shall submit to the Contracting Officer the final record drawings, consisting of two sets each of corrected electronic (CADD
and.pdf) files on hard drive; verification that the electronic (CADD and .pdf) files have been loaded and work on the designated computer systems and are error- and virus-free; the approved preliminary drawings; and all required reproduced items. The corrected.pdf files shall be compiled into sets so that the files can be viewed and printed by the Adobe PDF reader; sets shall be bookmarked by sheet number and title. All electronic CADD files will become the property of the Government.

1.3.3 Post-Record Drawing Work

In event the Contractor accomplishes additional work which changes the as-built conditions of the facility after submission of the record drawings, the Contractor shall furnish revised and/or additional drawings (electronic (CADD and.pdf) files), as required to depict as-built conditions. Incorporate revised and additional files into the completed sets of contract record electronic (CADD and .pdf) files. The requirements for these additional drawings and files, will be the same as for the record drawings included in the original submission.

1.3.4 Payment for Final Record Drawings

If there is no separate contract line item (CLIN) for as-built drawings, the Government will withhold the amount of $35,000, or 1% of the present construction value up tp $500,000, whichever is the greater, until the final as-built drawing submittal has been approved by the Government.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan; G

Warranty Tags

Spare Parts Data; G

SD-08 Manufacturer's Instructions

Posted Instructions

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

As-Built Record of Equipment and Materials; G

Certification of EPA Designated Items; G

Certification Of USDA Designated Items; G

1.5 SPARE PARTS DATA

Submit copies of the Spare Parts Data list.

a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

b. At acceptance of commissioning, ensure the required stock level is supplied as indicated in subparagraph a for maintenance and repair activities through the facilities warranty period. Provision of spare parts does not relieve the Contractor of responsibilities listed under the contract guarantee provisions.

1.6 WARRANTY MANAGEMENT

1.6.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant similar to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit of the warranty

management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representativel. The warranty management plan must include, but is not limited to, the following:

a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or suppliers involved.

b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.

c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.

d. As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:

(1) Name of item.
(2) Model and serial numbers.
(3) Location where installed.
(4) Name and phone numbers of manufacturers or suppliers.
(5) Names, addresses and telephone numbers of sources of spare parts.
(6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
(7) Cross-reference to warranty certificates as applicable.
(8) Starting point and duration of warranty period.
(9) Summary of maintenance procedures required to continue the warranty in force.
(10) Cross-reference to specific pertinent Operation and Maintenance manuals.
(11) Organization, names and phone numbers of persons to call for warranty service.
(12) Typical response time and repair time expected for various warranted equipment.

e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.

f. Procedure and status of tagging of equipment covered by warranties

longer than one year.

g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.6.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.

a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.

b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.6.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.6.4 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	
Address	
Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

PART 2 PRODUCTS

2.1 Equipment Data

List shall include on each item as applicable: Description, Manufacturer, Model or Catalog No., Serial No., Input (power voltage, BTU, etc.), Output (power, voltage, BTU, tons, etc.). Size or Capacity (tanks), and net inventory costs; any other data necessary to describe item and shall list all warrantors and warranty periods for each item of equipment.

2.2 CERTIFICATION OF EPA DESIGNATED ITEMS

Submit the Certification of EPA Designated Items as required by FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items and FAR 52-223-17 Affirmative Procurement of EPA designated items in Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor.

2.3 CERTIFICATION OF USDA DESIGNATED ITEMS

Submit the Certification of USDA Designated Items as required by FAR 52-223-1 Bio-based Product Certifications and FAR 52.223-2 Affirmative Procurement of Biobased Products Under Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current USDA standards for biobased materials content. The following exemptions may apply to the non-procurement of biobased content materials:

a. The product does not meet appropriate performance standards;

b. The product is not available within a reasonable time frame;

c. The product is not available competitively (from two or more sources);

d. The product is only available at an unreasonable price (compared with a comparable bio-based content product)."

Record each product used in the project that has a requirement or option of containing biobased content in accordance with SECTION 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING, noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, total value of biobased content, exemptions (a, b, c, or d, as indicated), and comments. Biobased content values may be determined by weight or volume percent, but must be consistent throughout.

PART 3 EXECUTION

3.1 RED ZONE MEETING

Approximately 80 percent through the period of performance and no later than 6 months before the anticipated Beneficial Occupancy Date (BOD) but prior to the final acceptance inspection of the last or only bid schedule item of work, the Contractor's Supervisor and Quality Control Manager and the Government's project delivery team will conduct what is know as the Red Zone Meeting to discuss the close-out process, to schedule the events and review responsibilities for actions necessary to produce a timely physical, as well as fiscal, project close-out. The Red Zone Meeting derives its name from the football term used to describe the team effort to move the ball the last 20 yards into the end zone. The close-out of a construction project sometimes can be equally as hard and most definitely requires the entire team's efforts. The ACO will chair the meeting. Exhibit One is a generic checklist.

3.2 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF contract drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies on site and red-lined PDF files. Submit As-Built Drawings 30 days prior to Beneficial Occupancy Date (BOD).

3.2.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

a. Use base colors of red, green, and blue. Color code for changes as follows:

(1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.

(2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.

(3) Additions (Green) - Added items, lettering in notes and leaders.

b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.

c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.

d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.

e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.

f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.

g. For deletions, cross out all features, data and captions that relate to that revision.

h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.

i. Indicate one of the following when attaching a print or sketch to a markup print:

1) Add an entire drawing to contract drawings

2) Change the contract drawing to show

3) Provided for reference only to further detail the initial design.

j. Incorporate all shop and fabrication drawings into the markup drawings.

3.2.2 As-Built Drawings Content

Show on the as-built drawings, but not limited to, the following information:

a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.

b. The location and dimensions of any changes within the building structure.

c. Layout and schematic drawings of electrical circuits and piping.

d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.

e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.

f. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.

g. Changes or Revisions which result from the final inspection.

h. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.

i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.

j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.

k. Changes in location of equipment and architectural features.

l. Modifications.

m. Actual location of anchors, construction and control joints, etc., in concrete.

n. Unusual or uncharted obstructions that are encountered in the contract work area during construction.

o. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.3 RECORD DRAWINGS

3.4 OPERATION AND MAINTENANCE MANUALS

Provide project operation and maintenance manuals. Provide four hard drives with electronic copies of the Operation and Maintenance Manual files and hard drives of the final Operation and Maintenance Manuals. Submit electronic copies to the Contracting Officer for approval within 30 calendar days of the Beneficial Occupancy Date (BOD). Update and resubmit files for final approval at BOD.

The Contractor shall be responsible for the preparation, coordination, execution and submittal of all Equipment Operating, Maintenance, and Repair manuals (O & M Manuals), including spare parts lists (with the names and PHONE NUMBERS of local suppliers), special tools, inventories of equipment manuals, and maintenance instructions, and shall conduct all training for operating and service personnel. Operation and maintenance manuals shall cover all system installations provided in this Contract and shall be in sufficient detail to facilitate normal maintenance and troubleshooting by persons with minimum experience with the installed equipment.

3.4.1 Submittal Requirements

All of the above listed items required in the technical specifications shall be fully developed and submitted to the Contracting Officer not less than 120 calendar days prior to the scheduled final acceptance inspection date and prior to scheduling training for operating and service personnel. The Contractor shall coordinate the content of each instruction period required in the technical specifications with the Contracting Officer's Representative prior to the actual start of the training period.

3.4.1.1 Field Training

Field Training is a requirement for substantial completion. The Contractor shall conduct a training course for the operating staff for each particular system.

3.4.1.2 Draft O & M Manuels

On those systems where complete and comprehensive operation and maintenance manuals cannot be fully developed until the system(s) is (are) checked, tested, and/or balanced, and the checking, testing, and/or balancing has not been done when submittals are required, a proposed draft of those system manual(s) shall be submitted electronically. The covers of draft O & M Manuals shall be labeled "DRAFT" in large (not less than font size 24), legible, printed letters. Submit fully developed O & M Manuals for approval after the systems have been checked, tested, and/or balanced but prior to the scheduled final acceptance inspection date.

3.4.1.3 Commencement of Warranty of Construction

Failure to submit all specified O & M manuals, spare parts listings, spare parts, special tools, inventories of installed property, and training video in a timely manner will be considered as delaying substantial completion of the work. Commencement of warranty under the Contract Clause WARRANTY OF CONSTRUCTION will not occur until all these items are delivered and approved by the Contracting Officer, but not earlier than the date of final acceptance of the work by the Government. When the O & M Manuals with drafts are approved they will not constitute a reason for delaying the start of the warranty period.

3.4.2 Government Possession of Work

The Government may take possession of any completed or partially completed work as provided for under Contract Clause entitled "USE AND POSSESSION PRIOR TO COMPLETION." If the installed equipment and/or systems thereto, have not been accepted by the Government due to the Contractor's failure to submit the above specified items, the Contractor shall operate and maintain such plant or system at no additional cost to the Government until such time that the specified items have been received, approved and any subsequent testing, check-out and/or training has been completed.

3.4.3 Payment

If there is no separate CLIN for O&M Manuals, the Government will withhold an amount representing $20,000,or 1% of the present construction value up to $500,000, whichever is the greater, as non-progressed work, until

submittal and approval of all O&M manuals are complete.

3.4.4 Preparation and Submission of Operation and Maintenance Manuals

This paragraph establishes general requirements for the preparation and submission of equipment operating, maintenance, and repair manuals as called for in the various sections of the specifications. Specific instruction(s) relating to a particular system or piece of equipment shall be incorporated into the manuals in accordance with the applicable technical specification. See Section 01 78 23 OPERATION AND MAINTENANCE DATA for additional information concerning the Operation and Maintenance data packages required by the technical sections.

3.4.4.1 General Requirements

Furnish permanent electronic copy of the final Equipment Operating, Maintenance, and Repair Manuals on hard drive along with the number of copies specified in part 3.7 Operation and Maintenance Manuals within this section. Documents on the hard drive shall be in portable document format (.pdf); hyperlinked, bookmarked, and searchable, all printed and graphic documents, drawings, and illustrations shall be legible and bookmarked.

3.4.4.2 Format

Bind hard copy manuals in durable, locking, 3-ring binders. Binders shall be white view-type binders with clear plastic overlays to allow insertion of title pages for binder identification. Use 8-1/2" x 11" sheets, except that larger sheets up to 11" x 17" may be used when double folded to this size and used as a pull-out. Documents which are larger than 11" x 17" shall be reduced to 11" x 17" for inclusion in the manuals except where this compromises legibility (for drawings that are to scale, add a graphic scale prior to reduction).Documents that cannot be reduced will be folded and inserted in plastic envelopes inserted in the binders so that the folded documents are securely bound into the binders. Loosely inserted documents or documents inserted into pockets in the inside covers of the binders shall not be acceptable.

Each binder shall be labeled on both cover and spine to indicate project name, project number, submitting contractor, date, general contents, volume number, and total number of volumes in set.

3.4.4.3 Equipment, Operating, Maintenace, and Repair Manuals

a. General

Provide separate manuals for each utility system as defined hereinafter. Provide the number of copies of the manuals specified above or in the applicable technical section. Include in the manuals, in separate sections, the following information for each item of equipment. These requirements may be supplemented by additional requirements specified in the technical sections:

(1) Performance sheets and graphs showing capacity data, efficiencies, electrical characteristics, pressure drops, and flow rates. Marked-up catalogs or catalog pages do not satisfy this requirement. Performance information shall be presented as concisely as possible and contain only data pertaining to equipment actually installed.

(2) Catalog cuts showing application information.

(3) Installation information showing minimum acceptable requirements.

(4) Operation and maintenance requirements. Include adequate illustrative material to identify and locate operating controls, indicating devices and locations of areas or items requiring maintenance.

(a) Describe, in detail, starting and stopping procedures for components, adjustments required to obtain optimum equipment performance, and corrective actions for malfunctions.

(b) Describe in the maintenance instructions the nature and frequency of routine maintenance and procedures to be followed. Indicate any special tools, materials, and test equipment that may be required

(5) Repair information including diagrams and schematics, guidance for diagnosing problems, and detailed instructions for making repairs. Provide troubleshooting information that includes a statement of the indication or symptom of trouble and the sequential instructions necessary. Include test hookups to determine the cause, special tools and test equipment, and methods for returning the equipment to operating conditions. Information may be in chart form or in tabular format with appropriate headings.

(6) Parts lists with names and addresses of closest parts supply agencies, the current unit prices, and the sources of supply. Include spare parts data for each different item of materials and equipment specified.

(7) Names and addresses of local manufacturers representatives.

b. Facility Heating Systems

Provide information for the following equipment: Air handling units (both heating only and heating and cooling).

b. Temperature Control and HVAC Distribution Systems

(1) Provide the information described for the following equipment: Air handling units.

(2) Provide all information described for the following equipment: Control components (sensors, controllers, adapters, and actuators), and flow measuring equipment.

c. Exterior Electrical Systems

Information shall be provided on the following equipment: Power transformers, relays, reclosers, breakers, and capacitor bank controls.

d. Interior Electrical Systems

Information shall be provided on the following equipment: Relays, motor control centers, switchgear, solid state circuit breakers, motor controller, and EPS lighting systems, control systems (wire diagrams and troubleshooting flow chart), and special grounding systems.

e. Energy Management and Control System

The maintenance manual shall include descriptions of maintenance for all equipment, including inspection, periodic preventative maintenance, fault diagnosis, and repair or replacement of defective components.

f. Fire Protection Systems

Information shall be provided on the following equipment: Alarm valves, manual valves, regulators, piping materials, sprinkler heads, nozzles.

g. Fire Detection Systems

The maintenance manual shall include description of maintenance for all equipment, including inspection, periodic preventive maintenance, fault diagnosis, and repair or replacement of defective components.

h. Plumbing Systems

Information shall be provided on the following equipment: Water heaters, valves, pressure regulators, backflow preventors, piping materials, and plumbing fixtures.

i. Miscellaneous Systems

Information shall be provided on the following: Communication and ADP systems, security and intrusion alarm, and other similar type special systems not otherwise specified.

3.5 CLEANUP

Leave premises "broom clean." Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site..

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

05/23

PART 1 GENERAL

1.1 REFERENCES
1.2 SUBMITTALS
1.3 MEETINGS
1.3.1 Pre-Construction Meeting
1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting
1.3.3 Submittal Coordination Meeting
1.3.4 Facility Turnover Meeting
1.4 FACILITY DATA WORKBOOK
1.5 OPERATION AND MAINTENANCE MANUAL MEDIA
1.5.1 CD or DVD Label and Disk Holder or Case
1.5.2 O&M Manual Tabbed Hard Copy
1.6 O&M MANUAL CONTENT
1.6.1 Part 1: Executive Summary
1.6.2 Part 2: Facility Design and Construction
1.6.2.1 General Facility and Systems Description
1.6.2.2 Contract Documents, RFP, Amendments, and Modifications
1.6.2.3 Room Inventory of Real Property and Finishes
1.6.3 Part 3: Facilities, Systems, and Assemblies Information
1.6.3.1 Organization
1.6.3.2 Related Specifications
1.6.3.3 Manufacturer's Operations and Maintenance Data
1.6.3.4 Approved Submittals and Certificates
1.6.3.5 Approved Coordination/Shop Drawings
1.6.4 Sequence of Operation for Operating Equipment
1.6.4.1 Safety Precautions and Hazards
1.6.4.2 Operator Prestart
1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures
1.6.4.4 Normal Operations
1.6.4.5 Emergency Operations
1.6.4.6 Operator Service Requirements
1.6.4.7 Environmental Conditions
1.6.4.8 Operating Log
1.6.4.9 Additional Requirements for Equipment Control Systems
1.6.4.10 Testing Equipment Information and Performance Data
1.6.5 Routine Maintenance Requirements
1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures
1.6.5.2 Lubrication Data
1.6.6 Repair Procedures
1.6.6.1 Troubleshooting Guides and Diagnostic Techniques
1.6.6.2 Wiring Diagrams and Control Diagrams
1.6.6.3 Removal and Replacement Instructions
1.6.6.4 Repair Work-Hours
1.6.6.5 Warranty Information

1.6.6.6 Extended Warranty Information
1.6.6.7 Construction Drawings and Utility Systems
1.6.6.8 Personnel Training Requirements
1.6.6.9 Contractor / Supplier Listing and Contact Information
1.6.7 Part 4: Facility Operations
1.6.7.1 Completed Facility Operating Plan
1.6.7.2 Testing Equipment and Special Tool Information
1.6.7.3 Testing and Performance Data
1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports
1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory
1.6.7.5.1 Maintenance Schedules
1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping
1.6.7.5.3 Janitorial and Cleaning Plans and Procedures
1.6.7.6 Utility Construction Drawings
1.6.7.6.1 Utility Schematic Diagrams
1.6.7.6.2 Enlarged Connection and Cutoff Plans
1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems
1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting
1.6.7.6.2.3 One-Line Diagrams and Meter Location of System
1.6.7.6.3 Spare Parts and Supply Lists
1.6.8 Part 5: Training
1.6.9 Part 6: Cx Project Report and TAB Report
1.6.10 Part 7: Regulatory Requirements
1.6.11 Part 8: Permits
1.6.12 Part 9: Operations and Maintenance Manual Approval
1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES
1.7.1 Package Quality
1.7.2 Data Package 1
1.7.3 Data Package 2
1.7.4 Data Package 3
1.7.5 Data Package 4
1.7.6 Data Package 5
1.7.7 Changes to Submittals

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 TRAINING
3.1.1 Training Plan
3.1.2 Training Content
3.1.3 Training Outline
3.1.4 Training Video Recording
3.1.5 Unresolved Questions from Attendees
3.1.6 Validation of Training Completion
3.1.7 Quality Control Coordination
3.2 SUBMITTAL SCHEDULING
3.2.1 Operation and Maintenance Manual, Progress Submittal
3.2.2 Operation and Maintenance Manual, Prefinal Submittal
3.2.3 Operation and Maintenance Manual, Final Submittal

-- End of Section Table of Contents --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA
05/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE GUIDELINE 1.4	(2019) Preparing Systems Manuals for Facilities

ASTM INTERNATIONAL (ASTM)

ASTM E1971	(2024) Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings
ASTM E2166	(2016; R 2023) Standard Practice for Organizing and Managing Building Data

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

Facility Data Workbook; G

Training Plan; G

Training Outline; G

Training Content; G

Operation And Maintenance Manual, Progress Submittal; G

Operation And Maintenance Manual, Prefinal Submittal; G

Operation And Maintenance Manual, Final Submittal; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

Training Plan; G

Construction Drawings And Utility Systems; G

1.3 MEETINGS

To assure that Operation and Maintenance Manual and Facility Data Workbook requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

1.3.1 Pre-Construction Meeting

At a minimum, discuss the following:

a. The requirement for Operation and Maintenance Manuals and Facility Data deliverables under this contract including coordination meetings

b. Processes and method of gathering Facility Data information during construction

c. Primary roles and responsibilities associated with the development and delivery of the Operation and Maintenance Manuals and Facility Data deliverables, and

d. Identify and agree upon a date and attendance list for the meetings described below:

1.3.2 Operation and Maintenance Manual and Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Pre-Construction Meeting prior to the submission of the Operation and Maintenance Manual Progress Submittal. Meeting attendance must include the Contractor's Operation and Maintenance Manual and Facility Data Workbook Preparer, the Contractor's Commissioning Manager (CxM), Quality Control Manager, the Commissioning Authority (CxA), the Government's Design Manager (DM), Contracting Officer's Representative, and Government's facility data reviewer. Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for Operation and Maintenance Manual and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of Operation and Maintenance Manual.

1.3.3 Submittal Coordination Meeting

Facilitate a meeting following submission and Government review of each design or progress submittal of the Operation and Maintenance Manual and Facility Data Workbook.

a. Include personnel from the Coordination meeting and any additional personnel identified.

b. The purpose of this meeting is to demonstrate ongoing compliance with the requirements identified in this specification. Discuss Government review comments and unresolved items preventing completion and

Government approval of the Operation and Maintenance Manuals.

c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting.

1.3.4 Facility Turnover Meeting

Include Operation and Maintenance Manual in Red Zone facility turnover meetings as specified in Section 01 30 00, ADMINISTRATIVE REQUIREMENTS.

1.4 FACILITY DATA WORKBOOK

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program as outlined in 01 78 24.00 10 FACILITY DATA REQUIREMENTS. As a minimum, provide Facility Data Workbook as a list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL MEDIA

Assemble Operation and Maintenance Manual into an electronically bookmarked file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete bookmarked operation and maintenance directory.

1.5.1 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

a. Building Number

b. Project Title

c. Activity and Location

d. Construction Contract Number

e. Prepared For: (Contracting Agency)

f. Prepared By: (Name, title, phone number and email address)

g. Include the disk content on the disk label

h. Date

i. Virus scanning program used

1.5.2 O&M Manual Tabbed Hard Copy

Provide a hard copy of the O&M manual upon completion of the project. Provide tabs for each section and subsection for ease of navigation by the user.

1.6 O&M MANUAL CONTENT

Organize the bookmarked Operation and Maintenance Manual into the following Parts in accordance with ASHRAE GUIDELINE 1.4, and as modified and detailed below. Word template for O&M Manual is available at: https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-78-23.

1.6.1 Part 1: Executive Summary

Provide a summary of the information found in the O&M manual including the purpose of the manual and a description of the manual's organization.

1.6.2 Part 2: Facility Design and Construction

1.6.2.1 General Facility and Systems Description

Provide an overview of the intent for design and use of the facility. Provide a PDF of the Construction Drawings prepared in accordance with 01 78 00 CLOSEOUT SUBMITTALS and bookmarked using the sheet title and sheet number. Include uncluttered 11 by 17 inches floor plans with room numbers, type or function of space, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code list and generally describe all the facility systems and any special building features (for example, HVAC Controls, Sprinkler Systems, Cranes, Elevators, and Generators). Include photographs marked up and labeled to show key operating components and the overall facility appearance.

1.6.2.2 Contract Documents, RFP, Amendments, and Modifications

Provide the contract construction documents complete, to include specifications, drawings, Request for Proposal, amendments, and modifications.

1.6.2.3 Room Inventory of Real Property and Finishes

Provide a list of installed equipment furnished under this contract. Include all information usually listed on manufacturer's name plate. Include, as applicable, the following information for each piece of equipment installed: description of item, all dimensions, location by room number, model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Real property includes, but is not limited to, floor coverings, wall surfaces, ceiling surfaces, windows, roofing, HVAC filters, plumbing fixtures, and lighting fixtures. Submit the final list 30 days after transfer of the completed facility.

Include spatial data defining actual net square footage and data of each room. Also include the room finish schedule including room names and numbers. Include schedules in the construction drawings in the room inventory. Add a column to each schedule to record what was provided by the contractor during construction. Provide a PDF of room inventory. Key the designations to the related area depicted on the contract drawings. List the following data:

RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA				
Description	Specification Section	Manufacturer and Catalog, Model, and Serial Number	Composition and Size	Where Used

1.6.3 Part 3: Facilities, Systems, and Assemblies Information

1.6.3.1 Organization

Bookmark information in this section using the current version of ASTM E2166Uniformat II, UFGS numbers, and document type as outlined in the example below. Bookmark/tab each item to the third level for easy navigation of the manual.

Example as shown in Table below:

PARTS AND SUBPART NUMBERING
3.1 B20 EXTERIOR CLOSURE (System)
3.1.1 B2030 EXTERIOR DOORS (Subsystem)
3.1.1.1 B2030110 GLAZED DOORS (Component)
3.1.1.1.1 Applicable specifications List in UFGS Format
3.1.1.1.2 Manufacturer's Operations and Maintenance Data
3.1.1.1.3 Approved Submittal
3.1.1.1.4 Coordination/Shop Drawings
3.1.1.1.5 Sequence of Operation for Operating Equipment
3.1.1.1.6 Testing Equipment Information and Performance Data
3.1.1.1.7 Routine Maintenance Requirements
3.1.1.1.8 Repair Procedures
3.1.1.1.9 Emergency Procedures & Locations of Applicable Controls
3.1.1.1.10 Warranties
3.1.1.1.11 Construction Drawings and Utility Systems
3.1.1.1.12 Contractor / Supplies Listing and Contact Information

1.6.3.2 Related Specifications

Reference each specification related to the subsystem in this section, and locate the actual specification section in Part 2 of the O&M Manual. List

specifications in table format as shown in the below example.

UFGS Number	Specification Title	Page Spec Begins in Part 2

1.6.3.3 Manufacturer's Operations and Maintenance Data

Provide a copy of all manufacturer specifications and cutsheets. Provide text-searchable, high-quality document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Provide Uniformat II Level 3 identification for D20, D30, D40 installed equipment. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same files. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master part catalog.

1.6.3.4 Approved Submittals and Certificates

Provide a copy of all submittals documented with the required approval as applicable for each UFGS specification listed in the table outlined in applicable specifications. Include copies of SD-07 Certificates submittals documented with the required approval, SD-08 Manufacturer's Instructions submittals documented with the required approval, and SD-10 Operation and Maintenance Data submittals documents with the required approval.

1.6.3.5 Approved Coordination/Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work. Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project. Drawings prepared by or for the Contractor to show how multiple systems and

interdisciplinary work will be coordinated.

1.6.4 Sequence of Operation for Operating Equipment

Provide record one-line diagrams for each floor, delineating mechanical equipment location within the building. Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.4.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard. Specify if any certifications or licenses are required to operate the equipment.

1.6.4.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.4.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.6.4.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.4.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.4.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.4.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary

operating records.

1.6.4.9 Additional Requirements for Equipment Control Systems

Provide Data Package 5 and the following for all control systems:

a. Provide a narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.

b. Submit complete controls equipment schedules, full as-built sequence of operations, wiring and logic diagrams, Input/Output Tables, equipment schedules, copies of checkout tests and calibrations performed by the Contractor (not Cx tests), and all associates information.

c. Full points list. Provide a listing of rooms with the following information for each room:

(1) Floor

(2) Room number

(3) Room name

(4) Air handler unit ID

(5) Reference drawing number

(6) Air terminal unit tag ID

(7) Heating or cooling valve tag ID

(8) Minimum cfm

(9) Maximum cfm

d. Full print out of all schedules and set points after testing and acceptance of the system.

e. Full as-built print out of software program.

f. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.4.10 Testing Equipment Information and Performance Data

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.5 Routine Maintenance Requirements

1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including requirements by type of activity. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

a. Define the anticipated time required to perform each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.

b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

1.6.5.2 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

a. A table showing recommended lubricants for specific temperature ranges and applications.

b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide procedural instructions for Oil Sampling for all equipment.

c. A Lubrication Schedule showing service interval frequency.

1.6.6 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards. Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials. Specify if any certifications or licenses are required to repair the equipment.

1.6.6.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical

malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.6.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.6.3 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required specialty tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.6.4 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.6.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Identify if replacement of a subassembly, attachment or accessory requires the entire assembly to be replaced. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.6.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.6.7 Construction Drawings and Utility Systems

The construction drawings are the final compilation of actual conditions reflected in the as-built redline markups. Provide construction drawings as outlined
in 01 78 00 CLOSEOUT SUBMITTALS.

Using Record Source Drawings, show and document details of the actual installation of the utility systems, annotate and highlight the Operation and Maintenance information. Provide the following drawings at a large enough scale to differentiate designated isolation units from surrounding

valves and switches.

1.6.6.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.6.9 Contractor / Supplier Listing and Contact Information

Provide a list that includes the name, address, telephone number, email and website of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.6.7 Part 4: Facility Operations

1.6.7.1 Completed Facility Operating Plan

Provide a plan that documents that procedures for the operation of systems and assemblies in the facility. The systems that should be included in the Operating Plan include, but are not limited to:

a. Electrical systems and equipment

b. Mechanical systems and equipment

c. Fire Protection systems and equipment

d. Control Systems and equipment

e. Architectural and Structural systems, fixtures, structures, and equipment

f. Vertical transportation such as elevators and escalators

1.6.7.2 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.7.3 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports

Compile and provide approved Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals.

1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory

1.6.7.5.1 Maintenance Schedules

Include recommended maintenance schedules for systems and equipment.

1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping

Include ongoing commissioning and optimization procedures and documentation to monitor and improve the performance of facility systems.

1.6.7.5.3 Janitorial and Cleaning Plans and Procedures

Include a copy of facility cleaning and janitorial plan with procedures and intended chemicals and equipment.

Provide environmentally friendly cleaning recommendations in accordance with ASTM E1971.

1.6.7.6 Utility Construction Drawings

1.6.7.6.1 Utility Schematic Diagrams

Provide a one-line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the 5 foot connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.

1.6.7.6.2 Enlarged Connection and Cutoff Plans

Provide enlarged floor plans and provide information between the 5 foot utilities connection point and where utilities connect to facility distribution. Enlarge floor plans/elevations of the rooms where the utility enters the building and indicate on these plans the locations of the main interiors and exterior connection and cutoff points for the utilities. Also enlarge floor plans/elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems

Provide in narrative format a description of the utility metering and monitoring systems. Include locations, function, and related systems.

1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting

Procedures for usage reporting and tracking in support of establishing and monitoring utility budgets and costs, and in developing annual energy reports.

1.6.7.6.2.3 One-Line Diagrams and Meter Location of System

Provide one-line diagrams and design drawings that highlight meter locations on the site.

1.6.7.6.3 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.8 Part 5: Training

Provide a copy of training plans used for each type of equipment along with training materials used, arranged in specification sequence. Provide a copy of training records, sign-in sheets, and agendas. Include training and documentation on the updating and continued use of the O&M Manual.

1.6.9 Part 6: Cx Project Report and TAB Report

Provide the final Cx Plan and complete Cx reports with evaluation and testing forms and records for each building system. Include relevant commissioned system assemblies test reports including installers checklists of assemblies. Provide all Cx Progress Reports, issues and resolutions logs with resolution or status of each item, and a list of any open items and seasonal or additional testing required.

1.6.10 Part 7: Regulatory Requirements

Provide information describing regulatory and policies compliance requirements or provide a reference to where it is stored.

1.6.11 Part 8: Permits

Provide information requiring frequently asked questions and associated answers or provide a reference to where it is stored.

1.6.12 Part 9: Operations and Maintenance Manual Approval

Provide a signed document stating that the project O&M Manual has been reviewed and confirming agreement with the approach it presents. Include contact information for the signer for coordination of any future changes.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. O&M Data Packages are one of the components of the O&M Manual. The information required in each type of data package follows:

1.7.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.7.2 Data Package 1

a. Safety precautions and hazards

b. Cleaning recommendations

c. Maintenance and repair procedures

d. Warranty information

e. Extended warranty information

f. Contractor information

g. Spare parts and supply list

1.7.3 Data Package 2

a. Safety precautions and hazards

b. Normal operations

c. Environmental conditions

d. Lubrication data

e. Preventive maintenance plan, schedule, and procedures

f. Cleaning recommendations

g. Maintenance and repair procedures

h. Removal and replacement instructions

i. Spare parts and supply list

j. Parts identification

k. Warranty information

l. Extended warranty information

m. Contractor information

1.7.4 Data Package 3

a. Safety precautions and hazards

b. Operator prestart

c. Startup, shutdown, and post-shutdown procedures

d. Normal operations

e. Emergency operations

f. Environmental conditions

g. Operating log

h. Lubrication data

i. Preventive maintenance plan, schedule, and procedures

j. Cleaning recommendations

k. Troubleshooting guides and diagnostic techniques

l. Wiring diagrams and control diagrams

m. Maintenance and repair procedures

n. Removal and replacement instructions

o. Spare parts and supply list

p. Product submittal data

q. O&M submittal data

r. Parts identification

s. Warranty information

t. Extended warranty information

u. Testing equipment and special tool information

v. Testing and performance data

w. Contractor information

x. Field test reports

1.7.5 Data Package 4

a. Safety precautions and hazards

b. Operator prestart

c. Startup, shutdown, and post-shutdown procedures

d. Normal operations

e. Emergency operations

f. Operator service requirements

g. Environmental conditions

h. Operating log

i. Lubrication data

j. Preventive maintenance plan, schedule, and procedures

k. Cleaning recommendations

l. Troubleshooting guides and diagnostic techniques

m. Wiring diagrams and control diagrams

n. Repair procedures

o. Removal and replacement instructions

p. Spare parts and supply list

q. Repair work-hours

r. Product submittal data

s. O&M submittal data

t. Parts identification

u. Warranty information

v. Extended warranty information

w. Personnel training requirements

x. Testing equipment and special tool information

y. Testing and performance data

z. Contractor information

aa. Field test reports

1.7.6 Data Package 5

a. Safety precautions and hazards

b. Operator prestart

c. Start-up, shutdown, and post-shutdown procedures

d. Normal operations

e. Environmental conditions

f. Preventive maintenance plan, schedule, and procedures

g. Troubleshooting guides and diagnostic techniques

h. Wiring and control diagrams

i. Maintenance and repair procedures

j. Removal and replacement instructions

k. Spare parts and supply list

l. Product submittal data

m. Manufacturer's instructions

n. O&M submittal data

o. Parts identification

p. Testing equipment and special tool information

q. Warranty information

r. Extended warranty information

s. Testing and performance data

t. Contractor information

u. Field test reports

v. Additional requirements for HVAC control systems

1.7.7 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS. Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be reviewed by the Commissioning Manager (CxM) and approved by the prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and . Include within the plan the following elements:

a. Equipment included in training

b. Intended audience

c. Location of training

d. Dates of training

e. Objectives

f. Outline of the information to be presented and subjects covered including description

g. Start and finish times and duration of training on each subject

h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)

i. Instructor names and instructor qualifications for each subject

j. List of texts and other materials to be furnished by the Contractor that are required to support training

k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The CxA is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.

b. Relevant health and safety issues.

c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.

d. Design intent.

e. Use of O&M Manual Files.

f. Review of control drawings and schematics.

g. Interactions with other systems.

h. Special maintenance and replacement sources.

i. Tenant interaction issues.

3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF) and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the in accordance with Section 01 45 00 QUALITY CONTROL.

3.2 SUBMITTAL SCHEDULING

3.2.1 Operation and Maintenance Manual, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide Operation and Maintenance Manual Files (Bookmarked PDF). Include the elements and portions of system construction completed up to this point. The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual Coordination Meeting. The Contractor's Commissioning Manager (CxM) shall review the Progress submittal prior to submission.

3.2.2 Operation and Maintenance Manual, Prefinal Submittal

Submit the 100 percent submittal of the Operation and Maintenance Prefinal Submittal to the Contracting Officer for approval within 90 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of Operation and Maintenance Progress submittal must be corrected prior to the Prefinal submission. The Prefinal Submittal must include Operation and Maintenance

Manual Files (Bookmarked PDF).

3.2.3 Operation and Maintenance Manual, Final Submittal

Submit completed Operation and Maintenance Manual Files (Bookmarked PDF). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal submittal, including the Field Verification, must be corrected prior to the Final submission.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 81 13.02

SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED v4 BD+C

4/23

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 DEFINITIONS
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 REFERENCES

PART 2 PRODUCTS

2.1 PERFORMANCE CRITERIA

PART 3 EXECUTION

3.1 SPECIAL REQUIREMENTS
3.2 ATTACHMENTS

ATTACHMENTS:

LEED v4 BD+C Project Checklist

-- End of Section Table of Contents --

SECTION 01 81 13.02
4/23

SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED v4 BD+C

PART 1 GENERAL

1.1 DESCRIPTION

a. This section describes requirements and procedures which are additive to and compatible with 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS, and necessary to contribute to sustainability certification under the LEED v4 BD+C for LEED v4 Building Design and Construction. Where language between LEED credit requirements and general sustainable design requirements conflict, Contractor to notify the Architect but pursue the more stringent requirement.

b. Refer to the U.S. Green Building Council LEED DD+C v4 checklist for New Construction for mandatory prerequisites.

c. A copy of the LEED v4 BD+C Project Checklist is attached at the end of this section.

1.2 RELATED WORK

Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS.

1.3 DEFINITIONS

a. Chain-of-Custody Certificates: Certificates signed by manufacturers certifying that wood used to make products was obtained from forests certified by an FSC-accredited certification body to comply with FSC STD 01 001, "FSC Principles and Criteria for Forest Stewardship." Certificates must include evidence that manufacturer is certified for chain of custody by an FSC-accredited certification body.

b. Regional Materials: Materials that have been extracted, harvested, or recovered, as well as manufactured, within 100 miles of project site. If only a fraction of a product or material is extracted or harvested or recovered and manufactured locally, then only that percentage (by weight) must contribute to regional value. The purchase location, distribution and all points of manufacture are now included in the definition of regional materials.

c. Biobased Material: Products which meet the SAN. Biobased raw materials must be tested using ASTM D6866 and be legally harvested, as defined by the exporting and receiving country. Exclude hide products, such as leather and other animal skin material.

d. Salvaged or reused materials: Construction materials recovered from existing buildings or construction sites and reused. Common salvaged materials include structural beams and post, flooring, doors, cabinetry, brick, and decorative items.

e. Refurbished materials: Products that could have been disposed of as solid waste. These products have completed their life cycle as consumer items and are then refurbished for reuse without substantial

alteration of their form. Refurbishing includes renovating, repairing, restoring or generally improving appearance, performance, quality, functionality, or value of a product.

f. Environmental Product Declaration (EPD): A compliant EPD is a third-party declaration with external verification; is product-specific (not industry-wide); conforms to ISO 14025, ISO 14040, ISO 14044, and CEN - 15804 or ISO 21930; and has at least a cradle-to-gate scope.

g. Health Product Declaration (HPD): A compliant HPD provides full disclosure of known hazards and demonstrates chemical inventory of the product to at least 0.1 percent (1000 ppm).

h. Extended producer responsibility: An environmental policy approach in which a producer’s responsibility for a product is extended to the postconsumer stage of the product’s lifecycle. An extended producer responsibility policy shifts responsibility (physically and/or economically, fully or partially) upstream toward the producer and away from municipalities; and creates incentives for producers to take into account environmental considerations when designing their products.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.02 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Erosion And Sedimentation Control Plan;S

Additional Inclusions In Sustainable Construction Progress Reports; G

SD-03 Product Data

Cut Sheets For All Plumbing Fixtures; G

Receipts For Salvaged And Refurbished Materials; S

Additional Product Submittals To Be Provided With Manufacturer Documentation:;S

Batt Insulation Products;S

SD-11 Closeout Submittals

Additional Closeout Submittals; G

Commissioning Reports; G

LEED Construction Phase Credit Documentation; G

1.5 SUBMITTAL INSTRUCTIONS

a. Additional Sustainability Action Plan requirements.

b. Erosion and Sedimentation Control Plan which conforms to erosion and sedimentation requirements of the 2012 EPA Construction General Permit or local standards and codes, whichever is more stringent. The plan must describe measures implemented to accomplish the following objectives:

(1) To prevent loss of soil during construction by stormwater runoff and wind erosion, including protecting topsoil by stockpiling for reuse.

(2) To prevent sedimentation of storm sewers or receiving streams.

(3) To prevent pollution of air with dust and particulate matter.

c. Additional inclusions in Sustainable Construction Progress Reports:

(1) Inspection report or date-stamped photos confirming compliance with the Erosion and Sedimentation Control Plan

(2) Provide updated information submittal using LEED Online forms and USGBC-provided spreadsheets for the following LEED Credits sought for Project.

d. Cut sheets for all plumbing fixtures used in the project.

e. Receipts for salvaged and refurbished materials used for Project, indicating sources and costs for salvaged and refurbished materials.

f. Additional Product Submittals to be provided with manufacturer documentation:

(1) Environmental Product Declarations.

(2) Corporate Sustainability Report.

(3) Evidence of manufacturer participation in an extended producer responsibility program.

(4) Health Product Declarations. Declare labels, or other third-party validated certifications of product health and ingredient inventory as acceptable to USGBC.

Regional Materials: Indicate location and distance from project site of material manufacturer and point of extraction, harvest, or recovery for each raw material. Include statement indicating cost for each regional material and fraction by weight that is considered regional. Show the purchase location, distribution and all points f manufacture.

(5) Biobased Materials: Indicate percent of biobased content. Include statement indicating cost of each relevant product and fraction by weight that is considered biobased material.

(6) Provide chain-of-custody certificates for products containing FSC certified wood. Include statement indicating cost for each

product with certified wood and fraction by weight that is certified wood.

(7) Construction and Waste management Calculator tracking recycling for the project. Provide documentation for all activities.

(8) USGBC approvedcalculator, including product information showing product information, MSDS, third party certifications, testing reports for low emitting materials.

(9) IAQ Management plan, including checklists, protection measures description, photographs of IAQ measures and records for filtration media.

(10) For paints and coatings applied on site within the weatherproofing membrane provide documentation of compliance with low-emitting materials requirements:

(a) The manufacturer or third-party certification must state the exposure scenario used to determine compliance. Claims of compliance for wet-applied products must state the amount applied in mass per surface area.

(b) Manufacturers' claims of compliance with the California Department of Public Health CDPH SECTION 01350 Standard Method v1.1-2010 must also state range of total VOCs after 14 days (336 hours), measured as specified in the CDPH SECTION 01350: 0.5 mg/cubic meter or less; between 0.5 and 5.0 mg/cubic meter; or 5.0 mg/cubic meter or more.

(c) VOC content for each product.

(12) For adhesives and sealants applied on site within the weatherproofing membrane provide documentation of compliance with low-emitting material requirements:

(a) The manufacturer or third-party certification must state the exposure scenario used to determine compliance. Claims of compliance for wet-applied products must state the amount applied in mass per surface area.

(b) Manufacturers' claims of compliance with the California Department of Public Health (CDPH) CDPH SECTION 01350 must also state the range of total VOCs after 14 days (336 hours), measured as specified in the CDPH SECTION 01350: 0.5 mg/cubic meter or less; between 0.5 and 5.0 mg/cubic meter; or 5.0 mg/cubic meter or more.

(c) VOC content for each product.

(13) For flooring installed on site within the weatherproofing membrane provide documentation of compliance with low-emitting material requirements:

(a) Manufacturers' claims or third-party certification of compliance with the California Department of Public Health CDPH SECTION 01350 Standard Method v1.1-2010 which states the exposure scenario used to determine compliance.

(14) Composite wood products: manufacturer documentation which confirms low formaldehyde emissions that meet the California Air Resources Board Composite Wood Products Airborne Toxic Control Measures (ATCM) for formaldehyde requirements for ultra-low-emitting formaldehyde (ULEF) resins or no added formaldehyde resins (NAUF).

(15) Ceiling, wall, thermal and acoustic insulation products:

(a) Manufacturers' claims or third-party certification of compliance with the California Department of Public Health CDPH SECTION 01350 which states the exposure scenario used to determine compliance.

g. Batt insulation products: manufacturer or third-party documentation which confirms the product contains no added formaldehyde.

(1) New furniture and furnishing products:

(a) Manufacturer or third-party documentation of testing in accordance with ANSI/BIFMA M7.1, and compliance with ANSI/BIFMA e3 Furniture Sustainability Standard, Sections 7.6.1 and 7.6.2 using either concentration modeling approach or emissions factor approach, and using the appropriate modeling scenario.

(b) Confirmation of modeling scenario used.

(2) Exterior applied adhesives, sealants, coatings, roofing, and waterproofing materials applied on site:

(a) Manufacturer documentation demonstrating compliance with VOC limits of California Air Resources Board CARB 93120 Suggested Control Measure (SCM) for Architectural Coatings and South Coast Air Quality Management District (SCAQMD) SCAQMD Rule 1168, effective July 1, 2005.

(b) A statement confirming that roofing installations did not use hot-mopped asphalt installation techniques.

(c) A statement confirming that parking lots and other paved surfaces did not use coal tar sealants.

(d) Additional VOC content requirements for wet-applied products. In addition to meeting the general requirements for VOC emissions (above), on-site wet-applied products must not contain excessive levels of VOCs, for the health of the installers and other tradesworkers who are exposed to these products. To demonstrate compliance, a product or layer must meet the following requirements, as applicable. Disclosure of VOC content must be made by the manufacturer. Any testing must follow the test method specified in the applicable regulation.

(e) All paints and coatings wet-applied on site must meet the applicable VOC limits of the California Air Resources Board (CARB) 2007, Suggested Control Measure (SCM) for Architectural Coatings, or the South Coast Air Quality Management District (SCAQMD) Rule 1113, effective June 3, 2011.
All adhesives and sealants wet-applied on site must meet the applicable chemical content requirements of SCAQMD Rule 1168, July

1, 2005, Adhesive and Sealant Applications, as analyzed by the methods specified in Rule 1168. The provisions of SCAQMD Rule 1168 do not apply to adhesives and sealants subject to state or federal consumer product VOC regulations.
For projects outside the U.S., all paints, coatings, adhesives, and sealants wet-applied on site must either meet the technical requirements of the above regulations, or comply with applicable national VOC control regulations, such as the European Decopaint Directive (2004/42/EC), the Canadian VOC Concentration Limits for Architectural Coatings, or the Hong Kong Air Pollution Control (VOC) Regulation.
If the applicable regulation requires subtraction of exempt compounds, any content of intentionally added exempt compounds larger than 1% weight by mass (total exempt compounds) must be disclosed.
If a product cannot reasonably be tested as specified above, testing of VOC content must comply with ASTM D2369-10; ISO 11890, part 1; ASTM D6886-03; or ISO 11890-2.
For projects in North America, methylene chloride and perchloroethylene may not be intentionally added in paints, coatings, adhesives, or sealants.

h. Additional Closeout Submittals

(1) LEED Online: Final completed LEED Online forms and associated required documentation uploaded to LEED Online for the following prerequisites and credits:

(a) Construction activity pollution prevention.

(b) Construction and demolition waste management planning.

(c) Building product disclosure and optimization – environmental product declaration.

(d) Building product disclosure and optimization – sourcing of raw materials.

(e) Building product disclosure and optimization – material ingredients.

(f) Construction and demolition waste management.

(g) Low-emitting materials.

(h) Construction indoor air quality management plan.

(i) Indoor air quality assessment.

(j) Innovation Credit: Low-mercury lamping.

(2) Report from testing and inspecting agency indicating results of indoor-air-quality testing and documentation showing compliance with indoor-air-quality testing procedures and requirements.

1.6 REFERENCES

ASTM INTERNATIONAL (ASTM)

ASTM D6866	(2022) Standard Test Methods Determining the Biobased Content of Solid, Liquid, and Gaseous Samples Using Radiocarbon Analysis

BIFMA INTERNATIONAL (BIFMA)

ANSI/BIFMA M7.1	(2011; R 2016) Test Method for Determining VOC Emissions from Office Furniture Systems, Components and Seating
ANSI/BIFMA e3	(2019) Furniture Sustainability Standard

CALIFORNIA AIR RESOURCES BOARD (CARB)

CARB 93120	(2007) Airborne Toxic Control Measure (ATCM) to Reduce Formaldehyde Emissions from Composite Wood Products

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350	(2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

NSF INTERNATIONAL (NSF)

NSF/ANSI 336	(2018) Sustainability Assessment For Commercial Furnishings Fabric

U.S. GREEN BUILDING COUNCIL (USGBC)

LEED v4 BD+C	LEED v4 Building Design and Construction

FOREST STEWARDSHIP COUNCIL (FSC)

FSC STD 01 001	(2015) Principles and Criteria for Forest Stewardship

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 14025	(2006) Environmental labels and declarations — Type III environmental declarations — Principles and procedures
ISO 14040	(2006) Environmental management — Life cycle assessment — Principles and framework
ISO 14044	(2006) Environmental management — Life cycle assessment — Requirements and guidelines
ISO 21930	(2017) Sustainability in buildings and civil engineering works — Core rules for environmental product declarations of

	construction products and services
ISO 26000	(2010) Guidance on Social Responsibility

CRADLE TO CRADLE PRODUCTS INNOVATION INSTITUTE (CTC)

Cradle to Cradle	Cradle to Cradle Certified Product Standard

GREENSCREEN FOR SAFER CHEMICALS

GreenScreen Benchmark	(2011) GreenScreen Benchmark - assess and classify hazards
GreenScreen List Translator	GreenScreen List Translator - identify chemicals of high concern.

REACH OPTIMIZATION

REACH	REACH Optimization material declaration

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)

CEN - 15804	(2012) Sustainability of construction works - Environmental product declarations- Core rules for the product category ofconstruction products

SUSTAINABLE AGRICULTURE NETWORK (SAN)

SAN	Sustainable Agriculture Standard

PART 2 PRODUCTS

2.1 PERFORMANCE CRITERIA

a. At least 20 permanently installed products from at least 5 different manufacturers must have one of the following:

(1) Publicly available, critically reviewed life-cycle assessment conforming to ISO 14044 that has at least a cradle to gate scope. For credit calculation purposes, this is valued at one quarter of a product.

(2) Environmental Product Declarations (EPD):

(a) Industry-wide (generic) EPD – Products with third-party certification (Type III), including external verification in which the manufacturer is explicitly recognized as a participant by the program operator are valued as one half of a product for the purposes of credit achievement calculation.

(b) Product-specific Type III EPD – Products with third-party certification (Type III), including external verification, in which the manufacturer is explicitly recognized as the participant by the program operator are valued as one whole product for the purposes of credit achievement calculation. OR

b. Products that comply with one of the criteria below account for at least 50 percent by cost of total value of permanently installed

products:

(1) Third party certified products that demonstrate impact reduction below industry average in at least three of the following categories:

(a) Global warming potential (greenhouse gases), in CO2e.

(b) Depletion of stratospheric ozone layer, in kg CFC-11.

(c) Acidification of land and water sources, in moles H+ or kg SO2.

(d) Eutrophication, in kg nitrogen or kg phosphate.

(e) Formation of tropospheric ozone, in kg NOx, kg O3 eq, or kg ethane.

(f) Depletion of nonrenewable energy resources, in MJ.

(2) For Credit achievement calculation, these products are valued at 100 percent of their cost for credit achievement calculations. Products sourced (extracted, manufactured, purchased) within 100 miles of project site are valued at 200 percent of their base contributing cost.

(3) Structure and enclosure materials may not constitute more than 30 percent of value of compliant building products.

c. At least 20 permanently installed products from at least 5 different manufacturers must have publicly released a report from their raw material suppliers which include raw material supplier extraction locations, a commitment to long-term ecologically responsible land use, a commitment to reducing environmental harms from extraction and manufacturing processes, and a commitment to meeting applicable standards or programs voluntarily that address responsible sourcing criteria.

(1) Products sourced from manufacturers with self-declared reports are valued as one half (1/2) of a product for credit achievement.

(2) Third-party verified corporate sustainability reports (CSR) which include environmental impacts of extraction operations and activities associated with the manufacturer’s product and the product’s supply chain, are valued as one whole product for credit achievement calculation. Acceptable CSR frameworks include the following:

(a) Global Reporting Initiative (GRI) Sustainability Report.

(b) Organization for Economic Co-operation and Development (OECD) Guidelines for Multinational Enterprises.

(c) U.N. Global Compact: Communication of Progress.

(d) ISO 26000: 2010 Guidance on Social Responsibility.

(e) USGBC approved program: Other USGBC approved programs meeting CSR criteria. OR

d. Products that meet at least one of the responsible extraction criteria below account for at least 25 percent by cost of total value of permanently installed products:

(1) Responsible extraction criteria:

(a) Extended producer responsibility. Products purchased from a manufacturer (producer) that participates in an extended producer responsibility program or is directly responsible for extended producer responsibility. Products meeting extended producer responsibility criteria are valued at 50 percent of their cost for the purposes of credit achievement calculation.

(b) Biobased materials. Biobased products must meet the Sustainable Agriculture Network's Sustainable Agriculture Standard. Biobased raw materials must be tested using ASTM D6866 and be legally harvested, as defined by the exporting and receiving country. Exclude hide products, such as leather and other animal skin material. Products meeting biobased materials criteria are valued at 100 percent of their cost for the purposes of credit achievement calculation.

(c) Wood products. Wood products must be certified by the Forest Stewardship Council or USGBC-approved equivalent. Products meeting wood products criteria are valued at 100 percent of their cost for the purposes of credit achievement calculation.

(d) Materials reuse. Reuse includes salvaged, refurbished, or reused products. Products meeting materials reuse criteria are valued at 100 percent of their cost for the purposes of credit achievement calculation.

(e) Recycled content. Recycled content is the sum of postconsumer recycled content plus one-half the pre-consumer recycled content, based on cost. Products meeting recycled content criteria are valued at 100 percent of their cost for the purposes of credit achievement calculation.

(f) USGBC approved program. Other USGBC approved programs meeting leadership extraction criteria.

(2) For credit achievement calculation, products sourced (extracted, manufactured, and purchased) within 100 miles of project site are valued at 200 percent of their base contributing cost. For credit achievement calculation, the base contributing cost of individual products compliant with multiple responsible extraction criteria is not permitted to exceed 100 percent its total actual cost (before regional multipliers) and double counting of single product components compliant with multiple responsible extraction criteria is not permitted and in no case is a product permitted to contribute more than 200 percent of its total actual cost.

(3) Structure and enclosure materials may not constitute more than 30 percent of value of compliant building products.

e. (Option 1) At least 20 permanently installed products from at least 5 different manufacturers must use any of the following programs to demonstrate chemical inventory of the product to at least 0.1 percent (1000 ppm).

(1) Manufacturer Inventory. The manufacturer has published complete content inventory for the product following these guidelines:

(a) A publicly available inventory of ingredients identified by name and Chemical Abstract Service Registration Number (CASRN).

(b) Materials defined as trade secret or intellectual property may withhold the name and CASRN but must disclose role, amount and GreenScreen benchmark, as defined in GreenScreen v1.2.

(c) Health Product Declaration. The end use product has a published, complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration open Standard.

(d) CRADLE TO CRADLE CERTIFIED. The end use product has been certified at the Cradle to Cradle v2 Basic level or Cradle to Cradle v3 Bronze level.

(e) Declare. The Declare product label must indicate that all ingredients have been evaluated and disclosed down to 1000 ppm.

(f) ANSI/BIFMA e3A Furniture Sustainability Standard. The documentation from the assessor or scorecard from BIFMA must demonstrate the product earned at least 3 points under 7.5.1.3 Advanced Level in e3 or 3 points under 7.4.1.3 Advanced Level in e3.

(g) Cradle to Cradle Material Health Certificate. The product has been certified at the Bronze level or higher and at least 90% of materials are assessed by weight.

(h) Product Lens Certification

(i) Facts - NSF/ANSI 336: Sustainability Assessment for Commercial Furnishings Fabric at any certification level

(j) USGBC-approved program. Other USGBC-approved programs meeting the material ingredient reporting criteria. OR

f. (Option 2) Products that document their material ingredient optimization using paths below for at least 25 percent by cost of total value of permanently installed products:

(1) GreenScreen Benchmark. Products that have fully inventoried chemical ingredients to 100 ppm that have no Benchmark 1 hazards:

(a) If any ingredients are assessed with the GreenScreen List Translator, value these products at 100 percent of cost.

(b) If ingredients have undergone a full GreenScreen Assessment, value these products at 150 percent of cost.

(2) Cradle to Cradle Certified. End use products are certified Cradle to Cradle. Products will be valued as follows:

(a) Cradle to Cradle v2 Gold: 100 percent of cost.

(b) Cradle to Cradle v2 Platinum: 150 percent of cost.

(c) Cradle to Cradle v3 Silver: 100 percent of cost.

(d) Cradle to Cradle v3 Gold or Platinum: 150 percent of cost.

(3) International Alternative Compliance Path – REACH Optimization. End use products and materials that do not contain substances that meet REACH criteria for substances of very high concern. If the product contains no ingredients listed on the REACH Authorization or Candidate list, value at 100 percent of cost.

(4) USGBC-approved program. Products that comply with USGBC approved building product optimization criteria. OR

g. (Option 3) Use building products for at least 25 percent, by cost, of total value of permanently installed products in Project that meet the following criteria:

(1) Products are sourced from product manufacturers who engage in validated and robust safety, health, hazard, and risk programs which at a minimum document at least 99 percent (by weight) of the ingredients used to make the building product or building material.

(2) Products are sourced from product manufacturers with independent third party verification of their supply chain that at a minimum verifies:

(a) Processes are in place to communicate and transparently prioritize chemical ingredients along the supply chain according to available hazard, exposure and use information to identify those that require more detailed evaluation.

(b) Processes are in place to identify, document, and communicate information on health, safety and environmental characteristics of chemical ingredients.

(c) Processes are in place to implement measures to manage health, safety and environmental hazard and risk of chemical ingredients.

(d) Processes are in place to optimize health, safety and environmental impacts when designing and improving chemical ingredients.

(e) Processes are in place to communicate, receive and evaluate chemical ingredient safety and stewardship information along the supply chain.

(f) Safety and stewardship information about the chemical ingredients is publicly available from all points along the supply chain.

(3) Products meeting Option 3 criteria are valued at 100 percent of their cost for the purposes of credit achievement calculation.

(4) For credit achievement calculation of options 2 and 3, products sourced (extracted, manufactured, purchased) within 100 miles of project site are valued at 200 percent of their base contributing

cost. For credit achievement calculation, the value of individual products compliant with either option 2 or 3 can be combined to reach the 25 percent threshold but products compliant with both option 2 and 3 may only be counted once.

(5) Structure and enclosure materials may not constitute more than 30 percent of value of compliant building products.

h. Low Pollutant-Emitting Materials additional requirements:

(1) Paints and coatings applied on site within the weatherproofing membrane must comply with the following criteria:

(a) 90 percent by volume of interior paints and coatings applied to walls, floors, and ceilings must be tested and determined compliant in accordance with California Department of Public Health (CDPH) CDPH SECTION 01350, using the applicable exposure scenario.

(b) 100 percent of products must meet the applicable VOC limits of the California Air Resources Board CARB 93120, Suggested Control Measure (SCM) for Architectural Coatings, or the South Coast Air Quality Management District (SCAQMD) SCAQMD Rule 1113:

1. Anti-Corrosive/Antirust Paints: 100 g/L.

2. Clear Wood Finish, Lacquer: 275 g/L.

3. Clear Wood Finish, Sanding Sealer: 275 g/L.

4. Clear Wood Finish, Varnish: 275 g/L.

5. Floor Coating: 50 g/L.

6. Interior Flat Paint, Coating or Primer: 50 g/L.

7. Interior Non-Flat Paint, Coating or Primer: 50 g/L.

8. Shellac, Clear: 730 g/L.

9. Shellac, Pigmented: 550 g/L.

10. Stain: 250 g/L.

11. Concrete Curing Compounds: 100 g/L.

12. Japans/Faux Finishing Coatings: 350 g/L.

13. Magnesite Cement Coatings: 450 g/L.

14. Waterproofing Sealers: 100 g/L.

15. Wood Preservatives: 350 g/L.

16. Low-Solids Coatings: 120 g/L.

(2) Each flooring element installed in the building interior which is not inherently non-emitting (stone, ceramic, powder-coated metals, plated or anodized metal, glass, concrete, clay brick, and

unfinished or untreated solid wood flooring) must be tested and determined compliant in accordance with CDPH SECTION 01350, using the applicable exposure scenario.

(3) Composite wood products within the weatherproofing membrane must be documented to have low formaldehyde emissions that meet the California Air Resources Board ATCM for formaldehyde requirements for ultra-low-emitting formaldehyde (ULEF) resins or no added formaldehyde resins (NAUF).

(4) Ceiling, wall, thermal and acoustic insulation products which are not inherently non-emitting (stone, ceramic, powder-coated metals, plated or anodized metal, glass, concrete, clay brick, and unfinished or untreated solid wood flooring) must be tested and determined compliant in accordance with CDPH SECTION 01350, using the applicable exposure scenario.

(5) Furniture: 90 percent, by cost, of new furniture and furnishing items must be tested in accordance with ANSI/BIFMA M7.1, and comply with ANSI/BIFMA e3 Furniture Sustainability Standard, Sections 7.6.1 (for half credit, by cost) or 7.6.2 (full credit, by cost) using either the concentration modeling approach or the emissions factor approach, using the appropriate modeling scenario.

PART 3 EXECUTION

3.1 SPECIAL REQUIREMENTS

a. Construction Indoor Air Quality Management additional requirement:

(1) Perform air quality testing.

(a) Conduct baseline indoor-air-quality testing, after construction ends and prior to occupancy, using testing protocols consistent with the EPA's "Compendium of Methods for the Determination of Air Pollutants in Indoor Air," and as additionally detailed in the LEED Reference Guide LEED v4 BD+C.

(b) Demonstrate that contaminant maximum concentrations listed below are not exceeded:

1. Formaldehyde: 27 ppb.

2. Particulates (PM10): 50 micrograms/cu. m.

3. Ozone (for buildings in EPA nonattainment areas): 0.075 ppm.

4. Total Volatile Organic Compounds (TVOC): 500 micrograms/cu. m.

5. Target chemicals listed in CDPH SECTION 01350, Table 4-1, except formaldehyde: CDPH SECTION 01350, Allowable Concentrations, Table 4-1.

6. Carbon Monoxide: 9 ppm and no greater than 2 ppm above outdoor levels.

(c) For each sampling point where maximum concentration limits are

exceeded, conduct additional flush-out with outside air and retest the specific parameter(s) exceeded to indicate the requirements are achieved. Repeat procedure until all requirements have been met. When retesting non-complying building areas, take samples from same locations as in the first test.

(d) Air-sample testing must be conducted as follows:

1. Measurements must be conducted prior to occupancy but during normal occupied hours, and with building ventilation system starting at the normal daily start time and operated at minimum outside air flow rate for occupied mode throughout duration of air testing.

2. Building must have all interior finishes installed including millwork, doors, paint, carpet, and acoustic tiles. Non-fixed furnishings such as workstations and partitions are encouraged, but not required, to be in place for testing.

3. Number of sampling locations varies depending on size of building and number of ventilation systems. For each portion of building served by a separate ventilation system, number of sampling points must not be less than one per 25,000 sq. ft. or for each contiguous floor area, whichever is larger, and must include areas with the least ventilation and greatest presumed source strength.

4. Air samples must be collected between 3 and 6 feet from the floor to represent the breathing zone of occupants, and over a minimum four-hour period.

3.2 ATTACHMENTS

LEED v4 BD+C Project Checklist

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 81 13

SUSTAINABLE CONSTRUCTION REQUIREMENTS

10/17

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 REFERENCE STANDARDS
1.4 DEFINITIONS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 QUALITY ASSURANCE
1.8 REFERENCES

PART 2 – PRODUCTS

2.1 PERFORMANCE CRITERIA

PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 01 81 13
10/17

SUSTAINABLE CONSTRUCTION REQUIREMENTS

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section describes general requirements and procedures to comply with federal mandates and U.S. Department of Veterans Affairs (VA) policies for sustainable construction.

b. The Design Professional has selected materials and utilized integrated design processes that achieve the Government’s objectives. Contractor is responsible to maintain and support these objectives in developing means and methods for performing work and in proposing product substitutions or changes to specified processes. Obtain approval from Contracting Officer for all changes and substitutions to materials or processes. Proposed changes must meet, or exceed, materials or processes specified.

1.2 RELATED WORK

a. Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

b. Section 01 74 19 CONSTRUCTION WASTE MANANGEMENT.

c. Section 01 81 13.02 SUSTAINABILITY CERTIFICATION REQUIREMENTS - LEED NC v4.f.

d. Section 01 91 00.01 DVA/ USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

1.3 REFERENCE STANDARDS

a. Carpet and Rug Institute Green Label Plus program.

b. U.S. Department of Agriculture BioPreferred program (USDA BioPreferred).

c. U.S. Environmental Protection Agency Comprehensive Procurement Guidelines (CPG).

d. U.S. Environmental Protection Agency WaterSense Program (EPA WaterSense).

e. U.S. Environmental Protection Agency ENERGY STAR Program (Energy Star).

f. U.S. Department of Energy Federal Energy Management Program (FEMP).

g. Green Electronic Council EPEAT Program (EPEAT).

h. VA Sustainable Design Manual.

1.4 DEFINITIONS

a. Recycled Content: Recycled content of materials is defined according to Federal Trade Commission Guides for the Use of Environmental Marketing Claims (16 CFR Part 260). Recycled content value of a material assembly is determined by weight. Recycled fraction of assembly is multiplied by cost of assembly to determine recycled content value.

(1) “Post-Consumer” material is defined as waste material generated by households or by commercial, industrial, and institutional facilities in their role as end users of the product, which can no longer be used for its intended purpose.

(2) “Pre-Consumer” material is defined as material diverted from waste stream during the manufacturing process. Excluded is reutilization of materials such as rework, regrind, or scrap generated in a process and capable of being reclaimed within the same process that generated it.

b. Biobased Products: Biobased products are derived from plants and other renewable agricultural, marine, and forestry materials and provide an alternative to conventional petroleum derived products. Biobased products include diverse categories such as lubricants, cleaning products, inks, fertilizers, and bioplastics.

c. Low Pollutant-Emitting Materials: Materials and products which are minimally odorous, irritating, or harmful to comfort and well-being of installers and occupants.

d. Volatile Organic Compounds (VOC): Chemicals that are emitted as gases from certain solids or liquids. VOCs include a variety of chemicals, some of which may have short- and long-term adverse health effects.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.02 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Sustainability Action Plan; S

Low Pollutant-Emitting Materials Tracking Spreadsheet; S

Construction Indoor Air Quality (Iaq) Management Plan; S

Sustainable Construction Progress Reports; S

Waste Diversion Management Plan; S

SD-03 Product Data

Product Submittals; S

SD-11 Closeout Submittals

Closeout Submittals; S

1.6 SUBMITTAL INSTRUCTIONS

a. All submittals to be provided by contractor to Contracting Officer.

b. Sustainability Action Plan:

(1) Submit documentation as required by this section; provide additional copies of typical submittals required under technical sections when sustainable construction requires copies of record submittals.

(2) Within 30 days after Preconstruction Meeting provide a narrative plan for complying with requirements stipulated within this section.

(3) Sustainability Action Plan must:

(a) Make reference to sustainable construction submittals defined by this section.

(b) Address all items listed under PERFORMANCE CRITERIA.

(c) Indicate individual(s) responsible for implementing the plan.

c. Low Pollutant-Emitting Materials Tracking Spreadsheet: Within 30 days after Preconstruction Meeting provide a preliminary Low Pollutant-Emitting Materials Tracking Spreadsheet. The Low Pollutant-Emitting Materials Tracking Spreadsheet must be an electronic file and include all materials on Project in categories described under Low Pollutant-Emitting Materials in Section 01 81 13.

d. Construction Indoor Air Quality (IAQ) Management Plan:

(1) Not more than 30 days after Preconstruction Meeting provide a Construction IAQ Management Plan as an electronic file including descriptions of the following:

(a) Instruction procedures for meeting or exceeding minimum requirements of ANSI/SMACNA 008, Chapter 3, including procedures for HVAC Protection, Source Control, Pathway Interruption, Housekeeping, and Scheduling.

(b) Instruction procedures for protecting absorptive materials stored on-site or installed from moisture damage.

(c) Schedule of submission of photographs of on-site construction IAQ management measures such as protection of ducts and on-site stored oil installed absorptive materials.

(d) Instruction procedures if air handlers must be used during construction, including a description of filtration media to be used at each return air grille.

(e) Instruction procedure for replacing all air-filtration media

immediately prior to occupancy after completion of construction, including a description of filtration media to be used at each air handling or air supply unit.

(f) Instruction procedures and schedule for implementing building flush-out.

e. Product Submittals:

(1) Recycled Content: Submit product data from manufacturer indicating percentages by weight of post-consumer and pre-consumer recycled content for products having recycled content (excluding MEP systems equipment and components).

(2) Biobased Content: Submit product data for products to be installed or used which are included in any of the USDA BioPreferred program’s product categories. Data to include percentage of biobased content and source of biobased material.

(3) Low Pollutant-Emitting Materials: Submit product data confirming compliance with relevant requirements for all materials on Project in categories described under Low Pollutant-Emitting Materials in 01 81 13.

(4) For applicable products and equipment, submit product documentation confirming Energy Star label, FEMP certification, EPA WaterSense, and/or EPEAT certification.

f. Sustainable Construction Progress Reports: Concurrent with each Application for Payment, submit a Sustainable Construction Progress Report every month to confirm adherence with Sustainability Action Plan.

(1) Include narratives of revised strategies for bringing work progress into compliance with plan and product submittal data.

(2) Include updated and current Low Pollutant-Emitting Materials Tracking Spreadsheet.

(3) Include construction waste tracking, in tons or cubic yards, including waste description, whether diverted or landfilled, hauler, and percent diverted for comingled quantities; and excluding land-clearing debris and soil. Provide haul receipts and documentation of diverted percentages for comingled wastes.

g. Closeout Submittals: Within 14 days after Substantial Completion provide the following:

(1) Final version of Low Pollutant-Emitting Materials Tracking Spreadsheet.

(2) Manufacturer’s cut sheets and product data highlighting the Minimum Efficiency Reporting Value (MERV) for filtration media installed at return air grilles during construction if permanently installed air handling units are used during construction.

(3) Manufacturer’s cut sheets and product data highlighting the Minimum Efficiency Reporting Value (MERV) for final filtration media in air handling units.

(4) Minimum 18 construction photographs including six photographs taken on three different occasions during construction of ANSI/SMACNA 008, Chapter 3 approaches employed, along with a brief description of each approach, documenting implementation of IAQ management measures, such as protection of ducts and on-site stored or installed absorptive materials.

(5) Flush-out Documentation:

(a) Product data for filtration media used during flush-out.

(b) Product data for filtration media installed immediately prior to occupancy.

(c) Signed statement describing building air flush-out procedures including dates when flush-out was begun and completed and statement that filtration media was replaced after flush-out.

1.7 QUALITY ASSURANCE

a. Preconstruction Meeting: After award of Contract and prior to commencement of Work, schedule and conduct meeting with Contracting Officer and Architect to discuss the Project Sustainable Action Plan content as it applies to submittals, project delivery, required Construction Indoor Air Quality (IAQ) Management Plan, and other Sustainable Construction Requirements. The purpose of this meeting is to develop a mutual understanding of the Sustainable Construction Requirements and coordination of contractor’s management of these requirements with the Contracting Officer and the Construction Quality Manager.

b. Construction Job Conferences: Status of compliance with Sustainable Construction Requirements of these specifications will be an agenda item at regular job meetings conducted during the course of work at the site.

1.8 REFERENCES

a. Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only. Comply with applicable provisions and recommendations of the following, except as otherwise shown or specified.

GREEN SEAL (GS)

GS-11	(2021) Paints, Coatings, Stains, and Sealers

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008	(2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350	(2017; Version 1.2) Standard Method for

	the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1113	(2016) Architectural Coatings
SCAQMD Rule 1168	(2022) Adhesive and Sealant Applications

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star	(1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP)

U.S. DEPARTMENT OF ENERGY FEDERAL ENERGY MANAGEMENT PROGRAM (FEMP)

PL-109-58	(1992; R 2005) Energy Efiicient Procument Requirements

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA WaterSense	(2006) WaterSense Water Efficiency Labeling System

PART 2 – PRODUCTS

2.1 PERFORMANCE CRITERIA

a. Construction waste diversion from landfill disposal must comprise at least 50 percent of total construction waste, excluding land clearing debris and soil. Alternative daily cover (ADC) does not qualify as material diverted from disposal.

b. Low Pollutant-Emitting Materials:

(1) Adhesives, sealants and sealant primers applied on site within the weatherproofing membrane must comply with VOC limits of SCAQMD Rule 1168:

(a) Flooring Adhesives and Sealants:

1. Indoor carpet adhesives: 50 g/L.

2. Wood Flooring Adhesive: 100 g/L.

3. Rubber Floor Adhesives: 60 g/L.

4. Subfloor Adhesives: 50 g/L.

5. Ceramic Tile Adhesives and Grout: 65 g/L.

6. Cove Base Adhesives: 50 g/L.

7. Multipurpose Construction Adhesives: 70 g/L.

8. Porous Material (Except Wood) Substrate: 50 g/L.

9. Wood Substrate: 30 g/L.

10. Architectural Non-Porous Sealant Primer: 250 g/L.

11. Architectural Porous Sealant Primer: 775 g/L.

12. Other Sealant Primer: 750 g/L.

13. Structural Wood Member Adhesive: 140 g/L.

14. Sheet-Applied Rubber Lining Operations: 850 g/L.

15. Top and Trim Adhesive: 250 g/L.

16. Architectural Sealant: 250 g/L.

17. Other Sealant: 420 g/L.

(b) Non-Flooring Adhesives and Sealants:

1. Drywall and Panel Adhesives: 50 g/L.

2. Multipurpose Construction Adhesives: 70 g/L.

3. Structural Glazing Adhesives: 100 g/L.

4. Metal-to-Metal Substrate Adhesives: 30 g/L.

5. Plastic Foam Substrate Adhesive: 50 g/L.

6. Porous Material (Except Wood) Substrate Adhesive: 50 g/L.

7. Wood Substrate Adhesive: 30 g/L.

8. Fiberglass Substrate Adhesive: 80 g/L.

9. Architectural Non-Porous Sealant Primer: 250 g/L.

10. Architectural Porous Sealant Primer: 775 g/L.

11. Other Sealant Primer: 750 g/L.

12. PVC Welding Adhesives: 510 g/L.

13. CPVC Welding Adhesives: 490 g/L.

14. ABS Welding Adhesives: 325 g/L.

15. Plastic Cement Welding Adhesives: 250 g/L.

16. Adhesive Primer for Plastic: 550 g/L.

17. Contact Adhesive: 80 g/L.

18. Special Purpose Contact Adhesive: 250 g/L.

19. Structural Wood Member Adhesive: 140 g/L.

20. Sheet Applied Rubber Lining Operations: 850 g/L.

21. Top and Trim Adhesive: 250 g/L.

22. Architectural Sealants: 250 g/L.

23. Other Sealants: 420 g/L.

(2) Aerosol adhesives applied on site within the weatherproofing membrane must comply with the following Green Seal GS-36.

(a) Aerosol Adhesive, General-Purpose Mist Spray: 65 percent VOCs by weight.

(b) Aerosol Adhesive, General-Purpose Web Spray: 55 percent VOCs by weight.

(c) Special-Purpose Aerosol Adhesive (All Types): 70 percent VOCs by weight.

(3) Paints and coatings applied on site within the weatherproofing membrane must comply with the following criteria:

(a) VOC content limits for paints and coatings established in Green Seal Standard GS-11.

(b) VOC content limit for anti-corrosive and anti-rust paints applied to interior ferrous metal substrates of 250 g/L established in Green Seal GS-11.

(c) Clear wood finishes, floor coatings, stains, primers, sealers, and shellacs applied to interior elements must not exceed VOC content limits established in SCAQMD Rule 1113.

(d) Comply with the following VOC content limits:

1. Anti-Corrosive/Antirust Paints: 250 g/L.

2. Clear Wood Finish, Lacquer: 550 g/L.

3. Clear Wood Finish, Sanding Sealer: 350 g/L.

4. Clear Wood Finish, Varnish: 350 g/L.

5. Floor Coating: 100 g/L.

6. Interior Flat Paint, Coating or Primer: 50 g/L.

7. Interior Non-Flat Paint, Coating or Primer: 150 g/L.

8. Sealers and Undercoaters: 200 g/L.

9. Shellac, Clear: 730 g/L.

10. Shellac, Pigmented: 550 g/L.

11. Stain: 250 g/L.

12. Clear Brushing Lacquer: 680 g/L.

13. Concrete Curing Compounds: 350 g/L.

14. Japans/Faux Finishing Coatings: 350 g/L.

15. Magnesite Cement Coatings: 450 g/L.

16. Pigmented Lacquer: 550 g/L.

17. Waterproofing Sealers: 250 g/L.

18. Wood Preservatives: 350 g/L.

19. Low-Solids Coatings: 120 g/L.

(4) Carpet installed in building interior must comply with one of the following:

(a) Meet testing and product requirements of the Carpet and Rug Institute Green Label Plus program.

(b) Maximum VOC concentrations specified in CDPH SECTION 01350, using office scenario at the 14 day time point.

(5) Each non-carpet flooring element installed in building interior which is not inherently non-emitting (stone, ceramic, powder-coated metals, plated or anodized metal, glass, concrete, clay brick, and unfinished or untreated solid wood flooring) must comply with one of the following:

(a) Meet requirements of the FloorScore standard as shown with testing by an independent third-party.

(b) Maximum VOC concentrations specified in CDPH SECTION 01350, using office scenario at 14 day time point.

(6) Composite wood and agrifiber products used within the weatherproofing membrane must contain no added urea-formaldehyde resins.

(7) Laminating adhesives used to fabricate on-site and shop-applied composite wood and agrifiber assemblies must not contain added urea-formaldehyde.

c. Recycled Content:

(1) Any products being installed or used that are listed on EPA Comprehensive Procurement Guidelines designated product list must meet or exceed the EPA’s recycled content recommendations. The EPA Comprehensive Procurement Guidelines categories include:

(a) Building insulation.

(b) Cement and concrete.

(c) Consolidated and reprocessed latex paint.

(d) Floor tiles.

(e) Flowable fill.

(f) Laminated paperboard.

(g) Modular threshold ramps.

(h) Nonpressure pipe.

(i) Patio blocks.

(j) Railroad grade crossing surfaces.

(k) Roofing materials.

(l) Shower and restroom dividers/partitions.

(m) Structural fiberboard.

(n) Nylon carpet and nylon carpet backing.

(o) Compost and fertilizer made from recovered organic materials.

(p) Hydraulic mulch.

(q) Lawn and garden edging.

(r) Plastic lumber landscaping timbers and posts.

(s) Park benches and picnic tables.

(t) Plastic fencing.

(u) Playground equipment.

(v) Playground surfaces.

(w) Bike racks.

d. Biobased Content:

(1) Materials and equipment being installed or used that are listed on the USDA BioPreferred program product category list must meet or exceed USDA’s minimum biobased content threshold. Refer to individual specification sections for detailed requirements applicable to that section.

(a) USDA BioPreferred program categories include:

1. Adhesive and Mastic Removers.

2. Carpets.

3. Cleaners.

4. Composite Panels.

5. Corrosion Preventatives.

6. Erosion Control Materials.

7. Dust Suppressants.

8. Fertilizers.

9. Floor Cleaners and Protectors.

10. Floor Coverings (Non-Carpet).

11. Glass Cleaners.

12. Hydraulic Fluids.

13. Industrial Cleaners.

14. Interior Paints and Coatings.

15. Mulch and Compost Materials.

16. Multipurpose Cleaners.

17. Multipurpose Lubricants.

18. Packaging Films.

19. Paint Removers.

20. Plastic Insulating Foam.

21. Pneumatic Equipment Lubricants.

22. Roof Coatings.

23. Wastewater Systems Coatings.

24. Water Tank Coatings.

25. Wood and Concrete Sealers.

26. Wood and Concrete Stains.

e. Materials, products, and equipment being installed which fall into a category covered by the EPA WaterSense program must be WaterSense-labeled or meet or exceed EPA WaterSense program performance requirements, unless disallowed for infection control reasons.

(1) EPA WaterSense categories include:

(a) Bathroom Faucets

(b) Commercial Toilets

(c) Irrigation Controllers

(d) Pre-Rinse Spray Valves

(e) Residential Toilets

(f) Showerheads

(g) Spray Sprinkler Bodies

(h) Urinals

f. Materials, products, and equipment being installed which fall into any of the following product categories must be Energy Star-labeled.

(1) Applicable Energy Star product categories as of 09/14/2017 include:

(a) Appliances:

1. Air Purifiers and Cleaners.

2. Clothes Dryers (Residential).

3. Clothes Washers (Commercial & Residential).

4. Dehumidifiers.

5. Dishwashers (Residential).

6. Freezers (Residential).

7. Refrigerators (Residential).

(b) Electronics and Information Technology:

1. Audio/Video Equipment.

2. Computers.

3. Data Center Storage.

4. Digital Media Player.

5. Enterprise Servers.

6. Imaging Equipment.

7. Monitors.

8. Professional Displays.

9. Set-Top and Cable Boxes.

10. Telephones.

11. Televisions.

12. Uninterruptible Power Supplies.

13. Voice over Internet Protocol (VoIP) Phones.

(c) Food Service Equipment (Commercial):

1. Dishwashers.
2. Fryers.
3. Griddles.
4. Hot Food Holding Cabinets.
5. Ice Makers.
6. Ovens.
7. Refrigerators and Freezers.
8. Steam Cookers.
9. Vending Machines.

(d) Heating and Cooling Equipment:

1. Air-Source Heat Pumps (Residential).
2. Boilers.
3. Ceiling Fans (Residential).
4. Central Air Conditioners (Residential).
5. Ductless Heating and Cooling (Residential).
6. Furnaces (Residential).
7. Water Heaters.
8. Geothermal Heat Pumps (Residential).
9. Light Commercial Heating and Cooling Equipment.
10. Room Air Conditioners (Residential).
11. Ventilation Fans (Residential).

(e) Other:

1. Decorative Light Strings.
2. Electric Vehicle Supply Equipment.
3. Laboratory-Grade Refrigerators and Freezers.
4. Light Bulbs.
5. Light Fixtures.
6. Pool Pumps.

7. Roof Products.

8. Water Coolers.

9. Windows, Doors, and Skylights.

g. Materials, products, and equipment being installed which fall into any of the following categories must be FEMP-designated. FEMP-designated (PL-109-58)product categories as of 09/14/2017 include:

(1) Boilers (Commercial).

(2) Dishwashers (Commercial).

(3) Electric Chillers, Air-Cooled (Commercial).

(4) Electric Chillers, Water-Cooled (Commercial).

(5) Exterior Lighting.

(6) Fluorescent Ballasts.

(7) Fluorescent Lamps, General Service.

(8) Ice Machines, Water-Cooled.

(9) Industrial Lighting (High/Low Bay).

(10) Light Emitting Diode (LED) Luminaires.

h. Electronic products and equipment being installed which fall into any of the following categories shall be EPEAT registered. Electronic products and equipment covered by EPEAT program as of 09/14/2017 include:

(1) Computers.

(2) Displays.

(3) Imaging Equipment.

(4) Televisions.

PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL

a. Construction Indoor Air Quality Management:

(1) During construction, meet or exceed recommended control measures of ANSI/SMACNA 008, Chapter 3.

(2) Protect stored on-site and installed absorptive materials from moisture damage.

(3) If permanently installed air handlers are used during construction, filtration media with a minimum efficiency reporting value (MERV) of 8 must be used at each return air grille, as determined by ASHRAE 52.2 (with errata but without addenda).

Replace all filtration media immediately prior to occupancy.

(4) Perform building flush-out as follows:

(a) After construction ends, prior to occupancy and with interior finishes installed, perform a building flush-out by supplying a total volume of 14000 cu. ft. of outdoor air per sq. ft. of floor area while maintaining an internal temperature of at least 60 degrees Fahrenheit and a relative humidity no higher than 60 percent. OR

(b) If occupancy is desired prior to flush-out completion, the space may be occupied following delivery of a minimum of 3500 cu. ft. of outdoor air per sq. ft. of floor area to the space. Once a space is occupied, it must be ventilated at a minimum rate of 0.30 cfm per sq. ft. of outside air or design minimum outside air rate determined until a total of 14000 cu. ft./sq. ft. of outside air has been delivered to the space. During each day of flush-out period, ventilation must begin a minimum of three hours prior to occupancy and continue during occupancy.

(5) Provide construction dust control to comply with SCAQMD Rule 403.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 86 25

CYBERSECURITY FOR FACILITY RELATED CONTROL SYSTEMS

PART 1 GENERAL

1.1 DEFINITIONS
1.2 SCOPE OF WORK
1.3 CONTRACTOR PERSONNEL REQUIREMENTS

PART 2 PRODUCTS

2.1 LEGACY PRODUCTS
2.2 UNIFIED CAPABILITIES REQUIREMENTS
2.3 NATIONAL DEFENSE AUTHORIZATION ACT

PART 3 EXECUTION

3.1 DESIGN
3.2 CONSTRUCTION

-- End of Section Table of Contents --

SECTION 01 86 25

CYBERSECURITY FOR FACILITY RELATED CONTROL SYSTEMS

PART 1 GENERAL

1.1 DEFINITIONS

Facility Related Control Systems (FRCS) are control systems which implement control of equipment and infrastructure to ensure the security and protection of real property; life safety and comfort of occupants; enhance efficiency and lower facility costs; monitor and record process data; and optimize operations.

1.2 SCOPE OF WORK

The Contractor must construct, and deliver all FRCS in accordance with Unified Facility Criteria (UFC) 4-010-06 Cybersecurity of Facility Related Control Systems (latest edition as of contract award). Contractor must be responsible for all submissions, deliverables, and functions as outlined in UFC 4-010-06 Chapter 5.

In accordance with UFC 4-010-06, Chapter 5, the Contractor must be responsible for delivery of an edited Unified Facility Guide Specification (UFGS) 25 05 11, Cybersecurity for Facility Related Control Systems, specification for each control system. The edited 25 05 11 shall be reviewed and approved by the Government prior to beginning construction.

In accordance with FAR 252.204-7012, Safeguarding Covered Defense Information and Cyber Incident Reporting, the Contractor shall implement adequate security measures to safeguard all covered defense information (CDI) and controlled unclassified information (CUI) associated with the project.

Prior to handling or transmitting any project-related information that may involve CDI or VA-sensitive data, the Contractor shall consult with the VA Office of Information and Technology (OIT) and/or the VA Cybersecurity Office to ensure compliance with applicable policies and procedures.

All cybersecurity controls shall meet or exceed the requirements of NIST SP 800-171, and any actual or suspected cyber incidents must be reported in accordance with FAR 252.204-7012 and VA procedures.

The Senior Resident Engineer (SRE) shall coordinate with OIT and/or VA Cybersecurity to verify compliance during inspections or audits involving information systems or data exchange.

All design submittals and deliverables will be reviewed and approved by the Government at all stages of the design.

For the purposes of design and construction, the impact ratings to be used for each system are provided in the table below. If the Contractor provides additional control systems beyond those anticipated by this RFP, Contractor must ensure those additional control systems are designed and constructed in accordance with UFC 4-010-06 and is responsible for coordinating via CO to obtain an approved impact rating for those

additional systems.

Building Control System	Confidentiality	Integrity	Availability
Elevators	L	L	L
Fire Detection/Alarm	L	L	M
Mass Notification System	L	L	M
ASHRAE (HVAC/Heating)	L	L	L
Emergency DAS	M	M	M
Electronic Security Systems (ESS)	M	M	M
Physical Access Control System (PASCS)	M	H	H
Instrusion Detection System (IDS)	L	M	L
Access Control System (ACS)	M	M	M
ESS Video Monitoring	M	M	M
Utility Monitoring and Control Systems (UMCS)	L	L	L
Electrical-Generator	L	M	M
Power Monitoring and Control Systems	L	L	L
Lighting COntrol Systems	L	L	L
Building Automation System (BAS)	L	L	L
Medical Gases	L	M	M
EMCS Systems	L	L	L
Public Address	M	M	M
Nurse Call System	M	H	H
Code Blue System	M	H	H
Pneeumatic Tube System (PTS)	M	M	M
Automated Guided Vehicle System (AGV)	M	M	M

Several of these control systems are expected to connect to or integrate with existing control systems at the installation. These requirements are defined in the related section for each system's primary discipline. Contractor must interface with the Base via CO to determine specific transport and configuration requirements and other items that may impact the ability to connect or integrate with these existing systems.

During construction, Contractor must execute all requirements of each approved 25 05 11 specification and shall be responsible for all construction submittals and functions. All construction submittals will be reviewed and approved by the Government.

1.3 CONTRACTOR PERSONNEL REQUIREMENTS

a. Cybersecurity Subject Matter Expert (SME)

Provide the services of a Cybersecurity SME. All certifications must be in effect and current at the time of the resume submission. The individual shall oversee all work within this section and the 25 05 11 specifications. The Cybersecurity SME must meet Information Assurance Manager Level II Certification requirements in accordance with DoD 8570.01 and DoD 8140.01 Information Workforce Improvement Program. The individual shall have experience with the DoD Risk Management Framework (RMF) process. Facility related control system (FRCS) and industrial control system (ICS) knowledge is required through demonstrated control system design, testing, and commissioning experience.

b. General Personnel Requirements

All Contractor Personnel requesting access to existing systems for the purpose of installing new control system devices/software, programming/configuring of control system(s) devices/software or modifying code for existing control system devices/software are required to have a favorable security background check and must complete and sign/submit the ICS SAAR-N (Systems Access Authorization Request) Forms (as needed).

Individuals developing or using Security Controls documentation, or other documents that are designated "Controlled Unclassified Information" (CUI), will be required to complete and submit a signed Non-Disclosure and Data Handling Agreement for CUI information and must submit completed DoD CUI Mandatory Training at https://www.dodcui.mil/Home/Training/ no less than 30 days prior to base access.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements unless otherwise noted.

2.1 LEGACY PRODUCTS

All provided equipment and software must be currently marketed products, not currently scheduled for end of life or obsolescence, to ensure system sustainability. Where older versions of software are provided due to incompatibility concerns between different versions, provide the Government with the means (licenses, install disks, and any other installation materials) to fully update the provided software to the latest marketed version.

2.2 UNIFIED CAPABILITIES REQUIREMENTS

Products which fall under the scope of the Unified Capabilities Requirements, such as active network infrastructure devices, must be listed in the Department of Defense Information Network (DoDIN) Approved Products List (APL) (https://aplits.disa.mil/apl).

2.3 NATIONAL DEFENSE AUTHORIZATION ACT

Contractor must review and comply with annual National Defense Authorization Act (NDAA) guidance on approved products and manufacturers and source restrictions.

PART 3 EXECUTION

3.1 DESIGN

The UFC 4-010-06 Paragraph 3-1.1 Steps 1 through 5, including a preliminary tailored 25 05 11 guide specification for each system, shall be completed for the 65 percent (or equivalent) design review for review and comment by the government. All design submittals and deliverables will be reviewed and approved by the Government at all stages of the design. All communications with the SO and ISSM/ISSO shall be through the USACE Contracting Officer Representative (CO).

3.2 CONSTRUCTION

UFGS Section 25 05 11 "Cybersecurity for Facility-Related Control Systems" and related materials shall be utilized and are found at the following location:
https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-1
Provide a separate edited specification 25 05 11 for each control system.

Each 25 05 11 is subject to government review and approval. Each 25 05 11 shall be fully executed during construction and all required submittals provided for subsequent government review and approval.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 91 00.01 25

DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS

05/19, CHG 2: 08/20

PART 1 GENERAL

1.1 COMMISSIONING DESCRIPTION
1.2 Related Work
1.3 Summary
1.4 Acronyms
1.5 DEFINITIONS
1.6 SYSTEMS TO BE COMISSIONED
1.7 COMMISSIONING TEAM
1.8 GOVERNMENT'S COMMISSIONING RESPONSIBILITIES
1.9 CONTRACTOR'S COMMISSIONING RESPONSIBILITIES
1.10 COMMISSIONING AGENT'S (CxA) RESPONSIBILITIES
1.11 COMMISSIONING DOCUMENTATION
1.12 SUBMITTALS
1.13 COMMISSIONING PROCESS
1.14 QUALITY ASSURANCE
1.15 COORDINATION

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

PART 3 EXECUTION

3.1 COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES
3.2 STARTUP, INITIAL CHECKOUT, AND PRE-FUNCTIONAL CHECKLISTS
3.3 DEFICIENCIES, NONCONFORMANCE, AND APPROVAL IN CHECKLISTS AND STARTUP
3.4 DDC SYSTEM TRENDING FOR COMMISSIONING
3.5 SYSTEMS FUNCTIONAL PERFORMANCE TESTING
3.6 DOCUMENTATION, NONCONFORMANCE AND APPROVAL OF TESTS
3.7 DEFERRED TESTING
3.8 OPERATION AND MAINTENANCE TRAINING REQUIREMENTS

-- End of Section Table of Contents --

SECTION 01 91 00.01 25

DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS
05/19, CHG 2: 08/20

PART 1 GENERAL

1.1 COMMISSIONING DESCRIPTION

a. This Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS shall form the basis of the construction phase commissioning process and procedures for VA projects with USACE as the Design and/or Construction agent. The Commissioning Agent (CxA) and Contractor's Commissioning Manager (CxM) shall add, modify, and refine the commissioning procedures, as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table and as approved by the Contracting Officer (CO), to suit field conditions and actual manufacturer's equipment, incorporate test data and procedure results, and provide detailed scheduling for all commissioning tasks.

b. Various sections of the project specifications require equipment startup, testing, and adjusting services. Requirements for startup, testing, and adjusting services specified in the Division 7, Division 21, Division 22, Division 23, Division 26, Division 27, Division 28, and Division 31 series sections of these specifications are intended to be provided in coordination with the commissioning services and are not intended to duplicate services. The Contractor shall coordinate the work required by individual specification sections with the commissioning services requirements specified herein.

c. Where individual testing, adjusting, or related services are required in the project specifications and not specifically required by this commissioning requirements specification, the specified services shall be provided and copies of documentation, as required by those specifications shall be submitted to the Contracting Officer and the CxA to be indexed for future reference.

d. Where training or educational services for VA are required and specified in other sections of the specifications, including but not limited to Division 7, Division 8, Division 21, Division 22, Division 23, Division 26, Division 27, Division 28, and Division 31 series sections of the specification, these services are intended to be provided in addition to the training and educational services specified herein.

e. Commissioning is a systematic process of verifying that the building systems perform interactively according to the construction documents and the VA's operational needs. The commissioning process shall encompass and coordinate the system documentation, equipment startup, control system calibration, testing and balancing, performance testing and training. Commissioning during the construction and post-occupancy phases is intended to achieve the following specific objectives according to the contract documents:

1. Verify that the applicable equipment and systems are installed in accordance with the contact documents and according to the manufacturer's recommendations.

2. Verify and document proper integrated performance of equipment and systems.

3. Verify that Operations & Maintenance documentation is complete.

4. Verify that all components requiring servicing can be accessed, serviced and removed without disturbing nearby components including ducts, piping, cabling or wiring.

5. Verify that the VA's operating personnel are adequately trained to enable them to operate, monitor, adjust, maintain, and repair building systems in an effective and energy-efficient manner.

6. Document the successful achievement of the commissioning objectives listed above.

f. The commissioning process does not take away from or reduce the responsibility of the Contractor to provide a finished and fully functioning product.

1.2 Related Work

a. UFGS Section 01 30 00, ADMINISTRATIVE REQUIREMENTS

b UFGS Section 01 32 01.10 PROJECT SCHEDULE

c. UFGS Section UFGS Section 01 33 00, SUBMITTAL PROCEDURES

d. UFGS Section Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS

e. UFGS Section 07 08 00.01 DVA/USACE PROJECTS FACILITY EXTERIOR CLOSURE COMMISSIONING.

f. UFGS Section 21 08 00 .01, DVA/USACE PROJECTS COMMISSIONING OF FIRE PROTECTION SYSTEMS.

g. UFGS Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

h. UFGS Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. UFGS Section 26 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF ELECTRICAL SYSTEMS.

j. UFGS Section 27 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF COMMUNICATIONS SYSTEMS.

k. UFGS Section 28 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

l. UFGS Section 33 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF SITE UTILITIES.

1.3 Summary

a. This Section includes general requirements that apply to implementation of commissioning without regard to systems, subsystems,

and equipment being commissioned.

b. The commissioning activities have been developed to support the Government requirements to meet guidelines for Federal Leadership in Environmental, Energy, and Economic Performance.

d. The commissioning activities have been developed to support the Green Buildings Initiative's Green Globes rating program and to support delivery of project performance in accordance with the Government requirements developed for the project.

1.4 Acronyms

List of Acronyms	
Acronym	**Meaning**
A/E	Architect / Engineer Design Team
AHJ	Authority Having Jurisdiction
ASHRAE	Association Society for Heating Air Condition and Refrigeration Engineers
BOD	Basis of Design
BSC	Building Systems Commissioning
A	Approve
ASI	Architect's Supplemental Instruction
CD	Construction Documents
CMMS	Computerized Maintenance Management System
CO	Contracting Officer (USACE)
COBie	Construction Operations Building Information Exchange
CPC	Construction Phase Commissioning
Cx	Commissioning
CxA	Commissioning Agent
CxM	Commissioning Manager
CxR	Commissioning Representative
D-B	Design-Build
D-B-B	Design-Bid-Build

List of Acronyms	
Acronym	**Meaning**
D	Design Development
DD	Design Development
DID	Design Intent Document
DPC	Design Phase Commissioning
DVA	Department of Veterans Affairs
FPT	Functional Performance Test
L	Lead
GBI-GG	Green Building Initiative - Green Globes
HVAC	Heating, Ventilation, and Air Conditioning
LEED	Leadership in Energy and Environmental Design
NC	Department of Veterans Affairs National Cemetery
NCA	Department of Veterans Affairs National Cemetery Administration
NEBB	National Environmental Balancing Bureau
O	Optional
O&M	Operations & Maintenance
OPR	Owner's Project Requirements
P	Participate
PFC	Pre-Functional Checklist
PFT	Pre-Functional Test
R	Review
RFI	Request For Information
RFP	Request For Proposal
RE	VA Resident Engineer
SD	Schematic Design
SO	Site Observation

List of Acronyms	
Acronym	**Meaning**
TAB	Test Adjust and Balance
USACE	United States Army Corps of Engineers
VA	Department of Veterans Affairs
VAMC	VA Medical Center
VA CFM	VA Office of Construction and Facilities Management
VACO	VA Central Office
VA PM	VA Project Manager
USGBC	United States Green Building Council

1.5 DEFINITIONS

Acceptance Phase Commissioning: Commissioning tasks executed after most construction has been completed, most Site Observations and Static Tests have been completed and Pre-Functional Testing has been completed and accepted. The main commissioning activities performed during this phase are verification that the installed systems are functional by conducting Systems Functional Performance tests and Owner Training.

Accuracy: The capability of an instrument to indicate the true value of a measured quantity.

Back Check: A back check is a verification that an agreed upon solution to a design comment has been adequately addressed in a subsequent design review.

Basis of Design (BOD): The Engineer's Basis of Design is comprised of two components: the Design Criteria and the Design Narrative, these documents record the concepts, calculations, decisions, and product selections used to meet the Owner's Project Requirements (OPR) and to satisfy applicable regulatory requirements, standards, and guidelines.

Benchmarks: Benchmarks are the comparison of a building's energy usage to other similar buildings and to the building itself. For example, ENERGY STAR Portfolio Manager is a frequently used and nationally recognized building energy benchmarking tool.

Building Information Modeling (BIM): Building Information Modeling is a parametric database which allows a building to be designed and constructed virtually in 3D, and provides reports both in 2D views and as schedules. This electronic information can be extracted and reused for pre-populating facility management CMMS systems. Building Systems Commissioning (BSC): NEBB acronym used to designate its commissioning program.

Calibrate: The act of comparing an instrument of unknown accuracy with

a standard of known accuracy to detect, correlate, report, or eliminate by adjustment any variation in the accuracy of the tested instrument.

CCTV: Closed circuit Television. Normally used for security surveillance and alarm detections as part of a special electrical security system.

COBie: Construction Operations Building Information Exchange (COBie) is an electronic industry data format used to transfer information developed during design, construction, and commissioning into the Computer Maintenance Management Systems (CMMS) used to operate facilities. See the Whole Building Design Guide website for further information (http://www.wbdg.org/resources/cobie.php)

Commissionability: Defines a design component or construction process that has the necessary elements that will allow a system or component to be effectively measured, tested, operated and commissioned.

Commissioning Agent (CxA): A qualified and independent Commissioning Professional with professional engineering registration (PE), active commissioning certification from a nationally recognized commissioning organization, and proven hands-on and verifiable experience in the commissioning of projects with equivalent scope. The CxA is employed by a nationally recognized, certified, and qualified commissioning firm, hired directly by government under a separate contract, who leads and administers the Cx process by managing the Cx team and overseeing the Commissioning Process, except as otherwise noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table in this document. Both the commissioning firm and the CxA are certified commissioning providers with experience and expertise in the commissioning of facilities of comparable scope and complexity. The CxA is responsible for government oversight of the commissioning process throughout all phases of the project. Where CxA is used in this document it means the Commissioning Agent, members of his staff or appointed members of the commissioning team. The CxA is collaborated by the CxM during construction period. (Note: Commissioning certification by a nationally recognized organization is required. Certifications from AABC Commissioning Group (ACG), Building Commissioning Association (BCA), or other equivalent commissioning authority would be deemed as acceptable.)

Commissioning Checklists: Lists of data or inspections to be verified to ensure proper system or component installation, operation, and function. Verification checklists are developed and used during all phases of the commissioning process to verify that the Owner's Project Requirements (OPR) is being achieved.

Commissioning Design Review: The commissioning design review is a collaborative review of the design professionals design documents for items pertaining to the following: owner's project requirements; basis of design; operability and maintainability (O&M) including documentation; functionality; training; energy efficiency, control systems' sequence of operations including building automation system features; commissioning specifications and the ability to functionally test the systems.

Commissioning Issue: A condition identified by the CxA, CxM, or other member of the Commissioning Team that adversely affects the commissionability, operability, maintainability, or functionality of a

system, equipment, or component. A condition that is in conflict with the Contract Documents and/or performance requirements of the installed systems and components. (See also - Commissioning Observation).

Commissioning Manager (CxM): A qualified individual with professional engineering registration (PE) and active commissioning certification from a nationally recognized commissioning organization, and proven hands-on and verifiable experience in the commissioning of projects with equivalent scope, appointed by the Construction General Contractor to manage the commissioning process on behalf of the Contractor. The CxM is employed by a nationally recognized, certified, and qualified commissioning firm, responsible for managing, scheduling, executing, and documenting commissioning activities during the construction period as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table in this document. The CxM is regularly employed in commissioning of total building systems and is an employee of a commissioning firm hired by the construction contractor. The commissioning firm is a first-tier subcontractor to the Prime Contractor. The CxM is supported by commissioning specialists employed by the commissioning firm. (Note: Commissioning certification by a nationally recognized organization is required. Certifications from AABC Commissioning Group (ACG), Building Commissioning Association (BCA), or other equivalent commissioning authority would be deemed as acceptable).

Commissioning Observation: An issue identified by the CxA, CxM, or other member of the Commissioning Team that does not conform to the project OPR, contract documents or standard industry best practices. (See also Commissioning Issue)

Commissioning Plan: A document that outlines the commissioning process, commissioning scope and defines responsibilities, processes, schedules, and the documentation requirements of the Commissioning Process.

Commissioning Process: A quality focused process for enhancing the delivery of a project. The process focuses upon verifying and documenting that the facility and all of its systems, components, and assemblies are planned, designed, installed, tested, can be operated, and maintained to meet the Owner's Project Requirements.

Commissioning Report: The final commissioning document which presents the commissioning process results for the project. Cx reports include an executive summary, the commissioning plan, issue log, correspondence, and all appropriate check sheets and test forms.

Commissioning Representative (CxR): An individual appointed by a sub-contractor to manage the commissioning process on behalf of the sub-contractor.

Commissioning Specifications: The contract documents that detail the objective, scope and implementation of the commissioning process as developed in the Commissioning Plan.

Commissioning Team: Individual team members whose coordinated actions are responsible for implementing the Commissioning Process.

Construction Phase Commissioning: All commissioning efforts executed

during the construction process after the design phase and prior to the Acceptance Phase Commissioning.

Contract Documents (CD): Contract documents include design and construction contracts, price agreements and procedure agreements. Contract Documents also include all final and complete drawings, specifications and all applicable contract modifications or supplements.

Construction Phase Commissioning (CPC): All commissioning efforts executed during the construction process after the design phase and prior to the Acceptance Phase Commissioning.

Coordination Drawings: Drawings showing the work of all trades that are used to illustrate that equipment can be installed in the space allocated without compromising equipment function or access for maintenance and replacement. These drawings graphically illustrate and dimension manufacturers' recommended maintenance clearances. On mechanical projects, coordination drawings include structural steel, ductwork, major piping and electrical conduit and show the elevations and locations of the above components.

Data Logging: The monitoring and recording of temperature, flow, current, status, pressure, etc. of equipment using stand-alone data recorders.

Deferred System Test: Tests that cannot be completed at the end of the acceptance phase due to ambient conditions, schedule issues or other conditions preventing testing during the normal acceptance testing period.

Deficiency: See "Commissioning Issue".

Design Criteria: A listing of the VA Design Criteria outlining the project design requirements, including its source. These are used during the design process to show the design elements meet the OPR.

Design Intent: The overall term that includes the OPR and the BOD. It is a detailed explanation of the ideas, concepts, and criteria that are defined by the owner to be important. The design intent documents are utilized to provide a written record of these ideas, concepts and criteria.

Design Narrative: A written description of the proposed design solutions that satisfy the requirements of the OPR.

Design Phase Commissioning (DPC): All commissioning tasks executed during the design phase of the project.

Environmental Systems: Systems that use a combination of mechanical equipment, airflow, water flow and electrical energy to provide heating, ventilating, air conditioning, humidification, and dehumidification for the purpose of human comfort or process control of temperature and humidity.

Executive Summary: A section of the Commissioning report that reviews the general outcome of the project. It also includes any unresolved issues, recommendations for the resolution of unresolved issues and all deferred testing requirements.

Functionality: This defines a design component or construction process which will allow a system or component to operate or be constructed in a manner that will produce the required outcome of the OPR.

Functional Test Procedure (FTP): A written protocol that defines methods, steps, personnel, and acceptance criteria for tests conducted on components, equipment, assemblies, systems, and interfaces among systems.

Industry Accepted Best Practice: A design component or construction process that has achieved industry consensus for quality performance and functionality. Refer to the current edition of the NEBB Design Phase Commissioning Handbook for examples.

Installation Verification: Observations or inspections that confirm the system or component has been installed in accordance with the contract documents and to industry accepted best practices.

Integrated System Testing: Integrated Systems Testing procedures entail testing of multiple integrated systems performance to verify proper functional interface between systems. Typical Integrated Systems Testing includes verifying that building systems respond properly to loss of utility, transfer to emergency power sources, re-transfer from emergency power source to normal utility source; interface between HVAC controls and Fire Alarm systems for equipment shutdown, interface between Fire Alarm system and elevator control systems for elevator recall and shutdown; interface between Fire Alarm System and Security Access Control Systems to control access to spaces during fire alarm conditions; and other similar tests as determined for each specific project. Issues Log: A formal and ongoing record of problems or concerns - and their resolution - that have been raised by members of the Commissioning Team during the course of the Commissioning Process.

LEAD: The "LEAD" Roles & Responsibilities designation represents the overall leadership and administration of the Cx process by managing the Cx team and overseeing and managing the Commissioning Process. The LEAD party responsibilities shall also include participation and review responsibilities as applicable.

Lessons Learned Workshop: A workshop conducted to discuss and document project successes and identify opportunities for improvements for future projects.

Maintainability: A design component or construction process that will allow a system or component to be effectively maintained. This includes adequate room for access to adjust and repair the equipment. Maintainability also includes components that have readily obtainable repair parts or service.

Manual Test: Testing using hand-held instruments, immediate control system readouts or direct observation to verify performance (contrasted to analyzing monitored data taken over time to make the 'observation').

Owner's Project Requirements (OPR): A written document that details the project requirements and the expectations of how the building and its systems will be used and operated. These include project goals,

measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information.

Peer Review: A formal in-depth review separate from the commissioning review processes. The level of effort and intensity is much greater than a typical commissioning facilitation or extended commissioning review. The VA usually hires an independent third-party (called the IDIQ A/E) to conduct peer reviews.

PARTICIPATE: The "PARTICIPATE" Roles & Responsibilities designation represents an active participation in the Cx process and its development by working closely and directly with the party designated as the LEAD, as well as review responsibility as applicable.

Precision: The ability of an instrument to produce repeatable readings of the same quantity under the same conditions. The precision of an instrument refers to its ability to produce a tightly grouped set of values around the mean value of the measured quantity.

Pre-Design Phase Commissioning: Commissioning tasks performed prior to the commencement of design activities that includes project programming and the development of the commissioning process for the project

Pre-Functional Checklist (PFC): A form used by the contractor to verify that appropriate components are onsite, correctly installed, set up, calibrated, functional and ready for functional testing.

Pre-Functional Test (PFT): An inspection or test that is done before functional testing. PFT's include installation verification and system and component start up tests.

Procedure or Protocol: A defined approach that outlines the execution of a sequence of work or operations. Procedures are used to produce repeatable and defined results.

Range: The upper and lower limits of an instrument's ability to measure the value of a quantity for which the instrument is calibrated.

Resolution: This word has two meanings in the Cx Process. The first refers to the smallest change in a measured variable that an instrument can detect. The second refers to the implementation of actions that correct a tested or observed deficiency.

Site Observation Visit: On-site inspections and observations made by the Commissioning Agent for the purpose of verifying component, equipment, and system installation, to observe contractor testing, equipment start-up procedures, or other purposes.

Site Observation Reports (SO): Reports of site inspections and observations made by the Commissioning Agent. Observation reports are intended to provide early indication of an installation issue which will need correction or analysis.

Special System Inspections: Inspections required by a local code authority prior to occupancy and are not normally a part of the commissioning process.

Static Tests: Tests or inspections that validate a specified static

condition such as pressure testing. Static tests may be specification or code initiated.

Start Up Tests: Tests that validate the component or system is ready for automatic operation in accordance with the manufactures requirements.

Systems Manual: A system-focused composite document that includes all information required for the owners operators to operate the systems.

Test Procedure: A written protocol that defines methods, personnel, and expectations for tests conducted on components, equipment, assemblies, systems, and interfaces among systems.

Testing: The use of specialized and calibrated instruments to measure parameters such as: temperature, pressure, vapor flow, air flow, fluid flow, rotational speed, electrical characteristics, velocity, and other data in order to determine performance, operation, or function.

Testing, Adjusting, and Balancing (TAB): A systematic process or service applied to heating, ventilating and air-conditioning (HVAC) systems and other environmental systems to achieve and document air and hydronic flow rates. The standards and procedures for providing these services are referred to as "Testing, Adjusting, and Balancing" and are described in the Procedural Standards for the Testing, Adjusting and Balancing of Environmental Systems, published by NEBB or AABC.

Thermal Scans: Thermographic pictures taken with an Infrared Thermographic Camera. Thermographic pictures show the relative temperatures of objects and surfaces and are used to identify leaks, thermal bridging, thermal intrusion, electrical overload conditions, moisture containment, and insulation failure.

Training Plan: A written document that details, in outline form the expectations of the operator training. Training agendas should include instruction on how to obtain service, operate, startup, shutdown and maintain all systems and components of the project.

Trending: Monitoring over a period of time with the building automation system.

Unresolved Commissioning Issue: Any Commissioning Issue that, at the time that the Final Report or the Amended Final Report is issued that has not been either resolved by the construction team or accepted by the DVA/USACE. Validation: The process by which work is verified as complete and operating correctly:

1. First party validation occurs when a firm or individual verifying the task is the same firm or individual performing the task.

2. Second party validation occurs when the firm or individual verifying the task is under the control of the firm performing the task or has other possibilities of financial conflicts of interest in the resolution (Architects, Designers, General Contractors and Third Tier Subcontractors or Vendors).

3. Third party validation occurs when the firm verifying the task is not associated with or under control of the firm performing or

designing the task.

Verification: The process by which specific documents, components, equipment, assemblies, systems, and interfaces among systems are confirmed to comply with the criteria described in the Owner's Project Requirements.

Warranty Phase Commissioning: Commissioning efforts executed after a project has been completed and accepted by the Owner. Warranty Phase Commissioning includes follow-up on verification of system performance, measurement and verification tasks and assistance in identifying warranty issues and enforcing warranty provisions of the construction contract.

Warranty Visit: A commissioning meeting and site review where all outstanding warranty issues and deferred testing is reviewed and discussed.

Whole Building Commissioning: Commissioning of building systems such as Building Envelope, HVAC, Electrical, Special Electrical (Fire Alarm, Security & Communications), Plumbing and Fire Protection as described in this specification.

1.6 SYSTEMS TO BE COMISSIONED

a. Commissioning of a system or systems specified for this project is part of the construction process. Documentation and testing of these systems, as well as training of the VA's Operation and Maintenance personnel, is required in cooperation with the VA and the Commissioning Agent.

b. The following systems will be commissioned as part of this project:

Systems To Be Commissioned	
System	**Description**
Building Exterior Closure	
Foundations (excluding structural)	Standard, special, slab-on-grade, vapor barriers, air barriers
Basements	Basement walls, crawl spaces, waterproofing, drainage
Roofing	Roof system (including parapet), roof openings (skylights, pipe chases, ducts, equipment curbs, etc.)

Systems To Be Commissioned	
System	**Description**
Note:	The emphasis on commissioning the above building envelope systems is on control of air flow, heat flow, noise, infrared, ultraviolet, rain penetration, moisture, durability, security, reliability, constructability, maintainability, and sustainability.
Equipment	
Parking Control Equipment	Barriers
Conveying Equipment	
Elevators	Interface with other systems (fire alarm, etc.) ASTM testing and certification by installing vendor
Fire Suppression	
Fire Pump	Fire Pump, jockey pump, fire pump controller/ATS
Fire Sprinkler Systems	Wet pipe system
Plumbing	
Domestic Water Distribution	Booster pumps, backflow preventers, potable water storage tanks
Domestic Hot Water Systems	Water heaters**, circulation pumps, point-of-use water heaters*
Sewerage Pump Systems	Sewage ejectors
Wastewater Pump Systems	Sump pumps
General Service Air Systems	Packaged compressor systems, air dryers, filtration
Medical Air Systems	Packaged medical air compressor units. Outlet certification, cross-connection verification
Medical Vacuum Systems	Packaged medical vacuum units, outlet certification, cross-connection verification
HVAC	

Systems To Be Commissioned	
System	**Description**
Noise and Vibration Control	Noise and vibration levels for critical equipment such as Air Handlers, Chillers, Cooling Towers, Boilers, Generators, etc. will be commissioned as part of the system commissioning
Direct Digital Control System	Operator Interface Computer, Operator Work Station (including graphics, point mapping, trends, alarms), Network Communications Modules and Wiring, Integration Panels. DDC Control panels will be commissioned with the systems controlled by the panel
Chilled Water System	Chillers (centrifugal, rotary screw, air-cooled), pumps (primary, secondary, variable primary), VFDs associated with chilled water system components, DDC Control Panels (including integration with Building Control System)
Condenser Water System	Cooling Towers, Heat Trace, Fluid Coolers, heat exchangers/economizers, pumps, VFDs associated with condenser water system components, DDC control panels.
Steam/Heating Hot Water System	Boilers, boiler feed water system, economizers/heat recovery equipment, condensate recovery, water treatment, boiler fuel system, controls, interface with facility DDC system.
HVAC Air Handling Systems	Air handling Units, packaged rooftop AHU, Outdoor Air conditioning units, humidifiers, DDC control panels
HVAC Ventilation/Exhaust Systems	General exhaust, toilet exhaust, laboratory exhaust, isolation exhaust, room pressurization control systems
HVAC Energy Recovery Systems	Heat Wheels, Heat Recovery Loops, AHU Integrated Heat Recovery
HVAC Terminal Unit Systems	VAV Terminal Units, CAV terminal units, fan coil units, fin-tube radiation, unit heaters
Decentralized Unitary HVAC Systems	Split-system HVAC systems, controls, interface with facility DDC
Unitary Heat Pump Systems	Water-source heat pumps, controls, interface with facility DDC
Facility Fuel Systems	Boiler fuel system, generator fuel system
Electrical	

Systems To Be Commissioned	
System	**Description**
Medium-Voltage Electrical Distribution Systems	Medium-Voltage Switchgear, Medium-Voltage Switches, Underground ductbank and distribution, Pad-Mount Transformers, Medium-Voltage Load Interrupter Switches,
Grounding & Bonding Systems	Witness 3rd party testing, review reports
Electric Power Monitoring Systems	Metering, sub-metering, power monitoring systems, PLC control systems
Electrical System Protective Device Study	Review reports, verify field settings consistent with Study
Secondary Unit Substations	Medium-voltage components, transformers, low-voltage distribution, verify breaker testing results (injection current, etc)
Low-Voltage Distribution System	Normal power distribution system, Life-safety power distribution system, critical power distribution system, equipment power distribution system, switchboards, distribution panels, panelboards, verify breaker testing results (injection current, etc)
Emergency Power Generation Systems	Generators, Generator paralleling switchgear, automatic transfer switches, PLC and other control systems
Lighting & Lighting Control** Systems	Emergency lighting, occupancy sensors, lighting control systems, architectural dimming systems, theatrical dimming systems, exterior lighting and controls
Lightning Protection System	Witness 3rd party testing, review reports
Communications	
Grounding & Bonding System	Witness 3rd party testing, review reports
Structured Cabling System	Witness 3rd party testing, review reports
Master Antenna Television System	Witness 3rd party testing, review reports
Public Address & Mass Notification Systems	Witness 3rd party testing, review reports
Intercom & Program Systems	Witness 3rd party testing, review reports

Systems To Be Commissioned	
System	**Description**
Nurse Call & Code Blue Systems	Witness 3rd party testing, review reports
Security Emergency Call Systems	Witness 3rd party testing, review reports
Duress Alarm Systems	Witness 3rd party testing, review reports
Electronic Safety and Security	
Grounding & Bonding	Witness 3rd party testing, review reports
Physical Access Control Systems	Witness 3rd party testing, review reports
Access Control Systems	Witness 3rd party testing, review reports
Security Access Detection Systems	Witness 3rd party testing, review reports
Video Surveillance System	Witness 3rd party testing, review reports
Electronic Personal Protection System	Witness 3rd party testing, review reports
Fire Detection and Alarm System	100% device acceptance testing, battery draw-down test, verify system monitoring, verify interface with other systems.
Site Utilities	
Water Utilities	City Water Service Entrance, Backflow Prevention, Pressure Control, Booster Pumps, Irrigation Systems
Sanitary Sewerage Utilities	City Sanitary Connection, Waste Treatment Systems
Storm Drainage Utilities	City Storm Water Connection, Site Storm Water Distribution
Energy Distribution Utilities	Connection to Third Party Energy (Steam, High Temp Hot Water, Chilled Water) Supply Systems, Metering, Pressure Control
Integrated Systems Tests	
Loss of Power Response	Loss of power to building, loss of power to campus, restoration of power to building, restoration of power to campus.

Systems To Be Commissioned	
System	**Description**
Fire Alarm Response	Integrated System Response to Fire Alarm Condition and Return to Normal
Table Notes	

1.7 COMMISSIONING TEAM

a. The commissioning team shall consist of, but not be limited to, representatives of Contractor, including Project Superintendent and subcontractors, installers, schedulers, suppliers, and specialists deemed appropriate by the Government, CxA, and CxM.

b. Members Appointed by Contractor:

1. Contractor's Commissioning Manager (CxM, See the CxM definition above): The designated person, company, or entity that plans, schedules and coordinates the commissioning activities for the construction team. See the Roles & Responsibilities table below for more detailed responsibilities.

2. Contractor's Commissioning Representative(s): Individual(s), each having authority to act on behalf of the entity he or she represents, explicitly organized to implement the commissioning process through coordinated actions.

c. Members Appointed by the Government:

1. Commissioning Agent (CxA, See the CxA definition above): The designated person, company, or entity that plans, schedules, and coordinates the commissioning team to implement the commissioning process. The Government will engage the CxA under a separate contract.

2. User: Representatives of the facility user and operation and maintenance personnel. See the Roles & Responsibilities table below for more detailed responsibilities.

3. A/E: Representative of the Architect and engineering design professionals.

1.8 GOVERNMENT'S COMMISSIONING RESPONSIBILITIES

a. Appoint an individual, company or firm to act as the Commissioning Agent. See the Roles & Responsibilities table for details.

b. Assign operation and maintenance personnel and schedule them to participate in commissioning team activities including, but not limited to, the following:

1. Coordination meetings.

2. Training in operation and maintenance of systems, subsystems, and equipment.

3. Testing meetings.

4. Witness and assist in Systems Functional Performance Testing.

c. Provide the Construction Documents, prepared by Architect and approved by the Government, to the CxA and CxM, and for use in managing the commissioning process, developing the commissioning plan, systems manuals, and reviewing the operation and maintenance training plan.

1.9 CONTRACTOR'S COMMISSIONING RESPONSIBILITIES

a. The Contractor shall assign a qualified Commissioning Manager (CxM, See the CxM definition above) to manage commissioning activities of the team, Contractor, and subcontractors, as indicated in the Roles & Responsibilities table.

b. The Contractor shall ensure that the commissioning responsibilities outlined in these specifications are included in all subcontracts and that subcontractors comply with the requirements of these specifications.

c. The Contractor shall ensure that each installing subcontractor shall assign representatives with expertise and authority to act on behalf of the subcontractor and schedule them to participate in and perform commissioning team activities including, but not limited to, the following:

1. Participate in commissioning coordination meetings.

2. Conduct operation and maintenance training sessions in accordance with approved training plans.

3. Verify that Work is complete and systems are operational according to the Contract Documents, including calibration of instrumentation and controls.

4. Evaluate commissioning issues and commissioning observations identified in the Commissioning Issues Log, field reports, test reports or other commissioning documents. In collaboration with entity responsible for system and equipment installation, recommend corrective action.

5. Review and comment on commissioning documentation.

6. Participate in meetings to coordinate Systems Functional Performance Testing.

7. Provide schedule for operation and maintenance data submittals, equipment startup, and testing to Commissioning Agent for incorporation into the commissioning plan.

8. Provide information to the CxA and CxM for developing commissioning plan.

9. Participate in training sessions for VA's operation and maintenance personnel.

10. Provide technicians who are familiar with the construction and operation of installed systems and who shall develop specific test procedures to conduct Systems Functional Performance Testing of installed systems.

1.10 COMMISSIONING AGENT'S (CxA) RESPONSIBILITIES

a. Organize and lead the commissioning team. See the Roles & Responsibilities table below for more detailed responsibilities.

b. Prepare the commissioning plan. See Paragraph 1.11-A of this specification Section for further information.

c. Review and comment on selected submittals from the Contractor for general conformance with the Construction Documents. Review and comment on the ability to test and operate the system and/or equipment, including providing gages, controls and other components required to operate, maintain, and test the system. Review and comment on performance expectations of systems and equipment and interfaces between systems relating to the Construction Documents.

d. At the beginning of the construction phase, coordinate with CxM for and participate in an initial construction phase coordination meeting for the purpose of reviewing the commissioning activities and establishing tentative schedules for operation and maintenance submittals; operation and maintenance training sessions; TAB Work; Pre-Functional Checklists, Systems Functional Performance Testing; and project completion.

e. Refer to the Roles & Responsibilities table for meetings that are expected to be coordinated and led by the CxA throughout the entire project. For meetings that CxA is not indicated to be the Lead, the CxA shall proactively and collaboratively participate and support the Lead individual, including but not limited to, coordination with CxM in preparation of and careful review of the meeting agenda and minutes prior to distribution. For meetings that CxA is identified as the Lead, the CxA shall convene commissioning team meetings for the purpose of coordination, communication, and conflict resolution; discuss status of the commissioning processes. Responsibilities include arranging for facilities, preparing agenda and attendance lists, and notifying participants. The Commissioning Agent shall prepare and distribute minutes to commissioning team members and attendees within five workdays of the commissioning meeting.

f. Observe construction and report progress, observations and issues to the CxM for distribution. This is not intended to substitute the CxM's own construction observation and progress reporting. Observe systems and equipment installation for adequate accessibility for maintenance and component replacement or repair, and for general conformance with the Construction Documents.

g. Coordinate with the CxM in preparation of the Project specific Pre-Functional Checklists and Systems Functional Performance Test

procedures.

h. Collaborate with the CxM for the coordination of the Systems Functional Performance Testing schedule with the Contractor.

i. Witness selected systems startups.

j. Verify selected Pre-Functional Checklists completed and submitted by the Contractor.

k. Witness Systems Functional Performance Testing and ensure it is properly documented.

l. Coordinate with and support the CxM to ensure test data, inspection reports, and certificates are properly compiled and are included in the systems manual and commissioning report.

m. Review and comment on operation and maintenance (O&M) documentation and systems manual outline for commissioning impact and compliance with the Contract Documents.

n. Review operation and maintenance training program developed by the Contractor. Verify training plans provide qualified instructors to conduct operation and maintenance training.

o. Coordinate with and support the CxM to ensure commissioning Field Observation Reports are prepared and documented.

p. Coordinate with and support the CxM in preparation of the Final Commissioning Report.

q. Return to the site at 10 months into the 12 month warranty period and review with the CxM and facility staff the current building operation and the condition of outstanding issues related to the original and seasonal Systems Functional Performance Testing. Also join the CxM in interview facility staff and identify problems or concerns they have operating the building as originally intended. Make suggestions for improvements and for recording these changes in the O&M manuals. Identify areas that may come under warranty or under the original construction contract. Collaborate with the CxM and assist facility staff in developing reports, documents and requests for services to remedy outstanding problems.

r. Coordinate with and support the CxM in assembling the final commissioning documentation, including the Final Commissioning Report and Addendum to the Final Commissioning Report.

1.11 COMMISSIONING DOCUMENTATION

a. Commissioning Plan: A document, prepared by the CxA during the pre-design and design phases, and tailored for the project and finalized by the CxM during the construction phase with close collaboration with and agreement by the CxA (Note: The changes to the plan by the CxM shall be considered to be minor and limited to ONLY those necessary for fine tuning the document to better align with the contractor activities such as coordination with the subcontractors and vendors), that outlines the preliminary schedule, allocation of resources, and documentation requirements of the commissioning process, and shall include, but is not limited, to the following:

1. Plan for delivery and review of submittals, systems manuals, and other documents and reports. Identification of the relationship of these documents to other functions and a detailed description of submittals that are required to support the commissioning processes. Submittal dates shall include the latest date approved submittals must be received without adversely affecting commissioning plan.

2. Description of the organization, layout, and content of commissioning documentation (including systems manual) and a detailed description of documents to be provided along with identification of responsible parties.

3. Identification of systems and equipment to be commissioned.

4. Schedule of Commissioning Coordination meetings.

5. Identification of items that must be completed before the next operation can proceed.

6. Description of responsibilities of commissioning team members.

7. Description of observations to be made.

8. Description of requirements for operation and maintenance training.

9. Schedule for commissioning activities with dates coordinated with overall construction schedule.

10. Process and schedule for documenting changes on a continuous basis to appear in Project Record Documents.

11. Process and schedule for completing prestart and startup checklists for systems, subsystems, and equipment to be verified and tested.

12. Preliminary Systems Functional Performance Test procedures.

b. Systems Functional Performance Test Procedures: The CxA will develop Systems Functional Performance Test Procedures for each system to be commissioned, including subsystems, or equipment and interfaces or interlocks with other systems. Systems Functional Performance Test Procedures will include a separate entry, with space for comments, for each item to be tested. Preliminary Systems Functional Performance Test Procedures will be provided to the Contracting Officer, CxM, Architect/Engineer, and Contractor for review and comment. During the construction phase and with close collaboration with the CxA, the CxM will finalize the Systems Functional Performance Test Procedures. The Systems Performance Test Procedure will include test procedures for each mode of operation and provide space to indicate whether the mode under test responded as required. Each System Functional Performance Test procedure, regardless of system, subsystem, or equipment being tested, shall include, but not be limited to, the following:

1. Name and identification code of tested system.

2. Test number.

3. Time and date of test.

4. Indication of whether the record is for a first test or retest following correction of a problem or issue.

5. Dated signatures of the person performing test and of the witness, if applicable.

6. Individuals present for test.

7. Observations and Issues.

8. Issue number, if any, generated as the result of test.

c. Pre-Functional Checklists: The CxA will prepare Pre-Functional Checklists. Pre-Functional Checklists shall be finalized by the CxM during the construction phase and completed and signed by the Contractor, verifying that systems, subsystems, equipment, and associated controls are ready for testing. The CxA and CxM will spot check Pre-Functional Checklists to verify accuracy and readiness for testing. Inaccurate or incomplete Pre-Functional Checklists shall be returned to the Contractor for correction and resubmission.

d. Test and Inspection Reports: The CxM will record test data, observations, and measurements on Systems Functional Performance Test Procedure, while the CxA will participate and assist the CxM in the entire effort. The report will also include recommendation for system acceptance or non-acceptance. Photographs, forms, and other means appropriate for the application shall be included with data. The CxM will compile test and inspection reports and test and inspection certificates and include them in systems manual and commissioning report.

e. Corrective Action Documents: The CxM will document corrective action taken for systems and equipment that fail tests. The documentation will include any recommended and/or completed corrective actions to systems and equipment and/or revisions to test procedures, if any. The CxA will witness and collaborate with the CxM to ensure that any retesting of systems and/or equipment requiring corrective action, as well as the retest results are properly documented.

f. Commissioning Issues Log: The CxM will prepare and maintain Commissioning Issues Log that describes Commissioning Issues and Commissioning Observations that are identified during the Commissioning process. The CxA will participate by reviewing the Commissioning Issues Log to ensure it is up to date and accurate. These observations and issues include, but are not limited to, those that are at variance with the Contract Documents. The Commissioning Issues Log will identify and track issues as they are encountered, the party responsible for resolution, progress toward resolution, and document how the issue was resolved. The Master Commissioning Issues Log will also track the status of unresolved issues.

1. Creating a Commissioning Issues Log Entry:

i. Identify the issue with unique numeric or alphanumeric identifier by which the issue may be tracked.

ii. Assign a descriptive title for the issue.

iii. Identify date and time of the issue.

iv. Identify test number of test being performed at the time of the observation, if applicable, for cross reference.

v. Identify system, subsystem, and equipment to which the issue applies.

vi. Identify location of system, subsystem, and equipment.

vii. Include information that may be helpful in diagnosing or evaluating the issue.

viii. Note recommended corrective action.

ix. Identify commissioning team member responsible for corrective action.

x. Identify expected date of correction.

xi. Identify person that identified the issue.

2. Documenting Issue Resolution:

i. Log date correction is completed or the issue is resolved.

ii. Describe corrective action or resolution taken. Include description of diagnostic steps taken to determine root cause of the issue, if any.

iii. Identify changes to the Contract Documents that may require action.

iv. State that correction was completed and system, subsystem, and equipment are ready for retest, if applicable.

v. Identify person(s) who corrected or resolved the issue.

vi.Identify person(s) verifying the issue resolution.

g. Final Commissioning Report: The CxM will document results of the commissioning process, including unresolved issues, and performance of systems, subsystems, and equipment. The CxA will participate and assist the CxM in the entire effort. The Commissioning Report will indicate whether systems, subsystems, and equipment have been properly installed and are performing according to the Contract Documents. This report will be used by the Government when determining that systems will be accepted. This report will be used to evaluate systems, subsystems, and equipment and will serve as a future reference document during VA occupancy and operation. It shall describe components and performance that exceed requirements of the Contract Documents and those that do not meet requirements of the Contract Documents. The commissioning report will include, but is not limited to, the following:

1. Lists and explanations of substitutions, compromises, variances with the Contract Documents, record of conditions; and, if

appropriate, recommendations for resolution. Design Narrative documentation maintained by the AE, with input from the CxA and CxM.

2. Commissioning plan.

3. Pre-Functional Checklists completed by the Contractor, with annotation of the CxA and CxM review and spot check.

4. Systems Functional Performance Test Procedures, with annotation of test results and test completion.

5. Commissioning Issues Log.

6. Listing of deferred and off season test(s) not performed, including the schedule for their completion.

H. Addendum to Final Commissioning Report: The CxM will prepare an Addendum to the Final Commissioning Report near the end of the Warranty Period. The CxA will participate and assist the CxM in the entire effort. The Addendum will indicate whether systems, subsystems, and equipment are complete and continue to perform according to the Contract Documents. The Addendum to the Final Commissioning Report shall include, but is not limited to, the following:

1. Documentation of deferred and off season test(s) results.

2. Completed Systems Functional Performance Test Procedures for off season test(s).

3. Documentation that unresolved system performance issues have been resolved.

4. Updated Commissioning Issues Log, including status of unresolved issues.

5. Identification of potential Warranty Claims to be corrected by the Contractor.

I. Systems Manual: The CxM will gather required information and compile the Systems Manual. The CxA will participate by reviewing the content and completeness. The Systems Manual will include, but is not limited to, the following:

1. Design Narrative, including system narratives, schematics, single-line diagrams, flow diagrams, equipment schedules, and changes made throughout the Project.

2. Reference to Final Commissioning Plan.

3. Reference to Final Commissioning Report.

4. Approved Operation and Maintenance Data as submitted by the Contractor.

1.12 SUBMITTALS

a. Preliminary Commissioning Plan Submittal: The CxA has prepared a Preliminary Commissioning Plan based on the final Construction

Documents. The Preliminary Commissioning Plan is included as an Appendix to this specification section. The Preliminary Commissioning Plan is provided for information only. It contains preliminary information about the following commissioning activities:

1. The Commissioning Team: A list of commissioning team members by organization.

2. Systems to be commissioned. A detailed list of systems to be commissioned for the project. This list also provides preliminary information on systems/equipment submittals to be reviewed by the Commissioning Agent; preliminary information on Pre-Functional Checklists that are to be completed; preliminary information on Systems Performance Testing, including information on testing sample size (where authorized by the Government).

3. Commissioning Team Roles and Responsibilities: Preliminary roles and responsibilities for each Commissioning Team member.

4. Commissioning Documents: A preliminary list of commissioning-related documents, include identification of the parties responsible for preparation, review, approval, and action on each document.

5. Commissioning Activities Schedule: Identification of Commissioning Activities, including but not limited to the following:

i. Submission and approval of the Commissioning Firm and Commissioning Specialists such as CxM and CxR.

ii. Submission and approval of the Testing, Adjusting, and Balancing (TAB) Firm and TAB Specialist.

iii. Submission of the Design Review Reports.

iv. Submission and approval of the Construction Phase Commissioning Plan.

v. Installation of permanent utilities (gas, water, electric, etc.).

vi. Building Envelope Construction.

vii. Submission and approval of the Building Envelope Inspection Checklists.

viii. Air Barrier Pressure Tests.

ix. Drainage and Vent, Building Sewers, Water Supply Systems and Backflow Prevention Assembly Tests.

x. Factory Acceptance Testing for each of the systems to be commissioned as required.

xi. Manufacturer's Equipment Start-Up for each of the systems to be commissioned.

xii. Potable Water System Flushing.

xiii. Operational Tests of the plumbing system.

xiv. Potable Water System Disinfection.

xv. Submission and approval of the TAB Schematic Drawings, Report Forms, and Procedures.

xvi. Submission and approval of Duct Air Leakage Test Procedures.

xvii. Duct Air Leakage Test Execution.

xviii. Submission and approval of the Final Duct Air Leakage Test Report.

xix. Testing, Adjusting, and Balancing (TAB) Field Work.

xx. Submission and approval of the TAB Report.

xxi. TAB Field Acceptance Testing.

xxii. Submission and approval of the Start-Up Testing Report for HVAC Controls and Instrumentation.

xxiii. Submission and approval of the Performance Verification Test Procedures for HVAC Controls and Instrumentation.

xxiv. Performance Verification Tests for HVAC Controls and Instrumentation.

xxv. Performance Verification Test Report for HVAC Controls and Instrumentation.

xxvi. Pre-Functional Checklist Submittal.

xxvii. Functional Performance Testing for each system to be commissioned, the expected duration and predecessors for the activity.

xxviii. Integrated Systems Tests.

xxix. Post-Test Deficiency Correction for each system to be commissioned.

xxx. Re-Testing.

xxxi. Endurance Tests.

xxxii. Training for each of the systems to be commissioned.

xxxiii. Systems Manual, Maintenance Plan, and Service Life Plan submission and approval.

xxxiv. Seasonal Testing.

xxxv. Phasing.

6. Pre-Functional Checklists: Preliminary Pre-Functional

Checklists for equipment, components, subsystems, and systems to be commissioned provide guidance on the level of detailed information the Contractor shall include on the final submission.

7. Systems Functional Performance Test Procedures: Preliminary step-by-step System Functional Performance Test Procedures to be used during Systems Functional Performance Testing. These Preliminary Systems Functional Performance procedures provide information on the level of testing rigor, and the level of Contractor support required during performance of system's testing.

b. Commissioning and Final Commissioning Plans Submittals: Based on the Final Construction Documents, the CxA will prepare the Commissioning Plan as described in this section. During the construction phase, the CxM shall review and finalize the Commissioning Plan. The changes to the plan shall be minimal in nature as needed to tailor the plan for better alignment with the Contractor activities. All changes made to the Commissioning Plan by the CxM shall be reviewed by the CxA and approved by the Contracting Officer. The CxM will submit three hard copies and three sets of electronic files of the Final Commissioning Plan.

c. Systems Functional Performance Test Procedure: The CxM will submit preliminary Systems Functional Performance Test Procedures to the CxA, and the Contracting Officer for review and comment. The CxA shall return review comments to the Contracting Officer and the CxM. The Contracting Officer will also return review comments to the CxM. The CxM will incorporate review comments and re-issue the revised preliminary Systems Functional Performance Test Procedures. Based on the revised preliminary Systems Functional Performance Test Procedures, the CxM in close collaboration with the CxA will prepare and issue the Final Systems Functional Test Procedures to be used in Systems Functional Performance Testing. This document shall be reviewed by all parties as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table, and approved by Contracting Officer.

d. Pre-Functional Checklists: Similarly, the CxM will submit preliminary Pre-Functional Checklists to be reviewed by the CxA, after collection and incorporation of the review comments from all parties as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table and with approval from Contracting Officer.

e. Test and Inspection Reports: The CxM will submit test and inspection reports to the DVA/USACE with copies to the CxA and other parties as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table.

f. Corrective Action Documents: The CxM will submit corrective action documents to the Contracting Officer with copies to the CxA and other parties as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table.

g. Preliminary Commissioning Report Submittal: The CxM will submit four electronic copies of the preliminary commissioning report. One electronic copy and four hard copies, with consolidated review comments, will be prepared by the CxM for redistribution amongst the parties, and preparation of the final submittal.

h. Final Commissioning Report Submittal: The CxM will submit four sets of electronically formatted information of the final commissioning report to the Contracting Officer. The final submittal will incorporate comments as directed by the Contracting Officer.

I. Data for Commissioning:

1. The CxM, through coordination with Contracting Officer and CxA, will request in writing from the Contractor specific information needed about each piece of commissioned equipment or system to fulfill requirements of the Commissioning Plan.

2. The CxM, through coordination with Contracting Officer and CxA, shall request further documentation as is necessary for the commissioning process or to support other Government data collection requirements, including Construction Operations Building Information Exchange (COBIE), Building Information Modeling (BIM), etc.

1.13 COMMISSIONING PROCESS

a. During the Pre-Design and Design Phases, the CxA will be responsible for the overall management of the commissioning process as well as coordinating scheduling of commissioning tasks with the Government and the other parties outside of this contract as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table. During the Construction and Warranty Phases, the CxM, in close coordination and collaboration with CxA, has the overall management of the commissioning process as well as coordinating and scheduling of commissioning tasks with the Government and the other parties outside of this contract as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table. The CxA is responsible for government oversight of the commissioning process throughout all phases of the project. As directed by the Contracting Officer, the Contractor shall incorporate Commissioning tasks, including, but not limited to, Systems Functional Performance Testing (including predecessors) with the Master Construction Schedule.
Spec writer's note: Coordinate the number of days listed in the following paragraphs with the Contracting Officer.

b. Within 30 days of contract award, the Contractor shall provide a certified commissioning professional throu
gh an independent commissioning firm, see the CxM definition above, to act as the Commissioning Manager (CxM) to manage and lead the commissioning effort on behalf of the Contractor. The Commissioning Manager shall be the single point of contact and communications for all commissioning related services by the Contractor. The credentials of the commissioning firm and CxM shall be reviewed and approved by the Contracting Officer.

c. Within 30 days of contract award, the Contractor shall ensure that each subcontractor designates specific individuals as Commissioning Representatives (CXR) to be responsible for commissioning related tasks. The Contractor shall ensure the designated Commissioning Representatives participate in the commissioning process as team members providing commissioning testing services, equipment operation, adjustments, and corrections if necessary. The Contractor shall ensure that all Commissioning Representatives shall have sufficient authority to direct their respective staff to provide the services required, and

to speak on behalf of their organizations in all commissioning related contractual matters.

1.14 QUALITY ASSURANCE

a. Instructor Qualifications: Factory authorized service representatives shall be experienced in training, operation, and maintenance procedures for installed systems, subsystems, and equipment.

b. Test Equipment Calibration: The Contractor shall comply with test equipment manufacturer's calibration procedures and intervals. Recalibrate test instruments immediately whenever instruments have been repaired following damage or dropping. Affix calibration tags to test instruments. Instruments shall have been calibrated within six months prior to use.

1.15 COORDINATION

a. Management: The CxM will coordinate the commissioning activities with the Contracting Officer, CxA and Contractor. The CxM will submit commissioning documents and information to the Contracting Officer, CxA and other parties as noted in COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table. All commissioning team members shall work together to fulfill their contracted responsibilities and meet the objectives of the contract documents.

b. Scheduling: The Contractor shall work with the CxM, CxA and Contracting Officer to incorporate the commissioning activities into the construction schedule. The CxM, with close coordination with CxA, will provide sufficient information (including, but not limited to, tasks, durations and predecessors) on commissioning activities to allow the Contractor to schedule commissioning activities. All parties shall address scheduling issues and make necessary notifications in a timely manner in order to expedite the project and the commissioning process. The Contractor shall update the Master Construction Schedule as directed by the Contracting Officer.

c. Initial Schedule of Commissioning Events: The CxA will provide the initial schedule of primary commissioning events in the Preliminary Commissioning Plan. The Commissioning Plan will provide a format for this schedule. As construction progresses, more detailed schedules will be developed by the Contractor with information from the CxM and other parties.

d. Commissioning Coordinating Meetings: The CxM will conduct periodic Commissioning Coordination Meetings of the commissioning team to review status of commissioning activities, to discuss scheduling conflicts, and to discuss upcoming commissioning process activities.

e. Pretesting Meetings: The CxM will conduct pretest meetings of the commissioning team to review startup reports, Pre-Functional Checklist results, Systems Functional Performance Testing procedures, testing personnel and instrumentation requirements.

f. Systems Functional Performance Testing Coordination: The Contractor shall coordinate testing activities to accommodate required quality assurance and control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate

testing and inspecting. The Contractor shall coordinate the schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

a. The Contractor shall provide all standard and specialized testing equipment required to perform Systems Functional Performance Testing. Test equipment required for Systems Functional Performance Testing will be identified in the detailed System Functional Performance Test Procedure prepared by the CxM.

b. Data logging equipment and software required to test equipment shall be provided by the Contractor.

c. All testing equipment shall be of sufficient quality and accuracy to test and/or measure system performance with the tolerances specified in the Specifications. If not otherwise noted, the following minimum requirements apply: Temperature sensors and digital thermometers shall have a certified calibration within the past year to an accuracy of 0.5 oC (1.0 oF) and a resolution of + or - 0.1 Degrees C (0.2 Degrees F). Pressure sensors shall have an accuracy of + or - 2.0 percent of the value range being measured (not full range of meter) and have been calibrated within the last year. All equipment shall be calibrated according to the manufacturer's recommended intervals and following any repairs to the equipment. Calibration tags shall be affixed or certificates readily available.

PART 3 EXECUTION

3.1 COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES

a. The following table outlines the roles and responsibilities for the Commissioning Team members during the Pre-Design Phase (D-B-B) and RFP Development Phase (D-B):

Pre-Design Phase (D-B-B)/RFP Development (D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Category	Task Description	CxA	CO	A/E	CxM	O&M	Notes

Pre-Design Phase (D-B-B)/RFP Development (D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Meetings	Pre-Design Kick-Off Meeting	P	P	L	N/A	P	
	Owner's Project Requirements (OPR) Meeting	P	P	L	N/A	P	
	Commissioning Planning Meetings	L	P	P	N/A	O	
Coordination	Coordinate with A/E, CO, AHJ, Vendors, etc. to ensure that Cx is incorporated into project planning and documents.	L	P	P	N/A	O	
Cx Plan & Spec	Draft Commissioning Plan	L	A	R	N/A	R	
Schedules	Preliminary Commissioning Schedule	L	A	R	N/A	R	
OPR and BOD	Prepare OPR	L	P	P	N/A	R	
	Prepare BOD	L	P	P	N/A	R	
Commissioning Documents	Identify systems to be commissioned.	L	P	P	N/A	R	
	Preliminary Acceptance Criteria	L	P	P	N/A	R	
	Commissioning Documents Preliminary Templates	L	P	P	N/A	R	

b. The following table outlines the roles and responsibilities for the Commissioning Team members during the Bid Phase (D-B):

Bid Phase		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Category	Task Description	CxA	CO	A/E	CxM	O&M	Notes
Pre-Bid Meeting	Organize, schedule and conduct meeting	P	L	P	N/A	O	

c. The following table outlines the roles and responsibilities for the Commissioning Team members during the Design Phase (D-B-B):

Design Phase (D-B-B)/RFP Development (D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Category	Task Description	CxA	CO	A/E	CxM	O&M	Notes
Meetings	Design Review Meetings	P	P	L	N/A	P	
	Controls Meeting	L	P	P	N/A	P	

Design Phase (D-B-B)/RFP Development (D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Coordination	Coordinate with A/E, CO, AHJ, Vendors, etc. to ensure that Cx interacts properly with other systems as needed to support the OPR and BOD	L	P	P	N/A	O	
Cx Plan & Spec	Commissioning Plan	L	A	R	N/A	R	
	Cx Specifications	L	A	R	N/A	R	
Schedules	Design Phase Preliminary Commissioning schedule	L	A	R	N/A	R	
OPR and BOD	Maintain OPR on behalf of Owner	P	A	L	N/A	P	
	Review BOD vs. OPR	L	A	P	N/A	P	
	Maintain BOD on behalf of Owner	P	A	L	N/A	R	

Design Phase (D-B-B)/RFP Development (D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Reviews	Focused Schematic (Concept)Designs (SDs)	P	L	P	N/A	P	
	Focused Schematic(Concept) Designs (SDs)Comments Resolution	P	L	P	N/A	P	
	Focused Design Developments (DDs)	P	L	P	N/A	P	
	Focused Design Developments (DDs)Comments Resolution	P	L	P	N/A	P	
	Focused Construction Documents (CDs)	P	L	P	N/A	P	
	Focused Construction Documents (CDs)Comments Resolution	P	L	P	N/A	P	
	Focused Pre-Final Construction Document	P	L	P	N/A	P	
	Focused Pre-Final Construction Document Comments Resolution	P	L	P	N/A	P	
	Focused Final Construction Document	P	L	P	N/A	P	
	Final Construction Document Comments Backcheck	P	L	P	N/A	P	

Design Phase (D-B-B)/RFP Development (D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Functional Test Protocols	Draft Pre-Functional Checklists (PFCs)	L	A	R	N/A	R	
	Draft System Functional Performance Tests (FPTs)	L	A	R	N/A	R	

d. The following table outlines the roles and responsibilities for the Commissioning Team members during the Design Phase (D-B):
e. The following table outlines the roles and responsibilities for the Commissioning Team members during the Bid and Pre-Construction Phase (D-B-B):

Bid and Pre-Construction Phase (D-B-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities							
Category	Task Description	CxA	CO	A/E	CxM	O&M	Notes
Pre-Bid Meeting	Organize, schedule and conduct meeting	P	L	P	P	O	
Pre-Construction Meeting	Organize, schedule and conduct meeting	P	L	P	P	O	

f. The following table outlines the roles and responsibilities for the Commissioning Team members during the Construction Phase:

Construction Phase(D-B-B & D-B)		CxA = Commissioning Agent						L = Lead
		CO = Contracting Officer						P = Participate
Commisssioning Roles & Responsibilities		A/E = Design Arch/Engineer						A = Approve
		CxM = Commissioning Manager						R = Review
		O&M = VA Facility O&M						O = Optional
Category	Task Description	CxA	CO	A/E	CxM	RE	O&M	Notes
Meetings	Construction Commissioning Kick Off meeting	P	P	P	L	P	P	
	Commissioning Meetings	P	P	P	L	P	O	
	Project Progress Meetings	P	P	P	L	P	O	
	Controls Meeting	P	P	P	L	P	P	
Coordination	Coordinate with Contractor, A/E, CO, AHJ, Vendors, etc. to ensure that Cx interacts properly with other systems as needed to support the OPR and BOD	P	P	P	L	P	P	
Cx Plan & Spec	Final Commissioning Plan	P	A	R	L	R	R	
Schedules	Duration Schedule for Commissioning Activities	P	A	R	L	R	N/A	

Construction Phase(D-B-B & D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M						L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities								
OPR and BOD	Maintain OPR on behalf of Owner	L	A	R	R	R	O	
	Maintain BOD/DID on behalf of Owner	P	A	L	P	R	O	
Document Reviews	TAB Plan Review	P	A	R	L	R	O	
	Submittal and Shop Drawing Review	P	A	R	L	R	O	
	Review Contractor Equipment Startup Checklists	P	A	O	L	R	O	
	Review Change Orders, ASIs, and RFIs	P	A	R	L	R	O	
Site Observations	Witness Factory Testing	P	A	O	L	O	O	
	Construction Observation Site Visits	P	A	O	L	P	P	
Functional Test Protocols	Finalize Pre-Functional Checklists	P	A	R	L	R	O	
	Finalize Functional Performance Test Protocols	P	A	R	L	R	O	

Construction Phase(D-B-B & D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M						L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities								
Technical Activities	Issues Resolution Meetings	P	A	P	L	P	O	
Reports and Logs	Status Reports	P	A	R	L	R	O	
	Maintain Commissioning Issues Log	p	A	R	L	R	O	

g. The following table outlines the roles and responsibilities for the Commissioning Team members during the Acceptance Phase:

Acceptance Phase (D-B-B & D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M						L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities								
Category	Task Description	CxA	CO	A/E	CxM	RE	O&M	Notes

Acceptance Phase (D-B-B & D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M						L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities								
Meetings	Commissioning Meetings	P	P	P	L	P	P	
	Project Progress Meetings	P	P	P	L	P	P	
	Pre-Test Coordination Meeting	P	P	P	L	P	P	
	Lessons Learned and Commissioning Report Review Meetings	P	P	P	L	P	P	
Coordination	Coordinate with Contractor, A/E, CO, AHJ, Vendors, etc. to ensure that Cx interacts properly with other systems as needed to support OPR and BOD	P	P	P	L	P	P	
Cx Plan	Maintain/Update Commissioning Plan	P	A	R	L	R	R	
Schedules	Prepare Functional Test Schedule	P	A	R	L	R	N/A	

Acceptance Phase (D-B-B & D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M					L = Lead P = Participate A = Approve R = Review O = Optional	
Commisssioning Roles & Responsibilities								
OPR and BOD	Maintain OPR on behalf of Owner	L	A	R	R	R	O	
	Maintain BOD/DID on behalf of Owner	P	A	L	P	R	O	
Document Reviews	Review Completed Pre-Functional Checklists	P	A	R	L	R	O	
	Pre-Functional Checklist Verification	P	A	R	L	R	O	
	Review Operations & Maintenance Systems Manuals	P	A	R	L	R	P	
	Training Plan Review	P	A	R	L	R	P	
	Warranty Review	R	P	R	L	R	P	
	Review TAB Report	R	A	R	L	R	P	
Site Observations	Construction Observation Site Visits	P	P	P	L	P	P	
	Witness Selected Equipment Startup	P	P	P	L	P	P	

Acceptance Phase (D-B-B & D-B)		CxA = Commissioning Agent CO = Contracting Officer A/E = Design Arch/Engineer CxM = Commissioning Manager O&M = VA Facility O&M						L = Lead P = Participate A = Approve R = Review O = Optional
Commisssioning Roles & Responsibilities								
Functional Test Protocols	TAB Verification	P	A	R	L	R	O	
	Systems Functional Performance Testing	P	A	P	L	P	P	
	Retesting	P	A	P	L	P	P	
Technical Activities	Issues Resolution Meetings	P	P	P	L	P	P	
	Systems Training	P	A	R	L	R	P	
Reports and Logs	Status Reports	R	A	R	L	R	O	
	Maintain Commissioning Issues Log	P	A	R	L	R	O	
	Final Commissioning Report	P	A	R	L	R	R	
	Prepare Systems Manuals	P	A	R	L	R	R	

h. The following table outlines the roles and responsibilities for the Commissioning Team members during the Warranty Phase:

Warranty Phase (D-B-B & D-B)		CxA = Commissioning Agent						L = Lead
		CO = Contracting Officer						P = Participate
Commisssioning Roles & Responsibilities		A/E = Design Arch/Engineer						A = Approve
		CxM = Commissioning Manager						R = Review
		O&M = VA Facility O&M						O = Optional
Category	Task Description	CxA	CO	A/E	CxM	RE	O&M	Notes
Meetings	Commissioning Meetings	P	P	O	L	P	P	
Document Reviews	Systems Monitoring Trended Data	P	A	O	L	R	P	
	Systems Manual Update	P	A	O	L	R	R	
Site Observations	Periodic Warranty Site Visits (4 & 9 Month)	P	P	O	L	P	P	
Functional Test Protocols	Deferred and/or seasonal Testing	P	P	O	L	P	P	
Technical Activities	Commissioning/Issu Resolution/Lessons Learned Meetings	P	P	O	L	P	P	
	Post-Occupancy Warranty Checkup and review of Significant Outstanding Issues	P	P	O	L	P	P	
Reports and Logs	Final Commissioning Report Amendment	P	A	R	L	R	R	
	Status Reports	R	A	R	L	R	O	
	Issues Logs Closure Report	P	A	R	L	R	O	

3.2 STARTUP, INITIAL CHECKOUT, AND PRE-FUNCTIONAL CHECKLISTS

a. The following procedures shall apply to all equipment and systems to be commissioned, according to Part 1, Systems to Be Commissioned.

1. Pre-Functional Checklists are important to ensure that the equipment and systems are hooked up and operational. These ensure

that Systems Functional Performance Testing may proceed without unnecessary delays. Each system to be commissioned shall have a full Pre-Functional Checklist completed by the Contractor prior to Systems Functional Performance Testing. No sampling strategies are used.

i. The Pre-Functional Checklist will identify the trades responsible for completing the checklist. The Contractor shall ensure the appropriate trades complete the checklists.

ii. The CxM will review completed Pre-Functional Checklists and field-verify the accuracy of the completed checklist using sampling techniques. The CxA and other parties as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table will also review the completed Pre-Functional Checklists for accuracy.

2. Startup and Initial Checkout Plan: The Contractor shall develop detailed startup plans for all equipment. The primary role of the Contractor in this process is to ensure that there is written documentation that each of the manufacturer recommended procedures have been completed. Parties responsible for startup shall be identified in the Startup Plan and in the checklist forms.

i. The Contractor shall develop the full startup plan by combining (or adding to) the checklists with the manufacturer's detailed startup and checkout procedures from the O&M manual data and the field checkout sheets normally used by the Contractor. The plan shall include checklists and procedures with specific boxes or lines for recording and documenting the checking and inspections of each procedure and a summary statement with a signature block at the end of the plan.

ii. The full startup plan shall at a minimum consist of the following items:

1) The Pre-Functional Checklists.

2) The manufacturer's standard written startup procedures copied from the installation manuals with check boxes by each procedure and a signature block added by hand at the end.

3) The manufacturer's normally used field checkout sheets.

iii. The CxM will submit the full startup plan to the Contracting Officer, CxA, Contractor, and other parties as noted in the COMMISSIONING PROCESS ROLES AND RESPONSIBILITIES table for review. Final approval will be by the DVA/USACE.

iv. The Contractor shall review and evaluate the procedures and the format for documenting them, noting any procedures that need to be revised or added.

3. Sensor and Actuator Calibration

i. All field installed temperature, relative humidity, CO2 and pressure sensors and gages, and all actuators (dampers and valves) on all equipment shall be calibrated using the methods described in Division 21, Division 22, Division 23, Division 26, Division 27, and Division 28 specifications.

ii. All procedures used shall be fully documented on the Pre-Functional Checklists or other suitable forms, clearly referencing the procedures followed and written documentation of initial, intermediate and final results.

4. Execution of Equipment Startup

i. Four weeks prior to equipment startup, the Contractor shall schedule startup and checkout with the Contracting Officer and Commissioning Agent. The performance of the startup and checkout shall be directed and executed by the Contractor.

ii. The CxM will observe the startup procedures for selected pieces of primary equipment. The CxA and other Government representative(s) may choose to participate in some or all of the startup procedures as observers.

iii. The Contractor shall execute startup and provide the Contracting Officer, CxM and CxA with a signed and dated copy of the completed startup checklists, and contractor tests.

iv. Only individuals that have direct knowledge and witnessed that a line item task on the Startup Checklist was actually performed shall initial or check that item off. It is not acceptable for witnessing supervisors to fill out these forms.

3.3 DEFICIENCIES, NONCONFORMANCE, AND APPROVAL IN CHECKLISTS AND STARTUP

a. The Contractor shall clearly list any outstanding items of the initial startup and Pre-Functional Checklist procedures that were not completed successfully, at the bottom of the procedures form or on an attached sheet. The procedures form and any outstanding deficiencies shall be provided to the CxM, CxA and Government within two days of completion.

b. The CxM and CxA will review the report and submit comments to the Contracting Officer. The CxM will work with the Contractor to correct and verify deficiencies or uncompleted items. The CxM will involve the CxA, Contracting Officer and others as necessary. The Contractor shall correct all areas that are noncompliant or incomplete in the checklists in a timely manner, and shall notify the Contracting Officer, CxM and CxA as soon as outstanding items have been corrected. The Contractor shall submit an updated startup report and a Statement of Correction on the original noncompliance report. When satisfactorily completed, the CxM in collaboration with CxA will recommend approval of the checklists and startup of each system to the Contracting Officer.

c. The Contractor shall be responsible for resolution of deficiencies as directed the Contracting Officer.

3.4 DDC SYSTEM TRENDING FOR COMMISSIONING

a. Trending is a method of testing as a standalone method or to augment manual testing. The Contractor shall trend any and all points of the system or systems at intervals specified below.

b. Alarms are a means to notify the system operator that abnormal

conditions are present in the system. Alarms shall be structured into three tiers - Critical, Priority, and Maintenance.

1. Critical alarms are intended to be alarms that require the immediate attention of and action by the Operator. These alarms shall be displayed on the Operator Workstation in a popup style window that is graphically linked to the associated unit's graphical display. The popup style window shall be displayed on top of any active window within the screen, including non DDC system software.

2. Priority level alarms are to be printed to a printer which is connected to the Operator's Work Station located within the engineer's office. Additionally, Priority level alarms shall be able to be monitored and viewed through an active alarm application. Priority level alarms are alarms which shall require reaction from the operator or maintenance personnel within a normal work shift, and not immediate action.

3. Maintenance alarms are intended to be minor issues which would require examination by maintenance personnel within the following shift. These alarms shall be generated in a scheduled report automatically by the DDC system at the start of each shift. The generated maintenance report will be printed to a printer located within the engineer's office.

c. The Contractor shall provide a wireless internet network in the building for use during controls programming, checkout, and commissioning. This network will allow project team members to more effectively program, view, manipulate and test control devices while being in the same room as the controlled device.

d. The Contractor shall provide graphical trending through the DDC control system of systems being commissioned. Trending requirements are indicated below and included with the Systems Functional Performance Test Procedures. Trending shall occur before, during and after Systems Functional Performance Testing. The Contractor shall be responsible for producing graphical representations of the trended DDC points that show each system operating properly during steady state conditions as well as during the System Functional Testing. These graphical reports shall be submitted to the Contracting Officer, CxM and CxA for review and analysis before, during dynamic operation, and after Systems Functional Performance Testing. The Contractor shall provide, but not limited to, the following trend requirements and trend submissions:

1. Pre-testing, Testing, and Post-testing - Trend reports of trend logs and graphical trend plots are required as defined by the CxA and CxM. The trend log points, sampling rate, graphical plot configuration, and duration will be dictated by the CxA and CxM. At any time during the Commissioning Process the CxA and CxM may recommend changes to aspects of trending as deemed necessary for proper system analysis. The Contractor shall implement any changes as directed by the Contracting Officer. Any pre-test trend analysis comments generated by the Commissioning Team should be addressed and resolved by the Contractor, as directed by the Contracting Officer, prior to the execution of Systems Functional Performance Testing.

2. Dynamic plotting - The Contractor shall also provide dynamic plotting during Systems Functional Performance testing at frequent intervals for points determined by the Systems Functional Performance Test Procedure. The graphical plots will be formatted and plotted at durations listed in the Systems Functional Performance Test Procedure.

3. Graphical plotting - The graphical plots shall be provided with a dual y-axis allowing 15 or more trend points (series) plotted simultaneously on the graph with each series in distinct color. The plots will further require title, axis naming, legend etc. all described by the Systems Functional Performance Test Procedure. If this cannot be sufficiently accomplished directly in the Direct Digital Control System then it is the responsibility of the Contractor to plot these trend logs in Microsoft Excel.

4. The following tables indicate the points to be trended and alarmed by system. The Operational Trend Duration column indicates the trend duration for normal operations. The Testing Trend Duration column indicates the trend duration prior to Systems Functional Performance Testing and again after Systems Functional Performance Testing. The Type column indicates point type: AI = Analog Input, AO = Analog Output, DI = Digital Input, DO = Digital Output, Calc = Calculated Point. In the Trend Interval Column, COV = Change of Value. The Alarm Type indicates the alarm priority; C = Critical, P = Priority, and M = Maintenance. The Alarm Range column indicates when the point is considered in the alarm state. The Alarm Delay column indicates the length of time the point must remain in an alarm state before the alarm is recorded in the DDC. The intent is to allow minor, short-duration events to be corrected by the DDC system prior to recording an alarm.

Dual-Path Air Handling Unit Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operationa Trend Duration**	**Testing Trend Duration**	**Alarm TYpe**	**Alarm Range**	**Alarm Delay**
OA Temperature	AI	15 Min	24 hours	3 days	N/A		
RA Temperature	AI	15 Min	24 hours	3 days	N/A		
RA Humidity	AI	15 Min	24 hours	3 days	P	>60% RH	10 min
Mixed Air Temp	AI	None	None	None	N/A		
SA Temp	AI	15 Min	24 hours	3 days	C	±5°F from SP	10 min
Supply Fan Speed	AI	15 Min	24 hours	3 days	N/A		
Return Fan Speed	AI	15 Min	24 hours	3 days	N/A		
RA Pre-Filter Status	AI	None	None	None	N/A		

Dual-Path Air Handling Unit Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operationa Trend Duration**	**Testing Trend Duration**	**Alarm TYpe**	**Alarm Range**	**Alarm Delay**
OA Pre-Filter Status	AI	None	None	None	N/A		
After Filter Status	AI	None	None	None	N/A		
SA Flow	AI	15 Min	24 hours	3 days	C	±10% from SP	10 min
OA Supply Temp	AI	15 Min	24 hours	3 days	P	±5°F from SP	10 min
RA Supply Temp	AI	15 Min	24 hours	3 days	N/A		
RA CHW Valve Position	AI	15 Min	24 hours	3 days	N/A		
OA CHW Valve Position	AI	15 Min	24 hours	3 days	N/A		
OA HW Valve Position	AI	15 Min	24 hours	3 days	N/A		
OA Flow	AI	15 Min	24 hours	3 days	P	10% from SP	5 min
RA Flow	AI	15 Min	24 hours	3 days	P	10% from SP	5 min
Initial UVC Intensity (%)	AI	None	None	None	N/A		
Duct Pressure	AI	15 Min	24 hours	3 days	C	25% from SP	6 min
CO2 Level	AI	15 Min	24 hours	3 days	P	10% from SP	10 min
Supply Fan Status	DI	COV	24 hours	3 days	C	Status <> Command	10 min
Return Fan Status	DI	COV	24 hours	3 days	C	Status <> Command	10 min
High Static Status	DI	COV	24 hours	3 days	P	True	1 min
Fire Alarm Status	DI	COV	24 hours	3 days	C	True	5 min
Freeze Stat Level 1	DI	COV	24 hours	3 days	C	True	10 min
Freeze Stat Level 2	DI	COV	24 hours	3 days	C	True	5 min

Dual-Path Air Handling Unit Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operationa Trend Duration**	**Testing Trend Duration**	**Alarm TYpe**	**Alarm Range**	**Alarm Delay**
Freeze Stat Level 3	DI	COV	24 hours	3 days	P	True	1 min
Fire/Smoke Damper Status	DI	COV	24 hours	3 days	P	Closed	1 min
Emergency AHU Shutdown	DI	COV	24 hours	3 days	P	True	1 min
Exhaust Fan Status	DI	COV	24 hours	3 days	C	Status <> Command	10 min
OA Alarm	DI	COV	24 hours	3 days	C	True	10 min
High Static Alarm	DI	COV	24 hours	3 days	C	True	10 min
UVC Emitter Alarm	DI	COV	24 hours	3 days	P	True	10 min
CO2 Alarm	DI	COV	24 hours	3 days	P	True	10 min
Power Failure	DI	COV	24 hours	3 days	P	True	1 min
Supply Fan Speed	AO	15 Min	24 hours	3 days	N/A		
Return Fan Speed	AO	15 Min	24 hours	3 days	N/A		
RA CHW Valve Position	AO	15 Min	24 hours	3 days	N/A		
OA CHW Valve Position	AO	15 Min	24 hours	3 days	N/A		
OA HW Valve Position	AO	15 Min	24 hours	3 days	N/A		
Supply Fan S/S	DO	COV	24 hours	3 days	N/A		
Return Fan S/S	DO	COV	24 hours	3 days	N/A		
Fire/Smoke Dampers	DO	COV	24 hours	3 days	N/A		
Exhaust Fan S/S	DO	COV	24 hours	3 days	N/A		
Exhaust Fan S/S	DO	COV	24 hours	3 days	N/A		

Dual-Path Air Handling Unit Trending and Alarms

Point	Type	Trend Interval	Operationa Trend Duration	Testing Trend Duration	Alarm TYpe	Alarm Range	Alarm Delay
Exhaust Fan S/S	DO	COV	24 hours	3 days	N/A		
AHU Energy	Calc	1 Hour	30 days	N/A	N/A		

Terminal Unit (VAV, CAV, etc.) Trending and Alarms

Point	Type	Trend Interval	Operationa Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Min	12 hours	3 days	P	±5°F from SP	
Air Flow	AI	15 Min	12 hours	3 days	P	±5°F from SP	
SA Temperature	AI	15 Min	12 hours	3 days	P	±5°F from SP	
Local Setpoint	AI	15 Min	12 hours	3 days	M	±10°F from SP	
Space Humidity	AI	15 Min	12 hours	3 days	P	> 60% RH	
Unoccupied Override	DI	COV	12 hours	3 days	M	N/A	
Refrigerator Alarm	DI	COV	12 hours	3 days	C	N/A	
Damper Position	AO	15 Minutes	12 hours	3 days	N/A		
Heating coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		

4-Pipe Fan Coil Trending and Alarms

Point	Type	Trend Interval	Operationa Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Minutes	12 hours	3 days	P	5F from SP	10 min
SA Temperature	AI	15 Minutes	12 hours	3 days	P	5F from SP	10 min
Pre-Filter Status	AI	None	None	None	M	> SP	1 hour

4-Pipe Fan Coil Trending and Alarms

Point	Type	Trend Interval	Operationa Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Water Sensor	DI	COV	12 hours	3 days	M	N/A	30 min
Cooling Coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		
Heating coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		
Fan Coil ON/OFF	DO	COV	12 hours	3 days	M	Status <> Command	30 min

2-Pipe Fan Coil Trending and Alarms

Point	Type	Trend Interval	Operationa Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Minutes	12 hours	3 days	P	5F from SP	10 min
SA Temperature	AI	15 Minutes	12 hours	3 days	P	5F from SP	10 min
Pre-Filter Status	AI	None	None	None	M	> SP	1 hour
Water Sensor	DI	COV	12 hours	3 days	M	N/A	30 min
Cooling Coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		
Heating coil Valve Position	AO	15 Minutes	12 hours	3 days	N/A		
Fan Coil ON/OFF	DO	COV	12 hours	3 days	M	Status <> Command	30 min

Unit Heater Trending and Alarms

Point	Type	Trend Interval	Operationa Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Space Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 min
Heating Valve Position	AO	15 Minutes	12 hours	3 days	N/A		

Unit Heater Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operational Trend Duration**	**Testing Trend Duration**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
Unit Heater ON/OFF	DO	COV	12 hours	3 days	M	Status <> Command	30 min
Steam and Condensate Pumps Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operational Trend Duration**	**Testing Trend Duration**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
Steam Flow (LB/HR)	AI	15 Minutes	12 hours	3 days	N/A		
Condensate Pump Run Hours	AI	15 Minutes	12 hours	3 days	N/A		
Water Meter (GPM)	AI	15 Minutes	12 hours	3 days	N/A		
Electric Meter (KW/H)	AI	15 Minutes	12 hours	3 days	N/A		
Irrigation Meter (GPM)	AI	15 Minutes	12 hours	3 days	N/A		
Chilled Water Flow (TONS)	AI	15 Minutes	12 hours	3 days	N/A		
Condensate Flow (GPM)	AI	15 Minutes	12 hours	3 days	N/A		
High Water Level Alarm	DI	COV	12 hours	3 days	C	True	5 min
Condensate Pump Start/Stop	DO	COV	12 hours	3 days	P	Status <> Command	10 min
Domestic Hot Water Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operational Trend Duration**	**Testing Trend Duration**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
Domestic Hot Water Trending and Alarms	AI	15 Minutes	12 hours	3 days	N/A		

Domestic Hot Water Trending and Alarms

Point	Type	Trend Interval	Operational Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Domestic HW Setpoint	AI	15 Minutes	12 hours	3 days	N/A		
Domestic HW Temperature	AI	15 Minutes	12 hours	3 days	C	> 135°F	10 Min
Domestic HW Temperature	AI	15 Minutes	12 hours	3 days	P	±5°F from SP	10 Min
Dom. Circ. Pump Status	DI	COV	12 hours	3 days	M	Status <> Command	30 min
Dom. Circ. Pump Start/Stop	DO	COV	12 hours	3 days	N/A		
Domestic HW Start/Stop	DO	COV	12 hours	3 days	N/A		

Hydronic Hot Water Trending and Alarms

Point	Type	Trend Interval	Operational Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
System HWS Temperature	AI	15 min	12 hours	3 days	C	±5°F from SP	10 Min
System HWR Temperature	AI	15 min	12 hours	3 days	M	±5°F from SP	300 Min
HX Entering Temperature	AI	15 min	12 hours	3 days	P	±5°F from SP	10 Min
HX Leaving Temperature	AI	15 min	12 hours	3 days	P	±5°F from SP	10 Min
System Flow (GPM)	AI	15 min	12 hours	3 days	N/A		
System Differential Pressure	AI	15 min	12 hours	3 days	P	±10% from SP	8 min
HW Pump Status	DI	COV	12 hours	3 days	C	Status <> Command	30 min

Hydronic Hot Water Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operational Trend Duration**	**Testing Trend Duration**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
HW Pump VFD Speed	AO	15 min	12 hours	3 days	N/A		
Steam Station 1/3 Control Valve Position	AO	15 min	12 hours	3 days	N/A		
Steam Station 2/3 Control Valve Position	AO	15 min	12 hours	3 days	N/A		
Steam Station Bypass Valve Position	AO	15 min	12 hours	3 days	N/A		
HW Pump Start/Stop	DO	COV	12 hours	3 days	N/A		
HWR Valve	DO	COV	12 hours	3 days	N/A		

Chilled Water System Trending and Alarms							
Point	**Type**	**Trend Inter**	**Operat Trend Durati**	**Testing Trend Duratio**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
Chiller Entering Temperature	AI	15 Minut	12 Hours	3 days	N/A		
Chiller Leaving Temperature	AI	15 Minut	12 Hours	3 days	P	±5°F from SP	10 Min
Chiller Flow	AI	15 Minut	12 Hours	3 days	N/A		
Chiller Percent Load	AI	15 Minut	12 Hours	3 days	N/A		
Chiller KW Consumption	AI	15 Minut	12 Hours	3 days	N/A		
Chiller Tonnage	AI	15 Minut	12 Hours	3 days	N/A		

Chilled Water System Trending and Alarms							
Point	**Type**	**Trend Inter**	**Operat Trend Durati**	**Testing Trend Duratio**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
Primary Loop Decoupler Flow	AI	15 Minut	12 Hours	3 days	N/A		
Primary Loop Flow	AI	15 Minut	12 Hours	3 days	N/A		
Primary Loop Supply Temperature	AI	15 Minut	12 Hours	3 days	N/A		
Secondary Loop Differential Pressure	AI	15 Minut	12 Hours	3 days	P	±5°F from SP	10 Min
Secondary Loop Flow	AI	15 Minut	12 Hours		N/A		
Secondary Loop Supply Temperature	AI	15 Minut	12 Hours	3 days	N/A		
Secondary Loop Return Temperature	AI	15 Minut	12 Hours	3 days	N/A		
Secondary Loop Tonnage	AI	15 Minut	12 Hours	3 days	N/A		
Primary Loop Pump Status	DI	COV	12 Hours	3 days	C	Status <> Command	30 min
Secondary Loop Pump Status	DI	COV	12 Hours	3 days	C	Status <> Command	30 min
Chiller Status	DI	COV	12 Hours	3 days	C	Status <> Command	30 min
Chiller Evaporator Iso-Valve	DI	COV	12 Hours	3 days	N/A		
Chiller Evaporator Flow Switch	DI	COV	12 Hours	3 days	N/A		
Chiller Unit Alarm	DI	COV	12 Hours	3 days	C	True	10 Min
Refrigerant Detector	DI	COV	12 Hours	3 days	C	True	10 Min
Refrigerant Exhaust Fan Status	DI	COV	12 Hours	3 days	M	Status <> Command	30 min
Emergency Shutdown	DI	COV	12 Hours	3 days	F	True	1 Min

Chilled Water System Trending and Alarms

Point	Type	Trend Inter	Operat Trend Durati	Testing Trend Duratio	Alarm Type	Alarm Range	Alarm Delay
Primary Loop Pump VFD Speed	AO	15 Minut	12 Hours	3 days	N/A		
Secondary Loop Pump VFD Speed	AO	15 Minut	12 Hours	3 days	N/A		
Primary Pump Start / Stop	DO	COV	12 Hours	3 days	N/A		
Secondary Pump Start / Stop	DO	COV	12 Hours	3 days	N/A		
Chiller Enable	DO	COV	12 Hours	3 days	N/A		
Chiller Iso-Valve Command	DO	COV	12 Hours	3 days	N/A		
Refrigerant Exhaust Fan Start / Stop	DO	COV	12 Hours	3 days	N/A		

Condenser Water System Trending and Alarms

Point	Type	Trend Interval	Operational Trend Duration	Testing Trend Duration	Alarm Type	Alarm Range	Alarm Delay
Chiller Condenser Entering Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Chiller Condenser Leaving Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Supply Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Return Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Basin Temp	AI	15 Minutes	12 Hours	3 days	P	< 45 oF	10 Min
Condenser Water Supply Temp	AI	15 Minutes	12 Hours	3 days	N/A		

Condenser Water System Trending and Alarms							
Point	**Type**	**Trend Interval**	**Operational Trend Duration**	**Testing Trend Duration**	**Alarm Type**	**Alarm Range**	**Alarm Delay**
Condenser Water Return Temp	AI	15 Minutes	12 Hours	3 days	N/A		
Outdoor Air Wet Bulb	AI	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Fan Status	DI	COV	12 Hours	3 days	P	Status <> Command	1 min
Cooling Tower Basin Heat	DI	COV	12 Hours	3 days	N/A		
Cooling Tower Heat Trace	DI	COV	12 Hours	3 days	N/A		
Chiller Isolation Valve	DI	COV	12 Hours	3 days	P	Status <> Command	1 min
Condenser Water Pump Status	DI	COV	12 Hours	3 days	P	Status <> Command	1 min
Chiller Condenser Bypass Valve	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Bypass Valve	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Fan Speed	AO	15 Minutes	12 Hours	3 days	N/A		
Cooling Tower Fan Start / Stop	DO	COV	12 Hours	3 days	N/A		
Condenser Water Pump Start / Stop	DO	COV	12 Hours	3 days	N/A		

e. The Contractor shall provide the following information prior to Systems Functional Performance Testing. Any documentation that is modified after submission shall be recorded and resubmitted to the Contracting Officer, CxM and CxA.

1. Point-to-Point checkout documentation;

2. Sensor field calibration documentation including system name, sensor/point name, measured value, DDC value, and Correction Factor.

3. A sensor calibration table listing the referencing the location of procedures to following in the O&M manuals, and the frequency at which calibration should be performed for all sensors, separated by system, subsystem, and type. The calibration requirements shall be submitted both in the O&M manuals and separately in a standalone document containing all sensors for inclusion in the commissioning documentation. The following table is a sample that can be used as a template for submission.

SYSTEM		
Sensor	**Calibration Frequency**	**O&M Calibration Procedure Reference**
Discharge air temperature	Once a year	Volume I Section D.3.aa
Discharge static pressure	Every 6 months	Volume II Section A.1.c

4. Loop tuning documentation and constants for each loop of the building systems. The documentation shall be submitted in outline or table separated by system, control type (e.g. heating valve temperature control); proportional, integral and derivative constants, interval (and bias if used) for each loop. The following table is a sample that can be used as a template for submission.

AIR HANDLING UNIT				
Control Reference	Proportional Constant	Integral Constant	Derivative Constant	Interval
Heating Valve Output	1000	20	10	2 sec.

3.5 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

a. This paragraph applies to Systems Functional Performance Testing of systems for all referenced specification Divisions.

b. Objectives and Scope: The objective of Systems Functional Performance Testing is to demonstrate that each system is operating according to the Contract Documents. Systems Functional Performance Testing facilitates bringing the systems from a state of substantial completion to full dynamic operation. Additionally, during the testing process, areas of noncompliant performance are identified and corrected, thereby improving the operation and functioning of the systems. In general, each system shall be operated through all modes of operation (seasonal, occupied, unoccupied, warm-up, cool-down, part- and full-load, fire alarm and emergency power) where there is a specified system response. The Contractor shall verify each sequence in the sequences of operation. Proper responses to such modes and conditions as power failure, freeze condition, low oil pressure, no flow, equipment failure, etc. shall also be tested.

c. Development of Systems Functional Performance Test Procedures: Before Systems Functional Performance Test procedures are written, the Contractor shall submit all requested documentation and a current list of change orders affecting equipment or systems, including an updated points list, program code, control sequences and parameters. Using the testing parameters and requirements found in the Contract Documents and approved submittals and shop drawings, the CxM will develop specific Systems Functional Test Procedures to verify and document proper operation of each piece of equipment and system to be commissioned. The Contractor shall assist the CxM in developing the Systems Functional Performance Test procedures as requested by the CxM i.e. by answering questions about equipment, operation, sequences, etc. Prior to execution, the CxM will provide a copy of the Systems Functional Performance Test procedures to the Contracting Officer, CxA, Architect/Engineer, and Contractor, who shall review the tests for feasibility, safety, equipment and warranty protection.

d. Purpose of Test Procedures: The purpose of each specific Systems Functional Performance Test is to verify and document compliance with the stated criteria of acceptance given on the test form. Representative test formats and examples are found in the Commissioning Plan for this project. (The Commissioning Plan is issued as a separate document and is available for review.) The test procedure forms developed by the CxM will include, but not be limited to, the following information:

1. System and equipment or component name(s)

2. Equipment location and ID number

3. Unique test ID number, and reference to unique Pre-Functional Checklists and startup documentation, and ID numbers for the piece of equipment

4. Date

5. Project name

6. Participating parties

7. A copy of the specification section describing the test requirements

8. A copy of the specific sequence of operations or other specified parameters being verified

9. Formulas used in any calculations

10. Required pretest field measurements

11. Instructions for setting up the test.

12. Special cautions, alarm limits, etc.

13. Specific step-by-step procedures to execute the test, in a clear, sequential and repeatable format

14. Acceptance criteria of proper performance with a Yes / No check box to

allow for clearly marking whether or not proper performance of each part of
the test was achieved.

15. A section for comments.

16. Signatures and date block for the CxM. A place for the Contractor to initial to signify attendance at the test.

e. Test Methods: Systems Functional Performance Testing shall be achieved by manual testing (i.e. persons manipulate the equipment and observe performance) and/or by monitoring the performance and analyzing the results using the control system's trend log capabilities or by standalone data loggers. The Contractor, CxM and CxA shall determine which method is most appropriate for tests that do not have a method specified.

1. Simulated Conditions: Simulating conditions (not by an overwritten value) shall be allowed, although timing the testing to experience actual conditions is encouraged wherever practical.

2. Overwritten Values: Overwriting sensor values to simulate a condition, such as overwriting the outside air temperature reading in a control system to be something other than it really is, shall be allowed, but shall be used with caution and avoided when possible. Such testing methods often can only test a part of a system, as the interactions and responses of other systems will be erroneous or not applicable. Simulating a condition is preferable. e.g., for the above case, by heating the outside air sensor with a hair blower rather than overwriting the value or by altering the appropriate setpoint to see the desired response. Before simulating conditions or overwriting values, sensors, transducers and devices shall have been calibrated.

3. Simulated Signals: Using a signal generator which creates a simulated signal to test and calibrate transducers and DDC constants is generally recommended over using the sensor to act as the signal generator via simulated conditions or overwritten values.

4. Altering Setpoints: Rather than overwriting sensor values, and when simulating conditions is difficult, altering setpoints to test a sequence is acceptable. For example, to see the Air Conditioning compressor lockout initiate at an outside air temperature below 12 C (54 F), when the outside air temperature is above 12 C (54 F), temporarily change the lockout setpoint to be 2 C (4 F) above the current outside air temperature.

5. Indirect Indicators: Relying on indirect indicators for responses or performance shall be allowed only after visually and directly verifying and documenting, over the range of the tested parameters, that the indirect readings through the control system represent actual conditions and responses. Much of this verification shall be completed during systems startup and initial checkout.

f. Setup: Each function and test shall be performed under conditions that simulate actual conditions as closely as is practically possible. The Contractor shall provide all necessary materials, system

modifications, etc. to produce the necessary flows, pressures, temperatures, etc. necessary to execute the test according to the specified conditions. At completion of the test, the Contractor shall return all affected building equipment and systems, due to these temporary modifications, to their pretest condition.

g. Sampling: No sampling is allowed in completing Pre-Functional Checklists. Sampling is allowed for Systems Functional Performance Test Procedures execution. The CxM and CxA will jointly determine the sampling rate. Functional Performance Testing must be performed on 100% of all plant equipment (chillers, cooling towers, exchangers, primary and secondary pumps, emergency generators, boilers and associated ancillary equipment, fuel delivery system, etc.), central vacuum pumps and air compressors, renewable energy systems (as applicable), and critical terminal equipment (isolation exhaust), and 20% of all terminal equipment (VAVs, fan coils, non-critical exhaust fans, etc.) If at any point, frequent failures are occurring and testing is becoming more troubleshooting than verification, the CxM and/or CxA may stop the testing and require the Contractor to perform and document a checkout of the remaining units, prior to continuing with Systems Functional Performance Testing of the remaining units.

h. Cost of Retesting: The cost associated with expanded sample System Functional Performance Tests shall be solely the responsibility of the Contractor. Any required retesting by the Contractor shall not be considered a justified reason for a claim of delay or for a time extension by the Contractor.

i. Coordination and Scheduling: The Contractor shall provide a minimum of seven days' notice to the CxM, CxA, and the Contracting Officer regarding the completion schedule for the Pre-Functional Checklists and startup of all equipment and systems. The CxM will schedule Systems Functional Performance Tests with the Contractor, CxA and Contracting Officer. The CxM will witness and document the Systems Functional Performance Testing of systems. The CxA and other Government representative(s) may choose to participate in some or all of the Systems Functional Performance Testing. The Contractor shall execute the tests in accordance with the Systems Functional Performance Test Procedure.

j. Testing Prerequisites: In general, Systems Functional Performance Testing will be conducted only after Pre-Functional Checklists have been satisfactorily completed. The control system shall be sufficiently tested and approved by the CxM, CxA, and the Contracting Officer before it is used to verify performance of other components or systems. The air balancing and water balancing shall be completed before Systems Functional Performance Testing of air-related or water-related equipment or systems are scheduled. Systems Functional Performance Testing will proceed from components to subsystems to systems. When the proper performance of all interacting individual systems has been achieved, the interface or coordinated responses between systems will be checked.

k. Problem Solving: The CxM and CxA will recommend solutions to problems found, however the burden of responsibility to solve, correct and retest problems is with the Contractor.

3.6 DOCUMENTATION, NONCONFORMANCE AND APPROVAL OF TESTS

a. Documentation: The CxM will witness, and document the results of all Systems Functional Performance Tests using the specific procedural forms developed by the CxA and CxM for that purpose. Prior to testing, the CxM will provide these forms to the CxA, Contracting Officer and the Contractor for review and approval. The Contractor shall include the filled out forms with the O&M manual data.

b. Nonconformance: The CxM will record the results of the Systems Functional Performance Tests on the procedure or test form. All items of nonconformance issues will be noted and reported to the Contracting Officer and CxA on Commissioning Field Reports and/or the Commissioning Master Issues Log.

1. Corrections of minor items of noncompliance identified may be made during the tests. In such cases, the item of noncompliance and resolution shall be documented on the Systems Functional Test Procedure.

2. Every effort shall be made to expedite the systems functional Performance Testing process and minimize unnecessary delays, while not compromising the integrity of the procedures. However, the CxM and CxA shall not be pressured into overlooking noncompliant work or loosening acceptance criteria to satisfy scheduling or cost issues, unless there is an overriding reason to do so by direction from the Contracting Officer.

3. As the Systems Functional Performance Tests progresses and an item of noncompliance is identified, the CxM shall discuss the issue with the Contractor, CxA and the Contracting Officer.

4. When there is no dispute on an item of noncompliance, and the Contractor accepts responsibility to correct it:

i. The CxM will document the item of noncompliance and the Contractor's response and/or intentions. The Systems Functional Performance Test then continues or proceeds to another test or sequence. After the day's work is complete, the CxM will submit a Commissioning Field Report to the CxA and Contracting Officer. The CxM will also note items of noncompliance and the Contractor's response in the Master Commissioning Issues Log. The Contractor shall correct the item of noncompliance and report completion to the Contracting Officer, CxM and CxA.

ii. The need for retesting will be evaluated by the CxM and CxA. The Contracting Officer will make the final determination for retesting based on input from CxM and CxA. If retesting is required, the CxM and the Contractor shall reschedule the test and the test shall be repeated.

5. If there is a dispute about item of noncompliance, regarding whether it is an item of noncompliance, or who is responsible:

i. The item of noncompliance shall be documented on the test form with the Contractor's response. The item of noncompliance with the Contractor's response shall also be reported on a Commissioning Field Report and on the Master Commissioning Issues Log.

ii. Resolutions shall be made at the lowest management level possible. Other parties are brought into the discussions as needed. Final interpretive and acceptance authority is with Contracting Officer.

iii. The CxM will document the resolution process.

iv. Once the interpretation and resolution have been decided, the Contractor shall correct the item of noncompliance, report it to the DVA/USCAE, CxM and CxA. The requirement for retesting will be determined by the CxM and CxA. If retesting is required, the CxM and the Contractor shall reschedule the test. Retesting shall be repeated until satisfactory performance is achieved.

c. Cost of Retesting: The cost to retest a System Functional Performance Test shall be solely the responsibility of the Contractor. Any required retesting by the Contractor shall not be considered a justified reason for a claim of delay or for a time extension by the Contractor.

d. Failure Due to Manufacturer Defect: If 10%, or three, whichever is greater, of identical pieces (size alone does not constitute a difference) of equipment fail to perform in compliance with the Contract Documents (mechanically or substantively) due to manufacturing defect, not allowing it to meet its submitted performance specifications, all identical units may be considered unacceptable by the Contracting Officer. In such case, the Contractor shall provide the Contracting Officer with the following:

1. Within one week of notification from the Contracting Officer, the Contractor shall examine all other identical units making a record of the findings. The findings shall be provided to the VA within two weeks of the original notice.

2. Within two weeks of the original notification, the Contractor shall provide a signed and dated, written explanation of the problem, cause of failures, etc. and all proposed solutions which shall include full equipment submittals. The proposed solutions shall not significantly exceed the specification requirements of the original installation.

3. The Contracting Officer shall determine whether a replacement of all identical units or a repair is acceptable.

4. Two examples of the proposed solution shall be installed by the Contractor and the Government shall be allowed to test the installations for up to one week, upon which the Contracting Officer will decide whether to accept the solution.

5. Upon acceptance, the Contractor shall replace or repair all identical items, at their expense and extend the warranty accordingly, if the original equipment warranty had begun. The replacement/repair work shall proceed with reasonable speed beginning within one week from when parts can be obtained.

6. After replacement of all identical items, commissioning of these replaced items shall be restarted to include verification of startup, field testing, pre-functional checklists, and functional

performance tests.

e. Approval: The CxM will note each satisfactorily demonstrated function on the test form. Formal approval of the Systems Functional Performance Test shall be made later after review by the CxM, CxA, and by the Contracting Officer. The CxM will evaluate each test and report to the CxA and Contracting Officer using a standard form. The Contracting Officer will give final approval on each test using the same form, and provide signed copies to the CxM, CxA and the Contractor.

3.7 DEFERRED TESTING

a. Unforeseen Deferred Systems Functional Performance Tests: If any Systems Functional Performance Test cannot be completed due to the building structure, required occupancy condition or other conditions, execution of the Systems Functional Performance Testing may be delayed upon approval of the Contracting Officer. These Systems Functional Performance Tests shall be conducted in the same manner as the seasonal tests as soon as possible.

b. Deferred Seasonal Testing: Deferred Seasonal Systems Functional Performance Tests are those that must be deferred until weather conditions are closer to the systems design parameters. The CxM and CxA will review systems parameters and recommend which Systems Functional Performance Tests should be deferred until weather conditions more closely match systems parameters. The Contractor shall review and comment on the proposed schedule for Deferred Seasonal Testing. The Contracting Officer will review and approve the schedule for Deferred Seasonal Testing. Deferred Seasonal Systems Functional Performances Tests shall be witnessed and documented by the CxM. Deferred Seasonal Systems Functional Performance Tests shall be executed by the Contractor in accordance with these specifications.

3.8 OPERATION AND MAINTENANCE TRAINING REQUIREMENTS

a. Training Preparation Conference: Before operation and maintenance training, the CxM will convene a training preparation conference to include Contracting Officer, VA's Resident Engineer, VA's Operations and Maintenance personnel, and the Contractor. The purpose of this conference will be to discuss and plan for Training and Demonstration of VA Operations and Maintenance personnel.

b. The Contractor shall provide training and demonstration as required by other Division 21, Division 22, Division 23, Division 26, Division 27, Division 28, and Division 31 sections. The Training and Demonstration shall include, but is not limited to, the following:

1. Review the Contract Documents.

2. Review installed systems, subsystems, and equipment.

3. Review instructor qualifications.

4. Review instructional methods and procedures.

5. Review training module outlines and contents.

6. Review course materials (including operation and maintenance

manuals).

7. Review and discuss locations and other facilities required for instruction.

8. Review and finalize training schedule and verify availability of educational materials, instructors, audiovisual equipment, and facilities needed to avoid delays.

9. For instruction that must occur outside, review weather and forecasted weather conditions and procedures to follow if conditions are unfavorable.

c. Training Module Submittals: The Contractor shall submit the following information to the CxM, CxA and Contracting Officer:

1. Instruction Program: Submit two copies of outline of instructional program for demonstration and training, including a schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module. At completion of training, submit two complete training manuals for VA's use.

2. Qualification Data: Submit qualifications for facilitator and/or instructor.

3. Attendance Record: For each training module, submit list of participants and length of instruction time.

4. Evaluations: For each participant and for each training module, submit results and documentation of performance-based test.

5. Demonstration and Training Recording:

i. General: Engage a qualified commercial photographer to record demonstration and training. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice. At beginning of each training module, record each chart containing learning objective and lesson outline.

ii. Video Format: Provide high quality color DVD color on standard size DVD disks.

iii. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to show area of demonstration and training. Display continuous running time.

iv. Narration: Describe scenes on video recording by audio narration by microphone while demonstration and training is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

v. Submit two copies within seven days of end of each training module.

6. Transcript: Prepared on 8-1/2-by-11-inch paper, punched and bound in heavy-duty, 3-ring, vinyl-covered binders. Mark

appropriate identification on front and spine of each binder. Include a cover sheet with same label information as the corresponding videotape. Include name of Project and date of videotape on each page.

d. Quality Assurance:

1. Facilitator Qualifications: A firm or individual experienced in training or educating maintenance personnel in a training program similar in content and extent to that indicated for this Project, and whose work has resulted in training or education with a record of successful learning performance.

2. Instructor Qualifications: A factory authorized service representative, complying with requirements in Division 01 Section "Quality Requirements," experienced in operation and maintenance procedures and training.

3. Photographer Qualifications: A professional photographer who is experienced photographing construction projects.

e. Training Coordination:

1. Coordinate instruction schedule with VA's operations. Adjust schedule as required to minimize disrupting VA's operations.

2. Coordinate instructors, including providing notification of dates, times, length of instruction time, and course content.

3. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by the Contracting Officer.

f. Instruction Program:

1. Program Structure: Develop an instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, and as follows:

i. Fire protection systems, including fire alarm, fire pumps, and fire suppression systems.

ii. Intrusion detection systems.

iii. Conveying systems, including elevators, wheelchair lifts, escalators, and automated materials handling systems.

iv. Medical equipment, including medical gas equipment and piping.

v. Laboratory equipment, including laboratory air and vacuum equipment and piping.

vi. Heat generation, including boilers, feedwater equipment, pumps, steam distribution piping, condensate return systems, heating hot water heat exchangers, and heating hot water distribution piping.

vii. Refrigeration systems, including chillers, cooling towers, condensers, pumps, and distribution piping.

viii. HVAC systems, including air handling equipment, air distribution systems, and terminal equipment and devices.

ix. HVAC instrumentation and controls.

x. Electrical service and distribution, including switchgear, transformers, switchboards, panelboards, uninterruptible power supplies, and motor controls.

xi. Packaged engine generators, including synchronizing switchgear/switchboards, and transfer switches.

xii. Lighting equipment and controls.

xiii. Communication systems, including intercommunication, surveillance, nurse call systems, public address, mass evacuation, voice and data, and entertainment television equipment.

xiv. Site utilities including lift stations, condensate pumping and return systems, and storm water pumping systems.

g. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participants are expected to master. For each module, include instruction for the following:

1. Basis of System Design, Operational Requirements, and Criteria: Include the following:

i. System, subsystem, and equipment descriptions.

ii. Performance and design criteria if Contractor is delegated design responsibility.

iii. Operating standards.

iv. Regulatory requirements.

v. Equipment function.

vi. Operating characteristics.

vii. Limiting conditions.

viii. Performance curves.

2. Documentation: Review the following items in detail:

i. Emergency manuals.

ii. Operations manuals.

iii. Maintenance manuals.

iv. Project Record Documents.

v. Identification systems.

vi. Warranties and bonds.

vii. Maintenance service agreements and similar continuing commitments.

3. Emergencies: Include the following, as applicable:

i. Instructions on meaning of warnings, trouble indications, and error messages.

ii. Instructions on stopping.

iii. Shutdown instructions for each type of emergency.

iv. Operating instructions for conditions outside of normal operating limits.

v. Sequences for electric or electronic systems.

vi. Special operating instructions and procedures.

4. Operations: Include the following, as applicable:

i. Startup procedures.

ii. Equipment or system break-in procedures.

iii. Routine and normal operating instructions.

iv. Regulation and control procedures.

v. Control sequences.

vi. Safety procedures.

vii. Instructions on stopping.

viii. Normal shutdown instructions.

ix. Operating procedures for emergencies.

x. Operating procedures for system, subsystem, or equipment failure.

xi. Seasonal and weekend operating instructions.

xii. Required sequences for electric or electronic systems.

xiii. Special operating instructions and procedures.

5. Adjustments: Include the following:

i. Alignments.

ii. Checking adjustments.

iii. Noise and vibration adjustments.

iv. Economy and efficiency adjustments.

6. Troubleshooting: Include the following:

i. Diagnostic instructions.

ii. Test and inspection procedures.

7. Maintenance: Include the following:

i. Inspection procedures.

ii. Types of cleaning agents to be used and methods of cleaning.

iii. List of cleaning agents and methods of cleaning detrimental to product.

iv. Procedures for routine cleaning

v. Procedures for preventive maintenance.

vi. Procedures for routine maintenance.

vii. Instruction on use of special tools.

8. Repairs: Include the following:

i. Diagnosis instructions.

ii. Repair instructions.

iii. Disassembly; component removal, repair, and replacement; and reassembly instructions.

iv. Instructions for identifying parts and components.

v. Review of spare parts needed for operation and maintenance.

h. Training Execution:

1. Preparation: Assemble educational materials necessary for instruction, including documentation and training module. Assemble training modules into a combined training manual. Set up instructional equipment at instruction location.

2. Instruction:

i. Facilitator: Engage a qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Contracting Officer for number of participants, instruction times, and location.

ii. Instructor: Engage qualified instructors to instruct VA's personnel to adjust, operate, and maintain systems, subsystems,

and equipment not part of a system.

1) The CxM will furnish an instructor to describe basis of system design, operational requirements, criteria, and regulatory requirements.

2) The Government may furnish an instructor to describe VA's operational philosophy.

3) The Contracting Officer will furnish the Contractor with names and positions of participants.

3. Scheduling: Provide instruction at mutually agreed times. For equipment that requires seasonal operation, provide similar instruction at start of each season. Schedule training with the Contracting Officer, CxA and the CxM with at least seven days' advance notice.

4. Evaluation: At conclusion of each training module, assess and document each participant's mastery of module by use of an oral, or a written, performance-based test.

5. Cleanup: Collect used and leftover educational materials and remove from Project site. Remove instructional equipment. Restore systems and equipment to condition existing before initial training use.

i. Demonstration and Training Recording:

1. General: Engage a qualified commercial photographer to record demonstration and training. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice. At beginning of each training module, record each chart containing learning objective and lesson outline.

2. Video Format: Provide high quality color DVD color on standard size DVD disks.

3. Recording: Mount camera on tripod before starting recording, unless otherwise necessary to show area of demonstration and training. Display continuous running time.

4. Narration: Describe scenes on videotape by audio narration by microphone while demonstration and training is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 41 00

DEMOLITION

08/22

PART 1 GENERAL

1.1 REFERENCES
1.2 PROJECT DESCRIPTION
1.2.1 Definitions
1.2.1.1 Demolition
1.2.1.2 Deconstruction
1.2.1.3 Demolition Plan
1.2.2 Demolition Plan
1.2.3 General Requirements
1.3 ITEMS TO REMAIN IN PLACE
1.3.1 Existing Construction Limits and Protection
1.3.2 Trees
1.3.3 Utility Service
1.3.4 Facilities
1.4 BURNING
1.5 AVAILABILITY OF WORK AREAS
1.6 SUBMITTALS
1.7 QUALITY ASSURANCE
1.7.1 Dust and Debris Control
1.8 PROTECTION
1.8.1 Traffic Control Signs
1.8.2 Protection of Personnel
1.9 EXISTING CONDITIONS

PART 2 PRODUCTS

2.1 FILL MATERIAL

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED
3.1.1 Structures
3.1.2 Utilities and Related Equipment
3.1.2.1 General Requirements
3.1.2.2 Disconnecting Existing Utilities
3.1.3 Paving and Slabs
3.1.4 Structural Steel
3.1.5 Miscellaneous Metal
3.1.6 Carpentry
3.1.7 Carpet
3.1.8 Acoustic Ceiling Tile
3.1.9 Air Conditioning Equipment
3.1.10 Cylinders and Canisters
3.1.11 Mechanical Equipment and Fixtures
3.1.11.1 Ducts

3.2 CONCURRENT EARTH-MOVING OPERATIONS
3.3 DISPOSITION OF MATERIAL
3.3.1 Title to Materials
3.3.2 Disposal of Ozone Depleting Substance (ODS)
3.3.2.1 Special Instructions
3.3.2.2 Fire Suppression Containers
3.3.3 Transportation Guidance
3.3.4 Unsalvageable and Non-Recyclable Material
3.4 CLEANUP
3.5 DISPOSAL OF REMOVED MATERIALS
3.5.1 Regulation of Removed Materials
3.5.2 Burning on Government Property
3.5.3 Removal from Government Property

-- End of Section Table of Contents --

SECTION 02 41 00

DEMOLITION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K	(2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 145	(1991; R 2012) Standard Specification for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes
AASHTO T 180	(2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6	(2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

CARPET AND RUG INSTITUTE (CRI)

CRI 104	(2015) Carpet Installation Standard for Commercial Carpet
CRI 105	(2015) Carpet Installation Standard for Residential Carpet

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25	(Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders; https://www.dla.mil/Portals/104/Documents/DispositionS /ddsr/docs/cylinderjointpub.pdf

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M	(2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures
MIL-STD-129	(2014; Rev R; Change 1 2018; Change 2 2019; Change 3 2023) Military Marking for Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 82	Protection of Stratospheric Ozone
49 CFR 173.301	Shipment of Compressed Gases in Cylinders and Spherical Pressure Vessels

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Deconstruction

Deconstruction is the process of taking apart a facility with the primary goal of preserving the value of all useful building materials.

1.2.1.3 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a Demolition Plan and submit proposed demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials, coordination with other work in progress, a disconnection schedule of utility services, and a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Coordinate with Waste Management Plan in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Provide procedures for safe conduct of the work

in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work includes demolition, deconstruction, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 17, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload pavements to remain. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.2 Trees

Protect trees within the project site which might be damaged during demolition or deconstruction, and which are indicated to be left in place, by a 6 foot high fence. Erect and secure fence a minimum of 5 feet from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer. All tree removal should be performed outside of the bat activity period.

1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide

approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished or deconstructed, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

The area of work of the non-permanent structure will be available at contract award. The work at the buildings to be removed will be available when the users transition out and the buildings are abandoned.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G

Deconstruction Plan; G

Existing Conditions

SD-07 Certificates

Notification; G

SD-11 Closeout Submittals

Receipts

1.7 QUALITY ASSURANCE

Submit timely notification of demolition projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the Regional Office of the United States Environmental Protection Agency (USEPA) State's environmental protection agency local air pollution control district/agency and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution.

1.8 PROTECTION

1.8.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Notify the Contracting Officer prior to beginning such work.

1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the structure being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officershowing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

2.1 FILL MATERIAL

a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill basements, voids, depressions or excavations resulting from demolition or deconstruction of structures.

b. Provide fill material conforming to the definition of satisfactory soil material as defined in ASTM D2487 . In addition, fill material must be free from roots and other organic matter, trash, debris, frozen materials, and stones larger than 2 inches in any dimension.

c. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as follows:

Soil classification	AASHTO M 145
Moisture-density relations	AASHTO T 180, Method B or D

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for recycling, or other disposal, as specified.

3.1.1 Structures

a. Building removal shall include complete removal of all hazardous and regulated materials, removal of all subgrade basements, foundations, internal and/or building inter-connected underground utilities and infrastructure. Remove sidewalks, curbs, gutters, and street light bases as indicated.

b. Building, or the remaining portions thereof, not exceeding 80 feet in height may be demolished by the mechanical method of demolition.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition or deconstruction work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.3 Paving and Slabs

Remove concrete and asphaltic concrete paving and slabs including aggregate base in it's entirety.Provide neat sawcuts at limits of pavement removal. Move, grind and store pavement and slabs designated to be recycled as directed by the Contracting Officer.

3.1.4 Structural Steel

Salvage for recycle structural steel, steel joists, girders, angles, plates, columns and shapes. Flame-cutting torches are permitted when other methods of dismantling are not practical.

3.1.5 Miscellaneous Metal

Recycle scrap metal as part of demolition and deconstruction operations. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility, in accordance with the Waste Management Plan.

3.1.6 Carpentry

Salvage for recycle lumber, millwork items, and finished boards.

3.1.7 Carpet

Remove existing carpet for reclamation in accordance with manufacturer recommendations and as follows. Remove used carpet in large pieces, roll tightly, and pack neatly in a container. Remove adhesive according to recommendations of the Carpet and Rug Institute (CRI). Provide adhesive removal solvents in compliance with CRI 104/CRI 105. Recycle removed carpet cushion.

3.1.8 Acoustic Ceiling Tile

Remove, neatly stack, and recycle acoustic ceiling tiles. Recycling may be available with manufacturer. Otherwise, give priority to a local recycling organization. Recycling is not required if the tiles contain or may have been exposed to asbestos material.

3.1.9 Air Conditioning Equipment

Remove air conditioning, refrigeration, and other equipment containing refrigerants without releasing chlorofluorocarbon refrigerants to the atmosphere in accordance with the Clean Air Act Amendment of 1990. Recover all refrigerants prior to removing air conditioning, refrigeration, and other equipment containing refrigerants and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.10 Cylinders and Canisters

Remove all fire suppression system cylinders and canisters and dispose of in accordance with the paragraph entitled "Disposal of Ozone Depleting Substance (ODS)."

3.1.11 Mechanical Equipment and Fixtures

3.1.11.1 Ducts

Classify removed duct work as scrap metal.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition or deconstruction work in areas occupied by structures to be demolished or deconstructed until all demolition and

deconstruction in the area has been completed and debris removed. Fill holes, open basements and other hazardous openings.

3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.2 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Remove recovered ODS from Government property and dispose of in accordance with 40 CFR 82. Dispose products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

3.3.2.1 Special Instructions

No more than one type of ODS is permitted in each container. Apply a warning/hazardous label to the containers in accordance with Department of Transportation regulations. Provide a tag with the following information on all cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS:

a. Activity name and unit identification code

b. Activity point of contact and phone number

c. Type of ODS and pounds of ODS contained

d. Date of shipment

e. National stock number (for information, call (804) 279-4525).

3.3.2.2 Fire Suppression Containers

Deactivate fire suppression system cylinders and canisters with electrical charges or initiators prior to shipment. Also, safety caps must be used to cover exposed actuation mechanisms and discharge ports on these special cylinders.

3.3.3 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Army Regulation 700-68, Naval Supply Instruction 4440.128C, Marine Corps Order 10330.2C, and Air Force Regulation 67-12), 49 CFR 173.301, and DOD 4000.25-1-M.

3.3.4 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable combustible material off the site.

3.4 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations as contractually specified in the Waste Management Plan.

3.5.2 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.5.3 Removal from Government Property

Transport waste materials removed from demolished and deconstructed structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19 TEMPORARY ENVIRONMENTAL CONTROLS.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 82 11

TRADITIONAL ASBESTOS ABATEMENT

1/21

PART 1 General

1.1 SUMMARY OF THE WORK
1.2 VARIATIONS IN QUANTITY
1.3 STOP ASBESTOS REMOVAL
1.4 DEFINITIONS
1.5 APPLICABLE CODES AND REGULATIONS
1.6 PROJECT COORDINATION
1.7 RESPIRATORY PROTECTION
1.8 WORKER PROTECTION
1.9 DECONTAMINATION FACILITIES
1.10 REFERENCES
1.11 SUBMITTALS
1.12 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT
2.2 MONITORING, INSPECTION AND TESTING
2.3 ASBESTOS HAZARD ABATEMENT PLAN
2.4 ENCAPSULANTS

PART 3 EXECUTION

3.1 REGULATED AREA PREPARATIONS
3.2 REMOVAL OF ACM
3.3 LOCKDOWN ENCAPSULATION
3.4 DISPOSAL OF ACM WASTE MATERIALS
3.5 PROJECT DECONTAMINATION
3.6 FINAL VISUAL INSPECTION AND AIR CLEARANCE TESTING
3.7 ABATEMENT CLOSEOUT AND CERTIFICATE OF COMPLIANCE

ATTACHMENTS:

Attachment C

Attachment A

Attachment B

-- End of Section Table of Contents --

SECTION 02 82 11
1/21

TRADITIONAL ASBESTOS ABATEMENT

NOTE: These specifications were derived from standard government specifications and standards. This document is provided for reference only and was provided by others. In the event of contact with any hazardous materials, contact an abatement specialist immedialtely and seek expert consultation and advise.

PART 1 General

1.1 SUMMARY OF THE WORK

a. Contract Documents and Related Requirements: Drawings, general provisions of the contract, including general and supplementary conditions and other Division 01 specifications, shall apply to the work of this section. The contract documents show or describe the work to be done under the contract and related requirements and conditions impacting the project. Related requirements and conditions include applicable codes and regulations, notices and permits, existing site conditions and restrictions on use of the site, requirements for partial owner occupancy during the work, coordination with other work and the phasing of the work. In the event the Asbestos Abatement Contractor discovers a conflict in the contract documents and/or requirements or codes, the conflict must be brought to the immediate attention of the Contracting Officer for resolution. Whenever there is a conflict or overlap in the requirements, the most stringent shall apply. Any actions taken by the Contractor without obtaining guidance from the Contracting Officer shall become the sole risk and responsibility of the Asbestos Abatement Contractor. All costs incurred due to such action are also the responsibility of the Asbestos Abatement Contractor.

b. Extent of Work:

(1) Below is a brief description of the estimated quantities of asbestos containing materials to be abated. These quantities are for informational purposes only and are based on the best information available at the time of the specification preparation. The Contractor shall satisfy themselves of the actual quantities to be abated. Nothing in this section may be interpreted as limiting the extent of work otherwise required by this contract and related documents.

(2) Removal, clean-up and disposal of asbestos containing materials (ACM) and asbestos/waste contaminated elements or debris in an appropriate regulated area for the following approximate quantities;

(
Approximately 130 linear feet (LF)of >6 inches diameter preformed pipe insulation in Building 16A.
Approximately 15 linear feet of >6 inches diameter preformed insulation in adjoining connecting areas of Building 16.
Approximately 34 fittings on fiberglass line 2 inches- 6 inches in diameter in Building 16A.

Approximately 10 fittings on fiberglass line 3 inches - 6 inches in diameter in adjoining connecting areas of Building 16.
Approximately 7 fittings on fiberglass line > 6 inches in diameter Building 16A.
Approximately 2 fittings on fiberglass line > 6 inches in diameter in adjoining connecting areas of Building 16.
Approximately 15 linear feet of hard white caulk on the vent in Building 16A.
Approximaterly 60 linear feet of window frame caulk in adjoining connecting areas of Building 16.
Approximately 15 linear feet of window glazing in adjoining connecting areas of Building 16.

c. Related Work:

(1) Section 07 84 00, FIRESTOPPING

(

d. Tasks:

(1) The work tasks are summarized briefly as follows:

(a) Pre-abatement activities including pre-abatement meeting(s), inspection(s), notifications, permits, submittal approvals, regulated area preparations, emergency procedures arrangements, and standard operating procedures for asbestos abatement work.

(b) Abatement activities including removal, encapsulation, enclosure, clean-up and disposal of ACM waste, recordkeeping, security, monitoring, and inspections.

(c) Cleaning and decontamination activities including final visual inspection, air monitoring and certification of decontamination.

e. Contractors Use of Premises:

(1) The Contractor and Contractor's personnel shall cooperate fully with the VA representative/consultant to facilitate efficient use of buildings and areas within buildings. The Contractor shall perform the work in accordance with the VA specifications, drawings, phasing plan and in compliance with any/all applicable Federal, State and Local regulations and requirements.

(2) The Contractor shall use the existing facilities in the building strictly within the limits indicated in contract documents as well as the approved VA Design and Construction Procedures. VA Design and Construction Procedures drawings of partially occupied buildings will show the limits of regulated areas; the placement of decontamination facilities; the temporary location of bagged waste ACM; the path of transport to outside the building; and the temporary waste storage area for each building/regulated area. Any variation from the arrangements shown on drawings shall be secured in writing from the VA representative through the pre-abatement plan of action. The following limitations of use shall apply to existing facilities shown on drawings:

1.2 VARIATIONS IN QUANTITY

The quantities and locations of ACM as indicated on the drawings and the extent of work included in this section are estimated which are limited by the physical constraints imposed by occupancy of the buildings and accessibility to ACM. Accordingly, minor variations (+/- 10 percent) in quantities of ACM within the regulated area are considered as having no impact on contract price and time requirements of this contract. Where additional work is required beyond the above variation, the contractor shall provide unit prices for newly discovered ACM and those prices shall be used for additional work required under the contractor.

1.3 STOP ASBESTOS REMOVAL

If the Contracting Officer; their field representative; the facility Safety Officer/Manager or their designee, or the VA Professional Industrial Hygienist/Certified Industrial Hygienist (VPIH/CIH) presents a verbal **Stop Asbestos Removal Order**, the Contractor/Personnel shall immediately stop all asbestos removal and maintain HEPA filtered negative pressure air flow in the containment and adequately wet any exposed ACM. If a verbal Stop Asbestos Removal Order is issued, the VA shall follow-up with a written order to the Contractor as soon as it is practicable. The Contractor shall not resume any asbestos removal activity until authorized to do so in writing by the VA Contracting Officer. A stop asbestos removal order may be issued at any time the VA Contracting Officer determines abatement conditions/activities are not within VA specification, regulatory requirements or that an imminent hazard exists to human health or the environment. Work stoppage will continue until conditions have been corrected to the satisfaction of the VA. Standby time and costs for corrective actions will be borne by the Contractor, including the VPIH/CIH time. The occurrence of any of the following events shall be reported immediately by the Contractor’s competent person to the VA Contracting Office or field representative using the most expeditious means (e.g., verbal or telephonic), followed up with written notification to the Contracting Officer as soon as practical. The Contractor shall immediately stop asbestos removal/disturbance activities and initiate fiber reduction activities if:

a. Airborne PCM analysis results equal to or greater than 0.01 f/cc above background levels inside the building but outside the regulated area;

b. breach or break in regulated area containment barrier(s);

c. less than –0.02 inch WCG pressure in the regulated area;

d. serious injury/death at the site;

c. fire/safety emergency at the site;

d. respiratory protection system failure;

e. power failure or loss or inadequate use of wetting agent;

f. any visible emissions observed outside the regulated area; or

g. failure to follow project specification requirements.

1.4 DEFINITIONS

a. General: Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Drawings must be recognized as diagrammatic in nature and not completely descriptive of the requirements indicated therein.

b. Glossary:

Abatement - Procedures to control fiber release from asbestos-containing materials. Includes removal, encapsulation, enclosure, demolition, and renovation activities related to asbestos containing materials (ACM).

Adequately wet - Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the ACM, then that material has not been adequately wetted.

Aerosol - Solid or liquid particulate suspended in air.

Aggressive method - Removal or disturbance of building material by sanding, abrading, grinding, or other method that breaks, crumbles, or disintegrates intact ACM.

Aggressive air sampling - EPA AHERA defined clearance sampling method using air moving equipment such as fans and leaf blowers to aggressively disturb and maintain in the air residual fibers after abatement.

AHERA - Asbestos Hazard Emergency Response Act. Asbestos regulations for schools issued in 1987.

Aircell - Pipe or duct insulation made of corrugated cardboard which contains asbestos.

Air monitoring - The process of measuring the fiber content of a known volume of air collected over a specified period of time. The NIOSH 7400 Method, Issue 3, Fifth Edition is used to determine the fiber levels in air. For personal samples, area air samples and clearance air testing using Phase Contrast Microscopy (PCM) analysis, the NIOSH Method 7402 (Issue 2, Fourth Edition) can be used when it is necessary to confirm fibers counted by PCM as being asbestos. The AHERA TEM analysis may be used for background, area samples and clearance samples when required by this specification, or at the discretion of the VPIH/CIH as appropriate.

Air sample filter - The filter used to collect fibers which are then counted. The filter is made of mixed cellulose ester (MCE) membrane for PCM (Phase Contrast Microscopy, 25 mm, 3-piece with 2 inches Static Extension Cowl, 0.8 micron pore size) and MCE for TEM (Transmission Electron Microscopy, 25 mm, 3-piece with 2 inches Static Extension Cowl, 0.45 micron pore size).

Amended water - Water to which a surfactant (wetting agent) has been added to increase the penetrating ability of the liquid.

Asbestos - Includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, actinolite asbestos, and any of these minerals that have been chemically treated or altered. Asbestos also includes PACM, as defined below.

Asbestos Hazard Abatement Plan (AHAP) - Asbestos work procedures required to be submitted by the contractor before work begins.

Asbestos-containing material (ACM) - Any material containing more than one percent of asbestos.

Asbestos contaminated elements (ACE) - Building elements such as ceilings, walls, lights, or ductwork that are contaminated with asbestos.

Asbestos-contaminated soil (ACS) – Soil found in the work area or in adjacent areas such as crawlspaces or pipe tunnels which is contaminated with asbestos-containing material debris and cannot be easily separated from the material.

Asbestos-containing waste (ACW) material - Asbestos-containing material or asbestos contaminated objects requiring disposal.

Asbestos Project Monitor – Some states require that any person conducting asbestos abatement air sampling, clearance inspections and clearance air sampling be licensed as an asbestos project monitor.

Asbestos waste decontamination facility - A system consisting of drum/bag washing facilities and a temporary storage area for cleaned containers of asbestos waste. Used as the exit for waste and equipment leaving the regulated area. In an emergency, it may be used to evacuate personnel.

Authorized person - Any person authorized by the VA, the Contractor, or government agency and required by work duties to be present in regulated areas.

Authorized visitor - Any person approved by the VA; the contractor; or any government agency representative having jurisdiction over the regulated area (e.g., OSHA, Federal and State EPA).

Barrier - Any surface that isolates the regulated area and inhibits fiber migration from the regulated area.

Containment Barrier - An airtight barrier consisting of walls, floors, and/or ceilings of sealed plastic sheeting which surrounds and seals the outer perimeter of the regulated area.

Critical Barrier - The barrier responsible for isolating the regulated area from adjacent spaces, typically constructed of 2-layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as doors, windows, penetrations or any other opening into the regulated area.

Primary Barrier – Plastic barriers placed over critical barriers and exposed directly to abatement work or to secondary barrier.

Secondary Barrier - Any additional plastic barriers used to isolate

and provide protection from debris during abatement work.

Breathing zone - The hemisphere forward of the shoulders with a radius of about (6 - 9 inches)from the worker's nose.

Bridging encapsulant - An encapsulant that forms a layer on the surface of the ACM.

Building/facility owner - The legal entity, including a lessee, which exercises control over management and recordkeeping functions relating to a building and/or facility in which asbestos activities take place.

Bulk testing - The collection and analysis of suspect asbestos containing materials.

Certified Industrial Hygienist (CIH) - A person certified in the comprehensive practice of industrial hygiene by the American Board of Industrial Hygiene.

Class I asbestos work - Activities involving the removal of Thermal System Insulation (TSI) and surfacing ACM and Presumed Asbestos Containing Material (PACM).

Class II asbestos work - Activities involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos-containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastic.

Clean room/Changing room - An uncontaminated room having facilities for the storage of employee's street clothing and uncontaminated materials and equipment.

Clearance sample - The final air sample taken after all asbestos work has been done and visually inspected. Performed by the VA's professional industrial hygiene consultant/Certified Industrial Hygienist (VPIH/CIH).

Closely resemble - The major workplace conditions which have contributed to the levels of historic asbestos exposure, are no more protective than conditions of the current workplace.

Competent person - In addition to the definition in 29 CFR 1926.32(f), one who is capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, as specified in 29 CFR 1926.32(f); in addition, for Class I and II work who is specially trained in a training course which meets the criteria of EPA's Model Accreditation Plan (40 CFR 763) for supervisor.

Contractor's Professional Industrial Hygienist (CPIH/CIH) - The asbestos abatement contractor's industrial hygienist. The industrial hygienist must meet the qualification requirements of a PIH and may report to a certified industrial hygienist (CIH).

Count - Refers to the fiber count or the average number of fibers greater than five microns in length with a length-to-width

(aspect) ratio of at least 3 to 1, per cubic centimeter of air.

Crawlspace – An area which can be found either in or adjacent to the work area. This area has limited access and egress and may contain asbestos materials and/or asbestos contaminated soil.

Decontamination area/unit - An enclosed area adjacent to and connected to the regulated area and consisting of an equipment room, shower room, and clean room, which is used for the decontamination of workers, materials, and equipment that are contaminated with asbestos.

Demolition - The wrecking or taking out of any load-supporting structural member and any related razing, removing, or stripping of asbestos products.

Disposal bag - Typically 6-mil thick sift-proof, dustproof, leak-tight container used to package and transport asbestos waste from regulated areas to the approved landfill. Each bag/container must be labeled/marked in accordance with EPA, OSHA and DOT requirements.

Disturbance – Asbestos Operations and Maintenance Activities (OSHA Class III) that disrupt the matrix of ACM or PACM, crumble or pulverize ACM or PACM, or generate visible debris from ACM or PACM. Disturbance includes cutting away small amounts of ACM or PACM, no greater than the amount that can be contained in one standard sized glove bag or waste bag in order to access a building component. In no event shall the amount of ACM or PACM so disturbed exceed that which can be contained in one glove bag or disposal bag, which shall not exceed 60 inches in length or width.

Drum - A rigid, impermeable container made of cardboard fiber, plastic, or metal which can be sealed in order to be sift-proof, dustproof, and leak-tight.

Employee exposure - The exposure to airborne asbestos that would occur if the employee were not wearing respiratory protection equipment.

Encapsulant - A material that surrounds or embeds asbestos fibers in an adhesive matrix and prevents the release of fibers.

Encapsulation - Treating ACM with an encapsulant.

Enclosure - The construction of an air tight, impermeable, permanent barrier around ACM to control the release of asbestos fibers from the material and also eliminate access to the material.

Equipment room - A contaminated room located within the decontamination area that is supplied with impermeable bags or containers for the disposal of contaminated protective clothing and equipment.

Fiber - A particulate form of asbestos, 5 microns or longer, with a length to width (aspect) ratio of at least 3 to 1.

Fibers per cubic centimeter (f/cc) - Abbreviation for fibers per cubic centimeter, used to describe the level of asbestos fibers in air.

Filter - Media used in respirators, vacuums, or other machines to remove particulate from air.

Firestopping - Material used to close the open parts of a structure in order to prevent a fire from spreading.

Friable asbestos containing material - Any material containing more than one (1) percent asbestos as determined using the method specified in 40 CFR 763, Polarized Light Microscopy, that, when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

Glovebag - Not more than a 60 x 60 inch impervious plastic bag-like enclosure affixed around an asbestos-containing material, with glove-like appendages through which materials and tools may be handled.

High efficiency particulate air (HEPA) filter – An ASHRAE MERV 17 filter capable of trapping and retaining at least 99.97 percent of all mono-dispersed particles of 0.3 micrometers in diameter.

HEPA vacuum - Vacuum collection equipment equipped with a HEPA filter system capable of collecting and retaining asbestos fibers.

Homogeneous area - An area of surfacing, thermal system insulation or miscellaneous ACM that is uniform in color, texture and date of application.

HVAC - Heating, Ventilation and Air Conditioning

Industrial hygienist (IH) - A professional qualified by education, training, and experience to anticipate, recognize, evaluate and develop controls for occupational health hazards. Meets definition requirements of the American Industrial Hygiene Association (AIHA).

Industrial hygienist technician (IH Technician) - A person working under the direction of an IH or CIH who has special training, experience, certifications and licenses required for the industrial hygiene work assigned. Some states require that an industrial hygienist technician conducting asbestos abatement air sampling, clearance inspection and clearance air sampling be licensed as an asbestos project monitor.

Intact - The ACM has not crumbled, been pulverized, or otherwise deteriorated so that the asbestos is no longer likely to be bound with its matrix.

Lockdown - Applying encapsulant, after a final visual inspection, on all abated surfaces at the conclusion of ACM removal prior to removal of critical barriers.

National Emission Standards for Hazardous Air Pollutants (NESHAP) - EPA's rule to control emissions of asbestos to the environment (40 CFR 61, Subpart M).

Negative initial exposure assessment - A demonstration by the employer which complies with the criteria in 29 CFR 1926.1101 (f)(2)(iii), that employee exposure during an operation is expected to be consistently below the PEL or Excursion Limit (EL).

Negative pressure - Air pressure which is lower than the surrounding area, created by exhausting air from a sealed regulated area through HEPA equipped filtration units. OSHA requires maintaining -0.02 inch water column gauge inside the negative pressure enclosure.

Negative pressure respirator - A respirator in which the air pressure inside the facepiece is negative during inhalation relative to the air pressure outside the respirator facepiece.

Non-friable ACM - Material that contains more than 1 percent asbestos but cannot be crumbled, pulverized, or reduced to powder by hand pressure.

Organic vapor cartridge - The type of cartridge used on air purifying respirators to remove organic vapor hazardous air contaminants.

Outside air - The air outside buildings and structures, including, but not limited to, the air under a bridge or in an open ferry dock.

Owner/operator - Any person who owns, leases, operates, controls, or supervises the facility being demolished or renovated or any person who owns, leases, operates, controls, or supervises the demolition or renovation operation, or both.

Penetrating encapsulant - Encapsulant that is absorbed into the ACM matrix without leaving a surface layer.

Permissible exposure limit (PEL) - The level of exposure OSHA allows for as an eight (8) hour time-weighted average (TWA). For asbestos fibers, the eight (8) hour time-weighted average PEL is 0.1 fibers per cubic centimeter (0.1 f/cc) of air and the 30-minute Excursion Limit (EL) is 1.0 fibers per cubic centimeter (1 f/cc).

Personal protective equipment (PPE) – equipment designed to protect user from injury and/or specific job hazard. Such equipment may include protective clothing, hard hats, safety glasses, fall protection, and respirators.

Personal sampling/monitoring - Representative air samples obtained in the breathing zone for one or more workers within the regulated area using a filter cassette and a calibrated air sampling pump to determine asbestos exposure.

Pipe tunnel – An area, typically located adjacent to mechanical spaces or boiler rooms in which the pipes servicing the heating system in the building are routed to allow the pipes to access heating elements. These areas may contain asbestos pipe insulation, asbestos fittings, debris or asbestos-contaminated soil.

Polarized light microscopy (PLM) - Light microscopy using dispersion staining techniques and refractive indices to identify and quantify the type of asbestos present in a bulk sample.

Polyethylene sheeting - Strong plastic barrier material 4 to 6-mils thick, semi-transparent, flame retardant per NFPA 241.

Positive/negative fit check - A method of verifying the seal of a

facepiece respirator by temporarily occluding the filters and breathing in (inhaling) and then temporarily occluding the exhalation valve and breathing out (exhaling) while checking for inward or outward leakage of the respirator, respectively.

Presumed ACM (PACM) - Thermal system insulation, surfacing, and flooring material installed in buildings prior to 1981. If the building owner has actual knowledge, or should have known through the exercise of due diligence that other materials are ACM, they too must be treated as PACM. The designation of PACM may be rebutted pursuant to 29 CFR 1926.1101 (k)(5).

Professional IH - An IH who meets the definition requirements of AIHA; meets the definition requirements of OSHA as a "Competent Person" at 29 CFR 1926.1101 (b); has completed two specialized EPA approved courses on management and supervision of asbestos abatement projects; has formal training in respiratory protection and waste disposal; and has a minimum of four projects of similar complexity with this project of which at least three projects serving as the supervisory IH. The PIH may be either the VA's PIH (VPIH/CIH) or Contractor's PIH (CPIH/CIH).

Project designer - A person who has successfully completed the training requirements for an asbestos abatement project designer as required by 40 CFR 763 Subpart E, Appendix C, Part I; (B)(5).

Assigned protection factor - A value assigned by OSHA/NIOSH to indicate the expected protection provided by each respirator class, when the respirator is properly selected and worn correctly. The number indicates the reduction of exposure level from outside to inside the respirator facepiece.

Qualitative fit test (QLFT) - A fit test using a challenge material that can be sensed by the wearer if leakage in the respirator occurs.

Quantitative fit test (QNFT) - A fit test using a challenge material which is quantified outside and inside the respirator thus allowing the determination of the actual fit factor.

Regulated area - An area established by the employer to demarcate where Class I, II, III asbestos work is conducted, and any adjoining area where debris and waste from such asbestos work may accumulate; and a work area within which airborne concentrations of asbestos exceed, or there is a reasonable possibility they may exceed the PEL.

Regulated ACM (RACM) - Friable ACM; Category I non-friable ACM that has become friable; Category I non-friable ACM that will be or has been subjected to sanding, grinding, cutting, or abrading or; Category II non-friable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of the demolition or renovation operation.

Removal - All operations where ACM, PACM and/or RACM is taken out or stripped from structures or substrates, including demolition operations.

Renovation - Altering a facility or one or more facility components in any way, including the stripping or removal of asbestos from a facility component which does not involve demolition activity.

Repair - Overhauling, rebuilding, reconstructing, or reconditioning of structures or substrates, including encapsulation or other repair of ACM or PACM attached to structures or substrates.

Shower room - The portion of the PDF where personnel shower before leaving the regulated area.

Supplied air respirator (SAR) - A respiratory protection system that supplies minimum Grade D respirable air per ANSI/Compressed Gas Association Commodity Specification for Air, CGA G-7.1.

Surfacing ACM - A material containing more than 1 percent asbestos that is sprayed, troweled on or otherwise applied to surfaces for acoustical, decorative, fireproofing and other purposes.

Surfactant - A chemical added to water to decrease water's surface tension thus making it more penetrating into ACM.

Thermal system ACM - A material containing more than 1 percent asbestos applied to pipes, fittings, boilers, breeching, tanks, ducts, or other structural components to prevent heat loss or gain.

Transmission electron microscopy (TEM) - A microscopy method that can identify and count asbestos fibers.

VA Professional Industrial Hygienist (VPIH/CIH) – The Department of Veterans Affairs Professional Industrial Hygienist must meet the qualifications of a PIH, and may report to a Certified Industrial Hygienist (CIH).

VA Representative - The VA official responsible for on-going project work.

VA Total – means a building or substantial part of the building is completely removed, torn or knocked down, bulldozed, flattened, or razed, including removal of building debris.

Visible emissions - Any emissions, which are visually detectable without the aid of instruments, coming from ACM/PACM/RACM/ACS or ACM waste material.

Waste/Equipment decontamination facility (W/EDF) – The area in which equipment is decontaminated before removal from the regulated area.

Waste generator - Any owner or operator whose act or process produces asbestos-containing waste material.

Waste shipment record - The shipping document, required to be originated and signed by the waste generator, used to track and substantiate the disposition of asbestos-containing waste material.

Wet cleaning - The process of thoroughly eliminating, by wet methods, any asbestos contamination from surfaces or objects.

c. Referenced Standards Organizations: See Section 01 42 19 REFERENCED

STANDARDS.

1.5 APPLICABLE CODES AND REGULATIONS

a. General Applicability of Codes, Regulations, and Standards:

(1) All work under this contract shall be done in strict accordance with all applicable Federal, State, and Local regulations, standards and codes governing asbestos abatement, and any other trade work done in conjunction with the abatement. All applicable codes, regulations and standards are adopted into this specification and will have the same force and effect as this specification.

(2) The most recent edition of any relevant regulation, standard, document or code shall be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirement(s) shall be utilized.

(3) Copies of all standards, regulations, codes and other applicable documents, including this specification and those listed in Section 1.5 shall be available at the worksite in the clean change area of the worker decontamination system and/or the Contractor’s on-site Field Office. These standards, regulations, codes and other applicable documents, including this specification and those listed in Section 1.5 may be made available electronically.

b. Asbestos Abatement Contractor Responsibility: The Asbestos Abatement Contractor (Contractor) shall assume full responsibility and liability for compliance with all applicable Federal, State and Local regulations related to any and all aspects of the asbestos abatement project. The Contractor is responsible for providing and maintaining training, accreditations, medical exams, medical records, personal protective equipment (PPE), respiratory protection, and respirator fit testing, as required by applicable Federal, State and Local regulations. The Contractor shall hold the VA and VPIH/CIH consultants harmless for any Contractor’s failure to comply with any applicable work, packaging, transporting, disposal, safety, health, or environmental requirement on the part of himself, his employees, or his subcontractors. The Contractor will incur all costs of the CPIH/CIH, including all sampling/analytical costs to assure compliance with OSHA/EPA/State/Local requirements related to failure to comply with the regulations applicable to the work.

c. Federal Requirements:

(1) Federal requirements which govern asbestos abatement include, but are not limited to, the following regulations:

(a) Occupational Safety and Health Administration (OSHA)

1. Title 29 CFR 1926.1101 - Construction Standard for Asbestos

2. Title 29 CFR 1926 Subpart E - Personal Protective Equipment and Life Saving Equipment

3. Title 29 CFR 1910.134 - Respiratory Protection

4. Title 29 CFR 1926 - Construction Industry Standards

5. Title 29 CFR 1926.33 - Access to Employee Exposure and Medical Records

6. Title 29 CFR 1926.59 same as 1910.1200 - Hazard Communication

7. Title 29 CFR 1926 Subpart C – General Safety and Health Provisions and Subpart D – Occupational Health and Environmental Controls

(b) Environmental Protection Agency (EPA):

1. 40 CFR 61 Subpart M - National Emission Standard for Hazardous Air Pollutants - Asbestos.

2. 40 CFR 763 - Asbestos Hazard Emergency Response Act (AHERA) and Asbestos Hazard Abatement Reauthorization Act (ASHARA)

(c) Department of Transportation (DOT)

1. Title 49 CFR 171 - 180 – Transportation

d. State Requirements:

(1) State requirements that apply to the asbestos abatement work, disposal, clearance, etc., include, but are not limited to, the following:
Connecticut General Statutes Title 19A - Public health and Well-Being Section 329 to 333
Connecticut General Statutes Title 20 - Professional and Occupational Licensing, Certification, Title Protection and Registration Section 435 to 442

e. Local Requirements:

(1) If local requirements are more stringent than federal or state standards, the local standards are to be followed.

f. Standards:

(1) Standards which govern asbestos abatement activities include, but are not limited to, the following:

(a) American National Standards Institute ASSP Z9.2 - Fundamentals Governing the Design and Operation of Local Exhaust Systems and ASSP Z88.2 - Practices for Respiratory Protection.

(b) Underwriters Laboratories UL 586 - UL Standard for Safety of HEPA Filter Units, 9th Edition; ANSI Approval 2017-12-19.

(2) Standards which govern encapsulation work include, but are not limited to the following:

(a) American Society for Testing and Materials International (ASTM)

(3) Standards which govern the fire and safety concerns in abatement work include, but are not limited to, the following:

(a) National Fire Protection Association NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

(b) NFPA 701 - Standard Methods for Fire Tests for Flame Resistant Textiles and Film.

(c) NFPA 101 - Life Safety Code

g. EPA Guidance Documents:

(1) EPA guidance documents which discuss asbestos abatement work activities are listed below. These documents are made part of this section by reference.

(2) Guidance for Controlling ACM in Buildings (Purple Book) EPA 560/5-85-024

(3) Asbestos Waste Management Guidance EPA 530-SW-85-007

(4) A Guide to Respiratory Protection for the Asbestos Abatement Industry EPA 560-OPTS-86-001

(5) Guide to Managing Asbestos in Place (Green Book) TS 799 20T July 1990

h. Notices:

(1) State and Local agencies: Send written notification as required by State and Local regulations prior to beginning any work on ACM as follows:

(2) Copies of notifications shall be submitted to the VA for the facility's records in the same time frame notification are given to EPA, State, and Local authorities.

i. Permits/Licenses: The contractor shall apply for and have all required permits and licenses to perform asbestos abatement work as required by Federal, State, and Local regulations.

j. Posting and Filing of Regulations: Maintain two (2) copies of applicable Federal, State, and Local regulations. Post one copy of each in the clean room at the regulated area where workers will have daily access to the regulations and keep another hard copy or electronic copy in the Contractor's office.

k. VA Responsibilities prior to commencement of work:

(1) Notify occupants adjacent to regulated areas of project dates and requirements for relocation, if needed. Arrangements must be made prior to starting work for relocation of desks, files, equipment, and personal possessions to avoid unauthorized access into the regulated area. **Note: Notification of adjacent personnel is required by OSHA in 29 CFR 1926.1101 (k) to prevent unnecessary or unauthorized access to the regulated area**.

(2) Submit to the Contractor results of background air sampling; including location of samples, person who collected the samples, equipment utilized, calibration data and method of analysis. During abatement, submit to the Contractor, results of bulk

material analysis and air sampling data collected during the course of the abatement. This information shall not release the Contractor from any responsibility for OSHA compliance.

l. Emergency Action Plan and Arrangements:

(1) An Emergency Action Plan shall be developed prior to commencing abatement activities and shall be agreed to by the Contractor and the VA. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.

(2) Emergency procedures shall be in written form and prominently posted in the clean room and equipment room of the decontamination unit. Everyone, prior to entering the regulated area, must read and sign these procedures to acknowledge understanding of the regulated area layout, location of emergency exits and emergency procedures.

(3) Emergency planning shall include written notification of police, fire, and emergency medical personnel of planned abatement activities; work schedule; layout of regulated area; and access to the regulated area, particularly barriers that may affect response capabilities.

(4) Emergency planning shall include consideration of fire, explosion, hazardous atmospheres, electrical hazards, slips/trips and falls, confined spaces, and heat stress illness. Written procedures for response to emergency situations shall be developed and employee training in procedures shall be provided.

(5) Employees shall be trained in regulated area/site evacuation procedures in the event of workplace emergencies.

(a) For non-life-threatening situations - employees injured or otherwise incapacitated shall be decontaminated following normal procedures with assistance from fellow workers, if necessary, before exiting the regulated area to obtain proper medical treatment.

(b) For life-threatening injury or illness, worker decontamination shall take least priority after measures to stabilize the injured worker, medical personnel shall remove them from the regulated area if back or neck injury is present, and secure proper medical treatment.

(6) Telephone numbers of any/all emergency response personnel shall be prominently posted in the clean room, along with the location of the nearest telephone.

(7) The Contractor shall provide verification of first aid/CPR training for personnel responsible for providing first aid/CPR. OSHA requires medical assistance within 3-4 minutes of a life-threatening injury/illness. Bloodborne Pathogen training shall also be verified for those personnel required to provide first aid/CPR.

(8) The Emergency Action Plan shall provide for a Contingency Plan in the event that an incident occurs that may require the modification of the standard operating procedures during

abatement. Such incidents include, but are not limited to, fire; accident; power failure; negative pressure failure; and supplied air system failure. The Contractor shall detail procedures to be followed in the event of an incident assuring that asbestos abatement work is stopped and wetting is continued until correction of the problem.

m. Pre-Construction Meeting:

(1) Prior to commencing the work, the Contractor shall meet with the VA Certified Industrial Hygienist (VPIH/CIH) to present and review, as appropriate, the items following this paragraph. The Contractor's Competent Person(s) who will be on-site shall participate in the pre-start meeting. The pre-start meeting is to discuss and determine procedures to be used during the project. At this meeting, the Contractor shall provide:

(a) Proof of Contractor licensing.

(b) Proof the Competent Person(s) is trained and accredited and approved for working in this State. Verification of the experience of the Competent Person(s) shall also be presented.

(c) A list of all workers who will participate in the project, including experience and verification of training and accreditation.

(d) A list of and verification of training for all personnel who have current first-aid/CPR training. A minimum of one person per shift must have adequate training.

(e) Current medical written opinions for all personnel working on-site meeting the requirements of 29 CFR 1926.1101 (m).

(f) Current fit-tests for all personnel wearing respirators on-site meeting the requirements of 29 CFR 1926.1101 (h) and Appendix C.

(g) A copy of the Contractor's Asbestos Hazard Abatement Plan. In these procedures, the following information must be detailed, specific for this project.

1. Regulated area preparation procedures;

2. Notification requirements procedure of Contractor as required in 29 CFR 1926.1101 (d) Multi-Employer Worksites;

3. Decontamination area set-up/layout and decontamination procedures for employees;

4. Abatement methods/procedures and equipment to be used;

5. Personal protective equipment to be used.

(h) At this meeting the Contractor shall provide all submittals as required.

(i) Procedures for handling, packaging and disposal of asbestos waste.

(j) Emergency Action Plan and Contingency Plan Procedures.

1.6 PROJECT COORDINATION

a. The following are the minimum administrative and supervisory personnel necessary for coordination of the work.

(1) Personnel:

(a) Administrative and supervisory personnel shall consist of a qualified Competent Person(s) as defined by OSHA in the Construction Standards and the Asbestos Construction Standard; Contractor Professional Industrial Hygienist and Industrial Hygiene Technicians. These employees are the Contractor's representatives responsible for compliance with these specifications and all other applicable requirements.

(b) Non-supervisory personnel shall consist of an adequate number of qualified personnel to meet the schedule requirements of the project. Personnel shall meet required qualifications. Personnel utilized on-site shall be pre-approved by the VA representative. A request for approval shall be submitted for any person to be employed during the project giving the person's name; last four digits of social security number; qualifications; accreditation card with color picture, if required by state; Certificate of Worker's Acknowledgment (Attachment C); and Affidavit of Medical Surveillance and Respiratory Protection and current Respirator Fit Test.

(c) Minimum qualifications for Contractor and assigned personnel are:

1. The Contractor has conducted within the last three (3) years, three (3) projects of similar complexity and dollar value as this project; has not been cited and penalized for serious violations of Federal (and State or Local as applicable) EPA and OSHA asbestos regulations in the past three (3) years; has adequate liability/occurrence insurance for asbestos work as required by the state; is licensed in applicable state; has adequate and qualified personnel available to complete the work; has comprehensive standard operating procedures for asbestos work; has adequate materials, equipment and supplies to perform the work.

2. The Competent Person has four (4) years of abatement experience of which two (2) years were as the Competent Person on the project; meets the OSHA definition of a Competent Person; has been the Competent Person on two (2) projects of similar size and complexity as this project within the past three (3) years; has completed EPA AHERA/OSHA/State/Local training requirements/accreditation(s) and refreshers; and has all required OSHA documentation related to medical and respiratory protection.

3. The Contractor Professional Industrial Hygienist/CIH (CPIH/CIH) shall have five (5) years of monitoring experience and supervision of asbestos abatement projects; has participated as senior IH on five (5) abatement projects, three (3) of which are similar in size and complexity as this project; has specialized EPA AHERA/OSHA training in asbestos abatement management,

respiratory protection, waste disposal and asbestos inspection; has completed the NIOSH 582 Course or equivalent, Contractor/Supervisor course; and has appropriate medical/respiratory protection records/documentation.

4. The Abatement Personnel shall have completed the EPA AHERA/OSHA abatement worker course; have training on the standard operating procedures of the Contractor; has one year of asbestos abatement experience within the past three (3) years of similar size and complexity; has applicable medical and respiratory protection documentation; has certificate of training/current refresher and State accreditation/license (Attachment C).

(d) All personnel shall be in compliance with OSHA construction safety training as applicable and submit certification.

1.7 RESPIRATORY PROTECTION

a. General - Respiratory Protection Program: The Contractor shall develop and implement a written Respiratory Protection Program (RPP) which is in compliance with OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.134. ASSP Z88.2 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH approved for asbestos abatement activities. The written RPP shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134(c) - Respiratory Protection Program.

b. Respiratory Protection Program Coordinator: The Respiratory Protection Program Coordinator (RPPC) must be identified and shall have two (2) years of experience coordinating RPP of similar size and complexity. The RPPC must submit a signed statement attesting to the fact that the program meets the above requirements.

c. Selection and Use of Respirators: The procedure for the selection and use of respirators must be submitted to the VA as part of the Contractor's qualifications. The procedure must be written clearly enough for workers to understand. A copy of the Respiratory Protection Program must be available in the clean room of the decontamination unit or in the onsite Contractor’s office, for reference by employees or authorized visitors.

d. Minimum Respiratory Protection: Shall be a ½-mask negative pressure air purifying respirator equipped with P100 filters, provided personal air samples in the workplace remain at or below 0.1 f/cc, determined as an 8-hour TWA. Full face powered air purifying respirator equipped with P100 filters shall be required until Contractor demonstrates that personal air samples are at or below 0.1 f/cc, determined as an 8-hour TWA. A higher level of respiratory protection shall be required, if fiber levels exceed 1 f/cc as an 8-hour TWA, inside the regulated work area. Respirator selection shall meet the requirements of 29 CFR 1926.1101(h) and 29 CFR 1910.134(d)(3)(i)(A) Table 1, except as indicated in this paragraph. Abatement personnel must have a respirator for their exclusive use.

e. Medical Written Opinion: No employee shall be allowed to wear a respirator unless a physician or other licensed health care professional has provided a written determination they are medically qualified to wear the class of respirator to be used on the project while wearing whole body impermeable garments and subjected to heat or

cold stress.

f. Respirator Fit Test: All personnel wearing respirators shall have a current qualitative/quantitative fit test which was conducted in accordance with 29 CFR 1910.134(f) and Appendix A. Quantitative fit tests shall be done for PAPRs which have been put into a motor/blower failure mode.

g. Respirator Fit Check: The Competent Person shall assure that the positive/negative pressure user seal check is done each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that prevents an effective facepiece to face seal as evidenced by failure of a user seal check shall preclude that person from entering the regulated area until resolution of the problem.

h. Maintenance and Care of Respirators: The Respiratory Protection Program Coordinator shall submit evidence and documentation showing compliance with 29 CFR 1910.134(h) Maintenance and Care of Respirators.

i. Supplied Air Systems: If a supplied air system is used, the system shall meet all requirements of 29 CFR 1910.134 and the ANSI/Compressed Gas Association (CGA) Commodity Specification for Air current requirements for Type 1 - Grade D breathing air. Low pressure systems are not allowed to be used on asbestos abatement projects. Supplied Air respirator use shall be in accordance with EPA/NIOSH publication EPA 560-OPTS-86-001 "A Guide to Respiratory Protection for the Asbestos Abatement Industry". The competent person on site will be responsible for the supplied air system to ensure the safety of the worker.

1.8 WORKER PROTECTION

a. Training of Abatement Personnel: Prior to beginning any abatement activity, all personnel shall be trained in accordance with OSHA 29 CFR 1926.1101(k)(9) and any additional State/Local requirements. Training must include, at a minimum, the elements listed at 29 CFR 1926.1101(k)(9)(viii). Training shall have been conducted by a third party, EPA/State approved trainer meeting the requirements of EPA 40 CFR 763 Appendix C (AHERA MAP). Initial training certificates and current refresher and accreditation proof must be submitted for each person working at the site.

b. Medical Examinations: Medical examinations meeting the requirements of 29 CFR 1926.1101(m) shall be provided for all personnel working in the regulated area, regardless of exposure levels. A current physician's written opinion as required by 29 CFR 1926.1101(m)(4) shall be provided for each person and shall include in the medical opinion that the person has been evaluated for working in a heat and cold stress environment while wearing personal protective equipment (PPE) and is able to perform the work without risk of material health impairment.

c. Regulated Area Entry Procedure: The Competent Person shall ensure that each time workers enter the regulated area; they remove ALL street clothes in the clean room of the decontamination unit and put on new disposable coveralls, head coverings, a clean respirator, and then proceed through the shower room to the equipment room where they put on non-disposable required personal protective equipment.

d. Decontamination Procedure: The Competent Person shall require all personnel to adhere to following decontamination procedures whenever they leave the regulated area.

(1) When exiting the regulated area, remove disposable coveralls, and ALL other clothes, disposable head coverings, and foot coverings or boots in the equipment room.

(2) Still wearing the respirator and completely naked, proceed to the shower. Showering is MANDATORY. Care must be taken to follow reasonable procedures in removing the respirator to avoid inhaling asbestos fibers while showering. The following procedure is required as a minimum:

(a) Thoroughly wet body including hair and face. If using a PAPR hold blower above head to keep filters dry.

(b) With respirator still in place, thoroughly decontaminate body, hair, respirator face piece, and all other parts of the respirator except the blower and battery pack on a PAPR. Pay particular attention to cleaning the seal between the face and respirator facepiece and under the respirator straps.

(c) Take a deep breath, hold it and/or exhale slowly, completely wetting hair, face, and respirator. While still holding breath, remove the respirator and hold it away from the face before starting to breathe.

(3) Carefully decontaminate the facepiece of the respirator inside and out. If using a PAPR, shut down using the following sequence: a) first cap inlets to filters; b) turn blower off to keep debris collected on the inlet side of the filter from dislodging and contaminating the outside of the unit; c) thoroughly decontaminate blower and hoses; d) carefully decontaminate battery pack with a wet rag being cautious of getting water in the battery pack thus preventing destruction. **(THIS PROCEDURE IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!)**

(4) Shower and wash body completely with soap and water. Rinse thoroughly.

(5) Rinse shower room walls and floor to drain prior to exiting.

(6) Proceed from shower to clean room; dry off and change into street clothes or into new disposable work clothing.

e. Regulated Area Requirements: The Competent Person shall meet all requirements of 29 CFR 1926.1101(o) and assure that all requirements for regulated areas at 29 CFR 1926.1101(e) are met. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator.

1.9 DECONTAMINATION FACILITIES

a. Description: Provide each regulated area with separate personnel decontamination facilities (PDF) and waste/equipment decontamination facilities (W/EDF). Ensure that the PDF are the only means of ingress

and egress to the regulated area and that all equipment, bagged waste, and other material exit the regulated area only through the W/EDF.

b. General Requirements: All personnel entering or exiting a regulated area must go through the PDF and shall follow the requirements at 29 CFR 1926.1101(j)(1) and these specifications. All waste, equipment and contaminated materials must exit the regulated area through the W/EDF and be decontaminated in accordance with these specifications. Walls and ceilings of the PDF and W/EDF must be constructed of a minimum of 3 layers of 6-mil opaque fire retardant polyethylene sheeting and be securely attached to existing building components and/or an adequate temporary framework. A minimum of 3 layers of 6-mil poly shall also be used to cover the floor under the PDF and W/EDF units. Construct doors so that they overlap and secure to adjacent surfaces. Weight inner doorway sheets with layers of duct tape or approved equivalent so that they close quickly after release. Put arrows on sheets so they show direction of travel and overlap. If the building adjacent area is occupied, construct a solid barrier on the occupied side(s) to protect the sheeting and reduce potential for non-authorized personnel entering the regulated area.

c. Temporary Facilities to the PDF and W/EDF: The Competent Person shall provide temporary water service connections to the PDF and W/EDF. Backflow prevention must be provided at the point of connection to the VA system. Water supply must be of adequate pressure and meet requirements of 29 CFR 1910.141(d)(3). Provide adequate temporary overhead electric power with ground fault circuit interruption (GFCI) protection. Provide a sub-panel equipped with GFCI protection for all temporary power in the clean room. Provide adequate lighting to provide a minimum of 50 foot candles in the PDF and W/EDF. Provide temporary heat, if needed, to maintain 70 degrees F throughout the PDF and W/EDF.

d. Personnel Decontamination Facility (PDF): The Competent Person shall provide a PDF consisting of shower room which is contiguous to a clean room and equipment room which is connected to the regulated area. The PDF must be sized to accommodate the number of personnel scheduled for the project. The shower room, located in the center of the PDF, shall be fitted with as many portable showers as necessary to insure all employees can complete the entire decontamination procedure within 15 minutes. The PDF shall be constructed of opaque poly for privacy. The PDF shall be constructed to eliminate any parallel routes of egress without showering.

(1) Clean Room: The clean room must be physically and visually separated from the rest of the building to protect the privacy of personnel changing clothes. The clean room shall be constructed of at least 3 layers of 6-mil opaque fire retardant poly to provide an air tight room. Provide a minimum of 2 - 3 foot wide 6-mil poly opaque fire retardant doorways. One doorway shall be the entry from outside the PDF and the second doorway shall be to the shower room of the PDF. The floor of the clean room shall be maintained in a clean, dry and sanitary condition. Shower overflow shall not be allowed into the clean room. Provide 1 storage locker per person. A portable fire extinguisher, minimum 10 pounds capacity, Type ABC, shall be provided in accordance with OSHA and NFPA 10. All persons entering the regulated area shall remove all street clothing in the clean room and dress in disposable protective clothing and respiratory protection. Any

person entering the clean room does so either from the outside with street clothing on or is coming from the shower room completely naked and thoroughly washed. Male/Females required to enter the regulated area shall be ensured of their privacy throughout the entry/exit process by posting guards at both entry points to the PDF so no male/female can enter or exit the PDF during his/her stay in the PDF.

(2) Shower Room: The Competent Person shall assure that the shower room is a completely water tight compartment to be used for the movement of all personnel from the clean room to the equipment room and for the showering of all personnel going from the equipment room to the clean room. Each shower shall be constructed so water runs down the walls of the shower and into a drip pan. Install a freely draining smooth floor on top of the shower pan. The shower room shall be separated from the rest of the building and from the clean room and equipment room using air tight walls made from at least 3 layers of 6-mil opaque fire retardant poly. The shower shall be equipped with a shower head and controls, hot and cold water, drainage, soap dish and continuous supply of soap, and shall be maintained in a sanitary condition throughout its use. The controls shall be arranged so an individual can shower without assistance. Provide a flexible hose shower head, hose bibs and all other items shown on Shower Schematic. Waste water will be pumped to a drain after being filtered through a minimum of a 100 micron sock in the shower drain; a 20 micron filter; and a final 5 micron filter. Filters will be changed a minimum of daily or more often as needed. Filter changes must be done in the shower to prevent loss of contaminated water. Hose down all shower surfaces after each shift and clean any debris from the shower pan. Residue is to be disposed of as asbestos waste.

(3) Equipment Room: The Competent Person shall provide an equipment room which shall be an air tight compartment for the storage of work equipment/tools, reusable personal protective equipment, except for a respirator and for use as a gross decontamination area for personnel exiting the regulated area. The Competent Person shall ensure that most gross contamination is removed from the outside of PPE, suits and respirators in the regulated work area, prior to entering the Equipment Room. The equipment room shall be separated from the regulated area by a minimum 3 foot wide door made with 2 layers of 6-mil opaque fire retardant poly. The equipment room shall be separated from the regulated area, the shower room and the rest of the building by air tight walls and ceiling constructed of a minimum of 3 layers of 6-mil opaque fire retardant poly. Damp wipe all surfaces of the equipment room after each shift change. Provide an additional loose layer of 6-mil fire retardant poly per shift change and remove this layer after each shift. If needed, provide a temporary electrical sub-panel equipped with GFCI in the equipment room to accommodate any equipment required in the regulated area.

(4) The PDF shall be as follows: Clean room at the entrance followed by a shower room followed by an equipment room leading to the regulated area. Each doorway in the PDF shall be a minimum of 2 layers of 6-mil opaque fire retardant poly.

See Attachment A

e. Waste/Equipment Decontamination Facility (W/EDF):

(1) The Competent Person shall provide a W/EDF consisting of a wash room, holding room, and clean room for removal of waste, equipment and contaminated material from the regulated area. Personnel shall not enter or exit the W/EDF except in the event of an emergency. Clean debris and residue in the W/EDF daily. All surfaces in the W/EDF shall be wiped/hosed down after each shift and all debris shall be cleaned from the shower pan. The W/EDF shall consist of the following:

(a) Wash Down Station: Provide an enclosed shower unit in the regulated area just outside the Wash Room as an equipment bag and container cleaning station.

(b) Wash Room: Provide a wash room for cleaning of bagged or containerized asbestos containing waste materials passed from the regulated area. Construct the wash room using 2 inches x 4 inches wood framing or approved equivalent and 3 layers of 6-mil fire retardant poly. Locate the wash room so that packaged materials, after being wiped clean, can be passed to the Holding Room. Doorways in the wash room shall be constructed of 2 layers of 6-mil fire retardant poly.

(c) Holding Room: Provide a holding room as a drop location for bagged materials passed from the wash room. Construct the holding room using 2 inches x 4 inches wood framing or approved equivalent and 3 layers of 6-mil fire retardant poly. The holding room shall be located so that bagged material cannot be passed from the wash room to the clean room unless it goes through the holding room. Doorways in the holding room shall be constructed of 2 layers of 6-mil fire retardant poly.

(d) Clean Room: Provide a clean room to isolate the holding room from the exterior of the regulated area. Construct the clean room using 2 inches x 4 inches wood framing or approved equivalent and 2 layers of 6-mil fire retardant poly. The clean room shall be located so as to provide access to the holding room from the building exterior. Doorways to the clean room shall be constructed of 2 layers of 6-mil fire retardant poly. When a negative pressure differential system is used, a rigid enclosure separation between the W/EDF clean room and the adjacent areas shall be provided.

(e) The W/EDF shall be as follows: Wash Room leading to a Holding Room followed by a Clean Room leading to outside the regulated area. See diagram Attachment B.

f. Waste/Equipment Decontamination Procedures: At the washdown station in the regulated area, thoroughly wet clean contaminated equipment and/or sealed polyethylene bags and pass into Wash Room after visual inspection. When passing anything into the Wash Room, close all doorways of the W/EDF, other than the doorway between the washdown station and the Wash Room. Keep all outside personnel clear of the W/EDF. Once inside the Wash Room, wet clean the equipment and/or bags. After cleaning and inspection, pass items into the Holding Room. Close all doorways except the doorway between the Holding Room and the Clean Room. Workers from the Clean Room/Exterior shall enter

the Holding Room and remove the decontaminated/cleaned equipment/bags for removal and disposal. At no time shall personnel from the clean side be allowed to enter the Wash Room.

1.10 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z9.2	(2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems
ASSP Z88.2	(2015) American National Standard Practices for Respiratory Protection

ASTM INTERNATIONAL (ASTM)

ASTM C732	(2017; R 2022 Standard Test Method for Aging Effects of Artificial Weathering on Latex Sealants
ASTM D522	(1993a; R 2008) Mandrel Bend Test of Attached Organic Coatings
ASTM D2794	(1993; R 2019) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E736	(2000; R 2011) Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members
ASTM E814	(2024) Standard Test Method for Fire Tests of Penetration Firestop Systems

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7.1	(2011) Commodity Specification for Air; 5th Edition

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable

	Fire Extinguishers
NFPA 101	(2024) Life Safety Code
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 701	(2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-282	(2015; Rev B; Notice 1 2020) Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 530-SW-85-007	(1985) Asbestos Waste Management Guidance
EPA 560/5-85-024	(1985) Guidance for Controlling Asbestos-Containing Materials in Buildings (Purple Book)
EPA 560-OPTS-86-001	(1986) A Guide to Respiratory Protection for the Asbestos Abatement Industry
TS 799 20T	(1990) Guide to Managing Asbestos in Place (Green Book)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.132	Personal Protective Equipment
29 CFR 1910.134	Respiratory Protection
29 CFR 1910.141	Sanitation
29 CFR 1910.1200	Hazard Communication
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.32	Definitions
29 CFR 1926.33	Access to Employee Exposure and Medical Records
29 CFR 1926.59	Hazard Communication
29 CFR 1926.1101	Asbestos
40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 763	Asbestos

49 CFR 171	General Information, Regulations, and Definitions

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH 582	Sampling and Evaluating Airborne Asbestos Fibers Coruses
NIOSH 7400	Airborne Fiber Count

UL SOLUTIONS (UL)

UL 586	(2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units
UL 1479	(2015; Reprint Apr 2024) Fire Tests of Through-Penetration Firestops

1.11 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. . When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Pre-Start Meeting Submittals; G

Materials And Equipment; G

Asbestos Hazard Abatement Plan; G

SD-03 Product Data

Encapsulants; G

SD-06 Test Reports

Submittals During Abatement; G

Monitoring, Inspection And Testing; G

SD-11 Closeout Submittals

Submittals At Completion Of Abatement; G

1.12 SUBMITTAL INSTRUCTIONS

a. Pre-Start Meeting Submittals:

(1) Submit to the VA a minimum of 14 days prior to the pre-start meeting the following for review and approval. Meeting this requirement is a prerequisite for the pre-start meeting for this

project:

(a) Submit a detailed work schedule for the entire project reflecting contract documents and the phasing/schedule requirements from the CPM chart.

(b) Submit a staff organization chart showing all personnel who will be working on the project and their capacity/function. Provide their qualifications, training, accreditations, and licenses, as appropriate. Provide a copy of the "Certificate of Worker's Acknowledgment (Attachment C)" and the "Affidavit of Medical Surveillance and Respiratory Protection" for each person.

(c) Submit Asbestos Hazard Abatement Plan developed specifically for this project, incorporating the requirements of the specifications, prepared, signed and dated by the CPIH/CIH.

(d) Submit the specifics of the materials and equipment to be used for this project with manufacturer names, model numbers, performance characteristics, pictures/diagrams, and number available for the following:

1. Supplied air system, negative air machines, HEPA vacuums, air monitoring pumps, calibration devices, pressure differential monitoring device and emergency power generating system.

2. Waste water filtration system, shower system, containment barriers.

3. Encapsulants, surfactants, hand held sprayers, airless sprayers, glovebags, and fire extinguishers.

4. Respirators, protective clothing, fall protection and other required personal protective equipment.

5. Fire safety equipment to be used in the regulated area.

(e) Submit the name, location, and phone number of the approved landfill; proof/verification the landfill is approved for ACM disposal; the landfill's requirements for ACM waste; the type of vehicle to be used for transportation; and name, address, and phone number of subcontractor, if used. Proof of asbestos training for transportation personnel shall be provided.

(f) Submit required notifications and arrangements made with regulatory agencies having regulatory jurisdiction and the specific contingency/emergency arrangements made with local health, fire, ambulance, hospital authorities and any other notifications/arrangements.

(g) Submit the name, location and verification of the laboratory and/or personnel to be used for analysis of air and/or bulk samples. Personal air monitoring must be done in accordance with OSHA 29 CFR 1926.1101(f) and Appendix A. Area or clearance air monitoring shall be conducted in accordance with EPA AHERA protocols.

(h) Submit qualifications verification: Submit the following evidence of qualifications. Make sure that all references are

current and verifiable by providing current phone numbers and documentation.

1. Asbestos Abatement Company: Project experience within the past 3 years; listing projects first most similar to this project: Project Name; Type of Abatement; Duration; Cost; Reference Name/Phone Number; Final Clearance; Completion Date

2. List of project(s) halted by owner, A/E, IH, regulatory agency in the last 3 years: Project Name; Reason; Date; Reference Name/Number; Resolution

3. List asbestos regulatory citations (e.g., OSHA), notices of violations (e.g., Federal and state EPA), penalties, and legal actions taken against the company including the company’s officers (including damages paid) in the last 3 years. Provide copies and all information needed for verification.

(i) Submit information on personnel: Provide a resume; address each item completely; copies of certificates, accreditations, and licenses. Submit an affidavit signed by the CPIH/CIH stating that all personnel submitted below have medical records in accordance with OSHA 29 CFR 1926.1101(m) and that the company has implemented a medical surveillance program and written respiratory protection program, and maintains recordkeeping in accordance with the above regulation. Submit the phone number and doctor/clinic/hospital used for medical evaluations.

1. CPIH/CIH and IH Technician: Name; years of abatement experience; list of projects similar to this one; certificates, licenses, accreditations for proof of AHERA/OSHA specialized asbestos training; professional affiliations; medical opinion; and current respirator fit test.

2. Competent Person(s)/Supervisor(s): Number; names; last four digits of social security numbers; years of abatement experience as Competent Person/Supervisor; list of similar projects in size/complexity as Competent Person/Supervisor; as a worker; certificates, licenses, accreditations; proof of AHERA/OSHA specialized asbestos training; maximum number of personnel supervised on a project; medical opinion (asbestos surveillance and respirator use); and current respirator fit test.

3. Workers: Numbers; names; last four digits of social security numbers; years of abatement experience; certificates, licenses, accreditations; training courses in asbestos abatement and respiratory protection; medical opinion (asbestos surveillance and respirator use); and current respirator fit test.

(j) Submit copies of State license for asbestos abatement; copy of insurance policy, including exclusions with a letter from agent stating in plain language the coverage provided and the fact that asbestos abatement activities are covered by the policy; copy of AHAPs incorporating the requirements of this specification; information on who provides your training, how often; who provides medical surveillance, how often; who performs and how is personal air monitoring of abatement workers conducted; a list of references of independent laboratories/IH's familiar with your air monitoring and standard operating procedures; and copies of

monitoring results of the five referenced projects listed and analytical methods used.

(k) Rented equipment must be decontaminated prior to returning to the rental agency.

(l) Submit, before the start of work, the manufacturer's technical data for all types of encapsulants, all SDS and application instructions.

b. Submittals During Abatement:

(1) The Competent Person shall maintain and submit a daily log at the regulated area documenting the dates and times of the following: purpose, attendees and summary of meetings; all personnel entering/exiting the regulated area; document and discuss the resolution of unusual events such as barrier breeching, equipment failures, emergencies, and any cause for stopping work; and representative air monitoring and results/TWA's/EL's. Submit this information daily to the VA's representative.

(2) The CPIH/CIH shall document and maintain the inspection and approval of the regulated area preparation prior to start of work and daily during work.

(a) Removal of any poly barriers.

(b) Visual inspection/testing by the CPIH/CIH or IH Technician prior to application of lockdown encapsulant.

(c) Packaging and removal of ACM waste from regulated area.

(d) Disposal of ACM waste materials; copies of Waste Shipment Records/landfill receipts to the VA's representative on a weekly basis.

c. Submittals at Completion of Abatement: The CPIH/CIH shall submit a project report consisting of the daily log book requirements and documentation of events during the abatement project including Waste Shipment Records signed by the landfill's agent. It will also include information on the containment and transportation of waste from the containment with applicable Chain of Custody forms. The report shall include a certificate of completion, signed and dated by the CPIH/CIH, in accordance with Attachment C (Attachment #1). All clearance and perimeter area samples must be submitted. The VA representative will retain the abatement report after completion of the project and provide copies of the abatement report to VAMC Office of Engineer and the Safety Office.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

a. General Requirements: Prior to the start of work, the contractor shall provide and maintain a sufficient quantity of materials and equipment to assure continuous and efficient work throughout the duration of the project. Work shall not start unless the following items have been delivered to the site and the CPIH/CIH has submitted verification to the VA's representative.

(1) All materials shall be delivered in their original package, container or bundle bearing the name of the manufacturer and the brand name (where applicable).

(2) Store all materials subject to damage off the ground, away from wet or damp surfaces and under cover sufficient enough to prevent damage or contamination. Flammable and combustible materials cannot be stored inside buildings. Replacement materials shall be stored outside of the regulated area until abatement is completed.

(3) The Contractor shall not block or hinder use of buildings by patients, staff, and visitors to the VA in partially occupied buildings by placing materials/equipment in any unauthorized location.

(4) The Competent Person shall inspect for damaged, deteriorating or previously used materials. Such materials shall not be used and shall be removed from the worksite and disposed of properly.

(5) Polyethylene sheeting for walls in the regulated area shall be a minimum of 4-mils. For floors and all other uses, sheeting of at least 6-mil shall be used in widths selected to minimize the frequency of joints. Fire retardant poly shall be used throughout.

(6) The method of attaching polyethylene sheeting shall be agreed upon in advance by the Contractor and the VA and selected to minimize damage to equipment and surfaces. Method of attachment may include any combination of moisture resistant duct tape, poly tape, furring strips, spray glue, staples, nails, screws, lumber and plywood for enclosures or approved equivalent procedures capable of sealing polyethylene to dissimilar finished or unfinished surfaces under both wet and dry conditions.

(7) Polyethylene sheeting utilized for the PDF shall be opaque white or black in color, 6-mil fire retardant poly.

(8) Installation and plumbing hardware, showers, hoses, drain pans, sump pumps and waste water filtration system shall be provided by the Contractor.

(9) An adequate number of HEPA vacuums, scrapers, sprayers, nylon brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the abatement project. All electrically operated hand tools, equipment, electric cords shall be connected to GFCI protection.

(10) Special protection for objects in the regulated area shall be detailed (e.g., plywood over carpeting or hardwood floors to prevent damage from scaffolds, water and falling material).

(11) Disposal bags – 2 layers of 6-mil poly for asbestos waste shall be pre-printed with labels, markings and address as required by OSHA, EPA and DOT regulations.

(12) The VA shall be provided an advance copy of the Safety Data Sheets (SDS) as required for all hazardous chemicals under 29 CFR 1910.1200 - Hazard Communication in the pre-start meeting submittal. Chlorinated compounds shall not be used with any spray adhesive, mastic remover or other product. Appropriate encapsulant(s) shall be provided.

(13) OSHA DANGER demarcation signs, as many and as required by 29 CFR 1926.1101(k)(7) shall be provided and placed by the Competent Person. All other posters and notices required by Federal and State regulations shall be posted in the Clean Room.

(14) Adequate and appropriate PPE for the project and number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d).

2.2 MONITORING, INSPECTION AND TESTING

a. General:

(1) Perform throughout abatement work monitoring, inspection and testing inside and around the regulated area in accordance with the OSHA requirements and these specifications. OSHA requires that the employee exposure to asbestos must not exceed 0.1 fiber per cubic centimeter (f/cc) of air, averaged over an 8-hour work shift. The CPIH/CIH is responsible for and shall inspect and oversee the performance of the Contractor IH Technician. The IH Technician shall continuously inspect and monitor conditions inside the regulated area to ensure compliance with these specifications. In addition, the CPIH/CIH shall personally manage air sample collection, analysis, and evaluation for personnel, regulated area, and adjacent area samples inside the building, but outside the regulated area. Inside the building, but outside the regulated area air samples shall be collected in the Clean Room of the PDF, at the approximate location of HEPA exhaust discharge, and at a minimum of three (3) locations in areas immediately outside the regulated work area to satisfy this specification. Additional inspection and testing requirements are also indicated in other parts of this specification.

(2) The VA will employ an independent industrial hygienist (VPIH/CIH) consultant and/or use its own IH to perform various services on behalf of the VA. The VPIH/CIH will perform the necessary monitoring, inspection, testing, and other support services to ensure that VA patients, employees, and visitors will not be adversely affected by the abatement work, and that the abatement work proceeds in accordance with these specifications, that the abated areas or abated buildings have been successfully decontaminated. The work of the VPIH/CIH consultant in no way relieves the Contractor from their responsibility to perform the work in accordance with contract/specification requirements, to perform continuous inspection, monitoring and testing for the safety of their employees, and to perform other such services as specified. The cost of the VPIH/CIH and their services will be borne by the VA except for any repeat of final inspection and testing that may be required due to unsatisfactory initial results. Any repeated final inspections and/or testing, if required, will be paid for by the Contractor.

(3) If fibers counted by the VPIH/CIH during abatement work, either inside or outside the regulated area, utilizing the NIOSH 7400 air monitoring method, exceed the specified respective limits, the Contractor shall stop work. The Contractor may request confirmation of the results by analysis of the samples by TEM. Request must be in writing and submitted to the VA's representative. Cost for the confirmation of results will be borne by the Contractor for both the collection and analysis of samples and for the time delay that may/does result for this confirmation. Confirmation sampling and analysis will be the responsibility of the CPIH/CIH with review and approval of the VPIH/CIH. An agreement between the CPIH/CIH and the VPIH/CIH shall be reached on the exact details of the confirmation effort, in writing, including such things as the number of samples, location, collection, quality control on-site, analytical laboratory, interpretation of results and any follow-up actions. This written agreement shall be co-signed by the IH's and delivered to the VA's representative.

b. Scope of Services of the VPIH/CIH Consultant:

(1) The purpose of the work of the VPIH/CIH is to: assure quality; adherence to the specification; resolve problems; prevent the spread of contamination beyond the regulated area; and assure clearance at the end of the project. In addition, their work includes performing the final inspection and testing to determine whether the regulated area or building has been adequately decontaminated. All air monitoring is to be done utilizing PCM/TEM. The VPIH/CIH will perform the following tasks:

(a) Task 1: Establish background levels before abatement begins by collecting background samples. Retain samples for possible TEM analysis.

(b) Task 2: Perform representative air monitoring, inspection, and testing outside the regulated area during actual abatement work to detect any faults in the regulated area isolation and any adverse impact on the surroundings from regulated area activities.

(c) Task 3: Perform unannounced visits to spot check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area and all aspects of the operation except personnel monitoring.

(d) Task 4: Provide support to the VA representative such as evaluation of submittals from the Contractor, resolution of conflicts, interpret data, etc.

(e) Task 5: Perform, in the presence of the VA representative, final inspection and testing of a decontaminated regulated area at the conclusion of the abatement to certify compliance with all regulations and VA requirements/specifications.

(f) Task 6: Issue certificate of decontamination (Attachment C) for each regulated area and project report.

(2) All documentation, inspection results and testing results

generated by the VPIH/CIH will be available to the Contractor for information and consideration. The Contractor shall cooperate with and support the VPIH/CIH for efficient and smooth performance of their work.

(3) The monitoring and inspection results of the VPIH/CIH will be used by the VA to issue any Stop Removal orders to the Contractor during abatement work and to accept or reject a regulated area or building as decontaminated.

c. Monitoring, Inspection and Testing by Contractor CPIH/CIH: The Contractor's CPIH/CIH is responsible for managing all monitoring, inspections, and testing required by these specifications, as well as any and all regulatory requirements adopted by these specifications. The CPIH/CIH is responsible for the continuous monitoring of all subsystems and procedures which could affect the health and safety of the Contractor's personnel. Safety and health conditions and the provision of those conditions inside the regulated area for all persons entering the regulated area is the exclusive responsibility of the Contractor/Competent Person. The person performing the personnel and area air monitoring inside the regulated area shall be an IH Technician, who shall be trained and shall have specialized field experience in sampling and analysis. The IH Technician shall have successfully completed a NIOSH 582 Course or equivalent and provide documentation. The IH Technician shall participate in the AIHA Asbestos Analysis Registry or participate in the Proficiency Analytical Testing program of AIHA for fiber counting quality control assurance. The IH Technician shall also be an accredited EPA AHERA/State Contractor/Supervisor and Building Inspector. The IH Technician shall have participated in five abatement projects collecting personal and area samples and have experience in substantially similar projects in size and scope. The analytical laboratory used by the Contractor to analyze the samples shall be AIHA accredited for asbestos PAT and approved by the VA prior to start of the project. A daily log shall be maintained by the CPIH/CIH or IH Technician, documenting all OSHA requirements for personal and area air monitoring for asbestos in 29 CFR 1926.1101(f), (g) and Appendix A. This log shall be made available to the VA representative and the VPIH/CIH upon request. The log will contain, at a minimum, information on personnel or area samples, other persons represented by the sample, the date of sample collection, start and stop times for sampling, sample volume, flow rate, and fibers/cc. The CPIH/CIH shall collect and analyze samples for each representative job being done in the regulated area, i.e., removal, wetting, clean-up, and load-out. No fewer than two (2) personal air samples or 25 percent of representative workforce per shift shall be collected, whichever is greater, in the regulated area; a minimum of three (3) area air samples at locations inside the building but immediately outside the regulated work area; one (1) area air sample shall be collected daily in the Clean Room of the PDF; and one (1) area air sample shall be collected daily at the approximate location of HEPA exhaust discharge. In addition to the continuous monitoring required, the CPIH/CIH will perform inspection and testing at the final stages of abatement for each regulated area as specified in the CPIH/CIH responsibilities. Additionally, the CPIH/CIH will monitor and record pressure readings within the containment daily with a minimum of two readings at the beginning and at the end of a shift, and submit the data in the daily report.

2.3 ASBESTOS HAZARD ABATEMENT PLAN

a. The Contractor shall have an established an Asbestos Hazard Abatement Plan (AHAP) in printed form and loose leaf folder consisting of simplified text, diagrams, sketches, and pictures that establish and explain clearly the procedures to be followed during all phases of the work by the Contractor's personnel. The AHAP must be modified as needed to address specific requirements of this project and the specifications. The AHAP shall be submitted for review and approval to the VA prior to the start of any abatement work. The minimum topics and areas to be covered by the AHAPs are:

(1) Minimum Personnel Qualifications

(2) Emergency Action Plan/Contingency Plans and Arrangements

(3) Security and Safety Procedures

(4) Respiratory Protection/Personal Protective Equipment Program and Training

(5) Medical Surveillance Program and Recordkeeping

(6) Regulated Area Requirements - Containment Barriers/Isolation of Regulated Area

(7) Decontamination Facilities and Entry/Exit Procedures (PDF and W/EDF)

(8) Negative Pressure Systems Requirements

(9) Monitoring, Inspections, and Testing

(10) Removal Procedures for ACM

(11) Removal of Contaminated Soil (if applicable)

(12) Encapsulation Procedures for ACM

(13) Disposal of ACM waste/equipment

(14) Regulated Area Decontamination/Clean-up

(15) Regulated Area Visual and Air Clearance

(16) Project Completion/Closeout

2.4 ENCAPSULANTS

a. Types of Encapsulants:

(1) The following four types of encapsulants, if used, must comply with performance requirements as stated in paragraph 2.5.2:

(a) Removal encapsulant - used as a wetting agent to remove ACM.

(b) Bridging encapsulant - provides a tough, durable coating on ACM.

(c) Penetrating encapsulant - penetrates/encapsulates ACM at least (1/2 inches).

(d) Lockdown encapsulant - seals microscopic fibers on surfaces after ACM removal.

b. Performance Requirements:

(1) Encapsulants shall meet the latest requirements of EPA; shall not contain toxic or hazardous substances; or solvents; and shall comply with the following performance requirements:

(a) General Requirements for all Encapsulants:

1. ASTM E84: Flame spread of 25; smoke emission of 50.

2. University of Pittsburgh Protocol: Combustion Toxicity; zero mortality.

3. ASTM C732: Accelerated Aging Test; Life Expectancy - 20 years.

4. ASTM E96/E96M: Permeability - minimum of 0.4 perms.

(b) Bridging/Penetrating Encapsulants:

1. ASTM E736: Cohesion/Adhesion Test - 24 kPa (50 pounds/square foot).

2. ASTM E119: Fire Resistance - 3 hours (Classified by UL for use on fibrous/cementitious fireproofing).

3. ASTM D2794: Gardner Impact Test; Impact Resistance - minimum 11.5 kg-mm (43 in/lb).

4. ASTM D522: Mandrel Bend Test; Flexibility - no rupture or cracking.

(c) Lockdown Encapsulants:

1. ASTM E119: Fire resistance - 3 hours (tested with fireproofing over encapsulant applied directly to steel member).

2. ASTM E736: Bond Strength - 48 kPa (100 pounds/square foot) (test compatibility with cementitious and fibrous fireproofing).

3. In certain situations, encapsulants may have to be applied to hot pipes/equipment. The encapsulant must be able to withstand high temperatures without cracking or creating any noxious gaseous or vapors during or after application.

c. Certificates of Compliance: The Contractor shall submit to the VA representative certification from the manufacturer indicating compliance with performance requirements for encapsulants when applied according to manufacturer recommendations.

PART 3 EXECUTION

3.1 REGULATED AREA PREPARATIONS

a. Site Security:

(1) Regulated area access is to be restricted only to authorized, trained/accredited and protected personnel. These may include the Contractor's employees, employees of Subcontractors, VA employees and representatives, State and Local inspectors, and any other designated individuals. A list of authorized personnel shall be established prior to commencing the project and be posted in the clean room of the decontamination unit.

(2) Entry into the regulated area by unauthorized individuals shall be reported immediately to the Competent Person by anyone observing the entry. The Competent Person shall immediately require any unauthorized person to leave the regulated area and then notify the VA Contracting Officer or VA Representative using the most expeditious means.

(3) A log book shall be maintained in the clean room of the decontamination unit. Anyone who enters the regulated area must record their name, affiliation, time in, and time out for each entry.

(4) Access to the regulated area shall be through a single decontamination unit. All other access (doors, windows, hallways, etc.) shall be sealed and demarcated as an asbestos regulated work area to prevent entry to or exit from the regulated area unless in the event of an emergency. The only exceptions for this requirement are the waste/equipment load-out area which shall be sealed except during the removal of containerized asbestos waste from the regulated area, and emergency exits. Emergency exits shall not be locked from the inside; however, they shall be sealed with poly sheeting and taped until needed. In any situation where exposure to high temperatures which may result in a flame hazard, fire retardant poly sheeting must be used.

(5) The Contractor's Competent Person shall control site security during abatement operations in order to isolate work in progress and protect adjacent personnel. A 24-hour security system shall be provided at the entrance to the regulated area to assure that all entrants are logged in/out and that only authorized personnel are allowed entrance.

(6) The Contractor will have the VA's assistance in notifying adjacent personnel of the presence, location and quantity of ACM in the regulated area and enforcement of restricted access by the VA's employees.

(7) The regulated area shall be locked during non-working hours and secured by VA Representative or Competent Person. The VA Police shall be informed of asbestos abatement regulated areas to provide security checks during facility rounds and emergency response.

b. Signage and Power Management:

(1) Post OSHA DANGER signs meeting the specifications of OSHA

29 CFR 1926.1101 at any location and approaches to the regulated area. Signs shall be posted at a distance sufficiently far enough away from the regulated area to permit any personnel to read the sign and take the necessary measures to avoid exposure. Additional signs will be posted following construction of the regulated area enclosure.

(2) Shut down and lock out/tag out electric power to the regulated area. Provide temporary power and lighting. Insure safe installation including GFCI of temporary power sources and equipment by compliance with all applicable electrical code and OSHA requirements for temporary electrical systems. Electricity shall be provided by the VA.

(3) Shut down and lock out/tag out heating, cooling, and air conditioning system (HVAC) components that are in, supply or pass through the regulated area. Investigate the regulated area and agree on pre-abatement condition with the VA's representative. Seal all intake and exhaust vents in the regulated area with duct tape and 2 layers of independently installed 6-mil poly. Also, seal any seams in system components that pass through the regulated area. Remove all contaminated HVAC system filters and place in labeled 6-mil polyethylene disposal bags for staging and eventual disposal as asbestos waste.

c. Negative Pressure Filtration System:

(1) The Contractor shall provide enough HEPA negative air machines to effect greater than or equal to - 0.02 inch WCG pressure differential inside the regulated work area. The Competent Person shall determine the number of units needed for the regulated area by dividing the cubic feet in the regulated area by 15 and then dividing that result by the cubic feet per minute (CFM) for each unit to determine the number of units needed to effect greater than or equal to - 0.02 inch WCG pressure inside the regulated work area. Provide standby units in the event of machine failure and/or emergency in an adjacent area. HEPA equipped negative air machines shall be discharged outside of the building a minimum of 30 feet from building make-up air, doors, open windows, patients, visitors and staff.

(2) NIOSH has done extensive studies and has determined that negative air machines typically operate at ~50 percent efficiency. The contractor shall consider this in their determination of number of units needed to provide greater than or equal to - 0.02 inch WCG pressure differential inside the regulated work area. The contractor shall use double the number of machines, based on their calculations, or submit proof their machines operate at stated capacities, at a 2 inches pressure drop across the filters.

(3) Design and Layout:

(a) Before start of work submit the design and layout of the regulated area and the negative air machines. The submittal shall indicate the number of, location of and size of negative air machines. The point(s) of exhaust, air flow within the regulated area, anticipated negative pressure differential, and supporting calculations for sizing shall be provided. In addition, submit the following:

1. Method of supplying power to the units and designation/location of the panels.

2. Description of testing method(s) for correct air volume and pressure differential.

3. If auxiliary power supply is to be provided for the negative air machines, provide a schematic diagram of the power supply and manufacturer's data on the generator and switch.

(4) Negative Air Machines (Hepa Units):

(a) Negative Air Machine Cabinet: The cabinet shall be constructed of steel or other durable material capable of withstanding potential damage from rough handling and transportation. The width of the cabinet shall be less than 30 inches in order to fit in standard doorways. The cabinet must be factory sealed to prevent asbestos fibers from being released during use, transport, or maintenance. Any access to and replacement of filters shall be from the inlet end. The unit must be on casters or wheels.

(b) Negative Air Machine Fan: The rating capacity of the fan must indicate the CFM under actual operating conditions. Manufacturer's typically use "free-air" (no resistance) conditions when rating fans. The fan must be a centrifugal type fan.

(c) Negative Air Machine Final Filter: The final filter shall be a HEPA filter. The filter media must be completely sealed on all edges within a structurally rigid frame. The filter shall align with a continuous flexible gasket material in the negative air machine housing to form an air tight seal. Each HEPA filter shall be certified by the manufacturer to have an efficiency of not less than 99.97 percent. Testing shall have been done in accordance with Military Standard MIL-STD-282 and Army Instruction Manual 136-300-175A. Each filter must bear a UL 586 label to indicate ability to perform under specified conditions. Each filter shall be marked with the name of the manufacturer, serial number, air flow rating, efficiency and resistance, and the direction of test air flow.

(d) Negative Air Machine Pre-filters: The pre-filters, which protect the final HEPA filter by removing larger particles, are required to prolong the operating life of the HEPA filter. Two stages of pre-filtration are required. A first stage pre-filter shall be a low efficiency type for particles 10 µm or larger. A second stage pre-filter shall have a medium efficiency effective for particles down to 5 µm or larger. Pre-filters shall be installed either on or in the intake opening of the NAM and the second stage filter must be held in place with a special housing or clamps.

(e) Negative Air Machine Instrumentation: Each unit must be equipped with a gauge to measure the pressure drop across the filters and to indicate when filters have become loaded and need to be changed. A table indicating the cfm for various pressure readings on the gauge shall be affixed near the gauge for reference or the reading shall indicate at what point the filters

shall be changed, noting cfm delivery. The unit must have an elapsed time meter to show total hours of operation.

(f) Negative Air Machine Safety and Warning Devices: An electrical/ mechanical lockout must be provided to prevent the fan from being operated without a HEPA filter. Units must be equipped with an automatic shutdown device to stop the fan in the event of a rupture in the HEPA filter or blockage in the discharge of the fan. Warning lights are required to indicate normal operation; too high a pressure drop across filters; or too low of a pressure drop across filters.

(g) Negative Air Machine Electrical: All electrical components shall be approved by the National Electrical Manufacturer's Association (NEMA) and Underwriters Laboratories (UL). Each unit must be provided with overload protection and the motor, fan, fan housing, and cabinet must be grounded.

(h) It is essential that replacement HEPA filters be tested using an “in-line” testing method, to ensure the seal around the periphery was not damaged during replacement. Damage to the outer HEPA filter seal could allow contaminated air to bypass the HEPA filter and be discharged to an inappropriate location. Contractor will provide written documentation of test results for negative air machine units with HEPA filters.

(5) Pressure Differential: The fully operational negative air system within the regulated area shall continuously maintain a pressure differential greater than or equal to - 0.02 inch WCG inside the regulated work area. Before any disturbance of any asbestos material, this shall be demonstrated to the VA by use of a pressure differential meter/manometer as required by OSHA 29 CFR 1926.1101(g)(5)(i). The Competent Person shall be responsible for providing, maintaining, and documenting the negative pressure and air changes as required by OSHA and this specification.

(6) Monitoring: The pressure differential shall be continuously monitored and recorded between the regulated area and the area outside the regulated area with a monitoring device that incorporates a strip chart recorder. The strip chart recorder shall become part of the project log and shall indicate at least greater than or equal to - 0.02 inch WCG pressure differential inside the regulated work area for the duration of the project.

(7) Auxiliary Generator: If the building is occupied during abatement, provide an auxiliary gasoline/diesel generator located outside the building in an area protected from the weather. In the event of a power failure of the general power grid and the VAMC emergency power grid, the generator must automatically start and supply power to a minimum of 50 percent of the negative air machines in operation.

(8) Supplemental Make-Up Air Inlets: Provide, as needed for proper air flow in the regulated area, in a location approved by the VA, openings in the plastic sheeting to allow outside air to flow into the regulated area. Auxiliary makeup air inlets must be located as far from the negative air machines as possible, off the floor near the ceiling, and away from the barriers that separate the

regulated area from the occupied clean areas. Cover the inlets with weighted flaps which will seal in the event of failure of the negative pressure system.

(9) Testing The System: The negative pressure system must be tested before any ACM is disturbed in any way. After the regulated area has been completely prepared, the decontamination units set up, and the negative air machines installed, start the units up one at a time. Demonstrate and document the operation and testing of the negative pressure system to the VA using smoke tubes and a negative pressure gauge. Verification and documentation of adequate negative pressure differential across each barrier must be done at the start of each work shift.

(10) Demonstration of the Negative Pressure Filtration System:

(a) The demonstration of the operation of the negative pressure system to the VA shall include, but not be limited to, the following:

1. Plastic barriers and sheeting move lightly in toward the regulated area.

2. Curtains of the decontamination units move in toward regulated area.

3. There is a noticeable movement of air through the decontamination units. Use the smoke tube to demonstrate air movement from the clean room to the shower room to the equipment room to the regulated area.

4. Use smoke tubes to demonstrate air is moving across all areas in which work is to be done. Use a differential pressure gauge to indicate a negative pressure of at > - 0.02 inch across every barrier separating the regulated area from the rest of the building. Modify the system as necessary to meet the above requirements.

(11) Use of the Negative Pressure Filtration System During Abatement Operations:

(a) Start units before beginning any disturbance of ACM occurs. After work begins, the units shall run continuously, maintaining 4 actual air changes per hour at a negative pressure differential of greater than - 0.02 inch water column gauge, for the duration of the work until a final visual clearance and final air clearance has been successfully completed.

(b) No negative air units shall be shut down at any time unless authorized by the VA Contracting Officer, verbally and in writing.

(c) Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area. After items have been pre-cleaned and decontaminated, they may be removed from the work area for storage until the completion of abatement in the work area.

(d) Abatement work shall begin at a location farthest from the

units and proceed towards them. If an electric failure occurs, the Competent Person shall stop all abatement work and immediately begin wetting all exposed asbestos materials for the duration of the power outage. Abatement work shall not resume until power is restored and all units are operating properly again.

(e) The negative air machines shall continue to run after all work is completed and until a final visual clearance and a final air clearance has been successfully completed for that regulated area.

(12) Dismantling The System: After completion of the final visual and final air clearance has been obtained by the VPIH/CIH, the units may be shut down. The unit exterior surfaces shall have been completely decontaminated; pre-filters are not to be removed and the units inlet/outlet sealed with 2 layers of 6-mil poly immediately after shut down. No filter removal shall occur at the VA site following successful completion of site clearance. OSHA/EPA/DOT asbestos labels shall be attached to the units.

d. Containment Barriers and Coverings in the Regulated Area:

(1) General: Seal off the perimeter to the regulated area to completely isolate the regulated area from adjacent spaces. All surfaces in the regulated area must be covered to prevent contamination and to facilitate clean-up. Should adjacent areas become contaminated as a result of the work, Contractor shall immediately stop work and clean up the contamination at no additional cost to the VA. Provide firestopping and identify all fire barrier penetrations due to abatement work as specified in Section 3.1.d.(8); FIRESTOPPING.

(2) Preparation Prior To Sealing: The Regulated Area: Place all tools, scaffolding, materials and equipment needed for working in the regulated area prior to erecting any plastic sheeting. All uncontaminated removable furniture, equipment and/or supplies shall be removed by the VA from the regulated area before commencing work. Any objects not scheduled for removal remaining in the regulated area shall be completely covered with 2 layers of 6-mil fire retardant poly sheeting and secured with duct tape. Lock out and tag out any HVAC/electrical systems in the regulated area.

(3) Controlling Access to The Regulated Area: Access to the regulated area is allowed only through the personnel decontamination facility (PDF). All other means of access shall be eliminated and OSHA DANGER demarcation signs posted as required by OSHA. If the regulated area is adjacent to, or within view of an occupied area, provide a visual barrier of 6-mil opaque fire retardant poly to prevent building occupant observation. If the adjacent area is accessible to the public, the barrier must be solid and capable of withstanding the negative pressure.

(4) Critical Barriers: Completely separate any operations in the regulated area from adjacent areas using 2 layers of 6-mil fire retardant poly and duct tape. Individually seal with 2 layers of 6-mil poly and duct tape all HVAC openings into the regulated area. Individually seal all lighting fixtures, clocks, doors, windows, convectors, speakers, or any other objects/openings in the regulated area. Heat must be shut off any objects covered

with poly.

(5) Primary Barriers:

(a) Cover the regulated area with two layers of 6-mil fire retardant poly on the floors and two layers of 4 mil, fire retardant poly on the walls, unless otherwise directed in writing by the VA representative. Floor layers must form a right angle with the wall and turn up the wall at least 12 inches. Seams must overlap at least 6 feet and must be spray glued and taped. Install sheeting so that layers can be removed independently from each other. Carpeting shall be covered with three layers of 6-mil poly. Corrugated cardboard sheets must be placed between the bottom and middle layers of poly. Mechanically support and seal with duct tape and glue all wall layers.

(b) If stairs and ramps are covered with 6-mil plastic, two layers must be used. Provide3/4 inch exterior grade plywood treads held in place with duct tape/glue on the plastic. Do not cover rungs or rails with any isolation materials.

(6) Secondary Barriers: A loose layer of 6-mil shall be used as a drop cloth to protect the primary layers from debris generated during the abatement. This layer shall be replaced as needed during the work.

(7) Extension of the Regulated Area: If the enclosure of the regulated area is breached in any way that could allow contamination to occur, the affected area shall be included in the regulated area and constructed as per this section. Decontamination measures must be started immediately and continue until air monitoring indicates background levels are met.

(8) Firestopping:

(a) Through penetrations caused by cables, cable trays, pipes, sleeves, conduits, etc. must be firestopped with a fire-rated firestop system providing an air tight seal.

(b) Firestop materials that are not equal to the wall or ceiling penetrated shall be brought to the attention of the VA Representative. The contractor shall list all areas of penetration, the type of sealant used, and whether or not the location is fire rated. Any discovery of penetrations during abatement shall be brought to the attention of the VA representative immediately. All walls, floors and ceilings are considered fire rated unless otherwise determined by the VA Representative or Fire Marshall.

(c) Any visible openings whether or not caused by a penetration shall be reported by the Contractor to the VA Representative for a sealant system determination. Firestops shall meet ASTM E814 and UL 1479 requirements for the opening size, penetrant, and fire rating needed.

e. Sanitary Facilities: The Contractor shall provide sanitary facilities for abatement personnel and maintain them in a clean and sanitary condition throughout the abatement project.

f. Personal Protective Equipment: Provide whole body clothing, head coverings, gloves and foot coverings and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132(d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.

g. Pre-Cleaning:

(1) The VA will provide water for abatement purposes. The Contractor shall connect to the existing VA system. The service to the shower(s) shall be supplied with backflow prevention.

(2) Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area. All workers performing pre-cleaning activities must don appropriate personal protective equipment (PPE), as specified throughout this document and as approved in the Contractor’s work plan. After items have been pre-cleaned and decontaminated, they may be removed from the work area for storage until the completion of abatement in the work area.

(3) Pre-clean all movable objects within the regulated area using a HEPA filtered vacuum and/or wet cleaning methods as appropriate. After cleaning, these objects shall be removed from the regulated area and carefully stored in an uncontaminated location. Drapes, clothing, upholstered furniture and other fabric items shall be disposed of as asbestos contaminated waste. Cleaning these asbestos contaminated items utilizing HEPA vacuum techniques and off-premises steam cleaning is very difficult and cannot guarantee decontamination. Carpeting will be disposed of prior to abatement if in the regulated area. If ACM floor tile is attached to the carpet while the Contractor is removing the carpet that section of the carpet will be disposed of as asbestos waste.

(4) Pre-clean all fixed objects in the regulated area using HEPA filtered vacuums and/or wet cleaning techniques as appropriate. Careful attention must be paid to machinery behind grills or gratings where access may be difficult but contamination may be significant. Also, pay particular attention to wall, floor and ceiling penetration behind fixed items. After pre-cleaning, enclose fixed objects with 2 layers of 6-mil poly and seal securely in place with duct tape. Objects (e.g., permanent fixtures, shelves, electronic equipment, laboratory tables, sprinklers, alarm systems, closed circuit TV equipment and computer cables) which must remain in the regulated area and that require special ventilation or enclosure requirements should be designated here along with specified means of protection. Contact the manufacturer for special protection requirements.

(5) Pre-clean all surfaces in the regulated area using HEPA filtered vacuums and/or wet cleaning methods as appropriate. Do not use any methods that would raise dust such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

h. Pre-Abatement Activities:

(1) Pre-Abatement Meeting: The VA representative, upon receipt, review, and substantial approval of all pre-abatement submittals and verification by the CPIH/CIH that all materials and equipment required for the project are on the site, will arrange for a pre-abatement meeting between the Contractor, the CPIH/CIH, Competent Person(s), the VA representative(s), and the VPIH/CIH. The purpose of the meeting is to discuss any aspect of the submittals needing clarification or amplification and to discuss any aspect of the project execution and the sequence of the operation. The Contractor shall be prepared to provide any supplemental information/documentation to the VA's representative regarding any submittals, documentation, materials or equipment. Upon satisfactory resolution of any outstanding issues, the VA's representative will issue a written order to proceed to the Contractor. No abatement work of any kind described in the following provisions shall be initiated prior to the VA written order to proceed.

(2) Pre-Abatement Construction and Operations:

(a) Perform all preparatory work for the first regulated area in accordance with the approved work schedule and with this specification.

(b) Upon completion of all preparatory work, the CPIH/CIH will inspect the work and systems and will notify the VA's representative when the work is completed in accordance with this specification. The VA's representative may inspect the regulated area and the systems with the VPIH/CIH and may require that upon satisfactory inspection, the Contractor's employees perform all major aspects of the approved AHAP(s), especially worker protection, respiratory systems, contingency plans, decontamination procedures, and monitoring to demonstrate satisfactory operation. The operational systems for respiratory protection and the negative pressure system shall be demonstrated for proper performance.

(c) The CPIH/CIH shall document the pre-abatement activities described above and deliver a copy to the VA's representative.

(d) Upon satisfactory inspection of the installation of and operation of systems, the VA's representative will notify the Contractor in writing to proceed with the asbestos abatement work in accordance with this specification and all applicable regulations.

(3) Pre-Abatement Inspections and Preparations:

(a) Before any work begins on the construction of the regulated area, the Contractor will:

1. Conduct a space-by-space inspection with an authorized VA representative and prepare a written inventory of all existing damage in those spaces where asbestos abatement will occur. Still or video photography may be used to supplement the written damage inventory. Document will be signed and certified as accurate by both parties.

2. A NESHAP (destructive) ACM inspection shall be conducted on all building structures that will be demolished. Ensure the following areas are inspected on the project: lay-in ceilings concealing ACM; ACM behind walls/windows from previous renovations; inside utility chases/walls; transite piping/ductwork/sheets; behind radiators; lab fume hoods; transite lab countertops; roofing materials; below window sills; water/sewer lines; electrical conduit coverings; crawlspaces (previous abatement contamination); flooring/mastic covered by carpeting/new flooring; exterior insulated wall panels; on underground fuel tanks; and steam line trench coverings. Verify during NESHAP survey that no other suspect ACM is present.

3. Ensure that all furniture, machinery, equipment, curtains, drapes, blinds, and other movable objects required to be removed from the regulated area have been cleaned and removed or properly protected from contamination.

4. If present and required, remove and dispose of carpeting from floors in the regulated area.

5. Inspect existing firestopping in the regulated area. Correct as needed.

3.2 REMOVAL OF ACM

a. Wetting ACM:

(1) Use amended water for the wetting of ACM prior to removal. The Competent Person shall assure the wetting of ACM meets the definition of "adequately wet" in the EPA NESHAP regulation and OSHA’s “wet methods” for the duration of the project. A removal encapsulant may be used instead of amended water with written approval of the VA's representative.

(2) Amended Water: Provide water to which a surfactant has been added to wet the ACM and reduce the potential for fiber release during disturbance of ACM. The mixture must be equal to or greater than the wetting provided by water amended by a surfactant consisting of one ounce of 50 percent polyoxyethylene ester and 50 percent polyoxyethylene ether mixed with 5 gallons (19L) of water.

(3) Removal Encapsulant: When authorized by VA, provide a removal encapsulant designed specifically for the removal of ACM. The material must, when used, result in adequate wetting of the ACM and retard fiber release during removal.

b. Secondary Barrier and Walkways:

(1) Install as a drop cloth a 6-mil poly sheet at the beginning of each work shift where removal is to be done during that shift. Completely cover floors and any walls within 10 feet of the area where work is to done. Secure the secondary barrier with duct tape to prevent it from moving or debris from getting behind it. Remove the secondary barrier at the end of the shift or as work in the area is completed. Keep residue on the secondary barrier wetted. When removing, fold inward to prevent spillage and place in a disposal bag.

(2) Install walkways using 6-mil black poly between the regulated area and the decontamination facilities (PDF and W/EDF) to protect the primary layers from contamination and damage. Install the walkways at the beginning of each shift and remove at the end of each shift.

c. Wet Removal of ACM:

(1) Adequately and thoroughly wet the ACM to be removed prior to removal with amended water or when authorized by VA, removal encapsulant to reduce/prevent fiber release to the air. Adequate time must be allowed for the amended water or removal encapsulant to saturate the ACM. Abatement personnel must not disturb dry ACM. Use a fine spray of amended water or removal encapsulant. Saturate the material sufficiently to wet to the substrate without causing excessive dripping. The material must be sprayed repeatedly/continuously during the removal process in order to maintain adequately wet conditions. Removal encapsulants must be applied in accordance with the manufacturer's written instructions. Perforate or carefully separate, using wet methods, an outer covering that is painted or jacketed in order to allow penetration and wetting of the material. Where necessary, carefully remove covering while wetting to minimize fiber release. **In no event shall dry removal occur except when authorized in writing by the VPIH/CIH and VA when a greater safety hazard (e.g., electricity) is present**.

(2) If ACM does not wet well with amended water due to composition, coating or jacketing, remove as follows:

(a) Mist work area continuously with amended water whenever necessary to reduce airborne fiber levels.

(b) Remove saturated ACM in small sections. Do not allow material to dry out. As material is removed, bag material, while still wet into disposal bags. Twist the bag neck tightly, bend over (gooseneck) and seal with a minimum of three tight wraps of duct tape. Clean /decontaminate the outside of the bag of any residue and move to washdown station adjacent to W/EDF.

(c) Fireproofing or Architectural Finish on Scratch Coat: Spray with a fine mist of amended water or removal encapsulant. Allow time for saturation to the substrate. Do not over saturate causing excess dripping. Scrape material from substrate. Remove material in manageable quantities and control falling to staging or floor. If the falling distance is over 20 feet, use a drop chute to contain material through descent. Remove residue remaining on the scratch coat after scraping is done using a stiff bristle hand brush. If a removal encapsulant is used, remove residue completely before the encapsulant dries. Periodically re-wet the substrate with amended water as needed to prevent drying of the material before the residue is removed from the substrate.

(d) Fireproofing or Architectural Finish on Wire Lath: Spray with a fine mist of amended water or removal encapsulant. Allow time to completely saturate the material. Do not over saturate causing excess dripping. If the surface has been painted or otherwise coated, cut small holes as needed and apply amended water or

removal encapsulant from above. Cut saturated wire lath into 2 feet x 6 feet sections and cut hanger wires. Roll up complete with ACM, cover in burlap and hand place in disposal bag. Do not drop to floor. After removal of lath/ACM, remove any overspray on decking and structure using stiff bristle nylon brushes. Depending on hardness of overspray, scrapers may be needed for removal.

(e) Pipe/Tank/Vessel/Boiler Insulation: Remove the outer layer of wrap while spraying with amended water in order to saturate the ACM. Spray ACM with a fine mist of amended water or removal encapsulant. Allow time to saturate the material to the substrate. Cut bands holding pre-formed pipe insulation sections. Slit jacketing at the seams, remove and hand place in a disposal bag. Do not allow dropping to the floor. Remove molded fitting insulation/mud in large pieces and hand place in a disposal bag. Remove any residue on pipe or fitting with a stiff bristle nylon brush. In locations where pipe fitting insulation is removed from fibrous glass or other non-asbestos insulated straight runs of pipe, remove fibrous material at least 6 inches from the point it contacts the ACM.

d. Wet Removal of Amosite:

(1) The following areas shown on drawings indicate locations of amosite ACM which will require local exhaust ventilation and collection as described below, in addition to wet removal. Provide specific description /locations/ drawings.

(2) Provide local exhaust ventilation and collection systems to assure collection of amosite fibers at the point of generation. A 12" flexible rigid non-collapsing duct shall be shall be located no more than 2 feet from any scraping/brushing activity. Primary filters must be replaced every 30 minutes on the negative air machines. Each scraping/brushing activity must have a negative air machine devoted to it. For pre-molded pipe insulation or cutting wire lathe attach a 4 feet square flared end piece on the intake of the duct. Support the duct horizontally at a point 2 feet below the work to effect capture. One person in the crew shall be assigned to operate the duct collection system on a continual basis.

(3) Amosite does not wet well with amended water. Submit full information/documentation on the wetting agent proposed prior to start for review and approval by the VPIH/CIH and VA Contracting Officer. Insure that the material is worked on in small sections and is thoroughly and continuously wetted. Package as soon as possible while wet. Remove as required.

e. Removal of ACM/Dirt Floors and Other Special Procedures:

(1) Major Abatement on Dirt Floors: When working on dirt floors, remove all visible asbestos debris using wet methods after set-up of PDF, W/EDF, negative air systems as required. Perform work and decontaminate/clean-up; perform lockdown as needed and complete work as required in these specifications. The asbestos contaminated soil (ACS) shall be removed and/or enclosed.

(2) Options for abatement of asbestos contaminated soil include:

Removal of top 6 inches of soil; encapsulate the soil using shotcrete or other spray applied concrete materials. Considerations for which option to be used will be made by the VA representative. Factors which may affect which option to be used may include: access to the work area; height of the area (such as is there sufficient height to use concrete materials in the area, etc.). Soils covered with permanent barriers **MUST HAVE PERMANENT SIGNAGE INSTALLED TO WARN AGAINST PENETRATION ASSOCIATED WITH POTENTIAL DISTURBANCE OF ASBESTOS.**

(a) Remove ACS as shown on drawings to a minimum depth of 6 using wet methods. After wetting with amended water to minimize dust, shovel dirt into disposal bags. The CPIH/CIH shall closely monitor work conditions and take appropriate action to protect workers from over exposure to asbestos and heat stress. The minimum number of air changes per hour shall be six using negative air machines. Use of special vacuum truck equipped with HEPA filtration to remove soil is an acceptable option.

(b) Enclosure of ACS using a concrete layer of 4 inches over the entire surface may also be done. Thoroughly dampen soil first with amended water before pouring concrete. Personnel shall be proficient in concrete finishing as well as asbestos trained.

(3) Crawlspaces/Pipe Tunnels: When working in crawlspaces or pipe tunnels, remove all visible asbestos debris using wet methods (if possible) after set-up of PDF, W/EDF, and after establishing negative air systems as required. Perform work and decontaminate/clean-up; perform lockdown as needed and complete work as required in these specifications. The asbestos contaminated soil (ACS) shall be removed and/or enclosed. Clearance requirements include confirmation sampling of affected soil by Polarized Light Microscopy (PLM). Clearance sampling requirements are specified in Sections 3.6.4 3.6.5 and 3.6.6.

(4) Options for abatement of asbestos contaminated soil include: Removal of top 6 inches of soil; encapsulate the soil using shotcrete or other spray applied concrete materials. Considerations for which option to be used will be made by the VA representative. Factors which may affect which option to be used may include: access to the work area; height of the area (such as is there sufficient height to use concrete materials in the area, etc.).

3.3 LOCKDOWN ENCAPSULATION

a. General: Lockdown encapsulation is an integral part of the ACM removal. At the conclusion of ACM removal and before removal of the primary barriers, the contractor shall encapsulate all surfaces with a bridging encapsulant.

b. Delivery and Storage: Deliver materials to the job site in original, new and unopened containers bearing the manufacturer's name and label as well as the following information: name of material, manufacturer's stock number, date of manufacture, thinning instructions, application instructions and the SDS for the material.

c. Worker Protection: Before beginning work with any material for which an SDS has been submitted, provide workers with any required personal

protective equipment. The required personal protective equipment shall be used whenever exposure to the material might occur. In addition to OSHA/specification requirements for respiratory protection, a paint pre-filter and an organic vapor cartridge, at a minimum, shall be used in addition to the HEPA filter when an organic solvent based encapsulant or other solvent is used. The CPIH/CIH shall be responsible for provision of adequate respiratory protection. Note: Flammable and combustible encapsulants or solvents shall not be used, unless authorized in writing by the VA.

d. Encapsulation of Scratch Coat Plaster or Piping:

(1) Apply two coats of lockdown encapsulant to the scratch coat plaster or piping after all ACM has been removed. Apply in strict accordance with the manufacturer's instructions. Any deviation from the instructions must be approved by the VA's representative in writing prior to commencing the work.

(2) Apply the lockdown encapsulant with an airless sprayer at a pressure and using a nozzle orifice as recommended by the manufacturer. Apply the first coat while the scratch coat is still damp from the asbestos removal process, after passing the visual inspection. If the surface has been allowed to dry, wet wipe or HEPA vacuum prior to spraying with encapsulant. Apply a second coat over the first coat in strict conformance with the manufacturer's instructions. Color the lockdown encapsulant and contrast the color in the second coat so that visual confirmation of completeness and uniform coverage of each coat is possible. Adhere to the manufacturer's instructions for coloring. At the completion of the encapsulation, the surface must be a uniform third color produced by the mixture.

e. Sealing Exposed Edges: Seal edges of ACM exposed by removal work which is inaccessible, such as a sleeve, wall penetration, etc., with two coats of bridging encapsulant. Prior to sealing, permit the exposed edges to dry completely to permit penetration of the bridging encapsulant. Apply in accordance with 3.3.4 (B).

3.4 DISPOSAL OF ACM WASTE MATERIALS

a. General: Dispose of waste ACM and debris which is packaged in accordance with these specifications, in accordance with OSHA, EPA, State, Local and DOT requirements. The landfill requirements for packaging must also be met. Transport will be in compliance with 49 CFR 171-180 regulations. Disposal shall be done at an approved landfill. Disposal of non-friable ACM shall be done in accordance with applicable regulations.

b. Procedures:

(1) The VA must be notified at least 24 hours in advance of any waste removed from the containment.

(2) Asbestos waste shall be packaged and moved through the W/EDF into a covered transport container in accordance with procedures is this specification. Waste shall be double-bagged and wetted with amended water prior to disposal. Wetted waste can be very heavy. Bags shall not be overfilled. Bags shall be evacuated with HEPA vacuum and securely sealed to prevent accidental opening and/or

leakage. The top shall be tightly twisted and goose necked prior to tightly sealing with at least three wraps of duct tape. Ensure that unauthorized persons do not have access to the waste material once it is outside the regulated area. All transport containers must be covered at all times when not in use. NESHAP signs must be on containers during loading and unloading. Material shall not be transported in open vehicles. If drums are used for packaging, the drums shall be labeled properly and shall not be re-used.

(3) Waste Load Out: Waste load out shall be done in accordance with the procedures in W/EDF Decontamination Procedures. Sealed waste bags shall be decontaminated on exterior surfaces by wet cleaning and HEPA vacuuming before being placed in the second waste bag and sealed, which then must also be wet wiped and HEPA vacuumed.

(4) Asbestos waste with sharp edged components, i.e., nails, screws, lath, strapping, tin sheeting, jacketing, metal mesh, etc., which might tear poly bags shall be wrapped securely in burlap before packaging and, if needed, use a poly lined fiber drum as the second container, prior to disposal.

3.5 PROJECT DECONTAMINATION

a. General:

(1) The entire work related to project decontamination shall be performed under the close supervision and monitoring of the CPIH/CIH.

(2) If the asbestos abatement work is in an area which was contaminated prior to the start of abatement, the decontamination will be done by cleaning the primary barrier poly prior to its removal and cleanings of the surfaces of the regulated area after the primary barrier removal.

(3) If the asbestos abatement work is in an area which was uncontaminated prior to the start of abatement, the decontamination will be done by cleaning the primary barrier poly prior to its removal, thus preventing contamination of the building when the regulated area critical barriers are removed.

b. Regulated Area Clearance: Clearance air testing and other requirements which must be met before release of the Contractor and re-occupancy of the regulated area space are specified in Final Testing Procedures.

c. Work Description: Decontamination includes the clearance air testing in the regulated area and the decontamination and removal of the enclosures/facilities installed prior to the abatement work including primary/critical barriers, PDF and W/EDF facilities, and negative pressure systems.

d. Pre-Decontamination Conditions:

(1) Before decontamination starts, all ACM waste from the regulated area shall be collected and removed, and the loose 6-mil layer of poly removed while being adequately wetted with amended water and disposed of along with any gross debris generated by the work.

(2) At the start of decontamination, the following shall be in place:

(a) Primary barriers consisting of 2 layers of 6-mil poly on the floor and 4 mil poly on the walls.

(b) Critical barriers consisting of 2 layers of 6-mil poly which is the sole barrier between the regulated area and openings to the rest of the building or outside.

(c) Decontamination facilities for personnel and equipment in operating condition and the negative pressure system in operation.

e. First Cleaning: Carry out a first cleaning of all surfaces of the regulated area including items of remaining poly sheeting, tools, scaffolding, ladders/staging by wet methods and HEPA vacuuming. Do not use dry dusting/sweeping/air blowing methods. Use each surface of a wetted cleaning cloth one time only and then dispose of as contaminated waste. Continue this cleaning until there is no visible residue from abated surfaces or poly or other surfaces. Remove all filters in the air handling system and dispose of as ACM waste in accordance with these specifications. The negative pressure system shall remain in operation during this time. Additional cleaning(s) may be needed as determined by the CPIH/VPIH/CIH.

f. Pre-Clearance Inspection and Testing: The CPIH/CIH and VPIH/CIH will perform a thorough and detailed visual inspection at the end of the cleaning to determine whether there is any visible residue in the regulated area. If the visual inspection is acceptable, the CPIH/CIH will perform pre-clearance sampling using aggressive clearance as detailed in 40 CFR 763 Subpart E (AHERA) Appendix A (III)(B)(7)(d). If the sampling results show values below 0.01 f/cc, then the Contractor shall notify the VA's representative of the results with a brief report from the CPIH/CIH documenting the inspection and sampling results and a statement verifying that the regulated area is ready for lockdown encapsulation. The VA reserves the right to utilize their own VPIH/CIH to perform a pre-clearance inspection and testing for verification.

g. Lockdown Encapsulation of Abated Surfaces: With the express written permission of the VA's representative, perform lockdown encapsulation of all surfaces from which asbestos was abated in accordance with the procedures in this specification. Negative pressure shall be maintained in the regulated area during the lockdown application.

3.6 FINAL VISUAL INSPECTION AND AIR CLEARANCE TESTING

a. General: Notify the VA representative 24 hours in advance for the performance of the final visual inspection and testing. The final visual inspection and testing will be performed by the VPIH/CIH starting after the final cleaning.

b. Final Visual Inspection: Final visual inspection will include the entire regulated area, the PDF, all poly sheeting, seals over HVAC openings, doorways, windows, and any other openings. If any debris, residue, dust or any other suspect material is detected, the final cleaning shall be repeated at no additional cost to the VA. Dust/material samples may be collected and analyzed at no additional cost to the VA at the discretion of the VPIH/CIH to confirm visual findings. When the regulated area is visually clean the final testing

can be done.

c. Final Air Clearance Testing:

(1) After an acceptable final visual inspection by the VPIH/CIH and VA Representative, the VPIH/CIH will perform the final clearance testing. Air samples will be collected and analyzed in accordance with procedures for AHERA in this specification. If work is less than 260 lf/160 sf/35 cf, 5 PCM samples shall be collected for clearance and a minimum of two field blank. If work is equal to or more than 260 lf/160 sf/35 cf, AHERA TEM sampling shall be performed for clearance. TEM analysis shall be done in accordance with procedures for EPA AHERA presented in this specification. If the release criteria are not met, the Contractor shall repeat the final cleaning and continue decontamination procedures until clearance is achieved. **All Additional inspection and testing costs will be borne by the Contractor**.

(2) If release criteria are met, proceed to perform the abatement closeout and to issue the certificate of completion (Attachment C) in accordance with these specifications.

d. Final Air Clearance Procedures:

(1) Contractor's Release Criteria: Work in a regulated area is complete when the regulated area is visually clean and airborne fiber levels have been reduced to or below 0.01 f/cc as measured by the AHERA PCM protocol and < 70 AHERA asbestos structures per square millimeter (s/mm2) by AHERA TEM. No averaging of results will be used for this project. All five (5) TEM samples inside the regulated area shall be at or below 70 asbestos s/mm2 to satisfy the project final clearance criteria.

(2) Air Monitoring and Final Clearance Sampling: To determine if the elevated airborne fiber counts encountered during abatement operations have been reduced to the specified level, the VPIH/CIH will secure samples and analyze them according to the following procedures:

(a) Fibers Counted: “Fibers” referred to in this section shall be either all fibers regardless of composition as counted in the NIOSH 7400 PCM method or asbestos fibers counted using the AHERA TEM method.

(b) Aggressive Sampling: All final air testing samples shall be collected using aggressive sampling techniques except where soil is not encapsulated or enclosed. Samples will be collected on 0.8 micron MCE filters for PCM analysis and 0.45 micron MCE for TEM. A minimum of 3850 Liters of air using calibrated sampling pumps shall be collected for PCM samples and a minimum of 1200 Liters of air using calibrated sampling pumps shall be collected for TEM clearance samples. Before pumps are started, initiate aggressive air mixing sampling as detailed in 40 CFR 763 Subpart E (AHERA) Appendix A (III)(B)(7)(d). Air samples will be collected in areas subject to normal air circulation away from corners, obstructed locations, and locations near windows, doors, or vents. After air sampling pumps have been shut off, circulating fans shall be shut off. The negative pressure system shall continue to operate.

(c) Final clearance for soil that is not encapsulated, samples will be collected on 0.8 micron MCE filters for PCM analysis and 0.45µ MCE filters for TEM. A minimum of 3850 Liters of air using calibrated sampling pumps shall be collected for PCM samples and a minimum of 1200 Liters of air using calibrated sampling pumps shall be collected for TEM clearance samples. Air clearance of work areas where contaminated soil has been removed is in addition to the requirement for clearance by bulk sample analysis discussed within these specifications. There will be no aggressive air sampling for the clearance of soil due to the fact that aggressive air sampling may overload the cassettes.

(d) Random bulk samples shall be collected from areas of soil which have been abated to ensure that the soil has been properly decontaminated. The total number of samples to be collected from the soil areas shall be; less than 1000 square feet of soil – 3 samples; greater than 1000 to less than 5000 square feet of soil – 5 samples; and greater than 5000 square feet of soil – 7 samples. The soil samples shall be collected in a statistically random manner and shall be analyzed by PLM method. The clearance level to determine the soil clean is less than 1 percent asbestos by weight as analyzed by PLM method. If this level is achieved, the soil areas shall be considered clear. If the levels are greater than 1 percent asbestos, the areas shall be re-cleaned until the sample results are less than 1 percent.

e. Clearance Sampling Using PCM – Less than 260LF/160SF:

(1) The VPIH/CIH will perform clearance samples as indicated by the specification.

(2) The NIOSH 7400 PCM method will be used for clearance sampling with a minimum collection volume of 3850 Liters of air. A minimum of 5 PCM clearance samples shall be collected. All samples must be equal to or less than 0.01 f/cc to clear the regulated area.

(3) Random bulk samples shall be collected from areas of soil which have been abated to ensure that the soil has been properly decontaminated. The total number of samples to be collected from the soil areas shall be; less than 1000 square feet of soil – 3 samples; greater than 1000 to less than 5000 square feet of soil – 5 samples; and greater than 5000 square feet of soil – 7 samples. The soil samples shall be collected in a statistically random manner and shall be analyzed by PLM method. The clearance level to determine the soil clean is less than 1 percent asbestos by weight as analyzed by PLM method. If this level is achieved, the soil areas shall be considered clear. If the levels are greater than 1 percent asbestos, the areas shall be re-cleaned until the sample results are less than 1 percent.

f. Clearance Sampling Using Tem – Equal to or more than 260LF/160SF: TEM

(1) Clearance requires 13 samples be collected; 5 inside the regulated area; 5 outside the regulated area; and 3 field blanks.

(2) The TEM method will be used for clearance sampling with a minimum collection volume of 1200 Liters of air. A minimum of 13 clearance samples shall be collected. All samples must be equal to or less than 70 AHERA structures per square millimeter (s/mm2)

AHERA TEM, no averaging of results for this specific project.

g. Laboratory Testing of PCM Clearance Samples: The services of an AIHA accredited laboratory will be employed by the VA to perform analysis for the PCM air samples. The accredited laboratory shall be successfully participating in the AIHA Proficiency Analytical Testing (PAT) program. Samples will be sent daily by the VPIH/CIH so that verbal/faxed reports can be received within 24-36 hours. A complete record, certified by the laboratory, of all air monitoring tests and results will be furnished to the VA’s representative and the Contractor.

h. Laboratory Testing of Tem Samples: Samples shall be sent by the VPIH/CIH to a NIST NVLAP accredited laboratory for analysis by TEM. The laboratory shall be successfully participating in the NIST Airborne Asbestos Analysis (TEM) program. Verbal/faxed results from the laboratory shall be available within 24-36 hours after receipt of the samples. A complete record, certified by the laboratory, of all TEM results shall be furnished to the VA's representative and the Contractor.

i. Laboratory Testing of Bulk Samples: Samples shall be sent by the VPIH/CIH or CPIH/CIH to a NIST NVLAP accredited laboratory for analysis by PLM. The laboratory shall successfully participate in the NIST NVLAP Bulk Asbestos Analysis (PLM) program. Verbal/faxed results from the laboratory shall be available within 24-36 hours after receipt of the samples. A complete record, certified by the laboratory, of all PLM and/or TEM results shall be furnished to the VA's representative and the Contractor.

3.7 ABATEMENT CLOSEOUT AND CERTIFICATE OF COMPLIANCE

a. Completion of Abatement Work:

(1) After thorough decontamination, seal negative air machines with 2 layers of 6-mil poly and duct tape to form a tight seal at the intake/outlet ends before removal from the regulated area. Complete asbestos abatement work upon meeting the regulated area visual and air clearance criteria and fulfilling the following:

(a) Remove all equipment and materials from the project area.

(b) Dispose of all packaged ACM waste as required.

(c) Repair or replace all interior finishes damaged during the abatement work, as required.

(d) Fulfill other project closeout requirements as required in this specification.

b. Certificate of Completion By Contractor (Attachment C): The CPIH/CIH shall complete and sign the "Certificate of Completion" in accordance with Attachment C (Attachmetn #1) at the completion of the abatement and decontamination of the regulated area.

c. Work Shifts: All work shall generally be done during administrative hours (8:00 AM to 4:30 PM) Monday - Friday excluding Federal Holidays. Any change in the work schedule must be approved in writing by the VA Representative.

d. Re-Insulation: If required as part of the contract, replace all asbestos containing insulation/fire-proofing with suitable non-asbestos material. Provide SDS's for all replacement materials in advance of installation for VA approval. Refer to Section 23 07 11, HVAC, PLUMBING, AND BOILER PLANT INSULATION.

-- End of Section --

02 82 11,02 82 13.13, 02 82 13.19, 02 82 13.41 – Attachment C
02 82 13.21, 02 82 13.31 – Attachment A

ATTACHMENT #1
CERTIFICATE OF COMPLETION

DATE: _____ VA Project #: _____

PROJECT NAME: _____ Abatement Contractor: _____

VAMC/ADDRESS: _____

1. I certify that I have personally inspected, monitored and supervised the abatement work of (specify regulated area or Building):
   which took place from / / to / /
2. That throughout the work all applicable requirements/regulations and the VA's specifications were met.
3. That any person who entered the regulated area was protected with the appropriate personal protective equipment and respirator and that they followed the proper entry and exit procedures and the proper operating procedures for the duration of the work.
4. That all employees of the Abatement Contractor engaged in this work were trained in respiratory protection, were experienced with abatement work, had proper medical surveillance documentation, were fit-tested for their respirator, and were not exposed at any time during the work to asbestos without the benefit of appropriate respiratory protection.
5. That I performed and supervised all inspection and testing specified and required by applicable regulations and VA specifications.
6. That the conditions inside the regulated area were always maintained in a safe and healthy condition and the maximum fiber count never exceeded 0.5 f/cc, except as described below.
7. That all abatement work was done in accordance with OSHA requirements and the manufacturer’s recommendations.

CPIH/CIH Signature/Date: _____

CPIH/CIH Print Name: _____

Abatement Contractor Signature/Date: _____

Abatement Contractor Print Name: _____

ATTACHMENT #2
CERTIFICATE OF WORKER'S ACKNOWLEDGMENT

PROJECT NAME:_____ DATE:_____

PROJECT ADDRESS:_____

ABATEMENT CONTRACTOR'S NAME:_____

WORKING WITH ASBESTOS CAN BE HAZARDOUS TO YOUR HEALTH. INHALING ASBESTOS HAS BEEN LINKED WITH VARIOUS TYPES OF CANCERS. IF YOU SMOKE AND INHALE ASBESTOS FIBERS, YOUR CHANCES OF DEVELOPING LUNG CANCER IS GREATER THAN THAT OF THE NON-SMOKING PUBLIC.

Your employer's contract with the owner for the above project requires that: You must be supplied with the proper personal protective equipment including an adequate respirator and be trained in its use. You must be trained in safe and healthy work practices and in the use of the equipment found at an asbestos abatement project. You must receive/have a current medical examination for working with asbestos. These things shall be provided at no cost to you. By signing this certificate of worker's acknowledgement you are indicating to the owner that your employer has met these obligations.

RESPIRATORY PROTECTION: I have been trained in the proper use of respirators and have been informed of the type of respirator to be used on the above indicated project. I have a copy of the written Respiratory Protection Program issued by my employer. I have been provided for my exclusive use, at no cost, with a respirator to be used on the above indicated project.

TRAINING COURSE: I have been trained by a third party, State/EPA accredited trainer in the requirements for an AHERA/OSHA Asbestos Abatement Worker training course, 32-hours minimum duration. I currently have a valid State accreditation certificate. The topics covered in the course include, as a minimum, the following:

- Physical Characteristics and Background Information on Asbestos
- Potential Health Effects Related to Exposure to Asbestos
- Employee Personal Protective Equipment
- Establishment of a Respiratory Protection Program
- State of the Art Work Practices
- Personal Hygiene
- Additional Safety Hazards
- Medical Monitoring
- Air Monitoring
- Relevant Federal, State and Local Regulatory Requirements, Procedures, and Standards
- Asbestos Waste Disposal

MEDICAL EXAMINATION: I have had a medical examination within the past 12 months which was paid for by my employer. This examination included: health history, occupational history, pulmonary function test, and may have included a chest x-ray evaluation. The physician issued a positive written opinion after the examination.

Signature:_____

Printed Name:_____

Social Security Number:_____

Witness:_____

ATTACHMENT #3
AFFIDAVIT OF MEDICAL SURVEILLANCE, RESPIRATORY PROTECTION AND TRAINING/ACCREDITATION

VA PROJECT NAME AND NUMBER:_____

VA MEDICAL FACILITY:_____

ABATEMENT CONTRACTOR'S NAME AND ADDRESS:_____

1. I verify that the following individual

Name:_____ Social Security Number:_____ who is proposed to be employed in asbestos abatement work associated with the above project by the named Abatement Contractor, is included in a medical surveillance program in accordance with 29 CFR 1926.1101(m), and that complete records of the medical surveillance program as required by 29 CFR 1926.1101(m)(n) and 29 CFR 1910.20 are kept at the offices of the Abatement Contractor at the following address.

Address:_____

2. I verify that this individual has been trained, fit-tested and instructed in the use of all appropriate respiratory protection systems and that the person is capable of working in safe and healthy manner as expected and required in the expected work environment of this project.

3. I verify that this individual has been trained as required by 29 CFR 1926.1101(k). This individual has also obtained a valid State accreditation certificate. Documentation will be kept on-site.

4. I verify that I meet the minimum qualifications criteria of the VA specifications for a CPIH.

Signature of CPIH/CIH:_____ Date:_____

Printed Name of CPIH/CIH:_____

Signature of Contractor:_____ Date:_____

Printed Name of Contractor:_____

ATTACHMENT #4
ABATEMENT CONTRACTOR/COMPETENT PERSON(S) REVIEW AND ACCEPTANCE OF THE VA'S ASBESTOS SPECIFICATIONS

VA Project Location:_____

VA Project #:_____

VA Project Description:_____

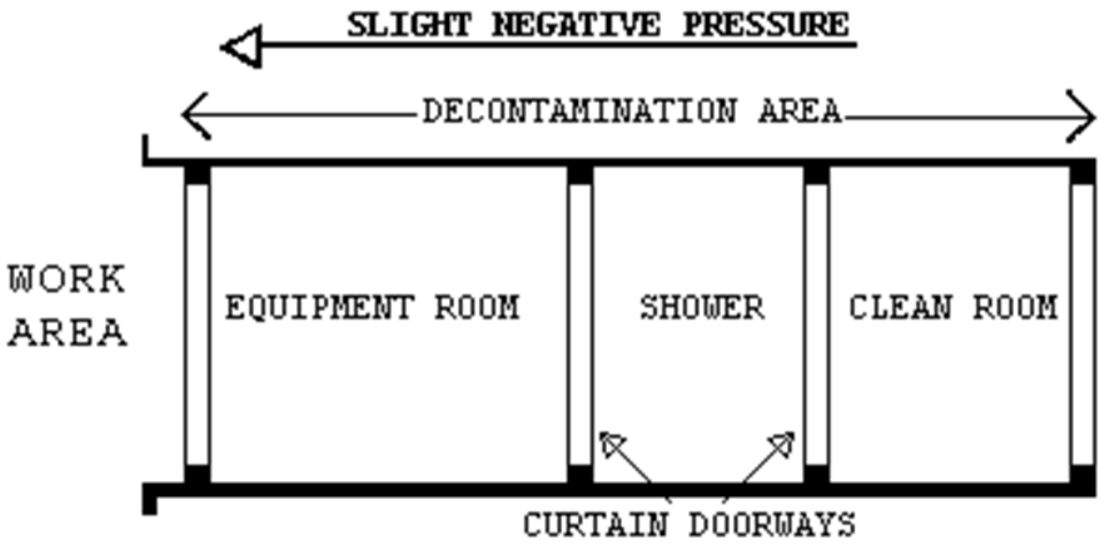
1. This form shall be signed by the Asbestos Abatement Contractor Owner and the Asbestos Abatement Contractor's Competent Person(s) prior to any start of work at the VA related to this Specification. If the Asbestos Abatement Contractor's/Competent Person(s) has not signed this form, they shall not be allowed to work on-site.

2 I, the undersigned, have read VA's Asbestos Specification regarding the asbestos abatement requirements. I understand the requirements of the VA's Asbestos Specification and agree to follow these requirements as well as all required rules and regulations of OSHA/EPA/DOT and State/Local requirements. I have been given ample opportunity to read the VA's Asbestos Specification and have been given an opportunity to ask any questions regarding the content and have received a response related to those questions. I do not have any further questions regarding the content, intent and requirements of the VA's Asbestos Specification.

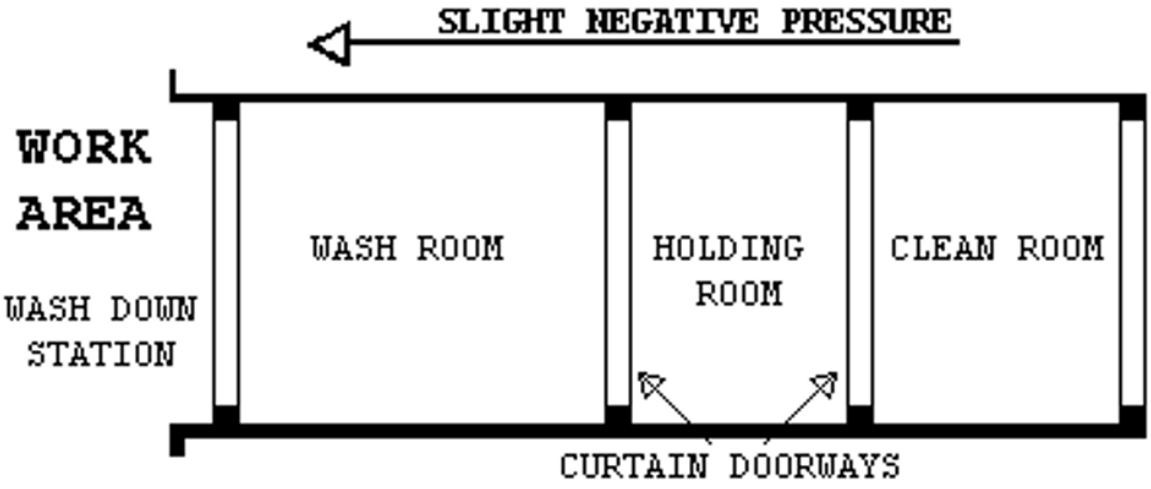
3. At the conclusion of the asbestos abatement, I will certify that all asbestos abatement work was done in accordance with the VA's Asbestos Specification and all ACM was removed properly and no fibrous residue remains on any abated surfaces.

Abatement Contractor Owner's Signature_____Date_____

02 82 11 , 02 82 13.13, 02 82 13.19, 02 82 13.41 – Attachment A



02 82 11 , 02 82 13.13,02 82 13.19, 02 82 13.41 – Attachment B



SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 82 13.13

GLOVEBAG ASBESTOS ABATEMENT

1/21

PART 1 GENERAL

1.1 SUMMARY OF WORK
1.2 VARIATIONS IN QUANTITY
1.3 STOP ASBESTOS REMOVAL
1.4 DEFINITIONS
1.5 APPLICABLE CODES AND REGULATIONS
1.6 PROJECT COORDINATION
1.7 RESPIRATORY PROTECTION
1.8 WORKER PROTECTION
1.9 DECONTAMINATION FACILITIES
1.10 REFERENCES
1.11 SUBMITTALS
1.12 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS, MATERIALS AND EQUIPMENT

2.1 MATERIALS AND EQUIPMENT
2.2 CONTAINMENT BARRIERS AND COVERINGS IN THE REGULATED AREA
2.3 MONITORING, INSPECTION AND TESTING
2.4 ASBESTOS HAZARD ABATEMENT PLAN
2.5 ENCAPSULANTS
2.6 CERTIFICATES OF COMPLIANCE

PART 3 EXECUTION

3.1 REGULATED AREA PREPARATIONS
3.2 REMOVAL OF PIPING ACM
3.3 GLOVEBAG REMOVAL PROCEDURES
3.4 LOCKDOWN ENCAPSULATION
3.5 DISPOSAL OF ACM WASTE MATERIALS
3.6 PROJECT DECONTAMINATION
3.7 FINAL VISUAL INSPECTIONS AND AIR CLEARANCE TESTING
3.8 ABATEMENT CLOSEOUT AND CERTIFICATE OF COMPLIANCE

ATTACHMENTS:

Attachment A

Attachment B

Attachment C

-- End of Section Table of Contents --

SECTION 02 82 13.13
1/21

GLOVEBAG ASBESTOS ABATEMENT

NOTE: These specifications were derived from standard government specifications and standards. This document is provided for reference only and was provided by others. In the event of contact with any hazardous materials, contact an abatement specialist immedialtely and seek expert consultation and advise.

PART 1 GENERAL

1.1 SUMMARY OF WORK

a. Contract Documents and Related Requirements: Drawings, general provisions of the contract, including general and supplementary conditions and other Division 01 specifications, shall apply to the work of this section. The contract documents show the work to be done under the contract and related requirements and conditions impacting the project. Related requirements and conditions include applicable codes and regulations, notices and permits, existing site conditions and restrictions on use of the site, requirements for partial owner occupancy during the work, coordination with other work and the phasing of the work. In the event the Asbestos Abatement Contractor discovers a conflict in the contract documents and/or requirements or codes, the conflict must be brought to the immediate attention of the Contracting Officer for resolution. Whenever there is a conflict or overlap in the requirements, the most stringent shall apply. Any actions taken by the Contractor without obtaining guidance from the Contracting Officer shall become the sole risk and responsibility of the Asbestos Abatement Contractor. All costs incurred due to such action are also the responsibility of the Asbestos Abatement Contractor.

b. Extent of Work: Below is a brief description of the estimated quantities of asbestos containing materials to be abated by the Glovebag method. These quantities are for informational purposes only and are based on the best information available at the time of the specification preparation. The Contractor shall satisfy himself as to the actual quantities to be abated. Nothing in this section may be interpreted as limiting the extent of work otherwise required by this contract and related documents.

(1) Removal, clean-up and disposal of ACM piping and fittings and asbestos contaminated elements in an appropriate regulated area in the following approximate quantities:

Approximately 130linear feet of >6 inches diameter preformed pipe insulation in Building 16A.
Approximately 15 linear feet of >6 inches diameter preformed insulation in adjoining connecting areas of Building
16
Approximately 34 fittings,on fiberglass line 2 inches – 6 inches

in diameter in Building 16A.
Approximately 10 fittings on fiberglass line 3 inches - 6 inches in diameter in adjoinging connecting areas of Building 16.
Apprximately 7 fittings on fiberglass line >6 inches in diameter in Building 16A.
Apprximately 2 fittings on fiberglass line >6 inches in diameter in adjoinging connecting areas of Building 16.
Approximately 15 linear feet of hard white caulk on the vent in Building 16A.
Approximately 60 linear feet of window frame caulk in adjoinging connecting areas of Building 16.
Approximately 15 linear feet of window glazing in adjoinging connecting areas of Builing 16.

b. Related Work:

(1) Section 07 84 00, FIRESTOPPING

c. TASKS:

(1) The work tasks are summarized briefly as follows:

(a) Pre-abatement activities including pre-abatement meeting(s), inspection(s), notifications, permits, submittal approvals, work-site preparations, emergency procedures arrangements, and Asbestos Hazard Abatement Plans for Glovebag asbestos abatement work.

(b) Abatement activities including removal, clean-up and disposal of ACM waste, recordkeeping, security, monitoring, and inspections.

(c) Cleaning and decontamination activities including final visual inspection, air monitoring and certification of decontamination.

(d) Abatement Contractor Use of Premises:

(2) The Contractor and Contractor's personnel shall cooperate fully with the VA Representative/consultant to facilitate efficient use of buildings and areas within buildings. The Contractor shall perform the work in accordance with the VA specifications, drawings, phasing plan and in compliance with any/all applicable Federal, State and Local regulations and requirements.

(3) The Contractor shall use the existing facilities in the building strictly within the limits indicated in contract documents as well as the approved VA Design and Construction Procedures. VA Design and Construction Procedures drawings of partially occupied buildings will show the limits of regulated areas; the placement of decontamination facilities; the temporary location of bagged waste ACM; the path of transport to outside the building; and the temporary waste storage area for each building/regulated area. Any variation from the arrangements shown on drawings shall be secured in writing from the VA Representative through the pre-abatement plan of action. The following limitations of use

shall apply to existing facilities shown on drawings.

1.2 VARIATIONS IN QUANTITY

The quantities and locations of ACM as indicated on the drawings and the extent of work included in this section are estimated, which are limited by the physical constraints imposed by occupancy of the buildings and accessibility to ACM. Accordingly, minor variations (+/- 10 percent) in quantities of ACM within the regulated area are considered as having no impact on contract price and time requirements of this contract. Where additional work is required beyond the above variation, the contractor shall provide unit prices for newly discovered ACM and those prices shall be used for additional work required under the contractor.

1.3 STOP ASBESTOS REMOVAL

If the Contracting Officer; their field representative; the facility Safety Officer/Manager or their designee, or the VA Professional Industrial Hygienist/Certified Industrial Hygienist (VPIH/CIH) presents a verbal Stop Asbestos Removal Order, the Contractor/Personnel shall immediately stop all asbestos removal and maintain HEPA filtered negative pressure air flow in the containment and adequately wet any exposed ACM. If a verbal Stop Asbestos Removal Order is issued, the VA shall follow-up with a written order to the Contractor as soon as it is practicable. The Contractor shall not resume any asbestos removal activity until authorized to do so in writing by the VA Contracting Officer. A stop asbestos removal order may be issued at any time the VA Contracting Officer determines abatement conditions/activities are not within VA specification, regulatory requirements or that an imminent hazard exists to human health or the environment. Work stoppage will continue until conditions have been corrected to the satisfaction of the VA. Standby time and costs for corrective actions will be borne by the Contractor, including the VPIH/CIH time. The occurrence of any of the following events shall be reported immediately by the Contractor’s competent person to the VA Contracting Office or field representative using the most expeditious means (e.g., verbal or telephonic), followed up with written notification to the Contracting Officer as soon as practical. The Contractor shall immediately stop asbestos removal/disturbance activities and initiate fiber reduction activities if:

a. Airborne PCM analysis results equal to or greater than 0.01 f/cc above background levels inside the building but outside the regulated area;

b. breach or break in regulated area containment barrier(s);

c. less than –0.02 inch WCG pressure in the regulated area;

d. serious injury/death at the site;

e. fire/safety emergency at the site;

f. respiratory protection system failure;

g. power failure or loss or inadequate use of wetting agent; or

h. any visible emissions observed outside the regulated area; or

i. failure to follow project specification requirements.

1.4 DEFINITIONS

a. General: Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Drawings must be recognized as diagrammatic in nature and not completely descriptive of the requirements indicated therein.

b. Glossary:

Abatement - Procedures to control fiber release from asbestos-containing materials. Includes removal, encapsulation, enclosure, demolition, and renovation activities related to asbestos containing materials (ACM).

Aerosol - Solid or liquid particulate suspended in air.

Adequately wet - Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the ACM, then that material has not been adequately wetted.

Aggressive method - Removal or disturbance of building material by sanding, abrading, grinding, or other method that breaks, crumbles, or disintegrates intact ACM.

Aggressive air sampling - EPA AHERA defined clearance sampling method using air moving equipment such as fans and leaf blowers to aggressively disturb and maintain in the air residual fibers after abatement.

AHERA - Asbestos Hazard Emergency Response Act. Asbestos regulations for schools issued in 1987.

Aircell - Pipe or duct insulation made of corrugated cardboard which contains asbestos.

Air monitoring - The process of measuring the fiber content of a known volume of air collected over a specified period of time. The NIOSH 7400 Method is used to determine the fiber levels in air. For personal samples, area air samples and clearance air testing using Phase Contrast Microscopy (PCM) analysis, the NIOSH 7402 can be used when it is necessary to confirm fibers counted by PCM as being asbestos. The AHERA TEM (EPA 560/5-89-001) analysis may be used for background, area samples and clearance samples when required by this specification, or at the discretion of the VPIH/CIH as appropriate.

Air sample filter - The filter used to collect fibers which are then counted. The filter is made of mixed cellulose ester membrane (MCE) for PCM (Phase Contrast Microscopy, 25 mm, 3-piece with 2 inches Static Extension Cowl, 0.8 micron pore size) and MCE for TEM (Transmission Electron Microscopy, 25 mm, 3-piece with 2 inches Static Extension Cowl, 0.45 micron pore size).

Amended water - Water to which a surfactant (wetting agent) has been added to increase the penetrating ability of the liquid.

Asbestos - Includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, actinolite asbestos, and any of these minerals that have been chemically treated or altered. Asbestos also includes PACM, as defined below.

Asbestos Hazard Abatement Plan (AHAP) - Asbestos work procedures required to be submitted by the contractor before work begins.

Asbestos-containing material (ACM) - Any material containing more than one percent of asbestos.

Asbestos contaminated elements (ACE) - Building elements such as ceilings, walls, lights, or ductwork that are contaminated with asbestos.

Asbestos-contaminated soil (ACS) – Soil found in the work area or in adjacent areas such as crawlspaces or pipe tunnels which is contaminated with asbestos-containing material debris and cannot be easily separated from the material.

Asbestos-containing waste (ACW) material - Asbestos-containing material or asbestos contaminated objects requiring disposal.

Asbestos Project Monitor – Some States require that any person conducting asbestos abatement air sampling, clearance inspections and clearance air sampling be licensed as an asbestos project monitor.

Asbestos waste decontamination facility - A system consisting of drum/bag washing facilities and a temporary storage area for cleaned containers of asbestos waste. Used as the exit for waste and equipment leaving the regulated area. In an emergency, it may be used to evacuate personnel.

Authorized person - Any person authorized by the VA, the Contractor, or government agency and required by work duties to be present in regulated areas.

Authorized visitor - Any person approved by the VA; the contractor; or any government agency representative having jurisdiction over the regulated area (e.g., OSHA, Federal and State EPA).

Barrier - Any surface that isolates the regulated area and inhibits fiber migration from the regulated area.

Containment Barrier - An airtight barrier consisting of walls, floors, and/or ceilings of sealed plastic sheeting which surrounds and seals the outer perimeter of the regulated area.

Critical Barrier - The barrier responsible for isolating the regulated area from adjacent spaces, typically constructed of 2-layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as doors, windows, penetrations or any other opening into the regulated area.

Primary Barrier – Plastic barriers placed over critical barriers and exposed directly to abatement work or to secondary barrier.

Secondary Barrier - Any additional plastic barriers used to isolate and provide protection from debris during abatement work.

Breathing zone - The hemisphere forward of the shoulders with a radius of about 6 - 9 inches from the worker's nose.

Bridging encapsulant - An encapsulant that forms a layer on the surface of the ACM.

Building/facility owner - The legal entity, including a lessee, which exercises control over management and recordkeeping functions relating to a building and/or facility in which asbestos activities take place.

Bulk testing - The collection and analysis of suspect asbestos containing materials.

Certified Industrial Hygienist (CIH) - A person certified in the comprehensive practice of industrial hygiene by the American Board of Industrial Hygiene.

Class I asbestos work - Activities involving the removal of Thermal System Insulation (TSI) and surfacing ACM and Presumed Asbestos Containing Material (PACM).

Class II asbestos work - Activities involving the removal of ACM which is not thermal system insulation or surfacing material. This includes, but is not limited to, the removal of asbestos-containing wallboard, floor tile and sheeting, roofing and siding shingles, and construction mastic.

Clean room/Changing room - An uncontaminated room having facilities for the storage of employee's street clothing and uncontaminated materials and equipment.

Clearance sample - The final air sample taken after all asbestos work has been done and visually inspected. Performed by the VA's Professional Industrial Hygiene Consultant/Certified Industrial Hygienist (VPIH/CIH).

Closely resemble - The major workplace conditions which have contributed to the levels of historic asbestos exposure, are no more protective than conditions of the current workplace.

Competent person - In addition to the definition in 29 CFR 1926.32(f), one who is capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, as specified in 29 CFR 1926.32(f); in addition, for Class I and II work who is specially trained in a training course which meets the criteria of EPA's Model Accreditation Plan (40 CFR 763) for supervisor.

Contractor's Professional Industrial Hygienist (CPIH/CIH) - The asbestos abatement contractor's industrial hygienist. The industrial hygienist must meet the qualification requirements of a PIH and may report to a certified industrial hygienist (CIH).

Count - Refers to the fiber count or the average number of fibers

greater than five microns in length with a length-to-width (aspect) ratio of at least 3 to 1, per cubic centimeter of air.

Crawlspace – An area which can be found either in or adjacent to the work area. This area has limited access and egress and may contain asbestos materials and/or asbestos contaminated soil.

Decontamination area/unit - An enclosed area adjacent to and connected to the regulated area and consisting of an equipment room, shower room, and clean room, which is used for the decontamination of workers, materials, and equipment that are contaminated with asbestos.

Demolition - The wrecking or taking out of any load-supporting structural member and any related razing, removing, or stripping of asbestos products.

Disposal bag - Typically 6-mil thick sift-proof, dustproof, leak-tight container used to package and transport asbestos waste from regulated areas to the approved landfill. Each bag/container must be labeled/marked in accordance with EPA, OSHA and DOT requirements.

Disturbance - Asbestos Operations and Maintenance Activities (OSHA Class III) that disrupt the matrix of ACM or PACM, crumble or pulverize ACM or PACM, or generate visible debris from ACM or PACM. Disturbance includes cutting away small amounts of ACM or PACM, no greater than the amount that can be contained in one standard sized glove bag or waste bag, in order to access a building component. In no event shall the amount of ACM or PACM so disturbed exceed that which can be contained in one glove bag or disposal bag, which shall not exceed 60 inches in length or width.

Drum - A rigid, impermeable container made of cardboard fiber, plastic, or metal which can be sealed in order to be sift-proof, dustproof, and leak-tight.

Employee exposure - The exposure to airborne asbestos that would occur if the employee were not wearing respiratory protection equipment.

Encapsulant - A material that surrounds or embeds asbestos fibers in an adhesive matrix and prevents the release of fibers.

Encapsulation - Treating ACM with an encapsulant.

Enclosure - The construction of an air tight, impermeable, permanent barrier around ACM to control the release of asbestos fibers from the material and also eliminate access to the material.

Equipment room - A contaminated room located within the decontamination area that is supplied with impermeable bags or containers for the disposal of contaminated protective clothing and equipment.

Fiber - A particulate form of asbestos, 5 microns or longer, with a length to width (aspect) ratio of at least 3 to 1.

Fibers per cubic centimeter (f/cc) - Abbreviation for fibers per cubic centimeter, used to describe the level of asbestos fibers in air.

Filter - Media used in respirators, vacuums, or other machines to remove particulate from air.

Firestopping - Material used to close the open parts of a structure in order to prevent a fire from spreading.

Friable asbestos containing material - Any material containing more than one (1) percent asbestos as determined using the method specified 40 CFR 763, Polarized Light Microscopy, that, when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

Glovebag - Not more than a 60 x 60 inch impervious plastic bag-like enclosure affixed around an asbestos-containing material, with glove-like appendages through which materials and tools may be handled.

High efficiency particulate air (HEPA) filter – An ASHRAE MERV 17 filter capable of trapping and retaining at least 99.97 percent of all mono-dispersed particles of 0.3 micrometers in diameter.

HEPA vacuum - Vacuum collection equipment equipped with a HEPA filter system capable of collecting and retaining asbestos fibers.

Homogeneous area - An area of surfacing, thermal system insulation or miscellaneous ACM that is uniform in color, texture and date of application.

HVAC - Heating, Ventilation and Air Conditioning

Industrial hygienist (IH) - A professional qualified by education, training, and experience to anticipate, recognize, evaluate and develop controls for occupational health hazards. Meets definition requirements of the American Industrial Hygiene Association (AIHA).

Industrial hygienist technician (IH Technician) - A person working under the direction of an IH or CIH who has special training, experience, certifications and licenses required for the industrial hygiene work assigned. Some States require that an industrial hygienist technician conducting asbestos abatement air sampling, clearance inspection and clearance air sampling be licensed as an asbestos project monitor.

Intact - The ACM has not crumbled, been pulverized, or otherwise deteriorated so that the asbestos is no longer likely to be bound with its matrix.

Lockdown - Applying encapsulant, after a final visual inspection, on all abated surfaces at the conclusion of ACM removal prior to removal of critical barriers.

National Emission Standards for Hazardous Air Pollutants (NESHAP) - EPA's rule to control emissions of asbestos to the environment (40 CFR 61, Subpart M).

Negative initial exposure assessment - A demonstration by the employer which complies with the criteria in 29 CFR 1926.1101 (f)(2)(iii), that employee exposure during an operation is expected to be consistently below the PEL or Excursion Limit (EL).

Negative pressure - Air pressure which is lower than the surrounding area, created by exhausting air from a sealed regulated area through HEPA equipped filtration units. OSHA requires maintaining -0.02 inch water column gauge inside the negative pressure enclosure.

Negative pressure respirator - A respirator in which the air pressure inside the facepiece is negative during inhalation relative to the air pressure outside the respirator facepiece.

Non-friable ACM - Material that contains more than 1 percent asbestos but cannot be crumbled, pulverized, or reduced to powder by hand pressure.

Organic vapor cartridge - The type of cartridge used on air purifying respirators to remove organic vapor hazardous air contaminants.

Outside air - The air outside buildings and structures, including, but not limited to, the air under a bridge or in an open ferry dock.

Owner/operator - Any person who owns, leases, operates, controls, or supervises the facility being demolished or renovated or any person who owns, leases, operates, controls, or supervises the demolition or renovation operation, or both.

Penetrating encapsulant - Encapsulant that is absorbed into the ACM matrix without leaving a surface layer.

Permissible exposure limit (PEL) - The level of exposure OSHA allows for an 8-hour time weighted average. For asbestos fibers, the eight (8) hour time-weighted average PEL is 0.1 fibers per cubic centimeter (0.1 f/cc) of air and the 30-minute Excursion Limit (EL) is 1.0 fibers per cubic centimeter (1 f/cc).

Personal protective equipment (PPE) – equipment designed to protect user from injury and/or specific job hazard. Such equipment may include protective clothing, hard hats, safety glasses, fall protection, and respirators.

Personal sampling/monitoring - Representative air samples obtained in the breathing zone for one or more workers within the regulated area using a filter cassette and a calibrated air sampling pump to determine asbestos exposure.

Pipe tunnel – An area, typically located adjacent to mechanical spaces or boiler rooms in which the pipes servicing the heating system in the building are routed to allow the pipes to access heating elements. These areas may contain asbestos pipe insulation, asbestos fittings, debris or asbestos-contaminated soil.

Polarized light microscopy (PLM) - Light microscopy using dispersion staining techniques and refractive indices to identify and quantify the type of asbestos present in a bulk sample.

Polyethylene sheeting - Strong plastic barrier material 4 to 6-mils thick, semi-transparent, flame retardant per NFPA 241.

Positive/negative fit check - A method of verifying the seal of a

facepiece respirator by temporarily occluding the filters and breathing in (inhaling) and then temporarily occluding the exhalation valve and breathing out (exhaling) while checking for inward or outward leakage of the respirator respectively.

Presumed ACM (PACM) - Thermal system insulation, surfacing, and flooring material installed in buildings prior to 1981. If the building owner has actual knowledge, or should have known through the exercise of due diligence that other materials are ACM, they too must be treated as PACM. The designation of PACM may be rebutted pursuant to 29 CFR 1926.1101 (k)(5).

Professional IH - An IH who meets the definition requirements of AIHA; meets the definition requirements of OSHA as a "Competent Person" at 29 CFR 1926.1101 (b); has completed two specialized EPA approved courses on management and supervision of asbestos abatement projects; has formal training in respiratory protection and waste disposal; and has a minimum of four projects of similar complexity with this project of which at least three projects serving as the supervisory IH. The PIH may be either the VA’s PIH (VPIH/CIH) or Contractor’s PIH (CPIH/CIH).

Project designer - A person who has successfully completed the training requirements for an asbestos abatement project designer as required by 40 CFR 763 Subpart E, Appendix C, Part I; (B)(5).

Assigned Protection factor - A value assigned by OSHA/NIOSH to indicate the expected protection provided by each respirator class, when the respirator is properly selected and worn correctly. The number indicates the reduction of exposure level from outside to inside the respirator facepiece.

Qualitative fit test (QLFT) - A fit test using a challenge material that can be sensed by the wearer if leakage in the respirator occurs.

Quantitative fit test (QNFT) - A fit test using a challenge material which is quantified outside and inside the respirator thus allowing the determination of the actual fit factor.

Regulated area - An area established by the employer to demarcate where Class I, II, III asbestos work is conducted, and any adjoining area where debris and waste from such asbestos work may accumulate; and a work area within which airborne concentrations of asbestos exceed, or there is a reasonable possibility they may exceed the PEL.

Regulated ACM (RACM) - Friable ACM; Category I non-friable ACM that has become friable; Category I non-friable ACM that will be or has been subjected to sanding, grinding, cutting, or abrading or; Category II non-friable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of the demolition or renovation operation.

Removal - All operations where ACM, PACM and/or RACM is taken out or stripped from structures or substrates, including demolition operations.

Renovation - Altering a facility or one or more facility components in any way, including the stripping or removal of asbestos from a facility component which does not involve demolition activity.

Repair - Overhauling, rebuilding, reconstructing, or reconditioning of structures or substrates, including encapsulation or other repair of ACM or PACM attached to structures or substrates.

Shower room - The portion of the PDF where personnel shower before leaving the regulated area.

Supplied air respirator (SAR) - A respiratory protection system that supplies minimum Grade D respirable air per ANSI/Compressed Gas Association Commodity Specification for Air, CGA G-7.1.

Surfacing ACM - A material containing more than 1 percent asbestos that is sprayed, troweled on or otherwise applied to surfaces for acoustical, decorative, fireproofing and other purposes.

Surfactant - A chemical added to water to decrease water's surface tension thus making it more penetrating into ACM.

Thermal system ACM - A material containing more than 1 percent asbestos applied to pipes, fittings, boilers, breeching, tanks, ducts, or other structural components to prevent heat loss or gain.

Transmission electron microscopy (TEM) - A microscopy method that can identify and count asbestos fibers.

VA Professional Industrial Hygienist (VPIH/CIH) – The Department of Veterans Affairs Professional Industrial Hygienist must meet the qualifications of a PIH, and may report to a Certified Industrial Hygienist (CIH).

VA Representative - The VA official responsible for on-going project work.

VA Total – means a building or substantial part of the building is completely removed, torn or knocked down, bulldozed, flattened, or razed, including removal of building debris.

Visible emissions - Any emissions, which are visually detectable without the aid of instruments, coming from ACM/PACM/RACM/ACS or ACM waste material.

Waste/Equipment decontamination facility (W/EDF) – The area in which equipment is decontaminated before removal from the regulated area.

Waste generator - Any owner or operator whose act or process produces asbestos-containing waste material.

Waste shipment record - The shipping document, required to be originated and signed by the waste generator, used to track and substantiate the disposition of asbestos-containing waste material.

Wet cleaning - The process of thoroughly eliminating, by wet methods, any asbestos contamination from surfaces or objects.

1.5 APPLICABLE CODES AND REGULATIONS

a. General Applicability of Codes, Regulations, and Standards:

(1) All work under this contract shall be done in strict accordance with all applicable Federal, State, and Local regulations, standards and codes governing asbestos abatement, and any other trade work done in conjunction with the abatement. All applicable codes, regulations and standards are adopted into this specification and will have the same force and effect as this specification.

(2) The most recent edition of any relevant regulation, standard, document or code shall be in effect. Where conflict among the requirements or with these specification, exists, the most stringent requirement(s) shall be utilized.

(3) Copies of all standards, regulations, codes and other applicable documents, including this specification and those listed in Section 1.5 shall be available at the worksite in the clean change area of the worker decontamination system and/or the Contractor’s on-site Field Office. These standards, regulations, codes and other applicable documents, including this specification and those listed in Section 1.5 may be made available electronically.

b. Asbestos Abatement Contractor Responsibility: The Asbestos Abatement Contractor (Contractor) shall assume full responsibility and liability for compliance with all applicable Federal, State and Local regulations related to any and all aspects of the asbestos abatement project. The Contractor is responsible for providing and maintaining training, accreditations, medical exams, medical records, personal protective equipment (PPE), respiratory protection, and respirator fit testing, as required by applicable Federal, State and Local regulations. The Contractor shall hold the VA and VPIH/CIH consultants harmless for any Contractor’s failure to comply with any applicable work, packaging, transporting, disposal, safety, health, or environmental requirement on the part of himself, his employees, or his subcontractors. The Contractor will incur all costs of the CPIH/CIH, including all sampling/analytical costs to assure compliance with OSHA/EPA/State/Local requirements related to failure to comply with the regulations applicable to the work.

c. Federal Requirements: Federal requirements which govern some aspect of asbestos abatement include, but are not limited to, the following regulations.

(1) Occupational Safety and Health Administration (OSHA)

(a) Title 29 CFR 1926.1101 - Construction Standard for Asbestos

(a) Title 29 CFR 1926 Subpart E - Personal Protective Equipment and Life Saving Equipment

(b) Title 29 CFR 1910.134 - Respiratory Protection

(c) Title 29 CFR 1926 - Construction Industry Standards

(d) Title 29 CFR 1926.33 - Access to Employee Exposure and Medical

Records

(e) Title 29 CFR 1926.59 same as 29 CFR 1910.1200 - Hazard Communication

(f) Title 29 CFR 1926 Subpart C – General Safety and Health Provisions and Subpart D – Occupational Health and Environmental Controls

(2) Environmental Protection Agency (EPA)

(a) 1. 40 CFR 61 Subpart M - National Emission Standard for Hazardous Air Pollutants - Asbestos

(a) 2. 40 CFR 763.80 - Asbestos Hazard Emergency Response Act (AHERA) and Asbestos Hazard Abatement Reauthorization Act (ASHARA)

(3) Department of Transportation (DOT)

(a) Title 49 CFR 171 - 180 – Transportation

d. State Requirements:

(1) State requirements that apply to the asbestos abatement work, disposal, clearance, etc., include, but are not limited to, the following:
Connecticut General Statutes Title 19A - Public Health and Well-Being Section 329 to 333
Connecticut General Statutes Title 20 - Professional and Occupational Licensing,Certification, Title Protection and Registration Section 435 to 442

e. Local Requirements:

(1) If Local requirements are more stringent than Federal or State standards, the local standards are to be followed.

f. Standards:

(1) Standards which govern asbestos abatement activities include, but are not limited to, the following:

(a) American National Standards Institute (ANSI/ASSP) ASSP Z9.2 -2018 - Fundamentals Governing the Design and Operation of Local Exhaust Systems and ASSP Z88.2 - Practices for Respiratory Protection.

(b) Underwriters Laboratories (UL) UL 586-2009 - UL Standard for Safety of HEPA filter Units, 9th Edition; ANSI Approval 2017-12-19.

(2) Standards which govern encapsulation work include, but are not limited to, the following:

(a) American Society for Testing and Materials International (ASTM).

(3) Standards which govern the fire and safety concerns in abatement work include, but are not limited to, the following:

(a) National Fire Protection Association (NFPA) NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

(b) NFPA 701 - Standard Methods for Fire Tests for Flame Resistant Textiles and Film.

(c) NFPA 101 - Life Safety Code.

g. EPA Guidance Documents:

(1) EPA guidance documents which discuss asbestos abatement work activities are listed below. These documents are made part of this section by reference. Guidance for Controlling ACM in Buildings (Purple Book) EPA 560/5-85-024.

(2) Asbestos Waste Management Guidance EPA 530-SW-85-007.

(3) A Guide to Respiratory Protection for the Asbestos Abatement Industry EPA 560-OPTS-86-001.

(4) Guide to Managing Asbestos in Place (Green Book) TS 799 20T July 1990.

h. Notices:

(1) State and Local agencies: Send written notification as required by state and local regulations including the local fire department prior to beginning any work on ACM as follows:

(2) Copies of notifications shall be submitted to the VA for the facility's records in the same time frame notification are given to EPA, State, and Local authorities.

i. Permits/Licenses: The contractor shall apply for and have all required permits and licenses to perform asbestos abatement work as required by Federal, State, and Local regulations prior to beginning any work on ACM as follows.

j. Posting and Filing of Regulations: Maintain two (2) copies of applicable Federal, State, and Local regulations. Post one copy of each at the regulated area where workers will have daily access to the regulations and keep another copy in the Contractor's office.

k. VA Responsibilities Prior to Commencement of Work:

(1) Notify occupants adjacent to regulated areas of project dates and requirements for relocation, if needed. Arrangements must be made prior to starting work for relocation of desks, files, equipment, and personal possessions to avoid unauthorized access into the regulated area. Note: **Notification of adjacent personnel is required by OSHA in 29 CFR 1926.1101 (k) to prevent unnecessary or unauthorized access to the regulated area.**

(2) Submit to the Contractor results of background air sampling; including location of samples, person who collected the samples, equipment utilized, calibration data and method of analysis. During abatement, submit to the Contractor, results of bulk material analysis and air sampling data collected during the

course of the abatement. This information shall not release the Contractor from any responsibility for OSHA compliance.

l. Emergency Action Plan and Arrangements:

(1) An Emergency Action Plan shall be developed by prior to commencing abatement activities and shall be agreed to by the Contractor and the VA. The Plan shall meet the requirements of 29 CFR 1926 Subpart C, Standard 29 CFR 1926.35 Employee Emergency Action Plans.

(2) Emergency procedures shall be in written form and prominently posted in the clean room and equipment room of the decontamination unit. Everyone, prior to entering the regulated area, must read and sign these procedures to acknowledge understanding of the regulated area layout, location of emergency exits and emergency procedures.

(3) Emergency planning shall include written notification of police, fire, and emergency medical personnel of planned abatement activities; work schedule; layout of regulated area; and access to the regulated area, particularly barriers that may affect response capabilities.

(4) Emergency planning shall include consideration of fire, explosion, hazardous atmospheres, electrical hazards, slips/trips and falls, confined spaces, and heat stress illness. Written procedures for response to emergency situations shall be developed and employee training in procedures shall be provided.

(5) Employees shall be trained in regulated area/site evacuation procedures in the event of workplace emergencies.

(a) For non-life-threatening situations - employees injured or otherwise incapacitated shall be decontaminated following normal procedures with assistance from fellow workers, if necessary, before exiting the regulated area to obtain proper medical treatment.

(b) For life-threatening injury or illness, worker decontamination shall take least priority after measures to stabilize the injured worker, medical personnel shall remove them from the regulated area if back or neck injury is present, and secure proper medical treatment.

(6) Telephone numbers of any/all emergency response personnel shall be prominently posted in the clean room, along with the location of the nearest telephone.

(7) The Contractor shall provide verification of first aid/CPR training for personnel responsible for providing first aid/CPR. OSHA requires medical assistance within 3-4 minutes of a life-threatening injury/illness. Bloodborne Pathogen training shall also be verified for those personnel required to provide first aid/CPR.

(8) The Emergency Action Plan shall provide for a Contingency Plan in the event that an incident occurs that may require the modification of the Asbestos Hazard Abatement Plans during abatement. Such incidents include, but are not limited to, fire;

accident; power failure; negative pressure failure; and supplied air system failure. The Contractor shall detail procedures to be followed in the event of an incident assuring that asbestos abatement work is stopped and wetting is continued until correction of the problem.

m. Pre-Construction Meeting:

(1) Prior to commencing the work, the Contractor shall meet with the VPIH/CIH to present and review, as appropriate, the items following this paragraph. The Contractor's Competent Person(s) who will be on-site shall participate in the pre-start meeting. The pre-start meeting is to discuss and determine procedures to be used during the project. At this meeting, the Contractor shall provide:

(a) Proof of Contractor licensing.

(b) Proof the Competent Person is trained and accredited and approved for working in this State. Verification of the experience of the Competent Person shall also be presented.

(c) A list of all workers who will participate in the project, including experience and verification of training and accreditation.

(d) A list of and verification of training for all personnel who have current first-aid/CPR training. A minimum of one person per shift must have adequate training.

(e) Current medical written opinions for all personnel working on-site meeting the requirements of 29 CFR 1926.1101 (m).

(f) Current fit-tests for all personnel wearing respirators on-site meeting the requirements of 29 CFR 1926.1101 (h) and Appendix C.

(g) A copy of the Contractor's Asbestos Hazard Abatement Plan. In these procedures, the following information must be detailed, specific for this project. A copy of the Contractor's Asbestos Hazard Abatement Plan (AHAP) for Class I Glovebag Asbestos Abatement. In these procedures, the following information must be detailed, specific for this project.

1. Regulated area preparation procedures;

2. Notification requirements procedure of Contractor as required in 29 CFR 1926.1101 (d) Multi-Employer Worksites;

3. If required, decontamination area set-up/layout and decontamination procedures for employees;

4. Glovebag abatement methods/procedures and equipment to be used; and

5. Personal protective equipment to be used

(2) At this meeting the Contractor shall provide all submittals as required.

(3) Procedures for handling, packaging and disposal of asbestos waste.

(4) Emergency Action Plan and Contingency Plan Procedures.

1.6 PROJECT COORDINATION

a. The following are the minimum administrative and supervisory personnel necessary for coordination of the work.

(1) Personnel

(a) Administrative and supervisory personnel shall consist of a qualified Competent Person(s) as defined by OSHA in the Construction Standards and the Asbestos Construction Standard; Contractor Professional Industrial Hygienist and Industrial Hygiene Technicians. These employees are the Contractor's representatives responsible for compliance with these specifications and all other applicable requirements.

(b) Non-supervisory personnel shall consist of an adequate number of qualified personnel to meet the schedule requirements of the project. Personnel shall meet required qualifications. Personnel utilized on-site shall be pre-approved by the VA Representative. A request for approval shall be submitted for any person to be employed during the project giving the person's name; last four digits of social security number; qualifications; accreditation card with color picture if required by State; Certificate of Worker's Acknowledgment; and Affidavit of Medical Surveillance and Respiratory Protection and current Respirator Fit Test.

(c) Minimum qualifications for Contractor and assigned personnel are:

1. The Contractor has conducted within the last three (3) years, three (3) projects of similar complexity and dollar value as this project; has not been cited and penalized for serious violations of Federal (and State or Local as applicable) EPA and OSHA asbestos regulations in the past three (3) years; has adequate liability/occurrence insurance for asbestos work as required by the State; is licensed in applicable State; has adequate and qualified personnel available to complete the work; has comprehensive Asbestos Hazard Abatement Plans (AHAPs) for asbestos work; and has adequate materials, equipment and supplies to perform the work.

2. The Competent Person has four (4) years of abatement experience of which two (2) years were as the Competent Person on the project; meets the OSHA definition of a Competent Person; has been the Competent Person on two (2) projects of similar size and complexity as this project within the past three (3) years; has completed EPA AHERA/OSHA/State/Local training requirements/accreditation(s) and refreshers; and has all required OSHA documentation related to medical and respiratory protection.

3. The Contractor Professional Industrial Hygienist/CIH (CPIH/CIH) shall have five (5) years of monitoring experience and supervision of asbestos abatement projects; has participated as senior IH on five (5) abatement projects, three (3) of which are

similar in size and complexity as this project; has specialized EPA AHERA/OSHA training in asbestos abatement management, respiratory protection, waste disposal and asbestos inspection; has completed the NIOSH 582 Course or equivalent, Contractor/Supervisor course; and has appropriate medical/respiratory protection records/documentation.

4. The Abatement Personnel shall have completed the EPA AHERA/OSHA abatement worker course; have training on the Asbestos Hazard Abatement Plans of the Contractor; has one year of asbestos abatement experience within the past three (3) years of similar size and complexity; has applicable medical and respiratory protection documentation; has certificate of training/current refresher and State accreditation/license.

(2) All personnel shall be in compliance with OSHA construction safety training as applicable and submit certification.

1.7 RESPIRATORY PROTECTION

a. General - Respiratory Protection Program: The Contractor shall develop and implement a written Respiratory Protection Program (RPP) which is in compliance with OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.134. ANSI Standard Z88.2-2015 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH approved for asbestos abatement activities. The written RPP shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134 (c) - Respiratory Protection Program.Respiratory Protection Program Coordinator: The Respiratory Protection Program Coordinator (RPPC) must be identified and shall have two (2) years of experience coordinating RPP of similar size and complexity. The RPPC must submit a signed statement attesting to the fact that the program meets the above requirements.

b. Selection and Use of Respirators: The procedure for the selection and use of respirators must be submitted to the VA as part of the Contractor's qualifications. The procedure must be written clearly enough for workers to understand. A copy of the Respiratory Protection Program must be available in the clean room of the decontamination unit or in the onsite Contractor’s office, for reference by employees or authorized visitors.

c. Minimum Respiratory Protection: Minimum respiratory protection shall be a ½-mask negative pressure air purifying respirator equipped with P100 filters, provided personal air samples in the workplace remain at or below 0.1 f/cc, determined as an 8-hour TWA. Full face powered air purifying respirator equipped with P100 filters shall be required until Contractor demonstrates that personal air samples are at or below 0.1 f/cc, determined as an 8-hour TWA. A higher level of respiratory protection shall be required, if fiber levels exceed 1 f/cc as an 8-hour TWA, inside the regulated work area. Respirator selection shall meet the requirements of 29 CFR 1926.1101 (h) and 29 CFR 1910.134 (d)(3)(i)(A) Table 1, except as indicated in this paragraph. Abatement personnel must have a respirator for their exclusive use.

d. Medical Written Opinion: No employee shall be allowed to wear a respirator unless a physician or other licensed health care professional has provided a written determination they are medically

qualified to wear the class of respirator to be used on the project while wearing whole body impermeable garments and subjected to heat or cold stress.

e. Respirator Fit Test: All personnel wearing respirators shall have a current quantitative fit test which was conducted in accordance with 29 CFR 1910.134 (f) and Appendix A. Fit tests shall be done for PAPR's which have been put into a failure mode.

f. Respirator Fit Check: The Competent Person shall assure that the positive/negative pressure user seal check is done each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that prevents an effective facepiece to face seal as evidenced by failure of a user seal check shall preclude that person from entering the regulated area until resolution of the problem.

g. Maintenance and Care of Respirators: The Respiratory Protection Program Coordinator shall submit evidence and documentation showing compliance with 29 CFR 1910.134 (h) maintenance and care of respirators.

1.8 WORKER PROTECTION

a. Training of Abatement Personnel: Prior to beginning any abatement activity, all personnel shall be trained in accordance with OSHA 29 CFR 1926.1101 (k)(9) and any additional State/Local requirements. Training must include, at a minimum, the elements listed at 29 CFR 1926.1101 (k)(9)(viii). Training shall have been conducted by a third party, EPA/State approved trainer meeting the requirements of EPA 40 CFR 763 Appendix C (AHERA MAP). Initial training certificates and current refresher and accreditation proof must be submitted for each person working at the site.

b. Medical Examinations: Medical examinations meeting the requirements of 29 CFR 1926.1101 (m) shall be provided for all personnel working in the regulated area, regardless of exposure levels. A current physician's written opinion as required by 29 CFR 1926.1101 (m)(4) shall be provided for each person and shall include in the medical opinion that the person has been evaluated for working in a heat and cold stress environment while wearing personal protective equipment (PPE) and is able to perform the work without risk of material health impairment.

c. Personal Protective Equipment: Provide whole body clothing, head coverings, foot coverings and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.

d. Regulated Area Entry Procedure: The Competent Person shall ensure that each time workers enter the regulated area; they remove ALL street clothes in the clean room of the decontamination unit and put on new disposable coveralls, head coverings, a clean respirator, and then proceed through the shower room to the equipment room where they put on non-disposable required personal protective equipment

e. Decontamination Procedure: The Competent Person shall require all personnel to adhere to following decontamination procedures whenever they leave the regulated area.

(1) When exiting the regulated area, remove disposable coveralls, and ALL other clothes, disposable head coverings, and foot coverings or boots in the equipment room.

(2) Still wearing the respirator and completely naked, proceed to the shower. Showering is MANDATORY. Care must be taken to follow reasonable procedures in removing the respirator to avoid inhaling asbestos fibers while showering. The following procedure is required as a minimum:

(a) Thoroughly wet body including hair and face. If using a PAPR hold blower above head to keep filters dry.

(b) With respirator still in place, thoroughly decontaminate body, hair, respirator face piece, and all other parts of the respirator except the blower and battery pack on a PAPR. Pay particular attention to cleaning the seal between the face and respirator facepiece and under the respirator straps.

(c) Take a deep breath, hold it and/or exhale slowly, completely wetting hair, face, and respirator. While still holding breath, remove the respirator and hold it away from the face before starting to breathe.

(3) Carefully decontaminate the facepiece of the respirator inside and out. If using a PAPR, shut down using the following sequence: a) first cap inlets to filters; b) turn blower off to keep debris collected on the inlet side of the filter from dislodging and contaminating the outside of the unit; c) thoroughly decontaminate blower and hoses; d) carefully decontaminate battery pack with a wet rag being cautious of getting water in the battery pack thus preventing destruction. **(THIS PROCEDURE IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!)**

(4) Shower and wash body completely with soap and water. Rinse thoroughly.

(5) Rinse shower room walls and floor to drain prior to exiting.

(6) Proceed from shower to clean room; dry off and change into street clothes or into new disposable work clothing.

f. Regulated Area Requirements: The Competent Person shall meet all requirements of 29 CFR 1926.1101 (o) and assure that all requirements for Class I Glovebag regulated areas at 29 CFR 1926.1101 (e), 29 CFR 1926.1101 (g)(1)(i)(ii)(iii), 29 CFR 1926.1101 (g)(5)(ii)(iii)(iv) are met. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator.

1.9 DECONTAMINATION FACILITIES

a. Description: Provide each regulated area with separate personnel decontamination facilities (PDF) and waste/equipment decontamination facilities (W/EDF). Ensure that the PDF are the only means of ingress

and egress to the regulated area and that all equipment, bagged waste, and other material exit the regulated area only through the W/EDF.

b. General Requirements: All personnel entering or exiting a regulated area must go through the PDF and shall follow the requirements at 29 CFR 1926.1101 (j)(1) and these specifications. All waste, equipment and contaminated materials must exit the regulated area through the W/EDF and be decontaminated in accordance with these specifications. Walls and ceilings of the PDF and W/EDF must be constructed of a minimum of 3-layers of 6-mil opaque fire retardant polyethylene sheeting and be securely attached to existing building components and/or an adequate temporary framework. A minimum of 3-layers of 6-mil poly shall also be used to cover the floor under the PDF and W/EDF units. Construct doors so that they overlap and secure to adjacent surfaces. Weight inner doorway sheets with layers of duct tape so that they close quickly after release. Put arrows on sheets so they show direction of travel and overlap. If the building adjacent area is occupied, construct a solid barrier on the occupied side(s) to protect the sheeting and reduce potential for non-authorized personnel entering the regulated area.

c. Temporary Facilities to the PDF and W/EDF: The Competent Person shall provide temporary water service connections to the PDF and W/EDF. Backflow prevention must be provided at the point of connection to the VA system. Water supply must be of adequate pressure and meet requirements of 29 CFR 1910.141(d)(3). Provide adequate temporary overhead electric power with ground fault circuit interruption (GFCI) protection. Provide a sub-panel equipped with GFCI protection for all temporary power in the clean room. Provide adequate lighting to provide a minimum of 50 foot candles in the PDF and W/EDF. Provide temporary heat, if needed, to maintain 70 degrees F throughout the PDF and W/EDF.

d. Personnel Decontamination Facility (PDF): The Competent Person shall provide a PDF consisting of shower room which is contiguous to a clean room and equipment room. The PDF must be sized to accommodate the number of personnel scheduled for the project. The shower room, located in the center of the PDF, shall be fitted with as many portable showers as necessary to insure all employees can complete the entire decontamination procedure within 15 minutes. The PDF shall be constructed of opaque poly for privacy. The PDF shall be constructed to eliminate any parallel routes of egress without showering.

(1) Clean Room: The clean room must be physically and visually separated from the rest of the building to protect the privacy of personnel changing clothes. The clean room shall be constructed of at least 3─layers of 6-mil opaque fire retardant poly to provide an air tight room. Provide a minimum of 2 - 3 foot wide 6-mil poly opaque fire retardant doorways. One doorway shall be the entry from outside the PDF and the second doorway shall be to the shower room of the PDF. The floor of the clean room shall be maintained in a clean, dry and sanitary condition. Shower overflow shall not be allowed into the clean room. Provide 1 storage locker per person. A portable fire extinguisher, minimum 10 pounds capacity, Type ABC, shall be provided in accordance with OSHA and NFPA 10. All persons entering the regulated area shall remove all street clothing in the clean room and dress in disposable protective clothing and respiratory protection. Any person entering the clean room does so either from the outside

with street clothing on or is coming from the shower room completely naked and thoroughly washed. Any person entering the regulated area to perform Glovebag removal work, in which a negative exposure assessment has been performed, shall don a double outer protective suit and respirator. Male/Females required to enter the regulated area shall be ensured of their privacy throughout the entry/exit process by posting guards at both entry points to the PDF so no male/female can enter or exit the PDF during his/her stay in the PDF.

(2) Shower Room: The Competent Person shall assure that the shower room is a completely water tight compartment to be used for the movement of all personnel from the clean room to the equipment room and for the showering of all personnel going from the equipment room to the clean room. Each shower shall be constructed so water runs down the walls of the shower and into a drip pan. Install a freely draining smooth floor on top of the shower pan. The shower room shall be separated from the rest of the building and from the clean room and equipment room using air tight walls made from at least 3-layers of 6-mil opaque fire retardant poly. The shower shall be equipped with a shower head and controls, hot and cold water, drainage, soap dish and continuous supply of soap, and shall be maintained in a sanitary condition throughout its use. The controls shall be arranged so an individual can shower without assistance. Provide a flexible hose shower head, hose bibs and all other items shown on Shower Schematic. Waste water will be pumped to a drain after being filtered through a minimum of a 100 micron sock in the shower drain; a 20 micron filter; and a final 5 micron filter. Filters will be changed a minimum of once per day or more often as needed. Filter changes must be done in the shower to prevent loss of contaminated water. Hose down all shower surfaces after each shift and clean any debris from the shower pan. Residue is to be disposed of as asbestos waste. The Competent Person shall provide a decontamination area at the outer perimeter of the regulated work area where the employees will decontaminate the outer protective suit and respirator by wet wiping and HEPA vacuuming.

(3) Equipment Room: The Competent Person shall provide an equipment room which shall be an air tight compartment for the storage of work equipment/tools, reusable personal protective equipment, except for a respirator and for use as a gross decontamination area for personnel exiting the regulated area. The equipment room shall be separated from the regulated area by a minimum 3 foot wide door made with 2-layers of 6-mil opaque fire retardant poly. The equipment room shall be separated from the regulated area, the shower room and the rest of the building by air tight walls and ceiling constructed of a minimum of 3-layers of 6-mil opaque fire retardant poly. Damp wipe all surfaces of the equipment room after each shift change. Provide an additional loose layer of 6-mil fire retardant poly per shift change and remove this layer after each shift. If needed, provide a temporary electrical sub-panel equipped with GFCI in the equipment room to accommodate any equipment required in the regulated area. The Competent Person shall provide a decontamination area at the outer perimeter of the regulated work area where the employees will decontaminate the outer protective suit and respirator by wet wiping and HEPA vacuuming.

(4) The PDF shall be as follows: Clean room at the entrance followed by a shower room followed by an equipment room leading to the regulated area. Each doorway in the PDF shall be a minimum of 2-layers of 6-mil opaque fire retardant poly.

(a) See Attachment A

e. Waste/Equipment Decontamination Facility (W/EDF)

(1) The Competent Person shall provide a W/EDF consisting of a wash room, holding room, and clean room for removal of waste, equipment and contaminated material from the regulated area. Personnel shall not enter or exit the W/EDF except in the event of an emergency. Clean debris and residue in the W/EDF daily. All surfaces in the W/EDF shall be wiped/hosed down after each shift and all debris shall be cleaned from the shower pan. The W/EDF shall consist of the following:

(a) Wash Down Station: Provide an enclosed shower unit in the regulated area just outside the Wash Room as an equipment bag and container cleaning station.

(b) Wash Room: Provide a wash room for cleaning of bagged or containerized asbestos containing waste materials passed from the regulated area. Construct the wash room using x 2 inches x 4 inches wood framing and 3-layers of 6-mil fire retardant poly. Locate the wash room so that packaged materials, after being wiped clean, can be passed to the Holding Room. Doorways in the wash room shall be constructed of 2-layers of 6-mil fire retardant poly.

(c) Holding Room: Provide a holding room as a drop location for bagged materials passed from the wash room. Construct the holding room using x 2 inches x 4 inches wood framing or approved equivalent and 3-layers of 6-mil fire retardant poly. The holding room shall be located so that bagged material cannot be passed from the wash room to the clean room unless it goes through the holding room. Doorways in the holding room shall be constructed of 2-layers of 6-mil fire retardant poly.

(d) Clean Room: Provide a clean room to isolate the holding room from the exterior of the regulated area. Construct the clean room using 2 inches x 4 inches wood framing or approved equivalent and 2-layers of 6-mil fire retardant poly. The clean room shall be located so as to provide access to the holding room from the building exterior. Doorways to the clean room shall be constructed of 2-layers of 6-mil fire retardant poly. When a negative pressure differential system is used, a rigid enclosure separation between the W/EDF clean room and the adjacent areas shall be provided.

(e) The W/EDF shall be as follows: Wash Room leading to a Holding Room followed by a Clean Room leading to outside the regulated area. See diagram.

See Attachment B

f. Waste/Equipment Decontamination Procedures: At the washdown station in the regulated area, thoroughly wet wipe/clean contaminated equipment and/or sealed polyethylene bags and pass into Wash Room

after visual inspection. When passing anything into the Wash Room, close all doorways of the W/EDF, other than the doorway between the washdown station and the Wash Room. Keep all outside personnel clear of the W/EDF. Once inside the Wash Room, wet clean the equipment and/or bags. After cleaning and inspection, pass items into the Holding Room. Close all doorways except the doorway between the Holding Room and the Clean Room. Workers from the Clean Room/Exterior shall enter the Holding Room and remove the decontaminated/cleaned equipment/bags for removal and disposal. At no time shall personnel from the clean side be allowed to enter the Wash Room.

1.10 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z9.2	(2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems
ASSP Z88.2	(2015) American National Standard Practices for Respiratory Protection

ASTM INTERNATIONAL (ASTM)

ASTM C732	(2017; R 2022 Standard Test Method for Aging Effects of Artificial Weathering on Latex Sealants
ASTM D522	(1993a; R 2008) Mandrel Bend Test of Attached Organic Coatings
ASTM D2794	(1993; R 2019) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E736	(2000; R 2011) Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members
ASTM E814	(2024) Standard Test Method for Fire Tests of Penetration Firestop Systems

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7.1	(2011) Commodity Specification for Air; 5th Edition

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers
NFPA 101	(2024) Life Safety Code
NFPA 241	(2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations
NFPA 701	(2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH 7400	Airborne Fiber Count
NIOSH 7402	Asbestos Fibers by Transmission Electron Microscopy (TEM)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 560/5-85-024	(1985) Guidance for Controlling Asbestos-Containing Materials in Buildings (Purple Book)
EPA 560/5-89-001	(1989) Guidelines for Conduting the AHERA TEM Clearance Test to Determine Completion of an Asbestos Abatement Project
EPA 560-OPTS-86-001	(1986) A Guide to Respiratory Protection for the Asbestos Abatement Industry
TS 799 20T	(1990) Guide to Managing Asbestos in Place (Green Book)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.132	Personal Protective Equipment
29 CFR 1910.134	Respiratory Protection
29 CFR 1910.141	Sanitation
29 CFR 1910.1200	Hazard Communication
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.33	Access to Employee Exposure and Medical Records

29 CFR 1926.35	Employee Emergency Action Plan
29 CFR 1926.59	Hazard Communication
29 CFR 1926.1101	Asbestos
40 CFR 61	National Emission Standards for Hazardous Air Pollutants
40 CFR 763	Asbestos
40 CFR 763.80	Asbestos Hazard Emergency Response Act (AHERA) and Asbestos Hazard Abatement Reauthorization Act (ASHARA)
49 CFR 171	General Information, Regulations, and Definitions

UL SOLUTIONS (UL)

UL 586	(2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units
UL 1479	(2015; Reprint Apr 2024) Fire Tests of Through-Penetration Firestops

1.11 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Pre-Start Meeting Submittals; G

Asbestos Hazard Abatement Plan ; G

SD-02 Shop Drawings

Containment Barriers And Coverings In The Regulated Area; G

SD-03 Product Data

Encapsulants; G

Materials And Equipment; G

SD-06 Test Reports

Submittals During Abatement; G

Monitoring, Inspection And Testing; G

SD-07 Certificates

Certificates Of Compliance; G

SD-11 Closeout Submittals

Submittals At Completion Of Abatement; G

1.12 SUBMITTAL INSTRUCTIONS

a. Pre-Start Meeting Submittals:

(1) Submit to the VA a minimum of 14 days prior to the pre-start meeting the following for review and approval. Meeting this requirement is a prerequisite for the pre-start meeting for this project:

(a) Submit a detailed work schedule for the entire project reflecting contract documents and the phasing/schedule requirements from the CPM chart.

(b) Submit a staff organization chart showing all personnel who will be working on the project and their capacity/function. Provide their qualifications, training, accreditations, and licenses, as appropriate. Provide a copy of the "Certificate of Worker's Acknowledgment" and the "Affidavit of Medical Surveillance and Respiratory Protection" for each person. See Attachment C.

(c) Submit Asbestos Hazard Abatement Plan developed specifically for this project, incorporating the requirements of the specifications, prepared, signed and dated by the CPIH/CIH.

(d) Submit the specifics of the materials and equipment to be used for this project with manufacturer names, model numbers, performance characteristics, pictures/diagrams, and number available for the following:

1. Supplied air system, negative air machines, HEPA vacuums, air monitoring pumps, calibration devices, pressure differential monitoring device and emergency power generating system.

2. Waste water filtration system, shower system, containment barriers.

3. Encapsulantys, surfactants, hand held sprayers, airless sprayers, Glovebas, and fire extinguishers.

4. Respirators, water filtration system, shower system, containment barriers equipment.

5. Fire safety equipment to be used in the regulated area.

(e) Submit the name, location, and phone number of the approved landfill; proof/verification the landfill is approved for ACM disposal; the landfill's requirements for ACM waste; the type of vehicle to be used for transportation; and name, address, and phone number of subcontractor, if used. Proof of asbestos training for transportation personnel shall be provided.

(f) Submit required notifications and arrangements made with regulatory agencies having regulatory jurisdiction and the specific contingency/emergency arrangements made with local health, fire, ambulance, hospital authorities and any other notifications/arrangements.

(g) Submit the name, location and verification of the laboratory and/or personnel to be used for analysis of air and/or bulk samples. Personal air monitoring must be done in accordance with OSHA 29 CFR 1926.1101(f) and Appendix A. Area or clearance air monitoring shall be conducted in accordance with EPA AHERA protocols.

(h) Submit qualifications verification: Submit the following evidence of qualifications. Make sure that all references are current and verifiable by providing current phone numbers and documentation.

1. Asbestos Abatement Company: Project experience within the past 3 years; listing projects first most similar to this project; Project Name; Type of Abatement; Duration; Cost; Reference Name/Phone Number; Final Clearance; and Completion Date.

2. List of project(s) halted by owner, A/E, IH, regulatory agency in the last 3 years: Project Name; Reason; Date; Reference Name/Number; and Resolution.

3. List asbestos regulatory citations (e.g., OSHA), notices of violations (e.g., Federal, State, Local NESHAP), penalties, and legal actions taken against the company including the company’s officers (including damages paid) in the last 3 years. Provide copies and all information needed for verification.

(i) Submit information on personnel: Provide a resume; address each item completely; copies of certificates, accreditations, and licenses. Submit an affidavit signed by the CPIH/CIH stating that all personnel submitted below have medical records in accordance with OSHA 29 CFR 1926.1101(m) and that the company has implemented a medical surveillance program and written respiratory protection program, and maintains recordkeeping in accordance with the above regulations. Submit the phone number and doctor/clinic/hospital used for medical evaluations.

1. CPIH/CIH and IH Technician: Name; years of abatement experience; list of projects similar to this one; certificates, licenses, accreditations for proof of AHERA/OSHA specialized asbestos training; professional affiliations; medical opinion; and current respirator fit test.

2. Competent Person(s)/Supervisor(s): Number; names; last four digits of social security numbers; years of abatement experience as Competent Person/Supervisor; list of similar projects in size/complexity as Competent Person/Supervisor; as a worker; certificates, licenses, accreditations; proof of AHERA/OSHA specialized asbestos training; maximum number of personnel supervised on a project; medical opinion (asbestos surveillance and respirator use); and current respirator fit test.

3. Workers: Numbers; names; last four digits of social security numbers; years of abatement experience; certificates, licenses, accreditations; training courses in asbestos abatement and respiratory protection; medical opinion (asbestos surveillance and respirator use); and current respirator fit test.

(j) Submit copies of State license for asbestos abatement; copy of insurance policy, including exclusions with a letter from agent stating in plain language the coverage provided and the fact that asbestos abatement activities are covered by the policy; copy of AHAP(s) incorporating the requirements of this specification; information on who provides your training, how often; who provides medical surveillance, how often; who performs and how is personal air monitoring of abatement workers conducted; a list of references of independent laboratories/IH's familiar with your air monitoring and Asbestos Hazard Abatement Plans; copies of monitoring results of the five referenced projects listed and analytical method(s) used.

(k) Rented equipment must be decontaminated prior to returning to the rental agency.

(l) Submit, before the start of work, the manufacturer's technical data for all types of encapsulants, all SDS, and application instructions.

b. Submittals During Abatement:

(1) The Competent Person shall maintain and submit a daily log at the regulated area documenting the dates and times of the following: purpose, attendees and summary of meetings; all personnel entering/exiting the regulated area; document and discuss the resolution of unusual events such as barrier breeching, equipment failures, emergencies, and any cause for stopping work; representative air monitoring and results/TWAs/ELs. Submit this information daily to the VA's Representative.

(2) The CPIH/CIH shall document and maintain the inspection and approval of the regulated area preparation prior to start of work and daily during work.

(a) Removal of any poly barriers and/or failure of negative pressure Glovebags.

(b) Visual inspection/testing by the CPIH/CIH or IH Technician prior to application of lockdown encapsulant.

(c) Packaging and removal of ACM waste from regulated area.

(d) Disposal of ACM waste materials; copies of Waste Shipment Records/landfill receipts to the VA's Representative on a weekly basis.

c. Submittals at Completion of Abatement: The CPIH/CIH shall submit a project report consisting of the daily log book requirements and documentation of events during the abatement project including Waste Shipment Records signed by the landfill's agent. It will also include information on the containment and transportation of waste from the containment with applicable Chain of Custody forms. The report shall

include a certificate of completion, signed and dated by the CPIH/CIH, in accordance with Attachment C (Attachment #1). All clearance and perimeter area samples must be submitted. The VA Representative will retain the abatement report after completion of the project and provide copies of the abatement report to VAMC Office of Engineer and the Safety Office.

PART 2 PRODUCTS, MATERIALS AND EQUIPMENT

2.1 MATERIALS AND EQUIPMENT

a. General Requirements (all abatement projects): Prior to the start of work, the contractor shall provide and maintain a sufficient quantity of materials and equipment to assure continuous and efficient work throughout the duration of the project. Work shall not start unless the following items have been delivered to the site and the CPIH/CIH has submitted verification to the VA's Representative.

(1) All materials shall be delivered in their original package, container or bundle bearing the name of the manufacturer and the brand name (where applicable).

(2) Store all materials subject to damage off the ground, away from wet or damp surfaces and under cover sufficient enough to prevent damage or contamination. Flammable and combustible materials cannot be stored inside buildings. Replacement materials shall be stored outside of the regulated area until abatement is completed.

(3) The Contractor shall not block or hinder use of buildings by patients, staff, and visitors to the VA in partially occupied buildings by placing materials/equipment in any unauthorized location.

(4) The Competent Person shall inspect for damaged, deteriorating or previously used materials. Such materials shall not be used and shall be removed from the worksite and disposed of properly.

(5) Polyethylene sheeting for walls in the regulated area shall be a minimum of 4-mils. For floors and all other uses, sheeting of at least 6-mils shall be used in widths selected to minimize the frequency of joints. Fire retardant poly shall be used throughout.

(6) The method of attaching polyethylene sheeting shall be agreed upon in advance by the Contractor and the VA and selected to minimize damage to equipment and surfaces. Method of attachment may include any combination of moisture resistant duct tape, poly tape, furring strips, spray glue, staples, nails, screws, lumber and plywood for enclosures or approved equivalent procedures capable of sealing polyethylene to dissimilar finished or unfinished surfaces under both wet and dry conditions.

(7) Polyethylene sheeting utilized for the PDF shall be opaque white or black in color, 6-mil fire retardant poly.

(8) Installation and plumbing hardware, showers, hoses, drain pans, sump pumps and waste water filtration system shall be provided by the Contractor.

(9) An adequate number of HEPA vacuums, scrapers, sprayers, nylon

brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the abatement project. All electrically operated hand tools, equipment, electric cords shall be connected to GFCI protection.

(10) Special protection for objects in the regulated area shall be detailed (e.g., plywood over carpeting or hardwood floors to prevent damage from scaffolds, water and falling material).

(11) Disposal bags – 2-layers of 6-mil poly for asbestos waste shall be pre-printed with labels, markings and address as required by OSHA, EPA and DOT regulations.

(12) The VA shall be provided an advance copy of the Safety Data Sheets (SDS) as required for all hazardous chemicals under OSHA 29 CFR 1910.1200 - Hazard Communication in the pre-project submittal. Chlorinated compounds shall not be used with any spray adhesive, mastic remover or other product. Appropriate encapsulant(s) shall be provided.

(13) OSHA DANGER demarcation signs, as many and as required by OSHA 29 CFR 1926.1101(k)(7) shall be provided and placed by the Competent Person. All other posters and notices required by Federal, State and Local regulations shall be posted in the Clean Room.

(14) Adequate and appropriate PPE for the project and number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d)

2.2 CONTAINMENT BARRIERS AND COVERINGS IN THE REGULATED AREA

a. General: Using critical barriers, seal off the perimeter to the regulated area to completely isolate the regulated area from adjacent spaces. All horizontal surfaces in the regulated area must be covered with 2-layers of 6-mil fire retardant poly to prevent contamination and to facilitate clean-up. Should adjacent areas become contaminated, immediately stop work and clean up the contamination at no additional cost to the Government. Provide firestopping and identify all fire barrier penetrations due to abatement work as specified in Section 2.2.g; FIRESTOPPING.

b. Preparation Prior to Sealing the Regulated Area: Place all tools, scaffolding, materials and equipment needed for working in the regulated area prior to erecting any plastic sheeting. Remove all uncontaminated removable furniture, equipment and/or supplies from the regulated area before commencing work, or completely cover with 2-layers of 6-mil fire retardant poly sheeting and secure with duct tape. Lock out and tag out any HVAC systems in the regulated area.

c. Controlling Access to the Regulated Area: Access to the regulated area is allowed only through the personnel decontamination facility (PDF), if required. All other means of access shall be eliminated and OSHA Danger demarcation signs posted as required by OSHA. If the

regulated area is adjacent to or within view of an occupied area, provide a visual barrier of 6-mil opaque fire retardant poly sheeting to prevent building occupant observation. If the adjacent area is accessible to the public, the barrier must be solid.

d. Critical Barriers: Completely separate any openings into the regulated area from adjacent areas using fire retardant poly at least 6-mils thick and duct tape. Individually seal with 2-layers of independently installed 6-mil poly and duct tape all HVAC openings into the regulated area. Individually seal all lighting fixtures, clocks, doors, windows, convectors, speakers, or any other objects in the regulated area. Heat must be shut off any objects covered with poly.

e. Secondary Barriers: A loose layer of 6-mil fire retardant poly shall be used as a drop cloth to protect the floor/horizontal surfaces from debris generated during the Glovebag abatement. This layer shall be replaced as needed during the work.

f. Extension of the Regulated Area: If the enclosure of the regulated area is breached in any way that could allow contamination to occur, the affected area shall be included in the regulated area and constructed as per this section. If the affected area cannot be added to the regulated area, decontamination measures must be started immediately and continue until air monitoring indicates background levels are met.

g. Firestopping:

(1) Through penetrations caused by cables, cable trays, pipes, sleeves must be firestopped with a fire-rated firestop system providing an air tight seal.

(2) Firestop materials that are not equal to the wall or ceiling penetrated shall be brought to the attention of the VA Representative. The Contractor shall list all areas of penetration, the type of sealant used, and whether or not the location is fire rated. Any discovery of penetrations during abatement shall be brought to the attention of the VA Representative immediately. All walls, floors and ceilings are considered fire rated unless otherwise determined by the VA Representative or Fire Marshall.

(3) Any visible openings whether or not caused by a penetration shall be reported by the Contractor to the VA Representative for a sealant system determination. Firestops shall meet ASTM E814 and UL 1479 requirements for the opening size, penetrant, and fire rating needed

2.3 MONITORING, INSPECTION AND TESTING

a. General:

(1) Perform throughout abatement work monitoring, inspection and testing inside and around the regulated area in accordance with the OSHA requirements and these specifications. OSHA requires that the employee exposure to asbestos must not exceed 0.1 fibers per cubic centimeter (f/cc) of air, averaged over an 8-hour work shift. The CPIH/CIH is responsible for and shall inspect and

oversee the performance of the Contractor IH Technician. The IH Technician shall continuously inspect and monitor conditions inside the regulated area to ensure compliance with these specifications. In addition, the CPIH/CIH shall personally manage air sample collection, analysis, and evaluation for personnel, regulated area, and adjacent area samples inside the building, but outside the regulated area. Inside the building, but outside the regulated area air samples shall be collected at the boundary of the regulated area and/or Clean Room of the PDF, at the approximate location of HEPA exhaust discharge (if used), and at a minimum of three (3) locations in areas immediately outside the regulated work area to satisfy this specification. Additional inspection and testing requirements are also indicated in other parts of this specification.

(2) The VA will employ an independent industrial hygienist (VPIH/CIH) consultant and/or use its own IH to perform various services on behalf of the VA. The VPIH/CIH will perform the necessary monitoring, inspection, testing, and other support services to ensure that VA patients, employees, and visitors will not be adversely affected by the abatement work, and that the abatement work proceeds in accordance with these specifications, that the abated areas or abated buildings have been successfully decontaminated. The work of the VPIH/CIH consultant in no way relieves the Contractor from their responsibility to perform the work in accordance with contract/specification requirements, to perform continuous inspection, monitoring and testing for the safety of their employees, and to perform other such services as specified. The cost of the VPIH/CIH and their services will be borne by the VA except for any repeat of final inspection and testing that may be required due to unsatisfactory initial results. Any repeated final inspections and/or testing, if required, will be paid for by the Contractor.

(3) If fibers counted by the VPIH/CIH during abatement work, either inside or outside the regulated area, utilizing the NIOSH 7400 air monitoring method, exceed the specified respective limits, the Contractor shall stop work. The Contractor may request confirmation of the results by analysis of the samples by TEM. Request must be in writing and submitted to the VA's Representative. Cost for the confirmation of results will be borne by the Contractor for both the collection and analysis of samples and for the time delay that may/does result for this confirmation. Confirmation sampling and analysis will be the responsibility of the CPIH/CIH with review and approval of the VPIH/CIH. An agreement between the CPIH/CIH and the VPIH/CIH shall be reached on the exact details of the confirmation effort, in writing, including such things as the number of samples, location, collection, quality control on-site, analytical laboratory, interpretation of results and any follow-up actions. This written agreement shall be co-signed by the IH's and delivered to the VA's Representative.

b. Scope of Services of the VPIH/CIH Consultant:

(1) The purpose of the work of the VPIH/CIH is to: Assure quality; resolve problems; and prevent the spread of contamination beyond the regulated area. In addition, their work includes performing the final inspection and testing to determine whether the

regulated area or building has been adequately decontaminated. All air monitoring is to be done utilizing PCM/TEM. The VPIH/CIH will perform the following tasks:

(a) Task 1: Establish background levels before abatement begins by collecting background samples. Retain samples for possible TEM analysis.

(b) Task 2: Perform representative air monitoring, inspection, and testing outside the regulated area during actual abatement work to detect any faults in the regulated area isolation and any adverse impact on the surroundings from regulated area activities.

(c) Task 3: Perform unannounced visits to spot check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area and all aspects of the operation except personnel monitoring.

(d) Task 4: Provide support to the VA Representative such as evaluation of submittals from the Contractor, resolution of unforeseen developments, etc.

(e) Task 5: Perform, in the presence of the VA Representative, final inspection and testing of a decontaminated regulated area or building at the conclusion of the abatement and clean-up work to certify compliance with all regulations and the VA requirements/specifications.

(f) Task 6: Issue certificate of decontamination for each regulated area or building and project report.

(2) All data, inspection results and testing results generated by the VPIH/CIH will be available to the Contractor for information and consideration. The Contractor shall cooperate with and support the VPIH/CIH for efficient and smooth performance of their work.

(3) The monitoring and inspection results of the VPIH/CIH will be used by the VA to issue any Stop Removal orders to the Contractor during abatement work and to accept or reject a regulated area or building as decontaminated.

(4) Monitoring, Inspection and Testing by Abatement Contractor CPIH/CIH: The Contractor’s CPIH/CIH is responsible for managing all monitoring, inspections, and testing required by these specifications, as well as any and all regulatory requirements adopted by these specifications. The CPIH/CIH is responsible for the continuous monitoring of all subsystems and procedures which could affect the health and safety of the Contractor’s personnel. Safety and health conditions and the provision of those conditions inside the regulated area for all persons entering the regulated area is the exclusive responsibility of the Contractor/Competent Person. The person performing the personnel and area air monitoring inside the regulated area shall be an IH Technician, who shall be trained and shall have specialized field experience in sampling and analysis. The IH Technician shall have successfully completed a NIOSH 582 Course or equivalent and provide documentation. The IH Technician shall participate in the AIHA Asbestos Analysis Registry or participate in the Proficiency

Analytical Testing program of AIHA for fiber counting quality control assurance. The IH Technician shall also be an accredited EPA AHERA/State Contractor/Supervisor and Building Inspector. The IH Technician shall have participated in five abatement projects collecting personal and area samples and have experience in substantially similar projects in size and scope. The analytical laboratory used by the Contractor to analyze the samples shall be AIHA accredited for asbestos PAT and approved by the VA prior to start of the project. A daily log shall be maintained by the CPIH/CIH or IH Technician, documenting all OSHA requirements for personal and area air monitoring for asbestos in 29 CFR 1926.1101 (f), (g) and Appendix A. This log shall be made available to the VA Representative and the VPIH/CIH upon request. The log will contain, at a minimum, information on personnel or area samples, other persons represented by the sample, the date of sample collection, start and stop times for sampling, sample volume, flow rate, and fibers/cc. The CPIH/CIH shall collect and analyze samples for each representative job being done in the regulated area, i.e., removal, wetting, clean-up, and load-out. No fewer than two (2) personal air samples or 25% of representative workforce per shift shall be collected, whichever is greater, in the regulated area; a minimum of three (3) area air samples at locations inside the building but immediately outside the regulated work area; one (1) area air sample shall be collected daily at the boundary of the regulated area and/or Clean Room of the PDF; and one (1) area air sample shall be collected daily at the approximate location of HEPA exhaust discharge, if used. In addition to the continuous monitoring required, the CPIH/CIH will perform inspection and testing at the final stages of abatement for each regulated area as specified in the CPIH/CIH responsibilities. Additionally, the CPIH/CIH will monitor and record pressure readings within the containment daily with a minimum of two readings at the beginning and at the end of a shift, and submit the data in the daily report.

2.4 ASBESTOS HAZARD ABATEMENT PLAN

a. The Contractor shall have established Asbestos Hazard Abatement Plan (AHAP) in printed form and loose leaf folder consisting of simplified text, diagrams, sketches, and pictures that establish and explain clearly the ways and procedures to be followed during all phases of the work by the Contractor's personnel. The AHAP must be modified as needed to address specific requirements of the project. The AHAP shall be submitted for review and approval prior to the start of any abatement work. The minimum topics and areas to be covered by the AHAP(s) are:

(1) Minimum Personnel Qualifications

(2) Contingency Plans and Arrangements

(3) Security and Safety Procedures

(4) Respiratory Protection/Personal Protective Equipment Program and Training

(5) Medical Surveillance Program and Recordkeeping

(6) Regulated Area Requirements for Glovebag Abatement

(7) Decontamination Facilities and Entry/Exit Procedures (PDF and W/EDF) or approved equivalent.

(8) Monitoring, Inspections, and Testing

(9) Removal Procedures for Piping ACM Using the Glovebag Method

(10) Disposal of ACM waste

(11) Regulated Area Decontamination/Clean-up

(12) Regulated Area Visual and Air Clearance

(13) Project Completion/Closeout

2.5 ENCAPSULANTS

a. Types of Encapsulants:

(1) The following four types of encapsulants must comply with performance requirements as stated in paragraph 2.6.2:

(a) Removal encapsulant - used as a wetting agent to remove ACM.

(b) Bridging encapsulant - provides a tough, durable coating on ACM.

(c) Penetrating encapsulant - penetrates/encapsulates ACM at least (1/2 inch).

(d) Lockdown encapsulant - seals microscopic fibers on surfaces after ACM removal.

b. Performance Requirements:

(1) Encapsulants shall meet the latest requirements of EPA; shall not contain toxic or hazardous substances; or solvents; and shall comply with the following performance requirements:

(a) General Requirements for all Encapsulants:

1. ASTM E84: Flame spread of 25; smoke emission of 50.

2. University of Pittsburgh Protocol: Combustion Toxicity; zero mortality.

3. ASTM C732: Accelerated Aging Test; Life Expectancy - 20 years.

4. ASTM E96/E96M: Permeability - minimum of 0.4 perms.

(b) Bridging/Penetrating Encapsulants:

1. ASTM E736: Cohesion/Adhesion Test - 24 kPa (50 pounds/square foot).

2. ASTM E119: Fire Resistance - 3 hours (Classified by UL for use on fibrous/cementitious fireproofing).

3. ASTM D2794: Gardner Impact Test; Impact Resistance - minimum 11.5 kg-mm (43 in/lb).

4. ASTM D522: Mandrel Bend Test; Flexibility - no rupture or cracking

(c) Lockdown Encapsulants:

1. ASTM E119: Fire resistance - 3 hours (tested with fireproofing over encapsulant applied directly to steel member).

2. ASTM E736: Bond Strength - 48 kPa (100 pounds/square foot) (test compatibility with cementitious and fibrous fireproofing).

3. In certain situations, encapsulants may have to be applied to hot pipes/equipment. The encapsulant must be able to withstand high temperatures without cracking or creating any noxious gaseous or vapors during or after application.

2.6 CERTIFICATES OF COMPLIANCE

a. The Contractor shall submit to the VA Representative certification from the manufacturer indicating compliance with performance requirements for encapsulants when applied according to manufacturer recommendations.

b. Recyclable Protective Clothing: If recyclable clothing is provided, all requirements of EPA, DOT and OSHA shall be met.

PART 3 EXECUTION

3.1 REGULATED AREA PREPARATIONS

a. Site Security:

(1) Regulated area access is to be restricted only to authorized, trained/accredited and protected personnel. These may include the Contractor's employees, employees of Subcontractors, VA employees and representatives, State and Local inspectors, and any other designated individuals. A list of authorized personnel shall be established prior to commencing the project and shall be posted in the clean room of the decontamination unit or in a designated area located immediately outside of the regulated area established for Glovebag removal activities.

(2) Entry into the regulated area by unauthorized individuals shall be reported immediately to the Competent Person by anyone observing the entry. The Competent Person shall immediately require any unauthorized person to leave the regulated area and then notify the VA Contracting Officer or VA Representative using the most expeditious means.

(3) A log book shall be maintained in the clean room of the decontamination unit or in a designated area located immediately outside of the regulated area established for Glovebag removal activities. Anyone who enters the regulated area must record their name, affiliation, time in, and time out for each entry.

(4) Access to the regulated area shall be through a single decontamination unit or in an area designated by the Competent Person for Glovebag removal activities. All other access (doors, windows, hallways, etc.) shall be sealed or locked to prevent entry to or exit from the regulated area. The only exceptions for this requirement are the waste/equipment load-out area which shall be sealed except during the removal of containerized asbestos waste from the regulated area, and emergency exits. Emergency exits shall not be locked from the inside; however, they shall be sealed with poly sheeting and taped until needed. In any situation where exposure to high temperatures which may result in a flame hazard, fire retardant poly sheeting must be used.

(5) The Contractor's Competent Person shall control site security during abatement operations in order to isolate work in progress and protect adjacent personnel. A 24 hour security system shall be provided at the entrance to the regulated area to assure that all entrants are logged in/out and that only authorized personnel are allowed entrance.

(6) The Contractor will have the VA's assistance in notifying adjacent personnel of the presence, location and quantity of ACM in the regulated area and enforcement of restricted access by the VA's employees.

(7) The regulated area shall be locked during non-working hours and secured by VA Representative or Competent Person. The VA Police shall be informed of asbestos abatement regulated areas to provide security checks during facility rounds and emergency response.

b. OSHA Danger Signs: Post OSHA DANGER signs meeting the specifications of OSHA 29 CFR 1926.1101 at any location and approaches to the regulated area where airborne concentrations of asbestos may exceed ambient background levels. Signs shall be posted at a distance sufficiently far enough away from the regulated area to permit any personnel to read the sign and take the necessary measures to avoid exposure. Additional signs will be posted following construction of the regulated area enclosure.

c. Shut Down - Lock Out Electrical: Shut down and lock out/tag out electric power to the regulated area. Provide temporary power and lighting. Insure safe installation including GFCI of temporary power sources and equipment by compliance with all applicable electrical code requirements and OSHA requirements for temporary electrical systems. Electricity shall be provided by the VA.

d. Shut Down - Lock Out HVAC: Shut down and lock out/tag out heating, cooling, and air conditioning system (HVAC) components that are in, supply or pass through the regulated area. Investigate the regulated area and agree on pre-abatement condition with the VA's Representative. Seal all intake and exhaust vents in the regulated area with duct tape and 2-ayers of independently installed 6-mil poly. Also, seal any seams in system components that pass through the regulated area. Remove all contaminated HVAC system filters and place in labeled 6-mil poly disposal bags for disposal as asbestos waste.

e. Containment Barriers and Coverings for the Regulated Area:

(1) General: Seal off any openings at the perimeter of the regulated area with critical barriers to completely isolate the regulated area and to contain all airborne asbestos contamination created by the abatement activities. Should the adjacent area past the regulated area become contaminated due to improper work activities, the Contractor shall suspend work inside the regulated area, continue wetting, and clean the adjacent areas in accordance with procedures described in these specifications. Any and all costs associated with the adjacent area cleanup shall not be borne by the VA.

(2) Preparation Prior to Sealing Off: Place all materials, equipment and supplies necessary to isolate the regulated area inside the regulated area. Remove all movable material/equipment as described above and secure all unmovable material/equipment as described above. Properly secured material/ equipment shall be considered to be outside the regulated area.

(3) Controlling Access to the Regulated Area: Access to the regulated area is allowed only through the personnel decontamination facility (PDF) or in an area designated by the Competent Person for Glovebag removal activities. All other means of access shall be eliminated and OSHA DANGER demarcation signs posted as required by OSHA. If the regulated area is adjacent to, or within view of an occupied area, provide a visual barrier of 6-mil opaque fire retardant poly to prevent building occupant observation. If the adjacent area is accessible to the public, the barrier must be solid and capable of withstanding the negative pressure.

(4) Critical Barriers: The regulated area must be completely separated from the adjacent area(s) and the outside by at least 2-layers of independently installed 6-mil fire retardant poly and duct tape/spray adhesive. Individually seal all supply and exhaust ventilation openings, lighting fixtures, clocks, doorways, windows, convectors, speakers, and other openings into the regulated area with 2-layers of 6-mil fire retardant poly, and taped securely in place with duct tape/spray adhesive. Critical barriers must remain in place until all work and clearances have been completed. Light fixtures shall not be operational during abatement. Auxiliary lighting shall be provided. If needed, provide plywood squares 6 inches x 6 inches x 3/8 inch () or approved equivalent, held in place with 6d smooth masonry/galvanized nail or approved equivalent driven through the center of the plywood square and duct tape on the poly so as to clamp the poly to the wall/surface. Locate plywood squares at each end, corner, and 4 feet (1200mm) maximum on centers

(5) Extension of the Regulated Area: If the regulated area barrier is breached in any manner that could allow the passage of asbestos fibers or debris, the Competent Person shall immediately stop work, continue wetting, and proceed to extend the regulated area to enclose the affected area as per procedures described in this specification. If the affected area cannot be enclosed, decontamination measures and cleanup shall start immediately. All personnel shall be isolated from the affected area until decontamination/cleanup is completed as verified by visual inspection and air monitoring. Air monitoring at completion must indicate background levels.

(6) Floor Barriers: All floors within 10 feet of Glovebag work shall be covered with 2-layers of 6-mil fire retardant poly. If no breach occurs during the Glovebag abatement operation, these layers of 6 mil fire retardant poly may be reused

f. Sanitary Facilities: The Contractor shall provide sanitary facilities for abatement personnel and maintain them in a clean and sanitary condition throughout the abatement project.

g. Pre-Cleaning:

(1) Pre-Cleaning Movable Objects:

(a) The VA will provide water for abatement purposes. The Contractor shall connect to the existing VA system. The service to the shower(s), if used, shall be supplied with backflow prevention.

(b) Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area. PPE must be donned by all workers performing pre-cleaning activities. After items have been pre-cleaned and decontaminated, they may be removed from the work area for storage until the completion of abatement in the work area.

(c) Pre-clean all movable objects within the regulated area using a HEPA filtered vacuum and/or wet cleaning methods as appropriate. After cleaning, these objects shall be removed from the regulated area and carefully stored in an uncontaminated location.

(2) Pre-Cleaning Fixed Objects:

(a) Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area.

(b) Pre-clean all fixed objects in the regulated area using HEPA filtered vacuums and wet cleaning techniques as appropriate. Careful attention must be paid to machinery behind grills or gratings where access may be difficult but contamination may be significant. Also, pay particular attention to wall, floor and ceiling penetration behind fixed items. After pre-cleaning, enclose fixed objects with 2-layers of 6-mil poly and seal securely in place with duct tape. Objects (e.g., permanent fixtures, shelves, electronic equipment, laboratory tables, sprinklers, alarm systems, closed circuit TV equipment and computer cables) which must remain in the regulated area and that require special ventilation or enclosure requirements should be designated here along with specified means of protection. Contact the manufacturer for special protection requirements.

(3) Pre-Cleaning Surfaces in the Regulated Area:

(a) Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area.

(b) Pre-clean all surfaces in the regulated area using HEPA filtered vacuums and wet cleaning methods as appropriate. Do not use any methods that would raise dust such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

h. Pre-Abatement Activities:

(1) Pre-Abatement Meeting: The VA Representative, upon receipt, review, and substantial approval of all pre-abatement submittals and verification by the CPIH/CIH that all materials and equipment required for the project are on the site, will arrange for a pre-abatement meeting between the Contractor, the CPIH/CIH, Competent Person, the VA Representatives, and the VPIH/CIH. The purpose of the meeting is to discuss any aspect of the submittals needing clarification or amplification and to discuss any aspect of the project execution and the sequence of the operation. The Contractor shall be prepared to provide any supplemental information or documentation to the VA's Representative regarding any submittals, materials or equipment. Upon satisfactory resolution of any outstanding issues, the VA's representative will issue a written order to proceed to the Contractor. No abatement work of any kind described in the following provisions shall be initiated prior to the VA written order to proceed.

(2) Pre-Abatement Inspections and Preparations:

(a) Before any work begins on the construction of the regulated area, the Contractor will:

1. Conduct a space-by-space inspection with an authorized VA Representative and prepare a written inventory of all existing damage in those spaces where asbestos abatement will occur. Still or video photography may be used to supplement the written damage inventory. Document will be signed and certified as accurate by both parties.

2. The VA Representative, the Contractor, and the VPIH/CIH must be aware of AEQA 10-95 indicating the failure to identify asbestos in the areas listed as well as common issues when preparing specifications and contract documents. This is especially critical when demolition is planned, because AHERA surveys are non-destructive, and ACM may remain undetected. A NESHAP (destructive) ACM inspection should be conducted on all building structures that will be demolished. Ensure the following areas are inspected on the project: Lay-in ceilings concealing ACM; ACM behind walls/windows from previous renovations; inside utility chases/walls; transite piping/ductwork/sheets; behind radiators; lab fume hoods; transite lab countertops; roofing materials; below window sills; water/sewer lines; electrical conduit coverings; crawl spaces(previous abatement contamination); flooring/mastic covered by carpeting/new flooring; exterior insulated wall panels; on underground fuel tanks; and steam line trench coverings.

3. Ensure that all furniture, machinery, equipment, curtains, drapes, blinds, and other movable objects required to be removed from the regulated area have been cleaned and removed or properly

protected from contamination.

4. Inspect existing firestopping in the regulated area. Correct as needed.

(3) Pre-Abatement Construction and Operations:

(a) Perform all preparatory work for the first regulated area in accordance with the approved work schedule and with this specification.

(b) Upon completion of all preparatory work, the CPIH/CIH will inspect the work and systems and will notify the VA's Representative when the work is completed in accordance with this specification. The VA's Representative may inspect the regulated area and the systems with the VPIH/CIH and may require that upon satisfactory inspection, the Contractor's employees perform all major aspects of the approved AHAP(s), especially worker protection, respiratory systems, contingency plans, decontamination procedures, and monitoring to demonstrate satisfactory operation.

(c) The CPIH/CIH shall document the pre-abatement activities described above and deliver a copy to the VA's Representative.

(d) Upon satisfactory inspection of the installation of and operation of systems the VA's Representative will notify the Contractor in writing to proceed with the asbestos abatement work in accordance with this specification.

3.2 REMOVAL OF PIPING ACM

a. Wetting Materials:

(1) Use amended water for the wetting of ACM prior to removal. The Competent Person shall assure that the wetting of ACM meets the definition of "adequately wet" in the EPA NESHAP regulation and OSHA’s “wet methods” for the duration of the project. A removal encapsulant may be used instead of amended water with written approval of the VA's Representative.

(2) Amended Water: Provide water to which a surfactant has been added to wet the ACM and reduce the potential for fiber release during disturbance of ACM. The mixture must be equal to or greater than the wetting provided by water amended by a surfactant consisting of one ounce of 50 percent polyoxyethylene ester and 50 percent polyoxyethylene ether mixed with 5 gallons (19L) of water.

(3) Removal Encapsulant: Provide a penetrating encapsulant designed specifically for the removal of ACM. The material must, when used, result in adequate wetting of the ACM and retard fiber release during disturbance equal to or greater than the amended water described above in B.

b. Secondary Barrier and Walkways: Install as a drop cloth a 6-mil poly sheet at the beginning of each work shift where removal is to be done during that shift. Secure the drop cloth (6-mil poly sheet) with duct tape or approved equivalent to prevent it from moving or debris from getting behind it. Remove the drop cloth (6-mil poly sheet) at the

end of the shift or as work in the area is completed. Keep residue on the drop cloth (6-mil poly sheet) wetted. When removing, fold inward to prevent spillage and place in a disposal bag.

c. Wet Removal of ACM: Using acceptable Glovebag procedures, adequately and thoroughly wet the ACM to be removed prior to removal with amended water or when authorized by VA, removal encapsulant to reduce/prevent fiber release to the air. Adequate time must be allowed for the amended water or removal encapsulant to saturate the ACM. Abatement personnel must not disturb dry ACM. Use a fine spray of amended water or removal encapsulant. Saturate the material sufficiently to wet to the substrate without causing excessive dripping. The material must be sprayed repeatedly/continuously during the removal process in order to maintain adequately wet conditions. Removal encapsulants must be applied in accordance with the manufacturer's written instructions. Perforate or carefully separate, using wet methods, an outer covering that is painted or jacketed in order to allow penetration and wetting of the material. Where necessary, carefully remove covering while wetting to minimize fiber release.

3.3 GLOVEBAG REMOVAL PROCEDURES

a. General: All applicable OSHA requirements and Glovebag manufacturer’s recommendations shall be met during Glovebag removal operations. In cases where live steam lines are present, the lines must be shut down prior to any work being performed on the system. No abatement work shall be conducted on live, pressurized steam lines. The Contractor may choose to use a High Temperature Glovebag in which a temperature rating ranges from 300 degrees F to 700 degrees F on steam lines that have recently been shut down and remain at high temperature for some time. In the case where a Glovebag is not feasible, the Contractor will need to build a full negative pressure containment of sufficient size or work within a negative pressure mini-enclosure and follow all regulations as it pertains to removal. The Contractor shall provide enough HEPA negative air machines to continuously maintain a negative pressure differential of -0.02 inch water column gauge (WCG) inside the regulated work area relative to adjacent non-work building areas. OSHA 29 CFR 1926.1101 (g)(5)(i)(A)(2) also requires at least four (4) air changes per hour. Contractor shall increase air changes per hour as necessary to maintain volatile organic compounds below the applicable OSHA PEL. Contractor shall protect pipe insulation from being disturbed on either side of the Glovebag removal operations with a “candy stripe” layer of 6-mil poly sheet and duct tape, if Glove bag removal activities cause the piping to dislodge ACM during performance of their work.

(1) Mix the surfactant with water in the garden sprayer, following the manufacturer's directions.

(2) Have each employee put on a HEPA filtered respirator approved for asbestos and check the fit using the positive/negative fit check.

(3) Have each employee put on a disposable full-body suit. Remember, the hood goes over the respirator straps.

(4) Check closely the integrity of the glove bag to be used. Check all seams, gloves, sleeves, and glove openings. OSHA requires the bottom of the bag to be seamless.

(5) Check the pipe where the work will be performed. If it is damaged (broken lagging, hanging, etc.), wrap the entire length of the pipe in poly sheeting and "candy stripe" it with duct tape.

(6) Attach Glovebag with required tools per manufacturer’s instructions.

(7) Using the smoke tube and aspirator bulb, test 10 percent of Glovebags by placing the tube into the water porthole (two-inch opening to glove bag), and fill the bag with smoke and squeeze it. If leaks are found, they shall be taped closed using duct tape and the bag shall be retested with smoke.

(8) Insert the wand from the water sprayer through the water porthole.

(9) Insert the hose end from a HEPA vacuum into the upper portion of the glove bag.

(10) Wet and remove the pipe insulation.

(11) If the section of pipe is covered with an aluminum jacket, remove it first using the wire cutters to cut any bands and then use tin snips to remove the aluminum. It is important to fold the sharp edges in to prevent cutting the bag when placing it in the bottom.

(12) When the work is complete, spray the upper portion of the bag and move all residue into the bottom of the bag with the other waste material. Be very thorough. Use adequate water.

(13) Put all tools, after washing them off in the bag, in one of the sleeves of glove bag and turn it inside out, drawing it outside of the bag. Twist the sleeve tightly several times to seal it and tape it several tight turns with duct tape. Cut through the middle of the duct tape and remove the sleeve. Put the sleeve in the next glove bag or put it in a bucket of water to decontaminate the tools after cutting the sleeve open.

(14) Turn on the HEPA vacuum and collapse the bag completely. Remove the vacuum nozzle, seal the hole with duct tape, twist the bag tightly several times in the middle, and tape it to keep the material in the bottom during removal of the glove bag from the pipe.

(15) Slip a disposal bag over the glove bag (still attached to the pipe). Remove the tape securing the ends, and slit open the top of the glove bag and carefully fold it down into the disposal bag. Double bag and gooseneck waste materials.

b. Negative Pressure Glovebag Procedure:

(1) In addition to the above requirements, the HEPA vacuum shall be run continuously during the Glovebag procedure until completion at which time the Glovebag will be collapsed by the HEPA vacuum prior to removal from the pipe/component.

(2) The HEPA vacuum shall be attached and operated as needed to prevent collapse of the Glovebag during the removal process.

3.4 LOCKDOWN ENCAPSULATION

a. General: Lockdown encapsulation is an integral part of the ACM removal. At the conclusion of ACM removal and before removal of the primary barriers, all piping surfaces shall be encapsulated with a bridging encapsulant.

b. Sealing Exposed Edges: Seal edges of ACM exposed by removal work with two coats of encapsulant. Prior to sealing, permit the exposed edges to dry completely to permit penetration of the encapsulant.

3.5 DISPOSAL OF ACM WASTE MATERIALS

a. General: Dispose of waste ACM and debris which is packaged in accordance with these specifications, OSHA, EPA and DOT. The landfill requirements for packaging must also be met. Transport will be in compliance with 49 CFR 171 - 180 regulations. Disposal shall be done at an approved landfill. Disposal of non-friable ACM shall be done in accordance with applicable regulations.

b. Procedures:

(1) The VA must be notified at least 24 hours in advance of any waste removed from the containment

(2) Asbestos waste shall be packaged and moved through the W/EDF into a covered transport container in accordance with procedures in this specification. Waste shall be double-bagged and wetted with amended water prior to disposal. Wetted waste can be very heavy. Bags shall not be overfilled. Bags shall be securely sealed to prevent accidental opening and/or leakage. The top shall be tightly twisted and goose necked prior to tightly sealing with at least three wraps of duct tape. Ensure that unauthorized persons do not have access to the waste material once it is outside the regulated area. All transport containers must be covered at all times when not in use. OSHA Danger signs must be displayed during loading and unloading. Material shall not be transported in open vehicles. If drums are used for packaging, the drums shall be labeled properly and shall not be re-used.

(3) Waste Load Out: Waste load out shall be done in accordance with the procedures in W/EDF Decontamination Procedures. Sealed waste bags shall be decontaminated on exterior surfaces by wet cleaning and HEPA vacuuming before being placed in the second waste bag and sealed, which then must also be wet wiped and HEPA vacuumed.

(4) Asbestos waste with sharp edged components, i.e., nails, screws, lath, strapping, tin sheeting, jacketing, metal mesh, etc., which might tear poly bags shall be wrapped securely in burlap before packaging and, if needed, use a poly lined fiber drum as the second container, prior to disposal.

3.6 PROJECT DECONTAMINATION

a. General:

(1) The entire work related to project decontamination shall be performed under the close supervision and monitoring of the CPIH/CIH.

(2) If the asbestos abatement work is in an area which was contaminated prior to the start of abatement, the decontamination will be done by cleaning the primary poly barrier prior to its removal and cleanings of the surfaces of the regulated area after the primary barrier removal.

(3) If the asbestos abatement work is in an area which was uncontaminated prior to the start of abatement, the decontamination will be done by cleaning the primary poly barrier prior to its removal, thus preventing contamination of the building when the regulated area critical barriers are removed.

b. Regulated Area Clearance: Air testing and other requirements which must be met before release of the Contractor and re-occupancy of the regulated area space are specified in Final Testing Procedures.

c. Work Description: Decontamination includes the clearance air testing in the regulated area and the decontamination and removal of the enclosures/facilities installed prior to the abatement work including primary/critical barriers, PDF and W/EDF facilities, and negative pressure systems.

d. Pre-Decontamination Conditions:

(1) Before decontamination starts, all ACM waste from the regulated area shall be removed, all waste collected and removed, and the secondary barrier of poly removed and disposed of along with any gross debris generated by the work.

(2) At the start of decontamination, the following shall be in place.

(a) Critical barriers over all openings consisting of two layers of 6-mil poly which is the sole barrier between the regulated area and the rest of the building or outside.

(b) Decontamination facilities, if required for personnel and equipment in operating condition.

e. First Cleaning: Carry out a first cleaning of all surfaces of the regulated area including items of remaining poly sheeting, tools, scaffolding, ladders/staging by wet methods and HEPA vacuuming. Do not use dry dusting/sweeping/air blowing methods. Use each surface of a wetted cleaning cloth one time only and then dispose of as contaminated waste. Continue this cleaning until there is no visible residue from abated surfaces or poly or other surfaces. Remove all filters in the air handling system and dispose of as ACM waste in accordance with these specifications. The negative pressure system shall remain in operation during this time, if used. Additional cleaning may be needed as determined by the CPIH/VPIH/CIH.

f. Pre-Clearance Inspection and Testing: The CPIH/CIH and VPIH/CIH will perform a thorough and detailed visual inspection at the end of the cleaning to determine whether there is any visible residue in the regulated area. If the visual inspection is acceptable, the CPIH/CIH will perform pre-clearance sampling using aggressive clearance as detailed in 40 CFR 763 Subpart E (AHERA) Appendix A (III)(B)(7)(d). If the sampling results show values below 0.01 f/cc, then the Contractor shall notify the VA's Representative of the results with a

brief report from the CPIH/CIH documenting the inspection and sampling results and a statement verifying that the regulated area is ready for lockdown encapsulation. The VA reserves the right to utilize their own VPIH/CIH to perform a pre-clearance inspection and testing for verification.

g. Lockdown Encapsulation of Abated Surfaces: With the express written permission of the VA's Representative, perform lockdown encapsulation of all surfaces from which asbestos was abated in accordance with the procedures in this specification.

3.7 FINAL VISUAL INSPECTIONS AND AIR CLEARANCE TESTING

a. General: Notify the VA Representative 24 hours in advance for the performance of the final visual inspection and testing. The final visual inspection and testing will be performed by the VPIH/CIH after the final cleaning.

b. Final Visual Inspection: Final visual inspection will include the entire regulated area, the PDF, all poly sheeting, seals over HVAC openings, doorways, windows, and any other openings. If any debris, residue, dust or any other suspect material is detected, the final cleaning shall be repeated at no additional cost to the VA. Dust/material samples may be collected and analyzed at no additional cost to the VA at the discretion of the VPIH/CIH to confirm visual findings. When the regulated area is visually clean the final testing can be done.

c. Final Air Clearance Testing:

(1) After an acceptable final visual inspection by the VPIH/CIH and VA Representative, the VPIH/CIH will perform the final clearance testing. Air samples will be collected and analyzed in accordance with procedures for AHERA in this specification. If work is less than 260 lf/160 sf/35 cf, 5 PCM samples shall be collected for clearance and a minimum of two field blank. If work is equal to or more than 260 lf/160 sf/35 cf, AHERA TEM (EPA 560/5-89-001) sampling shall be performed for clearance. TEM analysis shall be done in accordance with procedures for EPA AHERA presented in this specification. If the release criteria are not met, the Contractor shall repeat the final cleaning and continue decontamination procedures until clearance is achieved. All additional inspection and testing costs will be borne by the Contractor.

(2) If release criteria are met, proceed to perform the abatement closeout and to issue the certificate of completion in accordance with these specifications.

d. Final Air Clearance Procedures:

(1) Contractor's Release Criteria: Work in a regulated area is complete when the regulated area is visually clean and airborne fiber levels have been reduced to or below 0.01 f/cc as measured by the AHERA PCM protocol, and < 70 AHERA asbestos structures per square millimeter (s/mm2) by AHERA TEM (EPA 560/5-89-001). No averaging of results will be used for this project. All five (5) TEM samples inside the regulated area shall be at or below 70 asbestos s/mm2 to satisfy the project final clearance criteria.

(2) Air Monitoring and Final Clearance Sampling: To determine if the elevated airborne fiber counts encountered during abatement operations have been reduced to the specified level, the VPIH/CIH will secure samples and analyze them according to the following procedures:

(a) Fibers Counted: "Fibers" referred to in this section shall be either all fibers regardless of composition as counted in the NIOSH 7400 PCM method or asbestos fibers counted using the AHERA TEM (EPA 560/5-89-001) method.

(b) Aggressive Sampling: All final air testing samples shall be collected using aggressive sampling techniques except where soil is not encapsulated or enclosed. Samples will be collected on 0.8 micron MCE filters for PCM analysis and 0.45 micron MCE for TEM. A minimum of 3850 Liters of air using calibrated sampling pumps shall be collected for PCM samples and a minimum of 1200 Liters of air using calibrated sampling pumps shall be collected for TEM clearance samples. Before pumps are started, initiate aggressive air mixing sampling as detailed in 40 CFR 763 Subpart E (AHERA) Appendix A (III)(B)(7)(d). Air samples will be collected in areas subject to normal air circulation away from corners, obstructed locations, and locations near windows, doors, or vents. After air sampling pumps have been shut off, circulating fans shall be shut off. The negative pressure system shall continue to operate.

e. Clearance Sampling Using PCM:

(1) The VPIH/CIH will perform clearance samples as indicated by the specification.

(2) The NIOSH 7400 PCM method will be used for clearance sampling with a minimum collection volume of 3850 Liters of air. A minimum of 5 PCM clearance samples shall be collected. All samples must be equal to or less than 0.01 f/cc to clear the regulated area.

f. Clearance Sampling Using TEM:

(1) Clearance requires 13 samples be collected; 5 inside the regulated area; 5 outside the regulated area; and 3 field blanks.

(2) The TEM method will be used for clearance sampling with a minimum collection volume of 1200 Liters of air. A minimum of 13 clearance samples shall be collected. All samples must be equal to or less than 70 AHERA structures per square millimeter (s/mm2) AHERA TEM (EPA 560/5-89-001), no averaging of results for this specific project.

g. Laboratory Testing of PCM Samples: The services of an AIHA accredited laboratory will be employed by the VA to perform analysis for the PCM air samples. The accredited laboratory shall be successfully participating in the AIHA Proficiency Analytical Testing (PAT) program. Samples will be sent daily by the VPIH/CIH so that verbal/faxed reports can be received within 24-36 hours. A complete record, certified by the laboratory, of all air monitoring tests and results will be furnished to the VA's Representative and the Contractor.

h. Laboratory Testing of Tem Samples: Samples shall be sent by the VPIH/CIH to a NIST NVLAP accredited laboratory for analysis by TEM. The laboratory shall be successfully participating in the NIST NVLAP Airborne Asbestos Analysis (TEM) program. Verbal/faxed results from the laboratory shall be available within 24-36 hours after receipt of the samples. A complete record, certified by the laboratory, of all TEM results shall be furnished to the VA's Representative and the Contractor.

3.8 ABATEMENT CLOSEOUT AND CERTIFICATE OF COMPLIANCE

a. Completion of Abatement Work: After thorough decontamination, seal negative air machines with 2-layers of 6-mil poly and duct tape to form a tight seal at the intake/outlet ends before removal from the regulated area. Complete asbestos abatement work upon meeting the regulated area visual and air clearance criteria and fulfilling the following:

(1) Remove all equipment, materials, and debris from the project area.

(2) Package and dispose of all asbestos waste as required. Dispose of waste ACM and debris which is packaged in accordance with these specifications, OSHA, EPA and DOT. The landfill requirements for packaging must also be met. Transport will be in compliance with 49 CFR 171 – 180 regulations.

(3) Repair or replace all interior finishes damaged during the abatement work.

(4) The VA will be notified of any waste removed from the containment prior to 24 hours.

(5) Fulfill other project closeout requirements as specified elsewhere in this specification.

b. Certificate of Completion by Contractor: The CPIH/CIH shall complete and sign the "Certificate of Completion" in accordance with Attachment C (Attachment #1) at the completion of the abatement and decontamination of the regulated area.

c. Work Shifts: All work shall generally be done during administrative hours (8:00 AM to 4:30 PM) Monday - Friday excluding Federal Holidays. Any change in the work schedule must be approved in writing by the VA Representative.

d. Re-Insulation: If required as part of the contract, replace all asbestos containing insulation with suitable non-asbestos material. Provide SDS for all replacement materials. Refer to Section 23 07 11, HVAC, PLUMBING, AND BOILER PLANT INSULATION.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 83 33.13

LEAD-BASED PAINT REMOVAL AND DISPOSAL

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 DEFINITIONS
1.4 REFERENCES
1.5 PRE-REMOVAL MEETINGS
1.6 SUBMITTALS
1.7 SUBMITTAL INSTRUCTIONS
1.8 QUALITY ASSURANCE
1.9 WARRANTY

PART 2 PRODUCTS

2.1 PAINT REMOVAL PRODUCTS
2.2 ACCESSORIES

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 WORK PROCEDURES
3.4 LEAD-BASED PAINT REMOVAL
3.5 SUBSTRATE SURFACE PREPARATION
3.6 FIELD QUALITY CONTROL
3.7 CLEANING AND DISPOSAL

-- End of Section Table of Contents --

SECTION 02 83 33.13
1/21

LEAD-BASED PAINT REMOVAL AND DISPOSAL

PART 1 GENERAL

1.1 SUMMARY

a. Section Includes:

(1) Removing and disposal of lead-based paint (LBP) at interior and exterior locations as indicated in existing conditions report.

1.2 RELATED WORK

a. Section ,02 82 13.13 GLOVEBAG ASBESTOS ABATEMENT: Hazardous Material Abatement.

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1.3 DEFINITIONS

a. Action Level: Employee exposure, without regard to use of respirator, to an airborne lead concentration of 30 micrograms(µ) per cubic meter (m3) of air determined as an 8-hour Time-Weighted Average (TWA). As used in this section, ”30 micrograms per cubic meter of air" refers to OSHA 29 CFR 1926.62 Lead in Construction Action Level (AL).

b. Area Monitoring: Sampling of lead concentrations within lead control area and inside physical boundaries which are representative of airborne lead concentrations which may reach breathing zone of personnel potentially exposed to lead.

c. Breathing Zone: Area within hemisphere, forward of shoulders, with 6 to 9 inches) radius and center at nose or mouth of employee.

d. Certified Industrial Hygienist (CIH): As used in this section, refers to an Industrial Hygienist Certified in the Comprehensive Practice of Industrial Hygiene by the American Board of Industrial Hygiene and Board for Global EHS Credentialing, employed by Contractor.

e. Change Rooms and Shower Facilities: Rooms within designated physical boundary around lead control area equipped with separate storage facilities for clean protective work clothing and equipment and for street clothes which prevents cross contamination.

f. Competent Person: Person capable of identifying lead hazards in work area and authorized by Contractor to take corrective action. Meets the OSHA definition of Competent Person.

g. Decontamination Room: Room for removal of contaminated personal protective equipment (PPE).

h. Eight-Hour Time Weighted Average (TWA): Airborne concentration of lead averaged over 8-hour workday to which an employee is exposed.

i. High Efficiency Particulate Air (HEPA) Filter Equipment: HEPA filtered vacuuming equipment with UL 586 filter system capable of collecting and retaining lead-contaminated paint dust. HEPA filter means 99.97 percent efficient against 0.012 mil) size particles.

j. Lead: Metallic lead, inorganic lead compounds, and organic lead soaps. Excluded from this definition are other organic lead compounds.

k. Lead Control Area: Enclosed area or structure with full containment to prevent spreading lead dust, paint chips, and debris from lead-based paint removal operations. Lead control area is isolated by physical boundaries to prevent unauthorized entry of personnel.

l. Lead Permissible Exposure Limit (PEL): Fifty micrograms per cubic meter of air (50 µg/m3) determined as an 8-hour TWA as determined by 29 CFR 1926.62. When employee is exposed for more than 8-hours per work day, determine PEL by the following formula. PEL micrograms/cubic meter (parts per million) of air = 400/No. of hrs. worked per day.

m. Personnel Monitoring: Sampling of lead concentrations within employee breathing zone to determine 8-hour time weighted average concentration according to 29 CFR 1926.62. Take samples that are representative of the various employee's work tasks.

n. Physical Boundary: Area physically roped or partitioned off around enclosed lead control area to limit unauthorized entry of personnel. As used in this section, "inside boundary" shall mean same as "outside lead control area."

1.4 REFERENCES

a. Comply with references to extent specified in this section.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
29 CFR 1910.134	Respiratory Protection
29 CFR 1910.1200	Hazard Communication
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.59	Hazard Communication
29 CFR 1926.62	Lead
40 CFR 260	Hazardous Waste Management System: General
40 CFR 261	Identification and Listing of Hazardous Waste
40 CFR 262	Standards Applicable to Generators of Hazardous Waste

40 CFR 263	Standards Applicable to Transporters of Hazardous Waste
40 CFR 264	Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 265	Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities
40 CFR 268	Land Disposal Restrictions
49 CFR 172	Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements
49 CFR 178	Specifications for Packagings

UL SOLUTIONS (UL)

UL 586	(2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

1.5 PRE-REMOVAL MEETINGS

a. Conduct pre-removal meeting minimum 30 days before beginning Work of this section.

(1) Required Participants:

(a) Contracting Officer's Representative.

(b) Certified Industrial Hygienist.

(c) Architect/Engineer.

(d) Inspection and Testing Agency.

(e) Contractor.

(f) Paint removal contractor.

(g) Other installers responsible for finishing resulting surfaces.

(2) Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(a) Respiratory protection program.

(b) Hazard communication program.

(c) Hazardous waste management plan.

(d) Safety and health regulation compliance.

(e) Employee training.

(f) Removal schedule.

(g) Removal sequence.

(h) Preparatory work.

(i) Protection before, during, and after removal.

(j) Removal.

(k) Inspecting and testing.

(l) Other items affecting successful completion.

(3) Document and distribute meeting minutes to participants to record decisions affecting installation.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval.information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING01 81 13.xx SUSTAINABLE CONSTRUCTION REQUIREMENTS.
Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G, USACE

Lead-Based Paint Removal Plan; G USACE

Respiratory Protection Program; G USACE

Hazard Communication Program; G, USACE

Hazardous Waste Management Plan; G, USACE

SD-03 Product Data

Manufacturer's Literature And Data; G, USACE

SD-06 Test Reports

Test Reports; G, USACE

SD-07 Certificates

Certificates; G, USACE

SD-11 Closeout Submittals

Record Documents; G, USACE

1.7 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Manufacturer's Literature and Data:

(1) Description of each product.

(a) Paint removal products.

(b) Vacuum filters.

(c) Respirators.

(2) Safety Data Sheet for each paint removal product.

(3) Installation instructions.

(a) Paint removal products.

c. Test Reports: Submit testing laboratory reports.

(1) Submit air monitoring results within three working days, signed by testing laboratory employee performing air monitoring, employee analyzing sample, and CIH.

d. Certificates: Certify completed lead training.

(1) Submit certificate for each employee signed and dated by CIH and employee stating employee was trained.

e. Qualifications: Substantiate qualifications comply with specifications.

(1) Paint removal contractor.

(2) Testing laboratory.

(a) Name, address, and telephone number.

(b) Current evidence of participation in American Industrial Hygiene Association (AIHA) Laboratory Accreditation Program (LAP), LLC, Environmental Lead Laboratory Accreditation Program (ELLAP).

(c) Copy of current AIHA accreditation certificate.

(3) Industrial hygienist.

(a) Name, address, and telephone number.

(b) Resume showing previous experience.

(c) Copy of current ABIH CIH certification.

(4) Paint disposal facility.

(a) Name, address, and telephone number.

(b) Current license or authorization to receive and dispose lead contaminated waste.

f. Record Documents:

(1) Completed and signed hazardous waste manifest from waste transporter.

(2) Toxicity Characteristic Leaching Procedure (TCLP) test results to determine if waste is hazardous.

(3) Paint disposal facility receipts and disposition reports.

(4) Certification of medical examinations.

(5) Medical Opinion that employee is qualified to wear a respirator, that employees has been trained and fit tested for the respirator.

(6) Employee training certification.

1.8 QUALITY ASSURANCE

a. Safety and Health Regulation Compliance:

(1) Comply with laws, ordinances, rules, and regulations of Federal, State, and Local authorities having jurisdiction regarding removing, handling, storing, transporting, and disposing lead waste materials.

(a) Comply with applicable requirements of 29 CFR 1926.62.

(b) Notify Contracting Officer's Representative and request resolution of conflicts between regulations and specified requirements before starting work.

(2) Comply with the following local laws, ordinances, criteria, rules and regulations regarding removing, handling, storing, transporting, and disposing lead-contaminated materials:

(a) Department of Public Health - Chapter 400c - Sections: 20-474-20-482

(b) Department of Public Health - Chapter 368a - Sections: 19a-109aa, 19-110, 19a-110a, 19a-111a-k

b. Paint Removal Contractor: Experienced contractor, registered or licensed by applicable state agency regulating lead-based paint removal.

c. Testing Laboratory: State certified independent testing laboratory experienced in airborne lead monitoring, testing, and reporting.

(1) Successful participant in American Industrial Hygiene Association (AIHA) Laboratory Accreditation Program (LAP), LLC, Environmental Lead Laboratory Accreditation Program (ELLAP).

d. Certified Industrial Hygienist: Certified as CIH by American Board of

Industrial Hygiene in comprehensive practice and responsible for:

(1) Certify Training.

(2) Review and approve lead-based paint removal plan for conformance to applicable referenced standards.

(3) Inspect lead-based paint removal work for conformance with approved plan.

(4) Direct monitoring.

(5) Ensure work is performed according to specifications.

(6) Ensure personnel and environment hazardous exposures are adequately controlled.

e. Paint Disposal Facility: State certified disposal facility qualified to receive and dispose lead-based paint.

f. Lead-based Paint Removal Plan:

(1) Submit detailed, site-specific plan describing lead-based paint removal procedures.

(2) Include sketch showing location, size, and details of lead control areas, decontamination rooms, change rooms, shower facilities, and mechanical ventilation system.

(3) Include eating, drinking, and restroom procedures, interface of trades, work sequencing, collected wastewater and paint debris disposal plan, air sampling plan, respirators, protective equipment, and detailed description of containment methods ensuring airborne lead concentrations do not exceed action level outside lead control area.

(a) Eating, drinking, and smoking are not acceptable within lead control area.

(4) Include air sampling, training and strategy, sampling methodology, frequency, duration, and qualifications of air monitoring personnel.

g. Respiratory Protection Program: Establish and implement program required by 29 CFR 1910.134 and 29 CFR 1926.62.

(1) Provide each employee negative pressure or other appropriate respirator.

(a) Respirator fit each employee's respirator at initial fitting and at least annually thereafter, as required by 29 CFR 1910.134 Respiratory Protection.

h. Hazard Communication Program: Establish and implement program required by 29 CFR 1910.1200 which is the same as 29 CFR 1926.59.

i. Hazardous Waste Management Plan: Establish and implement plan according to applicable requirements of Federal, State, and Local hazardous waste regulations including the following:

(1) Identification of hazardous wastes associated with work.

(2) Estimated quantities of generated and disposed waste.

(3) Names and qualifications of each contractor transporting, storing, treating, and disposing wastes. Include facility location and 24-hour point of contact. Provide two copies of State permit applications and EPA Identification numbers.

(4) Names and qualifications (experience and training) of personnel working on-site with hazardous wastes.

(5) List of required waste handling equipment including cleaning, volume reduction, and transport equipment.

(6) Spill prevention, containment, and cleanup contingency implementation measures.

(7) Work plan and schedule for waste containment, removal, and disposal with daily waste cleaned up and containerization.

(8) Hazardous waste disposal cost.

1.9 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 PAINT REMOVAL PRODUCTS

Biodegradable, non-toxic, capable of removing existing paint layers in one application, and acceptable to CIH.

2.2 ACCESSORIES

a. Waste Collection Drums: 49 CFR 178; Type 1A2, steel, removable head, 55 gal. capacity, capable of containing waste without loss.

b. Vacuum Cleaner: HEPA filtered type.

c. Scrapers:

(1) Metal type for use on metal, concrete, and masonry surfaces.

(2) Plastic type for use on wood, plaster, gypsum board, and other surfaces.

d. Rinse Water: Potable.

e. Cleaning Cloths: Cotton.

PART 3 EXECUTION

3.1 EXAMINATION

a. Before exposure to lead-contaminated dust, provide workers with comprehensive medical examination required by 29 CFR 1926.62 (j)

Medical Surveillance.

b. Maintain complete and accurate employee medical records according to 29 CFR 1910.1020.

c. Train each employee performing paint removal, disposal, and air sampling operations according to 29 CFR 1926.62.

(1) Certify training is completed before employee is permitted to work on project and enter lead control area.

3.2 PREPARATION

a. Protect existing work indicated to remain.

(1) Perform paint removal work without damaging and contaminating adjacent work.

(2) Restore damage and contamination to original condition.

b. Notify Contracting Officer 20days before starting paint removal work.

c. Lead Control Area Requirements:

(1) Establish lead control area by completely enclosing lead-based paint removal work area with containment screens . and barrier tapes.

(2) Contain removal operations using negative pressure full containment system with minimum one change room and HEPA filtered exhaust.

d. Boundary Requirements: Provide physical boundaries around lead control area by roping off area or providing curtains, portable partitions or other enclosures to ensure that airborne lead concentrations do not meet or exceed action level outside of lead control area.

e. Heating, Ventilating and Air Conditioning (HVAC) Systems: Shut down, lock out, and isolate HVAC systems supplying exhausting, and passing through lead control areas. Seal HVAC inlets and outlet within lead control area with 6-mil plastic sheet and tape. Tape seal seams in HVAC components passing through lead control area.

f. Change Room and Shower Facilities: Provide clean change rooms and shower facilities within physical boundary around lead control area according to 29 CFR 1926.62.

g. Mechanical Ventilation System:

(1) Provide ventilation system to control personnel exposure to lead using HEPA equipped negative air machines.

(2) Design, construct, install, and maintain HEPA filtered fixed local exhaust ventilation system according to ANSI Z9.2 and approved by CIH.

(3) Exhaust ventilation air to exterior wherever possible.

(4) When exhaust ventilation air must be recirculated into work area, provide HEPA filter with reliable back-up filter and controls to monitor lead concentration in return air and to bypass recirculation system automatically when system fails.

h. Personnel Protection: Provide and use required protective clothing and equipment within lead control area.

i. Warning Signs: Provide warning signs complying with 29 CFR 1926.62 at lead control area approaches. Locate signs so personnel read signs and take necessary precautions before entering lead control area.

3.3 WORK PROCEDURES

a. Remove lead-based paint according to approved lead-based paint removal plan.

(1) Perform work only in presence of CIH or Industrial Hygienist (IH) Technician under direction of CIH ensuring continuous inspection of work in progress and direction of air monitoring activities.

(2) Handle, store, transport, and dispose lead or and lead contaminated waste according to 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265. Comply with land disposal restriction notification requirements as required by 40 CFR 268.

b. Use procedures and equipment required to limit occupational and environmental lead exposure when lead-based paint is removed according to 29 CFR 1926.62.

c. Dispose removed paint and waste according to Environmental Protection Agency (EPA), federal, state, and local requirements.

d. Personnel Exiting Procedures:

(1) When personnel exit lead control area, comply with the following procedures:

(a) Vacuum exposed clothing surfaces.

(b) Remove protective clothing and equipment in decontamination room. Place clothing in approved impermeable disposal bag.

(c) Shower.

(d) Dress in clean clothes before leaving lead control area.

e. Monitoring - General:

(1) Monitor airborne lead concentrations according to 29 CFR 1910.1025by testing laboratory as directed by CIH.

(2) Take personal air monitoring samples on employees anticipated to have greatest exposure risk as determined by CIH. Additionally, take air monitoring samples on minimum 25 percent of work crew or minimum of two employees, whichever is greater, during each work shift.

(3) Submit results of air monitoring samples, signed by CIH, within 48

hours after taking air samples. Notify Contracting Officer's Representative immediately of lead exposure at or exceeding action level outside of lead control area.

f. Monitoring During Paint Removal:

(1) Perform personal and area monitoring during entire paint removal operation.

(2) Conduct area monitoring at physical boundary daily for each work shift to ensure unprotected personnel are not exposed above action level anytime.

(3) For outdoor operations, take at least one sample on each shift leeward of lead control area. When adjacent areas are contaminated, clean area of contamination and have CIH visually inspect and certify lead contamination is cleaned.

(4) Stop work when outside boundary lead levels meet or exceed action level. Notify Contracting Officer's Representative, immediately.

(5) Correct conditions causing increased lead concentration as directed by CIH.

(6) Review sampling data collected during work stoppage to determine if conditions require additional work method modifications as determined by CIH.

(7) Resume paint removal when approved by CIH.

3.4 LEAD-BASED PAINT REMOVAL

a. Remove paint within areas indicated on drawings completely exposing substrate. Minimize damage to substrate.

b. Comply with paint removal processes described lead paint removal plan.

c. Lead-Based Paint Removal: Select processes for each application to minimize work area lead contamination and waste.

3.5 SUBSTRATE SURFACE PREPARATION

a. Protect substrates from deterioration and contamination until refinished.

(1) Protect metal substrates from flash rusting.

b. Prepare and paint substrates according to Section 09 91 00, PAINTING.

3.6 FIELD QUALITY CONTROL

a. Field Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.

b. Perform sampling and testing for:

(1) Air monitoring.

(2) Lead-Based Paint.

3.7 CLEANING AND DISPOSAL

a. Cleaning:

(1) Maintain lead control area surfaces free of accumulating paint chips and dust. Confine dust, debris, and waste to work area.

(2) HEPA vacuum clean and wet wipe with detergent solution work area daily, at end of each shift, and when paint removal operation is complete.

b. CIH Certification: Certify in writing that inside and outside lead control area air monitoring samples are less than action level, employee respiratory protection was adequate, the work was performed according to 29 CFR 1926.62, and no visible accumulations of lead-based paint and dust remain on worksite.

(1) Do not remove lead control area or roped-off boundary and warning signs before Contracting Officer's Representative's receipt of CIH's certification.

(2) Re-clean areas showing dust or residual paint chips.

c. Testing: Where indicated and when directed by Contracting Officer's Representative, test lead-based paint residue and used abrasive according to 40 CFR 261 for hazardous waste.

d. Waste Collection:

(1) Collect lead-contaminated materials including waste, scrap, debris, bags, containers, equipment, and clothing, which may produce airborne lead contamination.

(2) Place lead contaminated materials in waste disposal drums. Label each drum identifying waste type according to 49 CFR 172 and date waste materials were first put into drum. Obtain and complete the Uniform Hazardous Waste Manifest forms. Comply with land disposal restriction notification requirements required by 40 CFR 268:

(3) Coordinate temporary storage location on project site with Contracting Officer's Representative.

e. Waste Disposal:

(1) Minimum 14 days before delivery, notify Contracting Officer's Representative and abatment contractor shall arrange for job site inspection of drums and manifests by paint disposal facility personnel.

(2) The abatement contractor will arrange hazardous wastes removal, transport and delivery to paint disposal facility to ensure drums do not remain on project site longer than 45 calendar days from drum label date.

f. Waste Disposal:

(1) Do not store hazardous waste drums in temporary storage location longer than 90 calendar days from drum label date.

(2) Remove, transport, and deliver drums to paint disposal facility.

(a) Obtain signed receipt including date, time, quantity, and description of materials received according to 40 CFR 262.

(b) Obtain final report of materials disposition after disposal completion.

(3) The waste disposal shall follow all state, regional and local laws, regulations and statutes.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 30 00

CAST-IN-PLACE CONCRETE

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 TESTING AGENCY FOR CONCRETE MIX DESIGN
1.4 TOLERANCES
1.5 REGULATORY REQUIREMENTS
1.6 SUBMITTALS
1.7 SUBMITTAL INSTRUCTIONS
1.8 DELIVERY, STORAGE, AND HANDLING
1.9 PRE-CONCRETE CONFERENCE
1.10 REFERENCES

PART 2 PRODUCTS

2.1 FORMS
2.2 MATERIALS
2.3 CONCRETE MIXES
2.4 BATCHING AND MIXING

PART 3 EXECUTION

3.1 FORMWORK
3.2 PLACING REINFORCEMENT
3.3 VAPOR BARRIER
3.4 SLABS RECEIVING RESILIENT COVERING
3.5 CONSTRUCTION JOINTS
3.6 EXPANSION JOINTS AND CONTRACTION JOINTS
3.7 3.7 PLACING CONCRETE
3.8 HOT WEATHER
3.9 COLD WEATHER
3.10 PROTECTION AND CURING
3.11 REMOVAL OF FORMS
3.12 CONCRETE SURFACE PREPARATION
3.13 CONCRETE FINISHES
3.14 SURFACE TREATMENTS
3.15 APPLIED TOPPING
3.16 RETAINING WALLS
3.17 PRECAST CONCRETE ITEMS

-- End of Section Table of Contents --

SECTION 03 30 00
1/21

CAST-IN-PLACE CONCRETE

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies cast-in-place structural concrete and materials and mixes for other concrete.

1.2 RELATED WORK

a. Section 01 45 29, TESTING LABORATORY SERVICES: Materials testing and inspection during construction.

b. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS

c. Section 32 05 23, CEMENT AND CONCRETE FOR EXTERIOR IMPROVEMENTS: Concrete roads, walks, and similar exterior site work.

1.3 TESTING AGENCY FOR CONCRETE MIX DESIGN

a. Testing agency for the trial concrete mix design retained and reimbursed by the Contractor and approved by Contracting Officer. For all other testing, refer to Section 01 45 29 Testing Laboratory Services.

b. Testing agency maintaining active participation in Program of Cement and Concrete Reference Laboratory (CCRL) of National Institute of Standards and Technology.

c. Testing agency shall furnish equipment and qualified technicians to establish proportions of ingredients for concrete mixes.

1.4 TOLERANCES

a. Formwork: ACI 117, except the elevation tolerance of formed surfaces before removal of shores is (+0 inch) and (-3/4 inch).

b. Reinforcement Fabricating and Placing: ACI 117, except that fabrication tolerance for bar sizes Nos. 10, 13, and 16 (Nos. 3, 4, and 5) (Tolerance Symbol 1 in Fig. 2.1(a), ACI 117) used as column ties or stirrups is (+0 inch) and (-1/2 inch) where gross bar length is less than (12 feet), or (+0 inch) and (-3/4 inch) where gross bar length is 12 feet or more.

c. Cross-Sectional Dimension: ACI 117, except tolerance for thickness of slabs 12 inches or less is (+3/4 inch) and (-1/4 inch inch). Tolerance of thickness of beams more than 12 inch but less than 3 feet is (+3/4 inch) and (-3/8 inch).

d. Slab Finishes: ACI 117, Section 4.5.6, F-number method in accordance with ASTM E1155, except as follows:

(1) Test entire slab surface, including those areas within 2 feet of

construction joints and vertical elements that project through slab surface.

(2) Maximum elevation change which may occur within 2 feet of any column or wall element is (1/4 inch).

(3) Allow sample measurement lines that are perpendicular to construction joints to extend past joint into previous placement no further than (5 feet).

1.5 REGULATORY REQUIREMENTS

a. ACI SP-66 – ACI Detailing Manual.

b. ACI 318 - Building Code Requirements for Reinforced Concrete.

c. ACI 301 – Standard Specifications for Structural Concrete.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Testing Agency For Concrete Mix Design; G, AE

Mock-Up; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-06 Test Reports

Mill Test Reports; G, AE

Test Report For Concrete Mix Designs; G, AE

SD-07 Certificates

Manufacturer's Certificates; G, AE

1.7 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, and SAMPLES. All items indicated below are required submittals requiring Contracting Officer’s Representative (Contracting Officer) review and approval.

b. Shop Drawings: Reinforcing steel: Complete shop drawings

c. Mill Test Reports:

(1) Reinforcing Steel.

(2) Cement.

d. Manufacturer's Certificates:

(1) Abrasive aggregate.

(2) Lightweight aggregate for structural concrete.

(3) Air-entraining admixture.

(4) Chemical admixtures, including chloride ion content.

(5) Waterproof paper for curing concrete.

(6) Liquid membrane-forming compounds for curing concrete.

(7) Non-shrinking grout.

(8) Liquid hardener.

(9) Waterstops.

(10) Expansion joint filler.

(11) Adhesive binder.

e. Testing Agency for Concrete Mix Design: Approval request including qualifications of principals and technicians and evidence of active participation in program of Cement and Concrete Reference Laboratory (CCRL) of National Institute of Standards and Technology

f. Test Report for Concrete Mix Designs: Trial mixes including water-cement fly ash ratio curves, concrete mix ingredients, and admixtures.

1.8 DELIVERY, STORAGE, AND HANDLING

a. Conform to ACI 304R. Store aggregate separately for each kind or grade, to prevent segregation of sizes and avoid inclusion of dirt and other materials.

b. Deliver cement in original sealed containers bearing name of brand and manufacturer, and marked with net weight of contents. Store in suitable watertight building in which floor is raised at least 1 foot above ground. Store bulk cement and fly ash in separate suitable bins.

c. Deliver other packaged materials for use in concrete in original sealed containers, plainly marked with manufacturer's name and brand, and protect from damage until used.

1.9 PRE-CONCRETE CONFERENCE

a. General: At least 15 days prior to submittal of design mixes, conduct

a meeting to review proposed methods of concrete construction to achieve the required results.

b. Agenda: Includes but is not limited to:

(1) Submittals.

(2) Coordination of work.

(3) Availability of material.

(4) Concrete mix design including admixtures.

(5) Methods of placing, finishing, and curing.

(6) Finish criteria required to obtain required flatness and levelness.

(7) Timing of floor finish measurements.

(8) Material inspection and testing.

c. Attendees: Include but not limited to representatives of Contractor; subcontractors involved in supplying, conveying, placing, finishing, and curing concrete; lightweight aggregate manufacturer; admixture manufacturers; Contracting Officer; Consulting Engineer; Department of Veterans Affairs retained testing laboratories for concrete testing and finish (F-number) verification.

d. Minutes of the meeting: Contractor shall take minutes and type and distribute the minutes to attendees within five days of the meeting.

1.10 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; R 2015) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 211.1	(1991; R 2009) Standard Practice for Selecting Proportions for Normal, Heavyweight and Mass Concrete
ACI 211.2	(1998; R 2004) Standard Practice for Selecting Proportions for Structural Lightweight Concrete
ACI 214R	(2019) Evaluation of Strength Test Results of Concrete
ACI 301	(2020) Specifications for Structural Concrete
ACI 304R	(2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete

ACI 305R	(2020) Guide to Hot Weather Concreting
ACI 306R	(2016) Guide to Cold Weather Concreting
ACI 308R	(2016) Guide to External Curing of Concrete
ACI 309R	(2005) Guide for Consolidation of Concrete
ACI 318	(2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)
ACI 347R	(2014; Errata 1 2017) Guide to Formwork for Concrete
ACI SP-66	(2004) ACI Detailing Manual

AMERICAN HARDBOARD ASSOCIATION (AHA)

AHA A135.4	(1995; R 2004) Basic Hardboard

ASTM INTERNATIONAL (ASTM)

ASTM A185/A185M	(2007) Standard Specification for Steel Welded Wire Reinforcement, Plain, for Concrete
ASTM A615/A615M	(2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A706/A706M	(2024) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement
ASTM A767/A767M	(2024) Standard Specification for Zinc-Coated (Galvanized) Steel Bars for Concrete Reinforcement
ASTM A775/A775M	(2022) Standard Specification for Epoxy-Coated Steel Reinforcing Bars
ASTM A996/A996M	(2016) Standard Specification for Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement
ASTM A1064/A1064M	(2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C31/C31M	(2025a) Standard Practice for Making and Curing Concrete Test Specimens in the Field

ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C39/C39M	(2024) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C94/C94M	(2024d) Standard Specification for Ready-Mixed Concrete
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C171	(2020) Standard Specification for Sheet Materials for Curing Concrete
ASTM C172	(2010) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	(2024a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C192/C192M	(2024) Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory
ASTM C231/C231M	(2024) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	(2024) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C309	(2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C330	(2023) Standard Specification for Lightweight Aggregates for Structural Concrete
ASTM C494/C494M	(2024) Standard Specification for Chemical Admixtures for Concrete
ASTM C618	(2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C666/C666M	(2015) Resistance of Concrete to Rapid Freezing and Thawing
ASTM C881/C881M	(2020a) Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete

ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM C1315	(2019) Standard Specification for Liquid Membrane-Forming Compounds Having Special Properties for Curing and Sealing Concrete
ASTM D6/D6M	(2011) Standard Test Method for Loss on Heating of Oil and Asphaltic Compounds
ASTM D297	(2015; R 2019) Rubber Products - Chemical Analysis
ASTM D412	(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D4263	(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
ASTM E1155	(2020) Standard Test Method for Determining Floor Flatness and Floor Levelness Numbers
ASTM F1249	(2020) Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor
ASTM F1869	(2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
ASTM G90	(2017) Performing Accelerated Outdoor Weathering of Nonmetallic Materials Using Concentrated Natural Sunlight

AMERICAN WELDING SOCIETY (AWS)

AWS D1.4/D1.4M	(2018) Structural Welding Code - Reinforcing Steel

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

CRSI DA4	Manual of Standard Practice; Concrete Reinforcing Steel Institute

U.S. ARMY CORPS OF ENGINEERS (USACE)

COE CRD-C 513	(1974) Corps of Engineers Specifications for Rubber Waterstops

COE CRD-C 572 (1974) Corps of Engineers Specifications for Polyvinylchloride Waterstops

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST PS 1 (2009) DOC Voluntary Product Standard PS 1-07, Structural Plywood

NIST PS 20 (2021) American Softwood Lumber Standard

U.S. DEPARTMENT OF DEFENSE (DOD)

SS-S-210A Sealing Compound, Preformed Plastic, for Expansion Joints and Pipe Joints

PART 2 PRODUCTS

2.1 FORMS

a. Wood: NIST PS 20 free from loose knots and suitable to facilitate finishing concrete surface specified; tongue and grooved.

b. Plywood: NIST PS 1 Exterior Grade B-B (concrete-form) (5/8 inch), or 3/4 inch thick for unlined contact form. B-B High Density Concrete Form Overlay optional.

d. Permanent Steel Form for Concrete Slabs: Corrugated, ASTM A653/A653M, Grade E, and Galvanized, ASTM A653/A653M, ASTM G90. Provide venting where insulating concrete fill is used.

f. Form Lining:

(1) Hardboard: AHA A135.4, Class 2 with one (S1S) smooth side)

(2) Plywood: Grade B-B Exterior (concrete-form) not less than 1/4 inch thick.

(3) Plastic, fiberglass, or elastomeric capable of reproducing the desired pattern or texture.

g. Concrete products shall comply with following standards for biobased materials:

Material Type	Percent by Weight
Concrete Penetrating Liquid	79 percent biobased material
Concrete form Release Agent	87 percent biobased material
Concrete Sealer	11 percent biobased material

The minimum-content standards are based on the weight (not the volume) of the material.

h. Form Ties: Develop a minimum working strength of (3000 pounds) when

fully assembled. Ties shall be adjustable in length to permit tightening of forms and not have any lugs, cones, washers to act as spreader within form, nor leave a hole larger than 3/4 inch diameter, or a depression in exposed concrete surface, or leave metal closer than (1 1/2 inches) to concrete surface. Wire ties not permitted. Cutting ties back from concrete face not permitted.

2.2 MATERIALS

a. Portland Cement: ASTM C150/C150M Type I or II.

b. Fly Ash: ASTM C618, Class C or F including supplementary optional requirements relating to reactive aggregates and alkalies, and loss on ignition (LOI) not to exceed 5 percent. Do not exceed more than 25 percent total cementitious content by weight.

c. Coarse Aggregate: ASTM C33/C33M.

(1) Size 67 or Size 467 may be used for footings and walls over 12 inches thick.

(2) Coarse aggregate for interior slabs on grade shall conform to the following:

(a) Dense or well graded aggregate.

1. Percent retained on each sieve below the top size and above the No. 100 sieve:

a) 8 to 18 percent for 1-1/2 inches top size.

b) 8 to 22 percent for 3/4 or 1 inch top size.

2. The above requirements may be deviated from based on locally available material.

a) One or two non-adjacent sieves sizes may fall outside of the limits set above.

b) Percent retained on two adjacent sieves sizes shall not be less than 5 percent of the above required.

c) Percent retained on three adjacent sieve sizes shall not be less than 8 percent of the above required.

d) When the percent retained on each of two adjacent sieve sizes is less than 8 percent the total percent retained on either of these sieves and the adjacent outside sieve should be at least 13 percent (for example, if both the No. 4 and No. 8 sieves have 6 percent retained on each item then: 1. the total retained on the 3/8 inch and No. 4 sieves should be at least 13 percent, and 2. the total retained on the No. 8 and No. 16 sieves should be at least 13 percent.

(3) Maximum size of coarse aggregates not more than one-fifth of narrowest dimension between sides of forms, one-third of depth of slabs, nor three-fourth of minimum clear spacing between

reinforcing bars.

d. Lightweight Aggregates for Structural Concrete: ASTM C330, Table 1. Maximum size of aggregate not larger than one-fifth of narrowest dimension between forms, nor three-fourth of minimum clear distance between reinforcing bars. Contractor to furnish certified report to verify that aggregate is sound and durable, and has a durability factor of not less than 80 based on 300 cycles of freezing and thawing when tested in accordance with ASTM C666/C666M.

e. Fine Aggregate: ASTM C33/C33M. Fine aggregate for applied concrete floor topping shall pass a (No. 4) sieve, 10 percent maximum shall pass a (No. 100) sieve.

f. Mixing Water: Fresh, clean, and potable.

g. Admixtures:

(1) Water Reducing Admixture: ASTM C494/C494M, Type A and not contain more chloride ions than are present in municipal drinking water.

(2) Water Reducing, Retarding Admixture: ASTM C494/C494M, Type D and not contain more chloride ions than are present in municipal drinking water.

(3) High-Range Water-Reducing Admixture (Superplasticizer): ASTM C494/C494M, Type F or G, and not contain more chloride ions than are present in municipal drinking water. Use of superplasticizer requires Contracting Officer approval.

(4) Non-Corrosive, Non-Chloride Accelerator: ASTM C494/C494M, Type C or E, and not contain more chloride ions than are present in municipal drinking water. Admixture manufacturer must have long-term non-corrosive test data from an independent testing laboratory of at least one year duration using an acceptable accelerated corrosion test method such as that using electrical potential measures.

(5) Air Entraining Admixture: ASTM C260/C260M.

(6) Microsilica: Use only with prior review and acceptance of the Contracting Officer. Use only in conjunction with high range water reducer.

(7) Prohibited Admixtures: Calcium chloride, thiocyanate or admixtures containing more than 0.05 percent chloride ions are not permitted.

(8) Certification: Written conformance to the requirements above and the chloride ion content of the admixture prior to mix design review.

h. Vapor Barrier: ASTM F1249, WVT 0.012 foot/hr. .

i. Reinforcing Steel: ASTM A615/A615M, or ASTM A996/A996M, deformed, grade as shown.

j. Welded Wire Fabric: ASTM A185/A185M.

k. Reinforcing Bars to be Welded: ASTM A706/A706M.

l. Galvanized Reinforcing Bars: ASTM A767/A767M.

m. Epoxy Coated Reinforcing Bars: ASTM A775/A775M.

n. Cold Drawn Steel Wire: ASTM A1064/A1064M.

o. Reinforcement for Metal Pan Stair Fill: 2 inch wire mesh, either hexagonal mesh at (1.5 pounds per square yard), or square mesh at (1.17 pounds per square yard).

p. Supports, Spacers, and Chairs: Types which will hold reinforcement in position shown in accordance with requirements of ACI 318 except as specified.

q. Expansion Joint Filler: ASTM D1751.

r. Sheet Materials for Curing Concrete: ASTM C171.

s. Liquid Membrane-forming Compounds for Curing Concrete: ASTM C309, Type I, with fugitive dye, and shall meet the requirements of ASTM C1315.Compound shall be compatible with scheduled surface treatment, such as paint and resilient tile, and shall not discolor concrete surface.

t. Abrasive Aggregate: Aluminum oxide grains or emery grits.

w. Moisture Vapor Emissions & Alkalinity Control Sealer: 100 percent active colorless aqueous siliconate solution concrete surface.

(1) ASTM C1315 Type 1 Class A, and ASTM C309 Type 1 Class A, penetrating product to have no less than 34 percent solid content, leaving no sheen, volatile organic compound (VOC) content rating as required to suite regulatory requirements. The product shall have at least a five (5) year documented history in controlling moisture vapor emission from damaging floor covering, compatible with all finish materials.

(2) MVE 15-Year Warranty:

(a) When a floor covering is installed on a below grade, on grade, or above grade concrete slab treated with Moisture Vapor Emissions & Alkalinity Control Sealer according to manufacturer’s instruction, sealer manufacturer shall warrant the floor covering system against failure due to moisture vapor migration or moisture-born contaminates for a period of fifteen (15) years from the date of original installation. The warranty shall cover all labor and materials needed to replace all floor covering that fails due to moisture vapor emission & moisture born contaminates.

y. Non-Shrink Grout:

(1) ASTM C1107/C1107M, pre-mixed, produce a compressive strength of at least 18 MPa at three days and (5000 psi) at 28 days. Furnish test data from an independent laboratory indicating that the grout when placed at a fluid consistency shall achieve 95 percent bearing under a 4 foot by 4 foot base plate.

(2) Where high fluidity or increased placing time is required, furnish test data from an independent laboratory indicating that the grout when placed at a fluid consistency shall achieve 95 percent under an 18 inch by 36 inch base plate.

z. Adhesive Binder: ASTM C881/C881M.

aa. Waterstops:

(1) Polyvinyl Chloride Waterstop: COE CRD-C 572.

(2) Rubber Waterstops: COE CRD-C 513.

(3) Bentonite Waterstop: Flexible strip of bentonite 1 inch by 3/4 inch), weighing (5.85 pounds per foot) composed of Butyl Rubber Hydrocarbon (ASTM D297), Bentonite (SS-S-210A) and Volatile Matter (ASTM D6/D6M).

(4) Non-Metallic Hydrophilic: Swellable strip type compound of polymer modified chloroprene rubber that swells upon contact with water shall conform to ASTM D412 as follows: Tensile strength 420 psi minimum; ultimate elongation 600 percent minimum. Hardness shall be 50 minimum on the type A durameter and the volumetric expansion ratio in in 70 deg water shall be 3 to 1 minimum.

bb. Porous Backfill: Crushed stone or gravel graded from(1 inch to 3/4 inch).

dd. Epoxy Joint Filler: Two component, 100 percent solids compound, with a minimum shore D hardness of 50.

ee. Bonding Admixture: Non-rewettable, polymer modified, bonding compound.

2.3 CONCRETE MIXES

a. Mix Designs: Proportioned in accordance with Section 5.3, "Proportioning on the Basis of Field Experience and/or Trial Mixtures" of ACI 318.

(1) If trial mixes are used, make a set of at least 6 cylinders in accordance with ASTM C192/C192M for test purposes from each trial mix; test three for compressive strength at 7 days and three at 28

days.

(2) Submit a report of results of each test series, include a detailed listing of the proportions of trial mix or mixes, including cement, fly ash, admixtures, weight of fine and coarse aggregate per (cubic yard) measured dry rodded and damp loose, specific gravity, fineness modulus, percentage of moisture, air content, water-cement ratio, and consistency of each cylinder in terms of slump. include dry unit weight of lightweight structural concrete.

(3) Prepare a curve showing relationship between water-cement -fly ash ratio at 7-day and 28-day compressive strengths. Plot each curve using at least three specimens.

(4) If the field experience method is used, submit complete standard deviation analysis.

b. Fly Ash Testing: Submit certificate verifying conformance with ASTM C618 initially with mix design and for each truck load of fly ash delivered from source. Submit test results performed within 6 months of submittal date. Notify the Contracting Officer immediately when change in source is anticipated.

(1) Testing Laboratory used for fly ash certification/testing shall participate in the Cement and Concrete Reference Laboratory (CCRL) program. Submit most recent CCRL inspection report.

c. After approval of mixes no substitution in material or change in proportions of approval mixes may be made without additional tests and approval of the Contracting Officer or as specified. Making and testing of preliminary test cylinders may be carried on pending approval of cement and fly ash, providing Contractor and manufacturer certify that ingredients used in making test cylinders are the same. The Contracting Officer may allow Contractor to proceed with depositing concrete for certain portions of work, pending final approval of cement and fly ash and approval of design mix.

d. Cement Factor: Maintain minimum cement factors in Table I regardless of compressive strength developed above minimums. Use Fly Ash as an admixture with maximum of 25 percent replacement by weight in all structural work. Increase this replacement to 40 percent for mass concrete, and reduce it to 10 percent for drilled piers and caissons. Fly ash shall not be used in high-early mix design.

TABLE I - CEMENT AND WATER FACTORS FOR CONCRETE

Concrete Strength		Non-Air-Entrained	Air-Entrained	
Min. 28 Day Comp. Str. (psi)	Min. Cement (lbs/c. yd)	Max. Water Cement Ratio	Min. Cement (lbs/c. yd)	Max. Water Cement Ratio
(5000)1,3	(630)	0.45	(650)	0.40
(4000)1,3	(550)	0.55	(570)	0.50
(3000)1,3	(470)	0.65	(490)	0.55

(3000)1,2	(500)	See 4 below	(520)	See 4 below

(1) If trial mixes are used, the proposed mix design shall achieve a compressive strength (1200 psi) in excess of f'c. For concrete strengths above (5000 psi), the proposed mix design shall achieve a compressive strength (1400 psi) in excess of f'c.

(2) Lightweight Structural Concrete. Pump mixes may require higher cement values.

(3) For concrete exposed to high sulfate content soils maximum water cement ratio is 0.44.

(4) Determined by Laboratory in accordance with ACI 211.1 for normal concrete or ACI 211.2 for lightweight structural concrete.

e. Maximum Slump: Maximum slump, as determined by ASTM C143/C143M with tolerances as established by ASTM C94/C94M, for concrete to be vibrated shall be as shown in Table II.

TABLE II - MAXIMUM SLUMP, (INCHES)

Type of Construction	Normal Weight Concrete	Lightweight Structural Concrete
Reinforced Footings and Substructure Walls	(3 inches)	(3 inches)
Slabs, Beams, Reinforced Walls, and Building Columns	(4 inches)	(4 inches)

f. Slump may be increased by the use of the approved high-range water-reducing admixture (superplasticizer). Tolerances as established by ASTM C94/C94M. Concrete containing the high-range-water-reducing admixture may have a maximum slump of (9 inches). The concrete shall arrive at the job site at a slump of 2 inches to 3 inches), and 3 inches to 4 inches for lightweight concrete. This should be verified, and then the high-range-water-reducing admixture added to increase the slump to the approved level.

g. Air-Entrainment: Air-entrainment of normal weight concrete shall conform with Table III. Air-entrainment of lightweight structural concrete shall conform with Table IV. Determine air content by either ASTM C173/C173M or ASTM C231/C231M.

TABLE III - TOTAL AIR CONTENT
FOR VARIOUS SIZES OF COARSE AGGREGATES (NORMAL CONCRETE)

Nominal Maximum Size of Total Air Content	Coarse Aggregate, (Inches) Percentage by Volume

(3/8 in) .6 to 10	(1/2 in).5 to 9
(3/4 in) .4 to 8	(1 in).3-1/2 to 6-1/2
(1 1/2 in) .3 to 6	

TABLE IV
AIR CONTENT OF LIGHTWEIGHT STRUCTURAL CONCRETE

Nominal Maximum size of Total Air Content	Coarse Aggregate, (Inches) Percentage by Volume
Greater than (3/8 in) 4 to 8	(3/8 in) or less 5 to 9

h. High early strength concrete, made with Type III cement or Type I cement plus non-corrosive accelerator, shall have a 7-day compressive strength equal to specified minimum 28-day compressive strength for concrete type specified made with standard Portland cement.

i. Lightweight structural concrete shall not weigh more than air-dry unit weight shown. Air-dry unit weight determined on 6 inch by 12 inch test cylinders after seven days standard moist curing followed by 21 days drying at (73.4 {UNCF0B1} 3 degrees Fahrenheit), and 50 (plus or minus 7) percent relative humidity. Use wet unit weight of fresh concrete as basis of control in field.

j. Concrete slabs placed at air temperatures below (50 degrees Fahrenheit) use non-corrosive, non-chloride accelerator. Concrete required to be air entrained use approved air entraining admixture. Pumped concrete, synthetic fiber concrete, architectural concrete, concrete required to be watertight, and concrete with a water/cement ratio below 0.50 use high-range water-reducing admixture (superplasticizer).

k. Durability: Use air entrainment for exterior exposed concrete subjected to freezing and thawing and other concrete shown or specified. For air content requirements see Table III or Table IV.

l. Enforcing Strength Requirements: Test as specified in Section 01 45 29, TESTING LABORATORY SERVICES, during the progress of the work. Seven-day tests may be used as indicators of 28-day strength. Average of any three 28-day consecutive strength tests of laboratory-cured specimens representing each type of concrete shall be equal to or greater than specified strength. No single test shall be more than (500 psi) below specified strength. Interpret field test results in accordance with ACI 214R. Should strengths shown by test specimens fall below required values, the Contracting Officer may require any one or any combination of the following corrective actions, at no additional cost to the Government:

(1) Require changes in mix proportions by selecting one of the other appropriate trial mixes or changing proportions, including cement

content, of approved trial mix.

(2) Require additional curing and protection.

(3) If five consecutive tests fall below 95 percent of minimum values given in Table I or if test results are so low as to raise a question as to the safety of the structure, the Contracting Officer may direct Contractor to take cores from portions of the structure. Use results from cores tested by the Contractor retained testing agency to analyze structure.

(4) If strength of core drilled specimens falls below 85 percent of minimum value given in Table I, the Contracting Officer may order load tests, made by Contractor retained testing agency, on portions of building so affected. Load tests in accordance with ACI 318 and criteria of acceptability of concrete under test as given therein.

(5) Concrete work, judged inadequate by structural analysis, by results of load test, or for any reason, shall be reinforced with additional construction or replaced, if directed by the Contracting Officer.

2.4 BATCHING AND MIXING

General: Concrete shall be "Ready-Mixed" and comply with ACI 318 and ASTM C94/C94M, except as specified. Batch mixing at the site is permitted. Mixing process and equipment must be approved by the Contracting Officer. With each batch of concrete, furnish certified delivery tickets listing information in Paragraph 16.1 and 16.2 of ASTM C94/C94M. Maximum delivery temperature of concrete is (100 degrees Fahrenheit). Minimum delivery temperature as follows:

Atmospheric Temperature	Minimum Concrete Temperature
(30 degrees to 40 degrees F)	(60 degrees F.)
(0 degrees to 30 degrees F.)	(70 degrees F.)

(1) Services of aggregate manufacturer's representative shall be furnished during the design of trial mixes and as requested by the Contracting Officer for consultation during batching, mixing, and placing operations of lightweight structural concrete. Services will be required until field controls indicate that concrete of required quality is being furnished. Representative shall be thoroughly familiar with the structural lightweight aggregate, adjustment and control of mixes to produce concrete of required quality. Representative shall assist and advise the Contracting Officer.

PART 3 EXECUTION

3.1 FORMWORK

a. General: Design in accordance with ACI 347R is the responsibility of the Contractor. The Contractor shall retain a registered Professional

Engineer to design the formwork, shores, and reshores.

(1) Form boards and plywood forms may be reused for contact surfaces of exposed concrete only if thoroughly cleaned, patched, and repaired and the Contracting Officer approves their reuse.

(2) Provide forms for concrete footings unless the Contracting Officer determines forms are not necessary.

(3) Corrugated fiberboard forms: Place forms on a smooth firm bed, set tight, with no buckled cartons to prevent horizontal displacement, and in a dry condition when concrete is placed.

b. Treating and Wetting: Treat or wet contact forms as follows:

(1) Coat plywood and board forms with non-staining form sealer. In hot weather, cool forms by wetting with cool water just before concrete is placed.

(2) Clean and coat removable metal forms with light form oil before reinforcement is placed. In hot weather, cool metal forms by thoroughly wetting with water just before placing concrete.

(3) Use sealer on reused plywood forms as specified for new material.

c. Size and Spacing of Studs: Size and space studs, wales and other framing members for wall forms so as not to exceed safe working stress of kind of lumber used nor to develop deflection greater than 1/270 of free span of member.

d. Unlined Forms: Use plywood forms to obtain a smooth finish for concrete surfaces. Tightly butt edges of sheets to prevent leakage. Back up all vertical joints solidly and nail edges of adjacent sheets to same stud with 6d box nails spaced not over 6 inches apart.

e. Lined Forms: May be used in lieu of unlined plywood forms. Back up form lining solidly with square edge board lumber securely nailed to studs with all edges in close contact to prevent bulging of lining. No joints in lining and backing may coincide. Nail abutted edges of sheets to same backing board. Nail lining at not over 8 inches on center along edges and with at least one nail to each square foot of surface area; nails to be 3d blued shingle or similar nails with thin flatheads.

g. Wall Form Ties: Locate wall form ties in symmetrically level horizontal rows at each line of wales and in plumb vertical tiers. Space ties to maintain true, plumb surfaces. Provide one row of ties within 6 inches above each construction joint. Space through-ties adjacent to horizontal and vertical construction joints not over 18 inches on center.

(1) Tighten row of ties at bottom of form just before placing concrete and, if necessary, during placing of concrete to prevent seepage of concrete and to obtain a clean line. Ties to be entirely removed shall be loosened 24 hours after concrete is placed and shall be pulled from least important face when removed.

(2) Coat surfaces of all metal that is to be removed with paraffin, cup grease or a suitable compound to facilitate removal.

h. Inserts, Sleeves, and Similar Items: Flashing reglets, steel strips, masonry ties, anchors, wood blocks, nailing strips, grounds, inserts, wire hangers, sleeves, drains, guard angles, forms for floor hinge boxes, inserts or bond blocks for elevator guide rails and supports, and other items specified as furnished under this and other sections of specifications and required to be in their final position at time concrete is placed shall be properly located, accurately positioned, and built into construction, and maintained securely in place.

(1) Locate inserts or hanger wires for furred and suspended ceilings only in bottom of concrete joists, or similar concrete member of overhead concrete joist construction.

(2) Install sleeves, inserts and similar items for mechanical services in accordance with drawings prepared specially for mechanical services. Contractor is responsible for accuracy and completeness of drawings and shall coordinate requirements for mechanical services and equipment.

(3) Do not install sleeves in beams, joists or columns except where shown or permitted by the Contracting Officer. Install sleeves in beams, joists, or columns that are not shown, but are permitted by the Contracting Officer, and require no structural changes, at no additional cost to the Government.

(4) Minimum clear distance of embedded items such as conduit and pipe is at least three times diameter of conduit or pipe, except at stub-ups and other similar locations.

(5) Provide recesses and blockouts in floor slabs for door closers and other hardware as necessary in accordance with manufacturer's instructions.

i. Construction Tolerances:

(1) Set and maintain concrete formwork to assure erection of completed work within tolerances specified and to accommodate installation of other rough and finish materials. Accomplish remedial work necessary for correcting excessive tolerances. Erected work that exceeds specified tolerance limits shall be remedied or removed and replaced, at no additional cost to the Government.

(2) Permissible surface irregularities for various classes of materials are defined as "finishes" in specification sections covering individual materials. They are to be distinguished from tolerances specified which are applicable to surface irregularities of structural elements.

3.2 PLACING REINFORCEMENT

a. General: Details of concrete reinforcement in accordance with ACI 318 unless otherwise shown.

b. Placing: Place reinforcement conforming to CRSI DA4, unless otherwise shown.

(1) Place reinforcing bars accurately and tie securely at intersections and splices with (16 gauge) black annealed wire. Use epoxy-coated tie wire with epoxy-coated reinforcing. Secure reinforcing bars against displacement during the placing of concrete by spacers, chairs, or other similar supports. Portions of supports, spacers, and chairs in contact with formwork shall be made of plastic in areas that will be exposed when building is occupied. Type, number, and spacing of supports conform to ACI 318. Where concrete slabs are placed on ground, use concrete blocks or other non-corrodible material of proper height, for support of reinforcement. Use of brick or stone supports will not be permitted.

(2) Lap welded wire fabric at least 1 1/2 mesh panels plus end extension of wires not less than 12 inches in structural slabs. Lap welded wire fabric at least 1/2 mesh panels plus end extension of wires not less than 6 inches in slabs on grade.

(3) Splice column steel at no points other than at footings and floor levels unless otherwise shown.

c. Spacing: Minimum clear distances between parallel bars, except in columns and multiple layers of bars in beams shall be equal to nominal diameter of bars. Minimum clear spacing is 1 inch or 1-1/3 times maximum size of coarse aggregate.

d. Splicing: Splices of reinforcement made only as required or shown or specified. Accomplish splicing as follows:

(1) Lap splices: Do not use lap splices for bars larger than Number 36 (Number 11). Minimum lengths of lap as shown.

(2) Welded splices: Splicing by butt-welding of reinforcement permitted providing the weld develops in tension at least 125 percent of the yield strength (fy) for the bars. Welding conform to the requirements of AWS D1.4/D1.4M. Welded reinforcing steel conform to the chemical analysis requirements of AWS D1.4/D1.4M.

(a) Submit test reports indicating the chemical analysis to establish weldability of reinforcing steel.

(b) Submit a field quality control procedure to insure proper inspection, materials and welding procedure for welded splices.

(c) Department of Veterans Affairs retained testing agency shall test a minimum of three splices, for compliance, locations selected by the Contracting Officer.

(3) Mechanical Splices: Develop in tension and compression at least 125 percent of the yield strength (fy) of the bars. Stresses of transition splices between two reinforcing bar sizes based on area of smaller bar. Provide mechanical splices at locations indicated. Use approved exothermic, tapered threaded coupling, or swaged and threaded sleeve. Exposed threads and swaging in the field not permitted.

(a) Initial qualification: In the presence of the Contracting Officer, make three test mechanical splices of each bar size proposed to be spliced. Department of Veterans Affairs retained

testing laboratory will perform load test.

(b) During installation: Furnish, at no additional cost to the Government, one companion (sister) splice for every 50 splices for load testing. Department of Veterans Affairs retained testing laboratory will perform the load test.

e. Bending: Bend bars cold, unless otherwise approved. Do not field bend bars partially embedded in concrete, except when approved by the Contracting Officer.

f. Cleaning: Metal reinforcement, at time concrete is placed, shall be free from loose flaky rust, mud, oil, or similar coatings that will reduce bond.

g. Future Bonding: Protect exposed reinforcement bars intended for bonding with future work by wrapping with felt and coating felt with a bituminous compound unless otherwise shown.

3.3 VAPOR BARRIER

a. Except where membrane waterproofing is required, interior concrete slab on grade shall be placed on a continuous vapor barrier.

(1) Place 4 inches of fine granular fill over the vapor barrier to act as a blotter for concrete slab.

(2) Vapor barrier joints lapped 6 inches and sealed with compatible waterproof pressure-sensitive tape.

(3) Patch punctures and tears.

3.4 SLABS RECEIVING RESILIENT COVERING

a. Slab shall be allowed to cure for 6 weeks minimum prior to placing resilient covering. After curing, slab shall be tested by the Contractor for moisture in accordance with ASTM D4263 or ASTM F1869. Moisture content shall be less than 3 pounds per 1000 sf prior to placing covering.

b. In lieu of curing for 6 weeks, Contractor has the option, at his own cost, to utilize the Moisture Vapor Emissions & Alkalinity Control Sealer as follows:

(1) Sealer is applied on the day of the concrete pour or as soon as harsh weather permits, prior to any other chemical treatments for concrete slabs either on grade, below grade or above grade receiving resilient flooring, such as, sheet vinyl, vinyl composition tile, rubber, wood flooring, epoxy coatings and overlays.

(2) Manufacturer’s representative will be on the site the day of concrete pour to install or train its application and document. He shall return on every application thereafter to verify that proper procedures are followed.

(a) Apply Sealer to concrete slabs as soon as final finishing operations are complete and the concrete has hardened sufficiently to sustain floor traffic without damage.

(b) Spray apply Sealer at the rate of (200 square feet) per gallon. Lightly broom product evenly over the substrate and product has completely penetrated the surface.

(c) If within two (2) hours after initial application areas are subjected to heavy rainfall and puddling occurs, reapply Sealer product to these areas as soon as weather condition permits.

3.5 CONSTRUCTION JOINTS

a. Unless otherwise shown, location of construction joints to limit individual placement shall not exceed 80 feet in any horizontal direction, except slabs on grade which shall have construction joints shown. Allow 48 hours to elapse between pouring adjacent sections unless this requirement is waived by the Contracting Officer.

b. Locate construction joints in suspended floors near the quarter-point of spans for slabs, beams or girders, unless a beam intersects a girder at center, in which case joint in girder shall be offset a distance equal to twice width of beam. Provide keys and inclined dowels as shown. Provide longitudinal keys as shown.

c. Place concrete for columns slowly and in one operation between joints. Install joints in concrete columns at underside of deepest beam or girder framing into column.

d. Allow 2 hours to elapse after column is cast before concrete of supported beam, girder or slab is placed. Place girders, beams, grade beams, column capitals, brackets, and haunches at the same time as slab unless otherwise shown.

e. Install polyvinyl chloride or rubber water seals, as shown in accordance with manufacturer's instructions, to form continuous watertight seal.

3.6 EXPANSION JOINTS AND CONTRACTION JOINTS

a. Clean expansion joint surfaces before installing premolded filler and placing adjacent concrete.

b. Install polyvinyl chloride or rubber water seals, as shown in accordance with manufacturer's instructions, to form continuous watertight seal.

c. Provide contraction (control) joints in floor slabs as indicated on the contract drawings. Joints shall be either formed or saw cut, to the indicated depth after the surface has been finished. Complete saw joints within 4 to 12 hours after concrete placement. Protect joints from intrusion of foreign matter.

3.7 3.7 PLACING CONCRETE

a. Preparation:

(1) Remove hardened concrete, wood chips, shavings and other debris from forms.

(2) Remove hardened concrete and foreign materials from interior

surfaces of mixing and conveying equipment.

(3) Have forms and reinforcement inspected and approved by the Contracting Officer before depositing concrete.

(4) Provide runways for wheeling equipment to convey concrete to point of deposit. Keep equipment on runways which are not supported by or bear on reinforcement. Provide similar runways for protection of vapor barrier on coarse fill.

b. Bonding: Before depositing new concrete on or against concrete which has been set, thoroughly roughen and clean existing surfaces of laitance, foreign matter, and loose particles.

(1) Preparing surface for applied topping:

(a) Remove laitance, mortar, oil, grease, paint, or other foreign material by sand blasting. Clean with vacuum type equipment to remove sand and other loose material.

(b) Broom clean and keep base slab wet for at least four hours before topping is applied.

(c) Use a thin coat of one part Portland cement, 1.5 parts fine sand, bonding admixture; and water at a 50: 50 ratio and mix to achieve the consistency of thick paint. Apply to a damp base slab by scrubbing with a stiff fiber brush. New concrete shall be placed while the bonding grout is still tacky.

c. Conveying Concrete: Convey concrete from mixer to final place of deposit by a method which will prevent segregation. Method of conveying concrete is subject to approval of the Contracting Officer.

d. Placing: For special requirements see Paragraphs, HOT WEATHER and COLD WEATHER.

(1) Do not place concrete when weather conditions prevent proper placement and consolidation, or when concrete has attained its initial set, or has contained its water or cement content more than 1 1/2 hours.

(2) Deposit concrete in forms as near as practicable in its final position. Prevent splashing of forms or reinforcement with concrete in advance of placing concrete.

(3) Do not drop concrete freely more than 10 feet for concrete containing the high-range water-reducing admixture (superplasticizer) or 5 feet for conventional concrete. Where greater drops are required, use a tremie or flexible spout (canvas elephant trunk), attached to a suitable hopper.

(4) Discharge contents of tremies or flexible spouts in horizontal layers not exceeding 20 inches in thickness, and space tremies such as to provide a minimum of lateral movement of concrete.

(5) Continuously place concrete until an entire unit between construction joints is placed. Rate and method of placing concrete shall be such that no concrete between construction joints will be deposited upon or against partly set concrete,

after its initial set has taken place, or after 45 minutes of elapsed time during concrete placement.

(6) On bottom of members with severe congestion of reinforcement, deposit 1 inch layer of flowing concrete containing the specified high-range water-reducing admixture (superplasticizer). Successive concrete lifts may be a continuation of this concrete or concrete with a conventional slump.

(7) Concrete on metal deck:

(a) Concrete on metal deck shall be minimum thickness shown. Allow for deflection of steel beams and metal deck under the weight of wet concrete in calculating concrete quantities for slab.

1. The Contractor shall become familiar with deflection characteristics of structural frame to include proper amount of additional concrete due to beam/deck deflection.

e. Consolidation: Conform to ACI 309R. Immediately after depositing, spade concrete next to forms, work around reinforcement and into angles of forms, tamp lightly by hand, and compact with mechanical vibrator applied directly into concrete at approximately 18 inch intervals. Mechanical vibrator shall be power driven, hand operated type with minimum frequency of 5000 cycles per minute having an intensity sufficient to cause flow or settlement of concrete into place. Vibrate concrete to produce thorough compaction, complete embedment of reinforcement and concrete of uniform and maximum density without segregation of mix. Do not transport concrete in forms by vibration.

(1) Use of form vibration shall be approved only when concrete sections are too thin or too inaccessible for use of internal vibration.

(2) Carry on vibration continuously with placing of concrete. Do not insert vibrator into concrete that has begun to set.

3.8 HOT WEATHER

Follow the recommendations of ACI 305R or as specified to prevent problems in the manufacturing, placing, and curing of concrete that can adversely affect the properties and serviceability of the hardened concrete. Methods proposed for cooling materials and arrangements for protecting concrete shall be made in advance of concrete placement and approved by the Contracting Officer.

3.9 COLD WEATHER

Follow the recommendations of ACI 306R or as specified to prevent freezing of concrete and to permit concrete to gain strength properly. Use only the specified non-corrosive, non-chloride accelerator. Do not use calcium chloride, thiocyantes or admixtures containing more than 0.05 percent chloride ions. Methods proposed for heating materials and arrangements for protecting concrete shall be made in advance of concrete placement and approved by the Contracting Officer.

3.10 PROTECTION AND CURING

Conform to ACI 308R: Initial curing shall immediately follow the finishing operation. Protect exposed surfaces of concrete from premature drying, wash by rain and running water, wind, mechanical injury, and excessively hot or cold temperatures. Keep concrete not covered with membrane or other curing material continuously wet for at least 7 days after placing, except wet curing period for high-early-strength concrete shall be not less than 3 days. Keep wood forms continuously wet to prevent moisture loss until forms are removed. Cure exposed concrete surfaces as described below. Other curing methods may be used if approved by the Contracting Officer.

(1) Liquid curing and sealing compounds: Apply by power-driven spray or roller in accordance with the manufacturer's instructions. Apply immediately after finishing. Maximum coverage (400 square feet per gallon) on steel troweled surfaces and (300 square feet per gallon) on floated or broomed surfaces for the curing/sealing compound.

(2) Plastic sheets: Apply as soon as concrete has hardened sufficiently to prevent surface damage. Utilize widest practical width sheet and overlap adjacent sheets (2 inches). Tightly seal joints with tape.

(3) Paper: Utilize widest practical width paper and overlap adjacent sheets (2 inches). Tightly seal joints with sand, wood planks, pressure-sensitive tape, mastic or glue.

3.11 REMOVAL OF FORMS

a. Remove in a manner to assure complete safety of structure after the following conditions have been met.

(1) Where structure as a whole is supported on shores, forms for beams and girder sides, columns, and similar vertical structural members may be removed after 24 hours, provided concrete has hardened sufficiently to prevent surface damage and curing is continued without any lapse in time as specified for exposed surfaces.

(2) Take particular care in removing forms of architectural exposed concrete to insure surfaces are not marred or gouged, and that corners and arises are true, sharp and unbroken.

b. Control Test: Use to determine if the concrete has attained sufficient strength and curing to permit removal of supporting forms. Cylinders required for control tests taken in accordance with ASTM C172, molded in accordance with ASTM C31/C31M, and tested in accordance with ASTM C39/C39M. Control cylinders cured and protected in the same manner as the structure they represent. Supporting forms or shoring not removed until strength of control test cylinders have attained at least 70 percent of minimum 28-day compressive strength specified. Exercise care to assure that newly unsupported portions of structure are not subjected to heavy construction or material loading.

c. Reshoring: Reshoring is required if superimposed load plus dead load of the floor exceeds the capacity of the floor at the time of loading. Reshoring accomplished in accordance with ACI 347R at no additional cost to the Government.

3.12 CONCRETE SURFACE PREPARATION

a. Metal Removal: Unnecessary metal items cut back flush with face of concrete members.

b. Patching: Maintain curing and start patching as soon as forms are removed. Do not apply curing compounds to concrete surfaces requiring patching until patching is completed. Use cement mortar for patching of same composition as that used in concrete. Use white or gray Portland cement as necessary to obtain finish color matching surrounding concrete. Thoroughly clean areas to be patched. Cut out honeycombed or otherwise defective areas to solid concrete to a depth of not less than (1 inch). Cut edge perpendicular to surface of concrete. Saturate with water area to be patched, and at least 6 inches surrounding before placing patching mortar. Give area to be patched a brush coat of cement grout followed immediately by patching mortar. Cement grout composed of one part Portland cement, 1.5 parts fine sand, bonding admixture, and water at a 50:50 ratio, mix to achieve consistency of thick paint. Mix patching mortar approximately 1 hour before placing and remix occasionally during this period without addition of water. Compact mortar into place and screed slightly higher than surrounding surface. After initial shrinkage has occurred, finish to match color and texture of adjoining surfaces. Cure patches as specified for other concrete. Fill form tie holes which extend entirely through walls from unexposed face by means of a pressure gun or other suitable device to force mortar through wall. Wipe excess mortar off exposed face with a cloth.

c. Upon removal of forms, clean vertical concrete surface that is to receive bonded applied cementitious application with wire brushes or by sand blasting to remove unset material, laitance, and loose particles to expose aggregates to provide a clean, firm, granular surface for bond of applied finish.

3.13 CONCRETE FINISHES

a. Vertical and Overhead Surface Finishes:

(1) Unfinished areas: Vertical and overhead concrete surfaces exposed in pipe basements, elevator and dumbwaiter shafts, pipe spaces, pipe trenches, above suspended ceilings, manholes, and other unfinished areas will not require additional finishing.

(2) Interior and exterior exposed areas to be painted: Remove fins, burrs and similar projections on surfaces flush, and smooth by mechanical means approved by the Contracting Officer, and by rubbing lightly with a fine abrasive stone or hone. Use ample water during rubbing without working up a lather of mortar or changing texture of concrete.

(3) Interior and exterior exposed areas finished: Give a grout finish of uniform color and smooth finish treated as follows:

(a) After concrete has hardened and laitance, fins and burrs removed, scrub concrete with wire brushes. Clean stained concrete surfaces by use of a hone stone.

(b) Apply grout composed of one part of Portland cement, one part

fine sand, smaller than a (No. 30) sieve. Work grout into surface of concrete with cork floats or fiber brushes until all pits, and honeycombs are filled.

(c) After grout has hardened slightly, but while still plastic, scrape grout off with a sponge rubber float and, about 1 hour later, rub concrete vigorously with burlap to remove any excess grout remaining on surfaces.

(d) In hot, dry weather use a fog spray to keep grout wet during setting period. Complete finish of area in same day. Make limits of finished areas at natural breaks in wall surface. Leave no grout on concrete surface overnight.

(4) Textured: Finish as specified. Maximum quantity of patched area (2 square feet) in each (1000 square feet) of textured surface.

b. Slab Finishes:

(1) Monitoring and Adjustment: Provide continuous cycle of placement, measurement, evaluation and adjustment of procedures to produce slabs within specified tolerances. Monitor elevations of structural steel in key locations before and after concrete placement to establish typical deflection patterns for the structural steel. Determine elevations of cast-in-place slab soffits prior to removal of shores. Provide information to the Contracting Officer and floor consultant for evaluation and recommendations for subsequent placements.

(2) Set perimeter forms to serve as screed using either optical or laser instruments. For slabs on grade, wet screeds may be used to establish initial grade during strike-off, unless the Contracting Officer determines that the method is proving insufficient to meet required finish tolerances and directs use of rigid screed guides. Where wet screeds are allowed, they shall be placed using grade stakes set by optical or laser instruments. Use rigid screed guides, as opposed to wet screeds, to control strike-off elevation for all types of elevated (non slab-on-grade) slabs. Divide bays into halves or thirds by hard screeds. Adjust as necessary where monitoring of previous placements indicates unshored structural steel deflections to other than a level profile.

(3) Place slabs monolithically. Once slab placement commences, complete finishing operations within same day. Slope finished slab to floor drains where they occur, whether shown or not.

(4) Use straightedges specifically made for screeding, such as hollow magnesium straightedges or power strike-offs. Do not use pieces of dimensioned lumber. Strike off and screed slab to a true surface at required elevations. Use optical or laser instruments to check concrete finished surface grade after strike-off. Repeat strike-off as necessary. Complete screeding before any excess moisture or bleeding water is present on surface. Do not sprinkle dry cement on the surface.

(5) Immediately following screeding, and before any bleed water appears, use a 10 foot wide highway straightedge in a cutting and filling operation to achieve surface flatness. Do not use bull

floats or darbys, except that darbying may be allowed for narrow slabs and restricted spaces.

(6) Wait until water sheen disappears and surface stiffens before proceeding further. Do not perform subsequent operations until concrete will sustain foot pressure with maximum of 1/4 inch indentation.

(7) Scratch Finish: Finish base slab to receive a bonded applied cementitious application as indicated above, except that bull floats and darbys may be used. Thoroughly coarse wire broom within two hours after placing to roughen slab surface to insure a permanent bond between base slab and applied materials.

(8) Float Finish: Slabs to receive unbonded toppings, steel trowel finish, fill, mortar setting beds, or a built-up roof, and ramps, stair treads, platforms (interior and exterior), and equipment pads shall be floated to a smooth, dense uniform, sandy textured finish. During floating, while surface is still soft, check surface for flatness using a 10 foot highway straightedge. Correct high spots by cutting down and correct low spots by filling in with material of same composition as floor finish. Remove any surface projections and re-float to a uniform texture.

(9) Steel Trowel Finish: Concrete surfaces to receive resilient floor covering or carpet, monolithic floor slabs to be exposed to view in finished work, future floor roof slabs, applied toppings, and other interior surfaces for which no other finish is indicated. Steel trowel immediately following floating. During final troweling, tilt steel trowel at a slight angle and exert heavy pressure to compact cement paste and form a dense, smooth surface. Finished surface shall be smooth, free of trowel marks, and uniform in texture and appearance.

(10) Broom Finish: Finish exterior slabs, ramps, and stair treads with a bristle brush moistened with clear water after surfaces have been floated. Brush in a direction transverse to main traffic. Match texture approved by the Contracting Officer from sample panel.

(11) Finished slab flatness (FF) and levelness (FL) values comply with the following minimum requirements:

(a) Areas covered with carpeting, or not specified otherwise in b. below:

1. Slab on Grade:

a) Specified overall value FF 25/FL 20

b) Minimum local value FF 17/FL 15

2. Level suspended slabs (shored until after testing) and topping slabs:

a) Specified overall value FF 25/FL 20

b) Minimum local value FF 17/FL 15

3. Unshored suspended slabs:

a) Specified overall value FF 25

b) Minimum local value FF 17

4. Level tolerance such that 80 percent of all points fall within a 3/4 inch envelope (+3/8 inch, -3/8 inch) from the design elevation.

(b) Areas that will be exposed, receive thin-set tile or resilient flooring, or roof areas designed as future floors:

1. Slab on grade:

a) Specified overall value FF 36/FL 20

b) Minimum local value FF 24/FL 15

2. Level suspended slabs (shored until after testing) and topping slabs

a) Specified overall value FF 30/FL 20

b) Minimum local value FF 24/FL 15

3. Unshored suspended slabs:

a) Specified overall value FF 30

b) Minimum local value FF 24

4. Level tolerance such that 80 percent of all points fall within a 3/4 inch envelope (+3/8 inch, -3/8 inch) from the design elevation.

(c) "Specified overall value" is based on the composite of all measured values in a placement derived in accordance with ASTM E1155.

(d) "Minimum local value" (MLV) describes the flatness or levelness below which repair or replacement is required. MLV is based on the results of an individual placement and applies to a minimum local area. Minimum local area boundaries may not cross a construction joint or expansion joint. A minimum local area will be bounded by construction and/or control joints, or by column lines and/or half-column lines, whichever is smaller.

(12) Measurements

(a) Department of Veterans Affairs retained testing laboratory will take measurements as directed by the Contracting Officer, to verify compliance with FF, FL, and other finish requirements. Measurements will occur within 72 hours after completion of concrete placement (weekends and holidays excluded). Make measurements before shores or forms are removed to insure the "as-built" levelness is accurately assessed. Profile data for above characteristics may be collected using a laser level or any Type II apparatus (ASTM E1155, "profileograph" or "dipstick").

Contractor's surveyor shall establish reference elevations to be used by Department of Veterans Affairs retained testing laboratory.

(b) Contractor not experienced in using FF and FL criteria is encouraged to retain the services of a floor consultant to assist with recommendations concerning adjustments to slab thicknesses, finishing techniques, and procedures on measurements of the finish as it progresses in order to achieve the specific flatness and levelness numbers.

(13) Acceptance/ Rejection:

(a) If individual slab section measures less than either of specified minimum local FF/FL numbers, that section shall be rejected and remedial measures shall be required. Sectional boundaries may be set at construction and contraction (control) joints, and not smaller than one-half bay.

(b) If composite value of entire slab installation, combination of all local results, measures less than either of specified overall FF/FL numbers, then whole slab shall be rejected and remedial measures shall be required.

(14) Remedial Measures for Rejected Slabs: Correct rejected slab areas by grinding, planing, surface repair with underlayment compound or repair topping, retopping, or removal and replacement of entire rejected slab areas, as directed by the Contracting Officer, until a slab finish constructed within specified tolerances is accepted.

3.14 SURFACE TREATMENTS

a. Use on exposed concrete floors and concrete floors to receive carpeting .

b. Liquid Densifier/Sealer: Apply in accordance with manufacturer's directions just prior to completion of construction.

c. Non-Slip Finish: Except where safety nosing and tread coverings are shown, apply non-slip abrasive aggregate to treads and platforms of concrete steps and stairs, and to surfaces of exterior concrete ramps and platforms. Broadcast aggregate uniformly over concrete surface at rate of application of (7.5 percent per square foot) of area. Trowel concrete surface to smooth dense finish. After curing, rub treated surface with abrasive brick and water to slightly expose abrasive aggregate.

3.15 APPLIED TOPPING

a. Separate concrete topping on floor base slab of thickness and strength shown. Topping mix shall have a maximum slump of 8 inches for concrete containing a high-range water-reducing admixture (superplasticizer) and 4 inches for conventional mix. Neatly bevel or slope at door openings and at slabs adjoining spaces not receiving an applied finish.

b. Placing: Place continuously until entire section is complete, struck off with straightedge, leveled with a highway straightedge or highway bull float, floated and troweled by machine to a hard dense finish.

Slope to floor drains as required. Do not start floating until free water has disappeared and no water sheen is visible. Allow drying of surface moisture naturally. Do not hasten by "dusting" with cement or sand.

3.16 RETAINING WALLS

a. Use air-entrained concrete.

b. Expansion and contraction joints, waterstops, weep holes, reinforcement and railing sleeves installed and constructed as shown.

c. Exposed surfaces finished to match adjacent concrete surfaces, new or existing.

d. Place porous backfill as shown.

3.17 PRECAST CONCRETE ITEMS

Precast concrete items, not specified elsewhere. Cast using (3000 psi) air-entrained concrete to shapes and dimensions shown. Finish to match corresponding adjacent concrete surfaces. Reinforce with steel for safe handling and erection.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 04 - MASONRY

SECTION 04 01 00

MAINTENANCE OF MASONRY

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 FIELD CONDITIONS
1.10 WARRANTY

PART 2 PRODUCTS

2.1 MATERIALS
2.2 PRODUCTS - GENERAL
2.3 REPLACEMENT MASONRY UNITS
2.4 MIXES
2.5 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 EXISTING MORTAR JOINTS
3.3 TUCK POINTING
3.4 MASONRY UNIT REPLACEMENT
3.5 JOINT TOOLING
3.6 CLEANING

-- End of Section Table of Contents --

SECTION 04 01 00
1/21

MAINTENANCE OF MASONRY

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

a. Repointing existing damaged masonry joints.

b. Replacing existing damaged masonry units.

1.2 RELATED WORK

Mortars for new masonry.

1.3 REFERENCES

Comply with references to extent specified in this section.

ASTM INTERNATIONAL (ASTM)

ASTM C67/C67M	(2024) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
ASTM C144	(2024) Standard Specification for Aggregate for Masonry Mortar
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C207	(2024) Standard Specification for Hydrated Lime for Masonry Purposes
ASTM C216	(2024) Standard Specification for Facing Brick (Solid Masonry Units Made from Clay or Shale)
ASTM C270	(2025) Standard Specification for Mortar for Unit Masonry
ASTM C295/C295M	(2019) Standard Guide for Petrographic Examination of Aggregates for Concrete

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Testing; G,

Mockups; G,

Installer Qualifications; G,

SD-03 Product Data

Description Of Each Product

Unit Replacement

SD-04 Samples

Pointing Mortar

SD-06 Test Reports

Preconstruction Test Results Of Existing Masonry Mortar And Units

Recommended Mortar Mix And Mortar Materials Sources

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, Submittal Procedures .

b. Manufacturer's Literature and Data:

(1) Description of each product.

c. Samples: Pointing Mortar: Molded, 6 inches long for each type, texture, and color.

d. Test reports:

(1) Preconstruction test results of existing masonry mortar and units.

(2) Recommended mortar mix and mortar materials sources.

1.6 QUALITY ASSURANCE

a. Installer Qualifications: Documented experience in completion of work, similar in design, material, and extent specified.

b. Preconstruction Testing:

(1) Existing Brick: according to ASTM C67/C67M.

(2) Existing Mortar: according to ASTM C295/C295M. Recommend mortar mix compatible with existing and mortar material sources required to match existing color and texture .

c. Mockups: Prepare mockup in size indicated on Drawings, demonstrating quality and aesthetics of tuck pointing masonry unit replacement and cleaning .

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store materials covered, protected from weather, and elevated above grade. Prevent contamination of aggregates.

b. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

Environment:

a. Cold Weather Requirements: Maintain mortar ingredients and substrate within temperature range between 40 degrees F and 120 degrees F when outside temperature is less than 40 degrees F.

b. Hot Weather Requirements: Protect mortar-joint from evaporation of moisture from mortar material. When required, provide adequately shaded work area.

1.10 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 MATERIALS

Mortar Components:

a. Hydrated Lime: ASTM C207, Type S.

b. Aggregate: ASTM C144.

c. Portland Cement: ASTM C150/C150M, Type I.

d. Water: Potable, free of substances that are detrimental to grout, masonry, and metal.

2.2 PRODUCTS - GENERAL

a. Basis of Design: See Schedule of Finishes in the drawing set and appendix binder..

b. Provide each product from one manufacturer and from one production run .

2.3 REPLACEMENT MASONRY UNITS

a. Face Brick:

(1) ASTM C216, matching existing .

(2) Efflorescence: Rated slight efflorescent when tested according to

ASTM C67/C67M.

b. Other Masonry Units: Match existing.

2.4 MIXES

Tuck Pointing Mortar: ASTM C270;

a. Type O .

2.5 ACCESSORIES

Soapless, non-acidic, detergent, specially prepared for cleaning brick stone, and/or concrete masonry.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage. Protect from mortar droppings and cleaning operations.

c. Remove existing fixtures and fittings concealing masonry joints to permit repointing and repair.

3.2 EXISTING MORTAR JOINTS

a. Cut out existing bed and head mortar joints, to uniform depth of 3/4 inches, or to sound mortar without damaging edges and faces of existing masonry units to remain.

b. Remove dust and debris from joints. Do not rinse when temperature is below freezing.

3.3 TUCK POINTING

a. Dampen joints immediately before tuck pointing. Allow masonry units to absorb surface water.

b. Tightly pack tuck pointing mortar into joints in thin layers, 1/4 inch thick, maximum.

c. Allow layer to become slightly hardened before applying next layer.

d. Pack final layer flush with surfaces of masonry units.

3.4 MASONRY UNIT REPLACEMENT

a. Cut out mortar joints surrounding masonry units requiring replacement.

(1) Remove existing masonry units creating opening for replacement masonry unit installation.

(2) Remove mortar, dust, and debris from opening perimeter surfaces.

(3) Prevent debris from falling into cavity.

b. Dampen surfaces of surrounding existing masonry before installing replacement masonry units.

(1) Allow existing masonry to absorb surface moisture before installing replacement units.

(2) Butter contact surfaces of existing masonry and replacement masonry units with mortar.

(3) Center replacement masonry units in opening and press into position.

(4) Remove excess mortar.

(5) Tuck point replacement masonry units to ensure full head and bed joints.

3.5 JOINT TOOLING

a. Tool repointed and replaced masonry joints when mortar becomes slightly hardened.

b. Produce smooth, compacted, joint matching existing .

3.6 CLEANING

a. Remove mortar splatter from exposed surfaces immediately.

b. Clean exposed masonry surfaces on completion.

c. Remove mortar droppings and other foreign substances from wall surfaces.

d. Wet surfaces with clean water.

e. Wash with cleaning agent.

f. Brush masonry surfaces with stiff fiber brushes while washing.

g. Immediately after washing, rinse with clean water. Remove traces of detergent, foreign streaks or stains.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 04 - MASONRY

SECTION 04 05 13

MASONRY MORTARING

2/17

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED REQUIREMENTS
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 WARRANTY

PART 2 PRODUCTS

2.1 MATERIALS
2.2 PRODUCTS - GENERAL
2.3 MIXES

PART 3 EXECUTION

3.1 PREPARATION
3.2 MIXING
3.3 MORTARING
3.4 FIELD QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 04 05 13
2/17

MASONRY MORTARING

PART 1 GENERAL

1.1 SUMMARY

Section Includes: Masonry mortar installed by other masonry sections.

1.2 RELATED REQUIREMENTS

a. Mortar used in Section:

(1) Section 04 01 00, MAINTENANCE OF MASONRY.

(2) Section 04 20 00, UNIT MASONRY.

b. Mortar Color: See Schedule of Finishes in the drawing set and appendix binder..

1.3 REFERENCES

Comply with references to extent specified in this section.

ASTM INTERNATIONAL (ASTM)

ASTM C40/C40M	(2020) Standard Test Method for Organic Impurities in Fine Aggregates for Concrete
ASTM C91/C91M	(2018) Standard Specification for Masonry Cement
ASTM C144	(2024) Standard Specification for Aggregate for Masonry Mortar
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C207	(2024) Standard Specification for Hydrated Lime for Masonry Purposes
ASTM C270	(2025) Standard Specification for Mortar for Unit Masonry
ASTM C595/C595M	(2024) Standard Specification for Blended Hydraulic Cements
ASTM C780	(2025) Standard Test Method for Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry
ASTM C979/C979M	(2024) Standard Specification for Pigments for Integrally Colored Concrete

ASTM C1329/C1329M	(2016) Standard Specification for Mortar Cement

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G,

Preconstruction Testing; G,

SD-03 Product Data

Manufacturer's Literature And Data; G,

SD-06 Test Reports

Test Reports; G,

Test Mortar And Materials Specified; G,

SD-07 Certificates

Certificates; G,

1.5 SUBMITTAL INSTRUCTIONS

a. Manufacturer's Literature and Data: Description of each product.

b. Test Reports: Certify each product complies with specifications.

(1) Mortar.

(2) Admixtures.

c. Certificates: Certify each product complies with specifications.

(1) Portland cement.

(2) Masonry cement.

(3) Mortar cement.

(4) Hydrated lime.

(5) Fine aggregate.

(6) Color admixture.

d. Qualifications: Substantiate qualifications comply with specifications. Testing laboratory.

1.6 QUALITY ASSURANCE

Preconstruction Testing:

a. Engage independent testing laboratory to tests and submit reports. Deliver samples to laboratory in number and quantity required for testing.

b. Test mortar and materials specified.

c. Mortar:

(1) Test for compressive strength and water retention according to ASTM C270.

(2) Minimum Mortar compressive strengths 28 days:

(a) Type M: 2,500 psi.

(b) Type S: 1,800 psi.

(c) Type N: 750 psi.

d. Non Staining Cement: Test for water soluble alkali. Water Soluble Alkali: Maximum 0.03 percent.

e. Sand: Test for deleterious substances, organic impurities, soundness and grading.

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store masonry materials under waterproof covers on planking clear of ground. Protect loose, bulk materials from contamination.

b. Protect products from damage during handling and construction operations.

1.9 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 MATERIALS

a. Hydrated Lime: ASTM C207, Type S.

b. Aggregate for Masonry Mortar: ASTM C144 and as follows:

(1) Light colored sand for mortar for laying face brick.

(2) White plastering sand meeting sieve analysis for mortar joints for pointing except that 100 percent passes No. 8 sieve, and maximum 5 percent retained on No. 16 sieve.

(3) Test sand for color value according to ASTM C40/C40M. Sand producing color darker than specified standard is unacceptable.

c. Blended Hydraulic Cement: ASTM C595/C595M, Type IS, IP.

d. Masonry Cement: ASTM C91/C91M. Type N, S, Or M. Use white masonry cement whenever white mortar is specified.

e. Mortar Cement: ASTM C1329/C1329M, Type N, S or M.

f. Portland Cement: ASTM C150/C150M, Type I. Use white Portland cement wherever white mortar is specified.

g. Pigments: ASTM C979/C979M; inorganic, inert, mineral pigments only, unaffected by atmospheric conditions, nonfading, alkali resistant, and water insoluble.

h. Water: Potable, free of substances that are detrimental to mortar, masonry, and metal.

2.2 PRODUCTS - GENERAL

a. Basis of Design: As indicated on drawings.

b. Provide each product from one manufacturer and from one production run .

2.3 MIXES

a. Pointing Mortar for New Work:

(1) For Cast Stone or Precast Concrete: Proportion by volume; one part white Portland cement, two parts white sand, and 1/5 part hydrated lime.

(2) Pointing Mortar for Glazed Structural Facing Tile: Proportion by volume: One part white Portland cement, two parts of graded white sand passing Number 50 sieve, and 1/8 part hydrated lime.

b. Tuck Pointing Mortar for Repair Work: Tuck pointing mortar specified in Section 04 01 00, MAINTENANCE OF MASONRY.

c. Masonry Mortar: ASTM C270. Admixtures:

(1) Do not use mortar admixtures, and color admixtures unless approved by Contracting Officer's Representative.

(2) Do not use antifreeze compounds.

d. Colored Mortar:

(1) Maintain uniform mortar color for exposed work, throughout.

(2) Match mortar color in approved sample or sample panel specified

in Section 04 20 00, UNIT MASONRY.

(3) Alteration Work Mortar Color: Match existing mortar unless specified otherwise on drawings.

e. Color Admixtures:

(1) Proportion as specified by manufacturer.

(2) For color, see drawings.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

3.2 MIXING

a. Measure ingredients by volume using known capacity container.

b. Mix for three to five minutes in a mechanically operated mortar mixer.

c. Mix water with dry ingredients in sufficient amount to provide a workable mixture which will adhere to vertical surfaces of masonry units.

d. Mortar Stiffened Because of Water Loss Through Evaporation:

(1) Re-temper by adding water to restore to proper consistency and workability.

(2) Discard mortar reaching initial set or unused within two hours of mixing.

e. Pointing Mortar:

(1) Mix dry ingredients with enough water to produce damp mixture of workable consistency retaining shape when formed into ball.

(2) Allow mortar to stand in dampened condition for 60 to 90 minutes.

(3) Add water to bring mortar to a workable consistency before use.

3.3 MORTARING

a. Type S Mortar: Use for masonry containing vertical reinforcing bars (non-engineered), masonry below grade and engineered reinforced unit masonry work.

b. Brick Veneer Over Frame Back Up Walls: Use Type S Portland cement-lime mortar.

c. Type N Mortar: Use for other masonry work.

d. Type N Mortar: Use for pointing items and tuck pointing specified.

3.4 FIELD QUALITY CONTROL

Field Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES. Take and test samples during progress of work according to ASTM C780.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 04 - MASONRY

SECTION 04 20 00

UNIT MASONRY

8/17

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED REQUIREMENTS
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 FIELD CONDITIONS
1.10 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 PRODUCTS - GENERAL
2.3 UNIT MASONRY PRODUCTS
2.4 ANCHORS, TIES, AND REINFORCEMENT
2.5 ACCESSORIES

PART 3 EXECUTION

3.1 INSTALLATION - GENERAL
3.2 INSTALLATION - ANCHORAGE
3.3 INSTALLATION - REINFORCEMENT
3.4 INSTALLATION - BRICK EXPANSION AND CMU CONTROL JOINTS
3.5 INSTALLATION - BUILDING EXPANSION AND SEISMIC JOINTS
3.6 INSTALLATION - ISOLATION JOINT
3.7 INSTALLATION - BRICKWORK
3.8 INSTALLATION - CONCRETE MASONRY UNITS
3.9 INSTALLATION - GLAZED STRUCTURAL FACING TILE
3.10 POINTING
3.11 GROUTING
3.12 PLACING REINFORCEMENT
3.13 INSTALLATION OF REINFORCED BRICK MASONRY
3.14 INSTALLATION OF REINFORCED CONCRETE UNIT MASONRY
3.15 CONSTRUCTION TOLERANCES
3.16 CLEANING AND REPAIR
3.17 FIELD QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 04 20 00
8/17

UNIT MASONRY

PART 1 GENERAL

1.1 SUMMARY

Section Includes: Concrete masonry unit (CMU) assemblies for:

a. Exterior walls.

b. Interior walls and partitions .

1.2 RELATED REQUIREMENTS

a. Sealants and Sealant Installation: Section 07 92 00, JOINT SEALANTS.

1.3 REFERENCES

a. Comply with references to extent specified in this Section.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 315	(2018) Guide to Presenting Reinforcing Steel Design Details
ACI 530.1/ASCE 6/TMS 602	(1992) Specifications for Masonry Structures

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A951/A951M	(2011) Standard Specification for Steel Wire for Masonry Joint Reinforcement
ASTM A1064/A1064M	(2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C55	(2023) Standard Specification for Concrete Building Brick
ASTM C62	(2023) Standard Specification for Building Brick (Solid Masonry Units Made from Clay or Shale)
ASTM C67/C67M	(2024) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
ASTM C90	(2024a) Standard Specification for

Loadbearing Concrete Masonry Units

ASTM C126 (2022) Standard Specification for Ceramic Glazed Structural Clay Facing Tile, Facing Brick, and Solid Masonry Units

ASTM C216 (2024) Standard Specification for Facing Brick (Solid Masonry Units Made from Clay or Shale)

ASTM C612 (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation

ASTM C744 (2021) Standard Specification for Prefaced Concrete and Calcium Silicate Masonry Units

ASTM D1056 (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber

ASTM D2240 (2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness

ASTM F1667/F1667M (2021a) Standard Specification for Driven Fasteners: Nails, Spikes, and Staples

AMERICAN WELDING SOCIETY (AWS)

AWS D1.4/D1.4M (2018) Structural Welding Code - Reinforcing Steel

BRICK INDUSTRY ASSOCIATION (BIA)

BIA TN 11B (1988) Guide Specifications for Brick Masonry, Part 3

U.S. DEPARTMENT OF DEFENSE (DOD)

FF-S-107C (2013) Screws, Tapping and Drive

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welders And Welding Procedures Qualifications; G, AE

Mockups; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Test Reports; G, AE

SD-07 Certificates

Certificates; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Submittal Drawings:

(1) Fabrication, bending, and placement of reinforcing bars. Comply with ACI 315. Show bar schedules, diagrams of bent bars, stirrup spacing, lateral ties and other arrangements and assemblies.

(2) Special masonry shapes, profiles, and placement.

(3) Masonry units for typical window and door openings, and, for special conditions as affected by structural conditions.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

d. Samples:

(1) Face brick: Sample panel, 8 inches by 16 inches,) showing full color range and texture of bricks, bond, and proposed mortar joints.

(2) Ceramic Glazed Facing Brick: Sample panel, 8 inches by 16 inches,) showing full color range and texture of bricks, bond, and proposed mortar joints.

(3) Concrete masonry units, when exposed in finish work.

(4) Anchors and Ties: Each type.

(5) Joint Reinforcing: 48 inches long each type.

(6) Glazed Structural Facing Tile: Clipped panels (triplicate) of four wall units with base units, showing color range, each color and texture.

e. Sustainable Construction Submittals:

Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

f. Test reports: Certify products comply with specifications. Ceramic

glazed facing brick.

g. Certificates: Certify products comply with specifications.

(1) Face brick.

(2) Solid and load-bearing concrete masonry units, including fire-resistant rated units.

(3) Ceramic glazed facing brick.

1.6 QUALITY ASSURANCE

a. Welders and Welding Procedures Qualifications: AWS D1.4/D1.4M.

b. Mockups:

(1) Before starting masonry, build a mockup panel minimum 6 feet by 6 feet) with 24 inch 90 degree return for outside corner.

(a) Use masonry units from random cubes of units delivered on site.

(b) Include structural backup, reinforcing, ties, and anchors.

(2) Mockup panel approved by Contracting Officer's Representative set workmanship and aesthetic quality for masonry work.

(3) Clean sample panel to test cleaning methods.

(4) Remove mockup panel when directed by Contracting Officer's Representative.

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store products above grade, protected from contamination.

b. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

Comply with ACI 530.1/ASCE 6/TMS 602.

1.10 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE

Prepare submittal documents including design calculations and drawings signed and sealed by registered design professional, licensed in state where work is located.

2.2 PRODUCTS - GENERAL

a. Basis of Design: See Schedule of Finishes in the drawing set and appendix binder.

b. Provide each product from one manufacturer .

c. Sustainable Construction Requirements:

(1) Brick Concrete Masonry Unit Recycled Content: 0-20% total recycled content, minimum. Select products with recycled content to achieve overall Project recycled content requirement.

(a) Post-Consumer Recycled Content: 0%, minimum.

(b) Pre-Consumer Recycled Content: 0-20%, minimum.

(2) Steel Recycled Content: 30 percent total recycled content, minimum.

2.3 UNIT MASONRY PRODUCTS

a. Brick:

(1) Face Brick:

(a) ASTM C216, Grade SW, Type FBS.

(b) Brick when tested according to ASTM C67/C67M: Classified slightly efflorescent or better.

(c) Size:

1. Modular.

2. Thin Brick: 1/2 inch thick with angle shapes for corners.

(2) Building Brick: ASTM C62, Grade MW for backup and interior work; Grade SW where in contact with earth.

(3) Ceramic Glazed Facing Brick: ASTM C126.

(4) One Face Exposed: Grade S, Type I.

(5) Two Faces Exposed: Grade S, Type II.

b. Concrete Masonry Units (CMU):

(1) Hollow and Solid Load-Bearing Concrete Masonry Units: ASTM C90.

(a) Unit Weight: Normal weight.

(b) Fire rated units for fire rated partitions.

(2) Sizes: Modular, 8 inches by 16 inches nominal face dimension; thickness as indicated on drawings.

(3) For molded faces used as a finished surface, use concrete masonry units with uniform fine to medium surface texture unless specified otherwise.

(4) Use bullnose concrete masonry units at corners exposed in finished work with 1 inch minimum radius rounded vertical exterior corners (bullnose units).

(5) Customized Units:

(a) Sound-Absorbing Units:

1. Vertical slots in face to core areas.

2. Acoustical absorption insert: Mineral fiber and metal septum, providing unit with NRC rating of 0.70.

(b) Split-Face Units:

1. Split-Rib Units: Rib shapes as indicated on drawings on exposed faces.

2. Ground Face Units: Ground Honed finish on exposed faces.

(c) Glazed Face Units: Facing conform to ASTM C744.

c. Concrete Brick: ASTM C55.

(2) Size: 8W, thickness as shown.

2.4 ANCHORS, TIES, AND REINFORCEMENT

a. Steel Reinforcing Bars: ASTM A615/A615M; Grade 60, deformed bars.

b. Joint Reinforcement:

(1) Form from wire complying with ASTM A951/A951M.

(2) Hot dipped galvanized after fabrication.

(3) Width of joint reinforcement 1.6 inches less than nominal thickness of masonry wall or partition.

(4) Cross wires welded to longitudinal wires.

(5) Joint reinforcement minimum 10 feet long, factory cut.

(6) Joint reinforcement with crimp formed drip is not acceptable.

(7) Maximum spacing of cross wires 16 inch to longitudinal wires.

(8) Ladder Design:

(a) Longitudinal wires deformed 0.20 inch diameter wire .

(b) Cross wires 0.16 inch diameter .

(9) Trussed Design:

(a) Longitudinal and cross wires minimum 0.16 inch nominal diameter.

(b) Longitudinal wires deformed.

(10) Multiple Wythes and Cavity Wall Ties:

(a) Longitudinal wires 0.16 inch, two in each wythe with ladder truss wires 0.16 inch overlay, welded to each longitudinal wire.

(b) Longitudinal wires 0.16 inch with U shape 0.16 inch rectangular ties extending into other wythe minimum 3 inches spaced 16 inches on center. Adjustable type with U shape tie designed to receive 0.16 inch pintle projecting into other wythe 3 inches min.

c. Adjustable Veneer Anchor for Framed Walls:

(1) Two piece, adjustable anchor and tie.

(2) Anchor and tie may be either loop or angle type; provide only one type throughout.

(3) Loop Type:

(a) Anchor: Screw-on galvanized steel anchor strap 0.11 inch by 3/4 inch wide by 9 inches long, with 0.35 inch offset and 4 inch adjustment. Provide 0.20 inch hole at each end for fasteners.

(b) Ties: Triangular tie, fabricated of 0.20 inch diameter galvanized cold drawn steel wire. Ties long enough to engage anchor and be embedded minimum 2 inches into bed joint of masonry veneer.

(4) Angle Type:

(a) Anchor: Minimum 16 gage thick galvanized steel angle shaped anchor strap. Provide hole in vertical leg for fastener. Provide hole near end of outstanding leg to suit upstanding portion of tie.

(b) Tie: Fabricate from 0.20 inch diameter galvanized cold drawn steel wire. Form "L" shape to be embedded minimum 2 inches into the bed joint of masonry veneer and provide upstanding leg to fit

through hole in anchor and be long enough to allow 2 inches of vertical adjustment.

d. Dovetail Anchors:

(1) Corrugated steel dovetail anchors formed of 0.06 inch thick by 1 inch wide galvanized steel, 3-1/2 inches long where used to anchor 4 inch nominal thick masonry units, 5-1/2 inches long for masonry units more than 4 inches thick.

(2) Triangular wire dovetail anchor 4 inch wide formed of 9 gage steel wire with galvanized steel dovetail insert. Anchor length to extend minimum 3 inches into masonry, 1 inch into 1-1/2 inch thick units.

(3) Form dovetail anchor slots from 0.02 inch thick galvanized steel (with felt or fiber filler).

e. Individual Ties:

(1) Rectangular ties: Form from 3/16 inch diameter galvanized steel rod to rectangular shape minimum 2 inches wide by sufficient length for ends of ties to extend within 1 inch of each face of wall. Ties that are crimped to form drip are not acceptable.

(2) Adjustable Cavity Wall Ties:

(a) Adjustable wall ties may be furnished at Contractor's option.

(b) Two piece type permitting up to 1-1/2 inch adjustment.

(c) Form ties from 3/16 inch diameter galvanized steel wire.

(d) Form one piece to rectangular shape 4-1/8 inches wide by length required to extend into bed joint 2 inches.

(e) Form other piece to 3 inch long by 3 inch wide shape, having 3 inch long bent section for engaging 4-1/8 inch wide piece to form adjustable connection.

f. Wall Ties, (Mesh or Wire):

(1) Mesh wall ties formed of ASTM A1064/A1064M, W0.5, 0.08 inch galvanized steel wire 1/2 inch by 1/2 inch mesh, 3 inches wide by 8 inches long.

(2) Rectangular wire wall ties formed of W1.4, 0.12 inch galvanized steel wire 2 inches wide by 8 inches long.

h. Adjustable Steel Column Anchor:

(1) Two piece anchor consisting of a 1/4 inch diameter steel rod to be welded to steel with offset ends, rod to permit 4 inch vertical adjustment of wire anchor.

(2) Triangular shaped wire anchor 4 inches wide formed from 3/16 inch

diameter galvanized wire, to extend minimum 3 inches into joints of masonry.

i. Adjustable Steel Beam Anchor:

(1) Z or C type steel strap, 1 1/4 inches wide, 1/8 inch thick.

(2) Flange hook minimum 1 1/2 inches long.

(3) Length to embed in masonry minimum 2 inches in 4 inch nominal thick masonry and 4 inches in thicker masonry.

(4) Bend masonry end minimum 1 1/2 inches.

j. Ridge Wall Anchors:

(1) Form from galvanized steel minimum wide by 3/16 inch thick by long, plus 2 inch bends.

(2) Other lengths as indicated on drawings.

2.5 ACCESSORIES

a. Shear Keys:

(1) Solid extruded cross-shaped section of rubber, neoprene, or polyvinyl chloride, with durometer hardness of approximately 80 when tested according to ASTM D2240, and minimum shear strength of 500 psi.

(2) Shear Key Dimensions: Nominal 2-3/4 inches by 5/16 inch for long flange, and 1-1/2 inches by 5/8 inch for short flange.

b. Weeps:

(1) Weep Hole Wicks: Glass fiber ropes, 3/8 inch minimum diameter, 12 inches long.

(2) Weep Tubing: Round, polyethylene, 3/8 inch diameter, 4 inches long.

(3) Weep Hole: Flexible PVC louvered configuration with rectangular closure strip at top.

c. Cavity Drain Material: Open mesh polyester sheets or strips to prevent mortar droppings from clogging the cavity.

d. Preformed Compressible Joint Filler:

(1) Thickness and depth to fill joint.

(2) Closed Cell Neoprene: ASTM D1056, Type 2, Class A, Grade 1, B2F1.

(3) Non-Combustible Type: ASTM C612, Type 5, Max. Temp. 1800 degrees F.

e. Box Board:

(1) Mineral Fiber Board: ASTM C612, Type 1.

(2) 1 inch thickness.

(3) Other spacing material having similar characteristics is acceptable subject to Contracting Officer's Representative's approval.

f. Masonry Cleaner:

(1) Detergent type cleaner selected for each type masonry.

(2) Acid cleaners are not acceptable.

(3) Use soapless type specially prepared for cleaning brick or concrete masonry as appropriate.

g. Fasteners:

(1) Concrete Nails: ASTM F1667/F1667M, Type I, Style 11, 3/4 inch minimum length.

(2) Masonry Nails: ASTM F1667/F1667M, Type I, Style 17, 3/4 inch minimum length.

(3) Screws: FF-S-107C, Type A, AB, SF thread forming or cutting.

h. Welding Materials: AWS D1.4/D1.4M, type to suit application.

PART 3 EXECUTION

3.1 INSTALLATION - GENERAL

a. Install products according to manufacturer's instructions and approved submittal drawings . When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.

b. Keep finish work free from mortar smears or spatters, and leave neat and clean.

c. Wall Openings:

(1) Fill hollow metal frames built into masonry walls and partitions solid with mortar as laying of masonry progresses.

(2) When items are not available when walls are built, prepare openings for subsequent installation.

d. Tooling Joints:

(1) Do not tool until mortar has stiffened enough to retain thumb print when thumb is pressed against mortar.

(2) Tool while mortar is soft enough to be compressed into joints and not raked out.

(3) Finish joints in exterior face masonry work with jointing tool, and provide smooth, water-tight concave joint unless specified otherwise.

(4) Tool Exposed interior joints in finish work concave unless specified otherwise.

e. Partition Height:

(1) Extend partitions minimum 4 inches above suspended ceiling or to overhead construction where no ceiling occurs.

(2) Extend following partitions to overhead construction.

(a) Full height partitions, and fire partitions and smoke partitions indicated on drawings.

(b) Both walls at expansion joints.

(c) Corridor walls.

(d) Walls at stairway and stair halls, elevators, dumbwaiters, trash and laundry chute shafts, and other vertical shafts.

(e) Walls at refrigerator space.

(f) Reinforced masonry partitions.

(3) Extend finished masonry partitions minimum 4 inches above suspended ceiling and continue with concrete masonry units or structural clay tile to overhead construction:

f. Lintels:

(1) Lintels are not required for openings less than 40 inches wide that have hollow metal frames.

(2) Openings 41 inches wide to 63 inches wide without structural steel lintel or frames, require lintel formed of concrete masonry lintel or bond beam units filled with grout and reinforced with one No. 16 (No. 5) rod top and bottom for each 4 inches of nominal thickness unless shown otherwise.

(3) Precast concrete lintels of 3,000 psi concrete, same thickness as partition, and with one No. 16 (No. 5) deformed bar top and bottom for each 4 inches of nominal thickness, is acceptable in lieu of reinforced CMU masonry lintels.

(4) Use steel lintels, for openings greater than 63 inches wide, brick masonry openings, and elevator openings unless shown otherwise.

(5) Doors having overhead concealed door closers require steel lintel, and pocket for closer box.

(6) Lintel Bearing Length: Minimum 4 inches at both ends.

(7) Build masonry openings or arches over wood or metal centering and supports when steel lintels are not used.

g. Wall, Furring, and Partition Units:

(1) Lay out field units to provide one-half running bond, unless indicated otherwise.

(2) Align head joints of alternate vertical courses.

(3) At sides of openings, balance head joints in each course on vertical center lines of openings.

(4) Minimum Masonry Unit Length: 4 inches.

(5) On interior partitions provide 1/4 inch open joint for caulking between existing construction, exterior walls, concrete work, and abutting masonry partitions.

(6) Use minimum 4 inches nominal thick masonry for free standing furring, unless indicated otherwise.

(7) Do not abut existing plastered surfaces except suspended ceilings with new masonry partitions.

h. Use minimum 4 inches nominal thick masonry for fireproofing steel columns unless indicated otherwise.

i. Before connecting new masonry with previously laid masonry, remove loosened masonry or mortar, and clean and wet work in place as specified under wetting.

j. When new masonry partitions start on existing floors, machine cut existing floor finish material down to concrete surface.

k. Structural Steel Encased in Masonry:

(1) Where structural steel is encased in masonry and voids between steel and masonry are filled with mortar, provide minimum 1 inch mortar free expansion space between masonry and steel by applying box board material to steel before masonry is laid.

(2) Do not install spacing material where steel is bearing on masonry or masonry is bearing on steel.

l. Chases:

(1) Do not install chases in masonry walls and partitions exposed to view in finished work, including painted or coated finishes on masonry.

(2) Masonry 4 inch nominal thick may have electrical conduits 1 inch or less in diameter when covered with soaps, or other finishes.

(3) Fill recess chases after installation of conduit, with mortar and finish flush.

(4) When pipes or conduits, or both occur in hollow masonry unit partitions retain minimum one web of hollow masonry units.

m. Wetting and Wetting Test:

(1) Test and wet brick and clay tile according to BIA TN 11B.

(2) Do not wet concrete masonry units or glazed structural facing tile before laying.

n. Temporary Formwork: Provide formwork and shores as required for temporary support of reinforced masonry elements.

o. Construct formwork to conform to shape, line and dimensions indicated on drawings. Make sufficiently tight to prevent mortar, grout, or concrete leakage. Brace, tie and support formwork as required to maintain position and shape during construction and curing of reinforced masonry.

p. Do not remove forms and shores until reinforced masonry members have hardened sufficiently to carry their own weight and other reasonable temporary construction loads.

q Minimum Curing Times Before Removing Shores and Forms:

(1) Girders and Beams: 10 days.

(2) Slabs: Seven days.

(3) Reinforced Masonry Soffits: Seven days.

3.2 INSTALLATION - ANCHORAGE

a. Veneer to Framed Walls:

(1) Install adjustable veneer anchors.

(2) Fasten anchor to stud through sheathing with self-drilling and tapping screw, one at both ends of loop type anchor.

(3) Space anchors maximum 16 inches on center vertically at each stud.

b. Veneer to Concrete Walls:

(1) Install dovetail slots in concrete vertically at 16 inches on centers.

(2) Locate dovetail anchors at 16 inch maximum vertical intervals.

(3) Anchor new masonry facing to existing concrete with adjustable cavity wall ties spaced at 16 inches maximum vertical intervals, and at 16 inches maximum horizontal intervals. Fasten ties to concrete with power actuated fasteners or concrete nails.

c. Masonry Facing to Backup and Cavity Wall Ties:

(1) Use individual ties for new work.

(2) Stagger ties in alternate courses, and space at 16 inches maximum vertically, and 16 inches horizontally.

(3) At openings, provide additional ties spaced maximum 36 inches apart vertically around perimeter of opening, and within 12 inches from edge of opening.

(4) Anchor new masonry facing to existing masonry with adjustable

cavity wall ties spaced at 16 inch maximum vertical intervals and at every second masonry unit horizontally. Fasten ties to masonry with masonry nails.

(5) Option: Install joint reinforcing for multiple wythes and cavity wall ties spaced maximum 16 inches vertically.

(6) Tie interior and exterior wythes of reinforced masonry walls together with individual ties. Provide ties at intervals maximum 16 inches on center horizontally, and 16 inches on center vertically. Lay ties in the same line vertically in order to facilitate vibrating of the grout pours.

d. Anchorage of Abutting Masonry:

(1) Anchor interior 4 inch thick masonry partitions to exterior masonry walls with wall ties. Space ties at 24 inches maximum vertical intervals. Extend ties 4 inches minimum into masonry.

(2) Anchor interior masonry bearing walls or interior masonry partitions over 4 inches thick to masonry walls with rigid wall anchors spaced at 16 inch maximum vertical intervals.

(3) Anchor abutting masonry walls and partitions to concrete with dovetail anchors. Install dovetail slots vertically in concrete at centerline of abutting wall or partition. Locate dovetail anchors at 16 inch maximum vertical intervals. Secure anchors to existing wall with two 3/8 inch by 3 inch expansion bolts or two power-driven fasteners.

(4) Anchor abutting interior masonry partitions to existing concrete and existing masonry construction, with adjustable wall ties. Extend ties minimum 4 inches into joints of new masonry. Fasten ties to existing concrete and masonry construction, with powder actuated drive pins, nail or other means that provides rigid anchorage. Install anchors at 16 inch maximum vertical intervals.

e. Masonry Furring:

(1) Anchor masonry furring less than 4 inches nominal thick to masonry walls or to concrete with adjustable wall ties or dovetail anchors.

(2) Space at maximum 16 inches on center in both directions.

f. Anchorage to Steel Beams or Columns:

(1) Use adjustable beam anchors on each flange.

(2) At columns weld steel rod to steel columns at 12 inch intervals, and place wire ties in masonry courses at 16 inches maximum vertically.

3.3 INSTALLATION - REINFORCEMENT

a. Joint Reinforcement:

(1) Install joint reinforcement in CMU wythe of combination brick and CMU, cavity walls, and single wythe concrete masonry unit walls or

partitions.

(2) Reinforcing is acceptable in lieu of individual ties for anchoring brick facing to CMU backup in exterior masonry walls.

(3) Locate joint reinforcement in mortar joints at 16 inch maximum vertical intervals.

(4) Additional joint reinforcement is required in mortar joints at both 8 inches and 16 inches above and below windows, doors, louvers and similar openings in masonry.

(5) Wherever brick masonry is backed up with stacked bond masonry, install multiple wythe joint reinforcement in every two courses of CMU backup, and in corresponding joint of facing brick.

b. Steel Reinforcing Bars:

(1) Install reinforcing bars in cells of hollow masonry units where required for vertical reinforcement and in bond beam units for horizontal reinforcement. Install in wall cavities of reinforced masonry walls where indicated on drawings.

(2) Bond Beams:

(a) Form Bond beams of load-bearing concrete masonry units filled with grout and reinforced with two No. 5 reinforcing bars unless shown otherwise. Do not cut reinforcement.

(b) Brake bond beams only at expansion joints and at control joints, if shown.

(3) Stack Bond:

(a) Locate additional joint reinforcement in vertical and horizontal joints as indicated on drawings.

(b) Anchor vertical reinforcement into foundation or wall or bond beam below.

(c) Provide temporary bracing for walls over 8 feet tall until permanent horizontal bracing is completed.

(4) Grout openings:

(a) Leave cleanout holes in double wythe walls during construction by omitting units at base of one side of wall.

(b) Locate 3 inches by 3 inches min. cleanout holes at location of vertical reinforcement.

(c) Keep grout space clean of mortar accumulation and debris. Clean as work progresses and immediately before grouting.

3.4 INSTALLATION - BRICK EXPANSION AND CMU CONTROL JOINTS

a. Provide brick expansion joint (EJ) and CMU control joints (CJ) where indicated on drawings.

b. Keep joint free of mortar and other debris.

c. Joints Occur In Masonry Walls:

(1) Install preformed compressible joint filler in brick wythe.

(2) Install cross shaped shear keys in concrete masonry unit wythe with preformed compressible joint filler on both sides of shear key.

d. Use standard notched concrete masonry units (sash blocks) made in full and half-length units where shear keys are used to create a continuous vertical joint.

e. Interrupt joint reinforcement at expansion and control joints.

f. Fill opening in exposed face of expansion and control joints with sealant as specified in Section 07 92 00, JOINT SEALANTS.

3.5 INSTALLATION - BUILDING EXPANSION AND SEISMIC JOINTS

a. Keep expansion and seismic joints open and free of mortar. Remove mortar and other debris.

b. Install non-combustible, compressible type joint filler to fill space completely except where sealant is shown on joints in exposed finish work.

c. Fill opening in exposed face of expansion and seismic joints with sealant as specified in Section 07 92 00, JOINT SEALANTS.

3.6 INSTALLATION - ISOLATION JOINT

a. Where full height walls and partitions lie parallel or perpendicular to and under structural beams and shelf angles, provide minimum 3/8 inch separation between walls and partitions and bottom of beams and shelf angles.

b. Insert continuous full width strip of non-combustible type compressible joint filler.

c. Fill opening in exposed face of isolation joints with sealant as specified in Section 07 92 00, JOINT SEALANTS.

3.7 INSTALLATION - BRICKWORK

a. Lay clay brick according to BIA TN 11B.

b. Laying:

(1) Lay brick in one-half running bond with bonded corners, unless indicated otherwise.

(2) Maintain bond pattern throughout.

(3) Do not use brick smaller than half-brick at any angle, corner, break, and jamb.

(4) Where length of cut brick is greater than one half length,

maintain vertical joint location.

(5) Lay exposed brickwork joints symmetrical about center lines of openings.

(6) Do not structurally bond multi-wythe brick walls, unless indicated on drawings.

(7) Before starting work, lay facing brick on foundation wall and adjust bond to openings, angles, and corners.

(8) Lay brick for sills with wash and drip.

(9) Build solid brickwork as required for anchorage of items.

c. Joints:

(1) Exterior And Interior Joint Widths: Lay for three equal joints in (8 inches) vertically, unless shown otherwise.

(2) Rake joints for pointing with colored mortar when colored mortar is not full depth.

(3) Arches:

(a) Flat arches (jack arches) lay with camber of 1 in 200 (1/16 inch per foot) of span.

(b) Face radial arches with radial brick with center line of joints on radial lines.

(c) Form Radial joints of equal width.

(d) Bond arches into backing with metal ties in every other joint.

d. Weep Holes:

(1) Install weep holes at 24 inches on center in bottom of vertical joints of exterior masonry veneer or cavity wall facing over foundations, bond beams, and other water stops in wall.

(2) Form weep holes using wicks made of mineral fiber insulation strips turned up 8 inches in cavity. Anchor top of strip to backup to securely hold in place.

(3) Install sand or pea gravel in cavity approximately 3 inches high between weep holes.

e. Solid Exterior Walls:

(1) Build with 4 inches of nominal thick facing brick, backed up with concrete masonry units cast-in-place concrete .

(2) Construct solid brick jambs minimum 0.81 inches wide at exterior wall openings and at recesses, except where exposed concrete unit backup is shown.

(3) Do not install full bonding headers.

(4) Parging:

(a) For solid masonry walls, lay backup to height of six brick courses, parge backup with 1/2 inch of mortar troweled smooth; then lay exterior wythe to height of backup.

(b) Make parging continuous over backup, and extend 6 inches onto adjacent concrete or masonry.

(c) Parge ends and backs for recesses in exterior walls to thickness of 1/2 inch.

(d) Parge inside surface of exterior walls to produce true even surface to receive insulation.

(5) Coordinate with building insulation for thickness of insulation and allowance of air space behind exterior wythe.

(6) In locations where hurricane driven rains are expected, install bituminous dampproofing on cavity side of inner wythe.

f. Cavity Walls:

(1) Keep air space clean of mortar accumulations and debris.

(2) Lay the interior wythe of the masonry wall full height where dampproofing air barrier is required on cavity face. Coordinate to install dampproofing air barrier before laying outer wythe.

(3) Insulated Cavity Type Exterior Walls:

(a) Install insulation against cavity face of inner masonry wythe.

(b) Place insulation between rows of ties or joint reinforcing. Adhere insulation to masonry surface with a bonding agent as recommended by insulation manufacturer.

(c) Lay outer masonry wythe up with air space between insulation and masonry units.

(4) Veneer Framed Walls:

(a) Build with 4 inches of face brick over sheathed stud wall with air space.

(b) Keep air space clean of mortar accumulations and debris.

3.8 INSTALLATION - CONCRETE MASONRY UNITS

a. Types and Uses:

(1) Provide special concrete masonry shapes as required, including lintel and bond beam units, sash units, and corner units. Provide solid concrete masonry units, where full units cannot be installed, or where needed for anchorage of accessories.

(2) Provide solid load-bearing concrete masonry units or grout cell of hollow units at jambs of openings in walls, where structural members impose loads directly on concrete masonry, and where shown.

(3) Provide rounded corner (bullnose) shapes at opening jambs in exposed work and at exterior corners.

(4) Do not install brick jambs in exposed finish work.

(5) Install concrete building brick only as filler in backup material where not exposed.

(6) Construct fire resistance in fire rated partitions meeting fire ratings indicated on drawings.

(7) Structural Clay Tile Units (Option):

(a) Structural clay tile units load-bearing or non-load bearing as required, may be installed in lieu of concrete masonry units, only, but not as an exposed surface, foundation walls or where otherwise noted.

(b) Set units as specified for concrete masonry units.

(c) Install brick or load-bearing structural clay tile units, with cores set vertically, and filled with grout where structural members impose concentrated load directly on structural clay tile masonry.

(8) Where lead-lined concrete masonry unit partitions terminate below underside of overhead floor or roof deck, fill remaining open space between top of partition and underside of overhead floor or roof deck, with standard concrete masonry units of same thickness as lead lined units.

b. Laying:

(1) Lay concrete masonry units with 3/8 inch joints, with a bond overlap of minimum 1/4 of unit length, except where stack bond is indicated on drawings.

(2) Do not wet concrete masonry units before laying.

(3) Bond external corners of partitions by overlapping alternate courses.

(4) Lay first course in a full mortar bed.

(5) Set anchorage items as work progress.

(6) Where ends of anchors, bolts, and other embedded items, project into voids of units, completely fill voids with mortar or grout.

(7) Provide 1/4 inch open joint for sealant between existing construction, exterior walls, concrete work, and abutting masonry partitions.

(8) Lay concrete masonry units with full face shell mortar beds and fill head joint beds for depth equivalent to face shell thickness.

(9) Lay concrete masonry units so cores of units, that are to be filled with grout, are vertically continuous with joints of cross

webs of such cores completely filled with mortar. Unobstructed core openings minimum 2 inches by 3 inches.

(10) Do not wedge masonry against steel reinforcing. Minimum 1/2 inch clear distance between reinforcing and masonry units.

(11) Install deformed reinforcing bars of sizes indicated on drawings.

(12) At time of placement, ensure steel reinforcement is free of loose rust, mud, oil, and other contamination capable of affecting bond.

(13) Place steel reinforcement at spacing indicated on drawings before grouting.

(14) Minimum clear distance between parallel bars: One bar diameter.

(15) Hold vertical steel reinforcement in place vertically by centering clips, caging devices, tie wire, or other approved methods.

(16) Support vertical bars near each end and at maximum 192 bar diameter on center.

(17) Splice reinforcement or attach reinforcement to dowels by placing in contact and securing with wire ties.

(18) Stagger splices in adjacent horizontal reinforcing bars. Lap reinforcing bars at splices a minimum of 40 bar diameters.

(19) Grout cells of concrete masonry units, containing reinforcing bars, solid as specified.

(20) Install cavity and joint reinforcement as masonry work progresses.

(21) Rake joints 1/4 to 3/8 inch deep for pointing with colored mortar when colored mortar is not full depth.

c. Waterproofing Parging:

(1) Parge earth side of concrete masonry unit basement walls with mortar applied in two coats, each coat 1/4 inch thick.

(2) Clean wall surfaces to receive parging of dirt, oil, or grease, and moisten before application of first coat.

(3) Roughen first coat when partially set, permit to hardened for 24 hours, and moisten before application of second coat.

(4) Keep second coat damp for minimum 48 hours.

(5) Thicken parging and round to form a cove at the junction of outside wall face and footing.

3.9 INSTALLATION - GLAZED STRUCTURAL FACING TILE

a. Lay facing tile in running bond unless indicated otherwise. Match existing adjacent bond and joints in alteration work.

b. Laying:

(1) Set facing tile units in full bed of mortar with ends buttered, and units shoved into place. Fill joints with mortar, and rake out 3/8 inch deep for pointing.

(2) Install clean units when set.

(3) Perform cutting and grinding with power-driven cutting saws and grinders.

(4) Cut or drill units to accommodate electrical outlets, plumbing fixtures, grab-bars, fittings, and equipment.

(5) Cove Base Units:

(a) Set base flush with finish floor.

(b) Form base course of two-face partitions of two units to required thickness.

(6) Lay out partitions enclosing pipes or conduits with thickness to provide 2 inch minimum coverage of pipes or conduits.

(7) Joints:

(a) 1/4 inch nominal width for new work, match existing for alteration work.

(b) Maximum variations in joint width 1/16 inch.

(c) Reinforce Two-Face partitions of 4 inch nominal thickness with continuous joint reinforcement, or wire mesh ties in joints at top of base, at top of glazed structural facing tile, and at three-course intervals.

3.10 POINTING

a. Fill joints with pointing mortar using rubber float trowel to apply mortar solidly into raked joints.

b. Wipe off excess mortar from joints of glazed masonry units with dry cloth.

c. Tool exposed joints to smooth concave joint.

d. At joints with existing work, match existing joint.

3.11 GROUTING

a. Preparation:

(1) Clean grout space of mortar droppings before placing grout.

(2) Close cleanouts.

(3) Install vertical solid masonry dams across grout space for full height of wall at intervals of maximum 30 feet. Do not bond dam units into wythes as masonry headers.

(4) Verify reinforcing bars are installed as indicated on drawings.

b. Placing:

(1) Place grout in grout space in lifts as specified.

(2) Consolidate each grout lift after free water has disappeared but before plasticity is lost.

(3) Do not slush with mortar or use mortar with grout.

(4) Interruptions:

(a) When grouting must be stopped for more than an hour, top off grout 1-1/2 inches below top of last masonry course.

(b) Grout from dam to dam on high lift method.

(c) Longitudinal run of masonry may be stopped off only by raking back one-half masonry unit length in each course and stopping grout 4 inches back of rake on low lift method.

c. Puddling Method:

(1) Consolidate by puddling with grout stick during and immediately after placing.

(2) Grout cores of concrete masonry units containing reinforcing bars solid as masonry work progresses.

d. Low Lift Method:

(1) Construct masonry to 5 feet maximum height before grouting.

(2) Grout in one continuous operation and consolidate grout by mechanical vibration and reconsolidate after initial water loss and settlement has occurred.

e. High Lift Method:

(1) Do not pour grout until masonry wall has cured minimum of 4 hours.

(2) Place grout in 5 feet maximum lifts.

(3) Exception:

(a) Where following conditions are met, place grout in 12.67 feet maximum lifts.

(b) Masonry has cured minimum of 4 hours.

(c) Grout slump is maintained between 10 and 11 inches.

(d) No intermediate reinforced bond beams are placed between top and bottom of grout lift.

(4) When vibrating succeeding lifts, extend vibrator 12 to 18 inches into preceding lift.

3.12 PLACING REINFORCEMENT

a. General: Clean reinforcement of loose rust, mill scale, earth, ice or other materials which will reduce bond to mortar or grout. Do not use reinforcement bars with kinks or bends not shown on drawings or approved submittal drawings, or bars with reduced cross-section due to excessive rusting or other causes.

b. Position reinforcement accurately at spacing indicated on drawings. Support and secure vertical bars against displacement. Install horizontal reinforcement as masonry work progresses. Where vertical bars are shown in close proximity, provide clear distance between bars of minimum one bar diameter or 1 inch, whichever is greater.

c. For columns, piers and pilasters, maintain clear distance between vertical bars as indicated on drawings, minimum 1.5 bar diameters or 1-1/2 inches, whichever is greater. Provide lateral ties as indicated on drawings.

d. Splice reinforcement bars only where indicated on drawings, unless approved by Contracting Officer's Representative. Provide lapped splices. In splicing vertical bars or attaching to dowels, lap ends, place in contact and wire tie.

e. Provide minimum lap as indicated on approved submittal drawings, or if not indicated, minimum 48 bar diameters.

f. Weld splices where indicated on drawings according to AWS D1.4/D1.4M.

g. Embed metal ties in mortar joints as work progresses, with minimum mortar cover of 5/8 inch on exterior face of walls and 1/2 inch at other locations.

h. Embed prefabricated horizontal joint reinforcement as work progresses, with minimum cover of 5/8 inch on exterior face of walls and 1/2 inch at other locations. Lap joint reinforcement minimum 6 inches at ends. Use prefabricated “L” and “T” sections to provide continuity at corners and intersections. Cut and bend joint reinforcement for continuity at returns, offsets, column fireproofing, pipe enclosures and other special conditions.

i. Anchoring: Anchor reinforced masonry work to supporting structure as indicated on drawings.

j. Anchor reinforced masonry walls at intersections with non-reinforced masonry.

3.13 INSTALLATION OF REINFORCED BRICK MASONRY

a. Mortar Jointing and Bedding:

(1) Pattern Bond: Lay exterior wythes in pattern bond shown, or if not shown, lay in 1/2 running bond with vertical joints in each course centered on units in courses above and below. Lay inner wythes (if any) with units in each wythe bonded by lapping minimum 2 inches. Bond and interlock each course of each wythe at corners and intersections. Do not use units with less than 4 inch nominal horizontal face dimension at corners or jambs.

(2) Lay exterior wythes with bed (horizontal) and head (vertical) joints between units completely filled with mortar. Top of bed joint mortar may be sloped toward center of walls. Butter ends of units with sufficient mortar to completely fill head joints and shove into place. Do not furrow bed joints or slush head joints. Remove any mortar fins which protrude into grout space.

(3) Maintain joint widths shown for head and bed joints, except for minor variations required to maintain pattern bond. If not shown, lay with 3/8 inch head and bed joints.

(4) Maintain joint widths shown for head and bed joints, but adjust thickness of bed joints, if required, to allow for minimum 1/4 inch thickness of mortar between reinforcement and masonry units, except 1/4 inch bars (if any) may be laid in 1/2 inch thick bed joints and 0.2 inch diameter or smaller wire reinforcing may be laid in 3/8 inch thick bed joints.

b. Two-Wythe Wall Construction: Lay both wythes as specified for exterior wythes. Maintain grout space (collar or continuous vertical joint between wythes) of width indicated, but adjust, if required, to provide grout space minimum 1/2 inch wider than the sum of the vertical and horizontal reinforcement bars shown in grout space. Do not parge or fill grout space with mortar.

c. Multi-Wythe Wall Construction: Where walls of three or more wythes are indicated, lay exterior wythes as specified. Maintain space between wythes as required to allow for laying of number of wythes of unit width shown with minimum grout space between wythes. Allow for minimum 3/4 inch grout between wythes if non-reinforced; if reinforced, allow for grout space minimum 1/2 inch wider than sum of vertical and horizontal reinforcement bars indicated in grout space. Place or float interior wythe units in grout poured between exterior wythes as work progresses. Position units to allow minimum 3/4 inch grout between ends and sides of adjacent units.

d. Limit extent of masonry construction to sections which do not exceed maximum grouting requirements. Provide temporary dams or barriers to control horizontal flow of grout at ends of wall sections. Build dams full height of grout pour. If masonry units are used, do not bond into permanent masonry wythes. Remove temporary dams after completion of grout pour.

e. Low-Lift Grouting:

(1) Use Low-Lift grouting technique with fine grout for the following:

(a) Two-wythe walls with grout space of 2 inch or less in width.

(b) Multi-wythe walls.

(c) Columns, piers or pilasters where masonry units are shown in core areas enclosed by exterior masonry units.

(2) At Contractor’s option, low-lift grouting technique is acceptable for reinforced masonry construction with grout spaces wider than 2 inches, except use coarse grout and place in lifts maximum 8 inches in height.

(3) Construct low-lift masonry by placing reinforcement, laying masonry units and pouring grout as work progresses.

(4) Place vertical reinforcement bars and supports before laying of masonry units. Extend above elevation of maximum pour height as required to allow for splicing. Horizontal reinforcement bars may be placed progressively with laying of masonry units.

(5) Limit grout pours as required to prevent displacement of masonry by grout pressures (blowout), but do not exceed 4 feet pour height.

(6) Lay masonry units before each grout pour, but do not construct more than 12 inches above maximum grout pour height in exterior wythe and 4 inches above in other exterior wythe. Provide metal wall ties if required to prevent blowouts.

(7) Consolidate immediately by rodding or puddling; do not use trowels. Place grout continuously; do not interrupt pouring of grout for more than one hour. If poured in lifts, place from center-to-center of masonry courses. Terminate pour 1 1/2 inches below top of highest course in pour.

f. High-Lift Grouting:

(1) High-Lift grouting technique is acceptable for the following masonry construction:

(a) Two-wythe walls with grout spaces of 2 1/2 inches or greater width.

(b) Columns, piers, or pilasters when no unit masonry fill is shown to be placed in reinforced grout space.

(2) Place reinforcement and support in proper position, before laying of masonry units, except if shown to be placed in mortar joints, place as masonry units are laid. Place horizontal bars in grout spaces on same side of vertical bars.

(3) Construct high-lift masonry by laying masonry to full height and width before placing grout. Provide cleanout holes in first course of masonry, and use high-pressure water jet stream to remove excess mortar from grout spaces, reinforcement bars and top surface of structural members which support wall. Clean grout spaces daily during construction of masonry.

(4) Walls: Omit every other masonry unit in first course of one wythe to provide cleanout holes. Tie wythes together with metal ties as indicated on drawings or as required by code, but provide minimum 9 gage wire ties spaced maximum 24 inches on center horizontally and 16 inches on center vertically for running pattern bond or 12 inches on center vertically for stack bond (if any).

(5) Columns, Piers and Pilasters: Omit every other masonry unit around perimeter of member to provide cleanout holes. Provide reinforcing bands placed in bed joints as the masonry work progresses. Provide bands of the size and vertical spacing show, or as required by code, but minimum 9 gage wire spaced 12 inches

on center vertically.

(6) Preparation of Grout Spaces: Before grouting, inspect and clean grout spaces. Remove dirt, dust, mortar droppings, loose pieces of masonry and other foreign materials from grout spaces. Clean reinforcement and adjust to proper positioning. Clean top surface of structural members supporting masonry to ensure bond. After cleaning and inspection, close cleanout holes with matching masonry units and brace closures to resist grout pressures.

(7) Do not place grout until entire height of masonry to be grouted has attained sufficient strength to resist grout pressure, but minimum 3 days curing time. Install shores and bracing, if required, before starting grouting operations.

(8) Place grout by pumping into grout spaces, unless alternate methods are acceptable to Contracting Officer's Representative.

(9) Install coarse grout. Rod or vibrate each grout lift during placing and after excess moisture has been absorbed, but before plasticity is lost. Do not penetrate or damage grout placed in previous lifts or pours.

(10) Limit grout pours to sections which can be completed in one working day with maximum one hour interruption of pouring operation. Limit pours so as not to exceed the capacity of masonry to resist displacement or loss of mortar bond due to grout pressures.

(11) Do not exceed 12 feet pour height.

(12) Do not exceed 25 feet horizontal pour dimension.

(13) Where pour height exceeds 4 feet, place grout in series of lifts not exceeding 4 feet height. Place each lift as continuous pouring operation. Allow minimum 30 minutes and maximum one hour between lifts of given pour.

(14) When more than one pour is required to complete each section of masonry, extend reinforcement beyond masonry as required for splicing. Pour grout to within 1-1/2 inches of top course of first pour. After grouted masonry is cured, remove temporary dams, and lay masonry units and place reinforcement for second pour section before grouting. Repeat sequence, if more pours are required.

3.14 INSTALLATION OF REINFORCED CONCRETE UNIT MASONRY

a. Do not wet concrete masonry units (CMU).

b. Lay CMU units with full-face shell mortar beds. Fill vertical head joints (end joints between units) solidly with mortar from face of unit to distance behind face equal to thickness of longitudinal face shells. Solidly bed cross-webs of starting courses in mortar. Maintain head and bed 3/8 inch joint widths.

c. Where solid CMU units are shown, lay with full mortar head and bed joints.

d. Walls:

(1) Pattern Bond: Lay CMU wall units in 1/2-running bond with vertical joints in each course centered on units in courses above and below, unless otherwise indicated. Bond and interlock each course at corners and intersections. Use special-shaped units where shown, and as required for corners, jambs, sash, control joints, lintels, bond beams and other special conditions.

(2) Maintain vertical continuity of core or cell cavities, which are to be reinforced and grouted, to provide minimum clear dimension indicated and to provide minimum clearance and grout coverage for vertical reinforcement bars. Keep cavities free of mortar. Solidly bed webs in mortar where adjacent to reinforced cores or cells.

(3) Where horizontally reinforced beams (bond beams) are indicated on drawings, use special units or modify regular units to allow for placement of continuous horizontal reinforcement bars. Place small mesh expanded metal lath or wire screening in mortar joints under bond beam courses over cores or cells of non-reinforced vertical cells, or provide units with solid bottoms.

e. Columns, Piers and Pilasters:

(1) Use CMU units of size, shape and number of vertical core spaces shown. If not shown, use units which provide minimum clearances and grout coverage for number and size of vertical reinforcement bars shown.

(2) Provide pattern bond shown, or if not shown, alternate head joints in vertical alignment.

(3) Where bonded pilaster construction is shown, lay wall and pilaster units together to maximum grout pour height specified.

f. Grouting:

(1) Use fine grout for filling spaces less than 4 inches in one or both horizontal directions.

(2) Use coarse grout for filling 4 inch spaces or larger in both horizontal directions.

(3) Grouting Technique: At Contractor’s option, use either low-lift or high-lift grouting techniques.

g. Low-Lift Grouting:

(1) Provide minimum clear dimension of 2 inches and clear area of 8 sq. inches in vertical cores to be grouted.

(2) Place vertical reinforcement before grouting of CMU. Extend above elevation of maximum pour height as required for splicing. Support in position at vertical intervals not exceeding 192 bar diameters nor 10 feet.

(3) Lay CMU to maximum pour height. Do not exceed 5 feet height, or if bond beam occurs below 5 feet height, stop pour 1-1/2 inches

below top of bond beam.

(4) Rod or vibrate grout during placing. Place grout continuously; do not interrupt pouring of grout for more than one hour. Terminate grout pours 1-1/2 inches below top course of pour.

(5) Bond Beams: Stop grout in vertical cells 1-1/2 inches below bond beam course. Place horizontal reinforcement in bond beams; lap at corners and intersections as indicated on drawings. Place grout in bond beam course before filling vertical cores above bond beam.

h. High-Lift Grouting:

(1) Do not use high-lift grouting technique for grouting of CMU unless minimum cavity dimension and area is 3 inches and 10 sq. inches, respectively.

(2) Provide cleanout holes in first course at vertical cells which are to be filled with grout.

(3) Use units with one face shell removed and provide temporary supports for units above, or use header units with concrete brick supports, or cut openings in one face shell.

(4) Construct masonry to full height of maximum grout pour before placing grout.

(5) Limit grout lifts to maximum height of 5 feet and grout pour to maximum height of 24 feet, for single wythe hollow concrete masonry walls, unless otherwise indicated.

(6) Place vertical reinforcement before grouting. Place before or after laying masonry units, to suit application. Tie vertical reinforcement to dowels at base of masonry where shown and thread CMU over or around reinforcement. Support vertical reinforcement at intervals not exceeding 192 bar diameters nor 10 feet.

(7) Where individual bars are placed after laying masonry, place wire loops extending into cells as masonry is laid and loosen before mortar sets. After insertion of reinforcement bar, pull loops and bar to proper position and tie free ends.

(8) Where reinforcement is prefabricated into cage units before placing, fabricate units with vertical reinforcement bars and lateral ties of the size and spacing indicated.

(9) Place horizontal beam reinforcement as masonry units are laid.

(10) Embed lateral tie reinforcement in mortar joints where indicated. Place as masonry units are laid, at vertical spacing shown.

(11) Where lateral ties are shown in contact with vertical reinforcement bars, embed additional lateral tie reinforcement in mortar joints. Place as indicated on drawings, or if not shown, provide as required to prevent grout blowout or rupture of CMU face shells, but provide minimum diameter 0.16 inch wire ties spaced 16 inches on center for members with 20 inches or less side dimensions, and 8 inches on center for members with side dimensions exceeding 20 inches.

(12) Preparation of Grout Spaces: Before grouting, inspect and clean grout spaces. Remove dust, dirt, mortar droppings, loose pieces of masonry and other foreign materials from grout spaces. Clean reinforcement and adjust to proper position. Clean top surface of structural members supporting masonry to ensure bond. After final cleaning and inspection, close cleanout holes and brace closures to resist grout pressures.

(13) Do not place grout until entire height of masonry to be grouted has attained sufficient strength to resist displacement of masonry units and breaking of mortar bond. Install shores and bracing, if required, before starting grouting operations.

(14) Limit grout pours to sections which can be completed in one working day with maximum one hour interruption of pouring operation. Place grout in lifts which do not exceed 5 feet. Allow minimum 30 minutes and maximum one hour between lifts. Mechanically consolidate each lift.

(15) Place grout in lintels or beams over openings in one continuous pour.

(16) Where bond beam occurs more than one course below top of pour, fill bond beam course to within 1 inch of vertically reinforced cavities, during construction of masonry.

(17) When more than one pour is required to complete a given section of masonry, extend reinforcement beyond masonry as required for splicing. Pour grout to within 1-1/2 inches of top course of first pour. After grouted masonry is cured, lay masonry units and place reinforcement for second pour section before grouting. Repeat sequence if more pours are required.

3.15 CONSTRUCTION TOLERANCES

a. Lay masonry units plumb, level and true to line within tolerances according to ACI 530.1/ASCE 6/TMS 602 and as follows:

b. Maximum variation from plumb:

(1) In 10 feet - 1/4 inch.

(2) In 20 feet - 3/8 inch.

(3) In 40 feet or more - 1/2 inch.

c. Maximum variation from level:

(1) In any bay or up to 20 feet - 1/4 inch.

(2) In 40 feet or more - 1/2 inch.

d. Maximum variation from linear building lines

(1) In any bay or up to 20 feet - 1/2 inch.

(2) In 40 feet or more - 3/4 inch.

e. Maximum variation in cross-sectional dimensions of columns and thickness of walls from dimensions shown:

(1) Minus 1/4 inch.

(2) Plus 1/2 inch.

f. Maximum variation in prepared opening dimensions:

(1) Accurate to minus 0 inch.

(2) Plus 1/4 inch.

3.16 CLEANING AND REPAIR

a. General:

(1) Clean exposed masonry surfaces on completion.

(2) Protect adjoining construction materials and landscaping during cleaning operations.

(3) Cut out defective exposed new joints to depth of approximately 3/4 inch and repoint.

(4) Remove mortar droppings and other foreign substances from wall surfaces.

b. Brickwork:

(1) First wet surfaces with clean water, then wash down with detergent solution. Do not use muriatic acid.

(2) Brush with stiff fiber brushes while washing, and immediately wash with clean water.

(3) Remove traces of detergent, foreign streaks, or stains of any nature.

c. Concrete Masonry Units:

(1) Immediately following setting, brush exposed surfaces free of mortar or other foreign matter.

(2) Allow mud to dry before brushing.

d. Glazed Structural Facing Tile or Brick Units:

(1) Clean as recommended manufacturer. Protect light colored mortar joints from discoloration during cleaning.

(2) Use on solid masonry walls.

(3) Prepare schedule of test locations.

3.17 FIELD QUALITY CONTROL

Water Penetration Testing:

a. Seven days before plastering or painting, in presence of Contracting Officer's Representative, test solid exterior masonry walls for water penetration.

b. Direct water on masonry for a period of one hour when wind velocity is less than five miles per hour.

c. Should moisture appear on inside of walls tested, make additional tests at other areas as directed by Contracting Officer's Representative.

d. Correct areas showing moisture on inside of walls, and repeat test at repaired areas, to ensure moisture penetration has been stopped.

e. Make water test at following locations:

(1) Sixteen places on Building Number 17.

(2) Eight places on other buildings.

(3) At Connecting Corridor make one test for each 150 lineal feet of exterior masonry walls.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 12 00

STRUCTURAL STEEL FRAMING

11/18

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED REQUIREMENTS
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 MATERIALS
2.3 PRODUCTS - GENERAL
2.4 FABRICATION
2.5 FINISHES
2.6 ACCESSORIES

PART 3 EXECUTION

3.1 ERECTION
3.2 FIELD PAINTING
3.3 FIELD QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 05 12 00
11/18

STRUCTURAL STEEL FRAMING

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Structural steel shapes, plates, and bars.

(2) Structural pipe.

(3) Bolts, nuts, and washers.

1.2 RELATED REQUIREMENTS

a. Materials Testing And Inspection During Construction: Section 01 45 29, TESTING LABORATORY SERVICES.

e. Steel Decking: Section 05 31 00, STEEL DECKING.

f. Composite Steel Deck: Section 05 36 00, COMPOSITE METAL DECKING.

g. Fireproofing: Section 07 81 00, APPLIED FIREPROOFING.

h. Steel Framing: Section 08 44 13, GLAZED ALUMINUM CURTAIN WALLS Section 13 34 19, METAL BUILDING SYSTEMS .

j. Steel Support: Section 10 13 00, DIRECTORIES.

k. Painting: Section 09 91 00, PAINTING.

1.3 REFERENCES

Comply with references to extent specified in this section.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325	(2017) Steel Construction Manual
AISC 303	(2016) Code of Standard Practice for Steel Buildings and Bridges
AISC 360	(2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.22.1	Type B Plain Washers

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A501/A501M	(2021) Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing
ASTM A992/A992M	(2022) Standard Specification for Structural Steel Shapes
ASTM F2329/F2329M	(2015; R 2023) Standard Specification for Zinc Coating, Hot-Dip, Requirements for Application to Carbon and Alloy Steel Bolts, Screws, Washers, Nuts, and Special Threaded Fasteners
ASTM F3125/F3125M	(2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength

MASTER PAINTERS INSTITUTE (MPI)

MPI 18	(2022) Primer, Zinc Rich, Organic

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.752	Guidelines For Establishing The Components Of A Site-Specific Erection Plan

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-05 Design Data

Delegated Design Drawings And Calculations; G, AE

SD-06 Test Reports

Test Reports; G, AE

SD-07 Certificates

Certificates; G, AE

SD-11 Closeout Submittals

Record Surveys; G,

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Submittal Drawings:

(1) Show size, configuration, and fabrication and installation details.

c. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

d. Test Reports: Certify products comply with specifications.

(1) Welders' qualifying tests.

e. Certificates: Certify each product complies with specifications.

(1) Structural steel.

(2) Steel connections.

(3) Welding materials.

(4) Shop coat primer paint.

f. Qualifications: Substantiate qualifications comply with specifications.

(1) Fabricator with project experience list .

(2) Installer with project experience list .

(3) Welders and welding procedures.

g. Delegated Design Drawings and Calculations: Signed and sealed by responsible Architect/Engineer.

(1) Connection calculations.

h. Record Surveys: Signed and sealed by responsible surveyor or engineer.

1.6 QUALITY ASSURANCE

a. Fabricator Qualifications: AISC Quality Certification participant designated as AISC Certified Plant, Category STD.

(1) Regularly fabricates specified products.

(2) Fabricated specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

b. Installer Qualifications: AISC Quality Certification Program participant designated as AISC-Certified Erector, Category ACSE.

(1) Regularly installs specified products.

(2) Installed specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

c. Before commencement of Work, ensure steel erector provides written notification required by 29 CFR 1926.752. Submit a copy of the notification to Contracting Officer's Representative.

d. Welders and Welding Procedures Qualifications: AWS D1.1/D1.1M.

1.7 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE

a. Delegated Design: Prepare submittal documents including design calculations and drawings signed and sealed by registered design professional, licensed in state where project is located.

b. Design structural steel framing connections complying with specified performance:

(1) Load Capacity: Resist loads indicated on drawings . Account for connection and member loads and eccentricities.

(a) Request additional design criteria when necessary to complete connection design.

(2) Configuration: Design and detail all connections for each member

size, steel grade and connection type to resist the loads and reactions indicated on the drawings or specified herein. Use details consistent with details shown on drawings, supplementing where necessary. The details shown on drawings are conceptual and do not indicate the required weld sizes or number of bolts unless specifically noted. Use rational engineering design and standard practice in detailing, accounting for all loads and eccentricities in both the connection and the members. Promptly notify the Contracting Officer Representative of any location where the connection design criteria is not clearly indicated. The design of all connections is subject to the review and acceptance of the Contracting Officer's Representative. Submit structural calculations prepared and sealed by a qualified engineer registered in the state where the project is located. Submit calculations for review before preparation of detail drawings.

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2.2 MATERIALS

a. W-Shapes:

(1) ASTM A992/A992M.

(2)

b. M S -Shapes:

(1) ASTM A36/A36M.

(2)

(3)

c. Channel and Angles:

(1) ASTM A36/A36M.

(2)

(3)

d. Plates and Bars:

(1) ASTM A36/A36M.

(2)

(3)

e. Hollow Structural Sections:

(1) ASTM A501/A501M.

f. Structural Pipe: ASTM A53/A53M, Grade B.

g. Bolts, Nuts and Washers: Galvanized for galvanized framing and plain finish for other framing .

(1) High-strength bolts, including nuts and washers: ASTM F3125/F3125M.

(2) Bolts and nuts, other than high-strength: ASTM A307, Grade A.

(3) Plain washers, other than those in contact with high-strength bolt heads and nuts: ASME B18.22.1.

h. Welding Materials: AWS D1.1/D1.1M, type to suit application.

2.3 PRODUCTS - GENERAL

a. Basis of Design: See schedule for finishes in teh drawing set.

b. Sustainable Construction Requirements:

(1) Steel Recycled Content: 30 percent total recycled content, minimum.

(2) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Paints and coatings.

2.4 FABRICATION

a. Fabricate structural steel according to Chapter M, AISC 360.

b. Shop and Field Connections:

(1) Weld connections according to AWS D1.1/D1.1M. Welds shall be made only by welders and welding operators who have been previously qualified by tests as prescribed in AWS D1.1/D1.1M to perform type of work required.

(2) High-Strength Bolts: High-strength bolts tightened to a bolt tension minimum 70 percent of their minimum tensile strength. Tightening done with properly calibrated wrenches, by turn-of-nut method or by use of direct tension indicators (bolts or washers). Tighten bolts in connections identified as slip-critical using Direct Tension Indicators. Twist-off torque bolts are not an acceptable alternate fastener for slip critical connections.

2.5 FINISHES

a. Shop Priming:

(1) Prime paint structural steel according to AISC 303, Section 6.

(a) Interstitial Space Structural Steel: Prime paint, unless indicated to receive sprayed on fireproofing.

b. Shop Finish Painting: Apply primer and finish paint as specified in Section 09 91 00, PAINTING.

c. Do not paint:

(1) Surfaces within (2 inches) of field welded joints.

(2) Surfaces indicated to be encased in concrete.

(3) Surfaces receiving sprayed on fireproofing.

(4) Beam top flanges receiving shear connector studs applied.

d. Structural Steel Galvanizing: ASTM A123/A123M, hot dipped, after fabrication. Touch-up after erection: Clean and wire brush any abraded and other spots worn through zinc coating, including threaded portions of bolts and welds and touch-up with galvanizing repair paint.

(1) Galvanize structural steel framing installed at exterior locations.

e. Bolts, Nuts, and Washers Galvanizing: ASTM F2329/F2329M, hot-dipped.

2.6 ACCESSORIES

a. General: Shop paint steel according to AISC 303, Section 6.

b. Finish Paint System: Primer and finish as specified in Section 09 91 00, PAINTING.

c. Galvanizing Repair Paint: MPI 18.

PART 3 EXECUTION

3.1 ERECTION

a. Erect structural steel according to AISC 303 and AISC 360.

b. Set structural steel accurately at locations and elevations indicated on drawings.

c. Maintain erection tolerances of structural steel within AISC 303 requirements.

(1) Pour Stop Elevation Tolerance: (1/4 inch), maximum, before concrete placement.

d. Weld and bolt connections as specified for shop connections.

3.2 FIELD PAINTING

a. After welding, clean and prime weld areas to match adjacent finish.

b. Touch-up primer damaged by construction operations.

c. Apply galvanizing repair paint to galvanized coatings damaged by construction operations.

d. Finish Painting: As specified in Section 09 91 00, PAINTING.

3.3 FIELD QUALITY CONTROL

Record Survey:

(1) Engage registered land surveyor or registered civil engineer as specified in Section 01 00 00, GENERAL REQUIREMENTS to perform survey.

(2) Measure and record structural steel framing plumbness, level, and alignment after completing bolting and welding and before installation of work supported by structural steel.

(3) Identify deviations from allowable tolerances specified in AISC 325.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 31 00

STEEL DECKING

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 MATERIALS
2.3 METAL ROOF DECK
2.4 FABRICATION
2.5 FINISHES
2.6 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 ERECTION

-- End of Section Table of Contents --

SECTION 05 31 00
1/21

STEEL DECKING

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Single pan fluted metal form deck supporting concrete fill as roof substrate.

(2) Corrugated metal form deck supporting concrete fill as roof substrate.

(3) Single pan fluted metal roof deck as roof substrate.

(4) Acoustic metal roof deck as roof substrate.

1.2 RELATED WORK
a. Section 05 21 00, STRUCTURAL STEEL FRAMING: Structural Steel Shapes.
b. Section 09 91 00, PAINTING: Finish Painting.

1.3 REFERENCES

Comply with references to extent specified in this section.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100	(2012) North American Specification for the Design of Cold-Formed Steel Structural Members

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS D1.8/D1.8M	(2016) Structural Welding Code—Seismic Supplement

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability,

	Solution Hardened, and Bake Hardenable
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials

UL SOLUTIONS (UL)

UL 580	(2006; Reprint Apr 2024) UL Standard for Safety Tests for Uplift Resistance of Roof Assemblies

1.4 SUBMITTALS

SD-01 Preconstruction Submittals

Qualifications; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-07 Certificates

Certificates; G, AE

SD-11 Closeout Submittals

Insurance Certification; G,

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. All items indicated below are required submittals requiring Contracting Officer’s Representative review and approval.

b. Submittal Drawings:

(1) Show layout, connections to supporting members, anchorage, sump pans, accessories, deck openings and reinforcements.

(2) Show similar information necessary for completing installation as shown and specified, including supplementary framing, ridge and valley plates, cant strips, cut openings, special jointing or other accessories.

(3) Show welding, side lap, closure, deck reinforcing and closure reinforcing details.

(4) Show openings required for work of other trades, including openings not shown on structural drawings. Indicate where temporary shoring is required to satisfy design criteria.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Show steel decking section properties and structural characteristics.

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

e. Certificates: Certify each product complies with specifications.

(1) Fire Resistance Product Listing: For each metal deck type and thickness supporting concrete slab or fill.

(2) Show steel decking is UL Listed for specified application.

f. Qualifications: Substantiate qualifications comply with specifications.

(1) Welders and welding procedures.

1.6 QUALITY ASSURANCE

b. Welders and Welding Procedures Qualifications: AWS D1.8/D1.8M.

1.7 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE

Design steel decking and accessories according to AISI S100.

(2) Wind Uplift Resistance and Corner Conditions: UL 580, Class 90 or higher UL Class required by wind loading in the location of the project.

(4) Fire Resistance: ASTM E119; as component of 1 hour rated roof assembly.

(6) Design side and end closures and attachment to supporting steel to safely support wet weight of concrete and construction loads.

(7) Cantilever Closure Deflection: (1/8 inch), maximum.

2.2 MATERIALS

a. Galvanized Steel Sheet: ASTM A653/A653M; G60 coating.

b. Painted Steel Sheet: ASTM A1008/A1008M, Grade C or D, shop primed.

c. Primer for Shop Painted Sheets: Manufacturer's standard primer (2 coats). When finish painting of steel decking is specified in Section 09 91 00, PAINTING primer coating shall be compatible with specified finish painting.

d. Steel Shapes: ASTM A36/A36M.

2.3 METAL ROOF DECK

a. Metal Roof Deck: UL Listed as metal roof deck panels.

(1) Steel decking of the type, depth, thickness, and section properties as shown.

b. Metal Form Deck - Type 1: Single pan fluted units as permanent form for reinforced concrete slabs.

(1) Depth and Thickness: As indicated on drawings.

(2) Material: Galvanized sheet steel.

d. Metal Roof Deck: Single pan fluted units with flat horizontal top surfaces as permanent support for superimposed loads.

(1) Deck Style:

(a) Wide Rib (Type B) deck.

(2) Depth and Thickness: As indicated on drawings.

(3) Material: Galvanized sheet steel.

f. Do not use steel deck for hanging supports of building components

including suspended ceilings, electrical light fixtures, plumbing, heating, or air conditioning pipes or ducts or electrical conduits.

g. Include integral system for steel decking units used for interstitial levels.

(1) Provide system suitable for simple point of attachment for light duty hanger devices.

(2) Provide system suitable to allow for flexibility for attaching hangers for support of suspended ceilings, electrical, plumbing, heating, or air conditioning items, weight not to exceed (10 psf).

(3) Provide a minimum spacing pattern of (12 inches) on centers longitudinally and (24 inches) on centers transversely.

(4) Maximum allowable load suspended from any hanger: (50 pounds).

(5) System consisting of fold-down type hanger tabs or lip hanger is acceptable.

2.4 FABRICATION

a. Fabricate steel decking in sufficient lengths to extend over 3 or more supports, except for interstitial levels.

(1) Cut metal deck units to proper length in shop.

b. Fabricate accessories required to complete installation of steel decking.

(1) Exposed to View: Fabricate from sheet steel matching metal decking.

(2) Concealed from View: Fabricate from galvanized sheet steel.

c. Sheet Metal Accessories:

(1) Metal Cover Plates: For end-abutting decking, to close gaps at changes in deck direction, columns, walls and openings.

(a) Sheet Steel: Minimum (0.04 inch) thick.

(2) Continuous Sheet Metal Edging: At openings, concrete slab edges and roof deck edges.

(a) Sheet Steel: Minimum (0.04 inch) thick.

(3) Metal Closure Strips: For openings between decking and other construction. Form to configurations required to provide tight-fitting closures at open ends of flutes and sides of decking.

(a) Sheet Steel: Minimum (0.04 inch) thick.

(4) Ridge and Valley Plates: Minimum (4 inch) wide ridge and valley plates where roof slope exceeds 1/24 (1/2 inch per foot).

(a) Sheet Steel: Minimum (0.04 inch) thick.

(5) Cant Strips: Provide bent metal 45 degree leg cant strips where indicated on the drawings. Fabricate cant strips with minimum (5 inch) face width.

(a) Sheet Steel: Minimum (0.03 inch) thick.

(6) Seat Angles for Deck: Provide where beam does not frame into column.

(7) Sump Pans for Roof Drains: Fabricated from single piece galvanized sheet steel with level bottoms and sloping sides to direct water flow to drain. Provide sump pans of adequate size to receive roof drains and with bearing flanges minimum (3 inches) wide. Recess pans minimum (1-1/2 inches) below roof deck surface, unless otherwise shown or required by deck configuration. Drain holes will be field cut.

(a) Sheet Steel: Minimum (0.06 inch) thick.

2.5 FINISHES

Shop prime painted sheet steel with two coats of primer.

2.6 ACCESSORIES

a. Primer: Manufacturer's standard primer compatible with finish painting specified in Section 09 91 00, PAINTING.

b. Welding Materials: AWS D1.1/D1.1M, type to suit application.

c. Galvanizing Repair Paint: MPI 18.

d. Touch-Up Paint: Match shop finish.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Remove contaminates from structural steel surfaces where steel decking will be welded.

d. Verify structural steel framing installation is completed, plumbed, and aligned with temporary bracing installed where required.

e. Coordinate with structural steel erector to prevent overloading of structural members when placing steel decking for installation.

3.2 ERECTION

a. Do not use floor deck units for storage or working platforms until permanently secured. Do not overload deck units once placed. Replace deck units that become damaged after erection and before casting concrete at no cost additional to the Government.

b. Place steel decking at right angles to supporting members with ends

located over supports.

c. Lap end joints (2 inches), minimum.

d. Fluted Form Deck Fastening:

(1) Fasten form deck to steel supporting members by welding.

(a) Welds: (5/8 inch) diameter puddle welds or elongated welds of equal strength.

(b) Weld Spacing: Maximum (12 inches) on center with minimum two welds per unit at each support.

(c) Where two units abut, fasten each unit individually to supporting steel framework.

(2) End Closure Fastening: Tack weld or self-tapping No. 8 or larger machine screws at (3 feet) on center.

(a) Longitudinal End Closure Fastening: Tack weld only.

(3) Weld side laps of adjacent decking units.

(a) Fastener Locations: Mid-span and maximum (3 feet) on center.

e. Corrugated Form Deck Fastening:

(1) Weld end laps of corrugated form deck units in valley of side lap and at middle of sheet.

(a) Weld Spacing: Maximum (15 inches) on center.

(2) Weld corrugated deck to intermediate supports in X-pattern. Weld in valley of side laps on every other support and in valley of center corrugation on remaining support.

(a) Weld Spacing: Maximum (30 inches) on center.

f. Roof Deck Fastening:

(1) Fasten decking to steel supporting members by welding.

(a) Welds: (5/8 inch) diameter puddle welds or elongated welds of equal strength.

(b) Weld Spacing: Maximum (12 inches) on center at every support. Use closer spacing where required for lateral force resistance by diaphragm action.

(2) Fasten split or partial decking panels to structure in every valley.

(3) Fasten decking to each supporting member at ribs where side laps occur.

(a) Power driven fasteners is acceptable in lieu of welding if strength equivalent to welding specified above is provided. Submit test data and design calculations verifying equivalent

design strength.

(4) Mechanically fasten decking side laps with self-tapping No. 8 or larger machine screws.

(a) Fastener Locations: Mid-span and maximum (3 feet) on center.

(5) Provide additional fastening necessary to comply with UL Listing for specified performance.

g. Cutting and Fitting:

(3) Coordinate openings for other penetrations shown on approved submittal drawings but not shown on Structural Drawings.

(a) Cut and reinforce required opening.

(4) Make cuts neat and trim using metal saw, drill or punch-out device. Cutting with torches is prohibited.

(5) Do not make cuts in the metal deck that are not shown on the approved metal decking submittal drawings.

(a) When additional openings are required, submit scaled drawing, locating required opening and other openings and supports in immediate area.

(b) Do not cut the opening until drawing is approved by Contracting Officer's Representative.

(c) Provide additional reinforcing and framing required for opening.

(d) Failure to comply with these requirements is cause for rejection of the work and removal and replacement of the affected steel decking.

(6) Opening Reinforcement: Provide additional metal reinforcement and closure pieces as required for strength, continuity of decking, and support of other work.

h. Touch up damaged factory finishes.

(1) Apply galvanizing repair paint to damaged galvanized surfaces.

(2) Apply touch up paint to damaged shop painted surfaces.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 36 00

COMPOSITE METAL DECKING

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 DESIGN REQUIREMENTS
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 REQUIREMENTS

PART 3 EXECUTION

3.1 ERECTION
3.2 CLEANING

-- End of Section Table of Contents --

SECTION 05 36 00
1/21

COMPOSITE METAL DECKING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies material and services required for installation of composite steel decking including shear connector studs and miscellaneous closures required to prepare deck for concrete placement as shown and specified.

1.2 RELATED WORK

a. Section 01 45 29, TESTING LABORATORY SERVICES: Materials testing and inspection during construction.

1.3 DESIGN REQUIREMENTS

a. Design steel decking in accordance with AISI S100, except as otherwise shown or specified.

b. Design steel decking to comply with SDI codes.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welders Qualifications; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

Shear Stud Layouts; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Test Report - Establishing Structural Characteristics; G, AE

Test Report - Stud Base Qualification; G, AE

SD-07 Certificates

Certification; G, AE

Manufacturers Certificates; G, AE

Power Actuated Tool Operator Certificates; G, AE

SD-08 Manufacturer's Instructions

Manufacturer's Written Recommendations; G, AE

Welding Power Setting Recommendation; G, AE

Manufacturer’s Catalog Data For Welding Equipment; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. All items indicated below are required submittals requiring Contracting Officer’s Representative review and approval.

c. Shop Drawings: Shop and erection drawings showing decking unit layout, connections to supporting members, and information necessary to complete the installation as shown and specified, including supplementary framing, cant strips, cut openings, special jointing or other accessories.

(1) Show welding, side lap, closure, deck reinforcing and closure reinforcing details.

(2) Show openings required for work of other trades, including openings not shown on structural drawings.

(3) Indicate where temporary shoring is required to satisfy design criteria.

d. Manufacturer's Literature and Data: Showing steel decking section properties and specifying required structural characteristics.

e. Manufacturer's written recommendations for:

(1) Shape of decking section.

(2) Cleaning of steel decking prior to concrete placement.

f. Test Report - Establishing structural characteristics of composite concrete and steel decking system.

g. Test Report - Stud base qualification.

h. Welding power setting recommendation by shear stud manufacturer.

i. Shear Stud Layouts: Submit drawings showing the quantity, pattern, spacing and configuration of shear studs for each beam and girder.

j. Certification: For each type and gauge of metal deck supporting concrete slab or fill, submit certification of specified fire ratings. Certify that units supplied are UL listed as a “Steel Floor and Form Unit”.

k. Manufacturers Certificates for deck units attesting compliance with specified requirements.

l. Submit manufacturer’s catalog data for Welding Equipment and Welding Rods and Accessories intended use.

m. Power Actuated Tool Operator Certificates.

n. Welders qualifications.

1.6 QUALITY ASSURANCE

a. Fire Safety

(1) Underwriters’ Label: Provide composite metal floor deck units listed in Underwriters’ Laboratories “Building Materials Directory”, with each deck unit bearing the UL label and marking for specific system detailed.

(3) Insurance Certification: Assist the Government in preparation and submittal of roof installation acceptance certification as may be necessary in connection with fire and extended coverage insurance.

b. Wind Storm Resistance: Provide roof construction assembly capable of withstanding an uplift pressure of90 pounds per square foot) when tested in accordance with the uplift pressure test described in the FM DS 1-28 or as described in the UL 580.

c. Deck Units: Provide deck units and accessory products from a manufacturer engaged in the manufacture of steel decking for more than three (3) years. Submit manufacturer’s certificates attesting that the decking material complies with the specified requirements.

d. Certification of Powder-Actuated Tool Operator: Manufacturer’s certificate attesting that the operators are authorized to use the low velocity powder-actuated tool.

e. Qualifications for Welding Work: Submit qualified welder qualifications in accordance with AWS D1.1/D1.1M or under an approved qualification test.

1.7 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only. Refer to the latest edition of referenced Standards and codes.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100	(2012) North American Specification for the Design of Cold-Formed Steel Structural Members

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon

Structural Steel

ASTM A108	(2024) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS D1.3/D1.3M	(2018) Structural Welding Code - Sheet Steel

FM GLOBAL (FM)

FM DS 1-28	Design Wind Loads

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-P-21035B	(1991; Rev B; Notice 2 2003; Notice 3 2021) Paint, High Zinc Dust Content, Galvanizing Repair (Metric)

UL SOLUTIONS (UL)

UL 580	(2006; Reprint Apr 2024) UL Standard for Safety Tests for Uplift Resistance of Roof Assemblies

PART 2 PRODUCTS

2.1 MATERIALS

a. Steel Decking and Flashings: ASTM A653/A653M, Structural Quality suitable for shear stud weld-through techniques.

b. Recycled Content of Steel Products: Combined recycled content not less than 75

c. Galvanizing: ASTM A653/A653M, G60 . Thickness not less than (0.014 inch) .

d. Shear connector studs: ASTM A108, Grades 1015-1020, yield (50,000 pound/square inch) minimum, tensile strength - (60,000 pounds/square inch) minimum, reduction of area 50 percent minimum.

(1) Provide studs of uniform diameter, with heads concentric and on same axis to shaft.

(2) Provide studs, after welding, free from substance or defect which would interfere with its function as a shear connector.

(3) Do not paint or galvanize studs.

(4) Provide size of studs as shown on drawings.

(5) Provide studs manufactured by a company normally engaged in the manufacturer of shear studs, and can furnish equipment suitable for weld-through installation of shear studs.

e. Galvanizing Repair Paint: Mil. Spec. MIL-P-21035B.

f. Miscellaneous Steel Shapes: ASTM A36/A36M.

g. Welding Electrode: E60XX minimum.

h. Sheet Metal Accessories: ASTM A653/A653M, galvanized, unless noted otherwise. Provide accessories of every kind required to complete the installation of metal decking in the system shown. Finish sheet metal items to match deck including, but not limited to, the following items:

(1) Metal Cover Plates: For end-abutting deck units, to close gaps at changes in deck direction, columns, walls and openings. Same quality as deck units but not less than (18 gauge) sheet steel.

(2) Continuous sheet metal edging: at openings and concrete slab edges. Same quality as deck units but not less than (18 gauge) steel. Side and end closures supporting concrete and their attachment to supporting steel to be designed by the manufacturer to safely support the wet weight of concrete and construction loads. The deflection of cantilever closures to be limited to a total of 1/8 inch maximum.

(3) Metal Closure Strips: For openings between decking and other construction, of not less than (18 gauge) sheet steel of the same quality as the deck units. Form to the configuration required to provide tight-fitting closures at open ends of flutes and sides of decking.

(4) Seat angles for deck: Where a beam does not frame into a column.

2.2 REQUIREMENTS

a. Steel decking depth, gauge, and section properties to be as shown on contract documents. Provide edges of deck with vertical interlocking male and female lip providing for a positive mechanical connection.

b. Fabricate deck units with integral embossments to provide mechanical bond with concrete slab. Deck units combined with concrete slab to be capable of supporting total design loads.

c. Provide integral system with single point of attachment for light duty hanger devices for flexibility for attaching hangers for support of acoustical, lathing, plumbing, heating, air conditioning electrical and similar items.

(1) Provide a minimum spacing pattern of (12 inches) on centers longitudinally and (24 or 36 inches) on centers transversely.

(2) Provide suspension system capable of safely supporting a maximum allowable load of (100 pounds) concentrated at one hanger attachment point.

(3) System may consist of fold-down type hanger tabs or a lip hanger.

PART 3 EXECUTION

3.1 ERECTION

a. Do not start installation of metal decking until corresponding steel framework has been plumbed, aligned and completed, and until temporary shoring, where required, has been installed.

(1) Remove oil, dirt, paint, ice, water and rust from steel surfaces to which metal decking will be welded.

b. Coordinate and cooperate with structural steel erector in locating decking bundles to prevent overloading of structural members.

c. Do not use floor deck units for storage or working platforms until permanently secured.

(1) Do not overload deck units once placed.

(2) Replace deck units that become damaged after erection and prior to casting concrete at no additional cost to the Government.

d. Erect steel deck in accordance with manufacturer's printed instructions.

e. Ship steel deck units in standard widths and fabricated to proper length.

f. Provide steel decking in sufficient lengths to extend over 3 or more spans, except where structural steel layout does not permit.

g. Place steel decking units on supporting steel framework and adjust to final position before being permanently fastened.

(1) Bring each unit to proper bearing on supporting beams.

(2) Place deck units in straight alignment for entire length of run of flutes and with close registration of flutes of one unit with those of abutting unit.

(3) 3. Maximum space between ends of abutting units is (1/2 inch). If space exceeds (1/2 inch), install closure plates.

h. Ceiling hanger loops, if provided, must be flattened or removed to obtain bearing of units on structural steel.

i. Fastening Deck Units:

(1) Fasten floor deck units to steel supporting members by not less than 5/8 inch diameter puddle welds or elongated welds of equal strength, spaced not more than 12 inches on center with a minimum of two welds per unit at each support. Where two units abut, fasten each unit individually to the supporting steel framework.

(2) Tack weld or use self-tapping No. 8 or larger machine screws at 3 feet on center for fastening end closures. Only use welds to attach longitudinal end closures.

(3) Weld side laps of adjacent floor deck units that span more than (5 feet). Fasten at midspan or 3 feet on center, whichever is smaller.

j. Weld in conformance to AWS D1.3/D1.3M and done by qualified experienced welding mechanics.

k. Clean and touch-up area and welds scarred during erection, and repair with zinc rich galvanizing repair paint.

(1) Paint touch-up is not required for welds or scars that are to be in direct contact with concrete.

l. Provide metal concrete stops at edges of deck.

m. Cutting and Fitting:

(1) Fabricate metal deck units to proper length prior to shipping.

(2) Field cutting by the metal deck erector is restricted to bevel cuts, notching to fit around columns and similar items, and cutting openings that are located and dimensioned on the structural drawings.

(3) Other penetrations shown on the approved metal deck shop drawings but not shown on the structural drawings are to be located, cut and reinforced.

(4) Make cuts and penetrations neat and trim using a metal saw, drill or punchout device; cutting with torches is prohibited.

(5) Do not make cuts in the metal deck that are not shown on the approved metal deck drawings.

(6) If an additional opening not shown on the approved shop drawings is required, submit a sketch, to scale, locating the required new opening and other openings and supports in the immediate area. Do not cut the opening until the sketch has been reviewed and accepted by the Contracting Officer Representative. Provide additional reinforcing or framing required for the opening at no additional cost to the Government.

(7) Reinforcement at Openings: Provide additional metal reinforcement and closure pieces as required for strength, continuity of decking and support of other work shown.

n. Install shear connector studs through previously installed metal deck in conformance to AWS D1.1/D1.1M, Section 7.

Exception: Install studs with automatically timed welding equipment and as specified below:

(1) Do not place welded wire reinforcing or other materials and equipment which will interfere with stud installation on steel deck until shear connector studs are installed.

(2) Clean steel deck sheets free of oil, rust, dirt, and paint. Release water in deck’s valley so that it does not become

entrapped between deck and beam. Clean and dry surface to which stud is to be welded.

(3) Rest metal deck tightly upon top flange of structural member with bottom of deck rib in full contact with top of beam flange.

(4) Weld studs only through a single thickness of deck. Place decking so that a butt joint is obtained. Place studs directly over beam web, where one row of studs are required.

(5) Provide ferrules specially developed for the weld-through technique, and appropriate for size of studs installed. Remove ferrules after welding.

(6) Submit report of successful test program for stud base qualification as required by AWS D1.1/D1.1M, Appendix K.

3.2 CLEANING

Clean deck in accordance with manufacturer's recommendation before concrete placement.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 40 00

COLD-FORMED METAL FRAMING

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 DESIGN REQUIREMENTS
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 WALL FRAMING
2.3 JOIST FRAMING
2.4 FRAMING ACCESSORIES
2.5 ANCHORS, CLIPS, AND FASTENERS
2.6 REQUIREMENTS

PART 3 EXECUTION

3.1 FABRICATION
3.2 ERECTION
3.3 TOLERANCES
3.4 FIELD REPAIR

-- End of Section Table of Contents --

SECTION 05 40 00
1/21

COLD-FORMED METAL FRAMING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies materials and services required for installation of cold-formed steel, including tracks and required accessories as shown and specified. This Section includes the following:

(1) Exterior load-bearing steel stud walls.

(2) Interior load-bearing steel stud walls.

(3) Exterior non-load-bearing steel stud curtain wall.

(4) Steel joists.

(5) Steel trusses.

1.2 RELATED WORK

a. Section 05 12 00, STRUCTURAL STEEL FRAMING: Structural steel framing.

b. Section 05 21 00, STEEL JOIST FRAMING: Open web steel joists.

c. Section 09 22 16, NON-STRUCTURAL METAL FRAMING: Non-load-bearing metal stud framing assemblies.

d. Section 09 29 00, GYPSUM BOARD: Gypsum board assemblies.

1.3 DESIGN REQUIREMENTS

a. Design steel in accordance with American Iron and Steel Institute Publication "Specification for the Design of Cold-Formed Steel Structural Members", except as otherwise shown or specified.

b. Structural Performance: Engineer, fabricate and erect cold-formed metal framing with the minimum physical and structural properties indicated.

c. Structural Performance: Engineer, fabricate, and erect cold-formed metal framing to withstand design loads within limits and under conditions required.

(1) Design Loads:

(a) Gravity, wind and seismic loading as indicated on the drawings or in this specification.

(b) Blast Loads:

1. Calculated based on the design charge weight, W1, located at the minimum standoff distance of 50-ft measured horizontally

from the building envelope, but is not to exceed the pressure / impulse pairs specified below (as an equivalent right-triangular load):

Ground Level & Level 01 Exterior Walls: 13.5 psi / 51 psi-msec

Level 02 Exterior Walls: 9.0 psi / 37 psi-msec

Parapet Walls (above Roof Level): 5.0 psi / 11.25 psi-msec

(2) Design framing systems to withstand design loads without deflections greater than the following:

(a) Exterior Load-Bearing Walls: Lateral deflection of 1/360 of the wall height.

(b) Interior Load-Bearing Walls: Lateral deflection of 1/240 of the wall height.

(c) Exterior Non-load-Bearing Curtain wall: Lateral deflection of 1/360 of the wall height.

(3) Design all exterior cold-formed metal framing to meet the considerations for Critical Facilities per Section 6.2 of the VA PSRDM.

(a) Cold-formed metal framing must be designed to resist the specified blast loads applied over the tributary area of the member while limiting deflections that result in ductility ratios less than or equal to: 1.75 for studs continuous across a support or connected at top and bottom, and 0.8 for slip track connections. Refer to the response limits for Moderate Damage per ASCE/SEI 59 for all other conditions.

(b) Connections of metal framing to their supports must be designed for a minimum static capacity equal to the equivalent static reactions (based on the ultimate flexural capacity of the element). Alternatively, connections for members responding elastically (ductility ratio less than 1) may be designed for the maximum dynamic reactions calculated from the blast analysis and increased by a factor of 1.5, but not larger than the equivalent static reactions. The capacity of connections must be calculated as the LRFD capacity including strength reduction factors (phi) per appropriate design codes for each material.

(4) Design framing systems to provide for movement of framing members without damage or overstressing, sheathing failure, connection failure, undue strain on fasteners and anchors, or other detrimental effects when subject to a maximum ambient temperature change (range) of (120 degrees F).

(5) Design framing system to accommodate deflection of primary building structure and construction tolerances, and to maintain clearances at openings.

(6) Design exterior non-load-bearing curtain wall framing to accommodate lateral deflection without regard to contribution of sheathing materials.

(7) Engineering Responsibility: Engage a fabricator who assumes undivided responsibility for engineering cold-formed metal framing by employing a qualified professional engineer to prepare design calculations, shop drawings, and other structural data.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor’s Structural Engineer; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-05 Design Data

Design Of The Light Gauge; G, AE

Blast Design Calculations; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. All items indicated below are required submittals requiring Contracting Officer’s Representative review and approval

b. Shop Drawings: Shop and erection drawings showing steel unit layout, connections to supporting members, and information necessary to complete installation as shown and specified.

c. Manufacturer's Literature and Data: Showing steel component sections and specifying structural characteristics.

d. Design of the light gauge for this project shall be provided by the contractor’s structural engineer for the loads shown on the construction documents. Submit signed and sealed calculations performed by a structural engineer with at least 5 years experience in the design of light gauge metal and registered in the state of the project. Calculations shall be submitted with plans elevations and details for review and approval.

e. Blast Design Calculations: Light Gauge Members and Connections

(1) Submit blast calculations for review and approval prepared by qualified blast consultant, with a minimum of 5 years experience in blast resistant design. Design submittals must include a blast narrative and supporting calculations and must be signed and

stamped by the professional engineer responsible for their preparation. Blast analysis calculations must be performed using the methods described in Chapters 6 & 7 of the VA PSRDM. Any alternative measures of demonstrating blast resistance must be approved by the design team blast engineer-of-record

f. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

1.6 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM C955	(2024) Standard Specification for Cold-Formed Steel Structural Framing Members
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM E488/E488M	(2022) Standard Test Methods for Strength of Anchors in Concrete Elements
ASTM E1190	(2021)Standard Test Methods for Strength of Power-Actuated Fasteners Installed in Structural Members

AMERICAN WELDING SOCIETY (AWS)

AWS D1.3/D1.3M	(2018) Structural Welding Code - Sheet Steel

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-P-21035B	(1991; Rev B; Notice 2 2003; Notice 3 2021) Paint, High Zinc Dust Content, Galvanizing Repair (Metric)

PART 2 PRODUCTS

2.1 MATERIALS

a. Sheet Steel for joists, studs and accessories 16 gauge and heavier: ASTM A653/A653M, structural steel, zinc coated CP60 , with a yield of (50 ksi) minimum.

b. Sheet Steel for joists, studs and accessories 18 gauge and lighter: ASTM A653/A653M, structural steel, zinc coated G60 , with a yield of (33 ksi) minimum.

c. Galvanizing Repair Paint: MIL-P-21035B.

d. Nonmetallic, Non-shrink Grout: Premixed, nonmetallic, noncorrosive, nonstaining grout containing selected silica sands, Portland cement, shrinkage-compensating agents, plasticizing and water-reducing agents, complying with ASTM C1107/C1107M, with fluid consistency and a 30 minute working time.

2.2 WALL FRAMING

a. Steel Studs: Complying with ASTM C955. Manufacturer’s standard C-shaped steel studs of web depth indicated, with lipped flanges, and complying with the following:

(1) Minimum Base-Steel Thickness(uncoated): (0.0538 inch)

(2) Flange Width: (2 inches)

(3) Web: Punched .

b. Steel Track: Manufacturer’s standard U-shaped steel track, unpunched, of web depths indicated, with straight flanges, and complying with the following:

(1) Design Uncoated-Steel Thickness: Matching steel studs.

(2) Flange Width: Manufacturer’s standard deep flange where indicated, standard flange elsewhere.

2.3 JOIST FRAMING

a. Steel Joists: Manufacturer’s standard C-shaped steel joists, unpunched, of web depths indicated, with lipped flanges, and complying with the following:

(1) Minimum Base-Steel Thickness: (0.0538 inch).

(2) Design Thickness: (0.0566 inch).

(3) Flange Width: (2 inches).

b. Steel Joist Track: Manufacturer’s standard U-shaped steel joist track, unpunched, of web depths indicated, with straight flanges, and complying with the following:

(1) Design Thickness: Matching steel joists.

(2) Flange Width: (2 inches).

2.4 FRAMING ACCESSORIES

a. Fabricate steel framing accessories of the same material and finish used for framing members, with a minimum yield strength of (33 ksi).

b. Provide accessories of manufacturer’s standard thickness and configuration, unless otherwise indicated, as follows:

(1) Supplementary framing.

(2) Bracing, bridging, and solid blocking.

(3) Web stiffeners.

(4) Gusset plates.

(5) Deflection track and vertical slide clips.

(6) Stud kickers and girts.

(7) Joist hangers and end closures.

(8) Reinforcement plates.

2.5 ANCHORS, CLIPS, AND FASTENERS

a. Steel Shapes and Clips: ASTM A36/A36M, zinc coated by the hot-dip process according to ASTM A123/A123M.

b. Cast-in-Place Anchor Bolts and Studs: ASTM A307, Grade A, zinc coated by the hot-dip process according to ASTM A153/A153M.

c. Expansion Anchors: Fabricated from corrosion-resistant materials, with capability to sustain, without failure, a load equal to 5 times the design load, as determined by testing per ASTM E488/E488M conducted by a qualified independent testing agency.

d. Power-Actuated Anchors: Fastener system of type suitable for application indicated, fabricated from corrosion-resistant materials, with capability to sustain, without failure, a load equal to 10 times the design load, as determined by testing per ASTM E1190 conducted by a qualified independent testing agency.

e. Mechanical Fasteners: Corrosion-resistant coated, self-drilling, self-threading steel drill screws. Low-profile head beneath sheathing, manufacturer’s standard elsewhere.

2.6 REQUIREMENTS

a. Welding in accordance with AWS D1.3/D1.3M

b. Furnish members and accessories by one manufacturer only.

PART 3 EXECUTION

3.1 FABRICATION

a. Framing components may be preassembled into panels. Panels shall be square with components attached.

b. Cut framing components squarely or as required for attachment. Cut framing members by sawing or shearing; do not torch cut.

c. Hold members in place until fastened.

d. Fasten cold-formed metal framing members by welding or screw fastening, as standard with fabricator. Wire tying of framing members is not permitted.

(1) Comply with AWS requirements AWS D1.3/D1.3M and procedures for welding, appearance and quality of welds, and methods used in correcting welding work.

(2) Locate mechanical fasteners and install according to cold-formed metal framing manufacturer’s instructions with screw penetrating joined members by not less than 3 exposed screw threads.

e. Where required, provide specified insulation in double header members and double jamb studs which will not be accessible after erection.

3.2 ERECTION

a. Handle and lift prefabricated panels in a manner as to not distort any member.

b. Securely anchor tracks to supports as shown.

c. At butt joints, securely anchor two pieces of track to same supporting member or butt-weld or splice together.

d. Plumb, align, and securely attach studs to flanges or webs of both upper and lower tracks.

e. All axially loaded members shall be aligned vertically to allow for full transfer of the loads down to the foundation. Vertical alignment shall be maintained at floor/wall intersections.

f. Install jack studs above and below openings and as required to furnish support. Securely attach jack studs to supporting members.

g. Install headers in all openings that are larger than the stud spacing in that wall.

h. Attach bridging for studs in a manner to prevent stud rotation. Space bridging rows as shown.

i. Studs in one piece for their entire length, splices will not be permitted.

j. Provide a load distribution member at top track where joist is not

located directly over bearing stud.

k. Provide joist bridging and web stiffeners at reaction points where shown.

l. Provide end blocking where joist ends are not restrained from rotation.

m. Provide an additional joist under parallel partitions, unless otherwise shown, when partition length exceeds one-half joist span and when floor and roof openings interrupt one or more spanning members.

n. Provide temporary bracing and leave in place until framing is permanently stabilized.

o. Do not bridge building expansion joints with cold-formed metal framing. Independently frame both sides of joints.

p. Fasten reinforcement plate over web penetrations that exceed size of manufacturer’s standard punched openings.

3.3 TOLERANCES

a. Vertical alignment (plumbness) of studs shall be within 1/960th of the span.

b. Horizontal alignment (levelness) of walls shall be within 1/960th of their respective lengths.

c. Spacing of studs shall not be more than 1/8 inch +/- from the designed spacing providing that the cumulative error does not exceed the requirements of the finishing materials.

d. Prefabricated panels shall be not more than 1/8 inch +/- out of square within the length of that panel.

3.4 FIELD REPAIR

Touch-up damaged galvanizing with galvanizing repair paint.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 50 00

METAL FABRICATIONS

8/18

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 QUALITY ASSURANCE
1.6 REFERENCES

PART 2 PRODUCTS

2.1 DESIGN CRITERIA
2.2 MATERIALS
2.3 HARDWARE
2.4 FABRICATION GENERAL
2.5 SUPPORTS
2.6 FRAMES
2.7 GUARDS
2.8 COVERS AND FRAMES FOR PITS AND TRENCHES
2.9 GRATINGS
2.10 LOOSE LINTELS
2.11 SHELF ANGLES
2.12 PLATE DOOR SILL
2.13 SAFETY NOSINGS
2.14 LADDERS
2.15 RAILINGS
2.16 CATWALKS
2.17 TRAP DOOR AND FRAMES WITH CEILING HATCH
2.18 SIDEWALK DOOR
2.19 SCREENED ACCESS DOORS AND FRAMES
2.20 STEEL COUNTER OR BENCH TOP FRAME AND LEGS
2.21 STEEL PIPE BOLLARD

PART 3 EXECUTION

3.1 INSTALLATION, GENERAL
3.2 INSTALLATION OF SUPPORTS
3.3 COVERS AND FRAMES FOR PITS AND TRENCHES
3.4 DOOR FRAMES
3.5 OTHER FRAMES
3.6 GUARDS
3.7 GRATINGS
3.8 STEEL LINTELS
3.9 SHELF ANGLES
3.10 PLATE DOOR SILL
3.11 SAFETY NOSINGS

3.12 LADDERS
3.13 RAILINGS
3.14 CATWALK AND PLATFORMS
3.15 SIDEWALK DOOR, TRAP DOORS, AND FRAMES
3.16 SCREENED ACCESS DOOR
3.17 STEEL COMPONENTS FOR MILLWORK ITEMS
3.18 INSTALLATION OF STEEL PIPE BOLLARD
3.19 CLEAN AND ADJUSTING

-- End of Section Table of Contents --

SECTION 05 50 00
8/18

METAL FABRICATIONS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies items and assemblies fabricated from structural steel shapes and other materials as shown and specified.

b. Items specified.

(1) Support for Wall and Ceiling Mounted Items: (SD055000-01, SD055000-02, SD102113-01, SD102600-01, SD123100-01 & SD123100-02)

(2) Frames:

(3) Guards

(4) Covers and Frames for Pits and Trenches.

(5) Gratings

(6) Loose Lintels

(7) Shelf Angles

(8) Gas Racks

(9) Plate Door Sill

(10) Safety Nosings

(11) Ladders

(12) Railings

(13) Catwalks and Platforms

(14) Trap Doors with Ceiling Hatch

(15) Sidewalk Access Doors

(16) Screened Access Doors

(17) Steel Counter or Bench Top Frame and Leg

(18) Steel Pipe Bollards

1.2 RELATED WORK

a. Railings attached to steel stairs: Section 05 51 00, METAL STAIRS.

b. Prime and finish painting: Section 09 91 00, PAINTING.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Setting Drawings;

SD-02 Shop Drawings

Shop Drawings;

Recycled Content; S

SD-03 Product Data

Manufacturer's Literature And Data;

Recycled Content; S

SD-05 Design Data

Design Calculations;

SD-07 Certificates

Manufacturer's Certificates;

1.4 SUBMITTAL INSTRUCTIONS

a. Furnish setting drawings and instructions for installation of anchors to be preset into concrete and masonry work, and for the positioning of items having anchors to be built into concrete or masonry construction.

b. Shop Drawings:

(1) Each item specified, showing complete detail, location in the project, material and size of components, method of joining various components and assemblies, finish, and location, size and type of anchors.

(2) Mark items requiring field assembly for erection identification and furnish erection drawings and instructions.

(3) Provide templates and rough-in measurements as required.

c. Manufacturer's Literature and Data:

Grating, each type	Floor plate
Trap door	Wheel guards
Ceiling hatch	Sidewalk Access door
Manhole Covers	Safety nosing

d. Design Calculations for specified live loads including dead loads.

e. Manufacturer's Certificates:

(1) Anodized finish as specified.

(2) Live load designs as specified.

1.5 QUALITY ASSURANCE

a. Each manufactured product shall meet, as a minimum, the requirements specified, and shall be a standard commercial product of a manufacturer regularly presently manufacturing items of type specified.

b. Each product type shall be the same and be made by the same manufacturer.

c. Assembled product to the greatest extent possible before delivery to the site.

d. Include additional features, which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product.

1.6 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.6.1 (2016) Wood Screws (Inch Series)

ASME B18.2.2 (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A48/A48M (2022) Standard Specification for Gray Iron Castings

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A269/A269M	(2024) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A391/A391M	(2015)Grade 80 Alloy Steel Chain
ASTM A525	(2007) Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process
ASTM A786/A786M	(2015; R 2021) Standard Specification for Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B456	(2017; R 2022) Standard Specification for Electrodeposited Coatings of Copper Plus Nickel Plus Chromium and Nickel Plus Chromium
ASTM B632/B632M	(2024) Standard Specification for Aluminum-Alloy Rolled Tread Plate
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)
ASTM D3656/D3656M	(2013) Standard Specification for Insect Screening and Louver Cloth Woven from Vinyl-Coated Glass Yarns
ASTM F436	(2011) Hardened Steel Washers
ASTM F468	(2023) Standard Specification for Nonferrous Bolts, Hex Cap Screws, and Studs for General Use
ASTM F593	(2024) Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs
ASTM F1667/F1667M	(2021a) Standard Specification for Driven Fasteners: Nails, Spikes, and Staples

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual
NAAMM AMP 501	(2006) Metal Finishes Manual- Finishes for Aluminum

NAAMM AMP 504 (2006) Metal Finishes Manual- Finishes for Carbon Steel and Iron

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 1 (2015) Solvent Cleaning

SSPC SP 2 (2024) Hand Tool Cleaning

SSPC SP 3 (2024) Power Tool Cleaning

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

RR-T-650E (1994) Treads, Metallic and Nonmetallic, Skid-Resistant

RR-G-1602D (1996) Grating, Metal, Other Than Bar Type (Floor, Except for Naval Vessels)

PART 2 PRODUCTS

2.1 DESIGN CRITERIA

a. In addition to the dead loads, design fabrications to support the following live loads unless otherwise specified.

b. Ladders and Rungs: (250 pounds) at any point.

c. Railings and Handrails: (200 pounds) in any direction at any point.

d. Floor Plates, Gratings, Covers, Trap Doors, Catwalks, and Platforms: (100 pounds per square foot).

e. Manhole Covers: (250 pounds per square foot).

2.2 MATERIALS

a. Structural Steel: ASTM A36/A36M.

b. Aluminum, Extruded: ASTM B221, Alloy 6063-T5 unless otherwise specified. For structural shapes use alloy 6061-T6 and alloy 6061-T4511.

c. Floor Plate:

(1) Steel ASTM A786/A786M.

d. Steel Pipe (Bollard): ASTM A53/A53M.

(1) Galvanized for exterior locations.

(2) Type S, Grade A unless specified otherwise.

(3) NPS (inside diameter) as shown.

e. Primer Paint: As specified in Section 09 91 00, PAINTING.

f. Stainless Steel Tubing: ASTM A269/A269M, type 302 or 304.

g. Modular Channel Units:

(1) Factory fabricated, channel shaped, cold formed sheet steel shapes, complete with fittings bolts and nuts required for assembly.

(2) Form channel within turned pyramid shaped clamping ridges on each side.

(3) Provide case hardened steel nuts with serrated grooves in the top edges designed to be inserted in the channel at any point and be given a quarter turn so as to engage the channel clamping ridges. Provide each nut with a spring designed to hold the nut in place.

(4) Factory finish channels and parts with oven baked primer when exposed to view. Channels fabricated of ASTM A525, G90 galvanized steel may have primer omitted in concealed locations. Finish screws and nuts with zinc coating.

(5) Fabricate snap-in closure plates to fit and close exposed channel openings of not more than 0.0125 inch thick stainless steel.

h. Grout: ASTM C1107/C1107M, pourable type.

i. Insect Screening: ASTM D3656/D3656M.

2.3 HARDWARE

a. Rough Hardware:

(1) Furnish rough hardware with a standard plating, applied after punching, forming and assembly of parts; galvanized, cadmium plated, or zinc-coated by electro-galvanizing process. Galvanized G-90 where specified.

(2) Use G90 galvanized coating on ferrous metal for exterior work unless non-ferrous metal or stainless is used.

b. Fasteners:

(1) Bolts with Nuts:

(a) ASME B18.2.2.

(b) ASTM A307 for (60,000 psi) tensile strength bolts.

(c) ASTM F468 for nonferrous bolts.

(d) ASTM F593 for stainless steel.

(2) Screws: ASME B18.6.1.

(3) Washers: ASTM F436, type to suit material and anchorage.

(4) Nails: ASTM F1667/F1667M, Type I, style 6 or 14 for finish work.

2.4 FABRICATION GENERAL

a. Material

(1) Use material as specified. Use material of commercial quality and suitable for intended purpose for material that is not named or its standard of quality not specified.

(2) Use material free of defects which could affect the appearance or service ability of the finished product.

b. Size:

(1) Size and thickness of members as shown.

(2) When size and thickness is not specified or shown for an individual part, use size and thickness not less than that used for the same component on similar standard commercial items or in accordance with established shop methods.

c. Connections

(1) Except as otherwise specified, connections may be made by welding, riveting or bolting.

(2) Field riveting will not be approved.

(3) Design size, number and placement of fasteners, to develop a joint strength of not less than the design value.

(4) Holes, for rivets and bolts: Accurately punched or drilled and burrs removed.

(5) Size and shape welds to develop the full design strength of the parts connected by welds and to transmit imposed stresses without permanent deformation or failure when subject to service loadings.

(6) Use Rivets and bolts of material selected to prevent corrosion (electrolysis) at bimetallic contacts. Plated or coated material will not be approved.

(7) Use stainless steel connectors for removable members machine screws or bolts.

d. Fasteners and Anchors

(1) Use methods for fastening or anchoring metal fabrications to building construction as shown or specified.

(2) Where fasteners and anchors are not shown, design the type, size, location and spacing to resist the loads imposed without deformation of the members or causing failure of the anchor or fastener, and suit the sequence of installation.

(3) Use material and finish of the fasteners compatible with the kinds of materials which are fastened together and their location in the finished work.

(4) Fasteners for securing metal fabrications to new construction only, may be by use of threaded or wedge type inserts or by anchors for welding to the metal fabrication for installation before the concrete is placed or as masonry is laid.

(5) Fasteners for securing metal fabrication to existing construction or new construction may be expansion bolts, toggle bolts, power actuated drive pins, welding, self drilling and tapping screws or bolts.

e. Workmanship

(1) General:

(a) Fabricate items to design shown.

(b) Furnish members in longest lengths commercially available within the limits shown and specified.

(c) Fabricate straight, true, free from warp and twist, and where applicable square and in same plane.

(d) Provide holes, sinkages and reinforcement shown and required for fasteners and anchorage items.

(e) Provide openings, cut-outs, and tapped holes for attachment and clearances required for work of other trades.

(f) Prepare members for the installation and fitting of hardware.

(g) Cut openings in gratings and floor plates for the passage of ducts, sumps, pipes, conduits and similar items. Provide reinforcement to support cut edges.

(h) Fabricate surfaces and edges free from sharp edges, burrs and projections which may cause injury.

(2) Welding:

(a) Weld in accordance with AWS.

(b) Welds shall show good fusion, be free from cracks and porosity and accomplish secure and rigid joints in proper alignment.

(c) Where exposed in the finished work, continuous weld for the full length of the members joined and have depressed areas filled and protruding welds finished smooth and flush with adjacent surfaces.

(d) Finish welded joints to match finish of adjacent surface.

(3) Joining:

(a) Miter or butt members at corners.

(b) Where frames members are butted at corners, cut leg of frame member perpendicular to surface, as required for clearance.

(4) Anchors:

(a) Where metal fabrications are shown to be preset in concrete, weld (1-1/4 by 1/8 inch steel strap anchors, 6 inches long with (one inch) hooked end, to back of member at 2 feet on center, unless otherwise shown.

(b) Where metal fabrications are shown to be built into masonry use (1-1/4 by 1/8 inch steel strap anchors, 10 inches long with 2 inch hooked end, welded to back of member at 2 feet on center, unless otherwise shown.

(5) Cutting and Fitting:

(a) Accurately cut, machine and fit joints, corners, copes, and miters.

(b) Fit removable members to be easily removed.

(c) Design and construct field connections in the most practical place for appearance and ease of installation.

(d) Fit pieces together as required.

(e) Fabricate connections for ease of assembly and disassembly without use of special tools.

(f) Joints firm when assembled.

(g) Conceal joining, fitting and welding on exposed work as far as practical.

(h) Do not show rivets and screws prominently on the exposed face.

(i) The fit of components and the alignment of holes shall eliminate the need to modify component or to use exceptional force in the assembly of item and eliminate the need to use other than common tools.

f. Finish:

(1) Finish exposed surfaces in accordance with NAAMM AMP 500 Metal Finishes Manual.

(2) Aluminum: NAAMM AMP 501.

(a) Mill finish, AA-M10, as fabricated, use unless specified otherwise.

(b) Clear anodic coating, AA-C22A41, chemically etched medium matte, with Architectural Class 1, 0.7 mils or thicker.

(c) Colored anodic coating, AA-C22A42, chemically etched medium matte with Architectural Class 1, 0.7 mils or thicker.

(d) Painted: AA-C22R10.

(3) Steel and Iron: NAAMM AMP 504.

(a) Zinc coated (Galvanized): ASTM A123/A123M, G90 unless noted otherwise.

(b) Surfaces exposed in the finished work:

1. Finish smooth rough surfaces and remove projections.

2. Fill holes, dents and similar voids and depressions with epoxy type patching compound.

3. Seal all holes in all galvanized HSS members penetrating roof membrane.

(c) Shop Prime Painting:

1. Surfaces of Ferrous metal:

a) Items not specified to have other coatings.

b) Galvanized surfaces specified to have prime paint.

c) Remove all loose mill scale, rust, and paint, by hand or power tool cleaning as defined in SSPC SP 2 and SSPC SP 3.

d) Clean of oil, grease, soil and other detrimental matter by use of solvents or cleaning compounds as defined in SSPC SP 1.

e) After cleaning and finishing apply one coat of primer as specified in Section 09 91 00, PAINTING.

2. Non ferrous metals: Comply with NAAMM AMP 500 series.

(4) Stainless Steel: NAAMM AMP 504 Finish No. 4.

(5) Chromium Plating: ASTM B456, satin or bright as specified, Service Condition No. SC2.

g. Protection:

(1) Insulate aluminum surfaces that will come in contact with concrete, masonry, plaster, or metals other than stainless steel, zinc or white bronze by giving a coat of heavy-bodied alkali resisting bituminous paint or other approved paint in shop.

(2) Spot prime all abraded and damaged areas of zinc coating which expose the bare metal, using zinc rich paint on hot-dip zinc coat items and zinc dust primer on all other zinc coated items.

2.5 SUPPORTS

a. General:

(1) Fabricate ASTM A36/A36M structural steel shapes as shown.

(2) Use clip angles or make provisions for welding hangers and braces to overhead construction.

(3) Field connections may be welded or bolted.

b. For Ceiling Hung Toilet Stall:

(1) Use a continuous steel channel above pilasters with hangers centered over pilasters.

(2) Make provision for installation of stud bolts in lower flange of

channel.

(3) Provide a continuous steel angle at wall and channel braces spaced as shown.

(4) Use threaded rod hangers.

(5) Provide diagonal angle brace where the suspended ceiling over toilet stalls does not extend to side wall of room.

(6) Provide supports for ceiling hung pilasters at dressing booths and entrance screen to toilet room similar to support for toilet stall pilasters.

c. For Wall Mounted Items:

(1) For items supported by metal stud partitions.

(2) Steel strip or hat channel minimum of 0.0598 inch thick.

(3) Steel strip minimum of 6 inches wide, length extending one stud space beyond end of item supported.

(4) Steel hat channels where shown. Flange cut and flatted for anchorage to stud.

(5) Structural steel tube or channel for grab bar at water closets floor to structure above with clip angles or end plates formed for anchors.

(6) Use steel angles for thru wall counters. Drill angle for fasteners at ends and not over 4 inches on center between ends.

2.6 FRAMES

a. Elevator Entrance Wall Opening.

(1) Fabricate of channel shapes, plates, and angles as shown.

(2) Weld or bolt head to jamb as shown.

(3) Weld clip angles to bottom of frame and top of jamb members extended to structure above for framed construction.

(a) Provide holes for anchors.

(b) Weld head to jamb members.

b. Channel Door Frames:

(1) Fabricate of structural steel channels of size shown.

(2) Miter and weld frames at corners.

(3) Where anchored to masonry or embedded in concrete, weld to back of frame at each jamb, 3/16 inch thick by (1-3/4 inch) wide steel strap anchors with ends turned (2 inches), and of sufficient length to extend at least 12 inches into wall. Space anchors 24 inches above bottom of frame and 24 inches o.c. to top of jamb.

Weld clip angles to bottom of jambs and provide holes for expansion bolts.

(4) Where anchored to concrete or masonry in prepared openings, drill holes at jambs for anchoring with expansion bolts. Weld clip angles to bottom of frame and provide holes for expansion bolt anchors as shown. Drill holes starting 24 inches above bottom of frame and 24 inches o.c. to top of jamb and at top of jamb. Provide pipe spacers at holes welded to channel.

(5) Where closure plates are shown, continuously weld them to the channel flanges.

(6) Weld continuous (inch) thick steel angles to the interior side of each channel leg at the head and jambs to form a caulking groove.

(7) Prepare frame for installation of hardware specified in Section 08 71 00, DOOR HARDWARE.

(a) Cut a slot in the lock jamb to receive the lock bolt.

(b) Where shown use continuous solid steel bar stops at perimeter of frame, weld or secure with countersunk machine screws at not more than 18 inches on center.

c. Frames for Breech Opening:

(1) Fabricate from steel channels, or combination of steel plates and angles to size and contour shown.

(2) Weld strap anchors on back of frame at not over 2 feet on centers for concrete or masonry openings.

d. Frames for Lead Lined Doors:

(1) Obtain accurate dimensions and templates from suppliers of lead lined doors, finish hardware, and hollow steel door frames.

(2) Fabricate as shown for use in connection with lead lined doors.

(3) Deliver assembled frames with removable shipping spreaders at top and bottom.

(4) Extend angles at jambs from floor to structural slab above. At floors of interstitial spaces, terminate jamb sections and provide anchors as shown.

(5) Continuously weld plates and reinforcements to frame members and head members of angle frames between jambs.

(6) Weld strap anchors, not over 24 inches on centers, to the back of angles for embedment in masonry or concrete unless shown otherwise.

(7) Type 15 Door Frames:

(a) Structural steel angle frames with plate or bar full height to heads. Extend reinforcing at hinge cutouts two inches beyond cutout.

(b) Fabricate top anchorage to beam side at mid height.

(c) Weld clip angles to both legs of angle at top and bottom.

(d) Drill clips and plates, at top and bottom for anchoring jamb angles with two 3/8 inch expansion bolts at each location.

(e) Cut rabbet for pivot hinges and lock strike.

2.7 GUARDS

a. Wall Corner Guards:

(1) Fabricate from steel angles and furnish with anchors as shown.

(2) Continuously weld anchor to angle.

b. Guard Angles for Overhead Doors:

(1) Cut away top portion of outstanding leg of angle and extend remaining portion of angle up wall.

(2) Weld filler piece across head of opening to jamb angles.

(3) Make provisions for fasteners and anchorage.

c. Channel Guard at Loading Platform:

(1) Fabricate from steel channel of size shown.

(2) Weld anchors to channels as shown.

(3) Drill channel for bumper anchor bolts.

d. Edge Guard Angles for Openings in slabs.

(1) Fabricate from steel angles of sizes and with anchorage shown.

(2) Where size of angle is not shown, provide (2 x 2 x 1/4 inch) steel angle with (1-1/4 x 3/16 inch) strap anchors, welded to back.

(3) Miter or butt angles at corners and weld.

(4) Use one anchor near end and three feet on centers between end anchors.

e. Wheel Guards:

(1) Construct wheel guards of not less than 5/8 inch thick cast iron.

(2) Provide corner type, with flanges for bolting to walls.

2.8 COVERS AND FRAMES FOR PITS AND TRENCHES

a. Fabricate covers to support live loads specified.

b. Galvanized steel members after fabrication in accordance with

ASTM A123/A123M, G-90 coating.

c. Steel Covers:

(1) Use 1/4 inch thick floor plate for covers unless otherwise shown. Use gratings where shown as specified in paragraph GRATINGS. Use smooth floor plate unless noted otherwise.

(2) Provide clearance at all sides to permit easy removal of covers.

(3) Make cutouts within 1/4 inch of penetration for passage of pipes and ducts.

(4) Drill covers for flat head countersunk screws.

(5) Make cover sections not to exceed (25 square feet) in area and (200 pounds) in weight.

(6) Fabricate trench cover sections not be over 3 feet long and if width of trench is more than 3 feet or over, equip one end of each section with an angle or "T" bar stiffener to support adjoining plate.

(7) Use two, 1/2 inch diameter steel bar flush drop handles for each cover section.

d. Cast Iron Covers

(1) Fabricate covers to support live loads specified.

(2) Fabricate from ASTM A48/A48M, cast-iron, 1/2 inch minimum metal thickness, cast with stiffeners as required.

(3) Fabricate as flush type with frame, reasonably watertight and be equipped with flush type lifting rings. Provide seals where watertight covers noted.

(4) Make covers in sections not over (200 pounds) except round covers.

e. Steel Frames:

(1) Form frame from structural steel angles as shown. Where not shown use (2-1/2 x 2-1/2 x 1/4 inch) angles for frame openings over 4 feet long and (2 ix 2 x 1/4 inch) for frame openings less than (4 feet).

(2) Fabricate intermediate supporting members from steel "T's" or angles; located to support cover section edges.

(3) Where covers are required use steel border bars at frames so that top of cover will be flush with frame and finish floor.

(4) Weld steel strap anchors to frame. Space straps not over 24 inches o.c., not shown otherwise between end anchors. Use (1/4 x 1 x 8 inches) with 2 inch bent ends strap anchors unless shown otherwise.

(5) Drill and tap frames for screw anchors where plate covers occur.

f. Cast Iron Frames:

(1) Fabricate from ASTM A48/A48M cast iron to shape shown.

(2) Provide anchors for embedding in concrete, spaced near ends and not over 24 inches apart.

2.9 GRATINGS

a. Fabricate gratings to support live loads specified and a concentrated load as specified.

b. Provide clearance at all sides to permit easy removal of grating.

c. Make cutouts in gratings with 1/4 inch minimum to (one inch) maximum clearance for penetrations or passage of pipes and ducts. Edge band cutouts.

d. Fabricate in sections not to exceed (25 square feet) in area and (200 pounds) in weight.

e. Fabricate sections of grating with end-banding bars.

f. Fabricate angle frames and supports, including anchorage as shown.

(1) Fabricate intermediate supporting members from "T's" or angles.

(2) Locate intermediate supports to support grating section edges.

(3) Fabricate frame to finish flush with top of grating.

(4) Locate anchors at ends and not over 24 inches o.c.

(5) Butt or miter, and weld angle frame at corners.

g. Steel Bar Gratings:

(1) Fabricate grating using steel bars, frames, supports and other members shown in accordance with Metal Bar Grating Manual.

(2) Galvanize steel members after fabrication in accordance with ASTM A123/A123M, G-90 for exterior gratings, gratings in concrete floors, and interior grating where specified.

(3) Interior gratings: Prime paint unless specified galvanized.

(4) Use serrated bars for exterior gratings and interior gratings.

h. Aluminum Bar Gratings:

(1) Fabricate grating and frame assembly from aluminum as shown in accordance with Metal Bar Grating Manual.

(2) Use (1 x 3/16 inch) minimum size bearing bars.

(3) Mill finish unless specified otherwise.

(4) Use serrated bars for exterior gratings and interior gratings in the following areas:

i. Plank Gratings:

(1) Conform to RR-G-1602D.

(2) Manufacturers standard widths, lengths and side channels to meet live load requirements.

(3) Galvanize exterior steel gratings ASTM A123/A123M, G-90 after fabrication.

(4) Fabricate interior steel gratings from galvanized steel sheet, ASTM A525, where bearing on concrete or masonry.

(5) Fabricate other interior grating from steel sheet and finish with shop prime paint. Prime painted galvanized sheet may be used.

j. Cast Iron Gratings:

(1) Fabricate gratings to support a live load of (500 pounds per square foot).

(2) Fabricate gratings and frames for gutter type drains from cast-iron conforming to ASTM A48/A48M.

(3) Fabricate gratings in section not longer than 4 feet or over (200 pounds) and fit so as to be readily removable.

2.10 LOOSE LINTELS

a. Furnish lintels of sizes shown. Where size of lintels is not shown, provide the sizes specified.

b. Fabricate lintels with not less than 6 inch bearing at each end for nonbearing masonry walls, and 8 inch bearing at each end for bearing walls.

c. Provide one angle lintel for each 4 inches of masonry thickness as follows except as otherwise specified or shown.

(1) Openings 2-1/2 feet to 6 feet - (4 x 3-1/2 x 5/16 inch).

(2) Openings 6 feet to 10 feet - (6 x 3-1/2 x 3/8 inch).

d. For 6 inch thick masonry openings 750 mm to 3000 mm (2-1/2 feet to 10 feet use one angle (6 x 3-1/2 x 3/8 inch).

e. Provide bearing plates for lintels where shown.

f. Weld or bolt upstanding legs of double angle lintels together with 3/4 inch bolts spaced at 12 inches on centers.

g. Insert spreaders at bolt points to separate the angles for insertion of metal windows, louver, and other anchorage.

h. Where shown or specified, punch upstanding legs of single lintels to suit size and spacing of anchor bolts.

i. Elevator Entrance:

(1) Fabricate lintel from plate bent to channel shape, and provide a minimum of 4 inch bearing each end.

(2) Cut away the front leg of the channel at each end to allow for concealment behind elevator hoistway entrance frame.

2.11 SHELF ANGLES

a. Fabricate from steel angles of size shown.

b. Fabricate angles with horizontal slotted holes for 3/4 inch bolts spaced at not over 3 feet on centers and within 12 inches of ends.

c. Provide adjustable malleable iron inserts for embedded in concrete framing.

2.12 PLATE DOOR SILL

a. Fabricate of checkered plate as detailed.

(1) Aluminum Plate: ASTM B632/B632M, 0.125 inch thick.

(2) Steel Plate: ASTM A786/A786M, 0.125 inch thick), galvanized G90.

b. Fabricate for anchorage with flat head countersunk bolts at each end and not over (12 inches), o.c.

2.13 SAFETY NOSINGS

a. RR-T-650E, Type C.

(1) Aluminum: Class 2, Style 2.

(2) Cast iron: Class 4.

b. Fabricate nosings for exterior use from cast aluminum, and nosings for interior use from either cast aluminum or cast iron. Use one Class throughout.

c. Fabricate nosings approximately 4 inches wide with not more than 3/8 inch nose.

d. Provide nosings with integral type anchors spaced not more than 4 inches from each end and intermediate anchors spaced approximately 15 inches on center.

e. Fabricate nosings to extend within 4 inches of ends of concrete stair treads except where shown to extend full width.

f. Fabricate nosings to extend full width between stringers of metal stairs and full width of door openings.

g. On curved steps fabricate to terminate at point of curvature of steps having short radius curved ends.

2.14 LADDERS

a. Steel Ladders:

(1) Fixed-rail type with steel rungs shouldered and headed into and welded to rails.

(2) Fabricate angle brackets of 2 inch wide by 1/2 inch thick steel; brackets spaced maximum of 4 feet apart and of length to hold ladder 7 inches from wall to center of rungs. Provide turned ends or clips for anchoring.

(3) Provide holes for anchoring with expansion bolts through turned ends and brackets.

(4) Where shown, fabricate side rails curved, twisted and formed into a gooseneck.

(5) Galvanize exterior ladders after fabrication, ASTM A123/A123M, G-90.

b. Aluminum Ladders:

(1) Fixed-rail type, constructed of structural aluminum, with mill finish.

(2) Fabricate side rails and rungs of size and design shown, with the rungs shouldered and headed into and welded to the rails.

(3) Where shown fabrication side rails curved, twisted and formed into gooseneck.

(4) Fabricate angle brackets at top and bottom and intermediate brackets where shown. Drill for bolting.

c. Ladder Rungs:

(1) Fabricate from (one inch) diameter steel bars.

(2) Fabricate so that rungs will extend at least 4 inches into wall with ends turned (2 inches), project out from wall (7 inches), be 16 inches wide and be designed so that foot cannot slide off end.

(3) Galvanized after fabrication, ASTM A123/A123M, G-90 rungs for exterior use and for access to pits.

2.15 RAILINGS

a. In addition to the dead load design railing assembly to support live load specified.

b. Fabrication General:

(1) Provide continuous welded joints, dressed smooth and flush.

(2) Standard flush fittings, designed to be welded, may be used.

(3) Exposed threads will not be approved.

(4) Form handrail brackets to size and design shown.

(5) Exterior Post Anchors.

(a) Fabricate tube or pipe sleeves with closed ends or plates as shown.

(b) Where inserts interfere with reinforcing bars, provide flanged fittings welded or threaded to posts for securing to concrete with expansion bolts.

(c) Provide heavy pattern sliding flange base plate with set screws at base of pipe or tube posts.

(6) Interior Post Anchors:

(a) Provide flanged fittings for securing fixed posts to floor with expansion bolts, unless shown otherwise.

(b) Weld or thread flanged fitting to posts at base.

(c) For securing removable posts to floor, provide close fitting sleeve insert or inverted flange base plate with stud bolts or rivets concrete anchor welded to the base plate.

(d) Provide sliding flange base plate on posts secured with set screws.

(e) Weld flange base plate to removable posts set in sleeves.

c. Handrails:

(1) Close free ends of rail with flush metal caps welded in place except where flanges for securing to walls with bolts are shown.

(2) Make provisions for attaching handrail brackets to wall, posts, and handrail as shown.

d. Steel Pipe Railings:

(1) Fabricate of steel pipe with welded joints.

(2) Number and space of rails as shown.

(3) Space posts for railings not over 6 feet on centers between end posts.

(4) Form handrail brackets from malleable iron.

(5) Fabricate removable sections with posts at end of section.

(6) Removable Rails:

(a) Provide "U" shape brackets at each end to hold removable rail as shown. Use for top and bottom horizontal rail when rails are joined together with vertical members.

(b) Secure rail to brackets with 3/8 inch stainless steel through bolts and nuts at top rail only when rails joined with vertical members.

(c) Continuously weld brackets to post.

(d) Provide slotted bolt holes in rail bracket.

(e) Weld bolt heads flush with top of rail.

(f) Weld flanged fitting to post where posts are installed in sleeves.

(7) Opening Guard Rails:

(a) Fabricate rails with flanged fitting at each end to fit between wall opening jambs.

(b) Design flange fittings for fastening with machine screws to steel plate anchored to jambs.

(c) Fabricate rails for floor openings for anchorage in sleeves.

(8) Gates:

(a) Fabricate from steel pipe as specified for railings.

(b) Fabricate gate fittings from either malleable iron or wrought steel.

(c) Hang each gate on suitable spring hinges of clamp on or through bolted type. Use bronze hinges for exterior gates.

(d) Provide suitable stops, so that gate will swing as shown.

(e) Provide padlock eyes where shown.

(9) Chains:

(a) Chains: ASTM A391/A391M, Grade 63, straight link style, normal size chain bar 5/16 inch diameter, eight links per (foot) and with boat type snap hook on one end, and through type eye bolt on other end.

(b) Fabricate eye bolt for attaching chain to pipe posts, size not less than 3/8 inch diameter.

(c) Fabricate anchor at walls, for engagement of snap hook of either a 3/8 inch diameter eye bolt or punched angle.

(d) Galvanize chain and bolts after fabrication.

e. Aluminum Railings:

(1) Fabricate from extruded aluminum.

(2) Use tubular posts not less than 0.125 inch wall thickness for exterior railings.

(3) Punch intermediate rails and bottom of top rails for passage of posts and machine to a close fit.

(4) Where shown use extruded channel sections for top rail with 1/2 inch thick top cover plates and closed ends.

(5) Fabricate brackets of extruded or wrought aluminum as shown.

(6) Fabricate stainless pipe sleeves with closed bottom at least six inches deep having internal dimensions at least 1/2 inch greater than external dimensions of posts where set in concrete.

f. Stainless Steel Railings:

(1) Fabricate from (1-1/2 inches) outside diameter stainless steel tubing, ASTM A269/A269M, having a wall thickness of (0.065 inch).

(2) Join sections by an internal connector to form hairline joints where field assembled.

(3) Fabricate with continuous welded connections.

(4) Fabricate brackets of stainless steel to design shown.

(5) Fabricate stainless steel sleeves at least 6 inches deep having internal dimensions at least 1/2 inch greater than external dimensions of post.

2.16 CATWALKS

a. Fabricate catwalks including platforms, railings, ladders, supports and hangers, and arrangement of members as shown on drawings.

b. Fabricate stairs as specified in Section 05 51 00, METAL STAIRS.

c. Fabricate steel ladders as specified under paragraph LADDERS unless shown otherwise.

d. Fabricate steel pipe railings as specified under paragraph RAILINGS.

e. Catwalk and platforms floor surfaces as shown.

(1) Steel gratings as specified under paragraph gratings, either bar or plank type.

(2) Steel floor plate.

(3) Aluminum floor plate.

f. Prime paint catwalk system.

2.17 TRAP DOOR AND FRAMES WITH CEILING HATCH

a. Design to support a live load as specified.

b. Frames:

(1) Fabricate steel angle frame to set in concrete slabs and design to set flush with finished concrete slab or curb. If not shown use (2-1/2 x 2-1/2 x 1/4 inch) angles.

(2) Miter steel angles at corners and weld together.

(3) Weld steel bar stops to vertical leg of frame, to support doors flush with the top of the frame.

(4) Weld steel strap anchors on each side not over 24 inches on center to the backs of the frames. If not shown use (1/4 x 2 x 8 inch) long straps with 2 inch bent ends.

(5) Form frames from steel angles with welded corners for reinforcing and bracing of well lining and support of ceiling hatch.

c. Covers:

(1) Use 1/4 inch thick steel floor plate.

(2) Where double leaf covers are shown, reinforce at meeting edges.

(3) Use wrought steel hinges with fixed brass pins.

(a) Weld to cover.

(b) Secure to frame with machine screws.

(4) Where ladders occur, install hinges on the side opposite the ladder.

(5) Provide two bar type drop handles, flush with cover when closed for each leaf.

d. Well Lining:

(1) Fabricate well linings, for access through concrete floor slabs and suspended ceilings, from hatch to ceiling hatch or ceiling openings.

(2) Use steel sheet and shapes of size and thickness as shown. If not shown use 0.0598 inch thick steel sheet.

(3) If not shown use (2 x 2 x 1/4 inch) angle braces from ceiling level on each side angled at 45 degrees to structure above.

(4) Use (1 x 1 x 1/8 inch) angle bottom flange trim welded to well lining where no ceiling hatch occurs.

e. Ceiling Hatch:

(1) Construct hatch with "T" or angle frame designed to support edge of ceiling and hatch, weld to well lining.

(2) Form hatch panels of 1/8 inch steel, 3/16 inch aluminum or 0.0359 inch thick steel of pan type construction with (one inch) of mineral fiber insulation between.170832

(3) Use counter balance device, hinges, latch, hangers and other accessories required for installation and operation of hatch with not over (20 pounds) of force.

(4) Fabricate panels flush and reinforced to remain flat.

(5) Locate hatch panel flush with frame.

f. Finish with baked on prime coat.

2.18 SIDEWALK DOOR

a. Use flush, watertight, gutter type design.

b. Cover fabricate of 1/4 inch thick, diamond pattern floor plate.

c. Use automatic lock hold open feature and be hung on two flush type heavy bronze hinges capable of 90 degree swing on each door leaf.

d. Equip with locking and latching device and lifting devices; operable and accessible from both sides of doors.

e. Doors removable without disturbing frame.

f. Provide gutters at all joints for drainage of water.

2.19 SCREENED ACCESS DOORS AND FRAMES

a. Galvanized ASTM A123/A123M, G-90 after fabrication.

b. Wall frame:

(1) Fabricate frame from steel angles or channels as shown.

(2) Continuously weld (1-1/2 x 1/2 inch) steel channel door stop to angle frame. Cut out lock strike opening in channel.

(3) Miter and weld channel frame at corners. Reinforce corner with 1/8 inch plate angle.

(4) Reinforce channel frame with (1/8 x 6 inch) long steel plate at channel back to cutout for latch. Cutout lock strike opening in channel face. Drill and tap for hinge anchorage.

(5) Drill jambs for 1/4 inch bolt anchors at top and bottom and not over 18 inches between top and bottom.

(6) Fabricate frame for door to sit flush with face of frame.

c. Doors

(1) Fabricate door using steel channel frame with 1/8 inch angle plate reinforcing at corners.

(2) Miter and weld corners.

(3) Fabricate lock box of 1/16 inch plate and weld to channel surround.

(4) Provide wire mesh constructed of 0.135 inch diameter galvanized steel wire crimped and woven into (1-1/2 inch) diamond mesh pattern. Fasten the wire mesh to door frames by bending the ends of each strand of wire over through channel clinched and welded to channel door frame.

(5) Weld steel plate back-bands to channel door frame at hinge stiles only.

d. Hardware:

(1) Install hinged door to fixed frame with two (2-1/2 inch) brass or bronze hinges.

(2) Install lock or latch specified in Section 08 71 00, DOOR HARDWARE in lockbox.

2.20 STEEL COUNTER OR BENCH TOP FRAME AND LEGS

a. Fabricate channel or angle frame with mitered and welded corners as shown.

b. Drill top of frame with (1/4inch) holes spaced 8 inches on center for securing countertop.

c. Fabricate legs of angle or pipe shapes and continuously weld to frame.

d. Finish frame with backed on enamel prime coat.

2.21 STEEL PIPE BOLLARD

Provide bollard in accordance with ASTM A53/A53M with dimensions as shown in
standard detail SD320523-04. Anchor posts in concrete and fill solidly with concrete with a minimum compressive strength of 2500psi.

PART 3 EXECUTION

3.1 INSTALLATION, GENERAL

a. Set work accurately, in alignment and where shown, plumb, level, free of rack and twist, and set parallel or perpendicular as required to line and plane of surface.

b. Items set into concrete or masonry.

(1) Provide temporary bracing for such items until concrete or masonry is set.

(2) Place in accordance with setting drawings and instructions.

(3) Build strap anchors, into masonry as work progresses.

c. Set frames of gratings, covers, corner guards, trap doors and similar items flush with finish floor or wall surface and, where applicable, flush with side of opening.

d. Field weld in accordance with AWS.

(1) Design and finish as specified for shop welding.

(2) Use continuous weld unless specified otherwise.

e. Install anchoring devices and fasteners as shown and as necessary for securing metal fabrications to building construction as specified. Power actuated drive pins may be used except for removable items and where members would be deformed or substrate damaged by their use.

f. Spot prime all abraded and damaged areas of zinc coating as specified

and all abraded and damaged areas of shop prime coat with same kind of paint used for shop priming.

g. Isolate aluminum from dissimilar metals and from contact with concrete and masonry materials as required to prevent electrolysis and corrosion.

h. Secure escutcheon plate with set screw.

3.2 INSTALLATION OF SUPPORTS

a. to structure.

(1) Secure angles or channels and clips to overhead structural steel by continuous welding unless bolting is shown.

(2) Secure supports to concrete inserts by bolting or continuous welding as shown.

(3) Secure supports to mid height of concrete beams when inserts do not exist with expansion bolts and to slabs, with expansion bolts. unless shown otherwise.

(4) Secure steel plate or hat channels to studs as detailed.

b. Ceiling Hung Toilet Stalls:

(1) Securely anchor hangers of continuous steel channel above pilasters to structure above.

(2) Bolt continuous steel angle at wall to masonry or weld to face of each metal stud.

(3) Secure brace for steel channels over toilet stall pilasters to wall angle supports with bolts at each end spaced as shown.

(4) Install diagonal angle brace where the suspended ceiling over toilet stalls does not extend to side wall of room.

(5) Install stud bolts in lower flange of channel before installing furred down ceiling over toilet stalls.

(6) Install support for ceiling hung pilasters at entrance screen to toilet room similar to toilet stall pilasters.

c. Supports for Wall Mounted items:

(1) Locate center of support at anchorage point of supported item.

(2) Locate support at top and bottom of wall hung cabinets.

(3) Locate support at top of floor cabinets and shelving installed against walls.

(4) Locate supports where required for items shown.

3.3 COVERS AND FRAMES FOR PITS AND TRENCHES

a. Set frame and cover flush with finish floor.

b. Secure plates to frame with flat head countersunk screws.

c. Set gratings loose in drainage trenches or over pits unless shown anchored.

3.4 DOOR FRAMES

a. Secure clip angles at bottom of frames to concrete slab with expansion bolts as shown.

b. Level and plumb frame; brace in position required.

c. At masonry, set frames in walls so anchors are built-in as the work progresses unless shown otherwise.

d. Set frames in formwork for frames cast into concrete.

e. Where frames are set in prepared openings, bolt to wall with spacers and expansion bolts.

3.5 OTHER FRAMES

a. Set frame flush with surface unless shown otherwise.

b. Anchor frames at ends and not over 18 inches on centers unless shown otherwise.

c. Set in formwork before concrete is placed.

3.6 GUARDS

a. Steel Angle Corner Guards:

(1) Build into masonry as the work progress.

(2) Set into formwork before concrete is placed.

(3) Set angles flush with edge of opening and finish floor or wall or as shown.

(4) At existing construction fasten angle and filler piece to adjoining construction with 5/8 inch diameter by 3 inch long expansion bolts 18 inches on center.

(5) Install Guard Angles at Edges of Trench Stairwell Openings in Slab Overhead Doors where shown.

b. Channel Guard at Top Edge of Concrete Platforms:

(1) Install in formwork before concrete is placed.

(2) Set channel flush with top of the platform.

c. Wheel Guards:

(1) Set flanges of wheel guard at least 2 inches into pavement.

(2) Anchor to walls as shown, expansion bolt if not shown.

3.7 GRATINGS

a. Set grating flush with finish floor; top of curb, or areaway wall. Set frame so that horizontal leg of angle frame is flush with face of wall except when frame is installed on face of wall.

b. Set frame in formwork before concrete is placed.

c. Where grating terminates at a wall bolt frame to concrete or masonry with expansion bolts unless shown otherwise.

d. Secure removable supporting members in place with stainless steel bolts.

e. Bolt gratings to supports.

3.8 STEEL LINTELS

a. Use lintel sizes and combinations shown or specified.

b. Install lintels with longest leg upstanding, except for openings in 6 inch masonry walls install lintels with longest leg horizontal.

c. Install lintels to have not less than 6 inch bearing at each end for nonbearing walls, and 8 inch bearing at each end for bearing walls.

3.9 SHELF ANGLES

a. Anchor shelf angles with 3/4 inch bolts unless shown otherwise in adjustable malleable iron inserts, set level at elevation shown.

b. Provide expansion space at end of members.

3.10 PLATE DOOR SILL

a. Install after roofing base flashing and counter flashing work is completed.

b. Set in sealant and bolt to curb.

3.11 SAFETY NOSINGS

a. Except as specified and where preformed rubber treads are shown or specified install safety nosings at the following:

(1) Exterior concrete steps.

(2) Door sills of areaway entrances curbs.

(3) Exposed edges of curbs of door sills at transformer and service rooms.

(4) Interior concrete steps, including concrete filled treads of metal stairs of service stairs.

b. Install flush with horizontal and vertical surfaces.

c. Install nosing to within 4 inches of ends of concrete stair treads,

except where shown to extend full width.

d. Extend nosings full width of door openings.

e. Extend nosings, full width between stringers of metal stairs, and terminate at point of curvature of steps having short radius curved ends.

3.12 LADDERS

a. Anchor ladders to walls and floors with expansion bolts through turned lugs or angle clips or brackets.

b. In elevator pits, set ladders to clear all elevator equipment where shown on the drawings.

(1) Where ladders are interrupted by division beams, anchor ladders to beams by welding, and to floors with expansion bolts.

(2) Where ladders are adjacent to division beams, anchor ladders to beams with bent steel plates, and to floor with expansion bolts.

c. Ladder Rungs:

(1) Set ladder rungs into formwork before concrete is placed. Build ladder rungs into masonry as the work progresses.

(2) Set step portion of rung 6 inches from wall.

(3) Space rungs approximately 12 inches on centers.

(4) Where only one rung is required, locate it 16 inches above the floor.

3.13 RAILINGS

a. Steel Posts:

(1) Secure fixed posts to concrete with expansion bolts through flanged fittings except where sleeves are shown with pourable grout.

(2) Install sleeves in concrete formwork.

(3) Set post in sleeve and pour grout to surface. Apply beveled bead of urethane sealant at perimeter of post or under flange fitting as specified in Section 07 92 00, JOINT SEALANTS on exterior posts.

(4) Secure removable posts to concrete with either machine screws through flanged fittings which are secured to inverted flanges embedded in and set flush with finished floor, or set posts in close fitting pipe sleeves without grout.

(5) Secure sliding flanged fittings to posts at base with set screws.

(6) Secure fixed flanged fittings to concrete with expansion bolts.

(7) Secure posts to steel with welds.

b. Aluminum Railing, Stainless Steel Railing, and Ornamental Railing Posts:

(1) Install pipe sleeves in concrete formwork.

(2) Set posts in sleeve and pour grout to surface on exterior locations and to within 1/4 inch of surface for interior locations except to where posts are required to be removable.

(3) Apply beveled bead of urethane sealant over sleeve at post perimeter for exterior posts and flush with surface for interior posts as specified in Section 07 92 00, JOINT SEALANTS.

c. Anchor to Walls:

(1) Anchor rails to concrete or solid masonry with machine screws through flanged fitting to steel plate.

(a) Anchor steel plate to concrete or solid masonry with expansion bolts.

(b) Anchor steel plate to hollow masonry with toggle bolts.

(2) Anchor flanged fitting with toggle bolt to steel support in frame walls.

d. Removable Rails:

(1) Rest rails in brackets at each end and secure to bracket with stainless steel bolts and nuts where part of a continuous railing.

(2) Rest rail posts in sleeves where not part of a continuous railing. Do not grout posts.

e. Gates:

(1) Hang gate to swing as shown.

(2) Bolt gate hinges to jamb post with clamp on or through bolts.

f. Chains:

(1) Eye bolt chains to pipe posts.

(2) Eye bolt anchoring at walls.

(a) Expansion bolt to concrete or solid masonry.

(b) Toggle bolt to hollow masonry of frame wall installed support.

g. Handrails:

(1) Anchor brackets for metal handrails as detailed.

(2) Install brackets within 12 inches of return of walls, and at evenly spaced intermediate points not exceeding 4 feet on centers unless shown otherwise.

(3) Expansion bolt to concrete or solid masonry.

(4) Toggle bolt to installed supporting frame wall and to hollow masonry unless shown otherwise.

3.14 CATWALK AND PLATFORMS

a. Expansion bolt members to concrete unless shown otherwise.

b. Bolt or weld structural components together including ladders and stairs to support system.

c. Weld railings to structural framing.

d. Bolt or weld walk surface to structural framing.

e. Smooth field welds and spot prime damaged prime paint surface.

f. Fasten removable members with stainless steel fasteners.

3.15 SIDEWALK DOOR, TRAP DOORS, AND FRAMES

a. Set frame flush with finished concrete slab or curb.

b. Secure well linings to structure with expansion bolts unless shown otherwise.

c. Bolt ceiling hatch to well lining angle brace and to angle iron frames near corners and 12 inches on centers with not less than 3/8 inch roundhead machine screws.

d. Coordinate sidewalk door drain connections with plumbing work.

3.16 SCREENED ACCESS DOOR

a. Set frame in opening so that clearance at jambs is equal and secure with expansion bolts.

b. Use shims at bolts to prevent deformation of frame members in prepared openings.

c. Set frame in mortar bed and build in anchors as the masonry work progresses.

d. Grout jambs solid with mortar.

e. Secure insect screen to inside of door with stainless steel fasteners on doors in exterior walls.

3.17 STEEL COMPONENTS FOR MILLWORK ITEMS

Coordinate and deliver to Millwork fabricator for assembly where millwork items are secured to metal fabrications.

3.18 INSTALLATION OF STEEL PIPE BOLLARD

Set bollards vertically in concrete piers. Compressive strength of concrete piers shall be 3000psi. For dimensions of concrete piers See standard detail SD320523-04.

3.19 CLEAN AND ADJUSTING

a. Adjust movable parts including hardware to operate as designed without binding or deformation of the members centered in the opening or frame and, where applicable, contact surfaces fit tight and even without forcing or warping the components.

b. Clean after installation exposed prefinished and plated items and items fabricated from stainless steel, aluminum and copper alloys, as recommended by the metal manufacture and protected from damage until completion of the project.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 51 00

METAL STAIRS

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 QUALITY ASSURANCE
1.6 REFERENCES

PART 2 PRODUCTS

2.1 DESIGN CRITERIA
2.2 MATERIALS
2.3 FABRICATION GENERAL
2.4 RAILINGS
2.5 CLOSED RISER STAIRS
2.6 INDUSTRIAL STAIRS

PART 3 EXECUTION

3.1 STAIR INSTALLATION
3.2 RAILING INSTALLATION
3.3 FIELD PRIME PAINTING

-- End of Section Table of Contents --

SECTION 05 51 00
1/21

METAL STAIRS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies steel stairs with railings.

b. Types:

(1) Closed riser stairs with concrete filled treads and platforms.

(2) Industrial stairs: Closed and open riser stairs.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Concrete fill for treads and platforms.

b. Section 05 50 00, METAL FABRICATIONS: Wall handrails and railings for other than steel stairs.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.

SD-01 Preconstruction Submittals

Fabrication Qualifications; G

Welding Qualifications; G

Installer Qualifications; G

SD-02 Shop Drawings

Shop Drawings; G

Fabrication General; G,

SD-03 Product Data

Sustainable Design Submittals; S

Railings; G

Closed Riser Stairs; G

Industrial Stairs; G

SD-05 Design Data

Calculations; G

1.4 SUBMITTAL INSTRUCTIONS

a. Shop Drawings: Show design, fabrication details, installation, connections, material, and size of members.

b. Sustainable Design Submittals, as described below:

(1) Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.

c. Fabrication qualifications.

(1) Installer qualifications.

(2) Calculations.

d. Welding qualifications.

1.5 QUALITY ASSURANCE

a. Fabricator: A firm with a minimum of three (3) years' experience in type of work required by this section. Submit fabricator qualifications.

b. Installer: A firm with a minimum of three (3) years' experience in type of work required by this section. Submit installer qualifications.

c. Calculations: Provide professionally prepared calculations and certification of performance of this work, signed and sealed by a Professional Engineer registered in the state where the work is located. Perform structural design of the stair including supports for the metal stair frame. Indicate how Design Criteria as specified have been incorporated into the design.

d. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M and AWS D1.3/D1.3M.

1.6 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by basic designation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B18.2.3.8M	(1981; R 2005) Metric Hex Lag Screws
ASME B18.6.1	(2016) Wood Screws (Inch Series)

ASME B18.6.3	(2024) Machine Screws, Tapping Screws, and Metallic Drive Screws (Inch Series)
ASME B18.6.5M	(2000; R 2010) Standard Specification for Metric Thread-Forming and Thread-Cutting Tapping Screws
ASME B18.6.7M	(1999; R 2010) Metric Machine Screws
ASME B18.22M	(1981; R 2017) Metric Plain Washers
ASME B18.21.1	(2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A48/A48M	(2022) Standard Specification for Gray Iron Castings
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A786/A786M	(2015; R 2021) Standard Specification for Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable

ASTM A1011/A1011M	(2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS D1.3/D1.3M	(2018) Structural Welding Code - Sheet Steel

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100	(2012) North American Specification for the Design of Cold-Formed Steel Structural Members

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101	(2024) Life Safety Code

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 25	(1997; E 2004) Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel, Type I and Type II

PART 2 PRODUCTS

2.1 DESIGN CRITERIA

a. Design stairs to support live load of (100 pound force/ square feet) and a concentrated load of 300 pound force) applied on an area of (4 square inch).

(1) Uniform and concentrated loads need not be assumed to act concurrently.

(2) Provide stair framing capable of withstanding stresses resulting from railing loads in addition to the loads specified above. Limit deflection of treads, platforms, and framing members to L/360 or (1/4 inch), whichever is less.

b. Provide structural design, fabrication and assembly in accordance with requirements of NAAMM Metal Stairs Manual, except as otherwise specified or shown.

c. Design Grating treads in accordance with NAAMM Metal Bar Grating Manual.

d. Design handrails and top rails of guards to support uniform load of not 50 pound force/feet) applied in any direction and a concentrated load of 200 pound force) applied in any direction. Uniform and concentrated loads need not be assumed to act concurrently.

e. Infill of guards to support concentrated load of 50 pound force) applied horizontally on an area of (one square feet).

f. Design fire stairs to conform to NFPA 101.

2.2 MATERIALS

a. Steel Pipe: ASTM A53/A53M, Standard Weight, zinc coated.

b. Steel Grating: Metal bar type grating NAAMM BG.

c. Sheet Steel: ASTM A1008/A1008M.

d. Structural Steel: ASTM A36/A36M.

e. Steel Floor Plate: ASTM A786/A786M.

f. Steel Decking: Form from zinc coated steel conforming to ASTM A653/A653M, with properties conforming to AISI S100 Specification for the Design of Cold-Formed Steel Structural Members.

g. Steel Plate: ASTM A1011/A1011M.

h. Iron Castings: ASTM A48/A48M, Class 30.

i. Malleable Iron Castings: ASTM A47/A47M.

j. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 30 percent.

2.3 FABRICATION GENERAL

a. Fasteners:

(1) Conceal bolts and screws wherever possible.

(2) Use countersunk heads on exposed bolts and screws with ends of bolts and screws dressed flush after nuts are set.

(3) Galvanized zinc-coated fasteners in accordance with ASTM A153/A153M and used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of steel stair items.

(4) Standard/regular hexagon-head bolts and nuts be conforming to ASTM A307, Grade A.

(5) Square-head lag bolts conforming toASME B18.2.3.8M, ASME B18.2.1.

(6) Machine screws cadmium-plated steel conforming to ASME B18.6.7M, ASME B18.6.3.

(7) Wood screws, flat-head carbon steel conforming to ASME B18.6.5M,

ASME B18.6.1.

(8) Plain washers, round, general-assembly-grade, carbon steel conforming to ASME B18.22M, ASME B18.21.1.

(9) Lockwashers helical spring, carbon steel conforming to ASME B18.2.1, ASME B18.2.3.8M.

b. Welding:

(1) Structural steel, AWS D1.1/D1.1M, and sheet steel, AWS D1.3/D1.3M.

(2) Where possible, locate welds on unexposed side.

(3) Grind exposed welds smooth and true to contour of welded member.

(4) Remove welding splatter.

c. Remove sharp edges and burrs.

d. Fit stringers to head channel and close ends with steel plates welded in place where shown.

e. Fit face stringer to newel post by tenoning into newel post, or by notching and fitting face stringer to side of newel where shown.

f. Shop Prime Painting: Shop prime steelwork with red oxide primer in accordance with SSPC Paint 25.

(1) Hot dip galvanize steelwork as indicated in accordance with ASTM A123/A123M. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

g. Form exposed work true to line and level with accurate angles and surfaces and straight sharp edges. Ease exposed edges to a radius of approximately (1/32 inch), and bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

h. Continuously weld corners and seams in accordance with the recommendations of AWS D1.1/D1.1M. Grind smooth exposed welds and flush to match and blend with adjoining surfaces.

i. Form exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use Phillips flathead (countersunk) screws or bolts.

j. Provide and coordinate anchorage of the type indicated with the supporting structure. Fabricate anchoring devices, space as indicated and required to provide adequate support for the intended use of the work.

k. Use hot-rolled steel bars for work fabricated for bar stock unless work is indicated or specified as fabricated from cold-finished or cold-rolled stock.

l. Soffit Clips: Provide clips with holes for attaching metal furring

for gypsum wallboard soffits.

2.4 RAILINGS

a. Fabricate railings, including handrails, from steel pipe.

(1) Connections may be standard fittings designed for welding, or coped or mitered pipe with full welds.

(2) Wall handrails are provided under Section 05 50 00, METAL FABRICATIONS.

b. Return ends of handrail to wall and close free end.

c. Provide standard terminal castings where fastened to newel.

d. Space intermediate posts not over 6 feet on center between end post or newel post.

e. Fabricate handrail brackets from cast malleable iron.

f. Provide standard terminal fittings at ends of post and rails.

2.5 CLOSED RISER STAIRS

a. Provide treads, risers, platforms, railings, stringers, headers and other supporting members.

b. Fabricate pans for treads and platforms, and risers from sheet steel. Fabricate pans for platforms from steel decking where shown.

c. Form risers with sanitary cove.

d. Fabricate stringers, headers, and other supporting members from structural steel.

e. Construct newel posts of steel tubing having wall thickness not less than (3/16-inch), with forged steel caps and drops.

2.6 INDUSTRIAL STAIRS

a. Provide treads, platforms, railings, stringers and other supporting members as shown.

b. Treads and platforms of checkered steel floor plate:

(1) Turn floor plate down to form nosing on treads and edge of platform at head of stairs.

(2) Support tread and platforms with angles welded to plate.

(3) Do not leave exposed fasteners on top of treads or platform surfaces.

(4) Provide flat sheet steel risers for stairs with steel plate treads where shown.

c. Treads and platforms of steel grating:

(1) Fabricate steel grating treads and platforms in accordance with requirements of NAAMM MBG 531.

(2) Provide end-banding bars, except where carrier angle are used at tread ends.

(3) Support treads by use of carrier plates or carrier angle. Use carrier plate end banding bars on exterior stairs.

(4) Provide abrasive nosing on treads and edge of platforms at head of stairs.

(5) Provide toe plates on platforms where shown.

PART 3 EXECUTION

3.1 STAIR INSTALLATION

a. Provide columns, hangers, and struts required to support the loads imposed.

b. Perform job site welding and bolting as specified for shop fabrication.

c. Set stairs and other members in position and secure to structure as shown.

d. Install stairs plumb, level and true to line.

e. Provide steel closure plate to fill gap between the stringer and surrounding wall. Weld and apply primer, ready to accept paint finish.

3.2 RAILING INSTALLATION

a. Install standard terminal fittings at ends of posts and rails.

b. Secure brackets, posts and rails to steel by welds, and to masonry or concrete with expansion sleeves and bolts, except secure posts at concrete by setting in sleeves filled with commercial non-shrink grout.

c. Set rails horizontal or parallel to rake of stairs to within (1/8-inch in 12 feet).

d. Set posts plumb and aligned to within (1/8-inch in 12 feet).

3.3 FIELD PRIME PAINTING

a. Touch-up abraded areas with same primer paint used for shop priming.

b. Touch up abraded galvanized areas.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

SECTION 06 10 00

ROUGH CARPENTRY

10/17

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 PRODUCT DELIVERY, STORAGE AND HANDLING
1.6 QUALITY ASSURANCE
1.7 GRADING AND MARKINGS
1.8 REFERENCES

PART 2 PRODUCTS

2.1 LUMBER
2.2 PLYWOOD
2.3 STRUCTURAL-USE PANELS
2.4 ROUGH HARDWARE AND ADHESIVES

PART 3 EXECUTION

3.1 INSTALLATION OF FRAMING AND MISCELLANEOUS WOOD MEMBERS

-- End of Section Table of Contents --

SECTION 06 10 00
10/17

ROUGH CARPENTRY

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies wood blocking, framing, sheathing, furring, nailers, sub-flooring, rough hardware, and light wood construction.

1.2 RELATED WORK

a. Sustainable design requirements: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Gypsum sheathing: Section 09 29 00, GYPSUM BOARD.

c. Cement board sheathing: Section 06 16 63, CEMENTITIOUS SHEATHING.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Postconsumer And Preconsumer Recycled Content

Volatile Organic Compounds Per Volume

Composite Wood Products

Manufacturer’s Literature And Data; G

SD-07 Certificates

Manufacturer’s Certificate For Unmarked Lumber; G

1.4 SUBMITTAL INSTRUCTIONS

a. Sustainable Design Submittals, as described below:

(1) Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.

(2) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

(3) For composite wood products, submit documentation indicating that product contains no added urea formaldehyde.

b. Shop Drawings showing framing connection details, fasteners, connections and dimensions.

c. Manufacturer's Literature and Data:

(1) Submit data for lumber, panels, hardware and adhesives.

(2) Submit data for wood-preservative treatment from chemical treatment manufacturer and certification from treating plants that treated materials comply with requirements. Indicate type of preservative used and net amount of preservative retained.

(3) Submit data for fire retardant treatment from chemical treatment manufacturer and certification by treating plant that treated materials comply with requirements. Include physical properties of treated materials based on testing by a qualified independent testing agency.

(4) For products receiving a waterborne treatment, submit statement that moisture content of treated materials was reduced to levels specified before shipment to project site.

d. Manufacturer's certificate for unmarked lumber.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

a. Protect lumber and other products from dampness both during and after delivery at site.

b. Pile lumber in stacks in such manner as to provide air circulation around surfaces of each piece.

c. Stack plywood and other board products so as to prevent warping.

d. Locate stacks on well drained areas, supported at least (6 inches) above grade and cover with well-ventilated sheds having firmly constructed over hanging roof with sufficient end wall to protect lumber from driving rain.

1.6 QUALITY ASSURANCE

Installer Qualifications: A firm with a minimum of three (3) years' experience in the type of work required by this Section.

1.7 GRADING AND MARKINGS

Any unmarked lumber or plywood panel for its grade and species will not be allowed on VA Construction sites for lumber and material not normally grade marked, provide manufacturer's certificates (approved by an American Lumber Standards approved agency) attesting that lumber and material meet the specified the specified requirements.

1.8 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in the text by basic designation only.

AMERICAN INSTITUTE OF TIMBER CONSTRUCTION (AITC)

ANSI/AITC A190.1	(2022) Product Standard for Structural Glued Laminated Timber

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)

ALSC PS 20	(2021) American Softwood Lumber Standard

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B18.6.1	(2016) Wood Screws (Inch Series)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products
EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

APA - THE ENGINEERED WOOD ASSOCIATION (APA)

APA E30	(2016) Engineered Wood Construction Guide
APA PS 1	(2009) Structural Plywood (with Typical APA Trademarks)

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness

ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM D2559	(2012a; E 2016) Standard Specification for Adhesives for Bonded Structural Wood Products for Use Under Exterior Exposure Conditions
ASTM D3498	(2019a) Standard Specification for Adhesives for Field-Gluing Wood Structural Panels (Plywood or Oriented Stand Board) to Wood Based Floor System Framing
ASTM F844	(2019; R 2024) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use
ASTM F1667/F1667M	(2021a) Standard Specification for Driven Fasteners: Nails, Spikes, and Staples

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA Book	(2015) AWPA Book of Standards
AWPA T1	(2024) Use Category System: Processing and Treatment Standard
AWPA U1	(2024) Use Category System: User Specification for Treated Wood

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-55615	(Basic; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Wood Screw and Lag Bolt Self-Threading Anchors

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

TRUSS PLATE INSTITUTE (TPI)

TPI 85	Metal Plate Connected Wood Trusses

U.S. NAVAL SEA SYSTEMS COMMAND (NAVSEA)

QPL-19140	(2024) Lumber and Plywood, Fire-Retardant TreatedPART 2 PRODUCTS

2.1 LUMBER

a. Unless otherwise specified, each piece of lumber must bear grade mark, stamp, or other identifying marks indicating grades of material, and

rules or standards under which produced.

(1) Identifying marks are to be in accordance with rule or standard under which material is produced, including requirements for qualifications and authority of the inspection organization, usage of authorized identification, and information included in the identification.

(2) Inspection agency for lumber approved by the Board of Review, American Lumber Standards Committee, to grade species used.

b. Structural Members: Species and grade as listed having design stresses as shown.

c. Lumber Other Than Structural:

(1) Unless otherwise specified, species graded under the grading rules of an inspection agency approved by Board of Review, American Lumber Standards Committee.

(2) Framing lumber: Minimum extreme fiber stress in bending of 1100 PSI.

(3) Furring, blocking, nailers and similar items 4 inches and narrower Standard Grade; and, members 6 inches and wider, Number 2 Grade.

(4) Board Sub-flooring: Shiplap edge, 1 inch thick, not less than 8 inches wide.

d. Sizes:

(1) Conforming to ALSC PS 20.

(2) Size references are nominal sizes, unless otherwise specified, actual sizes within manufacturing tolerances allowed by standard under which produced.

e. Moisture Content:

Maximum moisture content of wood products is to be as follows at the time of delivery to site.

(1) Boards and lumber 2 inches and less in thickness: 19 percent or less.

(2) Lumber over 2 inches thick: 25 percent or less.

f. Fire Retardant Treatment:

(1) Comply with QPL-19140.

(2) Treatment and performance inspection, by an independent and qualified testing agency that establishes performance ratings.

g. Preservative Treatment:

(1) Do not treat Heart Redwood and Western Red Cedar.

(2) Treat wood members and plywood exposed to weather or in contact with plaster, masonry or concrete, including framing of open roofed structures; sills, sole plates, furring, and sleepers that are less than 24 inches from ground; nailers, edge strips, blocking, crickets, curbs, cant, vent strips and other members provided in connection with roofing and flashing materials.

(3) Treat other members specified as preservative treated (PT).

(4) Preservative treat by the pressure method complying with AWPA Book use category system standards AWPA U1 and AWPA T1, except any process involving the use of Chromated Copper Arsenate (CCA) or other agents classified as carcinogenic for pressure treating wood is not permitted.

2.2 PLYWOOD

a. Comply with APA PS 1.

b. Bear the mark of a recognized association or independent inspection agency that maintains continuing control over quality of plywood which identifies compliance by veneer grade, group number, span rating where applicable, and glue type.

c. Sheathing:

(1) APA rated Exposure 1 or Exterior; panel grade CD or better.

(2) Wall sheathing:

(a) Minimum 11/32 inch thick with supports 16 inches on center and 15/32 inch thick with supports (24 inches) on center unless specified otherwise.

(b) Minimum 48 inches wide at corners without corner bracing of framing.

(3) Roof sheathing:

(a) Minimum 11/32 inch thick with span rating 24/0 or 15/32 inch thick with span rating for supports 16 inches on center unless specified otherwise.

(b) Minimum 19/32 inch thick or span rating of 40/20 or 23/32 inch thick or span rating of 48/24 for supports 24 inches on center.

d. Subflooring:

(1) Under finish wood flooring or underlayment:

(a) APA Rated sheathing, Exposure 1. panel grade CD.

(b) Minimum 19/32 inch thick with span rating 32/16 or greater for supports at 16 inches on center and 23/32 inch thick with span rating 48/24 for supports at 24 inches on center.

(2) Combination subflooring-underlayment under resilient flooring or carpet:

(a) APA Rated Stud-I-Floor Exterior or Exposure 1, T and G.

(b) Minimum 19/32 inch thick or greater, span rating 16, for supports at 16 inches on center; 23/32 inch thick or greater, span rating 24, for supports at 24 inches on center.

(c) Minimum 3/4-inch thick or greater, span rating 32, for supports at 32 inches on center; 1-1/8 inch thick, span rating 48 for supports at 48 inches on center.

e. Underlayment:

(1) APA rated Exposure 1 or Exterior, panel grade C-C Plugged.

(2) Minimum 1/4 inch thick or greater over plywood subflooring and 3/8 inch thick or greater over board subflooring, unless otherwise shown.

2.3 STRUCTURAL-USE PANELS

a. Comply with APA E30.

b. Bearing the mark of a recognized association or independent agency that maintains continuing control over quality of panel which identifies compliance by end use, Span Rating, and exposure durability classification.

c. Wall and Roof Sheathing: APA Rated sheathing panels, durability classification of Exposure 1 or Exterior Span Rating of 16/0 or greater for supports 16 inches on center and 24/0 or greater for supports 24 inches on center.

d. Subflooring:

(1) Under finish wood flooring or underlayment:

(a) APA rated sheathing panels, durability classification of Exposure 1 or Exterior.

(b) Span Rating of 24/16 or greater for supports 16 inches.

(2) Under resilient floor or carpet.

(a) APA rated combination subfloor-underlayment grade panels, durability classification of Exposure 1 or Exterior T and G.

(b) Span Rating of 16 or greater for supports 16 inches on center and 24 or greater for supports 24 inches on center.

e. Underlayment:

(1) APA rated Exposure l.

(2) Minimum 1/4 inch thick or greater over subfloor.

f. Wood "I" Beam Members:

(1) Size and Shape as indicated in Contract documents.

(2) Cambered and marked "TOP UP".

(3) Plywood webs: APA PS 1, minimum 3/8 inch thick, unless shown otherwise.

(4) Flanges: Kiln dried stress rated dense lumber minimum 1-1/2 inch thick, width as indicated on Contract documents.

(5) Plywood web fitted into flanges and joined with ASTM D2559 adhesive to form "I" beam section unless shown otherwise.

g. Laminated Veneer Lumber (LVL):

(1) Bonded jointed wood veneers with ASTM D2559 adhesive.

(2) Scarf jointed wood veneers with grain of wood parallel.

(3) Size as indicated on Contract documents.

2.4 ROUGH HARDWARE AND ADHESIVES

a. Anchor Bolts:

(1) ASME B18.2.1 and ASME B18.2.2 galvanized, 1/2 inch unless shown otherwise.

(2) Extend at least 8 inches into masonry or concrete with ends bent 2 inches.

b. Miscellaneous Bolts: Expansion Bolts: CID A-A-55615; lag bolt, long enough to extend at least 2-1/2 inches into masonry or concrete. Provide 1/2 inch bolt unless shown otherwise.

c. Washers

(1) ASTM F844.

(2) Provide zinc or cadmium coated steel or cast iron for washers exposed to weather.

d. Screws:

(1) Wood to Wood: ASME B18.6.1 or ASTM C1002.

(2) Wood to Steel: ASTM C954, or ASTM C1002.

e. Nails:

(1) Size and type best suited for purpose unless noted otherwise. Provide aluminum-alloy nails, plated nails, or zinc-coated nails, for nailing wood work exposed to weather and on roof blocking.

(2) ASTM F1667/F1667M:

(a) Common: Type I, Style 10.

(b) Concrete: Type I, Style 11.

(c) Barbed: Type I, Style 26.

(d) Underlayment: Type I, Style 25.

(e) Masonry: Type I, Style 27.

(f) Provide special nails designed for use with ties, strap anchors, framing connectors, joists hangers, and similar items. Nails not less than 1-1/4 inches long, 8d and deformed or annular ring shank.

f. Framing and Timber Connectors:

(1) Fabricate of ASTM A653/A653M, Grade A; steel sheet not less than 0.052 inch thick unless specified otherwise. Apply standard plating to steel timber connectors after punching, forming and assembly of parts.

(2) Framing Angles: Angle designed with bendable legs to provide three way anchors.

(3) Straps:

(a) Designed to provide wind and seismic ties with sizes as shown or specified.

(b) Strap ties not less than 1-1/4 inches wide.

(c) Punched for fastener.

(4) Metal Bridging:

(a) V shape deformed strap with not less than two nail holes at ends, designed to nail to top and side of framing member and bottom and side of opposite member.

(b) Not less than 3/4 by 5 inches bendable nailing flange on ends.

(c) Fabricated of 0.04 inch minimum thick sheet.

(5) Joist Hangers:

(a) Fabricated of 0.063 inch minimum thick sheet, U design unless shown otherwise.

(b) Heavy duty hangers fabricated of minimum 0.108 inch thick sheet, U design with bent top flange to lap over beam.

(6) Timber Connectors: Fabricated of steel to shapes indicated on Contract drawings.

(7) Joist Ties: Mild steel flats, 3/16 by 1-1/4 inch size with ends bent about 30 degrees from horizontal, and extending at least 16 inches onto framing. Punch each end for three (3) spikes.

(8) Wall Anchors for Joists and Rafters:

(a) Mild steel strap, 3/16 by 1-1/4 inch with wall ends bent 2 inches, or provide 3/8 by 5 inch pin through strap end built

into masonry.

(b) Strap long enough to extend onto three joists or rafters, and punched for spiking at each bearing.

(c) Strap not less than 4 inches embedded end.

(9) Joint Plates:

(a) Steel plate punched for nails.

(b) Steel plates formed with teeth or prongs for mechanically clamping plates to wood.

(c) Size for axial eccentricity, and fastener loads.

g. Adhesives:

(1) For field-gluing plywood to lumber framing floor or roof systems: ASTM D3498.

(2) For structural laminated Wood: ASTM D2559.

(3) Adhesives to have a VOC content of 70 g/L or less when calculated according to 40 CFR 59, (EPA Method 24).

PART 3 EXECUTION

3.1 INSTALLATION OF FRAMING AND MISCELLANEOUS WOOD MEMBERS

a. Conform to applicable requirements of the following:

(1) ANSI/AITC A190.1 Timber Construction Manual for heavy timber construction.

(2) APA E30 for installation of plywood or structural use panels.

(3) TPI 85 for metal plate connected wood trusses.

b. Fasteners:

(1) Nails.

(a) Nail in accordance with the Recommended Nailing Schedule where detailed nailing requirements are not specified in nailing schedule. Select nail size and nail spacing sufficient to develop adequate strength for the connection without splitting the members.

(b) Use special nails with framing connectors.

(c) For sheathing and subflooring, select length of nails sufficient to extend 1 inch into supports.

(d) Use 8d or larger nails for nailing through 1 inch thick lumber and for toe nailing 2 inch thick lumber.

(e) Use 16d or larger nails for nailing through 2 inch thick lumber.

(f) Select the size and number of nails in accordance with the Nailing Schedule except for special nails with framing anchors.

(g) Nailing Schedule; Using Common Nails:

1. Joist bearing on sill or girder, toe nail three 8d nails or framing anchor.

2. Bridging to joist, toe nail each end two 8d nails.

3. Ledger strip to beam or girder three 16d nails under each joint.

4. Subflooring or Sheathing:

(a) 6 inch wide or less to each joist face nail two 8d nails.

(b) Subflooring, more than 6 inches wide, to each stud or joint, face nail three 8d nails.

(c) Plywood or structural use panel to each stud or joist face nail 8d, at supported edges 6 inches on center and at intermediate supports 10 inches on center. When gluing plywood to joint framing increase nail spacing to 12 inches at supported edges and 20 inches o.c. at intermediate supports.

5. Sole plate to joist or blocking, through sub floor face nail 20d nails, 16 inches on center.

6. Top plate to stud, end nail two 16d nails.

7. Stud to sole plate, toe nail or framing anchor. Four 8d nails.

8. Doubled studs, face nail 16d at 24 inches on center.

9. Built-up corner studs 16d at 24 inches on center.

10. Doubled top plates, face nails 16d at 16 inches on center.

11. Top plates, laps, and intersections, face nail two 16d.

12. Continuous header, two pieces 16d at 16 inches on center along each edge.

13. Ceiling joists to plate, toenail three 8d or framing anchor.

14. Continuous header to stud, four 16d.

15. Ceiling joists, laps over partitions, face nail three 16d or framing anchor.

16. Ceiling joists, to parallel rafters, face nail three 16d.

17. Rafter to plate, toe nail three 8d or framing anchor. Brace 1 inch thick board to each stud and plate, face nail three 8d.

18. Built-up girders and beams 20d at 32 inches on center along each edge.

(2) Bolts:

(a) Fit bolt heads and nuts bearing on wood with washers.

(b) Countersink bolt heads flush with the surface of nailers.

(c) Embed in concrete and solid masonry or provide expansion bolts. Special bolts or screws designed for anchor to solid masonry or concrete in drilled holes may be used.

(d) Provide toggle bolts to hollow masonry or sheet metal.

(e) Provide bolts to steel over 0.112 inch, 11 gage in thickness. Secure wood nailers to vertical structural steel members with bolts, placed one at ends of nailer and 24 inch intervals between end bolts. Provide clips to beam flanges.

(3) Drill Screws to steel less than 0.112 inch thick.

(a) ASTM C1002 for steel less than 0.033 inch thick.

(b) ASTM C954 for steel over 0.033 inch thick.

(4) Power actuated drive pins may be provided where practical to anchor to solid masonry, concrete, or steel.

(5) Do not anchor to wood plugs or nailing blocks in masonry or concrete. Provide metal plugs, inserts or similar fastening.

(6) Screws to Join Wood:

(a) Where shown or option to nails.

(b) ASTM C1002, sized to provide not less than 1 inch penetration into anchorage member.

(c) Spaced same as nails.

(7) Installation of Timber Connectors:

(a) Fit wood to connectors and drill holes for fasteners so wood is not split.

c. Set sills or plates level in full bed of mortar on masonry or concrete walls.

(1) Space anchor bolts 4 feet on centers between ends and within 6 inches of end. Stagger bolts from side to side on plates over 7 inches in width.

(2) Provide shims of slate, tile or similar approved material to level wood members resting on concrete or masonry. Do not use wood shims or wedges.

(3) Closely fit, and set to required lines.

d. Cut notch, or bore passage of ducts wires, bolts, pipes, conduits and to accommodate other work. Repair or replace miscut, misfit or damaged work.

e. Blocking Nailers, and Furring:

(1) Install furring, blocking, nailers, and grounds where shown.

(2) Provide longest lengths practicable.

(3) Provide fire retardant treated wood blocking where shown at openings and where shown or specified.

(4) Layers of Blocking or Plates:

(a) Stagger end joints between upper and lower pieces.

(b) Nail at ends and not over 24 inches between ends.

(c) Stagger nails from side to side of wood member over 5 inches in width.

(5) Fabricate roof edge vent strips with 1/4 by 1/4 inch notches, 4 inches on center, aligned to allow for venting of insulating concrete and venting base sheet.

(6) Unless otherwise shown, provide wall furring 1 by 3 inch continuous wood strips installed plumb on walls, using wood shims where necessary so face of furring forms a true, even plane. Space furring not over 16 inches on centers, butt joints over bearings and rigidly secure in place. Anchor furring on 16 inches centers.

f. Floor and Ceiling Framing:

(1) Set with crown edge up.

(2) Keep framing at least 2 inches away from chimneys.

(3) Bear on not less than 4 inches on concrete and masonry, and 1-1/2 inches on wood and metal unless shown otherwise.

(4) Support joist, trimmer joists, headers, and beams framing into carrying members at same relative levels on joist hangers unless shown otherwise.

(5) Lap and spike wood joists together at bearing, or butt end-to-end with scab ties at joint and spike to plates. Scab tie lengths not less than 8 inches lap on joist ends. Install wood I beam joists as indicated in Contract documents.

(6) Frame openings with headers and trimmer joist. Double headers carrying more than two tail joists and trimmer joists supporting headers carrying more than one tail joist unless otherwise indicated in Contract documents.

(7) Drive nails through headers into joists using two nails for 2 inch by 6 inch; three nails for 2 inch by 8 inch and four nails for 2 inch by 10 inch and over in size.

(8) Install nearest joist to double headers and spike joist to both header members before trimmer joist is installed and secured together.

(9) Doubled joists under partitions parallel with floor joists. Fire cut joists built into masonry or concrete.

(10) Where joists run perpendicular to masonry or concrete, anchor every third joist to masonry or concrete with one metal wall anchor. Securely spike anchors with three nails to side of joist near its bottom.

(11) Anchor joists running parallel with masonry or concrete walls to walls with steel flats spaced not over 6 feet apart. Extend steel flats over at least three joists and into masonry 4 inches with ends turned 2 inches; bolt to concrete. Set top of flats flush with top of joists, and securely nail steel flats to each joist.

(12) Hook ties at steel framing over top flange of steel members.

(13) Nonbearing partitions running parallel with ceiling joists, install solid 2 inch thick bridging same depth as ceiling joists cut to fit snug between joists for securing top plate of partitions. Securely spike bridging to joists. Space 4 feet on center.

g. Bridging:

(1) Provide 1 inch by 3 inch lumber with ends beveled for slope. Option: Metal bridging may be provided in lieu of wood bridging.

(2) Install one row of bridging for joist spans over 8 feet, but less than 16 feet long; install two rows for spans over 16 feet long.

(3) Install an extra row of bridging between trimmer and next two (2) joists if header is more than 2 feet from end of trimmer or from regular row of bridging.

(4) Secure with two nails at ends.

(5) Leave bottom ends loose until after subflooring or roof sheathing is installed.

(6) Install single row of bridging at centerline of span and two rows at the third points of span unless otherwise shown.

h. Roof Framing:

(1) Set rafters with crown edge up.

(2) Form a true plane at tops of rafters.

(3) Valley, Ridge, and Hip Members:

(a) Size for depth of cut on rafters.

(b) Straight and true intersections of roof planes.

(c) Secure hip and valley rafters to wall plates by using framing connectors.

(d) Double valley rafters longer than the available lumber, with pieces lapped not less than 4 feet and spiked together.

(e) Butt joint and scab hip rafters longer than the available lumber.

(4) Spike to wall plate and to ceiling joists except when secured with framing connectors.

(5) Frame openings in roof with headers and trimmer rafters. Double headers carrying more than one rafter unless shown otherwise.

(6) Install 2 inch by 4 inch strut between roof rafters and ceiling joists at 4 feet on center unless shown otherwise.

i. Framing of Dormers:

(1) Frame as indicated in Contract documents, with top edge of ridge beveled to pitch of roof header.

(2) Set studs on doubled trimmer rafters.

(3) Double studs at corners of dormers.

(4) Double plate on studs and notch rafters over plate and bear at least 3 inches on plates.

(5) Frame opening to receive window frame or louver frame.

j. Partition and Wall Framing:

(1) Provide 2 inch by 4 inch studs spaced 16 inches on centers; unless otherwise indicated on Contract documents.

(2) Install double studs at openings and triple studs at corners.

(3) Installation of sole plate:

(a) Anchor plates of walls or partitions resting on concrete floors in place with expansion bolts, one near ends of piece and at intermediate intervals of not more than 4 feet or with power actuated drive pins with threaded ends of suitable type and size, spaced 2 feet on center unless shown otherwise.

(b) Nail plates to wood framing through subfloor as specified in nailing schedule.

(4) Headers or Lintels:

(a) Make headers for openings of two pieces of 2 inch thick lumber of size shown with plywood filler to finish flush with face of studs or solid lumber of equivalent size.

(b) Support ends of headers on top of stud cut for height of opening. Spike cut stud to adjacent stud. Spike adjacent stud to header.

(5) Provide double top plates, with members lapped at least 2 feet spiked together.

(6) Install intermediate cut studs over headers and under sills to maintain uniformity of stud spacing.

(7) Provide single sill plates at bottom of opening unless otherwise indicated in Contract documents. Toe nail to end stud, face nail to intermediate studs.

(8) Install 2 inch blocking for firestopping so that maximum dimension of any concealed space is not over 8 feet.

(9) Install corner bracing when plywood or structured use panel sheathing is not used.

(a) Let corner bracing into exterior surfaces of studs at an angle of approximately 45 degrees, extended completely over walls plates, and secured at bearing with two nails.

(b) Provide 1 inch by 4 inch corner bracing.

k. Rough Bucks:

(1) Install rough wood bucks at opening in masonry or concrete where wood frames or trim occur.

(2) Brace and maintain bucks plumb and true until masonry has been built around them or concrete cast in place.

(3) Cut rough bucks from 2 inch thick stock, of same width as partitions in which they occur and of width shown in exterior walls.

(4) Extend bucks full height of openings and across head of openings; fasten securely with anchors specified.

l. Subflooring:

(1) Subflooring may be either boards, structural-use panels, or plywood.

(2) Lay board subflooring diagonally, with close joints. Stagger end joints and make joints over supports. Bear each board on at least three supports.

(3) Provide a clearance of approximately 1/2 inch at masonry or concrete at walls.

(4) Apply plywood and structural-use panel subflooring with face grain or long dimension at right angles to the supports, with edges 1/4 inch apart at side joints, and 1/8 inch apart at end joints.

(5) Combination subfloor-underlayment:

(a) Space edges 1/8 inch apart.

(b) Provide a clearance of 1/4 inch at masonry on concrete at

walls.

(6) Stagger panel end joints and make over support.

m. Underlayment:

(1) Where finish flooring of different thickness is used in adjoining areas, provide underlayment of thickness required to bring finish-flooring surfaces into same plane.

(2) Apply to dry, level, securely nailed, clean, wood subfloor without any projections.

(3) Plywood and particle underlayment are to be glue-nailed to subfloor.

(4) Butt underlayment panels to a light contact with a 1/32 inch space between plywood or hardboard underlayment panels and walls, and approximately 3/8 inch between particleboard underlayment panels and walls.

(5) Stagger underlayment panel end joints with respect to each other and offset joints with respect to joints in the subfloor at least 2 inches.

(6) After installation, avoid traffic on underlayment and damage to the finish surface.

n. Sheathing:

(1) Provide plywood or structural-use panels for sheathing.

(2) Lay panels with joints staggered, with edge and ends 1/8 inch apart and nailed over bearings as specified.

(3) Set nails not less than 3/8 inch from edges.

(4) Install 2 inch by 4 inch blocking spiked between joists, rafters and studs to support edge or end joints of panels.

(5) Match and align sheathing which is an extension of work in place to existing.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES

SECTION 06 16 63

CEMENTITIOUS SHEATHING

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 REFERENCES
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY AND STORAGE
1.6 STORAGE AND HANDLING
1.7 WARRANTY

PART 2 PRODUCTS

2.1 PRODUCTS - GENERAL
2.2 SHEATHING
2.3 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 SHEATHING INSTALLATION
3.3 PROTECTION

-- End of Section Table of Contents --

SECTION 06 16 63
1/21

CEMENTITIOUS SHEATHING

PART 1 GENERAL

1.1 SUMMARY

Section Includes: Cement board sheathing at exterior framed wall construction.

1.2 REFERENCES

Comply with references to extent specified in this Section.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A118.9	(2019) American National Standard Specifications for Test Methods and Specifications for Cementitious Backer Units

ASTM INTERNATIONAL (ASTM)

ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness
ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM C1325	(2022; E 2023) Standard Specification for Fiber-Mat Reinforced Cementitious Backer Units

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G,

Sustainable Construction Requirements; S

SD-04 Samples

Samples;

SD-11 Closeout Submittals

Warranty;

1.4 SUBMITTAL INSTRUCTIONS

a. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

(3) Warranty.

b. Samples:

(1) Cement Board: (8 inches by 8 inches), minimum size.

(2) Fasteners: One of each type used.

c. Provide Sustainable Construction Requirements Provide in accordance with other Division 1 sections.

1.5 DELIVERY AND STORAGE

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.6 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.7 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Manufacturer's Warranty: Warrant sheathing against material and manufacturing defects. Warranty Period: 10 years.

PART 2 PRODUCTS

2.1 PRODUCTS - GENERAL

a. Provide each product from one manufacturer.

b. Sustainable Construction Requirements: Sheathing Recycled Content: Select products with recycled content to achieve overall Project recycled content requirement.

2.2 SHEATHING

Cement Boards: Meeting ANSI A118.9 and ASTM C1325.

a. Thickness: 5/8 inch .

b. Width: 48 inches, minimum.

2.3 ACCESSORIES

a. Steel Drill Screws: Corrosion-resistant, self-drilling.

(1) ASTM C1002, Type S for fastening to framing less than 33 mils thick.

(2) ASTM C954 for fastening to framing 33 mils thick and greater.

b. Joint Reinforcement: Alkali resistant tape as recommended by sheathing manufacturer.

c. Bonding Material: As recommended by sheathing manufacturer.

d. Air Barrier: As specified in Section 07 27 27, FLUID-APPLIED MEMBRANE AIR BARRIER, VAPOR RETARDING .

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Verify framing is plumb and level and in plane.

d. Correct substrate deficiencies.

3.2 SHEATHING INSTALLATION

Install products according to manufacturer's instructions.

a. Secure units to framing members with screws spaced maximum 8 inches o.c. and not closer than 1/2 inch from edge of unit.

b. Install screw heads without penetrating cement board surface.

c. Install sheathing with 1/4 inch gap where sheathing abuts masonry or similar materials to prevent wicking of moisture.

d. Install sheathing with 3/8 inch gap where non-load-bearing construction abuts structural elements or building expansion joints.

e. Horizontal Installation: Abut ends of boards over centers of studs. Stagger end joints minimum one stud spacing for adjacent boards. Fasten boards at perimeter and within field of board to each stud.

f. Vertical Installation: Install board vertical edges centered over studs. Abut ends and edges of each board with those of adjacent

boards. Fasten boards at perimeter and with fin field of board to each stud.

g. Apply bonding material to imbed tape and completely fill board joints, and gaps between each panel.

3.3 PROTECTION

a. Remove loose or spalling joint finish. Patch areas missing joint finish.

b. Replace broken or damaged boards.

c. Protect boards from moisture using temporary coverings until finishes are applied.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 01 50.19

PREPARATION FOR RE-ROOFING

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 PREINSTALLATION MEETINGS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 QUALITY ASSURANCE
1.8 FIELD CONDITIONS
1.9 WARRANTY

PART 2 PRODUCTS

2.1 MATERIALS

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 RE-ROOFING PREPARATION - GENERAL
3.4 OVERBURDEN REMOVAL
3.5 ROOFING MEMBRANE AND SELECTIVE ROOFING SYSTEM COMPONENT REMOVAL
3.6 DECK PREPARATION
3.7 TEMPORARY ROOFING
3.8 EXISTING MEMBRANE PREPARATION FOR NEW ROOFING
3.9 BASE FLASHING REMOVAL
3.10 RECOVER BOARD INSTALLATION
3.11 FIELD QUALITY CONTROL
3.12 DISPOSAL

-- End of Section Table of Contents --

SECTION 07 01 50.19
1/21

PREPARATION FOR RE-ROOFING

PART 1 GENERAL

1.1 SUMMARY

a. Section Includes:

(1) Roofing membrane and selective roofing system component removal for new roof membrane installation.

b. Existing Roofing System: TPO . System components include:

(1) Pavers and paver supports.

(2) Aggregate ballast.

(3) Roof insulation and drainage board.

(4) Aggregate surfacing.

(5) Roofing membrane.

(67) Cover board.

(7) Roof insulation.

(8) Vapor retarder.

(9) Substrate board.

1.2 RELATED WORK

a. Section 05 31 00, STEEL DECKING: Replacement Roof Deck.

b. Section 06 10 00, ROUGH CARPENTRY: Replacement Roof Deck and Parapet Sheathing.

c. Section 07 54 23, THERMOPLASTIC POLYOLEFIN (TPO) ROOFING: New Roofing System.

d. Section 07 60 00, FLASHING AND SHEET METAL: Sheet Metal Counterflashing.

1.3 REFERENCES

Comply with references to extent specified in this section.

SINGLE PLY ROOFING INDUSTRY (SPRI)

ANSI/SPRI FX-1	(2021) Standard Field Test Procedure for Determining the Withdrawal Resistance of Roofing Fasteners

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ASNT SNT-TC-1A	(2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

ASTM INTERNATIONAL (ASTM)

ASTM C208	(2022) Standard Specification for Cellulosic Fiber Insulating Board
ASTM C578	(2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation
ASTM C728	(2017a; R 2022) Standard Specification for Perlite Thermal Insulation Board
ASTM C1177/C1177M	(2024) Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing
ASTM C1153	(2010) Standard Practice for Location of Wet Insulation in Roofing Systems Using Infrared Imaging
ASTM C1278/C1278M	(2024) Standard Specification for Fiber-Reinforced Gypsum Panel
ASTM D4263	(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST PS 1	(2009) DOC Voluntary Product Standard PS 1-07, Structural Plywood
NIST DOC PS 2	Performace Standard for Wood Structural Panels

1.4 PREINSTALLATION MEETINGS

Conduct preinstallation meeting minimum 30 days before beginning Work of this section.

(1) Required Participants:

(a) Contracting Officer.

(b) Architect/Engineer.

(c) Inspection and Testing Agency.

(d) Contractor.

(e) Installer.

(f) Manufacturer's field representative.

(g) Other installers responsible for adjacent and intersecting

work, including mechanical and electrical equipment installers.

(2) Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(a) Removal and installation schedule.

(b) Removal and installation sequence.

(c) Preparatory work.

(d) Protection before, during, and after installation.

(e) Removal and installation.

(f) Temporary roofing including daily terminations.

(g) Transitions and connections to other work.

(h) Inspecting and testing.

(i) Other items affecting successful completion.

(3) Document and distribute meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Photographs; G

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

SD-06 Test Reports

Infrared Roof Moisture Survey Report; G

Field Inspection Reports; G

1.6 SUBMITTAL INSTRUCTIONS

ab. Submittal Drawings:

(1) Show size, configuration, and installation details.

b. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Description of temporary roof system and components.

(3) List of patching materials.

(4) Recover board fastening requirements.

(5) Temporary roofing installation instructions and removal instructions.

(6) Existing roofing warrantor's instructions.

c. Photographs: Document existing conditions potentially affected by roofing operations before work begins.

d. Field Inspection Reports:

(1) Certify warrantor inspected completed roofing and existing warranty remains in effect.

e. Infrared Roof Moisture Survey Report.

1.7 QUALITY ASSURANCE

Installer Qualifications:

(1) Same installer as Section 07 54 23, THERMOPLASTIC POLYOLEFIN (TPO) ROOFING.

(2) Licensed to perform asbestos abatement in Project jurisdiction when removal of asbestos-containing material is required.

(3) Approved by existing roofing system warrantor when work affects existing roofing system under warranty.

1.8 FIELD CONDITIONS

a. Building Occupancy: Perform work to minimize disruption to normal building operations.

(1) Verify occupants are evacuated from affected building areas when working on structurally impaired roof decking above occupied areas.

(2) Provide notice minimum 72 hours before beginning activities affecting normal building operations.

b. Existing Roofing Available Information:

(1) The following are available for Contractor reference:

(a) Roof moisture survey.

(b) Test cores analysis.

(c) Construction drawings and project manual.

(2) Examine available information before beginning work of this section.

c. Weather Limitations: Proceed with reroofing preparation only during dry weather conditions as specified for new roofing installation in Section 07 54 23, THERMOPLASTIC POLYOLEFIN (TPO) ROOFING Section.

(1) Remove only as much roofing in one day as can be made watertight in same day.

d. Hazardous materials are not expected in existing roofing system.

(1) Hazardous materials discovered during execution of the work will be removed by Government as work of a separate contract.

1.9 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Existing Warranties: Perform work to maintain existing roofing warranty in effect.

(1) Notify warrantor before beginning, and upon completion of reroofing.

(2) Obtain warrantor's instructions for maintaining existing warranty.

PART 2 PRODUCTS

2.1 MATERIALS

a. Patching Materials: Match existing roofing system materials.

b. Plywood Sheathing: See Section 06 10 00, ROUGH CARPENTRY.

c. Metal Flashing: See Section 07 60 00, FLASHING AND SHEET METAL.

d. Temporary Protection Materials:

(1) Expanded Polystyrene (EPS) Insulation: ASTM C578.

(2) Plywood: NIST PS 1, Grade CD Exposure 1-18.

(3) Oriented Strand Board (OSB): NIST DOC PS 2, Exposure 1.

e. Temporary Roofing System Materials: Contractor's option.

f. Recover Board: One of the following:

(1) Insulation: See Section 07 22 00, ROOF AND DECK INSULATION.

(2) Fiber Board: ASTM C208(2017) e2, Type II, fiber board; (1/2 inch) thick.

(3) Glass Mat Gypsum Board: ASTM C1177/C1177M, water-resistant; Type X, (5/8 inch) thick.

(4) Fiber Reinforced Gypsum Board: ASTM C1278/C1278M, water-resistant; (5/8 inch) thick.

(5) Perlite: ASTM C728; (3/4 inch) thick.

g. Fasteners: Type and size required by roof membrane manufacturer to resist wind uplift.

PART 3 EXECUTION

3.1 EXAMINATION

Infrared Roof Moisture Survey: Ground-based, walk-over type performed according to ASTM C1153(2015).

(1) Record the entire survey on DVD and provide one copy to Contracting Officer with report.

(2) Include in report thermograms of suspect areas and corresponding daytime photos of same locations.

(3) Conduct inspection by NDT test technician certified to at least Level 2 in Thermal/Infrared test method according to ASNT SNT-TC-1A.

(4) Mark out roof areas determined to be wet to indicate minimum areas to be removed.

3.2 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing roofing system indicated to remain.

(1) Cover roof membrane with temporary protection materials without impeding drainage.

(2) Limit traffic and material storage to protected areas.

(3) Maintain temporary protection until replacement roofing is completed.

c. Protect existing construction and completed work from damage.

d. Protect landscaping from damage.

e. Maintain access to existing walkways and adjacent occupied facilities.

f. Coordinate use of rooftop fresh air intakes with Contracting Officer to minimize effect on indoor air quality.

g. Ensure temporary protection materials are available for immediate use in case of unexpected rain.

h. Ensure roof drainage remains functional.

(1) Keep drainage systems clear of debris.

(2) Prevent water from entering building and existing roofing system.

i. Coordinate rooftop utilities remaining active during roofing work with Contacting Officer.

3.3 RE-ROOFING PREPARATION - GENERAL

Notify Contacting Officer of planned operations, daily.
(1) Identify location and extent of roofing removal.

(2) Request authorization to proceed.

3.4 OVERBURDEN REMOVAL

a. Remove aggregate ballast.

(1) Store aggregate ballast for reuse.

b. Remove loose aggregate from bituminous membrane surface.

c. Remove pavers and paver support.

(1) Store undamaged pavers and paver supports for reuse.

(2) Dispose of damaged pavers.

d. Remove insulation and drainage board from protected roofing membrane.

(1) Store insulation and drainage board for reuse.

3.5 ROOFING MEMBRANE AND SELECTIVE ROOFING SYSTEM COMPONENT REMOVAL

a. Remove existing roofing membrane, only, in locations and to extent indicated on drawings.

b. Visually inspect cover board, roof insulation, vapor retarder, and substrate board for moisture immediately after roof membrane removal.

(1) Coordinate with Contracting Officer to observe inspections.

(2) Identify wet roofing system components required to be removed.

(3) Mark roofing system removal locations and extents.

c. Remove wet roofing system components.

(1) Remove or cut-off roofing system fasteners when removals expose structural roof deck.

d. Patch selective roofing system removals immediately after inspection and repair.

e. Install patching materials to match existing roofing system.

f. Patch roofing membrane to maintain building watertight, unless new roofing membrane is installed same day as removal and repair.

3.6 DECK PREPARATION

a. Inspect structural roof deck after roofing system removal.

b. Concrete Roof Decks:

(1) Visually confirm concrete roof deck is dry.

(2) Perform moisture test according to ASTM D4263(2018) each day for each separate roof area.

(a) Proceed with roofing work only when moisture is not observed.

c. Steel Roof Decks:

(1) Visually inspect structural roof deck installation and fasteners.

(a) Notify Contracting Officer of unsuitable conditions and inadequate fastenings potentially affecting roof system performance.

(2) Secure roof deck with additional fastenings as indicated on drawings.

(3) Replace roof deck as indicated on drawings.

(a) Replacement Roof Deck: See Section 05 31 00, STEEL DECKING.

3.7 TEMPORARY ROOFING

a. Install temporary roofing to maintain building watertight.

b. Remove temporary roofing before installing new roofing.

c. Prepare temporary roofing to receive new roofing.

3.8 EXISTING MEMBRANE PREPARATION FOR NEW ROOFING

Remove existing roofing surface projections and irregularities. Produce smooth surface to receive recover boards.

(1) Broom clean existing surface.

3.9 BASE FLASHING REMOVAL

a. Expose base flashings to permit removal.

(1) Two-Piece Counterflashings: Remove cap flashing and store for reuse.

(2) Single Piece Counterflashings: Carefully bend counterflashing.

(3) Metal Copings: Remove decorative cap and store for reuse.

b. Remove existing base flashings.

(1) Clean substrates to receive new flashings.

c. Replace counterflashings damaged during removal.

(1) Counterflashings: See Section 07 60 00 FLASHING AND SHEET METAL.

d. Remove existing parapet sheathing and inspect parapet framing.

(1) Notify Contracting Officer of damaged framing.

e. Install plywood sheathing, (19/32 inch) thick.

3.10 RECOVER BOARD INSTALLATION

a. Install recover boards over existing roofing membrane with butted joints. Stagger end joints in adjacent rows.

b. Fasten recover boards to resist wind-uplift.

(1) Fastening Requirements: See Section 07 54 23, THERMOPLASTIC POLYOLEFIN (TPO) ROOFING.

(2) Uplift Resistance: Base on pull out resistance determined by specified field testing.

3.11 FIELD QUALITY CONTROL

a. Field Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.

(1) Fastener Pull Out Tests: ANSI/SPRI FX-1 (2016).

b. Existing Roofing System Warrantor Services:

(1) Inspect reroofing preparation and roofing installation to verify compliance with existing warranty conditions.

(2) Submit reports of field inspections, and supplemental instructions issued during inspections.

3.12 DISPOSAL

a. Collect waste materials in containers.

b. Remove waste materials from project site, regularly, to prevent accumulation.

c. Legally dispose of waste materials.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 08 00.01

DVA/USACE PROJECTS FACILITY EXTERIOR CLOSURE COMMISSIONING

8/19

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUMMARY
1.4 DEFINITIONS
1.5 COMMISSIONED SYSTEMS
1.6 SUBMITTALS
1.7 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS
3.2 PRE-FUNCTIONAL CHECKLISTS
3.3 CONTRACTORS TESTS
3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING
3.5 TRAINING OF VA PERSONNEL

-- End of Section Table of Contents --

SECTION 07 08 00.01
8/19

DVA/USACE PROJECTS FACILITY EXTERIOR CLOSURE COMMISSIONING

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 07 and Division 08.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS will manage the commissioning process.

1.2 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections

b. Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS

c. UFGS Section 01 33 00 SUBMITTAL PROCEDURES

1.3 SUMMARY

a. This Section includes requirements for commissioning the Facility exterior closure, related subsystems and related equipment. This Section supplements the general requirements specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for more details regarding commissioning processes and procedures, as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.5 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 07 and Division 08 is part of the construction process. Documentation and testing of these systems, as well as training of the VA’s Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and of Division 07 and 08, is required in cooperation with the DVA/USACE and the CxA and CxM..

b. The Facility exterior closure systems commissioning will include the systems listed in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS:

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Pre-Functional Checklists; G, DO

Contractors Tests; G, DO

Systems Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Training Agendas And Trainer Resumes; G, DO

Training Plans; G, DO

1.7 SUBMITTAL INSTRUCTIONS

a. The commissioning process requires review of selected Submittals that pertain to the systems to be commissioned. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the DVA/USACE prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00 SUBMITTAL PROCEDURES for further details.

b. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of the building envelope systems will require inspection of individual elements of the envelope construction throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and the Commissioning plan to schedule envelope inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare Pre-Functional Checklists to be used to document equipment installation.

The Contractor shall complete the checklists. Completed checklists shall be submitted to the DVA/USACE, CxA, and CxM for review. The CxM and CxA may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM will return the marked-up checklist to the Contractor for correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to SECTION 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other sections of Division 07 or Division 08 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than 7 calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the CxA and Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer, and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Division 07 and 08 Sections for additional Contractor training requirements.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 13 52

MODIFIED BITUMINOUS SHEET WATERPROOFING

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 REFERENCES
1.3 PREINSTALLATION MEETINGS
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 FIELD CONDITIONS
1.10 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION
2.2 PRODUCTS - GENERAL
2.3 BITUMINOUS SHEET
2.4 PROTECTION MATERIAL
2.5 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 WATERPROOFING INSTALLATION
3.4 FIELD QUALITY CONTROL
3.5 CLEANING
3.6 PROTECTION

-- End of Section Table of Contents --

SECTION 07 13 52
1/21

MODIFIED BITUMINOUS SHEET WATERPROOFING

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Modified bituminous sheet material used for exterior below grade waterproofing and split slab waterproofing.

1.2 REFERENCES

Comply with references to extent specified in this section.

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS UU-B-790	(Rev A; Notice 2) Building Paper Vegetable Fiber: (Kraft, Waterproofed, Water Repellent and Fire Resistant)

ASTM INTERNATIONAL (ASTM)

ASTM C578	(2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation
ASTM D41/D41M	(2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D4586/D4586M	(2007; R 2018) Asphalt Roof Cement, Asbestos-Free
ASTM D6380/D6380M	(2003; E 2013; R 2013) Standard Specification for Asphalt Roll Roofing (Organic Felt)

AMERICAN HARDBOARD ASSOCIATION (AHA)

AHA A135.4	(1995; R 2004) Basic Hardboard

1.3 PREINSTALLATION MEETINGS

a. Conduct preinstallation meeting at project site minimum 30 days before beginning Work of this section.

(1) Required Participants:

(a) Contracting Officer.

(b) Architect/Engineer.

(c) Inspection and Testing Agency.

(d) Contractor.

(e) Installer.

(f) Manufacturer's field representative.

(g) Other installers responsible for adjacent and intersecting work, including substrate and flashing installers.

(2) Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(a) Installation schedule.

(b) Installation sequence.

(c) Preparatory work.

(d) Protection before, during, and after installation.

(e) Installation.

(f) Terminations.

(g) Transitions and connections to other work.

(h) Inspecting and testing.

(i) Other items affecting successful completion.

(3) Document and distribute meeting minutes to participants to record decisions affecting installation.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

SD-04 Samples

Samples; G

SD-06 Test Reports

Test Reports; G

SD-07 Certificates

Certificates; G

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Submittal Drawings: Show size, configuration, and installation details.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

(3) Warranty.

d. Samples:

(1) Waterproofing and Flashing Sheet: (8 inch) square, each type and color.

(2) Insulation: (8 inch) square.

e. Test reports: Certify products comply with specifications.

f. Certificates: Certify products comply with specifications.

g. Qualifications: Substantiate qualifications comply with specifications:

(1) Installer with project experience list.

1.6 QUALITY ASSURANCE

Installer Qualifications:

(1) Regularly installs specified products.

(2) Installed specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight conditioned facility.

b. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

Environment:

(1) Product Temperature: Minimum (40 degrees F) for minimum 48 hours before installation.

(2) Weather Limitations: Install waterproofing only during dry current and forecasted weather conditions.

1.10 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Manufacturer's Warranty: Warrant waterproofing system against material and manufacturing defects and agree to repair any leak caused by a defect in the waterproofing system materials or workmanship of the installer.

(1) Warranty Period: 10 years.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Modified bituminous sheet material for exterior below grade and split slab waterproofing.

2.2 PRODUCTS - GENERAL

a. Provide each product from one manufacturer.

b. Sustainable Construction Requirements:

(1) Insulation Recycled Content: Rigid Foam: 9 percent total recycled content, minimum.

2.3 BITUMINOUS SHEET

a. Cold applied waterproofing membrane composed primarily of modified bituminous material prefabricated in sheet form designed for below grade exterior and split slab waterproofing. Sheet reinforced with fibers at manufacturer's option.

b. Thickness: Not less than (120 mils), plus or minus (5 mils), and bonded to (4 mil) thick plastic sheet.

c. Provide release sheet to prevent bonding of bituminous sheet to itself.

2.4 PROTECTION MATERIAL

a. Polystyrene Insulation: ASTM C578, Type I or VIII, (1/2 inch) minimum thickness.

b. Hardboard: AHA A135.4, Service Type, (1/4 inch) thick.

c. Waterproofed Building Paper: FS UU-B-790 Notice 2 v04-1992 , Type I, Grade C.

d. Roll Roofing: ASTM D6380/D6380M, Class S (smooth), Type III with minimum net mass per unit area of roofing, (51 lbs./100 sq. ft.).

2.5 ACCESSORIES

a. Patching Compound: Factory-prepared, non-shrinking, fast-setting, cementitious adhesive compound containing no ferrous metal or oxide.

b. Primer: ASTM D41/D41M.

c. Roof Cement: ASTM D4586/D4586M.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

(1) Concrete surfaces cured minimum time recommended by waterproofing manufacturer.

(2) Substrate to be dry as recommended by waterproofing manufacturer.

b. Protect existing construction and completed work from damage.

c. Correct substrate deficiencies.

(1) Fill voids, joints, and cracks with patching compound.

d. Clean substrates. Remove contaminants capable of preventing full adhesion.

e. Priming:

(1) Prime concrete and masonry surfaces.

(2) Application method, amount of primer and condition or primer before installation of bituminous sheet as recommended by primer manufacturer.

(3) Reprime when required according to manufacturer's instructions.

3.2 INSTALLATION - GENERAL

Install products according to manufacturer's instructions and approved submittal drawings.

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer consideration.

3.3 WATERPROOFING INSTALLATION

a. Bituminous Sheet Installation:

(1) Remove release sheet before application.

(2) Lay bituminous sheet from low point to high point so laps shed water.

(3) Treat expansion, construction and control joints and evident working cracks as expansion joints. Apply bituminous sheet in double thickness over joint by first applying a strip of bituminous sheet minimum (8 inches) wide, centered over joint.

(4) Lap seams minimum (2 inches).

(5) Lay succeeding sheet with laps, and roll or press into place.

(6) Repair misaligned or inadequately lapped seams according to manufacturer’s instructions.

(7) Seal seams and terminations according to sheet manufacturer’s instructions.

b. Corner Treatment:

(1) At inside and outside corners, apply double cover using an initial strip minimum (11 inches) wide, centered along axis of corner.

(2) Cover each strip completely by the regular application of bituminous sheet.

(3) Provide a fillet or cant on inside corners.

(4) Form cants using patching compound.

(5) Do not use wood, fiber, and insulating materials for cants.

c. Projection Treatment:

(1) Apply a double layer of bituminous sheet around pipes and similar projections at least (6 inches) wide.

(2) At drains, apply a bead of roof cement over a double layer of bituminous sheet under clamping rings.

d. Patching:

(1) Repair tears, punctures, air blisters, and inadequately lapped seams, according to manufacturer’s instructions before protection course is applied.

e. Permanent Protection:

(1) Vertical Surfaces:

(a) Install hardboard, polystyrene insulation, or roll roofing protection material.

(b) Extend protection full height from footing to top of backfill.

(c) If graded backfill is used, use roll roofing or hardboard.

f. Horizontal Surfaces:

(1) Install roll roofing protection under concrete wearing courses.

(2) Install roll roofing, hardboard, or polystyrene insulation under earth backfill.

(3) Where no concrete wearing course occurs or when surfaces will bear heavy traffic and will not immediately be covered with a wearing course, use protection specified for vertical surfaces.

g. Temporary Protection:

(1) When waterproofing materials are subjected to damage by sunlight and cannot be immediately protected as specified, protect waterproofing materials by waterproof building paper or suitable coating approved by manufacturer of waterproofing system used.

3.4 FIELD QUALITY CONTROL

a. Field Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.

b. Testing:

(1) Before any protection or wearing course is applied, test all horizontal applications of waterproofing with a minimum of (1 inch) head of water above highest point and leave for 24 hours.

(2) Mark leaks and repair when waterproofing is dry.

(3) Certify, to Contracting Officer, that water tests have been made and that areas tested were found watertight.

c. Inspection:

(1) Do not cover waterproofed surfaces by other materials or backfill until work is approved by Contracting Officer.

3.5 CLEANING

a. Remove excess adhesive before adhesive sets.

b. Clean exposed waterproofing surfaces. Remove contaminants and stains.

3.6 PROTECTION

a. Protect waterproofing from construction operations.

b. Remove protective materials immediately before acceptance.

c. Repair damage.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 18 13

PEDESTRIAN TRAFFIC COATINGS

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 TEST AREA
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 WARRANTY
1.7 DELIVERY AND STORAGE
1.8 ENVIRONMENTAL REQUIREMENTS
1.9 SAFETY REQUIREMENTS
1.10 QUALITY ASSURANCE
1.11 REFERENCES

PART 2 PRODUCTS

2.1 PRIMER
2.2 ELASTOMERIC BASE AND TOP COAT
2.3 AGGREGATE

PART 3 EXECUTION

3.1 SURFACE PREPARATION
3.2 WORK COORDINATION
3.3 APPLICATION
3.4 CLEANING
3.5 PROTECTION

-- End of Section Table of Contents --

SECTION 07 18 13
1/21

PEDESTRIAN TRAFFIC COATINGS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies a surface applied elastomeric and composition waterproofing type membrane suitable for light pedestrian traffic and recreation areas, but not intended for heavy industrial use.

1.2 RELATED WORK

a. Section 06 10 00, ROUGH CARPENTRY: Plywood underlayment.

b. Section 07 60 00, FLASHING AND SHEET METAL: Metal flashing.

1.3 TEST AREA

Before start of general application, apply the elastomeric coating as specified in a representative test area. Provide a test area not less than (100 square feet). The area to be covered by the coating to include all site conditions such as flashings bases, corners and projections through the coating. Provide test area in location determined by the Contracting Officer. After Contracting Officer approval, area will serve as an example for the remaining work.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G

SD-03 Product Data

Manufacturers Literature And Data; G

SD-04 Samples

Samples; G

SD-11 Closeout Submittals

Manufacturer Warranty; G

1.5 SUBMITTAL INSTRUCTIONS

a. Manufacturers Literature and Data: Each material, indicating compliance with specification requirements.

b. Samples: Each finish color on (4 by 8 inch substrate, layered to show each coat and finish.

c. Installer qualifications.

d. Manufacturer warranty.

1.6 WARRANTY

a. Construction Warranty: Comply with FAR clause 52.246-21 “Warranty of Construction”.

b. Manufacturer Warranty: Manufacturer shall warranty their pedestrian traffic coating for a minimum of ten (10) years from the date of installation and final acceptance by the Government. Submit manufacturer warranty.

1.7 DELIVERY AND STORAGE

a. Deliver materials to the site in original sealed containers, clearly marked with manufacturer's name and brand, and type of material.

b. Store materials in weathertight and dry storage facility. Protect from damage from handling, weather and construction operations before, during and after installation. Store materials at temperatures and under conditions recommended by the manufacturer.

1.8 ENVIRONMENTAL REQUIREMENTS

Do not proceed with application of materials when ambient temperature is less or greater than that recommended by the coating material manufacturer. Do not apply traffic coatings to damp or wet substrates, when relative humidity exceeds 85 percent, or when temperatures are less than (5 degrees F) above dew point.

1.9 SAFETY REQUIREMENTS

Keep products away from heat, sparks and flame. Do not permit use of spark-producing equipment during application of flammable products or where explosive fumes are present.

1.10 QUALITY ASSURANCE

a. Manufacturer’s Qualifications: Obtain products from single manufacturer or from sources recommended by manufacturer for use with pedestrian traffic coatings system and incorporated in manufacturer’s warranty.

b. Installer’s Qualifications: Work to be performed by installer having three (3) years’ experience for work relating to this section and approved in writing by traffic coating manufacturer.

1.11 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C794	(2022) Standard Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants
ASTM D412	(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D2240	(2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness
ASTM E96/E96M	(2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

PART 2 PRODUCTS

2.1 PRIMER

Type required by the elastomeric coating manufacturer.

2.2 ELASTOMERIC BASE AND TOP COAT

Provide elastomeric base and topcoat materials to meet or exceed the following requirements:

Property	Test Method	Base Coat	Top Coat
Tensile Strength	ASTM D412	300(psi)	(600 psi)
Elongation, percent (of original benchmark distance)	ASTM D412	550	200
Hardness, Shore A	ASTM D2240	20	45
Peel Strength (on concrete)	ASTM C794	(30 pounds), 100 percent cohesion	100 percent cohesion to base coat
Permeability	ASTM E96/E96M	12 metric perms	0.06 metric perms

2.3 AGGREGATE

Thoroughly washed, clean, medium-grained sharp indigenous stone granules, graded between (.04 inch) and (.16 inch) in size, and having a hardness of 6.5 or greater on the Moh's scale.

PART 3 EXECUTION

3.1 SURFACE PREPARATION

a. Apply to clean, dry surfaces. Smooth rough spots and tool marks.

b. Fill holes, depressions and cracks with fillers compatible with the coating material and recommended by the coating manufacturer.

c. Subsurface imperfections that telegraph through the finish coating surface are not acceptable.

3.2 WORK COORDINATION

To provide a watertight installation, coordinate this work with flashing and drains required to be installed before the coating work begins and be completed after the coating is in place.

3.3 APPLICATION

a. Prime surfaces to receive elastomeric waterproofing materials as recommended by the product manufacturer.

b. Where horizontal surfaces intersect vertical surfaces, provide a sealant type fillet as recommended by the manufacturer.

c. Apply elastomeric base coat at a rate that will ensure a dry film thickness of not less than (35 mils).

d. Uniformly disperse aggregate topping at the rate of (10 pounds/100 square foot); inter-bond and vulcanize granules within the fluid elastomeric topping in accordance with manufacturer's instructions.

e. Apply elastomeric top coat at a rate which will ensure a dry film thickness of (15 mils) for the top coat and a total combined dry film thickness for the base and top coats of not less than (50 mils).

f. Complete the base to a uniform established line as shown.

3.4 CLEANING

Remove smears of elastomeric material from other work.

3.5 PROTECTION

Protect the finished coating from traffic until the coating cures and afterward until final acceptance.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 21 13

THERMAL INSULATION

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY
1.7 STORAGE AND HANDLING
1.8 WARRANTY

PART 2 PRODUCTS

2.1 INSULATION - GENERAL
2.2 THERMAL INSULATION
2.3 ACOUSTICAL INSULATION
2.4 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 THERMAL INSULATION
3.4 ACOUSTICAL INSULATION
3.5 CLEANING
3.6 PROTECTION

-- End of Section Table of Contents --

SECTION 07 21 13
1/21

THERMAL INSULATION

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Thermal insulation.

(a) Board or block insulation at foundation perimeter.

(b) Batt or blanket insulation at exterior framed and furred walls.

(c) Board or block insulation at floor assemblies above unconditioned spaces.

(d) Board or block insulation at masonry cavity walls.

(e) Loose fill insulation at exterior hollow masonry walls.

(2) Acoustical insulation.

(a) Semi-rigid insulation at interior framed partitions.

(b) Batt and blanket insulation at interior framed partitions and ceilings .

(c) Board insulation at interior concrete and masonry partitions.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Adhesives VOC Limits.

b. Section 04 20 00, UNIT MASONRY: Insulation for Cavity Face of Masonry.

c. Section 07 21 23, LOOSE-FILL INSULATION: Loose Fill Insulation for Attic Floors.

d. Section 07 84 00, FIRESTOPPING: Safing Insulation.

1.3 REFERENCES

Comply with references to extent specified in this section.

ASTM INTERNATIONAL (ASTM)

ASTM C516	(2019) Standard Specification for Vermiculite Loose Fill Thermal Insulation
ASTM C549	(2023) Standard Specification for Perlite Loose Fill Insulation

ASTM C552	(2022) Standard Specification for Cellular Glass Thermal Insulation
ASTM C553	(2024) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
ASTM C578	(2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation
ASTM C591	(2022) Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation
ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C665	(2024) Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing
ASTM C728	(2017a; R 2022) Standard Specification for Perlite Thermal Insulation Board
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness
ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM F1667/F1667M	(2021a) Standard Specification for Driven Fasteners: Nails, Spikes, and Staples

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Construction Submittals

SD-11 Closeout Submittals

Environmental Product Declaration (EPD); S

Third-Party Certification Certificate, Assessment, Or Validation And Compliance Report; S

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Submittal Drawings:

(1) Show insulation type, thickness, and R-value for each location.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Adhesive indicating manufacturer recommendation for each application.

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

(2) Low Pollutant-Emitting Materials: Show volatile organic compound types and quantities.

1.6 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.7 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

c. Protect foam plastic insulation from UV exposure.

1.8 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 INSULATION - GENERAL

a. Insulation Thickness:

(1) Provide thickness required by R-value shown on drawings.

(2) Provide thickness indicated when R-value is not shown on drawings.

b. Insulation Types:

(1) Provide one insulation type for each application.

c. Sustainable Construction Requirements:

(1) Insulation Recycled Content:

(a) Polyisocyanurate/polyurethane rigid foam: 9 percent recovered material.

(b) Polyisocyanurate/polyurethane foam-in-place: 5 percent recovered material.

(c) Glass fiber reinforced: 6 percent recovered material.

(d) Phenolic rigid foam: 5 percent recovered material.

(e) Rock wool material: 75 percent recovered material.

(2) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Non-Flooring Adhesives and Sealants.

2.2 THERMAL INSULATION

a. Perimeter Insulation In Contact with Soil:

(1) Polystyrene Board: ASTM C578, Type IV, V, VI, VII, or IX.

(2) Cellular Glass Block: ASTM C552, Type I or IV.

b. Exterior Framing or Furring Insulation:

(1) Mineral Fiber: ASTM C665, Type II, Class C, Category I where concealed by thermal barrier.

(2) Mineral Fiber: ASTM C665, Type III, Class A at other locations.

c. Inside Face of Exterior Wall Insulation:

(1) Mineral Fiber Board: ASTM C612, Type IB or II.

(2) Perlite Board: ASTM C728.

(3) Cellular Glass Block: ASTM C552, Type I.

d. Floor Assemblies Above Unconditioned Spaces:

(1) Mineral Fiber Board: ASTM C612, Type IB or Type II.

(2) Perlite Board: ASTM C728.

(3) Cellular Glass Block: ASTM C552, Type I.

e. Masonry Cavity Wall Insulation:

(1) Mineral Fiber Board: ASTM C612, Type II, with vapor retarder facing; maximum permeance (0.5 perms).

(2) Polyurethane or Polyisocyanurate Board: ASTM C591, Type I, with vapor retarder facing; maximum permeance (0.5 perms).

(3) Polystyrene Board: ASTM C578, Type X.

(4) Perlite Board: ASTM C728.

(5) Cellular Glass Block: ASTM C552, Type I or IV.

f. Masonry Fill Insulation:

(1) Vermiculite Insulation: ASTM C516, Type II.

(2) Perlite Insulation: ASTM C549, Type IV.

2.3 ACOUSTICAL INSULATION

a. Semi Rigid, Batts and Blankets:

(1) Widths and lengths to fit tight against framing.

(2) Mineral Fiber boards: ASTM C553, Type II, flexible, or Type III, semi rigid FSK faced.

(a) Density: nominal 4.5 pound.

(3) Mineral Fiber Batt or Blankets: ASTM C665 unfaced.

(4) Maximum Surface Burning Characteristics: ASTM E84.

(a) Flame Spread Rating: 25.

(b) Smoke Developed Rating: 450.

b. Sound Deadening Board:

(1) Mineral Fiber Board: ASTM C612, Type IB.

(a) Thickness: (1/2 inch).

(2) Perlite Board: ASTM C728.

(a) Thickness: (1/2 inch).

2.4 ACCESSORIES

a. Fasteners:

(1) Staples or Nails: ASTM F1667/F1667M, zinc-coated, size and type to suit application.

(2) Screws: ASTM C954 or ASTM C1002, size and length to suit application with washer minimum (2 inches) diameter.

(3) Impaling Pins: Steel pins with head minimum (2 inches) diameter.

(a) Length: As required to extend beyond insulation and retain cap washer when washer is placed on pin.

(b) Adhesive: Type recommended by manufacturer to suit application.

b. Insulation Adhesive: Nonflammable type recommended by insulation manufacturer to suit application.

c. Tape: Pressure sensitive adhesive on one face.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Clean substrates. Remove contaminants capable of affecting subsequently installed product's performance.

3.2 INSTALLATION - GENERAL

a. Install products according to manufacturer's instructions and approved submittal drawings.

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's consideration.

b. Install insulation with vapor barrier facing the heated side, unless indicated otherwise.

c. Install board and block insulation with joints close and flush, in regular courses, and with end joints staggered.

d. Install batt and blanket insulation with joints tight. Fill framing voids completely. Seal penetrations, terminations, facing joints, facing cuts, tears, and unlapped joints with tape.

e. Fit insulation tight against adjoining construction and penetrations, unless indicated otherwise.

3.3 THERMAL INSULATION

a. Perimeter Insulation In Contact with Soil:

(1) Vertical insulation:

(a) Fill joints of insulation with same material used for bonding.

(b) Bond polystyrene board to surfaces with adhesive.

(c) Bond cellular glass insulation to surfaces with hot asphalt or adhesive cement.

(2) Horizontal insulation under concrete floor slab:

(a) Lay insulation boards and blocks horizontally on level, compacted and drained fill.

(b) Extend insulation from foundation walls towards center of building minimum (24 inches).

b. Exterior Framing or Furring Insulation:

(1) General:

(a) Open voids are not acceptable.

(b) Pack insulation around door frames and windows, in building expansion joints, door soffits, and other voids.

(c) Pack behind outlets, around pipes, ducts, and services encased in walls.

(d) Hold insulation in place with pressure sensitive tape.

(e) Lap facing flanges together over framing for continuous surface. Seal penetrations through insulation and facings.

(2) Metal Studs: Fasten insulation between metal studs, framing, and furring with pressure sensitive tape continuous along flanged edges.

(3) Wood Studs:

(a) Fasten insulation between wood studs or framing with nails or staples through flanged edges on face of stud.

(b) Space fastenings maximum (six inches) apart.

(4) Roof Rafters and Floor Joists: Friction fit insulation between framing to provide minimum (2 inch) air space between insulation and roof sheathing and subfloor.

(5) Ceilings and Soffits:

(a) Wood Framing:

1. Fasten blanket insulation between wood framing and joists with nails or staples through flanged edges of insulation.

2. Space fastenings maximum (6 inches) on center.

(b) Metal Framing:

1. Fasten insulation between metal framing with pressure sensitive tape continuous along flanged edges.

2. At metal framing and ceilings suspension systems, install insulation above suspended ceilings and metal framing at right angles to main runners and framing.

3. Tape insulation tightly together without gaps. Cover metal framing members with insulation.

(c) Ceiling Transitions:

1. In areas where suspended ceilings transition to structural ceiling, install blanket or batt insulation.

2. Extend insulation from suspended ceiling to underside of structure above.

3. Secure blanket and batt with continuous cleats to structure above.

c. Inside Face of Exterior Wall Insulation:

(1) Location: On interior face of solid masonry and concrete walls, beams, beam soffits, underside of floors, and to face of studs to support interior wall finish where indicated.

(2) Bond insulation to solid vertical surfaces with adhesive. Fill joints with adhesive cement.

(3) Fasten board insulation to face of studs with screws, nails or staples. Space fastenings maximum (12 inches) on center. Stagger fasteners at board joints. Install fasteners at each corner.

d. Floor Assemblies Above Unconditioned Spaces:

(1) Use impaling pins for attach insulation to underside of horizontal surfaces. Space fastenings as required to hold insulation in place and prevent sagging.

(a) Bond insulation with adhesive when separate vapor retarder is used.

e. Masonry Cavity Wall Insulation:

(1) Install insulation on exterior faces of concrete and masonry inner wythes of cavity walls.

(2) Bond polystyrene board to surfaces with adhesive.

(3) Bond polyurethane or polyisocyanurate board, and perlite board to surfaces with adhesive.

(4) Bond cellular glass insulation to surfaces with hot asphalt or adhesive cement.

(5) Fill insulation joints with same material used for bonding.

f. Masonry Fill Insulation:

(1) Pour fill insulation in masonry unit hollow cores from tops of walls, or from sill where windows or other openings occur.

(2) Pour in lifts of maximum (20 feet).

3.4 ACOUSTICAL INSULATION

a. General:

(1) Install insulation without voids.

(2) Pack insulation around door frames and windows, in building expansion joints, door soffits, and other voids.

(3) Pack behind outlets, around pipes, ducts, and services encased in walls.

(4) Hold insulation in place with pressure sensitive tape.

(5) Lap facer flanges together over framing for continuous surface. Seal all penetrations through the insulation and facers.

(6) Do not compress insulation below required thickness except where embedded items prevent required thickness.

b. Semi Rigid, Batts and Blankets:

(1) When insulation is not full thickness of cavity, adhere insulation to one side of cavity, maintaining continuity of insulation and covering penetrations or embedments.

(a) Wood Framing:

1. Fasten blanket insulation between wood framing and joists with nails or staples through flanged edges of insulation.

2. Space fastenings maximum (6 inches) on center.

(b) Metal Framing:

1. Fasten insulation between metal framing with pressure sensitive tape continuous along flanged edges.

2. At metal framing or ceilings suspension systems, install blanket insulation above suspended ceilings or metal framing at right angles to the main runners or framing.

3. Tape insulation tightly together so no gaps occur and metal framing members are covered by insulation.

c. Sound Deadening Board: Secure with adhesive to masonry and concrete walls and with screws to metal and wood framing. Secure sufficiently in place until subsequent cover is installed. Seal all cracks with caulking.

3.5 CLEANING

Remove excess adhesive before adhesive sets.

3.6 PROTECTION

a. Protect insulation from construction operations.

b. Repair damage.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 21 23

LOOSE-FILL INSULATION

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTUCTIONS
1.6 DELIVERY
1.7 STORAGE AND HANDLING
1.8 WARRANTY

PART 2 PRODUCTS

2.1 PRODUCTS - GENERAL
2.2 LOOSE-FILL INSULATION
2.3 ACCESSORIES

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 INSTALLATION
3.4 FIELD QUALITY CONTROL
3.5 CLEANING

-- End of Section Table of Contents --

SECTION 07 21 23
1/21

LOOSE-FILL INSULATION

PART 1 GENERAL

1.1 SUMMARY

Loose-fill insulation with vapor barrier at attic floors.

1.2 RELATED WORK

Loose-Fill Insulation for Masonry Walls.

1.3 REFERENCES

Comply with references to extent specified in this section.

ASTM INTERNATIONAL (ASTM)

ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C728	(2017a; R 2022) Standard Specification for Perlite Thermal Insulation Board
ASTM C739	(2024) Standard Specification for Cellulosic Fiber Loose-Fill Thermal Insulation
ASTM C764	(2019) Standard Specification for Mineral Fiber Loose-Fill Thermal Insulation
ASTM C1015	(2017) Standard Practice for Installation of Cellulosic and Mineral Fiber Loose-Fill Thermal Insulation
ASTM D4397	(2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 211	(2019) Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Construction Submittals

SD-04 Samples

Samples; G

1.5 SUBMITTAL INSTUCTIONS

a. Submittal Procedures: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Include copy of packaging label.

(3) Installation instructions.

c. Samples:

(1) Loose-fill insulation: (1 pint).

(2) Blocking: (6 inch) long.

(3) Vapor Retarder: (6 inch by 6 inch).

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

1.6 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

(1) Include referenced specification number, type and class as applicable, recommended method of installation (pneumatic or pouring), minimum net weight of insulation, coverage charts, R-values, and required warning statements.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.7 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.8 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 PRODUCTS - GENERAL

a. Provide insulation from one manufacturer.

b. Sustainable Construction Requirements:

(1) Insulation Recycled Content:

(2) Loose-Fill Insulation: 75 percent total recycled content, minimum.

(3) Mineral Fiber: 75 percent total recycled content, minimum.

(4) Perlite: 23 percent total post-consumer recycled content, minimum.

2.2 LOOSE-FILL INSULATION

a. Cellulosic or Wood Fiber Loose-fill: ASTM C739.

b. Mineral Fiber Loose-fill: ASTM C764, Type I or II.

2.3 ACCESSORIES

a. Blocking: Wood, metal, mineral fiber or perlite boards, or other materials approved by Contracting Officer.

(1) Mineral Fiber Board: ASTM C612, Type IB.

(2) Perlite Board: ASTM C728.

b. Vapor Retarder: ASTM D4397; (8 mil) thick polyethylene sheeting conforming.

(1) Tape: Self-adhesive type with maximum perm rating matching vapor retarder.

PART 3 EXECUTION

3.1 EXAMINATION

a. Examine locations receiving loose-fill insulation according to ASTM C1015. Identify the following:

(1) Defects in electrical fixtures, equipment, wiring, junction boxes, receptacles, and switches.

(2) Openings through which loose-fill insulation material may escape.

(3) Air ducts with unsealed joints.

b. Report unsatisfactory conditions to Contracting Officer.

c. Do not install insulation until unsatisfactory conditions are corrected.

3.2 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Install blocking in accordance ASTM C1015 and as indicated.

(1) Opening Blocking:

(a) Install blocking around ceiling access-panels, and vents if the level to which the unsettled insulation will be installed exceeds their height.

(b) Cover openings into attic with temporary blocking to prevent insulation from falling into opening, including spaces enclosed by blockings.

(2) Device Blocking: Install blocking (2 inches) above height of finished insulation installation to ensure devices requiring maintenance or servicing remain accessible after insulation is installed.

(3) Heat Producing Device Blocking: Minimum clearances for blocking around heat producing devices.

(a) Masonry Chimneys: (4 inches) from outside face of masonry servicing fired equipment operating at maximum (1500 degrees F).

(b) Other Chimneys, Vents, Breechings, and Stacks: Minimum clearances as required by NFPA 211.

3.3 INSTALLATION

a. Install products according to manufacturer's instructions and approved submittal drawings .

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's consideration.

b. Vapor Retarder:

(1) Where space is accessible, install vapor retarder below insulation. Do not install a vapor retarder over existing insulation.

(2) Install without tears, breaks, or ruptures interfering with vapor retarder effectiveness and continuity.

(3) Lap joints in vapor retarder to ensure continuous seal. Tape laps to retain vapor barrier in place.

(4) Use self-adhesive tape for sealing joints and holes in the vapor retarder.

c. Insulation:

(1) Install insulation according to ASTM C1015.

(2) Fit the attic side of access panels with perlite or mineral fiber insulation boards.

(3) Install insulation in sufficient depth to provide thermal value specified after insulation settlement.

(4) Pneumatic installation:

(a) Use lowest air pressure according to manufacturer's instructions.

(b) Do not blow insulation into electrical devices and vents opening into insulated spaces.

(5) Install insulation and allow natural settlement to final density. Do not tamp or rod insulation.

3.4 FIELD QUALITY CONTROL

According to ASTM C1015.

3.5 CLEANING

a. Remove excess insulation.

b. Remove temporary blockings over openings.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 22 00

ROOF AND DECK INSULATION

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 FIELD CONDITIONS
1.10 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 PRODUCTS - GENERAL
2.3 ADHESIVES
2.4 ROOF AND DECK INSULATION
2.5 INSULATION ACCESSORIES
2.6 ACCESSORIES

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 INSTALLATION - GENERAL
3.4 SUBSTRATE BOARD INSTALLATION
3.5 VAPOR RETARDER INSTALLATION
3.6 INSULATION INSTALLATION
3.7 COVER BOARD INSTALLATION

-- End of Section Table of Contents --

SECTION 07 22 00
1/21

ROOF AND DECK INSULATION

PART 1 GENERAL

1.1 SUMMARY

a. Section Includes:

b. Roof and deck insulation, substrate board, vapor retarder, and cover board on metal deck substrates ready to receive roofing or waterproofing membrane.

c. Repairs and alteration work to existing roof insulation.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Non-Flooring Adhesives and Sealants VOC Limits.

b. Section 06 10 00, ROUGH CARPENTRY: Wood Cants, Blocking, and Edge Strips.

1.3 REFERENCES

Comply with references to extent specified in this section.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7	(2017) Minimum Design Loads for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM C208	(2022) Standard Specification for Cellulosic Fiber Insulating Board
ASTM C552	(2022) Standard Specification for Cellular Glass Thermal Insulation
ASTM C726	(2024) Standard Specification for Mineral Wool Roof Insulation Board
ASTM C728	(2017a; R 2022) Standard Specification for Perlite Thermal Insulation Board
ASTM C1177/C1177M	(2024) Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing
ASTM C1278/C1278M	(2024) Standard Specification for Fiber-Reinforced Gypsum Panel
ASTM C1289	(2023a) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board

ASTM C1396/C1396M	(2024) Standard Specification for Gypsum Board
ASTM D41/D41M	(2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D312	(2000; R 2006) Standard Specification for Asphalt Used in Roofing
ASTM D1970/D1970M	(2019) Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection
ASTM D2178/D2178M	(2015a; R 2021) Standard Specification for Asphalt Glass Felt Used in Roofing and Waterproofing
ASTM D2822/D2822M	(2005; R 2011; E 2011) Standard Specification for Asphalt Roof Cement, Asbestos-Containing
ASTM D4586/D4586M	(2007; R 2018) Asphalt Roof Cement, Asbestos-Free
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM F1667/F1667M	(2021a) Standard Specification for Driven Fasteners: Nails, Spikes, and Staples

NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

NRCA RoofMan	(2020) The NRCA Roofing Manual

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST PS 1	(2009) DOC Voluntary Product Standard PS 1-07, Structural Plywood
NIST DOC PS 2	Performace Standard for Wood Structural Panels

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Construction Submittals

SD-04 Samples

Samples; G

SD-11 Closeout Submittals

Environmental Product Declaration (EPD); S

Third-Party Certification Certificate, Assessment, Or Validation And Compliance Report; S

1.5 SUBMITTAL INSTRUCTIONS

ab. Submittal Drawings:

(1) Show size, configuration, and installation details.

(a) Nailers, cants, and terminations.

(b) Layout of insulation showing slopes, tapers, penetrations, and edge conditions.

b. Manufacturer's Literature and Data:

(1) Description of each product.

c. Samples:

(1) Roof insulation, each type.

(2) Fasteners, each type.

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

(2) Biobased Content:

(a) Show type and quantity for each product.

(3) Low Pollutant-Emitting Materials:

(a) Show volatile organic compound types and quantities.

(b) Certify each composite wood and agrifiber product contain no added urea formaldehyde.

e. Qualifications: Substantiate qualifications meet specifications.

(1) Installer.

1.6 QUALITY ASSURANCE

Same installer as Division 07 roofing section installer.

1.7 DELIVERY

a. Comply with recommendations of NRCA RoofMan.

b. Deliver products in manufacturer's original sealed packaging.

c. Mark packaging, legibly. Indicate manufacturer's name or brand, type, and manufacture date.

d. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Comply with recommendations of NRCA RoofMan.

b. Store products indoors in dry, weathertight facility.

c. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

Install products when existing and forecasted weather permit installation according to manufacturer's instructions.

1.10 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Manufacturer's Warranty: Warrant substrate board, vapor retarder, insulation, and cover board against material and manufacturing defects for ten (10) years as part of Division 07 thermal and moisture protection.

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE

a. Insulation Thermal Performance:

(1) Overall Average R-Value: (R-33), minimum.

(2) Any Location R-Value: (R-10), minimum.

b. Fire and Wind Uplift Resistance: Provide roof insulation complying with requirements specified in Division 07 roofing section.

c. Insulation on Metal Decking: UL labeled indicating compliance with one of the following:

(1) UL Listed.

(2) Insulation Surface Burning Characteristics: When tested according to ASTM E84.

(a) Flame Spread Rating: 75 maximum.

(b) Smoke Developed Rating: 150 maximum.

2.2 PRODUCTS - GENERAL

a. Provide each product from one manufacturer.

b. Sustainable Construction Requirements:

(1) Insulation Recycled Content:

(a) Mineral Fiber: 75 percent total recycled content, minimum.

(b) Fiberglass: 20 percent total recycled content, minimum.

(c) Cellulose: 75 percent post-consumer recycled content, minimum.

(d) Perlite Composite Board: 23 percent post-consumer recycled content, minimum.

(e) Rigid Foam: 9 percent total recycled content, minimum.

(f) Glass Fiber Reinforced Rigid Foam: 6 percent total recycled content, minimum.

(2) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Non-flooring adhesives and sealants.

(b) Composite wood and agrifiber.

(3) Bio-Based Materials: Where applicable, provide products designated by USDA and meeting or exceeding USDA recommendations for bio-based content, and products meeting Rapidly Renewable Materials and certified sustainable wood content definitions; refer to www.biopreferred.gov.

2.3 ADHESIVES

a. Primer: ASTM D41/D41M.

b. Asphalt: ASTM D312, Type III or IV for vapor retarders and insulation.

c. Modified Asphaltic Insulation Adhesive: Insulation manufacturer's recommended modified asphaltic, asbestos-free, cold-applied adhesive formulated to adhere roof insulation to substrate or to another insulation layer.

d. Bead-Applied Urethane Insulation Adhesive: Insulation manufacturer's recommended bead-applied, low-rise, one- or multicomponent urethane adhesive formulated to adhere roof insulation to substrate or to another insulation layer.

e. Full-Spread Applied Urethane Insulation Adhesive: Insulation manufacturer's recommended spray-applied, low-rise, two-component urethane adhesive formulated to adhere roof insulation to substrate or to another insulation layer.

f. Roof Cement: Asbestos free, ASTM D2822/D2822M, Type I or Type II; or, ASTM D4586/D4586M, Type I or Type II.

2.4 ROOF AND DECK INSULATION

a. Roof and Deck Insulation, General: Preformed roof insulation boards approved by roofing manufacturer.

b. Polyisocyanurate Board Insulation: ASTM C1289, Type II, Class 1, Grade 2, faced with glass fiber reinforced cellulosic felt facers on both major surfaces of the core foam.

c. Cellular Glass Board Insulation: ASTM C552, Type IV, kraft-paper sheet faced.

d. Perlite Board Insulation: ASTM C728, expanded perlite particles, selected binders, and cellulosic fibers with surface treated to reduce bitumen absorption.

e. Cellulosic Fiber Board Insulation: ASTM C208, Type II, Grade 2 for single-ply roofing .

f. Tapered Roof Insulation System:

(1) Fabricate of mineral fiberboard, polyisocyanurate, perlite board, or cellular glass. Use only one insulation material for tapered sections. Use only factory-tapered insulation.

(2) Cut to provide high and low points with crickets and slopes as shown.

(3) Minimum thickness of tapered sections; (1-1/2 inch).

(4) Minimum slope 1/48 1/4 inch per 12 inches).

g. Composite Nail Base Insulated Roof Sheathing:

(1) Oriented-Strand-Board-Surfaced, Polyisocyanurate-Foam Sheathing: Polyisocyanurate thermal insulation ASTM C1289, Type V, insulation thickness as shown, with oriented strand board laminated to top surface.

(2) Oriented Strand Board: NIST PS 1, Exposure 1, (5/8 inch) thick.

(3) Bottom surface faced with felt facers.

2.5 INSULATION ACCESSORIES

a. Glass (Felt): ASTM D2178/D2178M, Type VI, heavy duty ply sheet.

b. Cants and Tapered Edge Strips:

(1) Wood Cant Strips: Refer to Section 06 10 00, ROUGH CARPENTRY.

(2) Insulation Cant Strips: ASTM C208, Type II, Grade 1, cellulosic-fiber insulation board.

(3) Tapered Edge Strips: 1/12 1 inch per 12 inches), from (0 inches), 12 inches to 18 inches) wide.

(a) Cellulosic Fiberboard: ASTM C208.

(b) Mineral Fiberboard: ASTM C726.

(c) Perlite Board: ASTM C728.

c. Vapor Retarder:

(1) Glass-Fiber Felts: ASTM D2178/D2178M, Type IV, asphalt impregnated.

(2) Self-Adhering Sheet Vapor Retarder: ASTM D1970/D1970M, minimum (40 mils) thick membrane of HDPE film fully coated with asphalt adhesive, or (30 to 40 mils) thick membrane of butyl rubber based adhesive backed by a layer of high density cross-laminated polyethylene; maximum permeance rating of (0.1 perms).

d. Substrate Board:

(1) Gypsum Board: ASTM C1396/C1396M, (5/8 inch) thick, Type X.

(2) Glass-Mat, Water-Resistant Gypsum Roof Board: ASTM C1177/C1177M, Type X, (5/8 inch) thick, factory primed.

(3) Cellulosic-Fiber-Reinforced, Water-Resistant Gypsum Roof Board: ASTM C1278/C1278M, (5/8 inch) thick.

(4) Perlite Board Insulation: ASTM C728, (3/4 inch) .

e. Cover Board:

(1) Glass-Mat, Water-Resistant Gypsum Roof Board: ASTM C1177/C1177M, (1/2 inch) thick, factory primed.

(2) Cellulosic-Fiber-Reinforced, Water-Resistant Gypsum Roof Board: ASTM C1278/C1278M, (5/8 inch) thick.

(3) Cellulosic-Fiber Insulation Board: ASTM C208, Type II, Grade 2, (1/2 inch) thick.

(4) Oriented Strand Board: NIST DOC PS 2, Exposure 1, (7/16 inch) thick.

2.6 ACCESSORIES

a. Fasteners: Corrosion-resistant carbon steel fasteners and galvalume-coated steel or plastic round plates for fastening substrate board and insulation to roof deck.

b. Nails: ASTM F1667/F1667M; type to suit application.

PART 3 EXECUTION

3.1 EXAMINATION

Comply with requirements of Division 07 roofing section.

3.2 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

3.3 INSTALLATION - GENERAL

a. Install products according to manufacturer's instructions.

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's consideration.

b. Comply with requirements of UL Product iQ for insulated steel roof deck.

c. Attach substrate board and other products to meet requirements of Division 07 roofing section.

3.4 SUBSTRATE BOARD INSTALLATION

Fasten substrate board to top flanges of steel decking to resist uplift pressures according requirements for specified roofing system.

(1) Locate the long dimension edge joints solidly bearing on top of decking ribs.

3.5 VAPOR RETARDER INSTALLATION

a. Vapor Retarder Installation, General:

(1) Install continuous vapor retarder on roof decks where indicated.

(2) At vertical surfaces, turn up vapor retarder to top of insulation or base flashing.

(3) Seal penetrations through vapor retarder with roof cement to prevent moisture entry from below.

b. Cast in Place Concrete Decks, Except Insulating Concrete:

(1) Prime deck as specified.

(2) Apply two plies of asphalt saturated felt mopped down to deck.

c. Precast Concrete Unit Decks Without Concrete Topping:

(1) Prime deck as specified.

(2) Apply two plies of asphalt saturated felt.

(3) Mop to deck, keeping bitumen (4 inches) away from joints of

precast units. Bridge joints with felt. Mop between plies as specified.

3.6 INSULATION INSTALLATION

a. Insulation Installation, General:

(1) Base Sheet: Where required by roofing system, install one lapped base sheet specified in Division 07 roofing section by mechanically fastening to roofing substrate before installation of insulation.

(2) Cant Strips: Install wood cant strips specified in Section 06 10 00 ROUGH CARPENTRY at junctures of roofing system with vertical construction.

(3) Use same insulation as existing for roof repair and alterations unless specified otherwise.

b. Insulation Thickness:

(1) Thickness of roof insulation shown on drawings is nominal. Provide thickness required to comply with specified thermal performance.

(2) Insulation on Metal Decks: Provide insulation in minimum thickness recommended by insulation manufacturer to span deck flutes. Support edges of insulation on metal deck ribs.

(3) When actual insulation thickness differs from drawings, coordinate alignment and location of roof drains, flashing, gravel stops, fascias and similar items.

(4) Where tapered insulation is used, maintain insulation thickness at high points and roof edges shown on drawings.

(a) Low Point Thickness: Minimum (1-1/2 inches).

(5) Use minimum two layers of insulation when required thickness is (2.7 inch) or greater.

c. Lay insulating units with close joints, in regular courses and with end joints staggered.

(1) Stagger joints between layers minimum (6 inches).

d. Lay units with long dimension perpendicular to the rolled (longitudinal) direction of the roofing felt.

e. Seal cut edges at penetrations and at edges against blocking with bitumen or roof cement.

f. Cut to fit tightly against blocking or penetrations.

g. Cover all insulation installed on the same day; comply with temporary protection requirements of Division 07 roofing section.

h. Installation Method:

(1) Adhered Insulation:

(a) Prime substrate as required.

(b) Set each layer of insulation firmly in solid mopping of hot asphalt.

(c) Set each layer of insulation firmly in ribbons of bead-applied insulation adhesive.

(d) Set each layer of insulation firmly in uniform application of full-spread insulation adhesive.

(2) Mechanically Fastened Insulation:

(a) Fasten insulation according to requirements in Division 07 roofing section.

(b) Fasten insulation to resist uplift pressures specified in Division 07 roofing section and ASCE 7.

(3) Mechanically Fastened and Adhered Insulation:

(a) Fasten first layer of insulation according to "Mechanically Fastened Insulation" requirements.

(b) Fasten each subsequent layer of insulation according to "Adhered Insulation" requirements.

3.7 COVER BOARD INSTALLATION

a. Install cover boards over insulation with long joints in continuous straight lines with staggered end joints.

b. Offset cover board joints from insulation joints (6 inches), minimum.

c. Secure cover boards according to "Mechanically Fastened Insulation" requirements.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 27 27

FLUID-APPLIED MEMBRANE AIR BARRIERS, VAPOR RETARDING

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED REQUIREMENTS
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 FIELD CONDITIONS
1.10 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 PRODUCTS - GENERAL
2.3 AIR BARRIER
2.4 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - AIR BARRIER
3.3 FIELD QUALITY CONTROL
3.4 CLEANING
3.5 PROTECTION

-- End of Section Table of Contents --

SECTION 07 27 27
1/21

FLUID-APPLIED MEMBRANE AIR BARRIERS, VAPOR RETARDING

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Fluid-applied vapor-retarding air barrier at exterior above grade wall assemblies.

(2) Connection to adjacent air barrier components providing a durable, continuous, full building air barrier.

1.2 RELATED REQUIREMENTS

a. General Quality Assurance and Quality Control Requirements: Section 01 45 29 TESTING LABORATORY SERVICES.

d. Masonry Unit Air Barrier Substrates: Section 04 20 00 UNIT MASONRY.

e. Flashing Components of Factory Finished Roofing and Wall Systems Air Barriers Requiring Air Barrier Transitions: Division 07 roofing and wall system sections.

f. Metal Flashing Requiring Air Barrier Transitions: Section 07 60 00 FLASHING AND SHEET METAL.

g. Joint Sealants: Section 07 92 00 JOINT SEALANTS.

h. Exterior Wall Openings Requiring Air Barrier Transitions: Division 08 sections for glazed aluminum curtain walls and louvers and vents.

ij. Wall Sheathings Air Barrier Substrates: Section 09 29 00 GYPSUM BOARD.

1.3 REFERENCES

Comply with references to extent specified in this section.

AIR BARRIER ASSOCIATION OF AMERICA (ABAA)

ABAA QAP	Quality Assurance Program

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM C1193	(2016) Standard Guide for Use of Joint Sealants
ASTM D412	(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic

	Elastomers - Tension
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E96/E96M	(2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM E783	(2002; R 2018) Standard Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors
ASTM E1186	(2022) Standard Practices for Air Leakage Site Detection in Building Envelopes and Air Barrier Systems
ASTM E2178	(2021a) Standard Test Method for Determining Air Leakage Rate and Calculation of Air Permeance of Building Materials
ASTM E2357	(2018) Standard Test Method for Determining Air Leakage Rate of Air Barrier Assemblies

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. \ Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Construction Submittals

SD-06 Test Reports

Test Reports; G

Installation Audit; G

SD-07 Certificates

Certificates; G

1.5 SUBMITTAL INSTRUCTIONS

a. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

b. Sustainable Construction Submittals:

(1) Low Pollutant-Emitting Materials:

(a) Show volatile organic compound types and quantities.

c. Test reports:

(1) Submit field inspection and test reports.

d. Certificates: Certify each product complies with specifications.

e. Qualifications: Substantiate qualifications comply with specifications.

(1) Manufacturer with project experience list.

(2) Installer .

(a) Certify installer approval by air barrier manufacturer.

f. Installation Audit:

(1) Submit audit report.

1.6 QUALITY ASSURANCE

a. Coordinate work with adjacent and related work to provide continuous, unbroken, durable air barrier system.

b. Manufacturer Qualifications:

(1) Regularly and presently manufactures specified products.

(2) Manufactured specified products with satisfactory service on five similar installations for minimum five years.

(3) Accreditation by ABAA.

c. Installer Qualifications:

(1) Regularly and presently installs specified products.

(2) Approved by manufacturer.

(3) Accredited by ABAA.

(4) Applicators certified according to ABAA Quality Assurance Program ABAA QAP.

(5) Applicators trained and certified by manufacturer of air barrier system.

(6) Full time on-site field supervisor has completed three projects of similar scope within last year.

(7) Field Supervisor: Holds Sealant, Waterproofing, and Restoration Institute (SWRI) Wall Coating Validation Program Certificate, or similar qualification acceptable to Contracting Officer.

(8) Field supervisor accredited by ABAA as Level 3 Accredited Installer.

d. Testing Agency Qualifications:

(1) Accredited by International Accreditation Service, Inc. or American Association for Laboratory Accreditation.

(2) Certified perform ABAA Quality Assurance Program ABAA QAP installer audits.

(3) Staff experienced in installation of specified system and qualified to perform observation and inspection specified and determine compliance with project requirements.

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight, conditioned facility.

b. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

a. Environment:

(1) Work Area Ambient Temperature Range: (40 to 90 degrees F) continuously, beginning 48 hours before installation.

b. Surface Requirements: visibly dry, and complying with manufacturer's instructions.

1.10 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

a. System Responsibility: the exterior wall walls consisting of curtainwall, insulated metal wall panels, aluminum fins, light gauge metal framing fluid-applied waterproofing shall be provided, engineered, and installed by a single exterior wall contractor responsible for structural integrity, delegated wind load design, waterproofing and thermal performance. The exterior wall contractor will provide a single warranty for the entire perimeter façade assembly.

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE

a. Air-Barrier Assembly Air Leakage: Maximum (0.04 cfm/square feet) of surface area at (1.57 psf) differential pressure when tested according to ASTM E2357.

b. Full Building Air Leakage: Refer to Section 01 45 29 TESTING LABORATORY SERVICES.

c. Provide full system of compatible materials under conditions of service and application required. Compatibility based on testing by material manufacturer.

d. Perform as continuous vapor retarding air barrier and moisture drainage plane.

e. Transition to adjacent flashings and discharge water to building exterior.

f. Accommodate substrate movement and seal expansion and control joints, construction material transitions, opening transitions, penetrations, and perimeter conditions without moisture deterioration and air leakage exceeding performance requirements.

2.2 PRODUCTS - GENERAL

a. Provide air barrier system components from one manufacturer.

2.3 AIR BARRIER

Fluid-Applied, Vapor-Retarding Membrane Air Barrier:

(1) Elastomeric, modified bituminous or synthetic polymer membrane.

(2) Air Permeance: ASTM E2178: (0.004 cfm/square feet) of surface area at (1.57 psf) differential pressure.

(3) Vapor Permeance: ASTM E96/E96M: Maximum (0.1 perms).

(4) Elongation: Ultimate, ASTM D412, Die C: 500 percent, minimum.

(5) Thickness: (40 mils) dry film thickness, applied in single continuous coat.

(6) Surface Burning Characteristics: When tested according to ASTM E84.

(a) Flame Spread Rating: 25 maximum.

(b) Smoke Developed Rating: 450 maximum.

2.4 ACCESSORIES

a. Primer: Waterborne primer complying with VOC requirements, recommended air barrier manufacturer to suit application.

b. Counterflashing Sheet: Modified bituminous, minimum (40 mils) thick, self-adhering composite sheet consisting of minimum (33 mils) of

rubberized asphalt laminated to polyethylene film.

c. Substrate Patching Material: Manufacturer's standard trowel-grade filler material.

d. Sprayed Polyurethane Foam Sealant: Foamed-in-place, (1.5 to 2.0 pcf) density, with maximum flame-spread index of 25 when tested according to ASTM E84.

e. Flexible Opening Transition: Cured low-modulus silicone extrusion with reinforcing ribs, sized to fit opening widths, designed for adhesion to or insertion into aluminum framing extrusions, and compatible with air barrier system materials and accessories.

f. Joint Sealant: ASTM C920, single-component, neutral-curing silicone; Class 100/50 (low modulus), Grade NS, Use NT related to exposure, approved by membrane air barrier manufacturer for adhesion and compatibility with membrane air barrier and accessories.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Correct substrate deficiencies:

(1) Remove projections and excess materials and fill voids with substrate patching material.

(2) Remove contaminants capable of affecting subsequently installed product's performance.

d. Prepare and treat substrate joints and cracks according to ASTM C1193 and membrane air barrier manufacturer's instructions.

3.2 INSTALLATION - AIR BARRIER

a. Install products according to manufacturer's instructions and approved submittals drawings.

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's consideration.

b. Install air barrier components according to requirements of ABAA Quality Assurance Program ABAA QAP.

c. Apply primer.

d. Install transition strips and accessory materials.

e. Seal air barrier to adjacent components of building air barrier system.

f. Install flexible opening transition at each opening perimeter. Extend transition onto each substrate minimum (3 inches).

(1) Fill gaps at perimeter of openings with foam sealant.

g. At penetrations, seal transition strips around penetrating objects with termination mastic.

(1) Fill gaps at perimeter of penetrations with sprayed polyurethane foam sealant.

h. At top of through-wall flashings, seal with continuous transition strip of manufacturer's recommended material to suit application.

i. Apply air barrier in full contact with substrate to produce continuous seal with transitions.

j. Apply fluid membrane in thickness recommended by manufacturer, and minimum specified thickness.

k. Leave air barrier exposed until tested and inspected and tested by Contracting Officer.

3.3 FIELD QUALITY CONTROL

a. Field Inspections and Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.

(1) Perform inspections and tests before concealing air barrier with subsequent work.

b. Inspections:

(1) Compatibility of materials within air barrier system and adjacent materials.

(2) Suitability of substrate and support for air barrier.

(3) Suitability of conditions under which air barrier is applied.

(4) Adequacy of substrate priming.

(5) Application and treatment of joints and edges of transition strips, flexible opening transitions, and accessory materials.

(6) Continuity and gap-free installation of air barrier, transition strips, and accessory materials.

c. Field Tests:

(1) Qualitative air-leakage testing according to ASTM E1186.

(2) Quantitative air-leakage testing according to ASTM E783.

d. Inspection and Test Frequency: Determined by installed air barrier surface area.

(1) Up to (10,000 square feet): One inspection.

(2) (10,001 - 35,000 square feet): Two inspections.

(3) (35,001 - 75,000 square feet): Three inspections.

(4) (75,001 - 125,000 square feet): Four inspections.

(5) (125,001 - 200,000 square feet): Five inspections.

(6) Over (200,000 square feet): Six inspections.

e. Submit inspection and test reports to Contracting Officer' within seven calendar days of completing inspection and test.

f. Audit:

(1) Provide installer and site inspection audit by ABAA.

(2) Coordinate scheduling of work and associated audit inspections.

(3) Cooperate with ABAA's testing agency. Allow access to work and staging areas.

(4) Notify ABAA in writing of schedule for Work of this Section to allow sufficient time for testing and inspection.

(5) Pay for site inspections by ABAA to verify conformance with the ABAA Quality Assurance Program ABAA QAP.

g. Defective Work: Correct deficiencies, make necessary repairs, and retest as required to demonstrate compliance with specified requirements.

3.4 CLEANING

a. Remove masking materials.

b. Clean spills and overspray using cleaning agents recommended by manufacturers of affected construction.

3.5 PROTECTION

a. Protect air barrier from construction operations.

b. Protect air barrier from exposure to UV light exposure exceeding manufacturer's recommendation.

c. Replace overexposed materials and retest.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 42 13

METAL WALL PANELS

08/25

PART 1 GENERAL

1.1 REFERENCES
1.2 DEFINITIONS
1.3 SUBMITTALS
1.4 QUALITY CONTROL
1.4.1 Pre-Installation Conference
1.4.1.1 Installation Drawings
1.4.1.2 Wind Load Design Analysis
1.4.2 Manufacturer's Technical Representative
1.4.3 Qualification of Manufacturer
1.4.3.1 Manufacturer's Certificates
1.4.4 Certified Qualification of Installation Contractor
1.4.5 Single Source
1.4.6 Manufacturer's Maintenance Instructions
1.5 DELIVERY, HANDLING, AND STORAGE
1.6 PROJECT CONDITIONS
1.6.1 Field Measurements
1.6.2 Weather Limitations
1.7 WARRANTY
1.7.1 20 Year Warranty for Labor and Material

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTIONS
2.1.1 Metal Wall Panel General Performance
2.1.2 Structural Performance
2.2 FABRICATION
2.2.1 Sheet Metal Accessories
2.3 MATERIALS
2.3.1 Aluminum Sheet Panels
2.3.2 Factory Color Finish
2.3.2.1 Metal Preparation
2.3.2.2 Prime Coating
2.3.2.3 Exterior Finish Coating
2.3.2.4 Interior Finish Coating
2.3.2.5 Color Chart
2.3.2.6 Physical Properties
2.3.3 Miscellaneous Metal Framing
2.3.3.1 Fasteners for Miscellaneous Metal Framing
2.3.4 Fasteners
2.3.4.1 Exposed Fasteners
2.3.4.2 Hidden Fasteners
2.3.4.3 Screws
2.3.4.4 Rivets
2.3.4.5 Attachment Clips

2.3.5 Accessories
2.3.5.1 Rubber Closure Strips
2.3.5.2 Metal Closure Strips
2.3.5.3 Sealants and Caulking
2.3.5.4 Shop-Applied Caulking
2.3.5.5 Field-Applied Caulking
2.3.5.6 Pressure Sensitive Tape
2.3.6 Sheet Metal and Extruded Flashing and Trim
2.3.7 Repair of Finish Protection

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 INSTALLATION
3.3.1 Wall Panel Installation
3.3.1.1 Aluminum Wall Panels
3.3.1.2 Anchor Clips
3.3.1.3 Metal Protection
3.3.1.4 Joint Sealers
3.3.2 Fastener Installation
3.3.3 Flashing, Trim, and Closure Installation
3.3.3.1 General Requirements
3.3.3.2 Metal Flashing
3.3.3.3 Closures
3.4 APPLICATION
3.4.1 Workmanship
3.4.2 Erection Tolerances
3.5 FIELD QUALITY CONTROL
3.5.1 Leakage Tests
3.5.2 Repairs to Finish
3.5.3 Paint-Finish Metal Siding
3.5.4 Construction Monitoring
3.5.4.1 Manufacturer's Inspection
3.5.4.1.1 Frequency
3.5.4.1.2 Field Inspection Report
3.6 CLEAN-UP AND DISPOSAL

-- End of Section Table of Contents --

SECTION 07 42 13

METAL WALL PANELS
08/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA ADM	(2020) Aluminum Design Manual
AA ASD1	(2024) Aluminum Standards and Data

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 800	(2016) Voluntary Specifications and Test Methods for Sealants

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

ANSI/AISC 341	(2022) Seismic Provisions for Structural Steel Buildings

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100	(2012) North American Specification for the Design of Cold-Formed Steel Structural Members

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22	(2022; Supp 1 2023; Supp 2 2023) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A463/A463M	(2025a) Standard Specification for Steel Sheet, Aluminum-Coated, by the Hot-Dip Process
ASTM A606/A606M	(2023) Standard Specification for Steel Sheet and Strip, High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, with Improved

	Atmospheric Corrosion Resistance
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A755/A755M	(2018; R 2024) Standard Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A924/A924M	(2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM D522/D522M	(2017; R 2021) Mandrel Bend Test of Attached Organic Coatings
ASTM D523	(2014; R 2018) Standard Test Method for Specular Gloss
ASTM D610	(2025) Standard Practice for Evaluating Degree of Rusting on Painted Steel Surfaces
ASTM D714	(2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints
ASTM D822	(2013; R 2018) Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings
ASTM D968	(2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive
ASTM D1056	(2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber

ASTM D1308	(2020) Standard Test Method for Effect of Household Chemicals on Clear and Pigmented Coating Systems
ASTM D1654	(2008; R 2016; E 2017) Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments
ASTM D1667	(2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)
ASTM D2244	(2025) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates
ASTM D2247	(2025) Standard Practice for Testing Water Resistance of Coatings in 100 Percent Relative Humidity
ASTM D2794	(1993; R 2024) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM D3359	(2017) Standard Test Methods for Rating Adhesion by Tape Test
ASTM D3363	(2022) Standard Test Method for Film Hardness by Pencil Test
ASTM D4214	(2023) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films
ASTM D4587	(2011; R 2019; E 2019) Standard Practice for Fluorescent UV-Condensation Exposures of Paint and Related Coatings
ASTM D5894	(2016) Standard Practice for Cyclic Salt Fog/UV Exposure of Painted Metal, (Alternating Exposures in a Fog/Dry Cabinet and a UV/Condensation Cabinet)
ASTM E72	(2022) Conducting Strength Tests of Panels for Building Construction
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E1592	(2025) Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference
ASTM G152	(2013; R 2021) Standard Practice for

	Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
ASTM G153	(2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

MBMA MBSM	(2018) Metal Building Systems Manual

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793	(2012) Architectural Sheet Metal Manual, 7th Edition

1.2 DEFINITIONS

Metal Wall Panel: Metal wall panels, attachment system components and accessories necessary for a complete weather-tight wall system.

Wall Panel Assembly: Assembly consisting of wall panels, support and attachment system, closures, weather barriers, and substrate that together comply with the assembly testing and performance requirements of this section.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualification of Manufacturer; G

Qualification of Installation Contractor; G

Sample Warranty; G

SD-02 Shop Drawings

Installation Drawings ; G

SD-03 Product Data

Recycled Content; S

Aluminum Sheet Panels; G

Factory Color Finish

Closure Materials

Pressure Sensitive Tape

Sealants and Caulking

Galvanizing Repair Paint

Enamel Repair Paint

Accessories

SD-04 Samples

Aluminum Sheet Panels; G

Fasteners; G

Metal Closure Strips; G

Color Chart and chips; CONTRACTOR TO PROVIDE GOVERNMENT WITH SAMPLES OF ALL SPECIFIED FINISH MATERIALS. GOVERMENT SHALL APPROVE ALL FINISHES PRIOR TO INSTALLATION.G

SD-05 Design Data

Wind Load Design Analysis; G

SD-06 Test Reports

Leakage Tests; G

Wind Load Tests; G

Coating Tests; G

Chalking Tests; G

Seismic Tests; G

SD-07 Certificates

Coil Stock; G

Fasteners; G

Galvanizing Repair Paint; G

Enamel Repair Paint; G

SD-08 Manufacturer's Instructions

Installation of Wall panels; G

SD-09 Manufacturer's Field Reports

Manufacturer's Field Inspection Reports; G

SD-11 Closeout Submittals

Warranty; G

Maintenance Instructions; G

20 Year "No Dollar Limit" Warranty for Labor and Material

Environmental Product Declaration (EPD); S

Third-Party Certification Certificate, Assessment, Or Validation And Compliance Report; S

1.4 QUALITY CONTROL

1.4.1 Pre-Installation Conference

Upon notification of submittal receipt and approval by the Contracting Officer; and prior to the commencement of the work, attend a pre-installation conference to review the following:

a. Drawings and Specifications.

b. Qualification of Installer, Qualification of Welders.

c. Sustainable acquisition

d. Approved Warranty

e. Sample wall panels, 12 inches long by actual panel width

f. Sample metal closure strips, 10 inches long of each type

g. Color charts and chips

h. Coatings and base metal tests, chalking tests

i. Construction schedule, availability of materials, Installer's personnel, equipment, and facilities required to progress with the work without delay.

j. Methods and procedures related to installation of wall panels, including manufacturer's written instructions. Explicitly identify in writing, differences between manufacturer's instructions and the specified requirements.

k. Support conditions for compliance with requirements, including alignment between and attachment to structural members.

l. Flashing, special siding details, wall penetrations, openings, and condition of other construction that affect metal wall panels.

m. Governing regulations and requirements for insurance, certificates, and tests and inspections if applicable.

n. Temporary protection requirements for metal wall panel assembly during and after installation.

o. Wall panel observation and repair procedures after metal wall panel installation. Provide detailed written instructions including copies

of Safety Data Sheets for maintenance and repair materials, and manufacturer's maintenance instructions.

p. Wall panel removal and reinstallation procedures for removable panel areas indicated on the Drawings. Provide detailed written instructions.

1.4.1.1 Installation Drawings

Provide installation shop drawings for wall panels, flashing, accessories, and anchorage systems showing completely dimensioned structural frame and erection layouts, openings in the wall, special framing details, and construction details at corners, building intersections and flashing, location and type of mastic and metal filler strips.

1.4.1.2 Wind Load Design Analysis

Include in the wind design analysis a wall plan delineating dimensions and attachment patterns for each zone. Wind design analysis is required to be prepared and sealed by Registered Professional Engineer in the geographic area where the construction takes place. As applicable, submit the following wind load design analysis data, to include, but not limited to:

wind speed

exposure category, co-efficient, importance factor

type of facility

negative pressures for each zone

methods and requirements of attachment

1.4.2 Manufacturer's Technical Representative

The representative is required to receive authorization from manufacturer to approve field changes and is to be thoroughly familiar with the products and installations in the geographical area where construction takes place.

1.4.3 Qualification of Manufacturer

Certify that metal wall panel system manufacturer has a minimum of five years experience in manufacturing metal wall system and accessory products.

Manufacturer is required to provide engineering services by an authorized Registered Professional Engineer; currently licensed in the State where construction takes place, having a minimum of 4 years experience as an engineer knowledgeable in wind load design analysis, protocols, and procedures per MBMA MBSM, "Metal Building Systems Manual"; ASCE 7-22, and ASTM E1592.

Provide certified engineering calculations, using the products submitted, for Wind load requirements in accordance with ASCE 7-22.

1.4.3.1 Manufacturer's Certificates

Also provide the following certifications from the manufacturer:

Coil Stock

Fasteners

Galvanizing Repair Paint

Enamel Repair Paint

Submit certification from coil stock manufacturer or supplier that the machinery used forms the provided coil stock without warping, waviness, or rippling that is not a part of the panel profile, and without damage, abrasion or marring of the finish coating.

Provide evidence that products used within this specification are manufactured in the United States.

1.4.4 Certified Qualification of Installation Contractor

The installation contractor is to be approved and certified by the metal wall panel manufacturer prior to beginning the installation of the metal wall panel system. Subcontracting by Certified Contractor for the metal wall panel work is not permitted.

1.4.5 Single Source

Obtain each type of metal wall panels, clips, closure materials, and other accessories from the standard products of the single source from a single manufacturer to operate as a complete system for the intended use.

1.4.6 Manufacturer's Maintenance Instructions

Provide manufacturer's detailed written instructions including copies of Safety Data Sheets for maintenance and repair materials.

1.5 DELIVERY, HANDLING, AND STORAGE

Deliver and protect package components, sheets, metal wall panels, and other manufactured items to prevent damage or deformation during transportation and handling. Unload, store, and erect metal wall panels in a manner to prevent bending, warping, twisting, and surface damage. Always lift and carry panels by the long edge to prevent twisting and damage to the panels.

Stack and store metal wall panels horizontally on platforms or pallets, covered with suitable weather-tight and ventilated covering to ensure dryness, with positive slope for drainage of water. Do not store metal wall panels in contact with other materials that might cause staining, denting, or other surface damage. Ensure panels are stored in a dry environment, and minimize the amount of handling to minimize opportunities for damaging panels. Retain strippable protective covering on metal wall panel until actual installation.

1.6 PROJECT CONDITIONS

1.6.1 Field Measurements

Verify locations of wall framing and opening dimensions by field measurements before metal wall panel fabrication and indicate measurements on Shop Drawings.

1.6.2 Weather Limitations

Proceed with installation preparation only when existing and forecasted weather conditions permit Work to proceed without water entering into wall system or building.

1.7 WARRANTY

Final Warranty provided is required to conform to the Sample Warranty as reviewed and approved by the Contracting Officer.

a. System Responsibility: the exterior walls consisting of curtainwall, insulated metal wall panels, aluminum fins, light gauge metal framing fluid-applied waterproofing shall be provided, engineered, and installed by a single exterior wall contractor responsible for structural integrity, delegated wind load design, waterproofing and thermal performance. The exterior wall contractor will provide a single warranty for the entire perimeter façade assembly.

1.7.1 20 Year Warranty for Labor and Material

Furnish manufacturer's warranty for the metal wall panel system. The warranty period is to be no less than 20 years from the date of Government acceptance of the work. Write warranty directly to the Government, commencing at time of Government's acceptance of the wall panel work. The warranty is to provide that if within the warranty period the metal wall panel system shows evidence of corrosion, perforation, rupture, or excess weathering due to deterioration of the wall panel system resulting from defective materials, then correction of the defective workmanship be the responsibility of the metal wall panel system manufacturer. Repairs that become necessary because of defective materials and workmanship while metal wall panel system is under warranty are to be performed within 72 hours after notification, unless additional time is approved by the Contracting Officer.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTIONS

Metal Plate Cladding Panels: Single-skin, concealed fastener formed metal wall panels used as exterior wall cladding and soffit panels, designated on the Drawings, including panel support and attachment system, flashing, trim, closures, and accessories required for a complete weathertight installation.

Single Skin Metal Cladding Panels: Single-skin, formed metal wall panels used as exterior cladding, including support and attachment system, flashing, trim, closures and accessories required for a complete visual screen without thermal or weather barriers. Provide 18" wide perforated flat profile panels as shown on the drawings, with a limited amount of 12" wide panels. Perforation densities and patterns are indicated on the drawings. Strict adherence to the panel layout (width, length, perforation density)shown on the drawings is required.

2.1.1 Metal Wall Panel General Performance

Comply with performance requirements, conforming to AISI S100, without failure due to defective manufacture, fabrication, installation, or other

defects in construction. Ensure wall panels and accessory components conform to the following standards:

ASTM A1008/A1008M

ASTM A123/A123M

ASTM A36/A36M

ASTM A653/A653M

ASTM A463/A463M for aluminum coated steel sheet

ASTM A606/A606M

ASTM A755/A755M for metallic coated steel sheet for exterior coil pre-painted applications.

ASTM A780/A780M for repair of damage or uncoated areas of hot-dipped galvanized coating.

ASTM A924/A924M for metallic coated steel sheet

ASTM D522/D522M for applied coatings

2.1.2 Structural Performance

Ensure maximum calculated fiber stress will not exceed the allowable value in the AISI or AA manuals; a one third overstress for wind is allowed. Midspan deflection under maximum design loads is limited to L/180. Contract drawings show the design wind loads and the extent and general assembly details of the metal siding. Provide design for members and connections not shown on the drawings. Provide siding panels and accessories products by the same manufacturer.

Provide metal wall panel assemblies complying with the load and stress requirements in accordance with ASTM E1592. Wind Load force due to wind action governs the design for panels. Wall systems and attachments are required to resist the wind loads as determined by ASTM E72 and ASCE 7-22 in the geographic area where the construction takes place, in pounds per square foot. Submit five copies of wind load tests and seismic tests to the Contracting Officer. Provide metal wall panel assembly for seismic conditions complying with the applicable requirements of ANSI/AISC 341.

2.2 FABRICATION

Unless approved otherwise, fabricate and finish metal wall panels and accessories at the factory, by manufacturer's standard procedures and processes and as necessary to fulfill indicated and specified performance requirements. Comply with indicated profiles and with dimensional and structural requirements. Submit manufacturer's data indicating recycled content percentages.

Provide panel profile, including major ribs and intermediate stiffening ribs, if any, for full length of panel. Fabricate metal wall panel side laps with factory-installed captive gaskets or separator strips that provide a tight seal and prevent metal-to-metal contact, in a manner that seals weather-tight and minimize noise from movements within panel assembly.

2.2.1 Sheet Metal Accessories

Fabricate flashing and trim to comply with recommendations in SMACNA 1793 that apply to the design, dimensions, metal, and other characteristics of item indicated:

a. Form exposed sheet metal accessories that are without excessive oil canning, buckling, and tool marks and that are true to line and levels indicated, with exposed edges folded back to form hems.

b. End Seams: fabricate nonmoving end seams with flat-lock seams. Form seams and seal with epoxy seam sealer. Rivet joints for additional strength.

c. Sealed Joints: form non-expansion but movable joints in metal to accommodate elastomeric sealant to comply with SMACNA 1793.

d. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on faces of accessories exposed to view.

e. Fabricate cleats and attachment devices of size and metal thickness recommended by SMACNA 1793 or by metal wall panel manufacturer for application, but not less than thickness of metal being secured.

2.3 MATERIALS

2.3.1 Aluminum Sheet Panels

Submit a sample Aluminum Sheet Panel, 12 inches long by actual panel width, of each panel proposed to be used on the project.

Roll-form aluminum wall panels to the specified profile, with fy = 80 ksi, .040 inches thickness and depth as indicated. Provide material plumb and true, and within the tolerances listed:

a. Aluminum Sheet conforming to ASTM B209, AA ASD1 and AA ADM.

b. Individual panels are to have continuous length conforming to the exact layout shown on drawings with no joints or seams and formed without warping, waviness, or ripples that are not part of the panel profile and free of damage to the finish coating system.

c. Provide panels with thermal expansion and contraction consistent with the type of system specified.

2.3.2 Factory Color Finish

Comply with NAAMM AMP 500 for recommendations for applying and designating finishes. Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved samples and are assembled or installed to minimize contrast.

All panels are to receive a factory-applied polyvinylidene fluoride finish consisting of a baked-on top-coat with a manufacturer's recommended prime coat conforming to the following:

2.3.2.1 Metal Preparation

Carefully prepare all metal surface for painting on a continuous process coil coating line by alkali cleaning, hot water rinsing, application of chemical conversion coating, cold water rinsing, sealing with acid rinse, and thorough drying.

2.3.2.2 Prime Coating

Apply a base coat of epoxy paint, specifically formulated to interact with the top-coat, to the prepared surfaces by roll coating to a dry film thickness of 0.20 plus 0.05 mils. Prime coat is to be oven cured prior to application of finish coat.

2.3.2.3 Exterior Finish Coating

Roll coat the finish coating over the primer by roll coating to dry film thickness of 0.80 plus 5 mils for a total dry film thickness of 1.00 plus 0.10 mils. Oven-cure finish coat.

2.3.2.4 Interior Finish Coating

Apply a wash-coat on the reverse side over the primer by roll coating to a dry film thickness of 0.30 plus 0.05 mils for a total dry film thickness of 0.50 plus 0.10 mils. Oven-cured the wash coat.

2.3.2.5 Color Chart

Provide exterior finish color as specified. Submit manufacturer's color charts and chips, approximately 4 by 4 inches, showing full range of colors, textures, and patterns available for wall panels with factory applied finishes.

2.3.2.6 Physical Properties

Coating is to conform to the industry and manufacturer's standard performance criteria as listed by the following certified test reports:

General:	ASTM D5894 and ASTM D4587
Abrasion:	ASTM D968
Adhesion:	ASTM D3359
Chalking:	ASTM D4214
Chemical Pollution:	ASTM D1308
Color Change and Conformity:	ASTM D2244
Creepage:	ASTM D1654
Cyclic Corrosion Test:	ASTM D5894
Flame Spread:	ASTM E84
Flexibility:	ASTM D522/D522M

Formability:	ASTM D522/D522M
Gloss at 60 and 85 degrees:	ASTM D523
Humidity:	ASTM D2247 and ASTM D714
Oxidation:	ASTM D610
Pencil Hardness:	ASTM D3363
Reverse Impact:	ASTM D2794
Salt Spray:	ASTM B117
Weatherometer:	ASTM G152, ASTM G153 and ASTM D822

2.3.3 Miscellaneous Metal Framing

Cold-formed metallic-coated steel sheet conforming to ASTM A653/A653M unless otherwise indicated.

2.3.3.1 Fasteners for Miscellaneous Metal Framing

Type, material, corrosion resistance, size, and sufficient length to penetrate the supporting member a minimum of 1 inch with other properties required to fasten miscellaneous metal framing members to supporting members and substrates in accordance with the wall panel manufacturer's and ASCE 7-22 requirements.

2.3.4 Fasteners

2.3.4.1 Exposed Fasteners

Provide corrosion resistant fasteners for wall panels, made of multi coated (zinc plus anti-corrosion coating), compatible with the sheet panel and flashing and of a type and size recommended by the manufacturer to meet the performance requirements and design loads.

Fasteners for accessories are to be the manufacturer's standard. Provide an integral metal washer matching the color of attached material with compressible sealing EPDM gasket approximately 3/32 inch thick.

2.3.4.2 Hidden Fasteners

Provide corrosion resistant fasteners recommended by the manufacturer to meet the performance requirements and design loads.

2.3.4.3 Screws

Screws to be corrosion resistant multi coated (zinc plus anti-corrosion coating), being the type and size recommended by the manufacturer to meet the performance requirements.

2.3.4.4 Rivets

Rivets to be closed-end type, corrosion resistant coated zinc-coated steel, aluminum, or Series 304 stainless steel where watertight connections are required.

2.3.4.5 Attachment Clips

Fabricate clips from steel hot-dipped galvanized in accordance with ASTM A653/A653M, Z275 G 90, or Series 300 stainless steel. Size, shape, thickness, and capacity as required meeting the insulation thickness and design load criteria specified.

2.3.5 Accessories

All accessories to be compatible with the metal wall panels. Sheet metal flashing, trim, metal closure strips, caps, and similar metal accessories to not be less than the minimum thickness specified for the wall panels. Exposed metal accessories/finishes to match the panels furnished, except as otherwise indicated. Provide molded foam rib, ridge, and other closure strips of non-absorbent closed-cell, solid-cell synthetic rubber, or pre-molded neoprene to match configuration of the panels.

2.3.5.1 Rubber Closure Strips

Provide closed-cell, expanded cellular rubber conforming to ASTM D1056 and ASTM D1667; extruded or molded to the configuration of the specified wall panel and in lengths supplied by the wall panel manufacturer.

2.3.5.2 Metal Closure Strips

Submit sample metal closure strips, 10 inches long of each type proposed to be used on the project.

Provide factory fabricated aluminum closure strips to be the same thickness, color, finish, and profile of the specified wall panel.

2.3.5.3 Sealants and Caulking

Provide approved gun type sealants for use in hand- or air-pressure caulking guns at temperatures above 40 degrees F or frost-free application at temperatures above 10 degrees F with minimum solid content of 85 percent of the total volume. Sealants are required to dry with a tough, durable surface skin which permit remaining soft and pliable underneath, providing a weather-tight joint. No migratory staining is permitted on painted or unpainted metal, stone, glass, vinyl, or wood.

Prime all joints receiving sealants with a compatible one-component or two-component primer as recommended by the wall panel manufacturer.

2.3.5.4 Shop-Applied Caulking

Sealant for shop-applied caulking is to be non-curing butyl compliant with AAMA 800 to ensure the sealant's plasticity at the time of field erection.

2.3.5.5 Field-Applied Caulking

Sealant for field-applied caulking is to be an approved gun grade, non-sag sealant with an initial maximum Shore A durometer hardness of 25, and conforming to ASTM C920, Type II. Color to match panel colors.

2.3.5.6 Pressure Sensitive Tape

Provide pressure sensitive tape sealant, 100 percent solid with a release paper backing; permanently elastic, non-sagging, non-toxic and non-staining as approved by the wall panel manufacturer.

2.3.6 Sheet Metal and Extruded Flashing and Trim

Shop fabricate sheet metal and extruded flashing and trim where practicable to comply with recommendations in SMACNA 1793 that apply to design, dimensions, metal, and other characteristics of item indicated. Obtain field measurements for accurate fit before shop fabrication.

Fabricate sheet metal flashing and trim without excessive oil canning, buckling, and tool marks and true to line and levels indicated, with exposed edges folded back to form hems.

2.3.7 Repair of Finish Protection

Repair paint for color finish enameled wall panel is required to be compatible paint of the same formula and color as the specified finish furnished by the wall panel manufacturer. Provide quarts of repair paint matching the specified wall panels.

PART 3 EXECUTION

3.1 EXAMINATION

Examine the following:

a. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal wall panel supports, and other conditions affecting performance of the Work.

b. Examine primary and secondary wall framing to verify that rafters, purlins, angles, channels, and other structural panel support members and anchorages have been installed within alignment tolerances required by metal wall panel manufacturer, UL, ASTM, ASCE 7-22 and as required for the geographical area where construction takes place.

c. Examine solid wall sheathing to verify that sheathing joints are supported by framing or blocking and that installation is within flatness tolerances required by metal wall panel manufacturer.

d. Examine roughing-in for components and systems penetrating metal wall panels to verify actual locations of penetrations relative to seam locations of metal wall panels before metal wall panel installation.

Submit to the Contracting Officer a written report, endorsed by Installer, listing conditions detrimental to performance of the Work. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

Clean substrates of substances harmful to insulation, including removing projections capable of interfering with insulation attachment. Install miscellaneous framing installation, including sub-purlins, girts, angles, furring, and other miscellaneous wall panel support members and anchorage

are to be according to metal wall panel manufacturer's written instructions.

3.3 INSTALLATION

3.3.1 Wall Panel Installation

Provide full length metal wall panels, from sill to eave or joint to joint as indicated, unless otherwise indicated or restricted by shipping limitations. Anchor metal wall panels and other components of the Work securely in place, with provisions for thermal and structural movement in accordance with MBMA MBSM. Work is to allow for thermal movement of the wall panel, movement of the building structure, and to provide permanent freedom from noise due to wind pressure. Erect sheets true and plumb and in exact alignment with the horizontal and vertical edges of the building, securely anchored, and with the indicated eave, and sill. Ensure substrate is plumb and level prior to installation.

Erect wall panel system in accordance with the approved erection drawings, the printed instructions, and safety precautions of the manufacturer. Sheets are not to be subjected to overloading, abuse, or undue impact. Do not apply bent, chipped, or defective sheets. Field cutting metal wall panels by torch is not permitted.

3.3.1.1 Aluminum Wall Panels

Use aluminum or stainless-steel fasteners for exterior surfaces and aluminum or galvanized steel fasteners for interior surfaces.

3.3.1.2 Anchor Clips

Anchor metal wall panels and other components of the Work securely in place, using manufacturer's approved fasteners according to manufacturers' written instructions.

3.3.1.3 Metal Protection

Where dissimilar metals contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with bituminous coating, by applying rubberized-asphalt underlayment to each contact surface, or by other permanent separation as recommended by metal wall panel manufacturer.

3.3.1.4 Joint Sealers

Install gaskets, joint fillers, and sealants where indicated and where required for weatherproof performance of metal wall panel assemblies. Provide types of gaskets, fillers, and sealants indicated or, if not indicated, types recommended by metal wall panel manufacturer.

3.3.2 Fastener Installation

Anchor metal wall panels and other components of the Work securely in place, using manufacturer's approved fasteners according to manufacturers' written instructions. Never overdrive fasteners and ensure fasteners are installed straight and not at an angle.

3.3.3 Flashing, Trim, and Closure Installation

3.3.3.1 General Requirements

Comply with performance requirements, manufacturer's written installation instructions, and SMACNA 1793. Provide concealed fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints, and seams to form permanently watertight and weather resistant.

Install sheet and extruded metal work as to form weather-tight construction without waves, warps, buckles, fastening stresses or distortion, and allow for expansion and contraction. Cutting, fitting, drilling, and other operations in connection with sheet metal required to accommodate the work of other trades is to be performed by sheet and extruded metal mechanics.

3.3.3.2 Metal Flashing

Install exposed metal flashing at building corners, sills and eaves, junctions between metal siding and walling. Exposed metal flashing to be the same material, color, and finish as the specified metal wall panel. Fasten flashing at a minimum of 8 inches on center, except where flashing is held in place by the same screws that secure covering sheets. Flashing is to be furnished in at least 8 foot lengths. Exposed flashing is to have 1 inch locked and blind-soldered end joints, and expansion joints at intervals of not more than 16 feet.

Exposed flashing and flashing subject to rain penetration to be bedded in the specified joint sealant. Isolate flashing which is in contact with dissimilar metals by means of the specified asphalt mastic material to prevent electrolytic deterioration. Form drips to the profile indicated, with the edge folded back 1/2 inch to form a reinforced drip edge.

3.3.3.3 Closures

Install metal closure strips at open ends of corrugated or ribbed pattern walls, and at intersection of wall and wall unless open ends are concealed with formed eave flashing; and in other required areas.

Install mastic closure strips at intersection of the wall with metal walling; top and bottom of metal siding; heads of wall openings; and in other required locations.

3.4 APPLICATION

3.4.1 Workmanship

Make lines, arises, and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections which might affect the application. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793. Provide sheet metal flashing in the angles formed where roof decks abut

walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight.

3.4.2 Erection Tolerances

Erect metal wall panels straight and true with plumb vertical lines correctly lapped and secured in accordance with the manufacturer's written instructions.

3.5 FIELD QUALITY CONTROL

3.5.1 Leakage Tests

Finished application of metal wall panels are to be subject to inspection and test for leakage by request of the Contracting Officer, Architect/Engineer. Conduct inspection and tests at no cost to the Government.

Inspection and testing are to be made promptly after erection to permit correction of defects and the removal and replacement of defective materials.

3.5.2 Repairs to Finish

Scratches, abrasions, and minor surface defects of finish may be repaired with the specified repair materials. Finished repaired surfaces are to be uniform and free from variations of color and surface texture.

Repaired metal surfaces that are not acceptable to the project requirements or Contracting Officer are to be immediately removed and replaced with new material.

3.5.3 Paint-Finish Metal Siding

Paint-finish metal siding will be tested for color stability by the Government during the manufacturer's specified guarantee period.

Remove and replace panels that indicate color changes, fading, or surface degradation, determined by visual examination, with new panels at no expense to the Government. Subject new panels to the specified tests for an additional year from the date of their installation.

3.5.4 Construction Monitoring

Make visual inspections as necessary to ensure compliance with specified requirements. Additionally, verify the following:

a. Materials comply with the specified requirements.

b. All materials are properly stored, handled, and protected from damage. Damaged materials are removed from the site.

c. Framing and substrates are in acceptable condition, in compliance with specification, prior to application of wall panels.

d. Panels are installed without buckles, ripples, or waves and in uniform alignment and modulus.

e. Side laps are formed, sealed, fastened, or seam locked as required.

f. The proper number, type, and spacing of attachment clips and fasteners are installed.

g. Installer adheres to specified and detailed application parameters.

h. Associated flashing and sheet metal are installed in a timely manner in accord with the specified requirements.

3.5.4.1 Manufacturer's Inspection

The metal wall panel manufacturer's technical representative must visit the work site to inspect ongoing work. Inspections are to include observing installation technique and verifying the quality of work-in-place for compliance with the manufacturer's instructions. Deficiencies identified by the manufacturer's technical representative must be corrected and re-inspected by the manufacturer's technical representative.

3.5.4.1.1 Frequency

The manufacturer's technical representative must visit the work site to inspect and document ongoing work a minimum of three separate occasions during the course of the installation. One visit must occur during the first 20 squares of installation, one at substantial completion of the metal panel work, and all others during different periods of installation. Notify the Contracting officer a minimum of five working days prior to each visit by the manufacturer's technical representative.

3.5.4.1.2 Field Inspection Report

Document inspection results in a report prepared and signed by the manufacturer's technical representative for each visit. Submit the report to the Contracting Officer with the contractor's daily Quality Control report. The manufacturer's field inspection report must include a description of ongoing work observed and whether the inspected work was satisfactory or unsatisfactory. The final report must include certification by the manufacturer's technical representative that the work was performed in accordance with the manufacturer's instructions and contains no deficiencies. Submit the final manufacturer's field inspection reports to the Contracting Officer within five working days of the final visit.

3.6 CLEAN-UP AND DISPOSAL

Clean all exposed sheet metal work at completion of installation. Remove metal shavings, filings, nails, bolts, and wires from work area. Remove grease and oil films, excess sealants, handling marks, contamination from steel wool, fittings and drilling debris and scrub the work clean. Exposed metal surfaces are to be free of dents, creases, waves, scratch marks, solder or weld marks, and damage to the finish coating.

Collect and place scrap/waste materials in containers. Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site; transport demolished materials from government property and legally dispose of them.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 42 63

FABRICATED WALL PANEL ASSEMBLIES

08/25

PART 1 GENERAL

1.1 REFERENCES
1.2 DEFINITIONS
1.3 DESCRIPTION OF FABRICATED WALL PANEL ASSEMBLY SYSTEM
1.3.1 Metal Wall Panel General Performance
1.3.2 Structural Performance
1.3.3 Air Infiltration
1.3.4 Water Penetration Under Static Pressure
1.3.5 Water Penetration Under Dynamic Pressure
1.4 SUBMITTALS
1.5 QUALITY ASSURANCE
1.5.1 Pre-Installation Conference
1.5.2 Manufacturer's Technical Representative
1.5.3 Qualification of Manufacturer
1.5.4 Qualification of Installer
1.5.4.1 Qualifications for Welding Work
1.5.5 Single Source
1.5.6 Surface-Burning Characteristics
1.5.7 Fire-Resistance Ratings
1.5.8 Fabrication
1.5.8.1 Sheet Metal Accessories
1.5.9 Finishes
1.6 DELIVERY, HANDLING, AND STORAGE
1.6.1 Metal Wall Panels
1.6.2 Foam-Plastic Insulation
1.6.3 Plastic Materials
1.7 PROJECT CONDITIONS
1.8 WARRANTY

PART 2 PRODUCTS

2.1 PANEL MATERIALS
2.1.1 Steel Sheet
2.1.2 Foam-Insulation Core Wall Panel
2.1.3 Finish
2.2 ALUMINUM FINS
2.3 MISCELLANEOUS METAL FRAMING
2.3.1 General
2.4 FASTENERS
2.4.1 General
2.4.2 Exposed Fasteners
2.4.3 Screws
2.4.4 Rivets
2.4.5 Attachment Clips
2.5 ACCESSORIES

2.5.1 General
2.5.2 Closure Materials
2.5.2.1 Rubber Closure Strips
2.5.2.2 Metal Closure Strips
2.5.3 Joint Sealants
2.5.3.1 Sealants and Caulking
2.5.3.2 Shop-Applied
2.5.3.3 Field-Applied
2.5.3.4 Tape Sealant
2.6 SHEET METAL FLASHING AND TRIM
2.6.1 Fabrication
2.7 REPAIR OF FINISH PROTECTION

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 WALL PANEL INSTALLATION
3.4 FASTENER INSTALLATION
3.5 FLASHING, TRIM AND CLOSURE INSTALLATION
3.5.1 General Requirements
3.5.2 Metal Flashing (sheet metal and extruded aluminum as shown in the architectural details)
3.5.3 Closures
3.5.4 Aluminum Fin Installation
3.6 WORKMANSHIP
3.7 ACCEPTANCE PROVISIONS
3.7.1 Erection Tolerances
3.7.2 Leakage Tests
3.7.3 Repairs to Finish
3.7.4 Paint-Finish Metal Siding
3.8 MANUFACTURER'S INSPECTION
3.8.1 Frequency
3.8.2 Field Inspection Report
3.9 CLEAN-UP AND DISPOSAL

-- End of Section Table of Contents --

SECTION 07 42 63

FABRICATED WALL PANEL ASSEMBLIES
08/25

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 501.1	(2017) Standard Test Method for Water Penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI D100	(2017) Cold-Formed Steel Design Manual
AISI S100	(2016) North American Specification for the Design of Cold-Formed Steel Structural Members

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22	(2022; Supp 1 2023; Supp 2 2023; Supp 3 2025) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN WELDING SOCIETY (AWS)

AWS A5.1/A5.1M	(2025) Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding
AWS D1.1/D1.1M	(2025) Structural Welding Code - Steel
AWS D1.2/D1.2M	(2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A463/A463M	(2025a) Standard Specification for Steel Sheet, Aluminum-Coated, by the Hot-Dip Process
ASTM A606/A606M	(2023) Standard Specification for Steel

	Sheet and Strip, High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, with Improved Atmospheric Corrosion Resistance
ASTM A653/A653M	(2025) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A755/A755M	(2018; R 2024) Standard Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A792/A792M	(2025) Standard Specification for Steel Sheet, 55% Aluminum-Zinc Alloy-Coated by the Hot-Dip Process
ASTM A924/A924M	(2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM C273/C273M	(2020) Standard Test Method for Shear Properties of Sandwich Core Materials
ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM D522/D522M	(2017; R 2021) Mandrel Bend Test of Attached Organic Coatings
ASTM D523	(2014; R 2018) Standard Test Method for Specular Gloss
ASTM D714	(2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints
ASTM D822/D822M	(2013; R 2018) Standard Practice for Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings
ASTM D968	(2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive

ASTM D1056	(2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM D1308	(2020) Standard Test Method for Effect of Household Chemicals on Clear and Pigmented Coating Systems
ASTM D1621	(2016; R2023) Standard Test Method for Compressive Properties of Rigid Cellular Plastics
ASTM D1622	(2020) Standard Test Method for Apparent Density of Rigid Cellular Plastics
ASTM D1667	(2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)
ASTM D2244	(2025) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates
ASTM D2247	(2025) Standard Practice for Testing Water Resistance of Coatings in 100 Percent Relative Humidity
ASTM D2794	(1993; R 2024) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM D3363	(2022) Standard Test Method for Film Hardness by Pencil Test
ASTM D4214	(2023) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films
ASTM D6226	(2021) Standard Test Method for Open Cell Content of Rigid Cellular Plastics
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E136	(2024c) Standard Test Method for Assessing Combustibility of Materials Using a Vertical Tube Furnace at 750 Degrees C
ASTM E283/E283M	(2019) Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen

ASTM E331	(2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
ASTM E1592	(2025) Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference
ASTM G152	(2013; R 2021) Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
ASTM G153	(2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

MBMA MBSM	(2018) Metal Building Systems Manual

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793	(2012) Architectural Sheet Metal Manual, 7th Edition

UL SOLUTIONS (UL)

UL Bld Mat Dir	(updated continuously online) Building Materials Directory

1.2 DEFINITIONS

Fabricated Wall Panel Assembly: Metal wall and liner panels, attachment system components, miscellaneous metal framing, thermal insulation, and accessories shop fabricated for a complete weather-tight wall system.

Insulated Metal Wall Panel (IMP): A factory-fabricated panel consisting of two steel facings permanently bonded to a rigid foam-plastic insulating core, forming a single-component wall panel that provides the finish, air barrier, water barrier, and thermal insulation of the exterior wall in one assembly.

1.3 DESCRIPTION OF FABRICATED WALL PANEL ASSEMBLY SYSTEM

Factory fabricated, foam-insulation-core metal wall panel assembles (insulated metal panels) for the exterior wall types indicated on the drawings, including insulated metal wall panels with factory applied fluoropolymer finish, concealed panel attachment clips, fasteners, and closures, and panel joint sealants, gaskets, and preformed flashing and

trim and aluminum fins integrated with the exterior wall assembly panel system.

1.3.1 Metal Wall Panel General Performance

Comply with performance requirements, conforming to AISI S100, without failure due to defective manufacture, fabrication, installation, or other defects in construction. Provide wall panels and accessory components conforming to the following standards:

ASTM A1008/A1008M

ASTM A123/A123M

ASTM A36/A36M

ASTM A653/A653M

ASTM A463/A463M for aluminum coated steel sheet.

ASTM A606/A606M

ASTM A755/A755M for metallic coated steel sheet for exterior coil pre-painted applications.

ASTM A780/A780M for repair of damage or uncoated areas of hot-dipped galvanized coating.

ASTM A924/A924M for metallic coated steel sheet.

ASTM C273/C273M

ASTM D522/D522M for applied coatings.

UL Bld Mat Dir

1.3.2 Structural Performance

Maximum calculated fiber stress must not exceed the allowable value in the AISI or AA manuals; a one third overstress for wind is allowed. Midspan deflection under maximum design loads is limited to L/180. Contract drawings show the design wind loads and the extent and general assembly details of the metal siding. Provide design for members and connections not shown on the drawings. Provide siding panels and accessories products by the same manufacturer.

Provide metal wall panel assemblies complying with the load and stress requirements in accordance with ASTM E1592. Wind Load force due to wind action governs the design for panels. Wall systems and attachments are to resist the wind loads as determined by ASCE 7-22 in the geographic area where the construction will take place, in pounds per square foot. Submit electronic copies of wind load tests and seismic tests to the Contracting Officer.

1.3.3 Air Infiltration

Conform to the air leakage limits through the wall assembly area when tested according to ASTM E283/E283M.

1.3.4 Water Penetration Under Static Pressure

No water penetration when tested according to ASTM E331.

1.3.5 Water Penetration Under Dynamic Pressure

No evidence of water leakage when tested according to AAMA 501.1.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualification of Manufacturer

Qualification of Installer

Qualifications for Welding Work

SD-02 Shop Drawings

Fabrication and Installation Drawings

Wall Panel Assemblies

Flashing and Accessories

Anchorage Systems

Removal And Reinstallation Procedures For Removable Panels

SD-03 Product Data

Recycled Content for Insulation; S

Manufacturer's Catalog Data

Closure Materials

Insulation

Tape Sealant

Sealants and Caulking

Rated Wall Assembly

Accessories

SD-04 Samples

Wall Panel Assemblies

Fasteners

Metal Closure Strips

Insulation

Manufacturer's Color Charts and ChipsCONTRACTOR TO PROVIDE GOVERNMENT WITH SAMPLES OF ALL SPECIFIED FINISH MATERIALS. GOVERMENT SHALL APPROVE ALL FINISHES PRIOR TO INSTALLATION.

SD-05 Design Data

Wind Design Analysis

SD-06 Test Reports

Leakage Tests

Wind Load Tests

Seismic Tests

Factory Color Finish Performance Requirements

SD-07 Certificates

Fasteners

Wall System Assembly Wind Load and Fire Rating Classification Listings

SD-08 Manufacturer's Instructions

Installation of Wall Panels

SD-09 Manufacturer's Field Reports

Manufacturer's Field Inspection Reports; G

SD-11 Closeout Submittals

Warranty

Safety Data Sheets

Environmental Product Declaration (EPD); S

Third-Party Certification Certificate, Assessment, Or Validation And Compliance Report; S

1.5 QUALITY ASSURANCE

1.5.1 Pre-Installation Conference

After submittals are received and approved but before wall panel and insulation work, including associated work, is performed, the Contracting Officer will hold a pre-siding conference to review the following:

a. The drawings, including Fabrication and Installation drawings, showing complete Wall Panel Assemblies, and specifications. Drawings are to indicate completely dimensioned structural frame and erection layouts, openings in the wall, special framing details, and construction

details at corners, building intersections and flashing, location and type of mastic and metal filler strips. Include details for the following for review:

(1) Flashing and accessories

(2) Anchorage systems

(3) Manufacturer's catalog data

(4) Factory Color Finish

(5) Manufacturer's color charts and chips

(6) Sub-girts and Formed Shapes

(7) Closure Materials, including metal closure strips

(8) Sample Aluminum Fins, 12-inches long, each type including attachment components

(9) Insulation

(10) Pressure Sensitive Tape

(11) Rated Wall Assembly test data

(12) Accessories

(13) Fasteners

b. Include samples of the following for Government review and approval:

(1) Manufacturer's color charts and chips, approximately 4 by 4 inches, showing full range of colors, textures and patterns available for wall panels with factory applied finishes.

(2) Wall Panel Assemblies, 12 inches long by actual panel width.

(3) Closure Materials, including metal closure strips, provide sample closure strips 10 inches long of each type.

(4) Insulation, provide samples approximately 8 by 11 inches.

c. Finalize construction schedule and verify availability of materials, Installer's personnel, equipment, and facilities needed to make progress and avoid delays.

d. Methods and procedures related to metal wall panel installation, including manufacturer's written instructions for Installation of Wall panels, and verification of wall system assembly wind load and fire rating classification listings. Include detailed application instructions and standard manufacturer drawings altered as required by these specifications. Explicitly identify in writing, differences between manufacturer's instructions and the specified requirements.

e. Support conditions for compliance with requirements, including alignment between and attachment to structural members. Provide details of wind design analysis including wind speed, exposure

category, co-efficient, importance factor, designates type of facility, negative pressures for each zone, methods and requirements of attachment. Include wall plan delineating dimensions and attachment patterns for each zone. Wind design analysis to be prepared and sealed by a Licensed Engineer with demonstrated project design experience in the geographic area where the construction will take place.

f. Flashing, special siding details, wall penetrations, openings, and condition of other construction that will affect metal wall panels.

g. Governing regulations and requirements for insurance, certificates, tests and inspections. Safety plan review must include applicable Safety Data Sheets for maintenance/repair materials.

h. Temporary protection requirements for metal wall panel assembly during and after installation.

i. Wall panel observation and repair procedures after metal wall panel installation. Include review of sample Aluminized Steel Repair Paint.

j. Sample Warranty.

1.5.2 Manufacturer's Technical Representative

The representative must have authorization from manufacturer to approve field changes and be thoroughly familiar with the products and installations in the geographical area where construction will take place.

1.5.3 Qualification of Manufacturer

Metal wall panel system manufacturer must have:

a. A minimum of 5 years experience in manufacturing metal wall system and accessory products.

b. Provide engineering services by an engineer currently licensed with experience on projects in the geographical area where construction will take place, having a minimum of 4 years experience as an engineer knowledgeable in wind load design analysis, protocols, and procedures for the MBMA MBSM; ASCE 7-22, and ASTM E1592.

c. Provide certified engineering calculations using the products submitted for:

(1) Wind load requirements in accordance with FM Wind Design Guide and ASCE 7-22.

1.5.4 Qualification of Installer

The installation Contractor must be approved and certified by the wall panel manufacturer prior to beginning the installation of the metal wall system.

1.5.4.1 Qualifications for Welding Work

Welding procedures must conform to AWS A5.1/A5.1M, AWS D1.1/D1.1M for steel or AWS D1.2/D1.2M for aluminum.

1.5.5 Single Source

Obtain each type of metal wall and liner panels, fins, clips, closures and other accessories from the standard products of the single source from a single manufacturer to operate as a complete system for the intended use.

1.5.6 Surface-Burning Characteristics

Provide metal wall panels having insulation core material with the following surface-burning characteristics as determined by testing identical products according to ASTM E84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

a. Flame-Spread Index: 25 or less.

b. Smoke-Developed Index: 450 or less.

1.5.7 Fire-Resistance Ratings

Where indicated, provide metal wall panels identical to those of assemblies tested for fire resistance in accordance with ASTM E119 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

Indicate design designations from UL's "Fire Resistance Directory" or from the listings of another qualified testing agency.
Combustion Characteristics: ASTM E136.

1.5.8 Fabrication

Fabricate and finish metal wall panels and accessories at the factory by manufacturer's standard procedures and processes and as necessary to fulfill indicated performance requirements. Comply with indicated profiles and with dimensional and structural requirements.

Provide panel profile, including major ribs and intermediate stiffening ribs, if any, for full length of panel. Fabricate metal wall panel side laps with factory-installed captive gaskets or separator strips that provide a tight seal and prevent metal-to-metal contact, in a manner that will seal weather-tight and minimize noise from movements within panel assembly.

1.5.8.1 Sheet Metal Accessories

Fabricate flashing and trim to comply with recommendations in SMACNA 1793 that apply to the design, dimensions, metal, and other characteristics of item indicated:

a. Form exposed sheet metal accessories that are without excessive oil canning, buckling, and tool marks and that are true to line and levels indicated, with exposed edges folded back to form hems.

b. End Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer. Rivet joints for additional strength.

c. Sealed Joints: Form non-expansion but movable joints in metal to accommodate elastomeric sealant to comply with SMACNA standards.

d. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on faces of accessories exposed to view.

e. Fabricate cleats and attachment devices of size and metal thickness recommended by SMACNA or by metal wall panel manufacturer for application, but not less than thickness of metal being secured.

1.5.9 Finishes

Comply with NAAMM AMP 500 for recommendations for applying and designating finishes.

Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

1.6 DELIVERY, HANDLING, AND STORAGE

1.6.1 Metal Wall Panels

Deliver and package components, sheets, metal wall panels, fins, and other manufactured items so as not to be damaged or deformed and protected during transportation and handling. Unload, store, and erect metal wall panels in a manner to prevent bending, warping, twisting, and surface damage.

Stack and store metal wall panels horizontally on platforms or pallets, covered with suitable weather-tight and ventilated covering to ensure dryness, with positive slope for drainage of water. Do not store metal wall panels in contact with other materials that might cause staining, denting, or other surface damage. Retain strippable protective covering on metal wall panel for period of metal wall panel installation.

1.6.2 Foam-Plastic Insulation

Protect foam-plastic insulation as follows:

a. Do not expose to sunlight, except to extent necessary for period of installation and concealment.

b. Protect against ignition at all times. Do not deliver foam-plastic insulation materials to Project site before installation time.

1.6.3 Plastic Materials

Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

1.7 PROJECT CONDITIONS

Weather Limitations: Proceed with installation preparation only when existing and forecasted weather conditions permit Work to proceed without water entering existing walling system or building.

Field Measurements: Verify locations of wall framing and opening dimensions by field measurements before metal wall panel fabrication and indicate measurements on Shop Drawings.

1.8 WARRANTY

Provide manufacturer's warranty for the metal wall panel system. The warranty period is to be no less than 20 years from the date of Government acceptance of the work. Write warranty directly to the Government, commencing at time of Government's acceptance of wall panel work. The warranty is to provide that if within the warranty period the metal wall panel system shows evidence of corrosion, perforation, rupture or excess weathering due to deterioration of the wall panel system resulting from defective materials or workmanship, correction of the defective workmanship is to be the responsibility of the metal wall panel system manufacturer. Repairs that become necessary because of defective materials and workmanship while metal wall panel system is under warranty are to be performed within 72 hours after notification, unless additional time is approved by the Contracting Officer.

System Responsibility: the exterior walls consisting of curtainwall, insulated metal wall panels, aluminum fins, light gauge metal framing fluid-applied waterproofing shall be provided, engineered, and installed by a single exterior wall contractor responsible for structural integrity, delegated wind load design, waterproofing and thermal performance. The exterior wall contractor will provide a single warranty for the entire perimeter façade assembly.

PART 2 PRODUCTS

2.1 PANEL MATERIALS

2.1.1 Steel Sheet

Roll-form steel wall and liner panels to the specified profile, with fy = 22 gauge and depth as indicated. Provide material plumb and true, and within the tolerances listed:

a. Aluminum-Zinc Alloy-coated Steel Sheet conforming to ASTM A792/A792M and AISI D100.

b. Individual panel lengths must match the exact layout shown on drawings wwithout any additional joints and be formed without warping, waviness, or ripples that are not part of the panel profile and free of damage to the finish coating system.

c. Provide panels with thermal expansion and contraction consistent with the type of system specified.

(1) Smooth, flat surface texture.

2.1.2 Foam-Insulation Core Wall Panel

Provide factory-assembled steel wall panel assemblies, continuously foamed-in-place (formed by injecting liquid polyisocyanurate or polyurethane foam insulation between two metal sheets, where it expands, cures), consisting of an exterior and interior metal face sheet chemically bonded to the rigid insulating core. Provide joints between panels designed to form weather-tight seals. Include accessories required for weather-tight installation. Provide 36" wide panels as shown on the architectural drawings, and conforming to the exact layout indicated.

a. Closed-Cell Content: 90 percent when tested according to ASTM D6226.

b. Density: 2.0 to 2.6 lb/cu. ft. when tested according to ASTM D1622.

c. Compressive Strength: Minimum 20 psi when tested according to ASTM D1621.

d. Shear Strength: 26 psi when tested according to ASTM C273/C273M.

2.1.3 Finish

All panels are to receive a factory-applied 70 percent resin polyvinylidene fluoride finish containing 100 percent inorganic pigments consisting of a baked-on top-coat with a manufacturer's recommended prime coat conforming to the following:

a. Metal Preparation: All metal is to have the surfaces carefully prepared for painting on a continuous process coil coating line by alkali cleaning, hot water rinsing, application of chemical conversion coating, cold water rinsing, sealing with acid rinse, and thorough drying.

b. Prime Coating: A base coat of epoxy paint, specifically formulated to interact with the top-coat, is to be applied to the prepared surfaces by roll coating to a dry film thickness of 1.00 mils. Oven cure the prime coat prior to application of finish coat.

c. Exterior Finish Coating: Apply the finish coating over the primer by roll coating to dry film thickness of 0.80 mils. Oven cure this exterior finish coat.

d. Clear Coating: Apply the clear coating over finish coating to a dry film thickness of 0.80 mils.

e. Interior Finish Coating: Apply a backer coat on the reverse side over 0.20 mils of primer by roll coating plus a backer coat to a dry film thickness of 0.30 mils for a total dry film thickness of 0.80 mils for the entire backer coat system. The backer coat must be oven-cured.

f. Color: The exterior finish chosen from the manufacturer's standard color chart.

g. Physical Properties: Coating is to conform to the industry and manufacturer's standard performance criteria as listed by the following certified test reports:

Chalking:	ASTM D4214
Color Change and Conformity:	ASTM D2244
Weatherometer:	ASTM G152, ASTM G153 and ASTM D822/D822M
Humidity:	ASTM D2247 and ASTM D714
Salt Spray:	ASTM B117

Chemical Pollution:	ASTM D1308
Gloss at 60:	ASTM D523
Pencil Hardness:	ASTM D3363
Reverse Impact:	ASTM D2794
Flexibility:	ASTM D522/D522M
Abrasion:	ASTM D968
Flame Spread:	ASTM E84

2.2 ALUMINUM FINS

Provide factory-fabricated aluminum fins coordinated with the curtainwall and metal wall panel assemblies as indicated on the drawings.
a. Base Metal: Aluminum sheet or extrusion conforming to ASTM B209 or AA standards, alloy and temper as recommended by manufacturer for exterior fins.
b. Thickness: Minimum as indicated on drawings, sized as required to meet structural design loads and maximum deflection of L/180 under service wind load.
c. Configuration: Fins to be continuous in length between supports where practicable, with profiles, dimensions, and edge conditions as indicated on the drawings, free of warping, waviness, or other defects.
d. Finish: Factory-applied polyvinylidene fluoride (PVDF) coating system to match adjacent metal wall panels, complying with Factory Color Finish requirements of this specification section.
e. Supports and Anchorage: Provide manufacturer's standard aluminum or stainless-steel brackets, clips, and hardware compatible with adjacent construction, designed to accommodate differential movement between fins, curtainwall, and wall panels without distress or noise.

2.3 MISCELLANEOUS METAL FRAMING

2.3.1 General

Cold-formed metallic-coated steel sheet conforming to ASTM A653/A653M and specified in Section 05 40 00 COLD-FORMED METAL FRAMING unless otherwise indicated.

2.4 FASTENERS

2.4.1 General

Type, material, corrosion resistance, size and sufficient length to penetrate the supporting member a minimum of 1 inch with other properties required to fasten miscellaneous metal framing members to substrates in accordance with the wall panel manufacturer's and ASCE 7-22 requirements.

2.4.2 Exposed Fasteners

Fasteners for wall panels to be corrosion resistant zinc-coated steel, compatible with the sheet panel and flashing and of a type and size recommended by the manufacturer to meet the performance requirements and

design loads. Fasteners for accessories to be the manufacturer's standard. Provide an integral metal washer that is compatible and matches the color of attached material with compressible sealing EPDM gasket approximately 3/32 inches thick.

2.4.3 Screws

Provide corrosion resistant zinc-coated steel, screws being the type and size recommended by the manufacturer to meet the performance requirements.

2.4.4 Rivets

Rivets to be closed-end type, corrosion resistant zinc-coated steel where watertight connections are required.

2.4.5 Attachment Clips

Fabricate clips from steel hot-dipped galvanized in accordance with ASTM A653/A653M. Size, shape, thickness and capacity as required meeting the insulation thickness and design load criteria specified.

2.5 ACCESSORIES

2.5.1 General

All accessories to be compatible with the metal wall panels. Do not provide sheet metal flashing, trim, metal closure strips, caps and similar metal accessories less than the minimum thickness specified for the wall panels. Exposed metal accessories/finishes to match the panels provided, except as otherwise indicated. Provide molded foam rib, ridge and other closure strips of non-absorbent closed-cell or solid-cell synthetic rubber or pre-molded neoprene to match configuration of the panels.

2.5.2 Closure Materials

2.5.2.1 Rubber Closure Strips

Closed-cell, expanded cellular rubber conforming to ASTM D1056 and ASTM D1667; extruded or molded to the configuration of the specified wall panel and in lengths supplied by the wall panel manufacturer.

2.5.2.2 Metal Closure Strips

Factory fabricated closure strips to be the samemetal, gauge/ thickness, color, finish and profile of the specified wall panel.

2.5.3 Joint Sealants

2.5.3.1 Sealants and Caulking

Sealants are to be an approved gun type for use in hand- or air-pressure caulking guns at temperatures above 40 degrees F (or frost-free application at temperatures above 10 degrees F) with minimum solid content of 85 percent of the total volume. Sealant is to dry with a tough, durable surface skin which permits it to remain soft and pliable underneath, providing a weather-tight joint. No migratory staining is permitted on painted or unpainted metal, stone, glass, vinyl, or wood.

Prime all joints to receive sealants with a compatible one-component or

two-component primer as recommended by the wall panel manufacturer.

2.5.3.2 Shop-Applied

Sealant for shop-applied caulking must be an approved gun grade, non-sag one-component polysulfide or silicone conforming to ASTM C920, and with a curing time to ensure the sealant's plasticity at the time of field erection.

2.5.3.3 Field-Applied

Sealant for field-applied caulking must be an approved gun grade, non-sag one-component polysulfide or two-component polyurethane with an initial maximum Shore A durometer hardness of 25, and conforming to ASTM C920. Color to match panel colors.

2.5.3.4 Tape Sealant

Pressure sensitive, 100 percent solid with a release paper backing; permanently elastic, non-sagging, non-toxic and non-staining as approved by the wall panel manufacturer.

2.6 SHEET METAL FLASHING AND TRIM

2.6.1 Fabrication

Shop fabricate sheet metal flashing and trim where practicable to comply with recommendations in SMACNA 1793 that apply to design, dimensions, metal, and other characteristics of item indicated. Obtain field measurements for accurate fit before shop fabrication.

Fabricate sheet metal flashing and trim without excessive oil canning, buckling, and tool marks and true to line and levels indicated, with exposed edges folded back to form hems.

2.7 REPAIR OF FINISH PROTECTION

Repair paint for color finish enameled wall panel must be compatible paint of the same formula and color as the specified finish provided by the wall panel manufacturer.

PART 3 EXECUTION

3.1 EXAMINATION

a. Examine substrates, areas, and conditions, with Installer present, for compliance with requirements for installation tolerances, metal wall panel supports, and other conditions affecting performance of the Work.

b. Examine primary and secondary wall framing to verify that rafters, purlins, angles, channels, and other structural panel support members and anchorages have been installed within alignment tolerances required by metal wall panel manufacturer, UL, ASTM, ASCE 7-22 and as required for the geographical area where construction will take place.

c. Examine solid wall sheathing to verify that sheathing joints are supported by framing or blocking and that installation is within flatness tolerances required by metal wall panel manufacturer.

d. Examine roughing-in for components and systems penetrating metal wall panels to verify actual locations of penetrations relative to seam locations of metal wall panels before metal wall panel installation.

e. Submit to the Contracting Officer a written report, endorsed by Installer, listing conditions detrimental to performance of the Work.

f. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 PREPARATION

a. Clean substrates of substances harmful to insulation, including removing projections capable of interfering with insulation attachment.

b. Miscellaneous Framing: Install sub-purlins, girts, angles, furring, and other miscellaneous wall panel support members and anchorage according to metal wall panel manufacturer's written instructions.

3.3 WALL PANEL INSTALLATION

Provide metal wall panels of full length to conform to the exact layout indicated on the drawings. Anchor metal wall panels and other components of the Work securely in place, with provisions for thermal and structural movement in accordance with MBMA Metal Building Systems Manual (MBMA MBSM).

a. Steel Wall Panels: Use stainless-steel fasteners for exterior surfaces and galvanized steel fasteners for interior surfaces.

b. Anchor Clips: Anchor metal wall panels and other components of the Work securely in place, using manufacturer's approved fasteners according to manufacturer's written instructions.

c. Metal Protection: Where dissimilar metals will contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with bituminous coating, by applying rubberized-asphalt underlayment to each contact surface, or by other permanent separation as recommended by metal wall panel manufacturer.

d. Joint Sealers: Install gaskets, joint fillers, and sealants where indicated and where required for weatherproof performance of metal wall panel assemblies. Provide types of gaskets, fillers, and sealants indicated or, if not indicated, types recommended by metal wall panel manufacturer.

e. Erect wall panel system in accordance with the approved erection drawings, the printed instructions and safety precautions of the manufacturer.

f. Sheets are not to be subjected to overloading, abuse, or undue impact. Do not apply bent, chipped, or defective sheets.

g. Erect sheets true and plumb and in exact alignment with the horizontal and vertical edges of the building, securely anchored, and with the indicated eave, and sill.

h. Work is to allow for thermal movement of the wall panel, movement of the building structure, and to provide permanent freedom from noise due to wind pressure.

i. Field cutting metal wall panels by torch is not permitted.

3.4 FASTENER INSTALLATION

Anchor metal wall panels and other components of the Work securely in place, using manufacturer's approved fasteners according to manufacturer's written instructions.

3.5 FLASHING, TRIM AND CLOSURE INSTALLATION

3.5.1 General Requirements

Provide extruded aluminum trim where indicated by profile and/or by note on the architectural drawings.

Comply with performance requirements, manufacturer's written installation instructions, and SMACNA 1793. Provide concealed fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints, and seams that will be permanently watertight and weather resistant.

Sheet metalwork is to be accomplished to form weather-tight construction without waves, warps, buckles, fastening stresses or distortion, and allow for expansion and contraction. Cutting, fitting, drilling, and other operations in connection with sheet metal required to accommodate the work of other trades is to be performed by sheet metal mechanics.

3.5.2 Metal Flashing (sheet metal and extruded aluminum as shown in the architectural details)

a. Exposed metal flashing is to be installed at building corners, sills and eaves, junctions between metal siding and walling.

b. Exposed metal flashing is to be the same material, color, and finish as the specified metal wall panel.Refer to the architectural details for exact color assignment of trim and flashing.

c. Flashing is to be fastened at not more than 8 inches on center, except where flashing is held in place by the same screws that secure covering sheets.

d. Provide flashing in minimum 8 foot lengths. Exposed flashing is to have 1 inch locked and blind-soldered end joints, and expansion joints at intervals of not more than 16 feet.

e. Exposed flashing and flashing subject to rain penetration to be bedded in the specified joint sealant.

f. Flashing which is in contact with dissimilar metals to be isolated by means of the specified asphalt mastic material to prevent electrolytic deterioration.

g. Drips to be formed to the profile indicated, with the edge folded back 1/2 inch to form a reinforced drip edge.

3.5.3 Closures

Install metal closure strips at open ends of corrugated or ribbed pattern

walls, and at intersection of wall and wall unless open ends are concealed with formed eave flashing; and in other required areas.

Install mastic closure strips at intersection of the wall with metal walling; top and bottom of metal siding; heads of wall openings; and in other required locations.

3.5.4 Aluminum Fin Installation

Install aluminum fins where indicated on the drawings, plumb, level, and true to line, in accordance with the fin manufacturer's written installation instructions and the approved shop drawings.
a. Coordinate installation of fins with curtainwall, insulated metal wall panels, and fluid-applied waterproofing to maintain the continuity of air, water, and thermal barriers.
b. Attach fins to the supporting structure using manufacturer's standard stainless-steel or aluminum fasteners and brackets, sized and spaced to resist design wind and seismic loads without exceeding allowable stresses or specified deflection limits.
c. Provide isolation between aluminum fins and dissimilar metals or corrosive substrates in accordance with Metal Protection paragraph herein.
d. Set fins to maintain required clearances from glazing, panels, and adjacent construction; provide uniform reveals and joint widths as indicated.

3.6 WORKMANSHIP

Make lines, arises, and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections which might affect the application. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight.

3.7 ACCEPTANCE PROVISIONS

3.7.1 Erection Tolerances

Erect metal wall panels straight and true with plumb vertical lines correctly lapped and secured in accordance with the manufacturer's written instructions. Horizontal lines must not vary more than 1/8 inch in 40 feet.

3.7.2 Leakage Tests

Inspect and test finished application of metal wall panels when directed to do so by the Contracting Officer. Conduct inspections and tests at no additional cost to the Government.

Perform inspection and testing promptly after erection to permit correction of defects and the removal and replacement of defective materials.

3.7.3 Repairs to Finish

Scratches, abrasions, and minor surface defects of finish may be repaired with the specified repair materials. Finished repaired surfaces must be uniform and free from variations of color and surface texture.

Repaired metal surfaces that are not acceptable to the project requirements are to be immediately removed and replaced with new material.

3.7.4 Paint-Finish Metal Siding

Paint-finish metal siding will be tested for color stability by the Contracting Officer during the manufacturer's specified guarantee period.

Remove and replace panels that indicate color changes, fading, or surface degradation, determined by visual examination, with new panels at no expense to the Government. New panels will be subject to the specified tests for an additional year from the date of their installation.

3.8 MANUFACTURER'S INSPECTION

The metal wall panel manufacturer's technical representative must visit the work site to inspect ongoing work. Inspections are to include observing installation technique and verifying the quality of work-in-place for compliance with the manufacturer's instructions. Deficiencies identified by the manufacturer's technical representative must be corrected and re-inspected by the manufacturer's technical representative.

3.8.1 Frequency

The manufacturer's technical representative must visit the work site to inspect and document ongoing work a minimum of three separate occasions during the course of the installation. One visit must occur during the first 20 squares of installation, one at substantial completion of the metal panel work, and all others during different periods of installation. Notify the Contracting officer a minimum of 5 working days prior to each visit by the manufacturer's technical representative.

3.8.2 Field Inspection Report

Document inspection results in a report prepared and signed by the manufacturer's technical representative for each visit. Submit the report to the Contracting Officer with the Contractor's daily Quality Control report. The manufacturer's field inspection report must include a description of ongoing work observed and whether the inspected work was satisfactory or unsatisfactory. The final report must include certification by the manufacturer's technical representative that the work was performed in accordance with the manufacturer's instructions and contains no deficiencies. Submit the final manufacturer's field inspection reports to the Contracting Officer within 5 working days of the final visit.

3.9 CLEAN-UP AND DISPOSAL

Clean all exposed sheet metal work at completion of installation. Remove metal shavings, filings, nails, bolts, and wires from work area. Remove grease and oil films, excess sealants, handling marks, contamination from

steel wool, fittings and drilling debris and scrub the work clean. Exposed metal surfaces to be free of dents, creases, waves, scratch marks, solder or weld marks, and damage to the finish coating.

Collect and place scrap/waste materials in containers. Promptly dispose of demolished materials. Do not allow demolished materials to accumulate on-site; transport demolished materials from Government property and legally dispose of them.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 54 23

THERMOPLASTIC POLYOLEFIN (TPO) ROOFING

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 PREINSTALLATION MEETINGS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 QUALITY ASSURANCE
1.8 DELIVERY
1.9 STORAGE AND HANDLING
1.10 FIELD CONDITIONS
1.11 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION
2.2 SYSTEM PERFORMANCE
2.3 PRODUCTS - GENERAL
2.4 TPO ROOFING MEMBRANE
2.5 MEMBRANE ACCESSORY MATERIALS
2.6 WALKWAY PADS
2.7 ROOF PAVERS
2.8 ACCESSORIES

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 TEMPORARY PROTECTION
3.4 INSTALLATION - GENERAL
3.5 ROOFING INSTALLATION
3.6 FLASHING INSTALLATION
3.7 WALKWAY PAD INSTALLATION
3.8 PAVER INSTALLATION
3.9 FIELD QUALITY CONTROL
3.10 CLEANING
3.11 PROTECTION

-- End of Section Table of Contents --

SECTION 07 54 23
1/21

THERMOPLASTIC POLYOLEFIN (TPO) ROOFING

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

Thermoplastic Polyolefin (TPO) sheet roofing adhered to roof deck.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Non-Flooring Adhesives and Sealants VOC Limits.

b. Section 07 01 50.19, PREPARATION FOR REROOFING: Preparation of Existing Membrane Roofs and Repair Areas.

c. Section 07 22 00, ROOF AND DECK INSULATION: Roof Insulation.

1.3 REFERENCES

Comply with references to extent specified in this section.

SINGLE PLY ROOFING INDUSTRY (SPRI)

ANSI/SPRI FX-1	(2021) Standard Field Test Procedure for Determining the Withdrawal Resistance of Roofing Fasteners

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7	(2017) Minimum Design Loads for Buildings and Other Structures

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP	(2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASTM INTERNATIONAL (ASTM)

ASTM C67/C67M	(2024) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
ASTM C140/C140M	(2022c) Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units
ASTM C578	(2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation
ASTM C1371	(2022) Determination of Emittance of

	Materials Near Room Temperature Using Portable Emissometers
ASTM C1549	(2016; R 2022) Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer
ASTM D1876	(2008; R 2015; E 2015) Standard Test Method for Peel Resistance of Adhesives (T-Peel Test)
ASTM D4263	(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
ASTM D4434/D4434M	(2015) Standard Specification for Poly(Vinyl Chloride) Sheet Roofing
ASTM D6878/D6878M	Thermoplastic Polyolefin Based Sheet Roofing
ASTM E408	(2013) Standard Test Methods for Total Normal Emittance of Surfaces Using Inspection-Meter Techniques
ASTM E1918	(2016) Standard Test Method for Measuring Solar Reflectance of Horizontal and Low-Sloped Surfaces in the Field
ASTM E1980	(2011; R 2019) Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces
NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)	
NRCA RoofMan	(2020) The NRCA Roofing Manual
NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)	
NIST PS 1	(2009) DOC Voluntary Product Standard PS 1-07, Structural Plywood
NIST PS 2	(2019) Performace Standard for Wood Structural Panels
U.S. DEPARTMENT OF ENERGY (DOE)	
Energy Star	(1992; R 2006) Energy Star Energy Efficiency Labeling System (FEMP)

1.4 PREINSTALLATION MEETINGS

Conduct pre-installation meeting at project site minimum 30 days before beginning Work of this section.

a. Required Participants:

(1) Contracting Officer's Representative.

(2) Architect/Engineer.

(3) Inspection and Testing Agency.

(4) Contractor.

(5) Installer.

(6) Manufacturer's field representative.

(7) Other installers responsible for adjacent and intersecting work, including roof deck, flashings, roof penetrations, roof accessories, utility penetrations, rooftop curbs and equipment.

b. Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(1) Installation schedule.

(2) Installation sequence.

(3) Preparatory work.

(4) Protection before, during, and after installation.

(5) Installation.

(6) Terminations.

(7) Transitions and connections to other work.

(8) Inspecting and testing.

(9) Other items affecting successful completion.

(10) Pullout test of fasteners.

(11) Material storage, including roof deck load limitations.

c. Document and distribute meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

Temporary Protection Plan; G

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Construction Submittals

SD-04 Samples

Samples; G

SD-06 Test Reports

Field Quality Control Reports; G

SD-07 Certificates

Certificates; G

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G

SD-11 Closeout Submittals

Environmental Product Declaration (EPD); S

Third-Party Certification Certificate, Assessment, Or Validation And Compliance Report; S

1.6 SUBMITTAL INSTRUCTIONS

a. Submittal Drawings:

(1) Roof membrane layout.

(2) Roofing membrane fastener pattern and spacing.

(3) Roofing membrane seaming and joint details.

(4) Roof membrane penetration details.

(5) Base flashing and termination details.

(6) Paver layout.

(7) Paver anchoring locations and details.

b. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Minimum fastener pullout resistance.

(3) Installation instructions.

(4) Warranty.

c. Samples:

(1) Roofing Membrane: (6 inch) square.

(2) Base Flashing: (6 inch) square.

(3) Fasteners: Each type.

(4) Roofing Membrane Seam: (12 inches) square.

d. Sustainable Construction Submittals:

(1) Solar Reflectance Index (SRI) for roofing membrane.

(2) Biobased Content:

(a) Show type and quantity for each product.

(3) Low Pollutant-Emitting Materials:

(a) Show volatile organic compound types and quantities.

(4) Energy Star label for roofing membrane.

e. Certificates: Certify products comply with specifications.

(1) Fire and windstorm classification.

(2) Energy performance requirements.

e. Qualifications: Substantiate qualifications comply with specifications.

(1) Installer, including supervisors with project experience list.

(2) Manufacturer's field representative with project experience list.

f. Field quality control reports.

g. Temporary protection plan. Include list of proposed temporary materials.

h. Operation and Maintenance Data:

(1) Maintenance instructions.

1.7 QUALITY ASSURANCE

a. Installer Qualifications:

(1) Approved by roofing system manufacturer as installer for roofing system with specified warranty.

(2) Regularly installs specified products.

(3) Installed specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

(4) Employs full-time supervisors experienced installing specified system and able to communicate with Contracting Officer's Representative and installer's personnel.

b. Manufacturer's Field Representative:

(1) Manufacturer's full-time technical employee or independent roofing inspector.

(2) Individual certified by Roof Consultants Institute as Registered Roof Observer.

1.8 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.9 STORAGE AND HANDLING

a. Comply with NRCA RoofMan storage and handling requirements.

b. Store products indoors in dry, weathertight facility.

c. Store adhesives according to manufacturer's instructions.

d. Protect products from damage during handling and construction operations.

e. Products stored on the roof deck must not cause permanent deck deflection.

1.10 FIELD CONDITIONS

Environment:

a. Product Temperature: Minimum (40 degrees F) for minimum 48 hours before installation.

b. Weather Limitations: Install roofing only during dry current and forecasted weather conditions.

1.11 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Manufacturer's Warranty: Warrant roofing system against material and manufacturing defects and agree to repair any leak caused by a defect in the roofing system materials or workmanship of the installer.

(1) Warranty Period: 25 years.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Thermoplastic Polyolefin (TPO) sheet roofing adhered to roof deck.

2.2 SYSTEM PERFORMANCE

Design roofing system complying with specified performance:

a. Load Resistance: ASCE 7; Design criteria: as indicated on Drawings.

(1)

b. Energy Performance:

(1) EPA Energy Star Listed for low-slope roof products.

(2) ASTM E1980; Minimum 78 Solar Reflectance Index (SRI).

(3) Minimum 0.70 initial solar reflectance and minimum 0.75 emissivity.

(4) Three-Year Aged Performance: Minimum 0.55 solar reflectance tested in according to ASTM C1549 or ASTM E1918, and minimum 0.75 thermal emittance tested in according to ASTM C1371 or ASTM E408.

a. Where tested aged values are not available:

(1) Calculate compliance adjusting initial solar reflectance according to ASHRAE 90.1 - IP.

(2) Provide roofing system with minimum 64 three-year aged Solar Reflectance Index calculated according to ASTM E1980 with (2.1 BTU/hour/square foot) convection coefficient.

2.3 PRODUCTS - GENERAL

a. Provide roof system components from one manufacturer.

b. Sustainable Construction Requirements:

(1) Solar Reflectance Index: 78 minimum.

(2) Biobased Content: Where applicable, provide products designated by USDA and meeting or exceeding USDA recommendations for bio-based content, and products meeting Rapidly Renewable Materials and certified sustainable wood content definitions; refer to www.biopreferred.gov.

(3) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Non-flooring adhesives and sealants.

2.4 TPO ROOFING MEMBRANE

TPO Sheet: ASTM D6878/D6878M, internally fabric or scrim reinforced, (60

mils) thick, with no backing .

2.5 MEMBRANE ACCESSORY MATERIALS

a. Sheet Flashing: Manufacturer's standard sheet flashing of same material, type, reinforcement, thickness, and color as TPO sheet membrane.

b. Factory Formed Flashings: Inside and outside corners, pipe boots, and other special flashing shapes to minimize field fabrication.

c. Bonding Adhesive: Manufacturer's standard, water based.

d. Metal Termination Bars: Manufacturer's standard, stainless-steel or aluminum, (1-inch wide by 1/8 inch thick) factory drilled for fasteners.

e. Battens: Manufacturer's standard, galvannealed or galvanized steel sheet, (1-inch wide by 0.05 inch thick), factory punched for fasteners.

f. Fasteners: Manufacturer's standard coated steel with metal or plastic plates, to suit application.

g. Primers, Sealers, T-Joint Covers, Lap Sealants, and Termination Reglets: As specified by roof membrane manufacturer.

h. Adhesive and sealant materials recommended by roofing system manufacturer for intended use, identical to materials utilized in approved listed roofing system, and compatible with roofing membrane.

2.6 WALKWAY PADS

Manufacturer's standard, slip-resistant rolls, minimum (3 feet) wide by (3/16 inch) thick.

2.7 ROOF PAVERS

Roof Pavers: Precast, normal weight, non-interlocking concrete units with ribbed bottom surface for drainage .

a. Weight: Minimum (15 pounds/square feet).

b. Compressive Strength: ASTM C140/C140M; minimum (8,000 psi).

c. Freeze Thaw: ASTM C67/C67M; maximum 1 percent mass loss.

d. Units of size, shape, and thickness as shown on drawings.

2.8 ACCESSORIES

Temporary Protection Materials:

a. Expanded Polystyrene (EPS) Insulation: ASTM C578.

b. Plywood: NIST PS 1, Grade CD Exposure 1.

c. Oriented Strand Board (OSB): NIST PS 2, Exposure 1.

PART 3 EXECUTION

3.1 EXAMINATION

a. Examine and verify substrate suitability with roofing Installer and roofing inspector present.

(1) Verify roof penetrations are complete, secured against movement, and firestopped .

(2) Verify roof deck is adequately secured to resist wind uplift.

(3) Verify roof deck is clean, dry, and in-plane ready to receive roofing system.

b. Correct unsatisfactory conditions before beginning roofing work.

3.2 PREPARATION

a. Complete roof deck construction before beginning roofing work:

(1) Curbs, blocking, edge strips, nailers, cants, and other components to which insulation, roofing, and base flashing is attached in place ready to receive insulation and roofing.

(2) Coordinate roofing membrane installation with flashing work and roof insulation work so insulation and flashing are installed concurrently to permit continuous roofing operations.

(3) Complete installation of flashing, insulation, and roofing in same day except for the area where temporary protection is required when work is stopped for inclement weather or end of work day.

b. Dry out surfaces including roof deck flutes, that become wet from any cause during progress of the work before roofing work is resumed. Apply materials to dry substrates, only.

c. Broom clean roof decks. Remove dust, dirt and debris.

d. Remove projections capable of damaging roofing materials.

e. Concrete Decks, except Insulating Concrete:

(1) Test concrete decks for moisture according to ASTM D4263 before installing roofing materials.

(2) Prime concrete decks. Keep primer back (4 inches) from precast concrete deck joints.

(3) Allow primer to dry before application of bitumen.

f. Insulating Concrete Decks:

(1) Allow to dry out minimum five days after installation before installing roofing materials.

(2) Allow additional drying time when precipitation occurs before installing roofing materials.

g. Poured Gypsum Decks: Dry out poured gypsum according to manufacturer's instructions before installing roofing materials.

h. Existing Membrane Roofs and Repair Areas:

(1) Comply with requirements in Section 07 01 50.19 PREPARATION FOR REROOFING.

3.3 TEMPORARY PROTECTION

a. Install temporary protection consisting of a temporary seal and water cut-offs at the end of each day's work and when work is halted for an indefinite period or work is stopped when precipitation is imminent.

b. Install temporary cap flashing over top of base flashings where permanent flashings are not in place to protect against water intrusion into roofing system. Securely anchor in place to prevent blow off and damage by construction activities.

c. Temporarily seal exposed insulation surfaces within roofing membrane.

(1) Apply temporary seal and water cut off by extending roofing membrane beyond insulation and securely embedding edge of the roofing membrane in (1/4 inch) thick by (2 inches) wide strip of temporary closure sealant. Weight roofing membrane edge with sandbags, to prevent displacement; space sandbags maximum (8 feet) on center.

(2) Direct water away from work. Provide drainage, preventing water accumulation.

(3) Check daily to ensure temporary seal remains watertight. Reseal open areas and weight down.

d. Before the work resumes, cut off and discard portions of roof membrane in contact with temporary seal.

(1) Cut minimum (6 inches) back from sealed edges and surfaces.

e. Remove sandbags and store for reuse.

3.4 INSTALLATION - GENERAL

a. Install products according to manufacturer's instructions and approved submittal drawings.

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.

b. Comply with NRCA RoofMan installation requirements.

c. Comply with structural drawings for uplift resistance.

d. Do not allow membrane and flashing to contact surfaces contaminated with asphalt, coal tar, oil, grease, or other substances incompatible with TPO.

3.5 ROOFING INSTALLATION

a. Install the membrane so the sheets run perpendicular to the long dimension of the insulation boards.

b. Begin installation at the low point of the roof and work towards the high point. Lap membrane shingled in water flow direction.

c. Position the membrane free of buckles and wrinkles.

d. Roll membrane out; inspect for defects as membrane is unrolled. Remove defective areas:

(1) Lap edges and ends of sheets (2 inches) or more as recommended by the manufacturer.

(2) Heat weld laps. Apply pressure as required. Seam strength of laps as required by ASTM D4434/D4434M.

(3) Check seams to ensure continuous adhesion and correct defects.

(4) Finish seam edges with beveled bead of lap sealant.

(5) Finish seams same day as membrane is installed.

(6) Anchor membrane perimeter to roof deck or parapet wall as indicated on drawings.

(7) Repair areas of welded seams where samples have been taken or marginal welds, bond voids, or skips occurs.

(8) Repair fishmouths and wrinkles by cutting to lay flat and installing patch over cut area extending (4 inches) beyond cut.

e. Membrane Perimeter Anchorage:

(1) Install batten at perimeter of each roof area, curb flashing, expansion joints and similar penetrations on top of roof membrane as indicated on drawings.

(2) Mechanically Fastening:

(a) Space fasteners maximum (12 inches) on center, starting (1 inch) from ends.

(b) When battens are cut, round edges and corners before installing.

(c) After mechanically fastening strip cover and seal strip with a (6 inch) wide roof membrane strip; heat weld to roof membrane and seal edges.

(d) At gravel stops turn roofing membrane down over front edge of the blocking, cant, or nailer. Secure roofing membrane to vertical portion of nailer; or, if required by the membrane manufacturer, with fasteners spaced maximum (6 inches) on centers.

(e) At parapet walls intersecting building walls and curbs, secure roofing membrane to structural deck with fasteners (6 inches) on

centers or as shown in NRCA RoofMan.

f. Adhered System:

(1) Apply bonding adhesive in quantities required by roof membrane manufacturer.

(2) Fold sheet back on itself, clean and coat the bottom side of the membrane and the top of substrate with adhesive. Do not coat the lap joint area.

(3) After adhesive has set according to adhesive manufacturer's instruction, roll roofing membrane into adhesive minimizing voids and wrinkles.

(4) Repeat for other half of sheet.

g. Mechanically Fastened System Installation:

(1) Secure roofing membrane to structural deck with fasteners through battens to achieve specified wind uplift performance.

(a) Drill pilot holes for fasteners installed into cast-in-place concrete. Drill hole minimum (3/8 inch) deeper than fastener penetration.

(2) When fasteners are installed within membrane laps, locate battens minimum (1/2 inch) from the edge of sheets.

(3) Apply lap sealant under battens and anchor to deck while lap sealant is still fluid. Cover fastener head with fastener sealer.

(4) Where fasteners are installed over roofing membrane after seams are welded, cover fasteners with minimum (8 inch) diameter TPO membrane cap centered over fasteners. Where battens are used cover battens with minimum (8 inch) wide TPO strip cap centered over batten. Splice caps to roofing membrane and finish edges with lap sealant.

3.6 FLASHING INSTALLATION

a. Install flashings same day as roofing membrane is installed. When flashing cannot be completely installed in one day, complete installation until flashing is watertight and provide temporary covers or seals.

b. Flashing Roof Drains:

(1) Install roof drain flashing as recommended by roofing membrane manufacturer.

(a) Coordinate to set the metal drain flashing in asphalt roof cement, holding cement back from the edge of the metal flange.

(b) Do not allow the roof cement to come in contact with TPO roofing membrane.

(c) Adhere roofing membrane to metal flashing with bonding adhesive.

(2) Turn down the metal drain flashing and roofing membrane into drain body. Install clamping ring and strainer.

c. Installing Base Flashing and Pipe Flashing:

(1) Install flashing sheet to pipes, wall or curbs to minimum (8 inches) above roof surfaces and extending roofing manufacturer's standard lap dimension onto roofing membranes.

(a) Adhere flashing with bonding adhesive.

(b) Form inside and outside corners of flashing sheet according to NRCA RoofMan. Form pipe flashing according to NRCA RoofMan.

(c) Lap ends roofing manufacturer's standard dimension.

(d) Heat weld flashing membranes together and flashing membranes to roofing membranes. Finish exposed edges with lap sealant.

(e) Install flashing membranes according to NRCA RoofMan.

(2) Anchor top of flashing to walls and curbs with fasteners spaced maximum (6 inches) on center. Use surface mounted fastening strip with sealant on ducts. Use pipe clamps on pipes or other round penetrations.

(3) Apply sealant to top edge of flashing.

d. Installing Building Expansion Joints:

(1) Install base flashing on curbs as specified.

(2) Coordinate installation with roof expansion joint system .

(3) Install flexible tubing 1-1/2 times the width of joint centered over joint. Cover tubing with flashing sheet adhered to base flashing and lapping base flashing roofing manufacturer's standard dimension. Finish edges of laps with sealant.

e. Repairs to Membrane and Flashings:

(1) Remove sections of roofing membrane or flashing that are creased, wrinkled, or fishmouthed.

(2) Cover removed areas, cuts and damaged areas with a patch extending (4 inches) beyond damaged, cut, or removed area. Heat weld to roofing membrane or flashing sheet. Finish edge of lap with lap sealant.

3.7 WALKWAY PAD INSTALLATION

a. Heat weld walkway sheet to roofing membrane at edges. Weld area (2 inches) wide by the entire length of the walkway sheet.

b. Finish edges of laps with lap sealant.

3.8 PAVER INSTALLATION

Install pavers as soon as roofing membrane is installed.

(1) Saw cut or core drill pavers for cut units.

(2) Install pavers with butt joints in running bond with minimum one half-length units at ends.

(a) Stagger end joints; generally locate joints near midpoint of adjacent rows, except where end joints occur in valleys. Miter end joints to fit in valleys.

(b) Cut to fit within (1/2 inch) of penetrations.

(3) Install interlocking connectors in channel units for complete tie in of units, including cut units. Use corner spacings for a distance of (4 feet) or more around roof drains, penetrations, and other vertical surfaces in the field of the roof area.

(a) Space connectors at (12 inches) on center at the corners for (10 foot) square area.

(b) Space connectors at (12 inches) on center at the perimeter for (6 foot) wide strip.

(c) Space connectors at (12 inches) on center in the field.

(d) Install pavers under the perimeter retainer as shown on drawings.

(4) Install strapping where shown.

(a) Limit strap lengths to a maximum of (30 feet).

(b) Install straps at corner connection to the perimeter retainer at approximate 45 degree angle at approximate (10 to 12 feet) from corner.

(c) Install straps on both sides of valleys, hips, and ridges, with cross straps spaced maximum (4 feet) on center between end straps.

(d) Install straps at the perimeter of penetrations more than two pavers in width or length.

(e) Anchor straps to each paver with two fasteners per unit.

(f) Pre-drill holes for fasteners in pavers.

3.9 FIELD QUALITY CONTROL

a. Field Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.

(1) Fastener Pull Out Tests: ANSI/SPRI FX-1; one test for every (2,500 square feet) of deck. Perform tests for each combination of fastener type and roof deck type before installing roof insulation.

(a) Test at locations selected by Contracting Officer's Representative.

(b) Do not proceed with roofing work when pull out resistance is less than manufacturer's required resistance.

(c) Test Results:

1. Repeat tests using different fastener type or use additional fasteners achieve pull out resistance required to meet specified wind uplift performance.

2. Patch cementitious deck to repair areas of fastener tests holes.

(2) Examine and probe roofing membrane and flashing seams in presence of Contracting Officer's Representative and Manufacturer's field representative.

(3) Probe seams to detect marginal bonds, voids, skips, and fishmouths.

(4) Cut (4 inch) wide by (12 inch) long samples through seams where directed by Contracting Officer's Representative.

(5) Cut one sample for every (1500 feet) of seams.

(6) Cut samples perpendicular to seams.

(7) Failure of samples to pass ASTM D1876 test will be cause for rejection of work.

(8) Repair areas where samples are taken and where marginal bond, voids, and skips occur.

(9) Repair fishmouths and wrinkles by cutting to lay flat. Install patch over cut area extending (4 inches) beyond cut.

b. Manufacturer Services:

(1) Inspect initial installation, installation in progress, and completed work.

(2) Issue supplemental installation instructions necessitated by field conditions.

(3) Prepare and submit inspection reports.

(4) Certify completed installation complies with manufacturer's instructions and warranty requirements.

3.10 CLEANING

a. Remove excess adhesive before adhesive sets.

b. Clean exposed roofing surfaces. Remove contaminants and stains to comply with specified solar reflectance performance .

3.11 PROTECTION

a. Protect roofing system from traffic and construction operations.

(1) Protect roofing system when used for subsequent work platform, materials storage, or staging.

(2) Distribute scaffolding loads to exert maximum 50 percent roofing system materials compressive strength.

b. Loose lay temporary insulation board overlaid with plywood or OSB.

(1) Weight boards to secure against wind uplift.

c. Remove protective materials immediately before acceptance.

d. Repair damage.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 60 00

FLASHING AND SHEET METAL

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 REFERENCES
1.4 PERFORMANCE REQUIREMENTS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 FLASHING AND SHEET METAL MATERIALS
2.2 FLASHING ACCESSORIES
2.3 SHEET METAL THICKNESS
2.4 FABRICATION, GENERAL
2.5 FINISHES
2.6 THROUGH-WALL FLASHINGS
2.7 BASE FLASHING
2.8 COUNTERFLASHING (CAP FLASHING OR HOODS)
2.9 HANGING GUTTERS
2.10 CONDUCTORS (DOWNSPOUTS)
2.11 INSULATED EXPANSION JOINT COVERS

PART 3 EXECUTION

3.1 INSTALLATION
3.2 THROUGH-WALL FLASHING
3.3 BASE FLASHING
3.4 COUNTERFLASHING (CAP FLASHING OR HOODS)
3.5 COPINGS
3.6 EXPANSION JOINT COVERS, INSULATED
3.7 HANGING GUTTERS
3.8 CONDUCTORS (DOWNSPOUTS)

-- End of Section Table of Contents --

SECTION 07 60 00
1/21

FLASHING AND SHEET METAL

PART 1 GENERAL

1.1 DESCRIPTION

Formed sheet metal work for wall and roof flashing, copings, roof edge metal, fasciae, drainage specialties, and formed expansion joint covers are specified in this section.

1.2 RELATED WORK

a. Section 07 54 23 THERMOPLASTIC POLYOLEFIN (TPO) ROOFING: Membrane base flashings and stripping.

j. Section 07 71 00 ROOF SPECIALTIES: Manufactured flashing, copings, roof edge metal, and fasciae.

k. Section 07 71 00, ROOF SPECIALTIES: Integral flashing components of manufactured roof specialties and accessories or equipment.

l. Section 07 72 00, ROOF ACCESSORIES: Integral flashing components of manufactured roof specialties and accessories or equipment.

m. Division 07 ROOFING AND WALL SYSTEM: Flashing components of factory finished roofing and wall systems.

n. Section 07 92 00, JOINT SEALANTS: Joint Sealants.

o. Section 09 91 00, PAINTING: Paint materials and application.

p. Division 22, PLUMBING: Integral flashing components of manufactured roof specialties and accessories or equipment.

q. Section 22 14 00, FACILITY STORM DRAINAGE: Flashing of Roof Drains.

r. Division 23 HVAC: Integral flashing components of manufactured roof specialties and accessories or equipment.

1.3 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only. Editions of applicable publications current on date of issue of bidding documents apply unless otherwise indicated.

ALUMINUM ASSOCIATION (AA)

AA-C22A41	Aluminum Chemically etched medium matte, with clear anodic coating, Class I Architectural, thick
AA-C22A42	Chemically etched medium matte, with integrally colored anodic coating, Class I

	Architectural, thick
AA-C22A44	Chemically etched medium matte with electrolytically deposited metallic compound, integrally colored coating Class I Architectural, thick finish

SINGLE PLY ROOFING INDUSTRY (SPRI)

ANSI/SPRI ES-1	(2003) Wind Design Standard for Edge Systems Used with Low Slope Roofing Systems

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605	(2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels
AAMA 621	(2002) Voluntary Specification for High Performance Organic Coatings on Coil Coated Architectural Hot Dipped Galvanized (HDG) and Zinc-Aluminum Coated Steel Substrates

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B370	(2022) Standard Specification for Copper Sheet and Strip for Building Construction
ASTM D173/D173M	(2003; R 2011; E 2012) Bitumen-Saturated Cotton Fabrics Used in Roofing and Waterproofing
ASTM D412	(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM D3656/D3656M	(2013) Standard Specification for Insect

	Screening and Louver Cloth Woven from Vinyl-Coated Glass Yarns
ASTM D4586/D4586M	(2007; R 2018) Asphalt Roof Cement, Asbestos-Free

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS A-A-1925	Shield, Expansion; (Nail Anchors)
FS UU-B-790	(Rev A; Notice 2) Building Paper Vegetable Fiber: (Kraft, Waterproofed, Water Repellent and Fire Resistant)

1.4 PERFORMANCE REQUIREMENTS

a. Wind Uplift Forces: Resist the following forces per FM Approvals 1-49:

(3) Wind Zone 2: (31 to 45 pound force/square foot): (90 pound force/square foot) perimeter uplift force, (120 pound force/square foot) corner uplift force, and (45 pound force/square foot) outward force.

b. Wind Design Standard: Fabricate and install copings and roof-edge flashings tested per ANSI/SPRI ES-1 to resist design pressure indicated on Drawings.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

SD-07 Certificates

Certificates; G

1.6 SUBMITTAL INSTRUCTIONS

a. Shop Drawings: For all specified items, including:

(1) Flashings

(2) Copings

(3) Gravel Stop-Fascia

(4) Gutter and Conductors

(5) Expansion joints

(6) Fascia-cant

b. Manufacturer's Literature and Data: For all specified items, including:

(1) Two-piece counterflashing

(2) Thru wall flashing

(3) Expansion joint cover, each type

(4) Nonreinforced, elastomeric sheeting

(5) Copper clad stainless steel

(6) Polyethylene coated copper

(7) Bituminous coated copper

(8) Copper covered paper

(9) Fascia-cant

c. Certificates: Indicating compliance with specified finishing requirements, from applicator and contractor.

PART 2 PRODUCTS

2.1 FLASHING AND SHEET METAL MATERIALS

a. Stainless Steel: ASTM A240/A240M, Type 302B, dead soft temper.

b. Copper ASTM B370, cold-rolled temper.

c. Bituminous Coated Copper: Minimum copper ASTM B370, weight not less than (3 oz/sf). Bituminous coating shall weigh not less than (6 oz/sf); or, copper sheets may be bonded between two layers of coarsely woven bitumen-saturated cotton fabric ASTM D173/D173M. Exposed fabric surface shall be crimped.

d. Copper Covered Paper: Fabricated of electro-deposit pure copper sheets ASTM B370, bonded with special asphalt compound to both sides of creped, reinforced building paper, FS UU-B-790, Type I, style 5, or to a three ply sheet of asphalt impregnated crepe paper. Grooves running along the width of sheet.

e. Polyethylene Coated Copper: Copper sheet ASTM B370, weighing (3 oz/sf) bonded between two layers of (two mil) thick polyethylene sheet.

f. Aluminum Sheet: ASTM B209, alloy 3003-H14 except alloy used for color anodized aluminum shall be as required to produce specified color. Alloy required to produce specified color shall have the same structural properties as alloy 3003-H14.

g. Galvanized Sheet: ASTM A653/A653M.

h. Nonreinforced, Elastomeric Sheeting: Elastomeric substances reduced to thermoplastic state and extruded into continuous homogenous sheet 0.056 inch thick. Sheeting shall have not less than (1,000 psi) tensile strength and not more than seven percent tension-set at 50 percent elongation when tested in accordance with ASTM D412. Sheeting shall show no cracking or flaking when bent through 180 degrees over a 1/32 inch diameter mandrel and then bent at same point over same size mandrel in opposite direction through 360 degrees at temperature of (-20 °F).

2.2 FLASHING ACCESSORIES

a. Solder: ASTM B32; flux type and alloy composition as required for use with metals to be soldered.

b. Rosin Paper: FS UU-B-790, Type I, Grade D, Style 1b, Rosin-sized sheathing paper, weighing approximately (6 pounds/100 square feet).

c. Bituminous Paint: ASTM D1187/D1187M, Type I.

d. Fasteners:

(1) Use copper, copper alloy, bronze, brass, or stainless steel for copper and copper clad stainless steel, and stainless steel for stainless steel and aluminum alloy. Use galvanized steel or stainless steel for galvanized steel.

(2) Nails:

(a) Minimum diameter for copper nails: (0.109 inch).

(b) Minimum diameter for aluminum nails (0.105 inch).

(c) Minimum diameter for stainless steel nails: 0.095 inch and annular threaded.

(d) Length to provide not less than 7/8 inch penetration into anchorage.

(3) Rivets: Not less than 1/8 inch diameter.

(4) Expansion Shields: FS A-A-1925.

e. Sealant: As specified in Section 07 92 00, JOINT SEALANTS for exterior locations.

f. Insect Screening: ASTM D3656/D3656M, 18 by 18 regular mesh.

g. Roof Cement: ASTM D4586/D4586M.

2.3 SHEET METAL THICKNESS

a. Except as otherwise shown or specified use thickness or weight of sheet metal as follows:

b. Concealed Locations (Built into Construction):

(1) Copper: (10 oz) minimum 0.013 inch thick.

(2) Stainless steel: 0.010 inch thick.

(3) Copper clad stainless steel: 0.010 inch thick.

(4) Galvanized steel: 0.021 inch thick.

c. Exposed Locations:

(1) Copper: (16 oz).

(2) Stainless steel: (0.015 inch).

(3) Copper clad stainless steel: (0.015 inch).

d. Thickness of aluminum or galvanized steel is specified with each item.

2.4 FABRICATION, GENERAL

a. Jointing:

(1) In general, copper, stainless steel and copper clad stainless steel joints, except expansion and contraction joints, shall be locked and soldered.

(2) Jointing of copper over (20 oz) weight or stainless steel over 0.018 inch thick shall be done by lapping, riveting and soldering.

(3) Joints shall conform to following requirements:

(a) Flat-lock joints shall finish not less than 3/4 inch wide.

(b) Lap joints subject to stress shall finish not less than (one inch) wide and shall be soldered and riveted.

(c) Unsoldered lap joints shall finish not less than 4 inches wide.

(4) Flat and lap joints shall be made in direction of flow.

(5) Edges of bituminous coated copper, copper covered paper, nonreinforced elastomeric sheeting and polyethylene coated copper shall be jointed by lapping not less than 4 inches in the direction of flow and cementing with asphalt roof cement or sealant as required by the manufacturer's printed instructions.

(6) Soldering:

(a) Pre tin both mating surfaces with solder for a width not less than (1 1/2 inches) of uncoated copper, stainless steel, and copper clad stainless steel.

(b) Wire brush to produce a bright surface before soldering lead coated copper.

(c) Treat in accordance with metal producers recommendations other sheet metal required to be soldered.

(d) Completely remove acid and flux after soldering is completed.

b. Expansion and Contraction Joints:

(1) Fabricate in accordance with the Architectural Sheet Metal Manual recommendations for expansion and contraction of sheet metal work in continuous runs.

(2) Space joints as shown or as specified.

(3) Space expansion and contraction joints for copper, stainless steel, and copper clad stainless steel at intervals not exceeding (24 feet).

(4) Space expansion and contraction joints for aluminum at intervals not exceeding (18 feet), except do not exceed 10 feet for gravel stops and fascia-cant systems.

(5) Fabricate slip-type or loose locked joints and fill with sealant unless otherwise specified.

(6) Fabricate joint covers of same thickness material as sheet metal served.

c. Cleats:

(1) Fabricate cleats to secure flashings and sheet metal work over 12 inches wide and where specified.

(2) Provide cleats for maximum spacing of 12 inch centers unless specified otherwise.

(3) Form cleats of same metal and weights or thickness as the sheet metal being installed unless specified otherwise.

(4) Fabricate cleats from 2 inch wide strip. Form end with not less than 3/4 inch wide loose lock to item for anchorage. Form other end of length to receive nails free of item to be anchored and end edge to be folded over and cover nail heads.

d. Edge Strips or Continuous Cleats:

(1) Fabricate continuous edge strips where shown and specified to secure loose edges of the sheet metal work.

(2) Except as otherwise specified, fabricate edge strips or minimum 0.050 inch thick aluminum.

(3) Use material compatible with sheet metal to be secured by the edge strip.

(4) Fabricate in 10 feet maximum lengths with not less than 3/4 inch loose lock into metal secured by edge strip.

(5) Fabricate Strips for fascia anchorage to extend below the supporting wood construction to form a drip and to allow the flashing to be hooked over the lower edge at least (3/4-inch).

(6) Fabricate anchor edge maximum width of 3 inches or of sufficient width to provide adequate bearing area to insure a rigid

installation using 0.0625 inch thick aluminum.

e. Drips:

(1) Form drips at lower edge of sheet metal counter-flashings (cap flashings), fascias, gravel stops, wall copings, by folding edge back 1/2 inch and bending out 45 degrees from vertical to carry water away from the wall.

(2) Form drip to provide hook to engage cleat or edge strip for fastening for not less than 3/4 inch loose lock where shown.

f. Edges:

(1) Edges of flashings concealed in masonry joints opposite drain side shall be turned up 1/4 inch to form dam, unless otherwise specified or shown otherwise.

(2) Finish exposed edges of flashing with a 1/4 inch hem formed by folding edge of flashing back on itself when not hooked to edge strip or cleat. Use 1/4 inch minimum penetration beyond wall face with drip for through-wall flashing exposed edge.

(3) All metal roof edges shall meet requirements of IBC, current edition.

g. Metal Options:

(1) Where options are permitted for different metals use only one metal throughout.

(2) Stainless steel may be used in concealed locations for fasteners of other metals exposed to view.

(3) Where copper gravel stops, copings and flashings will carry water onto cast stone, stone, or architectural concrete, or stainless steel.

2.5 FINISHES

a. Use same finish on adjacent metal or components and exposed metal surfaces unless specified or shown otherwise.

b. In accordance with NAAMM AMP 500, unless otherwise specified.

c. Finish exposed metal surfaces as follows, unless specified otherwise:

(1) Copper: Mill finish.

(2) Stainless Steel: Finish No. 2B or 2D.

(3) Aluminum:

(a) Clear Finish: AA-C22A41 medium matte, clear anodic coating, Class 1 Architectural, (0.7 mils) thick.

(b) Colored Finish: AA-C22A42 (anodized) or AA-C22A44 (electrolytically deposited metallic compound) medium matte, integrally colored coating, Class 1 Architectural, (0.7 mils)

thick. Dyes will not be accepted.

(c) Fluorocarbon Finish: AAMA 2605, high performance organic coating.

(d) Mill finish.

(4) Steel and Galvanized Steel:

(a) Finish painted under Section 09 91 00, PAINTING unless specified as prefinished item.

(b) Manufacturer's finish:

1. Baked on prime coat over a phosphate coating.

2. Baked-on prime and finish coat over a phosphate coating.

3. Fluorocarbon Finish: AAMA 621, high performance organic coating.

2.6 THROUGH-WALL FLASHINGS

a. Form through-wall flashing to provide a mechanical bond or key against lateral movement in all directions. Install a sheet having 1/16 inch deep transverse channels spaced four to every (one inch), or ribbed diagonal pattern, or having other deformation unless specified otherwise.

(1) Fabricate in not less than 8 feet lengths; 10 feet maximum lengths.

(2) Fabricate so keying nests at overlaps.

b. For Masonry Work When Concealed Except for Drip:

(1) Either copper, stainless steel, or copper clad stainless steel.

(2) Form an integral dam at least 3/16 inch high at back edge.

(3) Form exposed portions of flashing with drip, approximately 1/4 inch projection beyond wall face.

c. For Masonry Work When Exposed Edge Forms a Receiver for Counter Flashing:

(1) Use same metal and thickness as counter flashing.

(2) Form an integral dam at least 3/16 inch high at back edge.

(3) Form exposed portion as snap lock receiver for counter flashing upper edge.

d. For Flashing at Architectural Precast Concrete Panels or Stone Panels.

(1) Use plan flat sheet of stainless steel.

(2) Form exposed portions with drip as specified or receiver.

e. Window Sill Flashing and Lintel Flashing:

(1) Use either copper, stainless steel, copper clad stainless-steel plane flat sheet, or nonreinforced elastomeric sheeting, bituminous coated copper, copper covered paper, or polyethylene coated copper.

(2) Fabricate flashing at ends with folded corners to turn up 3/16 inch in first vertical masonry joint beyond masonry opening.

(3) Turn up back edge as shown.

(4) Form exposed portion with drip as specified or receiver.

f. Door Sill Flashing:

(1) Where concealed, use either (20 ounce) copper, 0.018 inch thick stainless steel, or 0.018 inch thick copper clad stainless steel.

(2) Where shown on drawings as combined counter flashing under threshold, sill plate, door sill, or where subject to foot traffic, use either (24 ounce) copper, 0.024 inch stainless steel, or 0.024 inch thick stainless steel.

(3) Fabricate flashing at ends to turn up 3/16 inch in first vertical masonry joint beyond masonry opening with folded corners.

2.7 BASE FLASHING

a. Use metal base flashing at vertical surfaces intersecting built-up roofing without cant strips or where shown.

(1) Use either copper, or stainless steel, thickness specified unless specified otherwise.

(2) When flashing is over 10 inches in vertical height or horizontal width use either (20 oz) copper or 0.018 inch stainless steel.

(3) Use stainless steel at aluminum roof curbs where flashing contacts the aluminum.

(4) Use either copper, or stainless steel at pipe flashings.

b. Fabricate metal base flashing up vertical surfaces not less than 8 inch nor more than (16 inch).

c. Fabricate roof flange not less than 4 inches wide unless shown otherwise. When base flashing length exceeds 8 feet form flange edge with 1/2 inch hem to receive cleats.

d. Form base flashing bent from strip except pipe flashing. Fabricate ends for riveted soldered lap seam joints. Fabricate expansion joint ends as specified.

e. Pipe Flashing: (Other than engine exhaust or flue stack)

(1) Fabricate roof flange not less than 4 inches beyond sleeve on all sides.

(2) Extend sleeve up and around pipe and flange out at bottom not less

than 1/2 inch and solder to flange and sleeve seam to make watertight.

(3) At low pipes 8 inch to 18 inch above roof:

(a) Form top of sleeve to turn down into the pipe at least (one inch).

(b) Allow for loose fit around and into the pipe.

(4) At high pipes and pipes with goosenecks or other obstructions which would prevent turning the flashing down into the pipe:

(a) Extend sleeve up not less than 12 inch above roofing.

(b) Allow for loose fit around pipe.

2.8 COUNTERFLASHING (CAP FLASHING OR HOODS)

a. Either copper or stainless steel, unless specified otherwise.

b. Fabricate to lap base flashing a minimum of 4 inches with drip:

(1) Form lock seams for outside corners. Allow for lap joints at ends and inside corners.

(2) In general, form flashing in lengths not less than 8 feet and not more than (10 feet).

(3) Two-piece, lock in type flashing may be used in-lieu-of one piece counter-flashing.

(4) Manufactured assemblies may be used.

(5) Where counterflashing is installed at new work use an integral flange at the top designed to be extended into the masonry joint or reglet in concrete.

(6) Where counterflashing is installed at existing work use surface applied type, formed to provide a space for the application of sealant at the top edge.

c. One-piece Counterflashing:

(1) Back edge turned up and fabricate to lock into reglet in concrete.

(2) Upper edge formed to extend full depth of masonry unit in mortar joint with back edge turned up (1/4 inch).

d. Two-Piece Counterflashing:

(1) Receiver to extend into masonry wall depth of masonry unit with back edge turned up 1/4 inch and exposed edge designed to receive and lock counterflashing upper edge when inserted.

(2) Counterflashing upper edge designed to snap lock into receiver.

e. Surface Mounted Counterflashing; one or two piece:

(1) Use at existing or new surfaces where flashing cannot be inserted in vertical surface.

(2) One piece fabricate upper edge folded double for (2 1/2 inches) with top 3/4 inch bent out to form "V" joint sealant pocket with vertical surface. Perforate flat double area against vertical surface with horizontally slotted fastener holes at 16 inch centers between end holes. Option: One piece surface mounted counter-flashing (cap flashing) may be used. Fabricate as detailed on Plate 51 of SMACNA Architectural Sheet Metal Manual.

(3) Two pieces: Fabricate upper edge to lock into surface mounted receiver. Fabricate receiver joint sealant pocket on upper edge and lower edge to receive counterflashing, with slotted fastener holes at 16 inch centers between upper and lower edge.

f. Pipe Counterflashing:

(1) Form flashing for water-tight umbrella with upper portion against pipe to receive a draw band and upper edge to form a "V" joint sealant receiver approximately 3/4 inch deep.

(2) Fabricate 4 inch over lap at end.

(3) Fabricate draw band of same metal as counter flashing. Use (24 oz) copper or 0.013 inch thick stainless steel or copper coated stainless steel.

(4) Use stainless steel bolt on draw band tightening assembly.

(5) Vent pipe counter flashing may be fabricated to omit draw band and turn down (one inch) inside vent pipe.

g. Where vented edge decks intersect vertical surfaces, form in one piece, shape to slope down to a point level with and in front of edge-set notched plank; then, down vertically, overlapping base flashing.

2.9 HANGING GUTTERS

a. Fabricate gutters of not less than the following:

(3) (0.032 inch) thick aluminum.

b. Fabricate hanging gutters in sections not less than 8 feet long, except at ends of runs where shorter lengths are required.

c. Building side of gutter shall be same height as exterior side.

d. Gutter Bead: Stiffen outer edge of gutter by folding edge over approximately 3/4 inch toward roof and down approximately 3/4 inch unless shown otherwise.

e. Gutter Spacers:

(1) Fabricate of same material and thickness as gutter.

(2) Fabricate (one inch) wide strap and fasten to gutters not over 36 inches on center.

(3) Turn back edge up (one inch) and lap front edge over gutter bead.

(4) Rivet and solder to gutter except rivet and seal to aluminum.

f. Outlet Tubes:

(1) Form outlet tubes to connect gutters to conductors of same metal and thickness as gutters extend into the conductor (3 inch). Flange upper end of outlet tube (1/2 inch).

(2) Lock and solder longitudinal seam except use sealant in lieu of solder with aluminum.

(3) Solder tube to gutter.Seal aluminum tube to gutter and rivet to gutter.

(4) Fabricate basket strainers of same material as gutters.

g. Gutter Brackets:

(1) Fabricate of same metal as gutter. Use the following:

(c) (1/4 by 1 inch) aluminum.

(2) Fabricate to gutter profile.

(3) Drill two 3/16 inch diameter holes in anchor leg for countersunk flat head screws.

2.10 CONDUCTORS (DOWNSPOUTS)

a. Fabricate conductors of same metal and thickness as gutters in sections approximately 10 feet long with 3/4 inch wide flat locked seams.

(1) Fabricate open face channel shape with hemmed longitudinal edges.

b. Fabricate elbows by mitering, riveting, and soldering except seal aluminum in lieu of solder. Lap upper section to the inside of the lower piece.

c. Fabricate conductor brackets or hangers of same material as conductor, 1/16 inch thick by (one inch) minimum width. Form to support conductors (one inch) from wall surface in accordance with Architectural Sheet Metal Manual for rectangular and round shapes.

d. Conductor Heads:

(1) Fabricate of same material as conductor.

(2) Fabricate conductor heads to not less than 10 inch wide by 8 inch deep by 8 inches from front to back.

(3) Form front and side edges channel shape not less than 1/2 inch wide flanges with edge hemmed.

(4) Slope bottom to sleeve to conductor or downspout at not less than 60 degree angle.

(5) Extend wall edge not less than (one inch) above front edge.

(6) Solder joints for water tight assembly.

(7) Fabricate outlet tube or sleeve at bottom not less than 2 inches long to insert into conductor.

2.11 INSULATED EXPANSION JOINT COVERS

a. Either type optional, use only one type throughout.

b. Types:

(1) Construct of two preformed, stainless steel strips, not less than 0.015 inch thick, mechanically and adhesively bonded to both sides of a 1/16 inch thick neoprene or butyl sheet, or to a (32 mil) thick reinforced chlorinated polyethylene sheet. Adhesively attach a 3/8 inch thick sheet of closed cell, neoprene foam insulation, to the underside of the neoprene, butyl, or chlorinated polyethylene sheet.

(2) Constructed of a 1/16 inch thick vinyl sheet, flanged at both sides with stainless steel strips not less than 0.015 inch thick. Vinyl sheet locked and encased by the stainless steel strip and prepunched for nailing. A 3/8 inch thick closed cell polyvinyl chloride foam insulating strip shall be heat laminated to the underside of the vinyl sheet between the stainless steel strips.

c. Expansion joint covers shall have factory fabricated mitered corners, crossing tees, and other necessary accessories. Furnish in the longest available lengths.

d. Metal flange of sufficient width to extend over the top of the curb and down curb sides 2 inches with hemmed edge for lock to edge strip.

PART 3 EXECUTION

3.1 INSTALLATION

General:

a. Install flashing and sheet metal items as shown in Sheet Metal and Air Conditioning Contractors National Association, Inc., publication, ARCHITECTURAL SHEET METAL MANUAL, except as otherwise shown or specified.

b. Apply Sealant as specified in Section 07 92 00, JOINT SEALANTS.

c. Apply sheet metal and other flashing material to surfaces which are smooth, sound, clean, dry and free from defects that might affect the application.

d. Remove projections which would puncture the materials and fill holes and depressions with material compatible with the substrate. Cover holes or cracks in wood wider than 1/4 inch with sheet metal compatible with the roofing and flashing material used.

e. Coordinate with masonry work for the application of a skim coat of

mortar to surfaces of unit masonry to receive flashing material before the application of flashing.

f. Apply a layer of 15 pound saturated felt followed by a layer of rosin paper to wood surfaces to be covered with copper. Lap each ply 2 inch with the slope and nail with large headed copper nails.

g. Confine direct nailing of sheet metal to strips 12 inch or less wide. Nail flashing along one edge only. Space nail not over 4 inches on center unless specified otherwise.

h. Install bolts, rivets, and screws where indicated, specified, or required in accordance with the SMACNA Sheet Metal Manual. Space rivets at 3 inch on centers in two rows in a staggered position. Use neoprene washers under fastener heads when fastener head is exposed.

i. Coordinate with roofing work for the installation of metal base flashings and other metal items having roof flanges for anchorage and watertight installation.

j. Nail continuous cleats on 3 inch on centers in two rows in a staggered position.

k. Nail individual cleats with two nails and bend end tab over nail heads. Lock other end of cleat into hemmed edge.

l. Install flashings in conjunction with other trades so that flashings are inserted in other materials and joined together to provide a water tight installation.

m. Where required to prevent galvanic action between dissimilar metal isolate the contact areas of dissimilar metal with sheet lead, waterproof building paper, or a coat of bituminous paint.

n. Isolate aluminum in contact with dissimilar metals others than stainless steel, white bronze or other metal compatible with aluminum by:

(1) Paint dissimilar metal with a prime coat of zinc-chromate or other suitable primer, followed by two coats of aluminum paint.

(2) Paint dissimilar metal with a coat of bituminous paint.

(3) Apply an approved caulking material between aluminum and dissimilar metal.

o. Paint aluminum in contact with or built into mortar, concrete, plaster, or other masonry materials with a coat of bituminous paint.

p. Paint aluminum in contact with absorptive materials that may become repeatedly wet with two coats of bituminous paint or two coats of aluminum paint.

q. Bitumen Stops:

(1) Install bitumen stops for built-up roof opening penetrations through deck and at formed sheet metal gravel stops.

(2) Nail leg of bitumen stop at 12 inch intervals to nailing strip at

roof edge before roofing material is installed.

3.2 THROUGH-WALL FLASHING

a. General:

(1) Install continuous through-wall flashing between top of concrete foundation walls and bottom of masonry building walls; at top of concrete floors; under masonry, concrete, or stone copings and elsewhere as shown.

(2) Where exposed portions are used as a counterflashings, lap base flashings at least 4 inches and use thickness of metal as specified for exposed locations.

(3) Exposed edge of flashing may be formed as a receiver for two piece counter flashing as specified.

(4) Terminate exterior edge beyond face of wall approximately 1/4 inch with drip edge where not part of counter flashing.

(5) Turn back edge up 1/4 inch unless noted otherwise where flashing terminates in mortar joint or hollow masonry unit joint.

(6) Terminate interior raised edge in masonry backup unit approximately (1 1/2 inch) into unit unless shown otherwise.

(7) Under copings terminate both edges beyond face of wall approximately 1/4 inch with drip edge.

(8) Lap end joints at least two corrugations, but not less than (4 inches). Seal laps with sealant.

(9) Where dowels, reinforcing bars and fastening devices penetrate flashing, seal penetration with sealing compound. Sealing compound is specified in Section 07 92 00, JOINT SEALANTS.

(10) Coordinate with other work to set in a bed of mortar above and below flashing so that total thickness of the two layers of mortar and flashing are same as regular mortar joint.

(11) Where ends of flashing terminate turn ends up 1 inch and fold corners to form dam extending to wall face in vertical mortar or veneer joint.

(12) Turn flashing up not less than 8 inch between masonry or behind exterior veneer.

(13) When flashing terminates in reglet extend flashing full depth into reglet and secure with lead or plastic wedges spaced 6 inch on center.

(14) Continue flashing around columns:

(a) Where flashing cannot be inserted in column reglet hold flashing vertical leg against column.

(b) Counterflash top edge with 3 inch wide strip of saturated cotton unless shown otherwise. Secure cotton strip with roof

cement to column. Lap base flashing with cotton strip (1 1/2 inch).

b. Flashing at Top of Concrete Foundation Walls Where concrete is exposed. Turn up not less than 8 inch high and into masonry backup mortar joint or reglet in concrete backup as specified.

c. Flashing at Top of Concrete Floors (except where shelf angles occur): Place flashing in horizontal masonry joint not less than 8 inch below floor slab and extend into backup masonry joint at floor slab (1 1/2 inch).

d. Flashing at Cavity Wall Construction: Where flashing occurs in cavity walls turn vertical portion up against backup under waterproofing, if any, into mortar joint. Turn up over insulation, if any, and horizontally through insulation into mortar joint.

e. Flashing at Veneer Walls:

(1) Install near line of finish floors over shelf angles or where shown.

(2) Turn up against sheathing.

(3) At stud framing, hem top edge 3/4 inch and secure to each stud with stainless steel fasteners through sheathing.

(4) At concrete backing, extend flashing into reglet as specified.

(5) Coordinate with installation of waterproofing or asphalt felt for lap over top of flashing.

f. Lintel Flashing when not part of shelf angle flashing:

(1) Install flashing full length of lintel to nearest vertical joint in masonry over veneer.

(2) Turn ends up (one inch) and fold corners to form dam and extend end to face of wall.

(3) Turn back edge up to top of lintel; terminate back edge as specified for back-up wall.

g. Window Sill Flashing:

(1) Install flashing to extend not less than 4 inch beyond ends of sill into vertical joint of masonry or veneer.

(2) Turn back edge up to terminate under window frame.

(3) Turn ends up (one inch) and fold corners to form dam and extend to face of wall.

h. Door Sill Flashing:

(1) Install flashing under bottom of plate sills of doors over curbs opening onto roofs. Extend flashing out to form counter flashing or receiver for counter flashing over base flashing. Set in sealant.

(2) Extend sill flashing 8 inch beyond jamb opening. Turn ends up one inch in vertical masonry joint, extend end to face of wall. Join to counter flashing for water tight joint.

(3) Where doors thresholds cover over waterproof membranes install sill flashing over water proof membrane under thresholds. Extend beyond opening to cover exposed portion of waterproof membrane and not less than 6 inch beyond door jamb opening at ends. Turn up approximately 1/4 inch under threshold.

i. Flashing at Masonry, Stone, or Precast Concrete Copings:

(1) Install flashing with drips on both wall faces unless shown otherwise.

(2) Form penetration openings to fit tight against dowel or other item with edge turned up. Seal penetrations with sealant.

3.3 BASE FLASHING

a. Install where roof membrane type base flashing is not used and where shown.

(1) Install flashing at intersections of roofs with vertical surfaces or at penetrations through roofs, to provide watertight construction.

(2) Install metal flashings and accessories having flanges extending out on top of the built-up roofing before final bituminous coat and roof aggregate is applied.

(3) Set flanges in heavy trowel coat of roof cement and nail through flanges into wood nailers over bituminous roofing.

(4) Secure flange by nailing through roofing into wood blocking with nails spaced 3 inch on centers or, when flange over 4 inch wide terminate in a 1/2 inch folded edge anchored with cleats spaced 8 inch on center. Secure one end of cleat over nail heads. Lock other end into the seam.

b. For long runs of base flashings install in lengths of not less than 8 feet nor more than (ten feet). Install a 3 inch wide slip type, loose lock expansion joint filled with sealant in joints of base flashing sections over 8 feet in length. Lock and solder corner joints at corners.

c. Extend base flashing up under counter flashing of roof specialties and accessories or equipment not less than (3 inch).

3.4 COUNTERFLASHING (CAP FLASHING OR HOODS)

a. General:

(1) Install counterflashing over and in conjunction with installation of base flashings, except as otherwise specified or shown.

(2) Install counterflashing to lap base flashings not less than (4 inch).

(3) Install upper edge or top of counterflashing not less than 9 inch above top of the roofing.

(4) Lap joints not less than (4 inch). Stagger joints with relation to metal base flashing joints.

(5) Use surface applied counterflashing on existing surfaces and new work where not possible to integrate into item.

(6) When fastening to concrete or masonry, use screws driven in expansion shields set in concrete or masonry. Use screws to wood and sheet metal. Set fasteners in mortar joints of masonry work.

b. One Piece Counterflashing:

(1) Where flashing is installed at new masonry, coordinate to insure proper height, embed in mortar, and end lap.

(2) Where flashing is installed in reglet in concrete insert upper edge into reglet. Hold flashing in place with lead wedges spaced not more than 8 inch apart. Fill joint with sealant.

(3) Where flashing is surface mounted on flat surfaces.

(a) When top edge is double folded anchor flat portion below sealant "V" joint with fasteners spaced not over 16 inch on center:

1. Locate fasteners in masonry mortar joints.

2. Use screws to sheet metal or wood.

(b) Fill joint at top with sealant.

(4) Where flashing or hood is mounted on pipe.

(a) Secure with draw band tight against pipe.

(b) Set hood and secure to pipe with a one by x (1 x 1/8 inch) bolt on stainless steel draw band type clamp, or a stainless worm gear type clamp.

(c) Completely fill joint at top with sealant.

c. Two-Piece Counterflashing:

(1) Where receiver is installed at new masonry coordinate to insure proper height, embed in mortar, and lap.

(2) Surface applied type receiver:

(a) Secure to face construction in accordance, with manufacturers' instructions.

(b) Completely fill space at the top edge of receiver with sealant.

(3) Insert counter flashing in receiver in accordance with fabricator or manufacturer's instructions and to fit tight against base flashing.

d. Where vented edge occur install so lower edge of counterflashing is against base flashing.

e. When counter flashing is a component of other flashing install as shown.

3.5 COPINGS

a. General:

(1) On walls topped with a wood plank, install a continuous edge strip on the front and rear edge of the plank. Lock the coping to the edge strip with a 3/4 inch loose lock seam.

(2) Where shown turn down roof side of coping and extend down over base flashing as specified for counter-flashing. Secure counter-flashing to lock strip in coping at continuous cleat.

(3) Install ends adjoining existing construction so as to form space for installation of sealants. Sealant is specified in Section 07 92 00, JOINT SEALANTS.

b. Aluminum Coping:

(1) Install with 1/4 inch joint between ends of coping sections.

(2) Install joint covers, centered at each joint, and securely lock in place.

3.6 EXPANSION JOINT COVERS, INSULATED

a. Install insulated expansion joint covers at locations shown on curbs not less than 8 inch high above roof surface.

b. Install continuous edge strips of same metal as expansion joint flange, nailed at not less than 3 inch centers.

c. Install insulated expansion joint covers in accordance with manufacturer's directions locking edges to edge strips.

3.7 HANGING GUTTERS

a. Hang gutters with high points equidistant from downspouts. Slope at not less than (1/16 inch per foot).

b. Lap joints, except for expansion joints, at least in the direction of flow. Rivet and seal or solder lapped joints.

c. Support gutters in brackets spaced not more than 24 inch on centers, brackets attached to facial or wood nailer by at least two screws or nails.

(3) For aluminum gutters use aluminum brackets or stainless steel brackets.

(4) Use brass or stainless steel screws.

d. Secure brackets to gutters in such a manner as to allow free movement of gutter due to expansion and contraction.

e. Gutter Expansion Joint:

(1) Locate expansion joints midway between outlet tubes.

(2) Provide at least a (one inch) expansion joint space between end baffles of gutters.

(3) Install a cover plate over the space at expansion joint.

(4) Fasten cover plates to gutter section on one side of expansion joint only.

(5) Secure loose end of cover plate to gutter section on other side of expansion joint by a loose-locked slip joint.

f. Outlet Tubes: Set bracket strainers loosely into gutter outlet tubes.

3.8 CONDUCTORS (DOWNSPOUTS)

a. Where scuppers discharge into downspouts install conductor head to receive discharge with back edge up behind drip edge of scupper. Fasten and seal joint. Sleeve conductors to gutter outlet tubes and fasten joint and joints between sections.

b. Set conductors plumb and clear of wall, and anchor to wall with two anchor straps, located near top and bottom of each section of conductor. Strap at top shall be fixed to downspout, intermediate straps and strap at bottom shall be slotted to allow not less than 1/2 inch movement for each 10 feet of downspout.

c. Install elbows, offsets and shoes where shown and required. Slope not less than 45 degrees.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 71 00

ROOF SPECIALTIES

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY CONTROL
1.4 PERFORMANCE REQUIREMENTS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 UNDERLAYMENT
2.3 SOLDER
2.4 COPINGS
2.5 EXTRUDED ALUMINUM GRAVEL STOPS AND FASCIAS
2.6 EXTRUDED ALUMINUM FASCIA-CANT SYSTEM
2.7 EXTRUDED ALUMINUM ROOF EXPANSION JOINT COVERS
2.8 FINISH

PART 3 EXECUTION

3.1 INSTALLATION
3.2 PROTECTION OF ALUMINUM
3.3 ADJUSTING
3.4 PROTECTION

-- End of Section Table of Contents --

SECTION 07 71 00
1/21

ROOF SPECIALTIES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies copings, gravel stops, fascias, and expansion joints.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

b. Section 07 21 13, THERMAL INSULATION: General Insulation.

c. Section 07 22 00, ROOF AND DECK INSULATION: Rigid Insulations for Roofing.

d. Section 07 92 00, JOINT SEALANTS: Sealant Material and Installation.

1.3 QUALITY CONTROL

a. Provide roof accessories that products of manufacturers regularly engaged in producing the kinds of products specified.

b. For each accessory type provide products made by the same manufacturer.

c. Assemble each accessory to the greatest extent possible before delivery to the site.

d. Provide each accessory with FM approval listing for class specified.

1.4 PERFORMANCE REQUIREMENTS

a. Provide roof accessories that withstand exposure to weather and resist thermal movement without failure, rattling, leaking, or fastener disengagement due to defective manufacture, fabrication, or installation.

b. Manufacture and install roof accessories to allow for thermal movements from ambient and surface temperature changes to prevent buckling, opening of joints, hole elongation, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects.

(1) Provide clips that resist rotation and avoid shear stress as a result of thermal movements.

(2) For design purposes, base provisions for thermal movement on assumed ambient temperature (range) from (0 degrees F), ambient to (180 degrees F).

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

SD-03 Product Data

Sustainable Design Submittals

Manufacturer's Literature And Data; G

SD-04 Samples

Samples

SD-07 Certificates

Certificates; G

1.6 SUBMITTAL INSTRUCTIONS

a. Sustainable Design Submittals, as described below:

(1) Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.

b. Samples: Representative sample panel of color-anodized aluminum not less than x (4 x 4 inches), except extrusions are to be of a width not less than section to be used. Submit sample that shows coating with integral color and texture. Include manufacturer's identifying label.

c. Shop Drawings: Each item specified showing design, details of construction, installation and fastenings.

d. Manufacturer's Literature and Data: Each item specified.

e. Certificates: Stating that aluminum has been given specified thickness of anodizing.

1.7 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A666	(2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B370	(2022) Standard Specification for Copper Sheet and Strip for Building Construction
ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM D1970/D1970M	(2019) Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection
ASTM D226/D226M	(2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing
ASTM D4869/D4869M	(2016a) Standard Specification for Asphalt-Saturated Organic Felt Underlayment Used in Steep Slope Roofing

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605	(2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic

PART 2 PRODUCTS

2.1 MATERIALS

a. Aluminum, Extruded:

ASTM B221M ASTM B221.

b. Aluminum Sheet: ASTM B209M ASTM B209.

c. Galvanized Sheet Steel: ASTM A653/A653M; G-90 coating.

d. Stainless-Steel Sheet: ASTM A240/A240M or ASTM A666, Type 304.

e. Copper Sheet: ASTM B370, cold-rolled copper sheet, H00 or H01 temper.

f. Recycled Content of Metal Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 30 percent.

g. Insulation: ASTM C612, Class 1 or 2.

h. Asphalt Coating: ASTM D1187/D1187M, Type I, quick setting.

2.2 UNDERLAYMENT

a. Self-Adhering Modified Bitumen Underlayment:

(1) Provide self-adhering modified bitumen membrane underlayment material in compliance with ASTM D1970/D1970M, suitable for use as underlayment for metal copings and fascias.

(2) Provide membrane resistant to cyclical elevated temperatures for extended period of time in high heat service conditions (stable after testing at (240 degrees F)).

(3) Provide membrane with integral non-tacking top surface of polyethylene film or other surface material to serve as separator between bituminous material and metal products to be applied above.

(4) Provide primer.

b. Felt Underlayment: Provide No. 30 asphalt saturated organic, non-perforated felt underlayment in compliance with ASTM D226/D226M, Type II, or ASTM D4869/D4869M.

c. Slip Sheet: Provide (5 pounds per 100 sf) rosin sized unsaturated building paper for slip sheet.

2.3 SOLDER

Copper Solder conforming to ASTM B32, lead-free solder Grade Sn50, 50 percent tin and 50 percent lead.

2.4 COPINGS

a. Fabricate of aluminum sheet not less than (0.125 (0.018 inch) thick;

stainless steel

b. Turn outer edges down each face of wall as shown on construction documents.

c. Maximum lengths of (10 feet).

d. Shop fabricate external and internal corners as one-piece assemblies with not less than 12 inch leg lengths.

e. Provide 4 inch wide 0.032 inch thick watertight joint covers.

f. Provide anchor gutter bar of 0.032 inch thick with anchor holes formed for underside of joint.

g. Provide concealed guttered splice plate of 0.032 inch thick with butyl or other resilient seal strips anchored to splice plate for underside of joint. Use galvanized steel anchor plate providing compression spring anchoring of coping cover.

h. Finish: Two-coat fluoropolymer Color as specified.

2.5 EXTRUDED ALUMINUM GRAVEL STOPS AND FASCIAS

a. Fabricate of aluminum not less than 0.078 inch thick.

b. Turn fascia down face of wall and up above roof as shown in construction documents.

c. Maximum lengths of (10-feet).

d. Shop fabricate external and internal corners as one (1)-piece assemblies with not less than 12 inch leg lengths.

e. Provide 4 inch wide 0.078 inch thick watertight joint covers with 6 inch wide 0.030 inch thick underside joint flashing.

f. Finish: Two-coat fluoropolymer Color as specified.

2.6 EXTRUDED ALUMINUM FASCIA-CANT SYSTEM

a. The fascia-cant system consists of three (3) pieces, an extruded aluminum fascia, a galvanized steel cant, and an aluminum compression clamp.

b. Furnish in stock lengths of not more than 10 feet long.

c. Form fascia from not less than 0.070 inch thick aluminum. Provide 4 inch wide (0.032-inch) thick concealed sheet aluminum joint cover plates in back of fascia.

d. Form cant strip from galvanized steel not less than (0.0299 inch) thick, to profile shown and design to hold lower edge of the fascia.

e. Form compression clamp of not less than 0.032 inch thick aluminum designed to hold the top edge of the fascia and the built-up flashing.

f. Internal and external corners:

(1) Factory fabricate and fully weld mitered joints.

(2) Furnish corner sections in manufacturers standard sizes with not less than 12 inch leg lengths.

g. Factory fabricated fascia sump assemblies.

(1) Fabricate sump assemblies with stainless steel cores and extruded aluminum cover to match fascia-cant.

(2) Provide stainless steel outlet, tube sized to suit downspout and solder to core to make watertight.

(3) Furnish sump assembly in 20 inch minimum lengths.

h. Factory fabricated scupper assemblies:

(1) Fabricate scupper assembly with extended plates to match fascia-cant in 20 inch minimum lengths.

(2) Extend outlet opening not less than 2 inches with drip edge.

(3) Fabricate with stainless steel core or sleeve to drain water from toe of cant and flash in to built-up roofing with 4 inch wide flange.

i. Finish on aluminum: Two-coat fluoropolymer Color as specified.

2.7 EXTRUDED ALUMINUM ROOF EXPANSION JOINT COVERS

a. Fabricate in 10 foot lengths with fastener openings slotting for expansion not over 24 inch centers.

b. Provide four-way expansion, for joint widths shown on construction documents.

c. Mill finish.

d. Form waterstop or moisture seals of continuous sheets of neoprene, not less than 0.032 inch thick.

e. Fabricate corners as one (1) piece assembly with mitered and welded joint and least dimension legs not less than 12 inches long.

f. Factory fabricate end caps and transitions to insure waterproof assembly.

g. Five (5) piece assembly:

(1) Roof expansion joint cover system consists of an extruded aluminum cover, extruded frame or curb vertical section, galvanized steel cant, and aluminum compression clamp counter flashing, complete with moisture seals. Form cover and vertical section from extruded aluminum, 0.080 inch minimum thickness with spring stainless steel tension or pivot bar.

(2) Form cant from galvanized steel not less than 0.029 inch thick formed to profile shown on construction documents.

(3) Form splice plates of not less than 0.032 inch thick aluminum sheet.

(4) Form counter flashing member of 0.050 inch thick sheet aluminum, secured with screws to the top edge of the vertical section and providing compression clamp over base flashing.

(5) Provide compression gasket separating cover from curb bearing.

h. Two (2) piece assembly:

(1) Roof expansion joint system consists of an extruded aluminum cover combination extruded aluminum frame or curb with integral adjustable counter flashing flange, and moisture seals.

(2) Form cover from extruded aluminum 0.078 inch minimum thickness.

(3) Form cover anchor system of stainless steel pivot bar.

(4) Form frame assembly of not less than 0.076 inch aluminum except for flashing portion.

(5) Provide compression gasket separating cover from curb at bearing.

2.8 FINISH

a. In accordance with NAAMM AMP 500-505.

b. Fluoropolymer Finishes: High performance organic coating. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

(1) Two-Coat Fluoropolymer: AAMA 2605. Fluoropolymer finish containing not less than 70 percent PVDF resin by weight in color coat.

PART 3 EXECUTION

3.1 INSTALLATION

a. Examine substrates, areas, and conditions, to verify actual locations, dimensions, and other conditions affecting performance of the Work.

b. Verify that substrate is sound, dry, smooth, clean, sloped for drainage where applicable, and securely anchored.

c. Underlayment Installation:

(1) Self-Adhering Sheet Underlayment:

(a) Apply primer as required by manufacturer.

(b) Comply with temperature restrictions of underlayment manufacturer for installation.

(c) Apply wrinkle free, in shingle fashion to shed water, and with end laps of not less than (6 inches) staggered 24 inches between courses.

(d) Overlap side edges not less than (3-1/2 inches). Roll laps with roller.

(e) Cover underlayment within 14 days.

(f) Apply continuously under copings and roof-edge fascias and gravel stops.

(g) Coordinate application of self-adhering sheet underlayment under roof specialties with requirements for continuity with adjacent air barrier materials.

(2) Felt Underlayment:

(a) Install with adhesive for temporary anchorage to minimize use of mechanical fasteners under roof specialties.

(b) Apply in shingle fashion to shed water, with lapped joints of not less than (2 inches).

(3) Slip Sheet:

(a) Install with tape or adhesive for temporary anchorage to minimize use of mechanical fasteners under roof specialties.

(b) Apply in shingle fashion to shed water, with lapped joints of not less than (2 inches).

d. Install roof accessories where indicated in construction documents.

e. Secure with fasteners in accordance with manufacture's printed installation instructions and approved shop drawings unless shown otherwise. Provide fasteners suitable for application, for metal types being secured and designed to meet performance requirements.

f. Where soldered joints are required, clean surfaces to be soldered, removing oils and foreign matter.

(1) Pre-tin edges of sheets to be soldered to a width of (1-1/2 inches).

(2) Reduce pre-tinning where pre-tinned surface would show in completed work.

(3) Do not use torches for soldering.

(4) Heat surfaces to receive solder and flow solder into joint.

(5) Fill joint completely.

(6) Completely remove flux and spatter from exposed surfaces.

g. Coordinate to install insulation where shown; see Section 07 21 13, THERMAL INSULATION and Section 07 22 00, ROOF AND DECK INSULATION.

h. Comply with section 07 92 00, JOINT SEALANTS to install sealants where required by manufactures installation instructions.

i. Coordinate with roofing work for installation of items in sequence to prevent water infiltration.

j. Gravel Stops and Fascias:

(1) Install gravel stops and fascia with butt joints with approximately 1/4 inch space for expansion.

(2) Over each joint provide cover plates of sheet aluminum, complete with concealed sheet aluminum flashing, centered under each joint.

(3) Provide lap cover plates and concealed flashing over the gravel stop and fascia not less than (4 inches).

(4) Extend concealed flashing over built-up roofing, embed in roof cement and turn down over face of blocking at roof edge.

k. Aluminum Coping:

(1) Install sections of coping with approximately (1/4-inch) space between ends of sections.

(2) Center joint gutter bar and covers at joints and lock in place.

(3) When snap-on system is installed ensure front and back edges are locked in place.

l. Fascia-Cant System:

(1) Install galvanized steel cant; coordinate with roofing work and after completion of roofing work install extruded aluminum fascia, concealed joint cover plate, and aluminum compression clamp, where shown in construction documents.

(2) Install system to allow for expansion and contraction with (1/4 inch) space between extruded aluminum members and galvanized steel cant as required by manufacturer of system.

(3) Offset joints in extruded aluminum members from galvanized steel cant joints.

m. Expansion Joint Covers:

(1) Install to terminate base flashing 8 inches above roof.

(2) Install moisture seals to drain water to outlets that do not permit water to enter building.

(3) Provide stainless steel screws when exposed.

(4) Three piece assembly:

(a) Install curb section with screws to wood blocking, allowing 1/4 inch at butt joints between sections with splice plate at joint.

(b) Install cant to wood blocking by nailing along horizontal flange every (6 inches), with galvanized roofing nails 1 inch long.

(c) After completion of base flashing install cap flashing and compression clamp and fasten to the curb or metal cant with

stainless steel self-tapping screws with neoprene washers under head spaced approximately 18 inches on center.

(d) Install expansion joint cover with a 1/4 inch wide end joints.

(e) Install over end joint a cover plate complete with concealed aluminum flashing, centered under each joint. Fabricate flashing to lap cover not less than (4 inches).

(5) Two piece assembly:

(a) Install curb section with screws allowing 1/4 inch space at end joints with splice plate at joint.

(b) After completion of base flashing bend down cap flashing flange and secure to blocking with screws.

(c) Install expansion joint cover with 1/4 inch wide space at end joints and tension bars at 24 inches on center.

(d) Install cover plates with formed aluminum flashing concealed and centered on joint. Flashing to lap cover not less than (4 inches).

3.2 PROTECTION OF ALUMINUM

a. Provide protection for aluminum against galvanic action wherever dissimilar materials are in contact, by painting the contact surfaces of the dissimilar material with two (2) coats of asphalt coating (complete coverage), or by separating the contact surfaces with a preformed neoprene tape having pressure sensitive adhesive coating on one (1) side.

b. Paint aluminum in contact with wood, concrete and masonry, or other absorptive materials, that may become repeatedly wet, with two (2) coats of asphalt coating.

3.3 ADJUSTING

Adjust expansion joints to close tightly and be watertight; insuring maximum allowance for building movement.

3.4 PROTECTION

Protect roof accessories from damage during installation and after completion of the work from subsequent construction.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 72 00

ROOF ACCESSORIES

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY ASSURANCE
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 ROOF HATCH (SCUTTLE)
2.3 EQUIPMENT SUPPORTS
2.4 LOW SILHOUETTE GRAVITY VENTILATORS
2.5 METAL GRATING ROOF WALKWAY SYSTEM
2.6 FINISH

PART 3 EXECUTION

3.1 INSTALLATION
3.2 PROTECTION OF ALUMINUM
3.3 ADJUSTING
3.4 PROTECTION

-- End of Section Table of Contents --

SECTION 07 72 00
1/21

ROOF ACCESSORIES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies roof hatches; equipment supports; gravity ventilators; and metal grating roof walkway system.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

b. Section 07 21 13, THERMAL INSULATION: General insulation.

c. Section 07 22 00, ROOF AND DECK INSULATION: Rigid insulations for roofing.

d. Section 07 92 00, JOINT SEALANTS: Sealant material and installation.

1.3 QUALITY ASSURANCE

a. Provide roof accessories that are the products of manufacturers regularly engaged in producing the kinds of products specified.

b. For each accessory type provide the same product made by the same manufacturer.

c. Assemble each accessory to the greatest extent possible before delivery to the site.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

SD-03 Product Data

Sustainable Design Submittals

Manufacturer's Literature And Data; G

SD-04 Samples

Samples; G

SD-07 Certificates

Certificates; G

1.5 SUBMITTAL INSTRUCTIONS

a. Sustainable Design Submittals, as described below:

(1) Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.

b. Samples: Submit representative sample panel of color anodized aluminum not less than (4 x 4 inches). For extrusions, submit width not less than section to be installed. Show coating with integral color and texture and include manufacturer's identifying label.

c. Shop Drawings: Each item specified showing design, details of construction, installation and fastenings.

d. Manufacturer's Literature and Data: Each item specified.

e. Certificates: Stating that aluminum has been given specified thickness of anodizing.

1.6 REFERENCES

The publications listed below form a part of this specification to the extend referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM C726	(2024) Standard Specification for Mineral Wool Roof Insulation Board
ASTM C1289	(2023a) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board
ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as

Protective Coatings for Metal

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500 (2006) Metal Finishes Manual

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605 (2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.23 (Nov 2016) Ladders

PART 2 PRODUCTS

2.1 MATERIALS

a. Aluminum, Extruded: ASTM B221M, ASTM B221.

b. Aluminum Sheet: ASTM B209M, ASTM B209.

c. Galvanized Sheet Steel: ASTM A653/A653M; G-90 coating.

d. Recycled Content of Metal Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 30 percent.

e. Asphalt Coating: ASTM D1187/D1187M, Type I, quick setting.

2.2 ROOF HATCH (SCUTTLE)

a. Performance Characteristics:

(1) Cover to be reinforced to support a minimum live load of (40 pounds per square foot) with a maximum deflection of 1/150th of the span or (20 pounds per square foot) wind uplift.

(2) Operation of the Cover: Smooth and easy with controlled operation throughout the entire arc of opening and closing.

(3) Operation of the Cover: Not affected by temperature.

(4) Entire Hatch: Weathertight with fully welded corner joints on cover and curb.

b. Shop fabricate from aluminum with mill finish.

c. Curb and Cover:

(1) Exterior facing: Minimum 0.09 inch thick sheet aluminum with mill finish.

(2) Interior facing: Minimum 0.04 inch thick sheet aluminum.

(3) Minimum of 2 inch thick polyisocyanurate insulation (ASTM C1289) with a U-value = 0.47 W/mK (R-value = 12) between facings of cover and over exterior face of curb.

(4) Form exterior curb facing with an integral 3 inch wide roof flange and cap flashing minimum 0.09 inch thick sheet aluminum.

(5) Make curb 12 inches above finish roof surface.

(6) Form cover to lap curb and cap flashing.

(7) Size opening as shown on construction documents.

(8) Finish: as indicated.; color as indicated.

d. Hardware:

(1) Provide spring snap latch with inside and outside operating handles and padlock hasp on inside. Provide two snap latches when hinge side is over 7 feet long. Bolt hardware into heavy gauge channel reinforcement welded to the underside of the cover and concealed within the insulation space.

(2) Provide heavy duty pintle hinges.

(3) Provide automatic hold open and operating arm with enclosed torsion or compression spring lifting mechanism.

(4) Latch Strike: Stamped component bolted or welded to the curb assembly.

(5) Automatically lock in the open position at not less than 70 degrees.

(6) Provide weather stripping at cover closure.

(7) Galvanize all hardware items.

e. Assembly:

(1) Shop assemble roof scuttle.

(2) Weld joints exposed to the weather and built into the roofing.

(3) Finish weld smooth where exposed.

f. Safety Accessories:

(1) Ladder Assist Post: Provide a telescoping tubular section that locks automatically when fully extended. Control upward and downward movement by a stainless steel spring balancing mechanism. Provide unit completely assembled with fasteners for securing to the ladder rungs in accordance with the manufacturer’s instructions.

(2) Safety Railing: Provide a fixed, attached to the roof hatch railing assembly including rails, clamps, fasteners, safety barrier at railing opening, and accessories required for a complete installation; complying with 29 CFR 1910.23 requirements.

2.3 EQUIPMENT SUPPORTS

a. Supported Load Capacity: as indicated.

b. Fabricate equipment supports from 0.0516 inch thick galvanized ASTM A653/A653M steel fabricate with welded corners and with seams joined by continuous water and air tight welds.

c. Equipment supports to be internally reinforced with angles (48 inches) on center.

d. Form exterior curb with integral base, and deck closures for curbs installed on steel decking.

e. Use galvanized steel liners for curbs having inside dimension over (12 inches).

f. Internally insulate with (1-1/2 inch) glass-fiber board insulation (ASTM C726).

g. Fabricate curb with a minimum height of 8 inches above roof surface.

h. Attach preservative treated wood nailers to top of curb. Provide 2 inch by 2 inch minimum nominal size on curb with openings and 2 inch thick, width of curb up to 12 inches on equipment support curbs.

i. Make size of supports suit size of equipment furnished, with height as shown on construction documents, but not less than 8 inches above roof surface.

j. Top of Equipment Supports: Level with pitch built into curb when deck slopes. Equip supports with water diverter or cricket on side that obstructs water flow.

k. Finish: as indicated; color as indicated.

2.4 LOW SILHOUETTE GRAVITY VENTILATORS

a. Fabricate base of 0.04 inch thick aluminum, and vent of 0.032 inch thick aluminum.

(1) Height not to exceed 12 inches above top of roof curb.

(2) Design ventilators to withstand (85 miles) per hour wind velocity.

(3) Provide ventilators with a removable 18 by 18 mesh by (0.11 inch) diameter aluminum wire cloth insect screen.

(4) Provide security grille where indicated on construction documents.

b. Construct damper of the same material as the ventilator and design to completely close opening or remain wide open. Hold damper in closed

position by a brass chain and catch. Extend chains 12 inches below and engage catch when damper is closed.

c. Finish: as indicated; color as indicated.

2.5 METAL GRATING ROOF WALKWAY SYSTEM

a. Provide metal grating roof walkway system consisting of prefabricated pans, of 14 gauge, galvanized (G-90 Coating) steel grating with slip resistant surface.

b. Grating units to be in 2 foot widths and in (10 to 12 foot long) sections as required.

c. Provide complete with support framing, brackets, connectors, nosings and other accessories as required for complete roof walkway system.

(1) Include support stands at minimum 5 feet on center to hold planks a minimum of 9 inches above roof surface.

(2) Provide wind restraint attachment to roof structure of size and spacing required to meet wind uplift requirements.

d. Include step units, nosings framing and connectors to provide changes in elevation as required. Comply with ASCE 7-16 and 29 CFR 1910.23.

e. Equip walkways with safety railings where required by 29 CFR 1910.23.

f. Provide neoprene rubber pads having a shore A hardness of 80 to 90-Durometer under each support, or bearing surface.

g. Finish: as indicated; color as indicated.

2.6 FINISH

a. In accordance with NAAMM AMP 500 Series.

f. Fluoropolymer Finish: High performance organic coating. Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturer’s written instructions.

(1) Two-Coat Fluoropolymer Finish: AAMA 2605. System consisting of primer and fluoropolymer color topcoat containing not less than 70 percent PVDF resin by weight.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install roof specialties where indicated on construction documents.

b. Secure with fasteners in accordance with manufacture's printed installation instructions and approved shop drawings unless shown otherwise.

c. Coordinate to install insulation where shown; see Section 07 21 13, THERMAL INSULATION and Section 07 22 00, ROOF AND DECK INSULATION.

d. Comply with section 07 92 00, JOINT SEALANTS to install sealants where

required by manufactures installation instructions require sealant.

e. Coordinate with roofing work for installation of items in sequence to prevent water infiltration.

(1) After completion of base flashing bend down cap flashing flange and secure to blocking with screws.

(2) Install expansion joint cover with 1/4 inch wide space at end joints and tension bars at 24 inches on center.

(3) Install cover plates with formed aluminum flashing concealed and centered on joint. Flashing to lap cover not less than (4 inches).

f. Equipment Supports: Do not anchor to insulating concrete or metal deck. Anchor only to building structure as per manufacturers recommendations.

3.2 PROTECTION OF ALUMINUM

a. Provide protection for aluminum against galvanic action wherever dissimilar materials are in contact, by painting the contact surfaces of the dissimilar material with two (2) coats of asphalt coating (complete coverage), or by separating the contact surfaces with a preformed neoprene tape having pressure sensitive adhesive coating on side.

b. Paint aluminum in contact with wood, concrete and masonry, or other absorptive materials, that may become repeatedly wet, with two coats of asphalt coating.

3.3 ADJUSTING

Adjust roof hatch hardware to operate freely and so that cover will operate without binding, close tightly at perimeter, and latch securely.

3.4 PROTECTION

Protect roof accessories from damage during installation and after completion of the work from subsequent construction.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 81 00

APPLIED FIREPROOFING

7/23

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 PRODUCT DELIVERY, STORAGE AND HANDLING
1.6 FIELD CONDITIONS
1.7 QUALITY ASSURANCE
1.8 REFERENCES

PART 2 PRODUCTS

2.1 SPRAYED-ON FIREPROOFING
2.2 ADHESIVE
2.3 SEALER
2.4 WATER
2.5 MECHANICAL BOND MATERIAL

PART 3 EXECUTION

3.1 EXAMINATION
3.2 APPLICATION
3.3 FIELD TESTS
3.4 PATCHING AND REPAIRING

-- End of Section Table of Contents --

SECTION 07 81 00
7/23

APPLIED FIREPROOFING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies spray-applied mineral fiber and cementitious coverings to provide fire resistance to interior structural steel members shown.

1.2 RELATED WORK

Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G

Testing Laboratory Accreditations; G

SD-03 Product Data

Sustainable Design Submittals

Manufacturer's Literature And Data; G

SD-07 Certificates

Certificates; G

SD-08 Manufacturer's Instructions

Miscellaneous; G

1.4 SUBMITTAL INSTRUCTIONS

a. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

b. Installer qualifications.

c. Testing laboratory accreditations.

d. Manufacturer's Literature and Data:

(1) Manufacturer's complete and detailed application instructions and specifications.

(2) Manufacturer's repair and patching instructions.

e. Certificates:

(1) Certificate from testing laboratory attesting fireproofing material and application method meet the specified fire ratings.

(a) List thickness and density of material required to meet fire ratings.

(b) Accompanied by complete test report and test record.

(2) Manufacturer's certificate indicating sprayed-on fireproofing material supplied under the Contract is same within manufacturing tolerance as fireproofing material tested.

f. Miscellaneous:

(1) Manufacturer's written approval of surfaces to receive sprayed-on fireproofing.

(2) Manufacturer's written approval of completed installation.

(3) Manufacturer's written approval of the applicators of fireproofing material.

1.5 PRODUCT DELIVERY, STORAGE AND HANDLING

a. Deliver to job-site in sealed containers marked and labeled to show manufacturer's name and brand and UL certification markings of compliance with the specified requirements.

b. Remove damaged or opened containers from the site.

c. Store the materials off the ground, under cover, away from damp surfaces.

d. Keep dry until ready for use.

e. Remove materials that have been exposed to water before installation from the site.

1.6 FIELD CONDITIONS

a. Temperature: Do not apply fireproofing when substrate or ambient temperature is below (40 degrees F) unless temporary protection and heat are provided to maintain temperature at or above stated value during application and for 24 hours before and after application.

b. Humidity: Maintain relative humidity levels within limits recommended by fireproofing manufacturer.

c. Ventilation: Provide ventilation to properly dry the fireproofing after application. Provide a minimum of four (4) air exchanges per hour by forced air circulation. When permitted by Contracting Officer ,

ventilate by natural circulation.

1.7 QUALITY ASSURANCE

a. Installer Qualifications: A firm or individual certified, licensed, or otherwise qualified by fireproofing manufacturer as experienced and with sufficient trained staff to install manufacturer’s products according to specified requirements. Submit manufacturer’s certification that each installer is trained and qualified to install the specified fireproofing. Submit evidence that each installer has a minimum of three (3) years’ experience and a minimum of four (4) installations using the specified fireproofing.

b. Testing Laboratory Accreditation Requirements: Construction materials testing laboratories must be accredited by a laboratory accreditation authority. Submit a copy of the Certificate of Accreditation and Scope of Accreditation.

c. Test for fire endurance in accordance with ASTM E119, for fire rating specified, in a nationally recognized laboratory.

d. Manufacturer's inspection and approval of surfaces to receive fireproofing.

e. Manufacturer's approval of fireproofing applications.

f. Manufacturer's approval of completed installation.

g. Manufacturer's representative is to observe and advise at the commencement of application, and is required to visit the site as required thereafter for the purpose of ascertaining proper application.

h. Pre-Application Test Area.

(1) Apply a test area consisting of a typical overhead fireproofing installation, including not less than 15 feet of beam and deck.

(a) Apply to one (1) column.

(b) Apply for the hourly ratings required in the construction documents.

(2) Install in location selected by the Contracting Officer, for approval by the representative of the fireproofing material manufacturer and the Contracting Officer.

(3) Perform Bond test for cohesive and adhesive strength in accordance with ASTM E736 for each applied fireproofing design used.

(4) Perform density test in accordance with ASTM E736 for each applied fireproofing design used.

(5) Do not proceed in other areas until installation of test area has been completed and approved.

(6) Keep approved installation area open for observation as criteria for sprayed-on fireproofing.

1.8 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C841	(2023) Installation of Interior Lathing and Furring
ASTM C847	(2018; R 2024) Standard Specification for Metal Lath
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E605	(1993; R 2011) Thickness and Density of Sprayed Fire-Resistive Material (SFRM) Applied to Structural Members
ASTM E736	(2000; R 2011) Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members
ASTM E759/E759M	(1992; R 2023) Standard Test Method for Effect of Deflection on Sprayed Fire-Resistive Material Applied to Structural Members
ASTM E760/E760M	(1992; R 2023) Standard Test Method for Effect of Impact on Bonding of Sprayed Fire-Resistive Material Applied to Structural Members
ASTM E761/E761M	(1992; R 2023) Standard Test Method for Compressive Strength of Sprayed Fire-Resistive Material Applied to Structural Members
ASTM E859/E859M	(2023) Standard Test Method for Air Erosion of Sprayed Fire-Resistive Materials (SFRMs) Applied to Structural Members
ASTM E937/E937M	(1992; R 2023) Standard Test Method for Corrosion of Steel by Sprayed Fire-Resistive Material (SFRM) Applied to Structural Members
ASTM E1042	(2022) Standard Classification for Acoustically Absorptive Materials Applied by Trowel or Spray

ASTM G21 (2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59 (1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

EPA Method 24 (2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide https://www.approvalguide.com/PART 2

PRODUCTS

2.1 SPRAYED-ON FIREPROOFING

a. ASTM E1042, Class (a), Category A.

(1) Type I, factory mixed cementitious materials with approved aggregate.

b. Materials containing asbestos are not permitted.

c. Fireproofing characteristics when applied in the thickness and density required to achieve the fire-rating specified.

	Characteristic	Test	Results
1.	Deflection	ASTM E759/E7	No cracking, spalling, or delamination when backing to which it is applied has a deflection up to (10 ft.)
2.	Corrosion-Resistance	ASTM E937/E9	No promotion of corrosion of steel.
3.	Bond Impact	ASTM E760/E7	No cracking, spalling, or delamination.
4.	Cohesion/Adhesion (Bond Strength)	ASTM E736	Minimum cohesive/adhesive strength of (200 lbf per sq. ft.) for protected areas. (400 lbf per sq. ft.) for exposed areas.
5.	Air Erosion	ASTM E859/E8	Maximum gain weight of the collecting filter (0.025 gm per sq. ft.).

6.	Compressive Strength	ASTM E761/E7(	Minimum compressive strength (1000 psf).
7.	Surface Burning Characteristics with adhesive and sealer to be used	ASTM E84	Flame spread 25 or less smoke developed 50 or less
8.	Fungi Resistance	ASTM G21	Resistance to mold growth when inoculated with aspergillus niger (28 days for general application)

2.2 ADHESIVE

a.adhesive may be an integral part of the material or applied separately to surface receiving fireproofing material.

2.3 SEALER

a. Surface burning characteristics as specified for fireproofing material.

b. Fungus resistant.

c. Sealer may be an integral part of the material or applied separately to the exposed surface. When applied separately use contrasting color pigmented sealer, white preferred.

d. VOC content: Product to comply with VOC content limits of authorities having jurisdiction and the following VOC limits when calculated according to 40 CFR 59, (EPA Method 24):

(1) Flat Paints and Coatings: 50 g/L.

(2) Nonflat Paints and Coatings: 150 g/L.

(3) Primers, Sealers, and Undercoaters: 200 g/L.

2.4 WATER

a. Clean, fresh, and free from organic and mineral impurities.

b. pH of 6.9 to 7.1.

2.5 MECHANICAL BOND MATERIAL

a. Expanded Metal Lath: ASTM C847, minimum weight of (1.7 pounds per square yard) or as required, according to fire-resistance designs indicated and fire proofing manufacturer’s written instructions.

b. Fasteners: ASTM C841.

c. Reinforcing Fabric: Glass- or carbon-fiber fabric of type, weight, and form required to comply with fire-resistance designs indicated; approved and provided by fireproofing manufacturer.

d. Reinforcing Mesh: Metallic mesh reinforcement of type, weight, and

form required to comply with fire-resistance design indicated; approved and provided by fireproofing manufacturer. Include pins and attachments.

PART 3 EXECUTION

3.1 EXAMINATION

a. Verify surfaces to receive fireproofing are clean and free of dust, soot, oil, grease, water soluble materials or any foreign substance which would prevent adhesion of the fireproofing material.

b. Verify hangers, inserts and clips are installed before the application of fireproofing material.

c. Verify ductwork, piping, and other obstructing material and equipment is not installed that will interfere with fireproofing installation.

d. Verify concrete work on steel decking and concrete encased steel is completed.

e. When applied in conjunction with roof structures or roof decks, verify that roofing, installation of rooftop HVAC equipment, and other related work are complete.

f. Verify temperature and enclosure conditions required by fire-proofing material manufacturer.

g. Conduct tests according to fireproofing manufacturer’s written instructions to verify that substrates are free of substances capable of interfering with bond. Submit test report.

3.2 APPLICATION

a. Do not start application until written approval has been obtained from manufacturer of fireproofing materials that surfaces have been inspected by the manufacturer or his representative, and are suitable to receive sprayed-on fireproofing.

b. Coordinate application of fireproofing material with other trades.

c. Cover other work and exterior openings subject to damage from fallout or overspray of fireproofing materials during application.

d. Application of Metal Lath:

(1) Apply to beam and columns having painted surfaces which fail ASTM E736 Bond Test requirements in pre-application test area.

(2) Apply to beam flanges (12-inches) or more in width.

(3) Apply to column flanges (16-inches) or more in width.

(4) Apply to beam or column web (16-inches) or more in depth.

(5) Tack weld or mechanically fasten on maximum of (12-inch) center.

(6) Lap and tie lath member in accordance with ASTM C841.

e. Mix and apply in accordance with manufacturer's instructions.

(1) Mechanically control material and water ratios.

(2) Apply adhesive and sealer, when not an integral part of the materials, in accordance with the manufacturer's instructions.

(3) Apply to density and thickness indicated in FM APP GUIDE unless specified otherwise. Test in accordance with ASTM E119.

(4) Minimum ASTM E605 applied dry density per (cubic foot) for the underside of the walk on deck (interstitial) hung purlin or beam and steel deck, columns in interstitial spaces and mechanical equipment rooms to be as follows:

(a) Factory mixed cemitious material - (22 lb. per cubic ft.).

(b) Provide materials with higher density of (40 lb. per cubic feet) in mechanical rooms and parking garages.

c. Complete application is to be completed in one area. Inspection and approval by Contracting Officer is required before removal of application equipment and proceeding with further work.

3.3 FIELD TESTS

a. The applied fireproofing to be tested by a Contractring Officer approved independent testing laboratory and paid for by the Contractor. Submit test reports documenting results of tests on the applied material in the project.

b. Contracting Officer will select area to be tested in specific bays on each floor using a geometric grid pattern. Apply test sample every (10,000 square feet) of floor area or two (2) for each floor, whichever produces the greatest number of test areas.

c. Test for thickness and density in accordance with ASTM E605. Areas showing thickness less than that required as a result of fire endurance test are not acceptable.

d. Areas showing less than required fireproofing characteristics are not suitable for the following field tests.

(1) Test for cohesion/adhesion: ASTM E736.

(2) Test for bond impact strength: ASTM E760/E760M.

3.4 PATCHING AND REPAIRING

a. Inspect after mechanical, electrical and other trades have completed work in contact with fireproofing material, but before sprayed material is covered by subsequent construction.

b. Perform corrective measures in accordance with fireproofing material manufacturer's recommendations.

(1) Respray areas requiring additional fireproofing material to provide the required thickness, and replace dislodged or removed material.

(2) Spray material for patching by machine directly on point to be patched, or into a container and then hand apply.

(3) Do not hand mix material.

c. Repair:

(1) Respray test and rejected areas.

(2) Patch fireproofing material which is removed or disturbed after approval.

d. Perform final inspection of sprayed areas after patching and repair.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 84 00

FIRESTOPPING

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY AND STORAGE
1.6 QUALITY ASSURANCE
1.7 REFERENCES

PART 2 PRODUCTS

2.1 FIRESTOP SYSTEMS
2.2 SMOKE STOPPING IN SMOKE PARTITIONS

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 INSTALLATION
3.4 CLEAN-UP
3.5 INSPECTIONS AND ACCEPTANCE OF WORK

-- End of Section Table of Contents --

SECTION 07 84 00
1/21

FIRESTOPPING

PART 1 GENERAL

1.1 DESCRIPTION

a. Provide UL or equivalent approved firestopping system for the closures of openings in walls, floors, and roof decks against penetration of flame, heat, and smoke or gases in fire resistant rated construction.

b. Provide UL or equivalent approved firestopping system for the closure of openings in walls against penetration of gases or smoke in smoke partitions.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

b. Section 07 95 13, EXPANSION JOINT COVER ASSEMBLIES: Expansion and seismic joint firestopping.

c. Section 07 81 00, APPLIED FIREPROOFING: Spray applied fireproofing.

d. Section 07 92 00, JOINT SEALANTS: Sealants and application.

e. Section 23 31 00, HVAC DUCTS AND CASINGS: Fire and smoke damper assemblies in ductwork.

f. Section 23 37 00, AIR OUTLETS AND INLETS: Fire and smoke damper assemblies in ductwork.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G

Inspector Qualifications; G

SD-03 Product Data

Sustainable Design Submittals

Manufacturers Literature, Data, And Installation Instructions; G

SD-05 Design Data

List Of Fm, Ul, Or Wh Classification Number; G

SD-06 Test Reports

Certified Laboratory Test Reports For Astm E814 Tests; G

SD-07 Certificates

Certificates From Manufacturer; G

1.4 SUBMITTAL INSTRUCTIONS

a. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

b. Installer qualifications.

c. Inspector qualifications.

d. Manufacturers literature, data, and installation instructions for types of firestopping and smoke stopping used.

e. List of FM, UL, or WH classification number of systems installed.

f. Certified laboratory test reports for ASTM E814 tests for systems not listed by FM, UL, or WH proposed for use.

g. Submit certificates from manufacturer attesting that firestopping materials comply with the specified requirements.

1.5 DELIVERY AND STORAGE

a. Deliver materials in their original unopened containers with manufacturer’s name and product identification.

b. Store in a location providing protection from damage and exposure to the elements.

1.6 QUALITY ASSURANCE

a. FM, UL, or WH or other approved laboratory tested products will be acceptable.

b. Installer Qualifications: A firm that has been approved by FM Global according to FM 4991 or been evaluated by UL and found to comply with UL’s “Qualified Firestop Contractor Program Requirements.” Submit qualification data.

c. Inspector Qualifications: Contractor to engage a qualified inspector to perform inspections and final reports. The inspector to meet the criteria contained in ASTM E699 for agencies involved in quality assurance and to have a minimum of two years’ experience in construction field inspections of firestopping systems, products, and assemblies. The inspector to be completely independent of, and divested from, the Contractor, the installer, the manufacturer, and the supplier of material or item being inspected. Submit inspector qualifications.

1.7 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E699	(2009) Standard Practice for Evaluation of Agencies Involved in Testing, Quality Assurance, and Evaluating of Building Components
ASTM E814	(2024) Standard Test Method for Fire Tests of Penetration Firestop Systems
ASTM E2174	(2024) Standard Practice for On-Site Inspection of Installed Firestop Systems
ASTM E2393	(2024) Standard Practice for On-Site Inspection of Installed Fire Resistive Joint Systems and Perimeter Fire Barriers

FM GLOBAL (FM)

FM 4991	(2013) Approval of Firestop Contractors

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products
EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

PART 2 PRODUCTS

2.1 FIRESTOP SYSTEMS

a. Provide either factory built (Firestop Devices) or field erected (through-Penetration Firestop Systems) to form a specific building system maintaining required integrity of the fire barrier and stop the passage of gases or smoke. Firestop systems to accommodate building movements without impairing their integrity.

b. Through-penetration firestop systems and firestop devices tested in accordance with ASTM E814 using the "F" or "T" rating to maintain the same rating and integrity as the fire barrier being sealed. "T" ratings are not required for penetrations smaller than or equal to 4 inches nominal pipe or (16 square inches) in overall cross sectional area.

c. Firestop sealants used for firestopping or smoke sealing to have the following properties:

(1) Contain no flammable or toxic solvents.

(2) Release no dangerous or flammable out gassing during the drying or curing of products.

(3) Water-resistant after drying or curing and unaffected by high humidity, condensation or transient water exposure.

(4) When installed in exposed areas, capable of being sanded and finished with similar surface treatments as used on the surrounding wall or floor surface.

(5) VOC Content: Firestopping sealants and sealant primers to comply with the following limits for VOC content when calculated according to 40 CFR 59, (EPA Method 24):

(a) Sealants: 250 g/L.

(b) Sealant Primers for Nonporous Substrates: 250 g/L.

(c) Sealant Primers for Porous Substrates: 775 g/L.

d. Firestopping system or devices used for penetrations by glass pipe, plastic pipe or conduits, unenclosed cables, or other non-metallic materials to have following properties:

(1) Classified for use with the particular type of penetrating material used.

(2) Penetrations containing loose electrical cables, computer data cables, and communications cables protected using firestopping systems that allow unrestricted cable changes without damage to the seal.

e. Maximum flame spread of 25 and smoke development of 50 when tested in accordance with ASTM E84. Material to be an approved firestopping material as listed in UL Fire Resistance Directory or by a nationally recognized testing laboratory.

f. FM, UL, or WH rated or tested by an approved laboratory in accordance with ASTM E814.

g. Materials to be nontoxic and noncarcinogen at all stages of application or during fire conditions and to not contain hazardous chemicals. Provide firestop material that is free from Ethylene Glycol, PCB, MEK, and asbestos.

h. For firestopping exposed to view, traffic, moisture, and physical damage, provide products that do not deteriorate when exposed to these conditions.

(1) For piping penetrations for plumbing and wet-pipe sprinkler systems, provide moisture-resistant through-penetration firestop systems.

(2) For floor penetrations with annular spaces exceeding (4 inches) or

more in width and exposed to possible loading and traffic, provide firestop systems capable of supporting the floor loads involved either by installing floor plates or by other means acceptable to the firestop manufacturer.

(3) For penetrations involving insulated piping, provide through-penetration firestop systems not requiring removal of insulation.

2.2 SMOKE STOPPING IN SMOKE PARTITIONS

a. Provide silicone sealant in smoke partitions as specified in Section 07 92 00, JOINT SEALANTS.

b. Provide mineral fiber filler and bond breaker behind sealant.

c. Sealants to have a maximum flame spread of 25 and smoke developed of 50 when tested in accordance with ASTM E84.

d. When used in exposed areas capable of being sanded and finished with similar surface treatments as used on the surrounding wall or floor surface.

PART 3 EXECUTION

3.1 EXAMINATION

a. Submit product data and installation instructions, as required by article, submittals, after an on-site examination of areas to receive firestopping.

b. Examine substrates and conditions with installer present for compliance with requirements for opening configuration, penetrating items, substrates, and other conditions affecting performance of firestopping. Do not proceed with installation until unsatisfactory conditions have been corrected.

3.2 PREPARATION

a. Remove dirt, grease, oil, laitance and form-release agents from concrete, loose materials, or other substances that prevent adherence and bonding or application of the firestopping or smoke stopping materials.

b. Remove insulation on insulated pipe for a distance of (6 inches) on each side of the fire rated assembly prior to applying the firestopping materials unless the firestopping materials are tested and approved for use on insulated pipes.

c. Prime substrates where required by joint firestopping system manufacturer using that manufacturer’s recommended products and methods. Confine primers to areas of bond; do not allow spillage and migration onto exposed surfaces.

d. Masking Tape: Apply masking tape to prevent firestopping from contacting adjoining surfaces that will remain exposed upon completion of work and that would otherwise be permanently stained or damaged by such contact or by cleaning methods used to remove smears from firestopping materials. Remove tape as soon as it is possible to do

so without disturbing seal of firestopping with substrates.

3.3 INSTALLATION

a. Do not begin firestopping work until the specified material data and installation instructions of the proposed firestopping systems have been submitted and approved.

b. Install firestopping systems with smoke stopping in accordance with FM, UL, WH, or other approved system details and installation instructions.

c. Install smoke stopping seals in smoke partitions.

3.4 CLEAN-UP

a. As work on each floor is completed, remove materials, litter, and debris.

b. Clean up spills of liquid type materials.

c. Clean off excess fill materials and sealants adjacent to openings and joints as work progresses by methods and with cleaning materials approved by manufacturers of firestopping products and of products in which opening and joints occur.

d. Protect firestopping during and after curing period from contact with contaminating substances or from damage resulting from construction operations or other causes so that they are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated firestopping immediately and install new materials to provide firestopping complying with specified requirements.

3.5 INSPECTIONS AND ACCEPTANCE OF WORK

a. Do not conceal or enclose firestop assemblies until inspection is complete and approved by the Contracting Officer.

b. Furnish service of approved inspector to inspect firestopping in accordance with ASTM E2393 and ASTM E2174 for firestop inspection, and document inspection results. Submit written reports indicating locations of and types of penetrations and type of firestopping used at each location; type is to be recorded by UL listed printed numbers.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 92 00

JOINT SEALANTS

4/22

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK (INCLUDING BUT NOT LIMITED TO THE FOLLOWING)
1.3 QUALITY ASSURANCE
1.4 CERTIFICATION
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 PROJECT CONDITIONS
1.8 DELIVERY, HANDLING, AND STORAGE
1.9 DEFINITIONS
1.10 WARRANTY
1.11 REFERENCES

PART 2 PRODUCTS

2.1 SEALANTS
2.2 COLOR
2.3 JOINT SEALANT BACKING
2.4 WEEPS
2.5 FILLER
2.6 PRIMER
2.7 CLEANERS-NON POROUS SURFACES

PART 3 EXECUTION

3.1 INSPECTION
3.2 PREPARATIONS
3.3 BACKING INSTALLATION
3.4 SEALANT DEPTHS AND GEOMETRY
3.5 INSTALLATION
3.6 FIELD QUALITY CONTROL
3.7 CLEANING

-- End of Section Table of Contents --

SECTION 07 92 00
4/22

JOINT SEALANTS

PART 1 GENERAL

1.1 DESCRIPTION

This section covers interior and exterior sealant and their application, wherever required for complete installation of building materials or systems.

1.2 RELATED WORK (INCLUDING BUT NOT LIMITED TO THE FOLLOWING)

a. Sustainable Design Requirements: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Masonry Control and Expansion Joint: Section 04 20 00, UNIT MASONRY.

c. Firestopping Penetrations: Section 07 84 00, FIRESTOPPING.

d. Glazing: Section 08 80 00, GLAZING.

e. Glazed Aluminum Curtain Wall: Section 08 44 13, GLAZED ALUMINUM CURTAIN WALLS.

1.3 QUALITY ASSURANCE

a. Installer Qualifications: An experienced installer with a minimum of three (3) years' experience and who has specialized in installing joint sealants similar in material, design, and extent to those indicated for this Project and whose work has resulted in joint-sealant installations with a record of successful in-service performance. Submit qualification.

b. Source Limitations: Obtain each type of joint sealant through one (1) source from a single manufacturer.

c. Product Testing: Obtain test results from a qualified testing agency based on testing current sealant formulations within a 12-month period.

(1) Testing Agency Qualifications: An independent testing agency qualified according to ASTM C1021.

(2) Test elastomeric joint sealants for compliance with requirements specified by reference to ASTM C920, and where applicable, to other standard test methods.

(3) Test elastomeric joint sealants according to SWRI's Sealant Validation Program for compliance with requirements specified by reference to ASTM C920 for adhesion and cohesion under cyclic movement, adhesion-in peel, and indentation hardness.

(4) Test other joint sealants for compliance with requirements indicated by referencing standard specifications and test methods.

d. Lab Tests: Submit samples of materials that will be in contact or affect joint sealants to joint sealant manufacturers for tests as follows:

(1) Adhesion Testing: Before installing elastomeric sealants, test their adhesion to protect joint substrates according to the method in ASTM C794 to determine if primer or other specific joint preparation techniques are required.

(2) Compatibility Testing: Before installing elastomeric sealants, determine compatibility when in contact with glazing and gasket materials.

(3) Stain Testing: Perform testing per ASTM C1248 on interior and exterior sealants to determine if sealants or primers will stain adjacent surfaces. No sealant work is to start until results of these tests have been submitted to the Contracting Officer and the Contracting Officer has given written approval to proceed with the work.

e. Preconstruction Field-Adhesion Testing: Before installing elastomeric sealants, field test their adhesion to joint substrates according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1.1 in ASTM C1193 or Method A, Tail Procedure, in ASTM C1521.

(1) Locate test joints where indicated in construction documents or, if not indicated, as directed by Contracting Officer.

(2) Conduct field tests for each application indicated below:

(a) Each type of elastomeric sealant and joint substrate indicated.

(b) Each type of non-elastomeric sealant and joint substrate indicated.

(3) Notify Contracting Officer seven (7) days in advance of dates and times when test joints will be erected.

(4) Arrange for tests to take place with joint sealant manufacturer’s technical representative present.

f. Mockups: Before installing joint sealants, apply elastomeric sealants as follows to verify selections and to demonstrate aesthetic effects and qualities of materials and execution:

(1) Joints in mockups of assemblies that are indicated to receive elastomeric joint sealants.

1.4 CERTIFICATION

Contractor is to submit to the Contracting Officer written certification that joints are of the proper size and design, that the materials supplied are compatible with adjacent materials and backing, that the materials will properly perform to provide permanent watertight, airtight or vapor tight seals (as applicable), and that materials supplied meet specified performance requirements.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G

SD-03 Product Data

Sustainable Design Submittals

Manufacturer's Literature And Data; G

SD-04 Samples

Cured Samples Of Exposed Sealants For Each Color; G

SD-07 Certificates

Contractor Certification; G

SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions For Each Product Used; G

SD-11 Closeout Submittals

Environmental Product Declaration (EPD); S

Third-Party Certification Certificate, Assessment, Or Validation And Compliance Report; S

1.6 SUBMITTAL INSTRUCTIONS

a. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

b. Installer qualifications.

c. Contractor certification.

d. Manufacturer's installation instructions for each product used.

e. Cured samples of exposed sealants for each color.

f. Manufacturer's Literature and Data:

(1) Primers

(2) Sealing compound, each type, including compatibility when different sealants are in contact with each other.

1.7 PROJECT CONDITIONS

a. Environmental Limitations:

(1) Do not proceed with installation of joint sealants under following conditions:

(a) When ambient and substrate temperature conditions are outside limits permitted by joint sealant manufacturer or are below (40 degrees F).

(b) When joint substrates are wet.

b. Joint-Width Conditions:

(1) Do not proceed with installation of joint sealants where joint widths are less than those allowed by joint sealant manufacturer for applications indicated.

c. Joint-Substrate Conditions:

(1) Do not proceed with installation of joint sealants until contaminants capable of interfering with adhesion are removed from joint substrates.

1.8 DELIVERY, HANDLING, AND STORAGE

a. Deliver materials in manufacturers' original unopened containers, with brand names, date of manufacture, shelf life, and material designation clearly marked thereon.

b. Carefully handle and store to prevent inclusion of foreign materials.

c. Do not subject to sustained temperatures exceeding (90 degrees F) or less than (40 degrees F).

1.9 DEFINITIONS

a. Definitions of terms in accordance with ASTM C717 and as specified.

b. Backing Rod: A type of sealant backing.

c. Bond Breakers: A type of sealant backing.

d. Filler: A sealant backing used behind a back-up rod.

1.10 WARRANTY

a. Construction Warranty: Comply with FAR clause 52.246-21 "Warranty of Construction".

b. Manufacturer Warranty: Manufacturer shall warranty their sealant for a minimum of five (5) years from the date of installation and final acceptance by the Government. Submit manufacturer warranty.

1.11 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C509	(2006; R 2021) Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material
ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C717	(2019) Standard Terminology of Building Seals and Sealants
ASTM C734	(2015; R 2024) Standard Test Method for Low-Temperature Flexibility of Latex Sealants After Artificial Weathering
ASTM C794	(2022) Standard Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants
ASTM C919	(2024) Standard Practice for Use of Sealants in Acoustical Applications
ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM C1021	(2008; R 2014) Standard Practice for Laboratories Engaged in Testing of Building Sealants
ASTM C1193	(2016) Standard Guide for Use of Joint Sealants
ASTM C1248	(2022) Test Method for Staining of Porous Substrate by Joint Sealants
ASTM C1330	(2023) Standard Specification for Cylindrical Sealant Backing for Use with Cold Liquid-Applied Sealants
ASTM C1521	(2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints
ASTM D217	(2021a) Standard Test Methods for Cone Penetration of Lubricating Grease
ASTM D1056	(2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59 (1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

PART 2 PRODUCTS

2.1 SEALANTS

a. Exterior Sealants:

(1) Vertical surfaces, provide non-staining ASTM C920, Type S or M, Grade NS, Class 25, , Use NT.

(2) Horizontal surfaces, provide ASTM C920, Type S or M, Grade P, Class 25, Use T.

(3) Provide location(s) of exterior sealant as follows:

(a) Joints formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Provide sealant at exterior surfaces of exterior wall penetrations.

(b) Metal to metal.

(c) Masonry to masonry or stone.

(d) Stone to stone.

(e) Cast stone to cast stone.

(f) Masonry expansion and control joints.

(g) Wood to masonry.

(h) Masonry joints where shelf angles occur.

(i) Voids where items penetrate exterior walls.

(j) Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.

b. Floor Joint Sealant:

(1) ASTM C920, Type S or M, Grade P, Class 25, , Use T.

(2) Provide location(s) of floor joint sealant as follows.

(a) Seats of metal thresholds exterior doors.

(b) Control and expansion joints in floors, slabs, ceramic tile, and walkways.

c. Interior Sealants:

(1) VOC Content of Interior Sealants: Sealants and sealant primers used inside the weatherproofing system are to comply with the following limits for VOC content when calculated according to 40 CFR 59, (EPA Method 24):

(a) Architectural Sealants: 250 g/L.

(b) Sealant Primers for Nonporous Substrates: 250 g/L.

(c) Sealant Primers for Porous Substrates: 775 g/L.

(2) Vertical and Horizontal Surfaces: ASTM C920, Type S or M, Grade NS, Class 25, , Use NT.

(3) Food Service: Use a Vinyl Acetate Homopolymer, or other low VOC, non-toxic sealant approved for use in food preparation areas.

(4) Provide location(s) of interior sealant as follows:

(a) Typical narrow joint , 1/4 inch or less at walls and adjacent components.

(b) Perimeter of doors, windows, access panels which adjoin concrete or masonry surfaces.

(c) Interior surfaces of exterior wall penetrations.

(d) Joints at masonry walls and columns, piers, concrete walls or exterior walls.

(e) Perimeter of lead faced control windows and plaster or gypsum wallboard walls.

(f) Exposed isolation joints at top of full height walls.

(g) Joints between bathtubs and ceramic tile; joints between shower receptors and ceramic tile; joints formed where nonplanar tile surfaces meet.

(h) Joints formed between tile floors and tile base cove; joints between tile and dissimilar materials; joints occurring where substrates change.

(i) Behind escutcheon plates at valve pipe penetrations and showerheads in showers.

d. Acoustical Sealant:

(1) Conforming to ASTM C919; flame spread of 25 or less; and a smoke developed rating of 50 or less when tested in accordance with ASTM E84. Acoustical sealant have a consistency of 250 to 310 when tested in accordance with ASTM D217; remain flexible and adhesive after 500 hours of accelerated weathering as specified in ASTM C734; and be non-staining.

(2) Provide location(s) of acoustical sealant as follows:

(a) Exposed acoustical joint at sound rated partitions.

(b) Concealed acoustic joints at sound rated partitions.

(c) Joints where item pass-through sound rated partitions.

2.2 COLOR

a. Sealants used with exposed masonry are to match color of mortar joints.

b. Sealants used with unpainted concrete are to match color of adjacent concrete.

c. Color of sealants for other locations to be light gray or aluminum, unless otherwise indicated in construction documents.

2.3 JOINT SEALANT BACKING

a. General: Provide sealant backings of material and type that are nonstaining; are compatible with joint substrates, sealants, primers, and other joint fillers; and are approved for applications indicated by sealant manufacturer based on field experience and laboratory testing.

b. Cylindrical Sealant Backings: ASTM C1330, of type indicated below and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance:

(1) Type C: Closed-cell material with a surface skin.

c. Elastomeric Tubing Sealant Backings: Neoprene, butyl, EPDM, or silicone tubing complying with ASTM D1056 or synthetic rubber (ASTM C509), nonabsorbent to water and gas, and capable of remaining resilient at temperatures down to (minus 26 degrees F). Provide products with low compression set and of size and shape to provide a secondary seal, to control sealant depth, and otherwise contribute to optimum sealant performance.

d. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint-filler materials or joint surfaces at back of joint where such adhesion would result in sealant failure. Provide self-adhesive tape where applicable.

2.4 WEEPS

Weep/Vent Products: Provide the following unless otherwise indicated or approved.

(1) Round Plastic Tubing: Medium-density polyethylene, (3/8-inch) OD by thickness of stone or masonry veneer.

2.5 FILLER

a. Mineral fiberboard: ASTM C612, Class 1.

b. Thickness same as joint width.

c. Depth to fill void completely behind back-up rod.

2.6 PRIMER

a. As recommended by manufacturer of caulking or sealant material.

b. Stain free type.

2.7 CLEANERS-NON POROUS SURFACES

Chemical cleaners compatible with sealant and acceptable to manufacturer of sealants and sealant backing material. Cleaners to be free of oily residues and other substances capable of staining or harming joint substrates and adjacent non-porous surfaces and formulated to promote adhesion of sealant and substrates.

PART 3 EXECUTION

3.1 INSPECTION

a. Inspect substrate surface for bond breaker contamination and unsound materials at adherent faces of sealant.

b. Coordinate for repair and resolution of unsound substrate materials.

c. Inspect for uniform joint widths and that dimensions are within tolerance established by sealant manufacturer.

3.2 PREPARATIONS

a. Prepare joints in accordance with manufacturer's instructions and SWRI (The Professionals’ Guide).

b. Clean surfaces of joint to receive caulking or sealants leaving joint dry to the touch, free from frost, moisture, grease, oil, wax, lacquer paint, or other foreign matter that would tend to destroy or impair adhesion.

(1) Clean porous joint substrate surfaces by brushing, grinding, blast cleaning, mechanical abrading, or a combination of these methods to produce a clean, sound substrate capable of developing optimum bond with joint sealants.

(2) Remove loose particles remaining from above cleaning operations by vacuuming or blowing out joints with oil-free compressed air. Porous joint surfaces include but are not limited to the following:

(a) Concrete.

(b) Masonry.

(c) Unglazed surfaces of ceramic tile.

(3) Remove laitance and form-release agents from concrete.

(4) Clean nonporous surfaces with chemical cleaners or other means that do not stain, harm substrates, or leave residues capable of interfering with adhesion of joint sealants. Nonporous surfaces include but are not limited to the following:

(a) Metal.

(b) Glass.

(c) Porcelain enamel.

(d) Glazed surfaces of ceramic tile.

c. Do not cut or damage joint edges.

d. Apply non-staining masking tape to face of surfaces adjacent to joints before applying primers, caulking, or sealing compounds.

(1) Do not leave gaps between ends of sealant backings.

(2) Do not stretch, twist, puncture, or tear sealant backings.

(3) Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.

e. Apply primer to sides of joints wherever required by compound manufacturer's printed instructions or as indicated by pre-construction joint sealant substrate test.

(1) Apply primer prior to installation of back-up rod or bond breaker tape.

(2) Use brush or other approved means that will reach all parts of joints. Avoid application to or spillage onto adjacent substrate surfaces.

3.3 BACKING INSTALLATION

a. Install backing material, to form joints enclosed on three sides as required for specified depth of sealant.

b. Where deep joints occur, install filler to fill space behind the backing rod and position the rod at proper depth.

c. Cut fillers installed by others to proper depth for installation of backing rod and sealants.

d. Install backing rod, without puncturing the material, to a uniform depth, within plus or minus 1/8 inch for sealant depths specified.

e. Where space for backing rod does not exist, install bond breaker tape strip at bottom (or back) of joint so sealant bonds only to two opposing surfaces.

3.4 SEALANT DEPTHS AND GEOMETRY

a. At widths up to (1/4 inch), sealant depth equal to width.

b. At widths over (1/4 inch), sealant depth 1/2 of width up to 1/2 inch maximum depth at center of joint with sealant thickness at center of joint approximately 1/2 of depth at adhesion surface.

3.5 INSTALLATION

a. General:

(1) Apply sealants and caulking only when ambient temperature is between

(40 degrees and 100 degrees F).

(2) Do not install polysulfide base sealants where sealant may be exposed to fumes from bituminous materials, or where water vapor in continuous contact with cementitious materials may be present.

(3) Do not install sealant type listed by manufacture as not suitable for use in locations specified.

(4) Apply caulking and sealing compound in accordance with manufacturer's printed instructions.

(5) Avoid dropping or smearing compound on adjacent surfaces.

(6) Fill joints solidly with compound and finish compound smooth.

(7) Tool exposed joints to form smooth and uniform beds, with slightly concave surface conforming to joint configuration per Figure 5A in ASTM C1193 unless shown or specified otherwise in construction documents. Remove masking tape immediately after tooling of sealant and before sealant face starts to “skin” over. Remove any excess sealant from adjacent surfaces of joint, leaving the working in a clean finished condition.

(8) Finish paving or floor joints flush unless joint is otherwise detailed.

(9) Apply compounds with nozzle size to fit joint width.

(10) Test sealants for compatibility with each other and substrate. Use only compatible sealant. Submit test reports.

(11) Replace sealant which is damaged during construction process.

b. Weeps: Place weep holes and vents in joints where moisture may accumulate, including at base of cavity walls, above shelf angles, at all flashing, and as indicated on construction documents.

(1) Use round plastic tubing to form weep holes.

(2) Space weep holes formed from plastic tubing not more than (16 inches) o.c.

(3) Trim tubing material used in weep holes flush with exterior wall face after sealant has set.

c. For application of sealants, follow requirements of ASTM C1193 unless specified otherwise. Take all necessary steps to prevent three-sided adhesion of sealants.

d. Interior Sealants: Where gypsum board partitions are of sound rated, fire rated, or smoke barrier construction, follow requirements of ASTM C919 only to seal all cut-outs and intersections with the adjoining construction unless specified otherwise.

(1) Apply a 1/4 inch minimum bead of sealant each side of runners (tracks), including those used at partition intersections with dissimilar wall construction.

(2) Coordinate with application of gypsum board to install sealant

immediately prior to application of gypsum board.

(3) Partition intersections: Seal edges of face layer of gypsum board abutting intersecting partitions, before taping and finishing or application of veneer plaster-joint reinforcing.

(4) Openings: Apply a 1/4 inch bead of sealant around all cutouts to seal openings of electrical boxes, ducts, pipes and similar penetrations. To seal electrical boxes, seal sides and backs.

(5) Control Joints: Before control joints are installed, apply sealant in back of control joint to reduce flanking path for sound through control joint.

3.6 FIELD QUALITY CONTROL

a. Field-Adhesion Testing: Field-test joint-sealant adhesion to joint substrates according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C1193 or Method A, Tail Procedure, in ASTM C1521.

(1) Extent of Testing: Test completed elastomeric sealant joints as follows:

(a) Perform 10 tests for first 1000 feet of joint length for each type of elastomeric sealant and joint substrate.

(b) Perform one test for each 1000 feet of joint length thereafter or one test per each floor per elevation.

b. Inspect joints for complete fill, for absence of voids, and for joint configuration complying with specified requirements. Record results in a field adhesion test log.

c. Inspect tested joints and report on following:

(1) Whether sealants in joints connected to pulled-out portion failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each type of product and joint substrate.

(2) Compare these results to determine if adhesion passes sealant manufacturer’s field-adhesion hand-pull test criteria.

(3) Whether sealants filled joint cavities and are free from voids.

(4) Whether sealant dimensions and configurations comply with specified requirements.

d. Record test results in a field adhesion test log. Include dates when sealants were installed, names of persons who installed sealants, test dates, test locations, whether joints were primed, adhesion results and percent elongations, sealant fill, sealant configuration, and sealant dimensions.

e. Repair sealants pulled from test area by applying new sealants following same procedures used to originally seal joints. Ensure that original sealant surfaces are clean and new sealant contacts original sealant.

f. Evaluation of Field-Test Results: Sealants not evidencing adhesive failure from testing or noncompliance with other indicated requirements, will be considered satisfactory. Remove sealants that fail to adhere to joint substrates during testing or to comply with other requirements. Retest failed applications until test results prove sealants comply with indicated requirements.

3.7 CLEANING

a. Fresh compound accidentally smeared on adjoining surfaces: Scrape off immediately and rub clean with a solvent as recommended by manufacturer of the adjacent material or if not otherwise indicated by the caulking or sealant manufacturer.

b. Leave adjacent surfaces in a clean and unstained condition.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

SECTION 07 95 13

EXPANSION JOINT COVER ASSEMBLIES

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 PREINSTALLATION MEETINGS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 QUALITY ASSURANCE
1.8 DELIVERY
1.9 STORAGE AND HANDLING
1.10 FIELD CONDITIONS
1.11 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION
2.2 SYSTEM PERFORMANCE
2.3 MATERIALS
2.4 PRODUCTS - GENERAL
2.5 FABRICATION
2.6 FINISHES
2.7 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION
3.3 CLEANING
3.4 PROTECTION

-- End of Section Table of Contents --

SECTION 07 95 13
1/21

EXPANSION JOINT COVER ASSEMBLIES

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

a. Prefabricated floor, wall, and ceiling building expansion joint assemblies.

(1) Metal plate covers at floor wall and ceiling joints.

(2) Elastomeric joint covers at wall and ceiling joints.

(3) Preformed elastomeric sealant joint at interior floor and wall control joints.

(4) Exterior wall joints.

1.2 RELATED WORK

a. Section 05 50 00, METAL FABRICATIONS: Steel Plate Expansion Joint Covers.

b. Section 07 60 00, FLASHING AND SHEET METAL: Sheet Metal Expansion Joint Seals.

c. Section 07 72 00, ROOF ACCESSORIES: Roof Expansion Joint Cover Assemblies.

1.3 REFERENCES

Comply with references to extent specified in this Section.

ALUMINUM ASSOCIATION (AA)

AA-C22A41	Aluminum Chemically etched medium matte, with clear anodic coating, Class I Architectural, 0.7-mil thick
AA-C22A42	Chemically etched medium matte, with integrally colored anodic coating, Class I Architectural, 0.7 mils thick
AA-C22A44	Chemically etched medium matte with electrolytically deposited metallic compound, integrally colored coating Class I Architectural, 0.7-mil thick finish

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605	(2020) Voluntary Specification, Performance Requirements and Test

Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7 (2017) Minimum Design Loads for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A240/A240M (2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A283/A283M (2024) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates

ASTM A786/A786M (2015; R 2021) Standard Specification for Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates

ASTM B36/B36M (2023) Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar

ASTM B121/B121M (2023) Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar

ASTM B209 (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B209M (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)

ASTM B221 (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes

ASTM B221M (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

ASTM B455/B455M Copper-Zinc-Lead Alloy (Leaded-Brass) Extruded Shapes

ASTM C864 (2005; R 2015) Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers

ASTM D1187/D1187M (1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal

ASTM E1399/E1399M (1997; R 2022) Standard Test Method for

	Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems
ASTM E1966	(2024) Standard Test Method for Fire-Resistive Joint Systems

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

1.4 PREINSTALLATION MEETINGS

Conduct preinstallation meeting at project site minimum 30 days before beginning Work of this Section.

a. Required Participants:

(1) Contracting Officer's Representative.

(2) Architect/Engineer.

(3) Inspection and Testing Agency.

(4) Contractor.

(5) Installer.

(6) Manufacturer's field representative.

(7) Other installers responsible for adjacent and intersecting work.

b. Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(1) Installation schedule.

(2) Installation sequence.

(3) Preparatory work.

(4) Protection before, during, and after installation.

(5) Installation.

(6) Terminations.

(7) Transitions and connections to other work.

(8) Other items affecting successful completion.

c. Document and distribute meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00

SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Construction Submittals

SD-04 Samples

Samples; G

SD-07 Certificates

Certificates; G

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G

1.6 SUBMITTAL INSTRUCTIONS

a. Submittal Drawings:

(1) Include large-scale details indicating profiles of each type of expansion joint cover, splice joints between joint sections, transitions to other assemblies, terminations, anchorages, fasteners, and relationship to adjoining work and finishes.

(2) Show size, configuration, and fabrication and installation details.

(3) Include composite drawings showing work specified in other Sections coordinated with expansion joints.

b. Manufacturer's Literature and Data:

(1) Description of each product specified.

(2) Show movement capability of each cover assembly and suitability of material used in exterior seals for ultraviolet exposure.

(3) Description of materials and finishes.

(4) Installation instructions.

c. Samples: Submit 300 mm (12 inch) long samples.

(1) Each type and color of metal finish for each required thickness and alloy.

(2) Each type and color of flexible seal.

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

(2) Low Pollutant-Emitting Materials:

(a) Identify volatile organic compound types and quantities.

e. Qualifications: Substantiate qualifications comply with specifications.

(1) Installer.

f. Certificates: Indicate products comply with specifications.

(1) Fire rated expansion joint cover assemblies.

g. Operation and Maintenance Data:

(1) Care instructions for each exposed finish product.

1.7 QUALITY ASSURANCE

Installer Qualifications:

(1) Regularly installs specified products.

(2) Installed specified products with satisfactory service on five similar installations for minimum five years.

1.8 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.9 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.10 FIELD CONDITIONS

Field Measurements: Verify field conditions affecting expansion joint cover assembly fabrication and installation. Show field measurements on Submittalhop Drawings.

Coordinate field measurement and fabrication schedule to avoid delay.

1.11 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

a. Provide joint cover assemblies that permit unrestrained movement of joint without disengagement of cover, and, where applicable, maintain moisture, watertight and fire-rated protection.

b. Provide templates to related trades for location of support and anchorage items.

2.2 SYSTEM PERFORMANCE

a. Design expansion joint cover assemblies complying with specified performance.

b. Joint Movement: ASTM E1399/E1399M.

(1) Nominal Joint Width: as indicated.

(2) Minimum Movement Capability: 25 percent.

(3) Movement Type: Thermal and wind .

c. Floor Joints: Live loads, including rolling loads.

(1) Load Resistance: ASCE 7; Design criteria as indicated on Drawings.

(2) Maximum Deflection: 1/360 of span, maximum.

d. Fire Rated Joints: ASTM E1399/E1399M, ASTM E1966, including hose stream test at full-rated period.

(1) Fire rating: Match adjacent floor, wall, and ceiling construction.

(2) System: Capable of anticipated movement while maintaining fire rating.

(3) Coverless Applications: Maintain fire rating without joint cover system.

2.3 MATERIALS

a. Stainless Steel: ASTM A240/A240M, Type 302 or 304.

b. Structural Steel Shapes: ASTM A36/A36M.

c. Steel Plate: ASTM A283/A283M, Grade C.

d. Rolled Steel Floor Plate: ASTM A786/A786M.

e. Aluminum:

(1) Extruded: ASTM B221M (ASTM B221), alloy 6063-T5, 6063-T6, or 6061-T6.

(2) Plate and Sheet: ASTM B209M (ASTM B209), alloy 6061-T6.

f. Bronze: Manufacturer's standard alloy.

(1) Extruded: ASTM B455/B455M.

(2) Plate: ASTM B121/B121M.

g. Brass: ASTM B36/B36M.

h. Elastomeric Sealant: As specified in Section 07 92 00, JOINT SEALANTS.

i. Elastomeric Seals:

(1) Flexible extruded polyvinyl chloride, meeting a Shore A hardness of 75 with UV stabilizer. Manufacturer's standard colors.

j. Thermoplastic Rubber:

(1) ASTM C864.

(2) Dense Neoprene or other material standard with expansion joint manufacturers having the same physical properties.

k. Compression Seals: Pre-compressed secondary sealant using preformed expanding foam sealant; open-cell polyurethane foam impregnated with polymer-modified acrylic adhesive.

l. Water Barrier Sheets: Neoprene or EPDM flexible sheet materials minimum 45 mils thick.

(1) Provide with drain tubes for horizontal applications.

m. Vinyl Invertor Sealant Waterstops: Manufacturer's standard shapes and grade.

n. Moisture Barrier: Fabric reinforced clear vinyl sheet material sized to accommodate opening.

o. Flexible Membrane: (60 mil) EPDM sheet, with manufacturer's standard support foam.

p. Fire Barrier: Labeled by an approved independent testing laboratory for fire resistance ratings indicated for maximum joint width.

(a) Thermal Insulation: Manufacturer's standard with factory cut miters and transitions.

(b) Fire Barrier Lengths:

1. Joint widths up to and including (6 inches): Maximum (50 feet) to minimize field splicing.

2. Other Joint widths: 10 foot with overlapping ends for field splicing.

q. Ceramic Blanket: Manufacturer's standard joint filler to achieve fire rating indicated.

r. Butyl Caulk Tape: Self adhering double sided butyl rubber sealant tape with easy-release silicone coated paper.

2.4 PRODUCTS - GENERAL

a. Basis of Design: As indicated on drawings.

b. Provide each product from one manufacturer.

(1) Provide ceiling and wall expansion joint cover assemblies design matching floor to wall and floor to floor expansion joint cover design.

(2) Provide expansion joint cover assembly designs, profiles, materials and configuration indicated, as required to accommodate joint size variations in adjacent surfaces, and anticipated movement.

c. Sustainable Construction Requirements:

(1) Steel Recycled Content: 30 percent total recycled content, minimum.

(2) Stainless Steel Recycled Content: 70 percent total recycled content, minimum.

(3) Aluminum Recycled Content: 80 percent total recycled content, minimum.

(4) Low Pollutant-Emitting Materials: Maximum VOC content by weight.

(a) Non-Flooring Adhesives and Sealants.

2.5 FABRICATION

a. Fabricate Expansion Joint Cover Assemblies:

(1) As complete assembly ready for installation.

(2) In longest practicable lengths to minimize number of end joints.

(3) With factory mitered corners where joint changes directions or abuts other materials.

(a) With closure materials and transition pieces, tee-joints, corners, curbs, cross-connections and other assemblies.

(4) Joints within enclosed spaces such as chase walls, include (0.04 inch) thick galvanized steel cover where conventional expansion joint cover is not used.

(5) Where floor slab is fire rated provide ceramic blanket at joints.

(6) Seal Strip: Factory-formed and bonded to metal frames and anchor members.

(7) Compression Seals: Fabricate from expanding foam as secondary seal and elastomeric sealant to sizes and profiles shown.

b. Floor-to-Floor Metal Plate Joints:

(1) Frames: Metal, continuous on both sides of joint designed to support cover plate.

(a) Flush Design: Seating surface and raised floor rim to accommodate adjacent flooring.

(b) Anchorage: Concealed bolt and steel anchors for embedment in concrete.

(2) Cover Plate: Metal, matching frames where exposed.

(a) Supported Load: (400 psf), minimum.

(b) Rattle-free due to traffic.

(3) Fillers: Resilient material between raised rim of frame and edge of cover plate, where shown.

(a) No gaps or bulges over full design range joint movement.

(4) Fire Barrier: As required for fire resistance rating.

(5) Water Stop: Manufacturer's standard, continuous, full length of joint.

(6) Seismic: As required by Code.

(7) Finishes: As indicated on drawings..

c. Floor-to-Wall Metal Plate Joints:

(1) Frames: Metal, continuous on floor side of joint only.

(a) Provide wall side frame where required by manufacturer's design.

(2) Cover Plates: Angle cover plates with countersunk flat-head exposed fasteners for securing cover plate to wall unless shown otherwise.

(a) Fastener Spacing: As recommended by manufacturer.

(3) Joint Design: Match adjacent floor to floor design.

(4) Fire Barrier: As required for fire resistance rating.

(5) Water Stop: Manufacturer's standard, continuous, full length of joint.

(6) Seismic: As required by Code.

(7) Finishes: As indicated on drawings..

d. Interior Wall Joint Cover Assemblies:

(1) Frame: Metal, surface mounted, concealed fastening to wall on one

sides of joint.

(2) Cover Plate: Metal, smooth surface, lap both sides of joint and permitting free movement on one side.

(a) Fabricate with concealed attachment of cover to frame when cover is in close contact with adjacent wall surface finish.

(b) Use angle cover plates at intersecting walls.

(3) Joint Design: Match adjacent floor to floor design.

(4) Fire Barrier: As required for fire resistance rating.

(5) Seismic: As required by Code.

(6) Finishes: As indicated on drawings.

e. Exterior Wall Joint Assemblies:

(1) Design seal for variable movement and prevention of water and air infiltration.

(2) Frame: Metal, concealed, for fastening to wall on one side of joint.

(3) Cover Plate: Metal, surface mounted, lap both sides of joint, permitting free movement on one side.

(a) Fabricate with concealed attachment of cover to frame for cover with cover in close contact with adjacent finish surfaces.

(b) Use angle cover plate at intersecting walls.

(4) Water Seal: Vinyl seal strip as secondary seal behind primary seal.

(5) Seismic: As required by Code.

(6) Finish: As indicated on drawings.

f. Extruded Thermoplastic Rubber Joint Assemblies:

(1) Frames: Aluminum, both sides of joint.

(2) Primary Seal: Flexible rubber on exposed face after frame installation with factory welded watertight miters and transitions.

(a) Anchor spaced at ends and not over (24 inches).

1. Variable movement extruded rubber primary seal designed to remain in aluminum frame, throughout movement of joint.

(b) Flush mounted seal minimum (0.12 inch) thick with dual movement grooves designed for plus or minus 50 percent, movement of joint width.

(c) Recessed front face seal minimum (0.12 inch) thick with no movement grooves, designed for plus or minus 50 percent movement

of joint width.

(d) Provide pantographic wind load supports, maximum (8 feet) on center to support seal systems of (12 inches) wide and greater.

(3) Secondary Seal: Continuous vinyl sheet seal.

(4) Finishes: As indicated on drawings.

g. Ceiling and Soffit Assemblies:

(1) Frames: Metal, continuous on both sides of joint, flush mounted with no exposed fasteners.

(2) Flexible Insert: Variable movement semi-rigid vinyl locked into frame.

(a) Face Style: Flush or accordion, as shown, to span joint width without sagging.

(3) Seismic: As required by Code.

(4) Finishes: As indicated on drawings.

h. Garage Floor Joint Cover Plate:

(1) Frame: Angle edge frame on both sides of joint, size as shown.

(a) Anchors: Stud bolts minimum (4 inches) long and(3/8 inch) diameter welded to angle spaced maximum (24 inches) on center.

(b) Drill and top one frame for cover plate fasteners.

(2) Cover Plate: Aluminum or steel cover plate minimum (3/8 inch) thick with edges beveled, smooth finish, drilled for countersunk fasteners at ends and maximum (24 inches) on center.

(3) Seismic: As required by Code.

(4) Finishes: As indicated on drawings.

i. Preformed Sealant Joint: Factory installed elastomeric sealant between extruded aluminum angle frame both sides.

(1) Frames: Extruded aluminum angle on both sides of joint.

(2) Filler: Elastomeric sealant.

(3) Anticipated movement: 25 percent maximum.

(4) Finishes: As indicated on drawings.

2.6 FINISHES

a. Carbon Steel: NAAMM AMP 500, Galvanized G90.

b. Stainless Steel: NAAMM AMP 500, No. 2B bright finish.

c. Aluminum Anodized Finish: NAAMM AMP 500.

(1) Clear Anodized Finish: AA-C22A41; Class I Architectural, (0.7 mil) thick.

(2) Color Anodized Finish: AA-C22A42 or AA-C22A44; Class I Architectural,(0.7 mil) thick.

d. Aluminum Paint Finish:

(1) Fluorocarbon Finish: AAMA 2605; 70 percent fluoropolymer resin, 2-coat system.

(2) Fluorocarbon Finish: AAMA 2605; 70 percent fluoropolymer resin, 2-coat system.

e. Bronze Finish: NAAMM AMP 500, M32 mechanical finish, directional textured, natural medium satin.

2.7 ACCESSORIES

a. General: Manufacturer's standard anchors, fasteners, set screws, spaces, protective coating, and filler materials, adhesive and other accessories required for installation.

b. Barrier Coating: ASTM D1187/D1187M.

c. Adhesives: Low pollutant-emitting, water based type recommended by adhered product manufacturer for each application.

d. Fasteners: Type and size recommended by expansion joint cover assembly manufacturer.

(1) Exterior Applications: Stainless steel.

(2) Fasteners for Aluminum: Stainless steel.

(3) Other Applications: Galvanized steel or stainless steel.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

(1) Provide items embedded in concrete and masonry in time for building into work without delaying work.

b. Protect existing construction and completed work from damage.

c. Apply barrier coating to aluminum surfaces in contact with dissimilar metals and cementitious materials to minimum(30 mils) dry film thickness.

3.2 INSTALLATION

a. Install products according to manufacturer's instructions and approved submittal drawings.

(1) When manufacturer's instructions deviate from specifications,

submit proposed resolution for Contracting Officer's Representative consideration.

b. Install anchorage devices and fasteners for securing expansion joint assemblies to in-place construction where anchors are not embedded in concrete and masonry.

(1) Secure with metal fasteners, type and size to suit application.

c. Perform cutting, drilling and fitting required for installation of expansion joint cover assemblies.

d. Install joint cover assemblies aligned and positioned in correct relationship to expansion joint opening and adjoining finished surfaces measured from established lines and levels.

(1) Allow for thermal expansion and contraction of metal to avoid buckling.

(2) Accommodate joint opening size at time of installation.

e. Set floor covers at elevations flush with adjacent finished flooring, unless shown otherwise.

f. Grout floor frames set in prepared recesses.

g. Locate wall, ceiling and soffit covers in continuous contact with adjacent surfaces. Secure with required accessories.

h. Locate anchors at interval recommended by manufacturer, but minimum (3 inches) from each end, and, maximum (24 inches) on centers.

i. Maintain continuity of expansion joint cover assemblies with end joints held to a minimum and metal members aligned mechanically using splice joints.

j. Cut and fit ends to accommodate thermal expansion and contraction of metal to avoid buckling of frames and cover plates.

k. Flush Metal Cover Plates:

(1) Secure flexible filler between frames to allow compression and expansion.

(2) Adhere flexible filler materials to frames with adhesive or pressure-sensitive tape as recommended by manufacturer.

l. Waterstops:

(1) Install in conjunction with floor joints, and where shown.

(2) Install continuously to prevent water damage to finish spaces.

(3) Seal waterstop to frames to prevent water leakage.

(4) Install drainage tubes from waterstops to discharge collected water in nearest plumbing air gap drain.

m. Fire Barriers:

(1) Install in compliance with tested assembly.

(2) Install at joints in floors and in fire rated walls.

(3) Use fire barrier sealant furnished with expansion joint assembly.

n. Apply sealant where required to prevent water and air infiltration.

o. Vertical Exterior Extruded Thermoplastic Rubber.

(1) Install side frames mounted on sealant or butyl caulk tape with appropriate anchors (24 inches) on center complete with secondary seal.

(2) Install primary seals retained in extruded aluminum side frames.

p. Extruded Thermoplastic Rubber or Seals:

(1) For straight sections, install preformed seals in continuous lengths.

(2) Vulcanize or heat-seal field spliced joints to provide watertight joints as recommended by manufacturer.

q. Preformed Elastomeric Sealant Joint:

(1) Locate joint directly over joints in wall and floor substrates.

(2) Fasten full length to substrate using construction adhesive.

(3) Install flush or slightly below finish material.

3.3 CLEANING

a. Remove excess adhesive before adhesive sets.

b. Clean exposed metal surfaces. Remove contaminants and stains.

3.4 PROTECTION

a. Cover floor joints with plywood where wheel traffic occurs before Substantial completion.

b. Remove protective covering when adjacent work areas are completed. Clean exposed surfaces in compliance with manufacture's printed instructions.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 11 13

HOLLOW METAL DOORS AND FRAMES

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 MATERIALS
2.3 PRODUCTS - GENERAL
2.4 HOLLOW METAL DOORS
2.5 HOLLOW METAL FRAMES
2.6
2.7 FABRICATION
2.8 FINISHES
2.9 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 FRAME INSTALLATION
3.4 DOOR INSTALLATION
3.5 CLEANING
3.6 PROTECTION

-- End of Section Table of Contents --

SECTION 08 11 13
1/21

HOLLOW METAL DOORS AND FRAMES

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

a. Hollow metal doors hung in hollow metal frames at interior and exterior locations.

b. Hollow metal door frames for borrowed lights at interior locations.

c. Glazed openings in hollow metal doors.

1.2 RELATED WORK

a. Section 05 50 00, METAL FABRICATIONS: Frames fabricated of structural steel.

b. Section 08 41 13, ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS: Aluminum frames entrance work.

c. Section 08 71 00, DOOR HARDWARE: Door Hardware:

d. Section 08 80 00, GLAZING: Glazing.

e. Card Readers and Biometric Devices: Section 28 13 00, PHYSICAL ACCESS CONTROL SYSTEM.

1.3 REFERENCES

Comply with references to extent specified in this Section.

STEEL DOOR INSTITUTE (SDI/DOOR)

SDI/DOOR A250.8	(2023) Specifications for Standard Steel Doors and Frames

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or

	Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM D3656/D3656M	(2013) Standard Specification for Insect Screening and Louver Cloth Woven from Vinyl-Coated Glass Yarns
ASTM E90	(2023) Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements

MASTER PAINTERS INSTITUTE (MPI)

MPI 18	(2022) Primer, Zinc Rich, Organic

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2025) Standard for Fire Doors and Other Opening Protectives
NFPA 105	(2025) Standard for Smoke Door Assemblies and Other Opening Protectives

UL SOLUTIONS (UL)

UL 10C	(2016; Reprint May 2021) UL Standard for Safety Positive Pressure Fire Tests of Door Assemblies
UL 1784	(2015) Standard for Air Leakage Tests of Door Assemblies and Other Opening

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G

Manufacturer Qualifications; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Sustainable Construction Submittals; S

SD-05 Design Data

Blast Design Calculations; G, AE

SD-06 Test Reports

Test Reports; G, AE

Inspection and Testing; G

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submittal Drawings: Show size, configuration, and fabrication and installation details.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Include schedule showing each door and frame requirements fire label and smoke control label for openings.

(3) Installation instructions.

d. Sustainable Construction Submittals:

Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

Environmental Product Declarations.

Health Product Declarations.

Third-party verified documentation for ingredient inventory and/or GWP.

e. Test reports: Certify each product complies with specifications. Provide signed record of all inspections and testing of installed fire door assemblies, including list of deficiencies in accordance with NFPA 80.

f. Qualifications: Substantiate qualifications comply with specifications. Manufacturer with project experience list.

g. Blast Design Calculations. Submit design calculations for exterior hollow metal door assemblies demonstrating anchorage is capable of withstanding ultimate loading conditions for doors as specified. Anchor calculations are to be performed by a qualified professional engineer.

1.6 QUALITY ASSURANCE

Manufacturer Qualifications:

a. Regularly manufactures specified products.

b. Manufactured specified products with satisfactory service on five similar installations for a minimum of five years. Project Experience List: Provide contact names and addresses for completed projects.

1.7 DELIVERY

a. Fasten temporary steel spreaders across the bottom of each door frame before shipment.

b. Deliver products in manufacturer's original sealed packaging.

c. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

d. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.9 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish

materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 SYSTEM PERFORMANCE

Design hollow metal doors and frames complying with specified performance:

a. Fire Doors and Frames: UL 10C; NFPA 80 labeled. Fire Ratings: See drawings.

b. Stair Doors: Temperature rise rated fire doors.

c. Smoke Control Doors and Frames: UL 1784; NFPA 80 labeled, maximum 3.0 cubic feet/minute/square foot at 0.10 inches water gauge pressure differential.

d. Sound Rated Doors and Frames: Minimum 45 sound transmission class (STC) when tested according to ASTM E90.

e. Thermal Transmittance: 0.36 U-value, maximum at exterior doors .

f. Thermal Resistance: 7 R-value, minimum at exterior doors.

g. Blast Resistant Doors: Door, Frame and Anchorage:

(1) Provide conventional, hollow metal doors with vertical stiffeners.

(2) Design framing anchors at perimeter of openings for the door edge reactions of 100 lbs/in for single doors and 170 lbs/in for double doors.

(3) Double doors are to be supported using either multiple point latches or mullions. For multiple point latches, design top and bottom latches to resist ultimate shear loads of 4 kips. For mullions, provide mullions with a moment of inertia of 2.39 in^4 and a section modulus of 1.60 in^3 as minimums. Design mullion anchors at floor slabs to resist ultimate shear loads of 8 kips.

(4) Glazing: Glazing shall meet the blast requirements shown in Section 08 80 00, GLAZING.

(5) Minimum gauge of metal used on blast resistant doors shall be 14 gauge.

2.2 MATERIALS

a. Stainless Steel: ASTM A240/A240M; Type 304 .

b. Sheet Steel: ASTM A1008/A1008M, cold-rolled.

c. Galvanized Sheet Steel: ASTM A653/A653M.

d. Insect Screening: ASTM D3656/D3656M, 18 by 18 aluminum wire mesh.

e. Aluminum Sheet: ASTM B209M (ASTM B209).

f. Aluminum Extrusions: ASTM B221M (ASTM B221).

2.3 PRODUCTS - GENERAL

a. Basis of Design: Refer to Schedule for Finishes on the drawings.

b. Provide hollow metal doors and frames from one manufacturer.

c. Sustainable Construction Requirements:

(1) Steel Recycled Content: 30 percent total recycled content, minimum.

(2) Stainless Steel Recycled Content: 70 percent total recycled content, minimum.

(3) Aluminum Recycled Content: 50 percent total recycled content, minimum.

2.4 HOLLOW METAL DOORS

a. Hollow Metal Doors: SDI/DOOR A250.8; 1-3/4 inches thick. See drawings for sizes and designs.

(1) Interior Doors: Level 3 and Physical Performance Level A, extra-heavy duty; Model 2, seamless at stairs.

(2) Exterior Doors: Level 4 and Physical Performance Level A, maximum heavy duty; Model 2, seamless.

b. Door Faces:

(1) Interior Doors: Sheet steel Galvanized sheet steel minimum Z120 or ZF120 (G40 or A40) Z180 or ZF180 (G60 or A60) Z275 (G90) coating.

(2) Exterior Doors: Galvanized sheet steel minimum Z180 or ZF180 (G60 or A60) coating.

c. Door Cores:

(1) Interior Doors: Kraft paper honeycomb or vertical steel stiffeners.

(2) Exterior Doors: Polystyrene or polyurethane.

(3) Fire Doors: Manufacturer's standard complying with specified fire rating performance.

2.5 HOLLOW METAL FRAMES

a. Hollow Metal Frames: SDI/DOOR A250.8; face welded See drawings for sizes and designs.

(1) Interior Frames:

(a) Level 1 Hollow Metal Doors: 0.042 inch thick.

(b) Level 3 Hollow Metal Doors: 0.053 inch thick.

(2) Interior Borrowed Light Frames: 0.051 inch thick.

b. Frame Materials:

(1) Interior Frames: Sheet steel Galvanized sheet steel minimum Z180 or ZF180 (G60 or A60).

(2) Exterior Frames: Galvanized sheet steel minimum Z180 or ZF180 (G60 or A60).

2.6

2.7 FABRICATION

a. Hardware Preparation: SDI/DOOR A250.8; for hardware specified in Section 08 71 00, DOOR HARDWARE.

b. Hollow Metal Door Fabrication:

(1) Close top edge of exterior doors flush and seal to prevent water intrusion.

(2) Fill spaces between vertical steel stiffeners with insulation.

c. Fire Doors:

(1) Close top and vertical edges flush.

(2) Apply steel astragal to active leaf at pair and double egress doors. Exception: Where vertical rod exit devices are specified for both leaves swinging in same direction.

(3) Fire Door Clearances: NFPA 80.

f. Sound Rated Doors:

(1) Seals: Integral spring type automatic door bottom seal.

(2) Fabricate vision panel cutouts and frames to receive double glazing as shown on drawings.

i. Hollow Metal Frame Fabrication:

(1) Fasten mortar guards to back of hardware reinforcements, except on lead-lined frames.

(2) Concealed Closers in Head Frame: Provide 0.042 inch thick steel removable stop sections for access to concealed face plates and control valves, except when cover plates are furnished with closer.

(3) Terminated Stops: SDI/DOOR A250.8

(4) Borrowed Light Frames:

(a) Provide integral stop on exterior, corridor, or secure side of door.

(b) Design rabbet width and depth to receive glazing material or

panel shown on drawings.

(5) Two Piece Frames:

(a) One piece unequal leg finished rough buck sub-frames as shown, drilled for anchor bolts.

(b) Unequal leg finished frames formed to fit subframes and secured to subframe legs with countersunk, flat head screws, spaced 12 inches on center at head and jambs on both sides.

(c) Preassemble at factory for alignment.

(6) Frame Anchors:

(a) Floor anchors:

1. Provide extension type floor anchors to compensate for depth of floor fills.

2. Provide 0.053 inch thick steel clip angles welded to jamb and drilled to receive floor fasteners.

3. Provide 2 inch by 2 inch by 3/8 inch clip angle for lead lined frames, drilled for floor fasteners.

4. Provide mullion 0.093 inch thick steel channel anchors, drilled for two floor fasteners and frame anchor screws.

5. Provide continuous 0.042 inch thick steel rough bucks drilled for floor fasteners and frame anchor screws for sill sections. Space floor bolts 24 inches on center.

(b) Jamb anchors:

1. Place anchors on jambs:

a) Near top and bottom of each frame.

b) At intermediate points at maximum 24 inches spacing.

2. Form jamb anchors from steel minimum 0.042 inch thick.

3. Anchors set in masonry: Provide adjustable anchors designed for friction fit against frame and extended into masonry minimum 10 inches. Provide one of following types:

a) Wire Loop Type: 3/16 inch diameter wire.

b) T-Shape type.

c) Strap and stirrup type: Corrugated or perforated sheet steel.

4. Anchors for stud partitions: Provide tabs for securing anchor to sides of studs. Provide one of the following:

a) Welded type.

b) Lock-in snap-in type.

5. Anchors for frames set in prepared openings:

a) Steel pipe spacers 1/4 inch inside diameter, welded to plate reinforcing at jamb stops, or hat shaped formed strap spacers 2 inches wide, welded to jamb near stop.

b) Drill jamb stop and strap spacers for 1/4 inch flat head bolts to pass through frame and spacers.

c) Two piece frames: Subframe or rough buck drilled for 1/4 inch bolts.

6. Anchors for observation windows and other continuous frames set in stud partitions.

a) Weld clip anchors to sills and heads of continuous frames over 4 feet long.

b) Space maximum 24 inches on centers.

7. Modify frame anchors to fit special frame and wall construction.

8. Provide special anchors where shown on drawings and where required to suit application.

j. Sound Rated Door Frames: Seals: Integral continuous gaskets on frames.

2.8 FINISHES

a. Steel and Galvanized Steel : SDI/DOOR A250.8; shop primed.

b. Stainless Steel: NAAMM AMP 500; No. 4 polished finish. Blend welds to match adjacent finish.

c. Finish exposed surfaces after fabrication.

d. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

2.9 ACCESSORIES

a. Primers: SDI/DOOR A250.8.

b Barrier Coating: ASTM D1187/D1187M.

c. Welding Materials: AWS D1.1/D1.1M, type to suit application.

d. Clips Connecting Members and Sleeves: Match door faces.

e. Fasteners: Galvanized steel.

(1) Metal Framing: Steel drill screws.

(2) Masonry and Concrete: Expansion bolts and power actuated drive

pins.

f. Anchors: Galvanized steel.

g. Galvanizing Repair Paint: MPI 18.

h. Insulation: Unfaced mineral wool.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Apply barrier coating to metal surfaces in contact with cementitious materials to minimum 30 mils dry film thickness.

3.2 INSTALLATION - GENERAL

Install products according to manufacturer's instructions and approved submittal drawings.

a. When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.

b. Install fire doors and frames according to NFPA 80. Upon completion of installation, fire door assemblies shall be inspected and tested in accordance with NFPA 80.

c. Install smoke control doors and frames according to NFPA 105.

3.3 FRAME INSTALLATION

a. Apply barrier coating to concealed surfaces of frames built into masonry.

b. Plumb, align, and brace frames until permanent anchors are set.

(1) Use triangular bracing near each corner on both sides of frames with temporary wood spreaders at midpoint.

(2) Use wood spreaders at bottom of frame when shipping spreader is removed.

(3) Where construction permits concealment, leave shipping spreaders in place after installation, otherwise remove spreaders when frames are set and anchored.

(4) Remove wood spreaders and braces when walls are built and jamb anchors are secured.

c. Floor Anchors:

(1) Anchor frame jambs to floor with two expansion bolts.

(a) Lead Lined Frames: Use 3/8 inch diameter bolts.

(b) Other Frames: Use 1/4 inch diameter bolts.

(2) Power actuated drive pins are acceptable to secure frame anchors to concrete floors.

d. Jamb Anchors:

(1) Masonry Walls:

(a) Embed anchors in mortar.

(b) Fill space between frame and masonry with grout or mortar as walls are built.

(2) Metal Framed Walls: Secure anchors to sides of studs with two fasteners through anchor tabs.

(3) Prepared Masonry and Concrete Openings:

(a) Direct Securement: 1/4 inch diameter expansion bolts through spacers.

(b) Subframe or Rough Buck Securement:

1. 1/4 inch diameter expansion bolts on 24 inch centers.

2. Power activated drive pins on 24 inches centers.

(c) Secure two-piece frames to subframe or rough buck with machine screws on both faces.

e. Frames for Sound Rated Doors: Fill frames with insulation.

f. Touch up damaged factory finishes.

(1) Repair galvanized surfaces with galvanized repair paint.

(2) Repair painted surfaces with touch up primer.

3.4 DOOR INSTALLATION

a. Install doors plumb and level.

b. Adjust doors for smooth operation.

c. Touch up damaged factory finishes.

(1) Repair galvanized surfaces with galvanized repair paint.

(2) Repair painted surfaces with touch up primer.

3.5 CLEANING

Clean exposed door and frame surfaces. Remove contaminants and stains.

3.6 PROTECTION

a. Protect doors and frames from traffic and construction operations.

b. Remove protective materials immediately before acceptance.

c. Repair damage.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 31 13

ACCESS DOORS AND FRAMES

06/25

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSRUCTIONS
1.6 DELIVERY
1.7 STORAGE AND HANDLING
1.8 FIELD CONDITIONS
1.9 WARRANTY

PART 2 PRODUCTS

2.1 MATERIALS
2.2 PRODUCTS - GENERAL
2.3 ACCESS DOORS, FIRE-RATED
2.4 ACCESS DOORS, FLUSH PANEL, NON-RATED
2.5 ACCESS DOOR, RECESSED PANEL
2.6 FABRICATION - GENERAL
2.7 FINISHES
2.8 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 ACCESS DOOR AND FRAME INSTALLATION
3.4 ADJUSTMENT

-- End of Section Table of Contents --

SECTION 08 31 13
06/25

ACCESS DOORS AND FRAMES

PART 1 GENERAL

1.1 SUMMARY

Section Includes: Access doors and panels installed in walls and ceilings.

1.2 RELATED WORK

a. Section 05 50 00, METAL FABRICATIONS: Wire Mesh and Screen Access Doors.

b. Section 08 71 00, DOOR HARDWARE: Lock Cylinders.

c. Section 09 91 00, PAINTING: Field Painting.

d. Section 14 21 23, ELECTRIC TRACTION PASSENGER ELEVATORS: Access door at top of elevator shaft.

e. Section 21 13 13, WET-PIPE SPRINKLER SYSTEMS: Access Doors for Control or Drain Valves.

f. Section 21 12 16, DRY-PIPE SPRINKLER SYSTEMS: Access Doors for Control or Drain Valves.

g. Section 22 40 00, PLUMBING FIXTURES: Access Doors for Plumbing Valves

h. Section 23 31 00, HVAC DUCTS AND CASINGS: Locations of Access Doors for Ductwork Cleanouts.

1.3 REFERENCES

Comply with references to extent specified in this section.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.3/D1.3M	(2018) Structural Welding Code - Sheet Steel

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM A666	(2023) Standard Specification for Annealed

	or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2025) Standard for Fire Doors and Other Opening Protectives
NFPA 252	(2022) Standard Methods of Fire Tests of Door Assemblies

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

UL SOLUTIONS (UL)

UL 10B	(2008; Reprint Oct 2024) Fire Tests of Door Assemblies
UL 263	(2011; Reprint Aug 2021) UL Standard for Safety Fire Tests of Building Construction and Materials

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Sustainable Construction Submittals; S

SD-06 Test Reports

Inspection And Testing; G

1.5 SUBMITTAL INSRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submittal Drawings: Show size, configuration, and fabrication and installation details.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

d. Sustainable Construction Submittals:

Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

e. Inspection and testing: Provide signed record of all inspections and testing of installed fire door assemblies, including list of deficiencies in accordance with NFPA 80.

1.6 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.7 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.8 FIELD CONDITIONS

Field Measurements: Verify field conditions affecting access door fabrication and installation. Show field measurements on Submittal Drawings. Coordinate field measurement and fabrication schedule to avoid delay.

1.9 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 MATERIALS

a. Steel Sheet: ASTM A1008/A1008M.

b. Galvanized Steel: ASTM A653/A653M.

c. Stainless Steel: ASTM A666; Type 302 or Type 304.

2.2 PRODUCTS - GENERAL

a. Basis of Design: Refer to the Schedule of Finishes on the drawings.

b. Provide each product from one manufacturer.

c. Sustainable Construction Requirements:

(1) Steel Access Doors Recycled Content: 30 percent total recycled content, minimum.

(2) Stainless Steel Access Doors Recycled Content: 70 percent total recycled content, minimum.

2.3 ACCESS DOORS, FIRE-RATED

a. Door Construction:

(1) Ceiling Access Door Construction: ASTM E119 or UL 263.

(2) Wall Access Doors: NFPA 252 or UL 10B.

b. Label: Class B opening according to UL 10B or test by another nationally recognized laboratory. 1 hour fire-rated or 1-1/2 hour fire-rated, as indicated on the drawings, with maximum temperature rise of 216 degrees F).

c. Door Panel: Minimum 0.0359 inch thick stainless steel sheet, with mineral-fiber insulation core, insulated sandwich type construction.

d. Frame: Minimum 0.0598 inch thick steel sheet, depth and configuration to suit material and construction type where installed.

(1) Frame Flange: Provide at units installed in concrete, masonry, or gypsum board.

(2) Exposed Joints in Flange: Weld and grind smooth.

e. Provide automatic closing device.

f. Hinge: Continuous stainless steel hinge with stainless steel pin.

g. Lock: Self-latching, mortise type with provision for fitting flush a standard screw-in type lock cylinder.

(1) Lock cylinder specified in Section 08 71 00, DOOR HARDWARE.

(2) Latch release device operable from inside of door.

h. Anchors for Fire-Rated Access Doors: Comply with requirements of applicable fire test.

2.4 ACCESS DOORS, FLUSH PANEL, NON-RATED

a. Door Panel:

(1) 0.06 inch thick stainless steel sheet.

(2) Reinforce to maintain flat surface.

b. Frame:

(1) 0.06 inch thick stainless steel sheet, depth and configuration to suit material and construction type where installed.

(2) Frame Flange: Provide at units installed in concrete, masonry, and gypsum board.

(3) Exposed Joints in Flange: Weld and grind smooth.

c. Hinge:

(1) Concealed spring hinge, 175 degrees of opening.

(2) Removable hinge pin to allow removal of door panel from frame.

d. Lock:

(1) Flush, screwdriver-operated cam lock.

e. Gaskets:

(1) Provide gasketing for an airtight installation.

(2) Provide gasketing in accordance with manufacturer's written recommendations.

2.5 ACCESS DOOR, RECESSED PANEL

a. Door Panel:

(1) 0.05 inch thick stainless steel sheet to form a 1 inch deep recessed pan to accommodate installation of acoustical units and other materials where shown in walls and ceiling.

(2) Reinforce to prevent sagging.

b. Frame:

(1) 0.06 inch thick steel sheet of depth and configuration to suit installation in suspension system of ceiling or wall framing.

(2) Extend sides of frame to protect edge of acoustical units when door panel is in open position.

(3) Provide shims, bushings, clips and other devices necessary for installation.

c. Hinge: Continuous stainless steel hinge with stainless steel pin, or concealed hinge.

d. Lock:

(1) Flush screwdriver-operated cam lock.

(2) Plastic sleeve or stainless steel grommet to protect hole made in acoustical unit for screwdriver access to lock.

e. Gaskets:

(1) Provide gasketing for an airtight installation.

(2) Provide gasketing in accordance with manufacturer's written recommendations.

2.6 FABRICATION - GENERAL

a. Size: Minimum 24 inches square door unless otherwise shown

b. Component Fabrication: Straight, square, flat and in same plane where required.

(1) Exposed Edges: Slightly rounded, without burrs, snags and sharp edges.

(2) Exposed Welds: Continuous, ground smooth.

(3) Welding: AWS D1.3/D1.3M.

c. Locks and Non-Continuous Hinges: Provide in numbers required to maintain alignment of door panel with frame. For fire-rated doors, provide hinges and locks as required by fire test.

d. Anchoring: Make provisions in frame for anchoring to adjacent construction. Provide anchors in size, number and location on four sides to secure access door to substrate. Provide anchors as required by fire test.

2.7 FINISHES

a. Steel Paint Finish: Powder-Coat Finish: Manufacturer's standard two-coat finish system consisting of the following:

(1) One coat primer.

(2) One coat thermosetting topcoat.

(3) Dry-film Thickness: 2 mils minimum.

(4) Color: Refer to Schedule for Finishes on the drawings.

b. Stainless Steel Exposed Surfaces: NAAMM AMP 500; No.06 Metal Finishes.

c. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

2.8 ACCESSORIES

Fasteners: Type and size recommended by access door manufacturer, to suit application.

a. Stainless Steel Access Doors: Stainless steel fasteners.

b. Other Access Doors: Stainless steel fasteners.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation. Verify access door locations and sizes provide required maintenance access to installed building services components.

b. Protect existing construction and completed work from damage.

3.2 INSTALLATION - GENERAL

a. Install products according to manufacturer's instructions and approved submittal drawings. When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.

b. Install access doors and panels permitting access to service valves, traps, dampers, cleanouts, and other mechanical, electrical and conveyor control items concealed in walls and partitions, and concealed above gypsum board and plaster ceilings.

c. Install fire-rated access door according to NFPA 80. Upon completion of installation, fire access door assemblies shall be inspected and tested in accordance with NFPA 80.

d. Install fire-rated doors in fire-rated partitions and ceilings.

e. Install flush access panels in partitions and in gypsum board and plaster ceilings.

3.3 ACCESS DOOR AND FRAME INSTALLATION

a. Wall Installations: Install access doors in openings with sides vertical.

b. Ceiling Installations: Install access doors parallel to ceiling suspension grid or room partitions.

c. Frames without Flanges: Install frame flush with surrounding finish surfaces.

d. Frames with Flanges: Overlap opening, with face uniformly spaced from finish surface.

e. Recessed Panel Access Doors: Install with face of surrounding materials flush with door panel installed finish.

f. Secure frames to adjacent construction with fasteners.

g. Install type, size and quantity of anchoring device suitable for material surrounding opening to maintain alignment, and resist displacement, during normal use of access door.

h. Field Painting Primed Access Doors: Comply with the requirements of

Section 09 91 00, PAINTING.

3.4 ADJUSTMENT

a. Adjust hardware so door panel opens freely.

b. Adjust door when closed so door panel is centered in frame.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 33 00

COILING DOORS

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY
1.7 STORAGE AND HANDLING
1.8 FIELD CONDITIONS
1.9 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION
2.2 SYSTEM PERFORMANCE
2.3 MATERIALS
2.4 PRODUCTS - GENERAL
2.5 COILING DOORS
2.6 ELECTRIC MOTOR OPERATORS
2.7 MANUAL OPERATORS
2.8 FINISHES
2.9 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 COILING DOOR INSTALLATION
3.4 FIELD QUALITY CONTROL
3.5 CLEANING
3.6 PROTECTION

-- End of Section Table of Contents --

SECTION 08 33 00
1/21

COILING DOORS

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

a. Coiling doors.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Paints and Coatings VOC Limits.

b. Section 08 71 00, DOOR HARDWARE: Lock Cylinders for Cylindrical Locks.

c. Refer to Schedule for Finishes on the drawings for finish and color.

d. DIVISION 26, ELECTRICAL: Electric Devices and Wiring.

e. DIVISION 28, ELECTRONIC SAFETY AND SECURITY: Electric Devices and Wiring.

1.3 REFERENCES

Comply with references to extent specified in this Section.

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum

	and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal

MASTER PAINTERS INSTITUTE (MPI)

MPI 18	(2022) Primer, Zinc Rich, Organic
MPI 76	(2016) Primer, Alkyd, Quick Dry, for Metal

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1	(2022) Standard for Industrial Control and Systems: General Requirements
NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA MG 1	(2021) Motors and Generators
NEMA ST 20	(2014) Dry-Type Transformers for General Applications

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 80	(2025) Standard for Fire Doors and Other Opening Protectives

UL SOLUTIONS (UL)

UL 325	(2017; Reprint Feb 2020) UL Standard for Safety Door, Drapery, Gate, Louver, and Window Operators and Systems

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Dat; G, AE

Sustainable Construction Submittals; S

SD-07 Certificates

Certificates; G, AE

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submittal Drawings:

(1) Show size, configuration, and fabrication and installation details.

(2) Wiring diagrams for motors and controls, including wiring diagram for door, showing electrical interlock of motor with manually operated dead lock,

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

(2) Low Pollutant-Emitting Materials:

Show volatile organic compound types and quantities.

e. Certificates: Indicate each product complies with specifications

Indicate oversize fire doors and hardware that are identical in design, material, and construction to doors complying with specified performance.

f. Operation and Maintenance Data:

(1) Care instructions for each exposed finish product.

(2) Start-up, maintenance, troubleshooting, emergency, and shut-down instructions for each operational product.

(3) Parts list.

1.6 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.7 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.8 FIELD CONDITIONS

Field Measurements: Verify field conditions affecting overhead coiling door fabrication and installation. Show field measurements on Submittal Drawings. Coordinate field measurement and fabrication schedule to avoid delay.

1.9 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Manufacturer's Warranty: Warrant coiling doors against material and manufacturing defects.

Warranty Period: Two years.

PART 2 PRODUCTS

Finishes and materials indicated on the drawings are intended to establish the required pattern, color, texture, appearance, and other salient characteristics. The contractor shall submit samples of all proposed exterior finish materials and furnishings for government review and approval. No exterior finishes or furnishings shall be installed until written approval has been obtained from the government. Refer to the specifications for additional requirements, including material performance criteria and submittal requirements.

2.1 SYSTEM DESCRIPTION

a. Coiling doors spring counter balanced, overhead type, inside face mounted with guides at jambs set back at adequate distance to provide clear opening.

b. Where doors exceeding 80 square feet are indicated as manually operated, provide design, fabrication, and installation permitting future electric motor operation.

c. Configure coiling door and grille combinations with grille located on interior or room side of opening.

2.2 SYSTEM PERFORMANCE

a. Design coiling doors complying with specified performance:

Operation Cycles: 20,000 minimum.

2.3 MATERIALS

a. Sheet Steel: ASTM A653/A653M; G90 galvanized coating.

b. Structural Steel: ASTM A36/A36M.

c. Stainless Steel: ASTM A240/A240M, Type 302 or 304.

d. Aluminum, Plate and Sheet: ASTM B209M (ASTM B209).

e. Aluminum, Extrusions: ASTM B221M (ASTM B221), alloy 6063-T5.

2.4 PRODUCTS - GENERAL

a. Basis of Design: Refer to the SCHEDULE FOR FINISHES in the drawings.

b. Provide coiling doors from one manufacturer.

c. Sustainable Construction Requirements:

(1) Steel Recycled Content: 30 percent total recycled content, minimum.

(2) Stainless Steel Recycled Content: 70 percent total recycled content, minimum.

(3) Aluminum Recycled Content: 80 total recycled content, minimum.

(4) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products: Paints and coatings.

2.5 COILING DOORS

a. Doors: Interlocking slats of galvanized steel in manufacturer's standard profile, except exterior door slat profile to be flat.

(1) Slat Thickness: As required to resist specified loads.

(a) Galvanized Steel: Minimum thickness:

1. Doors less than 15 feet) wide: 0.030 inch.

2. Doors from 15 feet 1 inchto 21 feet wide: 0.036 inch.

3. Doors wider than 21 feet: 0.048 inch.

4. Insulation: 7/8 inch (22 mm) thick fire retardant mineral wool, ASTM C665-95 or ASTM C612-93

(b) Aluminum: Minimum thickness:

1. Doors less than 15 feet wide: 0.040 inch.

2. Doors from 15 feet 1 inch to 21 feet wide: 0.057 inch.

3. Doors wider than 21 feet: 0.064 inch.

4. Insulation: 7/8 inch (22 mm) thick fire retardant mineral wool, ASTM C665-95 or ASTM C612-93

(2) Bottom Rail: Two continuous angles of galvanized steel, minimum 0.125 inch thick, to receive safety device.

b. Hoods: Formed to fit contour of end brackets.

(1) Material:

(2) Sheet Steel: Minimum 0.0239 inch thick.

(a) Aluminum Sheet: Minimum 0.040 inch thick.

(3) Reinforce at top and bottom edges with rolled beads, rods or angles. Provide intermediate supporting brackets for hoods greater than 12 feet long.

(4) Fasten hood to brackets with screws or bolts and provide for fastening to wall with bolts.

c. Counterbalance Assembly:

(1) Design door to coil on barrel supported at end of opening on brackets, counterbalanced by helical springs. Counterbalance minimum 80 percent of door weight at any position.

(2) Barrel: Steel pipe or commercial welded steel tubing of proper diameter and thickness for door size, to limit deflection with door rolled up, not to exceed 1 in 400 (0.03 inch per foot) of span. Close barrel ends with cast iron plugs, machined to fit the opening.

(3) Spring: Oil-tempered, helically wound spring rotating on grease-sealed ball bearing or roller bearing units, capable of producing sufficient torque to ensure easy operation of door from any position, and adjustable from exterior of counterbalance assembly without removing hood or motor operator.

d. Brackets: Steel plate forming end closure and support for hood and end of barrel assembly.

Screw end of barrel or shaft into bracket hubs fabricated of cast iron or steel. Equip bracket hubs or barrel plugs with pre-lubricated ball bearings, shielded or sealed.

e. Guides: Standard formed sections or angles of steel.

(1) Thickness: Minimum 3/16 inch.

(2) Profile: Channel of sufficient depth to retain door in place under the horizontal pressure specified, and prevent ends of door from slipping out of guides. Flare guides at top to facilitate door installation.

(3) Provide stops to limit door travel above top of guides.

(4) Provide aluminum guides with replaceable wear strips to prevent metal to metal contact.

(5) Provide mounting brackets for closure between guides and jambs.

f. Locking:

(1) Cylinder Locks: Accept standard screw in cylinders furnished under Section, 08 71 00, DOOR HARDWARE.

(2) Electrically Operated Doors : Provide manufacturer's standard cylinder dead locking device on inside, key operated from both sides, interlocked with motor to prevent motor from operating when locks are activated.

2.6 ELECTRIC MOTOR OPERATORS

a. Provide operators complete with electric motor, machine cut reduction gears, steel chain and sprockets, magnetic brake, overload protection, brackets, push button controls, limit switches, magnetic reversing contactor, and other accessories necessary for proper operation including emergency manual operator.

b. Design:

(1) Design operator so motor can be removed without disturbing limit-switch timing and without affecting emergency manual operators.

(2) Make provision for emergency manual operation by chain-gear mechanism.

(3) Arrange emergency manual operating mechanism so it can be immediately put into and out of operation from floor with electrical or mechanical device to disconnect motor from operating mechanism when emergency manual operating mechanism is engaged without affecting limit switch timing, in case of electrical power failure.

(4) Provide interlock with motor to prevent motor from operating when manual locks are activated.

c. Motors: NEMA MG 1; TENV open drip-proof, maximum 3600 rpm.

(1) Power Characteristics: as indicated on drawings.

(2) High starting torque, reversible type, of sufficient horsepower and torque output to operate door in both directions from any position, and produce door travel speed of 8 inches/second, minimum, and 12 inches/sec., maximum, without exceeding rated capacity.

d. Controls: NEMA ICS 1 and NEMA ICS 2.

(1) Enclosures: NEMA ICS 6, Type 12 or Type 4, except Contractor enclosures may be Type 1.

(2) Provide each motor with an enclosed, across-the-line type, magnetic reversing contactor, thermal overload protection, solenoid operated brake, limit switches, and remote control switches at locations shown.

(3) Provide key activated switches on exterior requiring constant pressure to operate.

(4) Provide three-button type, push button switch on interior, unless noted to be key activated, with buttons marked, OPEN, CLOSE, and STOP.

(a) Type: Fully guarded to prevent accidental operation.

(b) OPEN and STOP Buttons: Momentary contact type.

(c) CLOSE Button: Constant contact type.

(d) When the door is in motion, pressing STOP button causes door to stop instantly and remain in stop position. From stop position, pressing OPEN or CLOSE buttons will operate door.

(5) Provide field adjustable, limit switches to automatically stop doors at fully open and closed positions. Locate limit switches to be readily accessible for adjustment.

(6) Safety device:

(a) Provide UL 325, non-contact sensors for garage door operators with photoelectric sensors in opening of electrically operated doors to disrupt the door’s operation when an object lies in the field of view and safety switches that reverse the door to an open position when extra current is used by the motor when encountering an obstruction.

(b) Upon safety device and control system failure, immediately stop, reverse, and fully open doors and lock out electrical controls. Permit continued manual operation until electrical controls are repaired.

(c) Do not use safety device as limit switch.

(7) Transformer: NEMA ST 20.

Control transformer in power circuits as required to reduce Voltage on control circuits to 120 Volts or less.

(8) Electrical Components: Comply with NFPA 70.

Hazardous Locations: UL Listed for specific hazard indicated on drawings.

2.7 MANUAL OPERATORS

a. Hand Chain Operation:

(1) Galvanized, endless chain operating over sprocket and extending to within 3 feet of floor.

(2) Obtain reduction through permanently lubricated gearing connected by roller chain and sprocket drive.

(3) Opening and Closing Force: Maximum 25 pound force.

b. Crank Operation:

(1) Locate crank approximately 34 inches above floor.

(2) Connect vertical shaft, gear box, and gears to doors .

(3) Opening and Closing Force: Maximum 25 pound force.

2.8 FINISHES

a. Steel:

(1) Clean steel surfaces of scale, rust, oil, and grease.

(2) Apply light colored shop prime paint after fabrication.

(a) Non-Galvanized Steel: Treat to ensure maximum paint adhesion, and apply corrosion inhibitive primer.

(b) Galvanized Steel: Apply phosphate treatment and corrosion inhibitive primer.

b. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

2.9 ACCESSORIES

a. Galvanizing Repair Paint: MPI 18.

b. Alkyd Metal Primer: MPI 76.

c. Barrier Coating: ASTM D1187/D1187M.

d. Touch-Up Paint: Match shop finish.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

3.2 INSTALLATION - GENERAL

Install products according to manufacturer's instructions and approved submittal drawings under direct supervision of manufacturer's representative or trained personnel. Install fire-rated doors to comply with NFPA 80.

3.3 COILING DOOR INSTALLATION

a. Locate anchors and inserts for guides, brackets, motors, switches, hardware, and other accessories accurately.

b. Securely attach guides to adjoining construction with minimum 3/8 inch

diameter bolts, spaced maximum 24 inches on center.

c. Locate control switches where shown.

Locate control switches minimum (5 feet) above floor line, so operating personnel have complete view of door .

d. Install electric devices and wiring as specified in DIVISION 26, ELECTRICAL and DIVISION 28, ELECTRONIC SAFETY AND SECURITY.

e. Isolate aluminum in contact with dissimilar metal, concrete and masonry by painting with coat of bituminous paint.

f. Paint aluminum in contact with wood or other absorptive materials with barrier coating.

g. Touch up damaged factory finishes.

h. Lubricate and adjust units to operate freely.

i. Touch up damaged factory finishes.

(1) Galvanized Surfaces: Apply galvanizing repair paint.

(2) Primed Surfaces: Apply touch up paint.

3.4 FIELD QUALITY CONTROL

Test fire-rated door closing and reset device after test is successfully completed.

3.5 CLEANING

a. Clean exposed doors surfaces. Remove contaminants and stains.

b. Polish exposed stainless steel surfaces.

3.6 PROTECTION

a. Protect coiling doors from construction operations.

b. Remove protective materials immediately before acceptance.

c. Repair damage.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 41 13

ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED REQUIREMENTS
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 PREINSTALLATION MEETINGS
1.7 QUALITY ASSURANCE
1.8 DELIVERY, STORAGE AND HANDLING
1.9 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE
2.2 MATERIALS
2.3 PRODUCTS - GENERAL
2.4 FRAMES
2.5 STILE AND RAIL DOORS
2.6 FLUSH PANEL DOORS
2.7 COLUMN COVERS AND TRIM
2.8 FABRICATION
2.9 FINISHES
2.10 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 PROTECTION, CLEANING AND REPAIRING

-- End of Section Table of Contents --

SECTION 08 41 13
1/21

ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS

PART 1 GENERAL

1.1 SUMMARY

Section Includes: Aluminum-framed entrances and storefronts.

1.2 RELATED REQUIREMENTS

a. Door Finish and Color: Refer to Schedule for Finishes on the drawings.

b. Glass and Glazing: Section 08 80 00, GLAZING.

c. Hardware: Section 08 71 00, DOOR HARDWARE.

d. Aluminum Finish and Color: Refer to the SCHEDULE FOR FINISHES on the drawings.

1.3 REFERENCES

Comply with references to extent specified in this section.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605	(2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7	(2017) Minimum Design Loads for Buildings and Other Structures

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M	(2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

ASTM D1187/D1187M	(1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal
ASTM E283	(2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen
ASTM E330/E330M	(2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference
ASTM E331	(2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
ASTM E1886	(2019) Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials
ASTM E1996	(2017) Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes
ASTM F468	(2023) Standard Specification for Nonferrous Bolts, Hex Cap Screws, and Studs for General Use
ASTM F593	(2024) Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs

MASTER PAINTERS INSTITUTE (MPI)

MPI 18	(2022) Primer, Zinc Rich, Organic

NATIONAL FENESTRATION RATING COUNCIL (NFRC)

NFRC 500	(2010) Procedure for Determining Fenestration Product Condensation Resistance Valves

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the

following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preinstallation Meeting Minutes; G, AE

Qualifications; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Sustainable Construction Submittals; S

SD-04 Samples

Samples; G, AE

SD-05 Design Data

Delegated Design Drawings And Calculations; G, AE

SD-06 Test Reports

Test Reports; G, AE

SD-07 Certificates

Certificates; G, AE

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submittal Drawings: Minimum 1 to 2 (half size) scale.

(1) Show size, configuration, and fabrication and installation details.

(2) Show anchorage and reinforcement.

(3) Show interface and relationship to adjacent work, including thermal, air, and water barrier continuity.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Doors, each type.

(3) Entrance and Storefront construction.

(4) Installation instructions.

(5) Warranty.

d. Samples:

(1) Door Corner Section: Minimum18 x 18 inches for each specified door type, showing head rail and hinge stile, door closer reinforcement, internal reinforcement .

(2) Aluminum Anodized Finish: Two sample extrusions minimum 6 inches long for each specified color in sets of three showing maximum color range.

(3) Aluminum Paint Finish: Two sample extrusions minimum 6 inches long for each specified color.

e. Sustainable Construction Submittals: Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

f. Test reports: Certify each product complies with specifications.

g. Certificates: Certify each product complies with specifications. Certify anodized finish thickness.

h. Qualifications: Substantiate qualifications comply with specifications.

(1) Manufacturer with project experience list .

(2) Installer with project experience list .

(3) Welders and welding procedures.

i. Delegated Design Drawings and Calculations: Signed and sealed by responsible design professional.

(1) Show location and magnitude of loads applied to building structural frame.

(2) Identify deviations from details shown on drawings.

(3) Blast Design Calculations:

(a) Submit calculations for review and approval prepared by qualified blast consultant, with a minimum of 5 years of experience in design of blast resistant window systems, verifying storefront assembly including anchors comply with specified blast resistance performance. Include a blast narrative and supporting calculations that correspond with drawings. Blast resistant glazing must meet the blast performance requirements stated in Section 08 80 00. Blast resistant frames and mullions must meet the blast resistant requirements for Critical Facilities per section 6.3 of the VA PSDRM. Blast analysis must be performed using the dynamic analysis methods per Section 6.7.3 of the PSDRM. Alternatively, glazed aluminum framed systems may be certified through testing (Standard Test Method for Glazing Systems Subject to Airblast Loadings ASTM F1642) against the specified design blast load. The tested assembly must be demonstrated to be sufficiently similar in glazing layup, mullions, frames,

connections, and hardware to that being constructed for the Project.

j. Operation and Maintenance Data:

(1) Care instructions for each exposed finish product.

1.6 PREINSTALLATION MEETINGS

Conduct preinstallation meeting at project site minimum 30 days before beginning Work of this section.

a. Required Participants:

(1) Contracting Officer's Representative.

(2) Architect/Engineer.

(3) Contractor.

(4) Installer.

(5) Manufacturer's field representative.

b. Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(1) Installation schedule.

(2) Installation sequence.

(3) Preparatory work.

(4) Protection before, during, and after installation.

(5) Installation.

(6) Terminations.

(7) Transitions and connections to other work.

(8) Other items affecting successful completion.

c. Document and distribute preinstallation meeting minutes to participants to record decisions affecting installation.

1.7 QUALITY ASSURANCE

a. Manufacturer Qualifications:

(1) Regularly manufactures specified products.

(2) Manufactured specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

b. Installer Qualifications: Manufacturer authorized representative.

(1) Regularly installs specified products.

(2) Installed specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

c. Welders and Welding Procedures Qualifications: AWS D1.2/D1.2M.

1.8 DELIVERY, STORAGE AND HANDLING

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

d. Store products indoors in dry, weathertight facility.

e. Protect products from damage during handling and construction operations.

1.9 WARRANTY

a. Construction Warranty: FAR clause 52.246-21, "Warranty of Construction."

b. Manufacturer's Warranty: Warrant painted finish against material and manufacturing defects. Warranty Period: 20 years.

PART 2 PRODUCTS

Finishes and materials indicated on the drawings are intended to establish the required pattern, color, texture, appearance, and other salient characteristics. The contractor shall submit samples of all proposed exterior finish materials and furnishings for government review and approval. No exterior finishes or furnishings shall be installed until written approval has been obtained from the government. Refer to the specifications for additional requirements, including material performance criteria and submittal requirements.

2.1 SYSTEM PERFORMANCE

a. Delegated Design: Prepare submittal documents including design calculations and drawings signed and sealed by registered design professional, licensed in state where work is located.

(1) Minor deviations to details shown on drawings to accommodate manufacturer’s standard products may be accepted by Contracting Officer's Representative when deviations do not affect design concept and specified performance.

b. Design aluminum framed entrances and storefronts complying with specified performance:

(1) Wind and Seismic Load Resistance: ASCE 7; Design criteria as indicated on Drawings when tested according to ASTM E330/E330M.

(a) Wind Load: 30 psf positive and negative, minimum.

(b) Maximum Deflection: 1/175 of span, maximum with minimum 1.65 safety factor.

(2) Thermal Movement: Accommodate ambient temperature range of 120 degrees Fahrenheit.

(3) Blast Resistance:

(a) Blast resistant frames and mullions must be designed to resist a design blast load of 13.5 psi peak pressure and 51 psi-msec impulse (as an equivalent triangular load) applied over the tributary area of the member while sustaining maximum deformations no greater than L/40 (or 3 degrees support rotation). In addition, the frames and mullions must be designed to resist the balanced design load of 15.1 psi peak pressure and 57 psi-msec impulse (as an equivalent triangular load) while sustaining maximum deformations no greater than L/20 (or 6 degrees support rotation).

(b) Connection between framing members and connections of the frame members to the supporting structure must be designed to resist the applied blast loads and develop the maximum flexural capacity of the supported members.

(4) Windborne-Debris Impact Resistance: Pass ASTM E1886.

(a) Openings within 30 feet of Grade: ASTM E1996 large missile test.

(b) Other Openings: ASTM E1996 small missile test.

(5) Condensation Resistance: NFRC 500. Fixed Framing: 45 CRF, minimum.

(6) Water Resistance: ASTM E331; No uncontrolled penetration at 8 pounds/square foot, minimum, pressure differential.

(7) Fixed Framing Air Infiltration Resistance: ASTM E283; 0.06 cubic foot/minute/square foot, maximum at 6.24 pounds/square foot, minimum, pressure differential.

(8) Entrance Doors Air Infiltration Resistance: ASTM E283; maximum allowable at 1.57 pounds/square foot, minimum, pressure differential.

(a) Single Doors: 0.5 cubic foot/minute/square foot.

(b) Paired Doors: 1.2 cubic foot/minute/square foot.

2.2 MATERIALS

a. Aluminum:

(1) Sheet Metal: ASTM B209MASTM B209, minimum 0.063 inch thick.

(2) Extrusions: ASTM B221MASTM B221.

(a) Framing: Minimum 0.125 inch) wall thickness.

(b) Glazing Beads, Moldings, and Trim: Minimum 0.050 inch thick.

(3) Alloy 6063 temper T5 for doors, door frames, fixed glass sidelights storefronts .

(4) Alloy 6061 temper T6 for guide tracks for sliding doors and other extruded structural members.

(a) Color Anodized Aluminum: Provide aluminum alloy required to produce specified color.

b. Stainless Steel: ASTM A240/A240M; Type 302 or Type 304.

c. Thermal Break: Manufacturer standard low conductive material retarding heat flow in the framework, where insulating glass is scheduled.

2.3 PRODUCTS - GENERAL

a. Basis of Design: Refer to SCHEDULE FOR FINISHES on the drawings.

b. Provide aluminum framed entrances and storefronts from one manufacturer and from one production run .

c. Provide aluminum entrances, storefront, windows, curtain wall systems from same manufacturer.

d. Sustainable Construction Requirements:

(1) Aluminum Recycled Content: 50 percent total recycled content, minimum.

2.4 FRAMES

a. Framing Members: Extruded aluminum, thermally broken .

b. Stops: Provide integral fixed stops and glass rebates and snap-on removable stops.

c. Provide concealed screws, bolts and other fasteners.

d. Secure cover boxes to frames in back of lock strike cutouts.

2.5 STILE AND RAIL DOORS

a. Stiles and Rails: Extruded aluminum, thermally broken .

(1) Thickness: 1-3/4 inch.

(2) Stiles and Head Rails: 3-1/2 inches wide.

(3) Bottom Rails: 10 inches wide.

b. Single-Acting Doors:

(1) Bevel: 1/8 inch at lock, hinge, and meeting stile edges.

(2) Clearances: 1/16 inch at hinge stiles, 1/8 inch at lock stiles and top rails, and 3/16 inch at floors and thresholds.

c. Glass Rebates: Integral with stiles and rails.

d. Glazing Beads: Extruded aluminum, 0.050 inch thick. Integral with stiles and rails or applied type, snap-fit secured.

e. Stile and Rail Joints: Welded or interlocking dovetail joints between stiles and rails.

(1) Clamp door together through top and bottom rails with 3/8 inch primed steel tie rod extending into stiles, and having self-locking nut and washer at both ends.

(2) Reinforce stiles and rails to prevent door distortion when tie rods are tightened.

(3) Provide compensating spring-type washer under each nut for stress relief.

(4) Construct joints to remain rigid and tight when door is operated.

f. Weather-stripping: Removable, woven pile type (silicone-treated) weather-stripping attached to aluminum or vinyl holder.

(1) Make slots for applying weather-stripping integral with doors and door frame stops.

(2) Apply continuous weather-stripping to heads, jambs, bottom, and meeting stiles of doors and frames so doors swing freely and close positively.

2.6 FLUSH PANEL DOORS

a. Frames: Aluminum extrusions.

b. Doors: 1-3/4 inches thick.

(1) Door Edges and Internal Reinforcing: Extruded aluminum tubes, single piece full height and width, welded joints.

(2) Core: Manufacturer's standard non-combustible insulation.

(3) Faces: Aluminum sheet metal with internal impact reinforcement, laminated to the door edges and core.

2.7 COLUMN COVERS AND TRIM

a. Column Covers and Trim: Sheet aluminum fabrications shown from sheet aluminum of longest available lengths.

b. Provide concealed fasteners.

c. Provide aluminum stiffeners and supporting members shown on drawings and as required to maintain component integrity and shape.

2.8 FABRICATION

a. Form metal parts and fit and assemble joints, except joints designed to accommodate movement. Seal joints to resist air infiltration and water penetration.

b. Welding:

(1) Make welds without distorting and discoloring exposed surfaces.

(2) Clean and dress welds. Remove welding flux and weld spatter.

c. Prepare and reinforce doors and frames for hardware and accessories.

(1) Coordinate preparation with specified hardware. See Section 08 71 00, DOOR HARDWARE.

(2) Fabricate reinforcement from stainless steel plates.

(a) Hinge and pivot reinforcing: Minimum 0.179 inch thick.

(b) Lock Face, Flush Bolts, Concealed Holders, Concealed and Surface Mounted Closers Reinforcing: Minimum 0.104 inch thick.

(c) Other Surface Mounted Hardware Reinforcing: Minimum 0.059 inch thick.

(3) Where concealed hardware is specified, provide space, cutouts, and reinforcement for installation and secure fastening.

d. Factory assembled doors.

2.9 FINISHES

a. Aluminum Paint finish:

(1) Fluorocarbon Finish: AAMA 2605; 70 percent fluoropolymer resin, 2-coat 2-coat system.

b. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

2.10 ACCESSORIES

a. Dielectric Tape: Plastic, non-absorptive, with pressure sensitive adhesive; 7 to 10 mils thick.

b. Barrier Coating: ASTM D1187/D1187M.

c. Welding Materials: AWS D1.2/D1.2M, type to suit application.

d. Fasteners:

(1) Aluminum: ASTM F468, Alloy 2024.

(2) Stainless Steel: ASTM F593, Alloy Groups 1, 2 and 3.

e. Anchors: Aluminum or stainless steel; type to suit application.

f. Galvanizing Repair Paint: MPI 18.

g. Touch-Up Paint: Match shop finish.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

(1) Coordinate floor closer installation recessed into concrete slabs.

(2) Coordinate anchor installation built into masonry and concrete.

b. Protect existing construction and completed work from damage.

c. Clean substrates. Remove contaminants capable of affecting subsequently installed product's performance.

d. Apply dielectric tape or barrier coating to aluminum surfaces in contact with dissimilar metals and cementitious materials to minimum 30 mils dry film thickness.

3.2 INSTALLATION - GENERAL

a. Install products according to manufacturer's instructions and approved submittal drawings .

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.

b. Install aluminum framed entrances and storefronts plumb and true, in alignment and to lines shown on drawings.

c. Anchor frames to adjoining construction at heads, jambs and sills.

d. Provide concealed aluminum clips to connect adjoining frame sections.

e. Install door hardware and hang doors. See Section 08 71 00, DOOR HARDWARE.

f. Adjust doors and hardware uniform clearances and proper operation.

g. Touch up damaged factory finishes.

(1) Repair galvanized surfaces with galvanized repair paint.

(2) Repair painted surfaces with touch up primer.

h. Tolerances:

(1) Variation from Plumb, Level, Warp, and Bow: Maximum 1/8 inch in 10 feet).

(2) Variation from Plane: Maximum 1/8 inch in 12 feet); 1/4 inch) over total length.

(3) Variation from Alignment: Maximum 1/16 inch in-line offset and maximum 1/8 inch) corner offset.

(4) Variation from Square: Maximum 1/8 inch diagonal measurement differential.

3.3 PROTECTION, CLEANING AND REPAIRING

a. Clean exposed aluminum and glass surfaces. Remove contaminants and stains.

b. Protect aluminum-framed entrances and storefronts from construction operations.

c. Remove protective materials immediately before acceptance.

d. Repair damage.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 44 13

GLAZED ALUMINUM CURTAIN WALLS

4/20

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY ASSURANCE
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY, STORAGE AND HANDLING
1.7 PROJECT CONDITIONS
1.8 REFERENCES
1.9 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION
2.2 PERFORMANCE REQUIREMENTS
2.3 MATERIALS
2.4 FABRICATION
2.5 METAL FINISHES

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 INSTALLATION
3.4 ADJUSTING
3.5 CLEANING
3.6 FIELD QUALITY CONTROL
3.7 DEMONSTRATION, TESTING, AND ACCEPTANCE
3.8 PROTECTION

-- End of Section Table of Contents --

SECTION 08 44 13
4/20

GLAZED ALUMINUM CURTAIN WALLS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies glazed aluminum curtain wall system.

a. Thermally isolated, pressure equalized on interior.

b. Type: Stick system to include following:

(1) Glass .

(2) Integral reinforcing.

(3) Integral reinforcing for vertical aluminum fin supports.

(4) Aluminum T-bracket supports for vertical aluminum fins.

(5) Closures, trim, subsills and flashings.

(6) Fasteners, anchors, and related reinforcement.

1.2 RELATED WORK

a. Sustainable Design Requirements: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Structural Steel: Section 05 12 00, STRUCTURAL STEEL FRAMING.

c. Miscellaneous Metal Members: Section 05 50 00, METAL FABRICATIONS.

d. Vertical Aluminum Fins: Section 07 42 63, FABRICATED WALL PANEL ASSEMBLIES

d. Firestopping between Curtain Wall and Structure: Section 07 84 00, FIRESTOPPING.

(1) Sheet Metal Flashing and Trim: Section 07 60 00, FLASHING AND SHEET METAL.

e. Joint Sealants: Section 07 92 00, JOINT SEALANTS.

f. Aluminum and Glass Hinged Entry Doors and Storefront Construction: Section 08 41 13, ALUMINUM-FRAMED ENTRANCES AND STOREFRONTS.

g. Glazing: Section 08 80 00, GLAZING.

h. Finish Color: Refer to the SCHEDULE FOR FINISHES on the drawings.

i. Louvers: Section 08 90 00, LOUVERS AND VENTS.

1.3 QUALITY ASSURANCE

a. Qualifications:

Approval by Contracting Officer is required of products or service of proposed manufacturer, suppliers and installers, and will be based upon submission by Contractor of certification that:

(1) Manufacturers Qualifications: Manufacturer with five years continuous documented experience in design, fabrication, and installation of glazed aluminum curtain wall systems of similar type and for projects of equivalent size.

(2) Installer: Manufacturer approved in writing who has continuously installed glazed aluminum curtain walls systems of similar type and for projects of equivalent size for previous five years.

(3) Manufacturer is to provide technical field representation at project site, as a minimum, at start of project, during middle, towards end of project, and during field testing of field mockup panel.

(4) Manufacturers Professional Engineer Qualifications: A Professional Engineer who is legally qualified to practice in state where Project is located and who is experienced in providing engineering services of kind indicated. Engineering services are defined as those performed for installations of glazed aluminum curtain walls that are similar to those indicated for this Project in material, design, and extent.

(5) Testing Laboratory: Contractor is to retain AAMA accredited commercial testing laboratory to perform tests specified. Submit information regarding testing laboratory’s facilities and qualifications of technical personnel to perform testing specified in this Section.

(6) Product Options: Information on construction documents establishes requirements for aesthetic effects and performance characteristics of glazed aluminum curtain wall system. Aesthetic effects are indicated by dimensions, arrangements, alignment, and profiles of components and assemblies as they relate to sightlines, one another, and adjoining construction. Performance characteristics are indicated by criteria subject to verification by one (1) or more methods including preconstruction testing, field testing, or in-service performance.

Do not modify intended aesthetic effects. If modifications are proposed, submit comprehensive explanatory data for review.

(7) Qualification of Welders:

Welding is to be performed by certified welders qualified in accordance with AWS D1.2/D1.2M, using procedures, materials, and equipment of the type required for this work.

b. Mockup:

(1) Construct, at job site, full size typical wall unit which

incorporates horizontal and vertical joints, framing, window units, panels, glazing, vertical aluminum fins and brackets, sealants, and other accessories as detailed and specified. Mock-up wall unit location, size and design are to be as indicated on construction documents. Orient mockup to be facing full sun when constructed. Refer to section 01 43 30 MOCKUPS for specific requirements.

(2) Performance Test

(a) Conduct performance test of mockup after approval of visual aspects has been obtained. Testing is to be performed on mockup according to requirements in “Field Quality Control” Article.

(b) Refer to Performance Requirements and Field Quality Control Articles, this section, for testing requirements.

(3) Approved Mock-up

(a) After completion and approval of performance test results of job site mockup, as directed by Contracting Officer, approved mock-up panel is to be used as minimum standard of comparison for entire curtain wall system.

c. Pre-Installation Conference

Prior to starting installation of glazed curtain wall system schedule conference with Contracting Officer to demonstrate the following:

(1) Clear understanding of construction documents.

(2) Onsite inspection and acceptance of structural and pertinent structural details relating to curtain wall system.

(3) Coordination of work of various trades involved. Conference is to be attended by Contractor; personnel directly responsible for installation of curtain wall system, flashing and sheet metal work, firestopping system and curtain wall manufacturer and their technical field representatives. Conflicts are to be resolved and confirmed in writing.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welders Qualifications; G, AE

Testing Laboratory Qualifications; G, AE

Testing Laboratory's Facilities And Qualifications Of Technical Personnel; G, AE

Manufacturers Qualifications; G, AE

Mockup; G, AE

Installer; G, AE

Technical Field Representation; G, AE

Manufacturers Professional Engineer Qualifications; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Sustainable Design Submittals; G, AE

Manufacturer's Literature And Product Data; G, AE

Glass; G, AE

Product Options; G, AE

SD-04 Samples

Samples; G, AE

SD-05 Design Data

Design Data; G, AE

Professionally Prepared Calculations; G, AE

SD-06 Test Reports

Factory Test Reports; G, AE

SD-07 Certificates

Manufacturer's Certificates; G, AE

SD-09 Manufacturer's Field Reports

Manufacturer’s Field Reports; G, AE

SD-10 Operation and Maintenance Data

Replacement Glazing Instructions

SD-11 Closeout Submittals

Manufacturer Warranty; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. In accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

(2) Percentage of recycled content.

c. Manufacturer's Literature and Product Data:

(1) Manufacturer’s standard details and fabrication methods.

(2) Data on finishing, components, and accessories.

(3) Instructions: Submit descriptive literature, detail specifications, performance test data and instructions for installation, and adjustments.

(4) Recommendations for maintenance and cleaning of exterior surfaces.

d. Shop Drawings:

(1) Show elevations of glazed curtain wall system at 1/4 inch scale, metal gages, details of construction, methods of anchorage, flashing and coping details, glazing details, firestopping assemblies at edge of slabs and details of installation. Show interfaces and relationships to work of other trades and continuity with adjacent thermal, weather, air and vapor barriers.

(2) Submit for curtain wall system, accessories , and mock-up. Tentative approval of drawings is to be received before fabrication of mock-up. Final approval of drawings is to be deferred pending approval of mock-up and accessories.

(3) Operation and Maintenance Manuals

(a) Submit cleaning and maintenance instructions.

e. Samples:

(1) Submit pairs of samples of each specified color and finish on 12-inch long section by width of each tubular, or extruded shape section or 12-inch by 12-inch wide sections of sheet shapes.

(2) Submit corner section of framing members showing fasteners, panels, glazing methods, glazing materials, and weather-stripping. Submit one sample minimum 12 inches by 12 inches. In lieu of submitting separate samples for corner section, intermediate section, and panel, one composite sample incorporating all components and features listed may be submitted.

(3) Where normal color variations are anticipated, include two or more units of each sample indicating extreme limits of color variations.

f. Glass:

(1) Specified in Section 08 80 00, GLAZING.

g. Quality Assurance Submittals:

(1) Design Data:

(a) Submit structural and thermal calculations for complete wall assembly, including attachment of aluminum T-bracket supports for vertical aluminum fins. Structural calculations and design shop drawings signed and sealed by a Professional Engineer (PE).

(b) Submit blast design calculations prepared by qualified blast consultant with a minimum of 5 years of experience in design of blast resistant window systems. Include a blast narrative and supporting calculations that correspond with drawings. Blast resistant glazing must meet the blast performance requirements stated in Section 08 80 00. Blast resistant frames and mullions must meet the blast resistant design requirements for Critical Facilities per Section 6.3 of the VA PSRDM. Blast analysis must be performed using the dynamic analysis methods per Section 6.7.3 of the PSRDM.

(2) Factory Test Reports:

Test Reports: Submit certified test reports, for each of following listed tests, from a qualified independent testing laboratory showing that glazed aluminum curtain wall system assembly has been tested in accordance with specified test procedures and complies with performance characteristics as indicated by manufacturer’s testing procedures. Submit factory tests required except that where a curtain wall system or component of similar type, size, and design as specified for this project has been previously tested within last year, under conditions specified herein, resulting test reports may be submitted in lieu of listed testing. Submit appropriate testing reports for specific tests indicated below:

(a) Deflection and structural tests.

(b) Water penetration tests.

(c) Air infiltration tests.

(d) Delamination tests.

(e) Thermal conductance tests.

(f) Blast Testing: When certifying blast resistance through testing (ASTM F1642), submit test report demonstrating sufficiently similar glazing layup, mullions, frames, connections, and hardware compared to assembly being installed.

h. Manufacturer's Certificates:

Submit Certificates of Compliance, with specification requirements, for the following:

(1) Metal extrusions.

(2) Metal accessories.

(3) Statement(s) that aluminum has been given specified thickness of anodizing or organic coating finish.

(4) Statement(s) indicating manufacturers and installers conform with qualifications as specified.

(5) Submit list (minimum of five of equivalent project size installations for both manufacturer and installer.

i. Manufacturer’s Field Reports:

Submit field reports of manufacturer’s field representative observations of curtain wall installation indicating observations made during inspection at beginning of project, during middle of installation and at conclusion of project.

j. Welders: Submit welders qualifications as specified.

k. Testing Laboratory: Submit Testing Laboratory qualifications.

1.6 DELIVERY, STORAGE AND HANDLING

a. Refer to AAMA CW-10 for care and handling of architectural aluminum from shop to site.

b. Prior to packaging for shipment from factory, mark wall components to correspond with shop and erection drawings and their placement location and erection sequence.

c. Prior to shipment from factory, place knocked-down lineal curtain wall members in cardboard containers and cover finished surfaces of members with protective covering of adhesive paper, waterproof tape, or strippable plastic. Do not cover metal surfaces that will be in contact with sealants after installation.

d. Inspect materials delivered to site for damage; unload and store with ventilation, free from heavy dust, not subject to combustion products or sources of water, and to permit easy access for inspection and handling. Sealing and caulking compounds, including handling, is to be in accordance with requirements of Section 07 92 00, JOINT SEALANTS.

1.7 PROJECT CONDITIONS

Where glazed aluminum curtain wall systems are indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on shop drawings. Coordinate fabrication schedule with construction progress to avoid delaying Work.

1.8 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referred to in text by basic designation only.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA MCWM-1	(1989) Metal Curtain Wall Manual
AAMA CW-10	(2015) Care and Handling of Architectural Aluminum from Shop to Site

AAMA TIR A11	(2015) Maximum Allowable Deflection of Framing Systems for Building Cladding Components at Design Wind Loads
AAMA 501	(2015) Methods of Test for Exterior Walls
AAMA 503	(2014) Field Testing of Newly Installed Storefronts, Curtain Walls and Sloped Glazing Systems
AAMA 2605	(2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7	(2017) Minimum Design Loads for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A193/A193M	(2024a) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B211/B211M	(2023) Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)
ASTM B316/B316M	(2020) Standard Specification for Aluminum and Aluminum-Alloy Rivet and Cold-Heading Wire and Rods

ASTM C794	(2022) Standard Test Method for Adhesion-in-Peel of Elastomeric Joint Sealants
ASTM C1193	(2016) Standard Guide for Use of Joint Sealants
ASTM C1363	(2024) Standard Test Method for Thermal Performance of Building Materials and Envelope Assemblies by Means of a Hot Box Apparatus
ASTM C1521	(2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E330/E330M	(2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference
ASTM E331	(2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
ASTM E413	(2022) Classification for Rating Sound Insulation
ASTM E783	(2002; R 2018) Standard Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors
ASTM E1105	(2015; R 2023) Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M	(2014; Errata 1 2014; Errata 2 2020) Structural Welding Code - Aluminum

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-C-18480	(1982; Rev B; Notice 2 2009; Notice 3 2022) Coating Compound, Bituminous, Solvent, Coal-Tar Base

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 25	(1997; E 2004) Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel, Type I and Type II
SSPC Paint 20	(2019) Zinc-Rich Primers (Type I, Inorganic, and Type II, Organic)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

1.9 WARRANTY

a. Construction Warranty: Comply with FAR clause 52.246-21, “Warranty of Construction”.

b. Manufacturer Warranty: Manufacturer shall warranty their glazed aluminum curtain wall system for a minimum of five years from date of installation and final acceptance by the Government. Submit manufacturer warranty.

PART 2 PRODUCTS

Finishes and materials indicated on the drawings are intended to establish the required pattern, color, texture, appearance, and other salient characteristics. The contractor shall submit samples of all proposed exterior finish materials and furnishings for government review and approval. No exterior finishes or furnishings shall be installed until written approval has been obtained from the government. Refer to the specifications for additional requirements, including material performance criteria and submittal requirements.

2.1 SYSTEM DESCRIPTION

a. Design Requirements:

(1) Curtain Wall System: Tubular aluminum sections with thermal break condition self-supporting framing, factory prefinished, vision glass, ; related flashings, anchorage, attachment devices, reinforcement and aluminum T-bracket supports for vertical aluminum shading fins.

(2) System Assembly: Site assembled.

(3) Maximum wall framing member deflection, in a direction normal to plane of wall: 1/175 of its clear span or 3/4 inch, whichever is less, when designed in accordance with requirements of AAMA TIR A11 and tested in accordance with ASTM E330/E330M.

(4) Maximum wall framing member deflection when a gypsum wallboard surface is affected: 1/360 of span.

(5) Maximum Framing Member Permanent Deformation: 0.2 percent of its clear span when tested in accordance with ASTM E330/E330M for a

minimum test period of 10 seconds at 1.5 times design wind pressures indicated as part of structural drawing wind load requirements.

(6) Aluminum Framing Size: 2-1/2" wide X 10-1/2" deep.

b. No glass breakage, or damage to fasteners, hardware or accessories is permitted due to deformation design requirements indicated and from the attachment of the vertical aluminum fins.

(a) Provide system complete with framing, mullions, trim, fasteners, anchors, accessories, concealed auxiliary members, and attachment devices for securing wall to structure as specified or indicated. Unless noted otherwise, comply with AAMA MCWM-1.

(b) Obtain all components of curtain wall system, including framing entrances from single manufacturer.

(c) Fully coordinate system accessories directly incorporated and adjacent to contiguous related work and ensure materials compatibility, deflection limitations, thermal movements, and clearances and tolerances as indicated or specified. Coordinate continuity with adjacent thermal, weather, air and vapor barriers.

(d) Provide system with adequate allowances for expansion and contraction of components and fastenings to prevent buckling damage, joint seal failure, glass breakage, undue stress on fastenings or other detrimental effects. For design purposes, base provisions for thermal movement on assumed ambient temperature range of from 0 degrees F to 120 degrees F.

(e) Provide wall system to accommodate tolerances in building frame and other contiguous work as indicated or specified.

c. Calculations: Submit professionally prepared calculations to indicate how design requirements for structural loading, thermal, and other performance criteria have been satisfied and will include the attachment of the vertical aluminum fins to the curtain wall system.

2.2 PERFORMANCE REQUIREMENTS

a. Delegated Design: Engage a qualified Professional Engineer, to design glazed aluminum curtain walls.

b. Conform with system performance requirements specified.

c. Provide curtain wall components tested in accordance with requirements below and meeting performance requirements specified:

(1) System Design: Design and size components to withstand dead loads and live loads caused by positive and negative wind loads acting normal to plane of wall as calculated in accordance with ASCE 7. , as measured in accordance with ASTM E330/E330M.

(2) Seismic Loads: Design and size components to withstand seismic loads and sway displacement as calculated in accordance with code. Provide with the following tolerances.

(a) Phase I: 3 stroke cycles using.005 x the story height – no

damage or failure.

(b) Phase II: 3 stroke cycles using.010 x the story height – no damage or failure.

(3) Water Penetration:

(a) No water penetration is to occur when wall is tested in accordance with ASTM E331 at a differential static test pressure of 20 percent of inward acting design wind pressure as indicated on structural drawings, but not less than 10 psf.

(b) Make provision in wall construction for adequate drainage to outside of water leakage or condensation that occurs within outer face of wall. Leave drainage and weep openings in members and wall open during test.

(c) All field water penetration testing will be performed prior to vertical aluminum fin installation.

(4) Air Infiltration: Test glazed aluminum curtain wall system according to AAMA 503, which requires testing according to ASTM E783.

(a) Static-Air-Differential: 6.24 lbf/sq. ft. minimum.

(b) Air Leakage: 0.06 cfm/sq. ft. of surface maximum.

(c) All field air infiltration testing will be performed prior to vertical aluminum fin installation.

(5) Deflections Test: ASTM E330/E330M, Procedure B:

(a) No member is to deflect in a direction parallel to plane of wall, when carrying its full design load, more than an amount which will reduce edge cover or glass bite below 75 percent of design dimension. No member after deflection under full design load, is to have a clearance between itself and top of panel, glass, sash, or other part immediately below it less than 1/8 inch; clearance between member and an operable window or door is to be minimum 1/16 inch.

(7) Thermal Conductance Tests: ASTM C1363.

The thermal transmittance of opaque panels are not to exceed a U-value, Btu/hr./sq. ft./ degree F, as required and indicated on construction documents for exterior wall system, when tested in accordance with ASTM C1363. Average calculated thermal transmittance of complete wall assembly including panels, windows, and all other components are not to exceed a U-value of

(8) Sound Attenuation Through Wall System (Exterior to Interior):

STC 50 , measured in accordance with ASTM E413.

(9) Blast Resistance (PSRDM):

(a) Blast resistant frames and mullions must be designed to resist a design blast load of 13.5 psi peak pressure and 51 psi-msec

impulse (as an equivalent triangular load) applied over the tributary area of the member while sustaining maximum deformations no greater than L/40 (or 3 degrees support rotation). In addition, the frames and mullions must be designed to resist the balanced design load of 15.1 psi peak pressure and 57 psi-msec impulse (as an equivalent triangular load) while sustaining maximum deformations no greater than L/20 (or 6 degrees support rotation).

(b) Connections between framing members and connections of the frame members to the supporting structure must be designed to resist the applied blast loads and develop the maximum flexural capacity of the supported members, including the loads imposed by the vertical aluminum fins.

2.3 MATERIALS

a. Extruded Aluminum Framing Members: ASTM B221M (ASTM B221); 6063-T5 extruded aluminum for non-structural components or 6063-T6 extruded aluminum for structural members; temper and alloy as recommended by manufacturer.

b. Sheet Aluminum: ASTM B209M (ASTM B209); 6065-T5 temper and alloy as recommended by manufacturer.

(1) Formed flashing and closures: Minimum 0.062 inch thick aluminum, in finish as selected.

(2) Extruded sill members: Minimum 0.062 inch thick aluminum, in finish as selected.

c. Steel Sections: ASTM A36/A36M.

d. Primer: TS TT-P-645; red, for shop application and field touch-up.

e. Fasteners:

(1) For Exterior Cap Retainers: ASTM A193/A193M B8 300 series, stainless steel screws.

(2) For Framework Connections: ASTM B211/B211M 2024-T4 aluminum, ASTM A193/A193M B8 300 series, stainless steel, and ASTM B316/B316M aluminum rivets, as required by connection.

(3) For Anchoring Glazed Aluminum Curtain Wall to Support Structure: ASTM A307 zinc plated steel fasteners.

f. Shims: Metal or plastic.

g. Joint Sealants and Accessories:

(1) In accordance with requirements specified in Section 07 92 00, JOINT SEALANTS.

(2) Structural Flush Glazed Joints: High performance silicone sealant applied in accordance with manufacturer's recommendations.

(3) Non-structural Flush Glazed Joints and Weather Seal Joints: Silicone sealants applied in accordance with manufacturer's recommendations.

(4) Sealants used inside the weatherproofing system are to have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, (EPA Method 24).

(5) Comply with recommendations of sealant manufacturer for specific sealant selections.

(6) Provide only sealants that have been tested per ASTM C794 to exhibit adequate adhesion to samples of glass and metal equivalent to those required for project.

(7) Exposed Metal to Metal Joints: Silicone sealant selected from manufacturer’s standard colors.

h. Glazing Materials:

(1) As specified under Section 08 80 00, GLAZING.

(2) Glazing Gaskets:

(a) Exterior: Continuous EPDM gaskets at each glass and spandrel panel.

(b) Interior: Continuous, closed cell PVC foam sealant tape, sealed at corners.

(3) Glass Sizes and Clearances:

(a) Accommodate up to 1-1/4 inch glazing.

(b) Sizes indicated are nominal. Verify actual sizes required by measuring frames. Coordinate dimensions for glass and glass holding members to meet applicable minimum clearances as recommended by glass manufacturer. Do not nip glass to remove flares or to reduce oversized dimensions. All cutting is to occur in factory.

(4) Glass Setting Materials:

(a) Provide head bead and drive wedge required for glass installation to suit curtain wall system in accordance with manufacture’s recommendations.

i. Louvers:

(1) As specified under Section 08 90 00, LOUVERS AND VENTS.

j. Louver Screening:

(1) As specified under Section 08 90 00, LOUVERS AND VENTS.

k. Firestopping: Refer to Section 07 84 00, FIRESTOPPING for requirements.

2.4 FABRICATION

a. Curtain wall components and aluminum T-brackets are to be of materials and thickness indicated in construction documents. Details indicated are representative of required design and profiles. Maintain

sightlines. Unless specifically indicated or specified otherwise, methods of fabrication and assembly are to be at discretion of curtain wall manufacturer. Perform fitting and assembling of components in shop to maximum extent practicable. Anchorage devices are to permit adjustment in three directions. No exposed fasteners are permitted.

b. Joints: Joints exceeding +1/16" are to be mechanically fastened.

c. Ventilation and Drainage: Direct water leakage to exterior by means of concealed drainage system and weeps. Flashings and other materials used internally are to be non-staining, noncorrosive, and non-bleeding.

d. Protection and Treatment of Metals:

(1) Remove from metal surfaces lubricants used in fabrication and clean off other extraneous material before leaving shop.

(2) Provide protection against galvanic action wherever dissimilar metals are in contact, except in case of aluminum in permanent contact with galvanized steel, zinc, stainless steel, or relatively small areas of white bronze. Paint contact surfaces with one coat bituminous paint conforming to MIL-C-18480 or apply appropriate caulking material or nonabsorptive, noncorrosive, and non-staining tape or gasket between contact surfaces.

e. Metal sills and Closures: Fabricate accessories, spandrel panels, trim closures of sizes and shapes indicated from similar materials and finish as specified for wall system.

f. Concealed Interior Mullion Reinforcing: ASTM A36/A36M steel shapes as required for strength and mullion size limitations, hot dip galvanized after fabrication in accordance with ASTM A123/A123M.

g. Metal Spandrel Panels: Manufacturer's standard laminated aluminum-faced panels of thickness indicated, flat with no deviations in plane exceeding 1/16 inch in 24 inches or 1/8 inch over entire panel. Provide with edge flanges:

(1) Face Sheets: 0.024-inch minimum thickness finished to match system framing. Texture: Smooth

(2) Concealed Back Sheets: Aluminum or galvanized steel in manufacturer's standard thickness.

h. Stabilizer Sheets:

(1) 1/2 inch thick gypsum board.

(2) Gypsum board at inner face and hardboard at outer face as required to provide surface burning characteristics when tested according to ASTM E84 as follows:

(a) Flame Spread: 25.

(b) Smoke Developed: 450

i. Panel Core Material:

(1) Rigid, closed-cell, polyisocyanurate thermal insulation: R-Value:

R11

(2) Edge Configuration: Sealed.

2.5 METAL FINISHES

a. In accordance with NAAMM AMP 500 series.

b. High-Performance Organic Finish: Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

(1) Three -coat fluoropolymer finish complying with AAMA 2605 and containing not less than 70 percent PVDF resin by weight in both color coat and clear topcoat. Color as specified in the Schedule for Finishes on the drawings.

c. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

d. Shop and Touch-Up Primer for Steel Components: SSPC Paint 25 zinc oxide.

e. Touch-Up Primer for galvanized Steel Surfaces: SSPC Paint 20 zinc rich.

f. Concealed Steel Items: Galvanized in accordance with ASTM A123/A123M to 2.0 oz./sq. ft. Primed with iron oxide paint.

g. Apply one coat of bituminous paint to concealed aluminum and steel surfaces one one coat(s) in contact with cementitious or dissimilar materials.

PART 3 EXECUTION

3.1 EXAMINATION

a. Prior to installation of glazed curtain wall system, arrange for representative(s) of manufacturer to examine structure and substrate to determine that they are properly prepared, and ready to receive glazed curtain wall work included herein.

b. Verifying Conditions and Adjacent Surfaces: After establishment of lines and grades and prior to system installation examine supporting structural elements. Verify governing dimensions, including floor elevations, floor to floor heights, minimum clearances between curtain wall and structural frames, and other permissible dimensional tolerances in building frame.

3.2 PREPARATION

a. Take field dimensions and examine condition of substrates, supports, and other conditions under which work of this section is to be performed to verify that work may properly commence. Do not proceed with installation until unsatisfactory conditions have been corrected.

b. Contact between aluminum and dissimilar metals are to receive a

protective coating of bituminous paint for prevention of electrolytic action and corrosion.

3.3 INSTALLATION

a. Install and erect glazed curtain wall system and all components, including the reinforcement and aluminum T-brackets for the vertical aluminum fins in accordance with written directions of curtain wall manufacturer. Match profiles, sizes, and spacing indicated on approved shop drawings.

b. Bench Marks and Reference Points: Establish and permanently mark bench marks for elevations and building line offsets for alignment at convenient points on each floor level. Should any error or discrepancy be discovered in location of marks, stop erection work in that area until discrepancies have been corrected.

c. Ensure that drainage system operates properly in accord with AAMA 501 procedures.

d. Do not proceed with structural silicone work when metal temperature is below 32 degrees F.

e. Isolate between aluminum and dissimilar metals with protective coating or plastic strip to prevent electrolytic corrosion.

f. Install aluminum T-bracket supports for the vertical aluminum shading fins per manufacturer's written instructions. Refer to section 07 42 63 FABRICATED WALL PANEL ASSEMBLIES for specific instructions.

g. Install glazed aluminum curtain wall system so as to maintain a virtually flat face cap, with no visible bowing.

h. Install entire system so that fasteners are not visible.

i. Tolerances:

(1) Maximum variation from plane or location shown on approved shop drawings: per 1/8 inch per12 feet of length up to not more than 1/2 inch in any total length.

(2) Maximum offset from true alignment between two (2) identical members abutting end to end in line: 1/32 inch.

(3) Sealant Space Between Curtain Wall Mullion and Adjacent Construction: Maximum of 3/4 inch and minimum of 1/4 inch.

j. Joint Sealants:

(1) Joint Sealants: Are to be in accordance with requirements of Section 07 92 00, JOINT SEALANTS.

(2) Surfaces to be primed and sealed are to be clean, dry to touch, free from frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter. Enclose joints on three sides. Clean out grooves to proper depth. Joint dimensions are to conform to approved detail drawings with a tolerance of plus 1/8 inch. Do not apply compound unless ambient temperature is between 40 and 90 degrees F. Clean out loose particles and mortar just before

sealing. Remove protective coatings or coverings from surfaces in contact with sealants before applying sealants or tapes. Solvents used to remove coatings are to be of type that leave no residue on metals.

(3) Match approved sample. Force compound into grooves with sufficient pressure to fill grooves solidly. Sealing compound is to be uniformly smooth and free of wrinkles and, unless indicated otherwise, is to be tooled and left sufficiently convex to result in a flush joint when dry. Do not trim edges of sealing material after joints are tooled. Mix only amount of multi-component sealant which can be installed within four hours, but at no time is this amount exceed 5 gallons.

(4) Apply primer to masonry, concrete, wood, and other surfaces as recommended by sealant manufacturer. Do not apply primer to surfaces which will be exposed after sealant work is completed.

(5) Tightly pack backing in bottom of joints which are over 1/2 inch in depth with specified backing material to depth indicated in construction documents. Roll backing material of hose or rod stock into joints to prevent lengthwise stretching.

(6) Install bond preventive material at back or bottom of joint cavities in which no backstop material is required, covering full width and length of joint cavities.

(7) Remove compound smears from surfaces of materials adjacent to sealed joints as work progresses. Use masking tape on each side of joint where texture of adjacent material will be difficult to clean. Remove masking tape immediately after filling joint. Scrape off fresh compound from adjacent surfaces immediately and rub clean with solvent approved by sealant and curtain wall manufacturers. Upon completion of sealing, remove remaining smears, stains, and other soiling, and leave work in clean neat condition.

k. Glass:

(1) Refer to Section 08 80 00, GLAZING, and drawings for glass types. Install in accordance with manufacturer's recommendations as modified herein.

(2) Before installing glass, inspect sash and frames to receive glass for defects such as dimensional variations, glass clearances, open joints, or other conditions that will prevent satisfactory glass installation. Do not proceed with installation until defects have been corrected.

(3) Clean sealing surfaces at perimeter of glass and sealing surfaces of rebates and stop beads before applying glazing compound, sealing compound, glazing tape, or gaskets.

(4) Use only approved solvents and cleaning agents recommended by compound or gasket manufacturer and by curtain wall manufacturer.

(5) Provide sashes designed for outside glazing.

(6) Provide continuous snap in glazing beads to suit glass as

specified.

(7) Insulating and tempered glass, and glass of other types that exceed 100 united inches in size: Provide void space at head and jamb to allow glass to expand or move without exuding sealant. Provide perimeter frames and ventilator sections with glazing rebates for unobstructed glazing surface 3/4 inch in height. Glazing rebate surfaces must be sloped to shed water.

(8) Provide adequate means to weep incidental water and condensation away from sealed edges of insulated glass units and out of wall system. Provide weeping of lock-strip gaskets in accordance with recommendation of glass manufacturer.

l. Metal Copings:

(1) Refer to Section 07 60 00, FLASHING AND SHEET METAL for requirements of metal copings when they are not a part of glazed curtain wall system work.

(2) Coordinate curtain wall installation with metal coping detail on construction documents. Provide watertight seal to meet criteria set forth in this section regarding air and water penetration.

m. Vertical Aluminum Shading Fins:

(1) Refer to Section 07 42 63, FABRICATED WALL PANEL ASSEMBLIES for requirements of vertical aluminum fins when they are not part of glazed curtain wall system work.

(2) Coordinate curtain wall installation with vertical aluminum fin details on the construction documents. Provide watertight seal to meet criteria set forth in this section regarding air and water penetration.

(3) Contractor shall provide detailed written replacement glazing instructions for the Owner's use regarding the required procedures when removing vertical aluminum fins prior to replacing glazing panels.

3.4 ADJUSTING

a. Adjust doors to provide a tight fit at contact points and operate easily.

b. Adjust weather-stripping to make even contact with surfaces.

c. Adjust operating hardware and moving parts.

3.5 CLEANING

a. Install curtain wall frame and associated metal to avoid soiling or smudging finish.

b. Clean metal surfaces promptly after installation, exercising care to avoid damage to coatings.

c. Remove excess glazing and sealant compounds, dirt, and other substances.

d. Follow recommendations of manufacturer in selection of cleaning agents. Do not use cleaning agents containing ammonia or other compounds that might damage finished metal surfaces.

e. Replace cracked, broken, and defective glass with new glass at no additional cost to Government. Just prior to final acceptance of curtain wall system clean glass surfaces on both sides, remove labels, paint spots, compounds, and other defacements, and clean metal fixed panels. Remove and replace components that cannot be cleaned successfully.

3.6 FIELD QUALITY CONTROL

a. Testing Agency: Engage an AAMA accredited commercial qualified independent testing and inspecting agency to perform field quality-control tests specified, and to prepare test reports: Submit information regarding testing laboratory's facilities and qualifications of technical personnel to Contracting Officer for approval.

b. Conduct field check test for water leakage on designated wall areas after erection to comply with AAMA MCWM-1. Conduct test on two wall areas, two bays wide by two stories high where directed. Conduct test and take necessary remedial action as directed by Contracting Officer.

c. Test Specimen:

(1) Test specimen is to include curtain wall assembly and construction. Test chamber is to be affixed to interior side of test specimen and test is to be conducted using negative static air pressure.

(2) Test specimens are to be selected by Contracting Officer after curtain wall system has been installed in accordance with construction documents.

d. Sealant Adhesion Tests: Test installed sealant, in presence of sealant manufacturer’s field representative, in a minimum of two areas and as follows:

(1) Test weatherseal sealant adhesion to joint substrates according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C1193 or Method A, Tail Procedure, in ASTM C1521.

e. Air Infiltration: Test glazed aluminum curtain wall system according to AAMA 503, which requires testing according to ASTM E783 and to values indicated below, whichever is more stringent.

(1) Field air leakage testing is not required for continuous curtain wall systems.

(2) Static-Air-Pressure Differential: 6.24 lbf/sq. ft. minimum.

(3) Air Leakage: 0.06 cfm/sq. ft. of surface maximum.

f. Water Penetration: Test glazed aluminum curtain wall system for compliance with requirements according to AAMA 503, which requires testing according to ASTM E1105.

(1) Uniform Static-Air-Pressure Difference: 20 percent of positive design wind load, but not less than (10 psf). No uncontrolled water is to be present.

g. Retesting:

(1) Should system fail field test, system may be modified or repaired, and retested.

(2) Should system fail second field test, system may be additionally modified or repaired, and retested.

(3) All modifications and repairs made to tested areas are to be recorded, and same modifications and repairs made to all system and adjacent construction on project.

(4) Should second test fail, Contracting Officer may require testing of additional areas of the curtain wall.

(5) Any retesting, if required, will be performed proir to installation nof the vertical aluminum fins.

h. Rejection:

(1) Failure of any of specimens to meet test requirements of third test is cause for rejection of wall system and adjacent construction on project.

3.7 DEMONSTRATION, TESTING, AND ACCEPTANCE

a. Acceptance: At completion of project, and as a condition of acceptance, all systems are to be operated for a period of 15 consecutive calendar days without breakdown.

3.8 PROTECTION

After installation, protect windows, and other exposed surfaces from disfiguration, contamination, contact with harmful materials, and from other construction hazards that will interfere with their operation, or damage their appearance or finish. Protection methods are to be in accordance with recommendations of product manufacturers or of respective trade association. Remove paper or tape factory applied protection immediately after installation. Clean surfaces of mortar, plaster, paint, smears of sealants, and other foreign matter to present neat appearance and prevent fouling of operation. In addition, wash with a stiff fiber brush, soap and water, and thoroughly rinse. Where surfaces become stained or discolored, clean or restore finish in accordance with recommendations of product manufacturer or respective trade association.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 71 00

DOOR HARDWARE

3/24

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 GENERAL
1.4 WARRANTY
1.5 MAINTENANCE MANUALS
1.6 SUBMITTALS
1.7 SUBMITTAL INSTRUCTIONS
1.8 DELIVERY AND MARKING
1.9 PREINSTALLATION MEETING
1.10 INSTRUCTIONS
1.11 REFERENCES

PART 2 PRODUCTS

2.1 BUTT HINGES
2.2 CONTINUOUS HINGES
2.3 OVERHEAD CLOSERS
2.4 DOOR STOPS
2.5 LOCKS AND LATCHES
2.6 ELECTRIC STRIKES
2.7 KEYS
2.8 KEY CABINET
2.9 ARMOR PLATES, KICK PLATES, MOP PLATES AND DOOR EDGING
2.10 EXIT DEVICES
2.11 FLUSH BOLTS (AUTOMATIC)
2.12 COORDINATORS
2.13 THRESHOLDS
2.14 WEATHERSTRIPS (FOR EXTERIOR DOORS)
2.15 MISCELLANEOUS HARDWARE
2.16 FINISHES
2.17 BASE METALS

PART 3 EXECUTION

3.1 HARDWARE HEIGHTS
3.2 INSTALLATION
3.3 FINAL INSPECTION
3.4 DEMONSTRATION
3.5 HARDWARE SETS

-- End of Section Table of Contents --

SECTION 08 71 00
3/24

DOOR HARDWARE

PART 1 GENERAL

1.1 DESCRIPTION

Door hardware and related items necessary for complete installation and operation of doors.

1.2 RELATED WORK

a. Caulking: Section 07 92 00 JOINT SEALANTS.

b. Application of Hardware: Section 08 11 13, HOLLOW METAL DOORS AND FRAMES and 08 36 13 SECTIONAL DOORS.

c. Painting: Section 09 91 00, PAINTING.

d. Card Readers: Section 28 13 00, PHYSICAL ACCESS CONTROL SYSTEMS.

e. Electrical: DIVISION 26, ELECTRICAL.

f. Fire Detection: Section 28 31 00, FIRE DETECTION AND ALARM.

1.3 GENERAL

a. All hardware shall comply with ABAAS, (Architectural Barriers Act Accessibility Standard) unless specified otherwise.

b. Provide rated door hardware assemblies where required by most current version of the ICC IBC.

c. Hardware for Labeled Fire Doors and Exit Doors: Conform to requirements of NFPA 80 for labeled fire doors and to NFPA 101 for exit doors, as well as to other requirements specified. Provide hardware listed by UL, except where heavier materials, large size, or better grades are specified herein under paragraph HARDWARE SETS. In lieu of UL labeling and listing, test reports from a nationally recognized testing agency may be submitted showing that hardware has been tested in accordance with UL test methods and that it conforms to NFPA requirements.

d. Hardware for application on metal and wood doors and frames shall be made to standard templates. Furnish templates to the fabricator of these items in sufficient time so as not to delay the construction.

e. The following items shall be of the same manufacturer, except as otherwise specified:

(1) Mortise locksets.

(2) Hinges for hollow metal and wood doors.

(3) Surface applied overhead door closers.

(4) Exit devices.

(5) Floor closers.

1.4 WARRANTY

Automatic door operators shall be subject to the terms of FAR Clause 52.246-21, except that the Warranty period shall be two years in lieu of one year for all items except as noted below:

a. Locks, latchsets, and panic hardware: five years.

b. Door closers and continuous hinges: 10 years.

1.5 MAINTENANCE MANUALS

In accordance with Section 01 00 00, GENERAL REQUIREMENTS Article titled "INSTRUCTIONS", furnish maintenance manuals and instructions on all door hardware. Provide installation instructions with the submittal documentation.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 23 SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES:

SD-01 Preconstruction Submittals

Ahc Certified Hardware Consultant; G,

Preinstallation Meeting; G,

Nationally Recognized Testing Agency; G,

SD-03 Product Data

Manufacturers' Literature; G,

SD-04 Samples

Samples; G,

SD-05 Design Data

Hardware Schedule; G,

Keying Chart; G,

SD-06 Test Reports

Test Reports; G,

SD-07 Certificates

Certificate Of Compliance; G,

SD-11 Closeout Submittals

Written Report; G,

1.7 SUBMITTAL INSTRUCTIONS

a. Submittals shall be in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. Submit six copies of the schedule per Section 01 33 23. Submit two final copies of the final approved schedules to VAMC Locksmith as record copies (VISN Locksmith if the VAMC does not have a locksmith).

b. Hardware Schedule: AHC certified hardware consultant to prepare and submit hardware schedule in the following form:

Hardware Item	Quantity	Size	Reference Publication Type No.	Finish	Mfr. Name and Catalog No.	Key Control Symbols	UL Mark (if fire rated and listed)	ANSI/BHMA Finish Designation

c. Samples and Manufacturers' Literature:

(1) Samples: All hardware items (proposed for the project) that have not been previously approved by Builders Hardware Manufacturers Association shall be submitted for approval. Tag and mark all items with manufacturer's name, catalog number and project number.

(2) Samples are not required for hardware listed in the specifications by manufacturer's catalog number, if the Contractor proposes to use the manufacturer's product specified.

d. Certificate of Compliance and Test Reports: Submit certificates that hardware conforms to the requirements specified herein. Certificates shall be accompanied by copies of reports as referenced. The testing shall have been conducted either in the manufacturer's plant and certified by an independent testing laboratory or conducted in an independent laboratory, within four years of submittal of reports for approval.

1.8 DELIVERY AND MARKING

Deliver items of hardware to job site in their original containers, complete with necessary appurtenances including screws, keys, and instructions. Tag one of each different item of hardware and deliver to Contracting Officer for reference purposes. Tag shall identify items by Project Specification number and manufacturer's catalog number. These items shall remain on file in Contracting Officer's office until all other similar items have been installed in project, at which time the Contracting Officer will deliver items on file to Contractor for

installation in predetermined locations on the project.

1.9 PREINSTALLATION MEETING

a. Convene a preinstallation meeting not less than 30 days before start of installation of door hardware. Require attendance of parties directly affecting work of this section, including Contractor and Installer, Architect, Project Engineer and VA Locksmith, Hardware Consultant, and Hardware Manufacturer’s Representative. Review the following:

(1) Inspection of door hardware.

(2) Job and surface readiness.

(3) Coordination with other work.

(4) Protection of hardware surfaces.

(5) Substrate surface protection.

(6) Installation.

(7) Adjusting.

(8) Repair.

(9) Field quality control.

(10) Cleaning.

1.10 INSTRUCTIONS

a. Hardware Set Symbols on Drawings: Except for protective plates, door stops, mutes, thresholds and the like specified herein, hardware requirements for each door are indicated on drawings by symbols. Symbols for hardware sets consist of letters (e.g., "HW") followed by a number. Each number designates a set of hardware items applicable to a door type.

b. Keying: Keying will match existing VAMC keying system. The key system shall be large format(ASSA ABLOY Accentra) removable core type as previously described. The key blanks shall be protected by a utility patent with a minimum seven years remaining on the patent from the start of construction, and protected by contract-controlled distribution. The manufacturer shall furnish code pattern listings in both paper and electronic formats so keys may be reproduced by code; provide electronic format in file type required by project’s key control software. The manufacturer shall design the new key system with the capacity to re-key the existing system and also provide for 25 percent expansion capability beyond this requirement. Submit a keying chart for approval showing proposed keying layout and listing expansion capacity.

(1) Keying information will be furnished to the Contractor by the Contracting Officer.

(2) Supply information regarding key control of cylinder locks to manufacturers of equipment having cylinder type locks. Notify Contracting Officer immediately when and to whom keys or keying

information is supplied. Return all such keys to the Contracting Officer.

1.11 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. In text, hardware items are referred to by series, types, etc., listed in such specifications and standards, except as otherwise specified.

ASTM INTERNATIONAL (ASTM)

ASTM E2180	(2018) Standard Test Method for Determining the Activity of Incorporated Antimicrobial Agent(s) in Polymeric or Hydrophobic Materials

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

ANSI/BHMA A156.1	(2021) Butts and Hinges
ANSI/BHMA A156.2	(2017) Bored and Preassembled Locks and Latches
ANSI/BHMA A156.3	(2020) Exit Devices
ANSI/BHMA A156.4	(2024) Door Controls - Closers
ANSI/BHMA A156.6	(2021) Architectural Door Trim
ANSI/BHMA A156.11	(2019) Cabinet Locks
ANSI/BHMA A156.13	(2022) Mortise Locks & Latches Series 1000
ANSI/BHMA A156.16	(2023) Auxiliary Hardware
ANSI/BHMA A156.18	(2020) Materials and Finishes
ANSI/BHMA A156.21	(2019) Thresholds
ANSI/BHMA A156.22	(2021) Gasketing
ANSI/BHMA A156.26	(2012) Continuous Hinges
ANSI/BHMA A156.31	(2013) Electric Strikes and Frame Mounted Actuators
ANSI/BHMA A156.36	(2010) Auxiliary Locks

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

ABAAS	Architectural Barriers Act Accessibility Standard

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2025) Standard for Fire Doors and Other Opening Protectives
NFPA 101	(2024) Life Safety Code

PART 2 PRODUCTS

2.1 BUTT HINGES

a. ANSI/BHMA A156.1. Provide only three-knuckle hinges, except five-knuckle where the required hinge type is not available in a three-knuckle version (e.g., some types of swing-clear hinges). The following types of butt hinges shall be used for the types of doors listed, except where otherwise specified:

(1) Exterior Doors: Type A2112/A5112 for doors 3 feet wide or less and Type A2111/A5111 for doors over 3 feet wide. Hinges for exterior outswing doors shall have non-removable pins. Hinges for exterior fire-rated doors shall be of stainless steel material.

(2) Interior Doors: Type A8112/A5112 for doors 3 feet wide or less and Type A8111/A5111 for doors over 3 feet wide. Hinges for doors exposed to high humidity areas (shower rooms, toilet rooms, kitchens, janitor rooms, etc. shall be of stainless steel material.

b. Provide quantity and size of hinges per door leaf as follows:

(1) Doors 4 feet to 7 feet 5 inches high: three hinges minimum.

(2) Doors up to 3 feet wide, standard weight: (4-1/2 inches x 4-1/2 inches) hinges.

(3) Doors over 3 feet to 3 feet 6 inches wide, standard weight: 5 inches x 4-1/2 inches.

c. See Articles "MISCELLANEOUS HARDWARE" and "HARDWARE SETS" for pivots and hinges other than butts specified above and continuous hinges specified below.

2.2 CONTINUOUS HINGES

a. ANSI/BHMA A156.26, Grade 1-600. Listed under Category N in BHMA's "Certified Product Directory."

b. General: Minimum 0.120-inch- thick, hinge leaves with minimum overall width of 4 inches; fabricated to full height of door and frame and to template screw locations; with components finished after milling and drilling are complete.

c. Continuous, Barrel-Type Hinges: Hinge with knuckles formed around a Teflon-coated 0.25-inch minimum diameter pin that extends entire length of hinge.

(1) Base Metal for Exterior Hinges: Stainless steel.

(2) Base Metal for Interior Hinges: Steel.

(3) Base Metal for Hinges for Fire-Rated Assemblies: Steel.

(4) Provide with non-removable pin (hospital tip option) at lockable outswing doors.

(5) Where required to clear adjacent casing, trim, and wall conditions and allow full door swing, provide wide throw hinges of minimum width required.

(6) Provide with manufacturer’s cut-outs for separate mortised power transfers and/or mortised automatic door bottoms where they occur.

(7) Where thru-wire power transfers are integral to the hinge, provide hinge with easily removable portion to allow easy access to wiring connections.

(8) Where models are specified that provide an integral wrap-around edge guard for the hinge edge of the door, provide manufacturer’s adjustable threaded stud and machine screw mechanism to allow the door to be adjusted within the wrap-around edge guard.

2.3 OVERHEAD CLOSERS

a. Conform to ANSI/BHMA A156.4, Grade 1.

b. Closers shall conform to the following:

(1) The closer shall have minimum 50 percent adjustable closing force over minimum value for that closer and have adjustable hydraulic back check effective between 60 degrees and 85 degrees of door opening.

(2) Where specified, closer shall have hold-open feature.

(3) Size Requirements: Provide multi-size closers, sizes 1 through 6, except where multi-size closer is not available for the required application.

(4) Material of closer body shall be forged or cast.

(5) Arm and brackets for closers shall be steel, malleable iron or high strength ductile cast iron.

(6) Where closers are exposed to the exterior or are mounted in rooms that experience high humidity, provide closer body and arm assembly of stainless steel material.

(7) Closers shall have full size metal cover; plastic covers will not be accepted.

(8) Closers shall have adjustable hydraulic back-check, separate valves for closing and latching speed, adjustable back-check positioning valve, and adjustable delayed action valve.

(9) Provide closers with any accessories required for the mounting application, including (but not limited to) drop plates, special soffit plates, spacers for heavy-duty parallel arm fifth screws, bull-nose or other regular arm brackets, longer or shorter arm

assemblies, and special factory templating. Provide special arms, drop plates, and templating as needed to allow mounting at doors with overhead stops and/or holders.

(10) Closer arms or backcheck valve shall not be used to stop the door from overswing, except in applications where a separate wall, floor, or overhead stop cannot be used.

(11) Provide parallel arm closers with heavy duty rigid arm.

(12) Where closers are to be installed on the push side of the door, provide parallel arm type except where conditions require use of top jamb arm.

(13) Provide all surface closers with the same body attachment screw pattern for ease of replacement and maintenance.

(14) All closers shall have a 1-1/2 inch minimum piston diameter.

2.4 DOOR STOPS

a. Conform to ANSI/BHMA A156.16.

b. Provide door stops wherever an opened door or any item of hardware thereon would strike a wall, column, equipment or other parts of building construction. For concrete, masonry or quarry tile construction, use expansion shields for mounting door stops.

c. Where cylindrical locks with turn pieces or pushbuttons occur, equip wall bumpers Type L02251 (rubber pads having concave face) to receive turn piece or button.

d. Provide floor stops (Type L02141 or L02161) in office areas; Type L02121 x 3 screws into floor elsewhere. Wall bumpers, where used, must be installed to impact the trim or the door within the leading half of its width. Floor stops, where used, must be installed within 4-inches of the wall face and impact the door within the leading half of its width.

e. Where drywall partitions occur, use floor stops, Type L02141 or L02161 in office areas, Type L02121 elsewhere.

f. Provide stop Type L02011, as applicable for exterior doors. At outswing doors where stop can be installed in concrete, provide stop mated to concrete anchor set in 3-inch core-drilled hole and filled with quick-setting cement.

g. Omit stops where floor mounted door holders are required and where automatic operated doors occur.

h. Provide appropriate roller bumper for each set of doors (except where closet doors occur) where two doors would interfere with each other in swinging.

i. Provide appropriate door mounted stop on doors in individual toilets where floor or wall mounted stops cannot be used.

j. Provide door stops on doors where combination closer magnetic holders are specified, except where wall stops cannot be used or where floor

stops cannot be installed within 4-inches of the wall.

k. Where the specified wall or floor stop cannot be used, provide concealed overhead stops (surface-mounted where concealed cannot be used).

2.5 LOCKS AND LATCHES

a. Conform to ANSI/BHMA A156.2. Locks and latches for doors 1-3/4 inch thick or over shall have beveled fronts. Lock cylinders shall have not less than seven pins Cylinder shall be removable by special key or tool. Construct all cores so that they will be interchangeable into the core housings of all mortise locks, rim locks, cylindrical locks, and any other type lock included in the Great Grand Master Key System. Disassembly of lever or lockset shall not be required to remove core from lockset. All locksets or latches on double doors with fire label shall have latch bolt with 3/4 inch throw, unless shorter throw allowed by the door manufacturer’s fire label. Provide temporary keying device or construction core to allow opening and closing during construction and prior to the installation of final cores. All new locks and latchsets shall match existing facility ASSA ABLOY Accentra hardware.

b. In addition to above requirements, locks and latches shall comply with following requirements:

(1) Mortise Lock and Latch Sets: Conform to ANSI/BHMA A156.13. Mortise locksets shall be series 1000, minimum Grade 2. All locksets and latchsets shall have lever handles fabricated from cast stainless steel. No substitute lever material shall be accepted. All locks and latchsets shall be furnished with 4-7/8-inch curved lip strike and wrought box. At outswing pairs with overlapping astragals, provide flat lip strip with 7/8-inch lip-to-center dimension. Lock function F02 shall be furnished with emergency tools/keys for emergency entrance. All lock cases installed on lead lined doors shall be lead lined before applying final hardware finish. Furnish armored fronts for all mortise locks. Where mortise locks are installed in high-humidity locations or where exposed to the exterior on both sides of the opening, provide non-ferrous mortise lock case.

(2) Auxiliary locks shall be as specified under hardware sets and conform to ANSI/BHMA A156.36.

2.6 ELECTRIC STRIKES

a. ANSI/BHMA A156.31 Grade 1.

b. General: Use fail-secure electric strikes at fire-rated doors.

2.7 KEYS

a. Stamp all keys with change number and key set symbol. Furnish keys in quantities as follows:

Locks/Keys	**Quantity**
Cylinder locks	2 keys each

Locks/Keys	Quantity
Cylinder lock change key blanks	100 each different key way
Master-keyed sets	6 keys each
Grand Master sets	6 keys each
Great Grand Master set	5 keys
Control key	2 keys

2.8 KEY CABINET

a. ANSI/BHMA A156.11. Provide key cabinet made of cold rolled, 0.0478 inch thick furniture steel electro-welded. Doors shall have "no sag" continuous brass-pin piano type hinge and be equipped with chrome plated locking door handles, hook cam and mechanical pushbutton door lock. Key Cabinet and Key Control System shall accommodate all keys for this project plus 25 percent. Provide minimum number of multiple cabinets where a single cabinet of largest size will not accommodate the required number of keys.

b. Key tags shall consist of two sets: Permanent self-locking and loan key snaphook type with tag colors as follows: Red fiber marker of the permanent self-locking type approximately 1-1/4 inch in diameter engraved with the legend "FILE KEY MUST NOT BE LOANED." Also furnish for each hook a white cloverleaf key marker with snap-hooks engraved with the legend "LOAN KEY."

c. The manufacturer of the lock cylinders and locks shall attach a key tag to keys of each lock cylinder and shall mark thereon the respective item number and key change number. Provide each group of keys in a key gathering envelope (supplied by Key Cabinet Manufacturer) in which the lock manufacturer shall include the following information: Item number, key change number and door number. The Contractor shall furnish the Key Cabinet Manufacturer the hardware and keying schedules and change keys.

d. The Key Cabinet Manufacturer shall set up a three-way cross index system, including master keys, listing the keys alphabetically, the hooks numerically and the key changes numerically on different colored index cards. Index cards shall be typewritten and inserted in a durable binder. Attach the keys to the two sets of numbered tags supplied with the cabinet. (The permanent tag and the loan key tag). Instruct the owner in proper use of the system. Install cabinet as directed by the Contracting Officer.

2.9 ARMOR PLATES, KICK PLATES, MOP PLATES AND DOOR EDGING

a. Conform to ANSI/BHMA A156.6.

b. Provide protective plates as specified below:

(1) Kick plates, mop plates and armor plates of metal, Type J100 series.

(2) Provide kick plates and mop plates where specified. Kick plates shall be 10 inches or 12 inches high. Mop plates shall be 6 inches high. Both kick and mop plates shall be minimum 0.050 inches thick. Provide kick and mop plates beveled on all 4 edges (B4E). On push side of doors where jamb stop extends to floor, make kick

plates 1-1/2 inches less than width of door, except pairs of metal doors which shall have plates 1 inch less than width of each door. Extend all other kick and mop plates to within 1/4 inch of each edge of doors. Kick and mop plates shall butt astragals. For jamb stop requirements, see specification sections pertaining to door frames.

2.10 EXIT DEVICES

a. Conform to ANSI/BHMA A156.3. Exit devices shall be Grade 1; type and function are specified in hardware sets. Provide flush with finished floor strikes for vertical rod exit devices in interior of building. Trim shall have cast satin stainless steel lever handles of design similar to locksets, unless otherwise specified. Provide key cylinders for keyed operating trim and, where specified, cylinder dogging.

b. Surface vertical rod panics shall only be provided less bottom rod; provide fire pins as required by exit device and door fire labels. Do not provide surface vertical rod panics at exterior doors.

c. Concealed vertical rod panics shall be provided less bottom rod at interior doors, unless lockable or otherwise specified; provide fire pins as required by exit device and door fire labels. Where concealed vertical rod panics are specified at exterior doors, provide with both top and bottom rods.

d. Where removable mullions are specified at pairs with rim panic devices, provide mullion with key-removable feature.

e. At non-rated openings with panic hardware, provide panic hardware with key cylinder dogging feature.

f. Exit devices for fire doors shall comply with Underwriters Laboratories, Inc., requirements for Fire Exit Hardware. Submit proof of compliance.

2.11 FLUSH BOLTS (AUTOMATIC)

a. Conform to ANSI/BHMA A156.3. Dimension of flush bolts shall conform to ANSI A115. Bolts shall conform to Underwriters Laboratories, Inc., requirements for fire door hardware. Flush bolts shall automatically latch and unlatch. Furnish dustproof strikes conforming to ANSI/BHMA A156.16 for bottom flushbolt. Face plates for dustproof strike shall be rectangular and not less than 1-1/2 by 3-1/2 inches.

b. At interior doors, provide auto flush bolts less bottom bolt, unless otherwise specified, except at wood pairs with fire-rating greater than 20 minutes; provide fire pins as required by auto flush bolt and door fire labels.

2.12 COORDINATORS

Conform to ANSI/BHMA A156.16. Coordinators, when specified for fire doors, shall comply with Underwriters Laboratories, Inc., requirements for fire door hardware. Coordinator may be omitted on exterior pairs of doors where either door will close independently regardless of the position of the other door. Coordinator may be omitted on interior pairs of non-labeled open where open back strike is used. Open back strike shall not be used on

labeled doors. Paint coordinators to match door frames, unless coordinators are plated. Provide bar type coordinators, except where gravity coordinators are required at acoustic pairs. For bar type coordinators, provide filler bars for full width and, as required, brackets for push-side surface mounted closers, overhead stops, and vertical rod panic strikes.

2.13 THRESHOLDS

a. Conform to ANSI/BHMA A156.21, mill finish extruded aluminum, except as otherwise specified. In existing construction, thresholds shall be installed in a bed of sealant with 1/4-20 stainless steel machine screws and expansion shields. In new construction, embed aluminum anchors coated with epoxy in concrete to secure thresholds. Furnish thresholds for the full width of the openings.

b. For thresholds at elevators entrances see other sections of specifications.

c. At exterior doors and any interior doors exposed to moisture, provide threshold with non-slip abrasive finish.

d. Provide with miter returns where threshold extends more than 0.5 inch beyond face of frame.

2.14 WEATHERSTRIPS (FOR EXTERIOR DOORS)

Conform to ANSI/BHMA A156.22. Air leakage shall not to exceed 0.50 CFM per foot of crack length (0.000774m3/s/m).

2.15 MISCELLANEOUS HARDWARE

a. Access Doors (including Sheet Metal, Screen and Woven Wire Mesh Types): Except for fire-rated doors and doors to Temperature Control Cabinets, equip each single or double metal access door with Lock Type E07213, conforming to ANSI/BHMA A156.11. Key locks as directed. Ship lock prepaid to the door manufacturer. Hinges shall be provided by door manufacturer.

b. Cylinders for Various Partitions and Doors: Key cylinders same as entrance doors of area in which partitions and door occur, except as otherwise specified. Provide cylinders to operate locking devices where specified for following partitions and doors:

(1) Folding doors and partitions.

(2) Wicket door (in roll-up door assemblies).

(3) Slide-up doors.

(4) Swing-up doors.

(5) Fire-rated access doors-Engineer's key set.

(6) Doors from corridor to electromagnetic shielded room.

(7) Day gate on vault door.

c. Mutes: Conform to ANSI/BHMA A156.16. Provide door mutes or door

silencers Type L03011 or L03021, depending on frame material, of white or light gray color, on each steel or wood door frame, except at fire-rated frames, lead-lined frames and frames for sound-resistant, lightproof and electromagnetically shielded doors. Furnish three mutes for single doors and two mutes for each pair of doors, except double-acting doors. Provide four mutes or silencers for frames for each Dutch type door. Provide two mutes for each edge of sliding door which would contact door frame.

2.16 FINISHES

a. Exposed surfaces of hardware shall have ANSI/BHMA A156.18, finishes as specified below. Finishes on all hinges, pivots, closers, thresholds, etc., shall be as specified below under "Miscellaneous Finishes." For field painting (final coat) of ferrous hardware, see Section 09 91 00, PAINTING.

b. 626 or 630: All surfaces on exterior and interior of buildings, except where other finishes are specified.

c. Miscellaneous Finishes:

(1) Hinges --exterior doors: 626 or 630.

(2) Hinges --interior doors: 652 or 630.

(3) Pivots: Match door trim.

(4) Door Closers: Factory applied paint finish. Dull or Satin Aluminum color.

(5) Thresholds: Mill finish aluminum.

(6) Cover plates for floor hinges and pivots: 630.

(7) Other primed steel hardware: 600.

d. Special Finish: Exposed surfaces of hardware for dark bronze anodized aluminum doors shall have oxidized oil rubbed bronze finish (dark bronze) finish on door closers shall closely match doors.

e. Anti-microbial Coating: All hand-operated hardware (levers, pulls, push bars, push plates, paddles, and panic bars) shall be provided with an anti-microbial/anti-fungal coating that has passed ASTM E2180 tests. Coating to consist of ionic silver (Ag+). Silver ions surround bacterial cells, inhibiting growth of bacteria, mold, and mildew by blockading food and respiration supplies.

2.17 BASE METALS

Apply specified U.S. Standard finishes on different base metals as following:

Finish	**Base Metal**
652	Steel

Finish	Base Metal
626	Brass or bronze
630	Stainless steel

PART 3 EXECUTION

3.1 HARDWARE HEIGHTS

a. For new buildings locate hardware on doors at heights specified below, with all hand-operated hardware centered within 34 inches to 48 inches, unless otherwise noted:

b. Hardware Heights from Finished Floor:

(1) Exit devices centerline of strike (where applicable) 40-5/16 inches.

(2) Locksets and latch sets centerline of strike 40-5/16 inches.

(3) Deadlocks centerline of strike 48 inches.

(4) Centerline of door pulls to be 40 inches.

(5) Push plates and push-pull shall be 50 inches to top of plate.

(6) Push-pull latch to be 40-5/16 inches to centerline of strike.

(7) Locate other hardware at standard commercial heights. Locate push and pull plates to prevent conflict with other hardware.

3.2 INSTALLATION

a. Closer devices, including those with hold-open features, shall be equipped and mounted to provide maximum door opening permitted by building construction or equipment. Closers shall be mounted on side of door inside rooms, inside stairs, and away from corridors. At exterior doors, closers shall be mounted on interior side. Where closers are mounted on doors they shall be mounted with hex nuts and bolts; foot shall be fastened to frame with machine screws. Door opening force shall not exeed 22.2 N (5 pounds) maximum in accordance with ABAAS.

b. Hinge Size Requirements:

Door Thickness	Door Width	Hinge Height
1-3/4 inch	3 feet and less	4-1/2 inches
1-3/4 inch	Over 3 feet but not more than 4 feet	(5 inches)
1-3/8 inch (hollow core wood doors)	Not over 4 feet	4-1/2 inches

c. Hinge leaves shall be sufficiently wide to allow doors to swing clear of door frame trim and surrounding conditions.

d. Where new hinges are specified for new doors in existing frames or existing doors in new frames, sizes of new hinges shall match sizes of existing hinges; or, Contractor may reuse existing hinges provided hinges are restored to satisfactory operating condition as approved by Contracting Officer. Existing hinges shall not be reused on door openings having new doors and new frames. Coordinate preparation for hinge cut-outs and screw-hole locations on doors and frames.

e. Hinges Required Per Door:

Door Description	**Number butts**
Doors 5 ft or less in height	2 butts
Doors over 5 ft high and not over 7 ft 6 in high	3 butts
Doors over 7 feet 6 inches high	4 butts

f. Fastenings: Suitable size and type and shall harmonize with hardware as to material and finish. Provide machine screws and lead expansion shields to secure hardware to concrete, ceramic or quarry floor tile, or solid masonry. Fiber or rawl plugs and adhesives are not permitted. All fastenings exposed to weather shall be of nonferrous metal.

g. After locks have been installed; show in presence of Contracting Officer that keys operate their respective locks in accordance with keying requirements. (All keys, Master Key level and above shall be sent Registered Mail to the Medical Center Director along with the bitting list. Also a copy of the invoice shall be sent to the Contracting Officer for his records.) Installation of locks which do not meet specified keying requirements shall be considered sufficient justification for rejection and replacement of all locks installed on project.

3.3 FINAL INSPECTION

Installer to provide letter to VA Resident/Project Engineer that upon completion, installer has visited the Project and has accomplished the following:

a. Re-adjust hardware.

b. Evaluate maintenance procedures and recommend changes or additions, and instruct VA personnel.

c. Identify items that have deteriorated or failed.

d. Submit written report identifying problems.

e. Contractor will inspect, test and adjust all overhead closers to meet specified closing pressures and requirements.

3.4 DEMONSTRATION

Demonstrate efficacy of mechanical hardware and electrical, and electronic hardware systems, including adjustment and maintenance procedures, to satisfaction of Resident/Project Engineer and VA Locksmith.

3.5 HARDWARE SETS

a. Following sets of hardware correspond to hardware symbols shown on drawings. Only those hardware sets that are shown on drawings will be required. Disregard hardware sets listed in specifications but not shown on drawings.

b. Hardware Consultant working on a project will be responsible for providing additional information regarding these hardware sets. The numbers shown in the following sets come from BHMA standards.

ELECTRIC HARDWARE ABBREVIATIONS LEGEND:
ADO = Automatic Door Operator
EMCH = Electro-Mechanical Closer-Holder
MHO = Magnetic Hold-Open (wall- or floor-mounted)

c. HARDWARE SETS

HW-1A

Each Door to Have:	RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Latchset	F01
1 Closer	C02011/C02021
1 Kick Plate	J102
1 Wall Stop	L02121 CONVEX
2 Sets Self-Adhesive Seals	R0Y164

DOOR POSITION SWITCH BY DIVISION 28

HW-2

Each Door to Have:	RATED/NON-RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Keyed Privacy Indicator Lock	F13 x OCCUPANCY INDICATOR
1 Closer	C02011/C02021
1 Kick Plate	J102
1 Mop Plate (@ Inswing Doors)	J103
1 Floor Stop	L02121 x 3 FASTENERS
1 Set Self-Adhesive Seals	R0Y154

STONE THRESHOLD BY OTHER TRADES.

HW-2D

Each Door to Have:	RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Privacy Lock	F02-MOD X OCCUPANCY INDICATOR
1 Closer	C02011/C02021
1 Kick Plate	J102
1 Mop Plate (@ Inswing Doors)	J103
1 Wall Stop	L02101 CONVEX
3 Silencers	L03011

STONE THRESHOLD BY OTHER TRADES.

HW-3E

Each Door to Have:	NON-RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Office Lock	F04
1 Floor Stop	L02121 x 3 FASTENERS
1 Set Self-Adhesive Seals	R0Y154
1 Coat Hook	L03121

OMIT COAT HOOK WHERE GLASS LITE PREVENTS INSTALLATION.

HW-5F

Each Door to Have:	RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Storeroom Lock	F07
1 Closer	C02011/C02021
1 Kick Plate	J101
1 Wall Stop	L02121 CONVEX
1 Set Self-Adhesive Seals	R0Y154

HW-5G

Each Door to Have:	NON-RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Storeroom Lock	F07
1 Closer With Stop Arm	C02011/C02021
1 Kick Plate	J102
3 Silencers	L03011

HW-6

Each Door to Have:	RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Exit Device	TYPE 1 F13 LEVER
1 Key Cylinder	TYPE AS REQUIRED
1 Closer	C02011/C02021
1 Floor Stop	L02121 x 3 FASTENERS
1 Set Self-Adhesive Seals	R0Y154

HW-11

Each Pair to Have:	RATED/NR
Hinges	QUANTITY & TYPE AS REQUIRED
1 Set Auto Flush Bolts	TYPE 25 LESS BOTTOM BOLT
1 Storeroom Lock	F07
1 Coordinator	TYPE 21A
1 Overlapping Astragal	R0Y634 x THRU-BOLTS
2 Closers	C02011/C02021
2 Kick Plates	J102 (@ STORAGE ROOMS ONLY)
2 Floor Stops	L02121 x 3 FASTENERS
3 Silencers	L03011

HW-12

Each Pair to Have:	RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Exit Device	TYPE 7 or 8 F01
1 Exit Device	TYPE 7 or 8 F08 LEVER
1 Key Cylinder	TYPE AS REQUIRED
1 Set Meeting Stile Astragals	R0Y834
2 Closers	C02011/C02021
2 Floor Stops	L02121 x 3 FASTENERS
1 Set Self-Adhesive Seals	R0Y154

HW-E3

Each Door to Have:	RATED
Hinges	QUANTITY & TYPE AS REQUIRED
1 Storeroom Lock	F13-MOD x RIGID OUTSIDE LEVER x KEY RETRACTS DEADBOLT AND LATCHBOLT
1 Latch Protector (outswing dr)	
1 Closer	C02011/C02021
1 Edge Guard (@ Wood Doors)	J208M / J211 (VERIFY), CUT: HARDWARE
1 Armor Plate	J101 x (0.125 INCH) THICKNESS
1 Overhead Holder	C01511-ADJUSTABLE

1 Threshold (outswing door)	J32120 x SILICONE GASKET
1 Threshold (inswing door)	ALUMINUM, PER ARCHITECTURAL DETAIL
1 Door Sweep	R0Y416
1 Set Frame Seals	R0Y164
1 Drip	R0Y976

HW-E5

Each Roll-up Door to Have:	NON-RATED
1 Cylinders	TYPE AS REQUIRED

BALANCE OF HARDWARE BY SECTION 08 33 00, COILING DOORS

HW-SH-3

Each AC, EL, REX, DPS Door to Have:	
Hinges	QUANTITY & TYPE AS REQUIRED
1 Transfer Hinge	4-WIRE TYPE AS REQUIRED
1 Electrified Lock	F07 (E01-REX, E06) 24VDC
1 Power Supply	REGULATED, FILTERED, 24VDC, AMPERAGE AS REQUIRED
1 Closer	C02011/C02021
1 Wall Stop	L02101 CONVEX
3 Silencers	L03011

120VAC POWER, CONDUIT, DOOR POSITION SWITCH AND WIRING BY DIVISION 26.
CARD READER BY DIVISION 28.

HW-SH-3A

Each Pair to Have:	
1 Storeroom Lock	F07
1 Electric Strike	E09311 (FAIL-SECURE), 24VDC
1 Power Supply	REGULATED, FILTERED, 24VDC, AMPERAGE AS REQUIRED

REMAINDER OF EXISTING HARDWARE TO REMAIN. 120VAC POWER, CONDUIT, DOOR POSITION SWITCH AND WIRING BY DIVISION 26. CARD READER BY DIVISION 28.

HW-SH-3F

Each AC, RR, EL, REX, DPSDoor to Have:	RATED
1 Continuous Transfer Hinge	X INTEGRAL HINGE GUARD CHANNEL X ADJUSTA-SCREWS x 4-THRUWIRE TRANSFER x IN-HINGE ACCESS PANEL
1 Electrified Lock	F13-MOD x RIGID OUTSIDE LEVER X NO INSIDE TURN X KEY RETRACTS LATCHBOLT AND DEADBOLT (E01-REX, E06) 24VDC
1 Power Supply	REGULATED, FILTERED, 24VDC, AMPERAGE AS REQUIRED
1 Closer with Stop Arm	C02011/C02021
1 Kick Plate	J101 x 1.275 MM (0.050 INCH) THICKNESS
1 Threshold (Outswing Door)	J32120 x SILICONE GASKET
1 Door Sweep	R0Y416
1 Set Frame Seals	R0Y164
1 Drip	R0Y976

120VAC POWER, CONDUIT, AND WIRING BY DIVISION 26.
CARD READER AND DOOR POSITION SWITCH BY DIVISION 28.

HW-SH-9

Each AC, EL, REX, DPS Pair to Have:	RATED
1 Continuous Hinge @ INACTIVE LEAF	
1 Continuous Transfer Hinge @ ACTIVE LEAF	x INTEGRAL HINGE GUARD CHANNEL x ADJUSTA-SCREWS x 4-THRUWIRE TRANSFER x IN-HINGE ACCESS PANEL
1 Transfer Hinge	4-WIRE TYPE AS REQUIRED
1 Set Auto Flush Bolts	TYPE 25
1 Dust Proof Strike	L04021
1 Electrified Lock	F07 (E01-REX, E06) 24VDC
1 Power Supply	REGULATED, FILTERED, 24VDC,AMPERAGE AS REQUIRED
1 Coordinator	TYPE 21A
1 Overlapping Astragal with THRU-BOLTS Self-Adhesive Seal	R0Y634 x R0Y154 x
2 Closers	C02011/C02021
2 Kick Plates	J102 (@ STORAGE ROOMS ONLY)
2 Floor Stops	L02121 x 3 FASTENERS
1 Threshold (Outswing Door)	J33220 x SILICONE GASKET
2 Door Sweep	R0Y416
1 Set Frame Seals	R0Y164
1 Drip	R0Y976

120VAC POWER, CONDUIT, AND WIRING BY DIVISION 26.
CARD READER AND DOOR POSITION SWITCH BY DIVISION 28.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 80 00

GLAZING

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 LABELS
1.4 PERFORMANCE REQUIREMENTS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 DELIVERY, STORAGE AND HANDLING
1.8 PROJECT CONDITIONS
1.9 WARRANTY
1.10 REFERENCES

PART 2 PRODUCT

2.1 GLASS
2.2 HEAT-TREATED GLASS
2.3 COATED GLASS
2.4 LAMINATED GLASS
2.5 SECURITY GLAZING ASSEMBLY
2.6 INSULATING GLASS UNITS
2.7 FIRE PROTECTION AND FIRE RESISTANCE GLAZING
2.8 GLAZING ACCESSORIES

PART 3 EXECUTION

3.1 EXAMINATION
3.2 PREPARATION
3.3 INSTALLATION – GENERAL
3.4 INSTALLATION - WET/DRY METHOD (PREFORMED TAPE AND SEALANT)
3.5 INSTALLATION - WET METHOD (SEALANT AND SEALANT)
3.6 INSTALLATION - INTERIOR WET/DRY METHOD (TAPE AND SEALANT)
3.7 INSTALLATION - INTERIOR WET METHOD (COMPOUND AND COMPOUND)
3.8 REPLACEMENT AND CLEANING
3.9 PROTECTION
3.10 MONOLITHIC GLASS SCHEDULE
3.11 LAMINATED GLASS SCHEDULE
3.12 INSULATING LAMINATED GLASS SCHEDULE (FORCE PROTECTION AND PHYSICAL SAFETY)
3.13 FIRE-PROTECTIVE AND FIRE-RESISTANCE GLAZING SCHEDULE

-- End of Section Table of Contents --

SECTION 08 80 00
1/21

GLAZING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies the following:

a. Glass.

b. Glazing materials and accessories for both factory and field glazed assemblies.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE DESIGN REQUIREMENTS: Sustainable Design Requirements.

b. Section 10 28 00, TOILET, BATH, AND LAUNDRY ACCESSORIES: Mirrors.

c. Section 08 44 13, GLAZED ALUMINUM CURTAIN WALLS Glazed Curtain Walls: Glazed Curtain Walls.

d. Refer to the SCHEDULE FOR FINISHES on the drawings: Color of reflective (metallic coated) glass.

e. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Junction and Switch Boxes.

f. Section 28 13 11, PHYSICAL ACCESS CONTROL SYSTEMS: Access Control Systems.

1.3 LABELS

a. Temporary labels:

(1) Provide temporary label on each light of glass identifying manufacturer or brand and glass type, quality and nominal thickness.

(2) Label in accordance with NFRC label requirements.

(3) Temporary labels are to remain intact until glass is approved by the Contracting Officer.

b. Permanent labels:

(1) Locate in corner for each pane.

(2) Label in accordance with ANSI Z97.1 and SGCC label requirements.

(a) Tempered glass.

(b) Laminated glass or have certificate for panes without permanent label.

(c) Organic coated glass.

(3) Fire rated glazing assemblies: Mark in accordance with ICC IBC.

1.4 PERFORMANCE REQUIREMENTS

a. General: Design glazing system consistent with guidance and practices presented in the GANA Glazing Manual, GANA Laminate Manual, and GANA Sealant Manual, as applicable to project. Installed glazing is to withstand applied loads, thermal stresses, thermal movements, building movements, permitted tolerances, and combinations of these conditions without failure, including loss or glass breakage attributable to defective manufacture, fabrication, or installation; failure of sealants or gaskets to remain watertight and airtight; deterioration of glazing materials; unsafe engagement of the framing system; deflections beyond specified limits; or other defects in construction.

b. Glazing Unit Design: Design glass, including engineering analysis meeting requirements of authorities having jurisdiction. Thicknesses listed are minimum. Coordinate thicknesses with framing system manufacturers.

(1) Design glass in accordance with ASTM E1300, and for conditions beyond the scope of ASTM E1300, by a properly substantiated structural analysis.

(2) Design Wind Pressures: As indicated on construction documents .

(3) Wind Design Data: As indicated on construction documents .

(4) Maximum Lateral Deflection: For glass supported on all four edges, limit center-of-glass deflection at design wind pressure to not more than the structural capacity of the glazing unit, the threshold at which frame engagement is no longer safely assured, 1/100 times the short-side length, or 0.75 inch , whichever is less.

c. Ballistic- and Blast- resistant glass or glazing assemblies:

(1) For blast-resistant and ballistic-resistant units comply with requirements in PSDRM and project-specific criteria provided on the drawings and specifications.

(2) Spall Resistance: Laminated glazing is not permitted to produce spall to interior (protected side) when impacted with scheduled ballistics.

(3) Tolerances:

(a) Outside dimensions: Overall outside dimensions (height and width) of laminated security glazing is to maintain tolerance of {UNCF0B1} $\pm$ 0.12 inch.

(b) Warpage: Out-of-flat (warpage or bowing) condition of laminates is not to exceed per lineal meter 0.10 inch per 3.3 lineal foot). The condition, if present, is to be localized to extent not greater than 0.03 inch for any 0.98 feet section.

(4) To demonstrate blast resistance for windows, perform analysis using window glazing blast analysis and design software developed by the US Government, such as SBEDS-W or WINGARD PE, which are capable of predicting the glass, film and laminate response when subjected to the blast loading environment. The dynamic analysis must use a glass probability of failure (POF) of 500 breaks per 1000. Glazing must be designed to achieve a glass Performance Condition 2 (i.e. glass may crack but must remain in the frame) in response to a design blast load of 13.5 psi peak pressure and 51 psi-msec impulse (as an equivalent triangular load). Alternatively, glazed assemblies may be certified through testing (Standard Test Method for Glazing Systems Subject to Airblast Loadings ASTM F1642) against the specified design blast load. The tested assembly must be demonstrated to be sufficiently similar in glazing layup, mullions, frames, connections, and hardware to that being constructed for the Project.

d. Windborne-Debris-Impact Resistance: Comply with enhanced-protection testing requirements in ASTM E1996 for project wind zone when tested according to ASTM E1886, based upon testing of specimens not less than the size required for project and utilizing installation method identical to that specified for project.

(1) Project Wind Zone: Wind Zone 3 .

(2) Large-Missile Test: For glazing located within 30 feet of grade.

(3) Small-Missile Test: For glazing located more than 30 feet above grade.

e. Building Enclosure Vapor Retarder and Air Barrier:

(1) Utilize the inner pane of multiple pane sealed units for the continuity of the air barrier and vapor retarder seal.

(2) Maintain a continuous air barrier and vapor retarder throughout the glazed assembly from glass pane to heel bead of glazing sealant.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Sustainable Design Submittals; S

Manufacturer's Literature And Data; G, AE

SD-04 Samples

Samples; G, AE

SD-06 Test Reports

Preconstruction Adhesion And Compatibility Test Report; G, AE

SD-07 Certificates

Manufacturer's Certificates; G, AE

SD-11 Closeout Submittals

Manufacturer Warranty; G, AE

1.6 SUBMITTAL INSTRUCTIONS

a. In accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

(2) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

c. Blast Analysis and Design Calculations when demonstrating blast resistance through analysis: Blast analysis and design calculations must be performed by a licensed professional engineer with a minimum of five years of professional experience in blast-resistant design.

d. Manufacturer's Certificates:

(1) Certificate stating that fire-protection and fire-resistive glazing units meet code requirements for fire-resistance-rated assembly and applicable safety glazing requirements.

(2) Certificate on solar heat gain coefficient when value is specified.

(3) Certificate on "R" value when value is specified.

(4) Certificate test reports confirming compliance with specified bullet resistive rating.

(5) Certificate that blast resistant glass meets the specified requirements when demonstrated through acceptable test method.

e. Manufacturer Warranty.

f. Manufacturer's Literature and Data:

(1) Glass, each kind required.

(2) Insulating glass units.

(3) Transparent (one-way vision glass) mirrors.

(4) Elastic compound for metal sash glazing.

(5) Glazing cushion.

(6) Sealing compound.

(7) Bullet resistive material.

(8) Plastic glazing material, each type required.

g. Samples:

(1) Size: 12 inches by 12 inches.

(2) Tinted glass.

(3) Reflective glass.

h. Preconstruction Adhesion and Compatibility Test Report: Submit glazing sealant manufacturer’s test report indicating glazing sealants were tested for adhesion to glass and glazing channel substrates and for compatibility with glass and other glazing materials.

1.7 DELIVERY, STORAGE AND HANDLING

a. Delivery: Schedule delivery to coincide with glazing schedules so minimum handling of crates is required. Do not open crates except as required for inspection for shipping damage.

b. Storage: Store cases according to printed instructions on case, in areas least subject to traffic or falling objects. Keep storage area clean and dry.

c. Handling: Unpack cases following printed instructions on case. Stack individual windows on edge leaned slightly against upright supports with separators between each.

d. Protect laminated security glazing units against face and edge damage during entire sequence of fabrication, handling, and delivery to installation location. Provide protective covering on exposed faces of glazing plastics, and mark inside as "INTERIOR FACE" or "PROTECTED FACE":

(1) Treat security glazing as fragile merchandise, and packaged and shipped in export wood cases with width end in upright position and blocked together in a mass. Storage and handling to comply with manufacturer's directions and as required to prevent edge damage or other damage to glazing resulting from effects of moisture, condensation, temperature changes, direct exposure to sun, other environmental conditions, and contact with chemical solvents.

(2) Protect sealed-air-space insulating glazing units from exposure to abnormal pressure changes, as could result from substantial changes in altitude during delivery by air freight. Provide temporary breather tubes which do not nullify applicable warranties on hermetic seals.

(3) Temporary protections: The glass front and polycarbonate back of glazing are to be temporarily protected with compatible, peelable, heat-resistant film which will be peeled for inspections and re-applied and finally removed after doors and windows are installed at destination. Since many adhesives will attack polycarbonate, the film used on exposed polycarbonate surfaces is to be approved and applied by manufacturer.

(4) Edge protection: To cushion and protect glass clad, and polycarbonate edges from contamination or foreign matter, the four edges are to be sealed the depth of glazing with continuous standard-thickness thermoplastic rubber tape. Alternatively, continuous channel shaped extrusion of thermoplastic rubber are to be used, with flanges extending into face sides of glazing.

(5) Protect "Constant Temperature" units including every unit where glass sheet is directly laminated to or directly sealed with metal-tube type spacer bar to polycarbonate sheet, from exposures to ambient temperatures outside the range of 16 to 24 degrees C (60 to 75 degrees F), during the fabricating, handling, shipping, storing, installation, and subsequent protection of glazing.

1.8 PROJECT CONDITIONS

Field measure openings before ordering tempered glass products to assure for proper fit of field measured products.

1.9 WARRANTY

a. Construction Warranty: Comply with the FAR clause 52.246-21 “Warranty of Construction”.

b. Manufacturer Warranty: Manufacturer shall warranty their glazing from the date of installation and final acceptance by the Government as follows. Submit manufacturer warranty.

(1) Bullet resistive plastic material to remain visibly clear without discoloration for 10 years.

(2) Insulating glass units to remain sealed for 10 years.

(3) Laminated glass units to remain laminated for five years.

(4) Polycarbonate to remain clear and ultraviolet light stabilized for five years.

(5) Insulating plastic to not have more than 6 percent decrease in light transmission and be ultraviolet light stabilized for ten years.

1.10 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 800	(2016) Voluntary Specifications and Test Methods for Sealants
AAMA 810.1	(2016) Expanded Cellular Glazing Tape

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z97.1	(2015) Safety Glazing Materials Used in Buildings - Safety Performance Specifications and Methods of Test

ASTM INTERNATIONAL (ASTM)

ASTM C542	(2005; R 2024) Standard Specification for Lock-Strip Gaskets
ASTM C864	(2005; R 2015) Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers
ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM C1036	(2021) Standard Specification for Flat Glass
ASTM C1048	(2018) Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass
ASTM C1172	(2024) Standard Specification for Laminated Architectural Flat Glass
ASTM C1376	(2015) Standard Specification for Pyrolytic and Vacuum Deposition Coatings on Flat Glass
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E1300	(2024) Standard Practice for Determining Load Resistance of Glass in Buildings
ASTM E1886	(2019) Standard Test Method for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Missile(s) and Exposed to Cyclic Pressure Differentials
ASTM E1996	(2017) Standard Specification for Performance of Exterior Windows, Curtain

	Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes
ASTM F1233	(2021) Standard Test Method for Security Glazing Materials and Systems

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

GLASS ASSOCIATION OF NORTH AMERICA (GANA)

GANA Glazing Manual	(2008) Glazing Manual
GANA Sealant Manual	(2008) Sealant Manual
GANA Laminate Manual	(2009) Laminated Glazing Reference Manual

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

INSULATING GLASS MANUFACTURERS ALLIANCE (IGMA)

IGMA TB-3001	(2001) Guidelines for Sloped Glazing
IGMA TM-3000	(1990; R 2016) North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial & Residential Use

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2025) Standard for Fire Doors and Other Opening Protectives
NFPA 252	(2022) Standard Methods of Fire Tests of Door Assemblies
NFPA 257	(2022) Standard on Fire Test for Window and Glass Block Assemblies

UL SOLUTIONS (UL)

UL 9	(2009; Reprint Oct 2024) Standard for Fire Tests of Window Assemblies
UL 263	(2011; Reprint Aug 2021) UL Standard for Safety Fire Tests of Building Construction and Materials
UL 752	(2023) UL Standard for Safety Bullet-Resisting Equipment

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

16 CFR 1201	Safety Standard for Architectural Glazing

PART 2 PRODUCT

Finishes and materials indicated on the drawings are intended to establish the required pattern, color, texture, appearance, and other salient characteristics. The contractor shall submit samples of all proposed exterior finish materials and furnishings for government review and approval. No exterior finishes or furnishings shall be installed until written approval has been obtained from the government. Refer to the specifications for additional requirements, including material performance criteria and submittal requirements.

2.1 GLASS

a. Provide minimum thickness stated and as additionally required to meet performance requirements. Provide minimum 1/4 inch thick glass units unless otherwise indicated.

b. Obtain glass units from single source from single manufacturer for each glass type.

c. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

d. Clear Glass:

(1) ASTM C1036, Type I, Class 1, Quality q3 .

e. Tinted Heat reflective and low emissivity coated glass:

(1) ASTM C1036, Type I, Class 2, Quality q3.

2.2 HEAT-TREATED GLASS

a. Roller Wave Limits for Heat-Treated Glass: Orient all roller wave distortion parallel to bottom surface of glazing, and provide units complying with the following limitations:

(1) Measurement Parallel to Line: Maximum peak to valley 0.008 inch.

(2) Measurement Perpendicular to Line: Maximum0.001 inch.

(3) Bow/Warp: Maximum 50 percent of bow and warp allowed by ASTM C1048.

b. Clear Heat Strengthened Glass:

(1) ASTM C1048, Kind HS, Condition A, Type I, Class 1, Quality q3.

c. Tinted Heat Strengthened Glass:

(1) ASTM C1048, Kind HS, Condition A, Type I, Class 2, Quality q3.

d. Clear Tempered Glass:

(1) ASTM C1048, Kind FT, Condition A, Type I, Class 1, Quality q3.

e. Tinted Tempered Glass.

(1) ASTM C1048, Kind FT, Condition A, Type I, Class 2, Quality q3.

2.3 COATED GLASS

a. Reflective-Coated Low-E Coated Tempered Glass:

(1) ASTM C1376 and ASTM C1048, Kind FT, Condition C, Type I, Class 1, Quality q3 with reflective metallic coating.

2.4 LAMINATED GLASS

a. Laminated Glass: ASTM C1172. Two or more lites of heat treated glass bonded with polyvinyl butyral, ionomeric polymer, or cast-in-place and cured-transparent-resin interlayer complying with interlayer manufacturer's written instructions. Minimum total laminated thickness of 1/4″ for blast resistant glazing.

b. Interlayer: Use min. 0.030 inch thick interlayer for vertical glazing unless otherwise indicated in construction documents .

c. Interlayer: Use 0.060 inch thick interlayer for:

(1) Horizontal or sloped glazing.

(2) Acoustical glazing.

(3) Assemblies requiring heat strengthened or fully tempered glass.

d. Interlayer: Use 0.090 inch thick interlayer where required to meet performance requirements.

e. Interlayer Color: Clear, unless otherwise indicated in construction documents .

2.5 SECURITY GLAZING ASSEMBLY

a. Provide ballistic level as scheduled in accordance with UL 752.

b. Forced Entry Resistance: As scheduled, in accordance with ASTM F1233.

c. Blast Resistance: Provide exterior glazing units that meet the specified blast performance requirements.

d. Laminated Glass Security Glazing Units: Fabricate from multiple lites of scheduled glass with polyvinyl butyral, ionomeric polymer, or cast-in-place and cured-transparent resin interlayers between the layers of glazing.

2.6 INSULATING GLASS UNITS

a. Provide factory fabricated, hermetically sealed glass unit consisting of two panes of glass separated by a dehydrated air space and comply with ASTM E2190. The exterior glass unit shall be fully tempered and the inner glass unit shall be laminated annealed at a minimum for all blast resistant glazing.

b. Assemble units using glass types specified in Insulating Glass Schedule and Blast Glazing assembly requirements

2.7 FIRE PROTECTION AND FIRE RESISTANCE GLAZING

a. Fire-Protection-Rated Glazing: Glazing units tested for use in fire door assemblies or fire windows, UL, ITS-WHI or equivalent listed and labeled by testing agency in accordance with IBC, for fire-protection ratings as indicated on construction documents , based upon positive-pressure testing per NFPA 257 or UL 9, and complying with NFPA 80.

(1) Hose-Stream Test: Units must comply, except units having fire-protection rating of 20 minutes.

(2) Temperature Rise Limitation: Units over100 square inch must comply with 450 degrees F limitation.

(3) Labeling: Permanently label fire-protection-rated glazing units in accordance with IBC.

(4) Safety Glazing: Comply with 16 CFR 1201, Category II.

(5) Fire-Protection-Rated Tempered Glass: For 20-minute fire-protection-rated door assemblies, of thickness scheduled.

(6) Fire-Protection-Rated Laminated Ceramic Glazing: Units made from two lites of clear, ceramic glass, 5/16 inch total thickness, for rating scheduled.

(7) Fire-Protection-Rated Laminated Glass with Intumescent Interlayers: Units made from multiple lites of uncoated, ultra-clear (low-iron) float glass, in intumescent interlayers, of thickness and rating scheduled.

b. Fire-Resistance-Rated Glazing: Glazing units tested for use in fire wall assemblies, UL, ITS-WHI or equivalent listed and labeled by testing agency in accordance with IBC for fire-resistance ratings of wall assemblies as indicated on construction documents , based upon testing according to NFPA 252 and ASTM E119 or UL 263.

(1) Labeling: Permanently label fire-resistance-rated glazing units in accordance with IBC.

(2) Safety Glazing: Comply with 16 CFR 1201, Category II.

(3) Fire-Resistance-Rated Laminated Glass with Intumescent Interlayers: Units made from multiple lites of uncoated, ultra-clear low-iron float glass, in intumescent interlayers, of thickness and rating scheduled.

(4) Fire-Resistance-Rated Double Glazing Units with Gel Fill: Units made from two lites of uncoated, fully tempered, ultra-clear (low-iron) float glass, with perimeter metal spacer and edge seal forming cavity filled with clear, fully transparent, heat-absorbing gel, of thickness and fire-protection rating scheduled.

2.8 GLAZING ACCESSORIES

a. As required to supplement the accessories provided with the items to be glazed and to provide a complete installation. Ferrous metal accessories exposed in the finished work are to have a finish that will not corrode or stain while in service. Fire rated glazing to be installed with glazing accessories in accordance with the manufacturer's installation instructions.

b. Setting Blocks: ASTM C864:

(1) Silicone type.

(2) Channel shape; having 1/4 inch internal depth.

(3) Shore A hardness of 80 to 90 Durometer.

(4) Block lengths: 2 inches except 4 to 6 inches for insulating glass.

(5) Block width: Approximately 1/16 inch less than the full width of the rabbet.

(6) Block thickness: Minimum 3/16 inch. Thickness sized for rabbet depth as required.

c. Spacers: ASTM C864:

(1) Channel shape having a 1/4 inch internal depth.

(2) Flanges not less 3/32 inch thick and web 1/8 inch thick.

(3) Lengths: 1 to 3 inches.

(4) Shore A hardness of 40 to 50 Durometer.

d. Glazing Tapes:

(1) Semi-solid polymeric based closed cell material exhibiting pressure-sensitive adhesion and withstanding exposure to sunlight, moisture, heat, cold, and aging.

(2) Shape, size and degree of softness and strength suitable for use in glazing application to prevent water infiltration.

(3) Complying with AAMA 800 for the following types:

(a) AAMA 810.1, Type 1, for glazing applications in which tape acts as the primary sealant.

(b) AAMA 810.1, Type 2, for glazing applications in which tape is used in combination with a full bead of liquid sealant.

e. Spring Steel Spacer: Galvanized steel wire or strip designed to position glazing in channel or rabbeted sash with stops.

f. Glazing Clips: Galvanized steel spring wire designed to hold glass in position in rabbeted sash without stops.

g. Glazing Gaskets: ASTM C864:

(1) Firm dense wedge shape for locking in sash.

(2) Soft, closed cell with locking key for sash key.

(3) Flanges may terminate above the glazing-beads or terminate flush with top of beads.

h. Lock-Strip Glazing Gaskets: ASTM C542, shape, size, and mounting as indicated.

i. Glazing Sealants: ASTM C920, silicone neutral cure:

(1) Type S.

(2) Class 25 or 50 as recommended by manufacturer for application.

(3) Grade NS.

(4) Shore A hardness of 25 to 30 Durometer.

(5) VOC Content: For sealants used inside the weatherproofing system, not more than 250 g/L or less when calculating according to 40 CFR 59, (EPA Method 24).

j. Neoprene, EPDM, or Vinyl Glazing Gasket: ASTM C864.

(1) Channel shape; flanges may terminate above the glazing channel or flush with the top of the channel.

(2) Designed for dry glazing.

k. Color:

(1) Color of glazing compounds, gaskets, and sealants used for aluminum color frames to match color of the finished aluminum and be nonstaining.

(2) Color of other glazing compounds, gaskets, and sealants which will be exposed in the finished work and unpainted are to be black, gray, or neutral color.

PART 3 EXECUTION

3.1 EXAMINATION

a. Verification of Conditions:

(1) Examine openings for glass and glazing units; determine they are proper size; plumb; square; and level before installation is started.

(2) Verify that glazing openings conform with details, dimensions and tolerances indicated on manufacturer is approved shop drawings.

b. Review for conditions which may adversely affect glass and glazing unit installation, prior to commencement of installation. Do not proceed with installation until unsatisfactory conditions have been corrected.

c. Verify that wash down of adjacent masonry is completed prior to erection of glass and glazing units.

3.2 PREPARATION

a. For sealant glazing, prepare glazing surfaces in accordance with GANA Sealant Manual.

b. Determine glazing unit size and edge clearances by measuring the actual unit to receive the glazing.

c. Shop fabricate and cut glass with smooth, straight edges of full size required by openings to provide GANA recommended edge clearances.

d. Verify that components used are compatible.

e. Clean and dry glazing surfaces.

f. Prime surfaces scheduled to receive sealants, as determined by preconstruction sealant-substrate testing.

3.3 INSTALLATION – GENERAL

a. Install in accordance with GANA Glazing Manual, GANA Sealant Manual, IGMA TB-3001, and IGMA TM-3000 unless specified otherwise.

b. Glaze in accordance with recommendations of glazing and framing manufacturers, and as required to meet the Performance Test Requirements specified in other applicable sections of specifications.

c. Set glazing without bending, twisting, or forcing of units.

d. Do not allow glass to rest on or contact any framing member.

e. Glaze doors and operable sash, in a securely fixed or closed and locked position, until sealant, glazing compound, or putty has thoroughly set.

g. Tempered Glass: Install with roller distortions in horizontal position unless otherwise directed.

j. Laminated Glass:

(1) Tape edges to seal interlayer and protect from glazing sealants.

(2) Do not use putty or glazing compounds.

k. Insulating Glass Units:

(1) Glaze in compliance with glass manufacturer's written instructions.

(2) When glazing gaskets are used, they are to be of sufficient size and depth to cover glass seal or metal channel frame completely.

(3) Do not use putty or glazing compounds.

(4) Do not grind, nip, cut, or otherwise alter edges and corners of fused glass units after shipping from factory.

(5) Install with tape or gunnable sealant in wood sash.

l. Fire Protective and Fire Resistance Glass:

(1) Wire Glass: Glaze in accordance with NFPA 80.

(2) Other fire protective and fire resistant glass: Glaze in accordance with manufacturer’s installation instructions and NFPA 80.

3.4 INSTALLATION - WET/DRY METHOD (PREFORMED TAPE AND SEALANT)

a. Cut glazing tape to length and set against permanent stops, 3/16 inch below sight line. Seal corners by butting tape and dabbing with butyl sealant.

b. Apply heel bead of butyl sealant along intersection of permanent stop with frame ensuring full perimeter seal between glass and frame to complete the continuity of the air and vapor seal.

c. Place setting blocks at 1/4 points with edge block no more than 6 inches from corners.

d. Rest glazing on setting blocks and push against tape and heel bead of sealant with sufficient pressure to achieve full contact at perimeter of pane or glass unit.

e. Install removable stops, with spacer strips inserted between glazing and applied stops, 1/4 inch below sight line. Place glazing tape on glazing pane or unit with tape flush with sight line.

f. Fill gap between glazing and stop with sealant to depth equal to bite of frame on glazing, but not more than 3/8 inch below sight line. Sealant type is to be compatible with glazing tape.

g. Apply cap bead of sealant along void between the stop and the glazing, to uniform line, flush with sight line. Tool or wipe sealant surface smooth.

3.5 INSTALLATION - WET METHOD (SEALANT AND SEALANT)

a. Place setting blocks at 1/4 points and install glazing pane or unit.

b. Install removable stops with glazing centered in space by inserting spacer shims both sides at 24 inch intervals, 1/4 inch below sight line.

c. Fill gaps between glazing and stops with sealant to depth of bite on glazing, but not more than 3/8 inch below sight line to ensure full contact with glazing and continue the air and vapor seal.

d. Apply sealant to uniform line, flush with sight line. Tool or wipe sealant surface smooth.

3.6 INSTALLATION - INTERIOR WET/DRY METHOD (TAPE AND SEALANT)

a. Cut glazing tape to length and install against permanent stops, projecting 1/16 inch above sight line.

b. Place setting blocks at 1/4 points with edge block no more than 6 inches from corners.

c. Rest glazing on setting blocks and push against tape to ensure full contact at perimeter of pane or unit.

d. Install removable stops, spacer shims inserted between glazing and applied stops at 24 inch intervals, 1/4 inch below sight line.

e. Fill gaps between pane and applied stop with sealant to depth equal to bite on glazing, to uniform and level line. Sealant type is to be compatible with glazing tape.

f. Trim protruding tape edge.

3.7 INSTALLATION - INTERIOR WET METHOD (COMPOUND AND COMPOUND)

a. Install glazing resting on setting blocks. Install applied stop and center pane by use of spacer shims at 24 inch centers, kept 1/4 inch below sight line.

b. Locate and secure glazing pane using glazers' clips.

c. Fill gaps between glazing and stops with glazing compound until flush with sight line. Tool surface to straight line.

3.8 REPLACEMENT AND CLEANING

a. Clean new glass surfaces removing temporary labels, paint spots, and defacement after approval by Contracting Officer.

b. Replace cracked, broken, and imperfect glass, or glass which has been installed improperly.

c. Leave glass, putty, and other setting material in clean, whole, and acceptable condition.

3.9 PROTECTION

Protect finished surfaces from damage during erection, and after completion of work. Strippable plastic coatings on colored anodized finish are not acceptable.

3.10 MONOLITHIC GLASS SCHEDULE

a. Glass Type MG# 01: Clear fully tempered float glass.

(1) Unit Thickness: 0.23 inch .

(2) Safety glazing label required.

3.11 LAMINATED GLASS SCHEDULE

a. Glass Type LG# 01: Clear laminated glass with two lites of fully tempered float glass.

(1) Minimum Thickness of Each Glass Lite: 0.12 inch or .23 inch, as indicated on the drawings.

(2) Interlayer Thickness: 0.060 inch.

(3) Safety glazing label required.

(4) Application: Interior acoustical glazing of units unless otherwise scheduled.

3.12 INSULATING LAMINATED GLASS SCHEDULE (FORCE PROTECTION AND PHYSICAL SAFETY)

a. Glass Type IL# 01: Reflective-coated, low-E coated, insulating laminated glass.

(1) Overall Unit Thickness: 1-1/4 inch .

(2) Outdoor Lite: Heat-strengthened float glass, except fully tempered float glass where indicated.

(a) Minimum Thickness of Outdoor Lite: 0.23 inch .

(b) Tint Color: Clear .

(c) Coating Type: Pyrolytic .

1. Coating Color: Silver.

2. Coating Location: Second surface.

(3) Interspace Content: Argon.

(4) Indoor Lite: Clear laminated glass with two lites of annealed float glass, except heat-strengthened float glass where required, and fully tempered float glass where indicated.

(a) Minimum Thickness of Each Glass Lite: 0.23 inch .

(b) Interlayer Thickness: 0.060 inch .

(5) Low-E Coating: Sputtered on second surface.

(6) Outdoor Visible Reflectance: 25 percent maximum.

(7) Visible Light Transmittance: 47 percent minimum.

(8) Solar Heat Gain Coefficient: .25 maximum.

(9) Safety glazing label required.

(10) Windborne debris-resistant glazing unit required.

(11) Blast Resistance: Provide units meeting the following:

(a) GP Value 2.

3.13 FIRE-PROTECTIVE AND FIRE-RESISTANCE GLAZING SCHEDULE

a. Glass Type FR# 01: Fire-protection-rated laminated ceramic glazing.

(1) Thickness: 0.39 inch.

(2) Rating: 60- minute.

(3) Application: Fire-protection-rated door and window assemblies.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 08 - OPENINGS

SECTION 08 90 00

LOUVERS AND VENTS

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 EXTERIOR WALL LOUVERS
2.3 CLOSURE ANGLES AND CLOSURE PLATES
2.4 WIRE GUARDS
2.5 BLANK-OFF PANELS
2.6 FINISH
2.7 PROTECTION

PART 3 EXECUTION

3.1 INSTALLATION
3.2 CLEANING AND ADJUSTING

-- End of Section Table of Contents --

SECTION 08 90 00
1/21

LOUVERS AND VENTS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies fixed and operable wall louvers, door louvers and wall vents.

1.2 RELATED WORK

a. Section 08 11 13, HOLLOW METAL DOORS AND FRAMES: Louvers in Steel Doors.

b. Section Refer to the SCHEDULE FOR FINISHES on the drawings: Color of finish.

c. Section 13 49 00, RADIATION PROTECTION: Louvers in lead lined wood doors.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Recycled Content For Aluminum Framing Materials; S

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Shop Drawings:

(1) Each type, showing material, finish, size of members, method of assembly, and installation and anchorage details.

c. Manufacturer's Literature and Data:

(1) Each type of louver and vent.

Recycled content for aluminum components

d. Color samples.

e. Blast Design Calculations: Louver System and Anchorage

(1) Submit calculations for review and approval prepared by qualified blast consultant, with a minimum of 5 years experience verifying louver assembly including anchors comply with specified blast resistance performance. Blast resistant louvers must meet the requirements for Critical Facilities per Section 6.6.2 of the VA PSDRM.

1.5 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-L	(2015) Laboratory Methods of Testing Louvers for Rating

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605	(2020) Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

ASTM D1187/D1187M (1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

PSRDM (2020; R 2025) Physical Security & Resiliency Design Manual

PART 2 PRODUCTS

Finishes and materials indicated on the drawings are intended to establish the required pattern, color, texture, appearance, and other salient characteristics. The contractor shall submit samples of all proposed exterior finish materials and furnishings for government review and approval. No exterior finishes or furnishings shall be installed until written approval has been obtained from the government. Refer to the specifications for additional requirements, including material performance criteria and submittal requirements.

2.1 MATERIALS

a. Aluminum, Extruded: ASTM B221M ASTM B221.

b. Stainless Steel: ASTM A240/A240M, Type 302B.

c. Galvanized Steel Sheet: ASTM A653/A653M; G90 min.

d. Carbon Steel and Sheet: ASTM A1008/A1008M (interior use louvers only).

e. Aluminum, Plate and Sheet: ASTM B209M (B209); alloy 3003 or 5005 with temper as required for forming.

f. Fasteners: Fasteners for securing louvers and wall vents to adjoining construction, except as otherwise specified or indicated in construction documents, to be toggle or expansion bolts of size and type as required for each specific type of installation and service condition.

(1) Where type, size, or spacing of fasteners is not shown or specified, submit shop drawings showing proposed fasteners, and method of installation.

(2) Fasteners for louvers, louver frames, and wire guards to be of stainless steel or aluminum with same finish as louvers.

(3) Fasteners for louvers, louver frames and wire guards within mental health areas to be non-removable/tamper-proof type.

g. Inorganic Zinc Primer: MPI No. 19.

h. Bituminous Coating: ASTM D1187/D1187M; cold applied asphalt mastic emulsion.

2.2 EXTERIOR WALL LOUVERS

a. General:

(1) Provide fixedtype louvers of size and design shown.

(2) Heads, sills and jamb sections are to have formed caulking slots or be designed to retain caulking. Head sections are to have exterior drip lip, and sill sections an integral water stop.

(3) Furnish louvers with sill extension or separate sill as shown.

(4) Frame is to be mechanically fastened or welded construction with welds dressed smooth and flush.

b. Performance Characteristics:

(1) Weather louvers are to have a minimum of 45.8 percent free area and to pass 1,000 fpm free area velocity at a pressure drop not exceeding 0.19 inch water gauge and carry not more than 0.01 ounces of water per one square foot of free area for 15 minutes when tested per AMCA 500-L.

(2) Louvers are to bear AMCA certified rating seals for air performance and water penetration ratings.

(3) Blast Resistance:

c. Louvers in exterior walls shall be blast resistant and meet the following criteria per the VA PDSRM:

(1) Design louvers to resist a design blast load of 13.5 psi peak pressure and 51 psi-msec impulse (as an equivalent triangular load) while sustaining a maximum deformation no greater than the response limits for Heavy Damage per ASCE 59. Alternatively, blast resistance may be demonstrated through dynamic testing. The materials and dimensions of tested configurations must reasonably represent the materials and dimensions provided for construction. Anchor louvers to the supporting structure for the collected blast reactions as determined by analysis or testing.

(2) In lieu of demonstrating blast resistance through analysis and/or testing, fabricate and install louvers to meet the following prescriptive requirements: Provide louvers with a minimum blade thickness as indicated below. For louver openings larger than 5.5-ft, provide mullions with a minimum wall thickness of 3-mm (0.125-in) and spaced so blade lengths do not exceed 5.5-ft long. Fabricate louver assembly so that connections between blades and louver framing (jambs, mullions, etc.) are capable of resisting a distributed ultimate load of 100 lbs/in. In addition, anchor the louver assembly to the supporting structure (walls, wall framing, etc.) to resist a distributed ultimate load of 100 lbs/in around the perimeter of the opening.

d. Aluminum Louvers:

(1) General: Frames, blades, sills and mullions (sliding interlocking type); 0.078-inch thick extruded 6063-T5 or –T52 aluminum. Blades to be drainable type and have reinforcing bosses.

(2) Louvers, fixed: Make frame sizes 1/2-inch smaller than openings. Single louvers frames are not to exceed 66 inches wide. When openings exceed 66 inches, provide twin louvers separated by mullion members.

(3) Louvers are to withstand the effects or gravity loads and the following wind loads and stresses within limits and under conditions indicated without permanent deformation of louver components, noise or metal fatigue caused by louver-blade rattle or flutter, or permanent damage to fasteners and anchors.

(a) Wind load acting inward or outward of not less than 30 pound per square foot.

e. Stainless Steel Louvers: Form stainless steel louvers using 0.063-inch thick sheet for frames, blades, sills and mullions.

(1) Provide louver with fixed 45 degree drainable blades with water baffle. Make overall frame size 1/2-inch less than opening, unless otherwise indicated in construction documents.

(2) Single louver sections are not to exceed 66 inches in width. For openings larger than 66 inches wide, provide multiple sections not larger than 66 inches wide separated by mullions.

f. Formed Steel Louvers: Form galvanized louvers using 0.059-inch thick sheet for frames, blades, sills and mullions.

(1) Provide louver with fixed 45 degree drainable blades with water baffle. Make overall frame size 1/2-inch less than opening, unless otherwise indicated in construction documents.

(2) Single louver sections are not to exceed 66 inches in width. For openings larger than 66 inches wide, provide multiple sections not larger than 66 inches side separated by mullions.

2.3 CLOSURE ANGLES AND CLOSURE PLATES

a. Fabricate from 0.078-inch thick stainless steel or aluminum.

b. Provide continuous closure angles and closure plates on inside head, jambs and sill of exterior wall louvers.

c. Secure angles and plates to louver frames with screws, and to masonry or concrete with fasteners as indicated in construction documents.

2.4 WIRE GUARDS

a. Provide wire guards on outside of all exterior louvers, except on exhaust air louvers.

b. Fabricate frames from 0.078-inch thick extruded or sheet aluminum designed to retain wire mesh.

c. Wire mesh to be woven from not less than 0.05-inch diameter stainless steel wire in1/2-inch square mesh.

d. Miter corners and join by concealed corner clips or locks extending not less than 2-1/4 inches into rails and stiles. Equip wire guards over 4 feet in height with a mid-rail constructed as specified for frame components.

e. Fasten frames to outside of louvers with aluminum or stainless steel

devices of same finish as louvers designed to allow removal and replacement without damage to the wire guard or the louver.

2.5 BLANK-OFF PANELS

a. Uninsulated panels attached with clips or screws as follows: Panel finish is to be same finish applied to louvers .

(1) Aluminum sheet for aluminum louvers, 0.050 inch minimum thickness.

(2) Galvanized-steel sheet for galvanized-steel louvers, 0.040 inch minimum.

(3) Stainless-steel sheet for stainless-steel louvers, not less than 0.038 inch minimum.

b. Insulated laminated panels consisting of an insulating core surfaced on back and front with metal sheets and attached to back of louver with clips on screws and gasketed or sealant sealed perimeter. Panel finish is to be same finish applied to louvers .

(1) Thickness: 1 inch .

(2) Aluminum sheet for aluminum louver 0.032 inch minimum.

(3) Galvanized-steel sheet for galvanized-steel louver 0.028 inch minimum.

(4) Stainless-steel sheet for stainless-steel louvers 0.031 inch minimum.

(5) Insulating Core: extruded-polystyrene foam.

2.6 FINISH

a. In accordance with NAAMM Metal Finishes Manual: AMP 500-505

b. Aluminum Louvers Air Intake Vents Wire Guards Blank Off Panels:

(1) Organic Finish: AAMA 2605 (Fluorocarbon coating) with total dry film thickness of not less than 1.2 mil, color as indicated in the SCHEDULE FOR FINISHES on the drawings.

c. Aluminum : Sand blasted satin finish.

d. Stainless Steel: Mechanical finish No. 4 in accordance with NAAMM AMP 500.

e. Galvanized Sheet Steel: Two-coat baked-enamel or powder-coat finish consisting of prime coat and thermosetting topcoat, with a minimum dry film thickness of 2 mils.

(1) Paint interior surfaces of lightproof louvers with two (2) additional finish shop coats of baked-on flat black enamel.

(2) Finish painting of exposed surfaces of shop primed louvers is specified in Section 09 91 00, PAINTING.

(3) Manufacturer finished louvers and blank-off panels are to have color as indicated in the SCHEDULE FOR FINISHES on the drawings.

f. Steel: Surfaces of steel work, for which no other finish is specified, are to be cleaned free from scale, rust, oil and grease, and then given a light colored prime paint after fabrication, except ferrous metals concealed in finished work. Paint all contact surfaces of assembled work (except welded contact surfaces) with an additional shop coat of similar paint.

g. Contractor to provide government with samples of all specified finish materials. Government shall approve all finishes prior to installation. Finishes indicated are to convey color, pattern, texture and salient characteristics only.

2.7 PROTECTION

a. Provide protection for aluminum against galvanic action wherever dissimilar materials are in contact, by painting the contact surfaces of the dissimilar material with a heavy coat of bituminous coating (complete coverage), or by separating the contact surfaces with a performed synthetic rubber tape having pressure sensitive adhesive coating on one side.

b. Isolate the aluminum from plaster, concrete and masonry by coating aluminum with zinc-chromate primer.

c. Protect finished surfaces from damage during fabrication, erection, and after completion of the work. Strippable plastic coating on organic finish is not approved.

PART 3 EXECUTION

3.1 INSTALLATION

a. Set work accurately, in alignment and where indicated in construction documents. Install plumb, level, free of rack and twist, and set parallel or perpendicular as required to line and plane of surface.

b. Furnish setting drawings and instructions for installation of anchors and for the positioning of items having anchors to be built into masonry construction. Provide temporary bracing for such items until masonry is set.

c. Provide anchoring devices and fasteners as shown and as necessary for securing louvers to building construction as specified. Power actuated drive pins may be used, except for removal items and where members would be deformed or substrate damaged by their use.

d. Set wall louvers in masonry walls during progress of the work. If wall louvers are not delivered to job in time for installation in prepared openings, make provision for later installation. Set in cast-in-place concrete in prepared openings.

3.2 CLEANING AND ADJUSTING

a. After installation, all exposed prefinished and plated items and all items fabricated from stainless steel and aluminum are to be cleaned as recommended by the manufacturer and protected from damage until

completion of the project.

b. All movable parts, including hardware, are to be cleaned and adjusted to operate as designed without binding or deformation of the members, so as to be centered in the opening of frame, and where applicable, to have all contact surfaces fit tight and even without forcing or warping the components.

c. Restore louvers and vents damaged during installation and construction so no evidence remains of corrective work. If results of restoration are unsuccessful, as determined by Contracting Officer damaged units and replace with new units.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 05 16

SUBSURFACE PREPARATION FOR FLOOR FINISHES

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY AND STORAGE
1.6 REFERENCES

PART 2 PRODUCTS

2.1 MOISTURE REMEDIATION COATING
2.2 CEMENTITIOUS SELF-LEVELING UNDERLAYMENT
2.3 CEMENTITIOUS TROWEL-APPLIED UNDERLAYMENT(NOT SUITABLE FOR RESINOUS FLOOR FINISHES)

PART 3 EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS
3.2 SURFACE PREPARATION
3.3 MOISTURE REMEDIATION COATING
3.4 CEMENTITOUS UNDERLAYMENT
3.5 PROTECTION
3.6 FIELD QUALITY CONTROL

-- End of Section Table of Contents --

SECTION 09 05 16
1/21

SUBSURFACE PREPARATION FOR FLOOR FINISHES

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies subsurface preparation requirements for areas to

b. receive the installation of applied and resinous flooring. This section includes testing concrete for moisture and pH, remedial floor coating for concrete floor slabs having unsatisfactory moisture or pH conditions, floor leveling and repair as required.

1.2 RELATED WORK

a. Section 07 92 00, JOINT SEALANTS.

b. Section 09 65 19, RESILIENT TILE FLOORING Section 09 67 23.20, RESINOUS EPOXY BASE WITH VINYL CHIP BROADCAST (RES-2)

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.xx SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Product Data; G, AE

SD-06 Test Reports

Test Data; G, AE

SD-07 Certificates

Written Approval Confirming Product Compatibility; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Written approval confirming product compatibility with subfloor material manufacturer and the flooring manufacturer

c. Product Data:

(1) Moisture remediation system

(2) Underlayment Primer

(3) Cementitious Self-Leveling Underlayment

(4) Cementitious Trowel-Applied Underlayment (Not suitable for resinous floor finishes)

d. Test Data:

(1) Moisture test and pH results performed by a qualified independent testing agency or warranty holding manufacturer’s technical representative.

1.5 DELIVERY AND STORAGE

a. Deliver materials in containers with labels legible and intact and grade-seals unbroken.

b. Store material to prevent damage or contamination.

1.6 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D4259	(2018) Standard Practice for Preparation of Concrete by Abrasion Prior to Coating Application
ASTM C109/C109M	(2024) Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or (50-mm) Cube Specimens)
ASTM D7234	(2012) Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers
ASTM E96/E96M	(2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM F710	(2022) Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring
ASTM F1869	(2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
ASTM F2170	(2019a) Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes
ASTM C348	(2021) Standard Test Method for Flexural

Strength of Hydraulic-Cement Mortars

ASTM C191 (2021) Standard Test Methods for Time of Setting of Hydraulic Cement by Vicat Needle

PART 2 PRODUCTS

2.1 MOISTURE REMEDIATION COATING

a. System Descriptions:

(1) High-solids, epoxy system designed to suppress excess moisture in concrete prior to an overlayment. For use under resinous products, VCT, tile and carpet where issues caused by moisture vapor are a concern.

b. Products: Subject to compliance with applicable fire, health, environmental, and safety requirements for storage, handling, installation, and clean up.

c. System Components: Verify specific requirements as systems vary by manufacturer. Verify build up layers and installation method. Verify compatibility with substrate. Use manufacturer’s standard components, compatible with each other and as follows:

(1) Liquid applied coating:

(a) Resin: epoxy.

(b) Formulation Description: Multiple component high solids.

(c) Application: Per manufacturer’s written installation requirements.

(d) Thickness: minimum 10 mils

d. Material Vapor Permeance: Application shall achieve a permeance rating of less than 0.1 perm in accordance with ASTM E96/E96M.

e. Maximum RH requirement: 100% testing in accordance with ASTM F2170.

Property	Test	Value
Tensile Strength	ASTM D638	4,400 psi
Volatile Organic Compound Limits (V.O.C.)	SCAMD Rule 1113 (Ammended 02/05/2016)	25 grams per liter
Permeance	ASTM E96	0.1 perms
Tensile Modulus	ASTM D638	1.9X105 psi
Percent Elongation	ASTM D638	12%
Cure Rate	Per manufacture’s Data	4 hours Tack free with 24hr recoat window
Bond Strength	ASTM D7234	100% bond to concrete failure

2.2 CEMENTITIOUS SELF-LEVELING UNDERLAYMENT

a. System Descriptions:

(1) High performance self-leveling underlayment resurfacer. Single component, self-leveling, cementitious material designed for easy application as an underlayment for all types of flooring materials. It is used for substrate repair and leveling.

b. Products: Subject to compliance with applicable fire, health, environmental, and safety requirements for storage, handling, installation, and clean up. Gypsum-based products are unacceptable.

c. System Characteristics:

(1) Wearing Surface: smooth

(2) Thickness: Per architectural drawings, ranging from feathered edge to 1", per application. Applications greater than 1" require additional 3/8" aggregate to mix or as recommended by manufacturer.

d. Underlayment shall be calcium aluminate cement-based, containing Portland cement. Gypsum-based products are unacceptable.

e. Compressive Strength: Minimum 4100 psi in 28 days in accordance with ASTM C109/C109M.

f. Flexural Strength: Minimum 1000 psi in 28 days in accordance with ASTM C348

g. Dry Time: Underlayment shall receive the application of moisture insensitive tile in 6 hours, floor coverings in 16 hours, and resinous flooring in 3-7 days.

h. Primer: compatible and as recommended by manufacturer for use over intended substrate

i. System Components: Manufacturer's standard components that are compatible with each other and as follows:

(1) Primer:

(a) Resin: copolymer

(b) Formulation Description: single component ready to use.

(c) Application Method: Squeegee and medium nap roller.

(d) All puddles shall be removed, and material shall be allowed to dry, 1-2 hours at 70F.

(e) Number of Coats: (1) one.

(2) Grout Resurfacing Base:

(a) Formulation Description: Single component, cementitious self-leveling high-early and high-ultimate strength grout.

(b) Application Method: colloidal mix pump, cam rake, spike roll.

1. Thickness of Coats: Per architectural scope, 1" lifts.

2. Number of Coats: More than one if needed.

(c) Aggregates: for applications greater than 1inch, require additional 3/8” aggregate to mix.

Property	Test	Value
Compressive Strength	ASTM C109/C109M	2,200 psi @ 24 hrs 3,000 psi @ 7 days
Initial set time Final Set time	ASTM C191	30-45 min. 1 to 1.5 hours
Bond Strength	ASTM D7234	100% bond to concrete failure

2.3 CEMENTITIOUS TROWEL-APPLIED UNDERLAYMENT(NOT SUITABLE FOR RESINOUS FLOOR FINISHES)

a. Underlayment shall be calcium aluminate cement-based, containing Portland cement. Gypsum-based products are unacceptable.

b. Compressive Strength: Minimum 4000 psi in 28 days

c. Trowel-applied underlayment shall not contain silica quartz (sand).

d. Dry Time: Underlayment shall receive the application of floor covering in 15-20 minutes.

PART 3 EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS

a. Maintain ambient temperature of work areas at not less than (60 degrees F), without interruption, for not less than 24 hours before testing and not less than three days after testing.

b. Maintain higher temperatures for a longer period of time where required by manufacturer's recommendation.

c. Do not install materials when the temperatures of the substrate or materials are not within 60-85 degrees F .

3.2 SURFACE PREPARATION

a. Existing concrete slabs with existing floor coverings:

(1) Conduct visual observation of existing floor covering for adhesion, water damage, alkaline deposits, and other defects.

(2) Remove existing floor covering and adhesives. Comply with local, state and federal regulations and the RFCI Recommended Work Practices for Removal of Resilient Floor Coverings, as applicable to the floor covering being removed.

b. Concrete shall meet the requirements of ASTM F710 and be sound, solid,

clean, and free of all oil, grease, dirt, curing compounds, and any substance that might act as a bond-breaker before application. As required prepare slab by mechanical methods. No chemicals or solvents shall be used.

c. General: Prepare and clean substrates according to flooring manufacturer's written instructions for substrate indicated.

d. Prepare concrete substrates per ASTM D4259 as follows:

(1) Dry abrasive blasting.

(2) Wet abrasive blasting.

(3) Vacuum-assisted abrasive blasting.

(4) Centrifugal-shot abrasive blasting.

(5) Comply with manufacturer's written instructions.

e. Repair damaged and deteriorated concrete according to flooring manufacturer's written recommendations.

f. Verify that concrete substrates are dry.

g. Perform anhydrous calcium chloride test, ASTM F1869. Proceed with application only after substrates have maximum moisture-vapor-emission rate of per flooring manufactures formal and project specific written recommendation.

h. Perform in situ probe test, ASTM F2170. Proceed with application only after substrates do not exceed a maximum potential equilibrium relative humidity per flooring manufacture’s formal and project specific written recommendation.

i. Provide a written report showing test placement and results.

j. Prepare joints in accordance with Section 07 92 00, JOINT SEALANTS and material manufacturer’s instructions.

k. Alkalinity: Measure surface pH in accordance with procedures provided in ASTM F710 or as outlined by qualified testing agency or flooring manufacturer’s technical representative.

l. Tolerances: Subsurface shall meet the flatness and levelness tolerance specified on drawings or recommended by the floor finish manufacturer. Tolerance shall also not to exceed 1/4” deviation in 10’. As required, install underlayment to achieve required tolerance.

m. Other Subsurface: For all other subsurface conditions, such as wood or metal, contact the floor finish or underlayment manufacturer, as appropriate, for proper preparation practices.

3.3 MOISTURE REMEDIATION COATING

a. Where results of relative humidity testing (ASTM F2170) exceed the requirements of the specified flooring manufacturer, apply remedial coating as specified to correct excessive moisture condition.

b. Prior to remedial floor coating installation mechanically prepare the concrete surface to provide a concrete surface profile in accordance with ASTM D4259.

c. Mix and apply moisture remediation coating in accordance with manufacturer’s instructions.

3.4 CEMENTITOUS UNDERLAYMENT

a. Install cementitious self-leveling underlayment as required to correct surface defects, floor flatness or levelness corrections to meet the tolerance requirements as or detailed on drawings, address non-moving cracks or joints, provide a smooth surface for the installation of floor covering, or meet elevation requirements detailed on drawings.

b. Mix and apply in accordance with manufacturer's instructions.

3.5 PROTECTION

Prior to the installation of the finish flooring, the surface of the underlayment should be protected from abuse by other trades by the use of plywood, tempered hardwood, or other suitable protection course

3.6 FIELD QUALITY CONTROL

Where specified, field sampling of products shall be conducted by a qualified, independent testing facility.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 22 16

NON-STRUCTURAL METAL FRAMING

6/18

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 TERMINOLOGY
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY, IDENTIFICATION, HANDLING AND STORAGE
1.7 REFERENCES

PART 2 PRODUCTS

2.1 PROTECTIVE COATING
2.2 STEEL STUDS AND RUNNERS (TRACK)
2.3 FURRING CHANNELS
2.4 FASTENERS, CLIPS, AND OTHER METAL ACCESSORIES
2.5 SUSPENDED CEILING SYSTEM FOR GYPSUM BOARD (OPTION)

PART 3 EXECUTION

3.1 INSTALLATION CRITERIA
3.2 INSTALLING STUDS
3.3 INSTALLING WALL FURRING FOR FINISH APPLIED TO ONE SIDE ONLY
3.4 INSTALLING SUPPORTS REQUIRED BY OTHER TRADES
3.5 INSTALLING SHAFT WALL SYSTEM
3.6 INSTALLING FURRED AND SUSPENDED CEILINGS OR SOFFITS
3.7 TOLERANCES

-- End of Section Table of Contents --

SECTION 09 22 16
6/18

NON-STRUCTURAL METAL FRAMING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies steel studs wall systems, shaft wall systems, ceiling or soffit suspended or furred framing, wall furring, fasteners, and accessories for the screw attachment of gypsum board, plaster bases or other building boards.

1.2 RELATED WORK

a. Load bearing framing: Section 05 40 00, COLD-FORMED METAL FRAMING.

b. Support for wall mounted items: Section 05 50 00, METAL FABRICATIONS.

c. Pull down tabs in steel decking: Section 05 36 00, COMPOSITE METAL DECKING.

d. Ceiling suspension systems for acoustical tile or panels and lay in gypsum board panels: Section 09 51 00, ACOUSTICAL CEILINGS Section 09 29 00, GYPSUM BOARD.

1.3 TERMINOLOGY

a. Description of terms shall be in accordance with ASTM C754, ASTM C11, ASTM C841 and as specified.

b. Underside of Structure Overhead: In spaces where steel trusses or bar joists are shown, the underside of structure overhead shall be the underside of the floor or roof construction supported by beams, trusses, or bar joists. In interstitial spaces with walk-on floors the underside of the walk-on floor is the underside of structure overhead.

c. Thickness of steel specified is the minimum bare (uncoated) steel thickness.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.xx SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE
Recycled Content; S

SD-06 Test Reports

Test Results; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Manufacturer's Literature and Data:

(1) Studs, runners and accessories.

(2) Hanger inserts.

(3) Channels (Rolled steel).

(4) Furring channels.

(5) Screws, clips and other fasteners.

(6) Recycled content.

(7) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

c. Shop Drawings:

(1) Typical ceiling suspension system.

(2) Typical metal stud and furring construction system including details around openings and corner details.

(3) Typical shaft wall assembly

(4) Typical fire rated assembly and column fireproofing showing details of construction same as that used in fire rating test.

d. Test Results: Fire rating test designation, each fire rating required for each assembly.

1.6 DELIVERY, IDENTIFICATION, HANDLING AND STORAGE

In accordance with the requirements of ASTM C754.

1.7 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A641/A641M (2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire

ASTM C11	(2023) Terminology Relating to Gypsum and Related Building Materials and Systems
ASTM C635/C635M	(2022) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings
ASTM C636/C636M	(2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels
ASTM C645	(2024) Standard Specification for Nonstructural Steel Framing Members
ASTM C754	(2020) Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products
ASTM C841	(2023) Installation of Interior Lathing and Furring
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness
ASTM E580/E580M	(2024a) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions

UL SOLUTIONS (UL)

UL 263	(2011; Reprint Aug 2021) UL Standard for Safety Fire Tests of Building Construction and Materials

PART 2 PRODUCTS

2.1 PROTECTIVE COATING

Galvanize steel studs, runners (track), rigid (hat section) furring channels, "Z" shaped furring channels, and resilient furring channels, with coating designation of G40 or equivalent.

2.2 STEEL STUDS AND RUNNERS (TRACK)

a. ASTM C645, modified for thickness specified and sizes as shown.

(1) Use C 645 steel, (0.0296-inch) minimum base-metal (30 mil).

(2) Runners same thickness as studs.

(3) Exception: Members that can show certified third party testing with gypsum board in accordance with ICC ES AC86 (Approved May 2012) need not meet the minimum thickness limitation or minimum

section properties set forth in ASTM C645. The submission of an evaluation report is acceptable to show conformance to this requirement. Use C 645 steel, 0.019 inch minimum base-metal (19 mil).

b. Provide not less than two cutouts in web of each stud, approximately 12 inches from each end, and intermediate cutouts on approximately (24-inch) centers.

c. Doubled studs for openings and studs for supporting concrete backer-board.

d. Studs 12 feet or less in length shall be in one piece.

e. Shaft Wall Framing:

(1) Conform to rated wall construction.

(2) C-H Studs or C-T Studs.

(3) E Studs.

(4) J Runners.

(5) Steel Jamb-Strut.

2.3 FURRING CHANNELS

a. Rigid furring channels (hat shape): ASTM C645.

b. Resilient furring channels:

(1) Not less than (0.0179-inch) thick bare metal.

(2) Semi-hat shape, only one flange for anchorage with channel web leg slotted on anchorage side, channel web leg on other side stiffens fastener surface but shall not contact anchorage surface other channel leg is attached to.

c. "Z" Furring Channels:

(1) Not less than (0.0179-inch)-thick base metal, with (1-1/4 inch) and (3/4-inch) flanges.

(2) Web furring depth to suit thickness of insulation.

d. Rolled Steel Channels: ASTM C754, cold rolled; or, ASTM C841, cold rolled.

2.4 FASTENERS, CLIPS, AND OTHER METAL ACCESSORIES

a. ASTM C754, except as otherwise specified.

b. For fire rated construction: Type and size same as used in fire rating test.

c. Fasteners for steel studs thicker than (0.033-inch) thick. Use ASTM C954 steel drill screws of size and type recommended by the manufacturer of the material being fastened.

d. Clips: ASTM C841 (paragraph 6.11), manufacturer’s standard items. Clips used in lieu of tie wire shall have holding power equivalent to that provided by the tie wire for the specific application.

e. Concrete ceiling hanger inserts (anchorage for hanger wire and hanger straps): Steel, zinc-coated (galvanized), manufacturers standard items, designed to support twice the hanger loads imposed and the type of hanger used.

f. Tie Wire and Hanger Wire:

(1) ASTM A641/A641M, soft temper, Class 1 coating.

(2) Gage (diameter) as specified in ASTM C754 or ASTM C841.

g. Attachments for Wall Furring:

(1) Manufacturers standard items fabricated from zinc-coated (galvanized) steel sheet.

(2) For concrete or masonry walls: Metal slots with adjustable inserts or adjustable wall furring brackets. Spacers may be fabricated from (0.0396-inch) thick galvanized steel with corrugated edges.

h. Power Actuated Fasteners: Type and size as recommended by the manufacturer of the material being fastened.

2.5 SUSPENDED CEILING SYSTEM FOR GYPSUM BOARD (OPTION)

a. Conform to ASTM C635/C635M, heavy duty, with not less than (1-3/8 inch) wide knurled capped flange face designed for screw attachment of gypsum board.

b. Wall track channel with (1-3/8 inch) wide flange.

PART 3 EXECUTION

3.1 INSTALLATION CRITERIA

a. Where fire rated construction is required for walls, partitions, columns, beams and floor-ceiling assemblies, the construction shall be same as that used in fire rating test.

b. Construction requirements for fire rated assemblies and materials shall be as shown and specified, the provisions of the Scope paragraph (1.2) of ASTM C754 and ASTM C841 regarding details of construction shall not apply.

3.2 INSTALLING STUDS

a. Install studs in accordance with ASTM C754, except as otherwise shown or specified.

b. Space studs not more than 24 inches on center.

c. Cut studs (1/4 to 3/8-inch) less than floor to underside of structure overhead when extended to underside of structure overhead.

d. Where studs are shown to terminate above suspended ceilings, provide bracing as shown or extend studs to underside of structure overhead.

e. Extend studs to underside of structure overhead for fire, rated partitions, smoke partitions, shafts, and sound rated partitions and insulated exterior wall furring.

g. Openings:

(1) Frame jambs of openings in stud partitions and furring with two studs placed back to back or as shown.

(2) Fasten back to back studs together with (3/8-inch) long Type S pan head screws at not less than (two feet) on center, staggered along webs.

(3) Studs fastened flange to flange shall have splice plates on both sides approximately (2 by 3 inches) screwed to each stud with two screws in each stud. Locate splice plates at 24 inches on center between runner tracks.

h. Fastening Studs:

(1) Fasten studs located adjacent to partition intersections, corners and studs at jambs of openings to flange of runner tracks with two screws through each end of each stud and flange of runner.

(2) Do not fasten studs to top runner track when studs extend to underside of structure overhead.

i. Chase Wall Partitions:

(1) Locate cross braces for chase wall partitions to permit the installation of pipes, conduits, carriers and similar items.

(2) Use studs or runners as cross bracing not less than (2-1/2 inches) wide.

j. Form building seismic or expansion joints with double studs back to back spaced (three inches) apart plus the width of the seismic or expansion joint.

k. Form control joint, with double studs spaced (1/2-inch) apart.

3.3 INSTALLING WALL FURRING FOR FINISH APPLIED TO ONE SIDE ONLY

a. In accordance with ASTM C754, or ASTM C841 except as otherwise specified or shown.

b. Wall furring-Stud System:

(1) Framed with (2-1/2 inch) or narrower studs, 24 inches on center.

(2) Brace as specified in ASTM C754 for Wall Furring-Stud System or brace with sections or runners or studs placed horizontally at not less than three foot vertical intervals on side without finish.

(3) Securely fasten braces to each stud with two Type S pan head screws at each bearing.

c. Direct attachment to masonry or concrete; rigid channels or "Z" channels:

(1) Install rigid (hat section) furring channels at 24 inches on center, horizontally or vertically.

(2) Install "Z" furring channels vertically spaced not more than 24 inches on center.

(3) At corners where rigid furring channels are positioned horizontally, provide mitered joints in furring channels.

(4) Ends of spliced furring channels shall be nested not less than (8 inches).

(5) Fasten furring channels to walls with power-actuated drive pins or hardened steel concrete nails. Where channels are spliced, provide two fasteners in each flange.

(6) Locate furring channels at interior and exterior corners in accordance with wall finish material manufacturers printed erection instructions. Locate "Z" channels within 4 inches of corner.

d. Installing Wall Furring-Bracket System: Space furring channels not more than 16 inches on center.

3.4 INSTALLING SUPPORTS REQUIRED BY OTHER TRADES

a. Provide for attachment and support of electrical outlets, plumbing, laboratory or heating fixtures, recessed type plumbing fixture accessories, access panel frames, wall bumpers, wood seats, toilet stall partitions, dressing booth partitions, urinal screens, chalkboards, tackboards, wall-hung casework, handrail brackets, recessed fire extinguisher cabinets and other items like auto door buttons and auto door operators supported by stud construction.

b. Provide additional studs where required. Install metal backing plates, or special metal shapes as required, securely fastened to metal studs.

3.5 INSTALLING SHAFT WALL SYSTEM

a. Conform to UL 263 for two-hour fire rating. Provide one hour fire rating Shaft wall as indicated on the drawings.

b. Position J runners at floor and ceiling with the short leg toward finish side of wall. Securely attach runners to structural supports with power driven fasteners at both ends and 24 inches on center.

c. After liner panels have been erected, cut C-H studs and E studs, from (3/8-inch) to not more than (1/2-inch) less than floor-to-ceiling height. Install C-H studs between liner panels with liner panels inserted in the groove.

d. Install full-length steel E studs over shaft wall line at

intersections, corners, hinged door jambs, columns, and both sides of closure panels.

e. Suitably frame all openings to maintain structural support for wall:

(1) Provide necessary liner fillers and shims to conform to label frame requirements.

(2) Frame openings cut within a liner panel with E studs around perimeter.

(3) Frame openings with vertical E studs at jambs, horizontal J runner at head and sill.

f. Elevator Shafts:

(1) Frame elevator door frames with (0.0341-inch) thick J strut or J stud jambs having (three-inch) long legs on the shaft side.

(2) Protrusions including fasteners other than flange of shaft wall framing system or offsets from vertical alignments more than (1/8-inch) are not permitted unless shown.

(3) Align shaft walls for plumb vertical flush alignment from top to bottom of shaft.

3.6 INSTALLING FURRED AND SUSPENDED CEILINGS OR SOFFITS

a. Install furred and suspended ceilings or soffits in accordance with ASTM C754 or ASTM C841 except as otherwise specified or shown for screw attached gypsum board ceilings and for plaster ceilings or soffits.

(1) Space framing at (16-inch) centers for metal lath anchorage.

(2) Space framing at (24-inch) centers for gypsum board anchorage.

b. New exposed concrete slabs:

(1) Use metal inserts required for attachment and support of hangers or hanger wires with tied wire loops for embedding in concrete.

(2) Furnish for installation under Division 3, CONCRETE.

(3) Suspended ceilings under concrete rib construction shall have runner channels at right angles to ribs and be supported from ribs with hangers at ends and at (48-inch) maximum intervals along channels. Stagger hangers at alternate channels.

c. Concrete slabs on steel decking composite construction:

(1) Use pull down tabs when available.

(2) Use power activated fasteners when direct attachment to structural framing can not be accomplished.

d. Where bar joists or beams are more than 48 inches apart, provide intermediate hangers so that spacing between supports does not exceed (48 inches). Use clips, bolts, or wire ties for direct attachment to

steel framing.

e. Existing concrete construction exposed or concrete on steel decking:

(1) Use power actuated fasteners either eye pin, threaded studs or drive pins for type of hanger attachment required.

(2) Install fasteners at approximate mid height of concrete beams or joists. Do not install in bottom of beams or joists.

f. Steel decking without concrete topping:

(1) Do not fasten to steel decking (0.0299-inch) or thinner.

(2) Toggle bolt to decking (0.0359-inch) or thicker only where anchorage to steel framing is not possible.

g. Installing suspended ceiling system for gypsum board (ASTM C635/C635M Option):

(1) Install only for ceilings to receive screw attached gypsum board.

(2) Install in accordance with ASTM C636/C636M.

(a) Install main runners spaced 48 inches on center.

(b) Install (four foot) tees not over 24 inches on center; locate for edge support of gypsum board.

(c) Install wall track channel at perimeter.

h. Installing Ceiling Bracing System:

(1) Construct bracing of (1-1/2 inch) channels for lengths up to 8 feet and 2 inch channels for lengths over 8 feet with ends bent to form surfaces for anchorage to carrying channels and over head construction. Lap channels not less than 2 feet at midpoint back to back. Screw or bolt lap together with two fasteners.

(2) Install bracing at an approximate 45 degree angle to carrying channels and structure overhead; secure as specified to structure overhead with two fasteners and to carrying channels with two fasteners or wire ties.

(3) Brace suspended ceiling or soffit framing in seismic areas in accordance with ASTM E580/E580M.

3.7 TOLERANCES

a. Fastening surface for application of subsequent materials shall not vary more than (1/8-inch) from the layout line.

b. Plumb and align vertical members within (1/8-inch).

c. Level or align ceilings within (1/8-inch).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 29 00

GYPSUM BOARD

4/20

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 TERMINOLOGY
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY, IDENTIFICATION, HANDLING AND STORAGE
1.7 ENVIRONMENTAL CONDITIONS
1.8 REFERENCES

PART 2 PRODUCTS

2.1 GYPSUM BOARD
2.2 GYPSUM SHEATHING BOARD
2.3 ACCESSORIES
2.4 FASTENERS
2.5 FINISHING MATERIALS AND LAMINATING ADHESIVE

PART 3 EXECUTION

3.1 INSTALLING GYPSUM BOARD
3.2 INSTALLING GYPSUM SHEATHING
3.3 CAVITY SHAFT WALL
3.4 FINISHING OF GYPSUM BOARD
3.5 REPAIRS
3.6 UNACCESSIBLE CEILINGS

-- End of Section Table of Contents --

SECTION 09 29 00
4/20

GYPSUM BOARD

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies installation and finishing of gypsum board.

1.2 RELATED WORK

a. Installation of steel framing members for walls, partitions, furring, soffits, and ceilings: Section 05 40 00, COLD-FORMED METAL FRAMING, and Section 09 22 16, NON-STRUCTURAL METAL FRAMING.

b. Sound deadening board: Section 07 21 13, THERMAL INSULATION.

c. Acoustical Sealants: Section 07 92 00, JOINT SEALANTS.

d. Gypsum base for veneer plaster: Section 09 26 00, VENEER PLASTERING.

e. Lead lined wallboard: Section 13 49 00, RADIATION PROTECTION.

f. Lay in gypsum board ceiling panels: Section 09 51 00, ACOUSTICAL CEILING.

1.3 TERMINOLOGY

a. Definitions and description of terms shall be in accordance with ASTM C11, ASTM C840, and as specified.

b. Underside of Structure Overhead: In spaces where steel trusses or bar joists are shown, the underside of structure overhead shall be the underside of the floor or roof construction supported by the trusses or bar joists.

c. "Yoked": Gypsum board cut out for opening with no joint at the opening (along door jamb or above the door).

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.xx SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-04 Samples

Samples; G, AE

SD-06 Test Reports

Test Results; G, AE

SD-07 Certificates

Certificates; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Manufacturer's Literature and Data:

(1) Cornerbead and edge trim.

(2) Finishing materials.

(3) Laminating adhesive.

(4) Gypsum board, each type.

(5) Sound barrier mullion trim cap system.

(6) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

c. Shop Drawings:

(1) Typical gypsum board installation, showing corner details, edge trim details and the like.

(2) Typical sound rated assembly, showing treatment at perimeter of partitions and penetrations at gypsum board.

(3) Typical shaft wall assembly.

(4) Typical fire rated assembly and column fireproofing, indicating details of construction same as that used in fire rating test.

(5) Sound barrier mullion trim cap system.

d. Samples:

(1) Cornerbead.

(2) Edge trim.

(3) Control joints.

(4) Sound barrier mullion trim cap system.

e. Test Results:

(1) Fire rating test, each fire rating required for each assembly.

(2) Sound rating test.

f. Certificates: Certify that gypsum board types, gypsum backing board types, cementitious backer units, and joint treating materials do not contain asbestos material.

1.6 DELIVERY, IDENTIFICATION, HANDLING AND STORAGE

In accordance with the requirements of ASTM C840.

1.7 ENVIRONMENTAL CONDITIONS

In accordance with the requirements of ASTM C840.

1.8 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C11	(2023) Terminology Relating to Gypsum and Related Building Materials and Systems
ASTM C475/C475M	(2017; R 2022) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board
ASTM C840	(2024) Standard Specification for Application and Finishing of Gypsum Board
ASTM C919	(2024) Standard Practice for Use of Sealants in Acoustical Applications
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness
ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM C1047	(2019) Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base
ASTM C1177/C1177M	(2024) Standard Specification for Glass Mat Gypsum Substrate for Use as Sheathing
ASTM C1178/C1178M	(2024) Standard Specification for Glass Mat Water-Resistant Gypsum Backing Panel

ASTM C1658/C1658M	(2019; E 2020) Standard Specification for Glass Mat Gypsum Panels
ASTM C1396/C1396M	(2024) Standard Specification for Gypsum Board

UL SOLUTIONS (UL)

UL Design U438	(1982) 2 Hour Wall Assembly

PART 2 PRODUCTS

2.1 GYPSUM BOARD

a. Gypsum Board: ASTM C1396/C1396M, Type X, 5/8 inch thick unless shown otherwise.

b. Coreboard or Shaft Wall Liner Panels.

(1) ASTM C1396/C1396M, Type X.

(2) ASTM C1658/C1658M: Glass Mat Gypsum Panels,

(3) Coreboard for shaft walls (12, 16, or 24 inches) wide by required lengths (one inch) thick with paper faces treated to resist moisture.

c. Water Resistant Gypsum Backing Board: ASTM C1178/C1178M, Type X, 5/8 inch thick.

d. Paper facings shall contain 100 percent post-consumer recycled paper content.

2.2 GYPSUM SHEATHING BOARD

a. ASTM C1396/C1396M, Type X, water-resistant core, 5/8 inch thick.

b. ASTM C1177/C1177M, Type X.

2.3 ACCESSORIES

a. ASTM C1047, except form of 0.015 inch thick zinc coated steel sheet or rigid PVC plastic.

b. Flanges not less than 7/8 inch wide with punchouts or deformations as required to provide compound bond.

2.4 FASTENERS

a. ASTM C1002 and ASTM C840, except as otherwise specified.

b. ASTM C954, for steel studs thicker than (0.33 inch).

c. Select screws of size and type recommended by the manufacturer of the material being fastened.

d. For fire rated construction, type and size same as used in fire rating test.

e. Clips: Zinc-coated (galvanized) steel; gypsum board manufacturer's standard items.

2.5 FINISHING MATERIALS AND LAMINATING ADHESIVE

ASTM C475/C475M and ASTM C840. Free of antifreeze, vinyl adhesives, preservatives, biocides and other VOC. Adhesive shall contain a maximum VOC content of 50 g/l.

PART 3 EXECUTION

a. Extend all layers of gypsum board from floor to underside of structure overhead on following partitions and furring:

(1) Two sides of partitions:

(a) Fire rated partitions.

(b) Smoke partitions.

(c) Sound rated partitions.

(d) Full height partitions shown (FHP).

(e) Corridor partitions.

(2) One side of partitions or furring:

(a) Inside of exterior wall furring or stud construction.

(b) Room side of room without suspended ceilings.

(c) Furring for pipes and duct shafts, except where fire rated shaft wall construction is shown.

(3) Extend all layers of gypsum board construction used for fireproofing of columns from floor to underside of structure overhead, unless shown otherwise.

b. In locations other than those specified, extend gypsum board from floor to heights as follows:

(1) Not less than 4 inches above suspended acoustical ceilings.

(2) At ceiling of suspended gypsum board ceilings.

(3) At existing ceilings.

3.1 INSTALLING GYPSUM BOARD

a. Coordinate installation of gypsum board with other trades and related work.

b. Install gypsum board in accordance with ASTM C840, except as otherwise specified.

c. Moisture and Mold–Resistant Assemblies: Provide and install moisture and mold-resistant glass mat gypsum wallboard products with

moisture-resistant surfaces complying with ASTM C1658/C1658M where shown and in locations which might be subject to moisture exposure during construction.

d. Use gypsum boards in maximum practical lengths to minimize number of end joints.

e. Bring gypsum board into contact, but do not force into place.

f. Ceilings:

(1) For single-ply construction, use perpendicular application.

(2) For two-ply assembles:

(a) Use perpendicular application.

(b) Apply face ply of gypsum board so that joints of face ply do not occur at joints of base ply with joints over framing members.

g. Walls (Except Shaft Walls):

(1) When gypsum board is installed parallel to framing members, space fasteners 12 inches on center in field of the board, and 8 inches on center along edges.

(2) When gypsum board is installed perpendicular to framing members, space fasteners 12 inches on center in field and along edges.

(3) Stagger screws on abutting edges or ends.

(4) For single-ply construction, apply gypsum board with long dimension either parallel or perpendicular to framing members as required to minimize number of joints except gypsum board shall be applied vertically over "Z" furring channels.

(5) For two-ply gypsum board assemblies, apply base ply of gypsum board to assure minimum number of joints in face layer. Apply face ply of wallboard to base ply so that joints of face ply do not occur at joints of base ply with joints over framing members.

(6) For three-ply gypsum board assemblies, apply plies in same manner as for two-ply assemblies, except that heads of fasteners need only be driven flush with surface for first and second plies. Apply third ply of wallboard in same manner as second ply of two-ply assembly, except use fasteners of sufficient length enough to have the same penetration into framing members as required for two-ply assemblies.

(7) No offset in exposed face of walls and partitions will be permitted because of single-ply and two-ply or three-ply application requirements.

(8) Installing Two Layer Assembly Over Sound Deadening Board:

(a) Apply face layer of wallboard vertically with joints staggered from joints in sound deadening board over framing members.

(b) Fasten face layer with screw, of sufficient length to secure

to framing, spaced 12 inches on center around perimeter, and 16 inches on center in the field.

(9) Control Joints ASTM C840 and as follows:

(a) Locate at both side jambs of openings if gypsum board is not "yoked". Use one system throughout.

(b) Not required for wall lengths less than (30 feet).

(c) Extend control joints the full height of the wall or length of soffit/ceiling membrane.

h. Acoustical or Sound Rated Partitions, Fire and Smoke Partitions:

(1) Cut gypsum board for a space approximately (1/8 to 1/4 inch) wide around partition perimeter.

(2) Coordinate for application of caulking or sealants to space prior to taping and finishing.

(3) For sound rated partitions, use sealing compound (ASTM C919) to fill the annular spaces between all receptacle boxes and the partition finish material through which the boxes protrude to seal all holes and/or openings on the back and sides of the boxes. STC minimum values as shown.

i. Electrical and Telecommunications Boxes:

(1) Seal annular spaces between electrical and telecommunications receptacle boxes and gypsum board partitions.

j. Accessories:

(1) Set accessories plumb, level and true to line, neatly mitered at corners and intersections, and securely attach to supporting surfaces as specified.

(2) Install in one piece, without the limits of the longest commercially available lengths.

(3) Corner Beads:

(a) Install at all vertical and horizontal external corners and where shown.

(b) Use screws only. Do not use crimping tool.

(4) Edge Trim (casings Beads):

(a) At both sides of expansion and control joints unless shown otherwise.

(b) Where gypsum board terminates against dissimilar materials and at perimeter of openings, except where covered by flanges, casings or permanently built-in equipment.

(c) Where gypsum board surfaces of non-load bearing assemblies abut load bearing members.

(d) Where shown.

(5) Sound Barrier Mullion Trim Cap System:

(a) Install mullion trim cap per manufacturer's written instructions.

(b) Finish: Organic Coatings System: Manufacturer's standard multi-coat superior performing organic coatings system complying with AAMA 2605, including at least 70 percent polyvinylidene fluoride (PVDF) resin, and at least 80 percent of aluminum extrusion and panels surfaces having minimum total dry film thickness (DFT) of 1.2 mils, 0.0012 inch (0.030 mm).

3.2 INSTALLING GYPSUM SHEATHING

a. Install in accordance with ASTM C840, except as otherwise specified or shown.

b. Use screws of sufficient length to secure sheathing to framing.

c. Space screws 3/8 inch from ends and edges of sheathing and 8 inches on center. Space screws a maximum of 8 inches on center on intermediate framing members.

d. Apply 2 foot by 8 foot sheathing boards horizontally with tongue edge up.

e. Apply (4 ft. by 8 ft. or 9 foot) gypsum sheathing boards vertically with edges over framing.

3.3 CAVITY SHAFT WALL

a. Coordinate assembly with Section 09 22 16, NON-STRUCTURAL METAL FRAMING, for erection of framing and gypsum board.

b. Conform to UL Design U438 or FM WALL CONSTRUCTION 12-2/HR (Nonbearing for two-hour fire rating.

c. Cut coreboard (liner) panels (one inch) less than floor-to-ceiling height, and erect vertically between J-runners on shaft side.

(1) Where shaft walls exceed 14 feet in height, position panel end joints within upper and lower third points of wall.

(2) Stagger joints top and bottom in adjacent panels.

(3) After erection of J-struts of opening frames, fasten panels to J-struts with screws of sufficient length to secure to framing staggered from those in base, spaced 12 inches on center.

d. Gypsum Board:

(1) Two hour wall:

(a) Erect base layer (backing board) vertically on finish side of wall with end joints staggered. Fasten base layer panels to studs with (one inch) long screws, spaced 24 inches on center.

(b) Use laminating adhesive between plies in accordance with UL or FM if required by fire test.

(c) Apply face layer of gypsum board required by fire test vertically over base layer with joints staggered and attach with screws of sufficient length to secure to framing staggered from those in base, spaced 12 inches on center.

(2) One hour wall with one layer on finish side of wall: Apply face layer of gypsum board vertically. Attach to studs with screws of sufficient length to secure to framing, spaced 12 inches on center in field and along edges.

(3) Where coreboard is covered with face layer of gypsum board, stagger joints of face layer from those in the coreboard base.

e. Treat joints, corners, and fasteners in face layer as specified for finishing of gypsum board.

f. Elevator Shafts:

(1) Protrusions including fasteners other than flange of shaft wall framing system or offsets from vertical alignments more than (1/8-inch) are not permitted unless shown.

(2) Align shaft walls for plumb vertical flush alignment from top to bottom of shaft.

3.4 FINISHING OF GYPSUM BOARD

a. Finish joints, edges, corners, and fastener heads in accordance with ASTM C840. Use Level 4 finish for al finished areas open to public view.

b. Before proceeding with installation of finishing materials, assure the following:

(1) Gypsum board is fastened and held close to framing or furring.

(2) Fastening heads in gypsum board are slightly below surface in dimple formed by driving tool.

c. Finish joints, fasteners, and all openings, including openings around penetrations, on that part of the gypsum board extending above suspended ceilings to seal surface of non decorated fire rated and sound rated gypsum board construction. After the installation of hanger rods, hanger wires, supports, equipment, conduits, piping and similar work, seal remaining openings and maintain the integrity of the construction. Sanding is not required of non decorated surfaces.

3.5 REPAIRS

a. After taping and finishing has been completed, and before decoration, repair all damaged and defective work, including nondecorated surfaces.

b. Patch holes or openings 1/2 inch or less in diameter, or equivalent size, with a setting type finishing compound or patching plaster.

c. Repair holes or openings over 1/2 inch diameter, or equivalent size, with 5/8 inch thick gypsum board secured in such a manner as to provide solid substrate equivalent to undamaged surface.

d. Tape and refinish scratched, abraded or damaged finish surfaces including cracks and joints in non decorated surface to provide fire protection equivalent to the fire rated construction and STC equivalent to the sound rated construction.

3.6 UNACCESSIBLE CEILINGS

At Mental Health and Behavioral Nursing Units, areas accessible to patients and not continuously observable by staff (e.g., patient bedrooms, day rooms), ceilings should be a solid material such as gypsum board. This will limit patient access. Access doors are needed to access electrical and mechanical equipment above the ceiling. These doors should be locked to prevent unauthorized access and secured to ceiling using tamper resistant fasteners.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 30 13

CERAMIC/PORCELAIN TILING

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY AND STORAGE
1.6 QUALITY ASSURANCE
1.7 WARRANTY
1.8 REFERENCES

PART 2 PRODUCTS

2.1 TILE
2.2 BACKER UNITS
2.3 JOINT MATERIALS FOR CEMENTITIOUS BACKER UNITS
2.4 FASTENERS
2.5 SETTING MATERIALS OR BOND COATS
2.6 GROUTING MATERIALS
2.7 PATCHING AND LEVELING COMPOUND
2.8 METAL DIVIDER STRIPS
2.9 WATER
2.10 CLEANING COMPOUNDS
2.11 FLOOR MORTAR BED REINFORCING
2.12 POLYETHYLENE SHEET

PART 3 EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS
3.2 ALLOWABLE TOLERANCE
3.3 SURFACE PREPARATION
3.4 CEMENTITIOUS BACKER UNITS
3.5 GLASS MAT WATER-RESISTANT BACKING BOARD
3.6 MARBLE
3.7 METAL DIVIDER STRIPS
3.8 CERAMIC TILE – GENERAL
3.9 CERAMIC TILE INSTALLED WITH PORTLAND CEMENT MORTAR
3.10 PORCELAIN TILE INSTALLED WITH LATEX PORTLAND CEMENT BONDING MORTAR
3.11 THIN SET CERAMIC AND PORCELAIN TILE INSTALLED WITH DRY-SET PORTLAND CEMENT AND LATEX-PORTLAND CEMENT MORTAR
3.12 THIN SET CERAMIC AND PORCELAIN TILE INSTALLED WITH ORGANIC ADHESIVE
3.13 THIN SET CERAMIC AND PORCELAIN TILE INSTALLED WITH CHEMICAL-RESISTANT BOND COAT
3.14 CERAMIC AND PORCELAIN TILE INSTALLED WITH ELASTOMERIC BOND COAT
3.15 GROUTING

3.16 MOVEMENT JOINTS
3.17 CLEANING
3.18 PROTECTION
3.19 TESTING FINISH FLOOR

-- End of Section Table of Contents --

SECTION 09 30 13
1/21

CERAMIC/PORCELAIN TILING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies interior ceramic, porcelain and quarry tile, marble thresholds and window stools, terrazzo divider strips, waterproofing membranes for thin-set applications, crack isolation membranes, and tile backer board.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

b. Section 07 92 00, JOINT SEALANTS: Sealing of Joints.

c. Section 07 95 13, EXPANSION JOINT COVER ASSEMBLIES: Preformed expansion joints in tile flooring.

d. SCHEDULE FOR FINISHES (on the drawings): Color, Texture, Pattern, and Size of Field Tile and Trim Shapes, and Color of Grout Specified.

e. Section 09 65 19, RESILIENT TILE FLOORING: Metal and Resilient Edge Strips at Joints with New Resilient Flooring.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G

SD-03 Product Data

Sustainable Design Submittals; S

Product Data; G

SD-04 Samples

Samples; G

SD-07 Certificates

Certification; G

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Sustainable Design Submittals as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

(2) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

c. Samples:

(1) Porcelain tile, each type, color, patterns and size.

(2) Wall tile, each color, size and pattern.

d. Product Data:

(1) Ceramic and porcelain tile, marked to show each type, size, and shape required.

(2) Chemical resistant mortar and grout (epoxy and furan).

(3) Cementitious backer unit.

(4) Dry-set portland cement mortar and grout.

(5) Divider strip.

(6) Elastomeric membrane and bond coat.

(7) Reinforcing tape.

(8) Leveling compound.

(9) Latex-portland cement mortar and grout.

(10) Commercial portland cement grout.

(11) Organic adhesive.

(12) Waterproofing isolation membrane.

(13) Fasteners.

e. Certification:

(1) Master grade certificate, ANSI A137.1.

(2) Manufacturer's certificates indicating that the following materials comply with specification requirements:

(a) Chemical resistant mortar and grout (epoxy and furan).

(b) Modified epoxy emulsion.

(c) Commercial portland cement grout.

(d) Cementitious backer unit.

(e) Dry-set portland cement mortar and grout.

(f) Elastomeric membrane and bond coat.

(g) Reinforcing tape.

(h) Latex-portland cement mortar and grout.

(i) Leveling compound.

(j) Organic adhesive.

(k) Waterproof isolation membrane.

(l) Factory back mounted tile documentation for suitability for application in wet area.

f. Installer Qualifications:

(1) Submit letter stating installer’s experience.

1.5 DELIVERY AND STORAGE

a. Deliver materials in containers with labels legible and intact and grade-seals unbroken.

b. Store material to prevent damage or contamination.

1.6 QUALITY ASSURANCE

a. Installers to be from a company specializing in performing installation of products specified and have a minimum of three (3) years’ experience.

b. Each type and color of tile to be provided from a single source.

c. Each type and color of mortar, adhesive, and grout to be provided from the same source.

1.7 WARRANTY

Comply with FAR clause 52.246-21, “Warranty of Construction”.

1.8 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in text by basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A10.20	(R2016) Safe Operating Practices for Tile, Terrazzo and Marble Work
ANSI A108.01	(2021a) Subsurfaces and Preparations by

Other Trades

ANSI A108.02	(2019) Materials, Environmental, and Workmanship
ANSI A108/A118/A136.1	(2024) American National Standard Specifications for the Installation of Ceramic Tile
ANSI A108.1A	(2017) Installation of Ceramic Tile in the Wet-Set Method with Portland Cement Mortar
ANSI A108.1B	(2017) Installation of Ceramic Tile on a Cured Portland Cement Mortar Setting Bed with Dry-Set or Latex-Portland Cement Mortar
ANSI A108.1C	(2021) Contractors Option; Installation of Ceramic Tile in the Wet-Set method with Portland Cement Mortar or Installation of Ceramic Tile on a Cured Portland Cement Mortar Setting Bed with Dry-Set or Latex-Portland Cement Mortar
ANSI A108.4	(2019) Ceramic Tile with Organic Adhesives or Water Cleanable Tile-Setting Epoxy Adhesive
ANSI A108.6	(2019) Ceramic Tile with Chemical Resistant, Water Cleanable Tile-Setting and Grout Epoxy
ANSI A108.10	(2017) Grout in Tilework
ANSI A108.13	(2021) Load Bearing, Bonded, Waterproof Membranes for Thin-Set Ceramic Tile and Dimension Stone
ANSI A118.1	(2019) Dry-Set Portland Cement Mortar
ANSI A118.3	(2021) Chemical Resistant, Water Cleanable Tile-Setting and –Grouting Epoxy and Water Cleanable Tile-Setting Epoxy Adhesive
ANSI A118.4	(2019) Modified Dry-Set Cement Mortar
ANSI A118.5	(2021) Chemical Resistant Furan Mortars and Grouts for Tile Installation
ANSI A118.6	(2019) American National Standard Specifications for Standard Cement Grouts for Tile Installation
ANSI A118.7	(2010) American National Standard Specifications for High Performance Cement Grouts for Tile Installation
ANSI A118.9	(2019) American National Standard Specifications for Test Methods and

	Specifications for Cementitious Backer Units
ANSI A118.10	(2019) Load Bearing, Bonded, Waterproof Membranes for Thin-Set Ceramic Tile and Dimension Stone Installation
ANSI A137.1	(2019) American National Standards Specifications for Ceramic Tile

ASTM INTERNATIONAL (ASTM)

ASTM A666	(2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM A1064/A1064M	(2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C109/C109M	(2024) Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or (50-mm) Cube Specimens)
ASTM C348	(2021) Standard Test Method for Flexural Strength of Hydraulic-Cement Mortars
ASTM C627	(2018) Evaluating Ceramic Floor Tile Installation Systems Using the Robinson-Type Floor Tester
ASTM C954	(2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness
ASTM C979/C979M	(2024) Standard Specification for Pigments for Integrally Colored Concrete
ASTM C1002	(2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs
ASTM C1027	(2009; R 2017) Standard Test Method for Determining Visible Abrasion Resistance of Glazed Ceramic Tile
ASTM C1127/C1127M	(2016) Standard Guide for Use of High Solids Content, Cold Liquid-Applied Elastomeric Waterproofing Membrane with an Integral Wearing Surface
ASTM C1178/C1178M	(2024) Standard Specification for Glass Mat Water-Resistant Gypsum Backing Panel

ASTM C1325	(2022; E 2023) Standard Specification for Fiber-Mat Reinforced Cementitious Backer Units
ASTM D1204	(2014; R 2020) Linear Dimensional Changes of Nonrigid Thermoplastic Sheeting or Film at Elevated Temperature
ASTM D2240	(2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness
ASTM D2497	(2018) Tolerances for Manufactured Organic-Base Filament Single Yarns
ASTM D3045	(1992; R 2010) Practice for Heat Aging of Plastics Without Load
ASTM D4397	(2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications
ASTM D5109	(2017) Standard Test Methods for Copper-Clad Thermosetting Laminates for Printed Wiring Boards (recommend deletion)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

TILE COUNCIL OF NORTH AMERICA (TCNA)

TCNA Hdbk	(2017) Handbook for Ceramic, Glass, and Stone Tile Installation
TCNA F122-14	Interior Floor Thin-Bed Method Over On-Ground Concrete With Waterproof Membrane
TCNA F122A-14	Above-Ground Concrete, Waterproof Membrane, Ceramic Tile

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

PART 2 PRODUCTS

2.1 TILE

a. Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples

of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

b. Comply with ANSI A137.1, Standard Grade, except as modified:

(1) Inspection procedures listed under the Appendix of ANSI A137.1.

(2) Abrasion Resistance Classification:

(a) Tested in accordance with values listed in Table 1, ASTM C1027.

(b) Class V, 12000 revolutions for floors in Corridors, Kitchens, Storage including Refrigerated Rooms

(c) Class IV, 6000 revolutions for remaining areas.

(3) Slip Resistant Tile for Floors:

(a) Coefficient of friction, when tested in accordance with ANSI A137.1 and measured per the TCNA Hdbk.

1. Equal to or greater than.42 for level interior tile floors that will be walked on when wet.

(4) Factory Blending: For tile with color variations, within the ranges selected during sample submittals blend tile in the factory and package so tile units taken from one (1) package show the same range in colors as those taken from other packages and match approved samples.

c. Porcelain Paver Tile: Nominal 5/16 inch thick, with cushion edges. Porcelain tile produced by the dust pressed method are to be made of approximately 50 percent feldspar; the remaining 50 percent is to be made up of various high-quality light firing ball clays yielding a tile with a water absorption rate of 0.5 percent or less and a breaking strength of between (390 to 400 pounds).

2.2 BACKER UNITS

a. Cementitious Backer Units:

(1) Use in showers or wet areas.

(2) Conform to ASTM C1325; Type A.

(3) Use in maximum lengths available to minimize end to end butt joints.

b. Glass Mat Water Resistant Backing Board:

(1) Use in showers or wet areas.

(2) Conform to ASTM C1178/C1178M.

(3) Use in maximum lengths available to minimize end to end butt joints.

2.3 JOINT MATERIALS FOR CEMENTITIOUS BACKER UNITS

a. Reinforcing Tape: Vinyl coated woven glass fiber mesh tape, open weave, 2 inches wide. Tape with pressure sensitive adhesive backing will not be permitted.

b. Tape Embedding Material: Latex-portland cement mortar complying with ANSI A108.01.

c. Joint material, including reinforcing tape, and tape embedding material, are to be as specifically recommended by the backer unit manufacturer.

2.4 FASTENERS

a. Screws for Cementitious Backer Units.

(1) Standard screws for gypsum board are not acceptable.

(2) Minimum 7/16 inch diameter head, corrosion resistant coated, with washers.

(3) ASTM C954 for steel 0.033 inch thick.

(4) ASTM C1002 for steel framing less than 0.0329 inch thick.

b. Washers: Galvanized steel, 1/2 inch minimum diameter.

2.5 SETTING MATERIALS OR BOND COATS

a. Conform to TCNA Hdbk for Ceramic Tile Installation.

b. Portland Cement Mortar: ANSI A108.02.

c. Latex-Portland Cement Mortar: ANSI A118.4.

(1) For wall applications, provide non-sagging, latex-portland cement mortar complying with ANSI A118.4.

(2) Prepackaged Dry-Mortar Mix: Factory-prepared mixture of portland cement; dry, redispersible, ethylene vinyl acetate additive; and other ingredients to which only water needs to be added at Project site.

d. Dry-Set Portland Cement Mortar: ANSI A118.1. For wall applications, provide non-sagging, latex-portland cement mortar complying with ANSI A118.1.

e. Organic Adhesives: ANSI A108/A118/A136.1, Type 1.

f. Chemical-Resistant Bond Coat:

(1) Epoxy Resin Type: ANSI A118.3.

(2) Furan Resin Type: ANSI A118.5.

g. Elastomeric Waterproofing Membrane and Bond Coat:

(1) TCNA F122-14 (on ground concrete) and TCNA F112A-14 (above ground

concrete).

(2) ANSI A118.10.

(3) One component polyurethane, liquid applied material having the following additional physical properties:

(a) Hardness: Shore "A" between 40-60.

(b) Elongation: Between 300-600 percent.

(c) Tensile strength: Between (40-60 pounds per square inch gauge).

(d) No volatile compounds (VOC).

(4) Coal tar modified urethanes are not acceptable.

h. Waterproofing Isolation Membrane:

(1) Sheet System TCNA F122-14 (on-ground concrete) and TCNA F122A-14 (above-ground concrete).

(2) Composite sheet consisting of ASTM D5109, Type II, Grade I Chlorinated Polyethylene (CM) sheet reinforced on both sides with a non-woven polyester fiber.

(3) Designed for use in wet areas as an isolation and positive waterproofing membranes for thin-set bonding of sheet to substrate and thin-set bonding of ceramic and porcelain tile or marble to sheet. Suited for both horizontal and vertical applications.

(4) Conform to the following additional physical properties:

Property	Units	Results	Test Method
Hardness Shore A	Points	70-80	ASTM D2240(10 Second Reading)
Shrinkage	Percent	5 maximum	ASTM D1204
Brittleness		No crack remains flexible at temperature -37 degrees C (-35 degrees F)	ASTM D2497 (1/2-inch) Mandrel Bend
Retention of Properties after Heat Aging	Percent of original	80 Tensile 80 Breaking 80 Elongation	ASTM D3045 , 90 degrees C (194 degrees F) for 168 hours

(5) Manufacturer’s standard sheet size with prefabricated or preformed inside and outside corners.

(6) Sheet manufacturer’s solvent welding liquid or xylene and edge sealant.

2.6 GROUTING MATERIALS

a. Coloring Pigments:

(1) Pure mineral pigments, lime proof and nonfading, complying with ASTM C979/C979M.

(2) Coloring pigments may only be added to grout by the manufacturer.

(3) Job colored grout is not acceptable.

(4) Use is required in Commercial Portland Cement Grout, Dry-Set Grout, and Latex-Portland Cement Grout.

b. Sand-Portland Cement Grout: ANSI A108.10, consisting of white or gray cement and white or colored aggregate as required to produce color indicated. Zero VOC content.

c. Standard Cement Grout: ANSI A118.6.

d. High Performance Tile Grout: ANSI A118.7 with a VOC content of 65 g/L or less when calculated according to 40 CFR 59 (EPA Method 24) .

(1) Polymer Type: Ethylene vinyl acetate or acrylic additive, in dry, redispersible form, prepackaged with other dry ingredients.

(2) Polymer Type: Acrylic resin or styrene-butadiene rubber in liquid-latex form for addition to prepackaged dry-grout mix.

e. Water-Cleanable Epoxy Grout: ANSI A118.3, with a VOC content of 65 g/L or less when calculated according to 40 CFR 59 (EPA Method 24).

(1) Provide product capable of withstanding continuous and intermittent exposure to temperatures of up to (140 and 212 degrees F), respectively, and certified by manufacturer for intended use.

2.7 PATCHING AND LEVELING COMPOUND

a. Portland cement base, polymer-modified, self-leveling compound, manufactured specifically for resurfacing and leveling concrete floors. Products containing gypsum are not acceptable.

b. Provide a patching and leveling compound with the following minimum physical properties:

(1) Compressive strength - (3500 psig) per ASTM C109/C109M.

(2) Flexural strength - (1000 psig) per ASTM C348 (28 day value).

(3) Tensile strength – (600 psi) per ANSI A118.7.

(4) Density – 1.9.

c. Capable of being applied in layers up to (1-1/2 inches) thick without fillers and up to 4 inches thick with fillers, being brought to a feather edge, and being trowelled to a smooth finish.

d. Primers, fillers, and reinforcement as required by manufacturer for application and substrate condition.

e. Ready for use in 48 hours after application.

2.8 METAL DIVIDER STRIPS

a. Terrazzo type divider strips.

b. Heavy top type strip with 3/16 inch wide top and (1 1/2 inch) long leg. Height to match tile and setting-bed thickness.

c. Embedded leg perforated and deformed for keying to mortar.

d. stainless-steel, ASTM A666, 300 Series exposed-edge material.

2.9 WATER

Clean, potable and free from salts and other injurious elements to mortar and grout materials.

2.10 CLEANING COMPOUNDS

a. Specifically designed for cleaning masonry and concrete and which will not prevent bond of subsequent tile setting materials including patching and leveling compounds and elastomeric waterproofing membrane and coat.

b. Materials containing acid or caustic Material are not acceptable.

2.11 FLOOR MORTAR BED REINFORCING

ASTM A1064/A1064M welded wire fabric without backing, MW3 x MW3 (2 x 2-W0.5 x W0.5).

2.12 POLYETHYLENE SHEET

a. Polyethylene sheet conforming to ASTM D4397.

b. Nominal thickness: (6 mils).

PART 3 EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS

a. Maintain ambient temperature of work areas at not less than (60 degrees F), without interruption, for not less than 24 hours before installation and not less than three (3) days after installation.

b. Maintain higher temperatures for a longer period of time where required by manufacturer's recommendation and ANSI Specifications for installation.

c. Do not install tile when the temperature is above (100 degrees F).

d. Do not install materials when the temperature of the substrate is below (60 degrees F).

e. Do not allow temperature to fall below (50 degrees F) after third day of completion of tile work.

3.2 ALLOWABLE TOLERANCE

a. Variation in plane of sub-floor, including concrete fills leveling compounds and mortar beds:

(1) Not more than 1/4 inch in 10 feet from required elevation where

portland cement mortar setting bed is used.

(2) Not more than 1/8 inch in 10 feet where dry-set portland cement, and latex-portland cement mortar setting beds and chemical-resistant bond coats are used.

b. Variation in Plane of Wall Surfaces:

(1) Not more than 1/4 inch in 8 feet from required plane where portland cement mortar setting bed is used.

(2) Not more than 1/8 inch in 8 feet where dry-set or latex-portland cement mortar or organic adhesive setting materials is used.

3.3 SURFACE PREPARATION

a. Cleaning New Concrete or Masonry:

(1) Chip out loose material, clean off all oil, grease dirt, adhesives, curing compounds, and other deterrents to bonding by mechanical method, or by using products specifically designed for cleaning concrete and masonry.

(2) Use self-contained power blast cleaning systems to remove curing compounds and steel trowel finish from concrete slabs where ceramic tile will be installed directly on concrete surface with thin-set materials.

(3) Steam cleaning or the use of acids and solvents for cleaning will not be permitted.

b. Patching and Leveling:

(1) Mix and apply patching and leveling compound in accordance with manufacturer's instructions.

(2) Fill holes and cracks and align concrete floors that are out of required plane with patching and leveling compound.

(a) Thickness of compound as required to bring finish tile system to elevation shown on construction documents.

(b) Float finish except finish smooth for elastomeric waterproofing.

(c) At substrate expansion, isolation, and other moving joints, allow joint of same width to continue through underlayment.

(3) Apply patching and leveling compound to concrete and masonry wall surfaces that are out of required plane.

(4) Apply leveling coats of material compatible with wall surface and tile setting material to wall surfaces, other than concrete and masonry that are out of required plane.

c. Mortar Bed for Slopes to Drains:

(1) Slope compound to drain where drains are shown on construction documents.

(2) Install mortar bed in depressed slab sloped to drains not less than (1/8 inch per foot).

(3) Allow not less than 2 inch depression at edge of depressed slab.

(4) Screed for slope to drain and float finish.

(5) Cure mortar bed for not less than seven (7) days. Do not use curing compounds or coatings.

(6) Perform flood test to verify mortar bed slopes to drain before installing tile. Contracting Officer to be present during flood test.

d. Additional preparation of concrete floors for tile set with epoxy, or furan-resin is to be in accordance with the manufacturer's printed instructions.

e. Cleavage Membrane:

(1) Install polythene sheet as cleavage membrane in depressed slab when waterproof membrane is not scheduled or indicated.

(2) Turn up at edge of depressed floor slab to top of floor.

f. Walls:

(1) In showers or other wet areas cover studs with polyethylene sheet.

(2) Apply patching and leveling compound to concrete and masonry surfaces that are out of required plane.

(3) Apply leveling coats of material compatible with wall surface and tile setting material to wall surfaces, other than concrete and masonry that are out of required plane.

(4) Apply metal lath to framing in accordance with ANSI A108/A118/A136.1:

(a) Use fasteners specified in paragraph "Fasteners." Use washers when lath opening is larger than screw head.

(b) Apply scratch and leveling coats to metal lath in accordance with ANSI A108.1C.

(c) Total thickness of scratch and leveling coats:

1. Apply 3/8 inch to 5/8 inch thick over solid backing.

2. (5/8 to 3/4 inch) thick on metal lath over studs.

3. Where wainscots are required to finish flush with wall surface above, adjust thickness required for flush finish.

(d) Apply scratch and leveling coats more than 3/4 inch thick in two (2) coats.

g. Existing Floors and Walls:

(1) Remove existing composition floor finishes and adhesive. Prepare surface by grinding, chipping, self-contained power blast cleaning or other suitable mechanical methods to completely expose uncontaminated concrete or masonry surfaces. Follow safety requirements of ANSI A10.20.

(2) Remove existing concrete fill or topping to structural slab. Clean and level the substrate for new setting bed and waterproof membrane or cleavage membrane.

(3) Where new tile bases are required to finish flush with plaster above or where they are extensions of similar bases in conjunction with existing floor tiles, cut channel in floor slab and expose rough wall construction sufficiently to accommodate new tile base and setting material.

3.4 CEMENTITIOUS BACKER UNITS

a. Remove polyethylene wrapping from cementitious backer units and separate to allow for air circulation. Allow moisture content of backer units to dry down to a maximum of 35 percent before applying joint treatment and tile.

b. Install in accordance with ANSI A118.9 except as specified otherwise.

c. Install units horizontally or vertically to minimize joints with end joints over framing members. Units with rounded edges; face rounded edge away from studs to form a “V” joint for joint treatment.

d. Secure cementitious backer units to each framing member with screws spaced not more than 8 inches on center and not closer than 1/2 inch from the edge of the backer unit or as recommended by backer unit manufacturer. Install screws so that the screw heads are flush with the surface of the backer unit.

e. Where backer unit joins shower pans or waterproofing, lap backer unit over turned up waterproof system. Install fasteners only through top one-inch of turned up waterproof systems.

f. Do not install joint treatment for seven (7) days after installation of cementitious backer unit.

g. Joint Treatment:

(1) Fill horizontal and vertical joints and corners with latex-portland cement mortar. Apply fiberglass tape over joints and corners and embed with same mortar.

(2) Leave 1/4 inch space for sealant at lips of tubs, sinks, or other plumbing receptors.

3.5 GLASS MAT WATER-RESISTANT BACKING BOARD

a. Install in accordance with manufacturer’s instructions. TCNA Systems W245-1.

b. Treat joints with tape and latex-portland cement mortar or adhesive.

3.6 MARBLE

a. Secure thresholds and stools in position with minimum of two stainless steel dowels.

b. Set in dry-set portland cement mortar or latex-portland cement mortar bond coat.

c. Set threshold to finish 1/2 inch above ceramic tile floor unless shown otherwise on construction documents, with bevel edge joint top flush with adjacent floor similar to TCNA detail TR611-14.

3.7 METAL DIVIDER STRIPS

a. Install metal divider strips in floor joints between ceramic and quarry tile floors and between tile floors and adjacent flooring of other materials where the finish floors are flush unless shown otherwise on construction documents.

b. Set divider strip in mortar bed to line and level centered under doors or in openings.

c. At preformed sealant joint: Refer to Section 07 95 13, EXPANSION JOINT COVER ASSEMBLIES.

(1) Comply with recommendations in TCNA for Vertical and Horizontal Joint Design Essentials. TCNA Systems EJ 171.

(a) Locate joint in tile surfaces directly above joint in sub-floor or where indicated when used with isolation membranes to allow off-setting of joint location from sub-floor joint.

(b) Fasten full length to sub-floor using a construction adhesive.

(c) Trowel setting material with full coverage over the entire leg.

(2) Set tile up against the joint ensuring that the top edge of the joint is flush or slightly below the top of the tile.

3.8 CERAMIC TILE – GENERAL

a. Comply with ANSI A108/A118/A136.1 series of tile installation standards applicable to methods of installation and TCNA Installation Guidelines.

b. Installing Mortar Beds for Floors:

(1) Install mortar bed in a manner that does not damage cleavage or waterproof membrane; (1-1/2 inch) minimum thickness.

(2) Install floor mortar bed reinforcing centered in mortar fill.

(3) Screed finish to level plane or slope to drains shown on construction documents, float finish.

(4) For thin set systems cure mortar bed not less than seven (7) days. Do not use curing compounds or coatings.

(5) For tile set with portland cement paste over plastic mortar bed

coordinate to set tile before mortar bed sets.

c. Setting Beds or Bond Coats:

(1) Where recessed or depressed floor slabs are filled with portland cement mortar bed, set ceramic mosaic floor tile in either portland cement paste over plastic mortar bed or latex-portland cement mortar over cured mortar bed except as specified otherwise, ANSI A108.1C, TCNA System F121-14 or F111-14.

(2) Set floor tile in elastomeric bond coat over elastomeric membrane per ANSI A108.13, TCNA System F122-14 where indicated on construction documents.

(3) Set wall tile installed over concrete or masonry in dry-set portland cement mortar, or latex-portland cement mortar, ANSI A108.1B and TCNA System W211-14, W221-14 or W222-14.

(4) Set wall tile installed over concrete backer board in latex-portland cement mortar, ANSI A108.1B.

(5) Set wall tile installed over portland cement mortar bed on metal lath base in portland cement paste over plastic mortar bed, or dry-set portland cement mortar or latex-portland cement mortar over a cured mortar bed, ANSI A108.1C, TCNA System W231-14, W241-14.

(6) Set tile over concrete in therapeutic pools in portland cement paste or dry set portland cement mortar, ANSI A108.1C, TCNA System P601MB-14.

(7) Set tile installed over gypsum board and gypsum plaster in organic adhesive, ANSI A108.1A, TCNA System W242-14.

(8) Set trim shapes in same material specified for setting adjoining tile.

d. Workmanship:

(1) Lay out tile work so that no tile less than one-half full size is used. Make all cuts on the outer edge of the field.

(2) Set tile firmly in place with finish surfaces in true planes. Align tile flush with adjacent tile unless shown otherwise on construction documents.

(3) Form intersections and returns accurately.

(4) Cut and drill tile neatly without marring surface.

(5) Cut edges of tile abutting penetrations, finish, or built-in items:

(a) Fit tile closely around electrical outlets, piping, fixtures and fittings, so that plates, escutcheons, collars and flanges will overlap cut edge of tile.

(b) Seal tile joints water tight as specified in Section 07 92 00, JOINT SEALANTS, around electrical outlets, piping fixtures and fittings before cover plates and escutcheons are set in place.

(6) Completed work is to be free from hollow sounding areas and loose, cracked or defective tile.

(7) Remove and reset tiles that are out of plane or misaligned.

(8) Floors:

(a) Extend floor tile beneath casework and equipment, except those units mounted in wall recesses.

(b) Align finish surface of new tile work flush with other and existing adjoining floor finish where indicated in construction documents.

(c) In areas where floor drains occur, slope tile to drains.

(d) Push and vibrate tiles over 8 inches square to achieve full support of bond coat.

(9) Walls:

(a) Cover walls and partitions, including pilasters, furred areas, and freestanding columns from floor to ceiling, or from floor to nominal wainscot heights as indicated in construction documents with tile.

(b) Finish reveals of openings with tile, except where other finish materials are indicated in construction documents.

(c)

(d) Finish wall surfaces behind and at sides of casework and equipment, except those units mounted in wall recesses, with same tile as scheduled for room proper.

(10) Joints:

(a) Keep all joints in line, straight, level, perpendicular and of even width unless shown otherwise on construction documents.

(b) Make joints 1/16 inch wide for glazed wall tile and mosaic tile work.

(c) Make joints in quarry tile work not less than 1/4 inch nor more than 3/8 inch wide. Finish joints flush with surface of tile.

(d) Make joints in paver tile, porcelain type; maximum (1/8 inch) wide.

(11) Back Buttering: For installations indicated below, obtain 100 percent mortar coverage by complying with applicable special requirements for back buttering of tile in referenced ANSI A108/A118/A136.1 series of tile installation standards:

(a) Tile wall installations in wet areas, including showers, tub enclosures, laundries and swimming pools.

(b) Tile installed with chemical-resistant mortars and grouts.

(c) Tile wall installations composed of tiles (8 by 8 inches) or larger.

(d) Exterior tile wall installations.

3.9 CERAMIC TILE INSTALLED WITH PORTLAND CEMENT MORTAR

a. Mortar Mixes for Floor, Wall and Base Tile (including Showers): ANSI A108.1A. except specified otherwise.

b. Installing Wall and Base Tile: ANSI A108.1A, except specified otherwise.

c. Installing Floor Tile: ANSI A108.1A, except as specified otherwise. Slope mortar beds to floor drains at a minimum of (1/8 inch per foot).

3.10 PORCELAIN TILE INSTALLED WITH LATEX PORTLAND CEMENT BONDING MORTAR

Due to the denseness of porcelain tile use latex portland cement bonding mortar that meets the requirements of ANSI A108.01. Mix bonding mortars in accordance with manufacturer’s instructions. Provide liquid ratios and comply with dwell times during the placement of bonding mortar and tile.

3.11 THIN SET CERAMIC AND PORCELAIN TILE INSTALLED WITH DRY-SET PORTLAND CEMENT AND LATEX-PORTLAND CEMENT MORTAR

a. Installation of Tile: ANSI A108.1B, except as specified otherwise.

b. Slope tile work to drains at not less than (1/8 inch per foot).

3.12 THIN SET CERAMIC AND PORCELAIN TILE INSTALLED WITH ORGANIC ADHESIVE

ANSI A108.4.

3.13 THIN SET CERAMIC AND PORCELAIN TILE INSTALLED WITH CHEMICAL-RESISTANT BOND COAT

a. Epoxy Resin Type: Install tile in accordance with Installation of Tile with Epoxy Mortar; ANSI A108.6.

b. Furan Resin Type: Proportion, mix and place in accordance with the manufacturer's printed instructions. Set tile in accordance with ANSI A108.8.

3.14 CERAMIC AND PORCELAIN TILE INSTALLED WITH ELASTOMERIC BOND COAT

a. Surface Preparation: Prepare surfaces as specified.

b. Installation of Elastomeric Membrane: ANSI A108.10 and TCNA F122-14 (on ground concrete) and TCNA F122A-14 (above-ground concrete).

(1) Prime surfaces, where required, in accordance with manufacturer's instructions.

(2) Install first coat of membrane material in accordance with manufacturer's instructions, in thickness of (30 to 50 mils).

(3) Extend material over flashing rings of drains and turn up vertical

surfaces not less than 4 inches above finish floor surface.

(4) When material has set, recoat areas with a second coat of elastomeric membrane material for a total thickness of (50 to 75 mils).

(5) After curing test for leaks with 1 inch of water for 24 hours.

c. Installation of Tile in Elastomeric Membrane:

(1) Spread no more material than can be covered with tile before material starts to set.

(2) Apply tile in second coat of elastomeric membrane material in accordance with the coating manufacturer's instructions in lieu at aggregate surfacing specified in ASTM C1127/C1127M. Do not install top coat over tile.

3.15 GROUTING

a. Grout Type and Location:

(1) Grout for glazed wall and base tile, paver tile and unglazed mosaic tile portland cement grout, latex-portland cement grout, dry-set grout, or commercial portland cement grout.

(2) Grout for quarry tile floor and base:

(a) Grout for floors of walk-in refrigerated rooms: Epoxy grout.

(b) Therapeutic pool areas: Portland cement grout.

(c) Grout for Kitchens:

1. Chemical-resistant grout as specified and recommended by manufacturer of bond coat.

2. Use only furan resin grout within 2 feet of ovens, steam kettles, water heaters, steam pipes, and in rooms.

3. Epoxy grout designed for equivalent heat resistance to furan resin grout may be used for furan resin grout.

(3) Grout for tile of therapeutic pools: Portland cement grout.

b. Workmanship:

(1) Install and cure grout in accordance with the applicable standard.

(2) Sand Portland Cement Grout: ANSI A108.10.

(3) Standard Cement Grout: ANSI A118.6.

(4) High Performance Grout: ANSI A118.7.

(5) Epoxy Grout: ANSI A108.6.

(6) Water-Cleanable Epoxy Grout: ANSI A118.3.

(7) Furan and Commercial Portland Cement Grout: ANSI A118.5 and in accordance with the manufacturer's printed instructions.

3.16 MOVEMENT JOINTS

a. Prepare tile expansion, isolation, construction and contraction joints for installation of sealant. Refer to Section 07 92 00, JOINT SEALANTS.

b. TCNA details EJ 171-14.

c. At expansion joints, rake out joint full depth of tile and setting bed and mortar bed. Do not cut waterproof or isolation membrane.

d. Rake out grout at joints between tile, and where indicated in construction documents not less than 1/4 inch deep.

3.17 CLEANING

a. Thoroughly sponge and wash tile. Polish glazed surfaces with clean dry cloths.

b. Methods and materials used are not permitted to damage or impair appearance of tile surfaces.

c. The use of acid or acid cleaners on glazed tile surfaces is prohibited.

d. Clean tile grouted with epoxy, furan and commercial portland cement grout and tile set in elastomeric bond coat as recommended by the manufacturer of the grout and bond coat.

3.18 PROTECTION

a. Keep traffic off tile floor, until grout and setting material is fully set and cured.

b. Where traffic occurs over tile floor is unavoidable, cover tile floor with not less than 3/8 inch thick plywood, wood particle board, or hardboard securely taped in place. Do not remove protective cover until time for final inspection. Clean tile of any tape, adhesive and stains.

3.19 TESTING FINISH FLOOR

a. Test floors in accordance with ASTM C627 to show compliance with codes 1 through 10.

b.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 51 00

ACOUSTICAL CEILINGS

12/18

PART 1 GENERAL

1.1 SUMMARY
1.2 RELATED REQUIREMENTS
1.3 REFERENCES
1.4 PREINSTALLATION MEETINGS
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 QUALITY ASSURANCE
1.8 DELIVERY
1.9 STORAGE AND HANDLING
1.10 FIELD CONDITIONS
1.11 WARRANTY

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION
2.2 SYSTEM PERFORMANCE
2.3 PRODUCTS - GENERAL
2.4 ACOUSTICAL UNITS
2.5 METAL SUSPENSION SYSTEM
2.6 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION - GENERAL
3.3 ACOUSTICAL UNIT INSTALLATION
3.4 CEILING SUSPENSION SYSTEM INSTALLATION
3.5 CEILING TREATMENT
3.6 CLEANING

-- End of Section Table of Contents --

SECTION 09 51 00
12/18

ACOUSTICAL CEILINGS

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Acoustical units.

(2) Metal ceiling suspension system for acoustical ceilings.

(3) Adhesive application.

1.2 RELATED REQUIREMENTS

a. Adhesive VOC Limits: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Color, pattern, and location of each type of acoustical unit: SCHEDULE FOR FINISHES (on the drawings).

c. Ceiling Suspension System: Section 09 22 16, NON-STRUCTURAL METAL FRAMING.

1.3 REFERENCES

Comply with references to extent specified in this section.

ASTM INTERNATIONAL (ASTM)

ASTM A641/A641M	(2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM C423	(2023; E 2024) Sound Absorption and Sound Absorption Coefficients by the Reverberation Room Method
ASTM C635/C635M	(2022) Standard Specification for Manufacture, Performance, and Testing of Metal Suspension Systems for Acoustical Tile and Lay-In Panel Ceilings
ASTM C636/C636M	(2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E119	(2024) Standard Test Methods for Fire Tests of Building Construction and Materials

ASTM E413	(2022) Classification for Rating Sound Insulation
ASTM E580/E580M	(2024a) Standard Practice for Installation of Ceiling Suspension Systems for Acoustical Tile and Lay-in Panels in Areas Subject to Earthquake Ground Motions
ASTM E1264	(2023) Standard Classification for Acoustical Ceiling Products

1.4 PREINSTALLATION MEETINGS

Conduct preinstallation meeting at project site minimum 30 days before beginning Work of this section.

(1) Required Participants:

(a) Contracting Officer's Representative.

(b) Architect/Engineer and Interior Designer.

(c) VA Interior Designer.

(d) Inspection and Testing Agency.

(e) Contractor.

(f) Installer.

(g) Manufacturer's field representative.

(h) Other installers responsible for adjacent and intersecting work, including sprinkler HVAC and lighting installers.

(2) Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(a) Installation schedule.

(b) Installation sequence.

(c) Preparatory work.

(d) Protection before, during, and after installation.

(e) Installation.

(f) Terminations.

(g) Transitions and connections to other work.

(h) Inspecting and testing.

(i) Other items affecting successful completion.

(3) Document and distribute meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Sustainable Construction Submittals; S

SD-04 Samples

Samples; G, AE

SD-07 Certificates

Certificates; G

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G

1.6 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submittal Drawings:

(1) Show size, configuration, and fabrication and installation details.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Ceiling suspension system indicating manufacturer recommendation for each application.

(3) Installation instructions.

(4) Warranty.

d. Samples:

(1) Acoustical units, (6 inches) in size, each type, including units

specified to match existing.

(a) Submit quantity required to show full color and texture range.

(2) Suspension system, trim and molding, (12 inches) long.

(3) Colored markers for access service.

(4) Approved samples may be incorporated into work.

e. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

(2) Biobased Content:

(a) Show type and quantity for each product.

(b) Show volatile organic compound types and quantities.

(3) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

f. Certificates: Certify each product complies with specifications.

(1) Acoustical units, each type.

g. Qualifications: Substantiate qualifications comply with specifications.

(1) Manufacturer with project experience list.

h. Operation and Maintenance Data:

(1) Care instructions for each exposed finish product.

1.7 QUALITY ASSURANCE

Manufacturer Qualifications:

(1) Regularly manufactures specified products.

(2) Manufactured specified products with satisfactory service on five similar installations for minimum five years.

(a) Project Experience List: Provide contact names and addresses for completed projects.

1.8 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.9 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight conditioned facility.

b. Protect products from damage during handling and construction operations.

1.10 FIELD CONDITIONS

Environment:

(1) Product Temperature: Minimum (70 degrees F) for minimum 48 hours before installation.

(2) Work Area Ambient Conditions: HVAC systems are complete, operational, and maintaining facility design operating conditions continuously, beginning 48 hours before installation until Government occupancy.

(3) Install products when building is permanently enclosed and when wet construction is completed, dried, and cured.

1.11 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Acoustical ceilings units on exposed grid suspension systems.

2.2 SYSTEM PERFORMANCE

a. Design product complying with specified performance:

(1) Maximum Deflection: 1/360 of span, maximum.

b. Surface Burning Characteristics: When tested according to ASTM E84.

(1) Flame Spread Rating: 25 maximum.

(2) Smoke Developed Rating: 450 maximum.

2.3 PRODUCTS - GENERAL

a. Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

b. Basis of Design: SCHEDULE FOR FINISHES on the drawings.

c. Provide acoustical units from one manufacturer.

(1) Provide each product exposed to view from one production run.

d. Provide suspension system from same manufacturer.

e. Sustainable Construction Requirements:

(1) Mineral Base Recycled Content: 65 percent, post-consumer recycled content, minimum. Select products with recycled content to achieve overall Project recycled content requirement.

(2) Steel Recycled Content: 30 percent total recycled content, minimum.

(3) Aluminum Recycled Content: 50 percent total recycled content, minimum.

(4) Biobased Content: 37 percent by weight biobased material, minimum.

(5) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Non-flooring adhesives and sealants.

2.4 ACOUSTICAL UNITS

a. General:

(1) Ceiling Panel and Tile: ASTM E1264, bio-based content according to USDA Bio-Preferred Product requirements.

(a) Mineral Fiber: (3/4 psf) weight, minimum.

(2) Classification: Provide type and form as follows:

(a) Type IV Units - Mineral base with membrane-faced overlay, Form 2 - Water felted, minimum (5/8 inch) thick. Apply poly (vinyl) chloride over paint coat.

(b) NRC (Noise Reduction Coefficient): ASTM C423, minimum 0.55 unless specified otherwise.

(c) CAC (Ceiling Attenuation Class): ASTM E413, 40-44 range unless specified otherwise.

(d) LR (Light Reflectance): Minimum 0.75.

(3) Lay-in panels: Sizes as indicated on Drawings, with square edges.

2.5 METAL SUSPENSION SYSTEM

a. General: ASTM C635/C635M, heavy-duty system, except as otherwise specified.

(1) Suspension System: Provide the following:

(a) Galvanized cold-rolled steel, bonderized.

(2) Main and Cross Runner: Use same construction Do not use lighter-duty sections for cross runners.

b. Exposed Grid Suspension System: Support of lay-in panels.

(1) Grid Width: (7/8 inch) minimum with (5/16 inch) minimum panel bearing surface.

(2) Molding: Fabricate from the same material with same exposed width and finish.

(3) Finish: Baked-on enamel flat texture finish.

(a) Color: To match adjacent acoustical units unless specified otherwise in SCHEDULE FOR FINISHES on the drawings.

c. Suspension System Support of Metal Type V, VI, and VII Tiles: Concealed grid type with runners for snap-in attachment of metal tile (pans).

d. Carrying Channels Secondary Framing: Cold-rolled or hot-rolled steel, black asphaltic paint finish, rust free.

(1) Weight per (per thousand linear feet), minimum:

Size		Cold-rolled		Hot-rolled	
	inches		pound		pound
	1-1/2		475		1120
	2		590		1260

f. Anchors and Inserts: Provide anchors or inserts to support twice the loads imposed by hangers.

(1) Hanger Inserts: Steel, zinc-coated (galvanized after fabrication).

(a) Nailing type option for wood forms:

1. Upper portion designed for anchorage in concrete and positioning lower portion below surface of concrete approximately (one inch).

2. Lower portion provided with minimum (5/16 inch) hole to permit attachment of hangers.

(b) Flush ceiling insert type:

1. Designed to provide a shell covered opening over a wire loop to permit attachment of hangers and keep concrete out of insert recess.

2. Insert opening inside shell approximately (5/8 inch) wide by (3/8 inch) high over top of wire.

3. Wire (3/16 inch) diameter with length to provide positive hooked anchorage in concrete.

g. Clips: Galvanized steel, designed to secure framing member in place.

h. Tile Splines: ASTM C635/C635M.

i. Wire: ASTM A641/A641M.

(1) Size:

(a) Wire Hangers: Minimum diameter (0.1055 inch).

(b) Bracing Wires: Minimum diameter (0.1350 inch).

2.6 ACCESSORIES

a. Adhesives: Low pollutant-emitting, water based type recommended by adhered product manufacturer for each application.

b. Perimeter Seal: Vinyl, polyethylene or polyurethane open cell sponge material, density of 1.3 plus or minus 10 percent, compression set less than 10 percent with pressure sensitive adhesive coating on one side.

(1) Thickness: As required to fill voids between back of wall molding and finish wall.

(2) Size: Minimum (3/8 inch) wide strip.

c. Access Identification Markers: Colored markers with pressure sensitive adhesive on one side, paper or plastic, (1/4 to 3/8 inch) diameter.

(1) Color Code: Provide the following color markers for service identification:

Color	Service
Red	Sprinkler System: Valves and Controls
Green	Domestic Water: Valves and Controls
Yellow	Chilled Water and Heating Water
Orange	Ductwork: Fire Dampers
Blue	Ductwork: Dampers and Controls
Black	Gas: Laboratory, Medical, Air and Vacuum

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

3.2 INSTALLATION - GENERAL

Install products according to manufacturer's instructions and approved submittal drawings.

(1) When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's Representative consideration.

3.3 ACOUSTICAL UNIT INSTALLATION

a. Applications:

(1) Cut acoustic units for perimeter borders and penetrations to fit tight against penetration for joint not concealed by molding.

b. Layout acoustical unit symmetrically, with minimum number of joints.

c. Installation:

(1) Install acoustic tiles after wet finishes have been installed and solvents have cured.

(2) Install lay-in acoustic panels in exposed grid with minimum (1/4 inch) bearing at edges on supports.

(a) Install tile to lay level and in full contact with exposed grid.

(b) Replace cracked, broken, stained, dirty, or tile.

(3) Markers:

(a) Install color coded markers to identify the various concealed piping, mechanical, and plumbing systems.

(b) Attach colored markers to exposed grid on opposite sides of the units providing access.

(c) Attach marker on exposed ceiling surface of upward access acoustical unit.

d. Touch up damaged factory finishes.

(1) Repair painted surfaces with touch up primer.

3.4 CEILING SUSPENSION SYSTEM INSTALLATION

a. General: Install according to ASTM C636/C636M.

(1) Use direct or indirect hung suspension system or combination of both.

(2) Support a maximum area of (16 sq. ft.) of ceiling per hanger.

(3) Prevent deflection in excess of 1/360 of span of cross runner and main runner.

(4) Provide additional hangers located at each corner of support components.

(5) Provide minimum (4 inch) clearance from the exposed face of the acoustical units to the underside of ducts, pipe, conduit, secondary suspension channels, concrete beams or joists; and steel beam or bar joist unless furred system is shown.

(6) Provide main runners minimum (48 inches) in length.

(7) Install hanger wires vertically. Angled wires are not acceptable except for seismic restraint bracing wires.

b. Direct Hung Suspension System: ASTM C635/C635M.

(1) Support main runners by hanger wires attached directly to the structure overhead.

(2) Maximum spacing of hangers, (4 feet) on centers unless interference occurs by mechanical systems. Use indirect hung suspension system where not possible to maintain hanger spacing.

c. Anchorage to Structure:

(1) Concrete:

(a) Install hanger inserts and wire loops required for support of hanger and bracing wire. Install hanger wires with looped ends through steel deck when steel deck does not have attachment device.

(b) Use eye pins or threaded studs with screw-on eyes in existing or already placed concrete structures to support hanger and bracing wire. Install in sides of concrete beams or joists at mid height.

(2) Steel:

(a) Install carrying channels for attachment of hanger wires.

1. Size and space carrying channels to support load within performance limit.

2. Attach hangers to steel carrying channels, spaced four feet on center, unless area supported or deflection exceeds the amount specified.

(b) Attach carrying channels to the bottom flange of steel beams spaced not (4 feet) on center before fireproofing is installed. Weld or use steel clips for beam attachment.

(c) Attach hangers to bottom chord of bar joists or to carrying channels installed between the bar joists when hanger spacing prevents anchorage to joist. Rest carrying channels on top of the bottom chord of the bar joists, and securely wire tie or clip to joist.

d. Indirect Hung Suspension System: ASTM C635/C635M.

(1) Space carrying channels for indirect hung suspension system maximum (4 feet) on center. Space hangers for carrying channels maximum (8 feet) on center or for carrying channels less than (4 feet) or center so as to insure that specified requirements are not exceeded.

(2) Support main runners by specially designed clips attached to carrying channels.

e. Seismic Ceiling Bracing System:

(1) Install according to ASTM E580/E580M.

(2) Connect bracing wires to structure above as specified for anchorage to structure and to main runner or carrying channels of suspended ceiling at bottom.

3.5 CEILING TREATMENT

a. Moldings:

(1) Install metal wall molding at perimeter of room, column, or edge at vertical surfaces.

(2) Install special shaped molding at changes in ceiling heights and at other breaks in ceiling construction to support acoustical units and to conceal their edges.

b. Perimeter Seal:

(1) Install perimeter seal between vertical leg of wall molding and finish wall, partition, and other vertical surfaces.

(2) Install perimeter seal to finish flush with exposed faces of horizontal legs of wall molding.

c. Existing ceiling:

(1) Where extension of existing ceilings occurs, match existing.

(2) Where acoustical units are salvaged and reinstalled or joined, use salvaged units within a space. Do not mix new and salvaged units within a space which results in contrast between old and new acoustic units.

(3) Comply with specifications for new acoustical units for new units required to match appearance of existing units.

d. Fire-Rated System:

(1) Total assembly, consisting of the ceiling suspension system, acoustical units, penetrations, structural components and floor or roof construction above, shall have a 1 hour fire rating based on tests conducted in conformance with ASTM E119.

(2) Provide concealed fire protection around penetrations in ceilings for electric and mechanical work, and other penetrations as required to maintain the integrity of the fire-rated assembly.

(3) Install fire rated ceiling systems to conform to tested assembly.

3.6 CLEANING

a. Remove excess adhesive before adhesive sets.

b. Clean exposed surfaces. Remove contaminants and stains.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 62 38

STATIC-CONTROL FLOORING

08/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES
1.2 SCHEDULING
1.3 SUBMITTALS
1.3.1 Samples
1.3.1.1 Static-Control Resilient Flooring
1.3.1.2 Operations and Maintenance Data
1.4 CERTIFICATIONS
1.4.1 Indoor Air Quality Certifications
1.4.1.1 Floor Covering Materials
1.4.1.2 Adhesives
1.5 EXTRA MATERIALS
1.6 QUALITY ASSURANCE
1.7 DELIVERY, STORAGE, AND HANDLING
1.7.1 Static-Control Resilient Flooring
1.8 ENVIRONMENTAL CONDITIONS
1.8.1 Static-Control Resilient Flooring
1.9 WARRANTY
1.9.1 Static-Control Resilient Flooring

PART 2 PRODUCTS

2.1 STATIC-CONTROL RESILIENT FLOORING
2.1.1 Static-Control Resilient Flooring
2.2 WALL BASE
2.2.1 Resilient Base
2.3 ADHESIVES
2.4 ACCESSORIES
2.5 ELECTRICAL GROUND CONNECTION
2.6 MANUFACTURER'S COLOR, PATTERN AND TEXTURE
2.7 FIRE RESISTANCE TESTING REQUIREMENTS

PART 3 EXECUTION

3.1 SURFACE PREPARATION
3.2 MOISTURE, ALKALINITY AND BOND TESTS
3.3 GENERAL INSTALLATION
3.4 INSTALLATION OF STATIC-CONTROL RESILIENT TILE FLOORING
3.5 INSTALLATION OF WALL BASE
3.5.1 Resilient Base
3.6 CLEANING AND PROTECTION
3.7 TESTING

-- End of Section Table of Contents --

SECTION 09 62 38

STATIC-CONTROL FLOORING
08/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E648	(2025) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source
ASTM F150	(2025) Standard Test Method for Electrical Resistance of Conductive and Static Dissipative Resilient Flooring
ASTM F1344	(2021a) Standard Specification for Rubber Floor Tile
ASTM F1869	(2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
ASTM F2170	(2019a) Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350	(2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers

CARPET AND RUG INSTITUTE (CRI)

CRI Green Label Plus	(2017) Green Label Plus Quality Manual

ELECTROSTATIC DISCHARGE ASSOCIATION (ESD)

ESD S6.1	(2019) Standard for the Protection of Electrostatic Discharge Susceptible Items – Grounding

GREEN SEAL (GS)

GS-36	(2013) Adhesives for Commercial Use

RESILIENT FLOOR COVERING INSTITUTE (RFCI)

FLOORSCORE	FLOORSCORE IAQ Certification

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS	SCS Global Services (SCS) Indoor Advantage

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168	(2022) Adhesive and Sealant Applications

UL SOLUTIONS (UL)

UL 2818	(2022) GREENGUARD Certification Program For Chemical Emissions For Building Materials, Finishes And Furnishings

1.2 SCHEDULING

Schedule static-control flooring work after any other work which would damage the finished surface of the flooring.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Recycled content for Conductive Rubber Tile; S

Accessories; G

Adhesives; G

Warranty

SD-04 Samples

Static-Control Resilient Flooring; G

Moldings; G

Special Treatment Materials; G

Accessories; G

SD-06 Test Reports

Fire Resistance

Moisture, Alkalinity and Bond

Testing

SD-07 Certificates

Indoor Air Quality for Conductive Rubber Tile; S

Indoor Air Quality for Static-Dissipative Rubber Tile; S

Indoor Air Quality for Adhesives; S

Qualifications of Applicator

SD-08 Manufacturer's Instructions

Static-Control Resilient Flooring; G

Accessories; G

SD-10 Operation and Maintenance Data

Static-Control Resilient Flooring; G

Accessories; G

1.3.1 Samples

1.3.1.1 Static-Control Resilient Flooring

Submit three samples of each indicated color and type of flooring, base, moldings, and accessories sized a minimum 2-1/2 by 4 inch.

1.3.1.2 Operations and Maintenance Data

a. Submit Data Package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

b. Submit three copies of manufacturer's maintenance instructions for each type of flooring material describing recommended type of cleaning equipment and materials, spotting and cleaning methods, and cleaning cycles.

1.4 CERTIFICATIONS

1.4.1 Indoor Air Quality Certifications

1.4.1.1 Floor Covering Materials

Provide Static-Dissipative Rubber Tile and wall base products certified to meet indoor air quality requirements by FLOORSCORE, UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold, CRI Green Label Plus or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide current product certification documentation from certification body.

1.4.1.2 Adhesives

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that

products meet the requirements of this Section. Provide current product certification documentation from certification body.

1.5 EXTRA MATERIALS

Provide extra material from same dye lot for future maintenance. Provide a minimum of 5 percent of total square yards of each flooring and base type, pattern, and color.

1.6 QUALITY ASSURANCE

The flooring manufacturer will approve the Qualifications of Applicator and certify that he/she has a minimum of 3 years of experience in the application of the materials to be used.

1.7 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the building site in original unopened containers bearing the manufacturer's name, style name, pattern color name and number, size, production run, project identification, handling instructions and related information. Observe ventilation and safety procedures specified in the Safety Data Sheets (SDS). Do not store flooring near materials that may off-gas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives.

1.7.1 Static-Control Resilient Flooring

Store materials in a clean, dry, secure, and well-ventilated area with ambient air temperature range as recommended by the manufacturer but not less than 68 degrees F or more than 85 degrees F. Stack materials according to manufacturer's recommendations. Protect materials from the direct flow of heat from hot-air registers, radiators and other heating fixtures and appliances.

1.8 ENVIRONMENTAL CONDITIONS

Provide temporary ventilation during work of this section.

1.8.1 Static-Control Resilient Flooring

Maintain areas in which resilient flooring is to be installed at a temperature range as recommended by the manufacturer but not less than 68 degrees F or more than 85 degrees F for 3 days before application, during application and 2 days after application, unless otherwise directed by the flooring manufacturer for the flooring being installed. Maintain a minimum temperature range as recommended by the manufacturer but not less than 55 degrees F thereafter for the duration of the contract. Provide adequate ventilation to remove moisture from area and to comply with regulations limiting concentrations of hazardous vapors.

1.9 WARRANTY

1.9.1 Static-Control Resilient Flooring

Provide manufacturer's standard performance guarantees or warranties including a five year wear warranty and ten year conductivity warranty.

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 STATIC-CONTROL RESILIENT FLOORING

2.1.1 Static-Control Resilient Flooring

Static-dissipative rubber tile conforming to ASTM F1344 Class 1 homogeneous, Type B (through mottled). Provide a hammered surface. Provide electrical resistance from surface to surface and surface to ground between 1,000,000 ohms (1.0 x 10 to the 6th) and 1,000,000,000 ohms (1.0 x 10 to the 9th) when tested in accordance with ASTM F150. Provide tile 24 inches square and 2.0mm thick.

Provide certification of indoor air quality for Static-Dissipative Rubber Tile.

2.2 WALL BASE

2.2.1 Resilient Base

Resilient base must conform to ASTM F1861 and match RB-1.

2.3 ADHESIVES

Provide conductive adhesive as recommended by the manufacturer of the static-control flooring. Provide adhesive for wall base as recommended by the wall base manufacturer.

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (use the office or classoom requirements, regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classoom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

2.4 ACCESSORIES

Use accessories recommended by the manufacturer of the flooring.

2.5 ELECTRICAL GROUND CONNECTION

Provide an electrical ground connection that meets the requirements of ESD S6.1. Connection between the static-control floor system and the external grounding system must be provided. Contact with the static-control floor system must be with conductive grounding strip and must have the greater of the following: a minimum contact area of 9

square inch or the dimensions recommended by the manufacturer. Provide the grounding conductor recommended by the manufacturer of the flooring. Connect and install the grounding conductor as recommend by the flooring manufacturer.

2.6 MANUFACTURER'S COLOR, PATTERN AND TEXTURE

Provide color, pattern and texture as indicated. Provide flooring in any one continuous area or replacement of damaged flooring in continuous area from same production run with same shade and pattern.

2.7 FIRE RESISTANCE TESTING REQUIREMENTS

Provide a minimum average critical radiant flux of 0.45 watts per square centimeter for flooring in corridors and exits when tested in accordance with ASTM E648.

PART 3 EXECUTION

3.1 SURFACE PREPARATION

Before any work under this section is begun, defects such as rough or scaling concrete, low spots, high spots, and uneven surfaces must be corrected, and damaged portions of concrete slabs must be repaired in accordance with flooring manufacturer's recommended instructions. Floor must be in a level plane with a maximum variation of 1/8 inch every 10 feet, except where indicated as sloped. Repair cracks and irregularities and prepare the subfloor in accordance with flooring manufacturer's recommended instructions. Curing and sealing compounds should not be used on concrete surfaces to receive flooring unless they have been tested and approved by the flooring manufacturer. In addition, remove paint, varnish, oils, release agents, sealers, waxes, and adhesives, as required by the flooring product in accordance with manufacturer's printed installation instructions. If a curing compound is required, it must be coordinated for compatibility with the flooring adhesive.

3.2 MOISTURE, ALKALINITY AND BOND TESTS

Determine the suitability of the concrete subfloor for receiving the flooring with regard to moisture content and pH level by moisture and alkalinity tests. Conduct moisture testing in accordance with ASTM F1869 or ASTM F2170, unless otherwise recommended by the flooring manufacturer. Conduct alkalinity testing as recommended by the flooring manufacturer. Determine the compatibility of the flooring adhesives to the concrete floors by a bond test in accordance with the flooring manufacturer's recommendations.

3.3 GENERAL INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

3.4 INSTALLATION OF STATIC-CONTROL RESILIENT TILE FLOORING

Install static-control resilient flooring, ground connections and accessories in accordance with the approved manufacturer's installation instructions. Tile lines and joints must be kept square, symmetrical, tight, and even. Tile at the perimeter of the area to be finished may vary as necessary to maintain full-size tiles in the field, but no

perimeter tile may be less than one-half the field tile size, except where irregular shaped rooms make it impossible. Tile must be cut, fitted, and scribed to walls, partitions, and projections after field flooring has been applied. Install grounding strips in accordance with manufacturer's installation instructions. Protect edges of flooring material meeting hard surface flooring with molding and install in accordance with the molding manufacturer's printed instructions.

3.5 INSTALLATION OF WALL BASE

3.5.1 Resilient Base

Install wall base in accordance with manufacturer's printed installation instructions. Prepare and apply adhesives in accordance with manufacturer's printed directions. Tighten base joints and make even with adjacent resilient flooring. Fill voids along the top edge of base at masonry walls with caulk. Roll entire vertical surface of base with hand roller, and press toe of base with a straight piece of wood to ensure proper alignment. Avoid excess adhesive in corners.

3.6 CLEANING AND PROTECTION

The flooring must be cleaned in accordance with the manufacturer's recommendations. Flooring must be protected by a covering of heavy-duty building paper before foot traffic is permitted. Lap and secure edges of kraft paper protection to provide a continuous cover. Boardwalks must be placed over flooring in areas where subsequent building operations might damage the floor. Remove and replace flooring that becomes loose, broken, or curled prior to acceptance, or flooring that does not conform to resistance requirements of ASTM F150.

3.7 TESTING

Test the flooring in accordance with and conform to the requirements of ESD S6.1.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 65 13

RESILIENT BASE AND ACCESSORIES

1/21

PART 1 GENERAL

1.1 SUMMARY
1.2 REFERENCES
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY
1.6 STORAGE AND HANDLING
1.7 FIELD CONDITIONS
1.8 WARRANTY

PART 2 PRODUCTS

2.1 PRODUCTS
2.2 RESILIENT BASE
2.3 PRIMER (FOR CONCRETE FLOORS)
2.4 LEVELING COMPOUND (FOR CONCRETE FLOORS)
2.5 ADHESIVES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION GENERAL
3.3 RESILIENT BASE INSTALLATION
3.4 CLEANING
3.5 PROTECTION

-- End of Section Table of Contents --

SECTION 09 65 13
1/21

RESILIENT BASE AND ACCESSORIES

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

(1) Resilient base (RB) adhered to interior walls and partitions.

1.2 REFERENCES

Comply with references to extent specified in this section.

ASTM INTERNATIONAL (ASTM)

ASTM D4259	(2018) Standard Practice for Preparation of Concrete by Abrasion Prior to Coating Application
ASTM F1861	(2021) Standard Specification for Resilient Wall Base

INTERNATIONAL CONCRETE REPAIR INSTITUTE (ICRI)

ICRI 310.2R	(2013) Selecting and Specifying Concrete Surface Preparation for Sealers, Coatings, Polymer Overlays, and Concrete Repair

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Sustainable Construction Submittals; S

SD-04 Samples

Samples; G, AE

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G,

1.4 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Adhesives and primers indicating manufacturer's recommendation for each application.

(3) Installation instructions.

c. Samples:

(1) Resilient Base: (6 inches) long, each type and color.

d. Sustainable Construction Submittals:

(1) Recycled Content: Identify post-consumer and pre-consumer recycled content percentage by weight.

(2) Low Pollutant-Emitting Materials:

(a) Rubber Flooring: Submit Floor Score label.

(b) Show volatile organic compound types and quantities.

(3) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

e. Operation and Maintenance Data:

(1) Care instructions for each exposed finish product.

1.5 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.6 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage when handling and during construction operations.

1.7 FIELD CONDITIONS

Environment:

(1) Product Temperature: Minimum (70 degrees F) for minimum 48 hours before installation.

(2) Work Area Ambient Temperature Range: (70 to 80 degrees F) continuously, beginning 48 hours before installation.

(3) Install products when building is permanently enclosed and when wet construction is completed, dried, and cured.

1.8 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 PRODUCTS

a. Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

b. Basis of Design: Refer to the drawing set.

c. Provide each product from one manufacturer and from one production run.

d. Provide resilient stair treads and sheet rubber flooring from same manufacturer.

e. Sustainable Construction Requirements:

(1) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Flooring Adhesives and Sealants.

2.2 RESILIENT BASE

a. Resilient Base: (1/8 inch) thick, (6 inches) high.

(1) Type: Rubber; use one type throughout.

(2) ASTM F1861, Type TP thermoplastic rubber, Group 2 - layered.

b. Applications:

(1) Style B - Cove.

2.3 PRIMER (FOR CONCRETE FLOORS)

Type recommended by adhesive manufacturer.

2.4 LEVELING COMPOUND (FOR CONCRETE FLOORS)

Provide products mixed with latex or polyvinyl acetate resins.

2.5 ADHESIVES

Low pollutant-emitting, water based type recommended by adhered product manufacturer for each application.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Correct substrate deficiencies.

(1) Fill cracks, pits, and depressions with leveling compound.

(2) Remove protrusions; grind high spots.

(3) Apply leveling compound to achieve (1/8 inch) in (10 feet) maximum surface variation.

d. Clean substrates. Remove contaminants capable of affecting subsequently installed product's performance.

(1) Mechanically clean concrete floor substrate according to ASTM D4259.

(2) Surface Profile: ICRI 310.2R.

e. Allow substrate to dry and cure.

f. Perform flooring manufacturer’s recommended bond, substrate moisture content, and pH tests.

3.2 INSTALLATION GENERAL

Install products according to manufacturer's instructions.

(1) When instructions deviate from specifications, submit proposed resolution for Contracting Officer consideration.

3.3 RESILIENT BASE INSTALLATION

a. Applications:

(1) Install resilient base in rooms scheduled on Drawings.

(2) Install resilient base on casework and locker toe spaces , and other curb supported fixed equipment.

(3) Extend resilient base into closets, alcoves, and cabinet knee spaces, and around columns within scheduled room.

b. Lay out resilient base with minimum number of joints.

(1) Length: (24 inches) minimum, each piece.

(2) Locate joints (6 inches) minimum from corners and intersection of

adjacent materials.

c. Installation:

(1) Apply adhesive uniformly for full contact between resilient base and substrate.

(2) Set resilient base with hairline butted joints aligned along top edge.

d. Factory form corners and end stops.

e. Roll resilient base ensuring complete adhesion.

3.4 CLEANING

a. Remove excess adhesive before adhesive sets.

b. Clean exposed resilient base and resilient stair tread surfaces. Remove contaminants and stains.

(1) Clean with mild detergent. Leave surfaces free of detergent residue.

c. Polish exposed resilient base to gloss sheen.

3.5 PROTECTION

a. Prohibit traffic on resilient stair treads 72 hours, minimum, after installation.

b. Protect products from construction traffic and operations.

(1) Cover flooring with reinforced kraft paper, and plywood or hardboard.

(2) Maintain protection until directed by Contracting Officer's Representative.

c. Replace damaged products and re-clean.

(1) Damaged Products include cut, gouged, scraped, torn, and unbonded products.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 65 19

RESILIENT TILE FLOORING

5/18

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY
1.6 STORAGE
1.7 QUALITY ASSURANCE
1.8 WARRANTY
1.9 REFERENCES

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS
2.2 LUXURY VINYL TILE
2.3 ADHESIVES
2.4 PRIMER FOR CONCRETE SUBFLOORS
2.5 LEVELING COMPOUND FOR CONCRETE FLOORS
2.6 POLISH AND CLEANERS
2.7 MOULDING

PART 3 EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS
3.2 SUBFLOOR TESTING AND PREPARATION
3.3 INSTALLATION
3.4 CLEANING AND PROTECTION
3.5 LOCATION

-- End of Section Table of Contents --

SECTION 09 65 19
5/18

RESILIENT TILE FLOORING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies the installation of luxury vinyl tile, rubber tile, and accessories required for a complete installation.

1.2 RELATED WORK

a. Sustainable Design Requirements: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Resilient Base: Section 09 65 13, RESILIENT BASE AND ACCESSORIES.

c. Subfloor Testing and Preparation: Section 09 05 16, SUBSURFACE PREPARATION FOR FLOOR FINISHES.

d. Color, Pattern and Texture for Resilient Tile Flooring and Accessories: as shown on the SCHEDULE FOR FINISHES in the drawings.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.xx SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G, AE

Mockup; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Sustainable Design Submittals; S

Manufacturer's Literature And Data; G, AE

SD-04 Samples

Samples; G, AE

SD-06 Test Reports

Test Reports; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Sustainable Design Submittals as described below:

(1) Volatile organic compounds per volume as described in PART 2 - PRODUCTS.

(2) Postconsumer and preconsumer recycled content as described in PART 2 - PRODUCTS.

(3) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Resilient material manufacturer’s recommendations for adhesives, underlayment, primers, and polish.

(3) Application, installation and maintenance instructions.

d. Samples:

(1) Tile: Each type, color, thickness and finish.

(2) Edge Strips: Each type, color, thickness and finish.

(3) Feature Strips: Each type, color, thickness and finish.

e. Shop Drawings:

(1) Layout of patterns as shown on the construction documents.

(2) Edge strip locations showing types and detail cross sections.

f. Test Reports:

(1) Abrasion resistance: Depth of wear for each tile type and color and volume loss of tile, certified by independent laboratory. Tested per ASTM F510/F510M.

(2) Moisture and pH test results as per Section 09 05 16, SUBSURFACE PREPARATION FOR FLOOR FINISHES.

1.5 DELIVERY

a. Deliver materials to the site in original sealed packages or containers, clearly marked with the manufacturer's name or brand, type and color, production run number and date of manufacture.

b. Materials from containers which have been distorted, damaged or opened prior to installation are not acceptable.

1.6 STORAGE

Store materials in a clean, dry, enclosed space off the ground, protected from harmful weather conditions and at temperature and humidity conditions recommended by the manufacturer. Protect adhesives from freezing. Store flooring, adhesives, and accessories in the spaces where they will be installed for at least 48 hours before beginning installation.

1.7 QUALITY ASSURANCE

a. Installer Qualifications: A company specializing in installation with minimum three (3) years' experience and employs experienced flooring installers who have retained, and currently hold, an INSTALL Certification, or a certification from a comparable certification program.

(1) Installers to be certified by INSTALL or a comparable certification program with the following minimum criteria:

(a) US Department of Labor approved four (4) year apprenticeship program, 160 hours a year.

(b) Career long training.

(c) Manufacturer endorsed training.

(d) Fundamental journeyman skills certification.

b. Mockup: Build floor tile mockup to verify selections made under sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

(1) Size: (100 sq. ft.) for each type, color, and pattern. Locations as indicated on construction documents.

(2) Contracting Officer approved mockup may become part of the completed Project if undisturbed at time of Substantial Completion.

c. Furnish product type materials from the same production run.

1.8 WARRANTY

Comply with FAR clause 52.246-21, "Warranty of Construction".

1.9 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D2047	(2017) Standard Test Method for Static Coefficient of Friction of Polish-Coated Floor Surfaces as Measured by the James Machine
ASTM D4078	(2002; R 2015) Water Emulsion Floor Polish

ASTM E648	(2023) Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source
ASTM E662	(2021a; E 2021) Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM F510/F510M	(2020) Resistance to Abrasion of Resilient Floor Coverings Using an Abrader with a Grit Feed Method
ASTM F710	(2022) Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring
ASTM F925	(2020) Test Method for Resistance to Chemicals of Resilient Flooring
ASTM F1700	(2024) Standard Specification for Solid Vinyl Floor Tile

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 PERFORMANCE REQUIREMENTS

a. Provide adhesives, underlayment, primers, and polish recommended by resilient floor material manufacturer.

b. Critical Radiant Flux: 0.45 watts per sq. cm or more, Class I, per ASTM E648.

c. Smoke Density: Less than 450 per ASTM E662.

d. Slip Resistance – Not less than 0.5 when tested with ASTM D2047.

2.2 LUXURY VINYL TILE

a. ASTM F1700, Class III, Printed Film Vinyl Tile, Type B.

b. Thickness: 0.098 inches.

c. Size: 18 x 48 inches.

d. Provide products with recycled content with not less than 30 percent.

e. Chemical Resistance: ASTM F925; pass.

2.3 ADHESIVES

Provide water resistant type adhesive for flooring, base and accessories as recommended by the manufacturer to suit substrate conditions. VOC content to be less than the 50 grams/L when calculated according to 40 CFR 59 (EPA Method 24). Submit manufacturer’s descriptive data, documentation stating physical characteristics, and mildew and germicidal characteristics.

2.4 PRIMER FOR CONCRETE SUBFLOORS

Provide in accordance with Section 09 05 16, SUBSURFACE PREPARATION FOR FLOOR FINISHES.

2.5 LEVELING COMPOUND FOR CONCRETE FLOORS

Provide cementitious products with latex or polyvinyl acetate resins in the mix in accordance with Section 09 05 16, SUBSURFACE PREPARATION FOR FLOOR FINISHES.

2.6 POLISH AND CLEANERS

a. Cleaners: As recommended in writing by floor tile manufacturer.

b. Polish: ASTM D4078.

2.7 MOULDING

a. Provide tapered mouldings of types as indicated on the construction documents for both edges and transitions of flooring materials specified. Provide vertical lip on moulding of maximum (1/4 inch). Provide bevel change in level between (1/4 and 1/2 inch) with a slope no greater than 1:2.

b. Fasteners for Aluminum Mouldings: Stainless steel of type required for substrate condition.

PART 3 EXECUTION

3.1 ENVIRONMENTAL REQUIREMENTS

a. Maintain flooring materials and areas to receive resilient flooring at a temperature above (68 degrees F) for three (3) days before application, during application and two (2) days after application, unless otherwise directly by the flooring manufacturer for the flooring being installed. Maintain a minimum temperature of (55 degrees F) thereafter. Provide adequate ventilation to remove

moisture from area and to comply with regulations limiting concentrations of hazardous vapors.

b. Do not install flooring until building is permanently enclosed and wet construction in or near areas to receive tile materials is complete, dry and cured.

3.2 SUBFLOOR TESTING AND PREPARATION

a. Prepare and test surfaces to receive resilient tile and adhesive as per Section 09 05 16, SUBSURFACE PREPARATION FOR FLOOR FINISHES.

b. Prepare concrete substrates in accordance with ASTM F710.

3.3 INSTALLATION

a. Install in accordance with manufacturer's instructions for application and installation unless specified otherwise.

b. Mix tile from at least two containers. An apparent line either of shades or pattern variance is not acceptable.

c. Tile Layout:

(1) If layout is not shown on construction documents, lay tile symmetrically about center of room or space with joints aligned.

(2) Vary edge width as necessary to maintain full size tiles in the field, no edge tile to be less than 1/2 the field tile size, except where irregular shaped rooms make it impossible.

(3) Place tile pattern in the same direction; do not alternate tiles unless specifically indicated in the construction documents to the contrary.

d. Application:

(1) Adhere floor tile to flooring substrates using a full spread of adhesive applied to substrate to produce a completed installation without open cracks, voids, raising and puckering at joints, telegraphing of adhesive spreader marks, and other surface imperfections.

(2) Scribe, cut, and fit floor tiles to butt neatly and tightly to vertical surfaces and permanent fixtures including built-in furniture, cabinets, pipes, outlets, and door frames.

(3) Extend floor tiles into toe spaces, door reveals, closets, and similar openings. Extend floor tiles to center of door openings.

(4) Roll tile floor with a minimum 100 pound roller.

e. Seal joints at pipes with sealants in accordance with Section 07 92 00, JOINT SEALANTS.

f. Installation of Edge Strips:

(1) Locate edge strips under center line of doors unless otherwise shown on construction documents.

(2) Set resilient edge strips in adhesive. Anchor metal edge strips with anchors and screws.

(3) Where tile edge is exposed, butt edge strip to touch along tile edge.

(4) Where thin set ceramic tile abuts resilient tile, set edge strip against floor file and against the ceramic tile edge.

3.4 CLEANING AND PROTECTION

a. Clean adhesive marks on exposed surfaces during the application of resilient materials before the adhesive sets. Exposed adhesive is not acceptable.

b. Keep traffic off resilient material for a minimum 72 hours after installation.

c. Clean flooring as recommended in accordance with manufacturer’s printed maintenance instructions and within the recommended time frame. As required by the manufacturer, apply the recommended number of coats and type of polish and/or finish in accordance with manufacturer’s written instructions.

d. When construction traffic occurs over tile, cover resilient materials with reinforced kraft paper properly secured and maintained until removal is directed by Contracting Officer. At entrances and where wheeled vehicles or carts are used, cover tile with plywood, hardboard, or particle board over paper, secured and maintained until removal is directed by Contracting Officer.

e. When protective materials are removed and immediately prior to acceptance, replace damaged tile and mouldings, re-clean resilient materials.

3.5 LOCATION

a. Unless otherwise indicated in construction documents, install tile flooring, under areas where casework, laboratory and pharmacy furniture and other equipment occur.

b. Extend tile flooring for room into adjacent closets and alcoves.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 67 23.20

RESINOUS (EPOXY BASE) WITH VINYL CHIP BROADCAST (RES-2)

12/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 QUALITY ASSURANCE
1.6 MATERIAL PACKAGING DELIVERY AND STORAGE
1.7 PROJECT CONDITIONS
1.8 WARRANTY
1.9 REFERENCES

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION FOR EPY-1 RESINOUS (EPOXY) VINYL CHIP FLAKE BROADCAST FLOORING SYSTEM
2.2 SUPPLEMENTAL MATERIALS
2.3 BASE CAP STRIP

PART 3 EXECUTION

3.1 INSPECTION
3.2 PROJECT CONDITIONS
3.3 INSTALLATION REQUIREMENTS
3.4 PREPARATION
3.5 APPLICATION
3.6 TOLERANCE
3.7 ENGINEERING DETAILS
3.8 CURING, PROTECTION AND CLEANING

-- End of Section Table of Contents --

SECTION 09 67 23.20
12/21

RESINOUS (EPOXY BASE) WITH VINYL CHIP BROADCAST (RES-2)

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies Resinous (Resinous epoxy base with vinyl chip flake broadcast) flooring with integral cove base:

(1) EPY-1 Resinous (epoxy) vinyl chip flake broadcast flooring system.

1.2 RELATED WORK

a. Concrete and Moisture Vapor Barrier: Section 03 30 00, CAST-IN-PLACE CONCRETE.

b. Substrate Preparation for Floor Finishes: Section 09 05 16.

c. Color and location of each type of resinous flooring: As indicated in the SCHEDULE FOR FINISHES in the drawings set. .

d. Floor Drains: Division 22, PLUMBING.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualification Data: For Installer; G, AE

Mockups; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G

Sustainable Submittal; S

SD-04 Samples

Samples; G, AE

SD-07 Certificates

Certifications And Approvals; G

SD-11 Closeout Submittals

Warranty; G

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Manufacturer's Literature and Data:

(1) Description of each product to be provided.

(2) Application and installation instructions.

(3) Maintenance Instructions: Submit manufacturer’s written instructions for recommended maintenance practices.

c. Qualification Data: For Installer.

d. Sustainable Submittal:

(1) Product data for products having recycled content, submit documentation indicating percentages by weight of post-consumer and pre-consumer recycled content.

(a) Include statements indicating costs for each product having recycled content.

(2) Product data for field applied, interior, paints, coatings, and primers, include printed statement of VOC content indicating compliance with environmental requirements.

e. Samples:

(1) Each color and texture specified in SCHEDULE FOR FINISHES on the drawings.

(2) Samples for verification: For each (color and texture) resinous flooring system required, 6 inches square, applied to a rigid backing by installer for this project.

(3) Sample showing construction from substrate to finish surface in thickness specified and color and texture of finished surfaces. Finished flooring must match the approved samples in color and texture.

f. Shop Drawings: Include plans, sections, component details, and attachment to other trades. Indicate layout of the following:

(1) Patterns.

(2) Edge configurations.

g. Certifications and Approvals:

(1) Manufacturer's certification of material and substrate compliance

with specification.

(2) Manufacturer's approval of installers.

(3) Contractor's certificate of compliance with Quality Assurance requirements.

h. Warranty: As specified in this section.

1.5 QUALITY ASSURANCE

a. Manufacture Certificate: Manufacture shall certify that a particular resinous flooring system has been manufactured and in use for a minimum of five (5) years.

b. Installer Qualifications: Engage an experienced installer (applicator) who is experienced in applying resinous flooring systems similar in material, design, and extent to those indicated for this project for a minimum period of five (5) years, whose work has resulted in applications with a record of successful in-service performance, and who is acceptable to resinous flooring manufacturer.

(1) Engage an installer who is certified in writing by resinous flooring manufacturer as qualified to apply resinous flooring systems indicated.

(2) Contractor shall have completed at least ten (10) projects of similar size and complexity. Include list of at least five (5) projects. List must include owner (purchaser); address of installation, contact information at installation project site; and date of installation.

(3) Installer's Personnel: Employ persons trained for application of specified product.

c. Source Limitations:

(1) Obtain primary resinous flooring materials including primers, resins, hardening agents, grouting coats and finish or sealing coats from a single manufacturer.

(2) Provide secondary materials, including patching and fill material, joint sealant, and repair material of type and from source recommended by manufacturer of primary materials.

d. Mockups: Apply mockups to verify selections made under sample submittals and to demonstrate aesthetic effects and establish quality standards for materials and execution.

(1) Apply full-thickness mockups on 48 inch square floor area selected by VA Contracting Officer.

(a) If applicable include 48 inch length of integral cove base.

(2) Approved mockups not damaged during the testing may become part of the completed work if undisturbed at time of Substantial Completion.

(3) Sign off from VA Contracting Officer on texture for slip

resistance and clean ability must be complete before installation of flooring system.

e. Pre-Installation Conference:

(1) Convene a meeting not less than thirty days prior to starting work.

(2) Attendance:

(a) Contractor

(b) VA Contracting Officer

(c) Manufacturer and Installer's Representative

(3) Review the following:

(a) Environmental requirements

1. Air and surface temperature

2. Relative humidity

3. Ventilation

4. Dust and contaminates

(b) Protection of surfaces not scheduled to be coated

(c) Inspect and discus condition of substrate and other preparatory work performed

(d) Review and verify availability of material; installer’s personnel, equipment needed

(e) Design and patterns and edge conditions.

(f) Performance of the coating with chemicals anticipated in the area receiving the resinous (urethane and epoxy mortar/cement) flooring system

(g) Application and repair

(h) Field quality control

(i) Cleaning

(j) Protection of coating systems

(k) One-year inspection and maintenance

(l) Coordination with other work

f. Manufacturer’s Field Services: Manufacturer’s representative shall provide technical assistance and guidance for surface preparation and application of resinous flooring systems.

g. Contractor Job Site Log: Contractor shall document daily; the work accomplished environmental conditions and any other condition event

significant to the long term performance of the urethane and epoxy mortar/cement flooring materials installation. The Contractor shall maintain these records for one year after Substantial Completion.

1.6 MATERIAL PACKAGING DELIVERY AND STORAGE

a. Deliver materials to the site in original sealed packages or containers, clearly marked with the manufacturer's name or brand, type and color, production run number and date of manufacture.

b. Protect materials from damage and contamination in storage or delivery, including moisture, heat, cold, direct sunlight, etc.

c. Maintain temperature of storage area between 60 and 80 degrees F .

d. Keep containers sealed until ready for use.

e. Do not use materials beyond manufacturer’s shelf life limits.

f. Package materials in factory pre-weighed and in single, easy to manage batches sized for ease of handling and mixing proportions from entire package or packages. No On site weighing or volumetric measurements are allowed.

1.7 PROJECT CONDITIONS

a. Environmental Limitations: Comply with resinous flooring manufacturer's written instructions for substrate temperature, ambient temperature, moisture, ventilation, and other conditions affecting resinous flooring application.

(1) Maintain material and substrate temperature between 65 and 85 degrees F during resinous flooring application and for not less than 24 hours after application.

(2) Concrete substrate shall be properly cured per referenced section 03 30 00, CAST-IN-PLACE CONCRETE. Standard cure time a minimum of 30 days. A vapor barrier must be present for concrete subfloors on or below grade.

(a) Resinous flooring applications where moisture testing resulting in readings exceeding limits as defined in this specification under part 3, section 3.4, paragraph B, shall employ an multiple component 15 mil thick system designed to suppress excess moisture in concrete.

(b) Application at a minimum thickness of 15 mils, over properly prepared concrete substrate as defined in section 3.4.

(c) Moisture uppression system must meet the design standards as follows:

Property	Test	Value
Tensile Strength	ASTM D638	4,400 psi
Volatile Organic Compound Limits (V.O.C.)	EPA & LEED	25 grams per liter

Permeance	ASTM E96/E96M @ 16mils/ 0.4mm on concrete	0.1 perms
Tensile Modulus	ASTM D638	1.9X105 psi
Percent Elongation	ASTM D638	12%
Cure Rate	Per manufactures Data	4 hours Tack free with 24hr recoat window
Bond Strength	ASTM D7234	100% bond to concrete failure

b. Lighting: Provide permanent lighting or, if permanent lighting is not in place, simulate permanent lighting conditions during resinous flooring application.

c. Close spaces to traffic during resinous flooring application and for not less than 24 hours after application, unless manufacturer recommends a longer period.

1.8 WARRANTY

a. Work subject to the terms of the Article “Warranty of Construction” FAR clause 52.246-21.

b. Warranty: Manufacture shall furnish a single, written warranty covering the full assembly (including substrata) for both material and workmanship for a extended period of three (3) full years from date of installation, or provide a joint and several warranty signed on a single document by manufacturer and applicator jointly and severally warranting the materials and workmanship for a period of three (3) full years from date of installation. A sample warranty letter must be included with bid package or bid may be disqualified.

1.9 REFERENCES

The publication listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C413	(2018; R 2023) Standard Test Method for Absorption of Chemical-Resistant Mortars, Grouts, and Monolithic Surfacings and Polymer Concretes
ASTM C531	(2018; R 2023) Standard Test Method for Linear Shrinkage and Coefficient of Thermal Expansion of Chemical-Resistant Mortars, Grouts, and Monolithic Surfacings, and Polymer Concretes
ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D790	(2017) Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials

ASTM D1308	(2002; R 2013) Effect of Household Chemicals on Clear and Pigmented Organic Finishes
ASTM D2240	(2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness
ASTM D4060	(2019) Abrasion Resistance of Organic Coatings by the Taber Abraser
ASTM D4226	(2020) Impact Resistance of Rigid (Poly-Vinyl Chloride) (PVC) Building Products
ASTM D4259	(2018) Standard Practice for Preparation of Concrete by Abrasion Prior to Coating Application
ASTM D7234	(2012) Standard Test Method for Pull-Off Adhesion Strength of Coatings on Concrete Using Portable Pull-Off Adhesion Testers
ASTM E96/E96M	(2024a) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM F1869	(2023) Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
ASTM F2170	(2019a) Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A326.3	(2021) Dynamic Coefficient of Friction (DCOF) of Hard Surface Flooring Materials

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION FOR EPY-1 RESINOUS (EPOXY) VINYL CHIP FLAKE BROADCAST FLOORING SYSTEM

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

a. System Descriptions:

(1) Monolithic, multi-component epoxy chemistry resinous flooring system. Primer with broadcast quartz aggregates, High performance

multi-component solvent free epoxy undercoat, Vinyl chip flake broadcast media in desired flake size (1/8", 1/4"). High performance multi component epoxy and solvent free sealers. System overall thickness 2-3mm.

b. Products: Subject to compliance with applicable fire, health, environmental, and safety requirements for storage, handling, installation, and clean up.

c. System Components: Verify specific requirements as systems vary by manufacturer. Verify build up layers of broadcast and installation method. Verify compatibility with substrate. Use manufacturer's standard components, compatible with each other and as follows:

(1) Primer with Broadcast quartz (primer coat):

(a) Resin: epoxy.

(b) Formulation Description: Multiple component high solids.

(c) Application Method: squeegee, back roll and broadcast.

(d) Thickness of coat(s): 10-20 mil.

(e) Number of Coats: One.

(f) Aggregates: Quartz broadcast into wet epoxy primer.

(2) Undercoat: (body coat)

(a) Resin: Epoxy.

(b) Formulation Description: Pigmented multi-component, high solids.

(c) Application Method: Notched squeegee and Back roll

(d) Number of Coats: One.

(e) Aggregates: vinyl chip flake broadcast into wet Undercoat.

(f) Thickness of coat(s): 30-60 mils.

(g) Number of Coats: One.

(3) Sealer coat:

(a) Resin: Epoxy.

(b) Formulation Description: Multiple component high solids, no solvent UV stable.

(c) Type/Finsh: Clear Gloss.

(d) Thickness of coat(s): 5-10 mils.

(e) Number of Coats: (2) two.

(f) Application: Squeegee and finish roll.

d. System Characteristics:

(1) Color and Pattern: As indicated in COLOR SCHEDULE on the drawings.

(2) Integral cove base: ½ inch radius epoxy mortar cove keyed into concrete substrate and or resinous flooring mortar system. No fillers integral cove base must be troweled in place with specified resinous mortar base.

(3) Overall System Thickness: Nominal 1/8”, .

(4) Finish: anti-slip resistant.

(5) Temperature Range: Systems vary by manufacturer; approximate range from a minimum of 45 to 150 degrees F.

e. Physical Properties:

(1) In accordance with ANSI A326.3 the DCOF (Dynamic Coefficient of Friction) shall be 0.42 or greater when the flooring surface is wet

(2) Physical Properties of flooring system when tested as follows

Property	Test	Value
Tensile Strength	ASTM D638	5,200 psi
Volatile Organic Compound limits (VOC)	EPA & LEED	Below 100 g/l
Flexural Strength	ASTM D790	4,000 psi
Water Absorption	ASTM C413	0.056%
Impact Resistance	ASTM D4226	> 160 in. lbs
Abrasion Resistance	ASTM D4060 CS-17	0.03 gm maximum weight loss
Thermal Coefficient of Linear Expansion	ASTM C531	17 x 10-6 in/in ºF
	ASTM D2240	85 to 90
Bond Strength	ASTM D7234	100% bond to concrete failure

f. Chemical Resistance in accordance ASTM D1308 - 02(2007) “Standard Test Method for Effect of Household Chemicals on Clear and Pigmented Organic Finishes”. ASTM International, West Conshohocken, PA, 2006, DOI: 10.1520/D1308-02R07, www.astm.org. No effect to the following exposures:

(1) Acetic acid (5 percent)

(2) Ammonium hydroxide (10 percent)

(3) Citric Acid (50 percent)

(4) Fatty Acid

(5) Motor Oil, 20W

(6) Hydrochloric acid (20 percent)

(7) Sodium Chloride

(8) Sodium Hypochlorite (10 percent)

(9) Sodium Hydroxide (30 percent)

(10) Sulfuric acid (25 percent)

(11) Urine, Feces

(12) Hydrogen peroxide (10 percent)

2.2 SUPPLEMENTAL MATERIALS

a. Textured Top Coat: Type recommended or produced by manufacturer of seamless resinous flooring system, slip resistance for desired final finish.

b. Joint Sealant: Type recommended or produced by resinous flooring manufacturer for type of service or joint conditioned indicated.

c. Waterproof Membrane: Type recommended or produced by manufacturer of resinous floor coatings for type of service and conditions as indicated in Drawings and/or specified.

d. Provide a chemical resistant epoxy novolac top-coat capable of resisting sustained temperatures up to (250 degrees F).

e. Crack Isolation Membrane: Type recommended or produced by manufacturer of resinous flooring for conditions asindicated in Drawingsand/or specified.

f. Patching and Fill Material: Resinous product of or approved by resinous coating manufacturer for application indicated. Resinous based materials only. Cementitious or single component product are not expectable.

2.3 BASE CAP STRIP

a. Zinc cove strip.

b. Shape for 2mm depth of base material, "J" or “L” configuration.

c. Finish:

(1) Finish exposed surfaces in accordance with NAAMM Metal Finishes Manual.

PART 3 EXECUTION

3.1 INSPECTION

a. Examine the areas and conditions where monolithic resinous system with integral base is to be installed with the VA Contracting Officer.

b. Moisture Vapor Emission Testing: Perform moisture vapor transmission testing in accordance with ASTM F1869 to determine the MVER of the

substrate prior to commencement of the work. See section 3.4, 3.

3.2 PROJECT CONDITIONS

a. Maintain temperature of rooms (air and surface) where work occurs, between 70- and 90-degrees F for at least 48 hours, before, during, and 24 hours after installation. Maintain temperature at least 70 degrees F during cure period.

b. Maintain relative humidity less than 75 percent.

c. Do not install materials until building is permanently enclosed and wet construction is complete, dry, and cured.

d. Maintain proper ventilation of the area during application and curing time period.

(1) Comply with infection control measures of the VA Medical Center.

3.3 INSTALLATION REQUIREMENTS

a. The manufacturer's instructions for application and installation shall be reviewed with the VA Contracting Officer for the seamless resinous (urethane and epoxy mortar) flooring system with integral cove base.

b. Substrate shall be approved by manufacture technical representative.

3.4 PREPARATION

a. General: Prepare and clean substrates according to resinous flooring manufacturer's written instructions for substrate indicated. Provide clean, dry, and neutral Ph substrate for resinous flooring application.

b. Concrete Substrates: Provide sound concrete surfaces free of laitance, glaze, efflorescence, curing compounds, form-release agents, dust, dirt, grease, oil, and other contaminants incompatible with resinous flooring.

(1) Prepare concrete substrates as follows:

(a) Shot-blast surfaces with an apparatus that abrades the concrete surface, contains the dispensed shot within the apparatus, and re circulates the shot by vacuum pickup.

(b) Comply with ASTM D4259 requirements, unless manufacturer's written instructions are more stringent.

(2) Repair damaged and deteriorated concrete according to resinous flooring manufacturer's written recommendations.

(3) Verify that concrete substrates are dry.

(a) Perform anhydrous calcium chloride test, ASTM F1869. Proceed with application only after substrates have maximum moisture-vapor-emission rate of 3 lb of water/1000 square feet in 24 hours. Per manufacturers recommendations.

(b) MVT threshold for monolithic resinous flooring shall not exceed 3 lbs/1000 square feet in a 24-hour period.

(c) When MVT emission exceeds this limit, apply manufacturer’s recommended vapor control primer or other corrective measures as recommended by manufacturer prior to application of flooring or membrane systems.

(d) Perform in situ probe test, ASTM F2170. Proceed with application only after substrates do not exceed a maximum potential equilibrium relative humidity of 85 percent.

(e) Provide a written report showing test placement and results.

(4) Verify that concrete substrates have neutral Ph and that resinous flooring will adhere to them. Perform tests recommended by manufacturer. Proceed with application only after substrates pass testing.

c. Resinous Materials: Mix components and prepare materials according to resinous flooring manufacturer's written instructions.

d. Use patching and fill material to fill holes and depressions in substrates according to manufacturer's written instructions.

e. Treat control joints and other nonmoving substrate cracks to prevent cracks from reflecting through resinous flooring according to manufacturer's written recommendations. Allowances should be included for flooring manufacturer recommended joint fill material, and concrete crack treatment.

f. Prepare wall to receive integral cove base :

(1) Verify wall material is acceptable for resinous flooring application, if not, install material (e.g. cement board) to receive base.

(2) Fill voids in wall surface to receive base, install undercoats (e.g. water proofing membrane, and/or crack isolation membrane) as recommended by resinous flooring manufacturer.

(3) Install base prior to flooring if required by resinous flooring manufacturer.

(4) Grind, cut or sand protrusions to receive base application.

3.5 APPLICATION

a. General: Apply components of resinous flooring system according to manufacturer's written instructions to produce a uniform, monolithic wearing surface of thickness indicated.

(1) Coordinate application of components to provide optimum adhesion of resinous flooring system to substrate, and optimum intercoat adhesion.

(2) Cure resinous flooring components according to manufacturer's written instructions. Prevent contamination during application and curing processes.

(3) At substrate expansion and isolation joints, provide joint in

resinous flooring to comply with resinous flooring manufacturer's written recommendations.

(a) Apply joint sealant to comply with manufacturer's written recommendations.

b. Apply Primer: over prepared substrate at manufacturer's recommended spreading rate for all areas to receive integrated cove base.

c. Apply cove base: Trowel to wall surfaces at a 1-inch radius, before applying flooring. Apply according to manufacturer's written instructions and details including those for taping, mixing, priming, and troweling, sanding, and top coating of cove base. Round internal and external corners.

d. Apply Primer: over prepared substrate at manufacturer's recommended spreading rate.

e. Trowel mortar base: Mix mortar material according to manufacturer's recommended procedures. Climatic and non-climatic resinous flooring systems may vary slightly on mode of application. Application should be based upon the following: Uniformly spread mortar over substrate using a specially designed screed box adjusted to manufacturer's recommended height. Metal trowel (hand or power) single mortar coat in thickness indicated for flooring system, grout to fill substrate voids. When cured, sand to remove trowel marks and roughness.

f. Broadcast: Immediately broadcast quartz silica aggregate into the primer using manufacturer's spray caster. Strict adherence to manufacturer's installation procedures and coverage rates is imperative.

g. Under Coat: Mix base material according to manufacturer's recommended procedures. Uniformly spread mixed material over previously primed substrate using manufacturer's installation tool. Roll material with strict adherence to manufacturer's installation procedures and coverage rates.

h. Broadcast: Immediately broadcast vinyl flakes into the body coat. Strict adherence to manufacturer's installation procedures and coverage rates is imperative.

i. First Sealer: Remove excess un-bonded flakes by lightly brushing and vacuuming the floor surface. Mix and apply sealer with strict adherence to manufacturer's installation procedures.

j. Second Sealer: Lightly sand first sealer coat. Mix and apply second sealer coat with strict adherence to manufacturer's installation procedures.

3.6 TOLERANCE

a. From line of plane: Maximum 1/8 inch in total distance of flooring and base. Broadcast resinous flooring system will contour substrate. Deviation and tolerance are subject to concrete tolerance.

b. From radius of cove: Maximum of 1/8 inch plus or 1/16-inch minus.

3.7 ENGINEERING DETAILS

a. Chase edges to “lock” the flooring system into the concrete substrate along lines of termination.

b. Penetration Treatment: Lap and seal resinous system onto the perimeter of the penetrating item by bridging over compatible elastomer at the interface to compensate for possible movement.

c. Trenches: Continue flooring system into trenches to maintain monolithic protection. Treat cold joints to assure bridging of potential cracks.

d. Treat floor drains by chasing the flooring system to lock in place at point of termination.

e. Treat control joints to bridge potential cracks and to maintain monolithic protection. Treat cold joints and construction joints to bridge potential cracks and to maintain monolithic protection on horizontal and vertical surfaces as well as horizontal and vertical interfaces.

f. Discontinue Resinous floor system at vertical and horizontal contraction and expansion joints by installing backer rod and compatible sealant after coating installation is completed. Provide sealant type recommended by manufacturer for traffic conditions and chemical exposures to be encountered.

3.8 CURING, PROTECTION AND CLEANING

a. Cure resinous flooring materials in compliance with manufacturer's directions, taking care to prevent contamination during stages of application and prior to completion of curing process.

b. Close area of application for a minimum of 24 hours.

c. Protect resinous flooring materials from damage and wear during construction operation.

(1) Cover flooring with kraft type paper.

(2) Optional 1/4 inch thick hardboard, plywood, or particle board where area is in foot or vehicle traffic pattern, rolling or fixed scaffolding and overhead work occurs.

d. Remove temporary covering and clean resinous flooring just prior to final inspection. Use cleaning materials and procedures recommended by resinous flooring manufacturer.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 91 00

PAINTING

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 DELIVERY AND STORAGE
1.6 QUALITY ASSURANCE
1.7 MOCK-UP PANEL
1.8 REGULATORY REQUIREMENTS
1.9 SAFETY AND HEALTH
1.10 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS:
2.2 PAINT PROPERTIES:
2.3 PLASTIC TAPE:
2.4 BIOBASED CONTENT

PART 3 EXECUTION

3.1 JOB CONDITIONS:
3.2 INSPECTION:
3.3 GENERAL WORKMANSHIP REQUIREMENTS:
3.4 SURFACE PREPARATION:
3.5 PAINT PREPARATION:
3.6 APPLICATION:
3.7 PRIME PAINTING:
3.8 EXTERIOR FINISHES:
3.9 INTERIOR FINISHES:
3.10 MECHANICAL AND ELECTRICAL WORK FIELD PAINTING SCHEDULE:
3.11 BUILDING AND STRUCTURAL WORK FIELD PAINTING:
3.12 IDENTITY PAINTING SCHEDULE:
3.13 PROTECTION CLEAN UP, AND TOUCH-UP:

-- End of Section Table of Contents --

SECTION 09 91 00

PAINTING

PART 1 GENERAL

1.1 DESCRIPTION

a. Work of this Section includes all labor, materials, equipment, and services necessary to complete the painting and finishing as shown on the construction documents and/or specified herein, including, but not limited to, the following:

(1) Prime coats which may be applied in shop under other sections.

(2) Prime painting unprimed surfaces to be painted under this Section.

(3) Painting items furnished with a prime coat of paint, including touching up of or repairing of abraded, damaged or rusted prime coats applied by others.

(4) Painting ferrous metal (except stainless steel) exposed to view.

(5) Painting galvanized ferrous metals exposed to view.

(6) Painting interior concrete block exposed to view.

(7) Painting gypsum drywall exposed to view.

(8) Painting of wood exposed to view, except items which are specified to be painted or finished under other Sections of these specifications. Back painting of all wood in contact with concrete, masonry or other moisture areas.

(9) Painting pipes, pipe coverings, conduit, ducts, insulation, hangers, supports and other mechanical and electrical items and equipment exposed to view.

(10) Painting surfaces above, behind or below grilles, gratings, diffusers, louvers lighting fixtures, and the like, which are exposed to view through these items.

(11) Painting includes shellacs, stains, varnishes, coatings specified, and striping or markers and identity markings.

(12) Incidental painting and touching up as required to produce proper finish for painted surfaces, including touching up of factory finished items.

(13) Painting of any surface not specifically mentioned to be painted herein or on construction documents, but for which painting is obviously necessary to complete the job, or work which comes within the intent of these specifications, is to be included as though specified.

1.2 RELATED WORK

a. Section 01 35 26, SAFETY REQUIREMENTS: Activity Hazard Analysis.

b. Section 01 81 13, SUSTAINABLE CONSTUCTION REQUIREMENTS: Sustainable Design Requirements.

c. Division 05 METALS: Shop prime painting of steel and ferrous metals.

d. Division 08 OPENINGS: Shop prime painting of steel and ferrous metals.

e. Refer to the drawings for type of Finish, Color, and Gloss Level of Finish Coat.

f. Division 10 SPECIALTIES: Shop prime painting of steel and ferrous metals.

g. Division 11 EQUIPMENT: Shop prime painting of steel and ferrous metals.

h. Division 12 FURNISHINGS: Shop prime painting of steel and ferrous metals.

i. Division 13 SPECIAL CONSTRUCTION: Shop prime painting of steel and ferrous metals.

j. Division 14 CONVEYING EQUIPMENT: Shop prime painting of steel and ferrous metals.

k. Division 21 FIRE SUPPRESSION: Shop prime painting of steel and ferrous metals.

l. Division 22 PLUMBING: Shop prime painting of steel and ferrous metals.

m. Division 23 HEATING; VENTILATION AND AIR-CONDITIONING: Shop prime painting of steel and ferrous metals.

n. Division 26 ELECTRICAL: Shop prime painting of steel and ferrous metals.

o. Division 27 COMMUNICATIONS: Shop prime painting of steel and ferrous metals.

p. Division 28 ELECTRONIC SAFETY AND SECURITY: Shop prime painting of steel and ferrous metals.

q. Division 32 EXTERIOR IMPROVEMENTS: Shop prime painting of steel and ferrous metals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.03 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Painter Qualifications; G, AE

Mock-Up Panel; G, AE

SD-03 Product Data

Sustainable Design Submittals; S

Manufacturer's Literature And Data; G, AE

SD-04 Samples

Sample Panels; G, AE

Sample Of Identity Markers If Used; G, AE

SD-07 Certificates

Manufacturers' Certificates; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Sustainable Design Submittals as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

c. Painter qualifications.

d. Manufacturer's Literature and Data:

(1) Before work is started, or sample panels are prepared, submit manufacturer's literature and technical data, the current Master Painters Institute (MPI) "Approved Product List" indicating brand label, product name and product code as of the date of contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use subsequent MPI "Approved Product List", however, only one (1) list may be used for the entire contract and each coating system is to be from a single manufacturer. All coats on a particular substrate must be from a single manufacturer. No variation from the MPI "Approved Product List" where applicable is acceptable.

e. Sample Panels:

(1) After painters' materials have been approved and before work is started, submit sample panels showing each type of finish and color specified.

(2) Panels to Show Color: Composition board, (4 x 10 inch).

(3) Panel to Show Transparent Finishes: Wood of same species and grain pattern as wood approved for use, (4 x 10 inch face)

minimum, and where both flat and edge grain will be exposed, (10 inches) long by sufficient size, (2 x 2 inch minimum or actual wood member to show complete finish.

(4) Attach labels to panel stating the following:

(a) Federal Specification Number or manufacturers name and product number of paints used.

(b) Product type and color.

(c) Name of project.

(5) Strips showing not less than 2 inch wide strips of undercoats and 4 inch wide strip of finish coat.

f. Sample of identity markers if used.

g. Manufacturers' Certificates indicating compliance with specified requirements:

(1) Manufacturer's paint substituted for Federal Specification paints meets or exceeds performance of paint specified.

(2) High temperature aluminum paint.

(3) Epoxy coating.

(4) Intumescent clear coating or fire-retardant paint.

(5) Plastic floor coating.

1.5 DELIVERY AND STORAGE

a. Deliver materials to site in manufacturer's sealed container marked to show following:

(1) Name of manufacturer.

(2) Product type.

(3) Batch number.

(4) Instructions for use.

(5) Safety precautions.

b. In addition to manufacturer's label, provide a label legibly printed as following:

(1) Federal Specification Number, where applicable, and name of material.

(2) Surface upon which material is to be applied.

(3) Specify Coat Types: Prime; body; finish; etc.

c. Maintain space for storage, and handling of painting materials and equipment in a ventilated, neat and orderly condition to prevent

spontaneous combustion from occurring or igniting adjacent items.

d. Store materials at site at least 24 hours before using, at a temperature between (45 and 85 degrees F).

1.6 QUALITY ASSURANCE

a. Qualification of Painters: Use only qualified journeyman painters for the mixing and application of paint on exposed surfaces. Submit evidence that key personnel have successfully performed surface preparation and application of coating on a minimum of three (3) similar projects within the past three (3) years.

b. Paint Coordination: Provide finish coats which are compatible with the prime paints used. Review other Sections of these specifications in which prime paints are to be provided to ensure compatibility of the total coatings system for the various substrates. Upon request from other subcontractors, furnish information on the characteristics of the finish materials proposed to be used, to ensure that compatible prime coats are used. Provide barrier coats over incompatible primers or remove and re-prime as required. Notify the Contracting Officer in writing of any anticipated problems using the coating systems as specified with substrates primed by others.

1.7 MOCK-UP PANEL

a. In addition to the samples specified herein to be submitted for approval, apply in the field, at their final location, each type and color of approved paint materials, applied 10 feet wide, floor to ceiling of wall surfaces, before proceeding with the remainder of the work, for approval by the Contracting Officer. Paint mock-ups to include one (1) door and frame assembly.

b. Finish and texture approved by Contracting Officer will be used as a standard of quality and workmanship for remainder of work.

c. Repaint individual areas which are not approved, as determined by the Contracting Officer, until approval is received.

1.8 REGULATORY REQUIREMENTS

Paint materials are to conform to the restrictions of the local Environmental and Toxic Control jurisdiction.

(1) Volatile Organic Compounds (VOC) Emissions Requirements: Field-applied paints and coatings that are inside the waterproofing system to not exceed limits of authorities having jurisdiction.

(2) Lead-Base Paint:

(a) Comply with Section 410 of the Lead-Based Paint Poisoning Prevention Act, as amended, and with implementing regulations promulgated by Secretary of Housing and Urban Development.

(b) Regulations concerning prohibition against use of lead-based paint in federal and federally assisted construction, or rehabilitation of residential structures are set forth in Subpart F, Title 24, Code of Federal Regulations, Department of Housing

and Urban Development.

(c) Do not use coatings having a lead content over 0.06 percent by weight of non-volatile content.

(3) Asbestos: Provide materials that do not contain asbestos.

(4) Chromate, Cadmium, Mercury, and Silica: Provide materials that do not contain zinc-chromate, strontium-chromate, Cadmium, mercury or mercury compounds or free crystalline silica.

(5) Human Carcinogens: Provide materials that do not contain any of the ACGIH 0116 and ACGIH 0100 confirmed or suspected human carcinogens.

(6) Use high performance acrylic paints in place of alkyd paints.

1.9 SAFETY AND HEALTH

a. Apply paint materials using safety methods and equipment in accordance with the following:

(1) Comply with applicable Federal, State, and local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis (AHA) as specified in Section 01 35 26, SAFETY REQUIREMENTS. The AHA is to include analyses of the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

b. Safety Methods Used During Paint Application: Comply with the requirements of SSPC PA Guide 10.

c. Toxic Materials: To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

(1) The applicable manufacturer’s Material Safety Data Sheets (MSDS) or local regulation.

(2) 29 CFR 1910.1000.

(3) ACGIH 0116 and ACGIH 0100, threshold limit values.

1.10 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0116	(2016) TLVs and BEIs
ACGIH 0100	(2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1	(2020) Scheme for the Identification of

Piping Systems

ASTM INTERNATIONAL (ASTM)

ASTM C28/C28M	(2010) Gypsum Plasters

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

TT-P-1411A	(1973) Paint, Copolymer-Resin, Cementitious (For Waterproofing Concrete and Masonry Walls) (CEP)
TT-F-322D	(1973) FILLER TWO-COMPONENT TYPE: FOR DENTS, CRACKS, SMALL-HOLES, AND BLOW HOLES

MASTER PAINTERS INSTITUTE (MPI)

MPI 1	(2012) Aluminum Paint
MPI 4	(2016) Interior/Exterior Latex Block Filler
MPI 9	(2016) Alkyd, Exterior Gloss (MPI Gloss Level 6)
MPI 11	(2016) Latex, Exterior Semi-Gloss, MPI Gloss Level 5
MPI 18	(2022) Primer, Zinc Rich, Organic
MPI 22	(2012) Aluminum Paint, High Heat (up to /1100° F)
MPI 45	(2016) Primer Sealer, Interior Alkyd
MPI 46	(2016) Undercoat, Enamel, Interior
MPI 47	(2016) Alkyd, Interior, Semi-Gloss (MPI Gloss Level 5)
MPI 48	(2016) Alkyd, Interior, Gloss (MPI Gloss Level 6-7)
MPI 50	(2015) Primer Sealer, Latex, Interior
MPI 54	(2016) Latex, Interior, Semi-Gloss (MPI Gloss Level 5)
MPI 68	(2016) Floor Paint, Latex, Gloss
MPI 77	(2015) Epoxy, Gloss
MPI 94	(2016) Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)
MPI 95	(2015) Primer, Quick Dry, for Aluminum
MPI 99	(2012) Sealer, Water Based, for Concrete Floors

MPI 101	(2016) Primer, Epoxy, Anti-Corrosive, for Metal
MPI 114	(2012) Latex, Interior, Gloss (MPI GLoss Level 6)
MPI 119	(2016) Latex, Exterior, Gloss (MPI Gloss Level 6)
MPI 134	(2015) Primer, Galvanized, Water Based
MPI 135	Non-Cementitious Galvanized Primer
MPI 138	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 2)
MPI 139	(2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 3)
MPI 141	(2016) Latex, Interior, High Performance Architectural, Semi-Gloss (MPI Gloss Level 5)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 2	(2018) Hand Tool Cleaning
SSPC SP 3	(2018) Power Tool Cleaning
SSPC PA Guide 10	(2008) Guide to Safety and Health Requirements

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000	(2017) Air Contaminants
40 CFR 59	National Volatile Organic Compound Emission Standards For Consumer and Commercial Products

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

PART 2 PRODUCTS

2.1 MATERIALS:

Conform to the coating specifications and standards referenced in PART 3. Submit manufacturer’s technical data sheets for specified coatings and solvents.

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.2 PAINT PROPERTIES:

a. Use ready-mixed (including colors), except two component epoxies, polyurethanes, polyesters, paints having metallic powders packaged separately and paints requiring specified additives.

b. Where no requirements are given in the referenced specifications for primers, use primers with pigment and vehicle, compatible with substrate and finish coats specified.

c. Provide undercoat paint produced by the same manufacturer as the finish coats. Use only thinners approved by the paint manufacturer and use only to recommended limits.

d. VOC Content: For field applications that are inside the weatherproofing system, paints and coating to comply with VOC content limits of authorities having jurisdiction and the following VOC content limits:

(1) Flat Paints and Coatings: 50 gram/liter.

(2) Non-flat Paints and Coatings: 150 gram/liter.

(3) Dry-Fog Coatings: 400 gram/liter.

(4) Primers, Sealers, and Undercoaters: 200 gram/liter.

(5) Anticorrosive and Antirust Paints applied to Ferrous Metals: 250 gram/liter.

(6) Zinc-Rich Industrial Maintenance Primers: 340 gram/liter.

(7) Pretreatment Wash Primers: 420 gram/liter.

(8) Shellacs, Clear: 730 gram/liter.

(9) Shellacs, Pigmented: 550 gram/liter.

e. VOC test method for paints and coatings is to be in accordance with 40 CFR 59 (EPA Method 24). Part 60, Appendix A with the exempt compounds’ content determined by Method 303 (Determination of Exempt Compounds) in the South Coast Air Quality Management District’s (SCAQMD) “Laboratory Methods of Analysis for Enforcement Samples” manual.

2.3 PLASTIC TAPE:

a. Pigmented vinyl plastic film in colors as indiicated on the drawings or specified.

b. Pressure sensitive adhesive back.

c. Widths as shown on construction documents.

2.4 BIOBASED CONTENT

a. Paint products shall comply with following bio-based standards for biobased materials:

Material Type	Percent by Weight
Interior Paint	20 percent biobased material
Interior Paint- Oil Based and Solvent Alkyd	67 percent biobased material
Exterior Paint	20 percent biobased material
Wood & Concrete Stain	39 percent biobased content
Polyurethane Coatings	25 percent biobased content
Water Tank Coatings	59 percent biobased content
Wood & Concrete Sealer- Membrane Concrete Sealers	11 percent biobased content
Wood & Concrete Sealer- Penetrating Liquid	79 percent biobased content

b. The minimum-content standards are based on the weight (not the volume) of the material.

PART 3 EXECUTION

3.1 JOB CONDITIONS:

a. Safety: Observe required safety regulations and manufacturer's warning and instructions for storage, handling and application of painting materials.

(1) Take necessary precautions to protect personnel and property from hazards due to falls, injuries, toxic fumes, fire, explosion, or other harm.

(2) Deposit soiled cleaning rags and waste materials in metal containers approved for that purpose. Dispose of such items off the site at end of each day’s work.

b. Atmospheric and Surface Conditions:

(1) Do not apply coating when air or substrate conditions are:

(a) Less than (5 degrees F) above dew point.

(b) Below (50 degrees F) or over (95 degrees F), unless specifically pre-approved by the Contracting Officer and the product manufacturer. Under no circumstances are application conditions to exceed manufacturer recommendations.

(c) When the relative humidity exceeds 85 percent; or to damp or wet surfaces; unless otherwise permitted by the paint manufacturer’s printed instructions.

(2) Maintain interior temperatures until paint dries hard.

(3) Do no exterior painting when it is windy and dusty.

(4) Do not paint in direct sunlight or on surfaces that the sun will warm.

(5) Apply only on clean, dry and frost-free surfaces except as follows:

(a) Apply water thinned acrylic and cementitious paints to damp (not wet) surfaces only when allowed by manufacturer's printed instructions.

(b) Concrete and masonry when permitted by manufacturer’s recommendations, dampen surfaces to which water thinned acrylic and cementitious paints are applied with a fine mist of water on hot dry days to prevent excessive suction and to cool surface.

(6) Varnishing:

(a) Apply in clean areas and in still air.

(b) Before varnishing vacuum and dust area.

(c) Immediately before varnishing wipe down surfaces with a tack rag.

3.2 INSPECTION:

Examine the areas and conditions where painting and finishing are to be applied and correct any conditions detrimental to the proper and timely completion of the work. Do not proceed with the work until unsatisfactory conditions are corrected to permit proper installation of the work.

3.3 GENERAL WORKMANSHIP REQUIREMENTS:

a. Application may be by brush or roller. Spray application only upon acceptance from the Contracting Officer in writing.

b. Furnish to the Contracting Officer a painting schedule indicating when the respective coats of paint for the various areas and surfaces will be completed. This schedule is to be kept current as the job progresses.

c. Protect work at all times. Protect all adjacent work and materials by suitable covering or other method during progress of work. Upon completion of the work, remove all paint and varnish spots from floors, glass and other surfaces. Remove from the premises all rubbish and accumulated materials of whatever nature not caused by others and leave work in a clean condition.

d. Remove and protect hardware, accessories, device plates, lighting fixtures, and factory finished work, and similar items, or provide in place protection. Upon completion of each space, carefully replace all removed items by workmen skilled in the trades involved.

e. When indicated to be painted, remove electrical panel box covers and doors before painting walls. Paint separately and re-install after all paint is dry.

f. Materials are to be applied under adequate illumination, evenly spread and flowed on smoothly to avoid runs, sags, holidays, brush marks, air bubbles and excessive roller stipple.

g. Apply materials with a coverage to hide substrate completely. When color, stain, dirt or undercoats show through final coat of paint, the surface is to be covered by additional coats until the paint film is of uniform finish, color, appearance and coverage, at no additional cost to the Government.

h. All coats are to be dry to manufacturer’s recommendations before applying succeeding coats.

i. All suction spots or “hot spots” in plaster after the application of the first coat are to be touched up before applying the second coat.

j. Do not apply paint behind frameless mirrors that use mastic for adhering to wall surface.

3.4 SURFACE PREPARATION:

a. General:

(1) The Contractor shall be held wholly responsible for the finished appearance and satisfactory completion of painting work. Properly prepare all surfaces to receive paint, which includes cleaning, sanding, and touching-up of all prime coats applied under other Sections of the work. Broom clean all spaces before painting is started. All surfaces to be painted or finished are to be completely dry, clean and smooth.

(2) See other sections of specifications for specified surface conditions and prime coat.

(3) Perform preparation and cleaning procedures in strict accordance with the paint manufacturer’s instructions and as herein specified, for each particular substrate condition.

(4) Clean surfaces before applying paint or surface treatments with materials and methods compatible with substrate and specified finish. Remove any residue remaining from cleaning agents used. Do not use solvents, acid, or steam on concrete and masonry. Schedule the cleaning and painting so that dust and other contaminants from the cleaning process will not fall in wet, newly painted surfaces.

(5) Maximum Moisture Content of Substrates: When measured with an electronic moisture meter as follows:

(a) Concrete: 12 percent.

(b) Fiber-Cement Board: 12 percent.

(c) Masonry (Clay and CMU’s): 12 percent.

(d) Gypsum Board: 12 percent.

c. Ferrous Metals:

(1) Remove oil, grease, soil, drawing and cutting compounds, flux and other detrimental foreign matter in accordance with SSPC SP 1 (Solvent Cleaning).

(2) Remove loose mill scale, rust, and paint, by hand or power tool cleaning, as defined in SSPC SP 2 (Hand Tool Cleaning) and SSPC SP 3 (Power Tool Cleaning). Where high temperature aluminum paint is used, prepare surface in accordance with paint manufacturer's instructions.

(3) Fill dents, holes and similar voids and depressions in flat exposed surfaces of hollow steel doors and frames, access panels, roll-up steel doors and similar items specified to have semi-gloss or gloss finish with TT-F-322D (Filler, Two-Component Type, For Dents, Small Holes and Blow-Holes). Finish flush with adjacent surfaces.

(a) Fill flat head countersunk screws used for permanent anchors.

(b) Do not fill screws of item intended for removal such as glazing beads.

(4) Spot prime abraded and damaged areas in shop prime coat which expose bare metal with same type of paint used for prime coat. Feather edge of spot prime to produce smooth finish coat.

(5) Spot prime abraded and damaged areas which expose bare metal of factory finished items with paint as recommended by manufacturer of item.

d. Zinc-Coated (Galvanized) Metal,0 Surfaces Specified Painted:

(1) Clean surfaces to remove grease, oil and other deterrents to paint adhesion in accordance with SSPC SP 1 (Solvent Cleaning).

(2) Spot coat abraded and damaged areas of zinc-coating which expose base metal on hot-dip zinc-coated items with MPI 18 (Organic Zinc Rich Coating). Prime or spot prime with MPI 134 (Waterborne Galvanized Primer) or MPI 135 (Non-Cementitious Galvanized Primer) depending on finish coat compatibility.

e. Masonry and Concrete:

(1) Clean and remove dust, dirt, oil, grease efflorescence, form release agents, laitance, and other deterrents to paint adhesion.

(2) Use emulsion type cleaning agents to remove oil, grease, paint and similar products. Use of solvents, acid, or steam is not permitted.

(3) Remove loose mortar in masonry work.

(4) Replace mortar and fill open joints, holes, cracks and depressions with new mortar specified in Section 04 05 13, MASONRY MORTARING.

Do not fill weep holes. Finish to match adjacent surfaces.

(5) Neutralize Concrete floors to be painted by washing with a solution of (3 pounds) of zinc sulfate crystals to (1 gallon) of water, allow to dry three (3) days and brush thoroughly free of crystals.

(6) Repair broken and spalled concrete edges with concrete patching compound to match adjacent surfaces as specified in Division 03, CONCRETE Sections. Remove projections to level of adjacent surface by grinding or similar methods.

f. Gypsum Board:

(1) Remove efflorescence, loose and chalking plaster or finishing materials.

(2) Remove dust, dirt, and other deterrents to paint adhesion.

(3) Fill holes, cracks, and other depressions with ASTM C28/C28M finished flush with adjacent surface, with texture to match texture of adjacent surface. Patch holes over (1-inch) in diameter as specified in Section for plaster or gypsum board.

3.5 PAINT PREPARATION:

a. Thoroughly mix painting materials to ensure uniformity of color, complete dispersion of pigment and uniform composition.

b. Do not thin unless necessary for application and when finish paint is used for body and prime coats. Use materials and quantities for thinning as specified in manufacturer's printed instructions.

c. Remove paint skins, then strain paint through commercial paint strainer to remove lumps and other particles.

d. Mix two (2) component and two (2) part paint and those requiring additives in such a manner as to uniformly blend as specified in manufacturer's printed instructions unless specified otherwise.

e. For tinting required to produce exact shades specified, use color pigment recommended by the paint manufacturer.

3.6 APPLICATION:

a. Start of surface preparation or painting will be construed as acceptance of the surface as satisfactory for the application of materials.

b. Unless otherwise specified, apply paint in three (3) coats; prime, body, and finish. When two (2) coats applied to prime coat are the same, first coat applied over primer is body coat and second coat is finish coat.

c. Apply each coat evenly and cover substrate completely.

d. Allow not less than 48 hours between application of succeeding coats, except as allowed by manufacturer's printed instructions, and approved by Contracting Officer.

e. Apply by brush or roller. Spray application for new or existing occupied spaces only upon approval by acceptance from Contracting Officer in writing.

(1) Apply painting materials specifically required by manufacturer to be applied by spraying.

f. Do not paint in closed position operable items such as access doors and panels, window sashes, overhead doors, and similar items except overhead roll-up doors and shutters.

3.7 PRIME PAINTING:

a. After surface preparation, prime surfaces before application of body and finish coats, except as otherwise specified.

b. Spot prime and apply body coat to damaged and abraded painted surfaces before applying succeeding coats.

c. Additional field applied prime coats over shop or factory applied prime coats are not required except for exterior exposed steel apply an additional prime coat.

d. Prime rabbets for stop and face glazing of wood, and for face glazing of steel.

f. Metals except boilers, incinerator stacks, and engine exhaust pipes:

(1) Steel and iron: MPI 95 (Fast Drying Metal Primer). Use MPI 101 (Cold Curing Epoxy Primer) where MPI 77 (Epoxy Cold Cured, Gloss finish is specified.

(2) Zinc-coated steel and iron: MPI 134 (Waterborne Galvanized Primer) .

(3) Machinery not factory finished: MPI 9 (Exterior Alkyd Enamel).

(4) Asphalt coated metal: MPI 1 (Aluminum Paint).

(5) Metal over (201 degrees F), Boilers, Incinerator Stacks, and Engine Exhaust Pipes: MPI 22 (High Heat Resistant Coating).

g. Gypsum Board:

(1) Surfaces scheduled to have MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) or MPI 114 (Interior Latex, Gloss) finish: Use MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) and MPI 114 (Interior Latex, Gloss) respectively.

(2) Primer: MPI 50 (Interior Latex Primer Sealer) except use MPI 45 (Interior Primer Sealer) or MPI 46 (Interior Enamel Undercoat) in shower and bathrooms.

j. Concrete Masonry Interior Surfaces of Ceilings and Walls:

(1) Use MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) or MPI 114 (Interior Latex, Gloss) except use two (2) coats where substrate has aged less than six (6) months.

(2) Use MPI 114 (Interior Latex, Gloss).

k. Concrete Floors: MPI 99 (Water-based Acrylic Curing and Sealing Compound).

3.8 EXTERIOR FINISHES:

a. Apply following finish coats where specified in SCHEDULE FOR FINISHES on the drawings.

Steel and Ferrous Metal , Including Tern:

(1) Two (2) coats of MPI 94 (Exterior Alkyd, Semi-Gloss) on exposed surfaces, except on surfaces over (201 degrees F).

(2) One (1) coat of MPI 22 (High Heat Resistant Coating) on surfaces over (290 degrees F) and on surfaces of , stacks engine exhaust pipes.

c. Machinery without factory finish except for primer: One (1) coat MPI 9 (Exterior Alkyd Enamel) .

e. Concrete Masonry Units and Concrete:

(1) General:

(a) Where specified in SCHEDULE FOR FINISHES on the drawings or shown.

(b) Mix as specified in manufacturer's printed directions.

(c) Do not mix more paint than can be used within four (4) hours after mixing. Discard paint that has started to set.

(d) Dampen warm surfaces above (75 degrees F) with fine mist of water before application of paint. Do not leave free water on surface.

(e) Cure paint with a fine mist of water as specified in manufacturer's printed instructions.

(2) Use two (2) coats of TT-P-1411A (Paint, Co-polymer-Resin, Cementitious), unless specified otherwise.

3.9 INTERIOR FINISHES:

a. Apply following finish coats over prime coats in spaces or on surfaces scheduled on the Drawings.

b. Metal Work:

(1) Apply to exposed surfaces.

(2) Omit body and finish coats on surfaces concealed after installation except electrical conduit containing conductors over 600 volts.

(3) Ferrous Metal, Galvanized Metal, and Other Metals Scheduled:

(a) Apply two (2) coats of MPI 47 (Interior Alkyd, Semi-Gloss) unless specified otherwise.

(b) Two (2) coats of MPI 48 (Interior Alkyd Gloss).

(c) Machinery: One (1) coat MPI 9 (Exterior Alkyd Enamel).

(d) Asphalt Coated Metal: One (1) coat MPI 1 (Aluminum Paint).

(e) Ferrous Metal over (290 degrees F): Boilers, Incinerator Stacks, and Engine Exhaust Pipes: One (1) coat MPI 22 (High Heat Resistant Coating.

c. Gypsum Board:

(1) One (1) coat of MPI 45 (Interior Primer Sealer) plus one (1) coat of MPI 139 (Interior High Performance Latex, MPI Gloss level 3).

(2) Two (2) coats of MPI 138 (Interior High Performance Latex, MPI Gloss Level 2).

(3) One (1) coat of MPI 45 (Interior Primer Sealer) plus one (1) coat of MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) or MPI 114 (Interior Latex, Gloss).

(4) One (1) coat of MPI 45 (Interior Primer Sealer) plus one (1) coat of MPI 48 (Interior Alkyd Gloss).

d. Masonry and Concrete Walls:

(1) Over MPI 4 (Interior/Exterior Latex Block Filler) on CMU surfaces.

(2) Two (2) coats of MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5). See the Drawings for specified finish.

(3) Two (2) coats of or MPI 114 (Interior Latex, Gloss). See the Drawings for specified finish.

e. Cement Board: One (1) coat of MPI 141 (Interior High Performance Latex, MPI Gloss Level 5 or MPI 114 (Interior Latex, Gloss). See the Drawings for specified finish.

f. Concrete Floors: One (1) coat of MPI 68 (Interior/ Exterior Latex Porch & Floor Paint, Gloss).

g. Miscellaneous:

(1) Apply where specified in SCHEDULE FOR FINISHES on the drawings.

(2) MPI 1 (Aluminum Paint): Two (2) coats of aluminum paint.

3.10 MECHANICAL AND ELECTRICAL WORK FIELD PAINTING SCHEDULE:

a. Field painting of mechanical and electrical consists of cleaning, touching-up abraded shop prime coats, and applying prime, body and finish coats to materials and equipment if not factory finished in space scheduled to be finished.

b. In spaces not scheduled to be finish painted in SCHEDULE FOR FINISHES in the drawins, paint as specified below.

c. Paint various systems specified in Division 02 – EXISTING CONDITIONS, Division 21 – FIRE SUPPRESSION, Division 22 - PLUMBING, Division 23 – HEATING, VENTILATION AND AIR-CONDITIONING, Division 26 - ELECTRICAL, Division 27 - COMMUNICATIONS, and Division 28 – ELECTRONIC SAFETY AND SECURITY.

d. Paint after tests have been completed.

e. Omit prime coat from factory prime-coated items.

f. Finish painting of mechanical and electrical equipment is not required when located in interstitial spaces, above suspended ceilings, in concealed areas such as pipe and electric closets, pipe basements, pipe tunnels, trenches, attics, roof spaces, shafts and furred spaces except on electrical conduit containing feeders 600 volts or more.

g. Omit field painting of items specified in “BUILDING AND STRUCTURAL WORK FIELD PAINTING”; “Building and Structural Work not Painted”.

h. Color:

(1) Paint items having no color specified in SCHEDULE FOR FINISHES in the drawins to match surrounding surfaces.

(2) Paint colors as specified in SCHEDULE FOR FINISHES in the drawins except for following:

(a) White: Exterior unfinished surfaces of enameled plumbing fixtures. Insulation coverings on breeching and uptake inside boiler house, drums and drum-heads, oil heaters, condensate tanks and condensate piping.

(b) Gray: Heating, ventilating, air conditioning and refrigeration equipment (except as required to match surrounding surfaces), and water and sewage treatment equipment and sewage ejection equipment.

(c) Aluminum Color: Ferrous metal on outside of boilers and in connection with boiler settings including supporting doors and door frames and fuel oil burning equipment, and steam generation system (bare piping, fittings, hangers, supports, valves, traps and miscellaneous iron work in contact with pipe).

(d) Federal Safety Red: Exposed fire protection piping, hydrants, post indicators, electrical conducts containing fire alarm control wiring, and fire alarm equipment.

(e) Federal Safety Orange: Entire lengths of electrical conduits containing feeders 600 volts or more.

(f) Color to match brickwork sheet metal covering on breeching outside of exterior wall of boiler house.

i. Apply paint systems on properly prepared and primed surface as follows:

(1) Exterior Locations:

(a) Apply two (2) coats of MPI 94 (Exterior Alkyd, Semi-gloss) to the following ferrous metal items:

Vent and exhaust pipes with temperatures under (201 degrees F), roof drains, fire hydrants, post indicators, yard hydrants, exposed piping and similar items.

(b) Apply two (2) coats of MPI 119 (Exterior Latex, High Gloss (acrylic)) to galvanized and zinc-copper alloy metal.

(c) Apply one (1) coat of MPI 22 (High Heat Resistant Coating), (1200 degrees F) to incinerator stacks, boiler stacks, and engine generator exhaust.

(2) Interior Locations:

(a) Apply two (2) coats of MPI 47 (Interior Alkyd, Semi-Gloss) to following items:

Metal under (201 degrees F) of items such as bare piping, fittings, hangers and supports.
Equipment and systems such as hinged covers and frames for control cabinets and boxes, cast-iron radiators, electric conduits and panel boards.
Heating, ventilating, air conditioning, plumbing equipment, and machinery having shop prime coat and not factory finished.

(b) Apply one (1) coat of MPI 50 (Interior Latex Primer Sealer) and one (1) coat of MPI 54 (Interior Latex, Semi-Gloss, MPI Gloss Level 5) on finish of insulation on boiler breeching and uptakes inside boiler house, drums, drumheads, oil heaters, feed water heaters, tanks and piping.

(c) Apply two (2) coats of MPI 22 (High Heat Resistant Coating) to ferrous metal surface over (290 degrees F) of following items:

Garbage and trash incinerator.
Medical waste incinerator.
Exterior of boilers and ferrous metal in connection with boiler settings including supporting members, doors and door frames and fuel oil burning equipment.
Steam line flanges, bare pipe, fittings, valves, hangers and supports over (290 degrees F).
Engine generator exhaust piping and muffler.

(d) Paint electrical conduits containing cables rated 600 volts or more using two (2) coats of MPI 94 (Exterior Alkyd, Semi-gloss) in the Federal Safety Orange color in exposed and concealed spaces full length of conduit.

(3) Other exposed locations:

(a) Metal surfaces, except aluminum, of cooling towers exposed to view, including connected pipes, rails, and ladders: Two (2) coats of MPI 1 (Aluminum Paint).

(b) Cloth jackets of insulation of ducts and pipes in connection with plumbing, air conditioning, ventilating refrigeration and

heating systems: One (1) coat of MPI 50 (Interior Latex Primer Sealer) and one (1) coat of MPI 11 (Exterior Latex Semi-Gloss .

3.11 BUILDING AND STRUCTURAL WORK FIELD PAINTING:

a. Painting and finishing of interior and exterior work except as specified here-in-after.

(1) Painting and finishing of new work including colors and gloss of finish selected is specified in drawing Finish Schedule.

(2) Painting of disturbed, damaged and repaired or patched surfaces when entire space is not scheduled for complete repainting or refinishing.

(3) Painting of ferrous metal and galvanized metal.

(4) Painting of wood with fire retardant paint exposed in attics, when used as mechanical equipment space (except shingles).

(5) Identity painting and safety painting.

b. Building and Structural Work not Painted:

(1) Prefinished items:

(a) Casework, doors, elevator entrances and cabs, metal panels, wall covering, and similar items specified factory finished under other sections.

(b) Factory finished equipment and pre-engineered metal building components such as metal roof and wall panels.

(2) Finished surfaces:

(a) Hardware except ferrous metal.

(b) Anodized aluminum, stainless steel, chromium plating, copper, and brass, except as otherwise specified.

(c) Signs, fixtures, and other similar items integrally finished.

(3) Concealed surfaces:

(a) Inside dumbwaiter, elevator and duct shafts, interstitial spaces, pipe basements, crawl spaces, pipe tunnels, above ceilings, attics, except as otherwise specified.

(b) Inside walls or other spaces behind access doors or panels.

(c) Surfaces concealed behind permanently installed casework and equipment.

(4) Moving and operating parts:

(a) Shafts, chains, gears, mechanical and electrical operators, linkages, and sprinkler heads, and sensing devices.

(b) Tracks for overhead or coiling doors, shutters, and grilles.

(5) Labels:

(a) Code required label, such as Underwriters Laboratories Inc., Intertek Testing Service or Factory Mutual Research Corporation.

(b) Identification plates, instruction plates, performance rating, and nomenclature.

(6) Galvanized metal:

(a) Exterior chain link fence and gates, corrugated metal areaways, and gratings.

(b) Gas Storage Racks.

(c) Except where specifically specified to be painted.

(7) Metal safety treads and nosings.

(8) Gaskets.

(9) Concrete curbs, gutters, pavements, retaining walls, exterior exposed foundations walls and interior walls in pipe basements.

(10) Structural steel encased in concrete, masonry, or other enclosure.

(11) Structural steel to receive sprayed-on fire proofing.

(12) Ceilings, walls, columns in interstitial spaces.

(13) Ceilings, walls, and columns in pipe basements.

3.12 IDENTITY PAINTING SCHEDULE:

a. Identify designated service in new buildings or projects with extensive remodeling in accordance with ASME A13.1, unless specified otherwise, on exposed piping, piping above removable ceilings, piping in accessible pipe spaces, interstitial spaces, and piping behind access panels. For existing spaces where work is minor match existing.

(1) Legend may be identified using snap-on coil plastic markers or by paint stencil applications.

(2) Apply legends adjacent to changes in direction, on branches, where pipes pass through walls or floors, adjacent to operating accessories such as valves, regulators, strainers and cleanouts a minimum of 40 feet apart on straight runs of piping. Identification next to plumbing fixtures is not required.

(3) Locate Legends clearly visible from operating position.

(4) Use arrow to indicate direction of flow using black stencil paint.

(5) Identify pipe contents with sufficient additional details such as temperature, pressure, and contents to identify possible hazard. Insert working pressure shown on construction documents where asterisk appears for High, Medium, and Low Pressure designations as follows:

(a) High Pressure - 60 psig and above.

(b) Medium Pressure - (15 to 59 psig).

(c) Low Pressure - 14 psig and below.

(d) Add Fuel oil grade numbers.

(6) Legend name in full or in abbreviated form as follows:

PIPING	COLOR OF EXPOSED PIPING	COLOR OF BACKGROUND	LEGEND LETTERS	ABBREVIATIONS
Blow-off	Green	White	Blow-off	
Boiler Feedwater	Green	White	Blr Feed	
A/C Condenser Water Supply	Green	White	A/C Cond Wtr Sup	
A/C Condenser Water Return	Green	White	A/C Cond Wtr Ret	
Chilled Water Supply	Green	White	Ch. Wtr Sup	
Chilled Water Return	Green	White	Ch. Wtr Ret	
Shop Compressed Air	Blue	White	Shop Air	
Air-Instrument Controls	Green	White	Air-Inst Cont	
Drain Line	Green	White	Drain	
Emergency Shower	Green	White	Emg Shower	
High Pressure Steam	Green	White	H.P.	
High Pressure Condensate Return	Green	White	H.P. Ret	
Medium Pressure Steam	Green	White	M. P. Stm	
Medium Pressure Condensate Return	Green	White	M.P. Ret	
Low Pressure Steam	Green	White	L.P. Stm	
Low Pressure Condensate Return	Green	White	L.P. Ret	
High Temperature Water Supply	Green	White	H. Temp Wtr Sup	
High Temperature Water Return	Green	White	H. Temp Wtr Ret	
Hot Water Heating Supply	Green	White	H. W. Htg Sup	
Hot Water Heating Return	Green	White	H. W. Htg Ret	
Gravity Condensate Return	Green	White	Gravity Cond Ret	
Pumped Condensate Return	Green	White	Pumped Cond Ret	
Vacuum Condensate Return	Green	White	Vac Cond Ret	
Fuel Oil - Grade (Diesel Fuel included under Fuel Oil)	Brown	White	Fuel Oil-Grade	
Boiler Water Sampling	Green	White	Sample	
Chemical Feed	Green	White	Chem Feed	
Continuous Blow-Down	Green	White	Cont. B D	
Pumped Condensate	Green	White	Pump Cond	
Pump Recirculating	Green	White	Pump-Recirc.	
Vent Line	Green	White	Vent	
Alkali	Orange	Black	Alk	
Bleach	Orange	Black	Bleach	
Detergent	Yellow	Black	Det	
Liquid Supply	Yellow	Black	Liq Sup	
Reuse Water	Yellow	Black	Reuse Wtr	
Cold Water (Domestic)	White	Green	White	C.W. Dom

Hot Water (Domestic)	
Supply	White YellowBlack H.W. Dom
Return	White YellowBlack H.W. Dom Ret
Tempered Water	White YellowBlack Temp. Wtr
Ice Water	
Supply	White Green White Ice Wtr
Return	White Green White Ice Wtr Ret
Reagent Grade Water	Green White RG
Reverse Osmosis	Green White RO
Sanitary Waste	Green White San Waste
Sanitary Vent	Green White San Vent
Storm Drainage	Green White St Drain
Pump Drainage	Green White Pump Disch
Chemical Resistant Pipe	
Waste	OrangeBlack Acid Waste
Vent	OrangeBlack Acid Vent
Atmospheric Vent	Green White ATV
Silver Recovery	Green White Silver Rec
Oral Evacuation	Green White Oral Evac
Fuel Gas	YellowBlack Gas
Fire Protection Water	
Sprinkler	Red Red White Auto Spr
Standpipe	Red Red White Stand
Sprinkler	Red Red White Drain
Hot Water Supply Dom./	
Solar Water	Green White H.W. Sup Dom/SW
Hot Water Return Dom./	
Solar Water	Green White H.W. Ret Dom/SW

(7) Electrical Conduits containing feeders over 600 volts, paint legends using 2 inch high black numbers and letters, showing the voltage class rating. Provide legends where conduits pass through walls and floors and at maximum 20 foot intervals in between. Use labels with yellow background with black border and words Danger High Voltage Class, 5000 .

(8) See Sections for methods of identification, legends, and abbreviations of the following:

(a) Conduits containing high voltage feeders over 600 volts: Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS / Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS.

b. Fire and Smoke Partitions:

(1) Identify partitions above ceilings on both sides of partitions except within shafts in letters not less than (2 1/2 inches) high.

(2) Stenciled message: "SMOKE BARRIER" or, "FIRE BARRIER" as applicable.

(3) Locate not more than 20 feet on center on corridor sides of partitions, and with a least one (1) message per room on room side of partition.

(4) Use semi-gloss paint of color that contrasts with color of

substrate.

c. Identify columns in pipe basements and interstitial space:

(1) Apply stenciled number and letters to correspond with grid numbering and lettering indicated on construction documents.

(2) Paint numbers and letters 4 inches high, locate (18 inches) below overhead structural slab.

(3) Apply on four (4) sides of interior columns and on inside face only of exterior wall columns.

(4) Color:

(a) Use black on concrete columns.

(b) Use white or contrasting color on steel columns.

3.13 PROTECTION CLEAN UP, AND TOUCH-UP:

a. Protect work from paint droppings and spattering by use of masking, drop cloths, removal of items or by other approved methods.

b. Upon completion, clean paint from hardware, glass and other surfaces and items not required to be painted of paint drops or smears.

c. Before final inspection, touch-up or refinished in a manner to produce solid even color and finish texture, free from defects in work which was damaged or discolored.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 10 - SPECIALTIES

SECTION 10 14 00

SIGNAGE

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY ASSURANCE
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY AND STORAGE
1.7 WARRANTY
1.8 REFERENCES

PART 2 PRODUCTS

2.1 SIGNAGE GENERAL
2.2 INTERIOR SIGN MATERIALS
2.3 INTERIOR SIGN TYPES
2.4 FABRICATION

PART 3 EXECUTION

3.1 INSTALLATION

-- End of Section Table of Contents --

SECTION 10 14 00
1/21

SIGNAGE

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies interior signage for room numbers, directional signs exterior signage, code required signs and temporary signs.

b. This section specifies exterior signage.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

b. Section 05 12 00, STRUCTURAL STEEL FRAMING: Structural Steel Supports.

d. Refer to SCHEDULE FOR FINISHES on the drawings.: Color and Finish of Interior Signs.c. DIVISION 26, ELECTRICAL Lighted EXIT signs for egress purposes are specified under and Electrical Work.

1.3 QUALITY ASSURANCE

a. Manufacturer’s Qualifications: Provide signage that is the product of one manufacturer, who has provided signage as specified for a minimum of three (3) years. Submit manufacturer’s qualifications.

b. Installer’s Qualifications: Minimum three (3) years’ experience in the installation of signage of the type as specified in this Section. Submit installer’s qualifications.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Manufacturer’s Qualifications; G, AE

Installer’s Qualifications; G, AE

SD-02 Shop Drawings

Sign Location Plan; G, AE

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature; G, AE

SD-04 Samples

Interior Sign Samples; G, AE

SD-05 Design Data

Sustainable Design Submittals; G, AE

Layout Patterns; G, AE

Structural Calculations; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES.

b. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

(2) Percentage of recycled content.

c. Interior Sign Samples: Sign panels and frames, with letters and symbols, for each sign type.

(1) Sign Panel, 8 x 10 inches, with letters.

(2) Color samples of each color, 6 x 6 inches. Show anticipated range of color and texture.

(3) Sample of typeface, arrow and symbols in a typical full size layout.

d. Manufacturer's Literature:

(1) Showing the methods and procedures proposed for the anchorage of the signage system to each surface type.

(2) Manufacturer’s printed specifications and maintenance instructions.

e. Sign Location Plan, showing location, type and total number of signs required.

f. Shop Drawings: Scaled for manufacture and fabrication of sign types. Identify materials, show joints, welds, anchorage, accessory items, mounting and finishes.

g. Full size layout patterns for dimensional letters.

h. Manufacturer’s qualifications.

i. Installer’s qualifications.

1.6 DELIVERY AND STORAGE

a. Deliver materials to job in manufacturer's original sealed containers with brand name marked thereon. Protect materials from damage.

b. Package to prevent damage or deterioration during shipment, handling, storage and installation. Maintain protective covering in place and in good repair until removal is necessary.

c. Deliver signs only when the site and mounting services are ready for installation work to proceed.

d. Store products in dry condition inside enclosed facilities.

1.7 WARRANTY

Comply with FAR clause 52.246-21, “Warranty of Construction”.

1.8 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1	(2017) Standard And Commentary Accessible and Usable Buildings and Facilities

ASTM INTERNATIONAL (ASTM)

ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B209M	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate (Metric)
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR 59	(1998) National Volatile Organic Compound Emission Standards For Consumer and Commercial Products
EPA Method 24	(2000) Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA Signage Design Manual	Signage Design Manual

PART 2 PRODUCTS

2.1 SIGNAGE GENERAL

a. Provide signs of type, size and design shown on the construction documents.

b. Provide signs complete with lettering, framing and related components for a complete installation.

c. Provide graphics items as completed units produced by a single manufacturer, including necessary mounting accessories, fittings and fastenings.

d. Do not scale construction documents for dimensions. Verify dimensions and coordinate with field conditions. Notify Contracting Officer of discrepancies or changes needed to satisfy the requirements of the construction documents.

e. Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.2 INTERIOR SIGN MATERIALS

a. Aluminum:

(1) Sheet and Plate: ASTM B209M (ASTM B209).

(2) Extrusions and Tubing: ASTM B221M (ASTM B221).

b. Cast Acrylic Sheet: Fed Spec MIL-PRF-8184F; Type II, class 1, Water white non-glare optically clear. Matt finish water white clear acrylic shall not be acceptable.

c. Polycarbonate: Fed Spec MIL-P-46144C; Type I, class 1.

d. Vinyl: Premium grade 0.004 inch thick machine cut, having a pressure sensitive adhesive and integral colors.

e. Adhesives:

(1) Adhesives for Field Application: Mildew-resistant, nonstaining adhesive for use with specific type of panels, sheets, or assemblies; and for substrate application; as recommended in writing by signage manufacturer.

(2) Adhesives to have VOC content of 50 g/L or less when calculated according to 40 CFR 59, (EPA Method 24).

f. Typography: Comply with VA Signage Design Manual.

(1) Type Style: Helvetica Medium and Helvetica Medium Condensed. Initial caps or all caps, as indicated in Sign Message Schedule .

(2) Arrow: Comply with graphic standards in construction documents.

(3) Letter spacing: Comply with graphic standards in construction documents.

(4) Letter spacing: Comply with graphic standards in construction documents.

(5) Provide text, arrows, and symbols in size, colors, typefaces and letter spacing shown in construction documents. Text shall be a true, clean, accurate reproduction of typeface(s). Text shown in construction documents is for layout purposes only; final text for signs is listed in Sign Message Schedule .

2.3 INTERIOR SIGN TYPES

a. Conform to the VA Signage Design Manual.

b. Provide component system as indicated on the drawings.

c. Component System Signs:

(1) Provide interior sign system as follows:

(a) Interchangeable system that allows for changes of graphic components of the installed sign, without changing sign in its entirety.

(b) Provide sign system comprised of following primary components: Rail Back: Horizontal rails, spaced to allow for uniform, modular sizing of sign types.

1. Rail Insert: Mount to back of Copy Panels to allow for attachment to Rail Back.

2. Copy Panels: Fabricate of ABS, phopolymer, acrylic, aluminum or stainless steel materials to allow for different graphic needs.

3. End Caps: Interlock to Rail Back to enclose and secure changeable Copy Panels.

4. Joiners and Accent Joiners: To connect separate Rail Backs together.

5. Top Accent Bars: To provide decorative trim cap that encloses the top of sign.

(c) Provide rail back, rail insert and end caps in anodized extruded aluminum.

(d) Provide signs in system that are convertible in the field to allow for enlargement from one (1) size to another in height and width through use of joiners or accent joiners, which connect rail back panels together blindly, providing a butt joint between copy panels. Connect accent joiners to rail backs with a visible 1/8"

horizontal rib, flush to the adjacent copy insert surfaces.

(e) Provide sign configurations as indicated on construction documents that vary in width from 9 inches to 80 inches), and have height dimensions of 2 inches, 3 inches, 6 inches, 9 inches and 12 inches. Height that can be increased beyond 12 inches, by repeating height module in full or in part.

(2) Provide rail back functions as internal structural member of sign. Fabricate of 6063T5-extruded aluminum, anodized black.

(a) Fabricate to accept an extruded aluminum or plastic insert on either side, depending upon sign type.

(b) Provide components that are convertible in field to allow for connection to other rail back panels.

(c) Provide mounting devices including wall mounting for screw-on applications, wall mounting with pressure sensitive tape and other mounting devices as needed.

(3) Provide rail insert functions as mounting device for copy panels on to the rail back. The rail insert mounts to the back of the copy panel with adhesive suitable for attaching particular copy insert material.

(a) Provide copy panels that slide or snap into the horizontal rail back.

(4) Provide copy panels that accept various forms of copy and graphics, and attach to the rail back with the rail insert. Provide copy panels fabricated of cast acrylic sheet.

(a) Provide copy panels that are interchangeable by sliding horizontally from either side of sign, and to other signs in system of equal or greater width or height.

(b) Provide materials that are cleanable without use of special chemicals or cleaning solutions.

(c) Copy Panel Materials.

1. ABS Inserts: .090 inches extruded ABS plastic core with .003 inches acrylic cap bonded during extrusion/texturing process.

a) Pressure bonded to extruded rail insert with adhesive.

b) Background Color: Integral or painted in acrylic lacquer.

c) Finished: Texture pattern.

2. Photopolymer Inserts: .125 inches phenolic photo polymer with raised copy etched to .0937 inches, bonded to an ABS plastic or extruded aluminum insert with adhesive.

a) a) Background Color: Painted, acrylic enamel.

3. Changeable Paper/ Insert Holder: Extruded insert holder

with integral rail insert for connection with structural back panel in 6063T5 aluminum with a black anodized finish.

a) Inserts into holder are paper with a clear .030 inches textured cover.

b) Background Color: Painted, acrylic lacquer.

4. Acrylic - .080 inches non-glare acrylic.

a) Pressure bonded to extruded rail insert using adhesive.

b) Background Color: Painted in acrylic lacquer or acrylic enamel.

5. Extruded 6063T5 aluminum with a black anodized finish insert holder with integral rail insert for connection with structural back panel to hold .030 inches textured polycarbonate insert and a sliding tile which mounts in the inset holder and slides horizontally.

(5) End Caps: Extruded using 6063T5 aluminum with a black anodized finish. End caps interlock with rail back with clips to form an integral unit, enclosing and securing the changeable copy panels, without requiring tools for assembly.

(a) Interchangeable to each end of sign and to other signs in signage system of equal height.

(b) Provide mechanical fasteners that can be added to the end caps that will secure it to rail back to make sign tamper resistant.

(6) Joiners: Extruded using 6063T5 aluminum with a black anodized finish. Rail joiners connect rail backs together blindly, providing a butt joint between copy inserts.

(7) Accent Joiners: Extruded using 6063T5 aluminum with a mirror polished finish. Connect joiner and rail backs together with a visible .125 inches horizontal rib, flush to the adjacent copy panel surfaces.

(8) Top Accent Rail: Extruded rail using 6063T5 aluminum with a mirror polished finish that provides a .125 inches high decorative trim cap. Cap butts flush to adjacent copy panel and encloses top of rail back and copy panel.

(9) Typography:

(a) Vinyl First Surface Copy (non-tactile): Applied vinyl copy.

(b) Subsurface Copy Inserts: Textured .030 inches clear polycarbonate face with subsurface applied vinyl copy.

1. Spray face back with paint and laminated to extruded aluminum carrier insert.

(c) Integral Tactile Copy Inserts: Phenolic photopolymer etched with .0937 inches raised copy.

(d) Silk-screened First Surface Copy (non-tactile): Injection molded or extruded ABS plastic or Aluminum insert with first surface applied enamel silk-screened copy.

d. Tactile Sign:

(1) Tactile sign made from a material that provides for letters, numbers and Braille to be integral with sign. Photopolymer etched metal, sandblasted phenolic or embossed material. Do not apply letters, numbers and Braille with adhesive.

(2) Numbers, letters and Braille to be raised 1/32 inches from the background surface. The draft of the letters, numbers and Braille to be tapered, vertical and clean.

(3) Braille Dots: Conform with ICC A117.1 for Braille position and layout; (a) Dot base diameter: .059 inches (b) Inter-dot spacing: .090 inches (c) Horizontal separation between cells: .241 inches (d) Vertical separation between cells: .395 inches

(4) Paint assembly specified color. After painting, apply white or other specified color to surface of the numbers and letters. Apply protective clear coat sealant to entire sign.

(5) Finish: Eggshell, 11 to 19 degree on a 60 degree glossmeter.

e. Provide cork or felt on bottom or mounting bracket when sign is mounted on counter or desk.

f. For ceiling mounted signs, provide mounting hardware on the sign that allows for sign disconnection, removal, reinstallation, and reconnection.

h. Dimensional Letters:

(1) Provide dimensional letters that are mill or laser cut acrylic in size and thickness indicated in construction documents.

(2) Provide draft of letters perpendicular to letters face.

(3) Fabricate letters with square corners, such as where a letter stem and bar intersect.

(4) Paint letters with acrylic polyurethane.

i. Temporary Interior Signs:

(1) Fabricated from 110 pound matte finished white paper cut to 4 inch wide by 12 inch long.

(a) Punched .125 inch hole with edge of hole spaced .5 inch in from edge and centered on 4 inch side.

(b) Reinforce hole on both sides with suitable material that prevents tie from pulling through hole.

(c) Ties: Steel wire 0.120 inch thick attached to tag with twist leaving 6 inch long free ends.

(2) Mark architectural room number on sign, with broad felt marker in clearly legible numbers or letters that identify room, corridor or space as shown on construction documents.

(3) Install temporary signs to rooms that have a room, corridor or space number. Attach to door frame, door knob or door pull.

(a) Doors that do not require signs are: corridor doors in corridor with same number, folding doors or partitions, toilet doors, bathroom doors within and between rooms, closet doors within rooms, communicating doors in partitions between rooms with corridor entrance doors.

(b) Replace and missing, damaged or illegible signs.

2.4 FABRICATION

a. Design interior signage components to allow for expansion and contraction for a minimum material temperature range of 100 degrees F, without causing buckling, excessive opening of joints or over stressing of adhesives, welds and fasteners.

b. Form work to required shapes and sizes, with true curve lines and angles. Provide necessary rebates, lugs and brackets for assembly of units. Provide concealed fasteners wherever possible.

c. Shop fabricate so far as practicable. Fasten joints flush to conceal reinforcement, or weld joints, where thickness or section permits.

d. Level and assemble contract surfaces of connected members so joints will be tight and practically unnoticeable, without applying filling compound.

e. Signs: Fabricate with fine, even texture to be flat and sound.

(1) Maintain lines and miters sharp, arises unbroken, profiles accurate and ornament true to pattern.

(2) Plane surfaces to be smooth, flat and without oil-canning, free of rack and twist.

(3) Maximum variation from plane of surface plus or minus 0.015 inches. Restore texture to filed or cut areas.

f. Finish extruded members to be free from extrusion marks. Fabricate square turns, sharp corners, and true curves.

g. Finish hollow signs with matching material on all faces, tops, bottoms and ends. Miter edge joints to give appearance of solid material.

h. Do not manufacture signs until final sign message schedule and location review has been completed by the Contracting Officer and forwarded to contractor.

i. Drill holes for bolts and screws. Mill smooth exposed ends and edges with corners slightly rounded.

j. Form joints exposed to weather to exclude water.

k. Movable Parts, Including Hardware: Cleaned and adjusted to operate as designed without binding or deformation of members. Center doors and covers in opening or frame.

(1) Align contact surfaces fit tight and even without forcing or warping components.

l. Pre-assemble items in shop to minimize field splicing and assembly. Disassemble units only as necessary for shipping and handling limitations. Clearly mark units for re-assembly and coordinated installation.

m. Prime painted surfaces as required. Apply finish coating of paint for complete coverage with no light or thin applications allowing substrate or primer to show.

(1) Finish surface smooth, free of scratches, gouges, drips, bubbles, thickness variations, foreign matter and other imperfections.

PART 3 EXECUTION

3.1 INSTALLATION

a. Locate signs as shown on the construction documents .

b. Where not otherwise indicated conform to the VA Signage Design Manual for installation requirements.

c. At each sign location there are no utility lines behind each sign location that will be affected by installation of signs.

(1) Correct and repair damage done to utilities during installation of signs at no additional cost to Government.

d. Provide inserts and anchoring devices which must be set in concrete or other material for installation of signs. Submit setting drawings, templates, instructions and directions for installation of anchorage devices, which may involve other trades.

e. Refer to Sign Message Schedule for mounting method. Mount signs in proper alignment, level and plumb according to the Sign Location Plan and the dimensions given on elevation and Sign Location Plans. When exact position, angle, height or location is not clear, contact Contracting Officer for resolution.

f. When signs are installed on glass, provide blank glass back up to be placed on opposite side of glass exactly behind sign being installed. Provide blank glass back that is the same size as sign being installed.

g. Touch up exposed fasteners and connecting hardware to match color and finish of surrounding surface.

h. At completion of sign installation, clean exposed sign surfaces. Clean and repair adjoining or adjacent surfaces that became soiled or damaged as a result of installation of signs.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 10 - SPECIALTIES

SECTION 10 22 13

WIRE MESH PARTITIONS

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY ASSURANCE
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 WARRANTY
1.7 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 NORMAL DUTY PARTITIONS
2.3 FABRICATION

PART 3 EXECUTION

3.1 INSTALLATION
3.2 ACCEPTANCE

-- End of Section Table of Contents --

SECTION 10 22 13
1/21

WIRE MESH PARTITIONS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies steel mesh partitions complete with doors, and hardware.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

1.3 QUALITY ASSURANCE

a. Manufacturer’s Qualifications: Manufacturer with three (3) years’ experience in providing items of types specified. Submit manufacturer’s qualifications.

b. Obtain wire mesh partitions from single manufacturer.

c. Installer’s Qualifications: Installers who have three (3) years’ experience in the installation of units required for this project. Submit installer’s qualifications.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Manufacturer’s Qualifications; G, AE

Installer’s Qualifications; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Sustainable Design Submittals; S

SD-05 Design Data

Erection Drawings And Specifications; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Sustainable Design Submittals, as described below:

(1) Postconsumer and preconsumer recycled content as specified in PART 2 - PRODUCTS.

c. Shop Drawings: Mesh partitions showing design, construction and materials.

d. Submit layout drawings with detailed erection drawings and specifications.

e. Manufacturer’s qualifications.

f. Installer’s qualifications.

1.6 WARRANTY

Comply with FAR clause 52.246-21 “Warranty of Construction”.

1.7 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation also.

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A500/A500M	(2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A510/A510M	(2020) Standard Specification for General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel, and Alloy Steel
ASTM A513/A513M	(2024) Standard Specification for Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength

Low-Alloy with Improved Formability,
Solution Hardened, and Bake Hardenable

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 MATERIALS

a. Steel Wire: ASTM A510/A510M.

b. Steel Plates, Channels, Angles, and Bars: ASTM A36/A36M.

c. Steel Sheet: Cold-rolled steel sheet, ASTM A1008/A1008M, Commercial Steel (CS), Type B.

d. Steel Pipe: ASTM A53/A53M, Schedule 40, unless another weight is indicated or required by structural loads.

e. Steel Tubing: ASTM A500/A500M, cold-formed structural-steel tubing or ASTM A513/A513M, Type 5, mandrel-drawn mechanical tubing.

f. Metallic-Coated Steel Sheet: ASTM A653/A653M, Commercial Steel (CS), type B; with G60 (Z180) zinc (galvanized) or A60 (ZF180) zinc-iron-alloy (galvannealed) coating designation.

g. Panel-to-Panel Fasteners: Manufacturer’s standard steel bolts, nuts, and washers.

h. Recycled Content of Steel Products: Post consumer plus one-half of preconsumer content not less than 30 percent.

2.2 NORMAL DUTY PARTITIONS

a. Woven Wire: 1-1/2 inch diamond mesh No. 10 gauge 0.1345 inch diameter uncoated steel crimped and woven.

b. Steel Shapes, Plates and Bars:

(1) Vertical Channel: 1-1/4 x 5/8 x 1/8 inch.

(2) Horizontal Channel: 1 x 5/8 inch.

(3) Center Reinforcing Bar: One (1) 1 x 1/2 x 1/8 inch channel with all wires woven through, or two (2) 1 x 3/8 x 1/8 inch channels bolted together with mesh in between.

(4) Corner Post: 1-1/4 x 1-1/4 x 1/8 inch angle.

(5) Capping Bar: 2-1/4 x 1 x 1/8 inch channel or 2 x 1/4 inch flat bar.

(6) Cast or forged adjustable floor shoes.

c. Doors and Service Window:

(1) Hinged Door:

(a) Frame: 1-1/4 x 1/2 x 1/8 inch channel, with a midpoint channel.

(b) Hardware: 1-1/2 pair butts NRP 4 inch. Pick proof mortise type lock, key operated outside, recessed knob inside.

(c) Miscellaneous: Provide sheet metal baffle at lock, continuous angle stop and flat bar closures.

(4) Finish:

(a) Hot dip galvanized.

2.3 FABRICATION

a. Woven wire clinched to frame, mortise and tenon joints. Frame units to be maximum 5 feet wide.

PART 3 EXECUTION

3.1 INSTALLATION

a. Erect the partition in accordance with the manufacturer’s shop drawings.

b. Secure top reinforcing channels with 1/4-inch "U" bolts, 28 inches on center.

c. Secure vertical posts with 1/4 inch bolts 12 to 15 inches on center, and anchor verticals to wall 15 inches on center, shim as required.

d. Provide floor shoes at each post and each corner, adjust to level, anchor to floor with two (2) anchors for each shoe.

e. Frame penetrations for building structure and mechanical/plumbing, openings with “U” cap terminations. Openings with unfinished wire mesh are not acceptable.

3.2 ACCEPTANCE

a. Repair or replace damaged parts, touch-up abraded paint with matching paint.

b. Install partitions level and firm. Adjust hardware to operate smoothly and latch securely.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 10 - SPECIALTIES

SECTION 10 26 00

WALL AND DOOR PROTECTION

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY ASSURANCE
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 DELIVERY AND STORAGE
1.7 WARRANTY
1.8 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 CORNER GUARDS
2.3 FASTENERS AND ANCHORS
2.4 FINISH

PART 3 INSTALLATION

3.1 STAINLESS STEEL CORNER GUARDS
3.2 STAINLESS STEEL WALL GUARDS
3.3 DOOR, DOOR FRAME PROTECTION AND HIGH IMPACT WALL COVERING

-- End of Section Table of Contents --

SECTION 10 26 00
1/21

WALL AND DOOR PROTECTION

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies wall guards, handrail/wall guard combinations, corner guards and door/door frame protectors .

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Sustainable Design Requirements.

b. Section 05 50 00, METAL FABRICATIONS: Structural Steel Corner Guards.

c. Section 08 71 00, DOOR HARDWARE: Armor plates and kick plates not specified in this section.

1.3 QUALITY ASSURANCE

a. Manufacturer’s Qualifications: Manufacturer with a minimum of three (3) years’ experience in providing items of type specified.

(1) Obtain wall and door protection from single manufacturer.

b. Installer’s Qualifications: Installers are to have a minimum of three (3) years’ experience in the installation of units required for this project.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Manufacturer’s Qualifications; G, AE

Installer’s Qualifications; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Sustainable Design Submittals; S

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Test Report; G

SD-11 Closeout Submittals

Manufacturer’s Warranty; G

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Sustainable Design Submittals, as described below:

(1) Volatile organic compounds per volume as specified in PART 2 - PRODUCTS.

(2) Percentage of recycled content.

c. Shop Drawings: show design and installation details.

d. Manufacturer's Literature and Data:

(1) Corner Guards.

(2) High Impact Wall covering.

e. Test Report: Showing that resilient material complies with specified fire and safety code requirements.

f. Manufacturer’s qualifications.

g. Installer’s qualifications.

h. Manufacturer’s warranty.

1.6 DELIVERY AND STORAGE

a. Deliver materials to the site in original sealed packages or containers marked with the name and brand, or trademark of the manufacturer.

b. Protect from damage from handling and construction operations before, during and after installation.

c. Store in a dry environment of approximately (70 degrees F) for at least 48 hours prior to installation.

1.7 WARRANTY

a. Construction Warranty: Comply with FAR clause 52.246-21 “Warranty of Construction”.

b. Manufacturer Warranty: Manufacturer shall warranty their wall and door protection for a minimum of five (5) years from date of installation and final acceptance by the Government. Submit manufacturer warranty.

1.8 REFERENCES

a. publications listed below form a part of this specification to extent referenced. publications are referenced in text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80	(2025) Standard for Fire Doors and Other Opening Protectives

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 MATERIALS

a. Stainless Steel: ASTM A240/A240M, Type 304.

2.2 CORNER GUARDS

a. Fabricate stainless steel corner guards of (.05 inch) thick material conforming to ASTM A240/A240M, Type 302 . Install corner guards as indicated on construction documents. Form corner guard to dimensions shown on construction documents.

2.3 FASTENERS AND ANCHORS

a. Provide fasteners and anchors as required for each specific type of installation.

b. Where type, size, spacing or method of fastening is not shown or specified in construction documents, submit shop drawings showing proposed installation details.

2.4 FINISH

a. Stainless Steel: In accordance with NAAMM AMP 500 finish Number 4.

PART 3 INSTALLATION

3.1 STAINLESS STEEL CORNER GUARDS

a. Mount guards on external corners of interior walls, partitions and columns as shown on construction documents.

b. Where corner guards are installed on walls, partitions or columns finished with plaster or ceramic tile, anchor corner guards as shown on construction documents. wired to metal studs expansion bolt to concrete or masonry with four (3/8-inch) diameter bolts, spaced 16 inches on centers Coat back surfaces of corner guards, where shown on construction documents, with a non-flammable, sound deadening material. Corner guards to overlap finish plaster surfaces.

c. Where corner guards are installed on exposed structural glazed facing tile units or masonry wall, partitions or columns, anchor corner guards as shown on the construction documents. Grout spaces solid between guards and backing with Portland cement and sand mortar.

d. Where corner guards are installed on gypsum board, clean surface and anchor guards with a neoprene solvent-type contact adhesive specifically manufactured for use on gypsum board construction. Remove excess adhesive from around edge of guard and allow curing undisturbed for 24 hours.

3.2 STAINLESS STEEL WALL GUARDS

Space brackets at not more than 3 feet on centers and anchor to the wall in accordance with manufacturer's installation instructions.

3.3 DOOR, DOOR FRAME PROTECTION AND HIGH IMPACT WALL COVERING

a. Surfaces to receive protection to be clean, smooth and free of obstructions.

b. Install protectors after frames are in place but preceding installation of doors in accordance with approved shop drawings and manufacturer’s specific instructions.

c. Apply with adhesive in controlled environment according to manufacturer’s recommendations.

d. Protection installed on fire rated doors and frames to be installed according to NFPA 80 and installation procedures listed in UL Building Materials Directory; or, equal listing by other approved independent testing laboratory establishing the procedures.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 10 - SPECIALTIES

SECTION 10 28 00

TOILET, BATH, AND LAUNDRY ACCESSORIES

1/21

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 REFERENCES
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 QUALITY ASSURANCE
1.7 DELIVERY
1.8 STORAGE AND HANDLING
1.9 WARRANTY

PART 2 PRODUCTS

2.1 MATERIALS
2.2 PRODUCTS - GENERAL
2.3 PAPER TOWEL DISPENSERS
2.4 TOILET TISSUE DISPENSERS
2.5 GRAB BARS
2.6 SHOWER CURTAIN RODS
2.7 CLOTHES HOOKS, ROBE OR COAT
2.8 TOWEL BARS
2.9 METAL FRAMED MIRRORS
2.10 MOP RACKS
2.11 FABRICATION - GENERAL
2.12 FINISH
2.13 ACCESSORIES

PART 3 EXECUTION

3.1 PREPARATION
3.2 INSTALLATION
3.3 CLEANING
3.4 PROTECTION

-- End of Section Table of Contents --

SECTION 10 28 00
1/21

TOILET, BATH, AND LAUNDRY ACCESSORIES

PART 1 GENERAL

1.1 DESCRIPTION

a. SUMMARY:

(1) Section Includes: Toilet and bath accessories at dressing rooms, toilets, baths, locker rooms and other areas indicated on drawings.

1.2 RELATED WORK

a. Refer to SCHEDULE FOR FINISHES on the drawings.
a. Section 09 30 13, CERAMIC/PORCELAIN TILING: Ceramic Toilet and Bath Accessories.

1.3 REFERENCES

a. Comply with references to extent specified in this section.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.6.4	(1998; R2005) Thread Forming and Thread Cutting Tapping Screws and Metallic Drive Screws, Inch Series

AMERICAN WELDING SOCIETY (AWS)

AWS D10.4	(1986; R 2000) Recommended Practices for Welding Austenitic Chromium-Nickel Stainless Steel Piping and Tubing

ASTM INTERNATIONAL (ASTM)

ASTM A269/A269M	(2024) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A312/A312M	(2022a) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A666	(2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM C1036	(2021) Standard Specification for Flat

	Glass
ASTM F446	(2018; R 2023) Standard Consumer Safety Specification for Grab Bars and Accessories Installed in the Bathing Area

U.S. DEPARTMENT OF DEFENSE (DOD)

FF-S-107C	(2013) Screws, Tapping and Drive

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500	(2006) Metal Finishes Manual

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS WW-P-541	(Rev E; Am 1; Notice 1; Notice 2) Plumbing Fixtures
A-A-3002	(1996)Mirror, Glass.

1.4 SUBMITTALS

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SD-01 Preconstruction Submittals

Qualifications; G, AE

SD-02 Shop Drawings

Submittal Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-04 Samples

Samples; G, AE

SD-07 Certificates

Certificates; G

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00, SUBMITTAL PROCEDURES.

b. Submittal Drawings:

(1) Show size, configuration, and fabrication, anchorage and installation details.

(2) Show mounting locations and heights.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

(3) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

d. Samples:

(1) Full sized, complete assembly of each product specified.

(2) Approved samples may be incorporated into project.

e. Certificates: Certify each product complies with specifications.

(1) Soap dispensers: Certify soap dispensers are fabricated of material that will not be affected by liquid soap, aseptic detergents, and hexachlorophene solutions.

f. Qualifications: Substantiate qualifications comply with specifications.

(1) Manufacturer with project experience list .

g. Operation and Maintenance Data:

(1) Care instructions for each exposed finish product.

1.6 QUALITY ASSURANCE

a. Manufacturer Qualifications:

(1) Regularly manufactures specified products.

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction

operations.

1.9 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 MATERIALS

a. Stainless Steel:

(1) Plate Or Sheet: ASTM A666, Type 304, 0.031 inch thick unless otherwise specified.

(2) Tubing: ASTM A269/A269M, Grade TP 304, seamless or welded.

(3) Pipe: ASTM A312/A312M; Grade TP 304.

b. Steel Sheet: ASTM A653/A653M, zinc-coated (galvanized) coating designation G90.

c. Glass:

(1) ASTM C1036, Type 1, Class 1, Quality q2, for mirrors, and for mirror doors in medicine cabinets.

(2) ASTM C1036, Type 1 Class 1 Quality q3, for shelves in medicine cabinets.

2.2 PRODUCTS - GENERAL

a. Provide each product from one manufacturer.

2.3 PAPER TOWEL DISPENSERS

a. Surface mounted type with sloping top.

b. Dispensing capacity for 300 sheets of any type of paper toweling.

c. Fabricate of stainless steel.

d. Provide door with continuous hinge at bottom, and spring tension cam lock or tumbler lock, keyed alike, at top, and refill sight slot in

front.

2.4 TOILET TISSUE DISPENSERS

a. Double roll surface mounted type.

b. Mount on continuous backplate.

c. Removable spindle ABS plastic or chrome plated plastic.

d. Wood rollers are not acceptable.

2.5 GRAB BARS

a. FS WW-P-541/8B, Type IV, bars, surface mounted, Class 2, grab bars and complying with ASTM F446.

b. Fabricate from stainless steel or nylon coated steel, use one type throughout project:

(1) Stainless steel: Grab bars, flanges, mounting plates, supports, screws, bolts, and exposed nuts and washers.

c. Mounting:

(3) Metal Toilet Dressing Partitions Mounted Grab Bars: Exposed type.

(4) Other Types and Locations: Concealed type.

d. Bars:

(1) Fabricate to 1-1/2 inch outside diameter.

(a) Stainless steel, minimum 0.05 inch thick.

(2) Fabricate in one continuous piece with ends turned toward walls.

(3) Continuously weld intermediate support to grab bar.

e. Flange for Concealed Mounting:

(1) Minimum 0.1 inch thick, maximum 3-1/8 inch diameter by 1/2 inch deep, with minimum three set screws for securing flange to back plate.

(2) Insert grab bar through center of flange and continuously weld perimeter of grab bar flush to back side of flange.

(3) In lieu of providing flange for concealed mounting, and back plate as specified, grab bar may be welded to back plate covered with flange.

f. Flange for Exposed Mounting:

(1) Minimum 3/16 inch thick, maximum 3-1/8 inch diameter.

(2) Insert grab bar through flange and continuously weld perimeter of grab bar flush to backside of flange.

(4) Where mounted on floor, provide four equally spaced holes, sized to accommodate 3/8 inch diameter bolts, maximum 3/8 inch from edge of flange.

g. Back Plates:

(1) Minimum 0.1046 inch thick metal.

(2) Fabricate in one piece, maximum 1/4 inch deep, with diameter sized to fit flange. Provide slotted holes to accommodate anchor bolts.

(3) Provide spreaders, through bolt fasteners, and cap nuts, where grab bars are mounted on partitions.

2.6 SHOWER CURTAIN RODS

a. Stainless steel tubing, minimum 0.050 inch wall thickness, 1-1/4 inch outside diameter.

b. Flanges, stainless steel rings, 2.6 inch minimum outside diameter, with 2 holes opposite each other for 1/4 inch stainless steel fastening bolts. Provide set screw within curvature of each flange for securing rod.

c. Intermediate Support: For rods over 72 inches long. Provide adjustable ceiling flanges with set screws, tubular hangers and stirrups.

2.7 CLOTHES HOOKS, ROBE OR COAT

a. Fabricate hook units from chromium plated brass with satin finish, or stainless steel, using 1/4 inch minimum thick stock, with edges and corners rounded smooth to thickness of metal, or 1/8 inch minimum radius.

b. Fabricate each unit as a double hook on a single shaft, integral with or permanently fastened to wall flange, provided with concealed fastenings.

c. Clothes Hooks Used In Mental Health And Behavioral Patient Care Units: Provide units free of anchor points and secured to the wall using tamper resistant hardware.

2.8 TOWEL BARS

a. FS WW-P-541/8B, Type IV, Bar, Surface mounted; Class 1, towel.

b. Stainless steel, or chromium plated copper alloy.

c. Bar Length: 18 and 24 inches as shown.

d. Finish brackets and supports to match bar.

2.9 METAL FRAMED MIRRORS

a. A-A-3002 metal frame; stainless steel .

b. Mirror Glass:

(1) Minimum 1/4 inch thick.

(2) Set mirror in a protective vinyl glazing tape.

c. Frames:

(1) Channel or angle shaped section with face of frame minimum 3/8 inch wide. Fabricate with square corners.

(2) Metal Thickness 0.035 inch.

(3) Filler:

(a) Where mirrors are mounted on walls having ceramic tile wainscots not flush with wall above, provide fillers contoured to conceal void between back of mirror and wall surface.

(b) Fabricate fillers from same material and finish as mirror frame.

(4) Attached Shelf for Mirrors:

(a) Fabricate shelf of same material and finish as mirror frame.

(b) Make shelf maximum 6 inches in depth, and extend full width of mirror.

(c) Close ends and front edge of shelf to same thickness as mirror frame width.

(d) Form shelf for aluminum framed mirror as integral part of bottom frame member.

(e) Form stainless steel shelf with concealed brackets to attach to mirror frame.

d. Back Plate:

(1) Fabricate backplate for concealed wall hanging from zinc-coated, or cadmium plated 0.036 inch thick sheet steel, die cut to fit face of mirror frame.

(2) Provide set screw type theft resistant concealed fastening system for mounting mirrors.

e. Mounting Bracket:

(1) Designed to support mirror tight to wall.

(2) Designed to retain mirror with concealed set screw fastenings.

f. Metal Framed Mirrors used in Mental Health and Behavioral Patient Care Units: Provide shatter proof glass or polished stainless steel units.

2.10 MOP RACKS

a. Minimum 40 inches long with five holders.

b. Clamps:

(1) Minimum of 0.05 inch thick stainless steel bracket retaining channel with hard rubber serrated cam; pivot mounted to channel.

(2) Clamps to hold handles from 1/2 inch minimum to 1-1/4 inch maximum diameter.

c. Support:

(1) Minimum 0.04 inch thick stainless steel hat shape channel to hold clamps away from wall as indicated.

(2) Drill wall flange for 1/8 inch fasteners above and below clamp locations.

d. Secure clamps to support with oval head machine screws or rivets into continuous reinforcing back of clamps.

2.11 FABRICATION - GENERAL

a. Welding, AWS D10.4.

b. Grind, dress, and finish welded joints to match finish of adjacent surface.

c. Form exposed surfaces from one sheet of stock, free of joints.

d. Provide steel anchors and components required for secure installation.

e. Form flat surfaces without distortion. Keep exposed surfaces free from scratches and dents. Reinforce doors to prevent warp or twist.

f. Isolate aluminum from dissimilar metals and from contact with building materials as required to prevent electrolysis and corrosion.

g. Hot-dip galvanized steel or stainless steel, anchors and fastening devices.

h. Shop assemble accessories and package with components, anchors, fittings, fasteners and keys.

i. Key items alike.

j. Provide templates and rough-in measurements.

k. Round and deburr edges of sheets to remove sharp edges.

2.12 FINISH

a. Stainless Steel: NAAMM AMP 500; No. 4 polished finish.

2.13 ACCESSORIES

a. Fasteners:

(1) Fasteners in Mental Health and Behavioral Patient Care Units: Tamper resistant hot-dipped galvanized or stainless steel.

(2) Exposed Fasteners: Stainless steel or chromium plated brass, finish to match adjacent surface.

(3) Concealed Fasteners:

(a) Shower, Bath Tubs, and High Moisture Areas: Stainless steel.

(b) Other Locations: Steel, hot-dipped galvanized.

(4) Toggle Bolts: For use in hollow masonry or frame construction.

(5) Sex bolts: For through bolting on thin panels.

(6) Expansion Shields: Lead or plastic for solid masonry and concrete substrate as recommended by accessory manufacturer to suit application.

(7) Screws:

(a) ASME B18.6.4.

(b) FF-S-107C Stainless steel Type A.

b. Adhesive: As recommended by manufacturer to suit application.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

(1) Verify blocking to support accessories is installed and located correctly.

b. Verify location of accessories with Contracting Officer's Representative.

c. Provide labor or prep as required for VA-furnished and contractor installed or VA-furnished and installed components.

3.2 INSTALLATION

a. Install products according to manufacturer's instructions and approved submittal drawings .

(1) When manufacturer's instructions deviate from specifications,

submit proposed resolution for Contracting Officer's Representative consideration.

b. Install grab bars according to ASTM F446.

c. Set work accurately, in alignment and where indicated, parallel or perpendicular as required to line and plane of surface. Install accessories plumb, level, free of rack and twist.

d. Toggle bolt to steel anchorage plates in frame partitions and hollow masonry.

e. Install accessories to function as designed. Perform maintenance service without interference with performance of other devices.

f. Position and install dispensers, and other devices in countertops, clear of drawers, permitting ample clearance below countertop between devices, and ready access for maintenance.

g. Align mirrors, dispensers and other accessories even and level, when installed in battery.

h. Install accessories to prevent striking by other moving, items or interference with accessibility.

3.3 CLEANING

After installation, clean toilet accessories according to manufacturer’s instructions.

3.4 PROTECTION

Protect accessories from damage until project completion.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 10 - SPECIALTIES

SECTION 10 44 13

FIRE EXTINGUISHER CABINETS

8/18

PART 1 GENERAL

1.1 REFERENCES
1.2 DESCRIPTION
1.3 RELATED WORK
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 REFERENCES

PART 2 PRODUCTS

2.1 FIRE EXTINGUISHER CABINET
2.2 FABRICATION
2.3 FINISH

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 10 44 13
8/18

FIRE EXTINGUISHER CABINETS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D4802	(2016) Standard Specification for Poly(Methyl Methacrylate) Acrylic Plastic Sheet

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10	(2022; ERTA 1 2021) Standard for Portable Fire Extinguishers

1.2 DESCRIPTION

This section covers recessed fire extinguisher cabinets.

1.3 RELATED WORK

a. Acrylic glazing: Section 08 80 00, GLAZING.

b. Field Painting: Section 09 91 00, PAINTING.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Manufacturer's Literature and Data: Fire extinguisher cabinet including installation instruction and rough opening required.

1.6 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D4802	(2016) Standard Specification for Poly(Methyl Methacrylate) Acrylic Plastic Sheet

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 FIRE EXTINGUISHER CABINET

Recessed type with flat trim of size and design shown.

2.2 FABRICATION

a. Form body of cabinet from 0.0359 inch thick sheet steel.

b. Fabricate door and trim from 0.0478 inch thick sheet steel with all face joints fully welded and ground smooth.

(1) Glaze doors with 1/4 inch thick ASTM D4802, clear acrylic sheet, Category B-1, Finish 1.

(2) Design doors to open 180 degrees.

(3) Provide continuous hinge, pull handle, and adjustable roller catch.

2.3 FINISH

a. Finish interior of cabinet body with baked-on semigloss white enamel.

b. Finish door, frame with manufacturer's standard baked-on prime coat suitable for field painting.

PART 3 EXECUTION

a. Install fire extinguisher cabinets in prepared openings and secure in accordance with manufacturer's instructions.

b. Install cabinet so that the extinguisher height within meets the requirements of NFPA 10

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 12 - FURNISHINGS

SECTION 12 24 00

WINDOW SHADES

8/17

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALITY ASSURANCE
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 WARRANTY
1.7 REFERENCES

PART 2 PRODUCTS

2.1 CLOTH SHADES
2.2 MATERIALS
2.3 FASTENINGS
2.4 FABRICATION

PART 3 EXECUTION

3.1 INSTALLATION
3.2 ADJUSTING
3.3 CLEANING AND PROTECTION
3.4 DEMONSTRATION

-- End of Section Table of Contents --

SECTION 12 24 00
8/17

WINDOW SHADES

PART 1 GENERAL

1.1 DESCRIPTION

This Section includes cloth shades, vertical blinds and venetian blinds. Provide window shades complete, including brackets, fittings and hardware.

1.2 RELATED WORK

a. Color of shade cloth as shown on the SCHEDULE FOR FINISHES on the drawings.

1.3 QUALITY ASSURANCE

a. Manufacturer's Qualification: Submit evidence that the manufacture has a minimum of three years' experience in providing item of type specified, and that the blinds have performed satisfactorily on similar installations. Submit qualifications.

b. Submit qualifications for installers who are trained and approved by manufacturer for installation of units provided.

c. Electrical Requirements:

(1) NFPA 70 Article 100.

(2) Listed and labeled in accordance with UL 325.

(3) Marked for intended use, and tested as a system.

(4) Individual testing of components is not acceptable in lieu of system testing.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G

SD-04 Samples

Samples; G, AE

SD-06 Test Reports

Fire Testing; G

SD-11 Closeout Submittals

Manufacturer’s Warranty; G

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Samples:

(1) Shade cloth, each type, 24 inch square, including cord and ring, showing color, finish and texture.

(2) Venetian blind slats, 12 inches long, including cord and tape, showing color and finish.

c. Manufacturer's literature and data; showing details of construction and hardware for:

(1) Cloth and window shades

(2) Percentage of recycled content.

(3) Environmental Product Declarations, Health Product Declarations, other third-party verified documentation for ingredient inventory and/or GWP.

d. Shop Drawings: Provide fabrication and installation details for cloth shades, including shade cloth materials, their orientation to rollers, and their seam and batten locations. Motor-Operated Shades: Include details of installation and diagrams for power, signal, and control wiring.

e. Fire Testing: Submit report of flame spread and smoke developed during product material tests by independent testing laboratory.

f. Manufacturer’s warranty.

1.6 WARRANTY

a. Construction Warranty: Comply with FAR clause 52.246-21, “Warranty of Construction”.

b. Manufacturer Warranty: Manufacturer shall warranty their window shades for a minimum of five years from date of installation and final acceptance by the Government. Submit manufacturer’s warranty.

1.7 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B221M	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 701	(2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

UL SOLUTIONS (UL)

UL 325	(2017; Reprint Feb 2020) UL Standard for Safety Door, Drapery, Gate, Louver, and Window Operators and Systems

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 CLOTH SHADES

Light-Filtering Shade Cloth: Woven fabric, stain and fade resistant.

a. Type: Single roll.

b. Weave: Basketweave.

c. Thickness: 0.034 inches.

d. Weight: 13.57 ounces per square yard.

e. Orientation on Shadeband: Up the bolt.

f. Openness Factor: 3 percent.

g. Fire-Test-Response Characteristics: Passes NFPA 701 small and large-scale vertical burn. Submit report for testing of shade cloth

materials identical to products provided.

h. Drive-End Location: As indicated on construction documents.

2.2 MATERIALS

a. Stainless Steel: ASTM A240/A240M.

b. Extruded Aluminum: ASTM B221M (ASTM B221).

c. Cords for Cloth roller shades: or #10 stainless steel chain having not less than 175 pounds breaking strength.

2.3 FASTENINGS

Zinc-coated or cadmium plated steel or stainless steel fastenings of length and type recommended by manufacturer. Except as otherwise specified, provide fastenings for installation with various structural materials as follows:

Type of Fastening	Structural Material
Wood screw	Wood
Tap screw	Metal
Case-hardened, self-tapping screw in pre-drilled hole	Solid masonry, concrete
Screw or bolt in expansion shields	Solid masonry, concrete
Toggle bolts	Hollow blocks, gypsum wallboard, plaster

2.4 FABRICATION

a. Fabricate cloth shades to fit measurements of finished openings obtained at site.

b. Cloth Shades: Rolling type, constructed of shade cloth mounted on rollers. Provide shade cloth with plain sides, and with hem at bottom to accommodate weight bar. Provide separate shades for each individual sash within opening. Provide shade length that exceeds height of window by 12 inches measured from head to sill, in addition to material required to make-up hem:

(1) Provide rollers with spindles, nylon bearings, tempered steel springs, and other related accessories required for positive action.

(2) Provide rollers of diameter and wall thicknesses required to accommodate operating mechanisms, weights, and widths of shadebands indicated without deflection.

(3) Provide rollers with permanently lubricated drive-end assemblies and idle-end assemblies designed to facilitate removal of shadebands for service.

(4) Secure shade cloth to rollers to prevent wrinkling or folding, and on line parallel to axis of rollers so that shade hangs plumb.

(5) Secure shade cloth with zinc-coated steel or stainless steel machine screws spaced not over 9 inches on centers.

(6) Do not attach shade cloth to rollers with tacks.

(7) Provide hem bar of extruded aluminum for entire width of shade band. Heat seal hem bar on all sides to prevent removal.

(8) Provide eyelets with clear openings large enough to accommodate cords, without cutting into cloth when set.

(9) Provide cords of sufficient length to permit shades to be drawn to bottom of opening with ends looped and held with cord rings. Attach cords to hems through metal eyelets in center of slats in bottom hems.

PART 3 EXECUTION

3.1 INSTALLATION

a. Measure openings before fabrication. Do not scale construction documents.

b. Cloth Shades: Mount window shades on end of face brackets, set on metal gussets, or casing of windows as required. Provide extension face brackets where necessary at mullions.

(1) Locate rollers in level position as high as practicable at heads of windows.

(2) Install shades to prevent infiltration of light over rollers.

(3) Where extension brackets are necessary for alignment of shades, provide metal lugs, and rigidly anchor lugs and brackets.

(4) Place brackets and rollers so that shades do not interfere with window and screen hardware.

(5) Mount shade to allow clearances for window operation hardware.

(6) Shade installation methods not specifically described, are subject to approval of Contracting Officer.

3.2 ADJUSTING

Adjust and shades to operate smoothly, free from binding or malfunction throughout entire operational range.

3.3 CLEANING AND PROTECTION

a. Clean shade surfaces after installation, according to manufacturer’s written instructions.

b. Provide final protection and maintain conditions that ensure that shades are without damage or deterioration at time of Substantial Completion.

c. Replace damaged shades that cannot be repaired, in a manner approved by Contracting Officer before time of Substantial Completion.

3.4 DEMONSTRATION

Furnish services of factory-authorized service representative to train maintenance personnel to adjust, operate, and maintain motorized shade operation systems.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 12 - FURNISHINGS

SECTION 12 36 00

WINDOW STOOLS

11/23

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 SUBMITTALS
1.4 SUBMITTAL INSTRUCTIONS
1.5 REFERENCES

PART 2 PRODUCTS

2.1 MATERIALS
2.2 COUNTERTOPS

PART 3 EXECUTION

3.1 INSTALLATION
3.2 PROTECTION AND CLEANING

-- End of Section Table of Contents --

SECTION 12 36 00
11/23

WINDOW STOOLS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies solid surfacing material window stools.

1.2 RELATED WORK

a. Color and patterns of solid surfacing material: See Schedule of Materials on the drawings.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 81 13.xx SUSTAINABLE CONSTRUCTION REQUIREMENTS. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-04 Samples

Samples; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Shop Drawings

(1) Show dimensions of section and method of assembly.

(2) Show details of construction at a scale of 1/2 inch to a foot.

c. Samples:

(1) 6 inch square samples each top.

1.5 REFERENCES

Publications listed below form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.

APA - THE ENGINEERED WOOD ASSOCIATION (APA)

APA PS 1	(2009) Structural Plywood (with Typical APA Trademarks)

ASTM INTERNATIONAL (ASTM)

ASTM D256	(2024) Standard Test Methods for Determining the Izod Pendulum Impact Resistance of Plastics
ASTM D570	(1998; E 2010; R 2010) Standard Test Method for Water Absorption of Plastics
ASTM D638	(2014) Standard Test Method for Tensile Properties of Plastics
ASTM D785	(2008; R 2015) Standard Test Method for Rockwell Hardness of Plastics and Electrical Insulating Materials
ASTM D4690	(2012; R 2018) Standard Specification for Urea-Formaldehyde Resin Adhesives
ASTM G21	(2015; R 2021; E 2021) Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3	(2005) Standard for High-Pressure Decorative Laminates

PART 2 PRODUCTS

Manufacturer information identified for interior finishes is provided as the basis of design and is not intended to restrict the use of equivalent products by other manufacturers. Proposed substitutions shall comply with the technical and performance requirements specified and are subject to owner approval. The contractor shall submit samples of all proposed finish materials for government review and approval. No finish materials shall be installed until written approval has been obtained from the government. Refer to the specifications for basis of design criteria and equivalent product requirements.

2.1 MATERIALS

a. Plywood: APA PS 1, Exterior type, veneer grade AC not less than five ply construction.

b. Adhesive

(1) For wood products: ASTM D4690, unextended urea resin or unextended melamine resin, phenol resin, or resorcinol resin.

(2) For Field Joints:

(a) Epoxy type, resistant to chemicals as specified for plastic laminate laboratory surfaces.

(b) Fungi resistant: ASTM G21, rating of 0.

c. Fasteners:

(1) Metals used for welding same metal as materials joined.

(2) Use studs, bolts, spaces, threaded rods with nuts or screws suitable for materials being joined with metal splice plates, channels or other supporting shape.

d. Solid Polymer Material:

(1) Filled Methyl Methacrylic Polymer.

(2) Performance properties required:

Property	Result	Test
Elongation	0.3% min.	ASTM D638
Hardness	90 Rockwell M	ASTM D785
Gloss (600 Gordon)	5-20	ANSI/NEMA LD 3
Color stability	No change	ANSI/NEMA LD 3 except 200 hour
Abrasion resistance	No loss of pattern Max wear depth 0.003 in - 10000 cycles	ANSI/NEMA LD 3
Water absorption weight (5 max)	24 hours 0.9	ASTM D570
Izod impact	0.25 ft-lb/in	ASTM D256 (Method A)
Impact resistance	No fracture	ANSI/NEMA LD 3 36" drop 2 lb. ball
Boiling water surface resistance	No visible change	ANSI/NEMA LD 3
High temperature resistance	Slight surface dulling	ANSI/NEMA LD 3

(3) Cast into sheet form and bowl form.

(4) Color throughout with subtle veining through thickness.

(5) Joint adhesive and sealer: Manufacturers silicone adhesive and sealant for joining methyl methacrylic polymer sheet.

(6) Bio-based products will be preferred.

2.2 COUNTERTOPS

a. Methyl Methacrylic Polymer Window Stools:

(1) Fabricate window stools of methyl methacrylic polymer cast sheet, 19-mm (3/4inch) thick.

(2) Fabricate window stools with eased edge at all exposed front edges. Finish all surfaces exposed to view, including bottom of sill overhang.

(3) Fabricate in one piece for full length from corner to corner up to 12 feet.

(4) Seal edges with adhesive sealant.

PART 3 EXECUTION

3.1 INSTALLATION

a. Before installing window stools verify that wall surfaces have been finished as specified.

b. Secure window stools to sill framing with metal fastening devices, or screws through pierced slots in rails.

(1) Where type, size or spacing of fastenings is not shown or specified, submit shop drawings showing proposed fastenings and method of installation.

(2) Use epoxy or silicone to fasten the epoxy resin countertops to the cabinets.

c. Scribe window stools to site conditions. do not allow more than 1/8" gap at all sides of window stool at final installation. Gap to be filled with 100% silicone sealant.

3.2 PROTECTION AND CLEANING

a. Tightly cover and protect against dirt, water, and chemical or mechanical injury.

b. Clean at completion of work.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 14 - CONVEYING EQUIPMENT

SECTION 14 21 00

TRACTION ELEVATOR - NEW

4/24

PART 1 GENERAL

1.1 DESCRIPTION
1.2 RELATED WORK
1.3 QUALIFICATIONS
1.4 REFERENCES
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS
1.7 WIRING DIAGRAMS
1.8 PERFORMANCE STANDARDS
1.9 WARRANTY
1.10 POWER SUPPLY
1.11 EMERGENCY POWER SUPPLY
1.12 ELEVATOR MACHINE ROOM AND MACHINE SPACE
1.13 HOISTWAY LIGHTING
1.14 TOOL CABINET

PART 2 PRODUCTS

2.1 MATERIALS
2.2 MANUFACTURED PRODUCTS
2.3 CONDUIT AND WIREWAY
2.4 CONDUCTORS
2.5 TRAVELING CABLES
2.6 CONTROLLER AND SUPERVISORY PANEL
2.7 MICROPROCESSOR CONTROL SYSTEM
2.8 MACHINE ROOM MONITOR
2.9 EMERGENCY POWER OPERATION
2.10 FIREFIGHTER’S SERVICE
2.11 INDEPENDENT SERVICE
2.12 SEISMIC REQUIREMENTS
2.13 ELEVATOR MACHINE BEAMS
2.14 TRACTION HOIST MACHINE
2.15 DEFLECTOR SHEAVES
2.16 HOIST ROPES
2.17 HOIST ROPE COMPENSATION
2.18 GOVERNOR ROPE
2.19 SPEED GOVERNOR
2.20 CAR SAFETY DEVICE
2.21 ASCENDING CAR OVERSPEED PROTECTION
2.22 CAR AND COUNTERWEIGHT BUFFERS
2.23 COUNTERWEIGHTS
2.24 GUIDE RAILS, SUPPORTS, AND FASTENINGS
2.25 NORMAL AND FINAL TERMINAL STOPPING DEVICES
2.26 CROSSHEAD DATA PLATE AND CODE DATA PLATE
2.27 WORKMAN'S LIGHTS AND OUTLETS

2.28 CARTOP OPERATING DEVICE
2.29 LEVELING DEVICE
2.30 EMERGENCY STOP SWITCHES
2.31 MAIN CAR OPERATING PANEL
2.32 CAR POSITION INDICATOR
2.33 AUDIO VOICE SYSTEM
2.34 AUTO DIAL TELEPHONE SYSTEM
2.35 CORRIDOR OPERATING DEVICES
2.36 DIGITAL CORRIDOR ARRIVAL LANTERN/POSITION INDICATOR
2.37 HOISTWAY ACCESS
2.38 HOISTWAY ENTRANCES: PASSENGER/SERVICE ELEVATORS
2.39 CAR AND COUNTERWEIGHT GUIDES
2.40 CAR FRAME: PASSENGER/SERVICE ELEVATORS
2.41 CAR PLATFORM: PASSENGER/SERVICE ELEVATORS
2.42 CAR ENCLOSURE: PASSENGER/SERVICE ELEVATORS
2.43 POWER DOOR OPERATORS: PASSENGER/SERVICE ELEVATORS

PART 3 EXECUTION

3.1 PREPARATION
3.2 ARRANGEMENT OF EQUIPMENT
3.3 WORKMANSHIP, INSTALLATION AND PROTECTION
3.4 CLEANING
3.5 PAINTING AND FINISHING
3.6 PRE-TESTS AND TESTS
3.7 INSTRUCTION OF VA PERSONNEL
3.8 GUARANTEE PERIOD OF SERVICE: MAINTENANCE SERVICE AND INSPECTIONS

-- End of Section Table of Contents --

SECTION 14 21 00
4/24

TRACTION ELEVATOR - NEW

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the engineering, furnishing and installation of the complete electric traction elevator system as described herein and as indicated on the Contract Drawings.

b. Items listed in the singular apply to each elevator in this specification except where noted.

d. Service/Passenger Elevator SE1 must be offset traction, microprocessor control system, AC VVVF motor control, power operated car and hoistway doors. Elevator must be Class “C3” loading rated as a minimum

ELEVATOR	SCHEDULE
Elevator Number	SE1
Overall Platform Size	6'-0" x 9'-5"
Clear Inside Platform	5'-4 1/2" x 8'-9"
Rated Load – lb	5000 lbs
Contract Speed - fpm	200 fpm
Total Travel – ft	67'-10"
Floors Served	G, 1, 2, 3, Roof
Number of Openings	4
Entrance Type & Size	4'-0" x 8'-0"
Type of Roping	1:1

1.2 RELATED WORK

a. Section 01 33 23 SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION (FAR 52.236-21) and, SPECIAL NOTES (VAAR 852.236-91), in GENERAL CONDITIONS.

b. Section 07 84 00, FIRESTOPPING: Sealing around penetrations to maintain the integrity of fire-rated construction.

c. Refer to SCHEDULE FOR FINISHES on drawings.

d. Section 13 05 41, SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS: Requirements for seismic restraint rof non-structural components.

e. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General electrical requirements that are common to more than one Section.

f. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW): Low Voltage power and lighting wiring.

g. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

h. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits for cables and wiring.

i. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Requirements for installing the over-current protective devices to ensure proper equipment and personnel protection.

j. Section 26 22 00, LOW-VOLTAGE TRANSFORMERS: Low voltage transformers.

k. Section 26 24 16, PANELBOARDS: Low voltage panelboards.

l. Section 26 43 13, TRANSIENT-VOLTAGE SURGE SUPPRESSION: Surge suppressors installed in panelboards.

m. Section 26 51 00, INTERIOR LIGHTING: Fixture and ballast type for interior lighting.

1.3 QUALIFICATIONS

a. Approval by the Contracting Officer (CO) is required for products and services of proposed manufacturers, suppliers, installers, and must be contingent upon submission of certificates by the Contractor stating the following:

(1) Elevator contractor is currently and regularly engaged in the installation of elevator equipment as one of his principal products.

(2) Elevator contractor must have five years of successful experience, trained supervisory personnel, and facilities to install elevator equipment specified herein.

(3) Elevator Mechanic (Installer) must have passed a Mechanic Examination approved by the U.S. Department of Labor and have technical qualifications of at least five years of experience in the elevator industry or 10,000 hours of field experience working in the elevator industry with technical update training. Apprentices must be actively pursuing Certified Elevator Mechanic status. Certification must be submitted for all workers employed in this capacity.

b. Welding at the project site must be performed by certified welders who have previously qualified by test as prescribed in American Welding Society Publications AWS D1.1/D1.1M to perform the type of work required. Welder Certificates must be submitted for all workers employed in this capacity. A welding or hot work permit is required for each day and must be obtained from the VAMC safety department. Request permit one day in advance.

c. Electrical work must be performed by a Licensed Master Electrician and Licensed Journeymen Electricians as requirements by NFPA 70. Certificates must be submitted for all workers employed in this capacity.

d. Approval will not be given to elevator contractors and manufacturers who have established on prior projects, either government, municipal, or commercial, a record of unsatisfactory elevator installations, have failed to complete awarded contracts within the Contract period, and do not have the requisite record of satisfactory performing elevator installations of similar type and magnitude.

e. Approval of Elevator Contractor’s equipment will be contingent on the equipment manufacturer providing factory training, engineering and technical support, including all manuals, wiring diagrams and tools necessary for adjusting, maintenance, repair, and testing of equipment to the VA for use by the VA’s designated Elevator Maintenance Service Provider.

f. Identifying an elevator maintenance service provider that must render services within one hour of receipt of notification, together with certification that the quantity and quality of replacement parts stock is enough to warranty continued operation of the elevator installation.

g. Equipment within a group of traction elevators must be the product of the same manufacturer.

h. The Contractor must provide and install safety devices that have been subjected to tests witnessed and certified by an independent professional testing laboratory that is not a subsidiary of the firm that manufactures supplies or installs the equipment.

1.4 REFERENCES

The publications listed below form a part of this specification. Elevator installation must meet the requirements of the latest editions published and adopted by the United States Department of Veterans Affairs (VA) on the date Contract is signed.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A17.1/CSA B44	(2022) Safety Code for Elevators and Escalators
ASME A17.2	(2023) Guide for Inspection of Elevators, Escalators, and Moving Walks

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A109/A109M	(2024) Standard Specification for Steel, Strip, Carbon (0.25 Maximum Percent), Cold-Rolled

ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM A666	(2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM A693	(2016; R 2022) Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2024) International Building Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA H-18-8	(2023) Seismic Design Requirements

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-59544	(2008; Notice 1 2013) Cable and Wire, Electrical (Power, Fixed Installation)
FS W-S-610	(1990) Splice Connectors
FS W-C-596F	(1995; Amd 1 1998; Amd 2 2001; Rev H 2014) Connector, Electrical, Power, General Specification for
FS RR-W-410	(2022; Rev J) Wire Rope and Strand

UL SOLUTIONS (UL)

UL 62	(2018) UL Standard for Safety Flexible Cords and Cables
UL 486A-486B	(2025) UL Standard for Safety Wire Connectors

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Samples;

Contract Qualifications And Certificates;

Method Of Fastening Panels To Steel Walls;

Welder Certificates;

SD-02 Shop Drawings

Shop Drawings;

SD-03 Product Data

Manufacturer Data;

Cut Sheets And Drawings;

Microprocessor Control System;

SD-05 Design Data

Complete Construction Drawings;

Complete Dimensioned Detail;

Dimensioned Drawings;

Mixing Of Manufactures;

Design Of Hall Pushbutton Fixtures;

Design And Location Of Access Switches;

SD-11 Closeout Submittals

Wiring Diagrams;

Warranty;

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Specification Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA and SAMPLES.

b. Before execution of work, furnish information to evidence full compliance with Contract requirements for proposed items. Such information must include, as required: Manufacturer's Name, Trade Names, Model or Catalog Number, Nameplate Data (size, capacity, and rating) and corresponding specification reference (Federal or project specification number and paragraph). All submitted drawings and related elevator material must be forwarded to the CO.

c. Shop Drawings:

(1) Complete scaled and dimensioned layout in plan and section view showing the arrangement of equipment and all details of each elevator unit specified including:

(a) Hoisting machines, controllers, power conversion devices,

governors, and all other components located in machine room.

(b) Car, counterweight, sheaves, supporting beams, guide rails, brackets, buffers, and size of car platform, car frame members, and other components located in hoistway.

(c) Rail bracket spacing and maximum vertical forces on guide rails in accordance with VA H-18-8 for Seismic Risk Zone 2 or greater.

(d) Reaction at points of support and buffer impact loads.

(e) Weight of principal parts.

(f) Top and bottom clearances and over travel of car and counterweight.

(g) Location of main line switch/shunt trip circuit breaker, switchboard panel, light switch, and feeder extension points in the machine room.

(2) Drawings of hoistway entrances and doors showing details of construction and method of fastening to the structural members of the building.

(a) If drywall construction is used to enclose hoistway, submit details of interface fastenings between entrance frames and drywall.

(b) Sill details including sill support.

d. Samples:

(1) One each of stainless steel, 3 in. x 5 in..

(2) One each of baked enamel, 3 in. x 5 in..

(3) One each of color floor covering.

(4) One each of protection pads, 3 in. x 5 in. if used.

(5) One each car and hoistway Braille plate sample.

(6) One each car and hall button sample.

(7) One each car and hall lantern/position indicator sample.

(8) One each wall and ceiling material finish sample.

(9) One each car lighting sample.

e. Manufacturer Data: Name of manufacturer, type or style designation, and applicable data of the following equipment must be shown on the elevator layouts:

(1) Hoisting Machine.

(2) Hoisting Machine Motor, HP and RPM ratings, Voltage, Starting and Full Load Ampere, and Number of Phases.

(3) Controller.

(4) Starters and Overload Current Protection Devices.

(5) Car and Counterweight Safety Devices; Type “B” safeties, Governor and Rope Gripper.

(6) Electric Door Operator; HP, RPM, Voltage, and Ampere rating of motor.

(7) Hoistway Door Interlocks.

(8) Car and Counterweight Buffers; maximum and minimum rated loads, maximum rated striking speed and stroke.

(9) Cab Ventilation Unit; HP rating and CFM rating.

(10) Hoist and Compensation Ropes, breaking strength, allowable working load, and actual working load.

f. Complete construction drawings of elevator car enclosure showing dimensioned details, fastenings to platform, car lighting, ventilation, ceiling framing, top exits and location of car equipment.

g. Complete dimensioned detail of vibration isolating foundations for traction hoisting machines.

h. Dimensioned drawings showing details of:

(1) All signal and operating fixtures.

(2) Car and counterweight roller guides.

(3) Hoistway door tracks, hangers, and sills.

(4) Door operator, infrared curtain units.

i. Cut sheets and drawings showing details of controllers and supervisory panels.

j. Furnish Contract Qualifications and Certificates as required under: Paragraph "QUALIFICATIONS".

1.7 WIRING DIAGRAMS

a. Provide three complete sets of paper and one electronic set field wiring and straight-line wiring diagrams showing all electrical circuits in the hoistway, machine room and fixtures. Install one set coated with an approved plastic sealer and mounted in the elevator machine room as directed by the Resident Engineer.

b. In the event field modifications are necessary during installation, diagrams must be revised to include all corrections made prior to and during the final inspection. Corrected diagrams must be delivered to the Resident Engineer/Contracting Officer within 30 days of final acceptance.

c. Provide the following information relating to the specific type of

microprocessor controls installed:

(1) Owner's information manual, containing job specific data on major components, maintenance, and adjustment.

(2) System logic description.

(3) Complete wiring diagrams needed for field troubleshooting, adjustment, repair and replacement of components. Diagrams must be base diagrams, containing all changes and additions made to the equipment during the design and construction period.

(4) Changes made during the construction and warranty period must be noted on the drawings in adequate time to have the finalized drawings reproduced for mounting in the machine room no later than six months prior to the expiration of the warranty period.

1.8 PERFORMANCE STANDARDS

a. The elevators must meet the highest standards of the industry and specifically the following:

(1) Contract speed is high speed in either direction of travel with rated capacity load in the elevator. Speed variation under all load conditions, regardless of direction of travel, must not vary more than 3 percent.

(2) The controlled rate of change of acceleration and retardation of the car must not exceed 0.1G per ft/s/s and the maximum acceleration and retardation must not exceed 0.2G per ft/s/s.

(3) Starting, stopping, and leveling must be smooth and comfortable without appreciable steps of acceleration and deceleration.

b. Passenger/ Service door operator must open the car door and hoistway door at 2.5 ft per second and close at 1 ft per second. Freight door operators must open and close at 1 ft per second.

c. Floor level stopping accuracy must be within .125 in. above or below the floor, regardless of load condition.

d. Noise and Vibration Isolation: All elevator equipment including their supports and fastenings to the building, must be mechanically and electrically isolated from the building structure to minimize objectionable noise and vibration transmission to car, building structure, or adjacent occupied areas of building.

e. Sound Isolation: Noise level relating to elevator equipment operation in machine room must not exceed 80 decibels. All decibel readings must be taken 3 ft off the floor and 3 ft from equipment.

f. Airborne Noise: Measured noise level of elevator equipment during operation must not exceed 50 decibels in elevator lobbies and 60 decibels inside car under any condition including door operation and car ventilation exhaust blower on its highest speed.

1.9 WARRANTY

a. Submit all labor and materials furnished regarding elevator system and

installation to terms of "Warranty of Construction" articles of FAR clause 52.246-21. The One-Year Warranty and Guarantee Period of Service must commence and run concurrent after final inspection, completion of performance test and upon acceptance of each elevator.

b. During warranty period if a device is not functioning properly in accordance with specification requirements, more maintenance than the Contract requires keeping device operational, device must be removed and a new device meeting all requirements must be installed as part of work until satisfactory operation of installation is obtained. Period of warranty must start anew for such parts from date of completion of each new installation performed, in accordance with foregoing requirements.

1.10 POWER SUPPLY

a. Power supply requirements in each machine room, see Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES, Electrical specifications and Electrical drawings.

b. Main Line Fused Disconnect Switch/Shunt Trip Circuit Breaker for each controller must be located inside the machine room at the strike jamb side of the machine room door and lockable in the “Off” position.

c. Provide Surge Suppressors to protect the elevator equipment.

1.11 EMERGENCY POWER SUPPLY

a. Emergency power supply, its starting means, transfer switch for transfer of elevator supply from normal to emergency power, two pair of conductors in a conduit from an auxiliary contact on the transfer switch (open or close contacts as required by Elevator Controller Manufacturer) to terminals in the group elevator controller and other related work must be provided by the Electrical Contractor.

b. Upon loss of normal power supply there must be a delay before transferring to emergency power of 10 seconds minimum to 45 seconds maximum, the delay must be accomplished through an adjustable timing device.

c. Prior to the return of normal power an adjustable timed circuit must be activated that will cause all elevators to remain at a floor if already there or stop and remain at the next floor if in flight. Actual transfer of power from emergency power to normal building power must take place after all elevators are stopped at a floor with doors open.

d. Car lighting circuits must be connected to the emergency power panel.

1.12 ELEVATOR MACHINE ROOM AND MACHINE SPACE

a. Provide a machine room that meets the requirements of ASME A17.1/CSA B44, NFPA 70 and ICC IBC. The elevator drive machine and elevator controller must be in the same code approved machine room.

b. Provide stairs and landing for access to the machine room. The landing must be large enough to accommodate full opening of the door plus 24 in.

c. Locate the shunt trip breaker/main line disconnect and machine room light switch on the lock side of the door inside the machine room.

d. Locate sprinkler pipes to provide 7 ft head clearance. Do not locate sprinkler heads, heat detectors, and smoke detectors directly over elevator equipment.

1.13 HOISTWAY LIGHTING

a. Provide lighting throughout the entire hoistway with 3-way switches at the top and bottom of the hoistway accessible from elevator hoistway entrance prior to entering the pit or stepping onto the car top.

b. Lighting must illuminate top of elevator cab when it is at the top floor and the pit when at the bottom floor.

1.14 TOOL CABINET

Provide a metal parts/tool cabinet, having two shelves and hinged doors. Cabinet must be 48 in. high, 30 in. wide and 18 in. deep.

PART 2 PRODUCTS

2.1 MATERIALS

a. Where stainless steel is specified, it must be corrosion resisting steel complying with ASTM A240/A240M, ASTM A666, and ASTM A693, Class 302 or 304, Condition A with Number 4 finish on exposed surfaces. Stainless steel must have the grain of belting in the direction of the longest dimension and surfaces must be smooth and without waves. During installation stainless-steel surfaces must be protected with suitable material.

b. Where cold rolled steel is specified, it must be low-carbon steel rolled to stretcher level standard flatness, complying with ASTM A109/A109M.

2.2 MANUFACTURED PRODUCTS

a. Materials, devices, and equipment furnished must be of current production by manufacturers regularly engaged in the manufacture of such items. The elevator equipment, including controllers, door operators, and supervisory system must be the product of manufacturers of established reputation, provided such items are capably engineered and produced under coordinated specifications to ensure compatibility with the total operating system.

b. Manufacturers of equipment assemblies which include components made by others must assume complete responsibility for the final assembled unit. Components must be compatible with each other and with the total assembly for the intended service.

c. Mixing of manufactures related to a single system or group of components must be identified in the submittals.

d. Key operated switches provided for this elevator installation must be provided with four (4) keys for each individual switch or lock. Provide different key tumblers for different switch and lock functions. Each key must have a tag bearing a stamped or etched legend

identifying its purpose.

2.3 CONDUIT AND WIREWAY

a. Provide new conduit and wireway. Install electrical conductors, except traveling cable, in rigid zinc-coated steel or aluminum conduit, electrical metallic tubing or metal wireways. Rigid conduit smaller than .75 in. or electrical metallic tubing smaller than .50 in. electrical trade size must not be used. All raceways completely embedded in concrete slabs, walls, or floor fill must be rigid steel conduit. Wireway (duct) must be installed in the hoistway and to the controller and between similar apparatus in the elevator machine room. Fully protect self-supporting connections, where approved, from abrasion or other mechanical injury. Flexible metal conduit not less than .375 in. electrical trade size may be used, not exceeding 18 in. in length unsupported, for short connections between risers and limit switches, interlocks, and for other applications permitted by NFPA 70.

b. All conduit terminating in steel cabinets, junction boxes, wireways, switch boxes, outlet boxes and similar locations must have approved insulation bushings. Install a steel lock nut under the bushings if they are constructed completely of insulating materials. Protect the conductors at ends of conduits not terminating in steel cabinets or boxes by terminal fittings having an insulated opening for the conductors.

c. Rigid conduit and EMT fittings using set screws or indentations as a means of attachment must not be used.

d. Connect motor or other items subject to movement, vibration or removal to the conduit or EMT systems with flexible, steel conduits.

2.4 CONDUCTORS

a. Conductors must be new stranded or solid coated annealed copper in accordance with CID A-A-59544 for Type RHW or THW. Where 16 and 18 AWG are permitted by NFPA 70, single conductors or multiple conductor cables in accordance with UL 62 for Type TF may be used provided the insulation of single conductor cable and outer jacket of multiple conductor cable is flame retardant and moisture resistant. Multiple conductor cable must have color or number coding for each conductor. Conductors for control boards must be in accordance with NFPA 70. Joints or splices are not permitted in wiring except at outlets. Tap connectors may be used in wireways provided they meet all UL requirements.

b. Provide new conduit and wiring between machine room, hoistway and fixtures.

c. All wiring must test free from short circuits or ground faults. Insulation resistance between individual external conductors and between conductors and ground must be a minimum of one megohm.

d. Where size of conductor is not given, voltage and amperes must not exceed limits set by NFPA 70.

e. Provide equipment grounding. Ground the conduits, supports, controller enclosure, motor, platform and car frame, and all other non-current conducting metal enclosures for electrical equipment in

accordance with NFPA 70. The ground wires must be copper, green insulated and sized as required by NFPA 70. Bond the grounding wires to all junction boxes, cabinets, and wire raceways.

f. Terminal connections for all conductors used for external wiring between various items of elevator equipment must be solderless pressure wire connectors in accordance with FS W-S-610. The Elevator Contractor may, at his option, make these terminal connections on #10 gauge or smaller conductors with approved terminal eyelets set on the conductor with a special setting tool, or with an approved pressure type terminal block. Terminal blocks using pierce-through serrated washers are not acceptable.

2.5 TRAVELING CABLES

a. All conductors to the car must consist of new flexible traveling cables conforming to the requirements of NFPA 70. Traveling cables must run from the junction box on the car directly to the controller in the machine room. Junction boxes on the car must be equipped with terminal blocks. Terminal blocks having pressure wire connectors of the clamp type that meet UL 486A-486B requirements for stranded wire may be used in lieu of terminal eyelet connections. Terminal blocks must have permanent indelible identifying numbers for each connection. Cables must be securely anchored to avoid strain on individual terminal connections. Flame and moisture resistant outer covering must remain intact between junction boxes. Abrupt bending, twisting and distortion of the cables must not be permitted.

b. Provide spare conductors equal to 10 percent of the total number of conductors furnished, but not less than five spare conductors in each traveling cable.

c. Provide shielded wires for the auto dial telephone system within the traveling cable, five pair shielded wires for card reader, Ethernet cables required for Wi-Fi and other systems, two pair 14-gauge wires for 110 Volt power supply, and wire for video display monitor if specified.

d. Traveling cables that contact the hoistway or elevator due to sway or change in position, provide shields or pads to the elevator and hoistway to prevent damage to the traveling cables.

e. Hardware cloth may be installed from the hoistway suspension point to the elevator pit to prevent traveling cables from rubbing or chafing and securely fastened and tensioned to prevent buckling. Hardware cloth is not required when traveling cable is hung against a flat wall.

2.6 CONTROLLER AND SUPERVISORY PANEL

a. UL/CSA Labeled Controller: Mount all assemblies, power supplies, chassis switches, and relays on a steel frame in a NEMA Type 1 General Purpose Enclosure. Cabinet must be securely attached to the building structure.

b. Properly identify each device on all panels by name, letter, or standard symbol which must be neatly stencil painted or decaled in an indelible and legible manner. Identification markings must be coordinated with identical markings used on wiring diagrams. The ampere rating must be marked adjacent to all fuse holders. All spare

conductors to controller and supervisory panel must be neatly formed, laced, and identified.

c. Controller must be provided with wiring and components for additional future travel if required.

2.7 MICROPROCESSOR CONTROL SYSTEM

a. Provide a new microprocessor control system with absolute position/speed feedback to control dispatching, signal functions, door operation, and VVVF Drive for hoist motor control. Complete details of the components and printed circuit boards, together with a complete operational description, must be submitted for approval. Add Regenerative Drive when economically advantages to the VA.

b. Controller manufacturer must provide factory training, engineering and technical support, including all manuals, wiring diagrams, and tools necessary for adjusting, maintenance, repair, and testing of equipment to the VA for use by the VA’s designated Elevator Maintenance Service Provider.

c. Microprocessor dispatching system must evaluate building traffic demand including number of elevators in service, hall calls, car calls, elevator position, direction of travel, load in elevator, door status and select an elevator to answer hall calls for least possible passenger wait times.

d. Provide means for weighing carload for each elevator. When load in a car reaches an adjustable predetermined level of the rated capacity, that car must bypass registered landing calls until the load in the car drops below the predetermined level. Calls bypassed in this manner must remain registered for the next car. The initial adjustment of the load weighting bypass setting must be 60 to 100 percent.

e. If weight in the car is not commensurate with the number of registered car calls, cancel car calls. Systems that employ either load weighing or door protective device for activation of this feature are acceptable.

2.8 MACHINE ROOM MONITOR

a. Provide a monitor in each machine room, separate monitors for each passenger elevator group, and each service elevator group. Provide a separate keyboard for each monitor.

b. The monitor must contain indicators to provide the following information:

(1) The floor where each elevator is currently located.

(2) The direction that each elevator is currently traveling or is scheduled to travel.

(3) The location and direction of currently registered hall calls.

(4) Elevators that are currently out of service.

(5) Elevators that are currently bypassing hall calls.

(6) Elevators that are currently engaged in passenger transfers.

(7) Operations program under which entire group is currently operating.

(8) Zone divisions of the entire group.

(9) Door positions.

(10) Status indication for elevators on independent service, car top inspection, fire service, earthquake protection, and activated stop switch and alarm bell.

c. The maintenance terminal must be suitable for all troubleshooting procedures related to the specific type or microprocessor installed on this project.

2.9 EMERGENCY POWER OPERATION

a. The control system for Elevator(s) must provide for the operation of at least one car per elevator group on emergency power upon failure of the normal power supply.

b. Auxiliary equipment on elevator controllers, wiring between associated elevator controllers and wiring between elevator controllers and remote selector panel as required to permit the elevators to operate as detailed, must be provided by the Elevator Contractor.

c. Upon loss of normal power supply there must be a delay before transferring to emergency power of 10 seconds minimum to 45 seconds maximum, the delay must be accomplished through an adjustable timing device. After adjusting delay, the associated elevators must function the same as on normal power :

d. Prior to the return of normal power an adjustable timer circuit must activate that will cause all cars to remain at a floor if already there or stop and remain at the next floor if in flight. Actual transfer of power from emergency power to normal building power must take place after all cars are stopped at a floor with their doors open.

2.10 FIREFIGHTER’S SERVICE

Provide Firefighter’s Service. Key switches must be barrel-key FEO-K1.

a. Main Floor:

b. Alternate Floor:

c. Verify main and alternate floors with RE/CO or authorized representative.

2.11 INDEPENDENT SERVICE

Provide an INDEPENDENT SERVICE key switch on the face of the main car operating panel that must have its positions marked "ON" and "OFF". When the switch is in the "ON" position, the car must respond only to calls

registered on its car dispatch buttons and must bypass all calls registered on landing push buttons. The car must start when a car call is registered, car call button or door close button is pressed, car and hoistway doors are closed, and interlock circuits are made. When switch is returned to "OFF" position, normal service must be resumed.

2.12 SEISMIC REQUIREMENTS

Meet the requirements of VA H-18-8.

2.13 ELEVATOR MACHINE BEAMS

Overhead beams must support machines and machinery in place to prevent movement under any conditions imposed in service.

2.14 TRACTION HOIST MACHINE

a. Provide geared/gearless traction machine with an AC motor, brake, drive sheave, and deflector sheave, mounted in proper alignment on an isolated bedplate.

b. Provide new hoist machine mounted direct drive, digital, closed-loop velocity encoder.

c. Drive sheaves must be free from cracks, sand holes, and other imperfections that would tend to injure the hoist ropes. Sheave must be turned smooth and true with rope grooves of proper design to insure maximum traction and maximum life of the hoist ropes.

d. Hoisting machine brake must be drum or disc type and must stop and hold the elevator with 125 percent of rated load.

2.15 DEFLECTOR SHEAVES

a. Provide new deflector sheaves with a metal guard mounted below the sheave and a guard to prevent ropes from jumping out of grooves. Securely fasten guard to sheave support beams.

b. Two-to-one idler sheaves on car and counterweight, if used, must be provided with metal guards that prevent foreign objects from falling between ropes and sheave grooves and to prevent ropes from jumping out of grooves.

c. Securely mount overhead sheaves on overhead beams in proper alignment with basement traction sheave, car and counterweight rope hitches or sheaves. Provide blocking beams where sheaves are installed on two or more levels.

2.16 HOIST ROPES

a. Provide elevator with the required number and size of ropes to ensure adequate traction and required safety factor. Hoisting ropes must be pre-formed 8x19 or 8x25 traction steel, conforming to FS RR-W-410 with minimum nominal diameter of .50 in..

b. Securely attach a corrosion resistant metal data tag to one hoisting rope fastening on top of the elevator.

2.17 HOIST ROPE COMPENSATION

a. Provide hoist rope compensation when required by elevator controller or drive machine manufacturer. Compensation must consist of a necessary number and size of encapsulated chains attached to the underside of car and counterweight frames.

(1) Provide guide(s) in pit to minimize chain sway.

(2) Provide take-up adjustment to compensate for hoist rope stretch.

(3) Pad areas where compensation may strike car or hoistway items.

2.18 GOVERNOR ROPE

a. New Governor Rope must be 6x19 or 8x19 wire rope, preformed traction steel, uncoated, fiber core, conforming to FS RR-W-410 with minimum nominal diameter of .375 in. having a minimum safety factor of 5. Tiller rope construction is not acceptable.

b. Under normal operation rope must run free and clear of governor jaws, rope guards, and other stationary parts.

c. Securely attach governor rope tag to governor rope releasing carrier.

2.19 SPEED GOVERNOR

a. Provide Centrifugal car driven governor to operate the car safety device . Governor must be complete with weighted pit tension sheave, governor release carrier and mounting base with protected cable sleeves.

b. Provide overspeed switch and speed reducing switches when required.

c. The governor rope clamping device must be designed to prevent appreciable damage to or deformation of the governor rope that results from the stopping action of the device operating the safety.

d. Provide metal guard over top of governor rope and sheaves.

e. Where the elevator travel does not exceed 100 feet, the weight tension sheave may be mounted on a pivoted steel arm in lieu of operating in steel guides.

2.20 CAR SAFETY DEVICE

Provide new “Type B Safeties” on the elevator.

2.21 ASCENDING CAR OVERSPEED PROTECTION

Provide a device, direct acting on the suspension means, to prevent ascending over speed and unintended motion away from the landing when the doors are not closed and locked.

2.22 CAR AND COUNTERWEIGHT BUFFERS

Provide buffer new buffers for each car and each counterweight. Securely fasten buffers and supports to the pit channels and in the alignment with striker plates on car and counterweight. Each installed buffer must have a

permanently attached metal plate indicating its stroke and load rating. Buffer anchorage must not puncture pit waterproofing.

2.23 COUNTERWEIGHTS

a. Elevator must be counterweighted with the weight of the car plus 40-50 percent of the rated capacity load as required by the controller manufacturer.

b. Furnish two tie rods with cotter pins and double nuts at top and bottom. Install counterweight retainer plates or other approved means on tie rods to prevent counterweight sub-weights from jumping and/or rattling. Both ends of tie-rods must be visible and accessible.

c. Provide new counterweight guard if required. If compensation chains are required remove counterweight guards.

2.24 GUIDE RAILS, SUPPORTS, AND FASTENINGS

a. Guide rails for car must be planed steel T-sections and weigh 18.5 lb/ft. Guide rails for counterweight must be planed steel T-sections and weigh15 lb/ft.

b. Securely fasten guide rails to the brackets or other supports by heavy duty steel rail clips.

c. Provide car and counterweight rail brackets and counterweight spreader brackets of sufficient size and design to secure substantial rigidity to prevent spreading or distortion of rails under any condition.

d. Guide rails must extend from channels on pit floor to within 3 in. of the underside of the concrete slab or grating at top of hoistway with a maximum deviation of .125 in. from plumb in all directions. Provide a minimum of .75 in. clearance between bottom of rails and top of pit channels.

e. Guide rail anchorages in pit must be made in a manner that will not reduce effectiveness of the pit waterproofing.

f. In the event inserts or bond blocks are required for the attachment of guide rails, the Contractor must furnish such inserts or bond blocks and must install them in the forms before the concrete is poured. Use inserts or bond blocks only in concrete or block work where steel framing is not available for support of guide rails. Expansion-type bolting for guide rail brackets will not be permitted.

g. Guide rails must be clean and free of any signs of rust, grease, or abrasion before final inspection. Paint the shank and base of the T-section with one field coat of manufacturer’s standard enamel.

2.25 NORMAL AND FINAL TERMINAL STOPPING DEVICES

a. Mount new terminal slowdown switches and direction limit switches on the elevator or in hoistway to reduce speed and bring car to an automatic stop at the terminal landings.

(1) Switches must function with load up to and including 125 percent of rated elevator capacity at contract speed during normal operation.

(2) Switches, when opened, must permit operation of elevator in reverse direction of travel.

b. Mount final terminal stopping switches in the hoistway.

(1) Switches must be positively opened should the car travel beyond the terminal direction limit switches.

(2) Switches must be independent of other stopping devices.

(3) Switches, when opened, must remove power from hoist motor, apply hoist machine brake and prevent operation of car in either direction.

2.26 CROSSHEAD DATA PLATE AND CODE DATA PLATE

a. Permanently attach a non-corrosive metal Data Plate to car crosshead.

b. Permanently attach a non-corrosive Code Data Plate to the controller.

2.27 WORKMAN'S LIGHTS AND OUTLETS

Provide new duplex GFCI protected receptacles and lamps with guards on top of each elevator car and beneath the platform. The receptacles must be in accordance with FS W-C-596F for Type D7, 2-pole, 3-wire grounded, rated for 15 amperes and 125 volts.

2.28 CARTOP OPERATING DEVICE

a. Provide a new cartop operating device.

b. The device must be activated by a toggle switch mounted in the device. The switch must be clearly marked "INSPECTION" and "NORMAL" on the faceplate, with .25 in. letters.

c. Movement of the elevator must be accomplished by the continuous pressure on a direction button and a safety button.

d. Provide an emergency stop switch, push to stop/pull to run.

e. Provide permanent identification for the operation of all components in the device.

f. The device must be permanently attached to the elevator crosshead on the side of the elevator nearest to the hoistway doors used for accessing the top of the car elevator.

2.29 LEVELING DEVICE

a. Elevator must be equipped with a new two-way leveling device to automatically bring the car to within .125 in. of exact level with the landing for which a stop is initiated regardless of load in car or direction of travel.

b. If the car stops short or travels beyond the floor, the leveling device, within its zone must automatically correct this condition and maintain the car within .125 in. of level with the floor landing regardless of the load carried.

2.30 EMERGENCY STOP SWITCHES

a. Provide an emergency stop switch, push to stop/pull to run, for each cartop device, pit, machine spaces, service panel and firefighter’s control panel inside the elevator. Mount stop switches in the pit adjacent to pit access door, at top of the pit ladder 48 in. above the bottom landing sill and 48 in. above the pit floor adjacent to the pit ladder.

b. Each stop switch must be red in color and must have "STOP" and "RUN" positions legibly and indelibly identified.

2.31 MAIN CAR OPERATING PANEL

a. Provide new main car operating panel in the car enclosure on the front return panel for passenger/service elevators. The top floor car call push button must not be more than 48 in. above the finished floor. "Vandal Resistant" car push buttons and indicator lights must be LED illuminated, round with a minimum diameter of 1 in.. Each button must contain an integral registration LED light which must illuminate upon registration of a call and must extinguish when that call is answered.

b. One-piece hinged front faceplate with edges beveled 15 degrees hinged must have the firefighter’s service panel recessed into the upper section and the service operation panel recessed into the lower section fitted with hinged doors. Doors must have concealed hinges, be in the same front plane as the faceplate and fitted with key operated locks. Secure the faceplate with stainless steel tamperproof screws.

c. All terminology and tactile symbols on the faceplate must be on square or rectangular plates recessed into the faceplate with its surface flush with the surface of the faceplate. Use .25 in. letters to identify all devices in the faceplate. The tactile symbols with contrasting background must be .50 in. high raised. .030 in. on the plate. Surface mounted plates are not acceptable.

d. The upper section must contain the following items in order listed from top to bottom:

(1) Elevator number, .50 in. high with black paint for contrast.

(2) Capacity plate information with black paint for contrast with freight loading class and number of passengers allowed.

(3) LED illuminated digital car position indicator with direction arrows.

(4) Emergency car lighting system consisting of a rechargeable battery, charger, controls and LED illuminated light fixture. The system must automatically provide emergency light in the car upon failure or interruption of the normal car lighting service, and function irrespective of the position of the light control switch in the car. The system must maintain a minimum illumination of 1.0 foot-candle when measured 48 in. above the car floor and approximately 12 in. in front of the car operating panel, for not less than four hours.

(5) Firefighter’s Emergency Operation Panel must be 66 in. minimum to 72 in. maximum to the top of the panel above finished floor.

(6) Firefighter’s Emergency Indicator Light must be round with a minimum diameter of 1 in..

(7) Medical Emergency key switch marked “MEDICAL EMERGENCY” with two positions labeled “ON” and “OFF” and Medical Emergency Indicator Light located next to the key switch must be round with a minimum diameter of 1 in.. Instruction for Medical Emergency operation must be engraved below the key switch and light.

(8) Key operated Independent Service Switch on the face of panel

(9) Provide a Door Hold Button on the faceplate above the call buttons. It must have “DOOR HOLD” indelibly marked on the button. Button must light when activated. When activated, the door must stay open for a maximum of one minute. To override hold timer, push a car call button or door close button.

(10) Provide round "Vandal Resistant" car push buttons and indicator lights, minimum diameter of 1 in. and LED illuminated, corresponding to the floors served. Car buttons must be legibly and indelibly identified by a floor number, letter, or symbol not less than .50 in. high in the face of the button or indicator.

(11) Door Open and Door Close buttons must be located below the car call buttons. They must have “OPEN” and “CLOSE” legibly and indelibly identified by letters in the face of the respective button. Door Open button must be located closest to the door jamb.

(12) Red Emergency Alarm button that must be located below the car operating buttons. Mount the emergency alarm button not lower than 35 in. above the finished floor. It must be connected to audible signaling devices. Provide audible signaling devices including the necessary wiring.

(13) Emergency Help push button must activate two-way communications by Auto Dial telephone system that is compatible with the VAMC’s telephone system. Help button must be LED illuminated and flash when call is acknowledged. Legibly and indelibly label the button “HELP” in the face of the button with .50 in. high letters.

e. The service operation panel in the lower section must contain the following items:

(1) Light switch labeled “LIGHTS” for controlling interior car lighting with its two positions marked “ON” and “OFF”.

(2) Inspection switch that will disconnect normal operation and activate hoistway access switches at terminal landings. Switch must be labeled “ACCESS ENABLE” with its two positions marked “ON” and “OFF”.

(3) Three-position switch labeled “FAN” with its positions marked “HIGH”, “LOW” and “OFF” for controlling car ventilating blower.

(4) Two-position, spring return, toggle switch or push button to test the emergency light and alarm device. It must be labeled “TEST EMERGENCY LIGHT AND ALARM”.

(5) Two-position toggle switch labeled "Independent Service" with its positions marked "ON" and “OFF”.

(6) Two-position emergency stop switch, when operated, must interrupt power supply and stop the elevator independently of regular operating devices. Emergency stop switch must be marked “PUSH TO STOP” and “PULL TO RUN”.

2.32 CAR POSITION INDICATOR

Provide an alpha-numeric digital car position indicator in the main car operating panel, consisting of numerals and arrows 2.5 in. high, to indicate position of car and direction of car travel. Locate position indicator at the top of the main car operating panel, illuminated by light emitting diodes.

2.33 AUDIO VOICE SYSTEM

Provide digitized audio voice system. Audio voice must announce floor designations, direction of travel, and special announcements. The voice announcement system must be a natural sounding human voice that receives messages and must comply with ADA requirements for audible car position indicators. The voice announcer must have two separate volume controls, one for the floor designations and direction of travel, and another for special announcements. The voice announcer must have a full range loudspeaker, located on top of the cab. The audio voice unit must contain the number of ports necessary to accommodate the number of floors, direction messages, and special announcements. Install voice announcer per manufacturer’s recommendations and instructions. The voice system must be the product of a manufacturer of established reputation. Provide manufacturer literature and list of voice messages.

a. Fire Service Message.

b. "Please do not block doors".

c. Provide special messages as directed by Contraction Officer.

2.34 AUTO DIAL TELEPHONE SYSTEM

a. Provide a complete, ADA compliant, auto dial communication system that is compatible with the VAMC's telephone system.

b. Two-way communication device must be in full compliance with ASME A17.1/CSA B44 Rule 2.27.1.1 (2019) or later editions.

2.35 CORRIDOR OPERATING DEVICES

a. Provide new corridor operating device faceplates must be .125 in. thick flat stainless steel with all edges beveled 15 degrees.

b. Corridor push button faceplates must be sized to accommodate corridor pictograph on faceplate. The centerline of the landing push buttons must be 42 in. above the corridor floor.

c. Elevator Corridor Call Station Pictograph must be engraved in the

faceplate.

d. Fasten corridor operating devices and signal device faceplates with stainless steel tamperproof screws.

e. All terminology and tactile symbols on the faceplate must be raised .030 inch with contrasting background, on square or rectangular plates recessed into the faceplate with its surface flush with the surface of the faceplate. The handicapped markings with contrasting background must be .5 in. high raised .030 in. on the plate, square or rectangular. Use .25 in. letters to identify all other devices in the faceplate. Surface mounted plates are not acceptable.

f. Provide one risers of landing call buttons for each elevator or group of elevators as shown on Contract Drawings.

g. "Vandal Resistant" hall buttons and indicator lights must be LED illuminated upon registration of a call and shall extinguish when that call is answered.

h. The direction of each button must be legibly and indelibly identified by arrows not less than .50 in. high in the face of each button. Provide a corresponding Braille plate on the left side of each button.

i. Hall call push buttons must not re-open the doors while the car and hoistway doors are closing at that floor, the call must be registered for the next available elevator. Calls registered must be canceled if closing doors are re-opened by means of "DOOR OPEN" button or infrared curtain unit.

j. Provide emergency power indicator light, medical emergency key switch and indicator light, fire service recall key switch and indicator light, firefighter's recall instruction, communication failure light, audible enunciator, and reset key switch in the fixture at the designated main floor.

k. Submit design of hall pushbutton fixtures for approval.

2.36 DIGITAL CORRIDOR ARRIVAL LANTERN/POSITION INDICATOR

a. Provide with new combination corridor arrival lantern/position indicator digital display mounted above hoistway entrances at each floor in healthcare facilities. For non-healthcare facilities provide combination fixtures only at main and alternate fire recall floors unless specified for all floors. Provide each terminal landing with "UP" or "DOWN", minimum 2.5 in. high digital arrow lanterns and each intermediate landing with "UP" and "DOWN" arrow lanterns. Each lens must be LED illuminated of proper intensity, shielded to illuminate individual lens only. The lenses in each lantern must be illuminated green to indicate "UP" travel and red to indicate "DOWN" travel. Lanterns must signal in advance of car arrival at the landing indicating the direction of travel. Corridor lanterns must not be illuminated when a car passes a floor without stopping. Each lantern must be equipped with an audible electronic chime which must sound once for "UPWARD" bound car and twice for "DOWNWARD" bound car. Audible signal must not sound when a car passes the floor without stopping. Provide adjustable sound level on audible signal. Car riding lanterns are not acceptable.

b. Install alpha-numeric digital position indicator between the arrival lanterns. Indicator faceplate must be stainless steel. Numerals must be not less than 2.5 in. high with direction arrows. Cover plates must be readily removable for re-lamping. The appropriate direction arrow must be illuminated during entire travel of car in corresponding direction.

2.37 HOISTWAY ACCESS

a. Provide new hoistway access switches for elevator at top terminal landing to permit access to top of car, and at bottom terminal landing to permit access to pit. Elevators with side slide doors, mount the access key switch 6 ft above the corridor floor in the wall next to the strike jamb.

b. Exposed portion of each access switch or its faceplate must have legible, indelible legends to indicate "UP", "DOWN" and "OFF" positions.

c. Each access switch must be a constant pressure cylinder type lock having not less than five pins or five stainless steel disc combination with key removable only when switch is in the "OFF" position.

d. Lock must not be operable by any other key which will operate any other lock or device used for any other purpose at the VA Medical Center.

e. Arrange the hoistway switch to initiate and maintain movement of the car. When the elevator is operated in the down direction from the top terminal landing, limit the zone of travel to a distance not greater than the top of the car crosshead level with the top floor. Submit design and location of access switches for approval.

f. Provide emergency access for all hoistway entrances, keyways for passenger and service elevators.

2.38 HOISTWAY ENTRANCES: PASSENGER/SERVICE ELEVATORS

a. Provide complete entrances with sills, sill supports, hangers, hanger supports, tracks, angle struts, unit frames, door panels, fascia plates, toe guards, hardware, bumpers, sight guards and wall anchors.

b. Provide one piece stainless steel sills grooved for door guides and recessed for fascia plates. Sills must have overall height of not less than .75 in. set true, straight and level, with hoistway edges plumb over each other, and top surfaces flush with finished floor. Hoistway entrance frames and sills must be grouted solid full length after installation.

c. Construct hanger supports of not less than .375 in. thick steel plate and bolted to strut angles.

d. Structural steel angles 3 in. x 3 in. x .375 in. must extend from top of sill to bottom of floor beam above and must be securely fastened at maximum 18 in. on center and at each end with two bolts.

e. Provide jambs and head soffits, of not less than 14-gauge stainless steel. Jambs and head soffits must be bolted/welded construction and

provided with three anchors each side. Side jambs must be curved. Radius of curvature must be 3.5 in.. Head jamb must be square and must overhang corridor face of side jambs by .25 in.. Rigidly fasten jambs and head soffits to building structure and grouted solid. After installation, protect jambs and head soffits to prevent damage to finish during construction.

f. Provide raised numerals or letters on cast, rear mounted plates for all openings. Numerals must be a minimum of 2 in. high, located on each side of entrance frame, with centerline of 5 ft above the landing sill. The number plates must contain Braille.

g. Provide unique car number on every elevator entrance at designated main and alternate fire service floor levels, minimum 3 in. in height.

h. Provide new stainless steel hoistway doors. Door panels must be flush hollow metal construction, not less than 1.25 in. thick, consisting of one continuous piece 16-gauge stainless steel on corridor side wrapped around the leading edge. Separate two plates by a sound-deadening material and reinforce by steel shapes welded to the plates at frequent intervals. Reinforce panels as required for installation of hangers, power-operating and door-opening devices. Top and bottom of door panels must have continuous stiffener channels welded in place. Reinforcement of the door panels must be a minimum of 0.04 in. in thickness and of the hat section type.

i. Hang doors on two-point suspension hangers having sealed ball-bearing sheaves not less than 3 in. in diameter, with non-metallic sound-reducing tires. Equip hangers with adjustable ball-bearing rollers to take upward thrust of panels. Provide two removable non-metallic door gibs of approved material and a separate fire gib at the bottom of each door panel. Gibs must be replaceable without removing of door panel. One door panel for each entrance must bear a BOCA label, Underwriters' label, or other labels may be furnished provided they are based on fire test reports and factory inspection procedures acceptable to the Contraction Officer. Fasten stainless steel sight guard of 14-gauge metal, extending full height of panel, to leading edge of fast speed panel of two-speed doors.

j. Door panels for each entrance must bear a BOCA label, Underwriters' label or labels from other accredited test laboratories may be furnished provided they are based on fire test reports and factory inspection procedures acceptable to the RE/CO or authorized representative.

k. Provide 14-gauge sheet steel fascia plates in hoistway to extend vertically from head of hanger support housing to sill above. Plates must be 3 in. wider than door opening of elevator and reinforced to prevent waves and buckles. Below bottom terminal landing and over upper terminal landing provide shear guards beveled back to and fastened to the wall.

l. Equip each hoistway entrance with new electrical/mechanical interlock, functioning as hoistway unit system, to prevent operation of elevator until all hositway doors are locked in the closed position unless car is operating in the leveling zone or hoistway access switch is used.

m. Wiring installed from the hoistway riser to each door interlock must be NFPA 70 type SF-2 or equivalent.

2.39 CAR AND COUNTERWEIGHT GUIDES

a. Provide new adjustable roller guides each assembled on a substantial metal base, to permit individual alignment to the guide rails.

b. Each guide must consisting of not less than three wheels, each with durable, resilient oil-resistant material tires rotating on ball bearings having sealed-in lubrication. Assemble rollers on a substantial metal base and mount to provide continuous spring pressure contact of all wheels with the corresponding rail surfaces under all conditions of loading and operation. Secure the roller guides at top and bottom on each side of car frame and counterweight frame. All mounting bolts must be fitted with nuts, flat washers, split lock washers and if required, beveled washers.

c. Provide sheet metal guards to protect rollers on top of car and counterweight.

d. Minimum diameter of car rollers must be 6 in.. The entire elevator car must be properly balanced to equalize pressure on all guide rollers. Cars must be balanced in post-wise and front-to-back directions. Test for this balanced condition must be witnessed at time of final inspection.

e. Minimum diameter of counterweight rollers must not be less than 4 in.. Properly balance counterweight frame to equalize pressure on all guide rollers. The Contractor must have the option of furnishing, for counterweight only, mechanically adjusted roller guide in lieu of spring-loaded roller guides as specified.

f. Equip car and counterweight with an auxiliary guiding device for each guide that must prevent the car or counterweight from leaving the rails if the normal guides fail. These auxiliary guides must not, during normal operation, touch the guiding surfaces of the rails. The auxiliary guides may be an extension of the normal guide shoe mounting plate if that plate is fabricated from hot rolled steel. The portion of the auxiliary guide that contacts the rail surfaces in the event of loss of the normal guides must be lined with an approved bearing material to minimize damage to the rail guiding surfaces.

2.40 CAR FRAME: PASSENGER/SERVICE ELEVATORS

Car frame must be constructed of channel stiles, crosshead, gussets, braces and cable hitch plate securely bolted and/or welded. The entire assembly must be constructed to withstand unequal loading of platform. Car frame members must be constructed to relieve the car enclosure of all strains.

2.41 CAR PLATFORM: PASSENGER/SERVICE ELEVATORS

a. The platform must be designed to withstand the forces developed under loading class specified. Provide car entrance with stainless-steel sill with machined or extruded door guide grooves and recessed for platform guard. Cover underside and all exposed edges of wood filled platform with sheet metal of not less than 26-gauge, with all exposed joints and edges folded under. Fire resistant paint is not acceptable. Platform must have flexible composition flooring not less than .125 in. thick. Adhesive material must be type recommended by

manufacturer of flooring. Lay flooring flush with threshold plate and base.

b. Provide a platform guard (toe guard) of not less than 12-gauge sheet-steel on the entrance side, extend 3 in. beyond each side of entrance jamb. Securely brace platform guard to car platform and bevel bottom edge at a 60-75-degree angle from horizontal. Square platform sill with the hoistway sills to provide clearance that must not exceed 1.25 in.

c. Provide ne isolation pads between the platform and platform frame. Use approved rubber pads or other approved material.

d. Provide adjustable diagonal brace rods to hold platform firmly within car suspension frame.

e. Balance car front to back and side to side. Provide balancing frame and weights, properly located, to achieve the required true balance.

f. Provide a bonding wire between frame and platform.

2.42 CAR ENCLOSURE: PASSENGER/SERVICE ELEVATORS

a. Car enclosure must have a minimum dome height inside the cab of 8 ft.

b. Securely fasten car enclosure to platform by through bolts located at intervals of not more than 18 in. running through an angle at the base of panels to underside of platform.

c. Front return wall panel, entrance columns, entrance head-jamb and transom must be 14-gauge stainless steel. Transom must be full width of cab. Side and rear walls must be constructed of 14-gauge cold rolled steel. Coat exterior of walls with mastic sound insulation material approximately .10 in. thick followed by a prime coat of paint.

d. Side and rear walls of passenger elevators may have raised panels covered in fire rated materials approved for use in elevator interior.

e. Side and rear walls of service elevators, up to the center line of the top handrail, must be covered with stainless steel. Side and rear walls to the ceiling must be covered with stainless steel applied directly to the cab walls or raised panels. Submit a method of fastening panels to steel walls.

f. Construct canopy of not less than 12-gauge steel.

g. Provide car top railings.

h. Provide a hinged top emergency exit cover. Exit must be unobstructed when open and must have mechanical stops on the cover. Provide an exit switch to prevent operation of the elevator when the emergency exit is open.

i. Provide duplex, GFCI protected receptacle in car. Locate flush-mounted receptacle on the centerline of the main car operating panel, 6 in. above the car floor.

j. Lighting for passenger/service elevators:

(1) Provide stainless steel hanging ceiling frame. Construct frame of .125 in. thick x 1.50 in. wide x 1.50 in. high “T” and “L” sections, divide ceiling into six panels.

(2) Provide LED illuminated car light fixtures above the ceiling panels. Maintain a minimum light level of 50-foot candles at 36 in. above the finished floor.

k. Optional lighting for service elevators:

(1) Provide car with indirect LED lamps mounted front to rear in lighting coves along each side of the cab ceiling, no hanging ceiling.

(2) Equip the lighting cove with asymmetrical reflectors having specular finish. Maintain a minimum light level of 50-foot candles 36 in. above finished floor at the car operating panels.

(3) Enclose the entire vertical space between the light trough outer edge and the cab canopy with approved opaque white or clear lumicite sheeting. Lumicite sheeting must be removable for cleaning and re-lamping.

l. Provide a blower unit arranged to exhaust through an opening in the canopy. Provide a stainless or chrome plated fan grill on the interior side of the opening. Provide screening over intake and exhaust end of blower. Provide 2-speed fan, with rated air displacement of 250 cfm and 400 cfm at respective speeds. Mount fan on top of car with rubber isolation to prevent transmission of vibration to car structure. Provide a 3-position switch to control the unit in the service panel.

m. Provide car enclosure with two sets of handrails with centerlines 30 in. and 42 in. above the car floor, circumference not to exceed 6 in.

(1) Locate handrails 1.50 in. from cab wall. Install handrails on side and rear walls . Conceal all handrail fastenings. Handrails must be removable from inside the car enclosure.

(2) Provide service elevators with flat stock handrails with the ends at the entrance turned back to the wall.

n. Provide passenger car with center opening horizontal sliding doors and service car with center or side opening horizontal sliding doors constructed the same as hoistway doors. Door panels must be not less than 16-gauge sheet steel and not less than 1.25 in. thick flush type construction with the car side and leading edge wrapped in one continuous piece of 16-gauge stainless steel. Top and bottom of door panels must have continuous stiffener channels welded in place. Reinforcement of the door panels must be approximately 0.04 in.in thickness and of the hat section type. Reinforce each door panel for hangers, interlock mechanism, opening and closing devices. Do not use hangers that are constructed integrally with the door panels.

o. Hang doors on two-point suspension hangers having sealed ball-bearing sheaves not less than 3 in. in diameter, with non-metallic sound-reducing tires. Equip hangers with adjustable ball-bearing rollers to take upward thrust of panels. Provide two non-metallic door gibs on each door panel and a separate fire gib. Gibs must be

replaceable without removing of door panel. One door panel for each entrance must bear a BOCA label, Underwriters' label, or other labels may be furnished provided they are based on fire test reports and factory inspection procedures acceptable to the Contracting Officer. Fasten stainless steel sight guard of 14-gauge metal, extending full height of panel, to leading edge of fast speed panel of two-speed doors.

P. Provide new hardware, tracks, gibs, separate fire gibs, gate switch, door drive clutch assemblies and door restrictor.

q. Provide one set of protective pads for service elevator of sufficient length to completely cover two sides, rear walls and front return of cab interior. Pads must consist of a minimum of .25 in. thick glass fiber insulation securely sewn between flame resistant vinyl coated coverings. Color of the covering must be approved by the RE/CO or their representative. Provide stainless steel pad buttons or hooks, spaced at intervals of not more than 18 in. to adequately support pads.

2.43 POWER DOOR OPERATORS: PASSENGER/SERVICE ELEVATORS

a. Provide a new high-speed heavy-duty door operator to automatically open the car and hoistway doors simultaneously when the car is level with the floor and automatically close the doors simultaneously at the expiration of the door-open time. Provide microprocessor door control with circuitry to constantly monitor and automatically adjust door operation based upon velocity, position, and motor current. Motor must be of the high-internal resistance type, capable of withstanding high currents resulting from stall without damage to door operator or motor. The door operator must open the car door and hoistway door simultaneously, at a speed of 2.5 ft per second. Closing speed of the doors must be 1 ft per second. Reversal of direction of the doors from the closing to opening operation, whether initiated by obstruction of the infrared curtain or the door "OPEN" button, must be accomplished within 1.50 in. maximum of door movement. Emphasis is placed on obtaining quiet interlock and door operation; smooth, fast, dynamic braking for door reversals, and stopping of the doors at extremes of travel.

b. Equip car doors with electric contact that prevents operation of car until doors are closed unless car is operating in leveling zone or hoistway access switch is used. Locate door contact to prevent its being tampered with from inside of car.

c. Car and hoistway doors must be manually operable in an emergency without disconnecting the power door operating equipment unless the car is outside the unlocking zone.

(1) Doors must not open by power unless the elevator is within the leveling zone.

(2) Provide infrared curtain unit. The device must cause the car and hoistway doors to reverse automatically to the fully open position should the unit be actuated while the doors are closing. Unit must function when the doors are not closed, except during firefighter’s operation.

d. Should the doors be prevented from closing for more than a predetermined adjustable interval of 15 to 30 seconds by operation of

the curtain unit, the doors must stay open, the audio voice message and a buzzer located on the car must sound only on automatic operation. **Do not provide door nudging.**

(1) If an obstruction of the doors does not activate the photo-electric door control device and prevent the doors from closing for more than a predetermined adjustable interval of 15 to 30 seconds, the doors must reverse to the fully open position and remain open until the “Door Close” button re-establishes the closing cycle.

e. Provide door "OPEN" and "CLOSE" buttons. When the door "OPEN" button is pressed and held, the doors, if in the open position, must remain open and if the doors are closing, they must stop, reverse and re-open. Momentary pressure of the door "CLOSE" button must initiate the closing of the doors prior to the expiration of the normal door open time.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine work of other trades on which the work of this Specification depends. Report defects to the RE/CO or authorized representative in writing that may affect the work of the elevator contractor.

b. Examine elevator hoistway openings for plumb, level, in line, and that elevator pit is proper size, waterproofed and drained with necessary access door and ladder.

c. Examine machine room for proper illumination, heating, ventilation, electrical equipment and beams are correctly located complete with access stairs and door.

d. If the Elevator Contractor requires changes in size or location of trolley beams, beam supports and trap doors, etc., to accomplish their work, changes must be justified, subject to approval of the CO, and include additional cost in their bid.

e. Work required prior to the completion of the elevator installation:

(1) Supply of electric feeder wires to the terminals of the elevator control panel, including circuit breaker.

(2) Provide light and GFCI outlets in the elevator pit and machine room.

(3) Furnish electric power for testing and adjusting elevator equipment.

(4) Furnish circuit breaker panel in machine room for car and hoistway lights and receptacles.

(5) Supply power for cab lighting and ventilation from an emergency power panel specified in Division 26, ELECTRICAL.

(6) Machine room enclosed and protected from moisture, with self-closing, self-locking door and access stairs.

(7) Provide fire extinguisher in machine room.

f. Provide to General Contractor for installation; inserts, anchors, bearing plates, brackets, supports and bracing including all setting templates and diagrams for placement.

3.2 ARRANGEMENT OF EQUIPMENT

Arrange equipment in machine room so that major equipment components can be removed for repair or replacement without dismantling or removing other equipment in the same machine room. Locate controller near and visible to its respective hoisting machine.

3.3 WORKMANSHIP, INSTALLATION AND PROTECTION

a. Installations must be performed by Certified Elevator Mechanics and Apprentices to best possible industry standards. Details of the installation must be mechanically and electrically correct. Materials and equipment must be new and without imperfections.

b. Recesses, cutouts, slots, holes, patching, grouting, refinishing to accommodate installation of equipment must be included in the Contractor's work. All new holes in concrete must be core drilled.

c. Structural members must not be cut or altered. Work in place that is damaged or defaced must be restored equal to original new condition.

d. Finished work must be straight, plumb, level and square with smooth surfaces and lines. All machinery and equipment must be protected against dirt, water, or mechanical injury. At completion, all work must be thoroughly cleaned and delivered in perfect unblemished condition.

e. Sleeves for conduit and other small holes must project 2 in. above concrete slabs.

f. Hoist cables that are exposed to accidental contact in the machine room and pit must be completely enclosed with 16-gauge sheet metal or expanded metal guards.

g. Exposed gears, sprockets, and sheaves must be guarded from accidental contact.

3.4 CLEANING

a. After completion of the installation and prior to final inspection, all equipment must be thoroughly cleaned of grease, oil, cement, plaster, dust and other debris.

b. Clean machine room and equipment.

c. Perform hoistway clean down.

d. Prior to final acceptance remove protective coverings from finished or ornamental surfaces. Clean and polish as required for type of material.

3.5 PAINTING AND FINISHING

a. All equipment, except specified as architectural finish, must be

painted one coat of approved color, conforming to manufacturer's standard.

b. Hoist machine and motor must be painted manufacturer's standard finish and color.

c. Controller, sheave, car frame and platform, counterweight, beams, rails and buffers except their machined surfaces, cams, brackets and all other uncoated ferrous metal items must be painted one factory primer coat or approved equal.

d. Stencil or apply decal floor designations not less than 4 in. high on hoistway doors, fascia or walls within door restrictor areas. The color of paint used must contrast with the color of the surfaces it is applied.

e. Elevator hoisting machine, controller, governor, main line switch/shunt trip circuit breaker, safety plank, and cross head of car must be identified by 4 in. high numerals and letters located as directed. Numerals must contrast with surrounding color and must be stenciled or decaled.

f. Hoistway Entrances of Passenger, and Service Elevators:

(1) Door panels, except stainless steel surfaces, must be given rust resistant treatment and a factory finish of one coat of baked-on primer and one factory finish coat of baked-on enamel.

(2) Fascia plates, top and bottom shear guards, dust covers, hanger covers, and other metalwork, including built-in or hidden work and structural metal, except stainless steel entrance frames and surfaces to receive baked enamel finish must be given one approved prime coat in the shop, and one field coat of paint of approved color.

g. Elevator Cabs for Passenger and Service Elevators:

(1) Interior and exterior steel surfaces must be given rust resistant treatment before finish is applied.

(2) Interior steel surfaces must be factory finished with one coat of paint of approved color.

(3) Give exterior faces of car doors one finish coat of paint of approved color.

h. Hoistway Entrances of Freight Elevators:

(1) Metal surfaces of doors and frames must receive shop prime coat.

(2) Finish painting, after installation, must be one coat of paint of approved color.

i. Elevator Cabs for Freight Elevators:

(1) Give interior of cab one prime coat and a minimum of one coat of paint of approved color.

(2) Give exterior of cab one prime coat and one finish coat of paint

of approved color.

(3) All surfaces of door frames, door panels, and cab interior surfaces that become damaged or marred must be restored to original condition before final acceptance of work.

3.6 PRE-TESTS AND TESTS

a. Pre-test the elevators and related equipment in the presence of the Resident Engineer/Contracting Officer or authorized representative for proper operation before requesting final inspection. Conduct final inspection at other than normal working hours, if required by RE/CO.

(1) The VA must obtain the services of an Independent Qualified Elevator Inspector, ANSI/ASME QEI-1 Certified, that must witness all tests. The QEI must utilize an Elevator Acceptance Inspection Form to record the results of inspection and all testing and to identify safety code and Contract deficiencies. Specific values must be provided for all tests required by ASME A17.1/CSA B44, ASME A17.2, and Contract Documents. After completion of inspection and testing, the QEI must sign a copy of the completed form and provide it to the CO. Within 14 days of the inspection, the QEI must also prepare a formal inspection report, including all test results and deficiencies. After successful completion of inspection and testing, the QEI must complete, sign and provide a certificate of compliance.

(2) Government must furnish electric power including necessary current for starting, testing, and operating machinery of each elevator.

(3) Contractor must furnish the following test instruments and materials on-site and at the designated time of inspection: properly marked certified test weights, voltmeter, amp probe, thermometers, direct reading tachometer, megohm meter, vibration meter, sound meter, light meter, stopwatch, and a means of two-way communication.

b. Inspect workmanship, equipment furnished, and installation for compliance with specification.

c. Balance Tests: The percent of counterbalance must be checked by placing test weights in car until the car and counterweight are equal in weight when located at the mid-point of travel. If the actual percent of counterbalance does not conform to the specification, the amount of counterweight must be adjusted until conformance is reached.

d. Full-Load Run Test: Elevator must be tested for a period of one-hour continuous run with full Contract load in the car. The test run must consist of the elevator stopping at every floor, in either direction of travel, for not less than five or more than ten seconds per floor.

e. Speed Test: The actual speed of the elevator must be determined in both directions of travel with full Contract load and no load in the elevator. Speed must be determined by applying a certified tachometer to the car hoisting ropes or governor rope. The actual measured speed of the elevator with all loads in either direction must be within 3 percent of specified rated speed. Full speed runs must be quiet and free from vibration and sway.

f. Temperature Rise Test: The temperature rise of the hoist motor must be determined during the full load test run. Temperatures must be measured using thermometers. Under these conditions, the temperature rise of the equipment must not exceed 50 degrees Centigrade above ambient temperature. Test must start when all machine room equipment is within 5 degrees Centigrade of the ambient temperature. Other tests for heat run on motors must be performed as prescribed by the Institute of Electrical and Electronic Engineers.

g. Car Leveling Test: Elevator car leveling devices must be tested for accuracy of leveling at all floors with no load in car and with Contract load in car, in both directions of travel. Accuracy of floor level must be within plus or minus .125 in. of level with landing floor for which the stop has been initiated regardless of load in car or direction of travel. The car leveling device must automatically correct over travel as well as under travel and must maintain the car floor within plus or minus .125 in. of level with the landing floor regardless of change in load.

h. Brake Test: The action of the brake must be prompt and a smooth stop must result in the up and down directions of travel with no load and rated load in the elevator. Down stopping must be tested with 125 percent of rated load in the elevator.

i. Insulation Resistance Test: The elevator's complete wiring system must be free from short circuits and ground faults and the insulation resistance of the system must be determined by use of megohm meter, at the discretion of the Elevator Inspector conducting the test.

j. Safety Devices: Car and counterweight safety devices must be tested.

k. Overload Devices: Test all overload current protection devices in the system at final inspection.

l. Limit Stops:

(1) The position of the car when stopped by each of the normal limit switches with no load and with Contract load in the car must be accurately measured.

(2) Final position of the elevator relative to the terminal landings must be determined when the elevator has been stopped by the final limits. The lower limit stop must be made with Contract load in the elevator. Elevator must be operated at inspection speed for both tests. Normal limit stopping devices must be inoperative for the tests.

m. Oil Buffer Tests: These tests must be conducted with operating device and limit stops inoperative and with Contract load in the elevator for the car buffer and with no load in the elevator for the counterweight buffer. Preliminary test must be made at the lowest (leveling) speed. Final tests must be conducted at Contract speed. Buffers must compress and return to the fully extended position without oil leakage.

n. Operating and Signal System: The elevator must be operated by the operating devices provided and the operation signals and automatic floor leveling must function in accordance with requirements specified. Starting, stopping and leveling must be smooth and

comfortable without appreciable steps of acceleration or deceleration.

o. Performance of the Elevator supervisory system must be witnessed and approved by the elevator inspector and a representative of the Resident Engineer.

p. Evidence of malfunction in any tested system or parts of equipment that occurs during the testing must be corrected, repaired, or replaced at no additional cost to the Government and the test repeated.

q. If equipment fails test requirements and a re-inspection is required, the Contractor must be responsible for the cost of re-inspection; salaries, transportation expenses, and per-diem expenses incurred by the elevator inspector and representative of the Resident Engineer.

3.7 INSTRUCTION OF VA PERSONNEL

a. Provide competent instruction to VA personnel regarding the operation of equipment and accessories installed under this Contract, for a period equal to one 8-hour day. Instruction must commence after completion of all work and at the time and place directed by the Resident Engineer/Contracting Officer or authorized representative.

b. Written instructions in triplicate relative to care, adjustments and operation of all equipment and accessories must be furnished and delivered to the RE/CO or authorized representative in independently bound folders. DVD recordings will also be acceptable. Written instructions must include correct and legible wiring diagrams, nomenclature sheet of all electrical apparatus including location of each device, complete and comprehensive sequence of operation, complete replacement parts list with descriptive literature, and identification and diagrams of equipment and parts. Information must also include electrical operation characteristics of all circuits, relays, timers, electronic devices and related characteristics for all rotating equipment.

c. Provide supplementary instruction for any new equipment that may become necessary because of changes, modifications or replacement of equipment or operation under requirements of paragraph entitled "Warranty of Construction”.

3.8 GUARANTEE PERIOD OF SERVICE: MAINTENANCE SERVICE AND INSPECTIONS

a. Provide complete maintenance service and inspections on each elevator installation of one year after completion and acceptance of each elevator in this specification by the Resident Engineer or Contracting Officer. This maintenance service must run concurrently with the warranty. Maintenance work must be performed by Certified Elevator Mechanics and Apprentices.

b. This Contract covers full maintenance including emergency call back service, inspections, and servicing the elevators listed in the schedule of elevators. The Elevator Contractor must perform the following:

(1) Bi-weekly systematic examination of equipment.

(2) During each maintenance visit the Elevator Contractor must clean, lubricate, adjust, repair and replace all parts as necessary to

keep the equipment in like new condition and proper working order.

(3) Furnishing all lubricant, cleaning materials, parts and tools necessary to perform the work required. Lubricants must be only those products recommended by the manufacturer of the equipment.

(4) Equalizing tension, shorten or renew hoisting ropes.

(5) As required, motors, controllers, selectors, leveling devices, operating devices, switches on cars and in hoistways, hoistway doors and car doors or gate operating device, interlock contacts, car and counterweight guides, car door sills, hangers for doors, car doors or gates, signal system, car safety device, governors, tension sheaves and buffers must be cleaned, lubricated and adjusted.

(6) Guide rails, overhead sheaves and beams, counterweight frames, and bottom of platforms must be cleaned every three months. Car tops and machine room floors must be cleaned monthly. Accumulated rubbish must be removed from the pits monthly. A general cleaning of the entire installation including all machine room equipment and hoistway equipment must be accomplished quarterly. Cleaning supplies and vacuum cleaner must be furnished by the Contractor.

(7) Maintain the performance standards set forth in this specification.

(8) The operational system must be maintained to the standards specified hereinafter including any changes or adjustments required to meet varying conditions of hospital occupancy.

(9) Maintain smooth starting, stopping, running and accurate leveling.

c. Maintenance service must not include the performance of work required because of improper use, accidents and negligence for which the Elevator Contractor is not directly responsible.

d. Provide 24-hour emergency call-back service that must consist of promptly responding to calls within two hours for emergency service should a shutdown or emergency develop between regular examinations. Overtime emergency call-back service must be limited to minor adjustments and repairs required to protect the immediate safety of persons and equipment in and about the elevator.

e. Service and emergency personnel must report to the Resident Engineer, Contracting Officer or authorized representative upon arrival at the hospital and again upon completion of the required work. A copy of the work ticket containing a complete description of the work performed must be given to the RE/CO or authorized representative.

f. The Elevator Contractor must maintain a logbook in the machine room. The log must list the date and time of all weekly examinations and all trouble calls. Each trouble call must be fully described including the nature of the call, necessary correction performed, or parts replaced.

g. Written “Maintenance Control Program” must be in place to maintain the equipment in compliance with ASME A17.1/CSA B44 section 8.6.

-- End of Section --